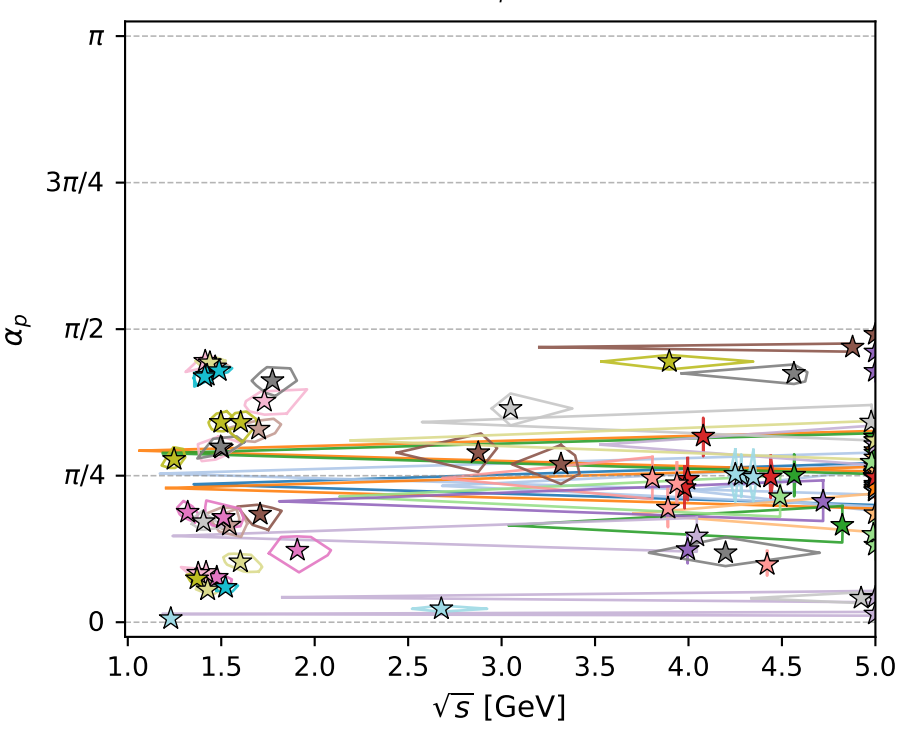
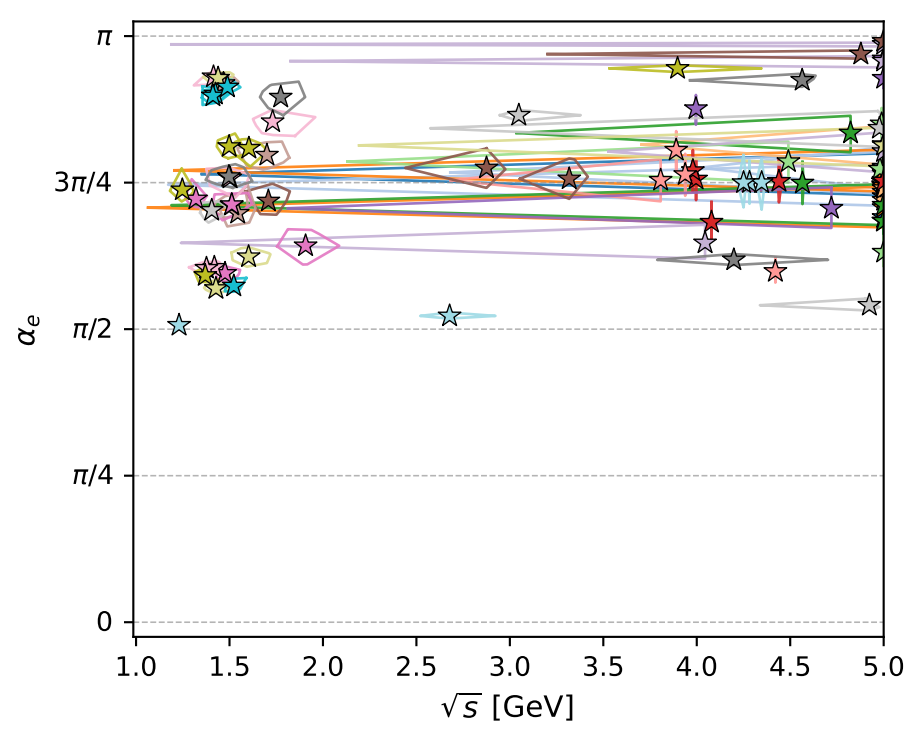
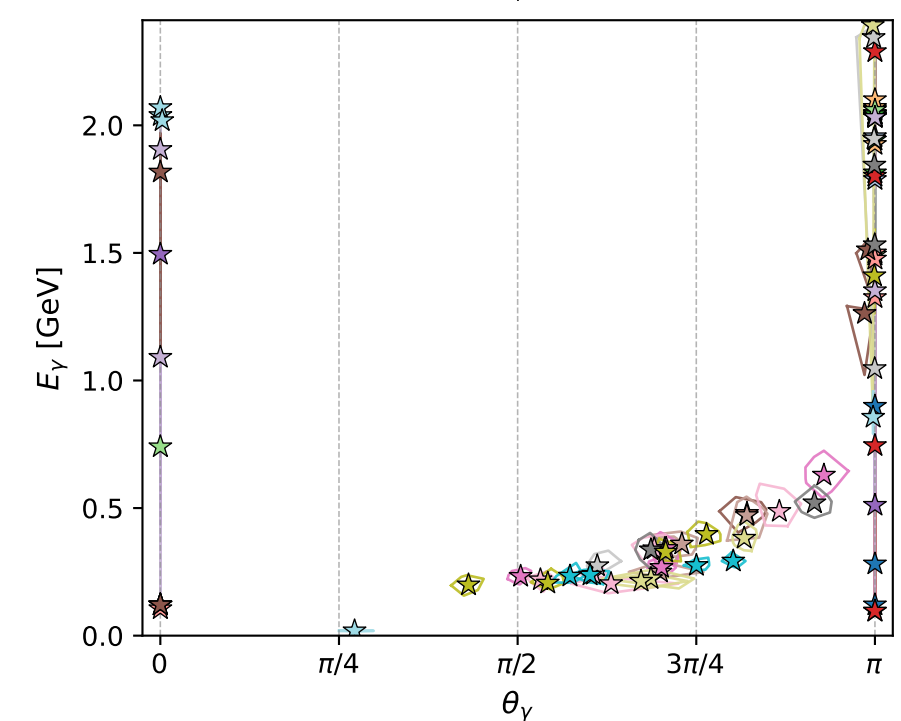
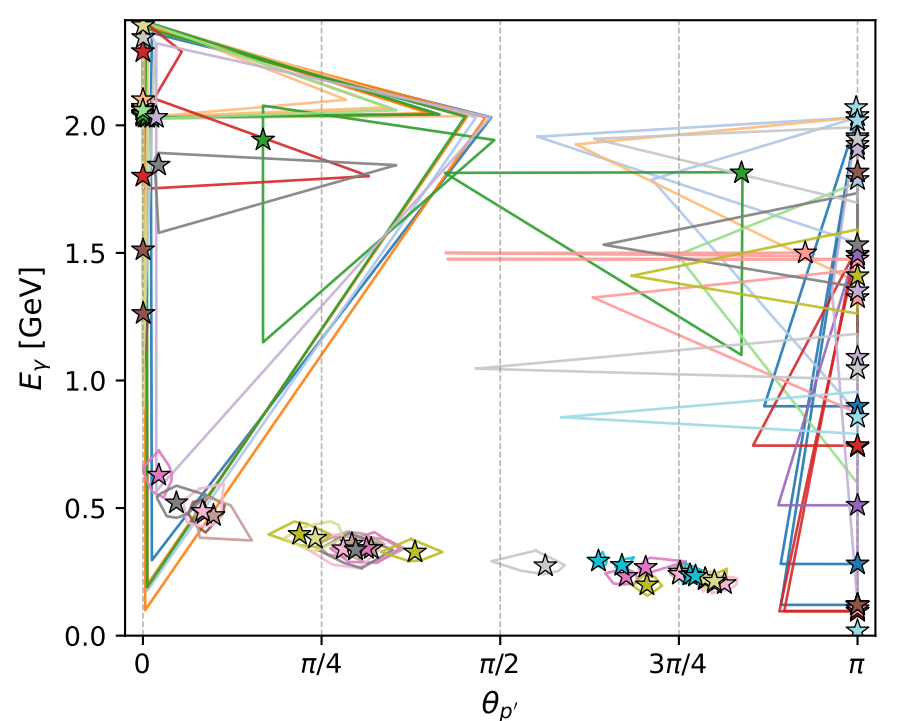
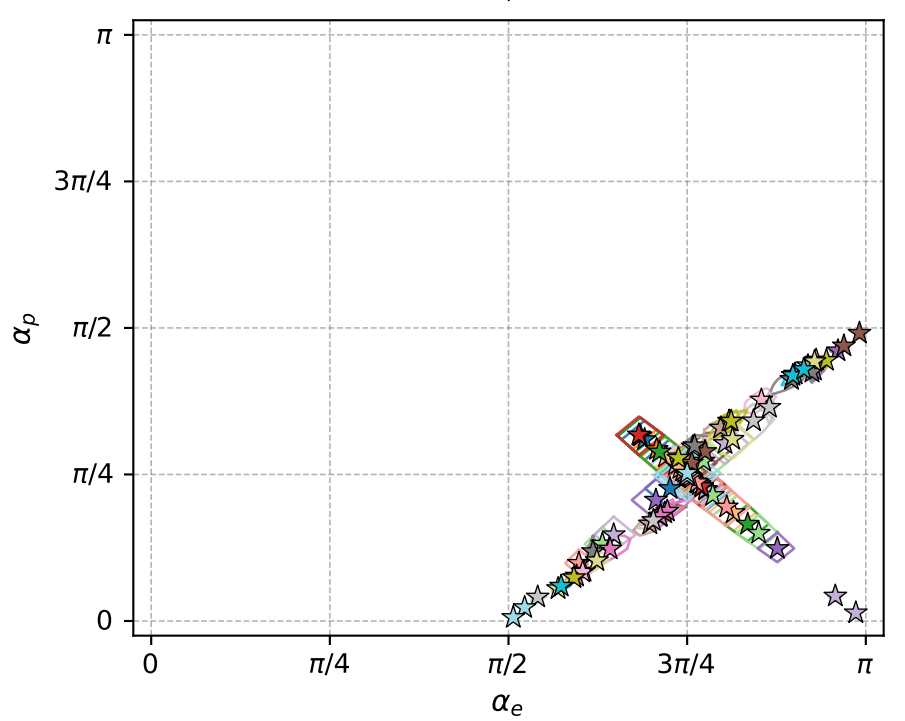
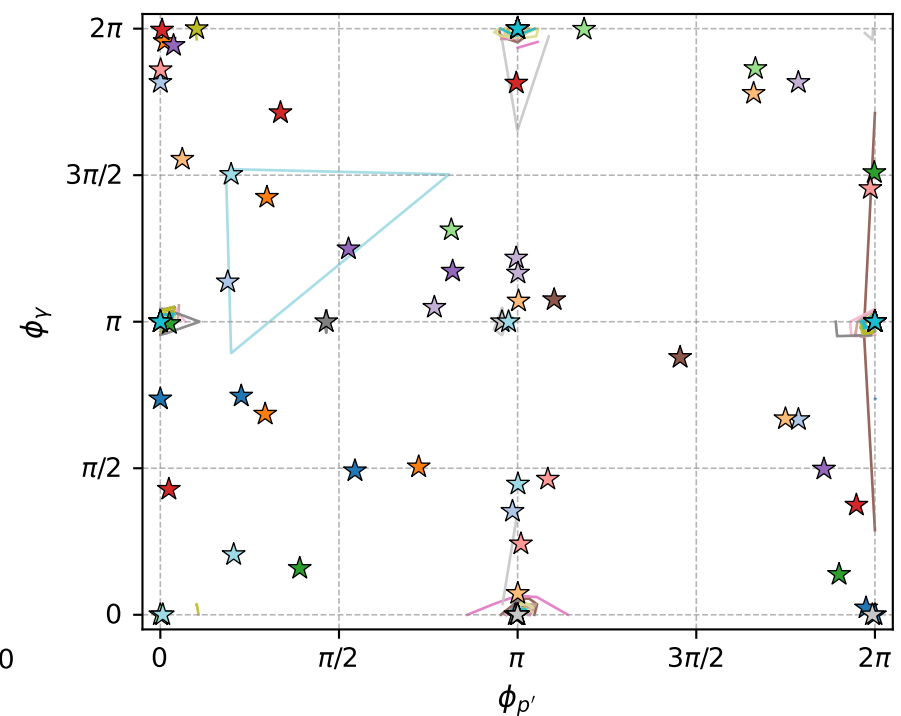
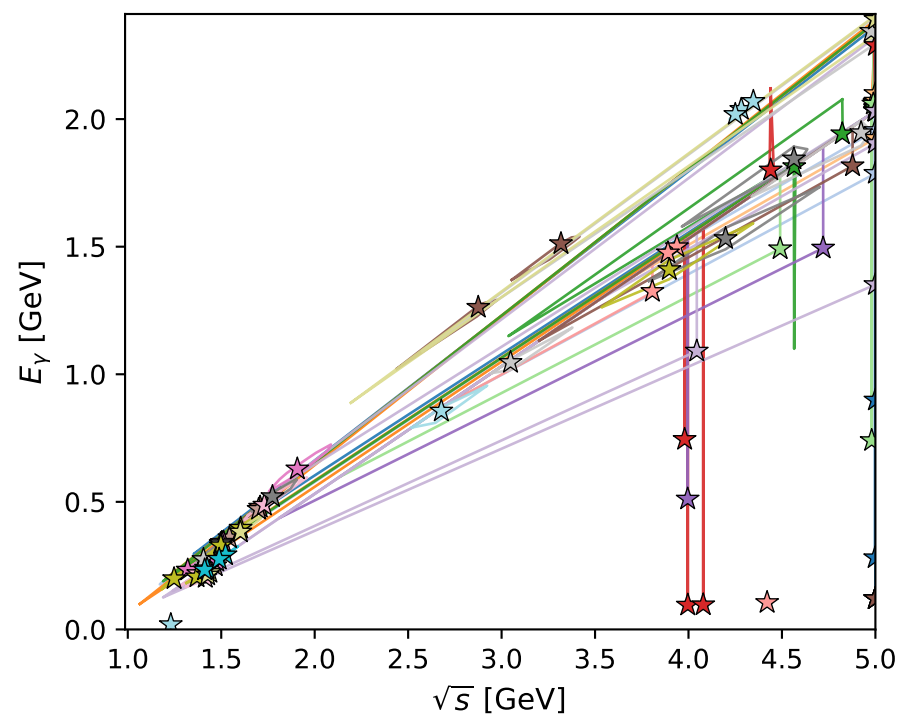
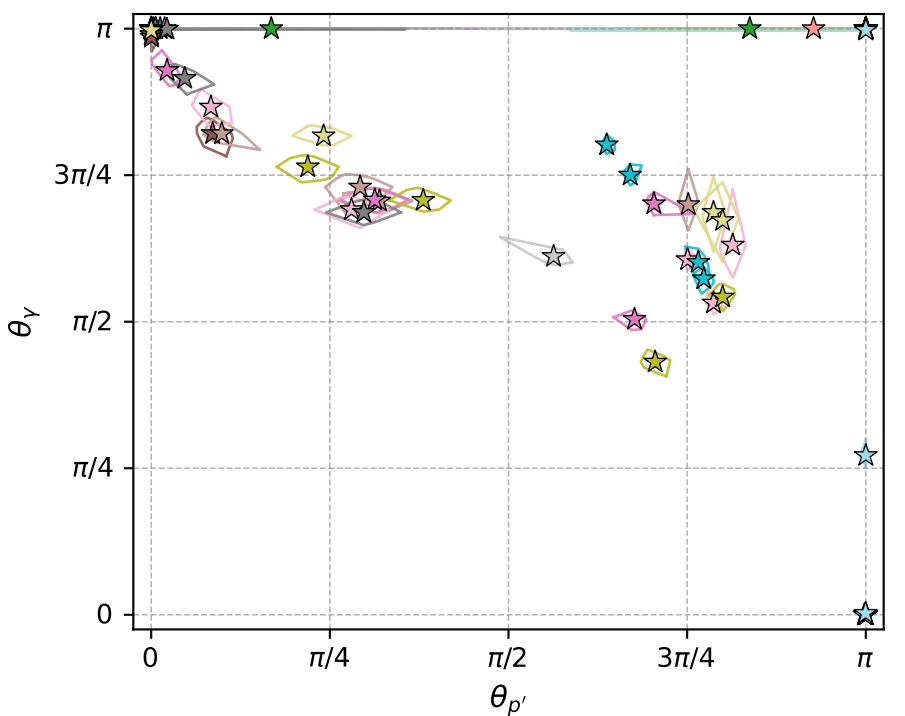


electron: polarization cluster P3; all local-minimum configurations and pairwise projections of their 8D contours



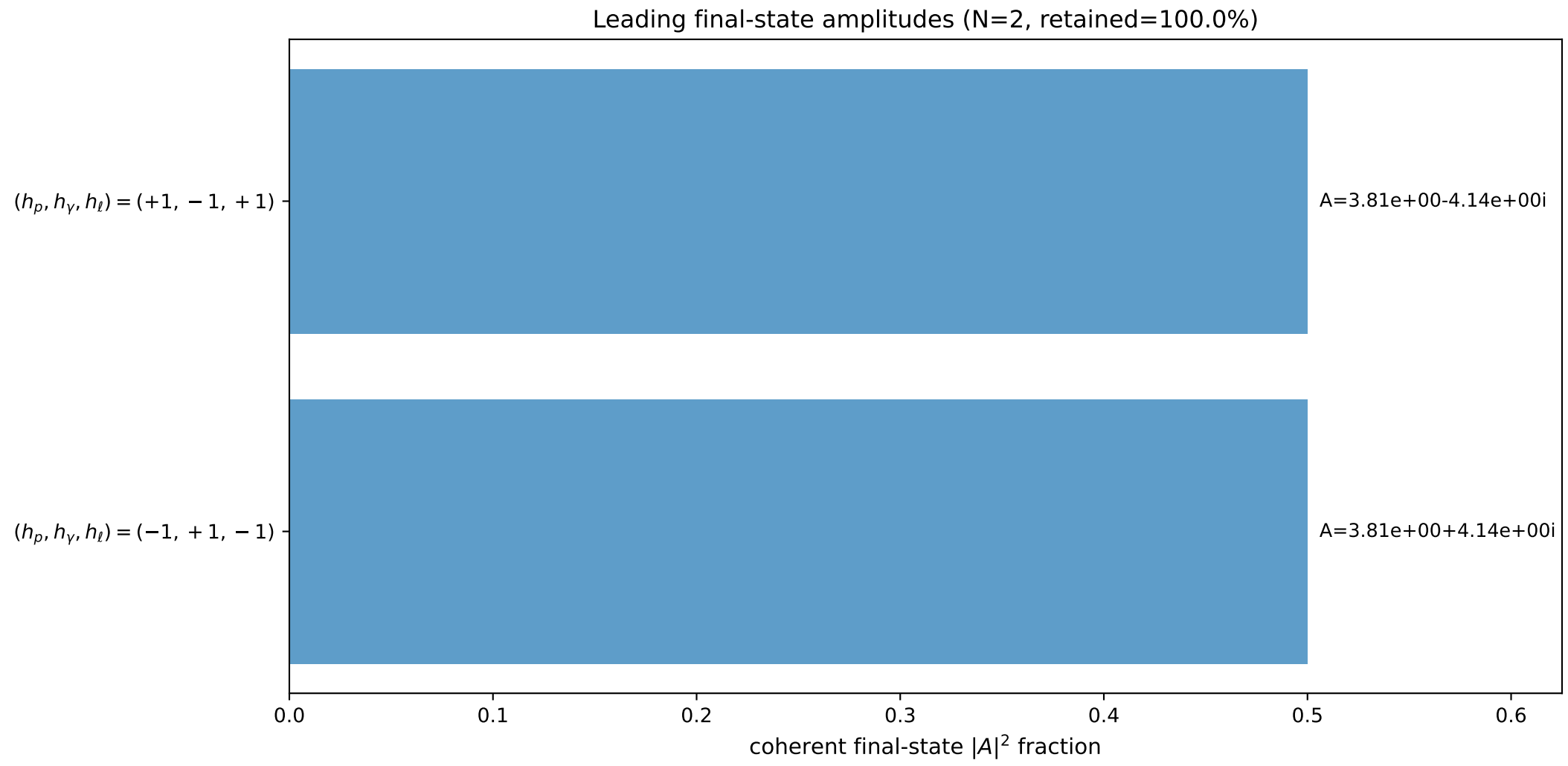
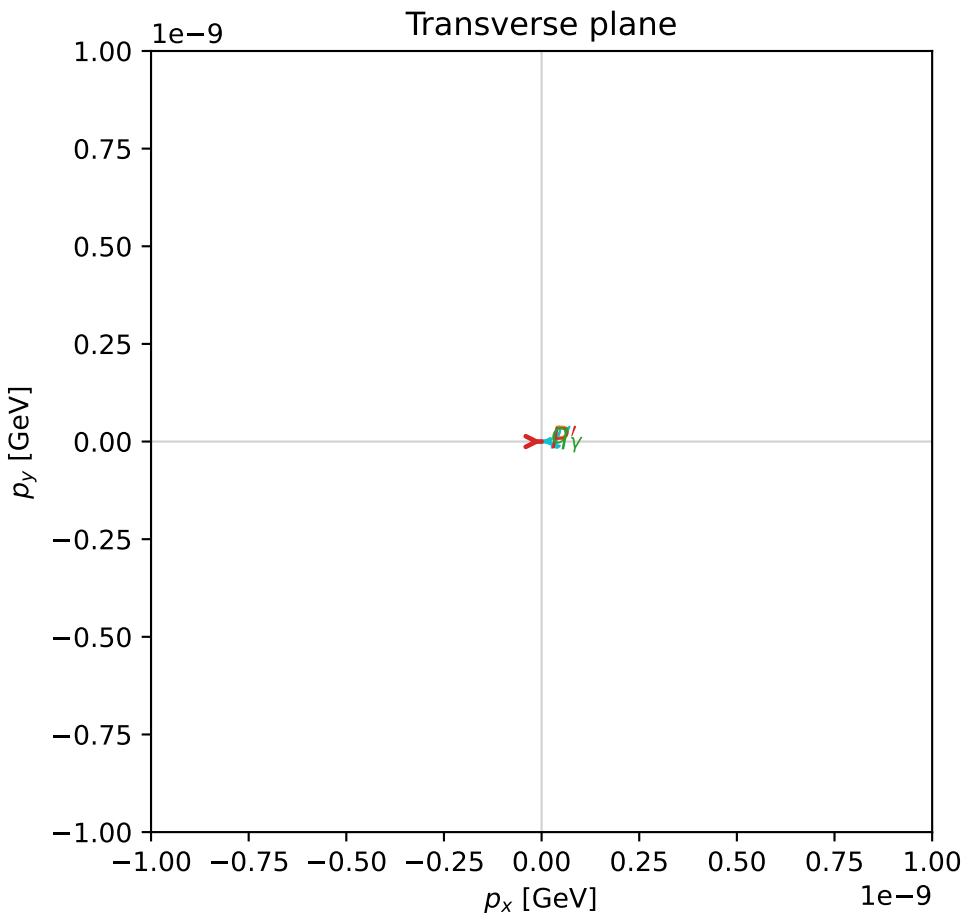
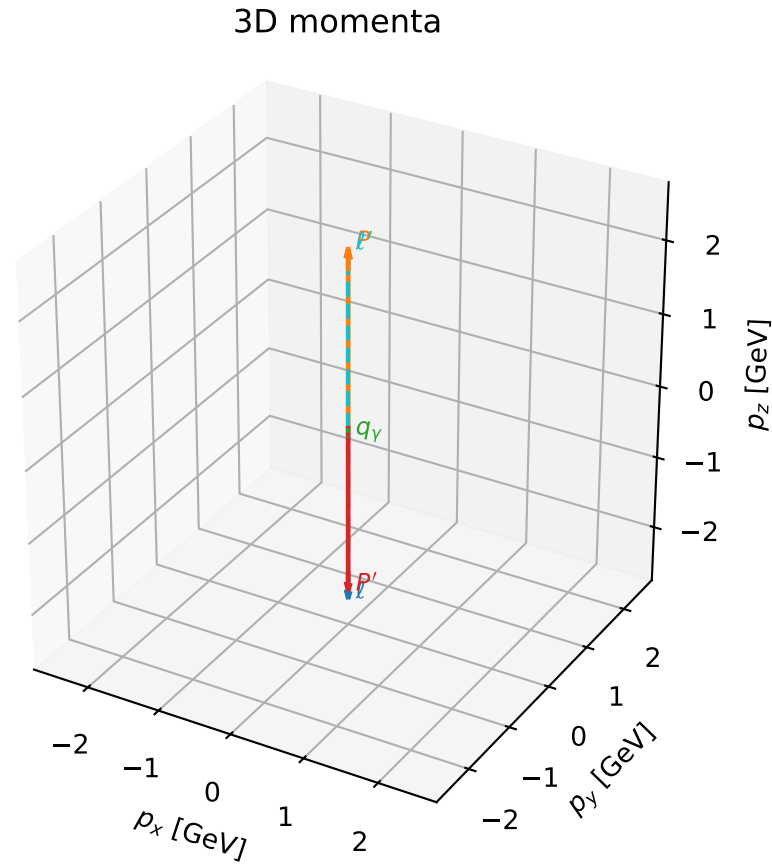
polarization cluster P3: α_e in $[\pi/2, \pi]$, $\alpha_p \sim \pi/4$
 87 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=256
 dGHZ range= $2.28069\text{e-}08$ to 0.03356521
 global-minimum offsets= $5.234334\text{e-}10$ to 0.03356518

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

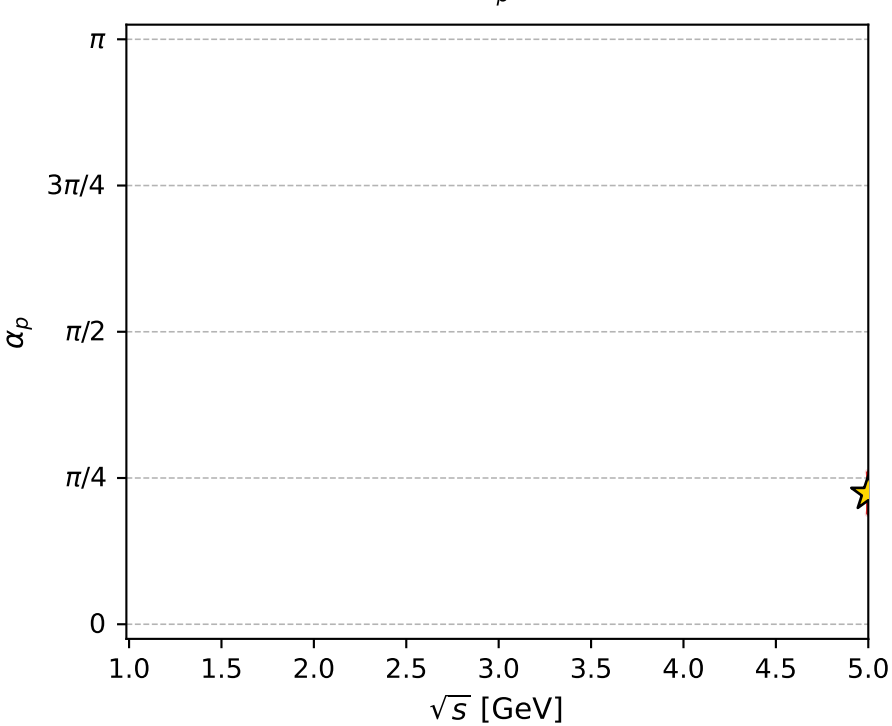
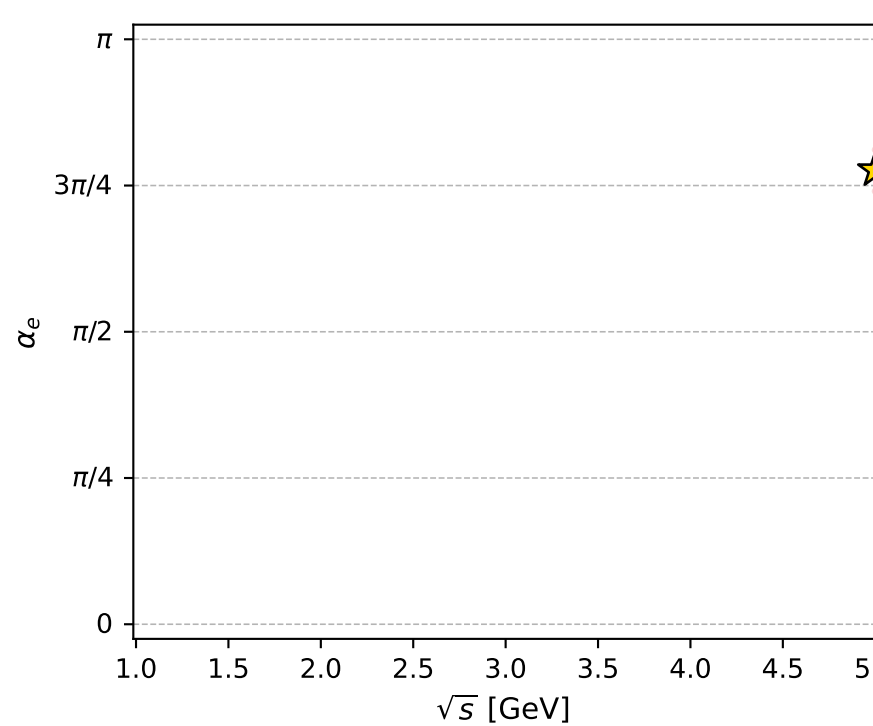
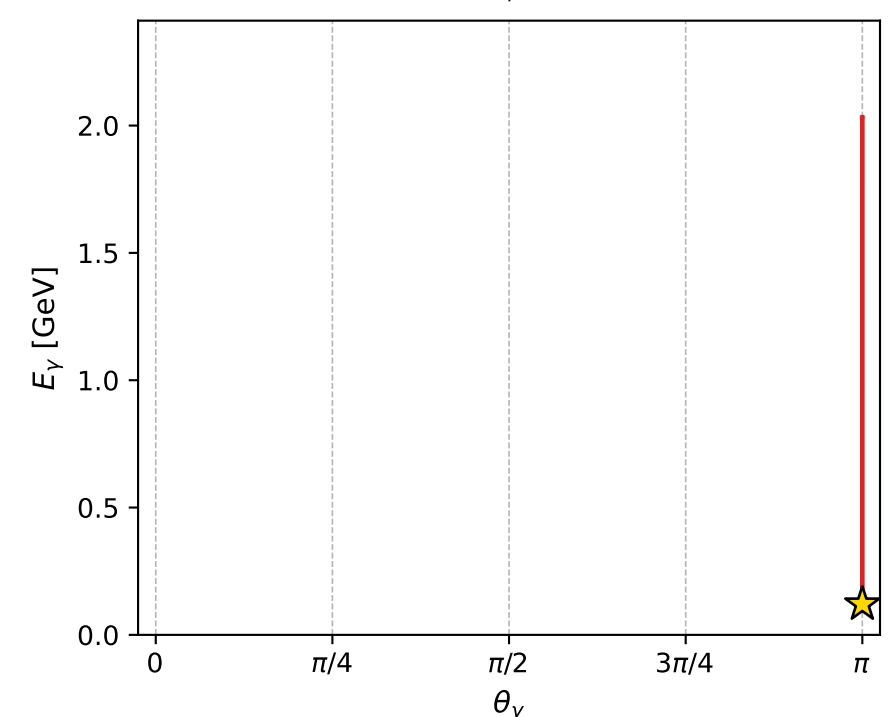
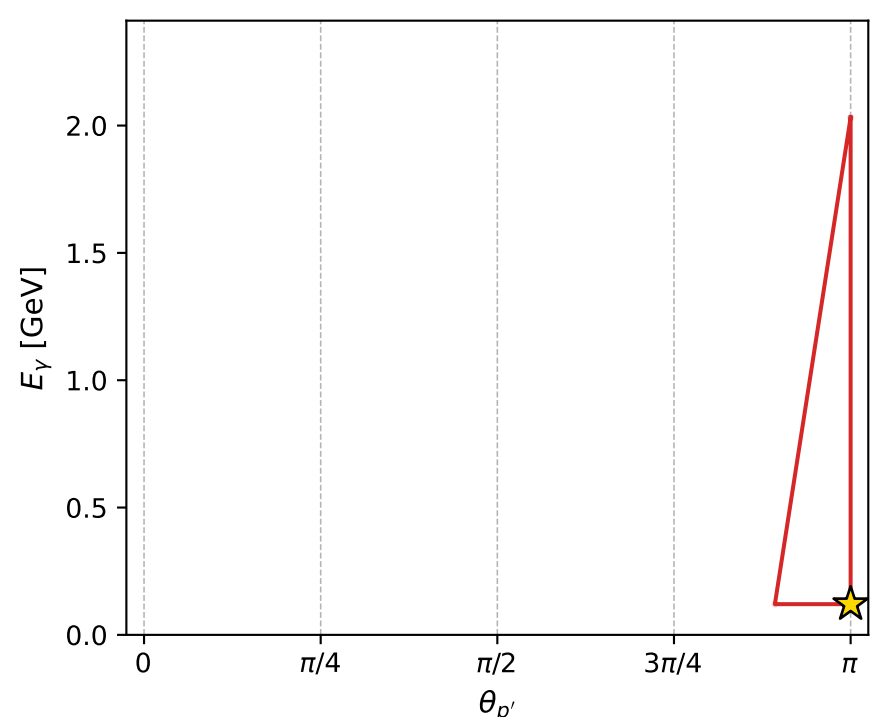
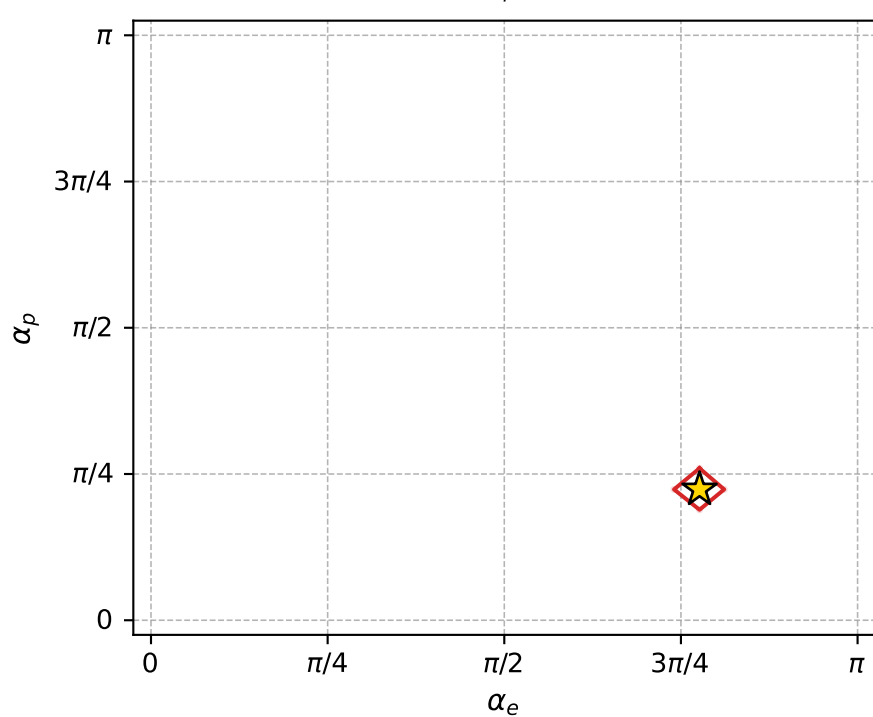
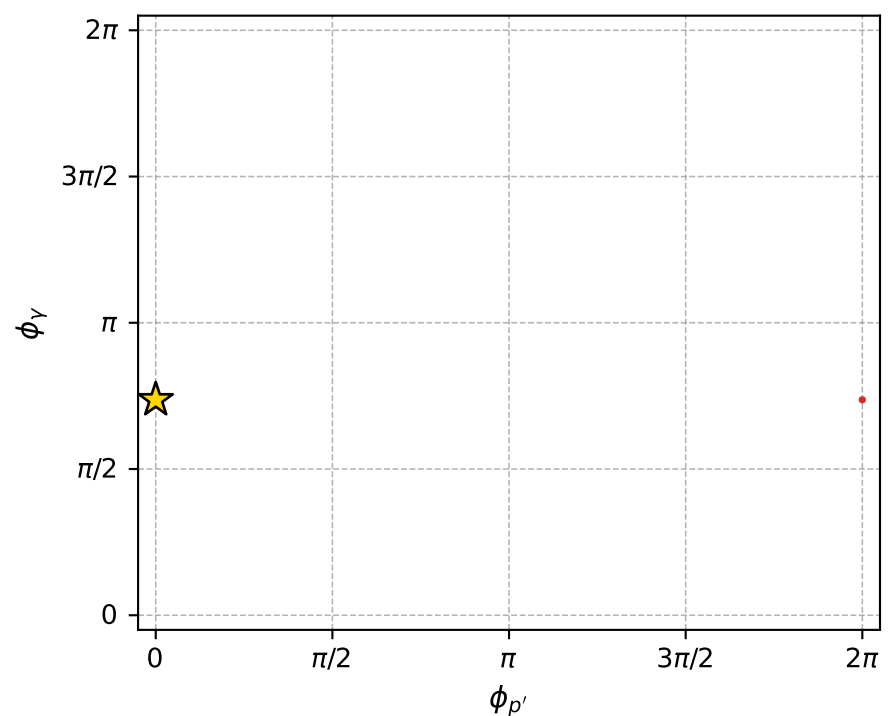
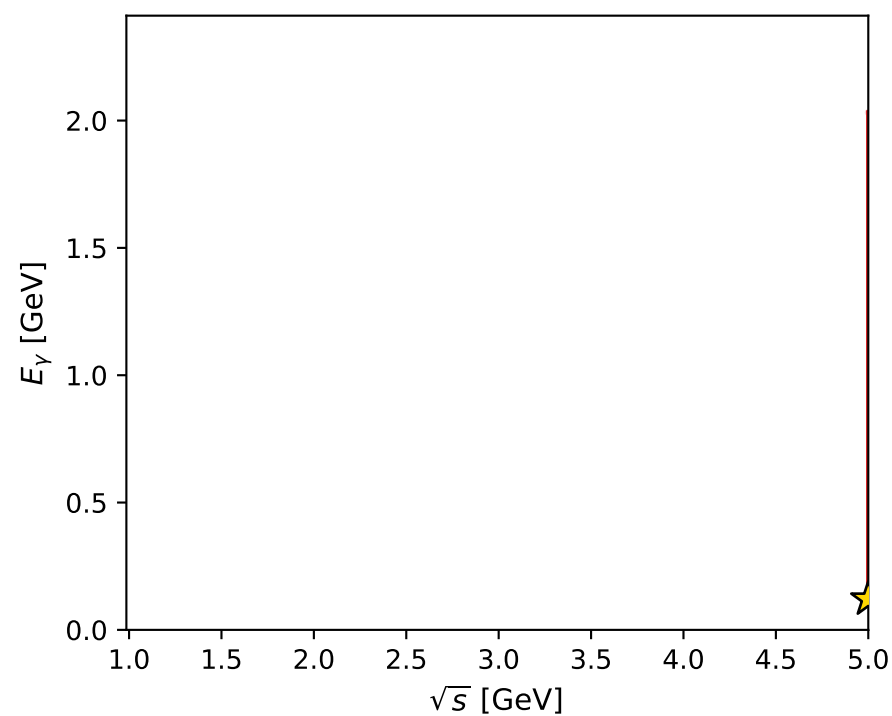
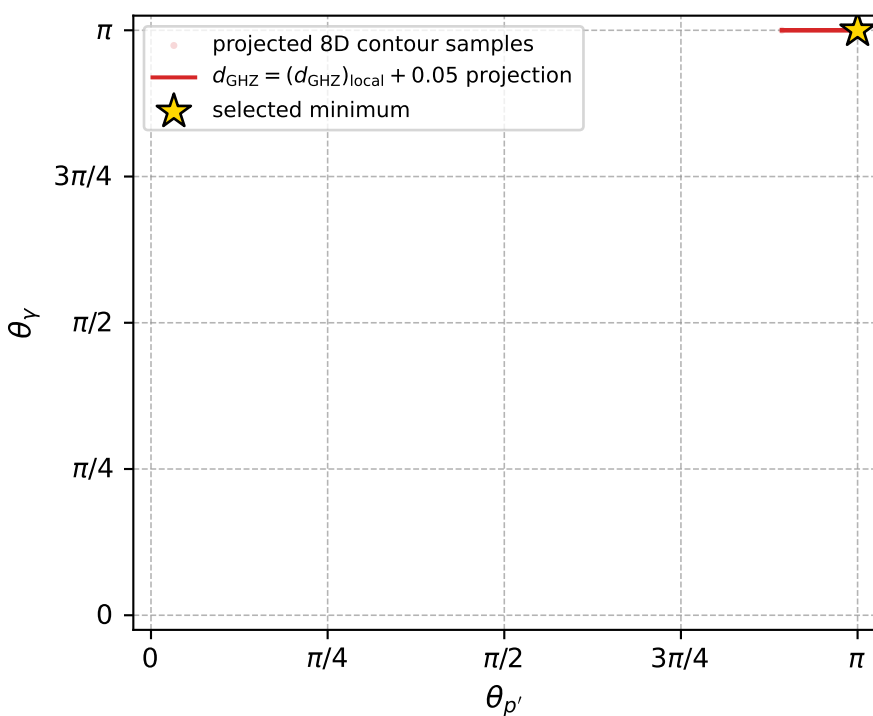
dghz_mixing_angles_polarization_03_local_minimum_1 $d_{\{\text{rm GHZ}\}}=2.28069\text{e-}08$
 region: polarization_cluster_03_local_minimum_1
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4389177$ rad
 incoming lepton: $-0.763116|+\rangle +0.646261|-\rangle$
 proton mixing angle: $\alpha_p=0.70266188$ rad
 incoming proton: $0.763125|+\rangle +0.646251|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=8.73394\text{e-}09$, $\phi_Y=2.31425$
 $E_Y=0.120598$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.18183$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924438$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4076$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_Y $E= 0.1206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1206$



electron: polarization cluster P3, local minimum 1; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.2806895\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000023$
 above global minimum = $5.2343336\text{e-}10$
 8D contour samples = 33

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.1205983$
 $\text{phi_p_out} = 8.733938\text{e-}09$
 $\text{phi_gamma_out} = 2.314255$
 $\text{alpha_e} = 2.438918$
 $\text{alpha_p} = 0.7026619$

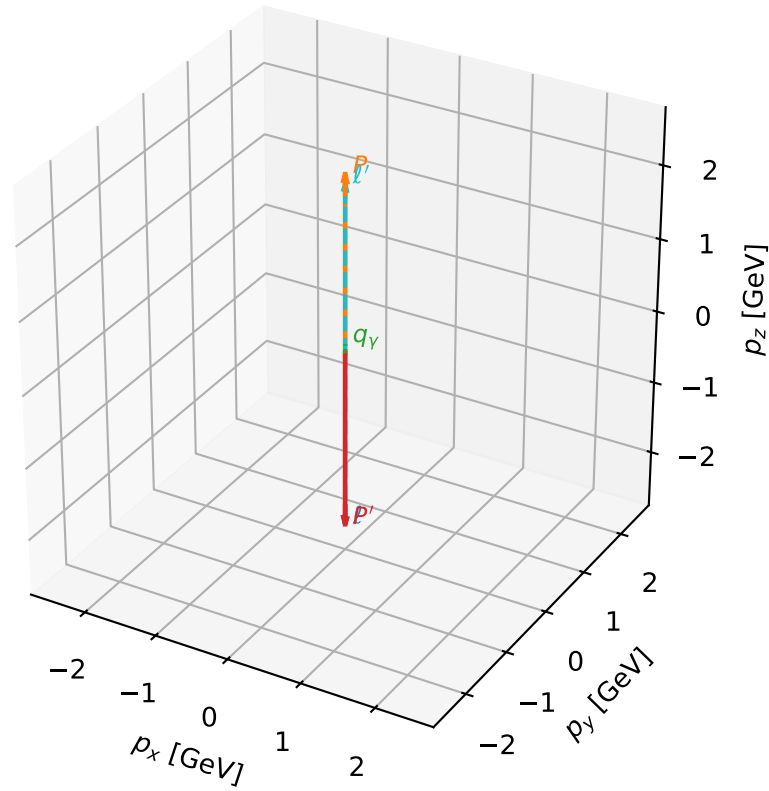
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_5 $d_{\{\text{rm GHZ}\}}=2.39007\text{e-}08$
region: polarization_cluster_03_local_minimum_5
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.2809494$ rad
incoming lepton: $-0.65195|+> +0.758262|->$
proton mixing angle: $\alpha_p=0.71016018$ rad
incoming proton: $0.758257|+> +0.651955|->$

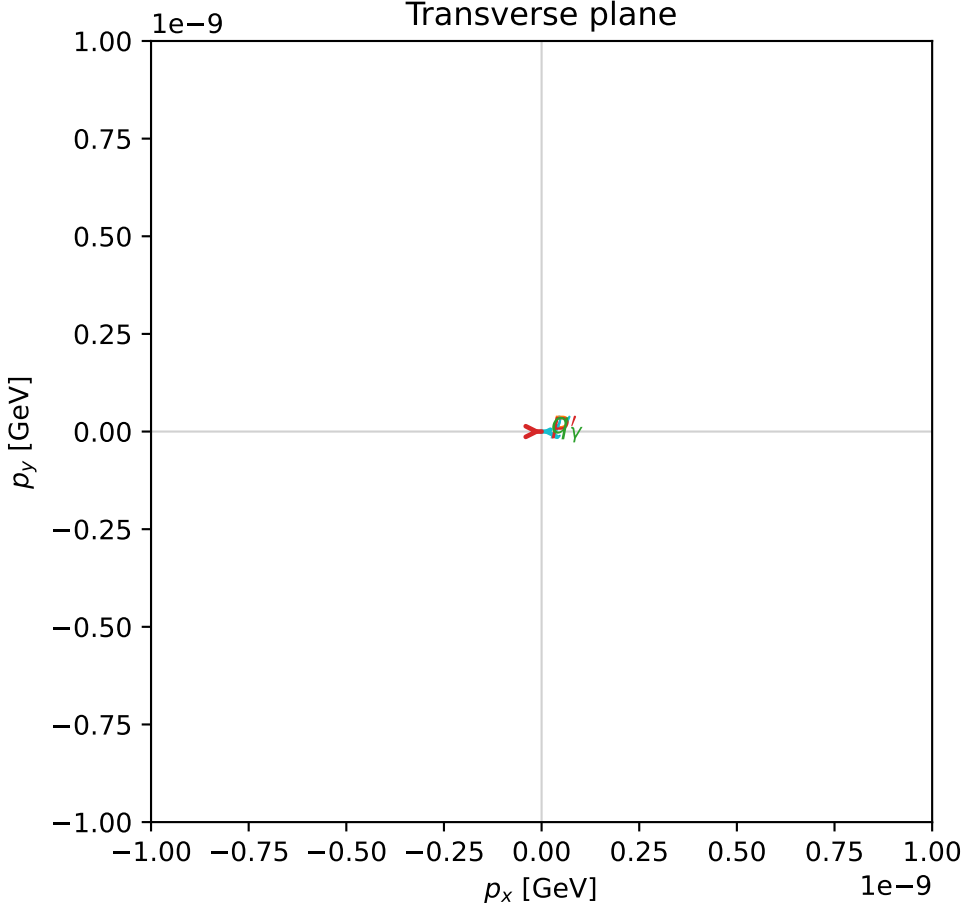
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=6.28319$, $\phi_Y=0$
 $E_Y=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

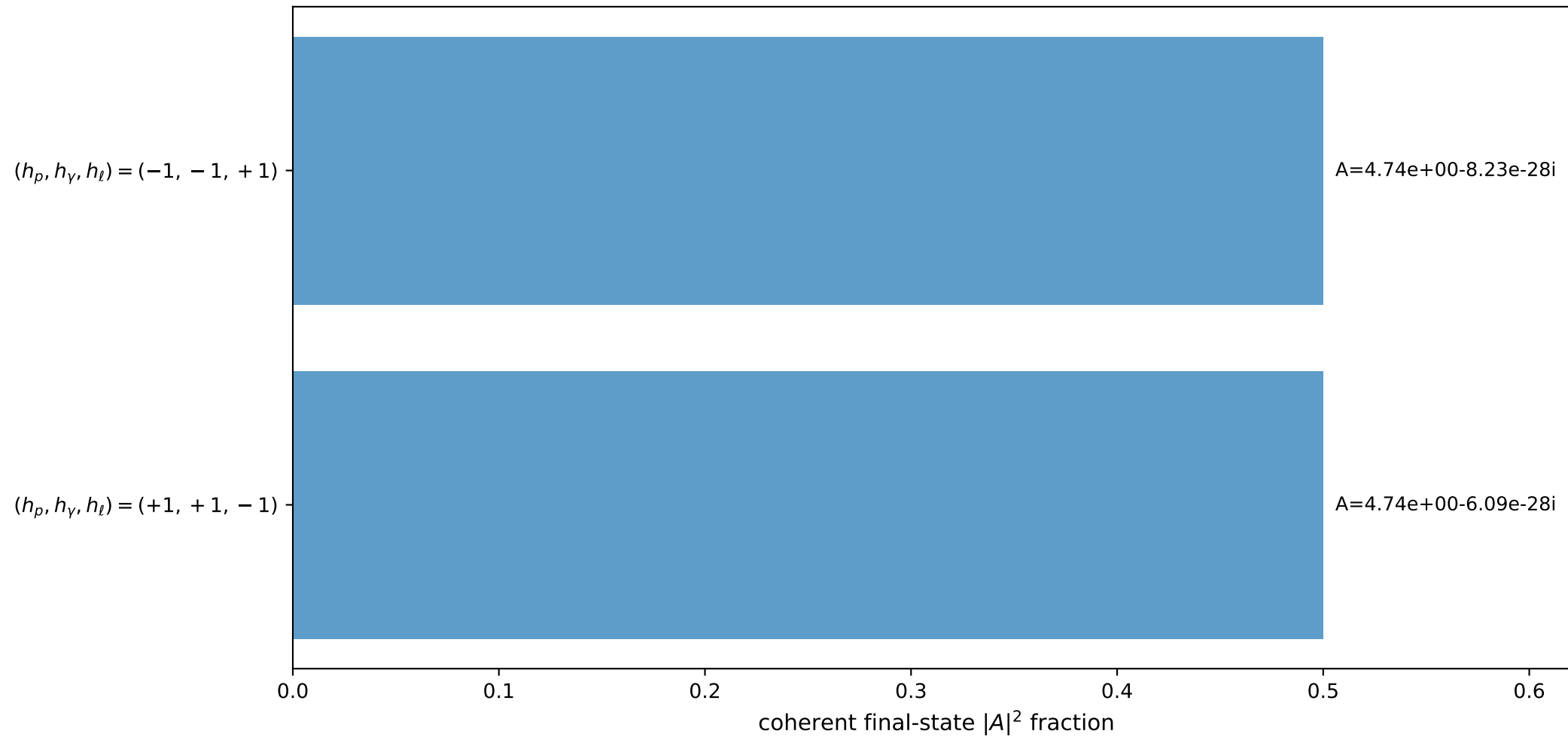
3D momenta



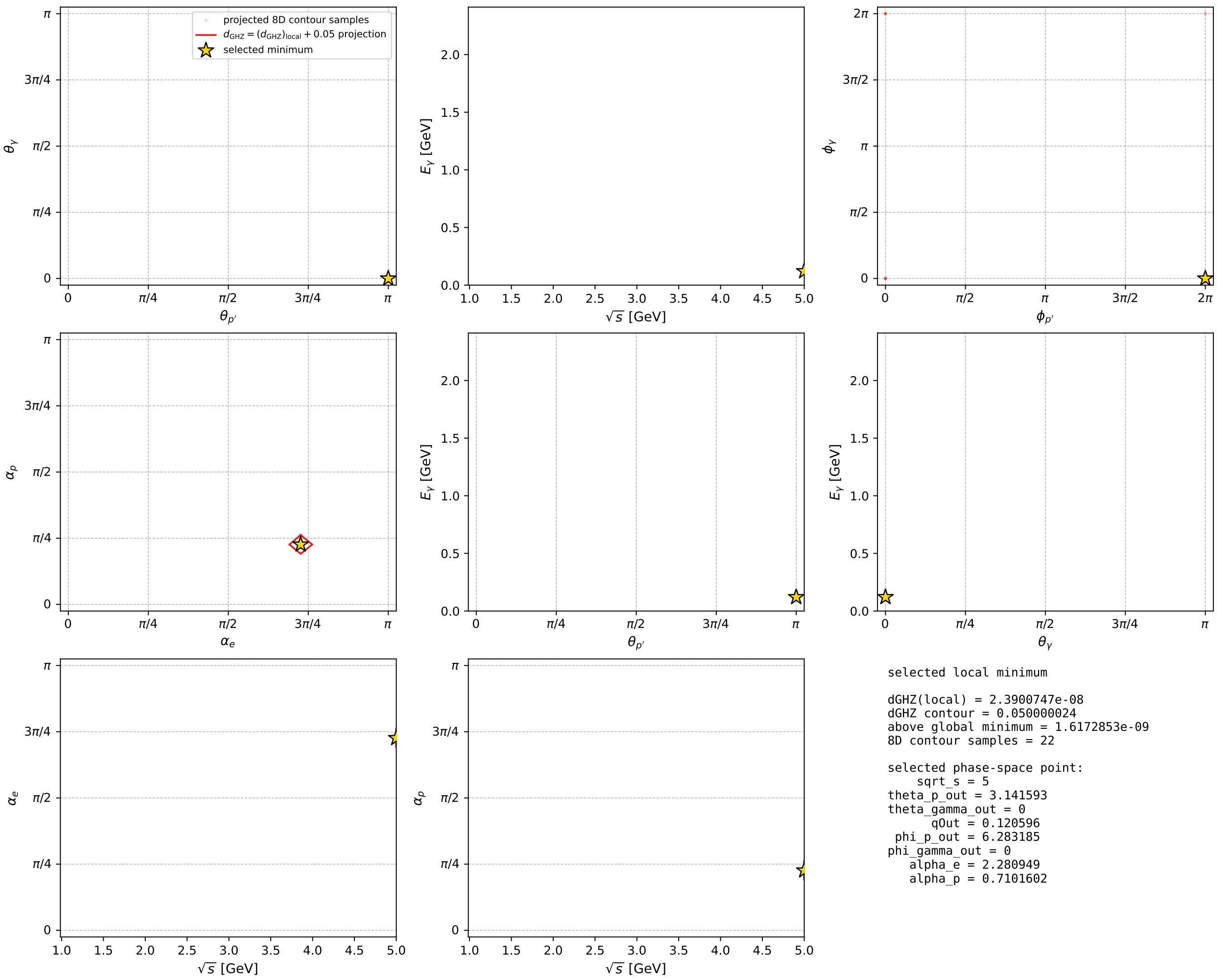
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 5; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

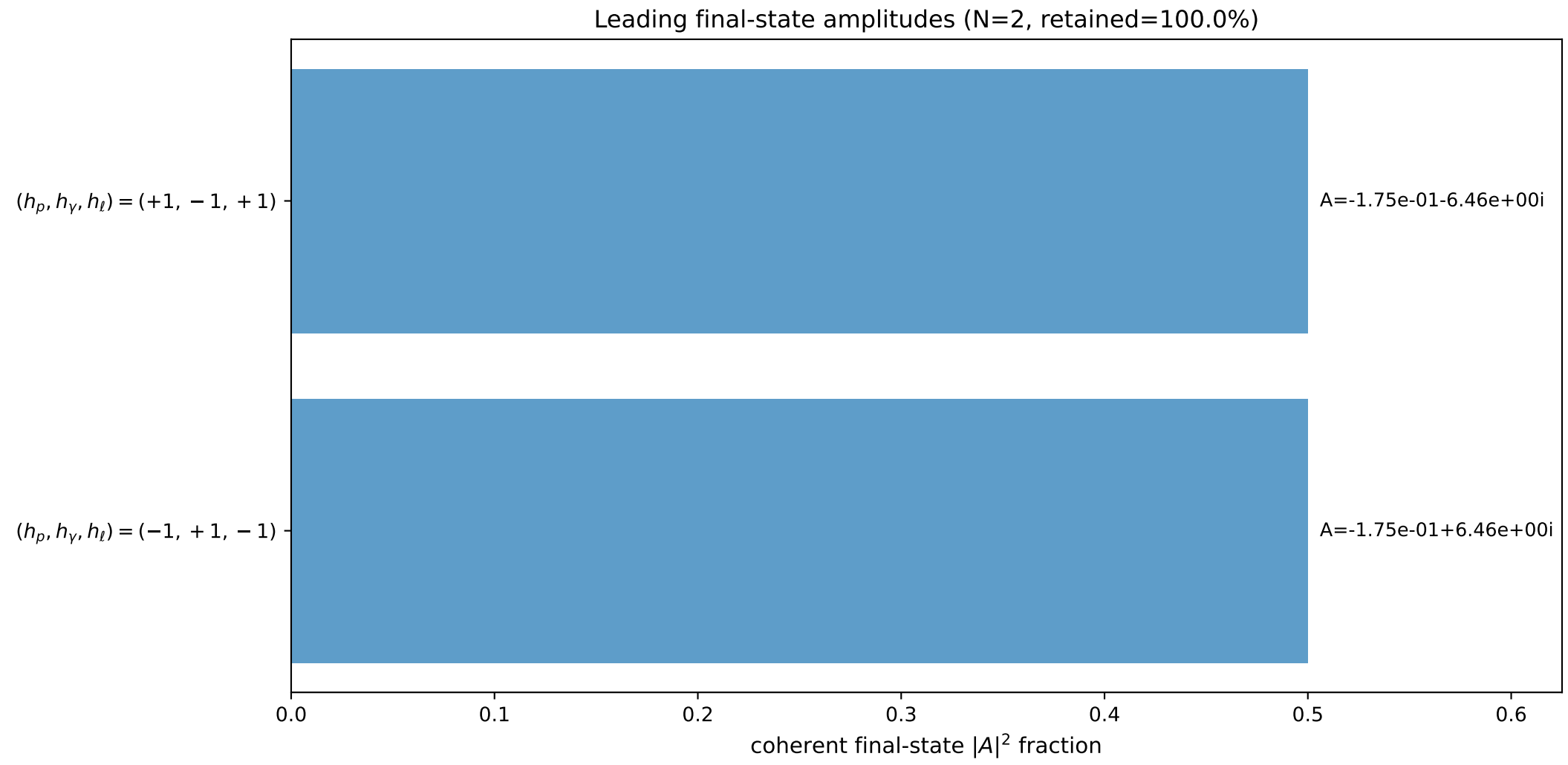
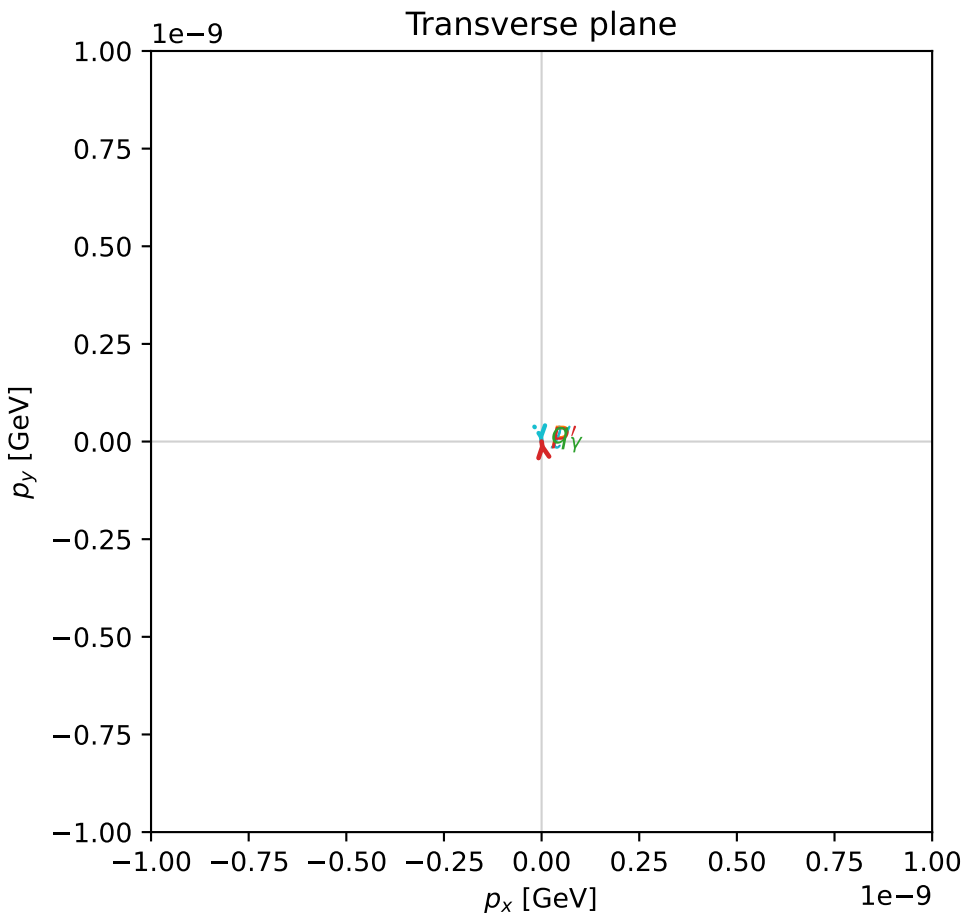
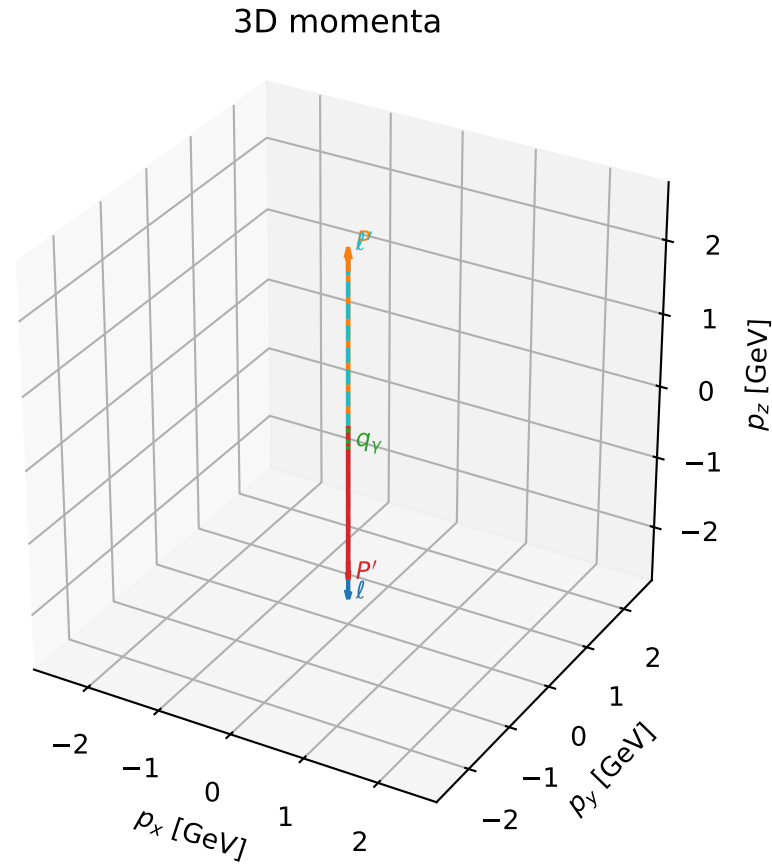


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

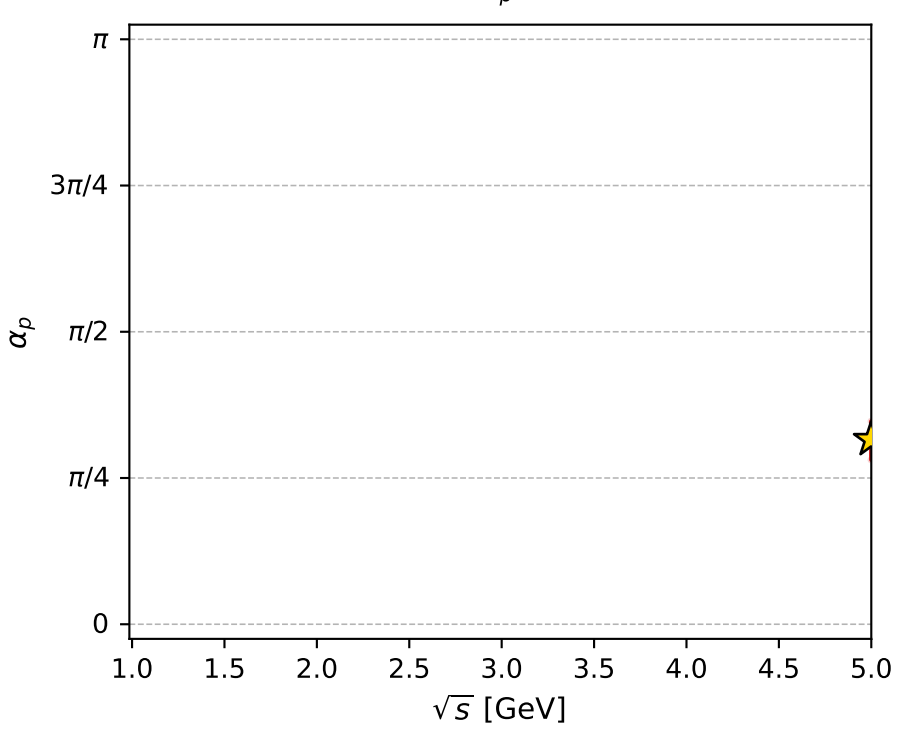
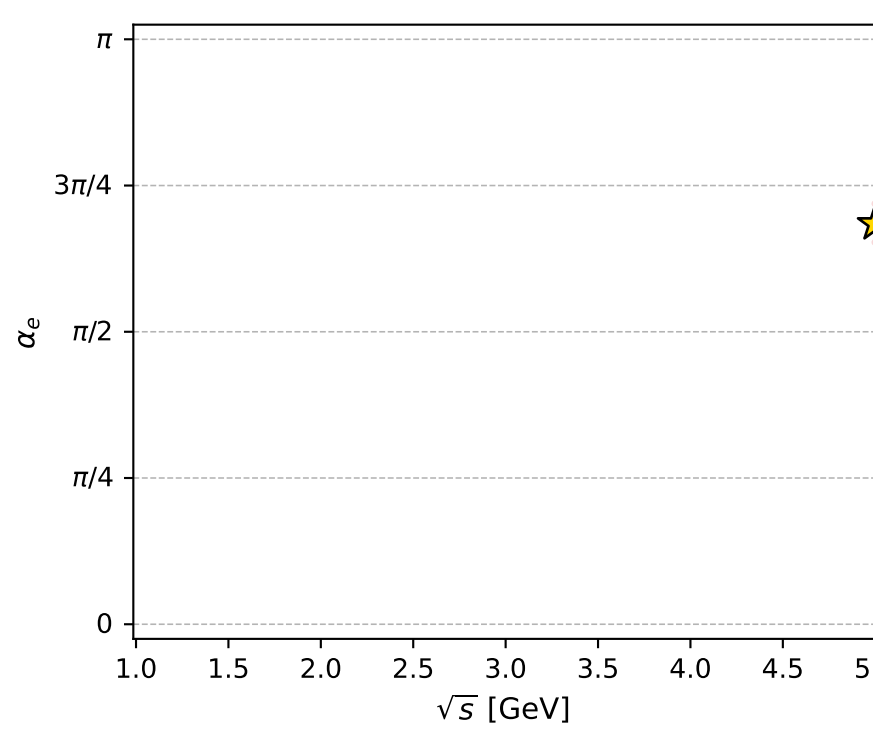
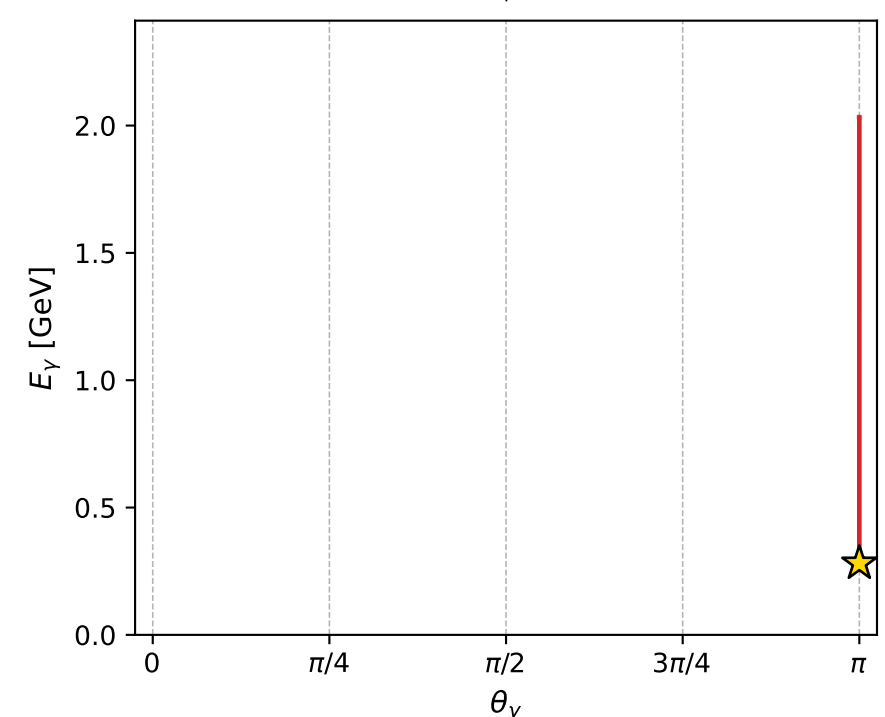
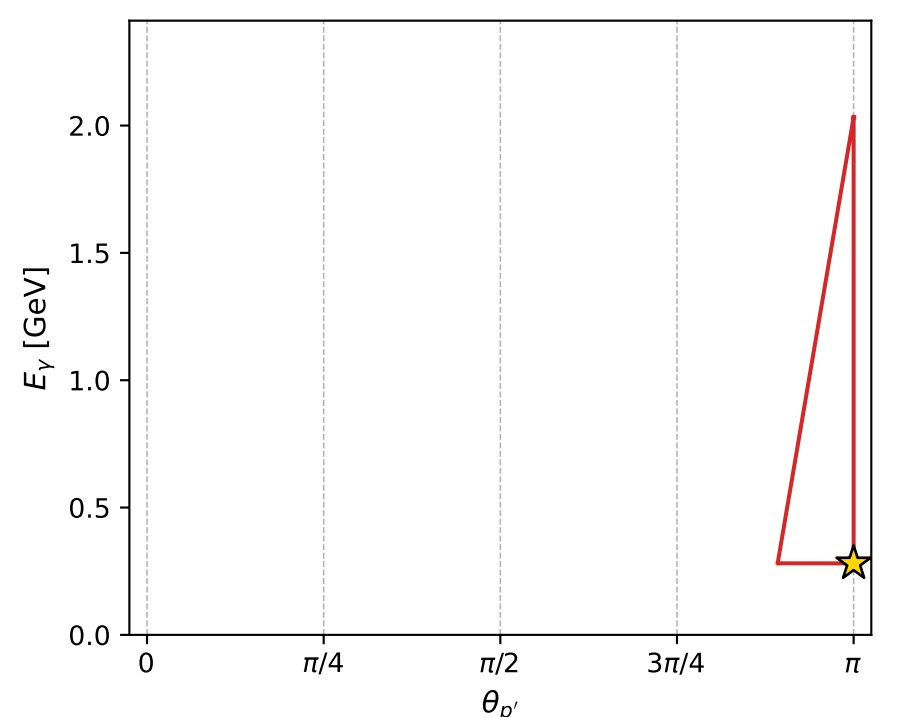
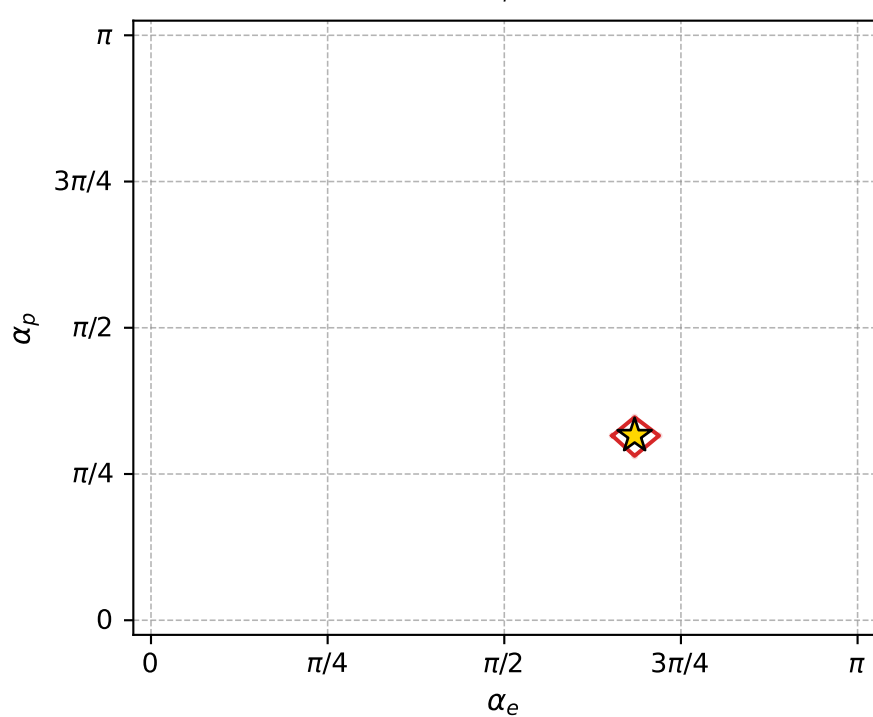
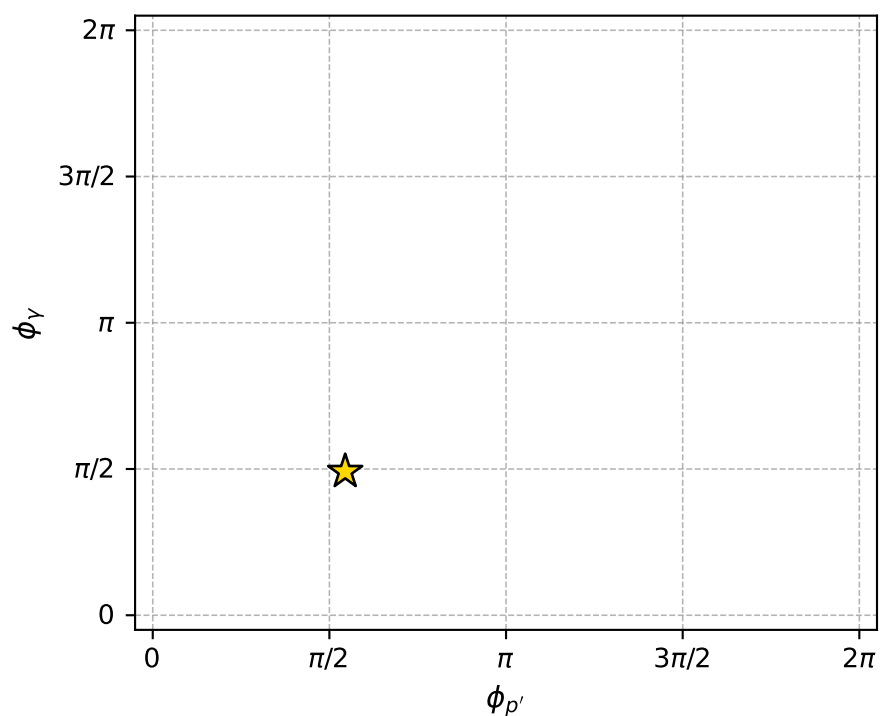
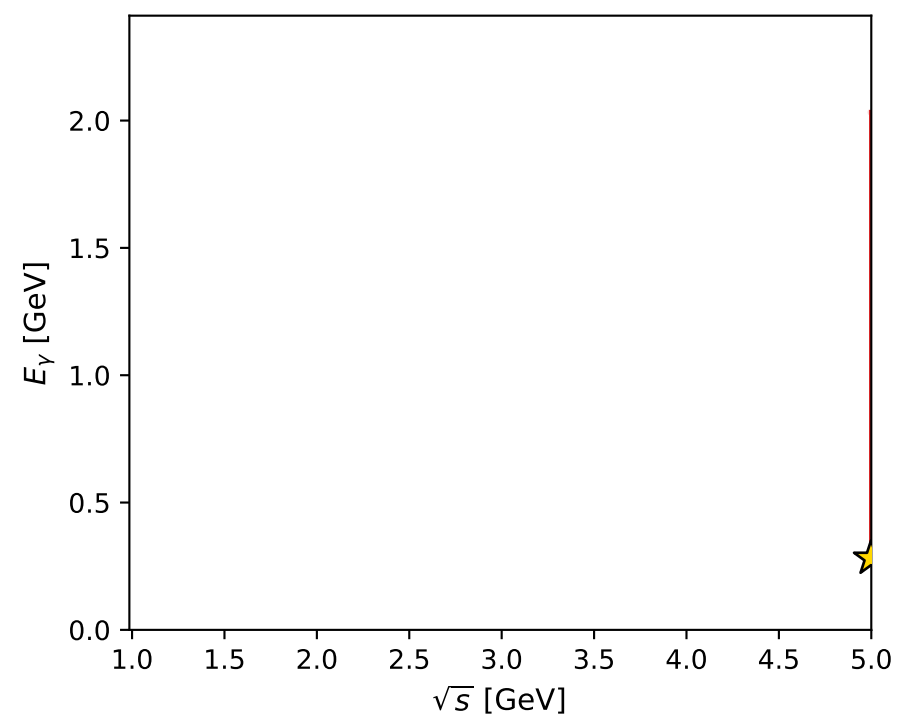
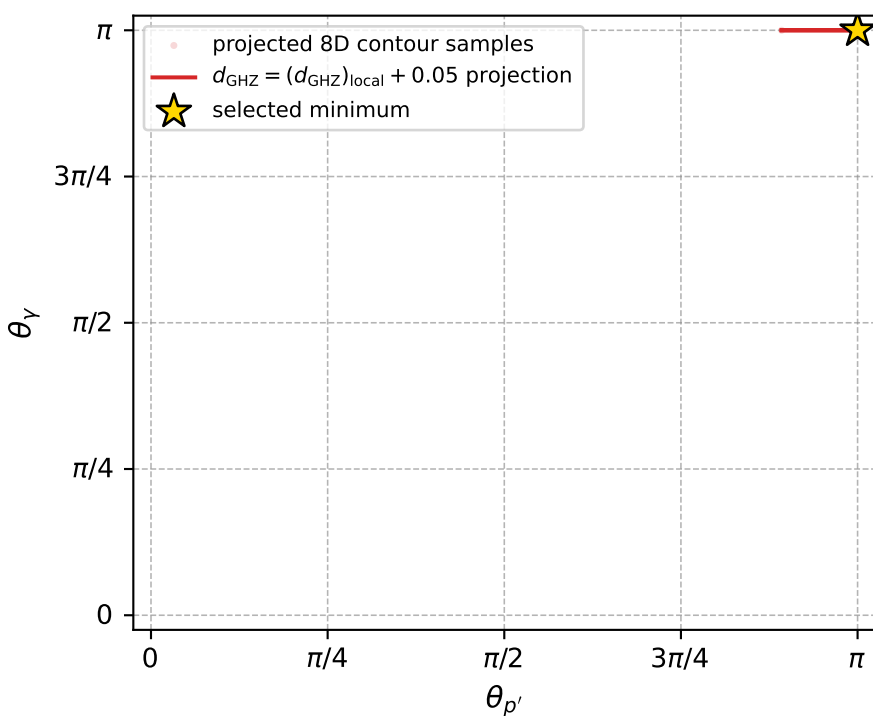
dghz_mixing_angles_polarization_03_local_minimum_12 d_{\rm GHZ}=2.46147e-08
region: polarization_cluster_03_local_minimum_12
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.1502889$ rad
incoming lepton: $-0.547599|+> +0.836741|->$
proton mixing angle: $\alpha_p=0.99129539$ rad
incoming proton: $0.547606|+> +0.836736|->$

$s=24.9869$, $\sqrt{s}=4.99869$, $|\vec{P}|=2.41134$, $|\vec{P}'|=2.11911$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=1.71126$, $\phi_Y=1.54366$
 $E_Y=0.281071$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.83329$
 $Q^2=23.1506$, $x_B=0.995208$, $t=-20.4521$, $W^2=0.991326$, $y=0.964951$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5874$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.4002$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4002$
 P' $E= 2.3174$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1191$
 q_Y $E= 0.2811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2811$



electron: polarization cluster P3, local minimum 12; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4614682\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $2.3312203\text{e-}09$
 8D contour samples = 73

selected phase-space point:

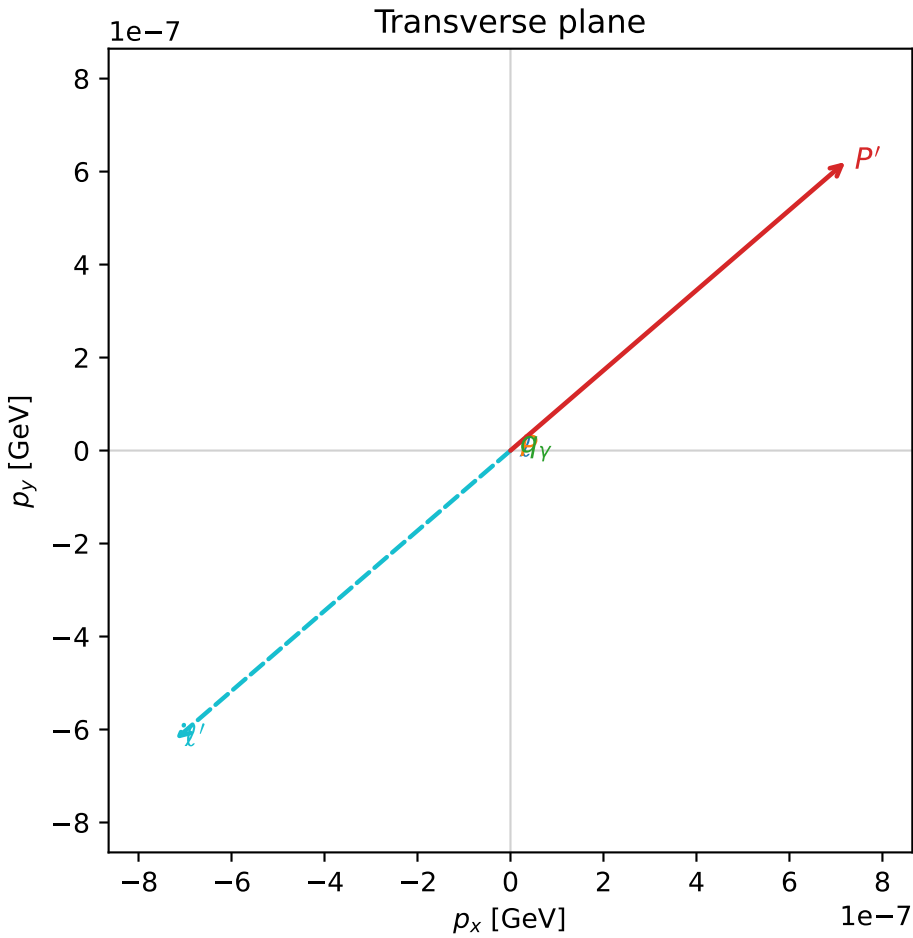
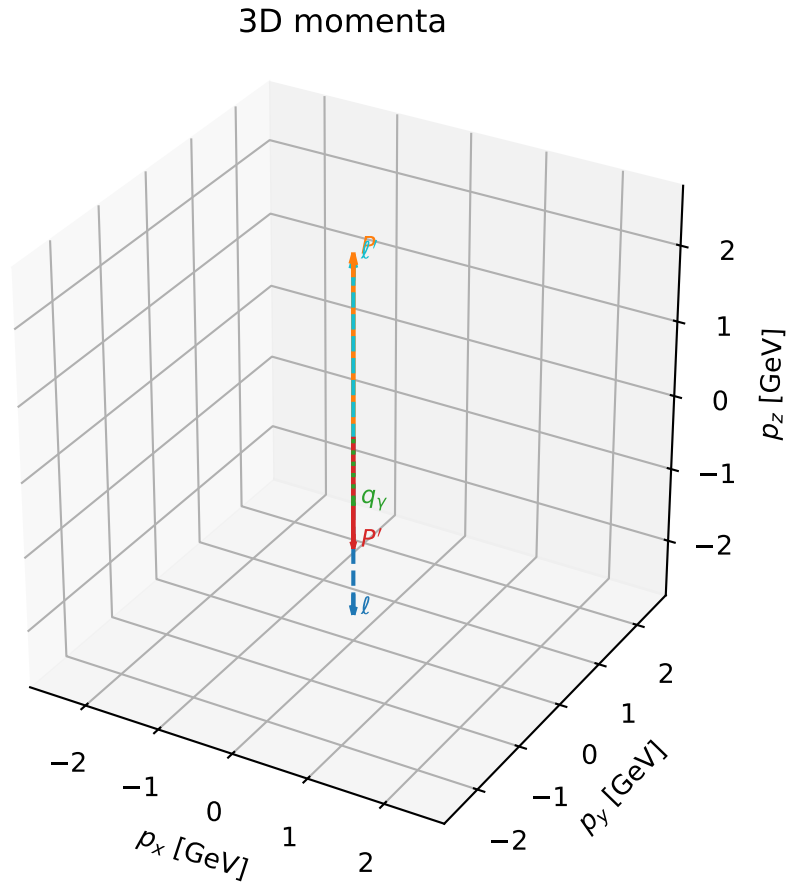
$\text{sqrt_s} = 4.998687$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.2810707$
 $\text{phi_p_out} = 1.711265$
 $\text{phi_gamma_out} = 1.543657$
 $\alpha_e = 2.150289$
 $\alpha_p = 0.9912954$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_13 $d_{\{\text{rm GHZ}\}}=2.46222\text{e-}08$
region: polarization_cluster_03_local_minimum_13
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.1691893$ rad
incoming lepton: $-0.563315|+\rangle +0.826242|-\rangle$
proton mixing angle: $\alpha_p=0.97239039$ rad
incoming proton: $0.563326|+\rangle +0.826235|-\rangle$

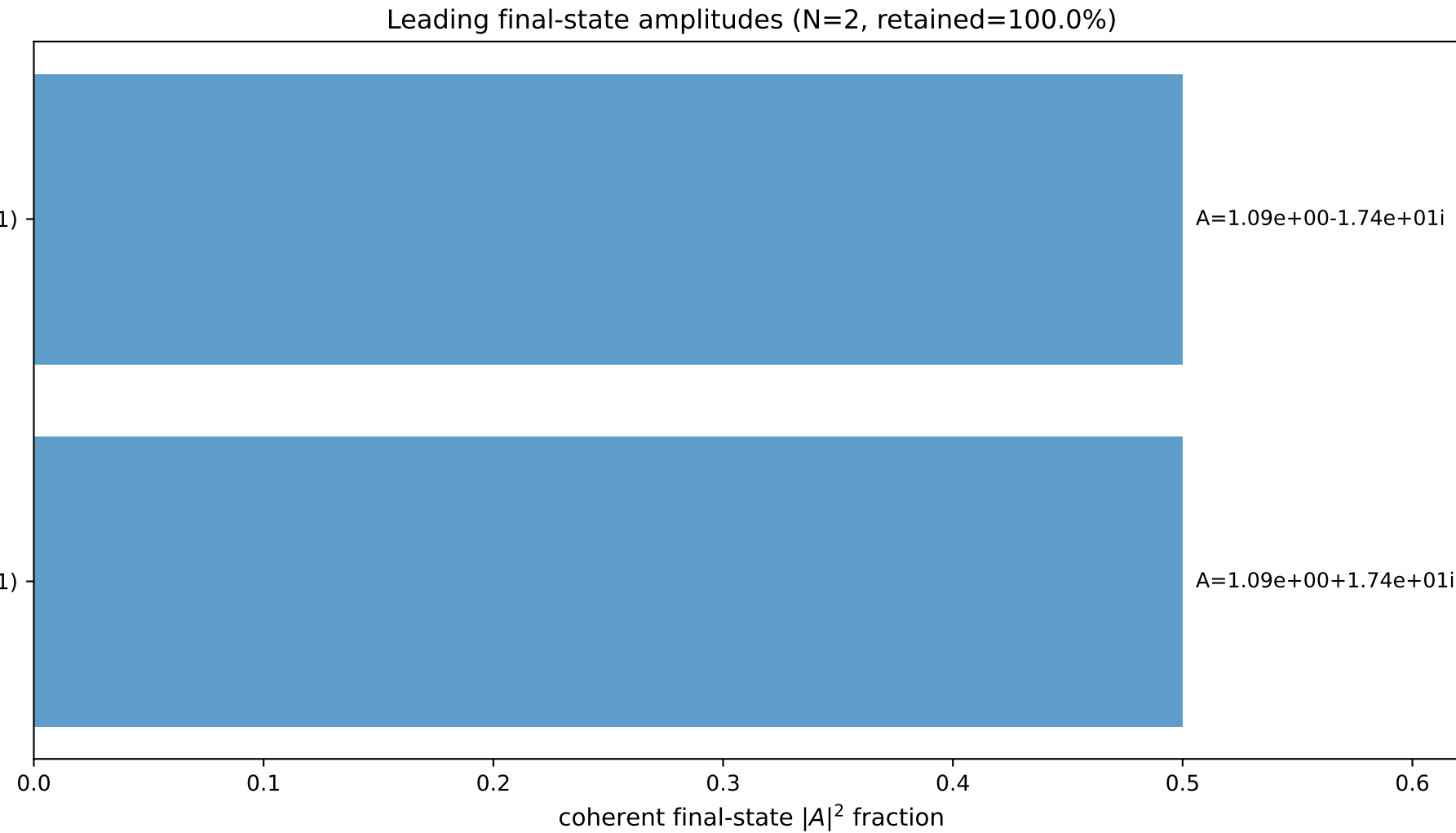
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.46356$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=0.711355$, $\phi_Y=2.34502$
 $E_Y=0.899044$, $\theta_{l'}=4.02607\text{e-}07$, $\phi_{l'}=3.85295$
 $Q^2=22.7946$, $x_B=0.978784$, $t=-14.2982$, $W^2=1.37394$, $y=0.965527$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.3626$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3626$
 P' $E= 1.7383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4636$
 q_Y $E= 0.8990$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.8990$

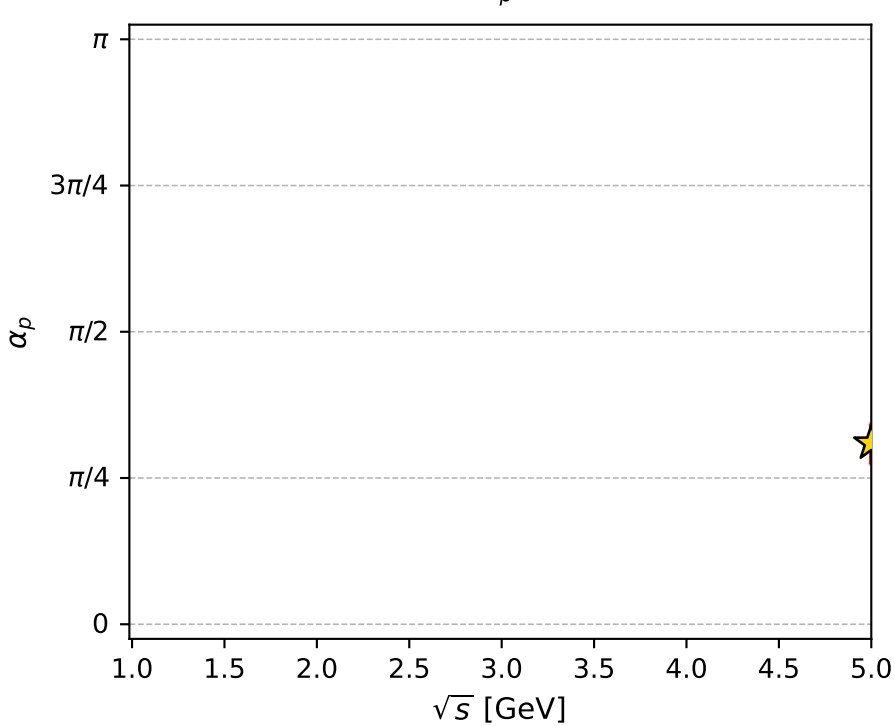
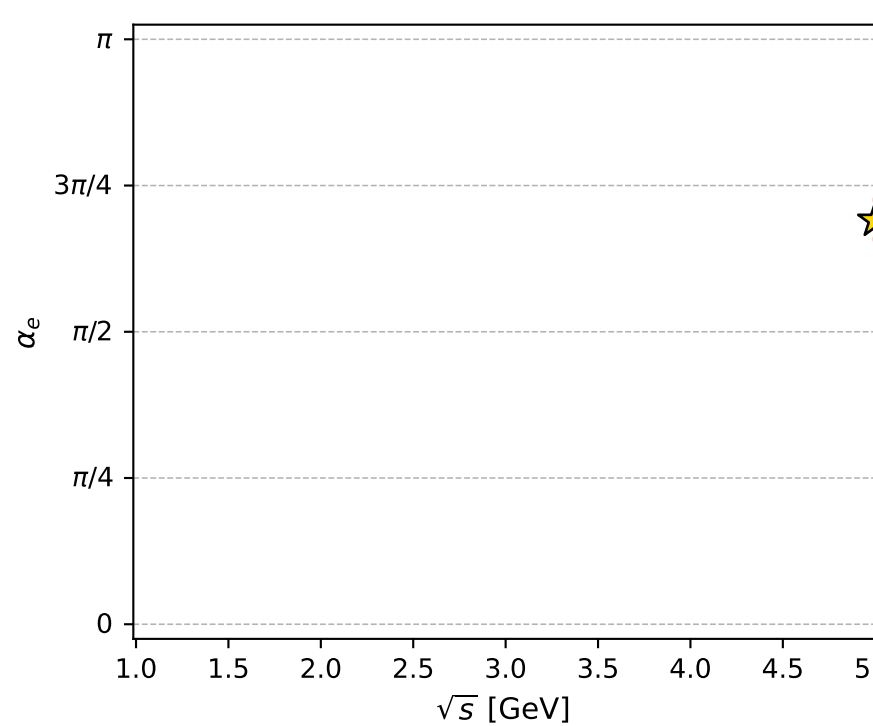
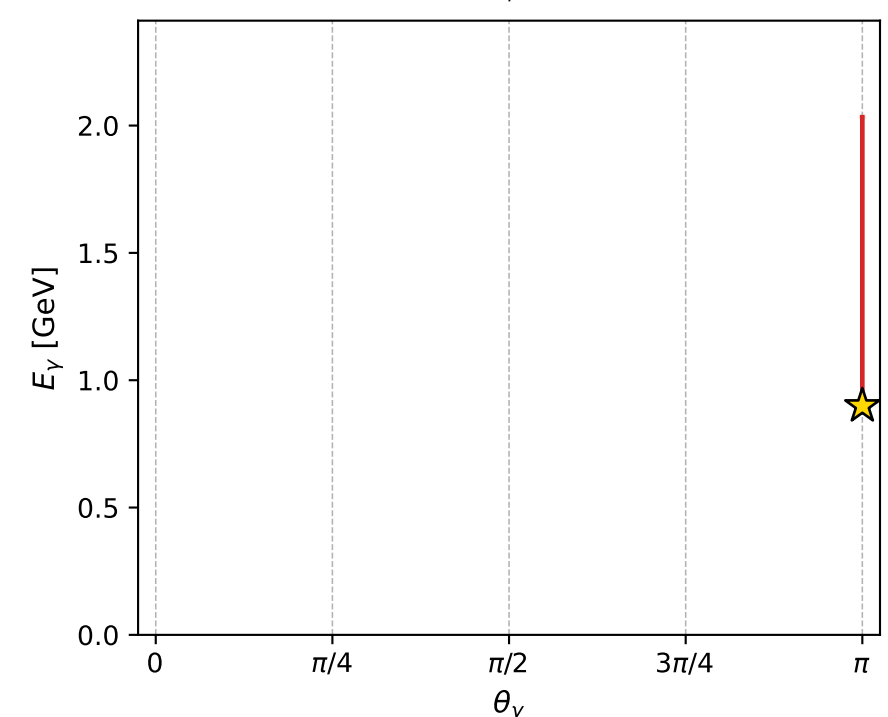
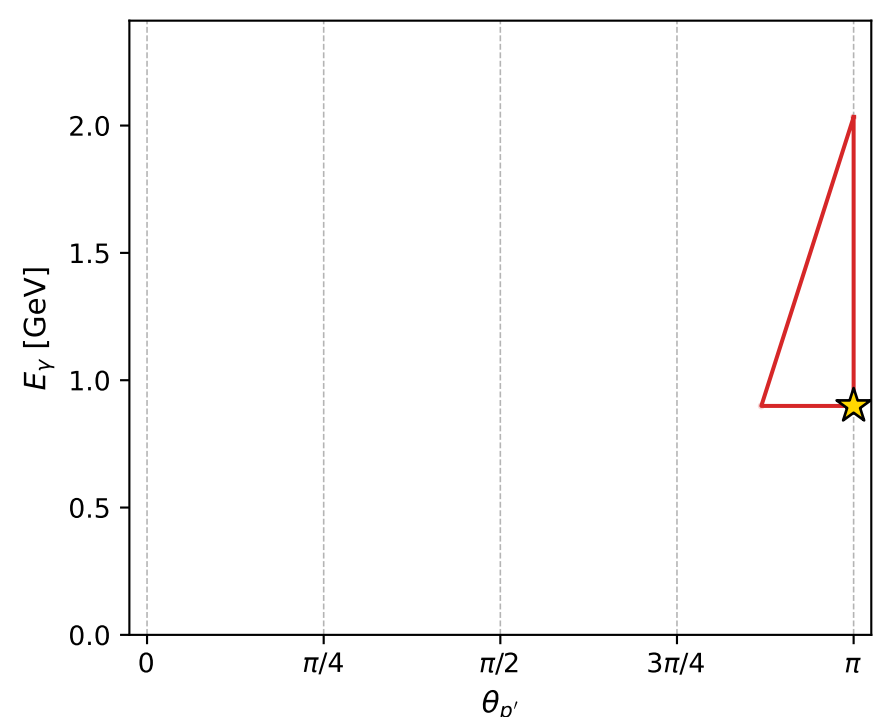
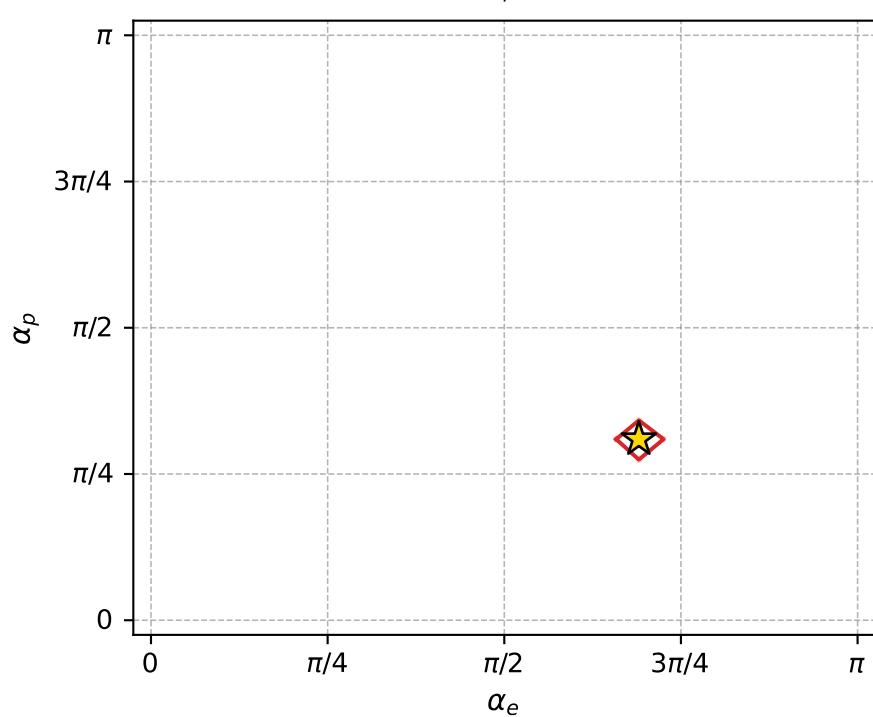
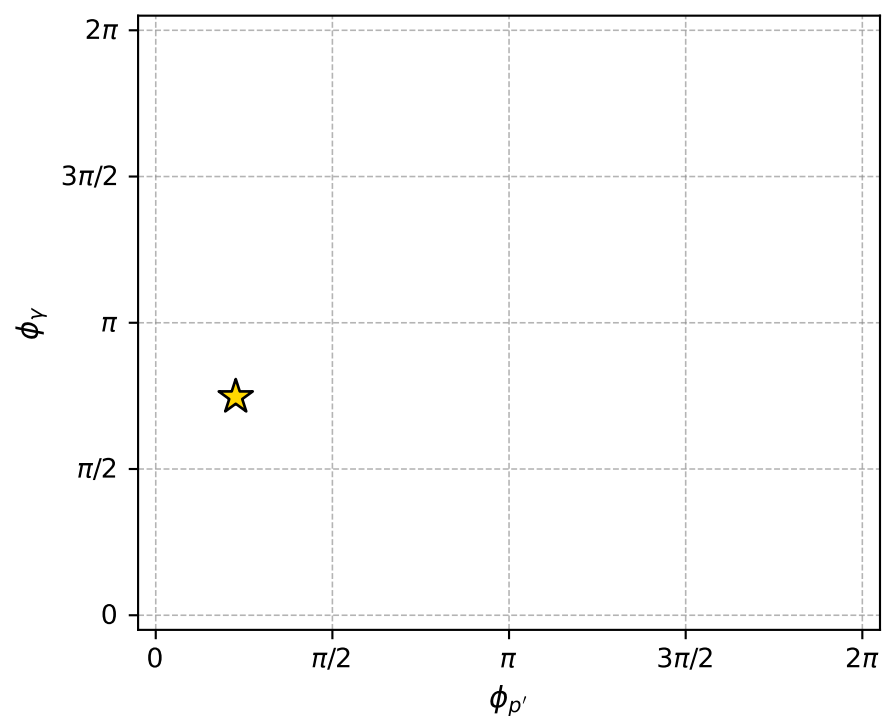
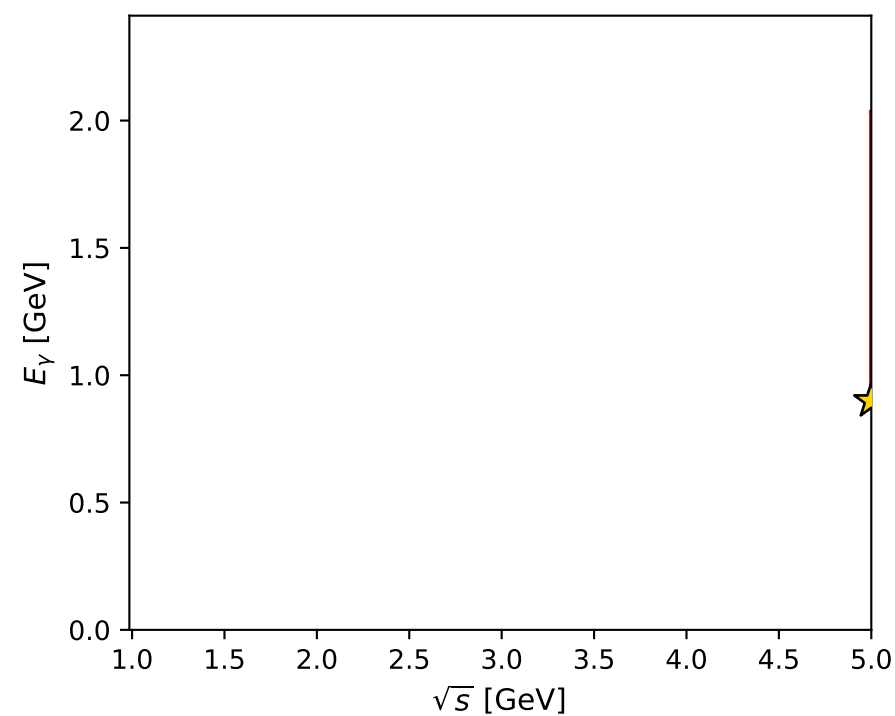
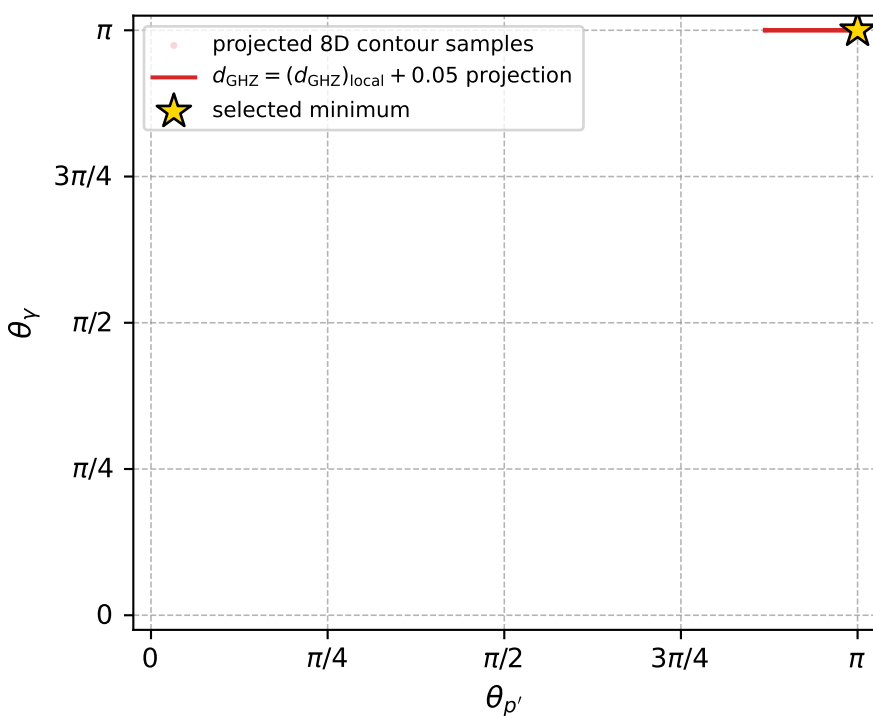


$(h_p, h_Y, h_l) = (+1, -1, +1)$

$(h_p, h_Y, h_l) = (-1, +1, -1)$



electron: polarization cluster P3, local minimum 13; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4622181\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $2.3387191\text{e-}09$
 8D contour samples = 30

selected phase-space point:

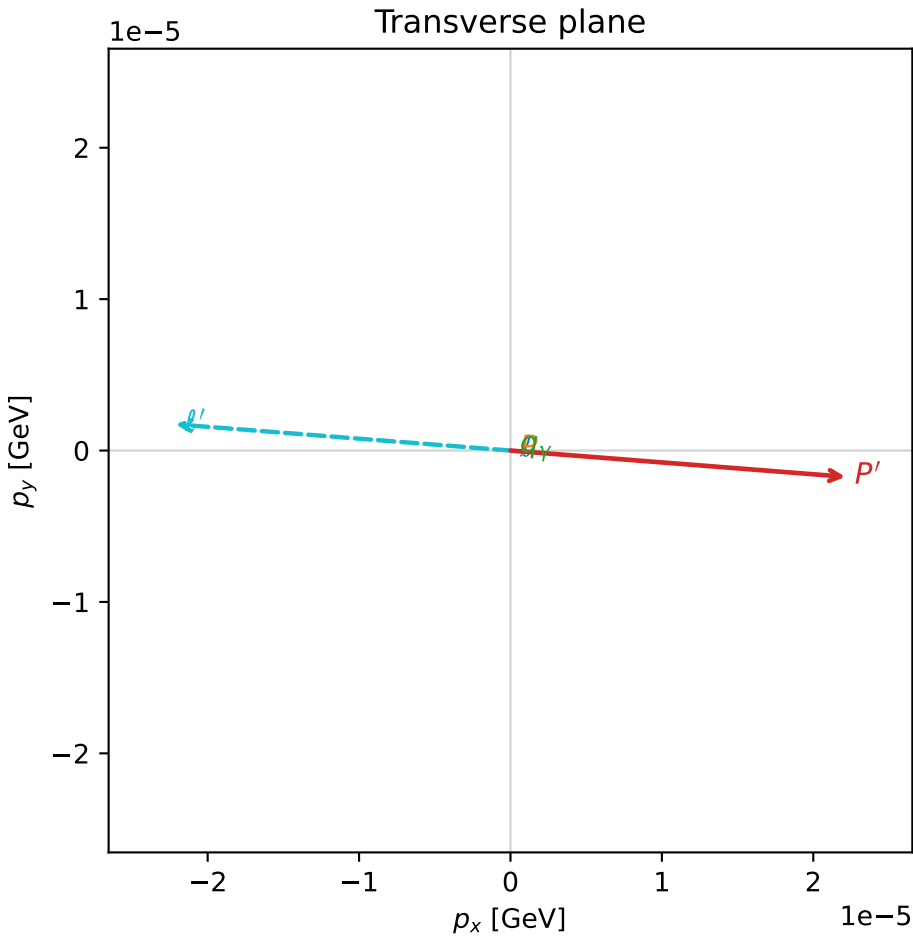
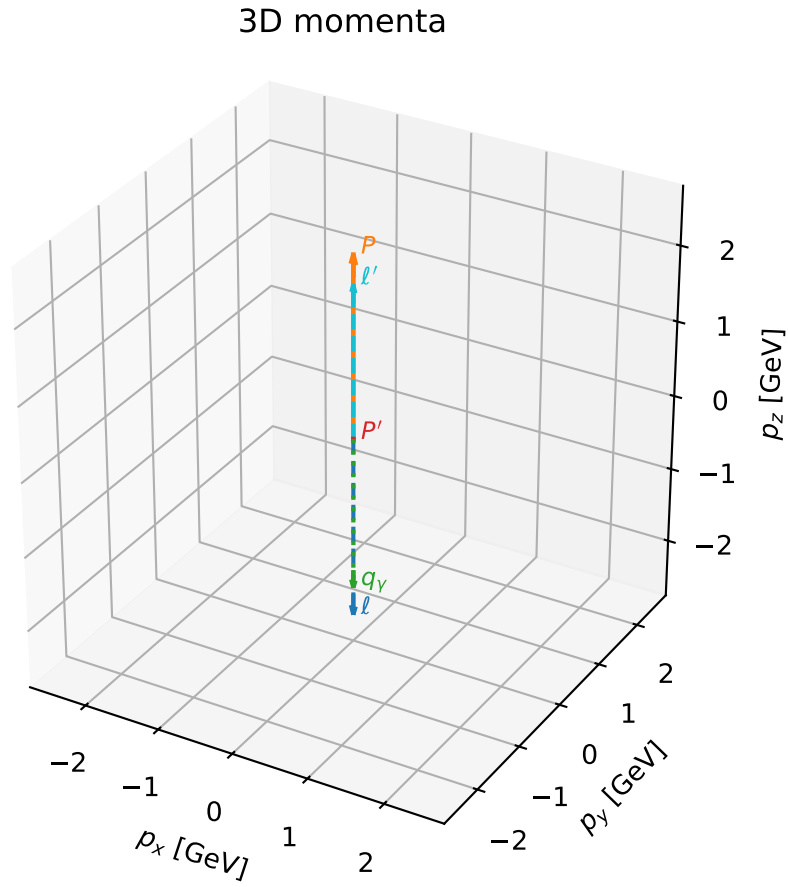
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141592$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.899044$
 $\text{phi_p_out} = 0.7113552$
 $\text{phi_gamma_out} = 2.345024$
 $\text{alpha_e} = 2.169189$
 $\text{alpha_p} = 0.9723904$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_15 $d_{\{\text{rm GHZ}\}}=2.47594\text{e-}08$
region: polarization_cluster_03_local_minimum_15
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.4020754$ rad
incoming lepton: $-0.738794|+\rangle +0.673931|-\rangle$
proton mixing angle: $\alpha_p=0.73950355$ rad
incoming proton: $0.738803|+\rangle +0.673921|-\rangle$

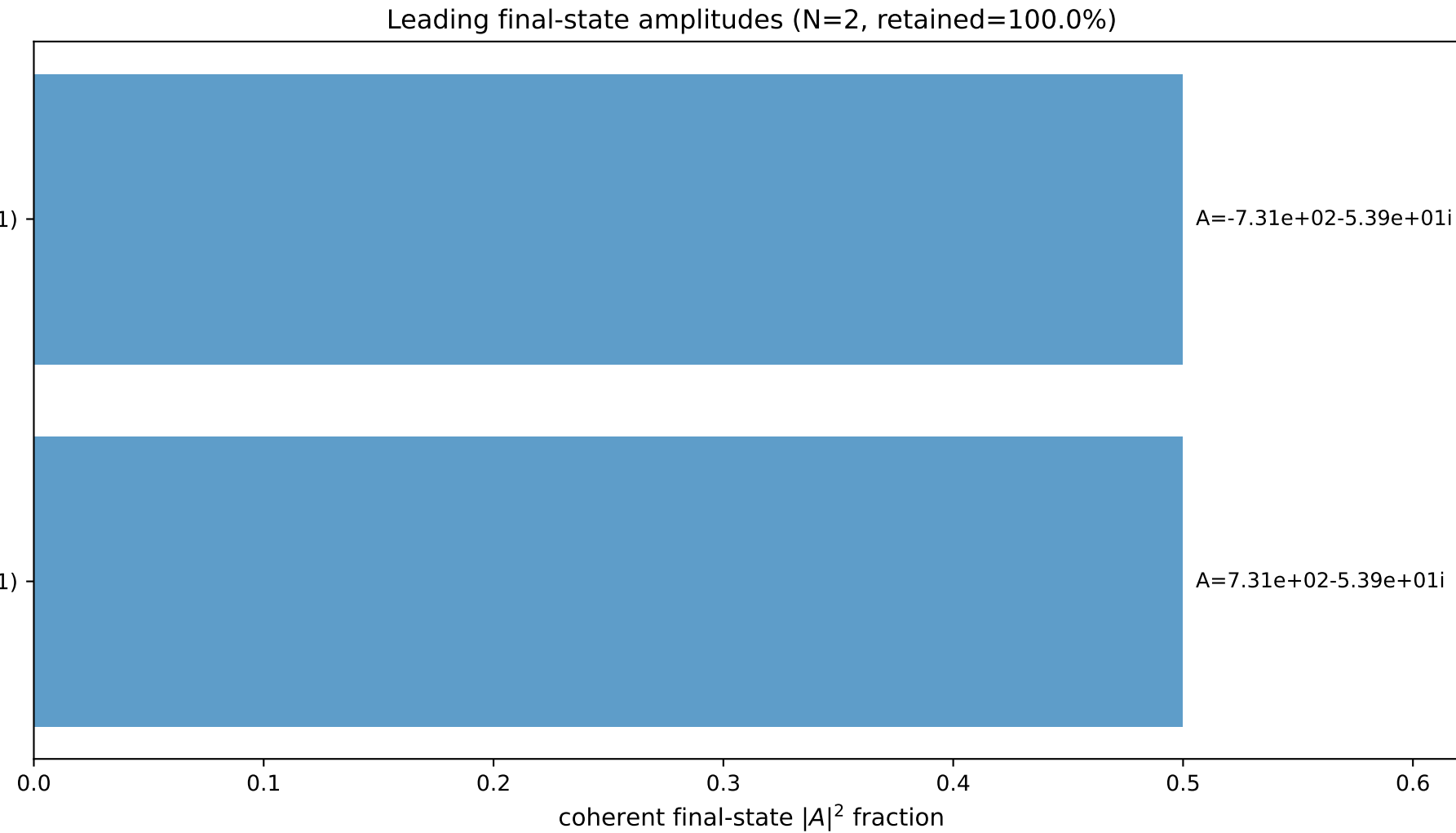
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.00056453$
 $\theta_{P'}=0.0393096$, $\theta_Y=3.14159$, $\phi_{P'}=6.20497$, $\phi_Y=0.0735511$
 $E_Y=2.03121$, $\theta_{\ell'}=1.09255\text{e-}05$, $\phi_{\ell'}=3.06338$
 $Q^2=19.5912$, $x_B=0.837086$, $t=-3.09252$, $W^2=4.69268$, $y=0.970368$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0306$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0306$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0312$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0312$

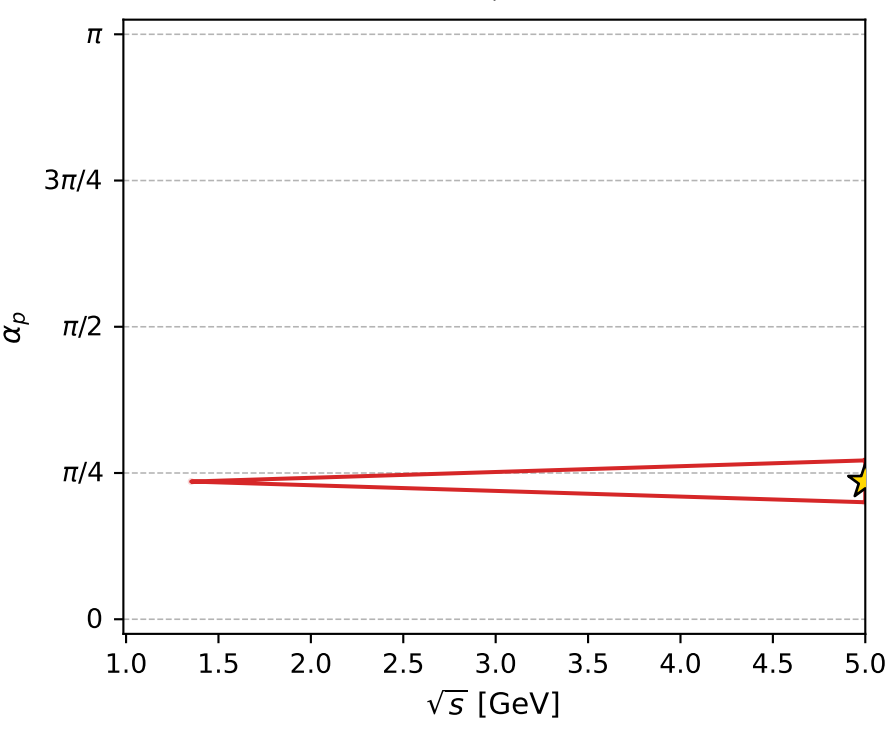
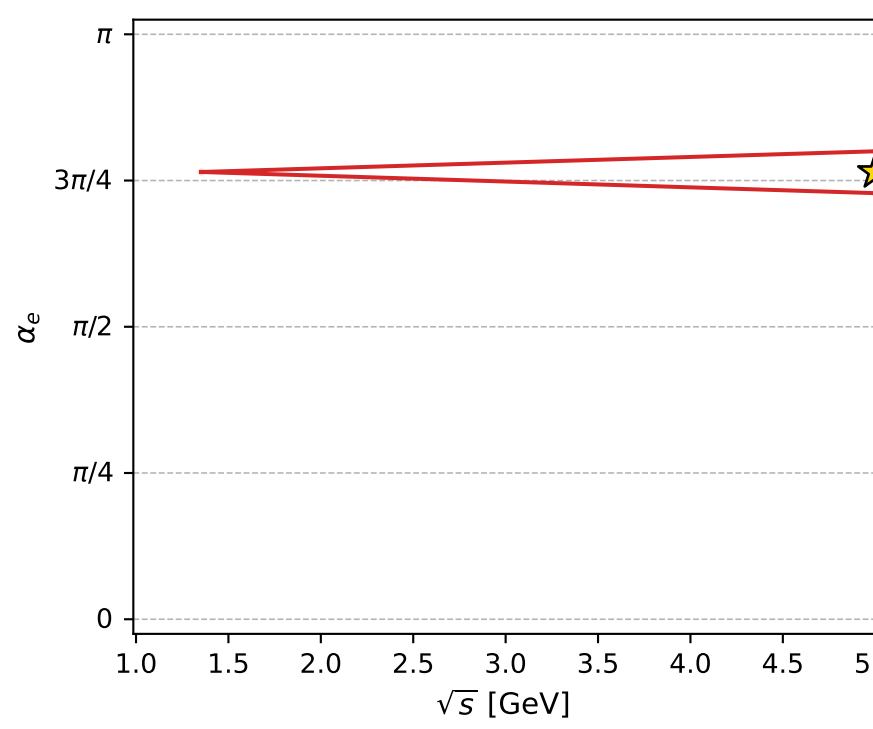
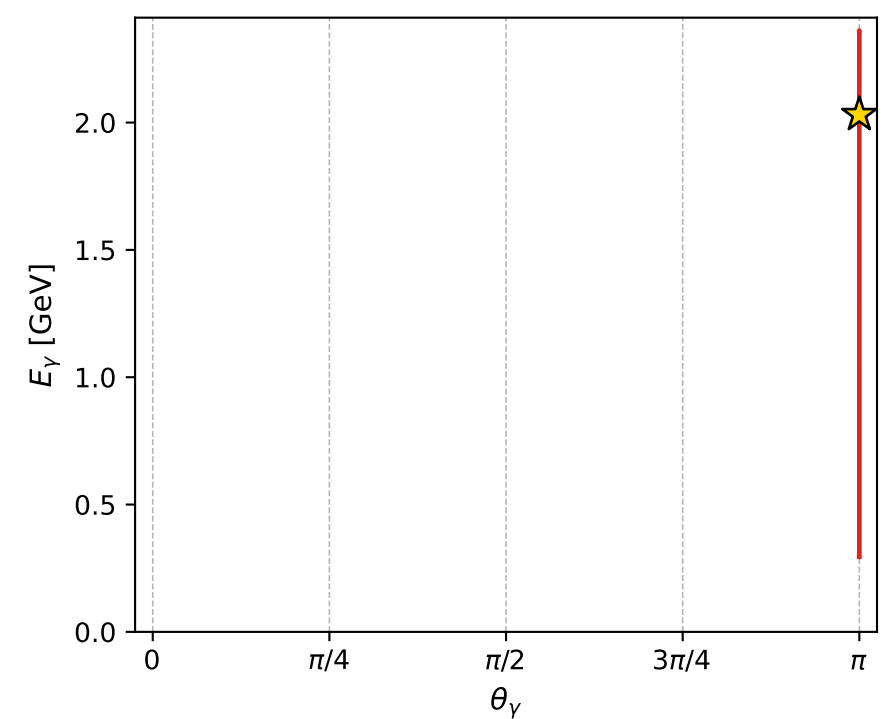
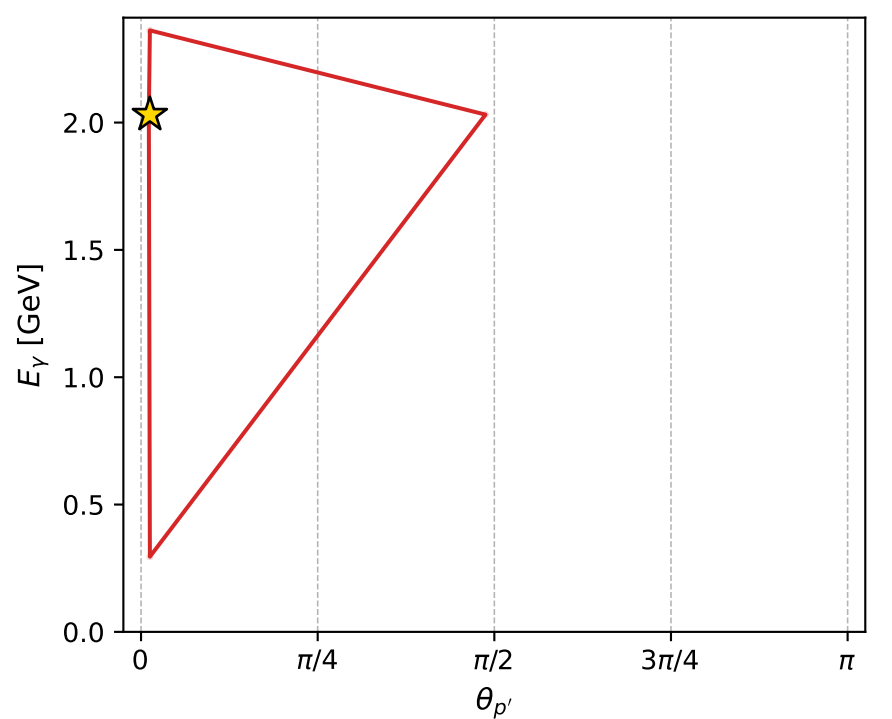
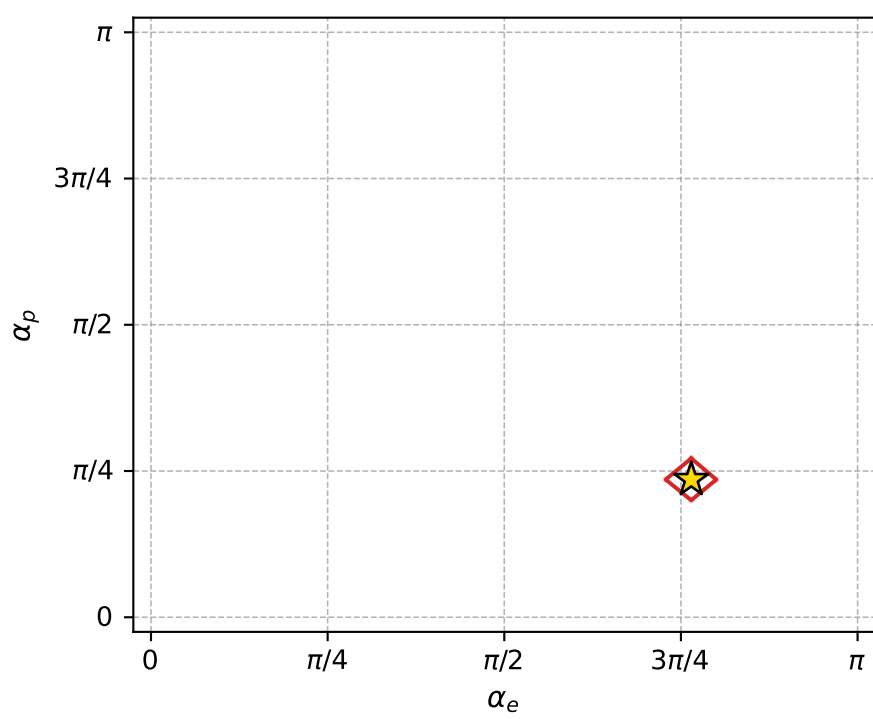
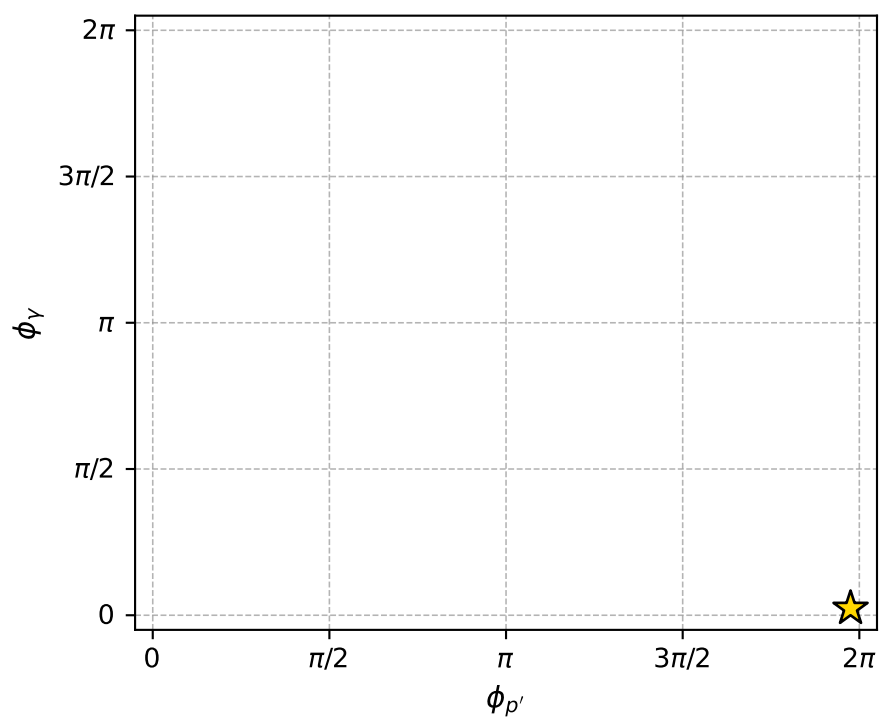
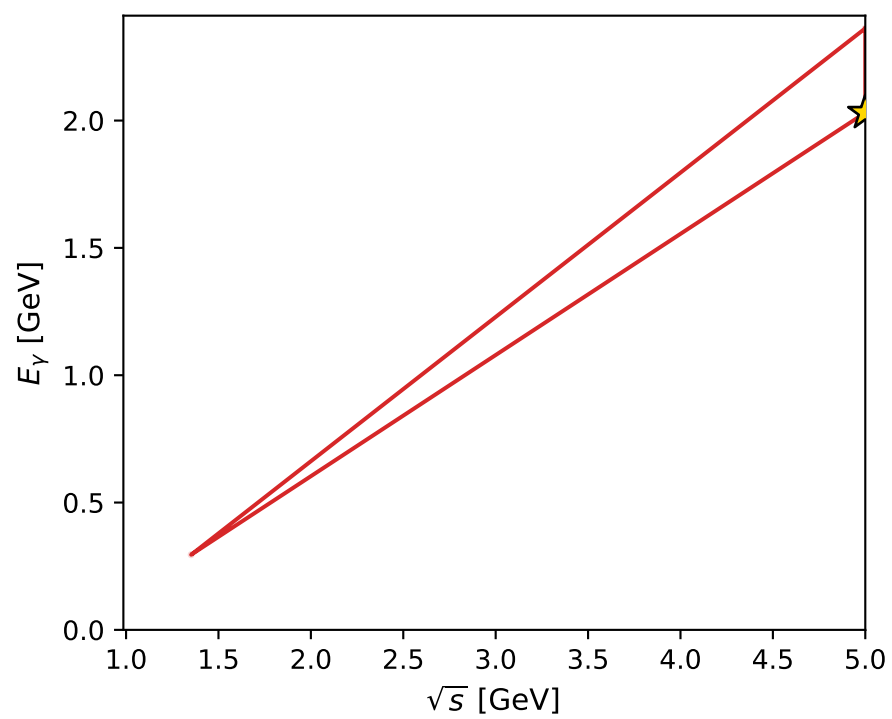
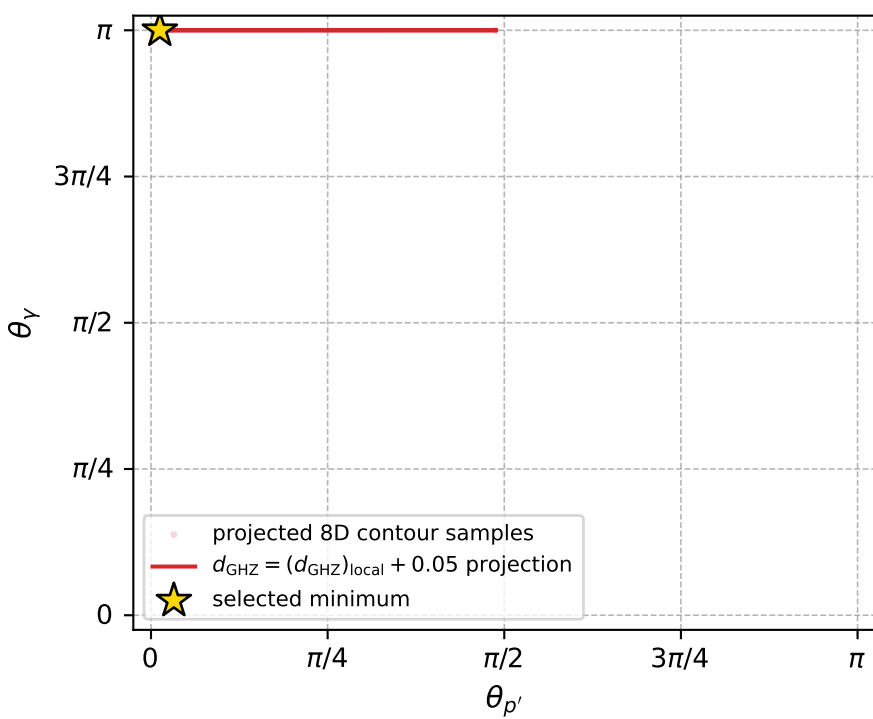


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 15; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4759392\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $2.4759297\text{e-}09$
 8D contour samples = 102

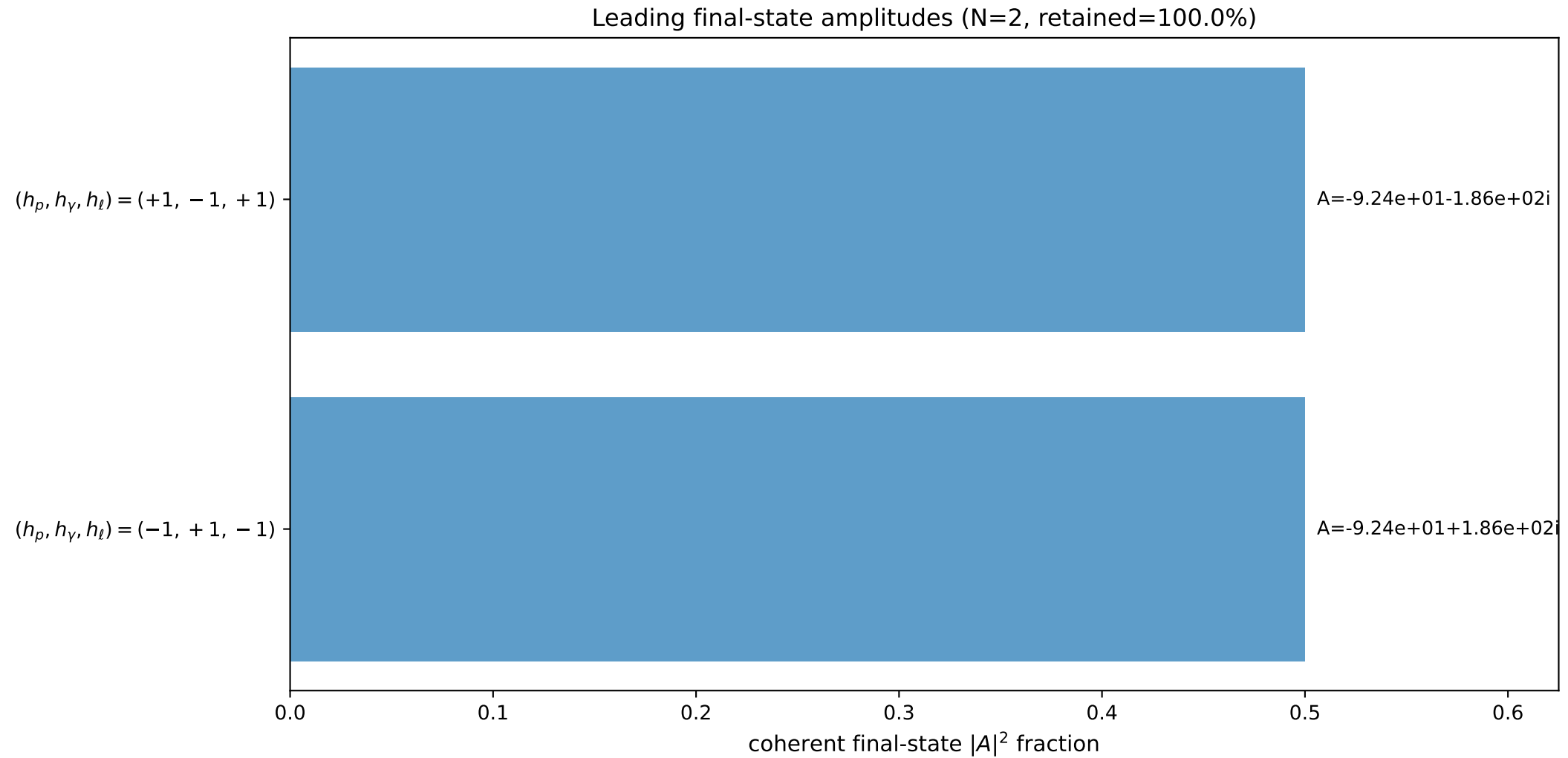
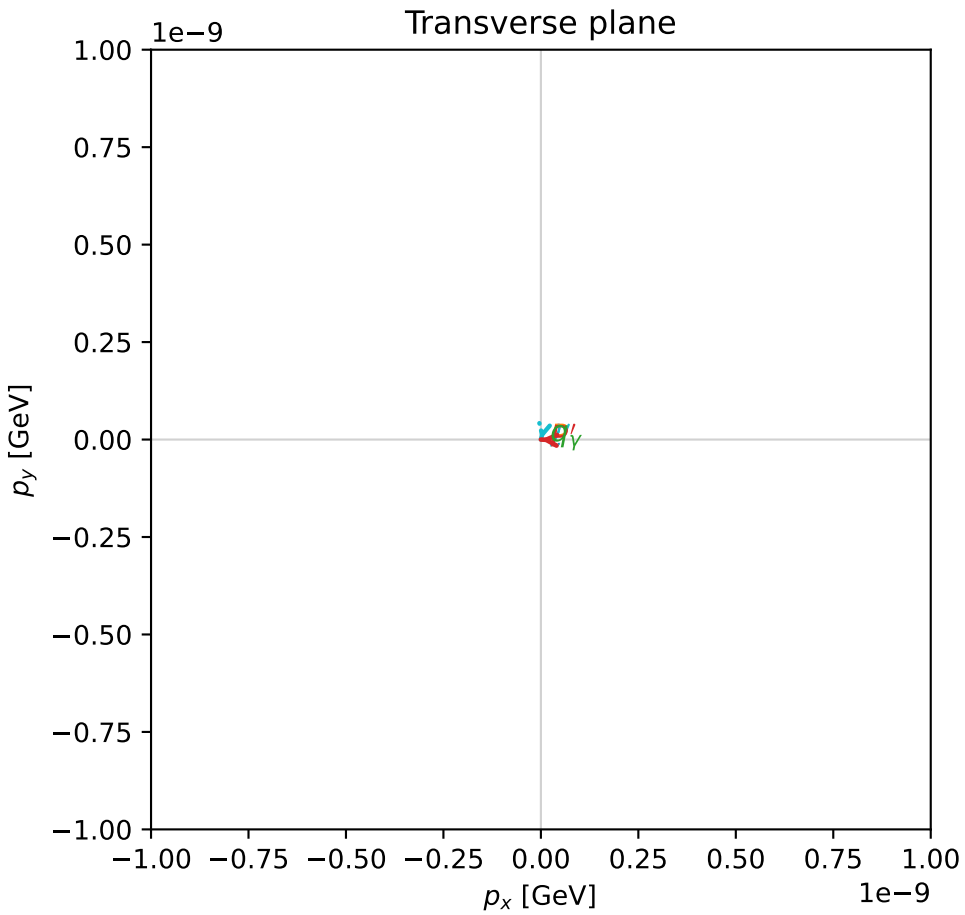
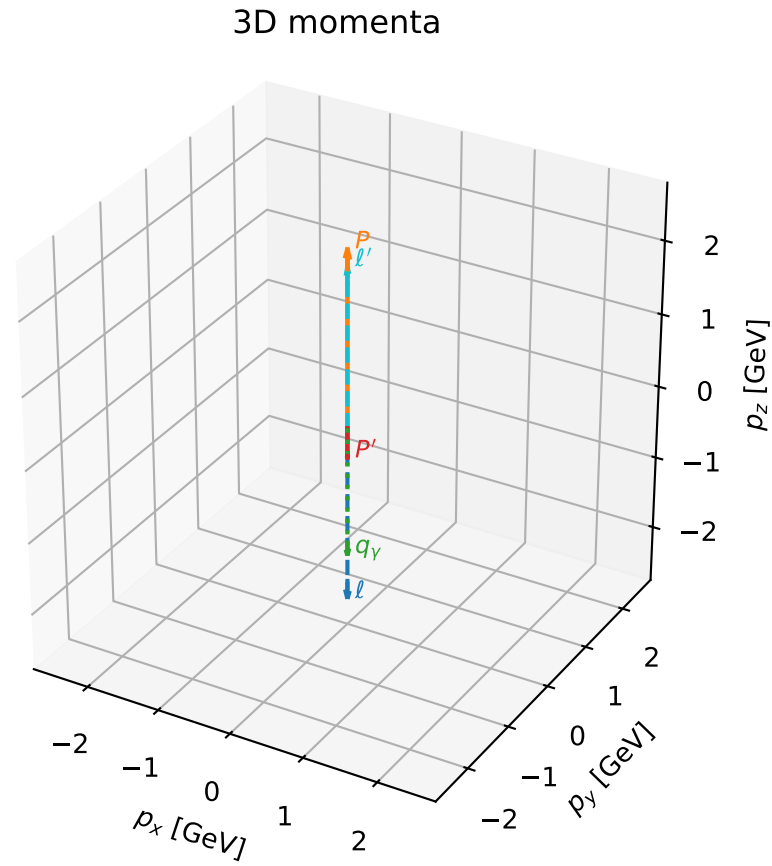
selected phase-space point:
 $\text{sqrt_s} = 4.999853$
 $\text{theta_p_out} = 0.03930963$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.031208$
 $\text{phi_p_out} = 6.204971$
 $\text{phi_gamma_out} = 0.07355107$
 $\alpha_e = 2.402075$
 $\alpha_p = 0.7395036$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

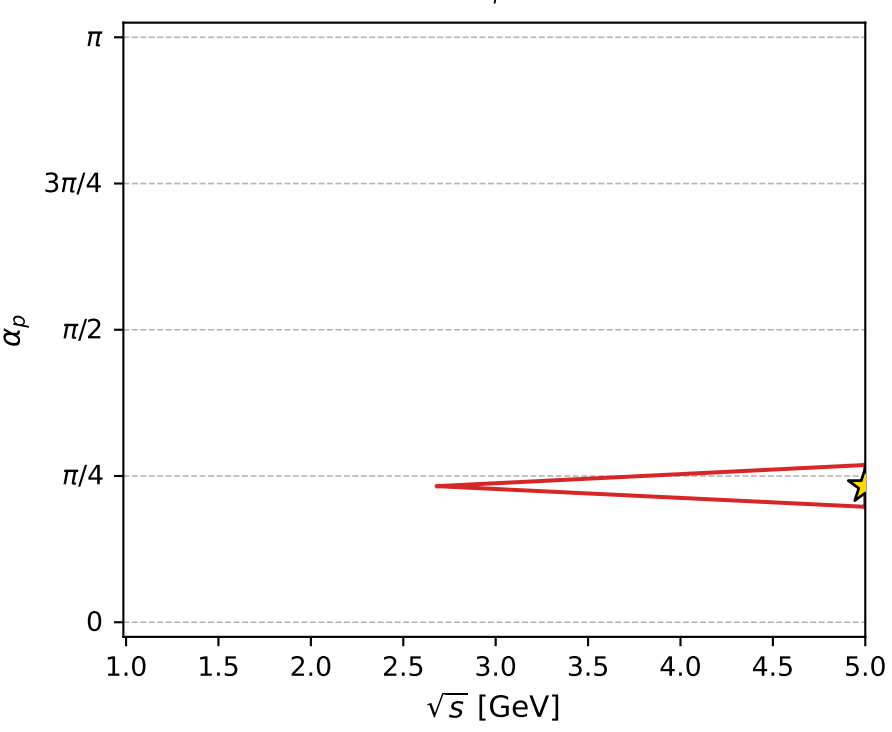
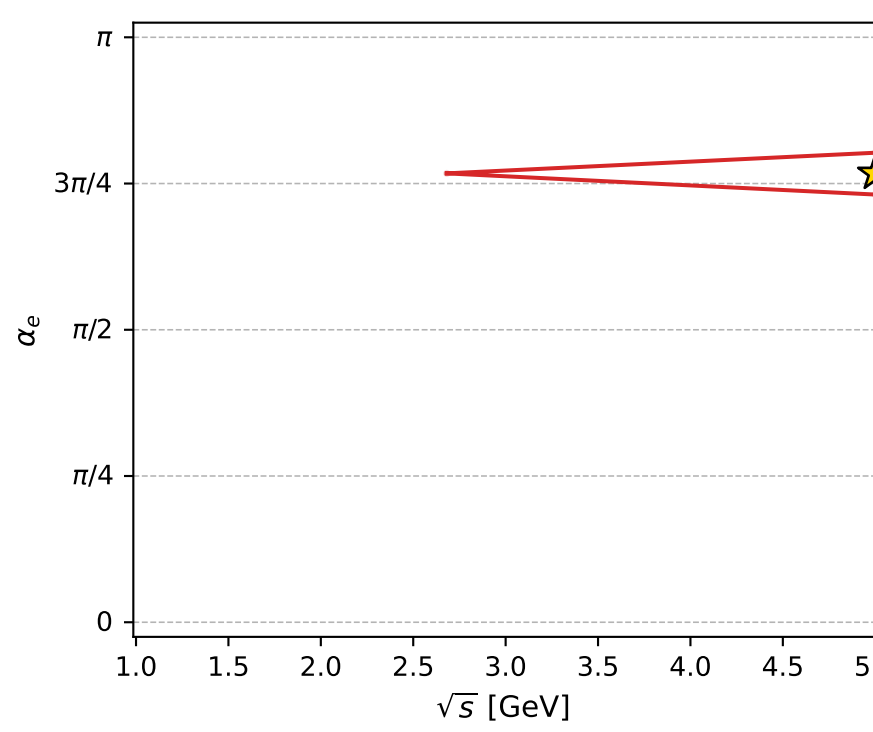
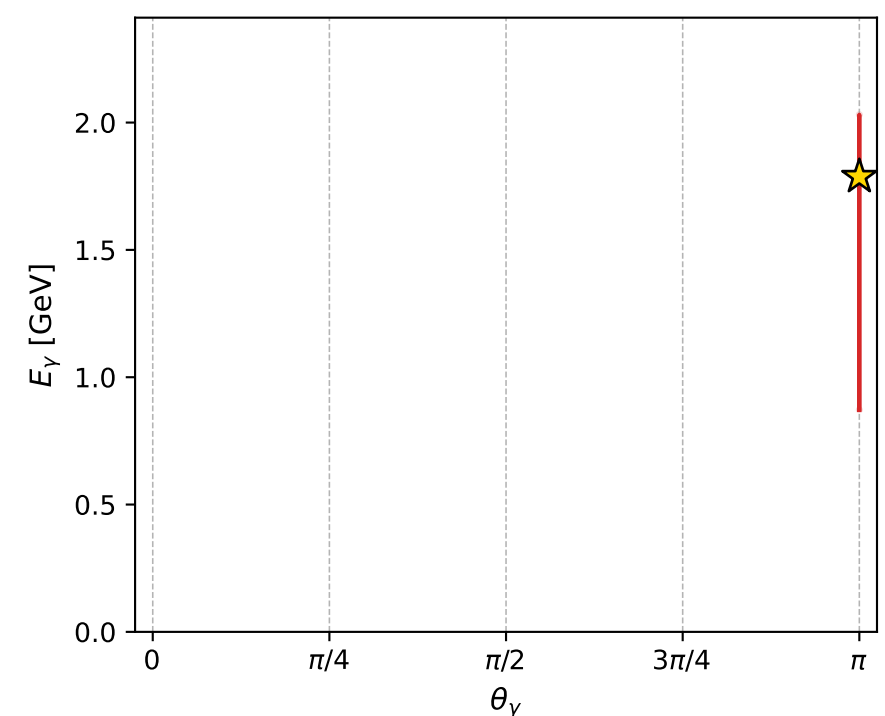
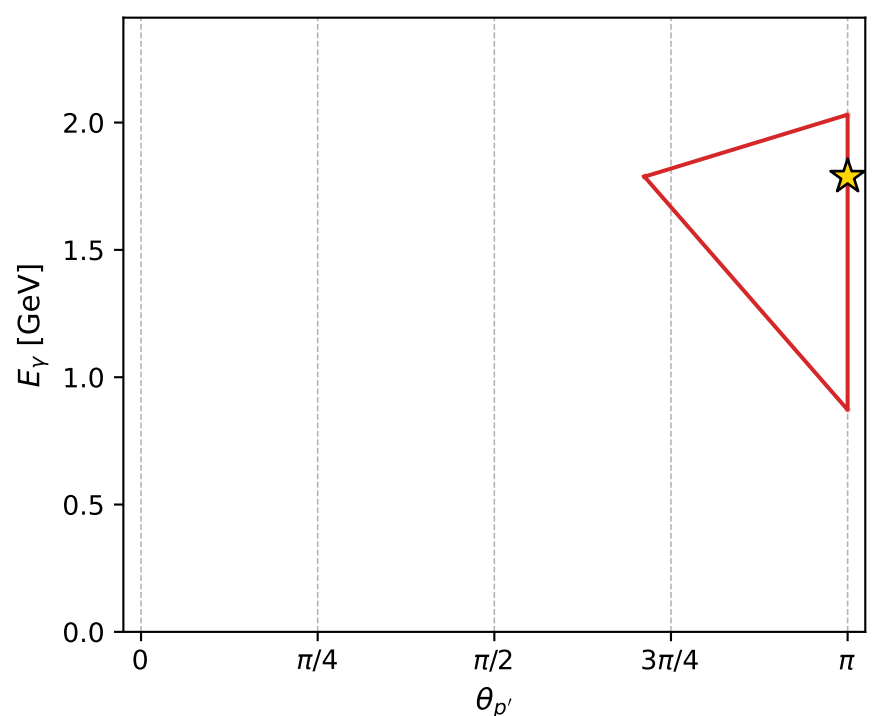
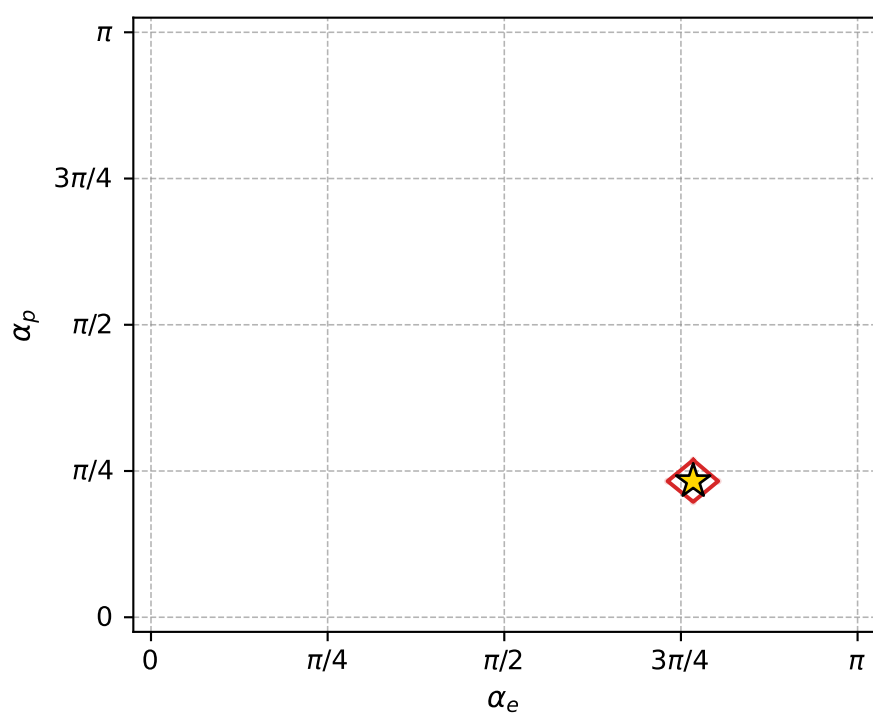
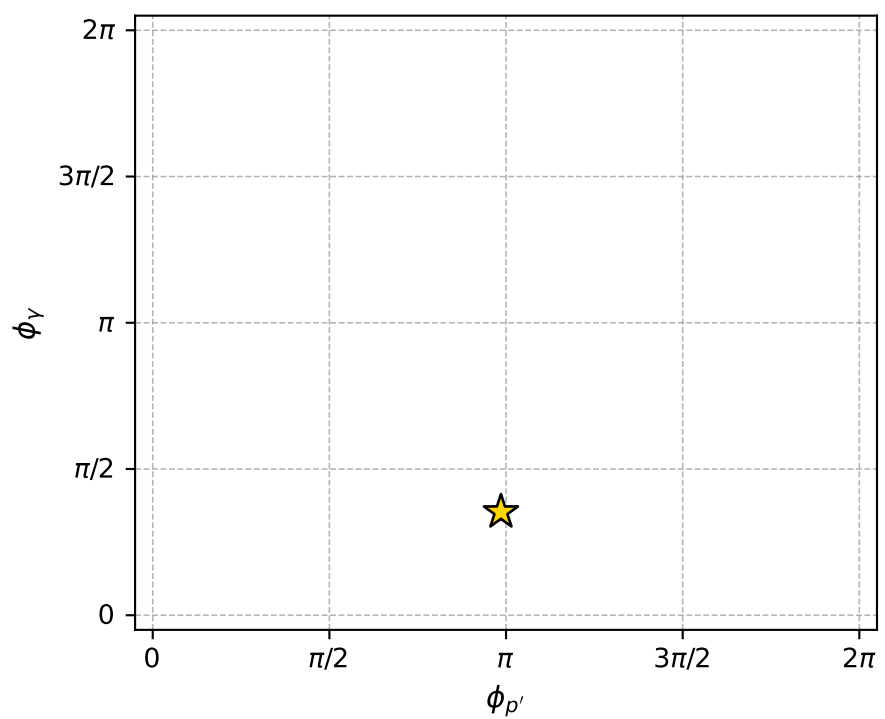
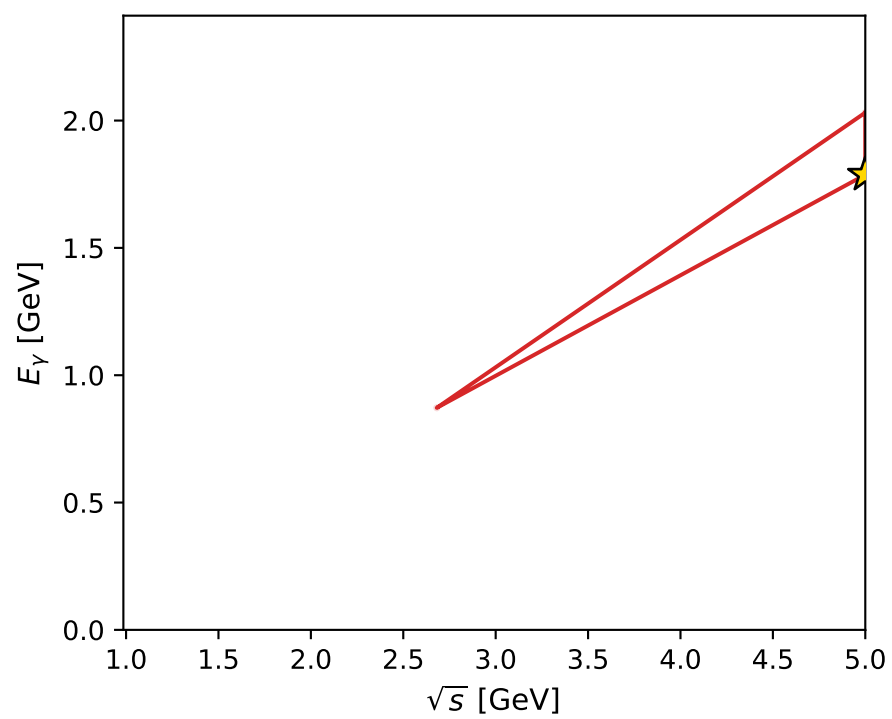
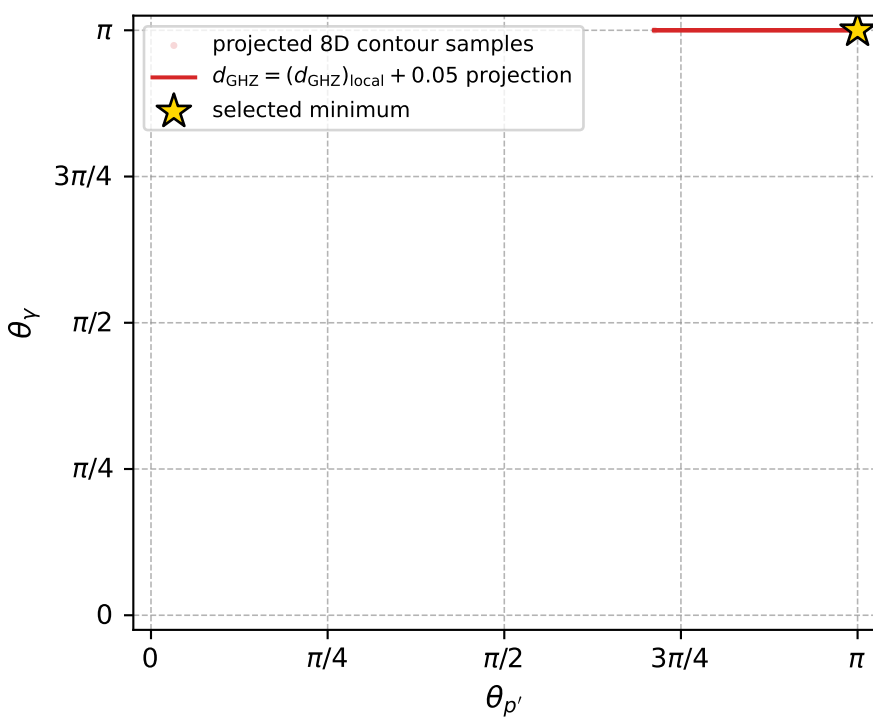
dghz_mixing_angles_polarization_03_local_minimum_17 d_{\rm GHZ}=2.4853e-08
region: polarization_cluster_03_local_minimum_17
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.4106629$ rad
incoming lepton: $-0.744554|+\rangle +0.667562|-\rangle$
proton mixing angle: $\alpha_p=0.73092649$ rad
incoming proton: $0.744556|+\rangle +0.66756|-\rangle$

$s=24.9975$, $\sqrt{s}=4.99975$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.403477$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=3.09629$, $\phi_Y=1.10925$
 $E_Y=1.78759$, $\theta_{l'}=0$, $\phi_{l'}=4.47422$
 $Q^2=21.1384$, $x_B=0.905421$, $t=-5.47151$, $W^2=3.08794$, $y=0.968025$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 l' $E= 2.1911$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1911$
 P' $E= 1.0211$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4035$
 q_Y $E= 1.7876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7876$



electron: polarization cluster P3, local minimum 17; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.485303\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $2.5695681\text{e-}09$
 8D contour samples = 55

selected phase-space point:

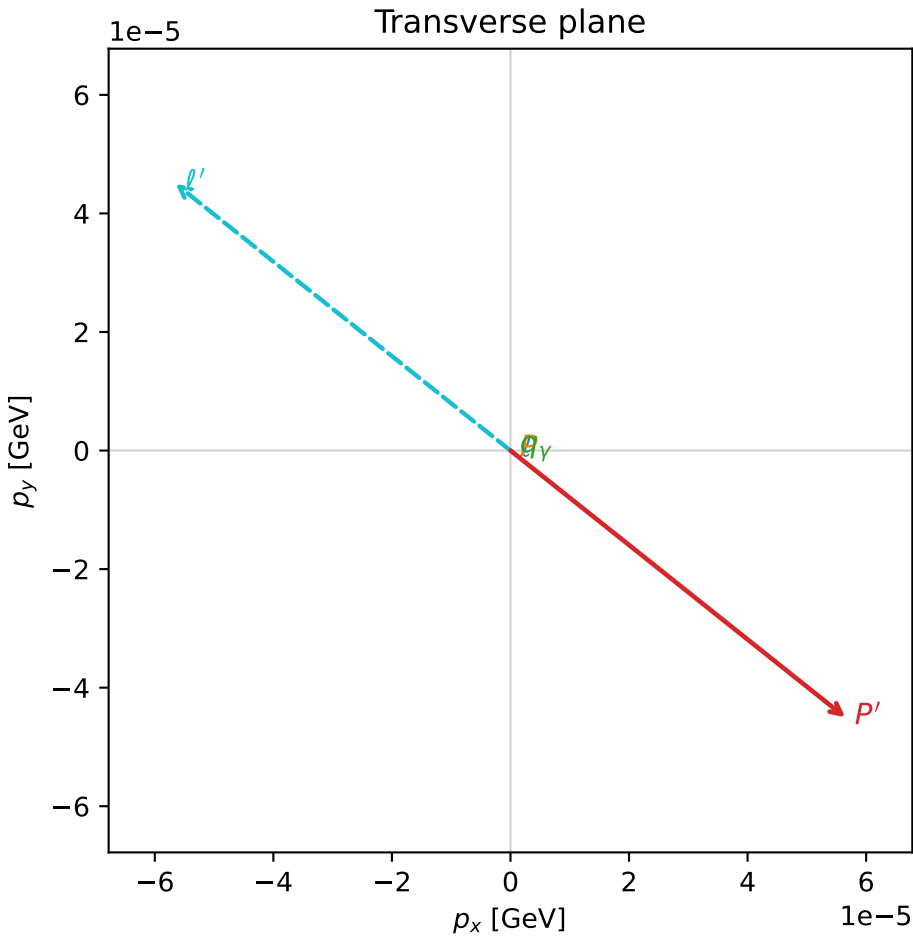
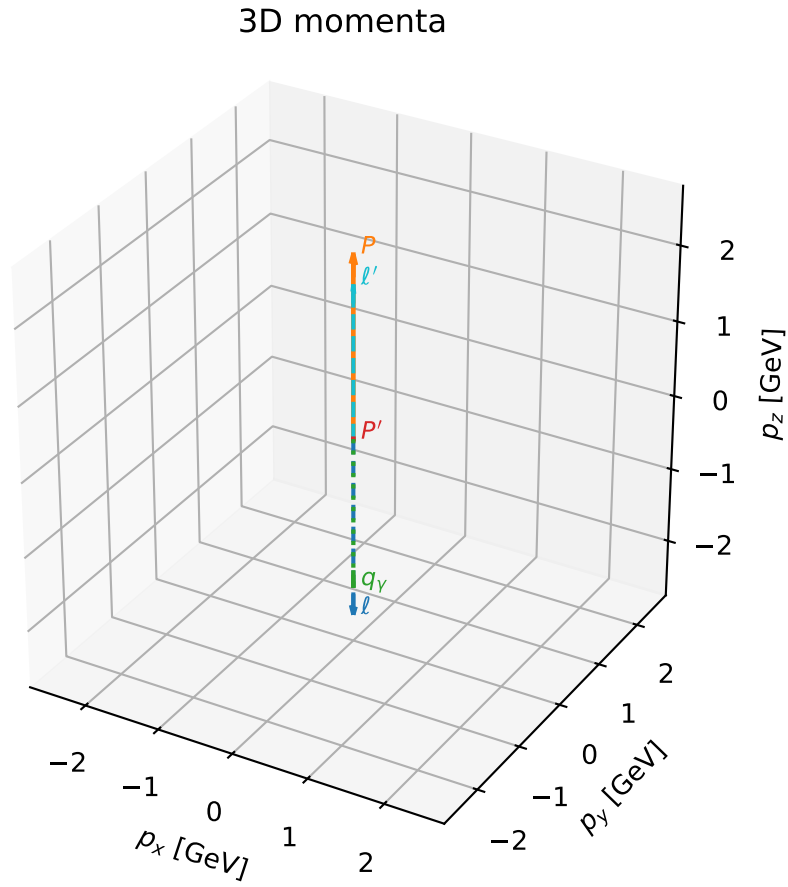
$\text{sqrt_s} = 4.999752$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.787589$
 $\text{phi_p_out} = 3.096293$
 $\text{phi_gamma_out} = 1.109251$
 $\text{alpha_e} = 2.410663$
 $\text{alpha_p} = 0.7309265$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_20 $d_{\{\text{rm GHZ}\}}=2.54044\text{e-}08$
region: polarization_cluster_03_local_minimum_20
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3449091$ rad
incoming lepton: $-0.699082|+\rangle +0.715042|-\rangle$
proton mixing angle: $\alpha_p=0.79667666$ rad
incoming proton: $0.699087|+\rangle +0.715037|-\rangle$

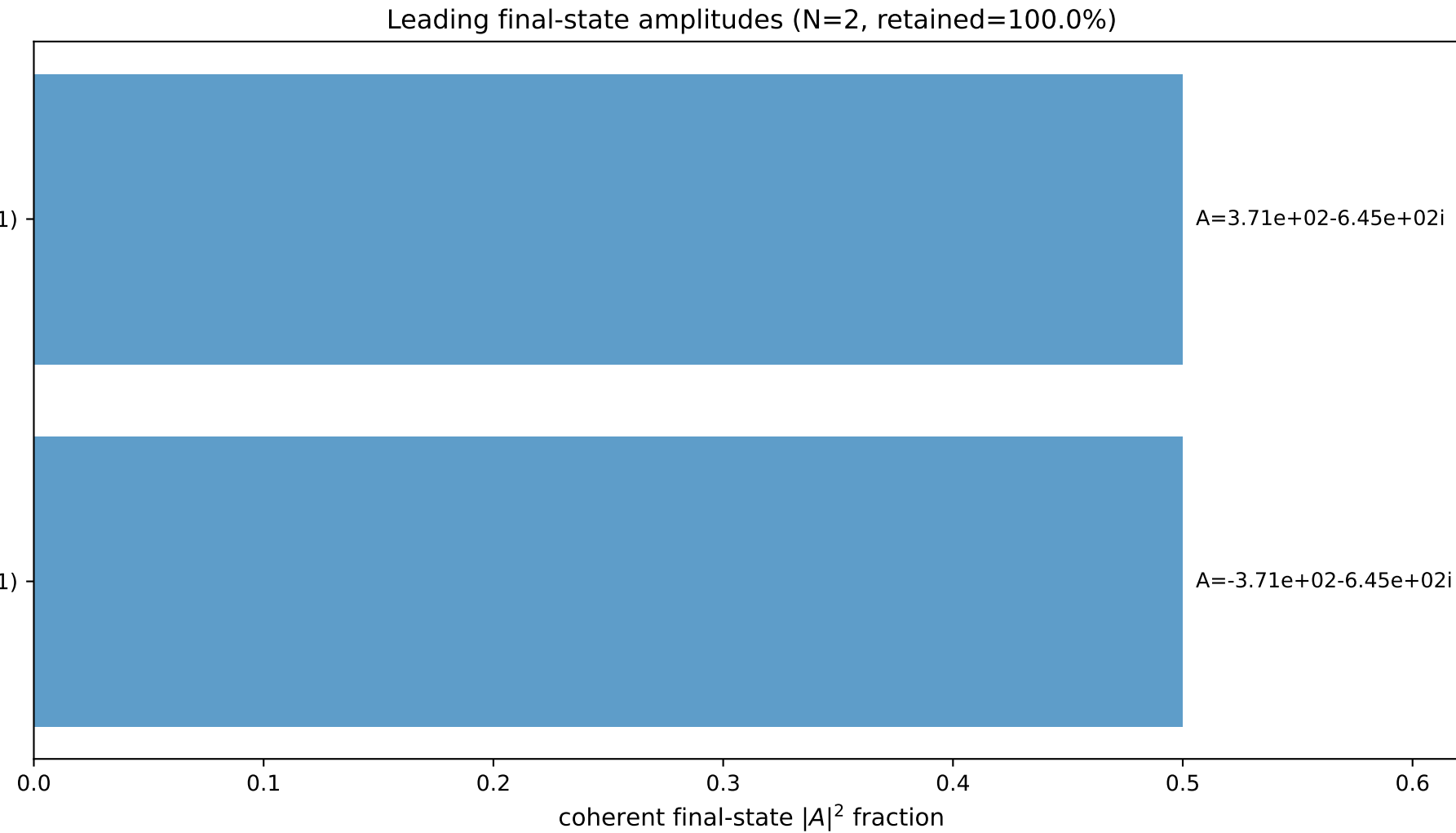
$s=24.9686$, $\sqrt{s}=4.99686$, $|\vec{P}|=2.41039$, $|\vec{P}'|=0.0036789$
 $\theta_{P'}=0.019635$, $\theta_Y=3.14159$, $\phi_{P'}=5.61047$, $\phi_Y=2.09314$
 $E_Y=2.03127$, $\theta_{\ell'}=3.56238\text{e-}05$, $\phi_{\ell'}=2.46887$
 $Q^2=19.5491$, $x_B=0.836335$, $t=-3.07484$, $W^2=4.70547$, $y=0.970358$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 ℓ' $E= 2.0276$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0037$
 q_Y $E= 2.0313$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0313$

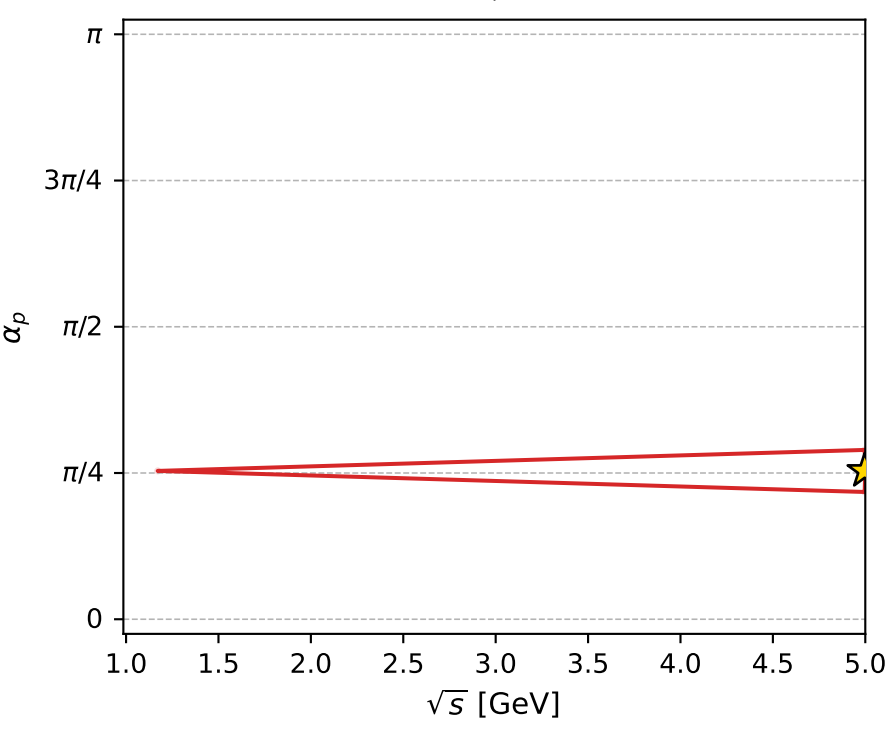
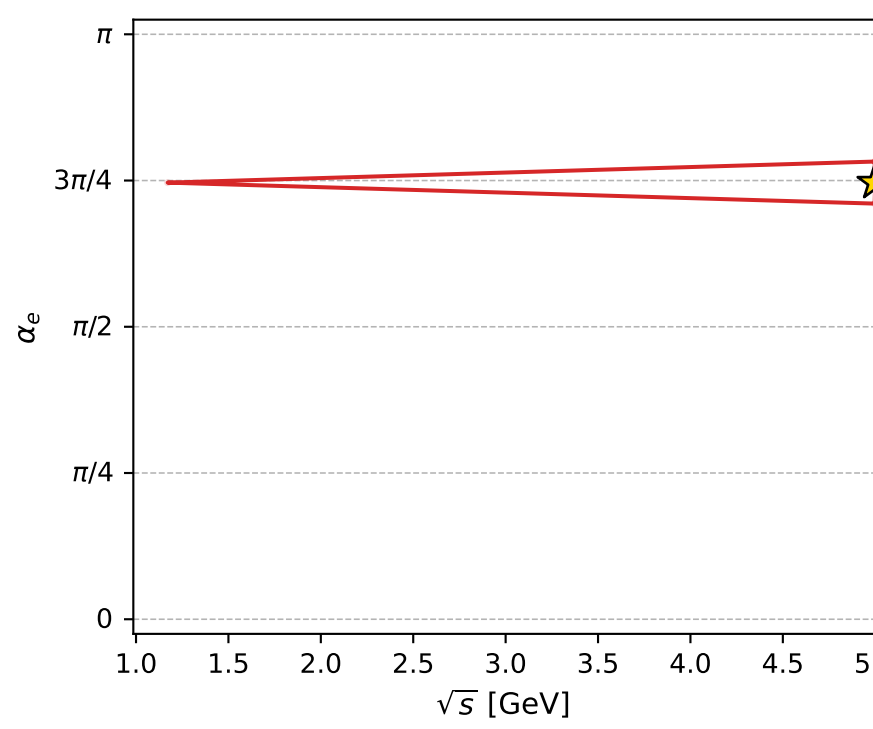
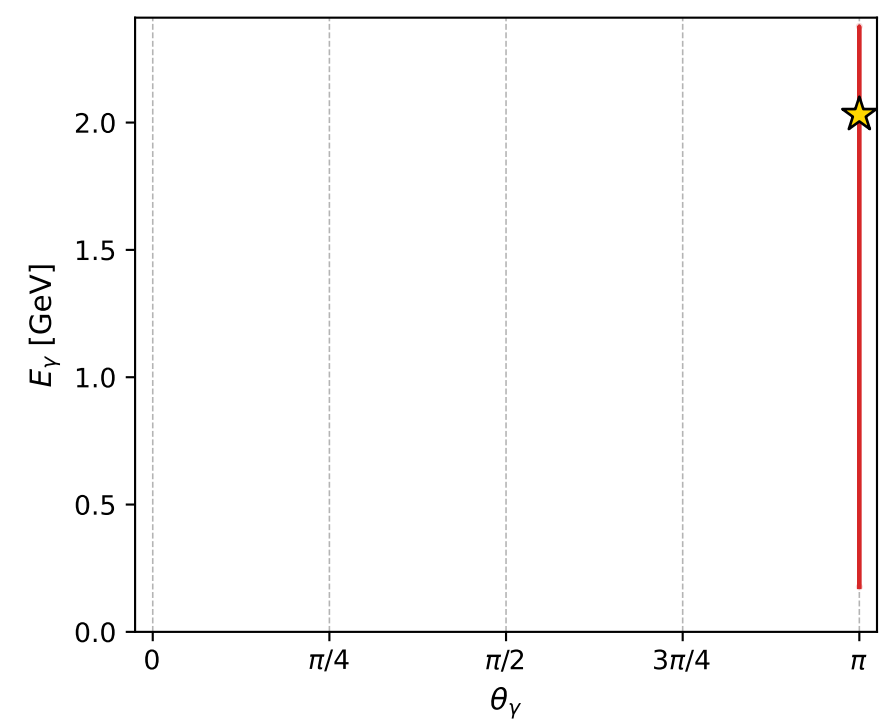
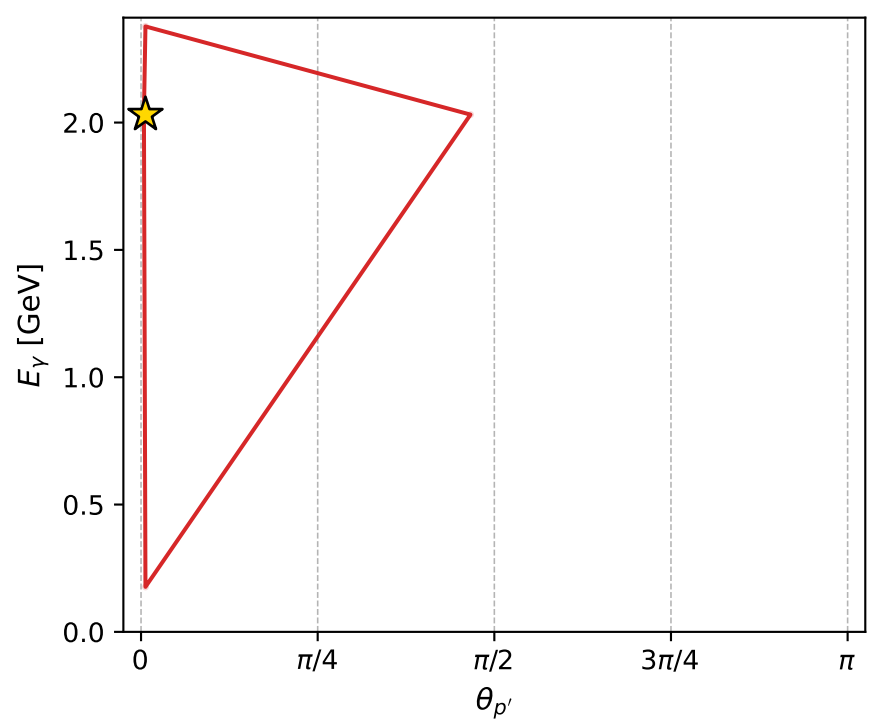
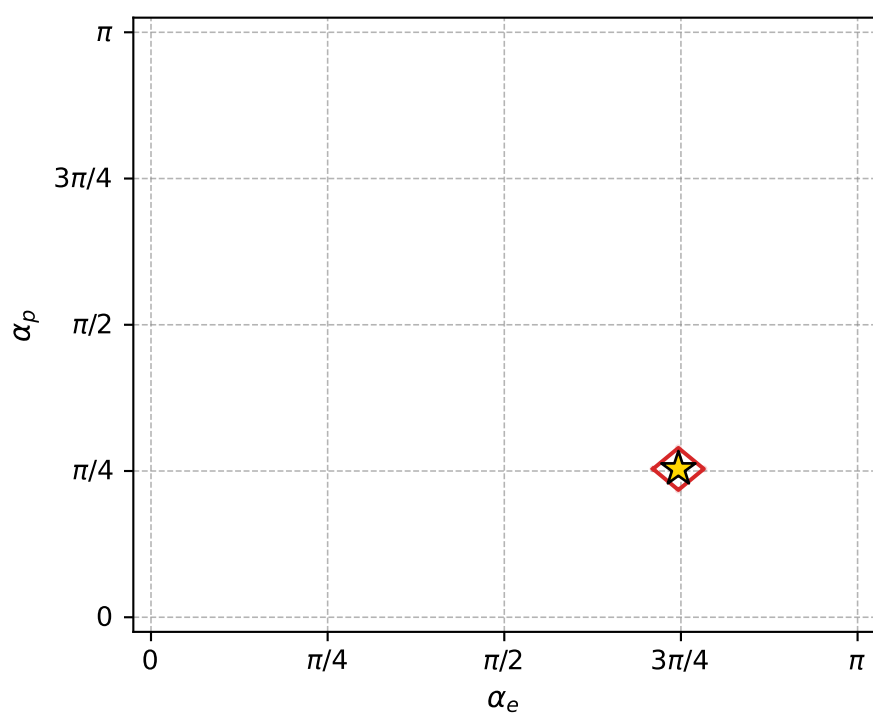
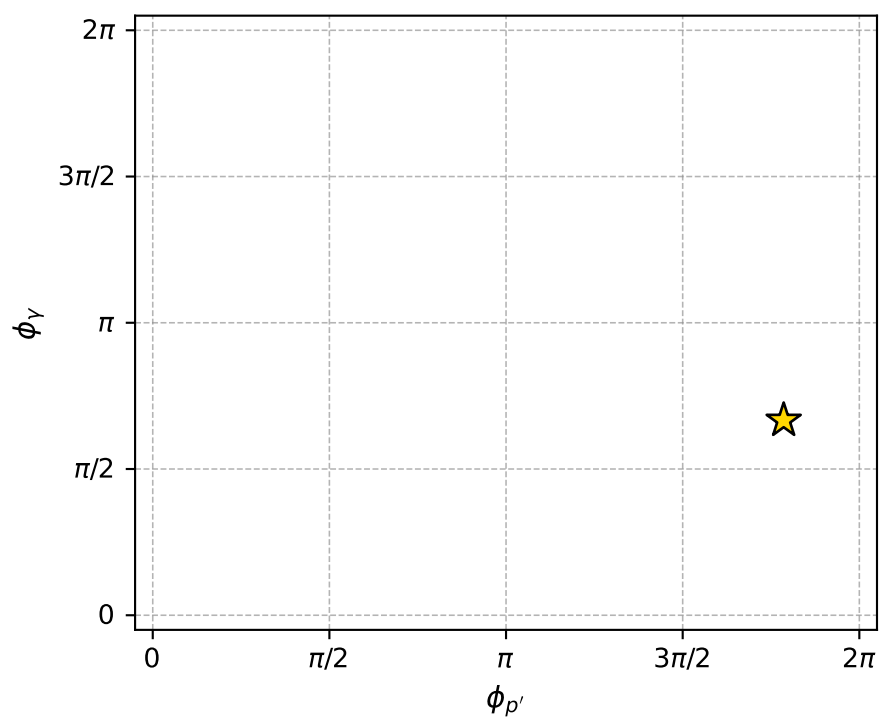
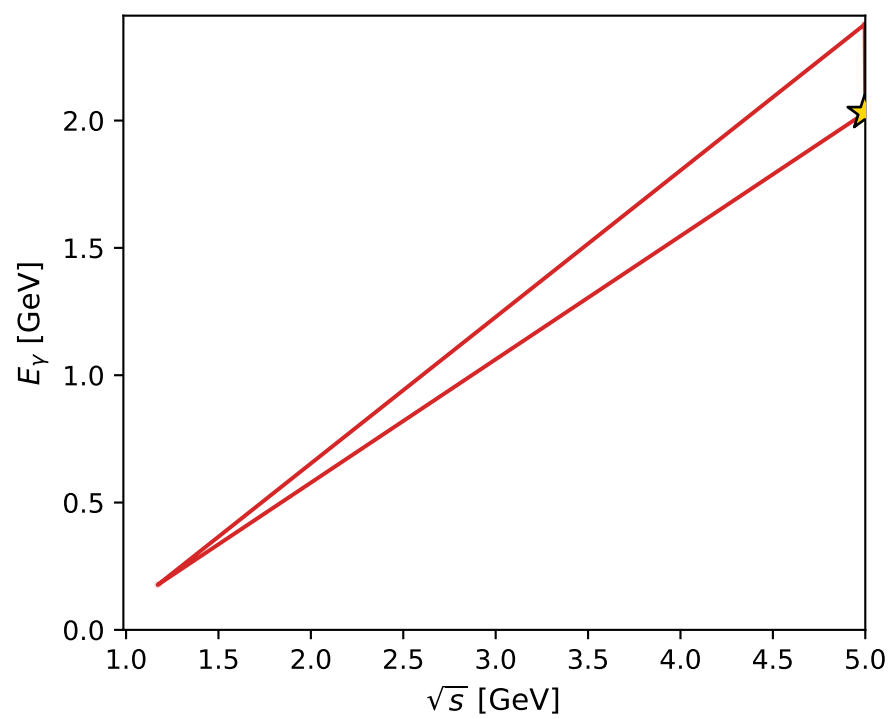
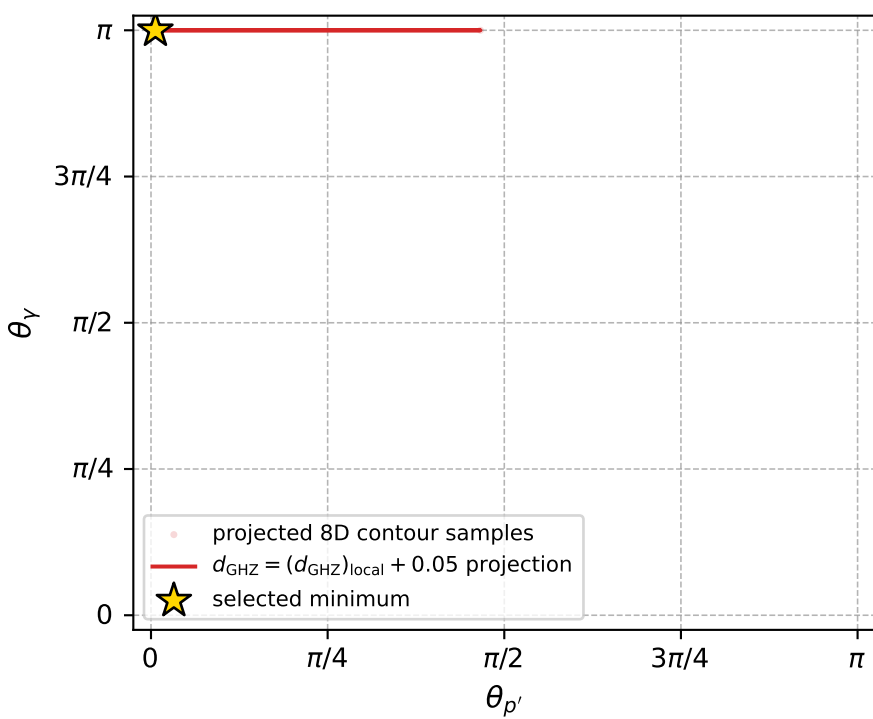


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 20; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5404386\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $3.120924\text{e-}09$
 8D contour samples = 123

selected phase-space point:
 $\text{sqrt_s} = 4.996864$
 $\text{theta_p_out} = 0.01963495$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.031267$
 $\text{phi_p_out} = 5.610466$
 $\text{phi_gamma_out} = 2.093138$
 $\text{alpha_e} = 2.344909$
 $\text{alpha_p} = 0.7966767$

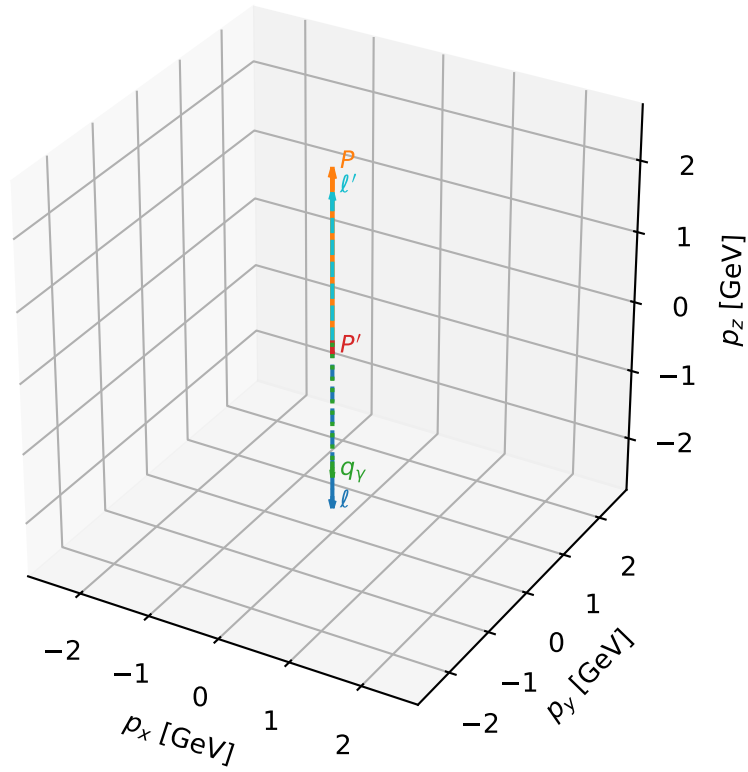
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_38 d_{\rm GHZ}=2.64248e-08
region: polarization_cluster_03_local_minimum_38
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.4362483$ rad
incoming lepton: $-0.761388|+> +0.648296|->$
proton mixing angle: $\alpha_p=0.70535267$ rad
incoming proton: $0.761383|+> +0.648302|->$

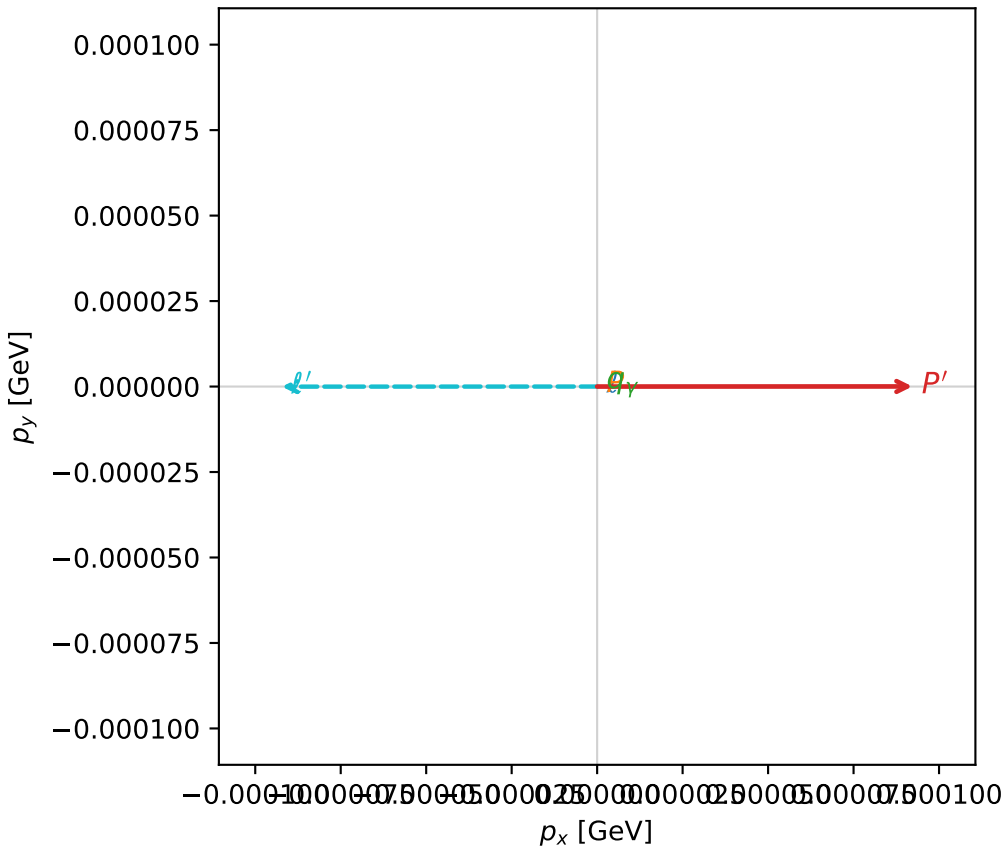
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.140649$
 $\theta_{p'}=3.14094$, $\theta_Y=3.14159$, $\phi_{p'}=0.000316874$, $\phi_Y=5.70698$
 $E_Y=1.95543$, $\theta_{\ell'}=4.39858e-05$, $\phi_{\ell'}=3.14191$
 $Q^2=20.2231$, $x_B=0.864884$, $t=-3.82814$, $W^2=4.03919$, $y=0.969416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0961$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0961$
 P' $E= 0.9485$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1406$
 q_Y $E= 1.9554$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9554$

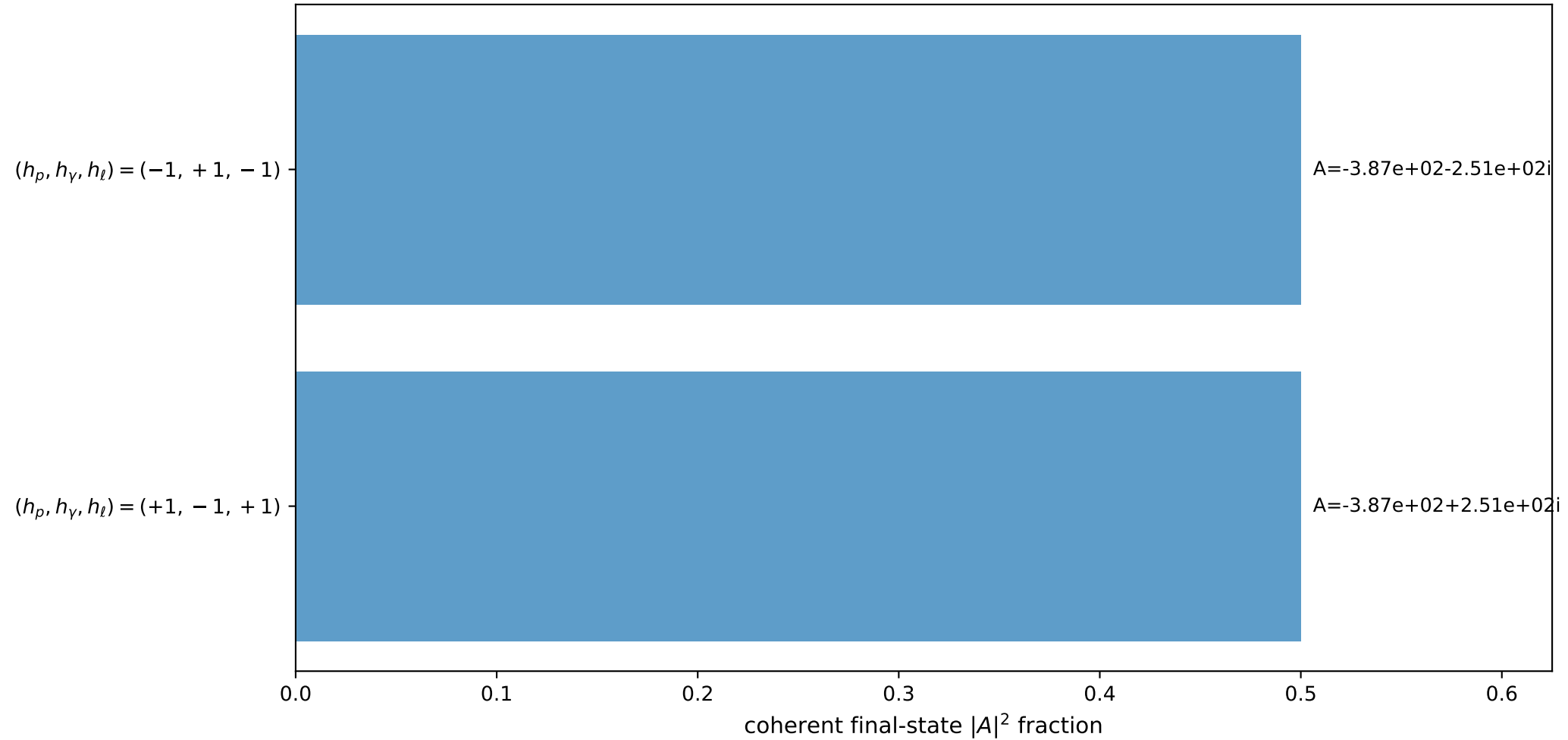
3D momenta



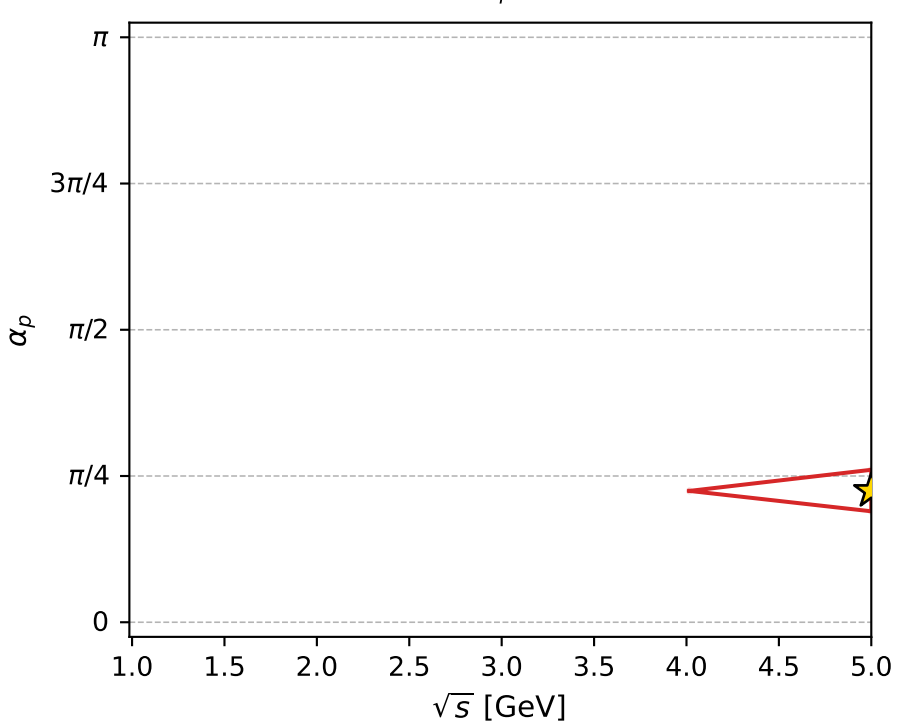
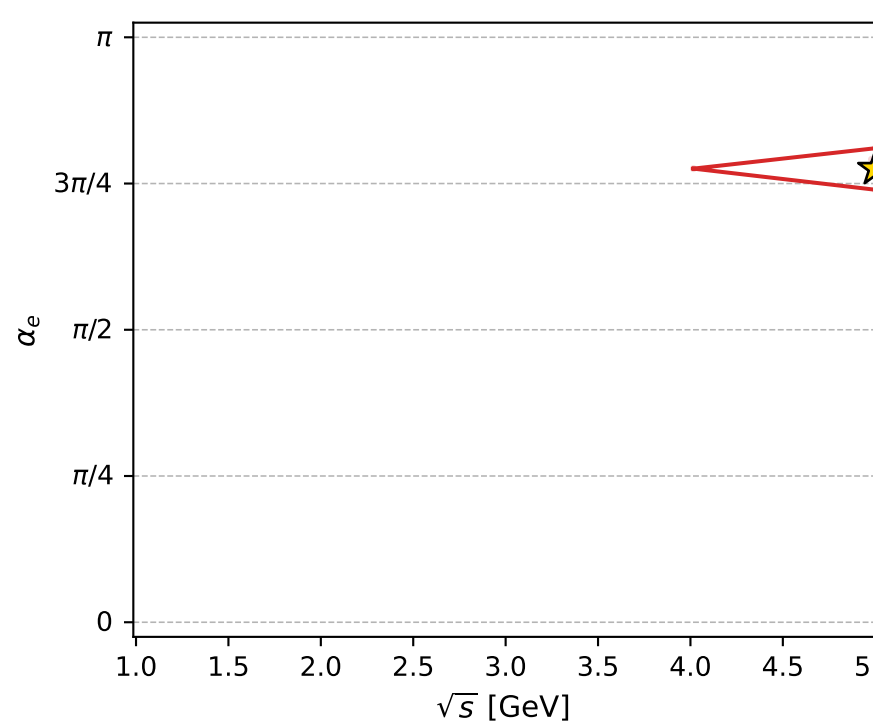
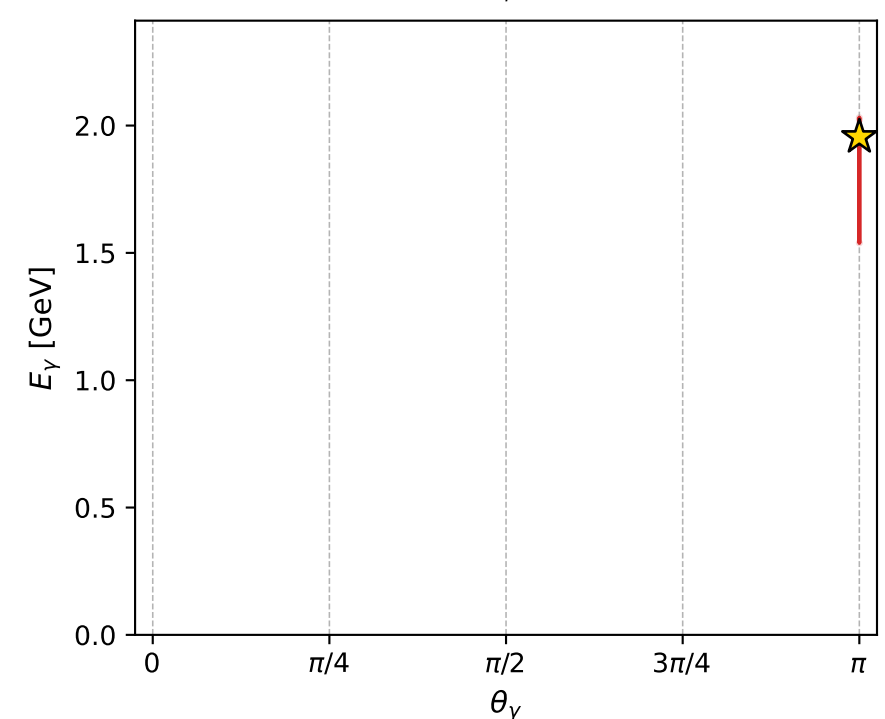
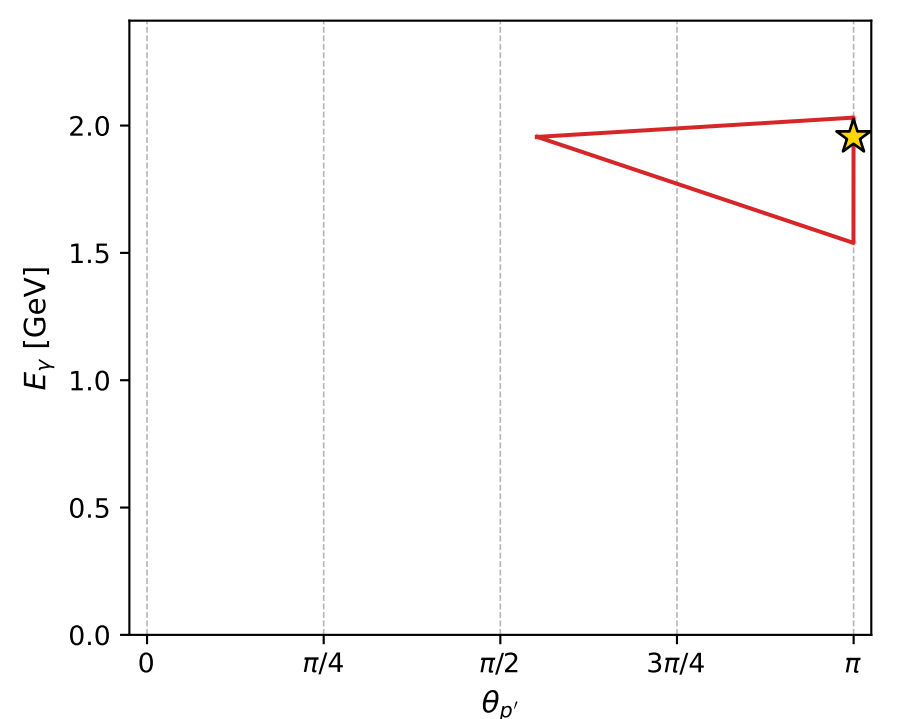
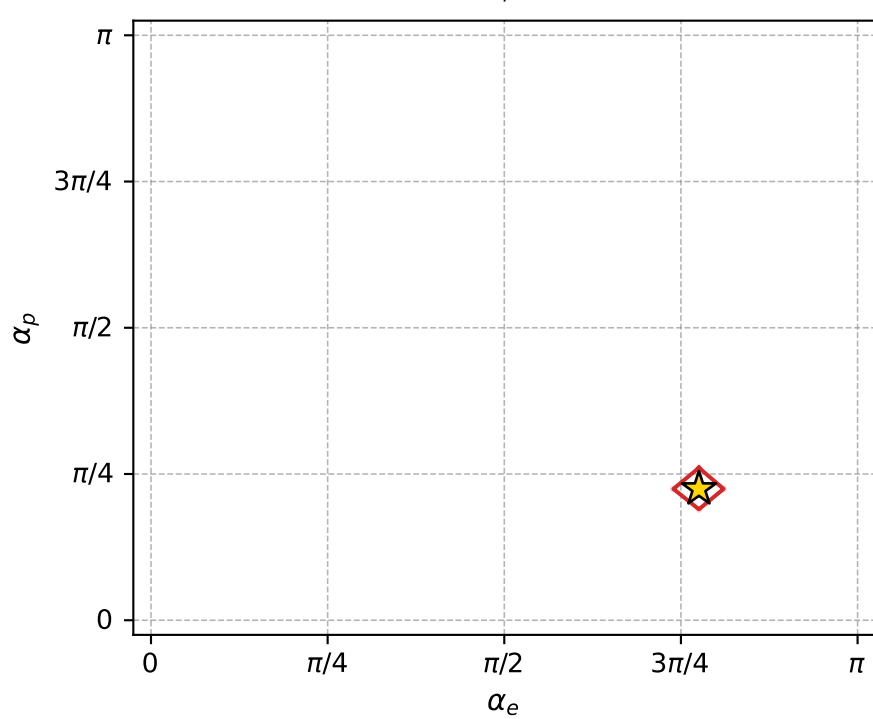
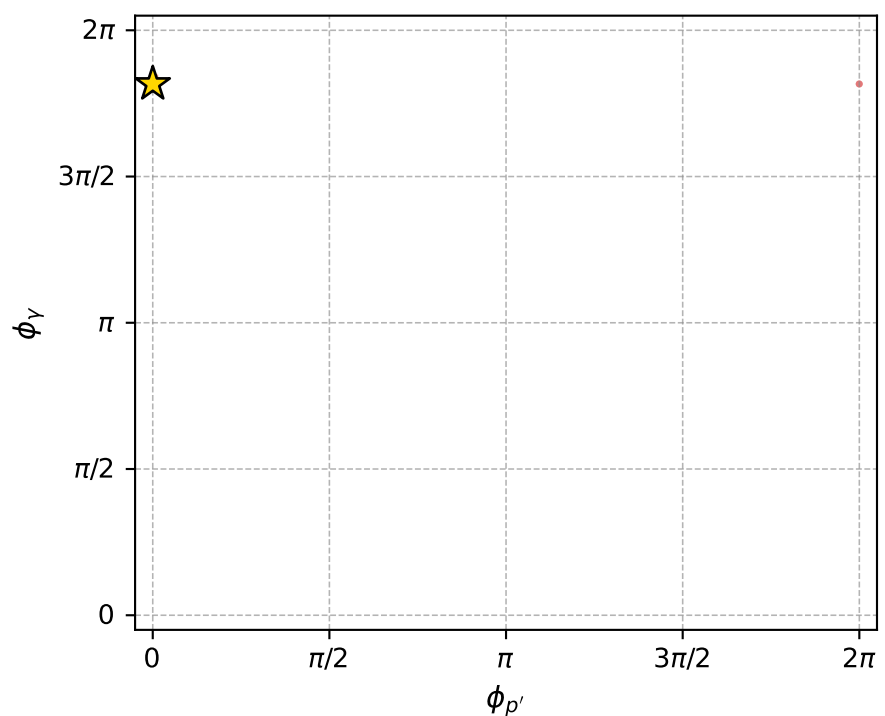
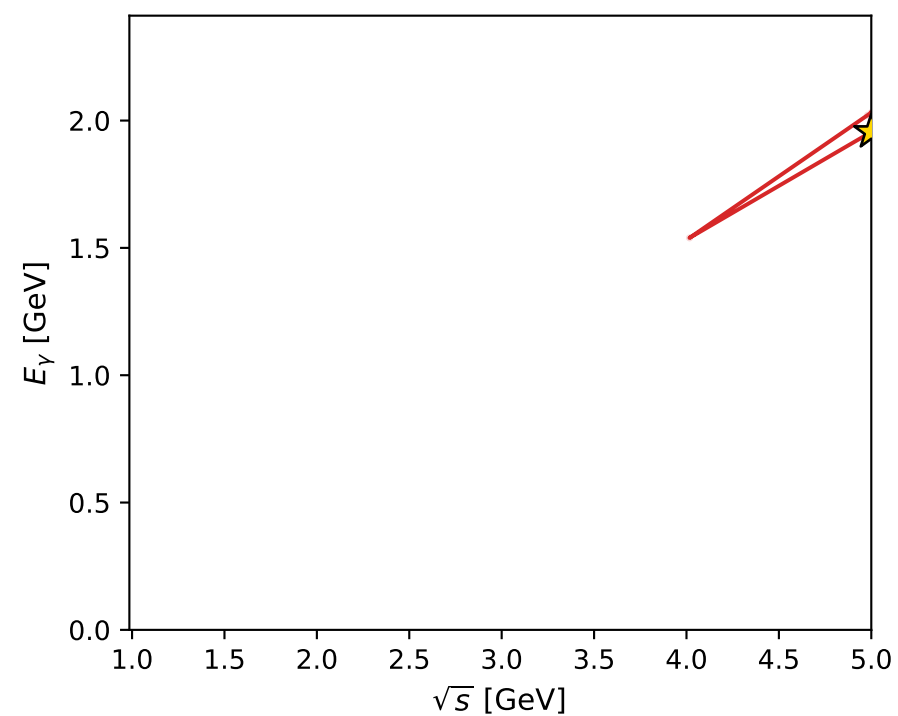
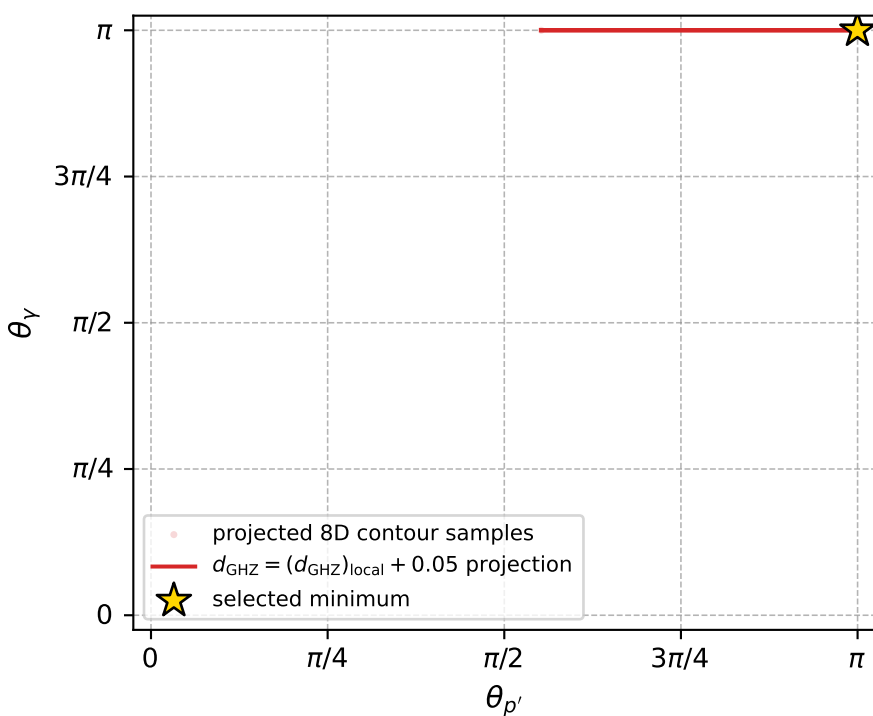
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 38; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6424831\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.1413691\text{e-}09$
 8D contour samples = 81

selected phase-space point:

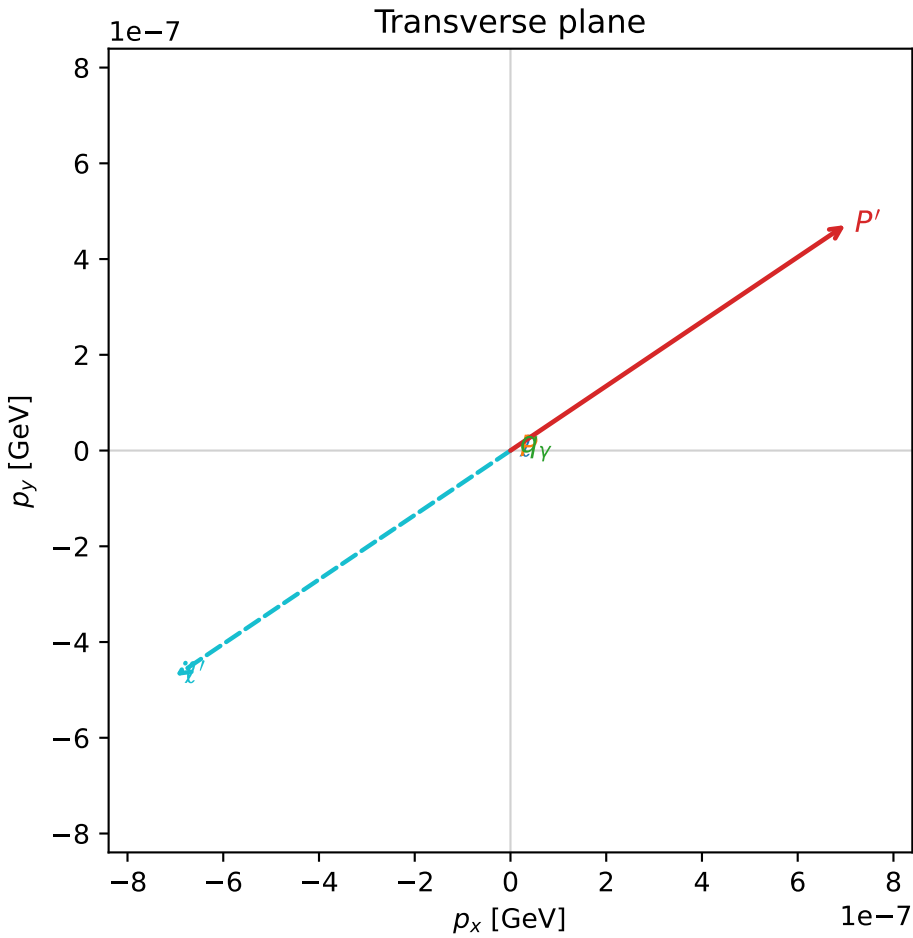
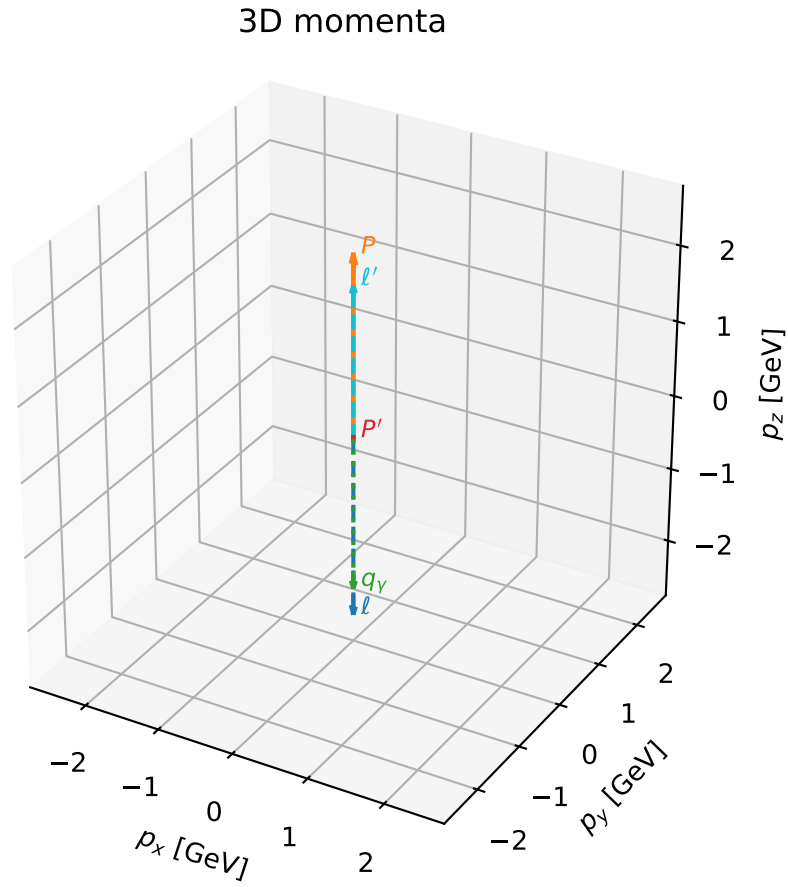
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.140937$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.955432$
 $\text{phi_p_out} = 0.0003168744$
 $\text{phi_gamma_out} = 5.706981$
 $\text{alpha_e} = 2.436248$
 $\text{alpha_p} = 0.7053527$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_43 $d_{\{\text{rm GHZ}\}}=2.6533\text{e-}08$
region: polarization_cluster_03_local_minimum_43
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3049352$ rad
incoming lepton: $-0.669948|+\rangle +0.742408|-\rangle$
proton mixing angle: $\alpha_p=0.83665018$ rad
incoming proton: $0.669953|+\rangle +0.742403|-\rangle$

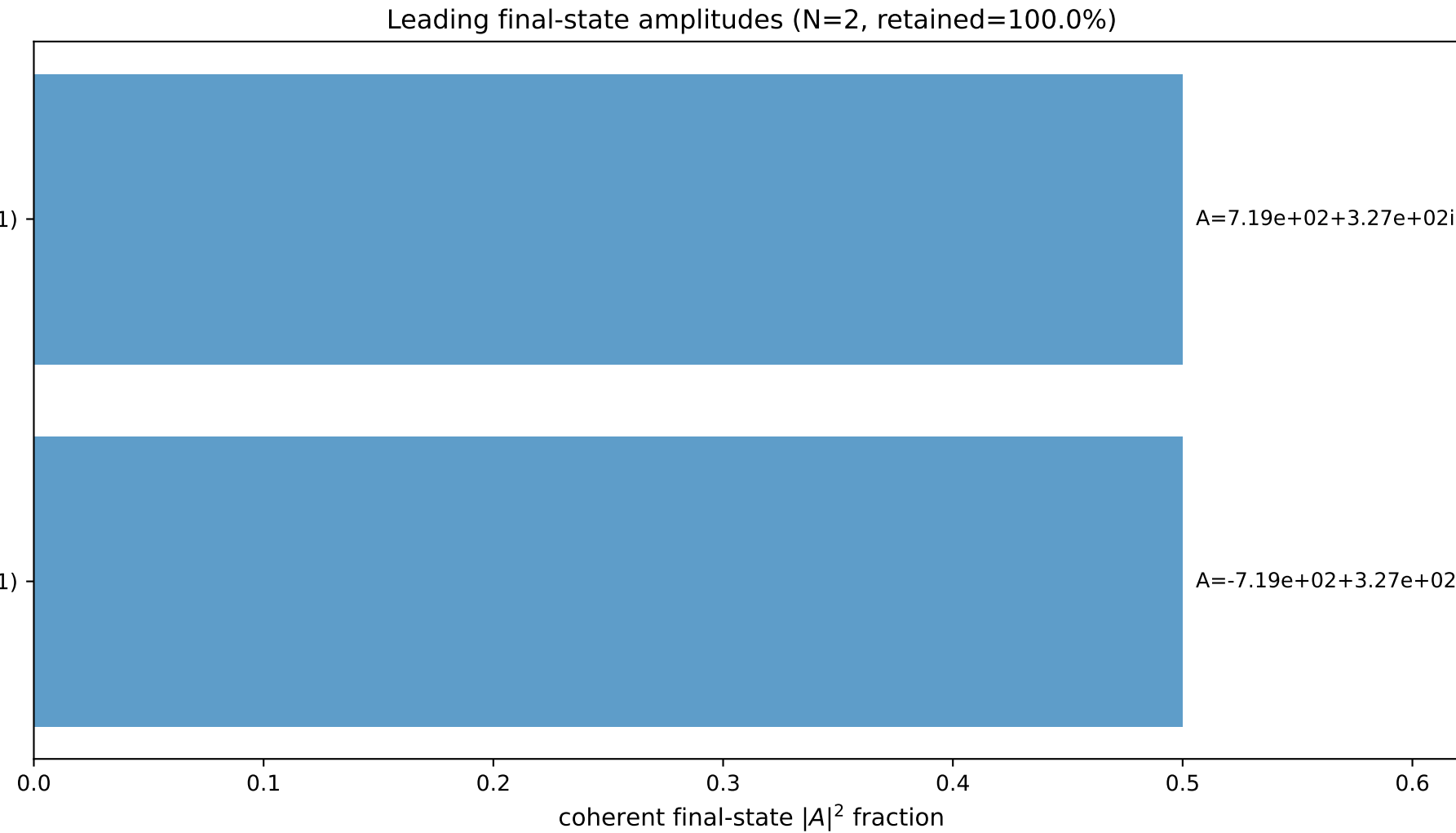
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.023742$
 $\theta_{P'}=3.55227\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=0.592674$, $\phi_Y=3.5688$
 $E_Y=2.04272$, $\theta_{\ell'}=4.1803\text{e-}07$, $\phi_{\ell'}=3.73427$
 $Q^2=19.4792$, $x_B=0.832104$, $t=-2.98239$, $W^2=4.81021$, $y=0.970541$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0190$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0190$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0237$
 q_Y $E= 2.0427$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0427$

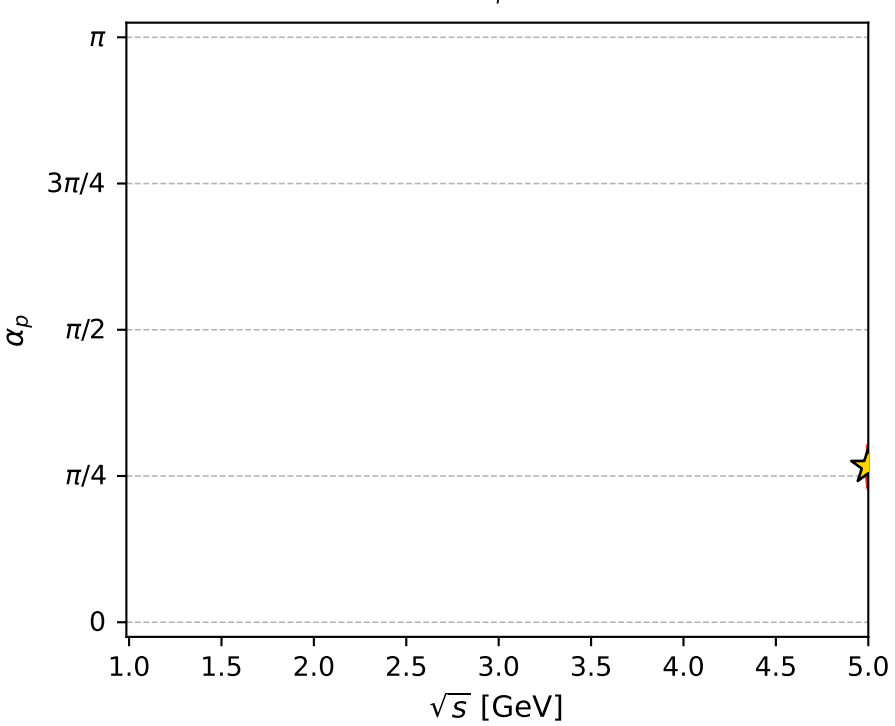
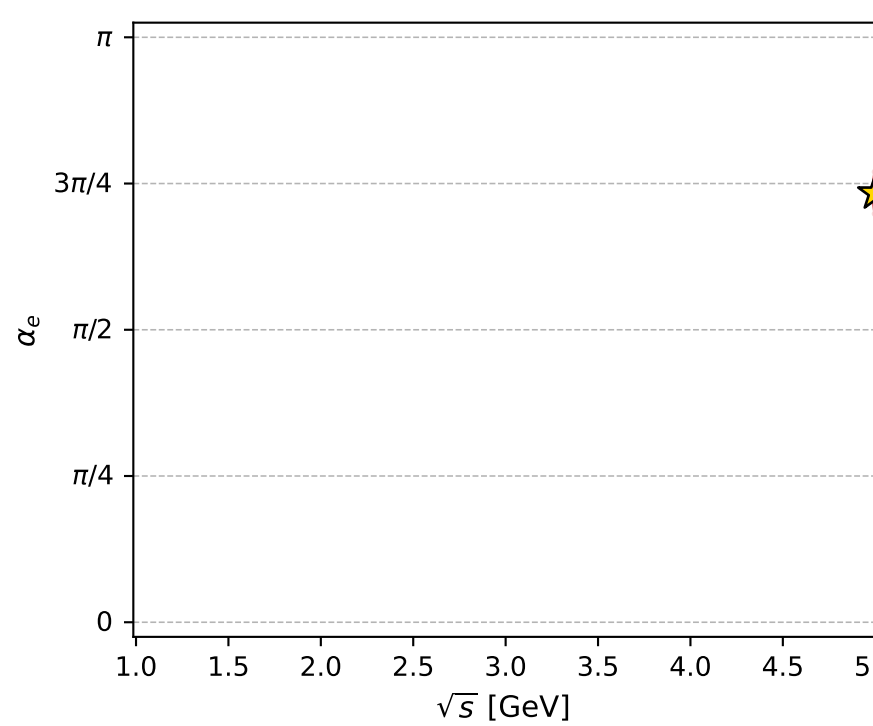
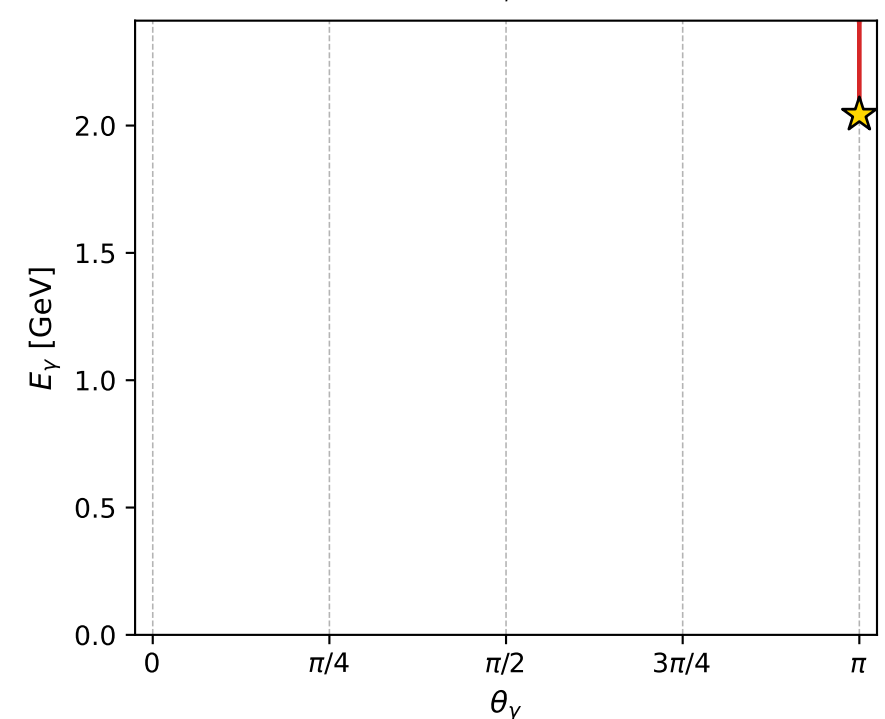
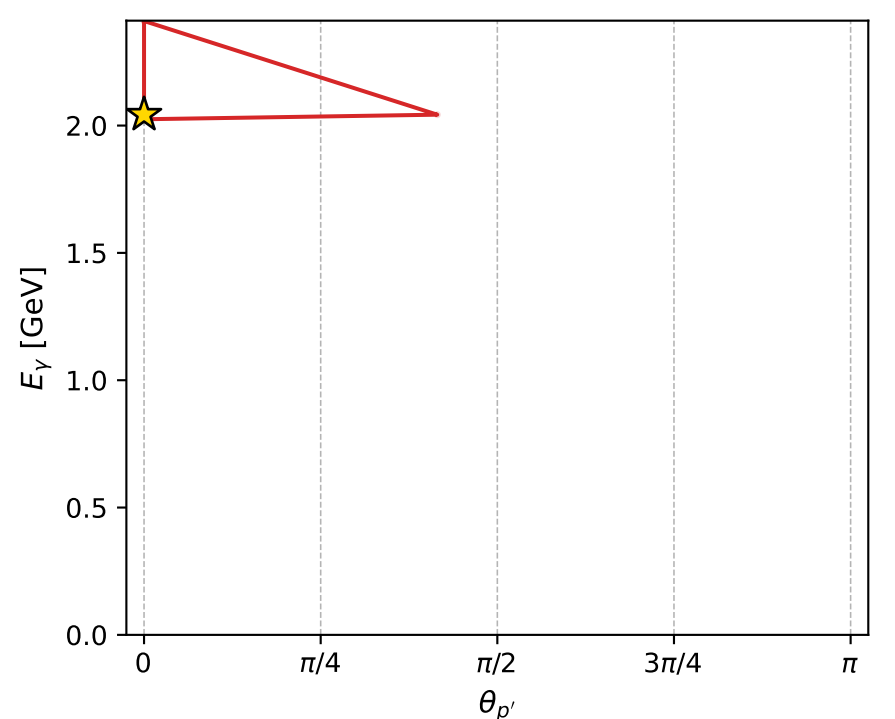
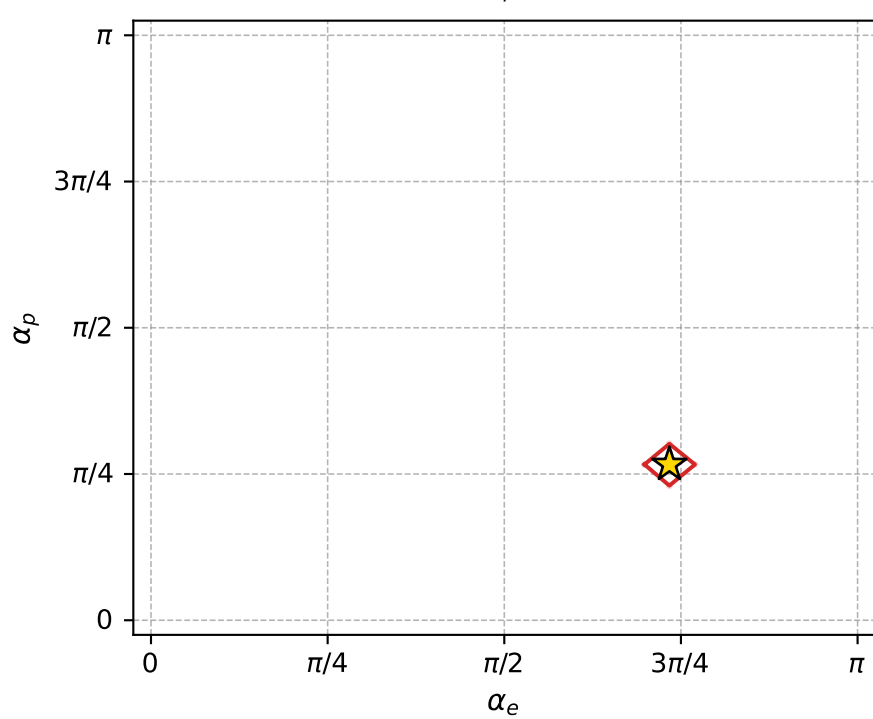
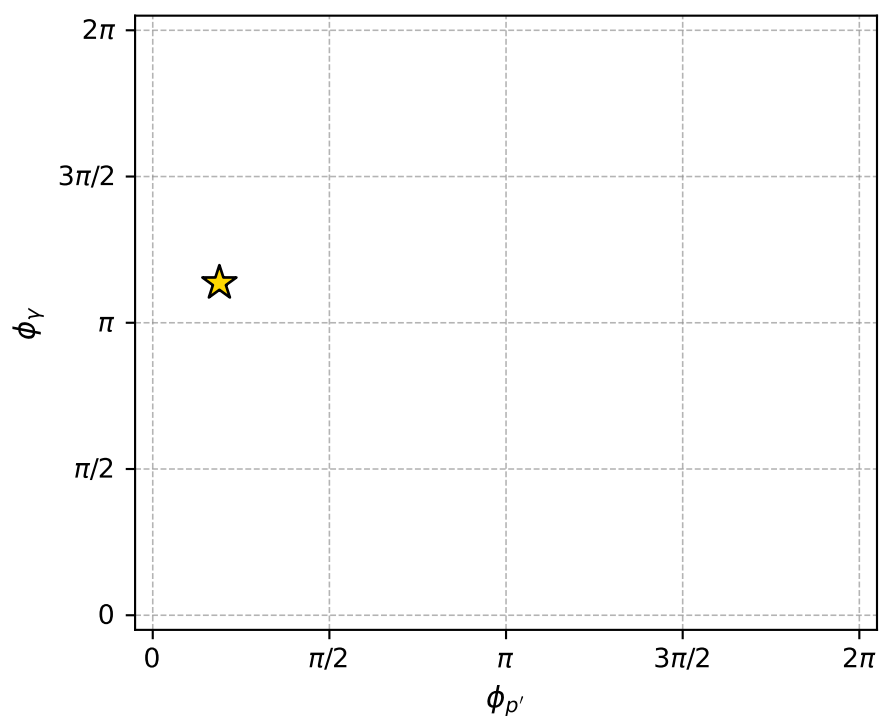
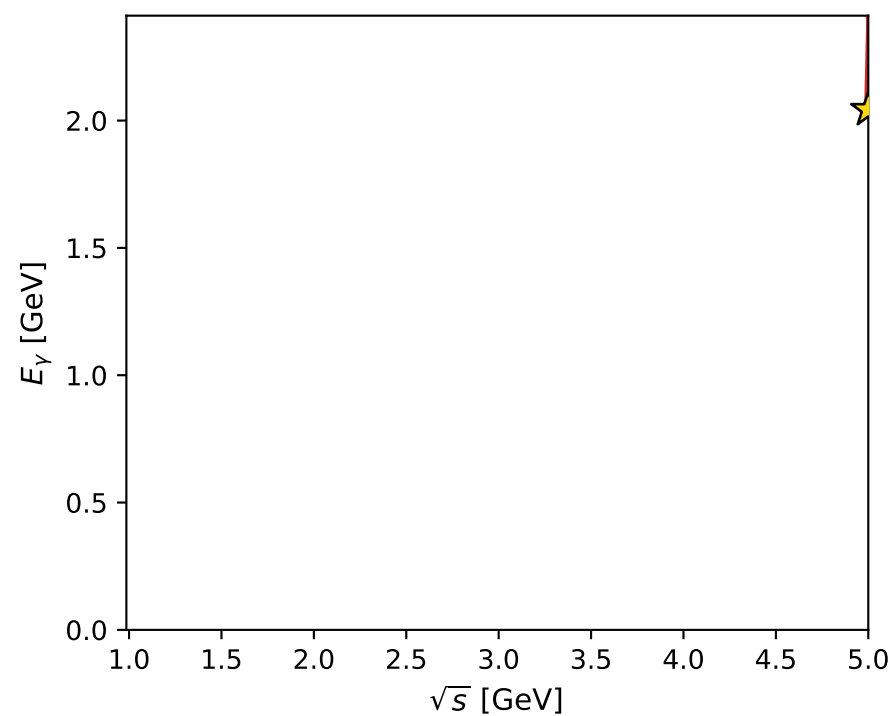
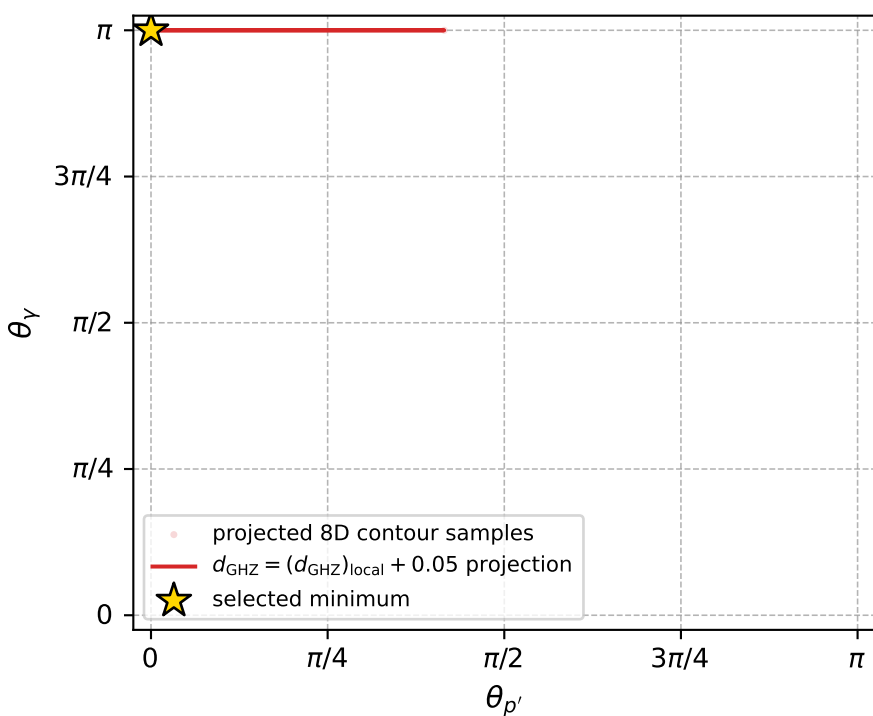


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P3; local minimum 43; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6532966\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.249504\text{e-}09$
 8D contour samples = 36

selected phase-space point:

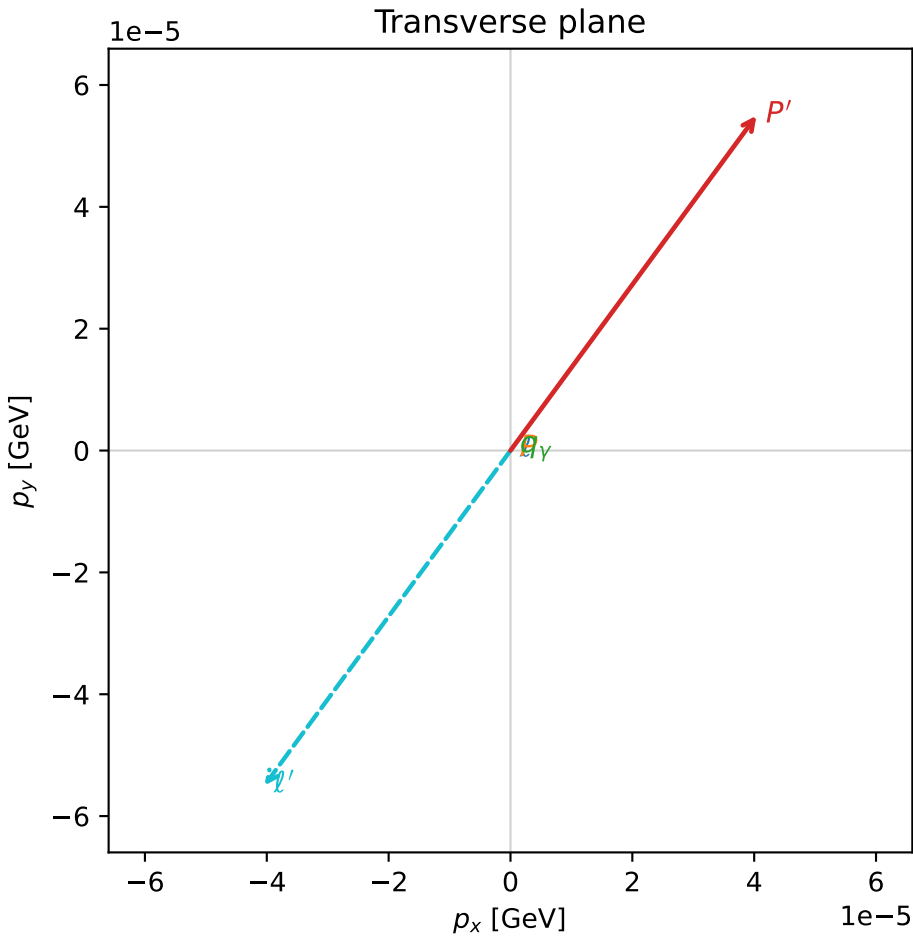
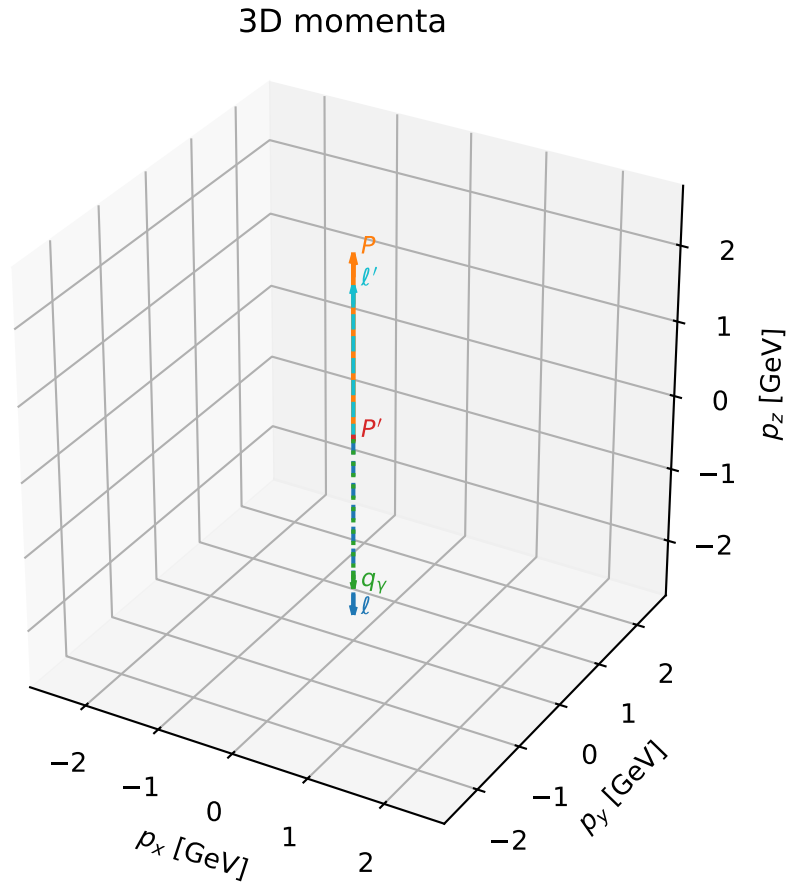
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.552267\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.042721$
 $\text{phi_p_out} = 0.5926742$
 $\text{phi_gamma_out} = 3.568799$
 $\text{alpha_e} = 2.304935$
 $\text{alpha_p} = 0.8366502$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_45 $d_{\{\text{rm GHZ}\}}=2.6597\text{e-}08$
region: polarization_cluster_03_local_minimum_45
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3367051$ rad
incoming lepton: $-0.693192|+\rangle +0.720753|-\rangle$
proton mixing angle: $\alpha_p=0.80488895$ rad
incoming proton: $0.693191|+\rangle +0.720754|-\rangle$

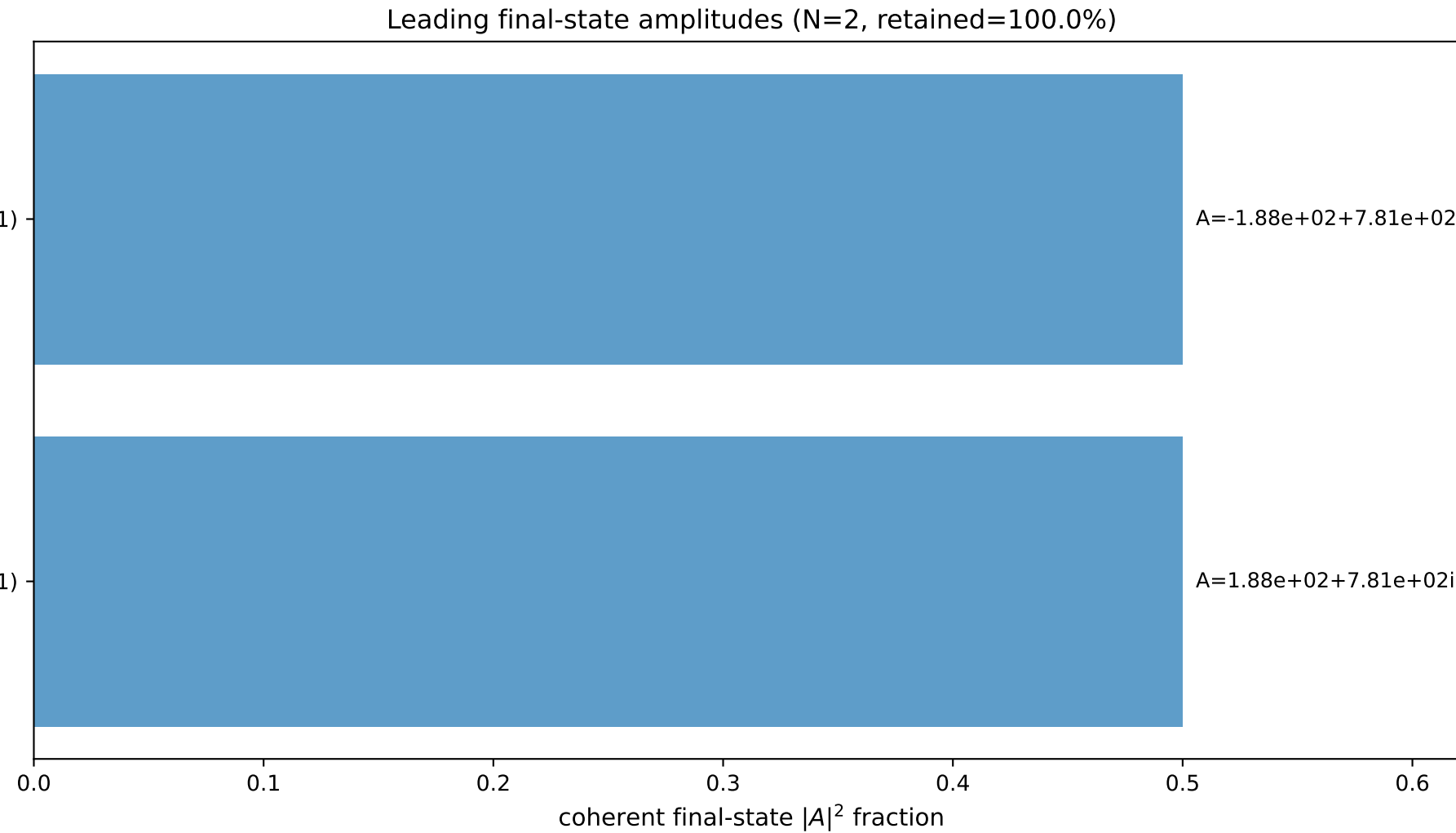
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0277887$
 $\theta_{P'}=0.00245437$, $\theta_Y=3.14159$, $\phi_{P'}=0.937225$, $\phi_Y=4.47598$
 $E_Y=2.04469$, $\theta_{\ell'}=3.38161\text{e-}05$, $\phi_{\ell'}=4.07882$
 $Q^2=19.4592$, $x_B=0.831222$, $t=-2.96345$, $W^2=4.831$, $y=0.970571$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0169$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0169$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0278$
 q_Y $E= 2.0447$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0447$

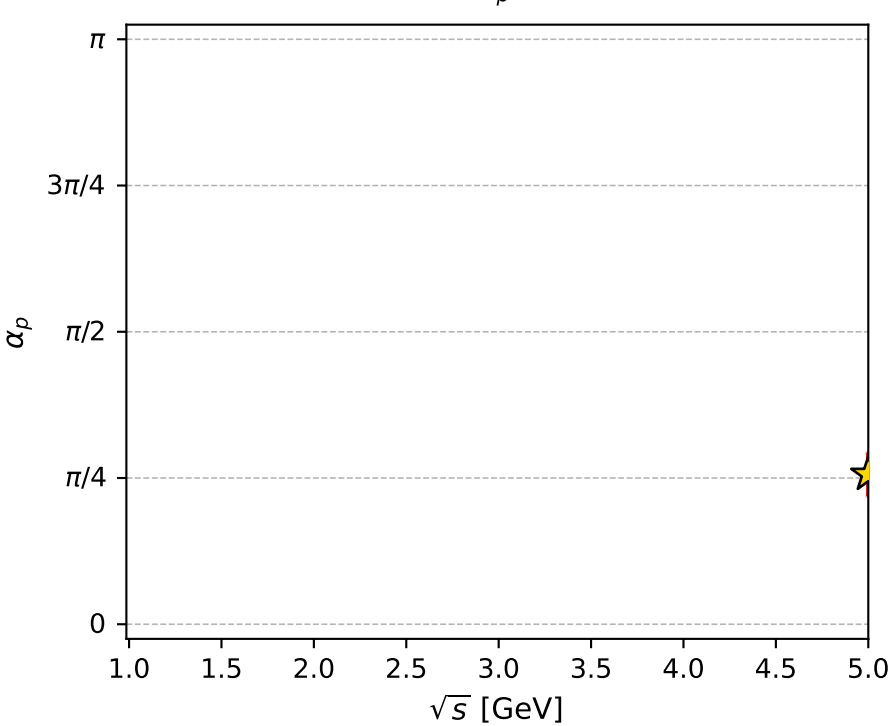
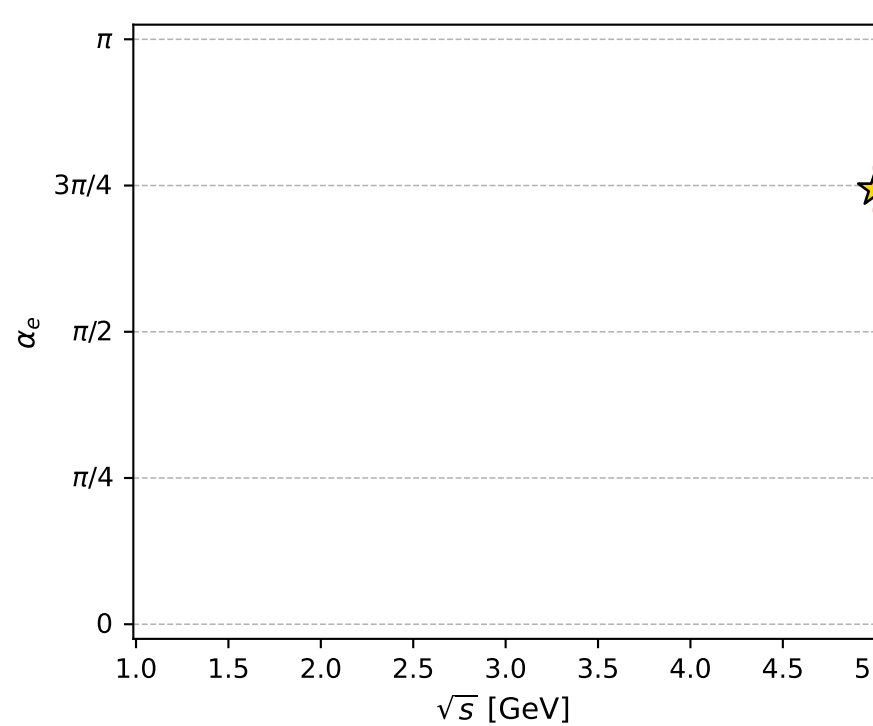
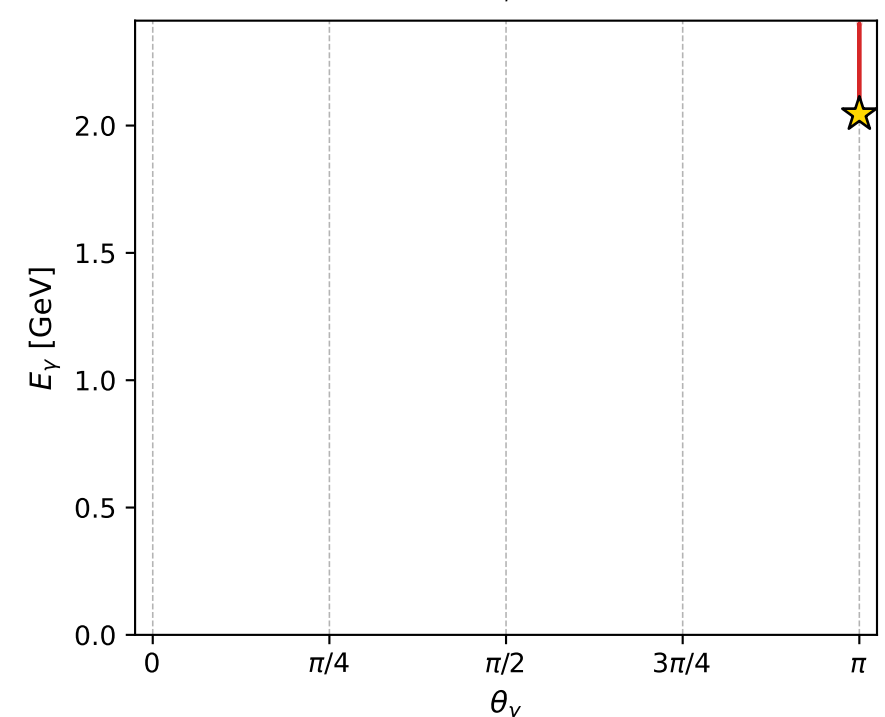
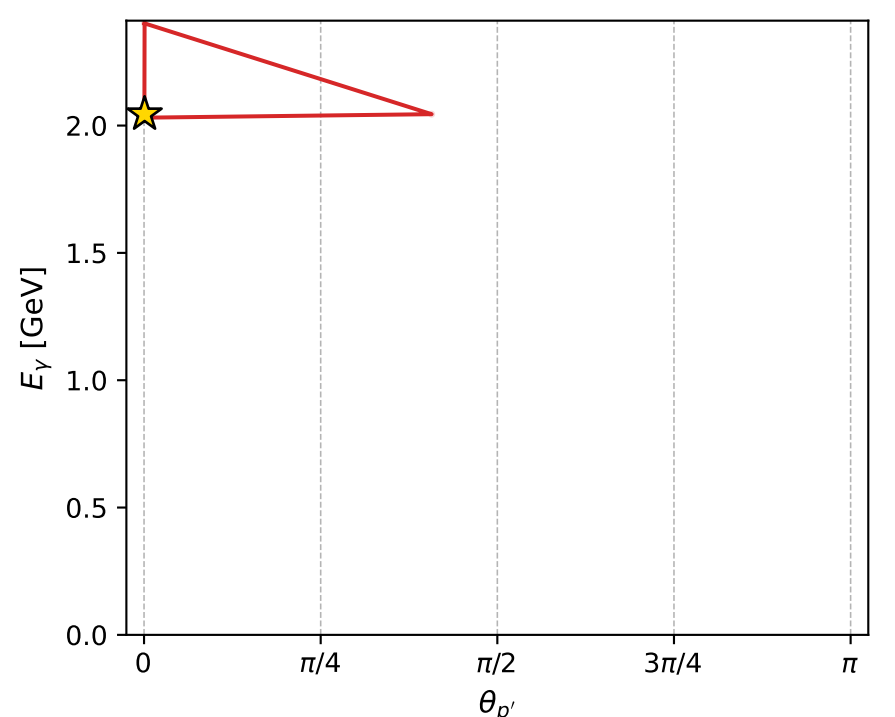
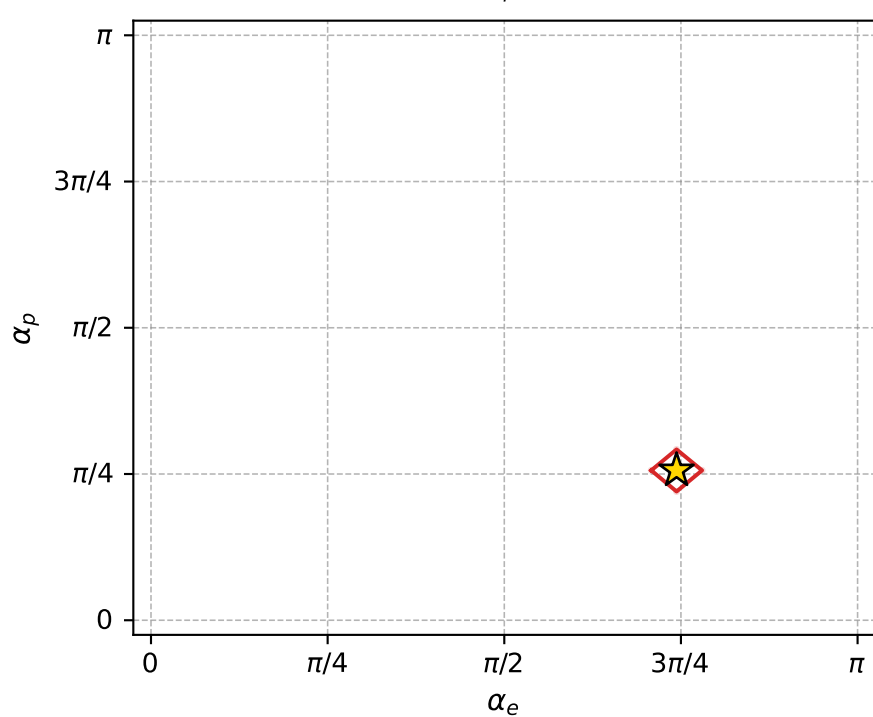
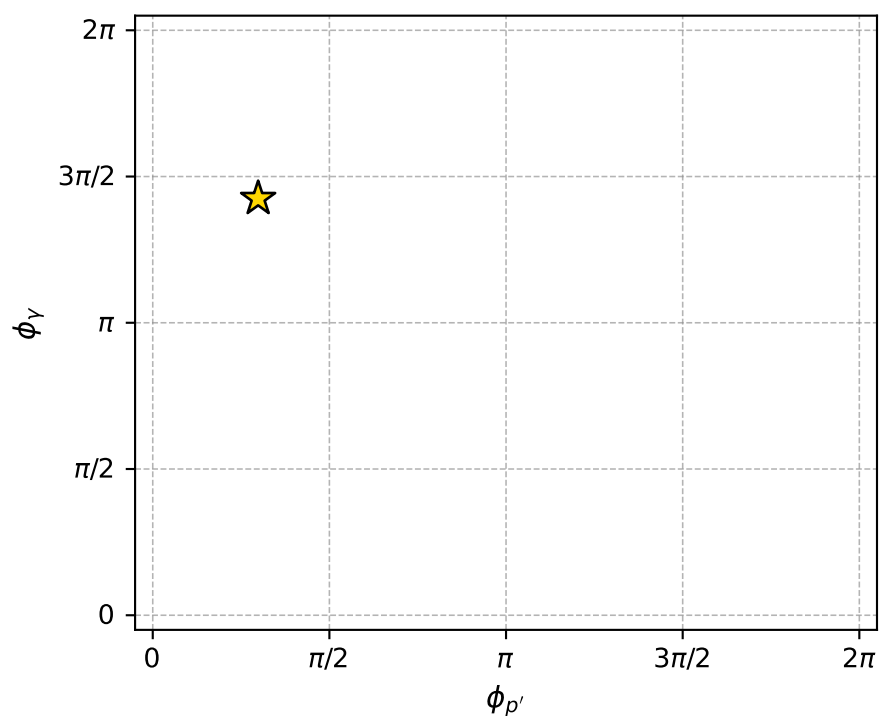
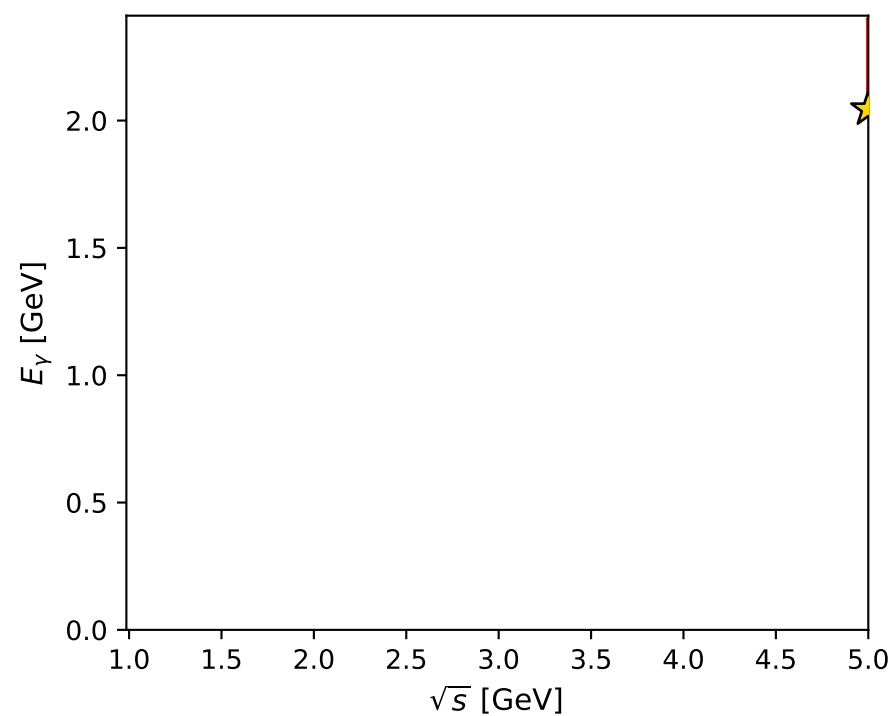
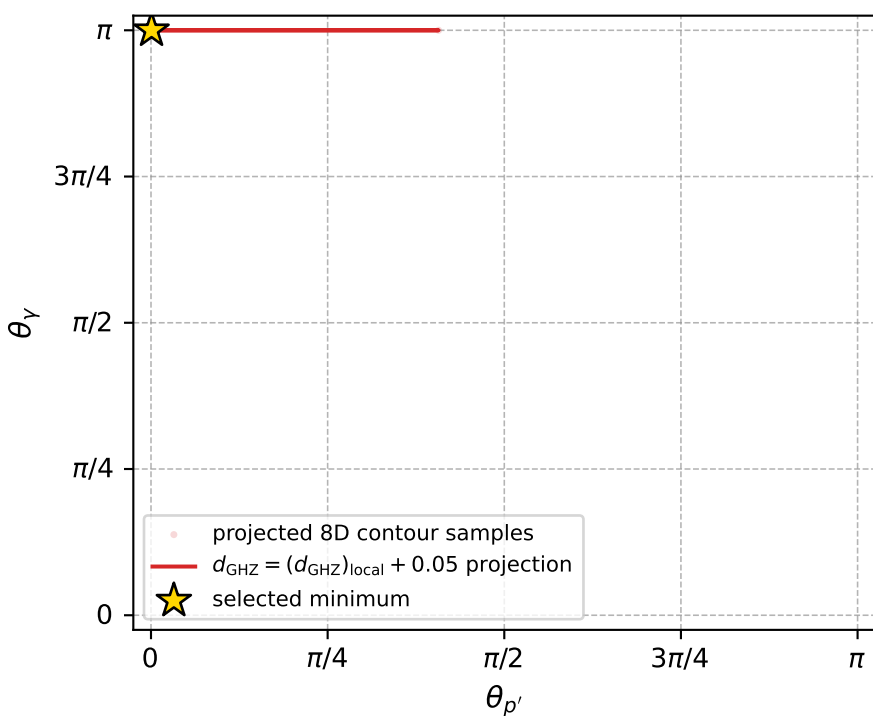


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 45; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.659703\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.3135677\text{e-}09$
 8D contour samples = 63

selected phase-space point:

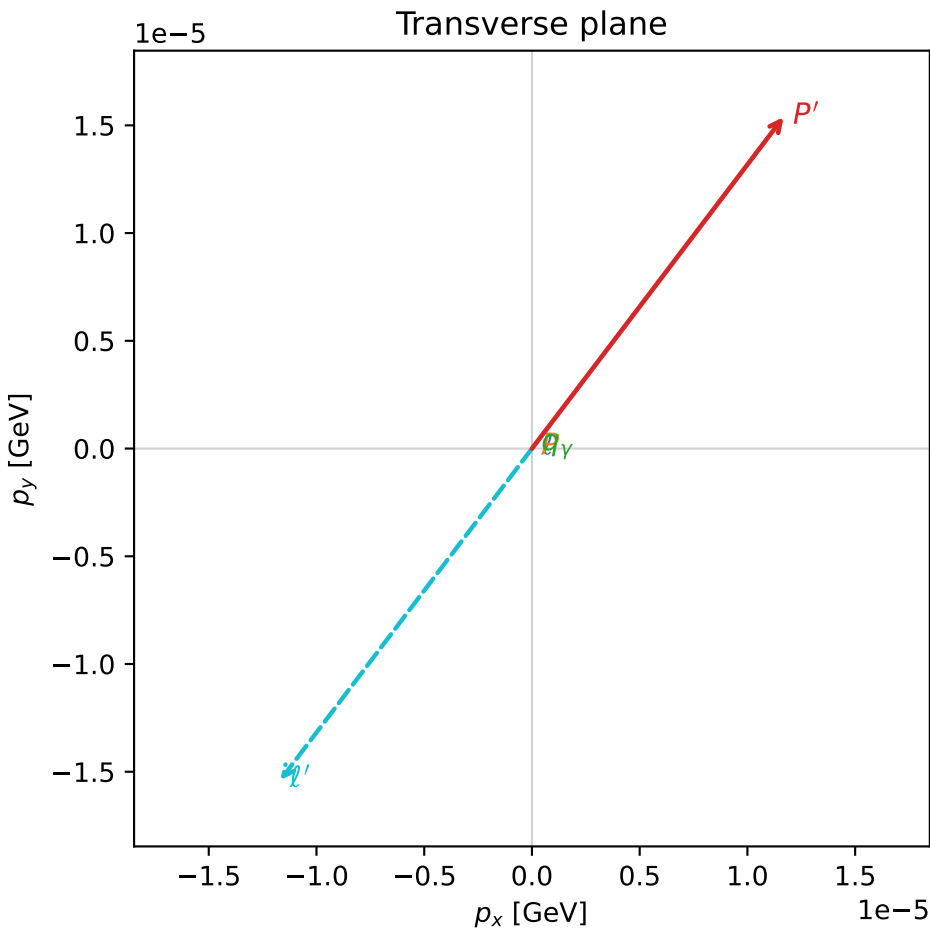
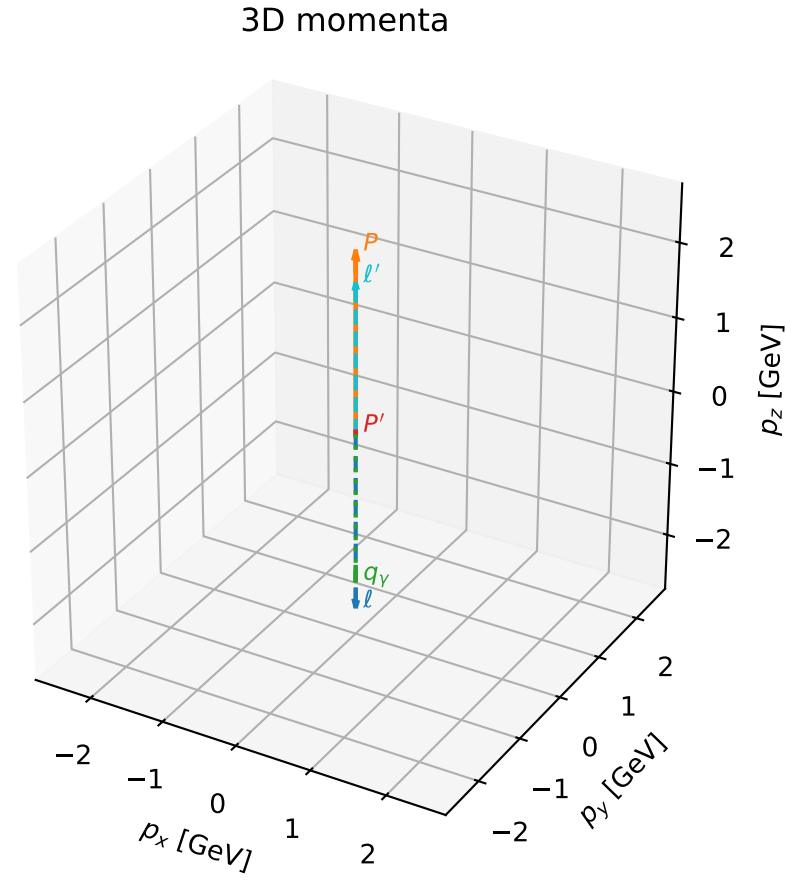
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.002454369$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.044688$
 $\text{phi_p_out} = 0.9372247$
 $\text{phi_gamma_out} = 4.475978$
 $\text{alpha_e} = 2.336705$
 $\text{alpha_p} = 0.804889$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_48 $d_{\{\text{rm GHZ}\}}=2.67266\text{e-}08$
 region: polarization_cluster_03_local_minimum_48
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3692897$ rad
 incoming lepton: $-0.716306|+\rangle +0.697787|-\rangle$
 proton mixing angle: $\alpha_p=0.772294$ rad
 incoming proton: $0.716312|+\rangle +0.69778|-\rangle$

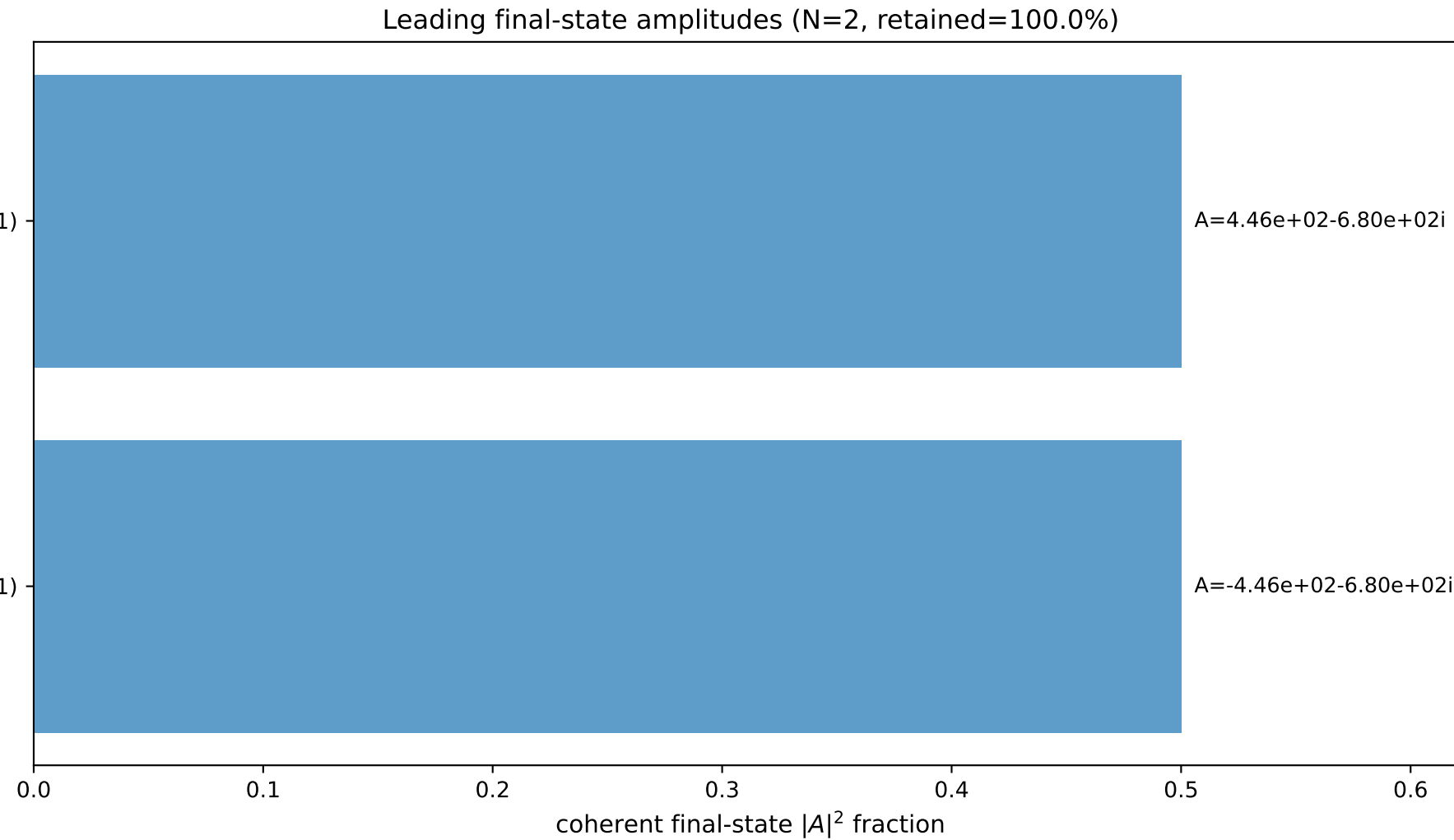
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0314813$
 $\theta_{P'}=0.000613592$, $\theta_Y=3.14159$, $\phi_{P'}=0.92167$, $\phi_Y=2.15105$
 $E_Y=2.04648$, $\theta_{\ell'}=9.58645\text{e-}06$, $\phi_{\ell'}=4.06326$
 $Q^2=19.4408$, $x_B=0.830413$, $t=-2.94624$, $W^2=4.85005$, $y=0.970599$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0150$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0150$
 P' $E= 0.9385$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0315$
 q_Y $E= 2.0465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0465$

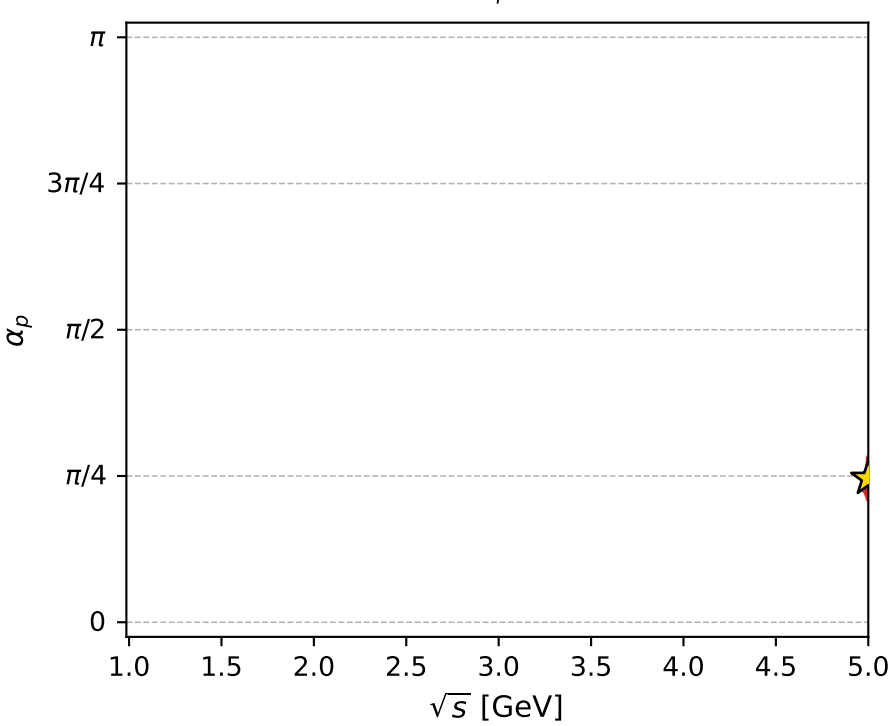
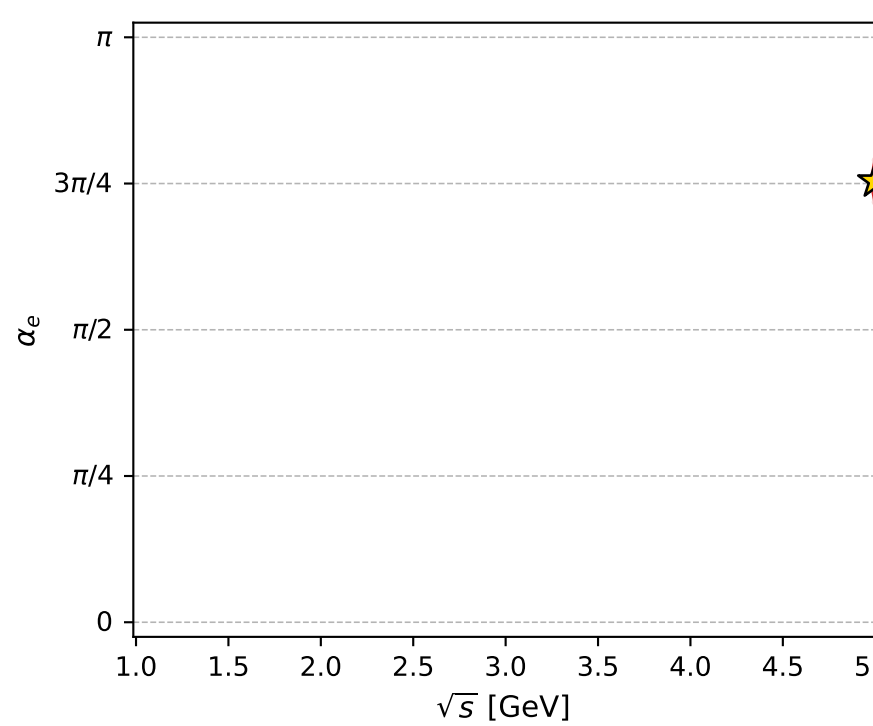
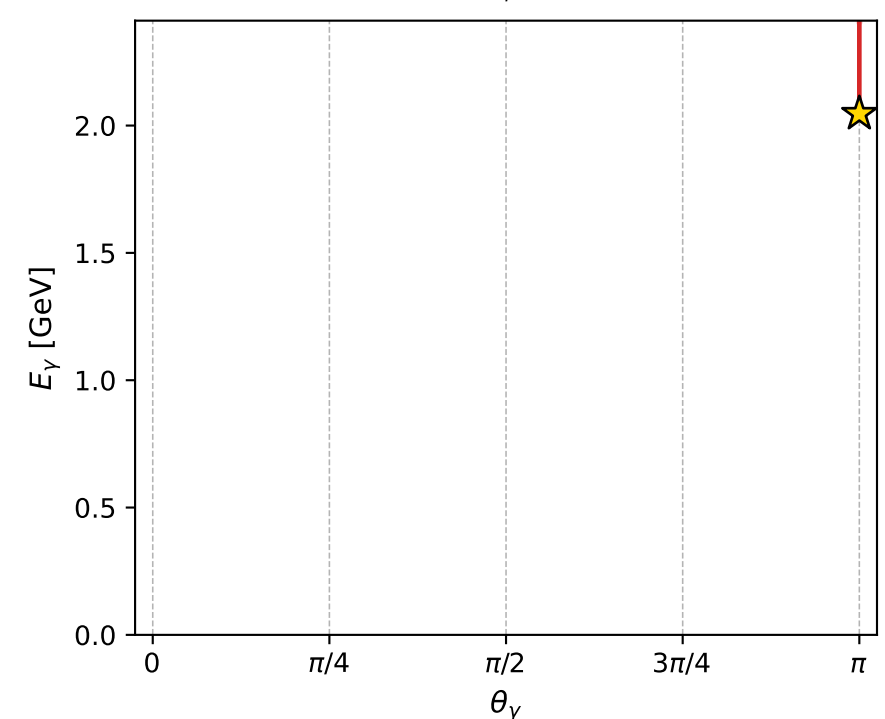
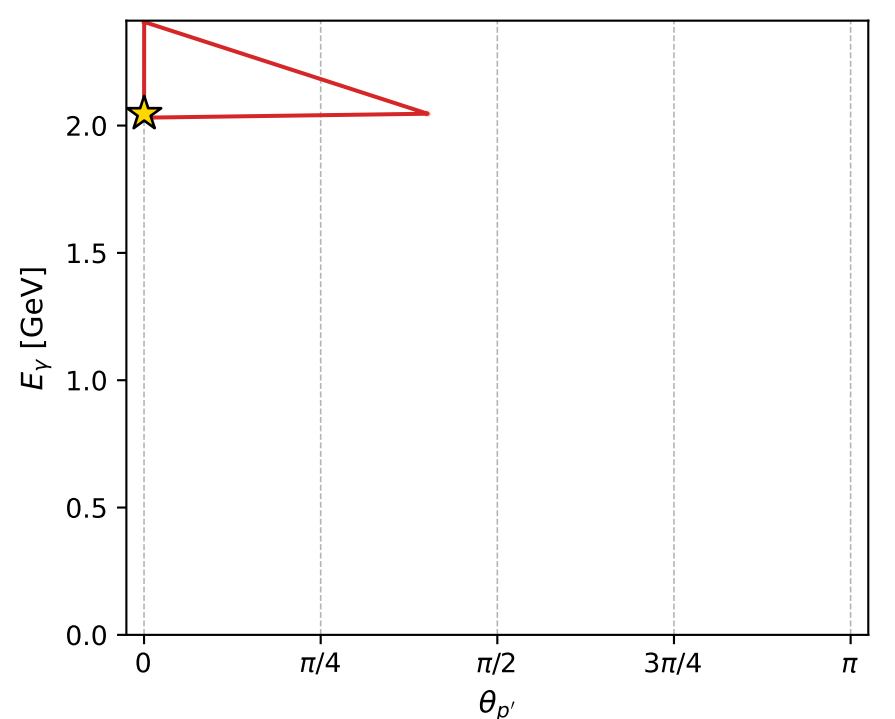
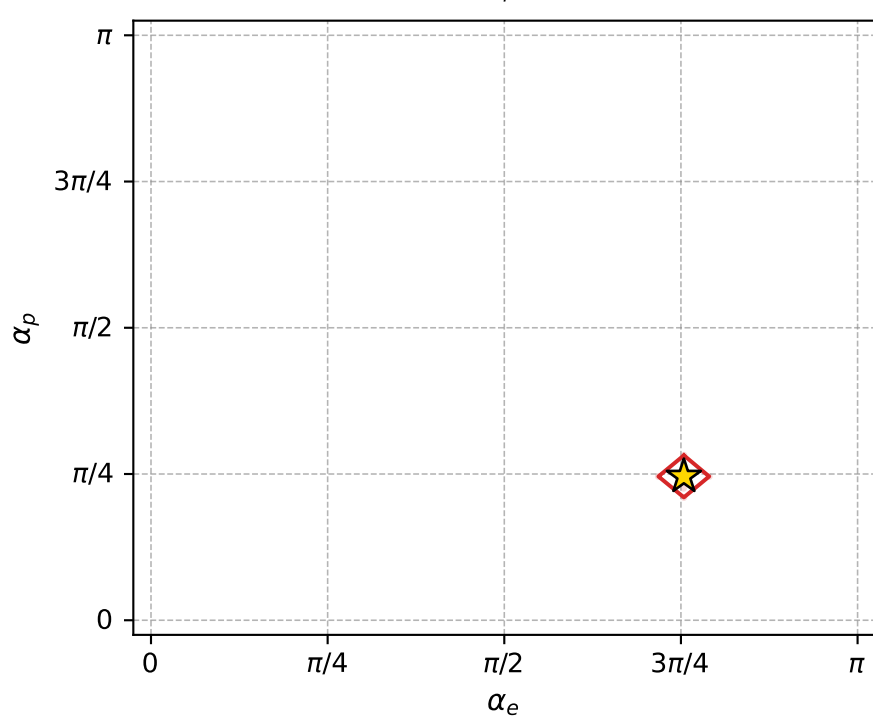
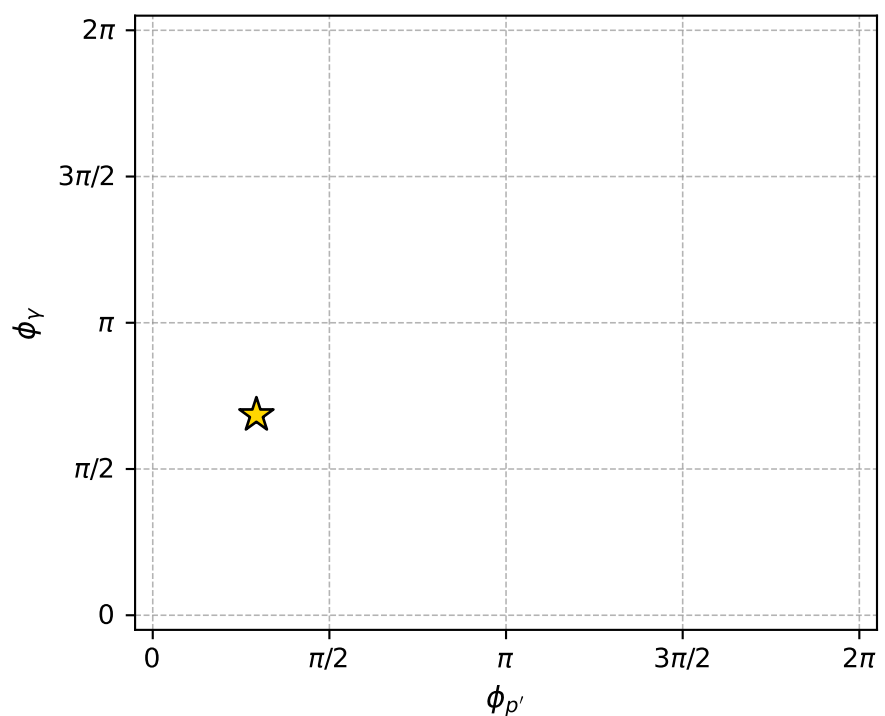
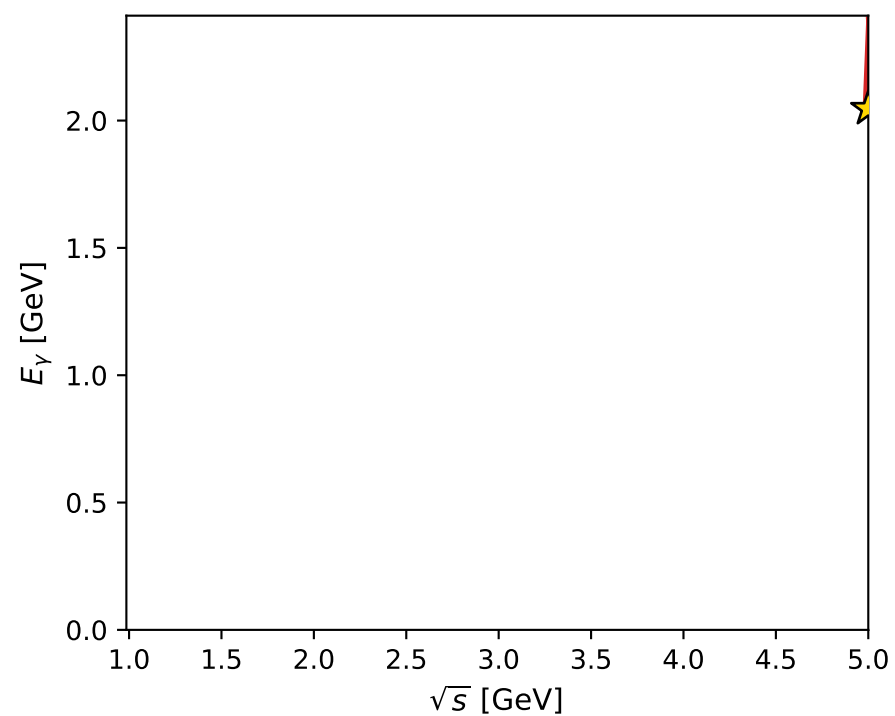
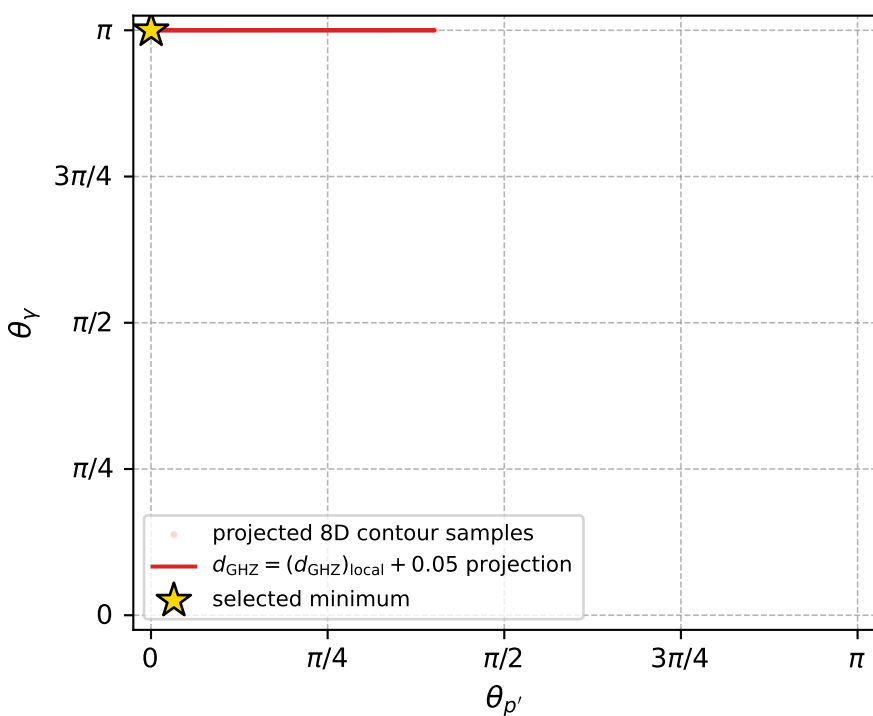


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 48; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6726606\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.4431437\text{e-}09$
 8D contour samples = 79

selected phase-space point:

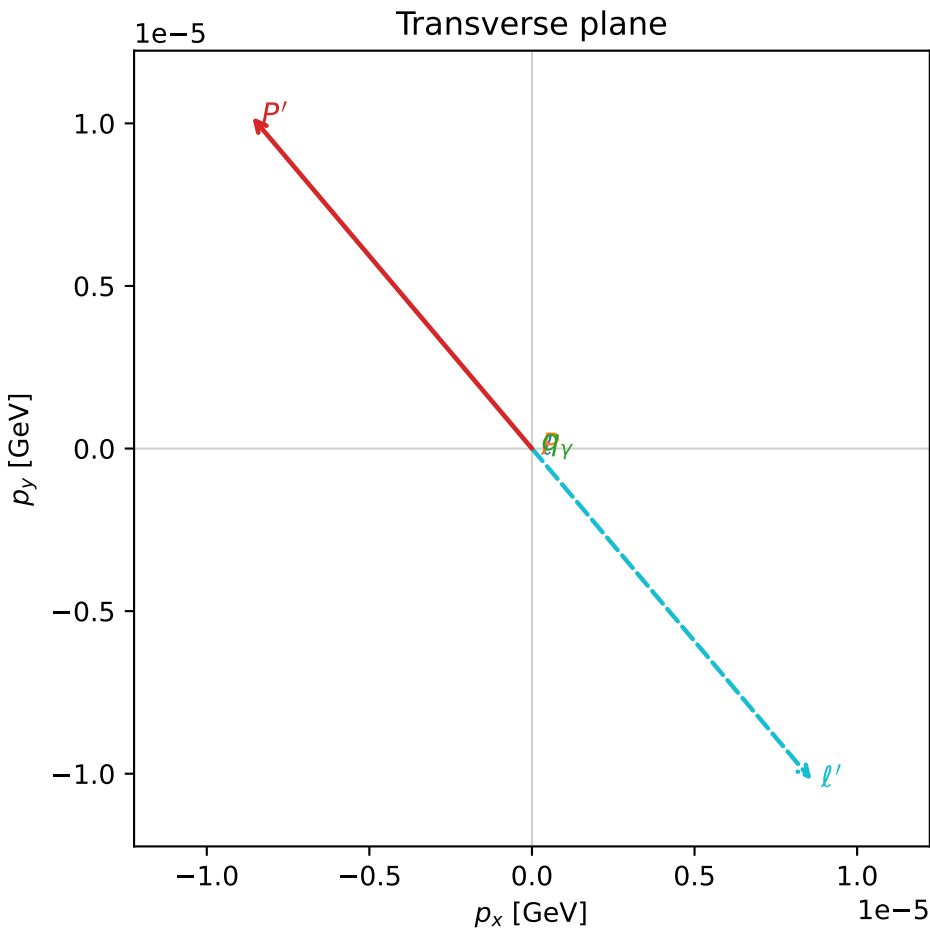
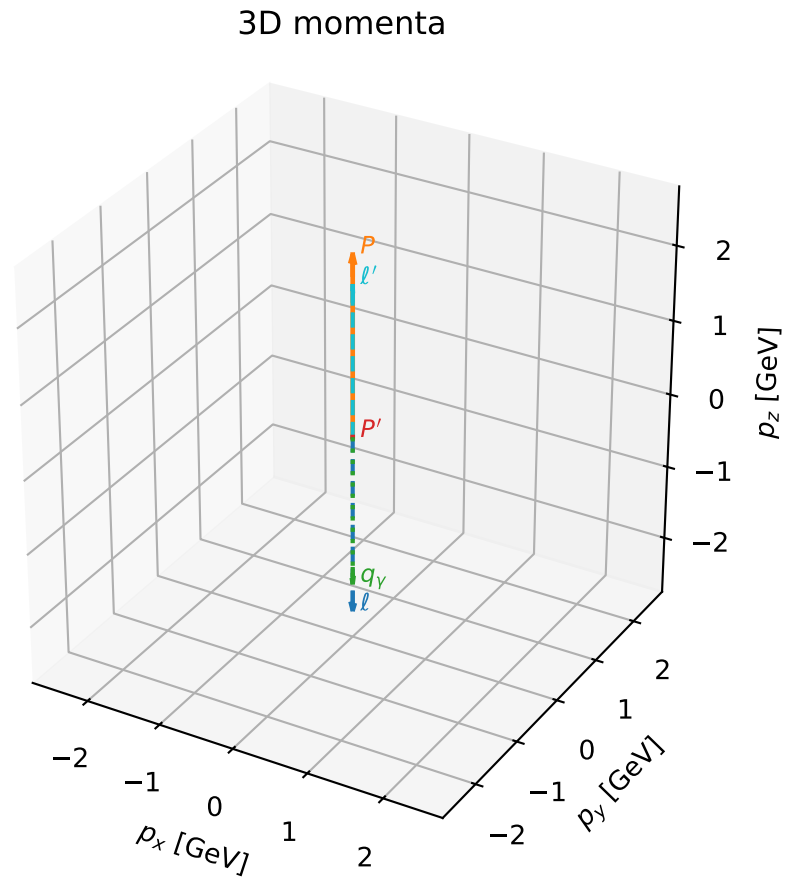
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0006135923$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.046477$
 $\text{phi_p_out} = 0.9216701$
 $\text{phi_gamma_out} = 2.151048$
 $\text{alpha_e} = 2.36929$
 $\text{alpha_p} = 0.772294$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_51 $d_{\{\text{rm GHZ}\}}=2.68417\text{e-}08$
 region: polarization_cluster_03_local_minimum_51
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2220181$ rad
 incoming lepton: $-0.606159|+\rangle +0.795344|-\rangle$
 proton mixing angle: $\alpha_p=0.91958237$ rad
 incoming proton: $0.606152|+\rangle +0.795349|-\rangle$

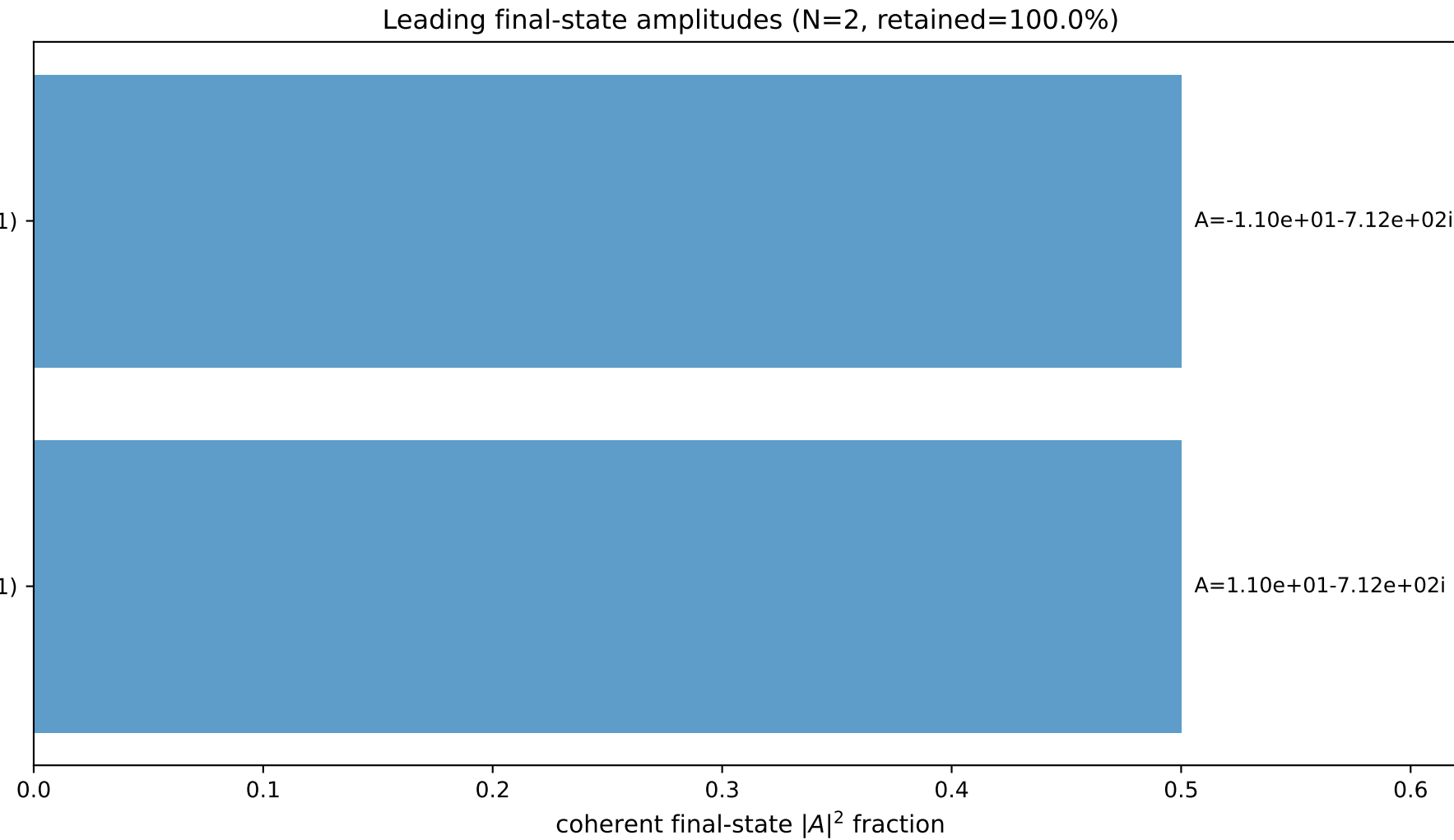
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.001359$
 $\theta_{P'}=0.00981748$, $\theta_Y=3.14159$, $\phi_{P'}=2.27173$, $\phi_Y=1.5862$
 $E_Y=2.03168$, $\theta_{\ell'}=6.57126\text{e-}06$, $\phi_{\ell'}=5.41332$
 $Q^2=19.5887$, $x_B=0.836921$, $t=-3.08882$, $W^2=4.6968$, $y=0.970376$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0303$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0303$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0014$
 q_Y $E= 2.0317$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0317$

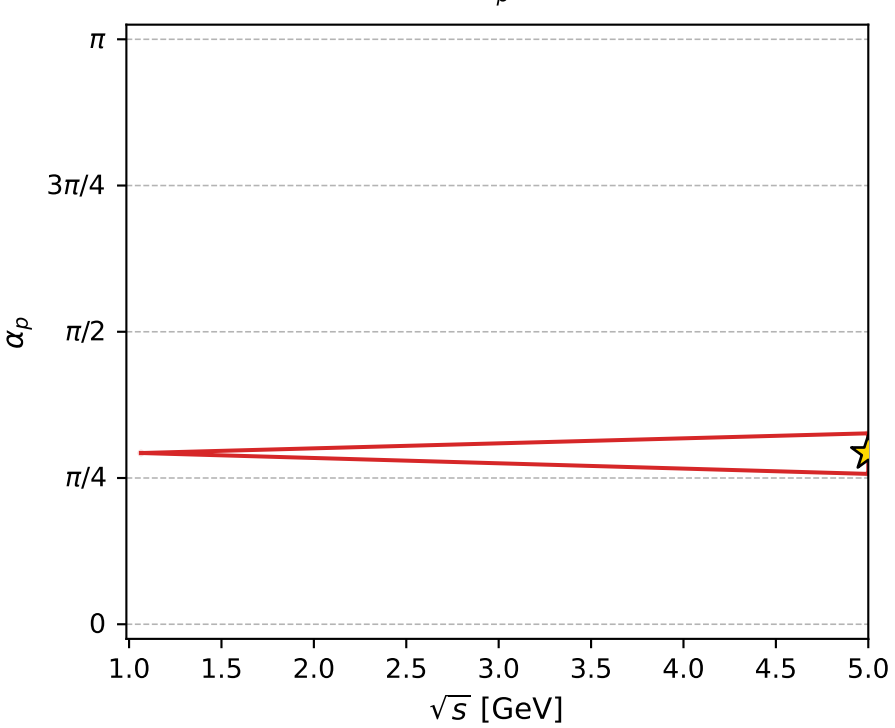
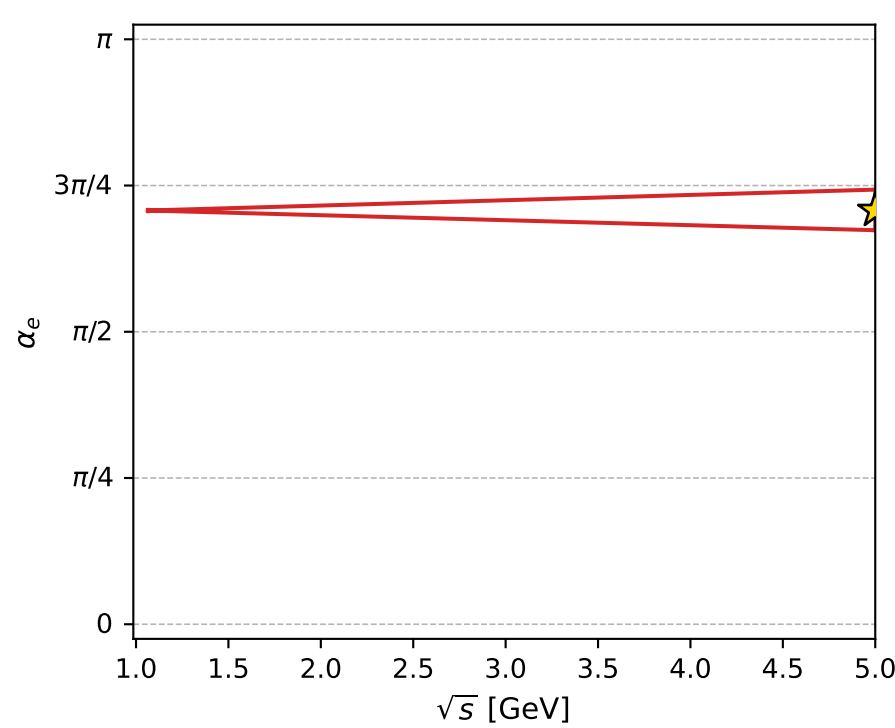
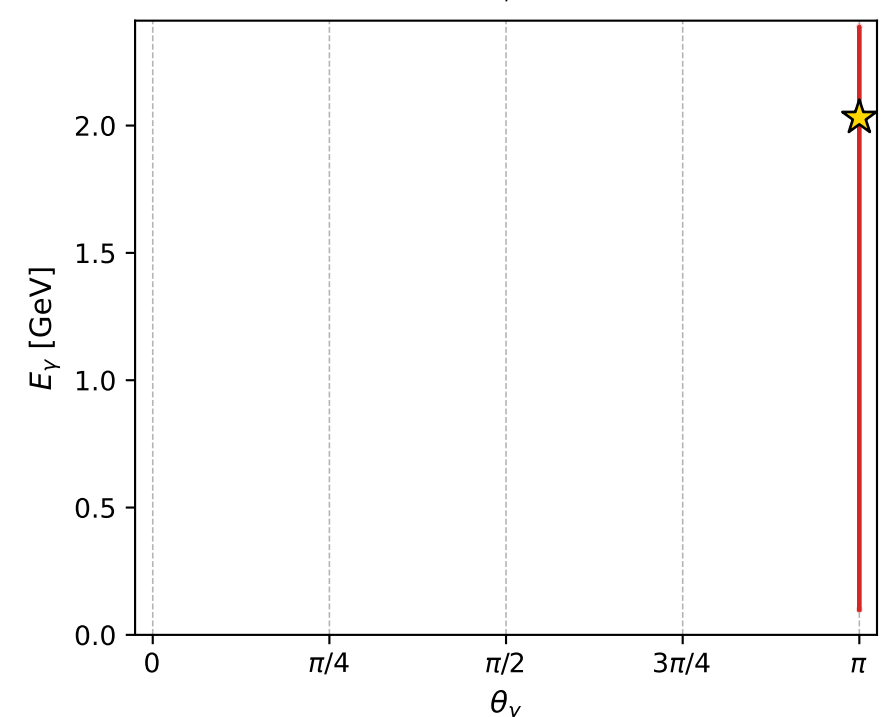
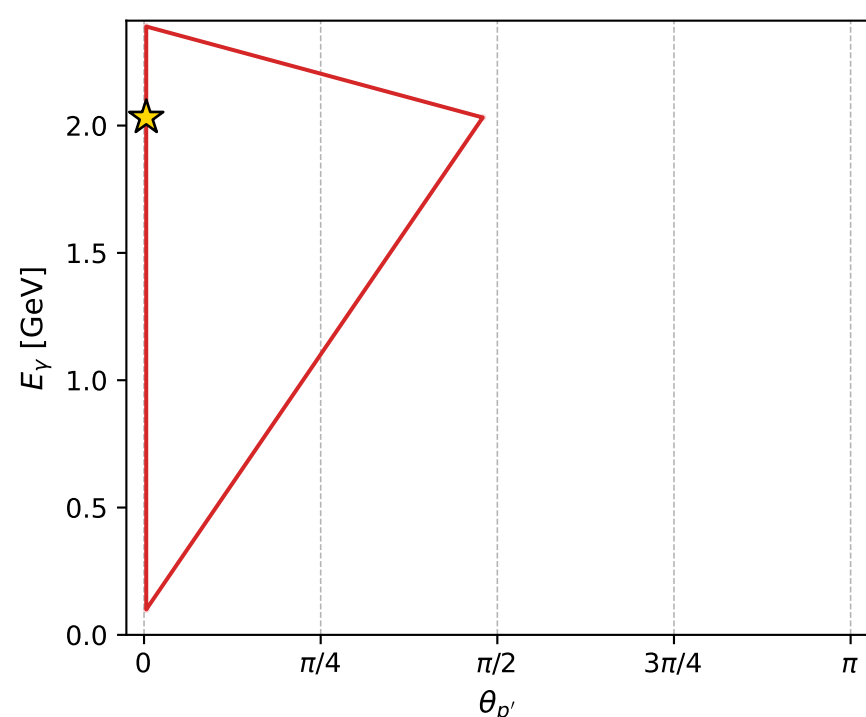
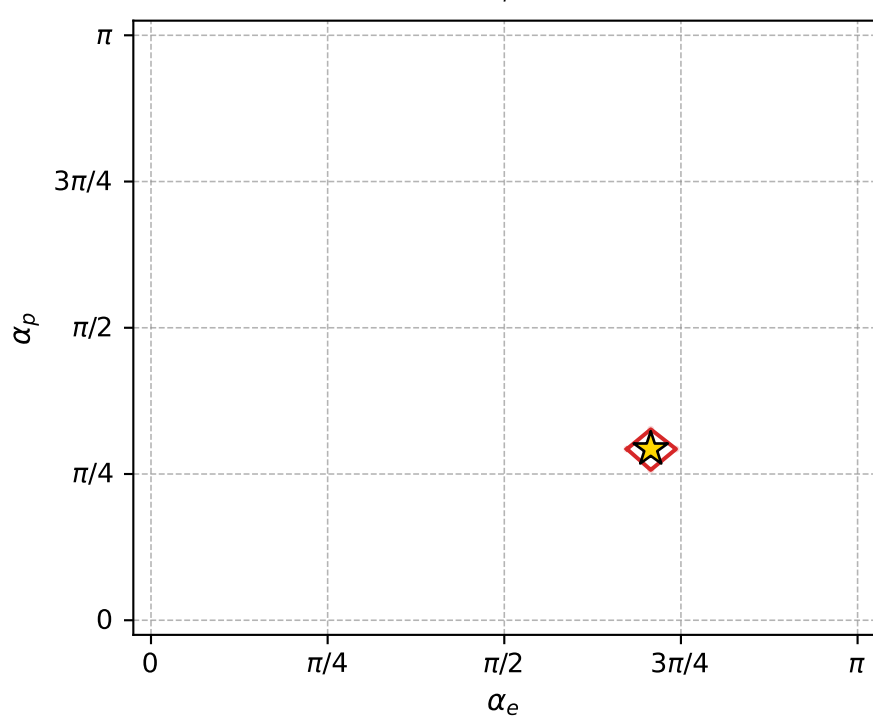
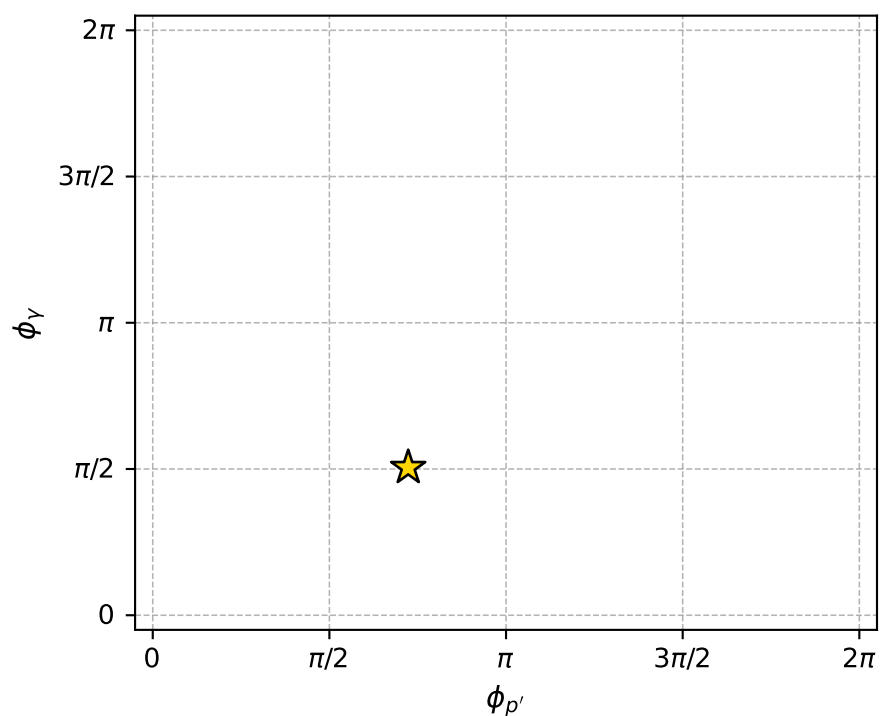
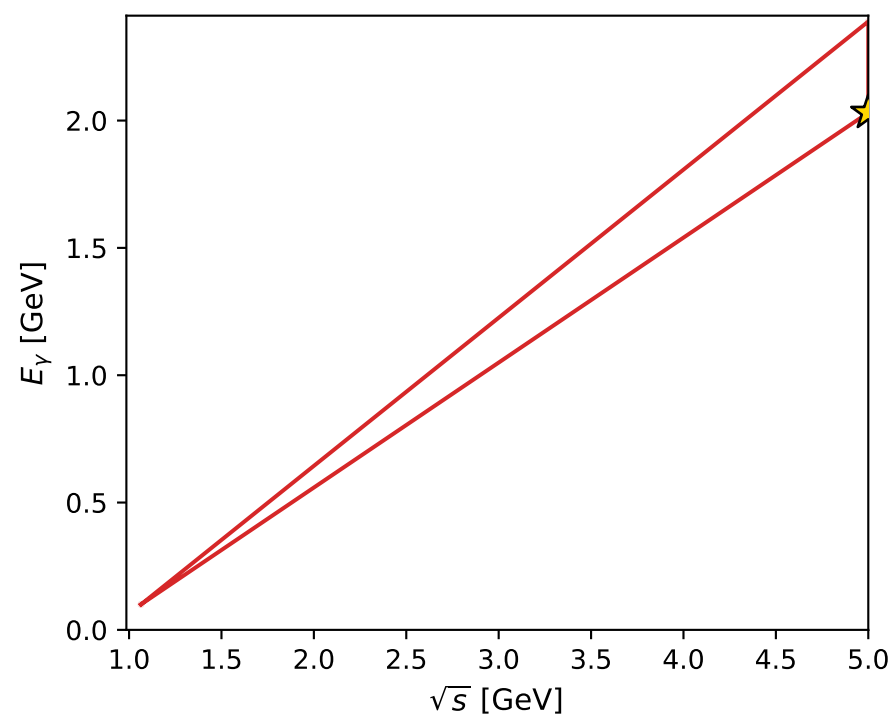
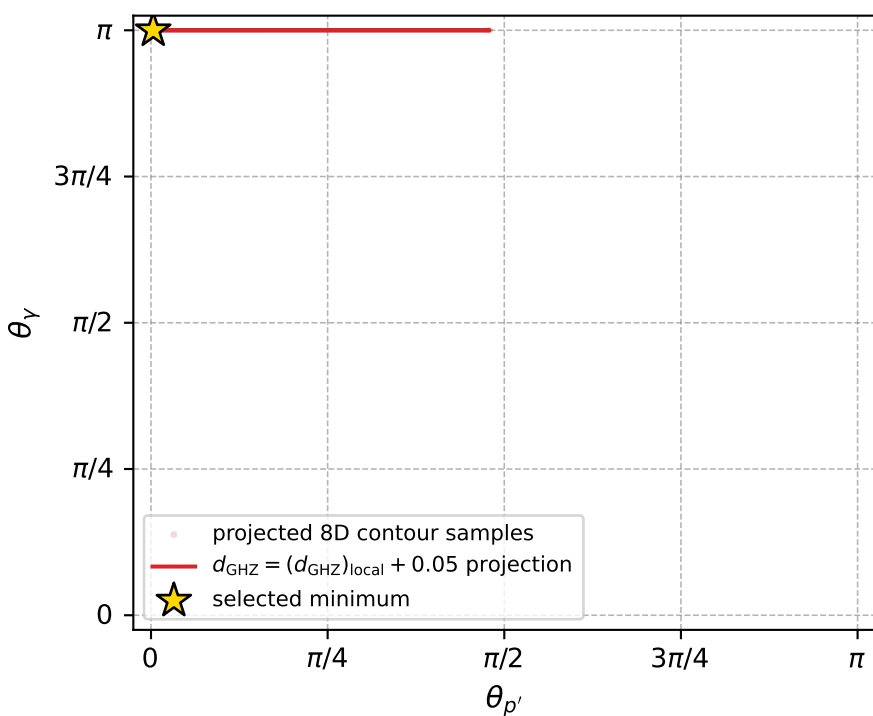


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 51; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6841654\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.5581925\text{e-}09$
 8D contour samples = 66

selected phase-space point:

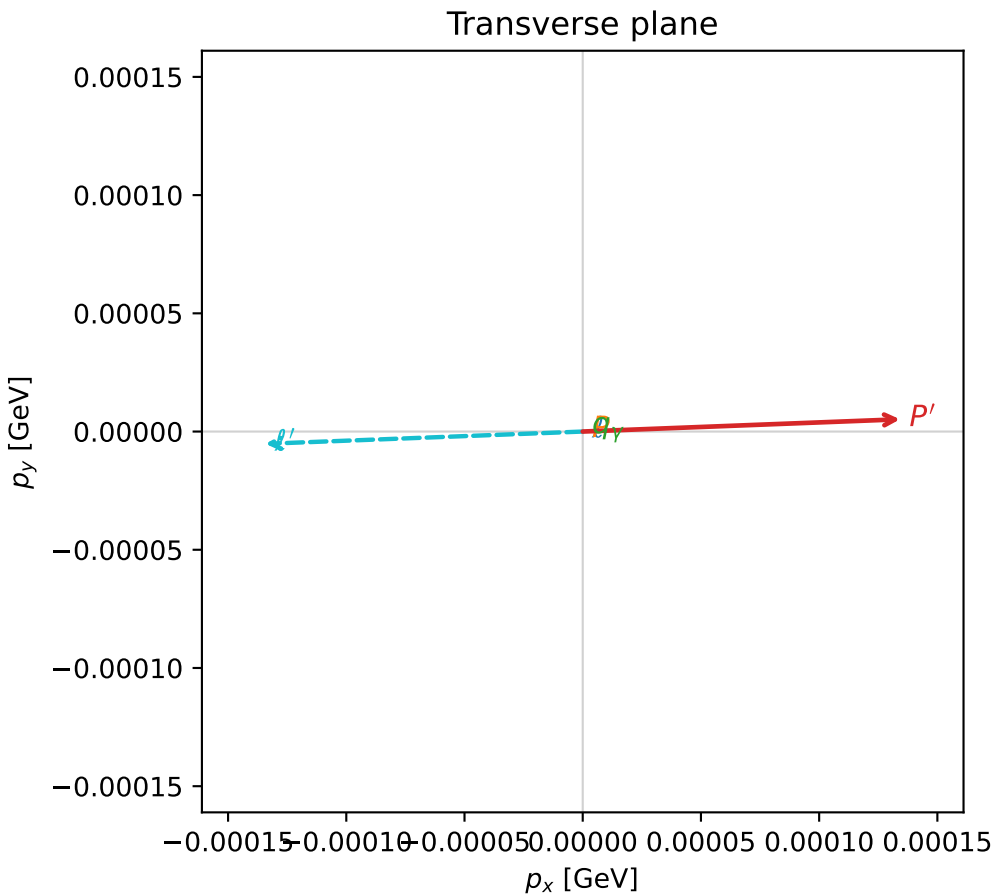
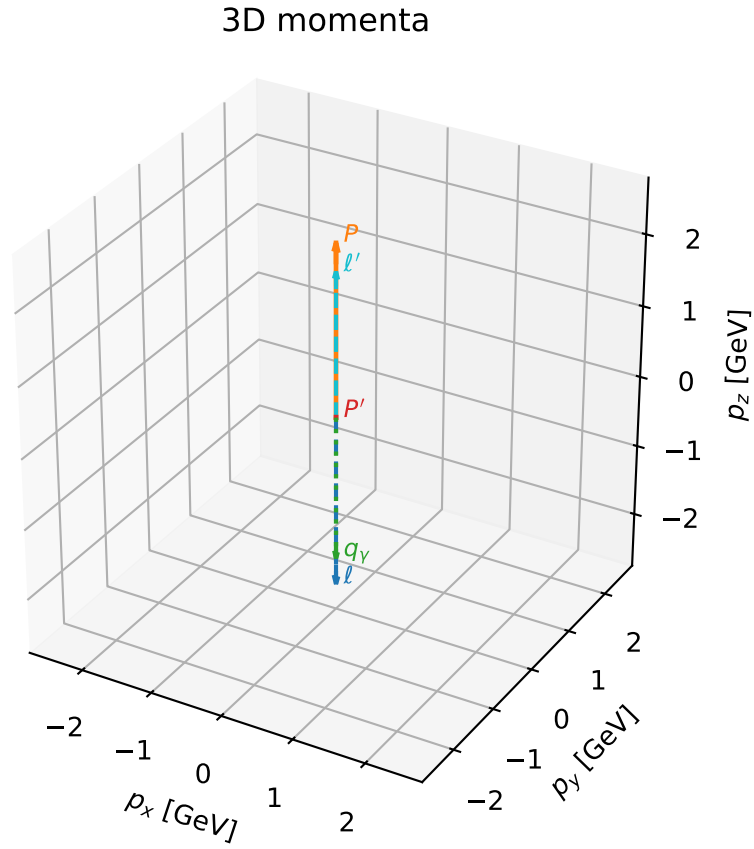
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.009817477$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.031679$
 $\text{phi_p_out} = 2.271731$
 $\text{phi_gamma_out} = 1.586203$
 $\text{alpha_e} = 2.222018$
 $\text{alpha_p} = 0.9195824$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_52 $d_{\{\text{rm GHZ}\}}=2.69014\text{e-}08$
 region: polarization_cluster_03_local_minimum_52
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4223145$ rad
 incoming lepton: $-0.752281|+\rangle +0.658842|-\rangle$
 proton mixing angle: $\alpha_p=0.71927442$ rad
 incoming proton: $0.752284|+\rangle +0.658839|-\rangle$

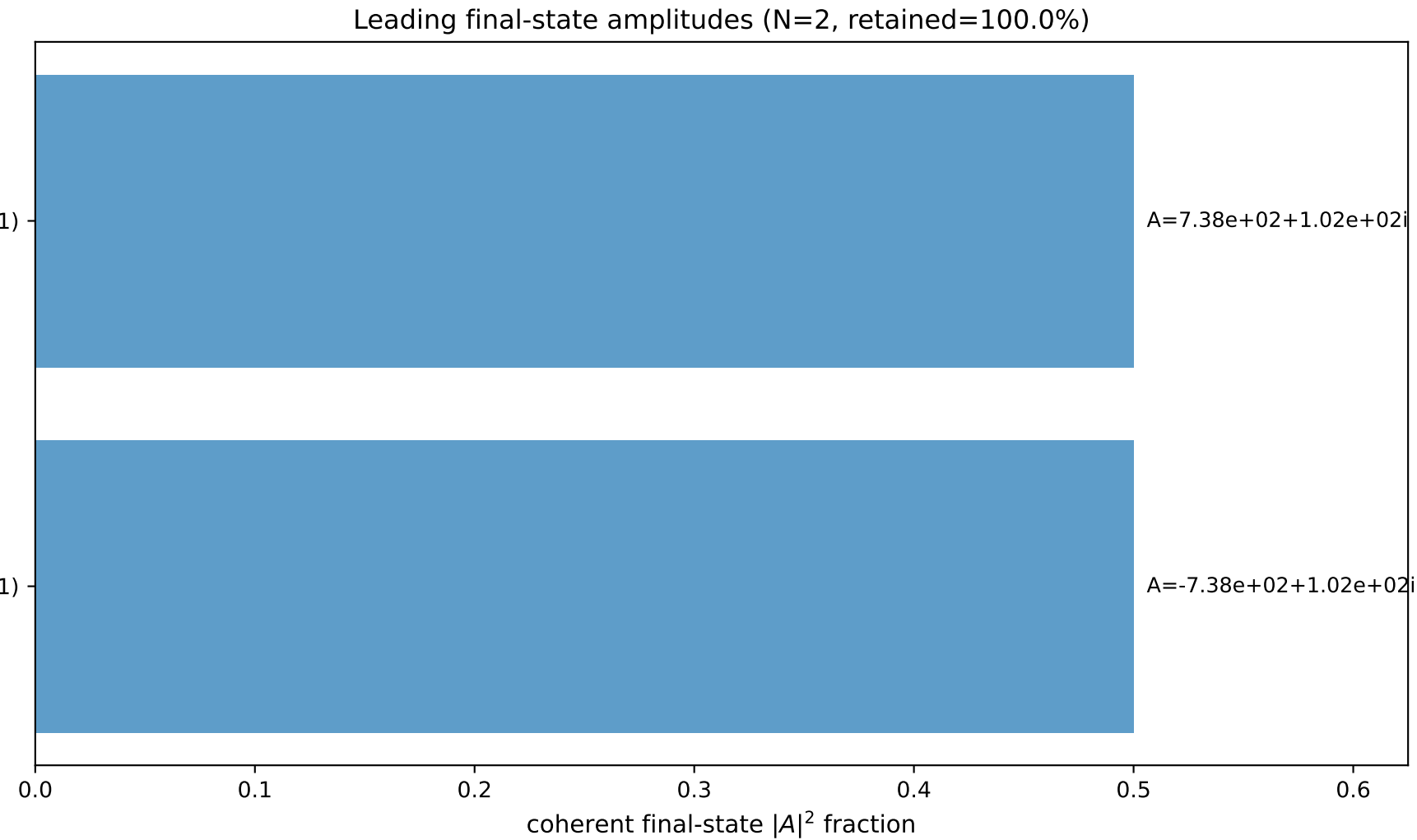
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00684209$
 $\theta_{p'}=0.0196328$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0386213$, $\phi_\gamma=6.14533$
 $E_\gamma=2.03441$, $\theta_l=6.62473\text{e-}05$, $\phi_l=3.18021$
 $Q^2=19.5621$, $x_B=0.835752$, $t=-3.0625$, $W^2=4.72433$, $y=0.970416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0276$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0068$
 q_γ $E= 2.0344$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0344$

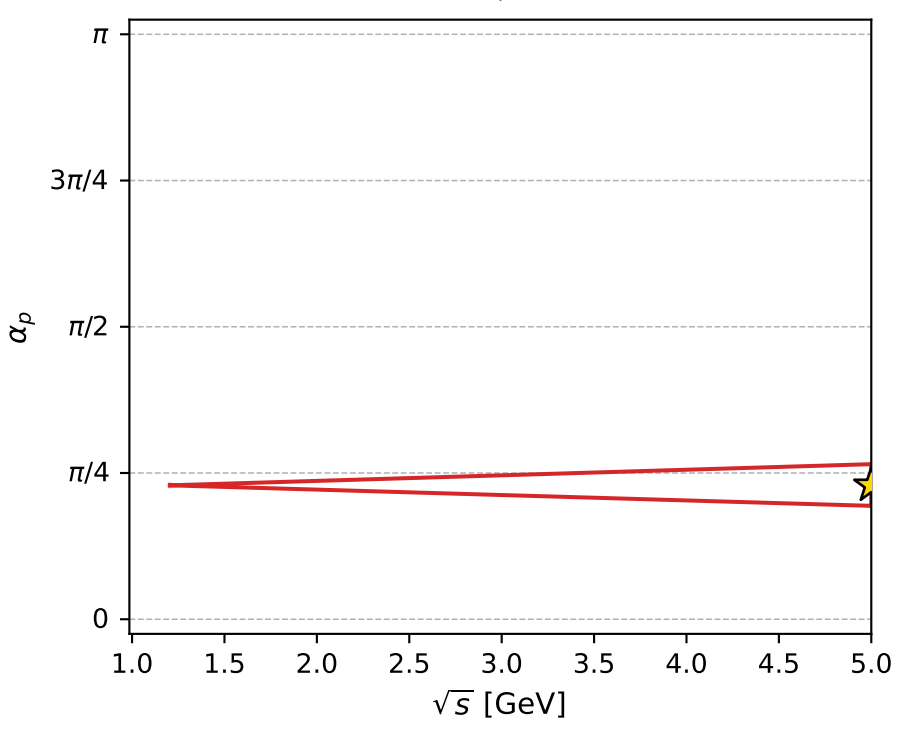
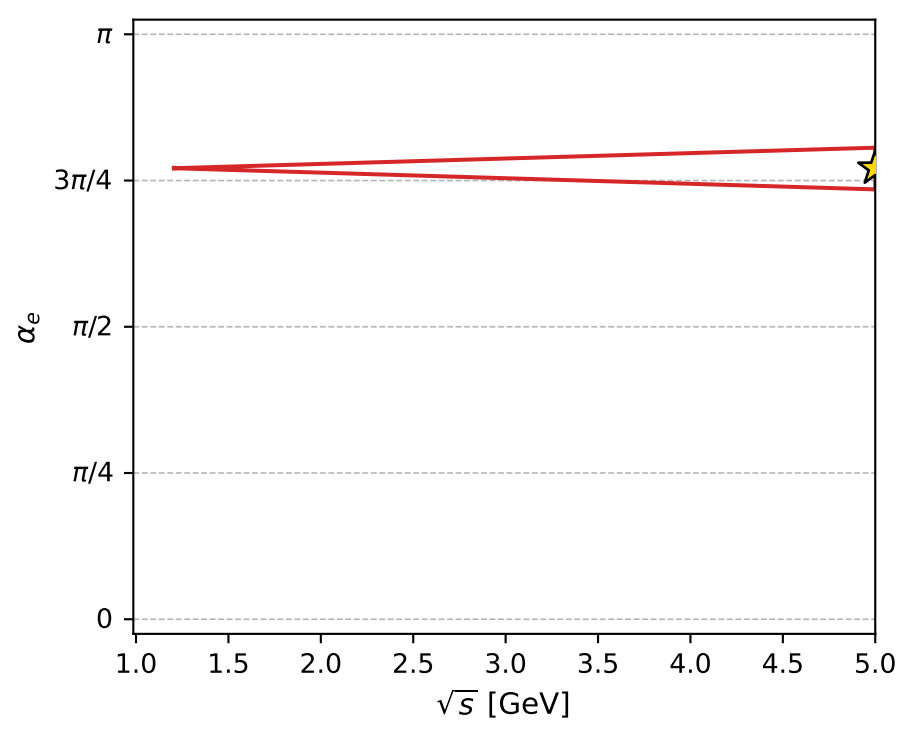
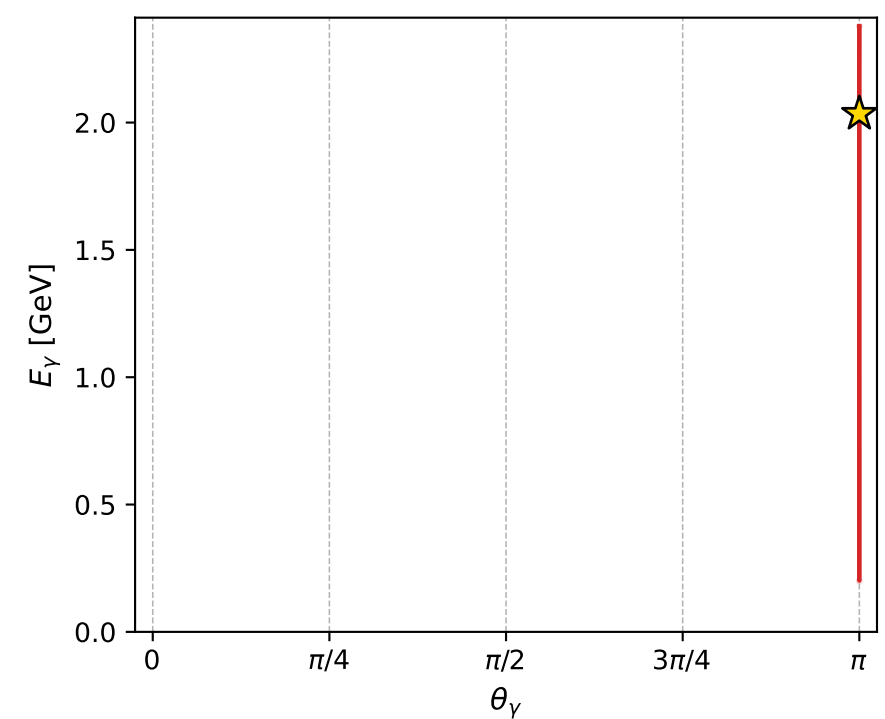
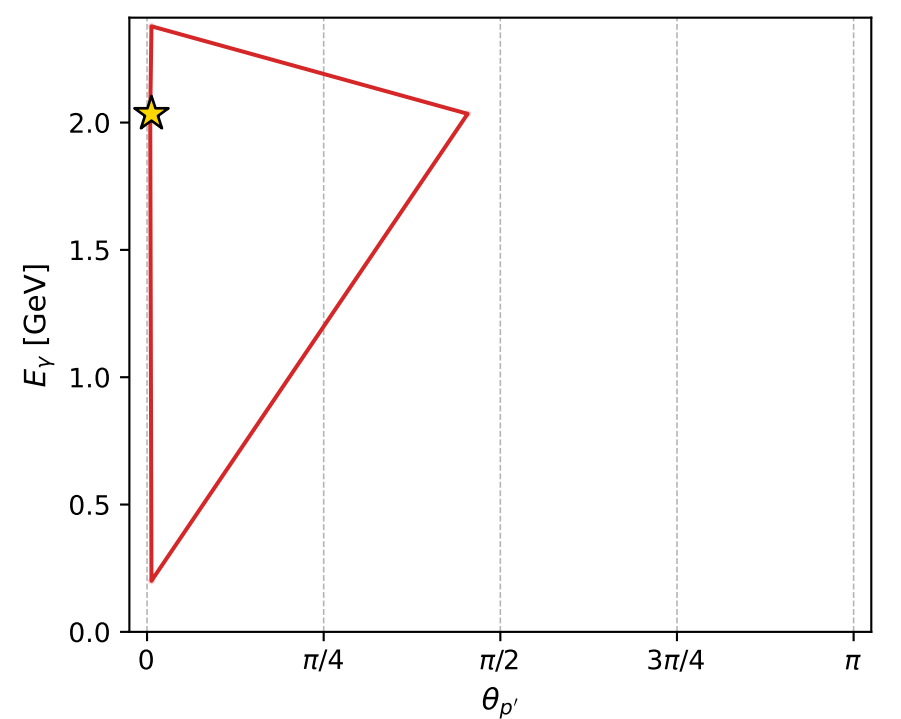
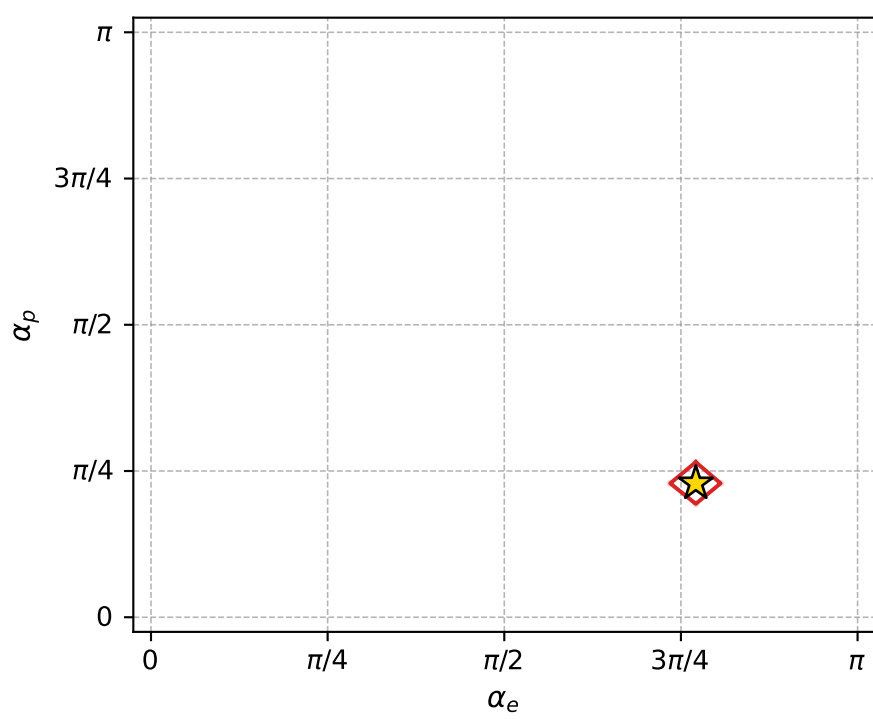
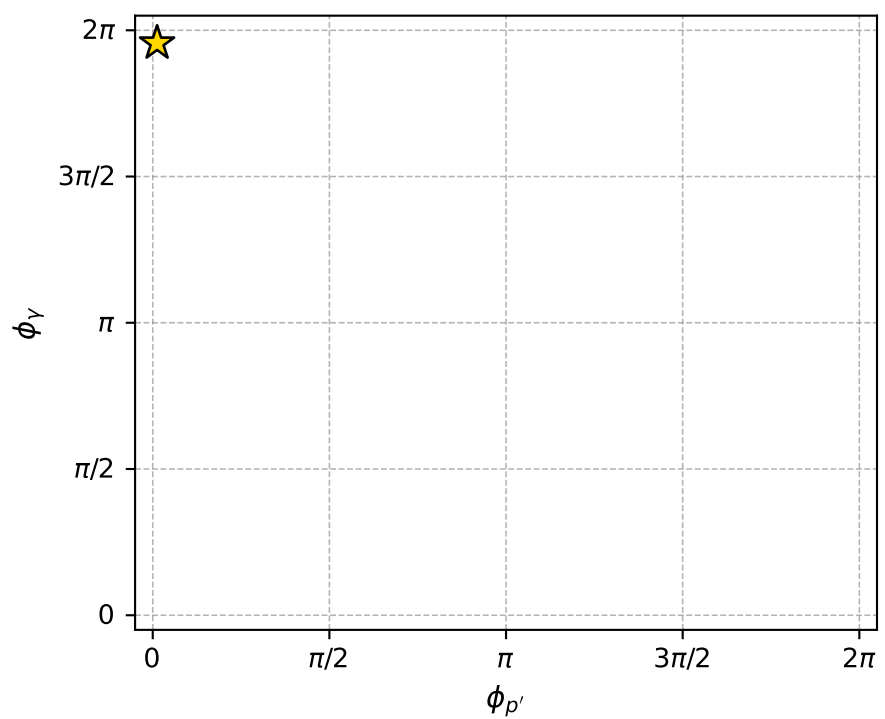
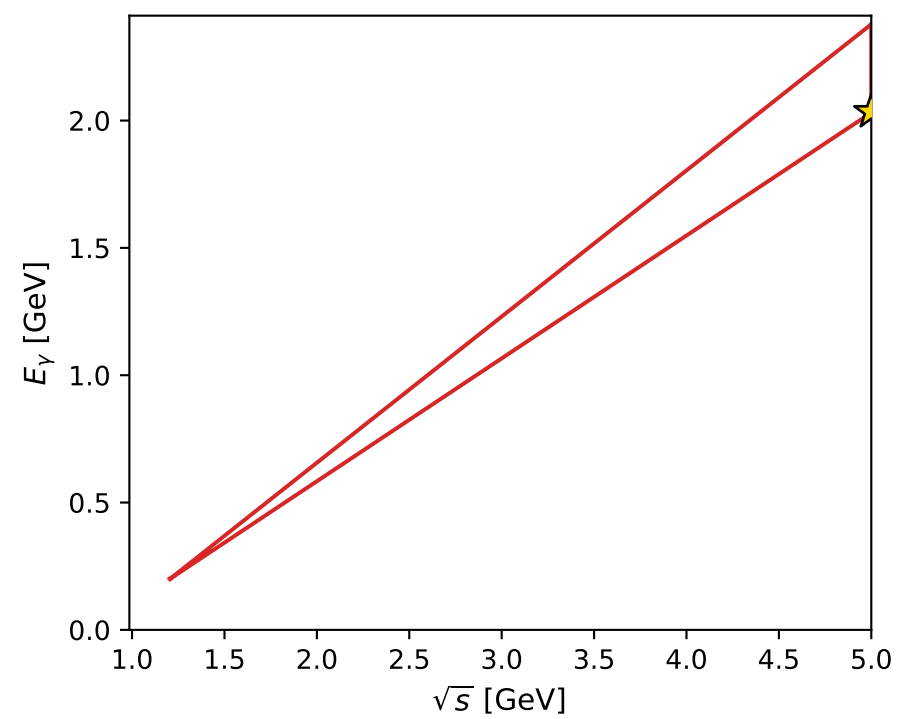
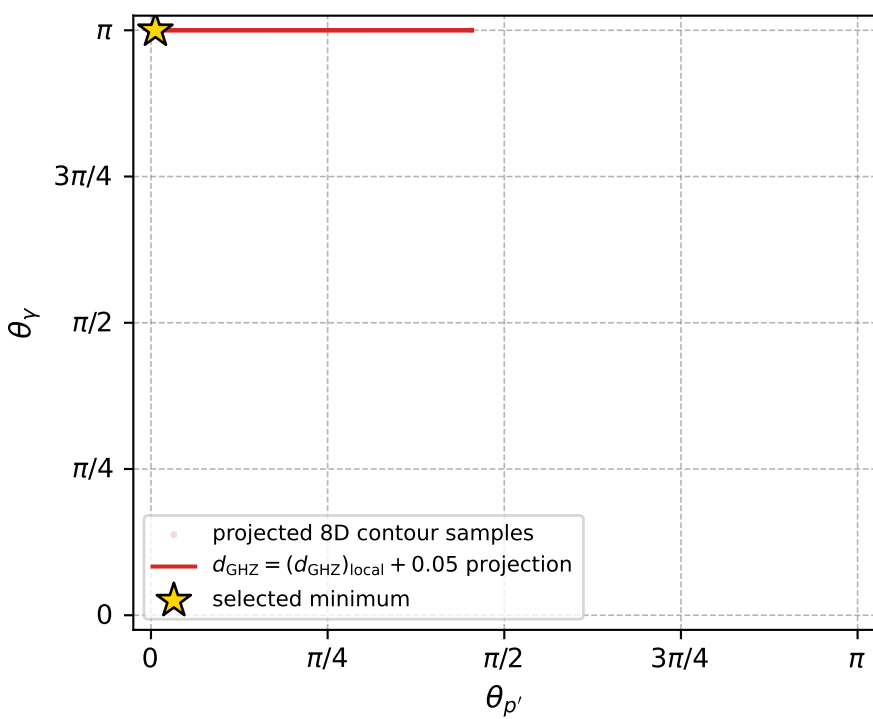


$(h_p, h_\gamma, h_l) = (+1, +1, -1)$

$(h_p, h_\gamma, h_l) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 52; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6901433\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.617971\text{e-}09$
 8D contour samples = 69

selected phase-space point:

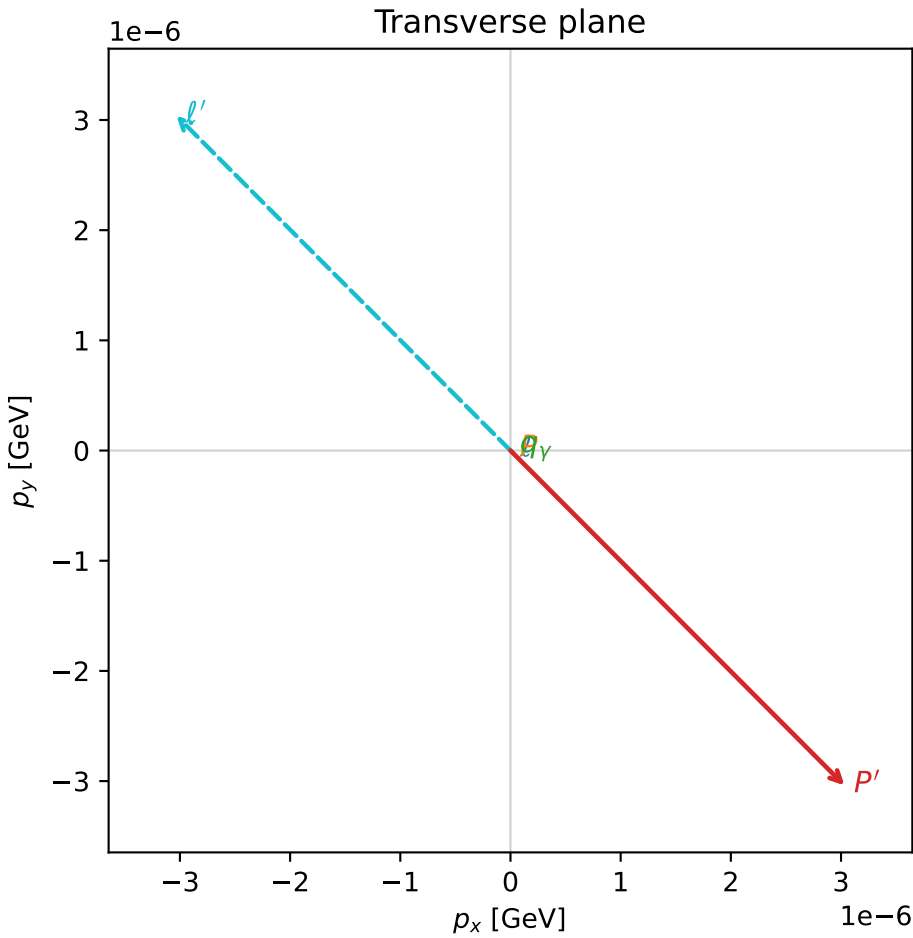
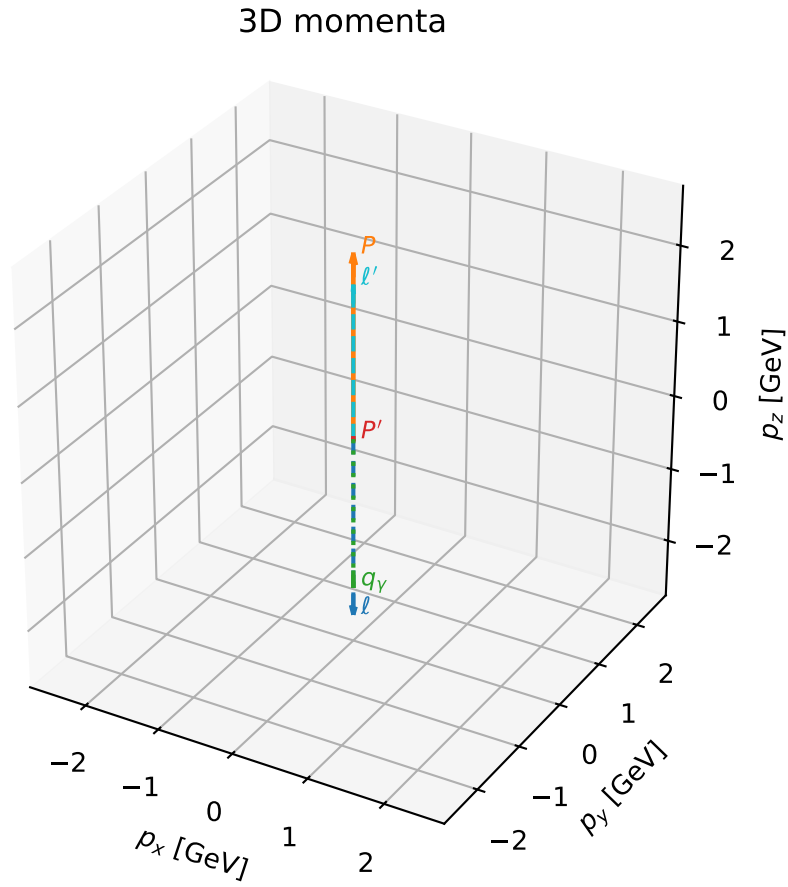
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.01963281$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.034408$
 $\text{phi_p_out} = 0.03862133$
 $\text{phi_gamma_out} = 6.145331$
 $\text{alpha_e} = 2.422314$
 $\text{alpha_p} = 0.7192744$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_53 $d_{\{\text{rm GHZ}\}}=2.69649\text{e-}08$
 region: polarization_cluster_03_local_minimum_53
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2615246$ rad
 incoming lepton: $-0.637099|+\rangle +0.770782|-\rangle$
 proton mixing angle: $\alpha_p=0.8800608$ rad
 incoming proton: $0.637104|+\rangle +0.770778|-\rangle$

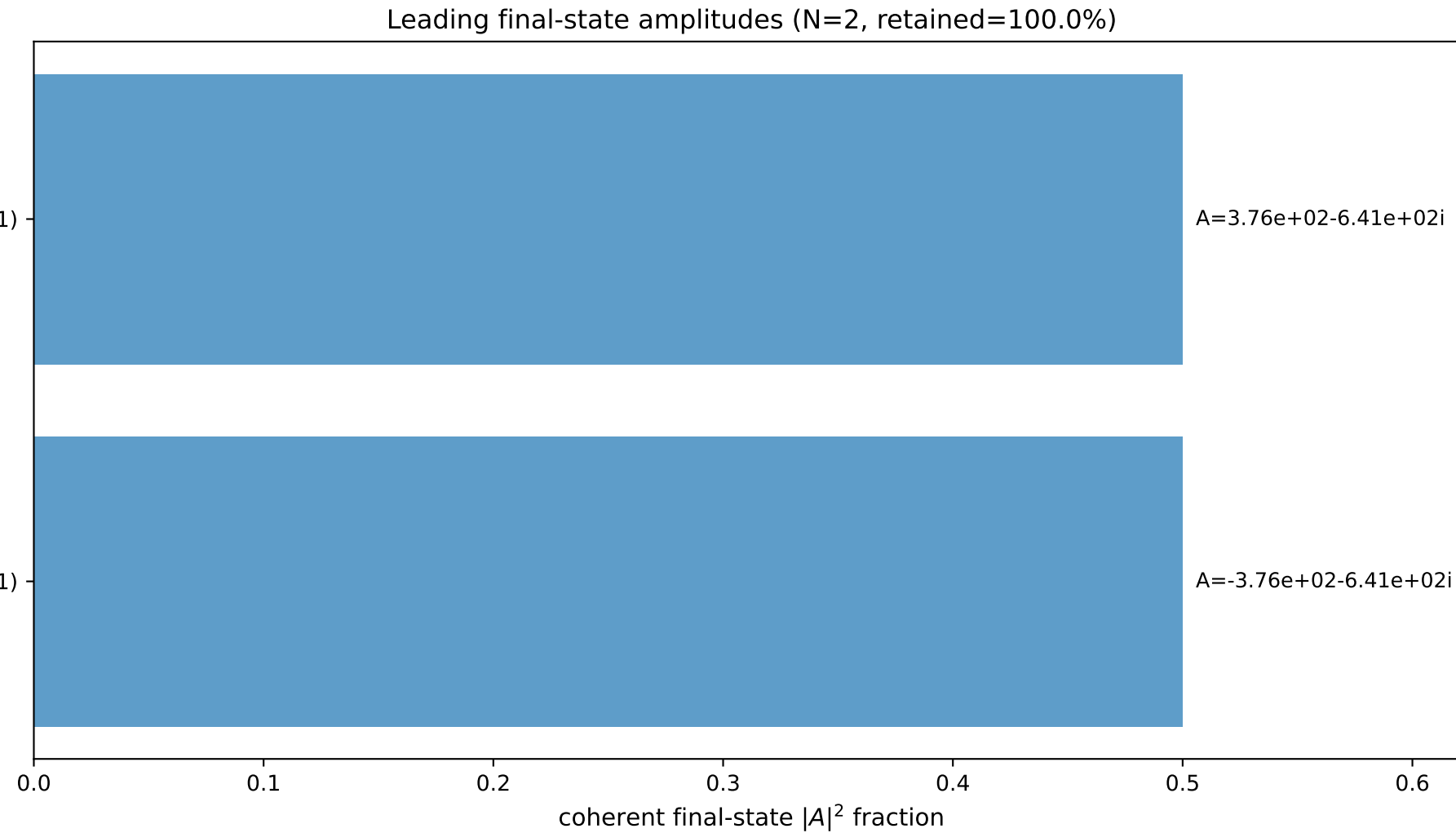
$s=24.9999$, $\sqrt{s}=4.99999$, $|\vec{P}|=2.41201$, $|\vec{P}'|=0.00908808$
 $\theta_{P'}=0.000472367$, $\theta_Y=3.14159$, $\phi_{P'}=5.49656$, $\phi_Y=2.10193$
 $E_Y=2.03552$, $\theta_{\ell'}=2.11843\text{e-}06$, $\phi_{\ell'}=2.35497$
 $Q^2=19.5511$, $x_B=0.83527$, $t=-3.05175$, $W^2=4.73565$, $y=0.970432$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0264$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0091$
 q_Y $E= 2.0355$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0355$

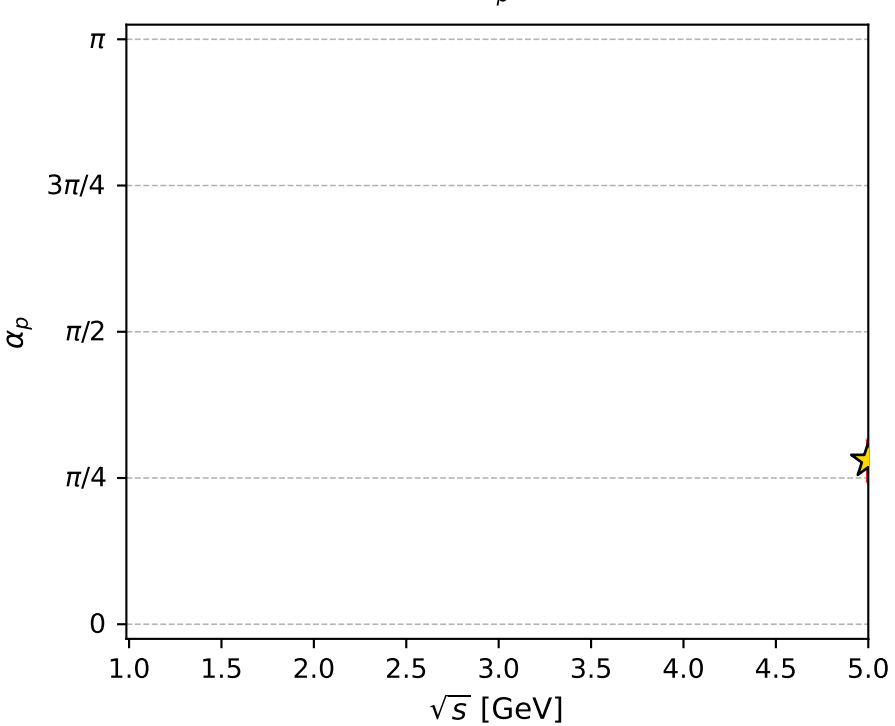
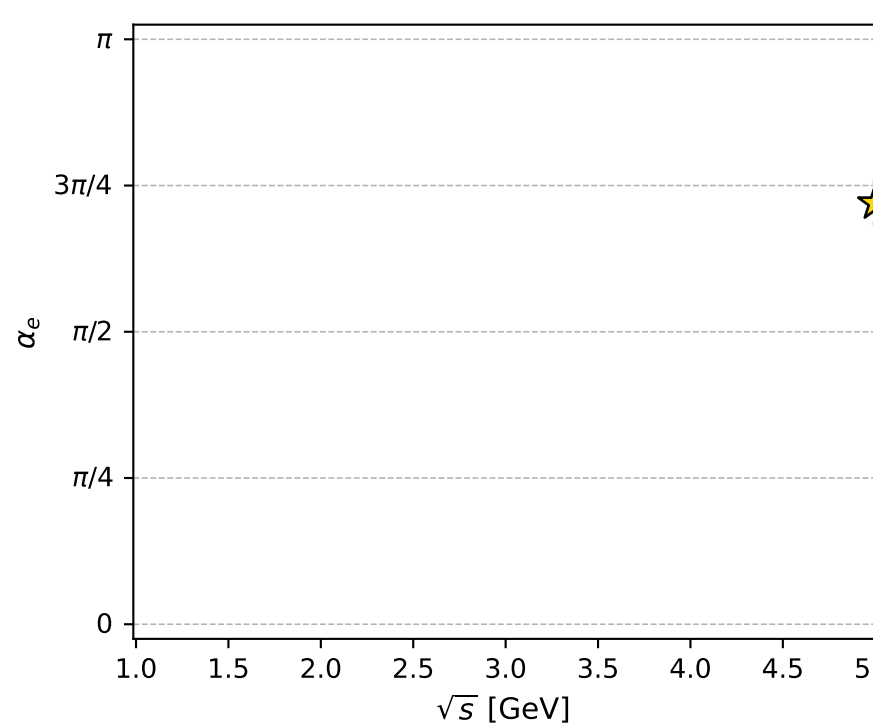
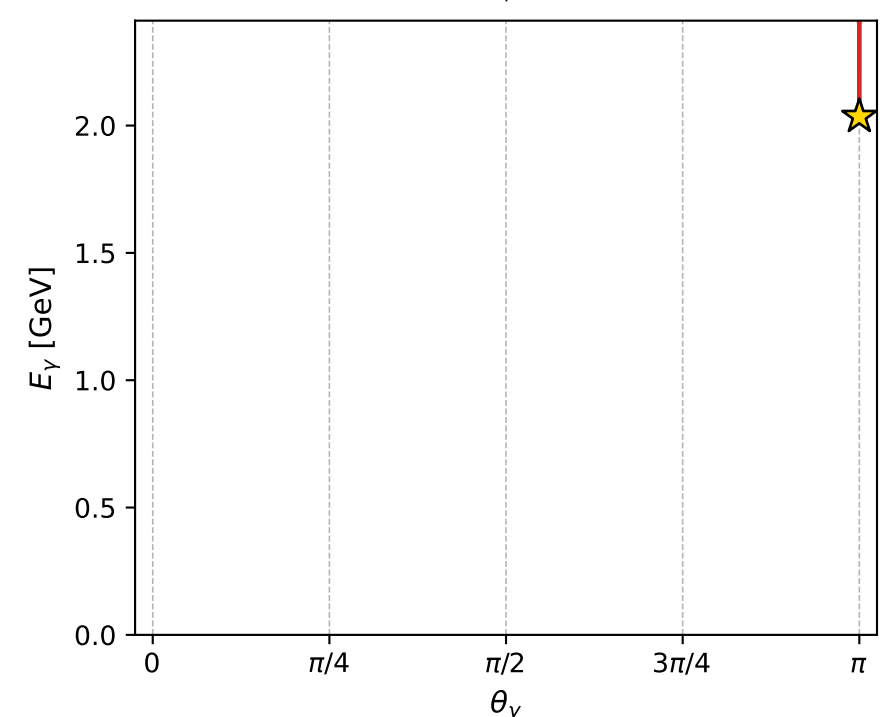
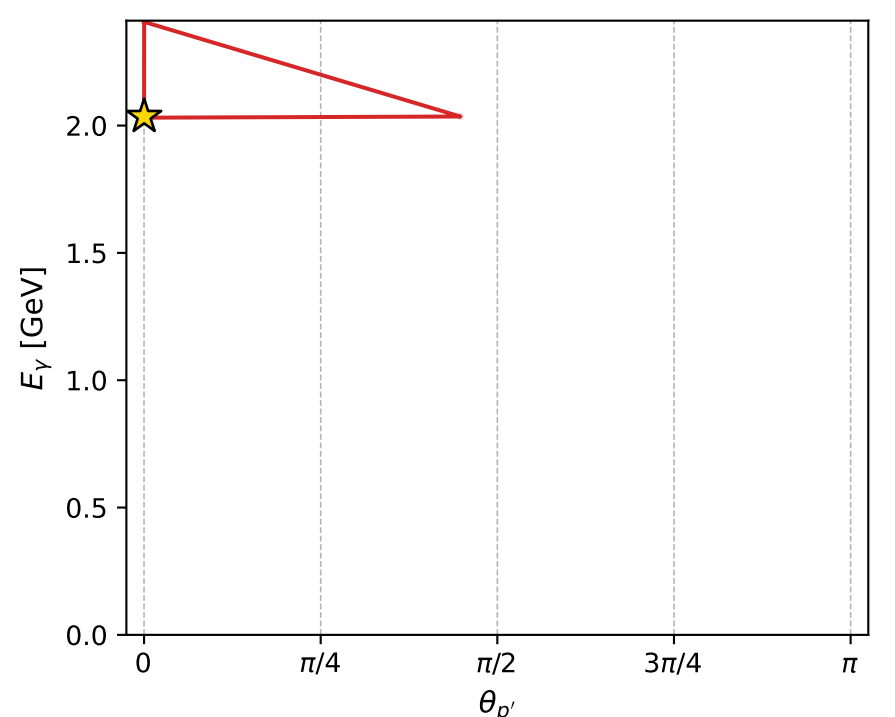
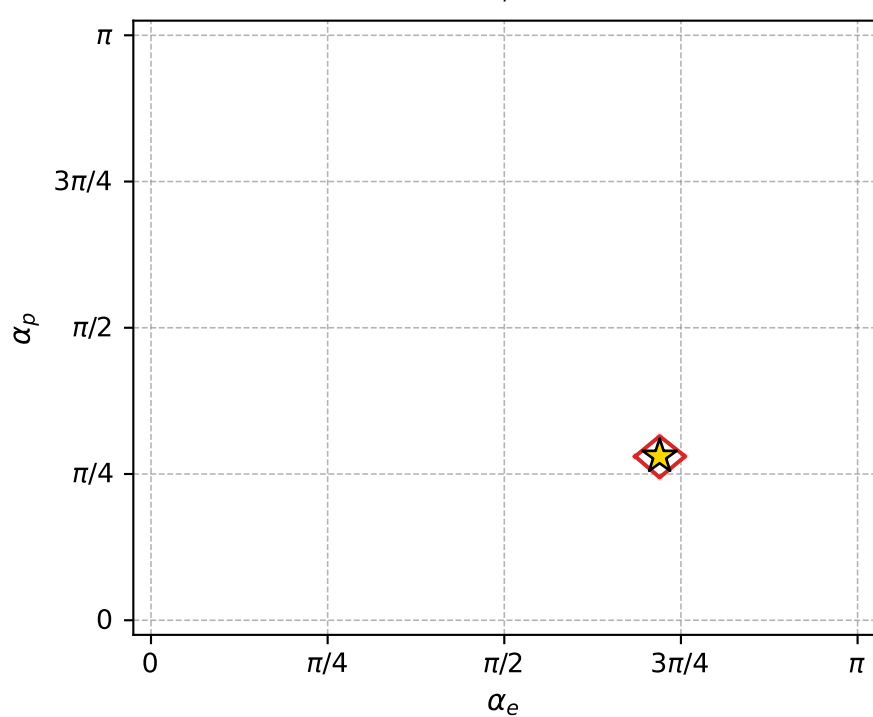
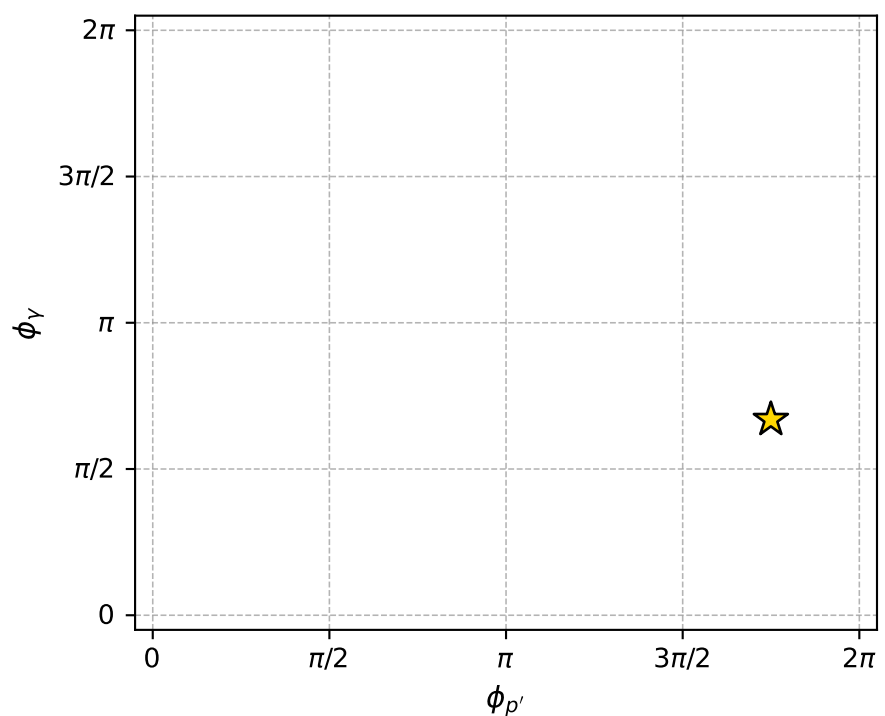
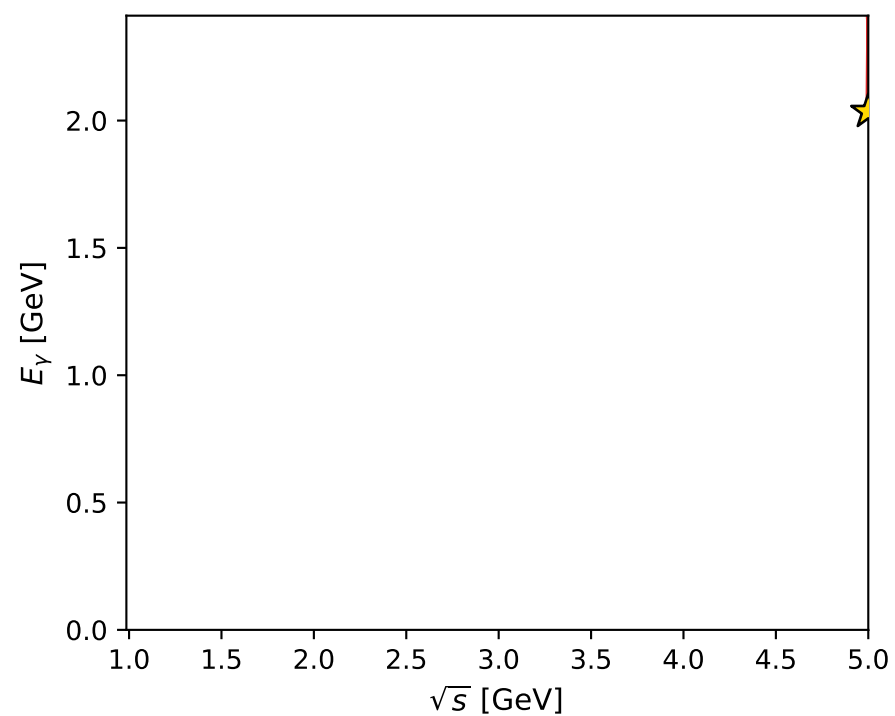
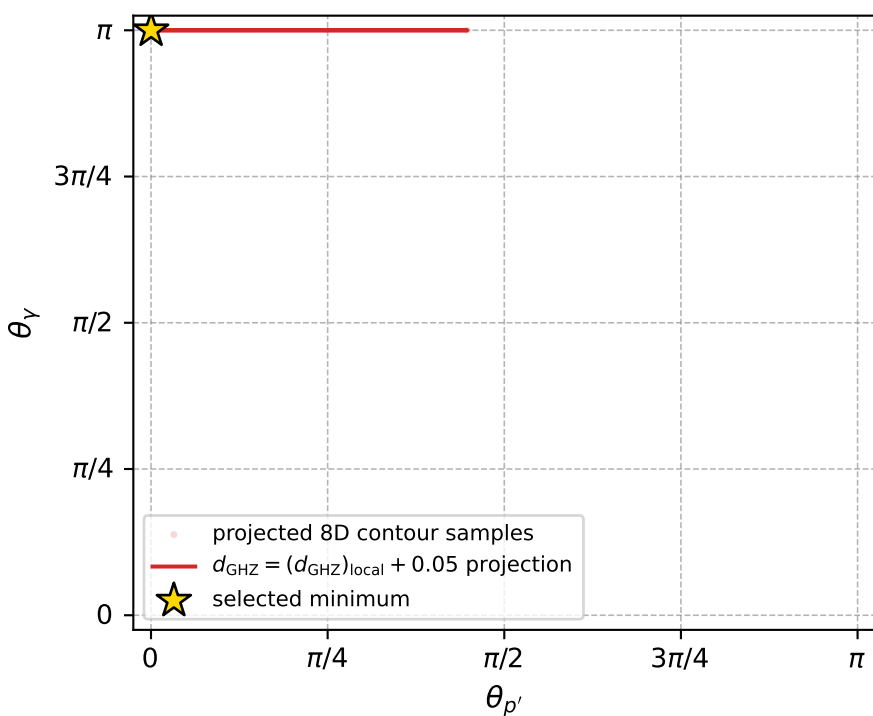


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 53; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.696492\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.6814577\text{e-}09$
 8D contour samples = 65

selected phase-space point:

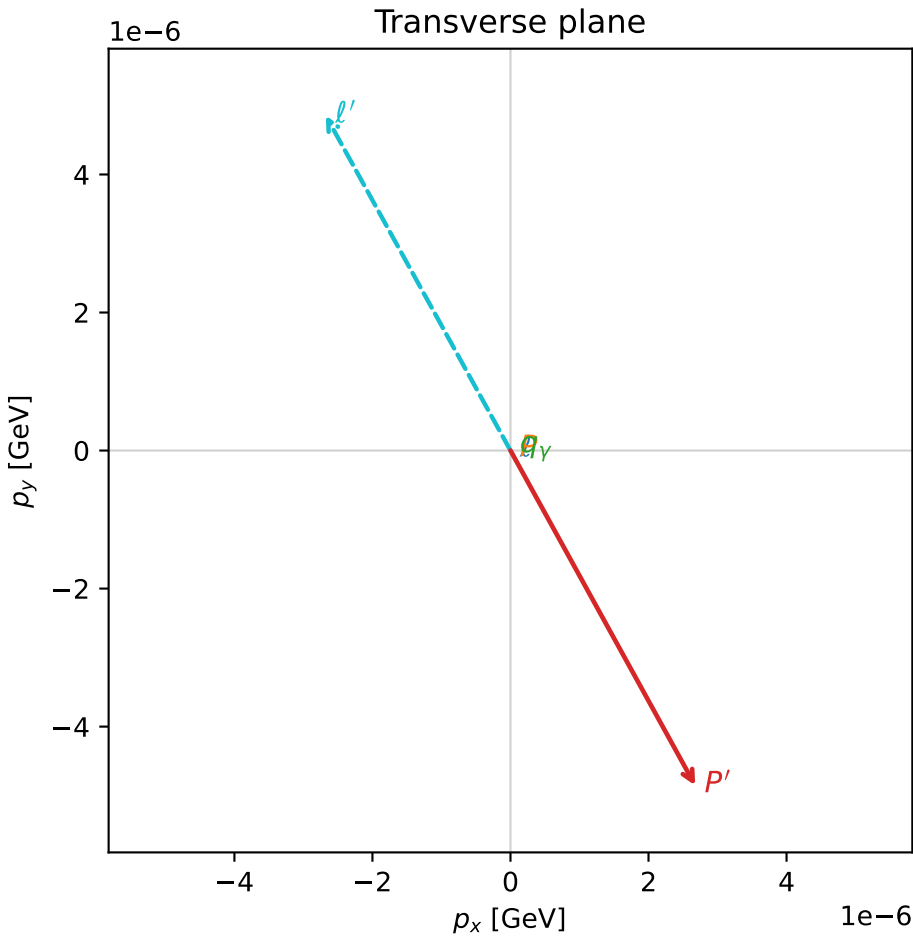
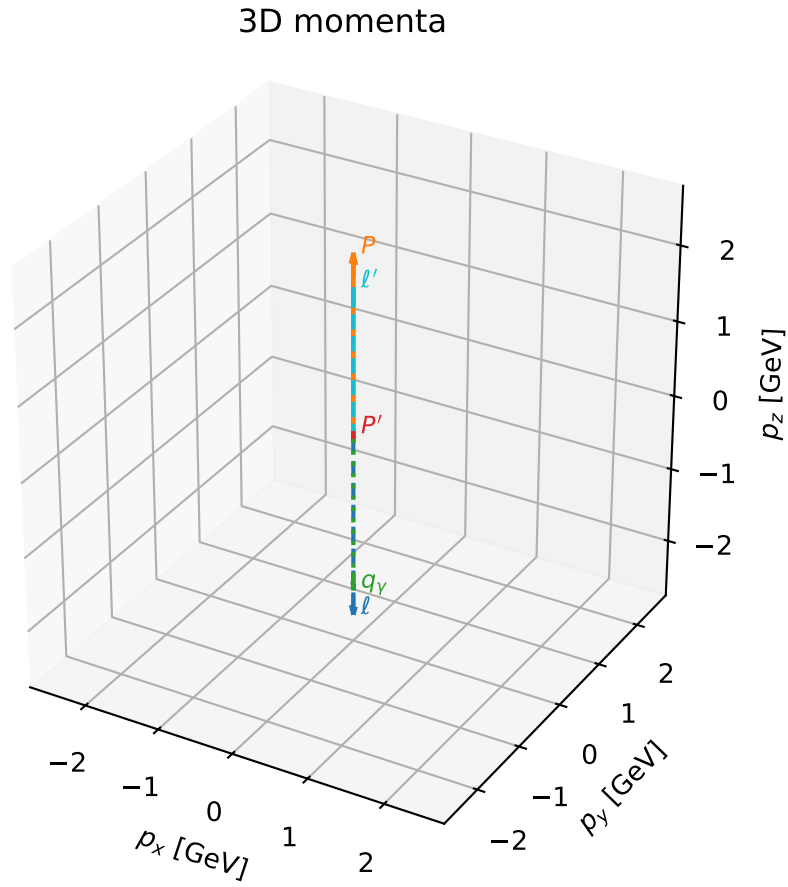
$\text{sqrt_s} = 4.999994$
 $\text{theta_p_out} = 0.0004723665$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.035519$
 $\text{phi_p_out} = 5.496559$
 $\text{phi_gamma_out} = 2.101933$
 $\text{alpha_e} = 2.261525$
 $\text{alpha_p} = 0.8800608$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_55 $d_{\{\text{rm GHZ}\}}=2.70614\text{e-}08$
region: polarization_cluster_03_local_minimum_55
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.360551$ rad
incoming lepton: $-0.710181|+\rangle +0.70402|-\rangle$
proton mixing angle: $\alpha_p=0.78104501$ rad
incoming proton: $0.710178|+\rangle +0.704022|-\rangle$

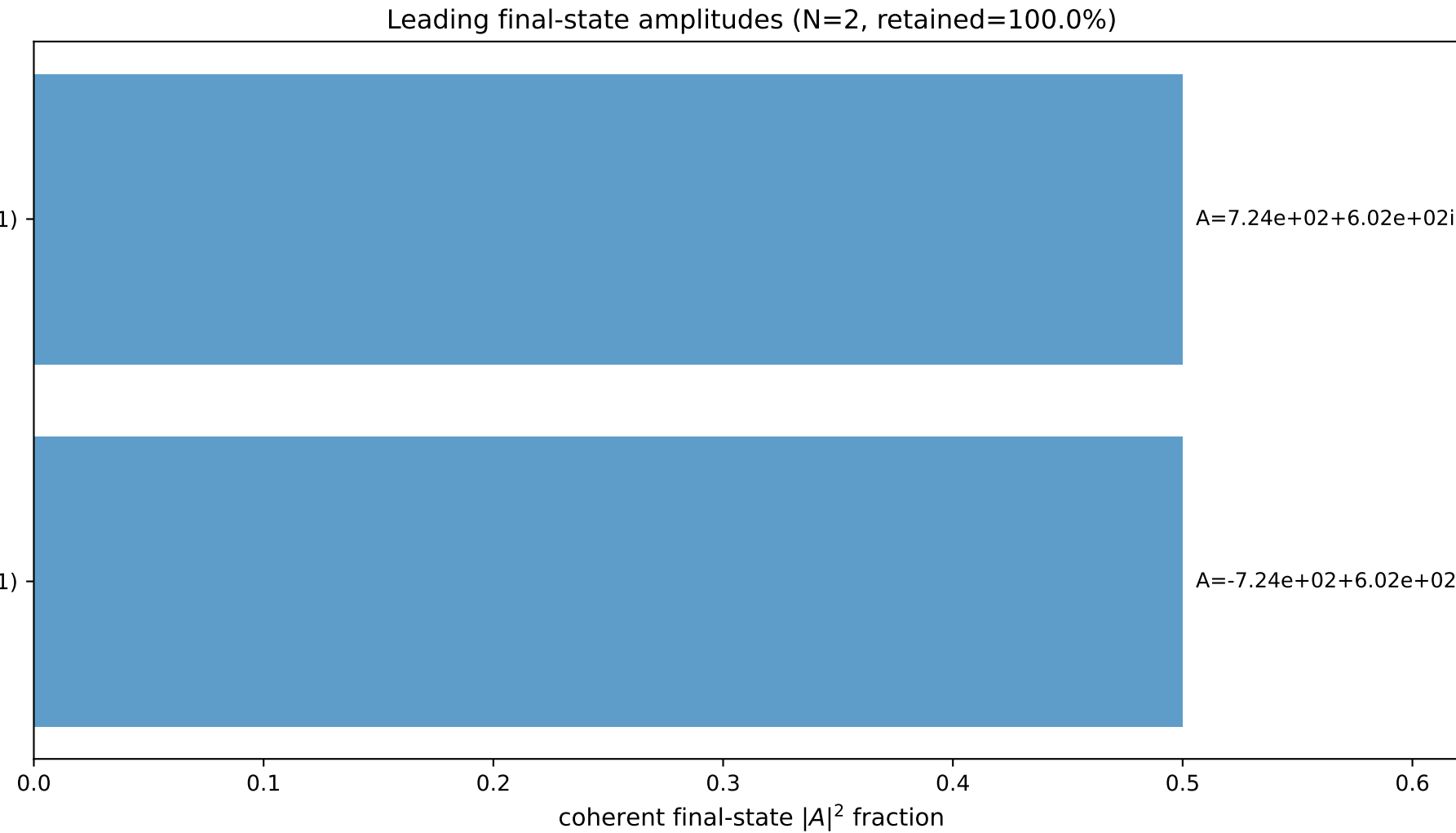
$s=24.9555$, $\sqrt{s}=4.99555$, $|\vec{P}|=2.40971$, $|\vec{P}'|=0.0757091$
 $\theta_{P'}=7.31897\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=5.21638$, $\phi_Y=5.58978$
 $E_Y=2.0651$, $\theta_{\ell'}=2.78532\text{e-}06$, $\phi_{\ell'}=2.07479$
 $Q^2=19.1755$, $x_B=0.820345$, $t=-2.74225$, $W^2=5.07927$, $y=0.970893$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4097$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4097$
 P $E= 2.5858$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4097$
 ℓ' $E= 1.9894$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9894$
 P' $E= 0.9411$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0757$
 q_Y $E= 2.0651$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0651$

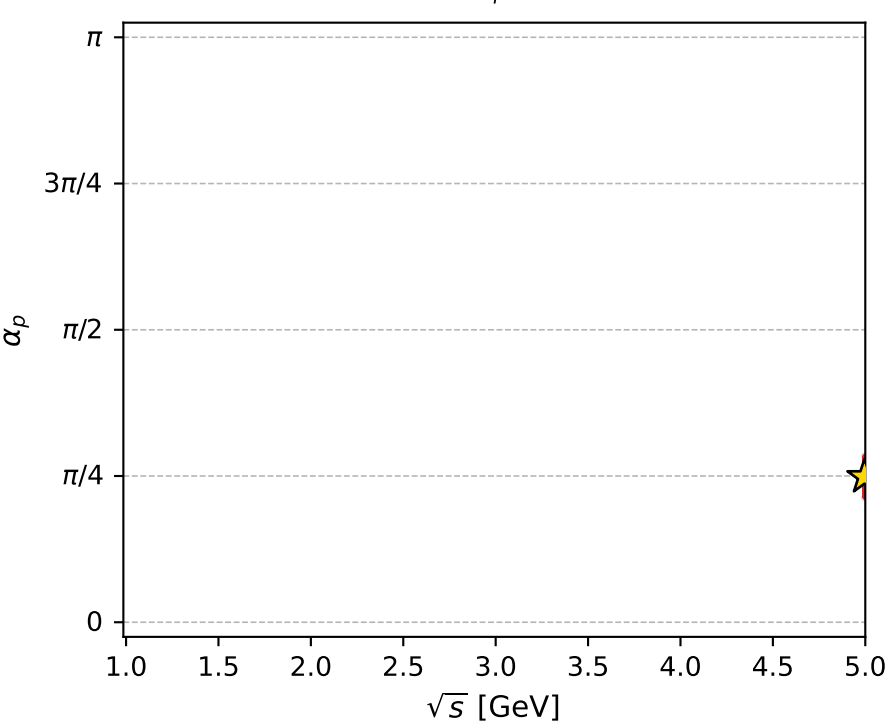
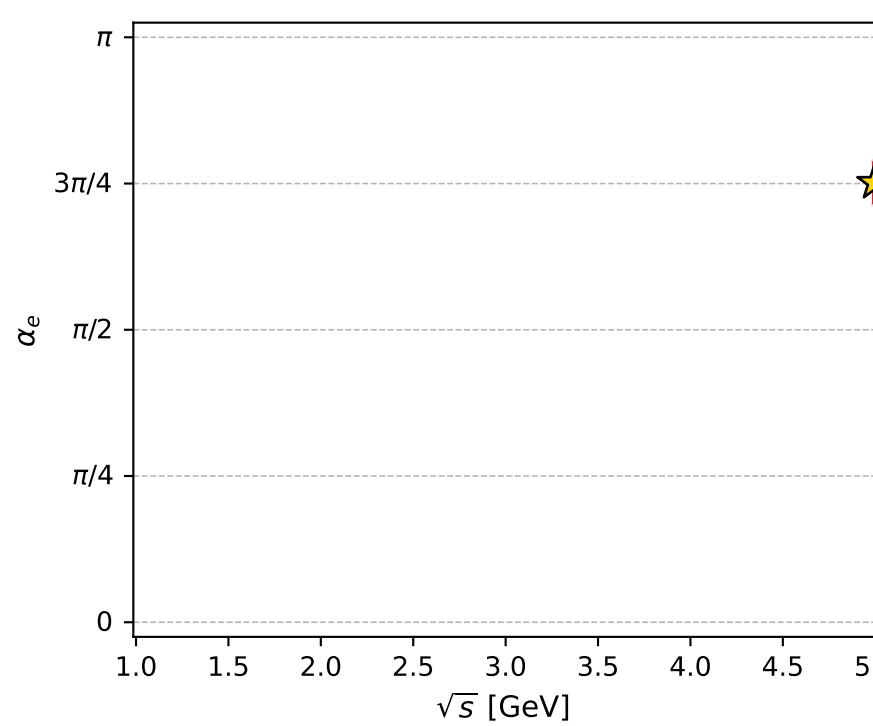
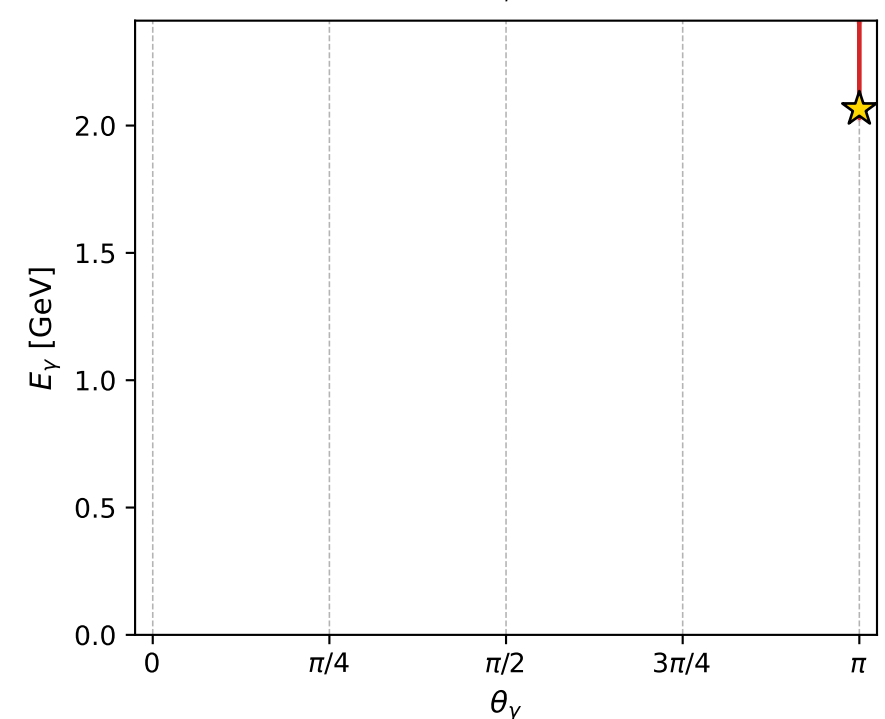
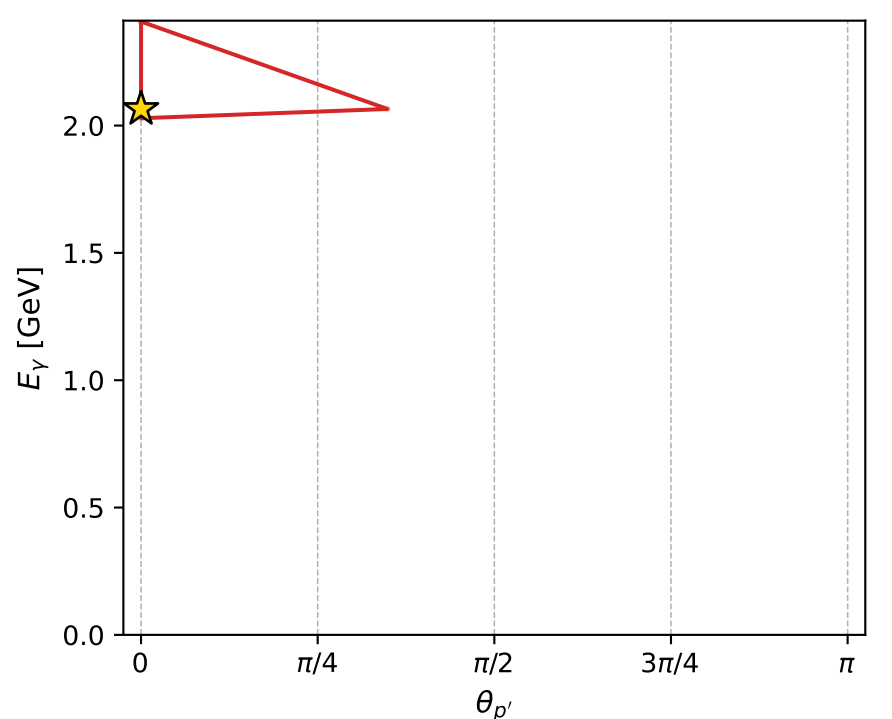
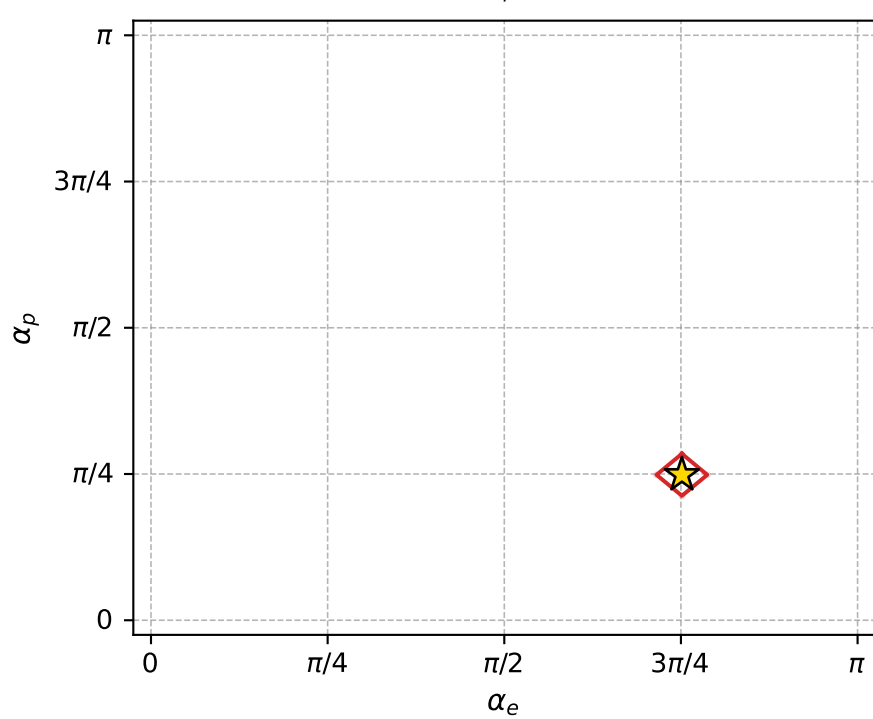
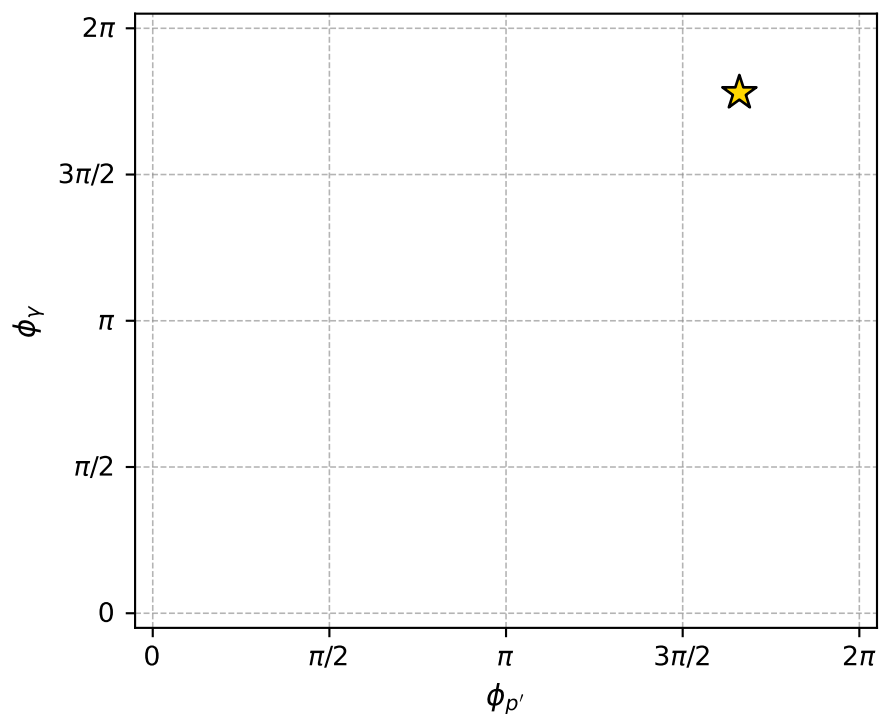
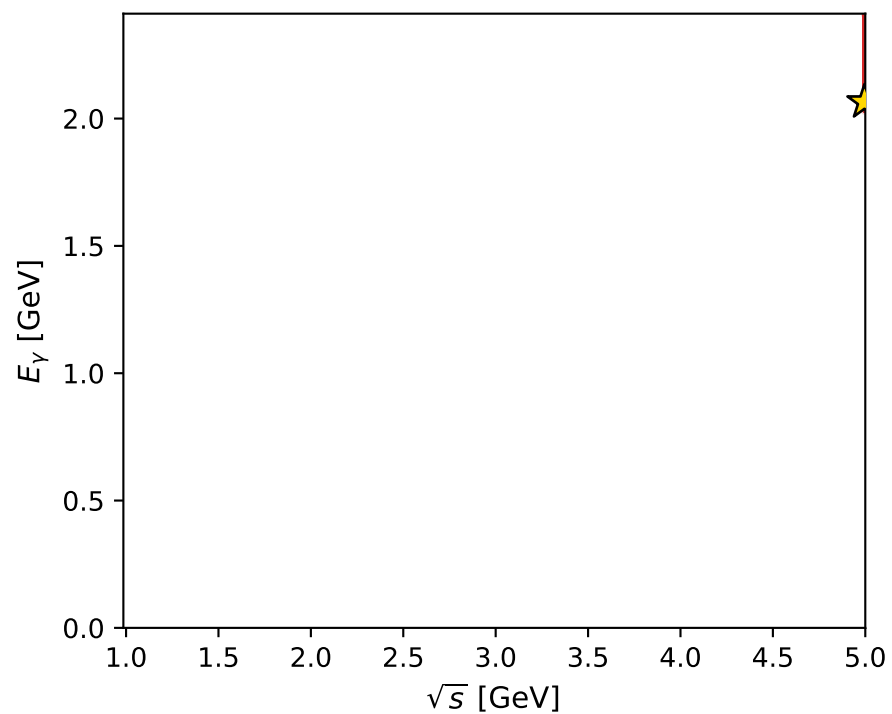
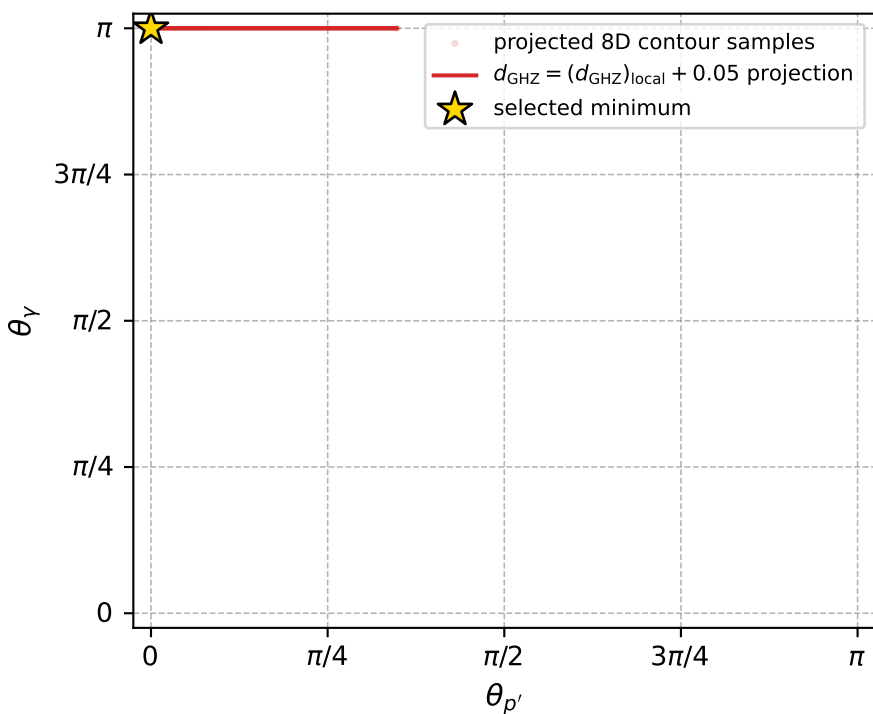


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 55; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7061397\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.7779355\text{e-}09$
 8D contour samples = 88

selected phase-space point:

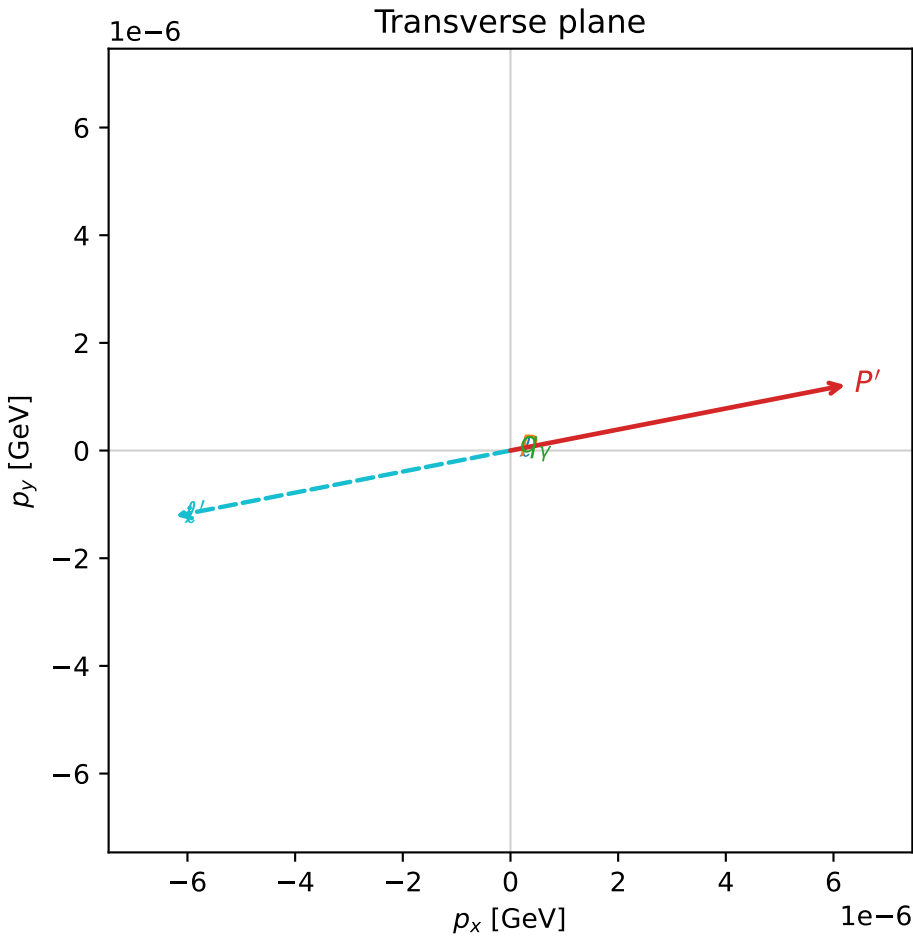
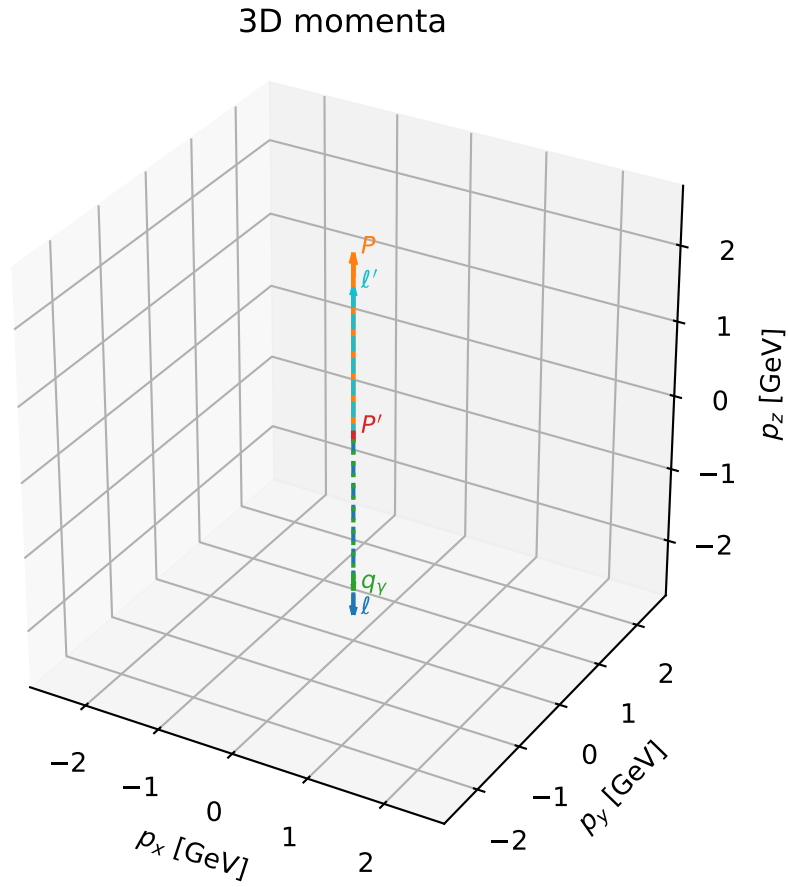
$\text{sqrt_s} = 4.99555$
 $\text{theta_p_out} = 7.318972\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.065104$
 $\text{phi_p_out} = 5.216379$
 $\text{phi_gamma_out} = 5.589779$
 $\text{alpha_e} = 2.360551$
 $\text{alpha_p} = 0.781045$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_56 $d_{\{\text{rm GHZ}\}}=2.70735\text{e-}08$
region: polarization_cluster_03_local_minimum_56
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3601707$ rad
incoming lepton: $-0.709913|+\rangle +0.70429|- \rangle$
proton mixing angle: $\alpha_p=0.78143038$ rad
incoming proton: $0.709907|+\rangle +0.704296|- \rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0826338$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=0.192921$, $\phi_Y=4.88356$
 $E_Y=2.0705$, $\theta_{\ell'}=3.18833\text{e-}06$, $\phi_{\ell'}=3.33451$
 $Q^2=19.1791$, $x_B=0.818899$, $t=-2.71555$, $W^2=5.12133$, $y=0.970995$

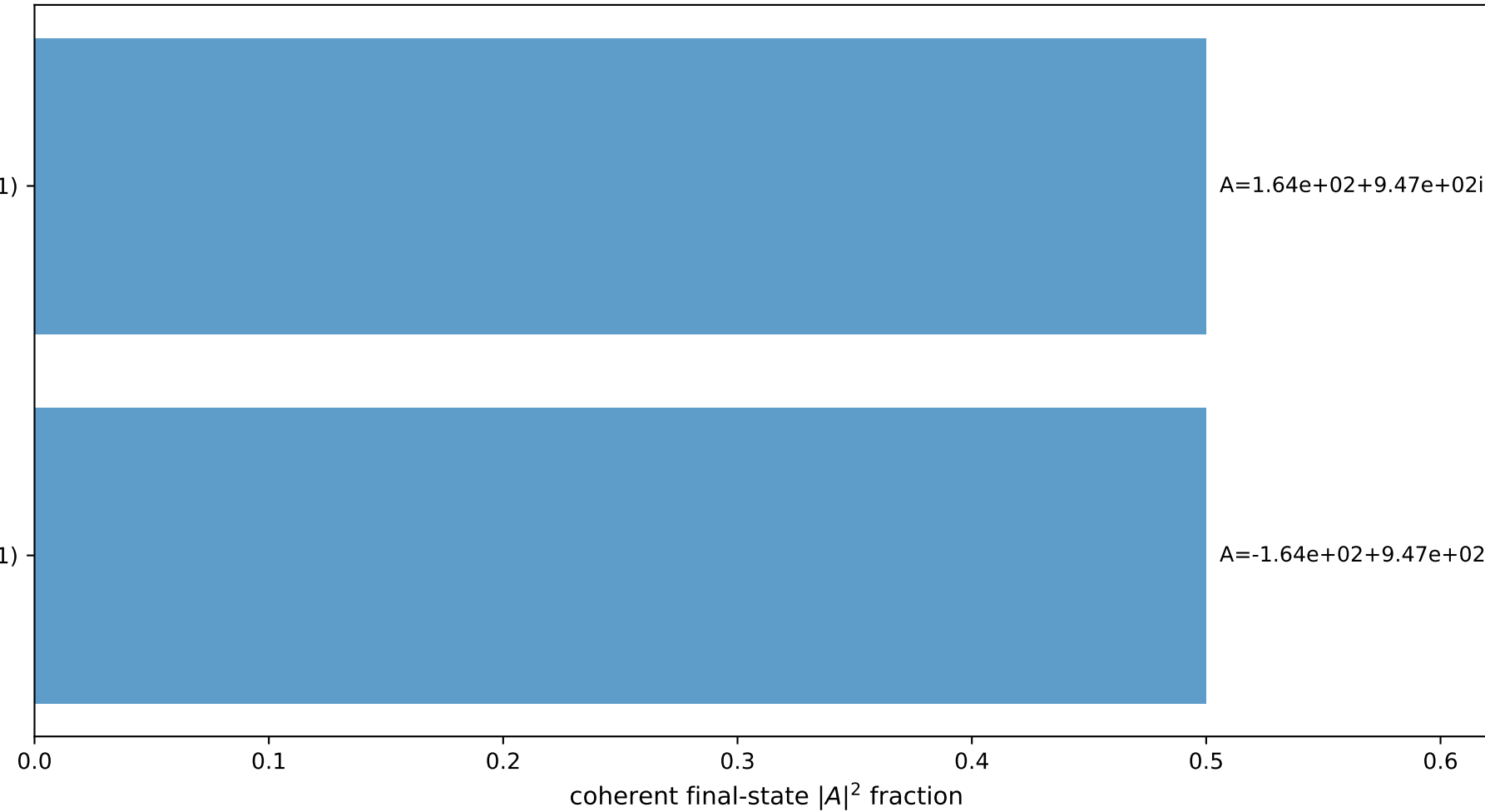
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9879$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9879$
 P' $E= 0.9416$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0826$
 q_Y $E= 2.0705$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0705$



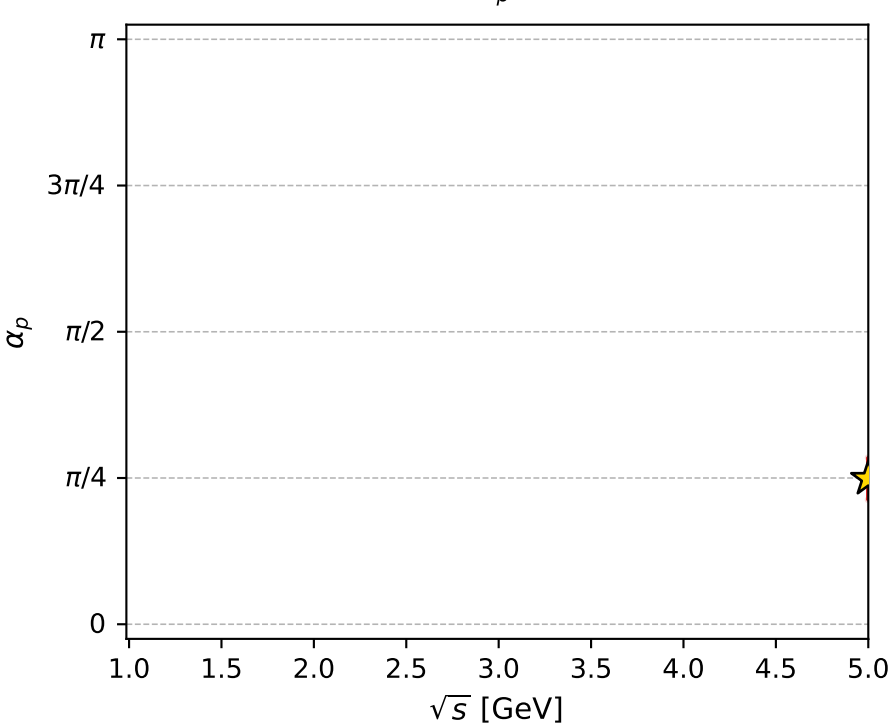
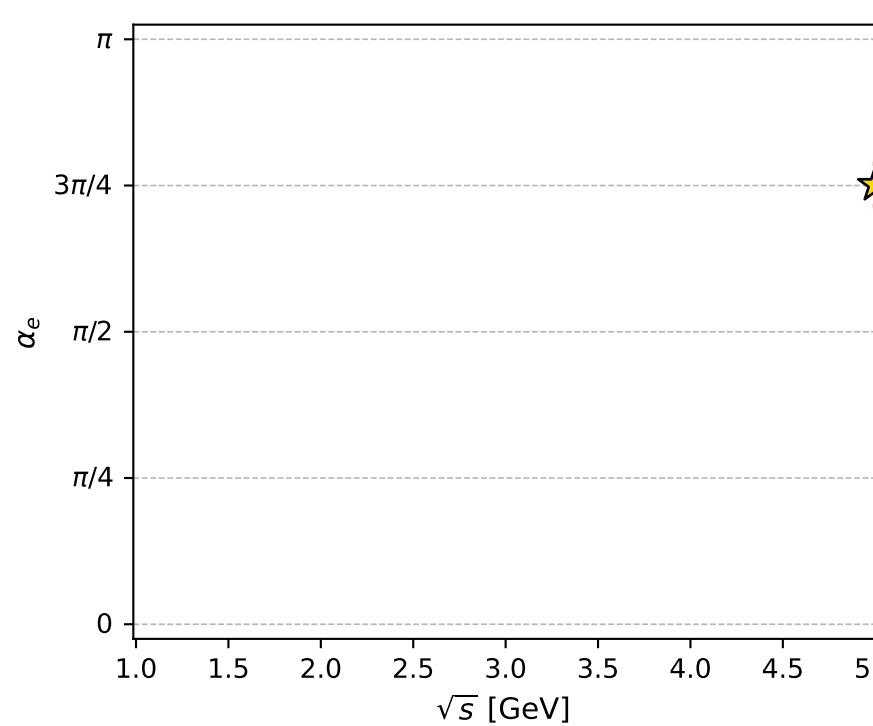
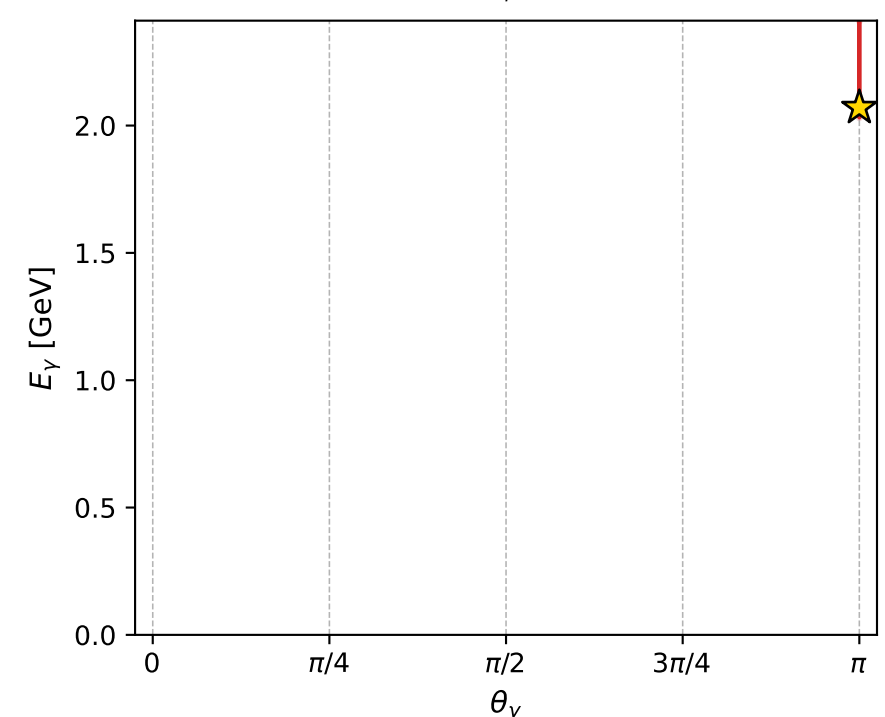
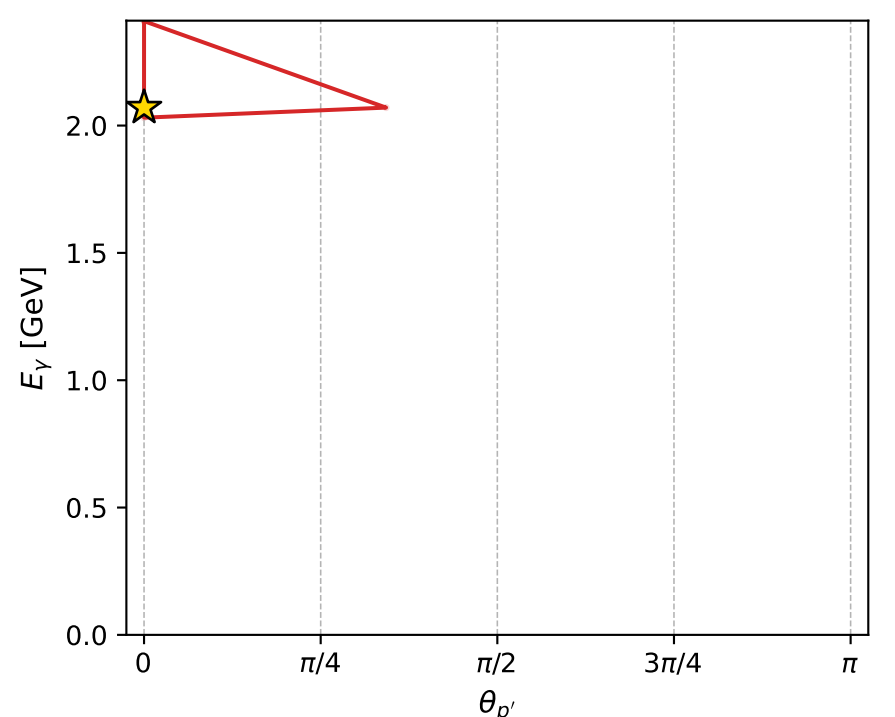
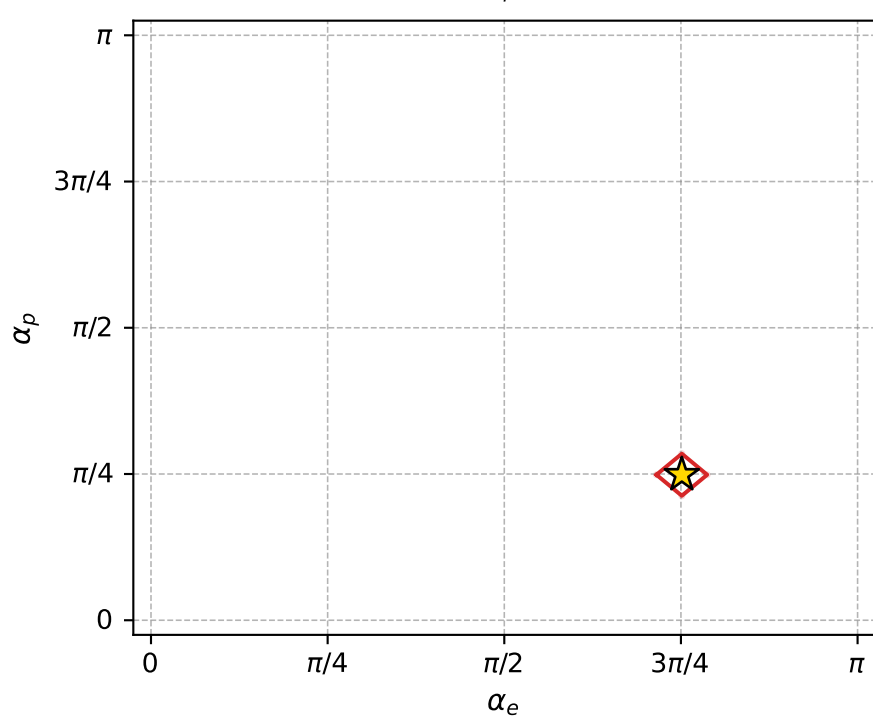
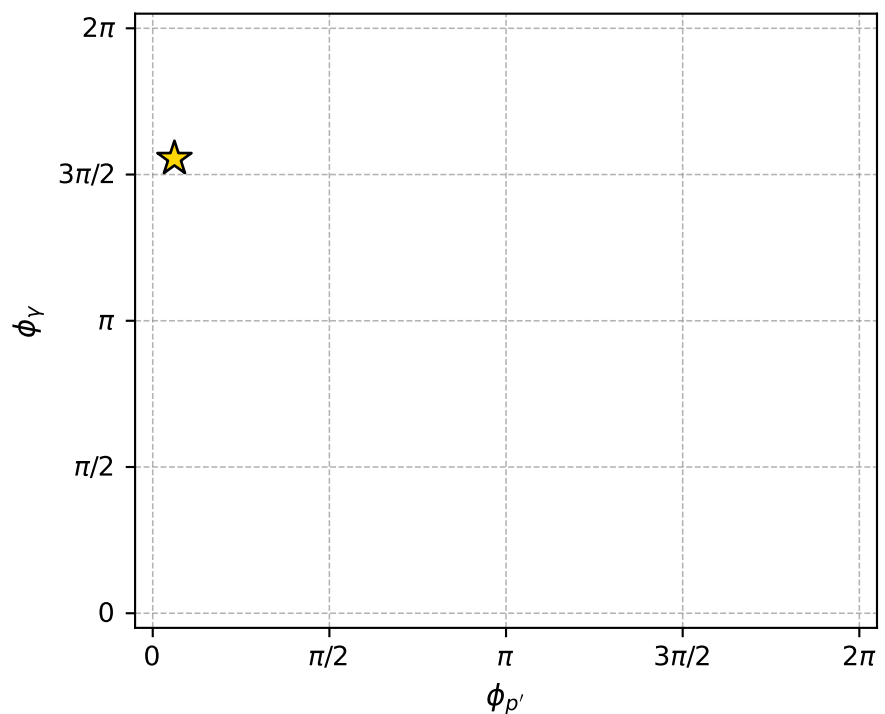
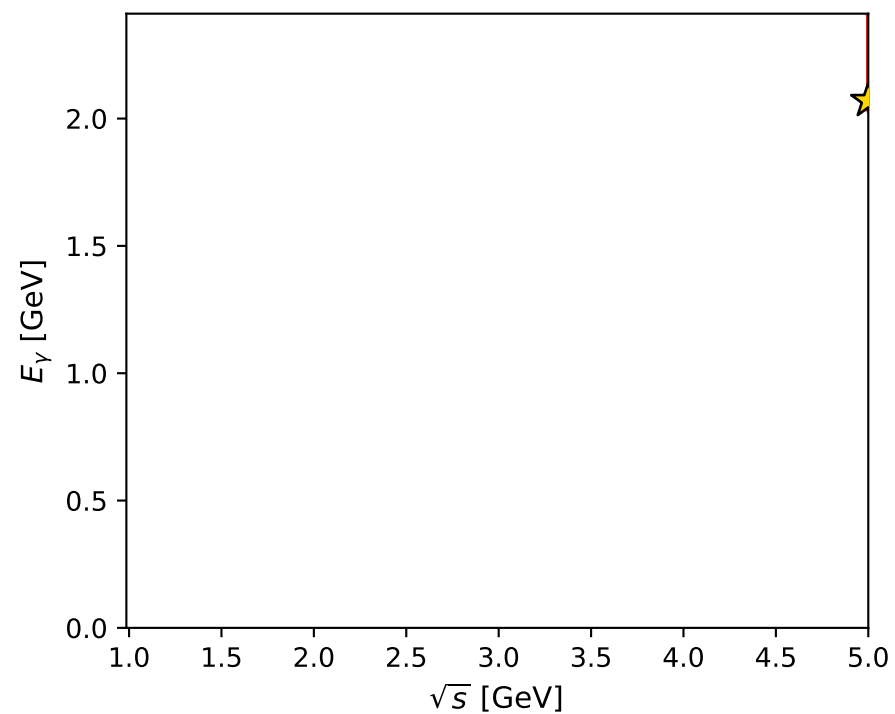
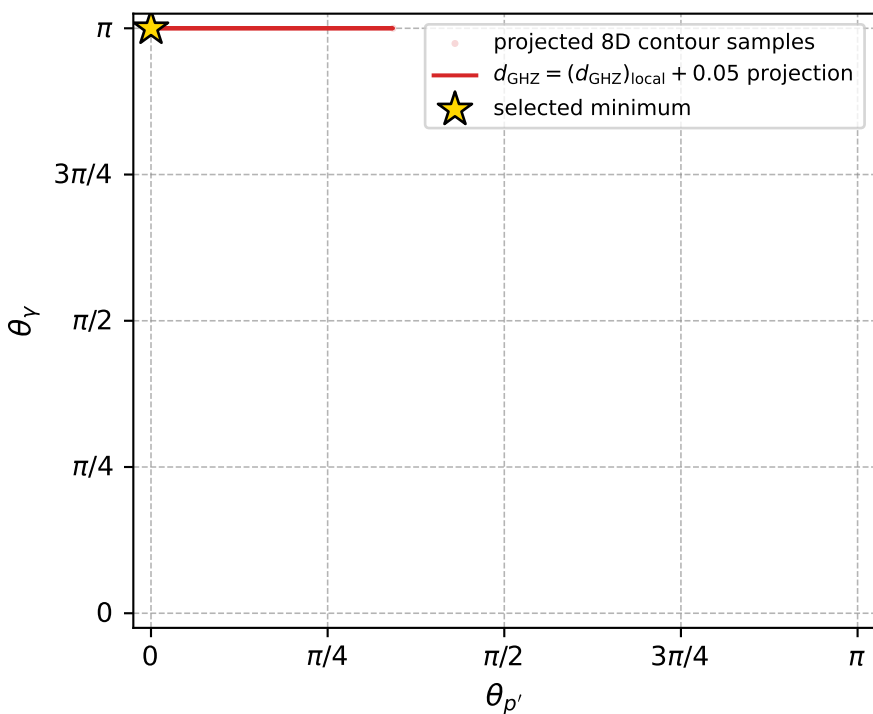
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 56; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7073481\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.7900189\text{e-}09$
 8D contour samples = 59

selected phase-space point:

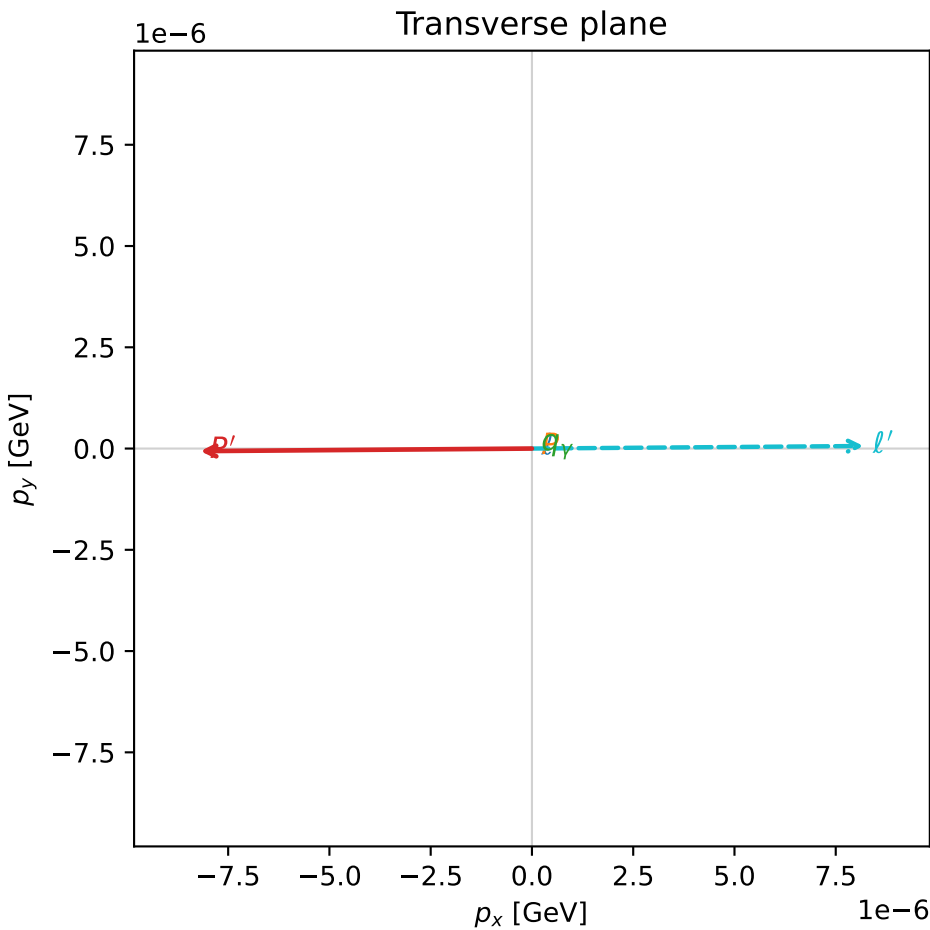
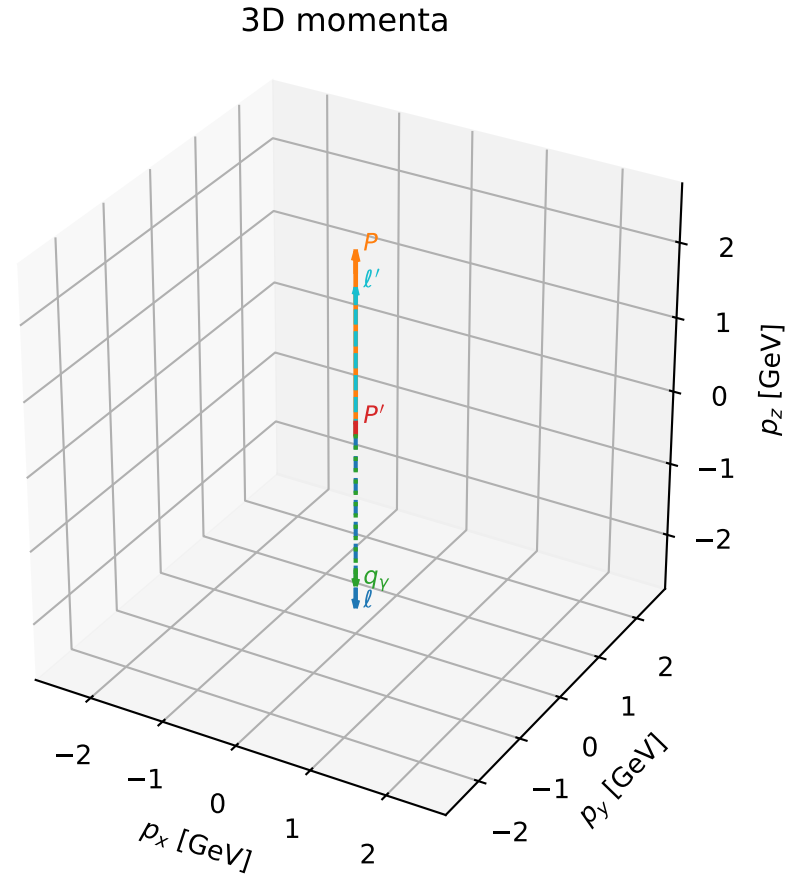
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 7.669904\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.0705$
 $\text{phi_p_out} = 0.1929208$
 $\text{phi_gamma_out} = 4.883562$
 $\text{alpha_e} = 2.360171$
 $\text{alpha_p} = 0.7814304$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_60 $d_{\{\text{rm GHZ}\}}=2.74796\text{e-}08$
 region: polarization_cluster_03_local_minimum_60
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3243577$ rad
 incoming lepton: $-0.68424|+\rangle +0.729257|-\rangle$
 proton mixing angle: $\alpha_p=0.8172232$ rad
 incoming proton: $0.684249|+\rangle +0.729249|-\rangle$

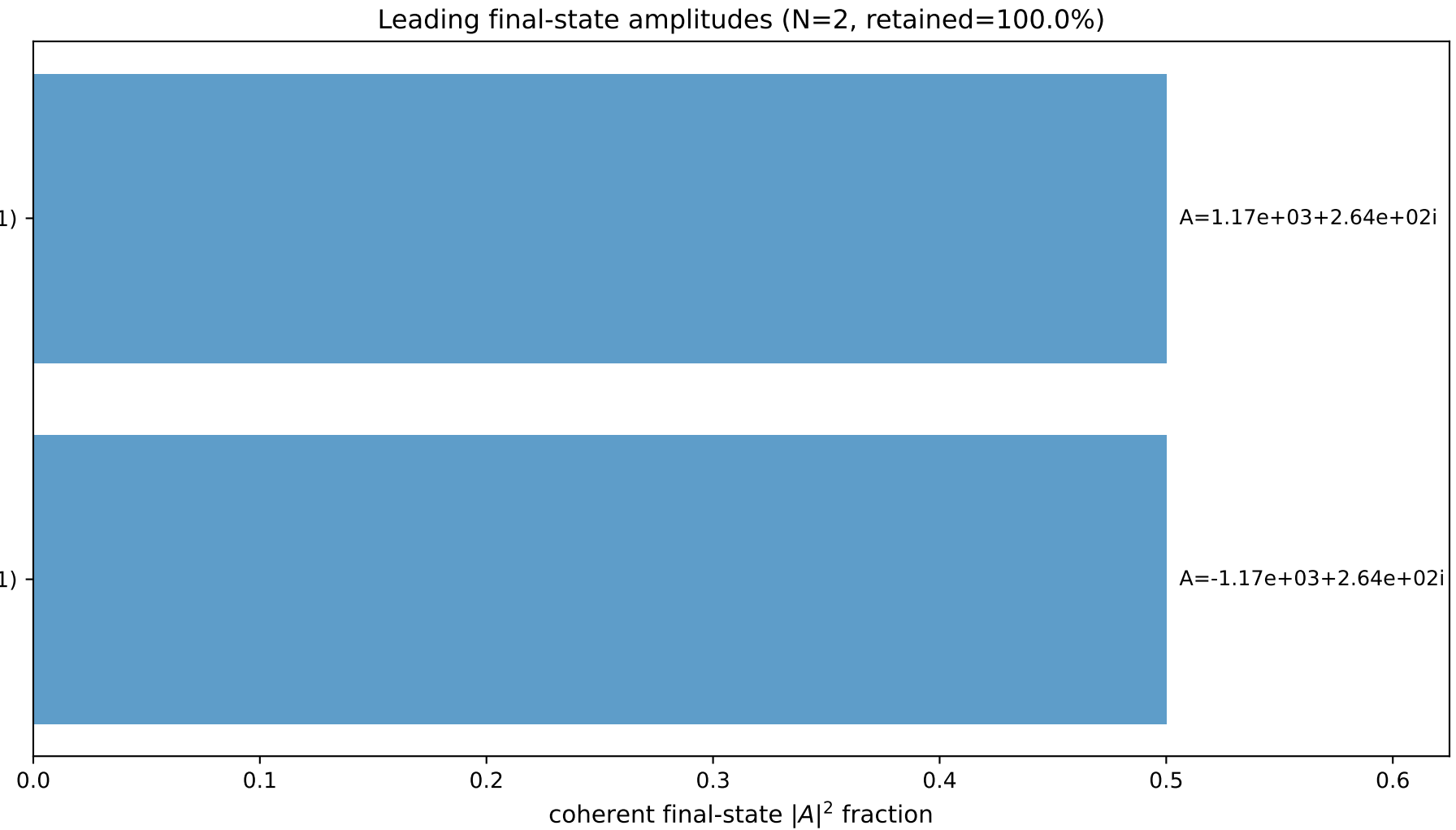
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.150619$
 $\theta_{P'}=5.43472\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=3.14933$, $\phi_Y=3.36397$
 $E_Y=2.1003$, $\theta_{\ell'}=4.19848\text{e-}06$, $\phi_{\ell'}=0.00773789$
 $Q^2=18.8107$, $x_B=0.802708$, $t=-2.43097$, $W^2=5.50317$, $y=0.971552$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9497$
 P' $E= 0.9500$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1506$
 q_Y $E= 2.1003$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1003$

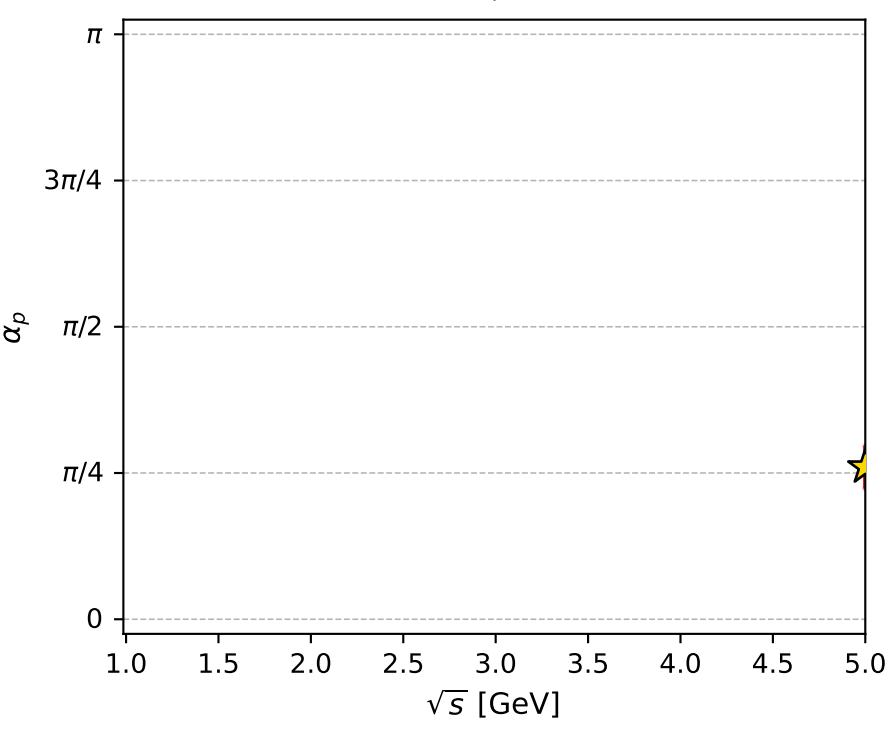
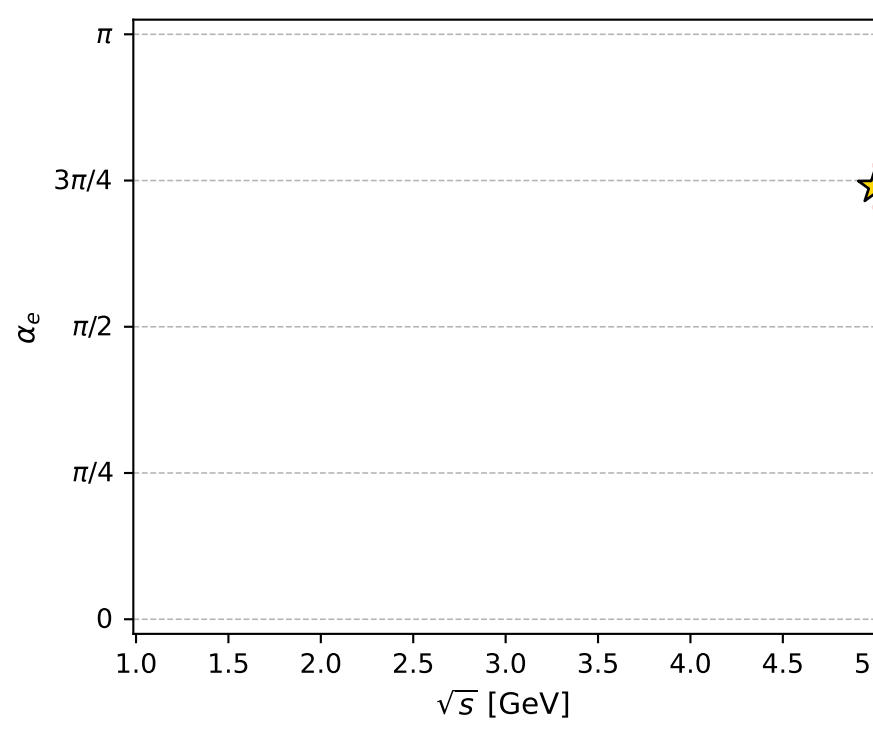
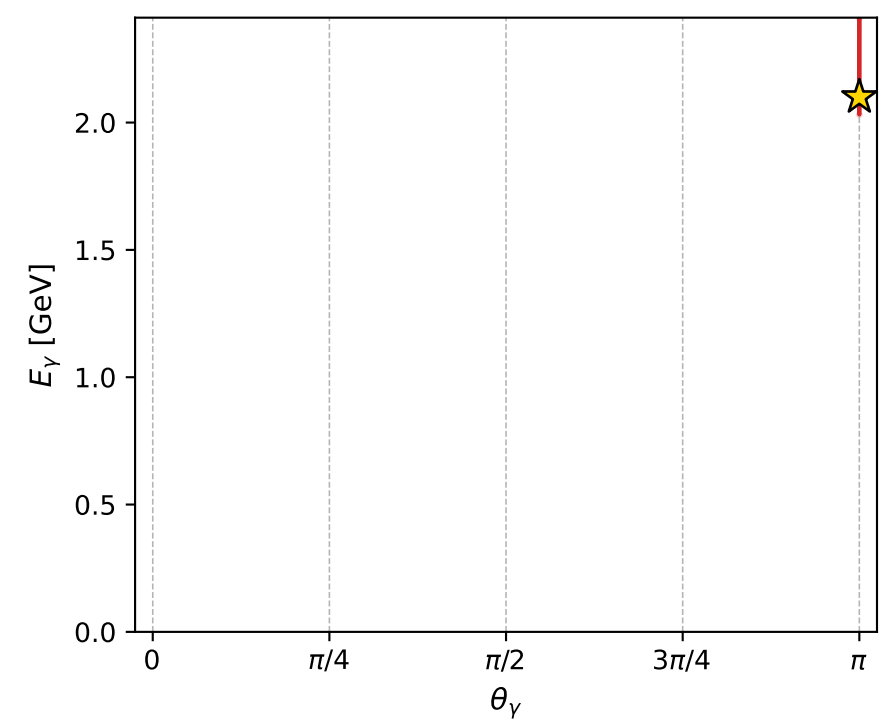
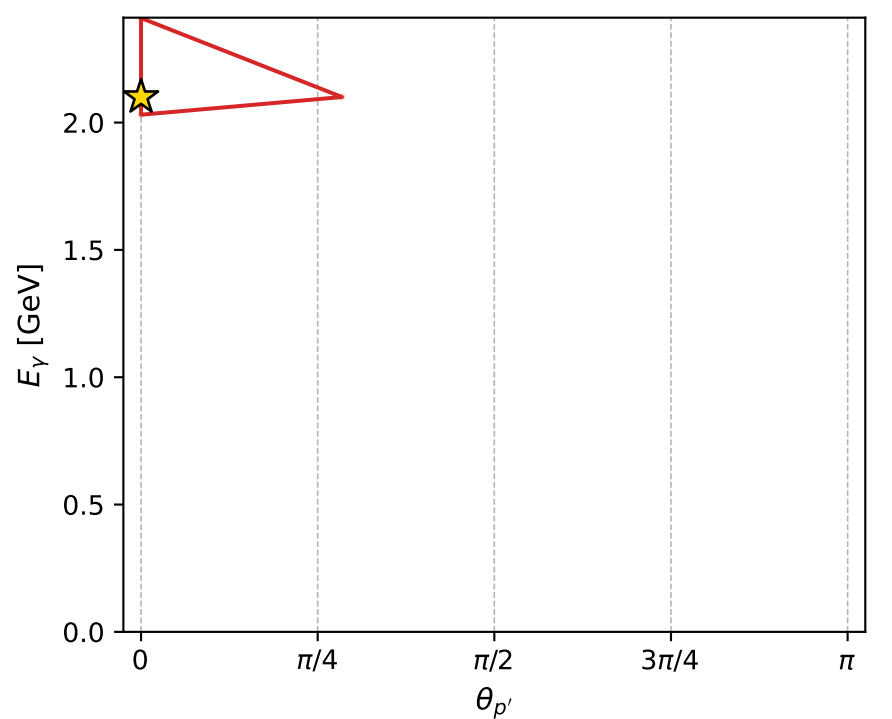
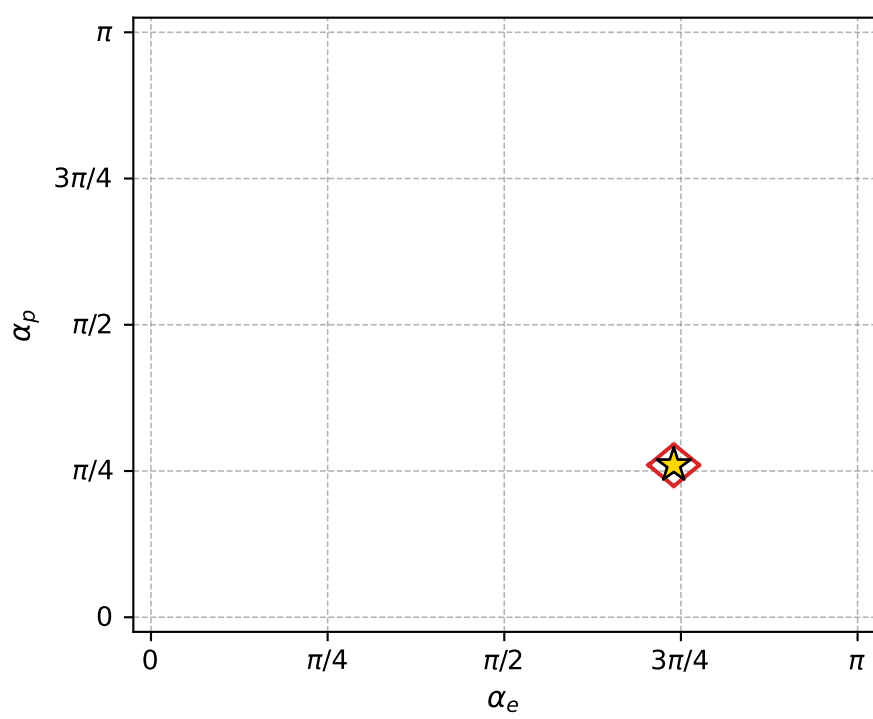
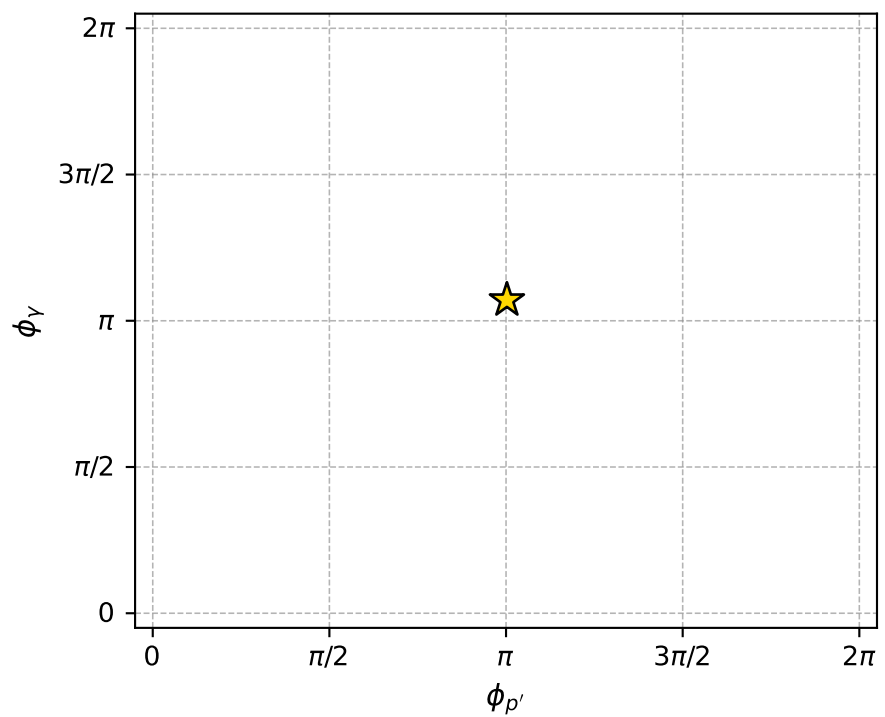
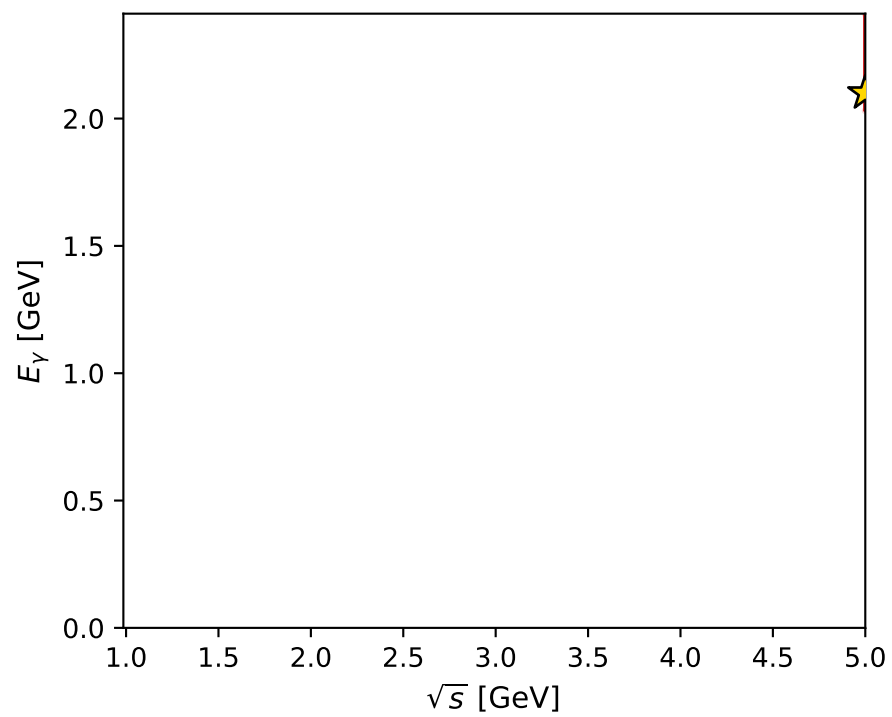
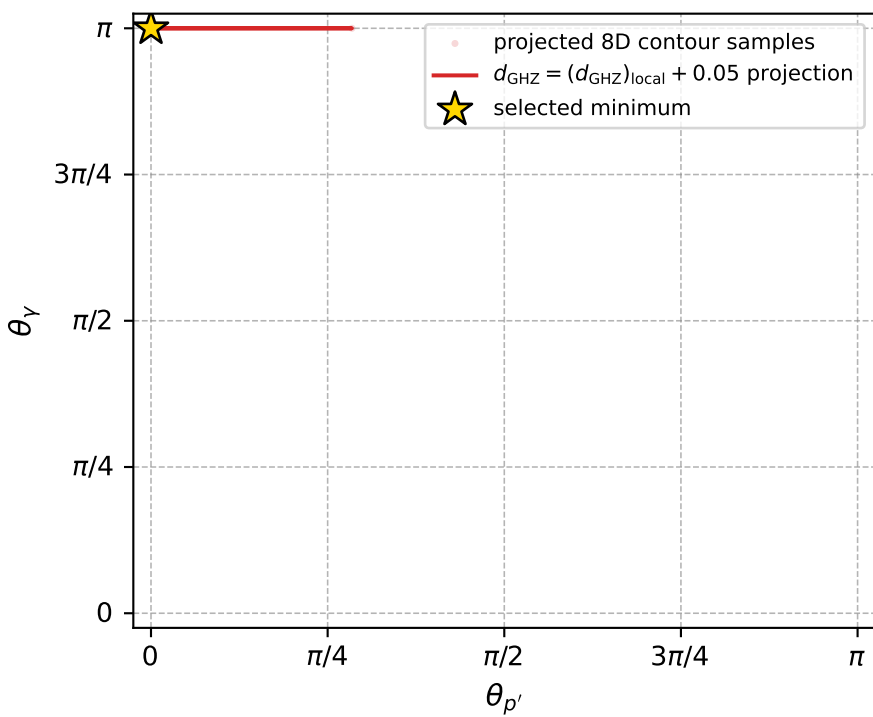


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 60; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7479612\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $5.1961503\text{e-}09$
 8D contour samples = 53

selected phase-space point:

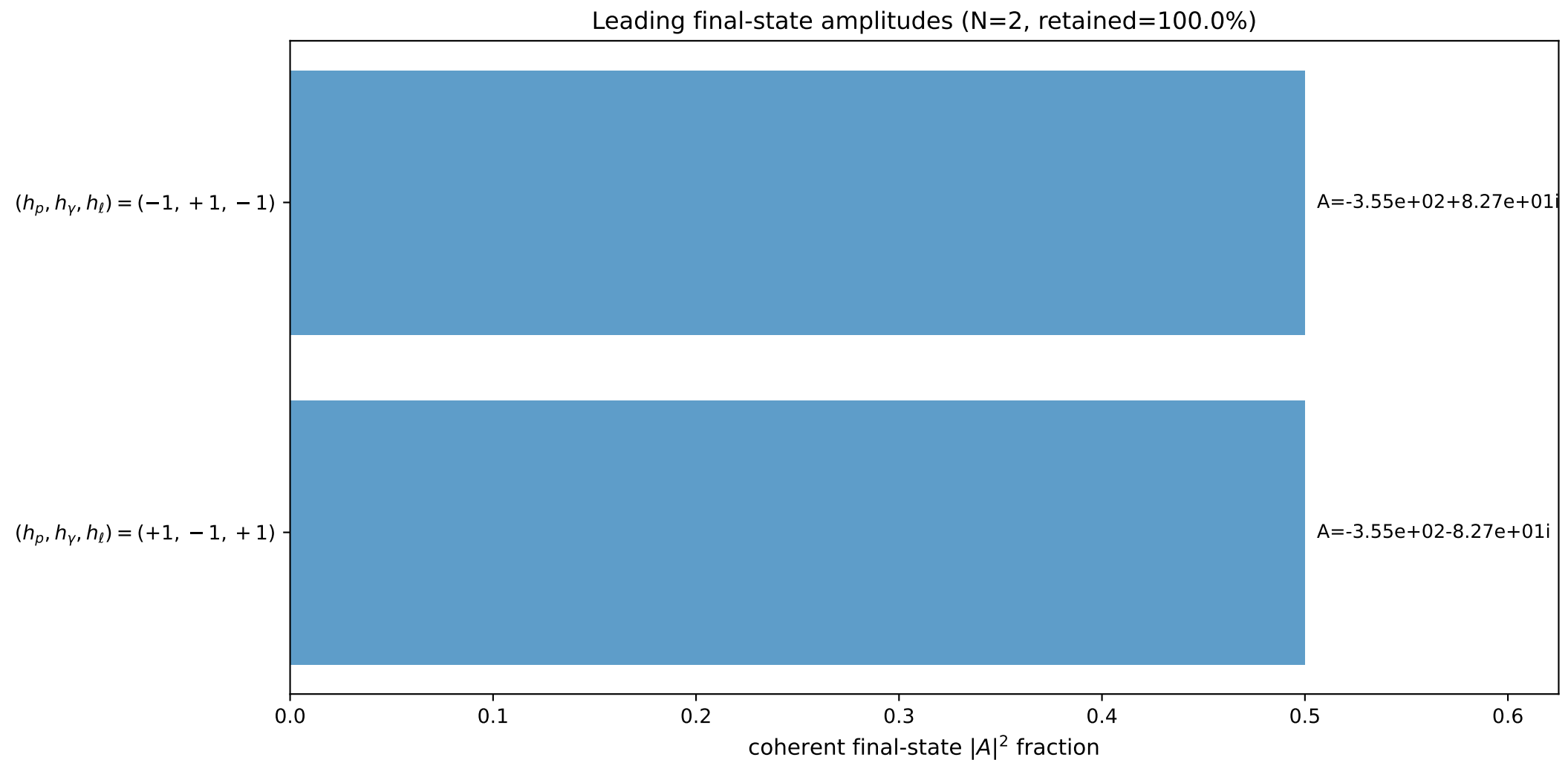
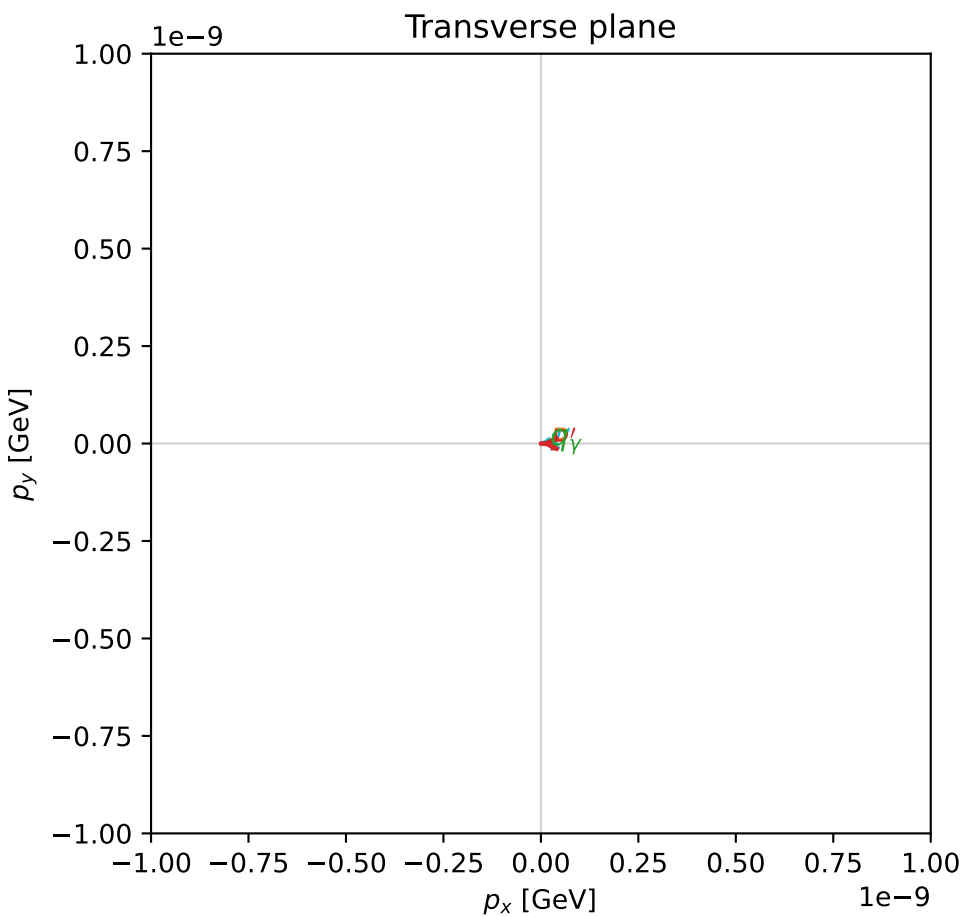
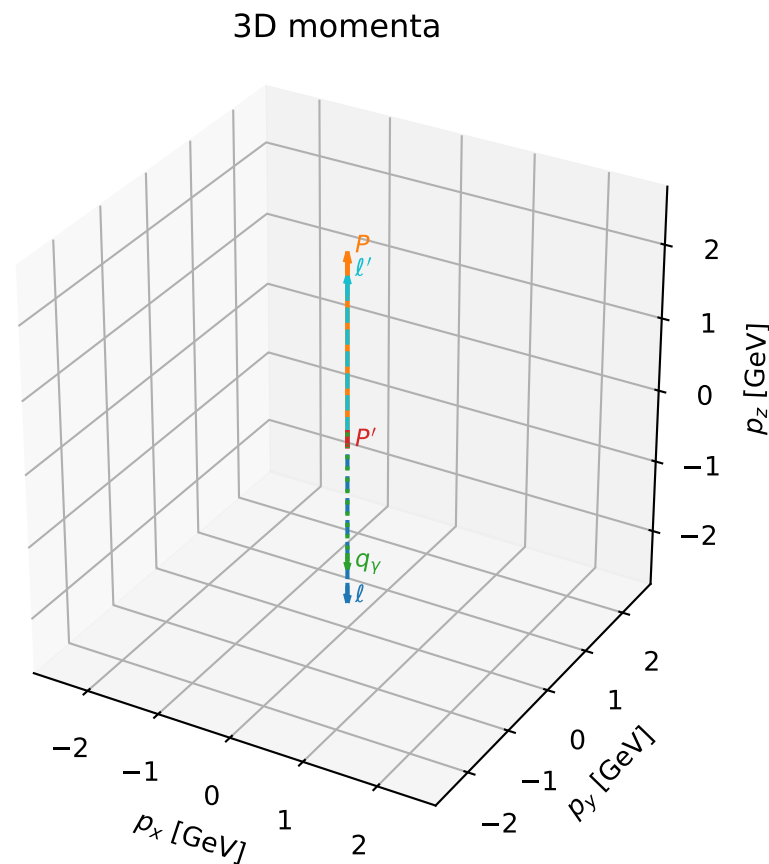
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 5.434724\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.100302$
 $\text{phi_p_out} = 3.149331$
 $\text{phi_gamma_out} = 3.363974$
 $\text{alpha_e} = 2.324358$
 $\text{alpha_p} = 0.8172232$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

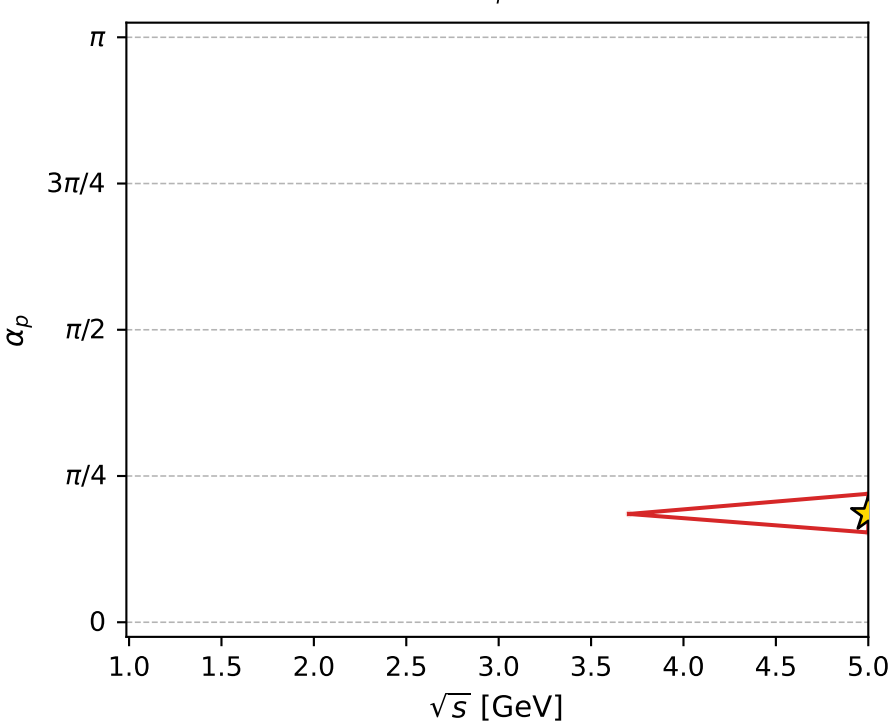
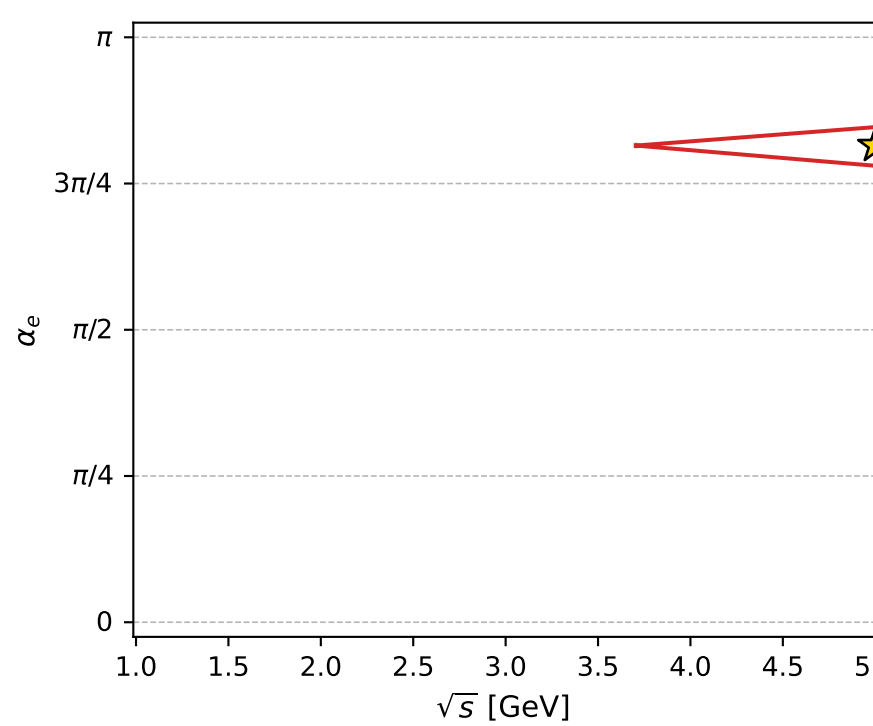
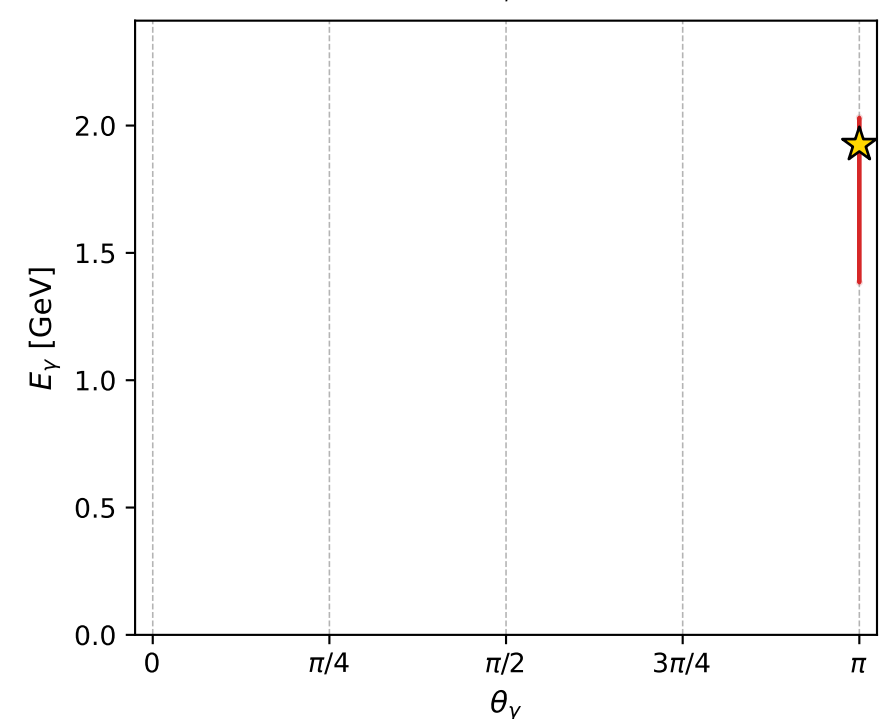
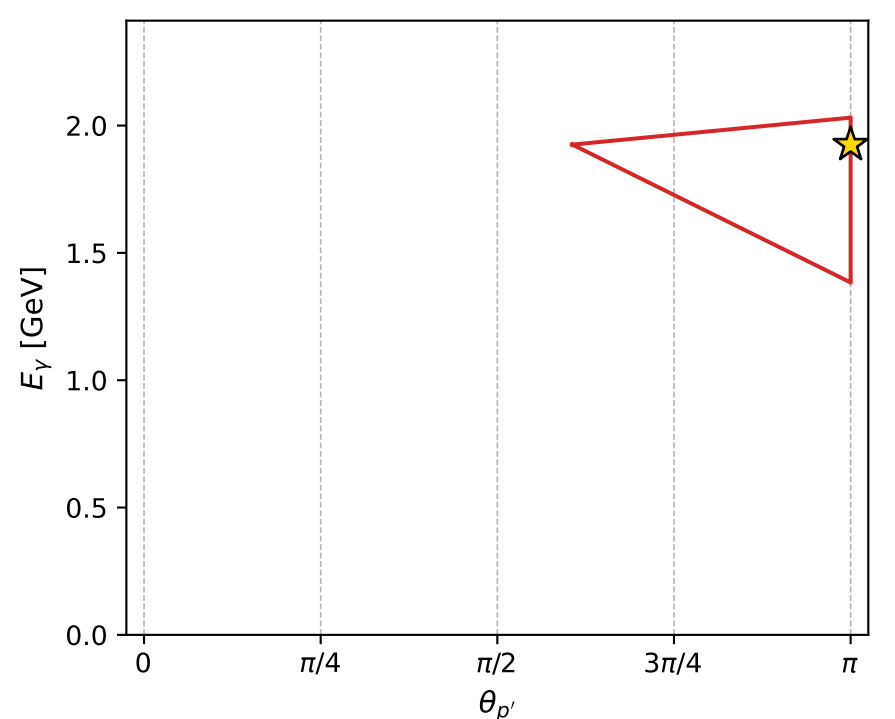
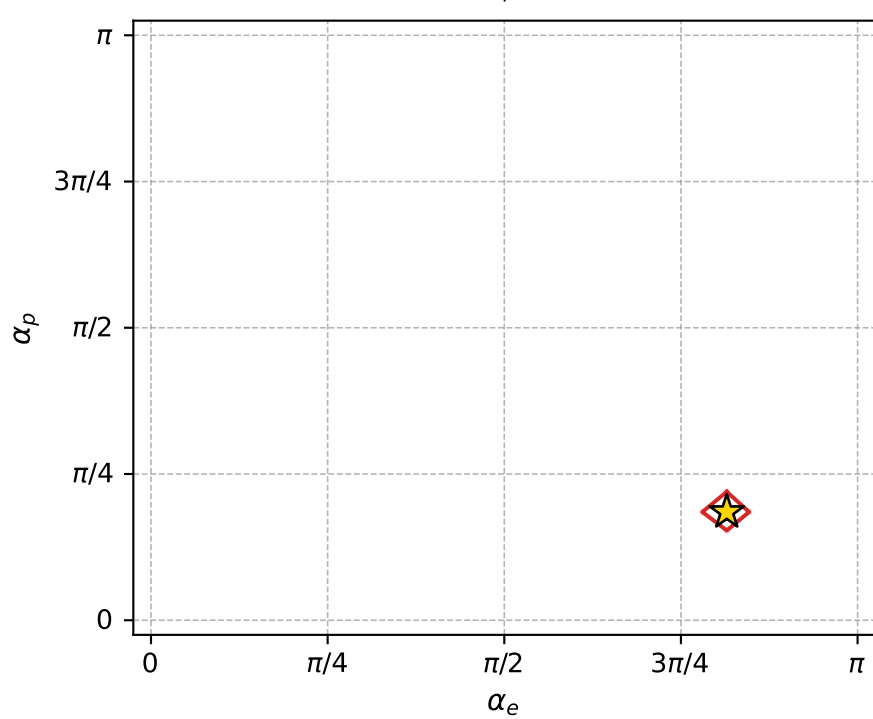
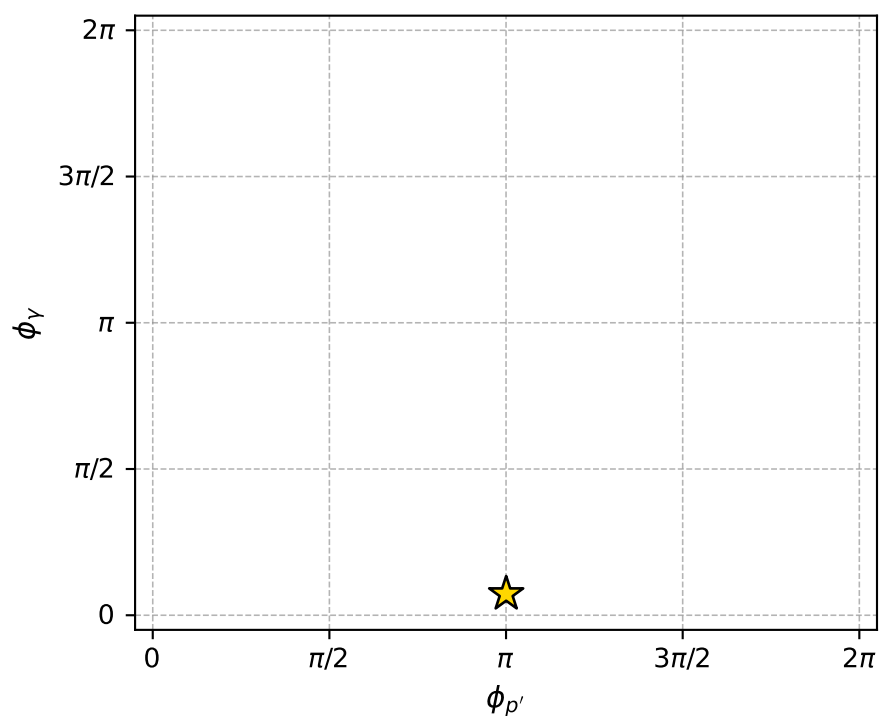
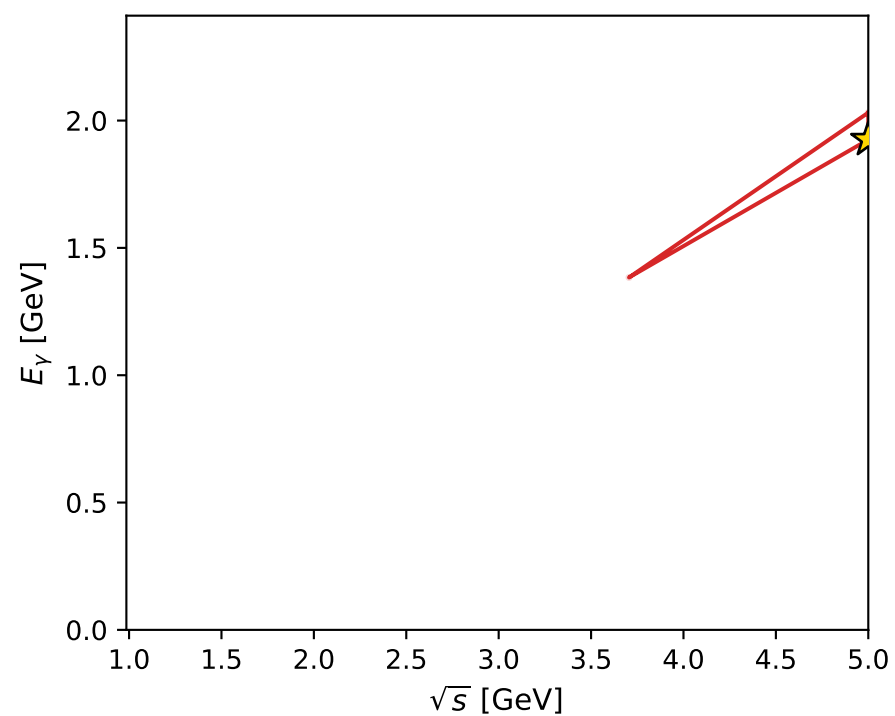
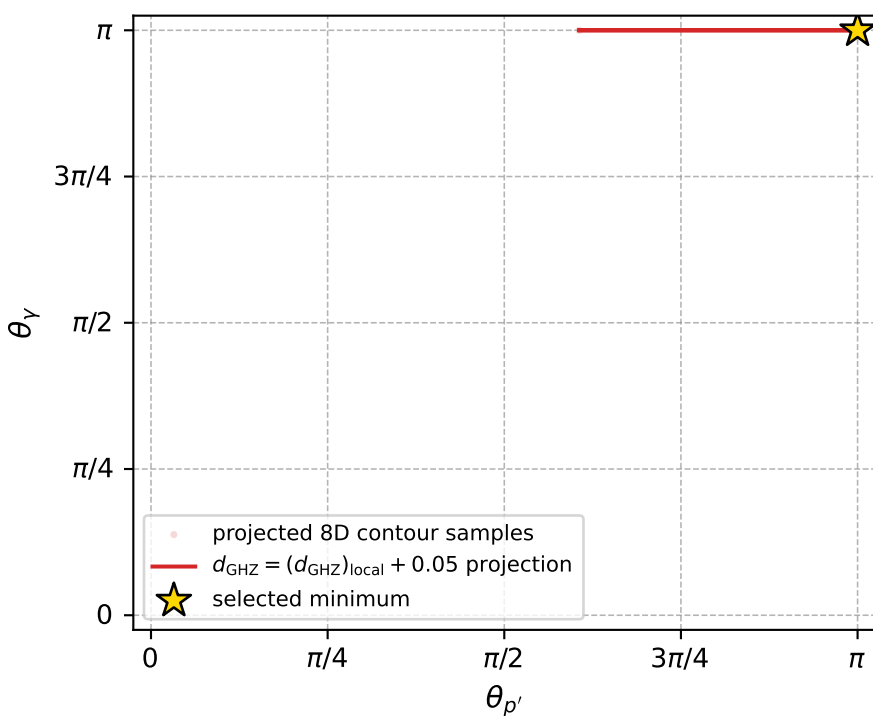
dghz_mixing_angles_polarization_03_local_minimum_66 $d_{\{\text{rm GHZ}\}}=2.78662\text{e-}08$
 region: polarization_cluster_03_local_minimum_66
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.5600686$ rad
 incoming lepton: $-0.835626|+\rangle +0.549298|-\rangle$
 proton mixing angle: $\alpha_p=0.58153866$ rad
 incoming proton: $0.835618|+\rangle +0.54931|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.192212$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=3.14281$, $\phi_Y=0.229158$
 $E_Y=1.92515$, $\theta_{l'}=0$, $\phi_{l'}=3.39574$
 $Q^2=20.4284$, $x_B=0.873944$, $t=-4.12349$, $W^2=3.8264$, $y=0.969106$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.1174$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1174$
 P' $E= 0.9575$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1922$
 q_Y $E= 1.9251$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9251$



electron: polarization cluster P3, local minimum 66; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.786623\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.5827682\text{e-}09$
 8D contour samples = 48

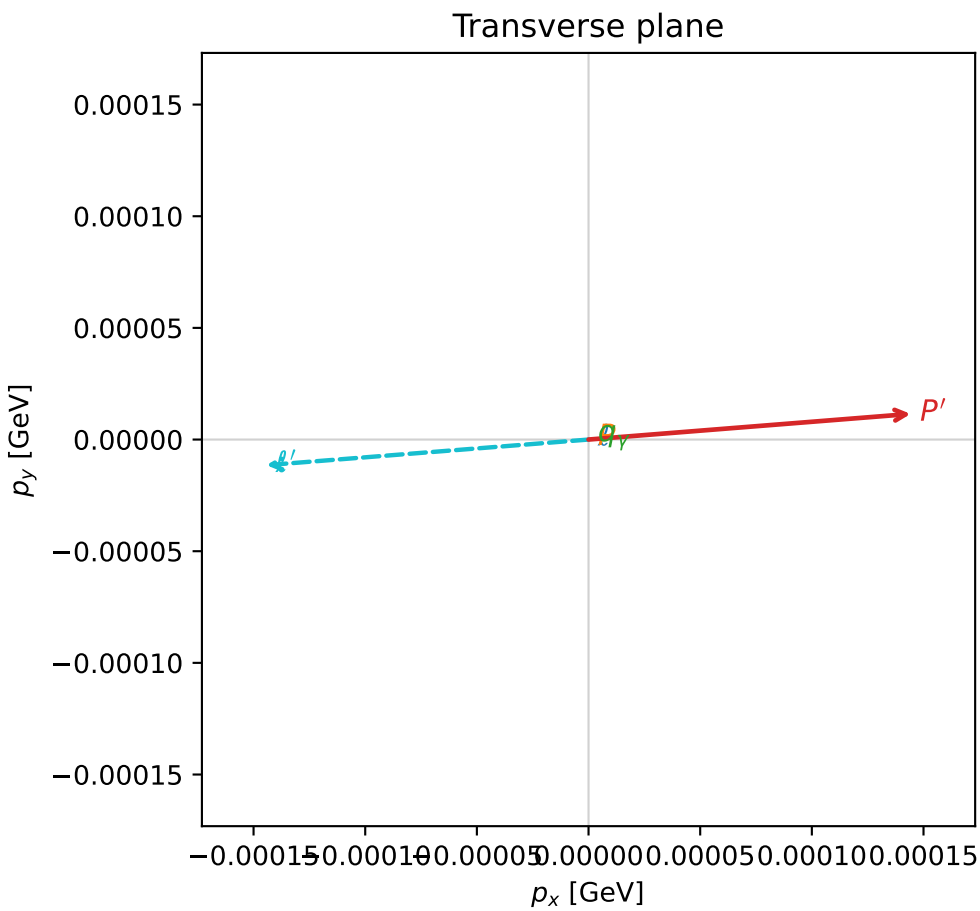
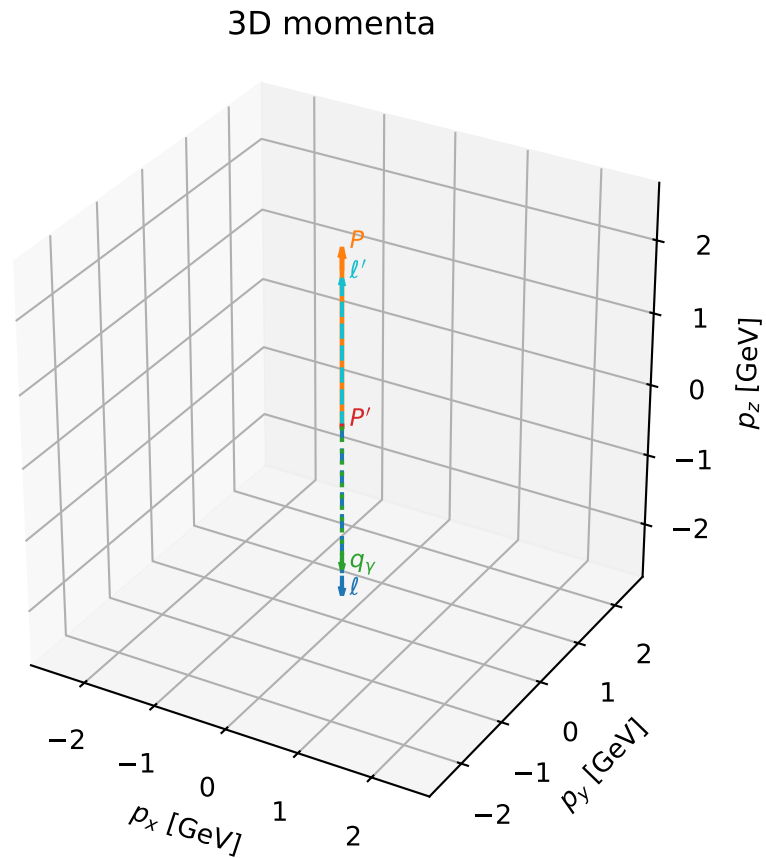
selected phase-space point:
 $\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.925149$
 $\text{phi_p_out} = 3.142815$
 $\text{phi_gamma_out} = 0.2291578$
 $\text{alpha_e} = 2.560069$
 $\text{alpha_p} = 0.5815387$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_67 $d_{\{\text{rm GHZ}\}}=2.80689\text{e-}08$
 region: polarization_cluster_03_local_minimum_67
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.235145$ rad
 incoming lepton: $-0.616546|+\rangle +0.787319|-\rangle$
 proton mixing angle: $\alpha_p=0.90642644$ rad
 incoming proton: $0.616563|+\rangle +0.787305|-\rangle$

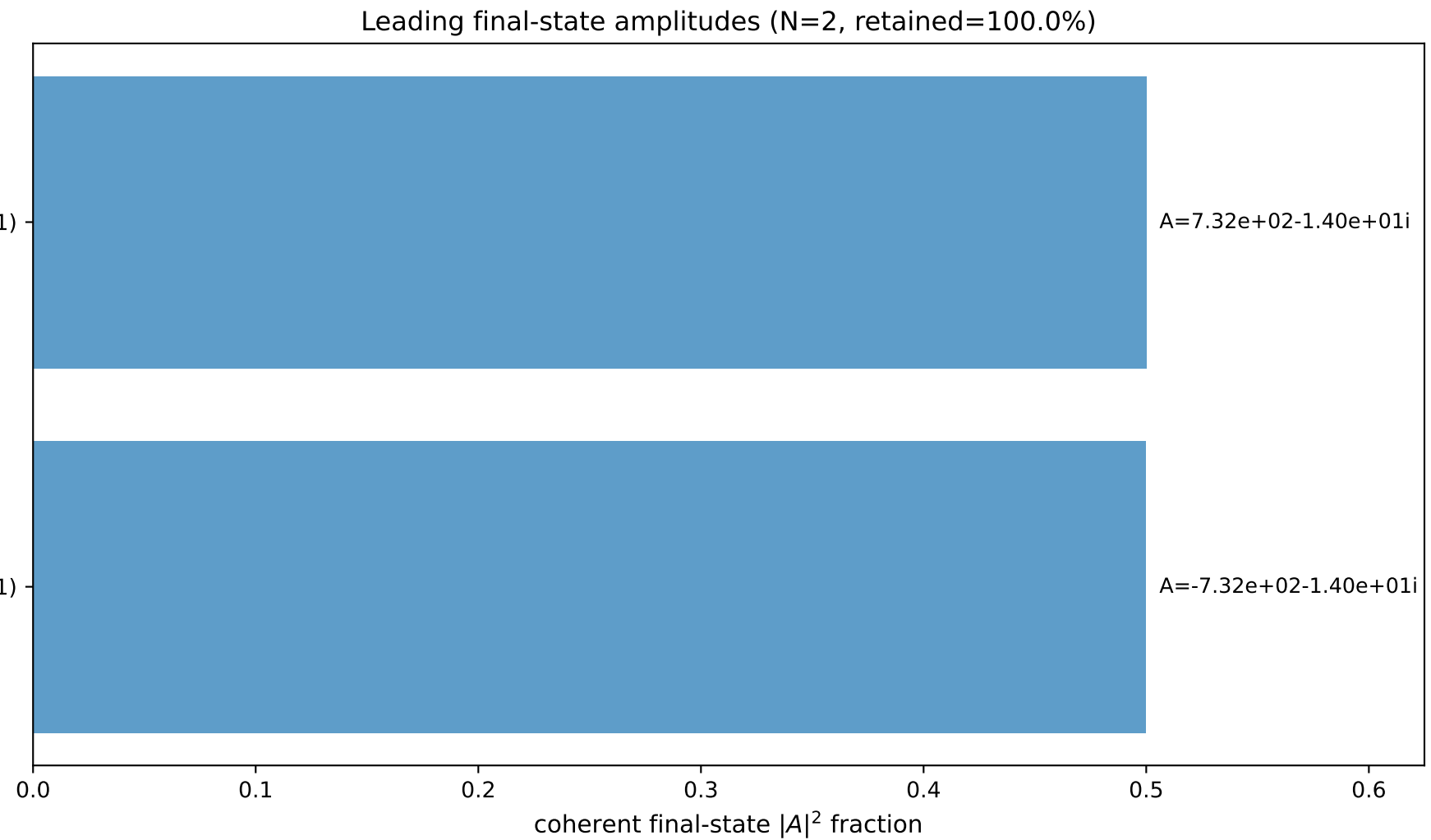
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0078299$
 $\theta_{P'}=0.0184851$, $\theta_Y=3.14159$, $\phi_{P'}=0.0794118$, $\phi_Y=3.1225$
 $E_Y=2.03482$, $\theta_{\ell'}=7.14003\text{e-}05$, $\phi_{\ell'}=3.221$
 $Q^2=19.556$, $x_B=0.835536$, $t=-3.05764$, $W^2=4.72917$, $y=0.970421$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0270$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0270$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0078$
 q_Y $E= 2.0348$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0348$

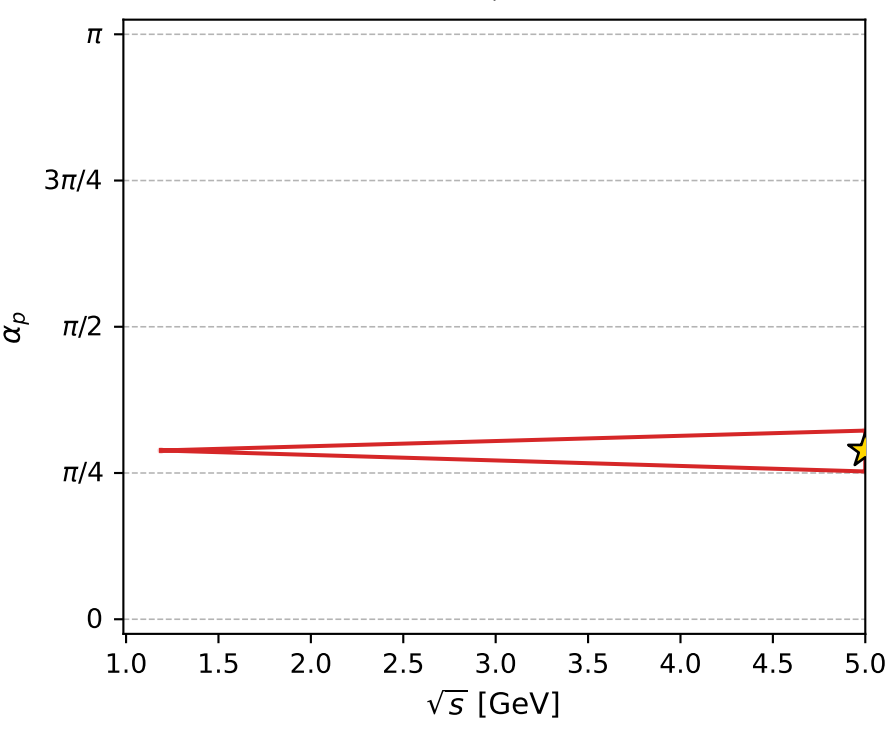
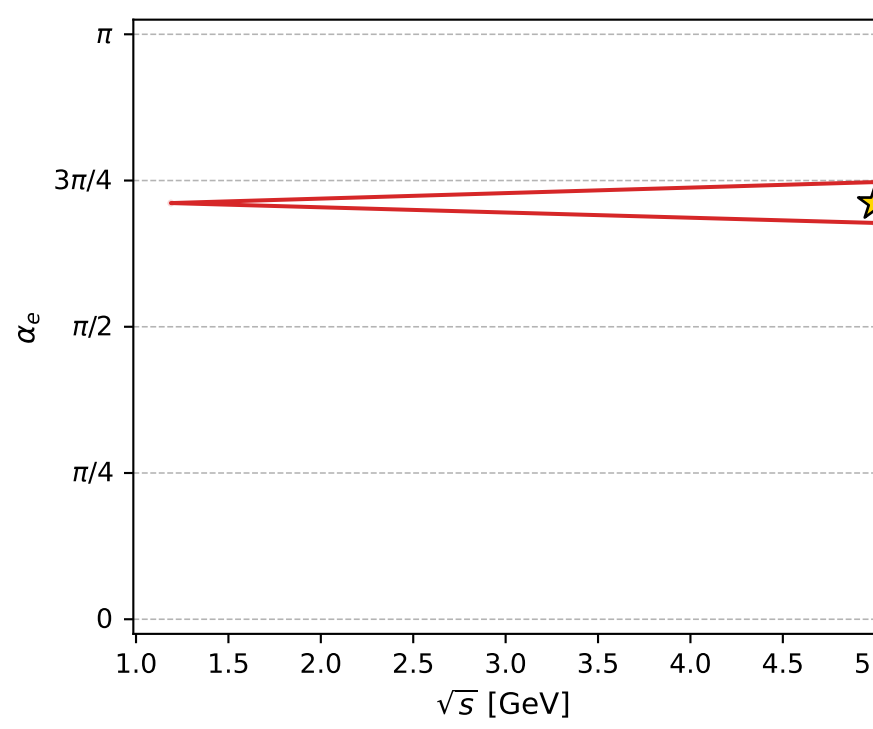
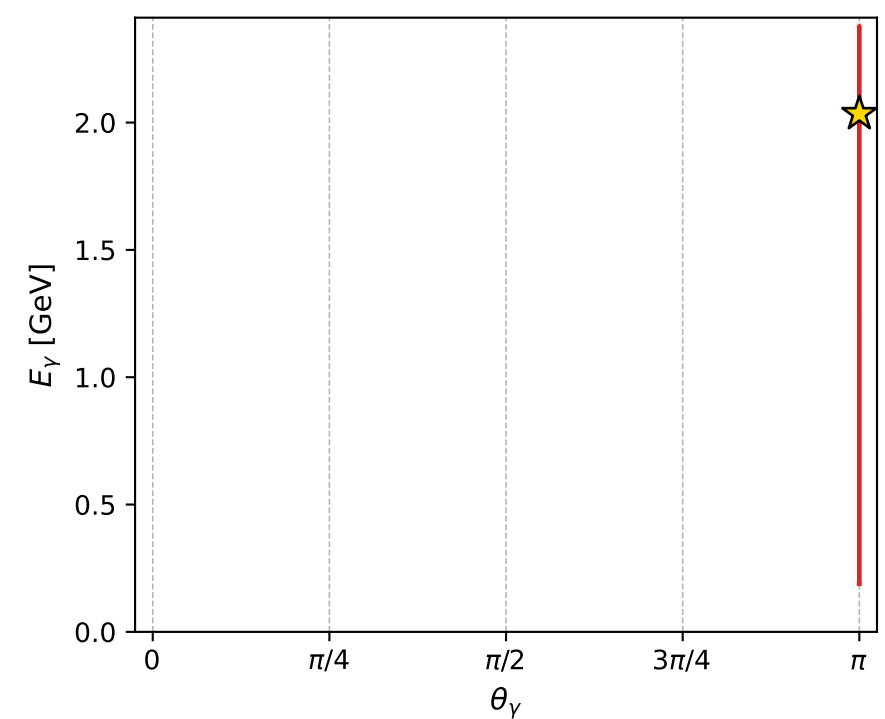
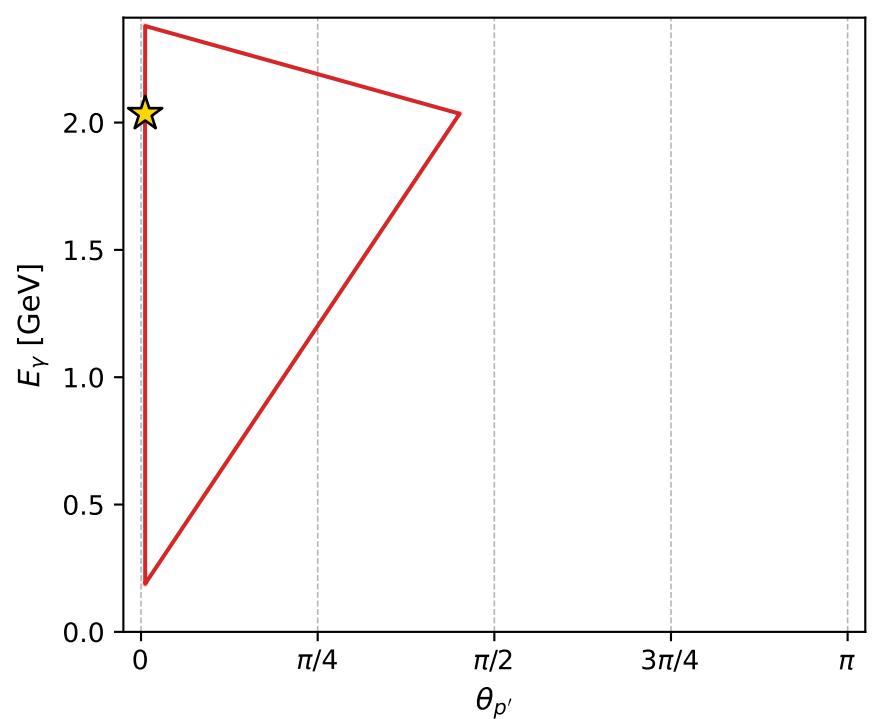
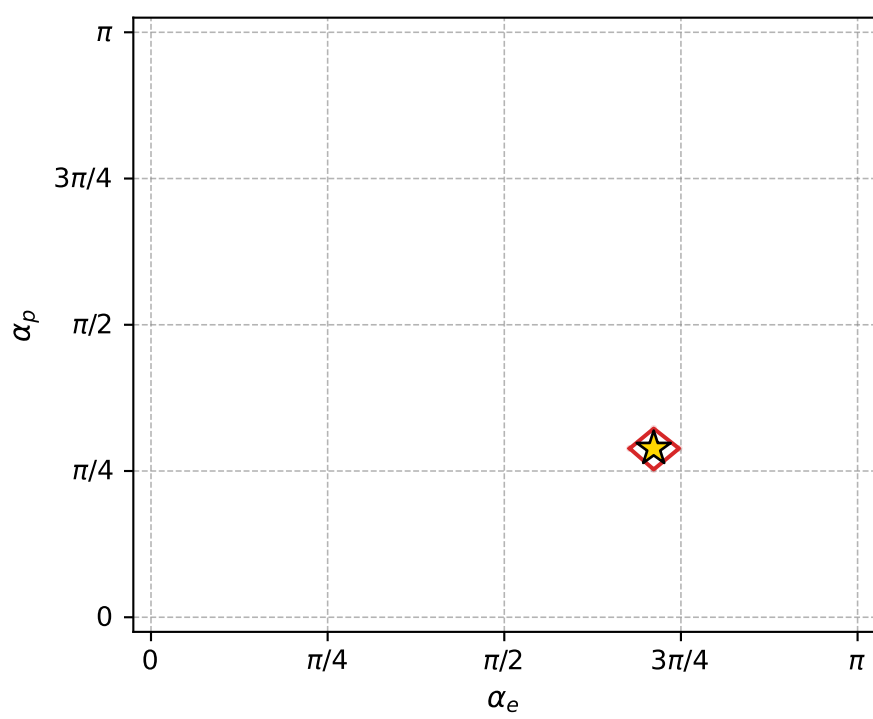
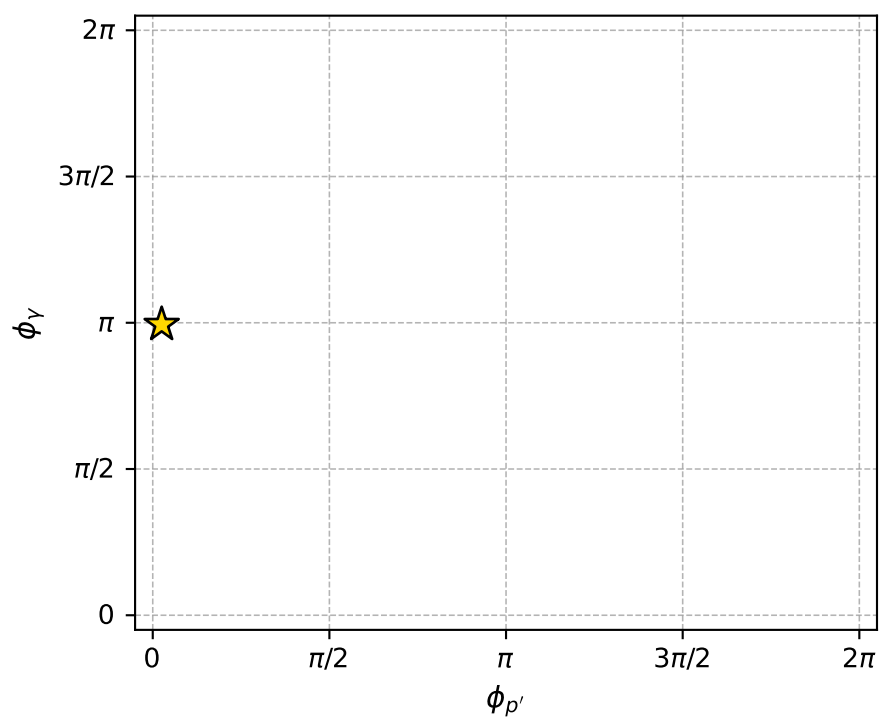
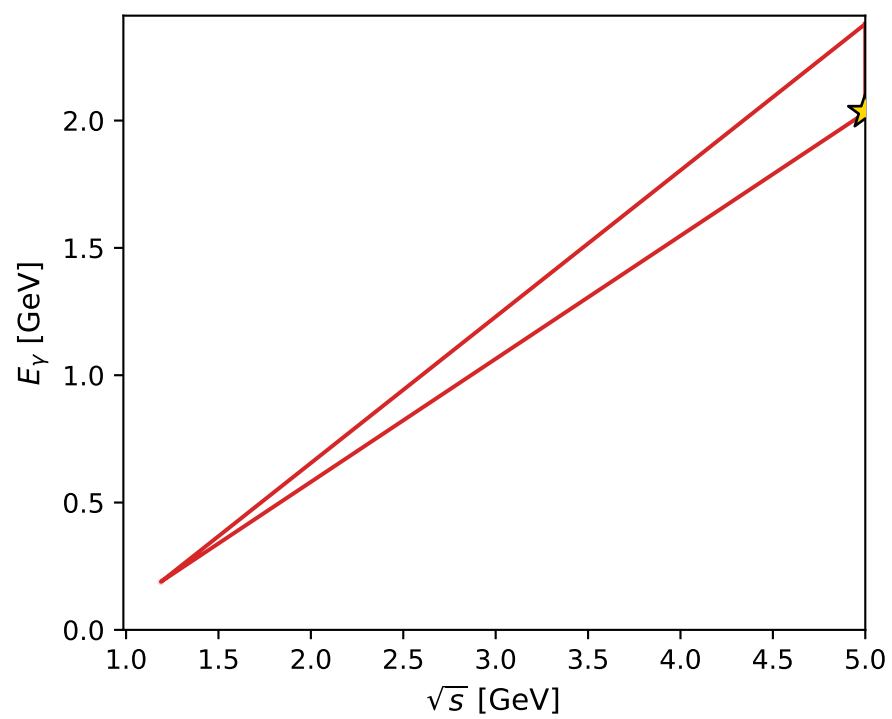
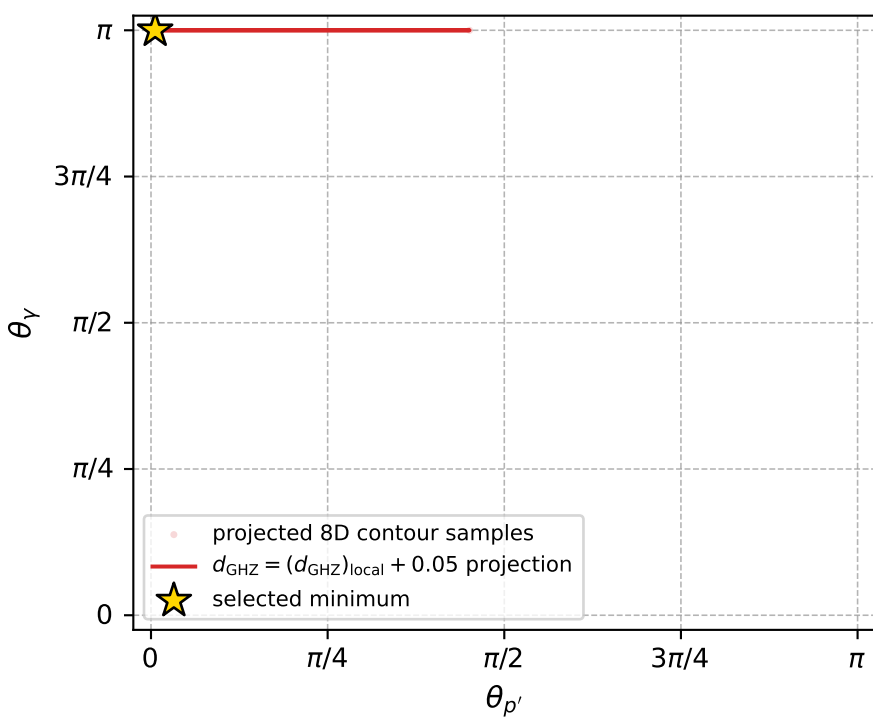


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 67; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.806889\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.7854282\text{e-}09$
 8D contour samples = 117

selected phase-space point:

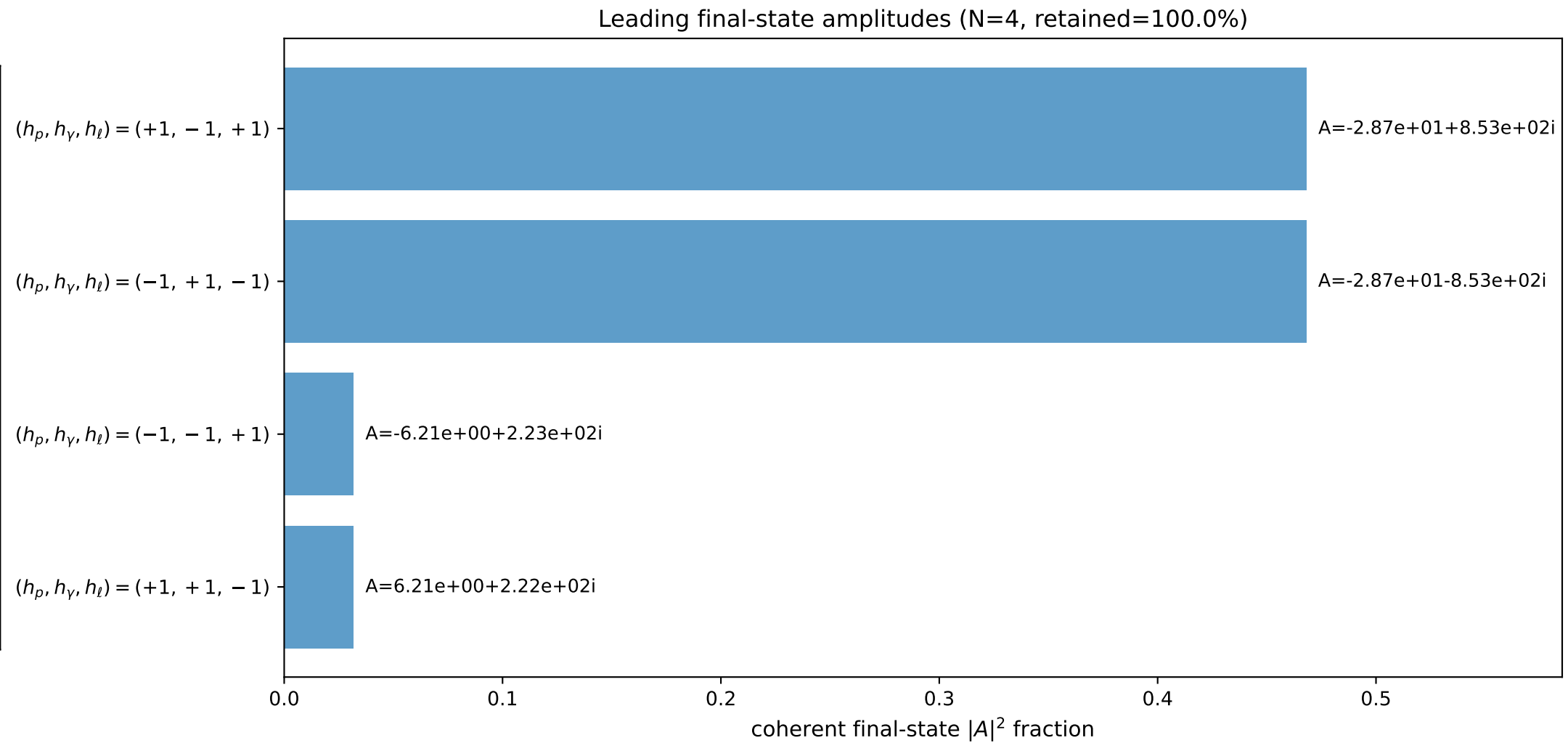
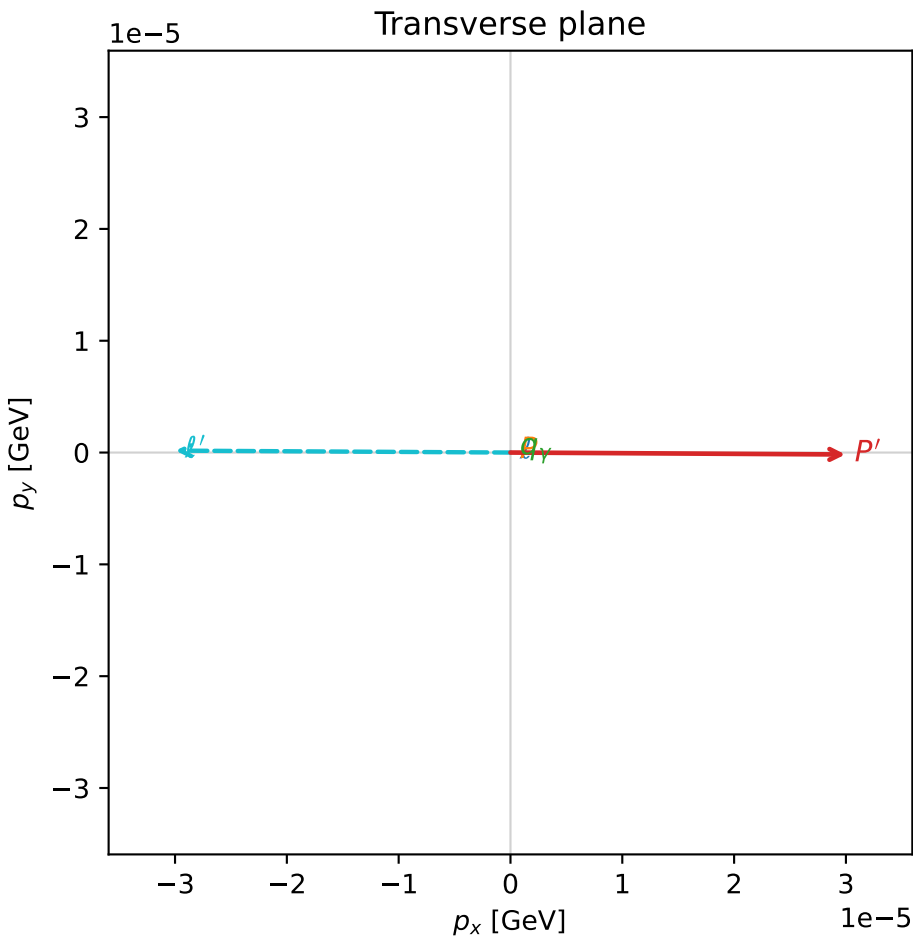
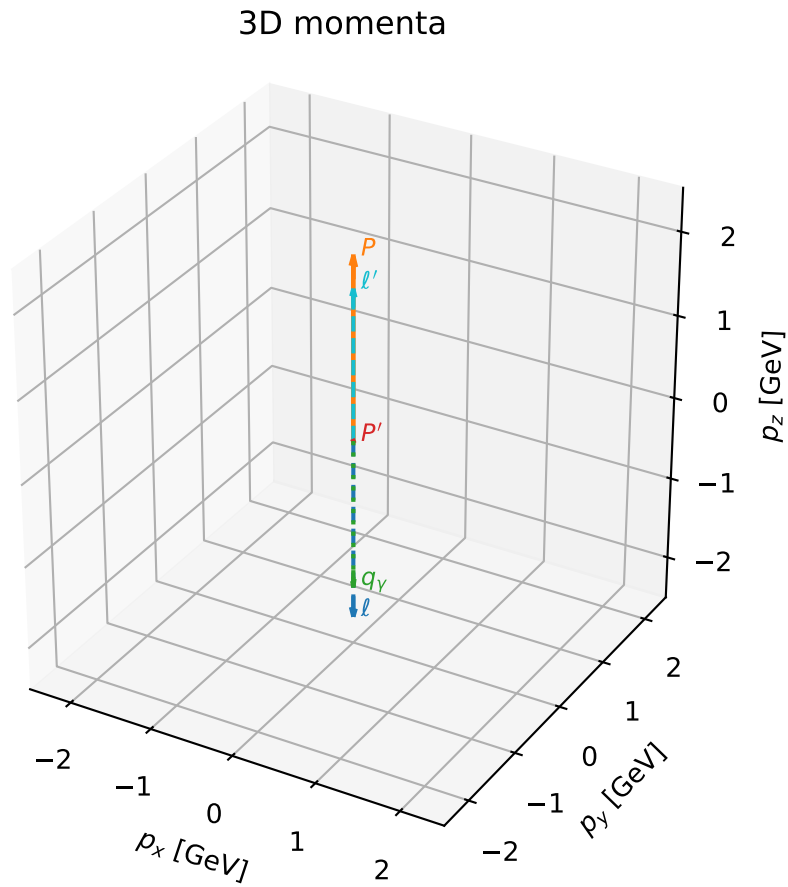
$\text{sqrt_s} = 4.999852$
 $\text{theta_p_out} = 0.01848508$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.034824$
 $\text{phi_p_out} = 0.07941184$
 $\text{phi_gamma_out} = 3.1225$
 $\text{alpha_e} = 2.235145$
 $\text{alpha_p} = 0.9064264$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

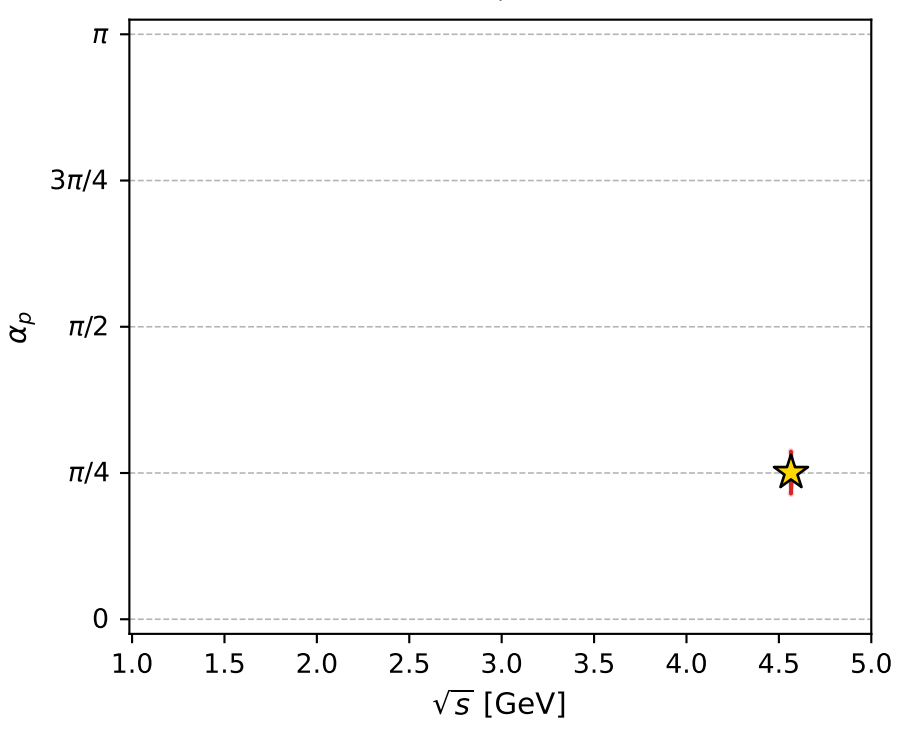
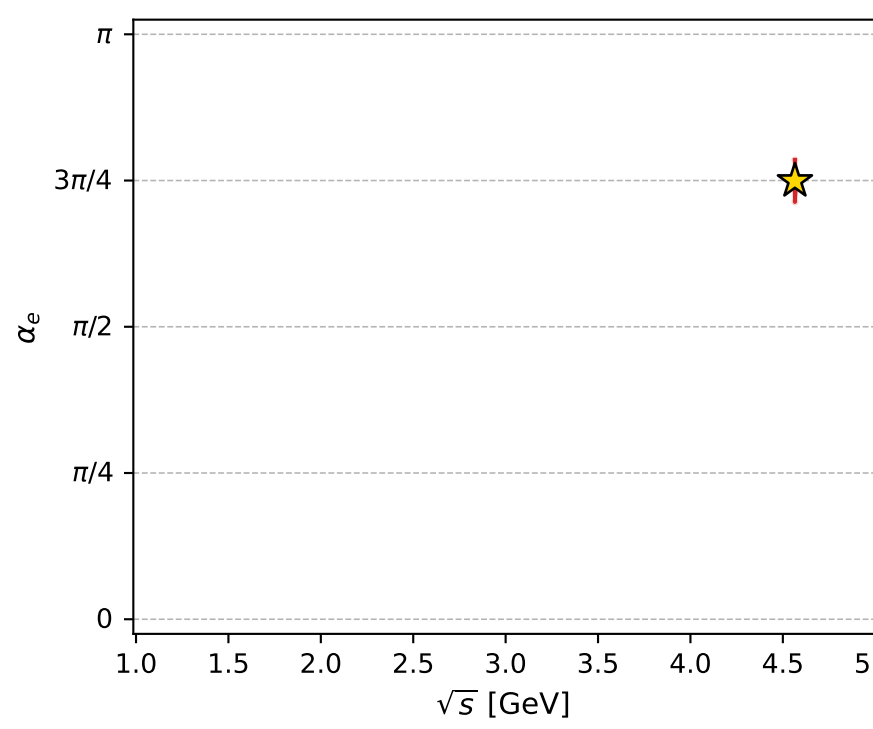
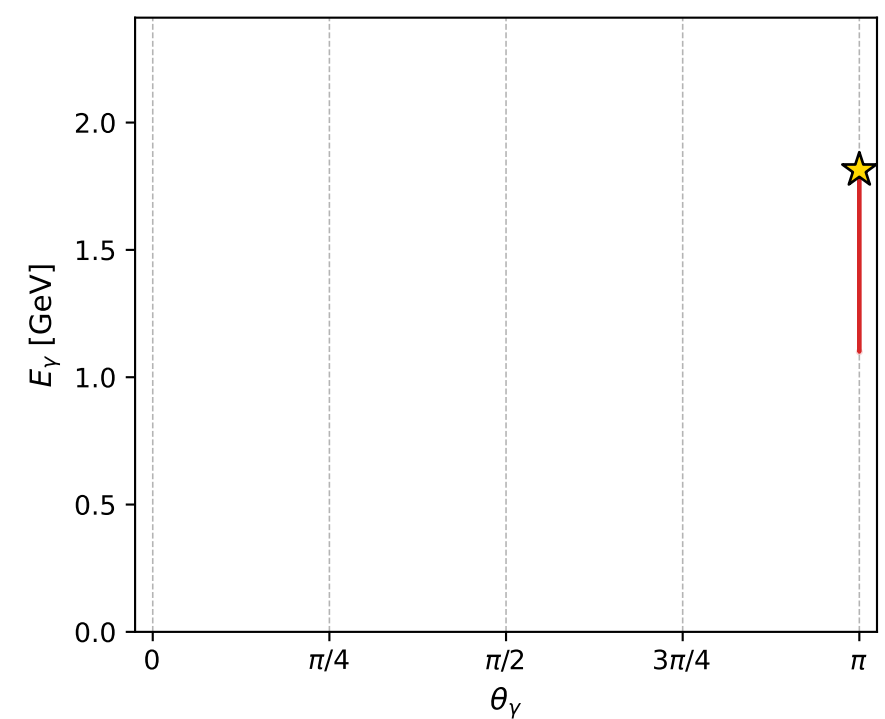
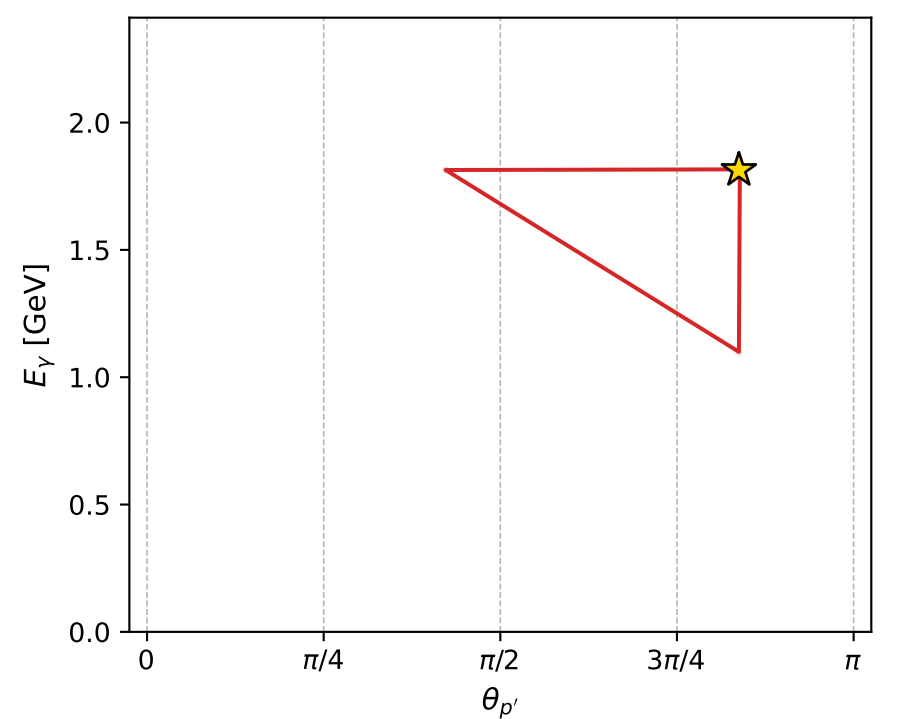
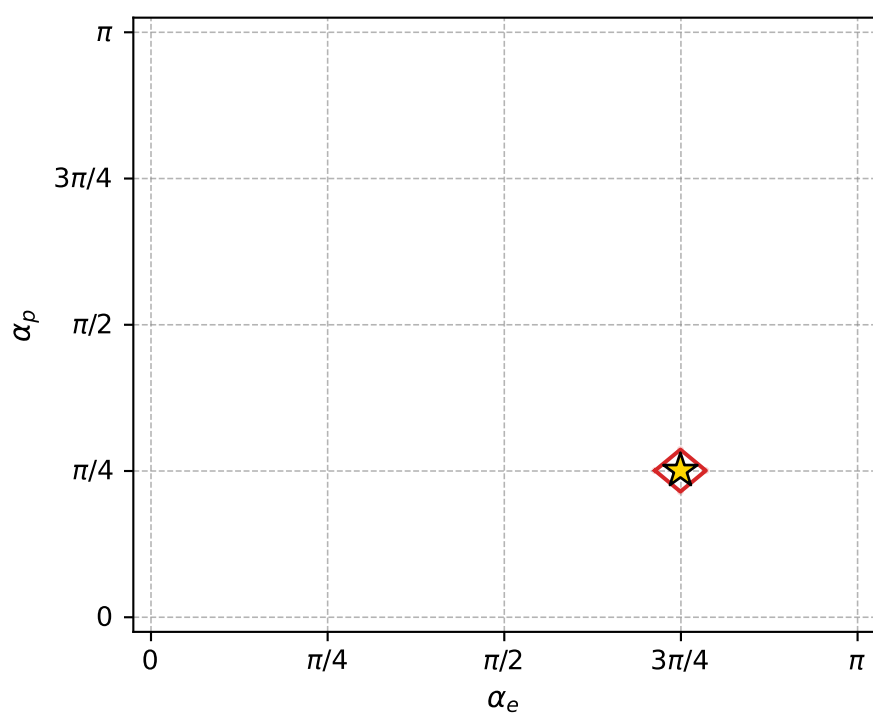
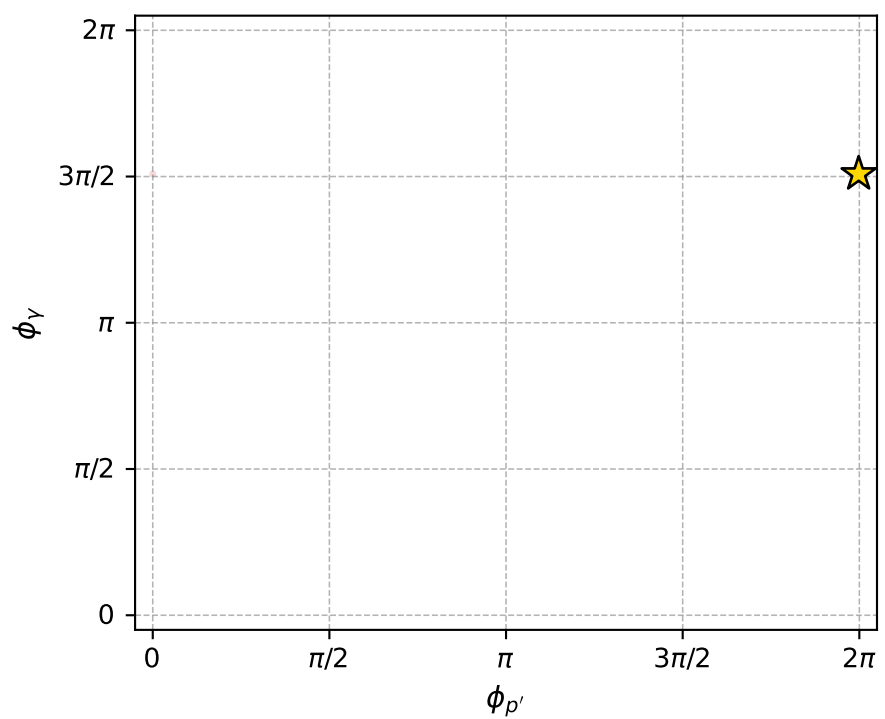
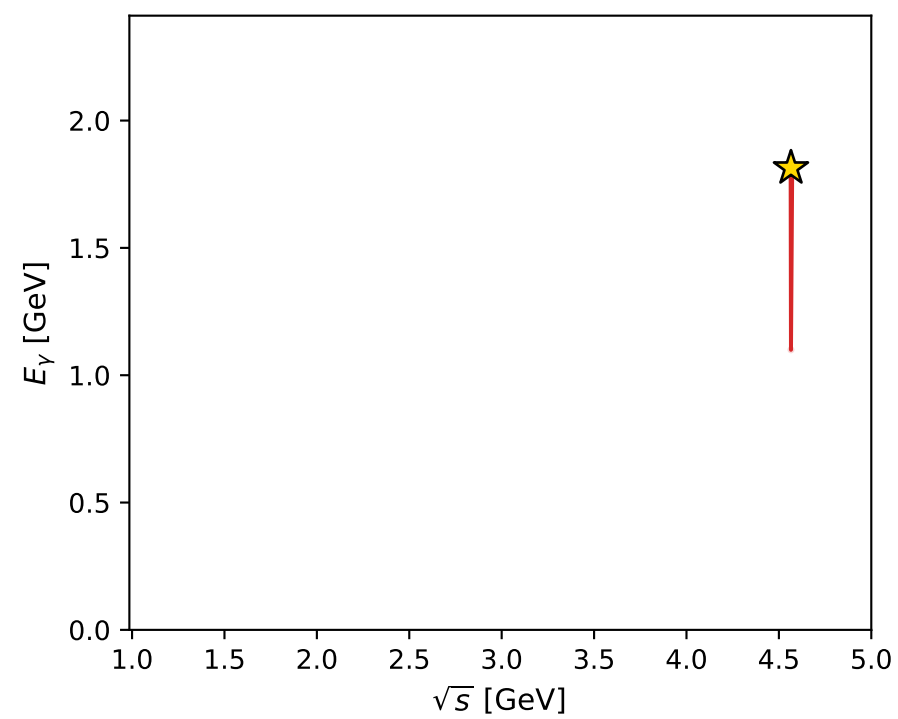
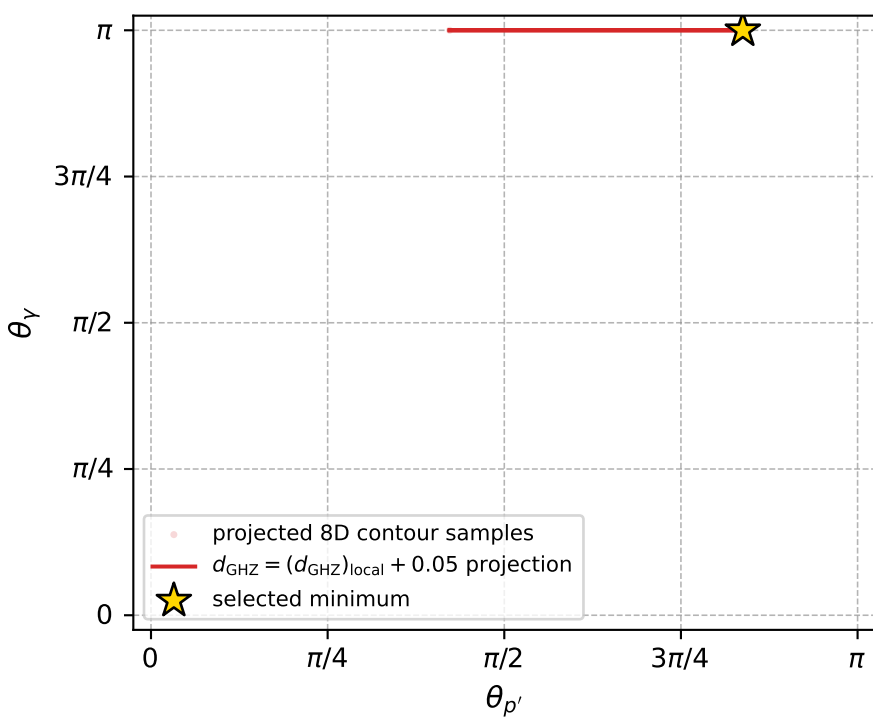
dghz_mixing_angles_polarization_03_local_minimum_68 $d_{\{\text{rm GHZ}\}}=2.82151\text{e-}08$
region: polarization_cluster_03_local_minimum_68
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3539097$ rad
incoming lepton: $-0.705489|+\rangle +0.708721|-\rangle$
proton mixing angle: $\alpha_p=0.78764165$ rad
incoming proton: $0.705519|+\rangle +0.708691|-\rangle$

$s=20.8429$, $\sqrt{s}=4.5654$, $|\vec{P}|=2.18634$, $|\vec{P}'|=6.13374\text{e-}05$
 $\theta_{P'}=2.63147$, $\theta_Y=3.14159$, $\phi_{P'}=6.27743$, $\phi_Y=4.74029$
 $E_Y=1.81367$, $\theta_{\ell'}=1.6513\text{e-}05$, $\phi_{\ell'}=3.13583$
 $Q^2=15.8617$, $x_B=0.823387$, $t=-2.70366$, $W^2=4.2821$, $y=0.964981$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1863$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1863$
 P $E= 2.3791$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1863$
 ℓ' $E= 1.8137$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8137$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0001$
 q_Y $E= 1.8137$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8137$



electron: polarization cluster P3, local minimum 68; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8215129\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.9316673\text{e-}09$
 8D contour samples = 142

selected phase-space point:

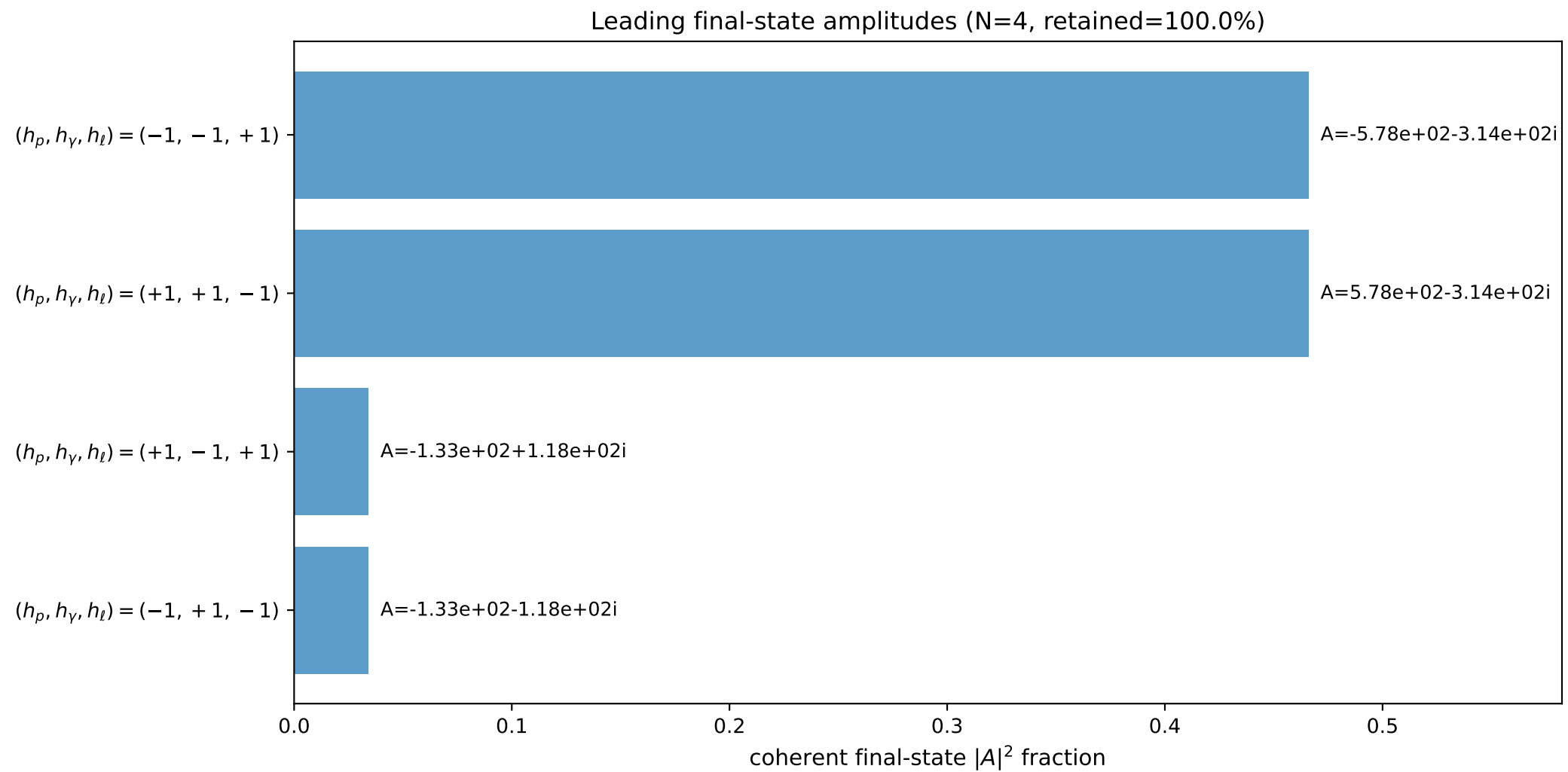
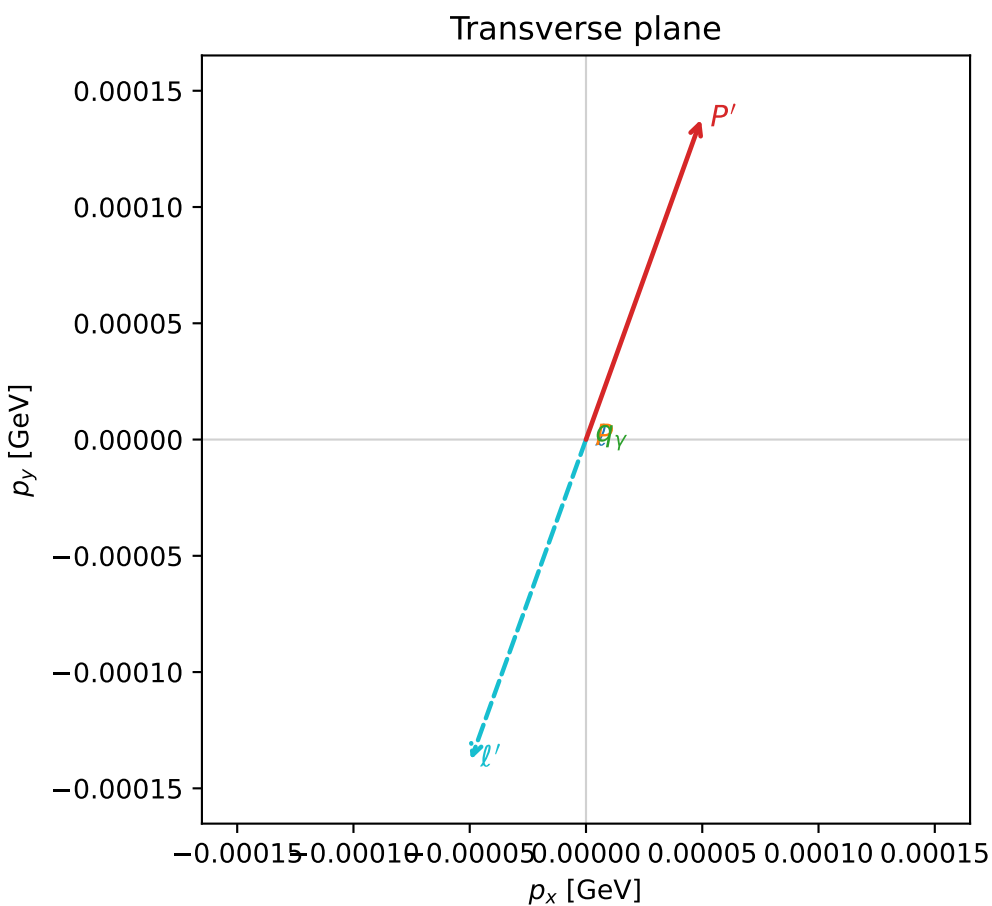
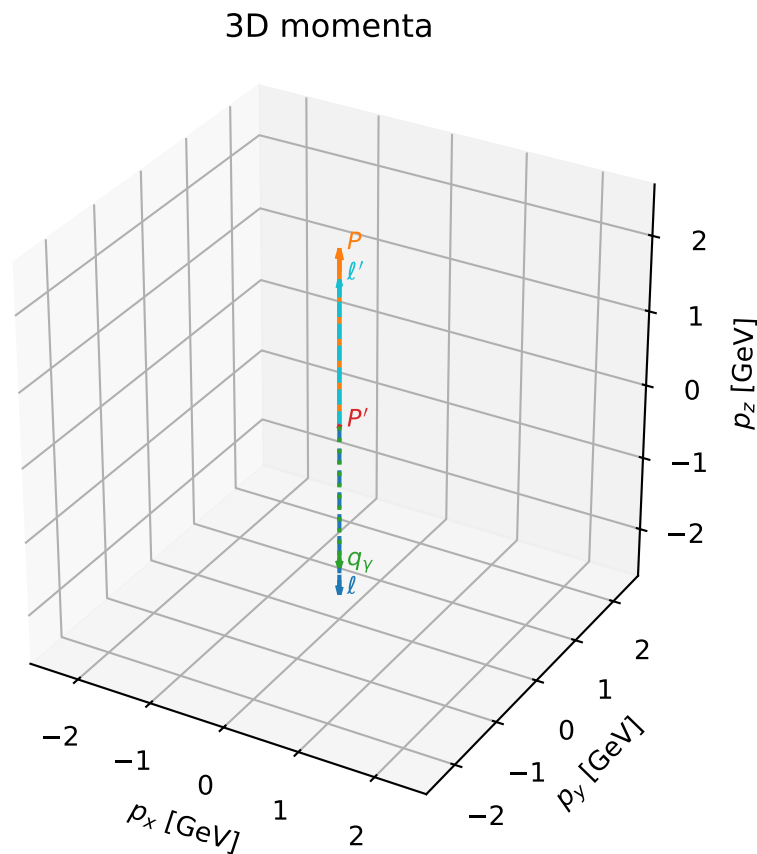
$\text{sqrt_s} = 4.565401$
 $\text{theta_p_out} = 2.63147$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.813674$
 $\text{phi_p_out} = 6.277426$
 $\text{phi_gamma_out} = 4.740295$
 $\alpha_e = 2.35391$
 $\alpha_p = 0.7876416$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

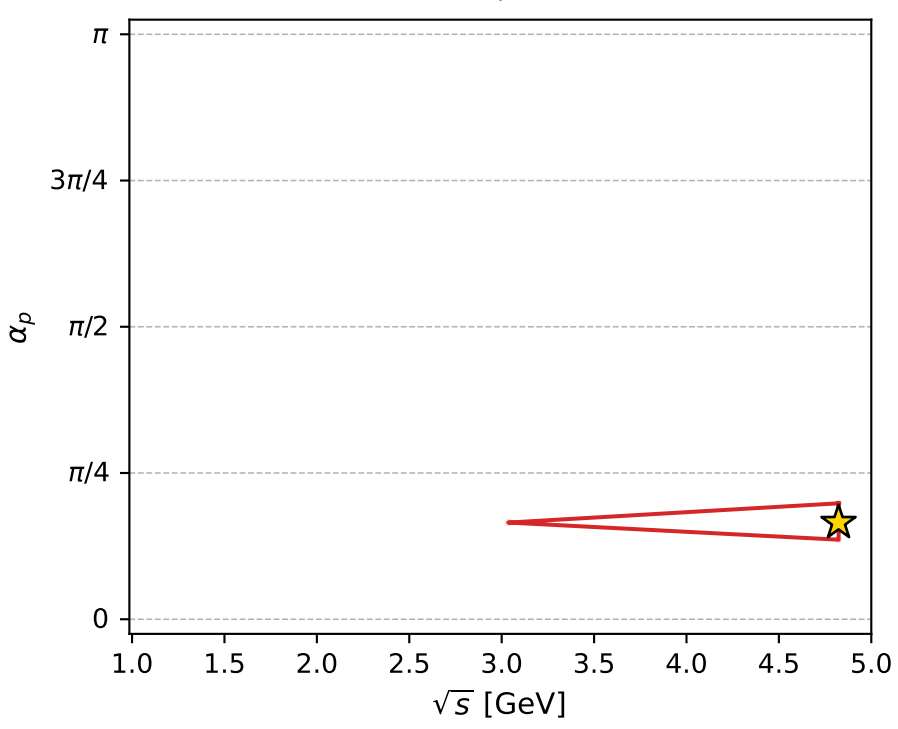
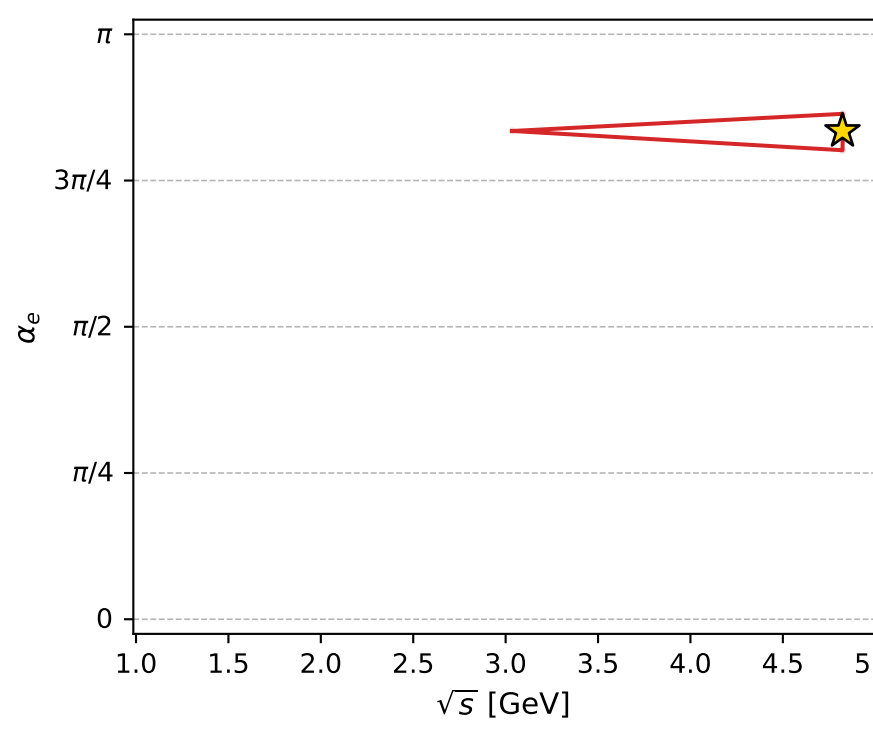
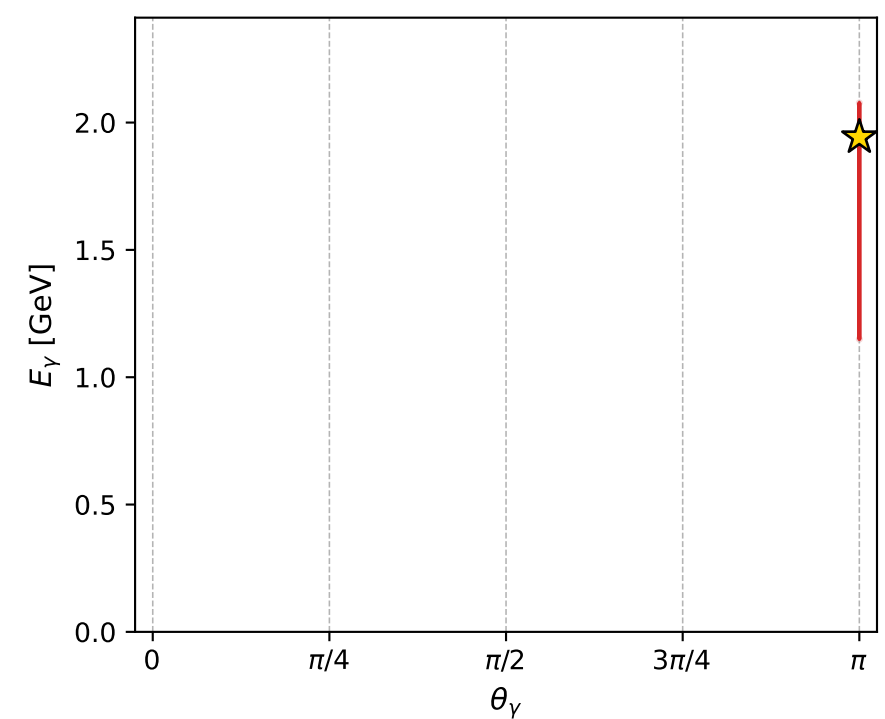
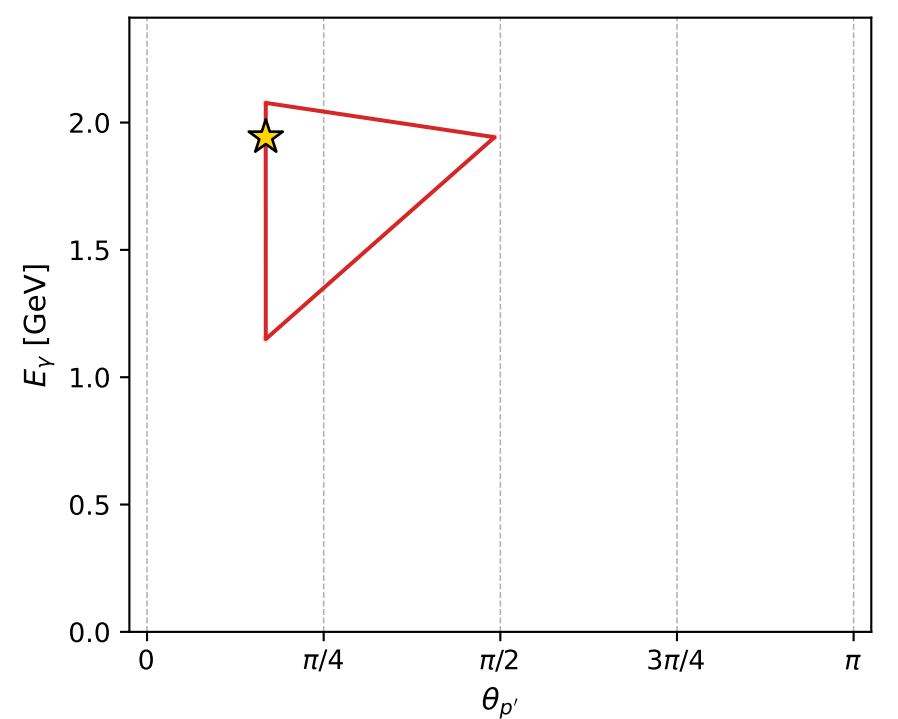
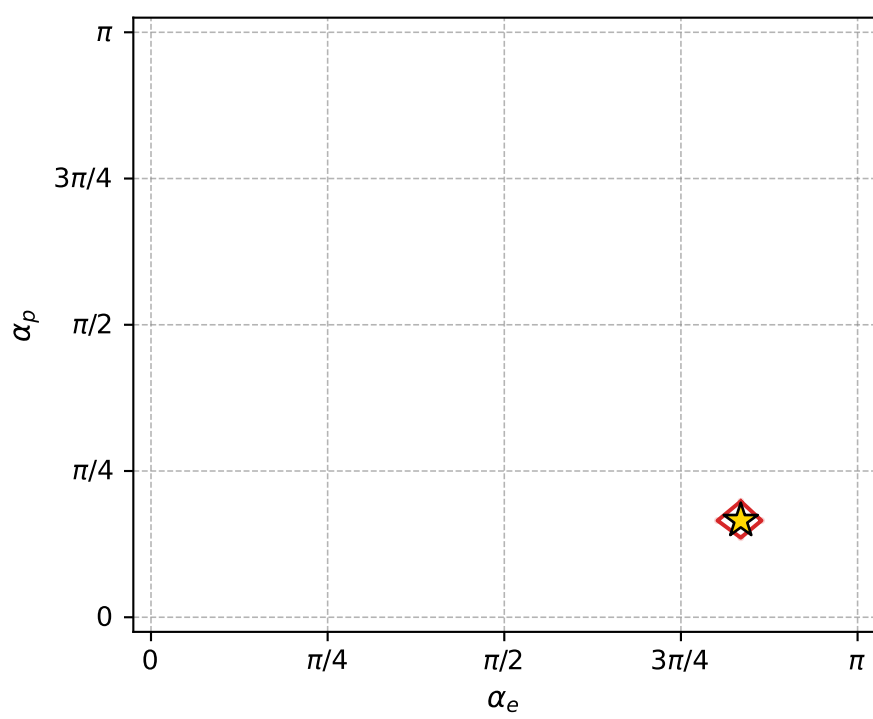
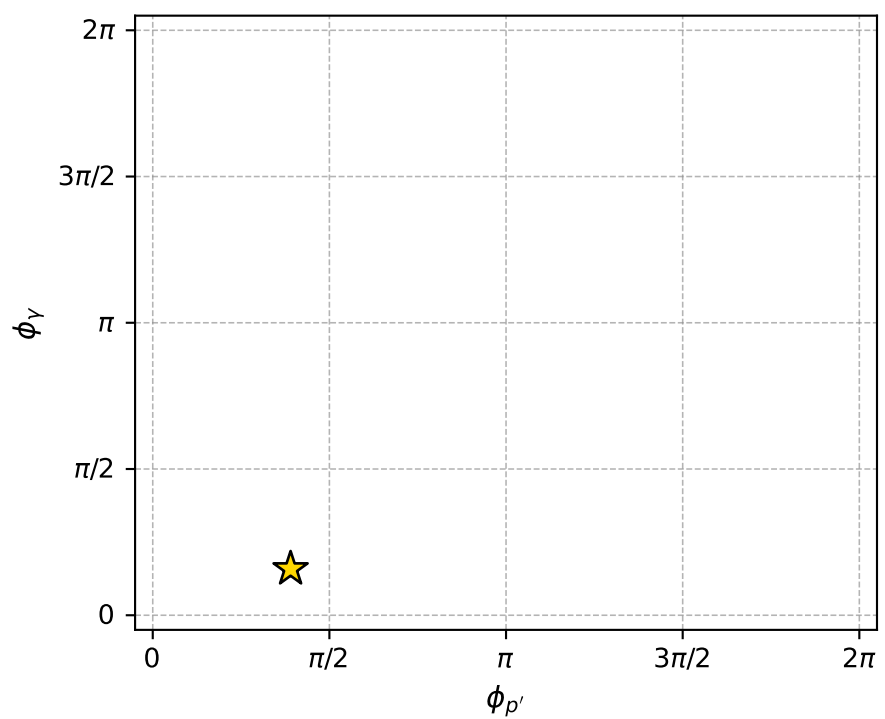
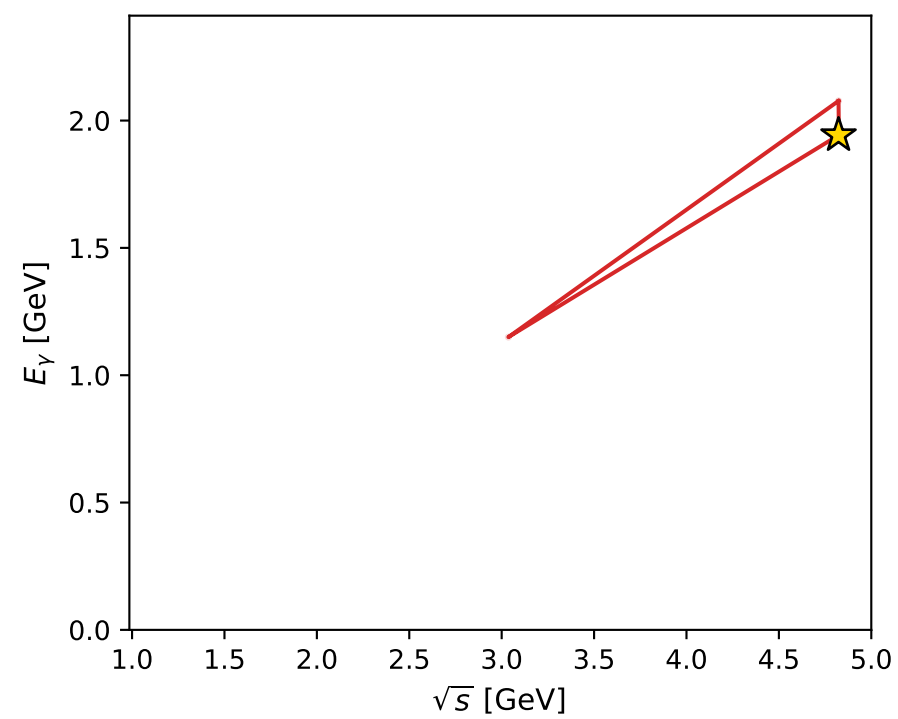
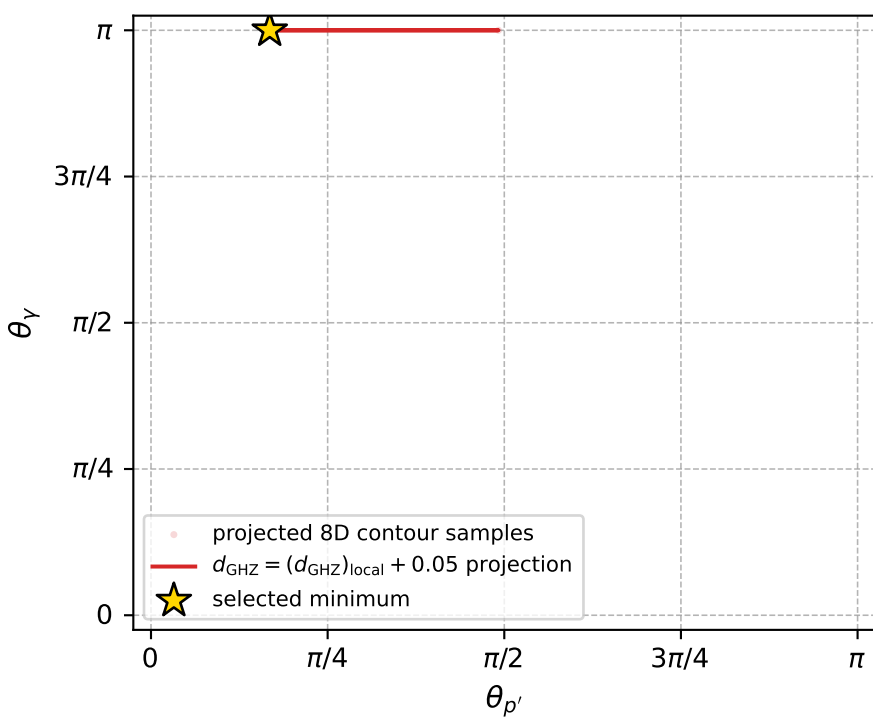
dghz_mixing_angles_polarization_03_local_minimum_70 $d_{\text{GHZ}}=2.84345\text{e-}08$
region: polarization_cluster_03_local_minimum_70
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.622295$ rad
incoming lepton: $-0.868168|+> +0.496271|->$
proton mixing angle: $\alpha_p=0.51928971$ rad
incoming proton: $0.868172|+> +0.496264|->$

$s=23.2589$, $\sqrt{s}=4.82275$, $|\vec{P}|=2.32016$, $|\vec{P}'|=0.000290339$
 $\theta_{P'}=0.527964$, $\theta_Y=3.14159$, $\phi_{P'}=1.22578$, $\phi_Y=0.497916$
 $E_Y=1.9425$, $\theta_{\ell'}=7.53075\text{e-}05$, $\phi_{\ell'}=4.36737$
 $Q^2=18.0253$, $x_B=0.831793$, $t=-2.93401$, $W^2=4.52495$, $y=0.968333$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3202$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3202$
 P $E= 2.5026$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3202$
 ℓ' $E= 1.9422$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 1.9422$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0003$
 q_Y $E= 1.9425$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9425$



electron: polarization cluster P3, local minimum 70; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8434488\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $6.1510261\text{e-}09$
 8D contour samples = 140

selected phase-space point:

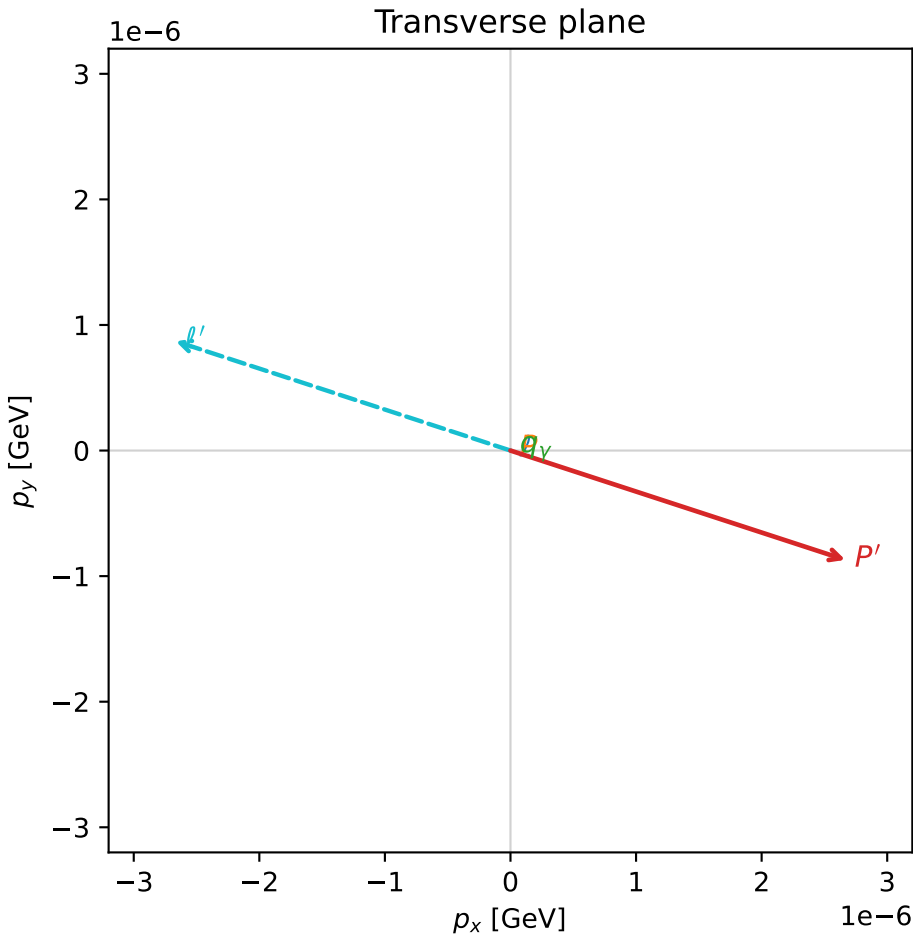
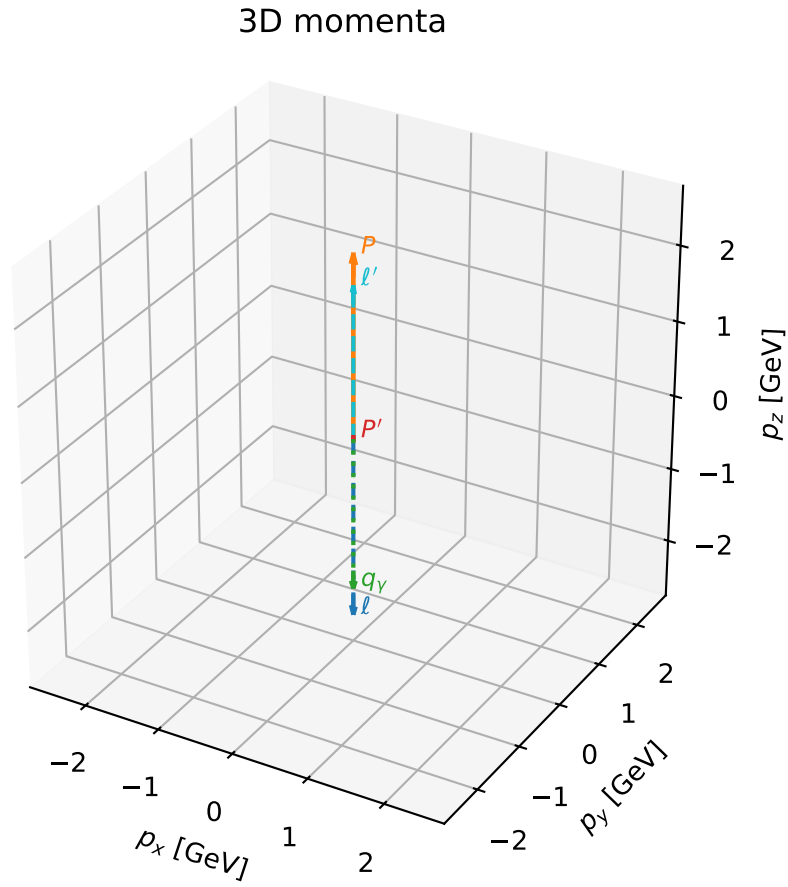
$\text{sqrt_s} = 4.822748$
 $\text{theta_p_out} = 0.5279643$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.942499$
 $\text{phi_p_out} = 1.225782$
 $\text{phi_gamma_out} = 0.4979155$
 $\alpha_e = 2.622295$
 $\alpha_p = 0.5192897$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_73 $d_{\{\text{rm GHZ}\}}=2.91539\text{e-}08$
region: polarization_cluster_03_local_minimum_73
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.1495745$ rad
incoming lepton: $-0.547002|+\rangle +0.837132|-\rangle$
proton mixing angle: $\alpha_p=0.99200753$ rad
incoming proton: $0.54701|+\rangle +0.837126|-\rangle$

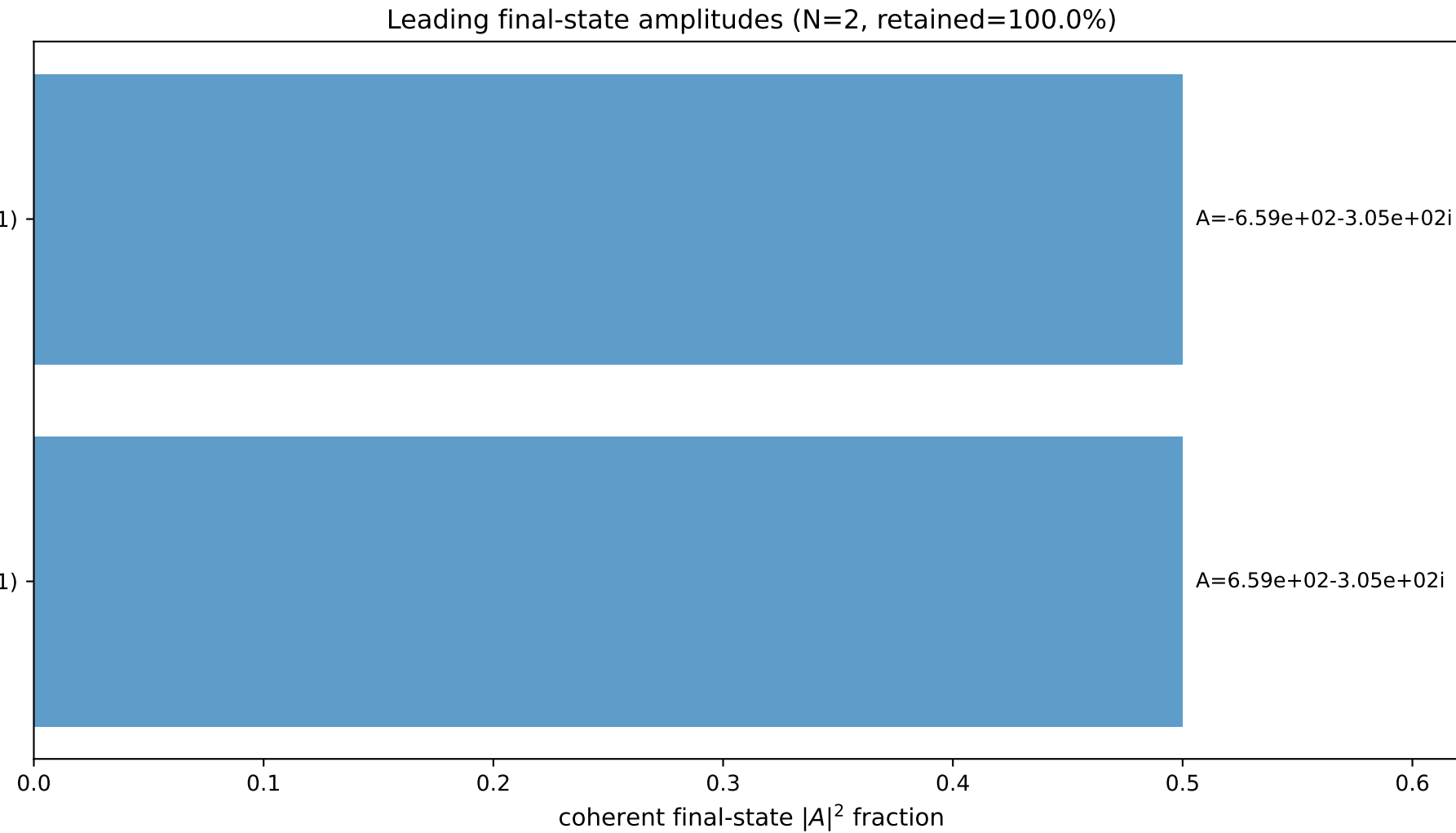
$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=0.0232778$
 $\theta_{P'}=0.000120512$, $\theta_Y=3.14159$, $\phi_{P'}=5.96763$, $\phi_Y=0.433102$
 $E_Y=2.04248$, $\theta_{\ell'}=1.38933\text{e-}06$, $\phi_{\ell'}=2.82604$
 $Q^2=19.4813$, $x_B=0.832205$, $t=-2.98455$, $W^2=4.8078$, $y=0.970537$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0192$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0192$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0233$
 q_Y $E= 2.0425$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0425$

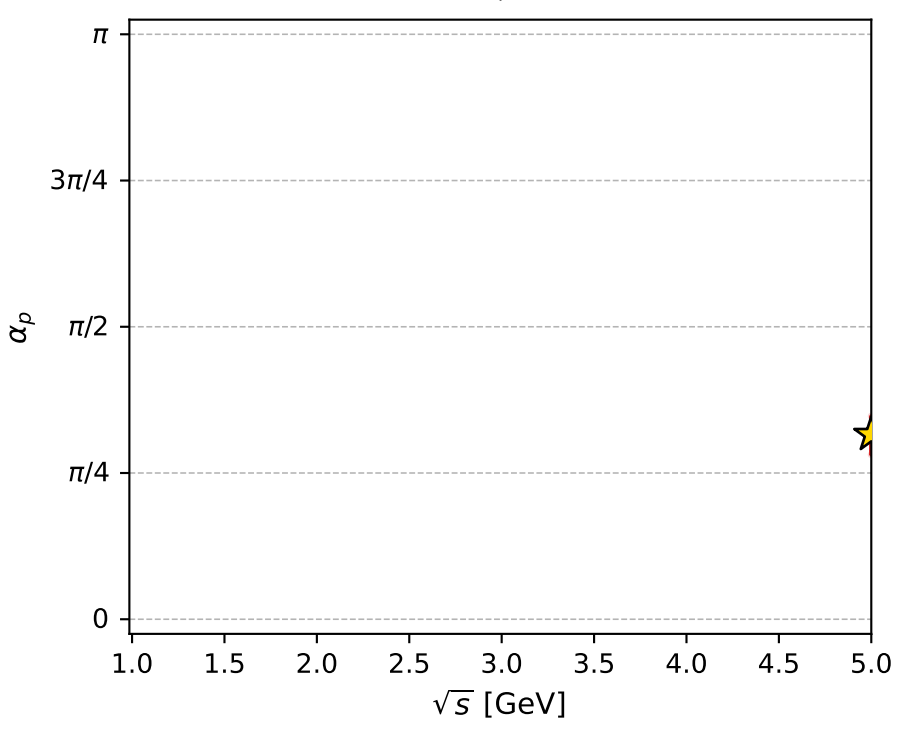
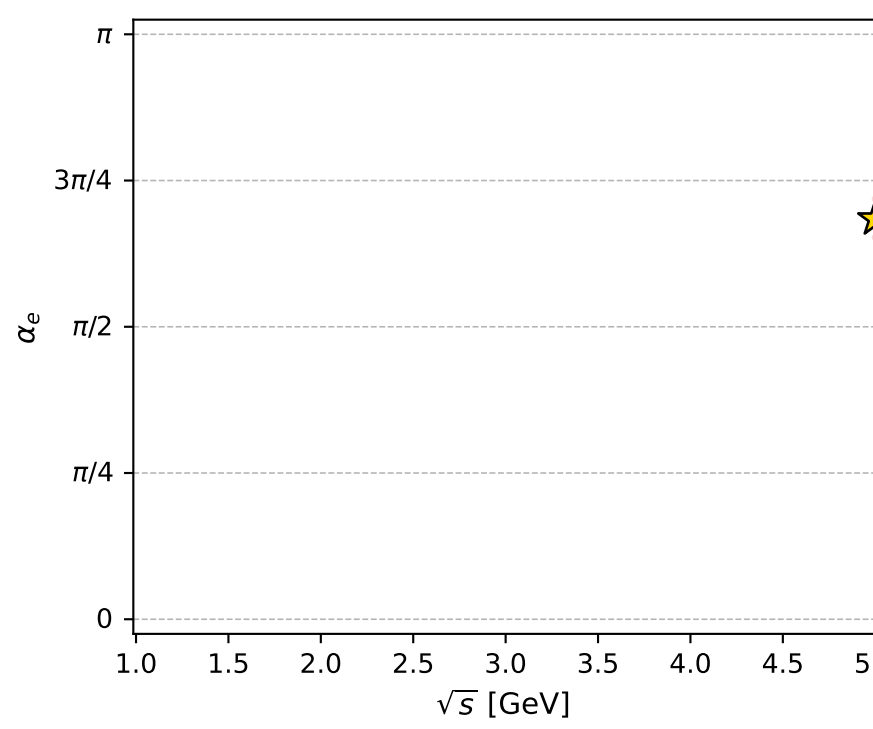
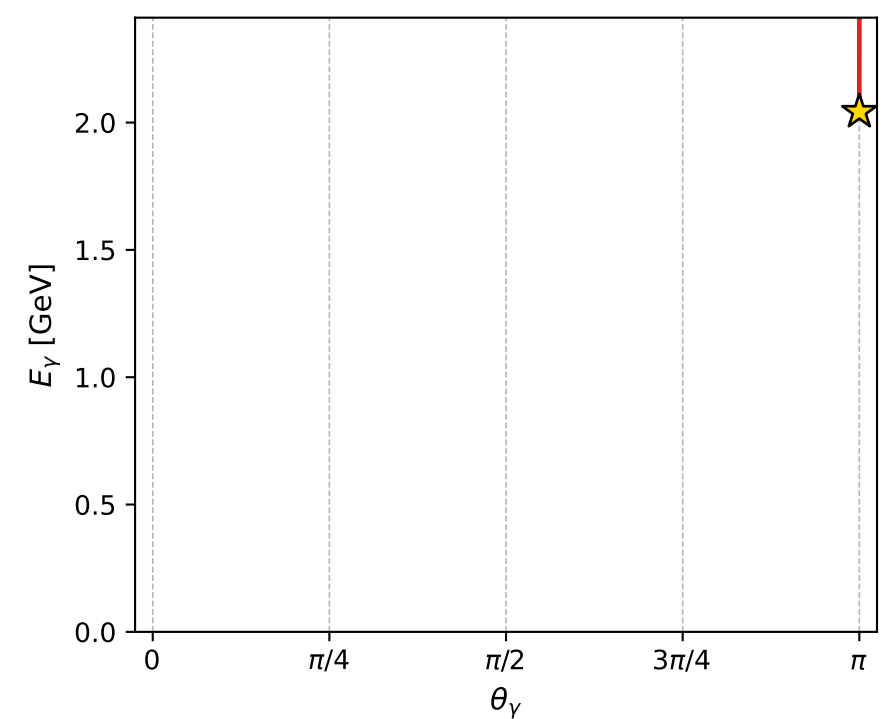
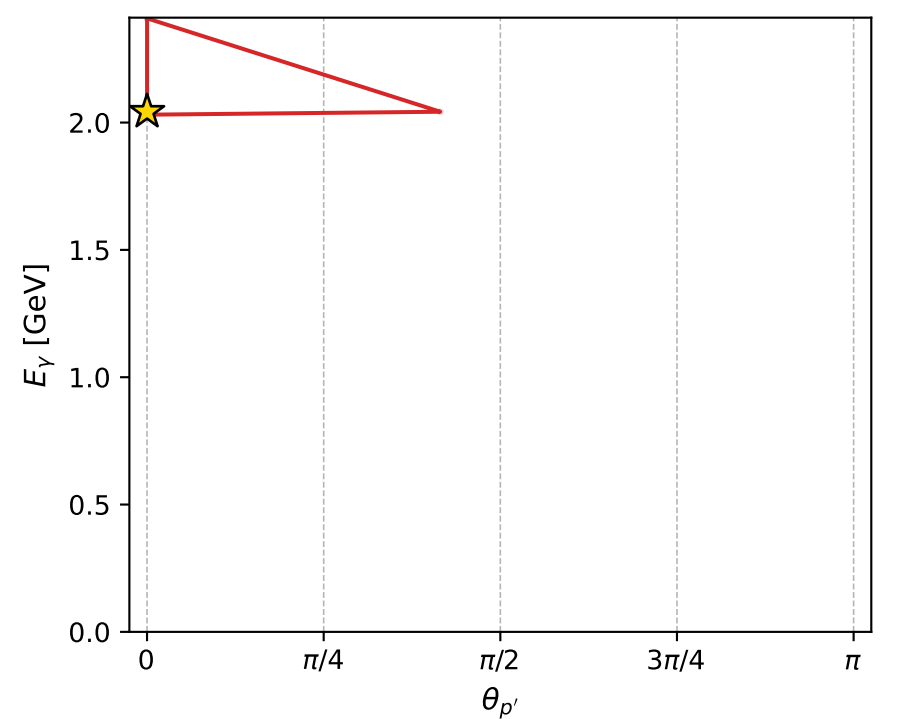
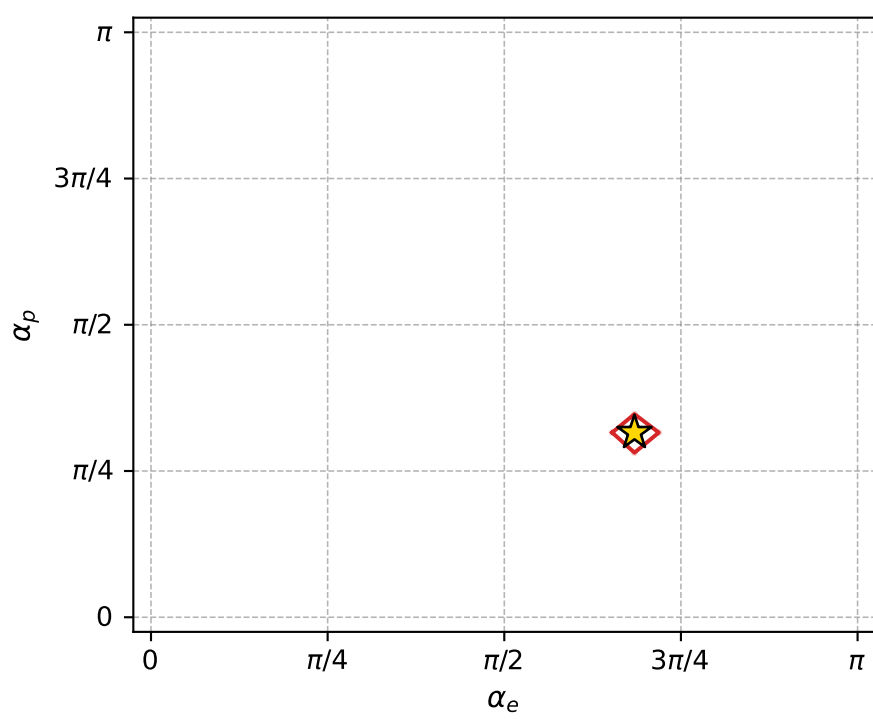
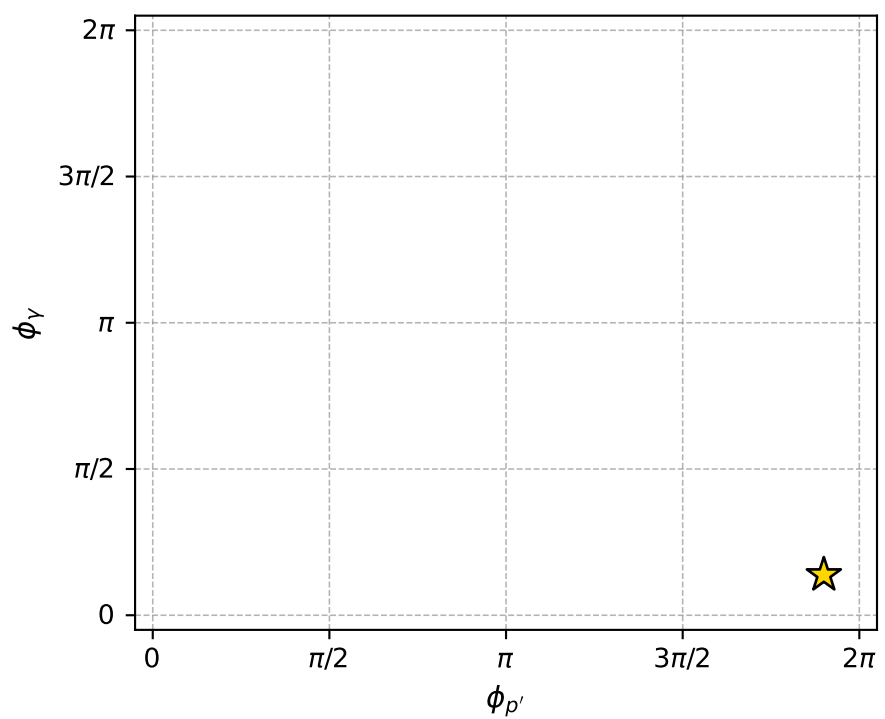
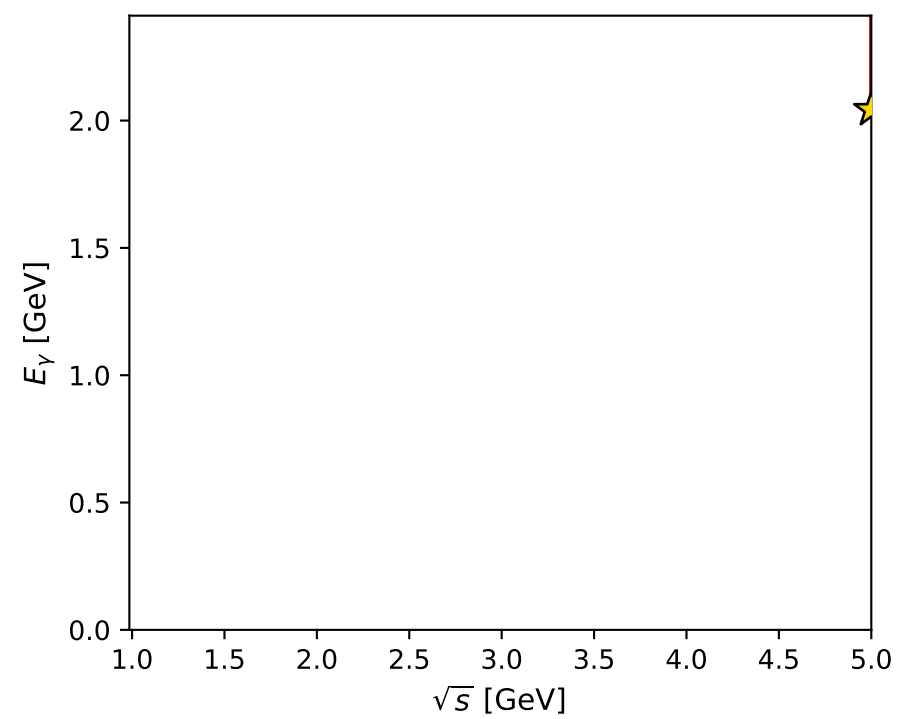
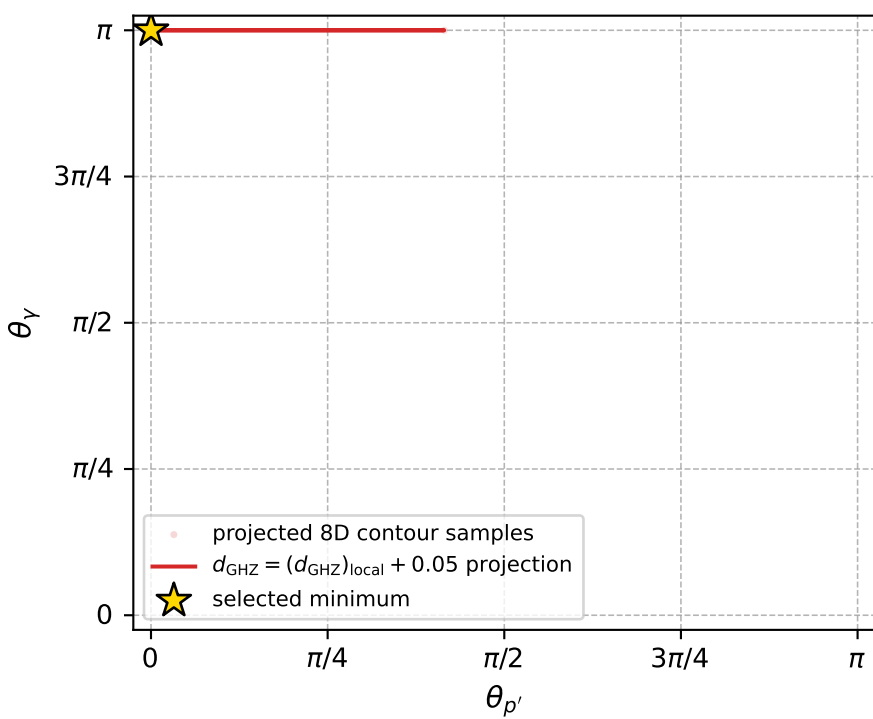


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P3, local minimum 73; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9153902\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000029$
 above global minimum = $6.8704403\text{e-}09$
 8D contour samples = 112

selected phase-space point:

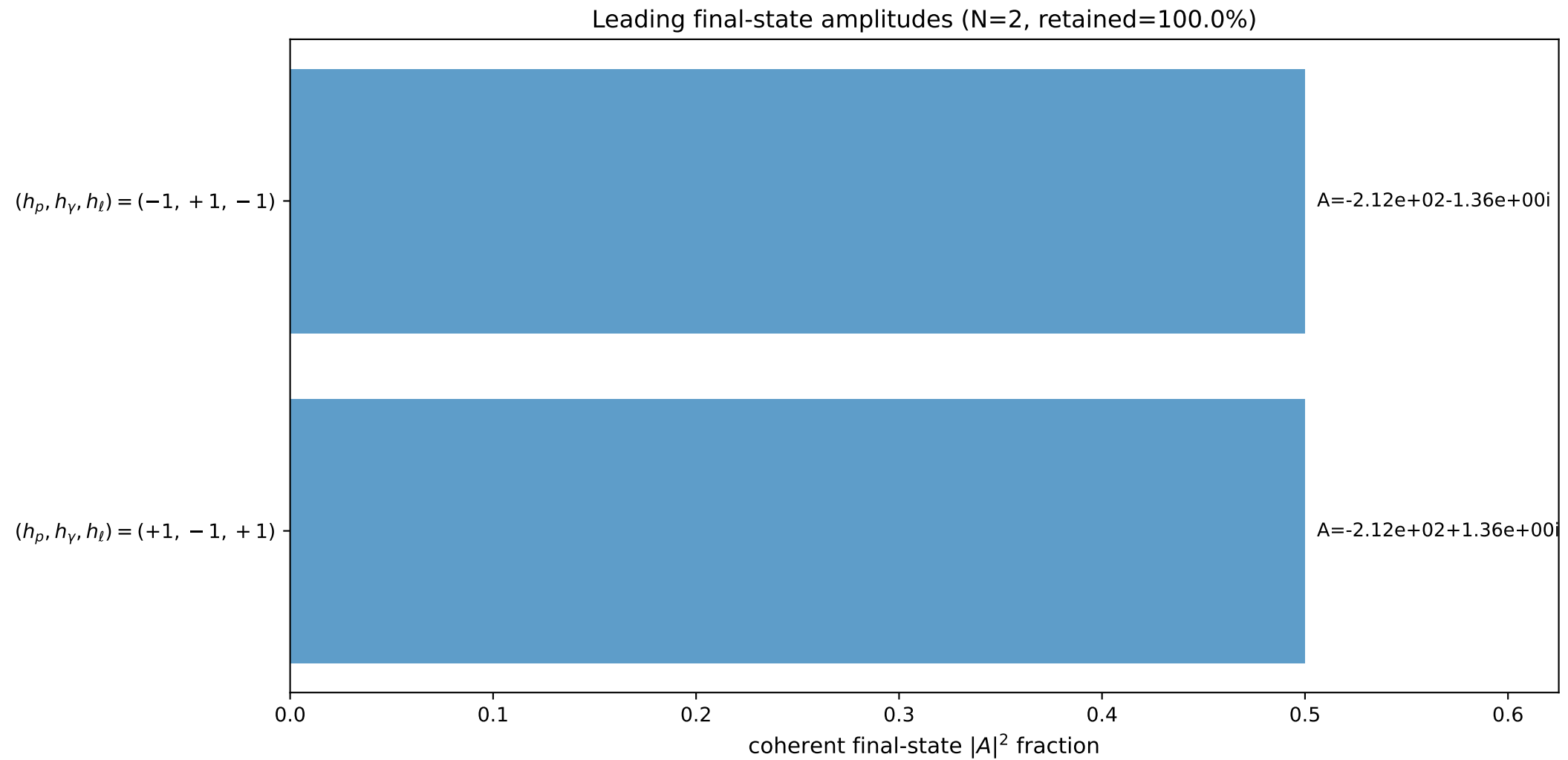
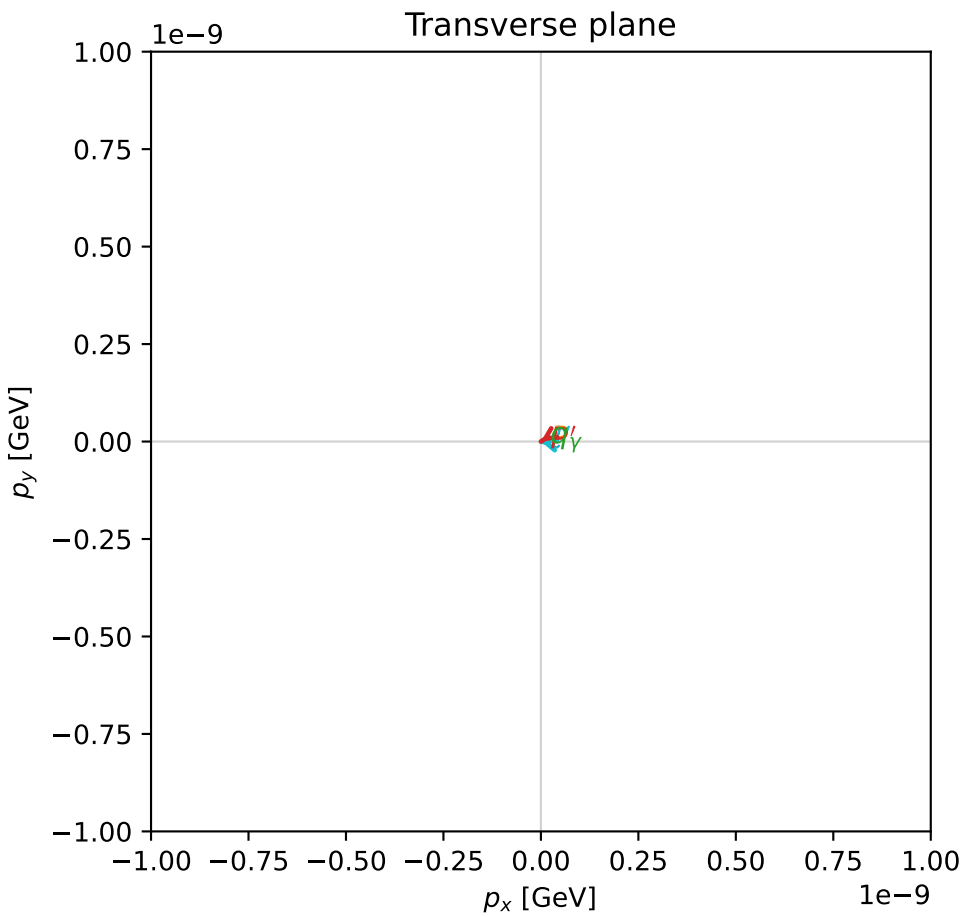
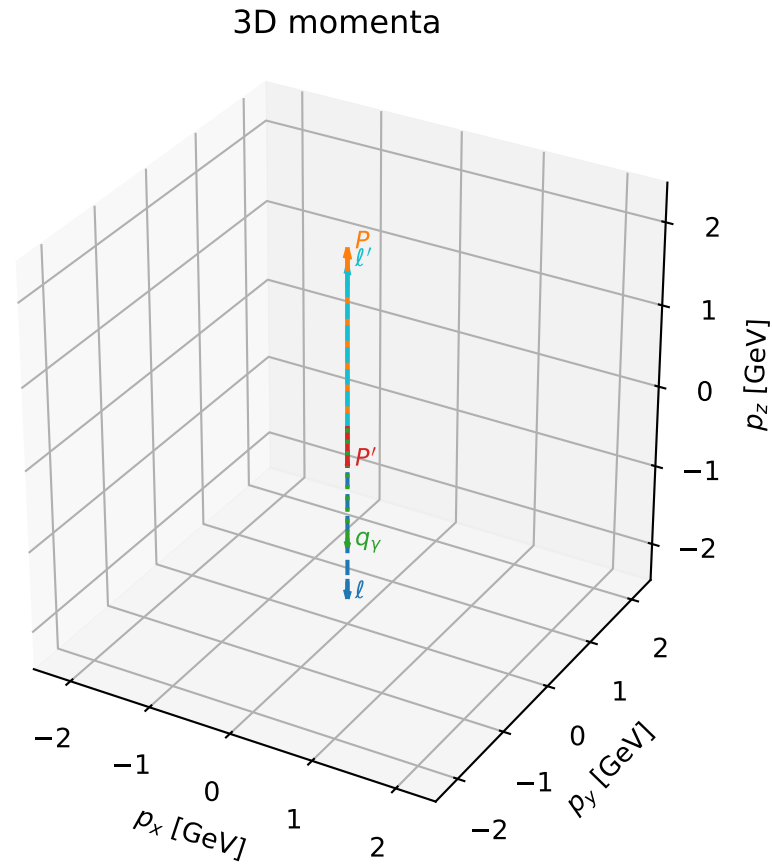
$\text{sqrt_s} = 4.99997$
 $\text{theta_p_out} = 0.0001205121$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.04248$
 $\text{phi_p_out} = 5.967632$
 $\text{phi_gamma_out} = 0.4331021$
 $\text{alpha_e} = 2.149575$
 $\text{alpha_p} = 0.9920075$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

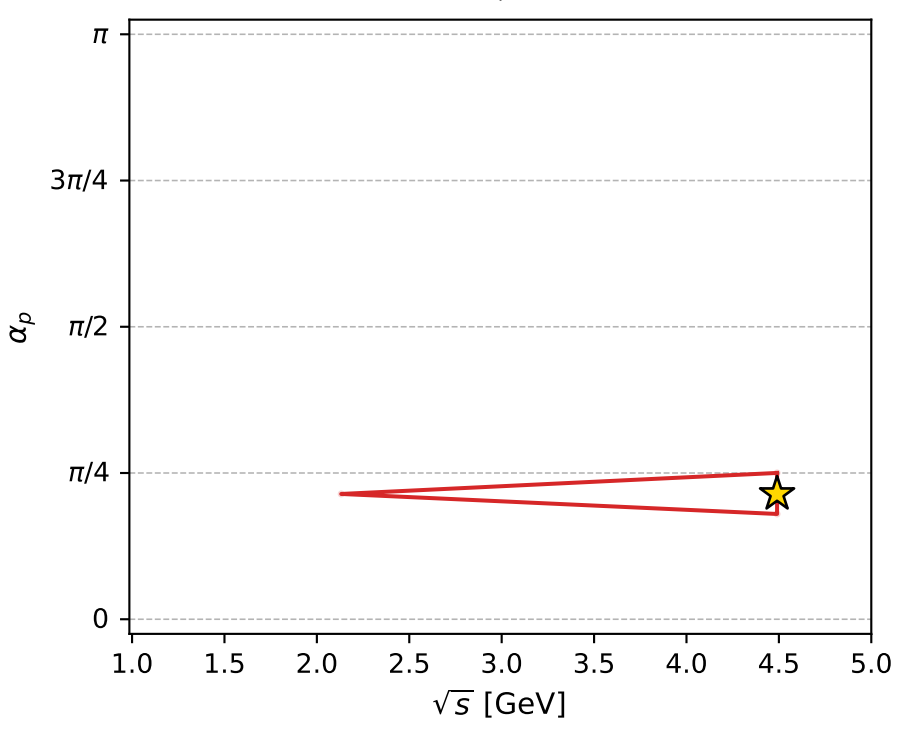
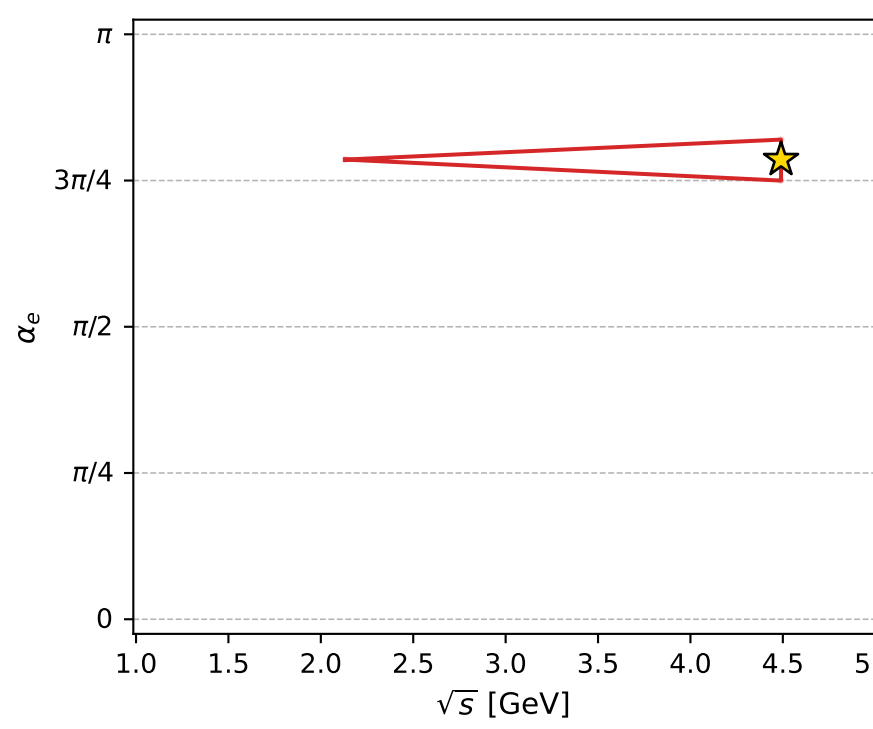
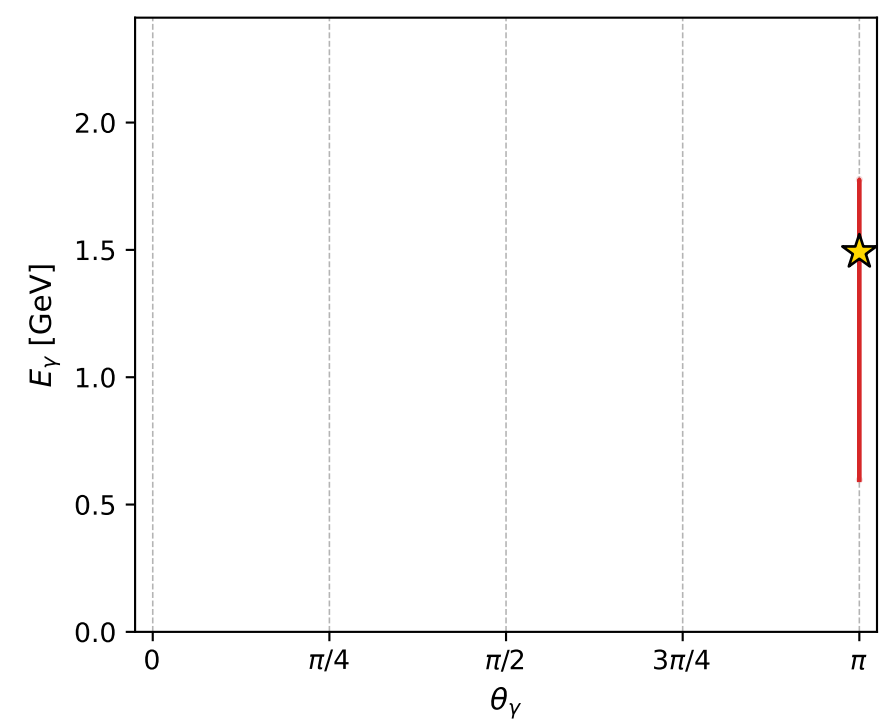
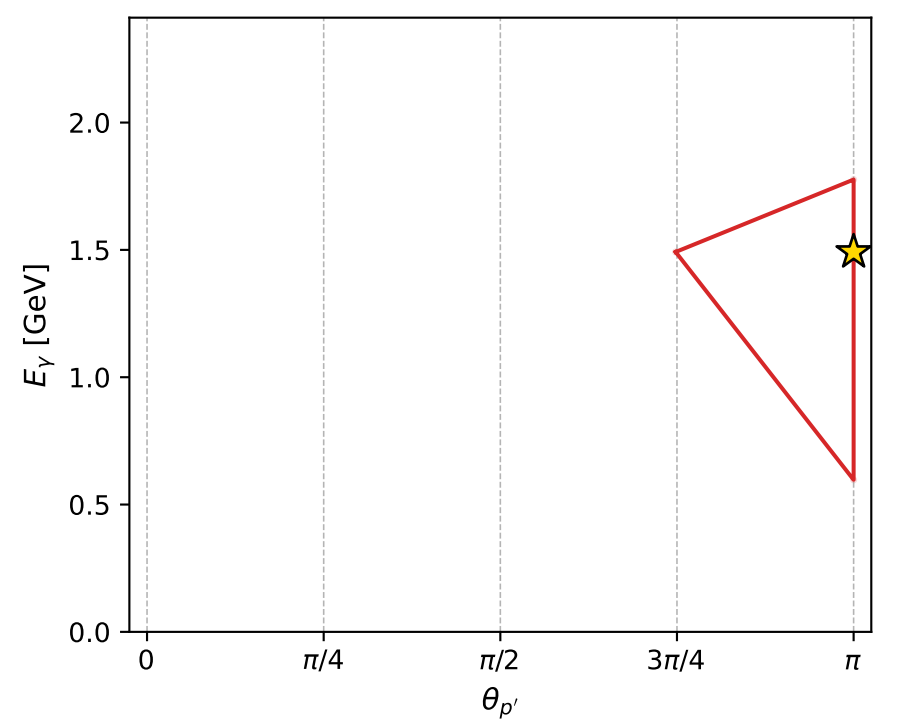
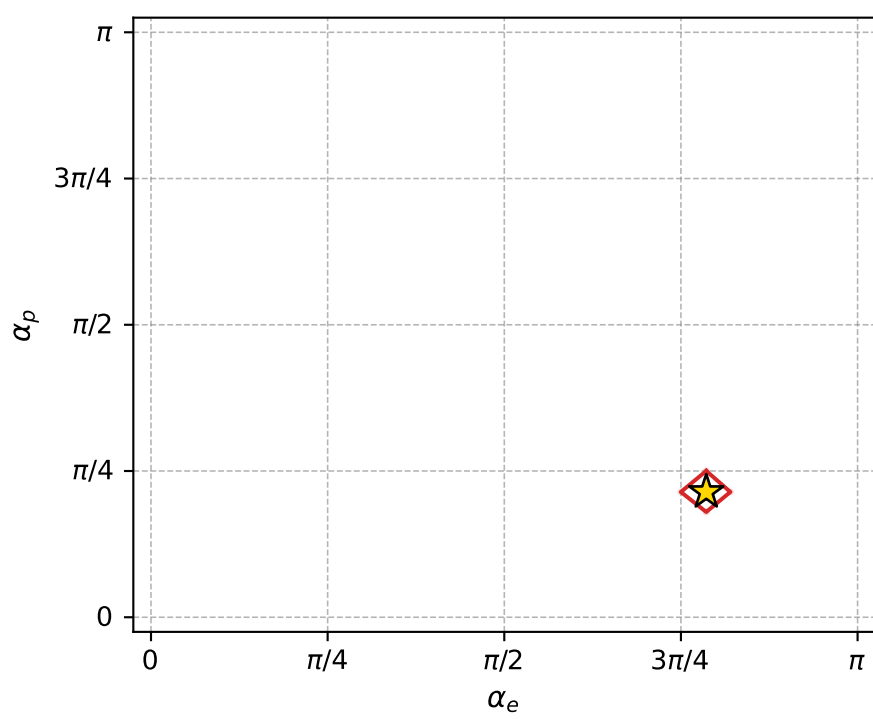
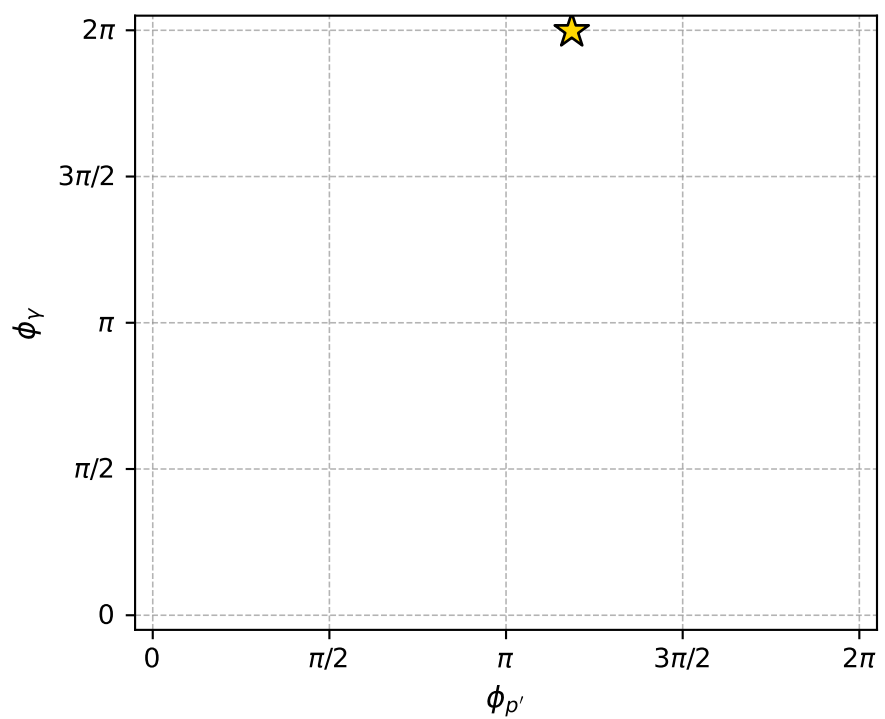
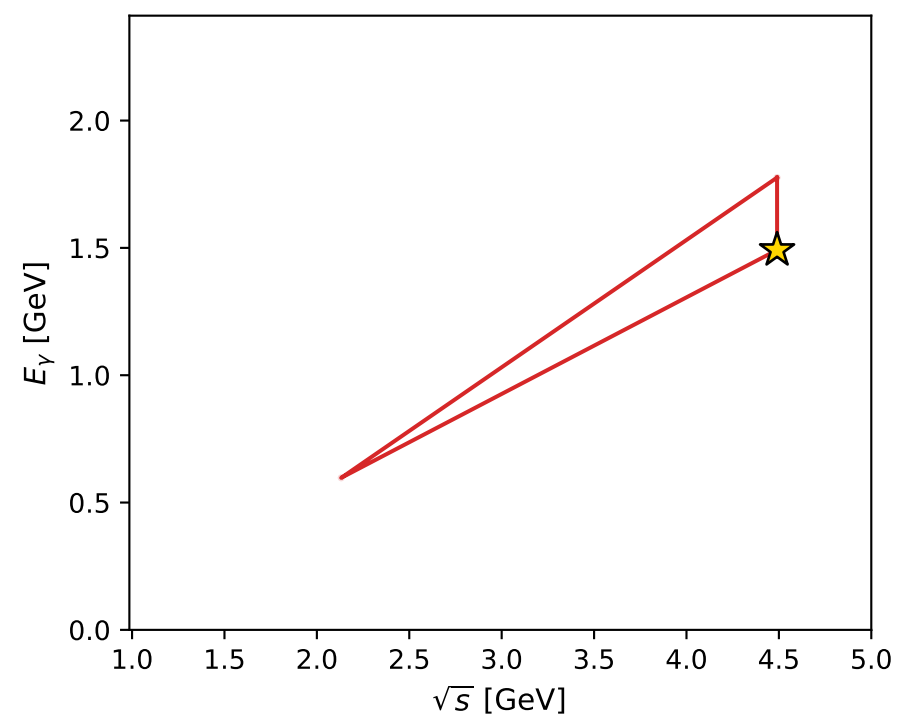
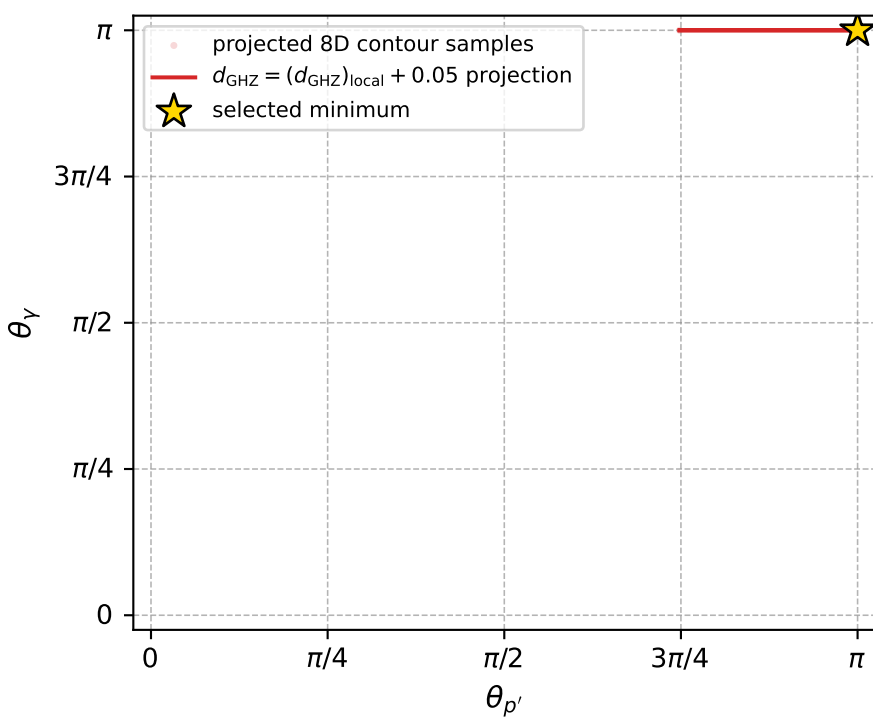
dghz_mixing_angles_polarization_03_local_minimum_87 $d_{\{\text{rm GHZ}\}}=3.19716\text{e-}08$
 region: polarization_cluster_03_local_minimum_87
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4687294$ rad
 incoming lepton: $-0.78204|+> +0.623228|->$
 proton mixing angle: $\alpha_p=0.67288163$ rad
 incoming proton: $0.782029|+> +0.623242|->$

$s=20.1618$, $\sqrt{s}=4.49019$, $|\vec{P}|=2.14712$, $|\vec{P}'|=0.461231$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=3.72593$, $\phi_Y=6.27676$
 $E_Y=1.49185$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.90749$
 $Q^2=16.774$, $x_B=0.905891$, $t=-5.1192$, $W^2=2.62242$, $y=0.960305$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1471$
 P $E= 2.3431$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1471$
 ℓ' $E= 1.9531$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9531$
 P' $E= 1.0453$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.4612$
 q_Y $E= 1.4918$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4918$



electron: polarization cluster P3, local minimum 87; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.1971613\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000032$
 above global minimum = $9.6881515\text{e-}09$
 8D contour samples = 61

selected phase-space point:

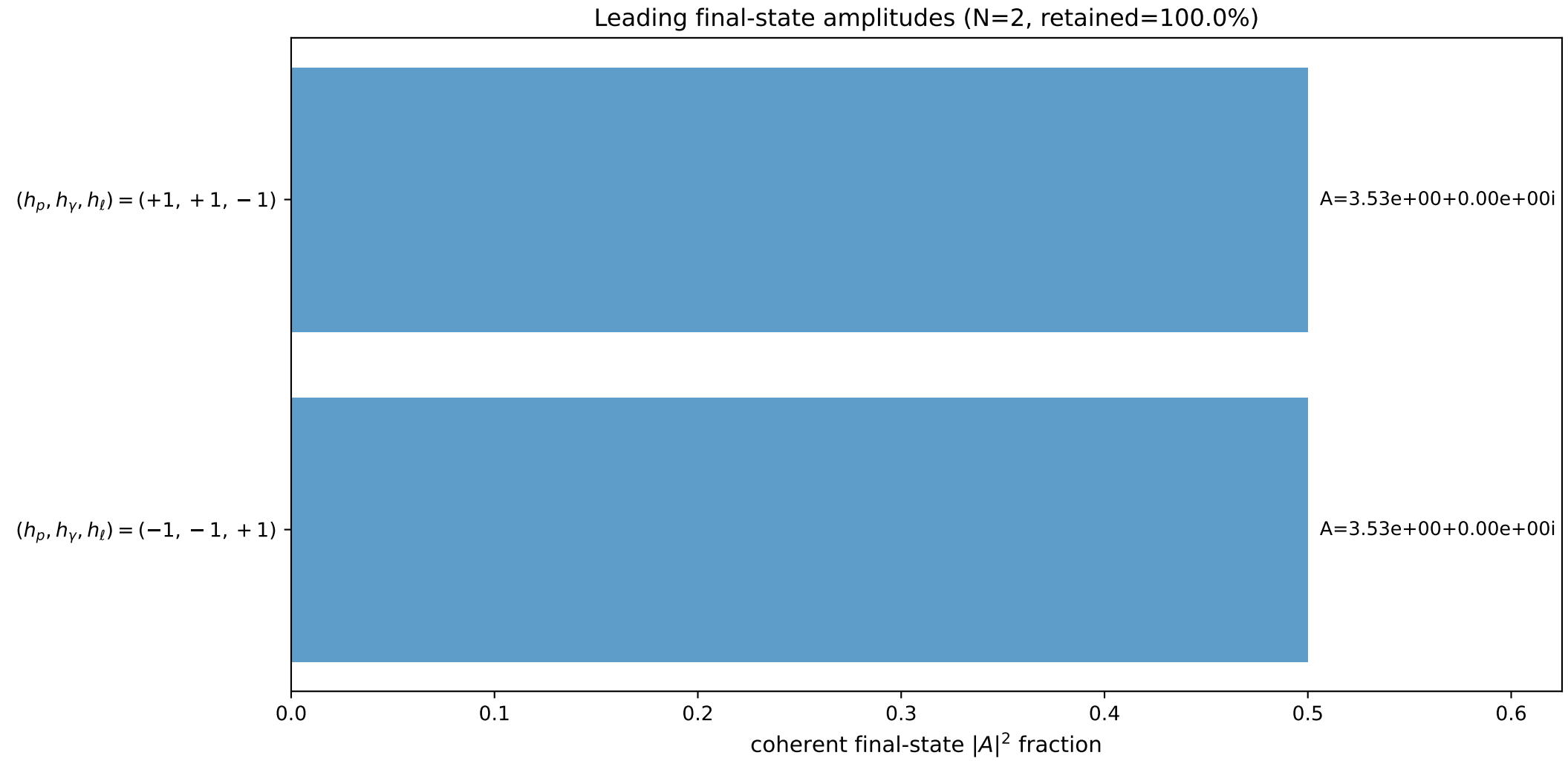
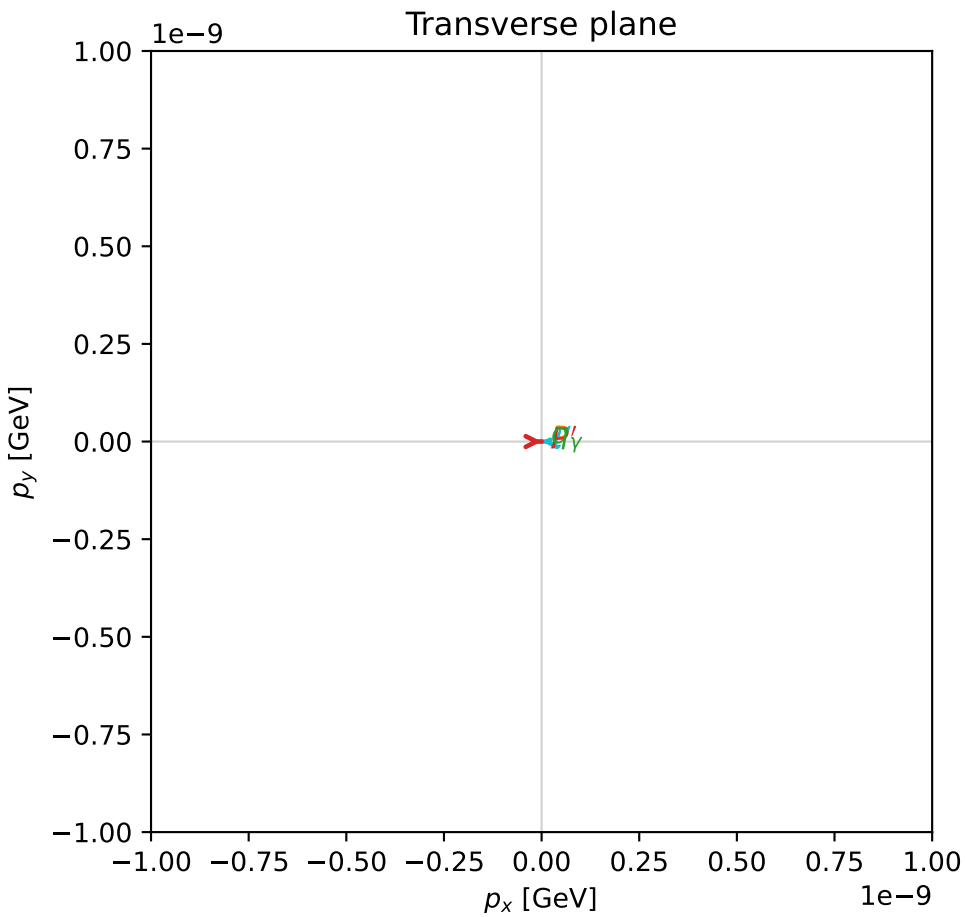
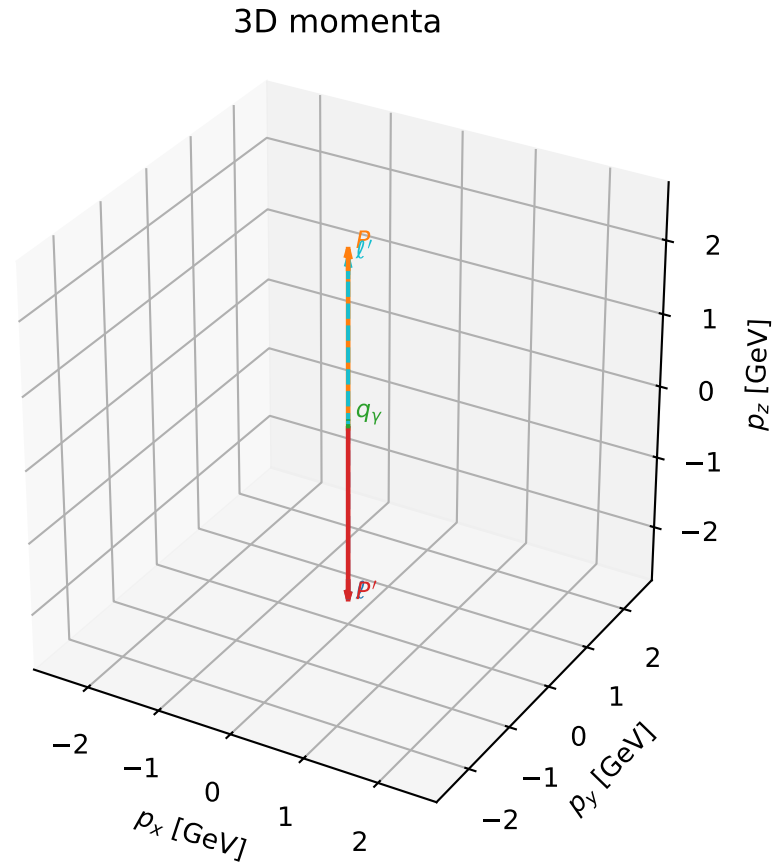
$\text{sqrt_s} = 4.490189$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.491847$
 $\text{phi_p_out} = 3.725934$
 $\text{phi_gamma_out} = 6.276764$
 $\text{alpha_e} = 2.468729$
 $\text{alpha_p} = 0.6728816$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

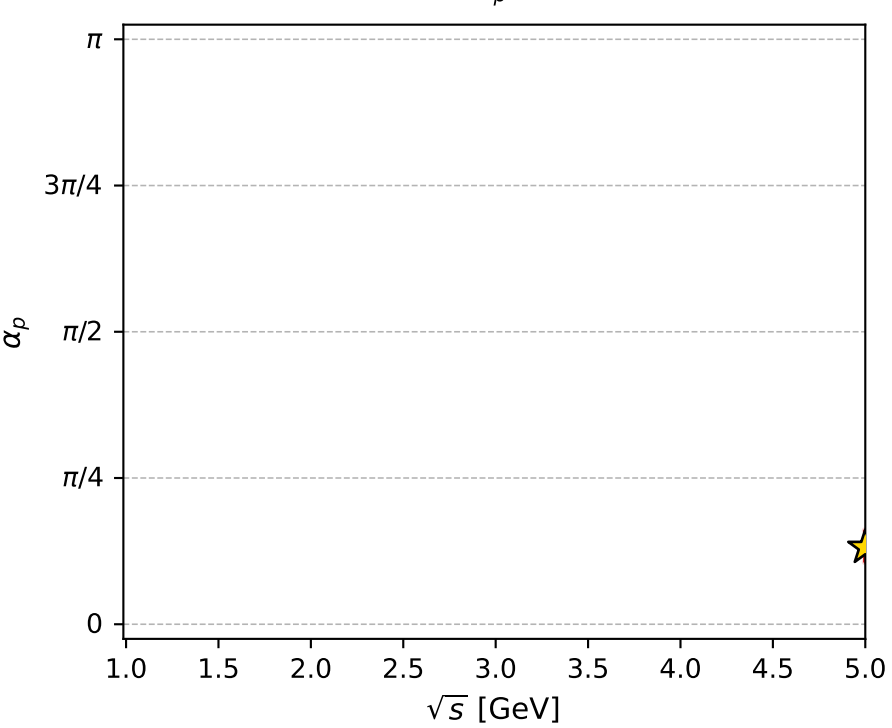
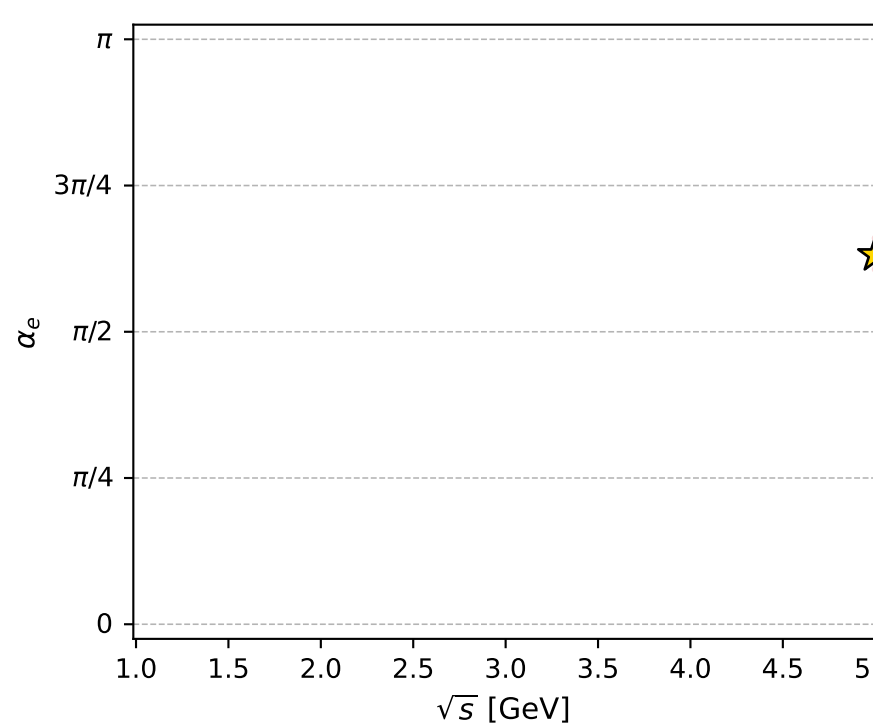
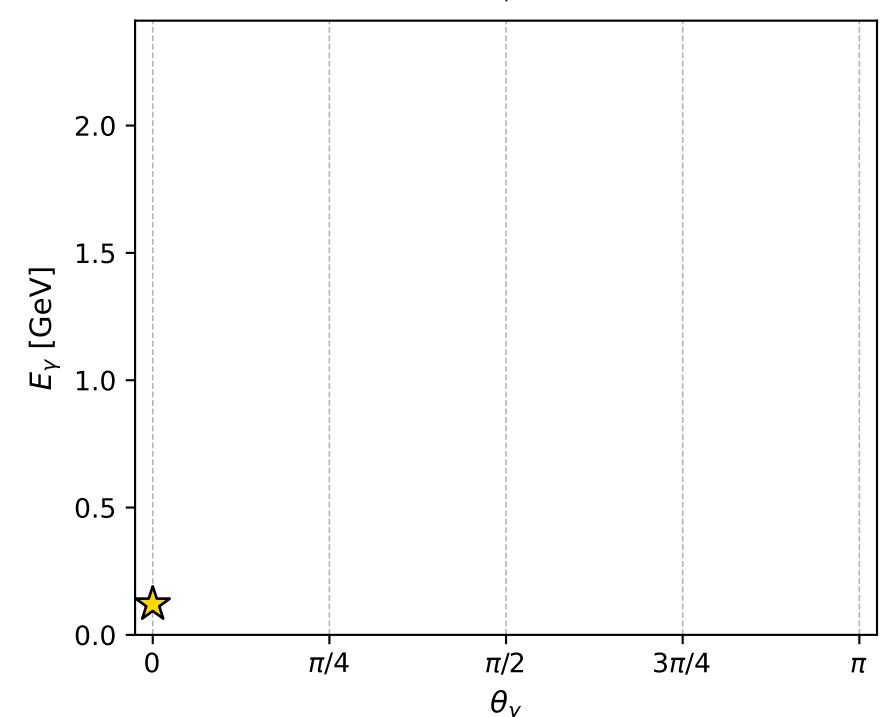
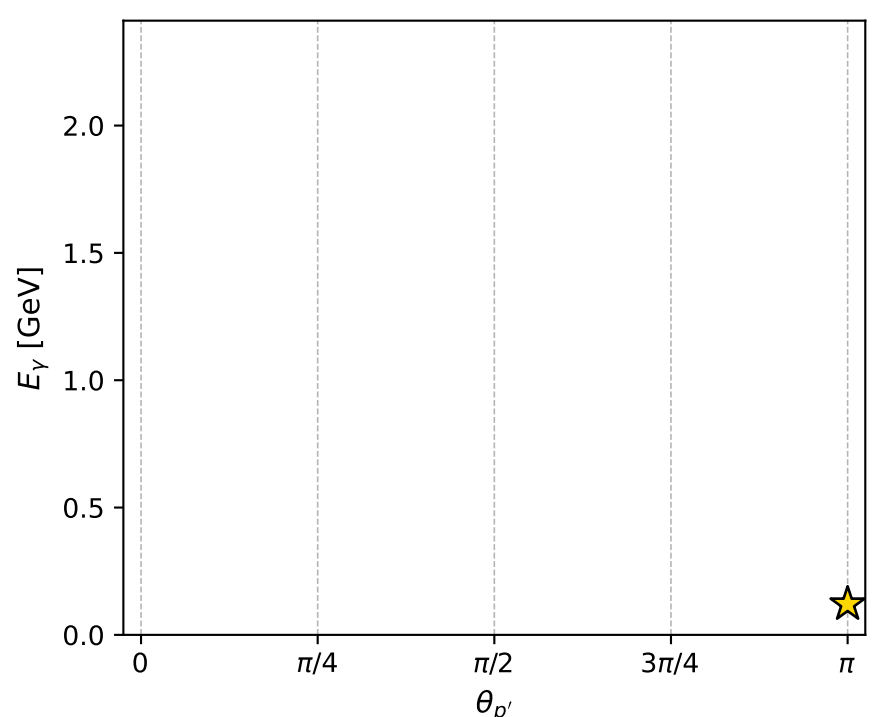
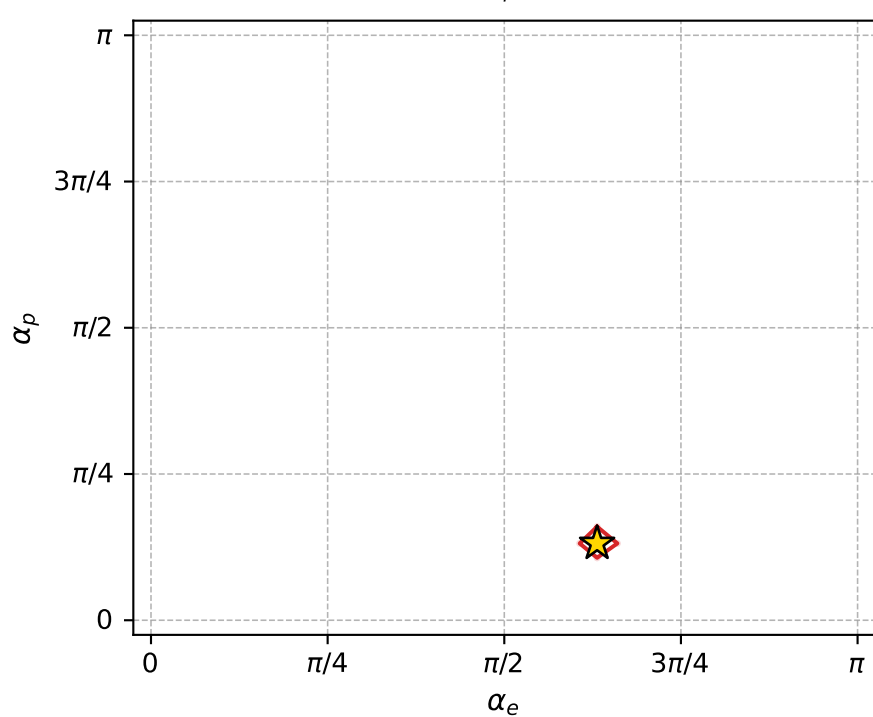
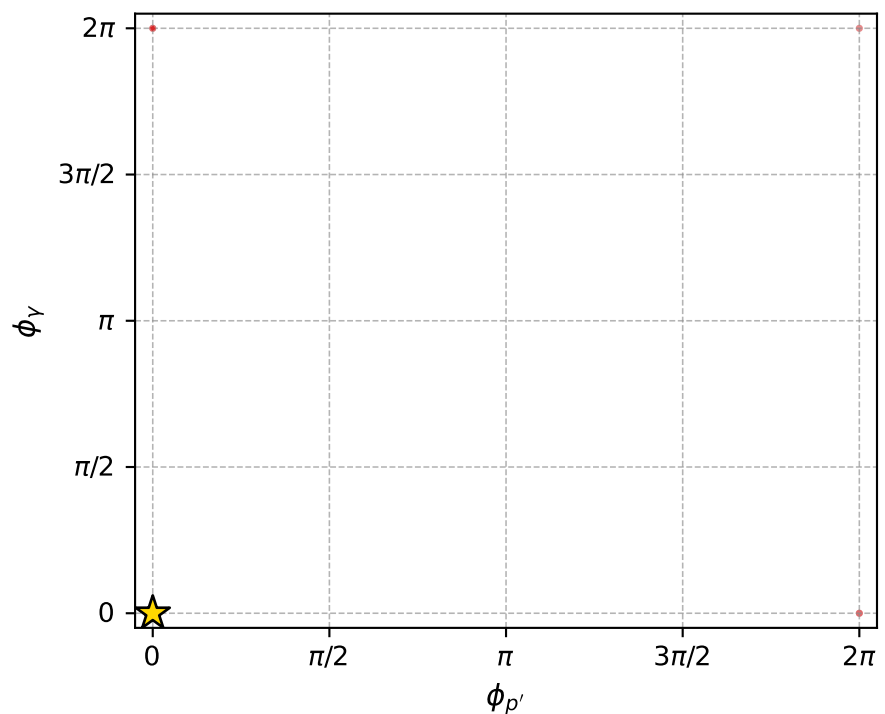
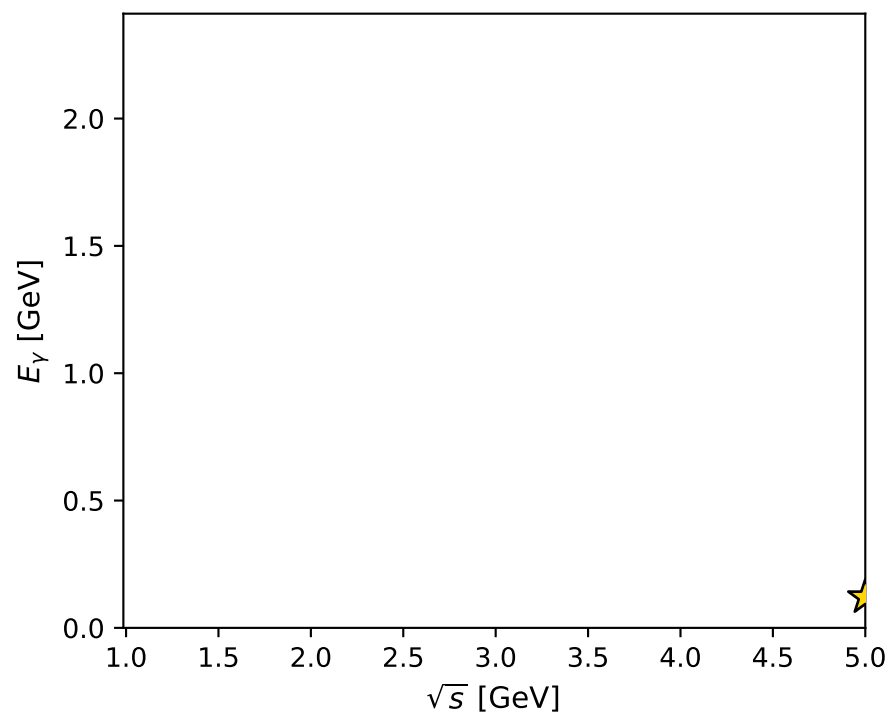
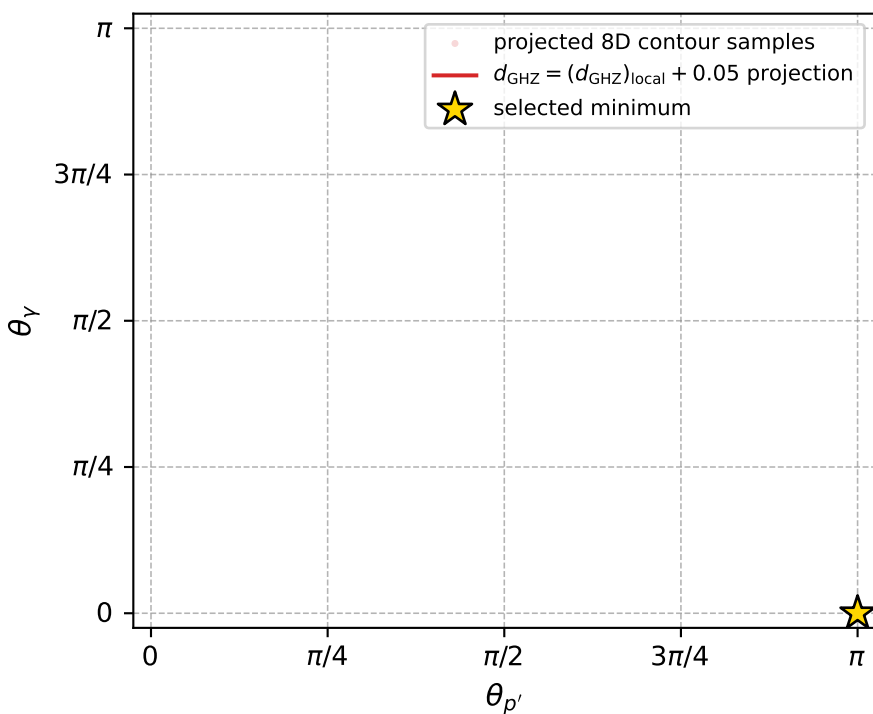
dghz_mixing_angles_polarization_03_local_minimum_89 d_{\rm GHZ}=3.21673e-08
region: polarization_cluster_03_local_minimum_89
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.9834494$ rad
incoming lepton: $-0.401041|+\rangle +0.91606|-\rangle$
proton mixing angle: $\alpha_p=0.41264296$ rad
incoming proton: $0.916064|+\rangle +0.401032|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0$, $\phi_Y=0$
 $E_Y=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$



electron: polarization cluster P3, local minimum 89; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.2167311\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000032$
 above global minimum = $9.8838496\text{e-}09$
 8D contour samples = 25

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 0.120596$
 $\text{phi_p_out} = 0$
 $\text{phi_gamma_out} = 0$
 $\text{alpha_e} = 1.983449$
 $\text{alpha_p} = 0.412643$

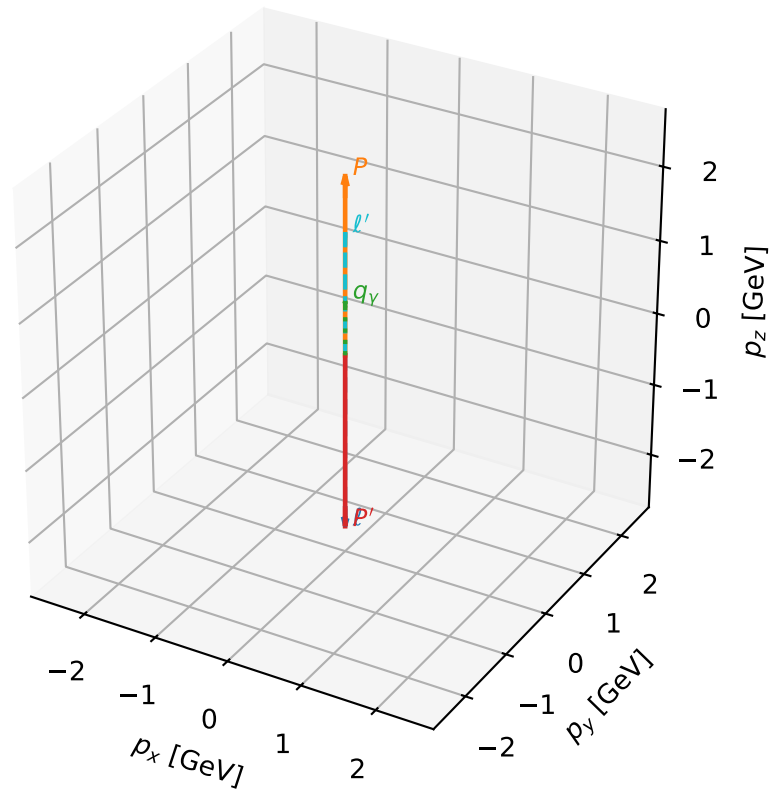
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_92 $d_{\{\text{rm GHZ}\}}=3.31027\text{e-}08$
 region: polarization_cluster_03_local_minimum_92
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4281603$ rad
 incoming lepton: $-0.75612|+\rangle +0.654433|-\rangle$
 proton mixing angle: $\alpha_p=0.85738407$ rad
 incoming proton: $0.654418|+\rangle +0.756133|-\rangle$

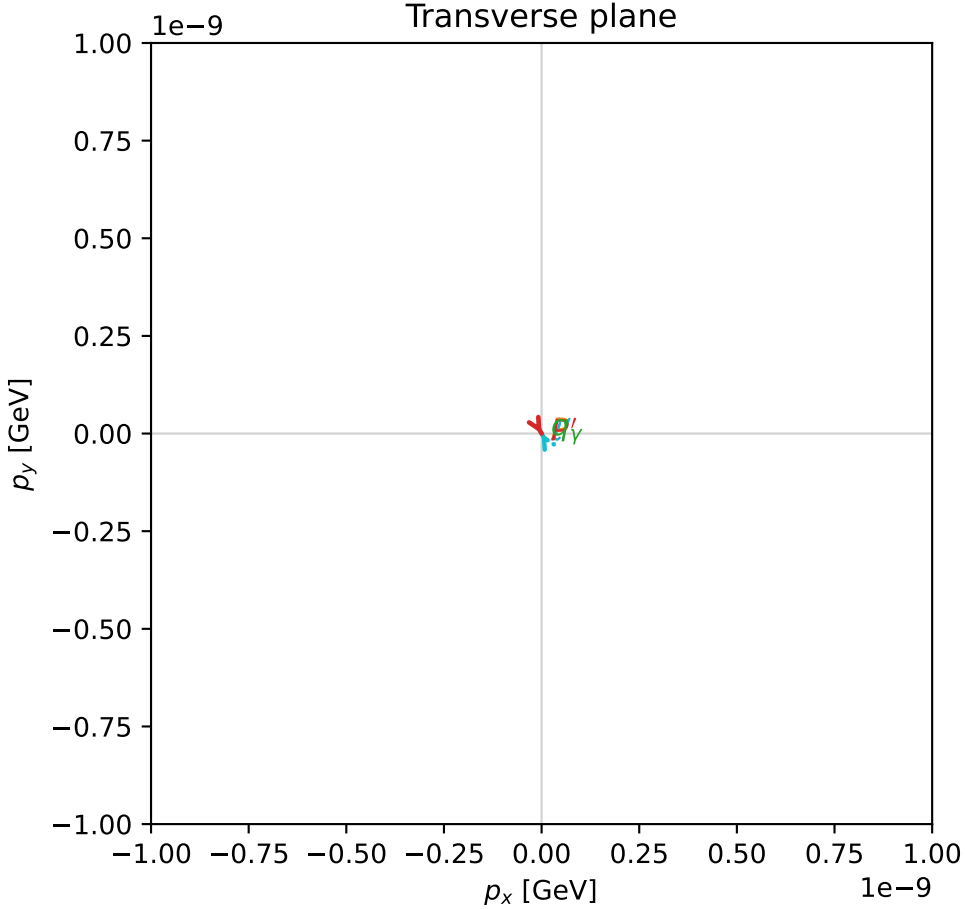
$s=24.8013$, $\sqrt{s}=4.98009$, $|\vec{P}|=2.40171$, $|\vec{P}'|=2.40171$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=5.23216$, $\phi_Y=5.8557$
 $E_Y=0.740299$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.09057$
 $Q^2=15.9609$, $x_B=0.684007$, $t=-23.0728$, $W^2=8.25336$, $y=0.975459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4017$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4017$
 P $E= 2.5784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4017$
 ℓ' $E= 1.6614$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6614$
 P' $E= 2.5784$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4017$
 q_Y $E= 0.7403$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7403$

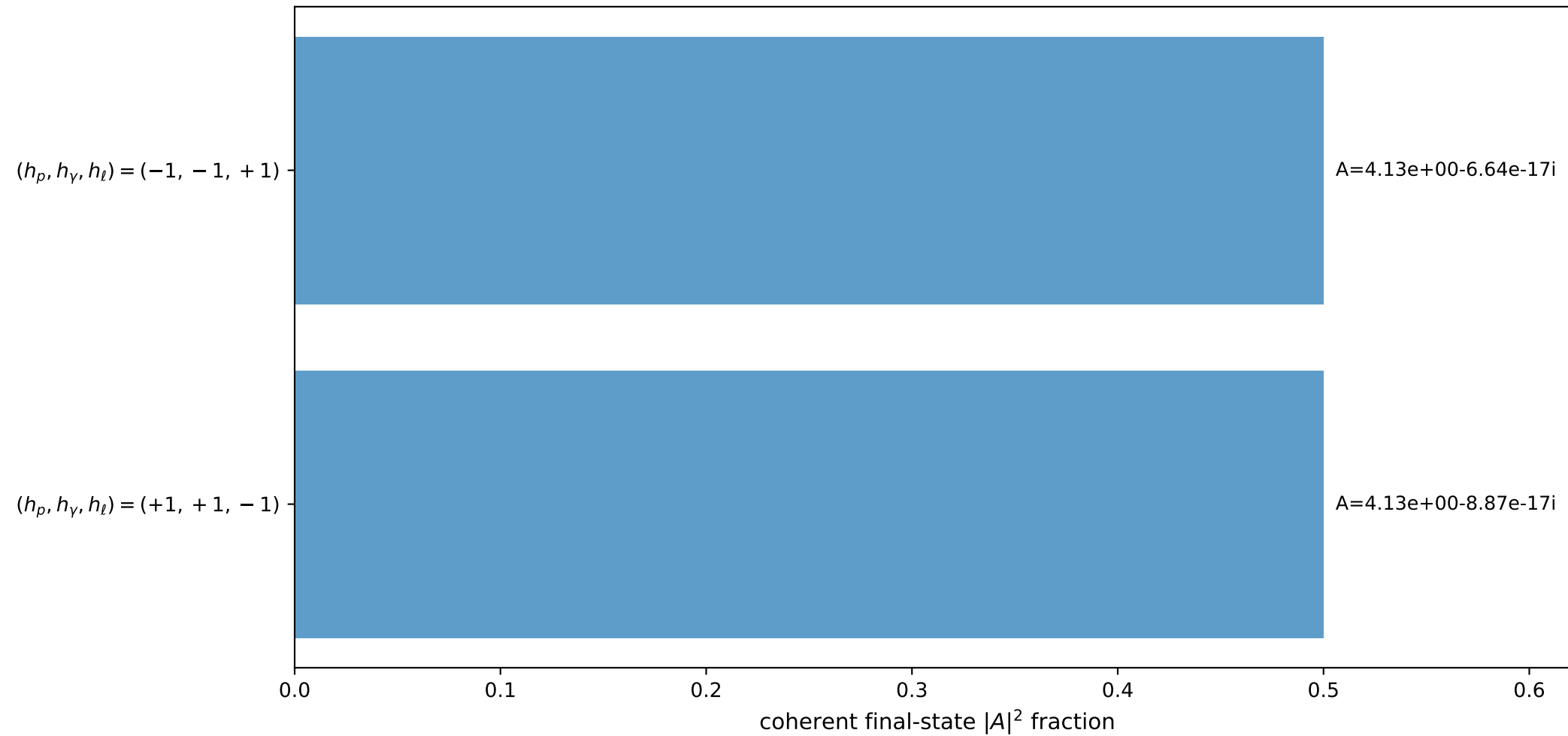
3D momenta



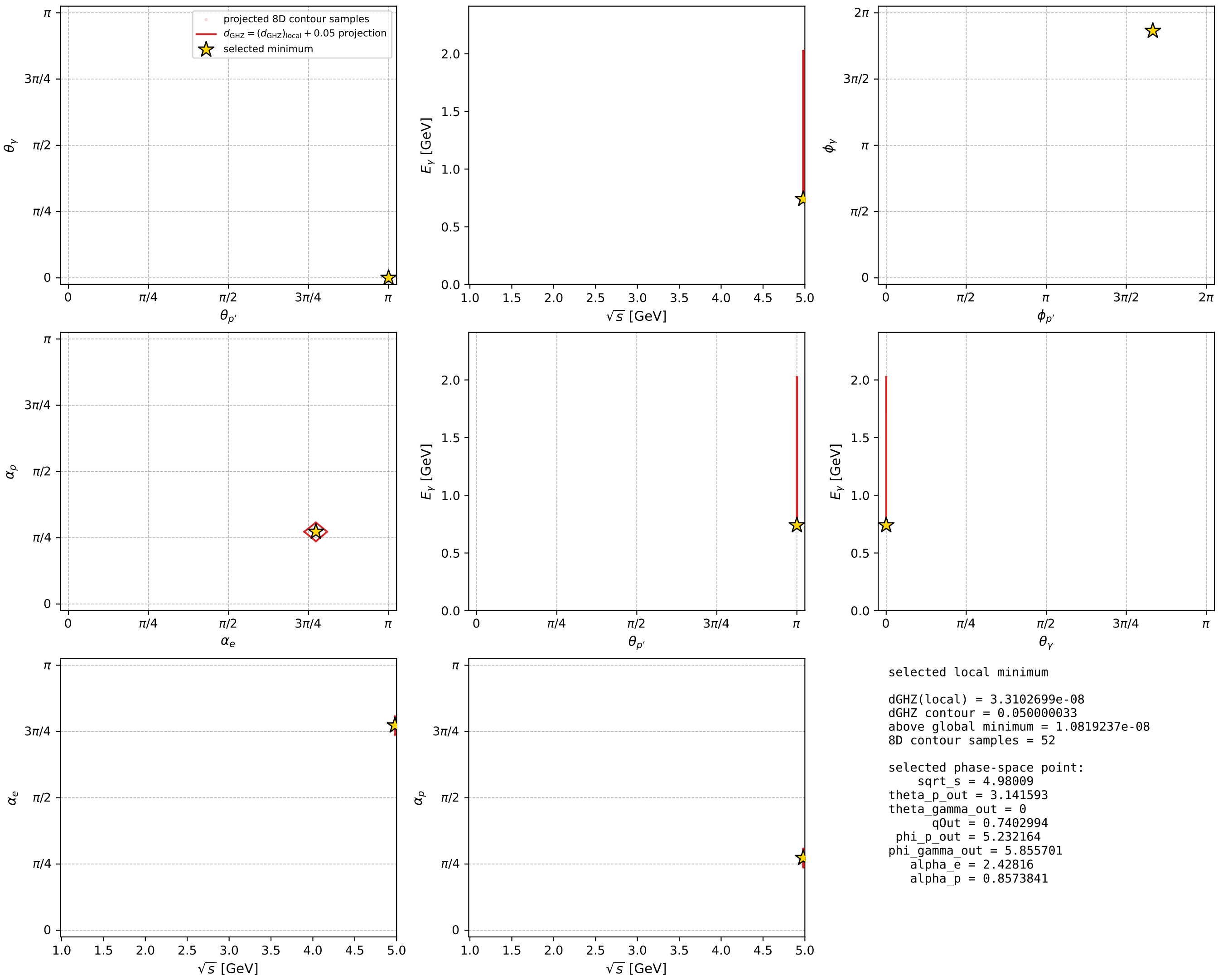
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 92; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

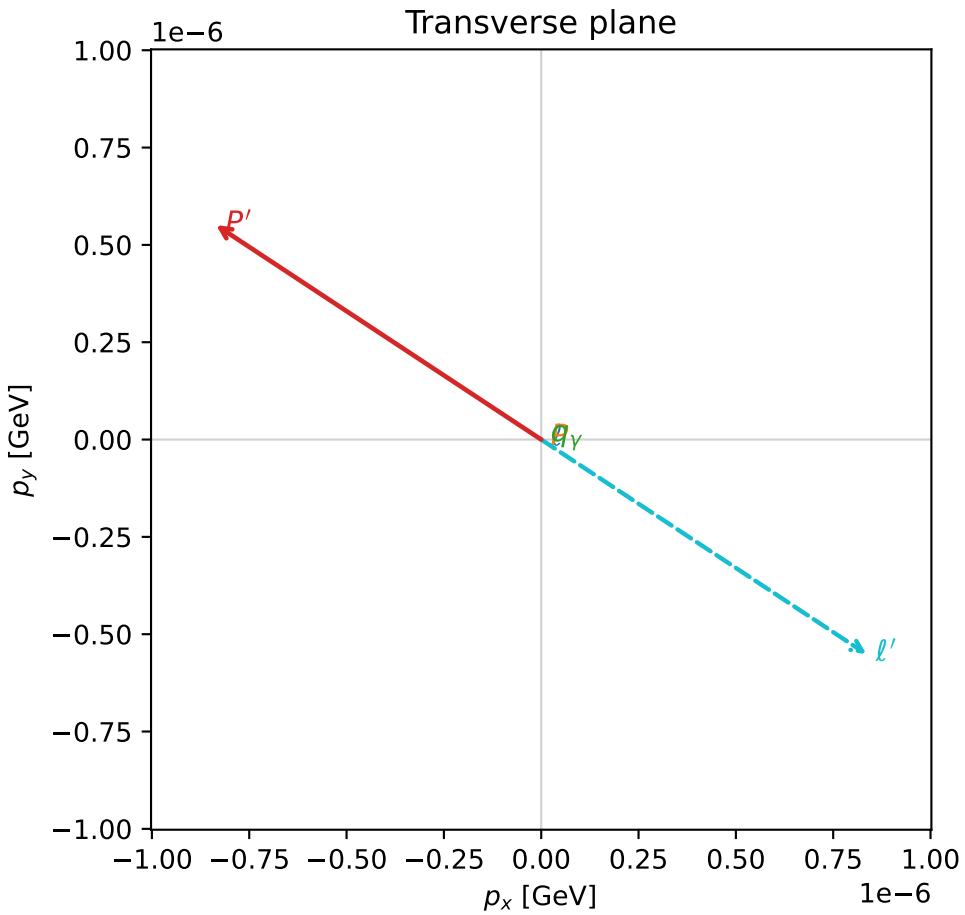
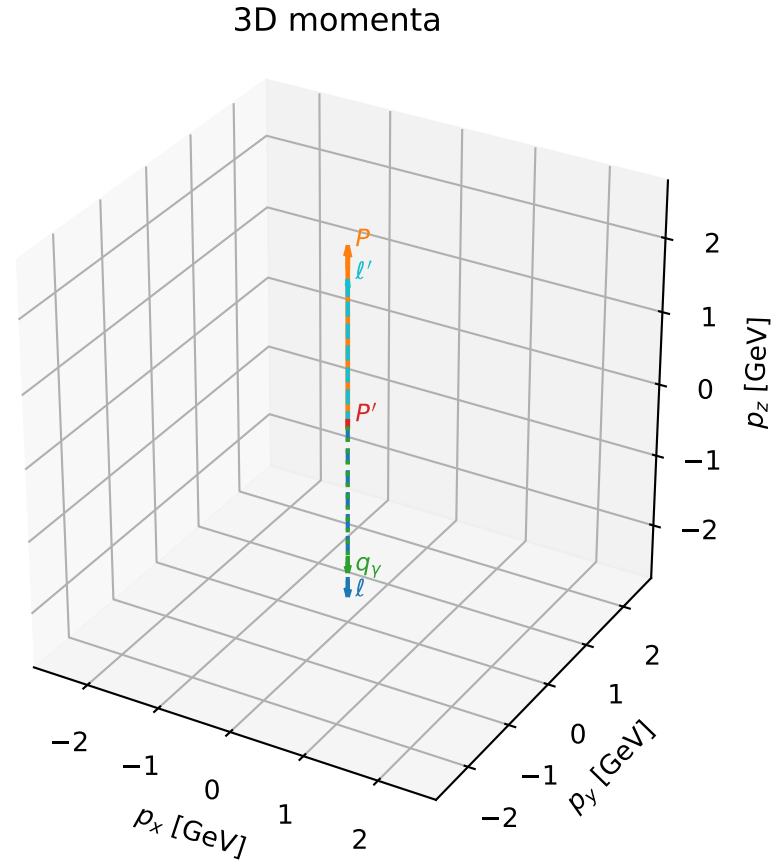


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_93 $d_{\{\text{rm GHZ}\}}=3.31851\text{e-}08$
 region: polarization_cluster_03_local_minimum_93
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.6709187$ rad
 incoming lepton: $-0.891263|+\rangle +0.453487|-\rangle$
 proton mixing angle: $\alpha_p=0.47067803$ rad
 incoming proton: $0.891261|+\rangle +0.453491|-\rangle$

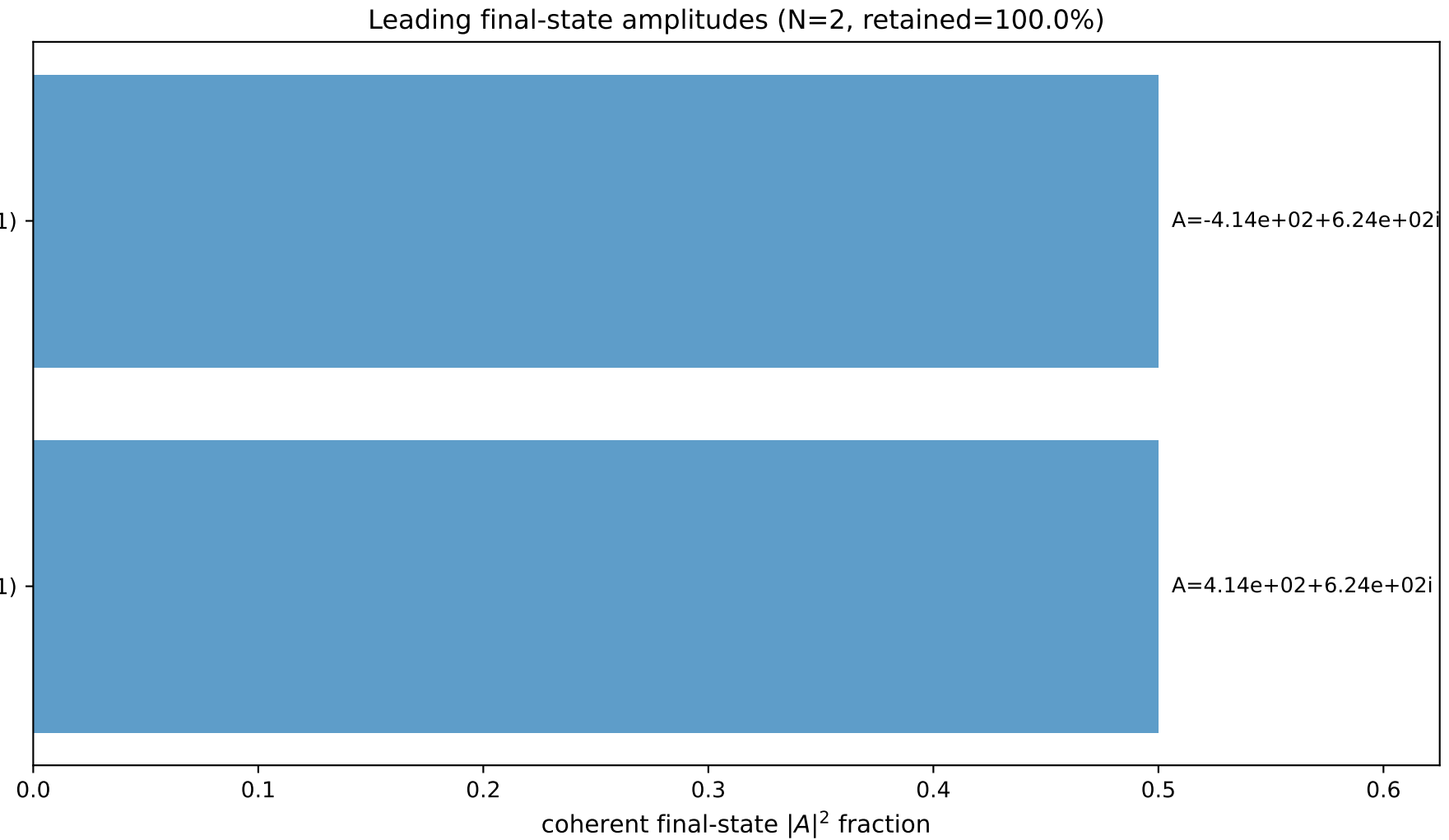
$s=24.8907$, $\sqrt{s}=4.98906$, $|\vec{P}|=2.40635$, $|\vec{P}'|=0.0697271$
 $\theta_{p'}=1.43494\text{e-}05$, $\theta_Y=3.14159$, $\phi_{p'}=2.55848$, $\phi_Y=4.12637$
 $E_Y=2.0591$, $\theta_{l'}=5.029\text{e-}07$, $\phi_{l'}=5.70007$
 $Q^2=19.1485$, $x_B=0.821501$, $t=-2.76326$, $W^2=5.04052$, $y=0.970777$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 P $E= 2.5827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4064$
 l' $E= 1.9894$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9894$
 P' $E= 0.9406$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0697$
 q_Y $E= 2.0591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0591$

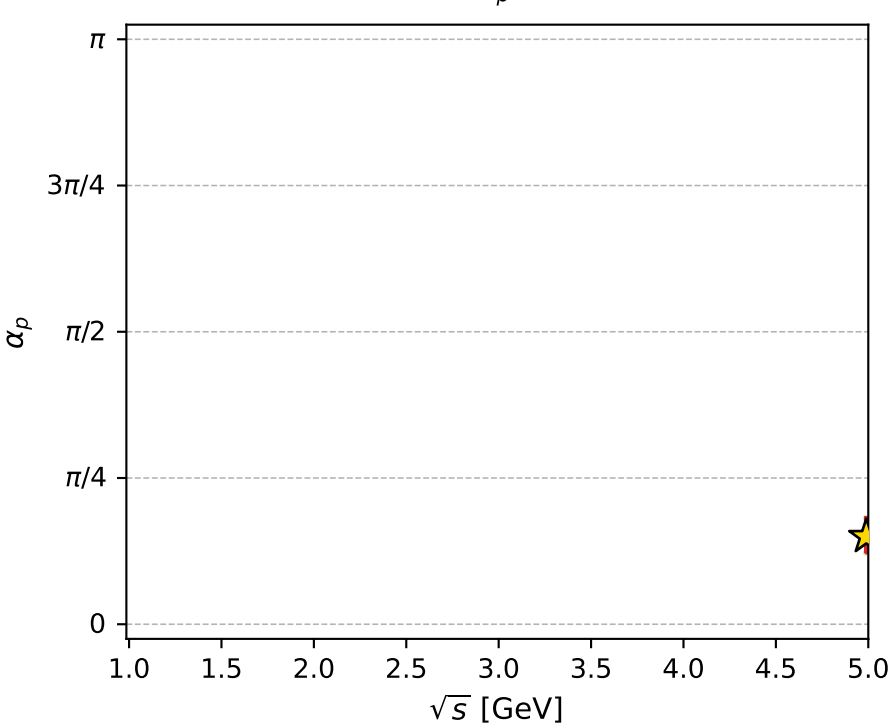
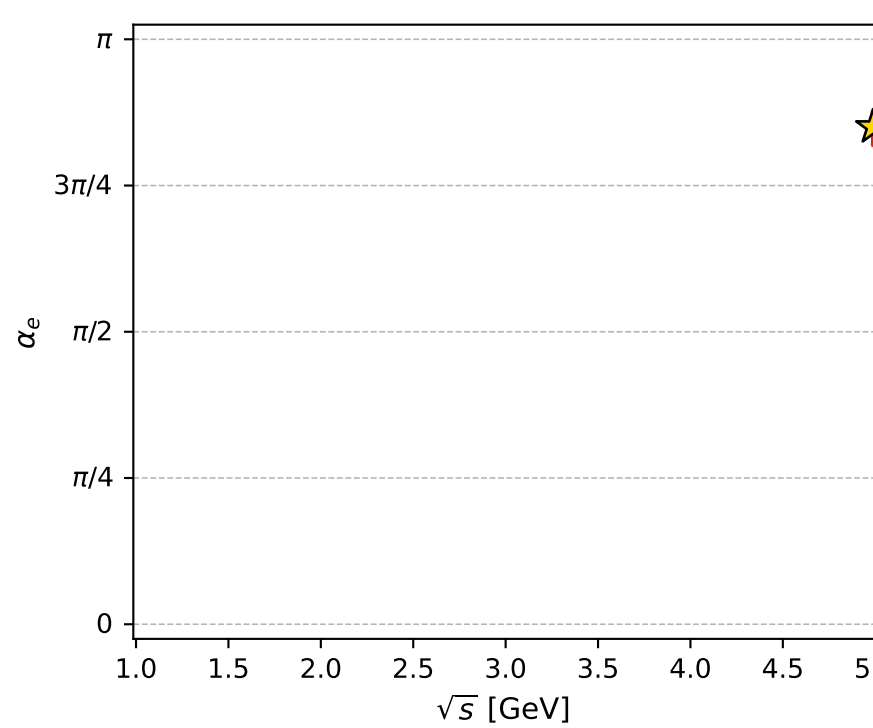
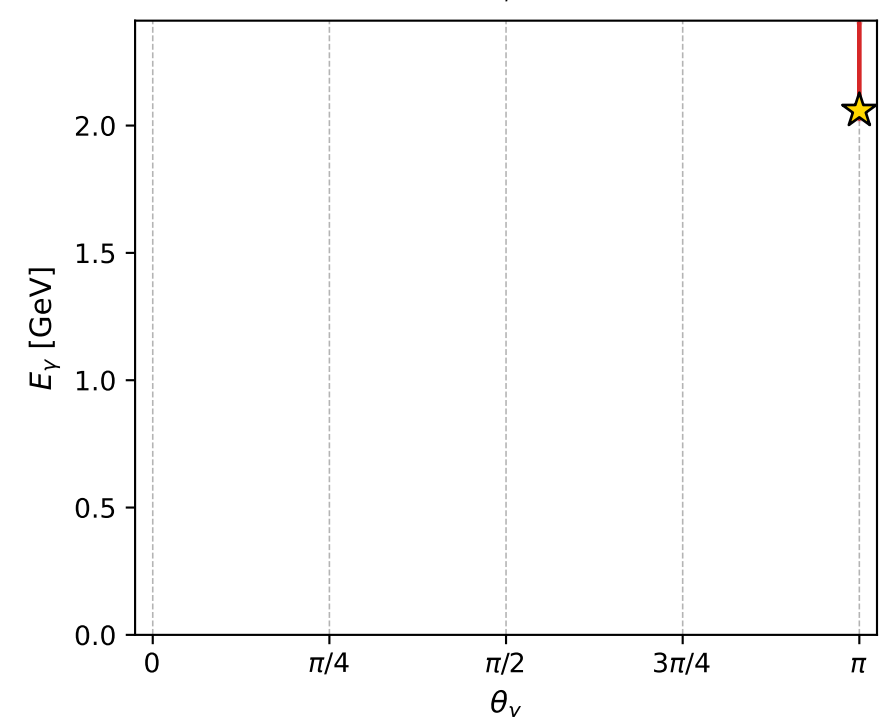
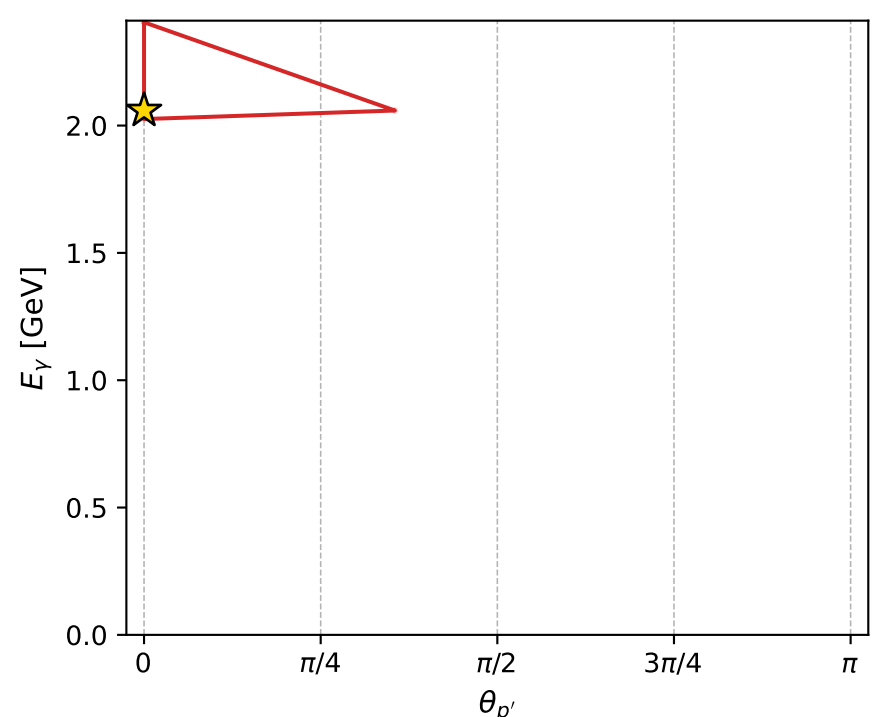
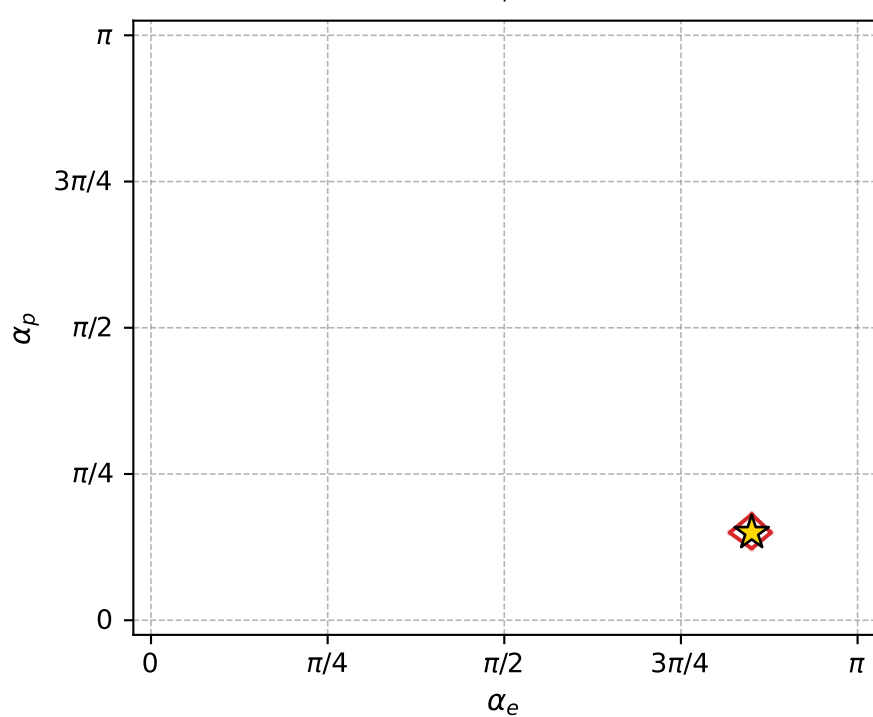
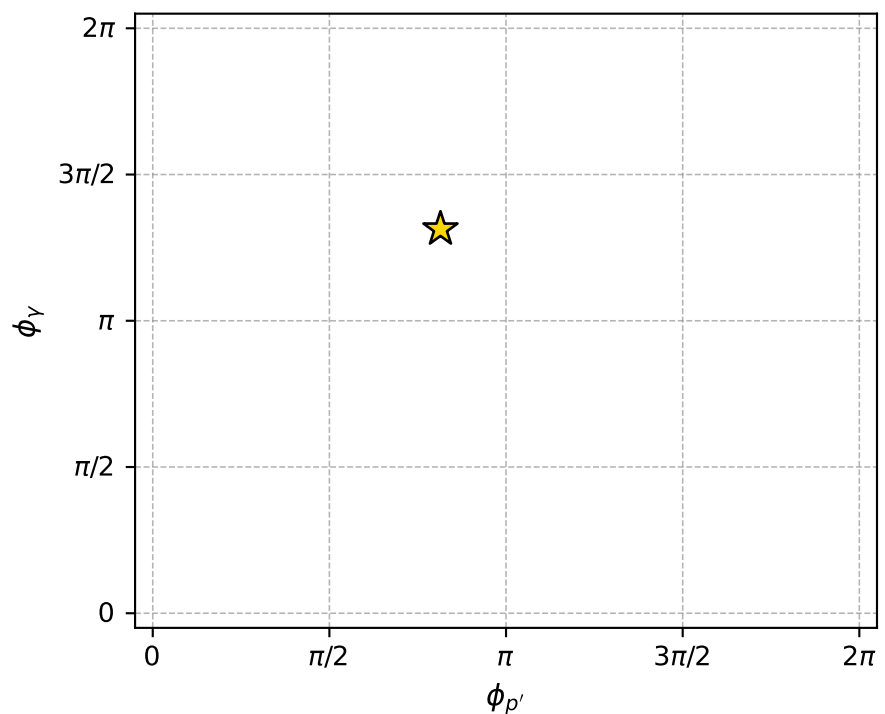
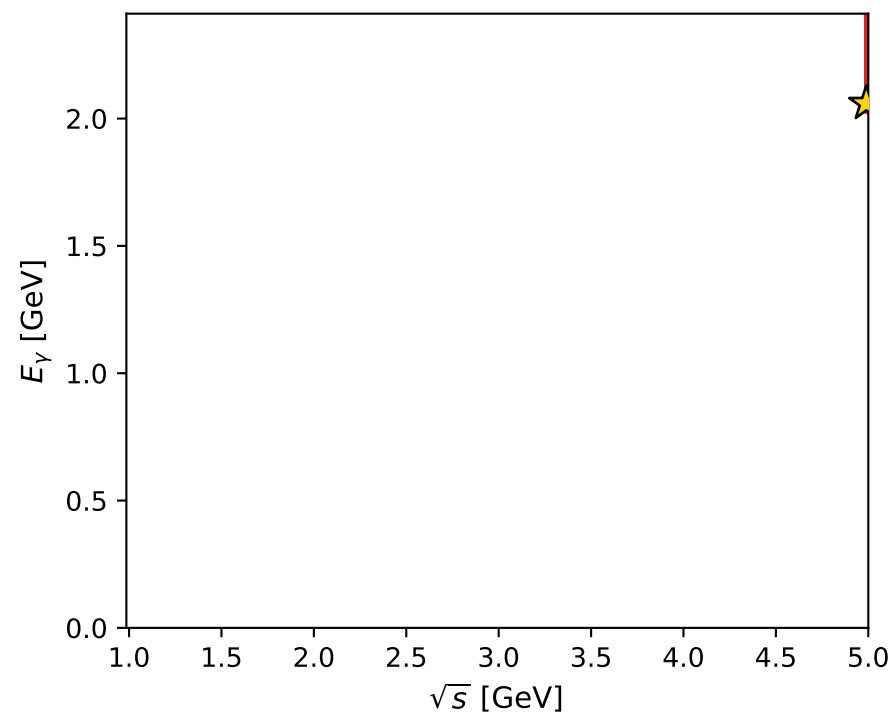
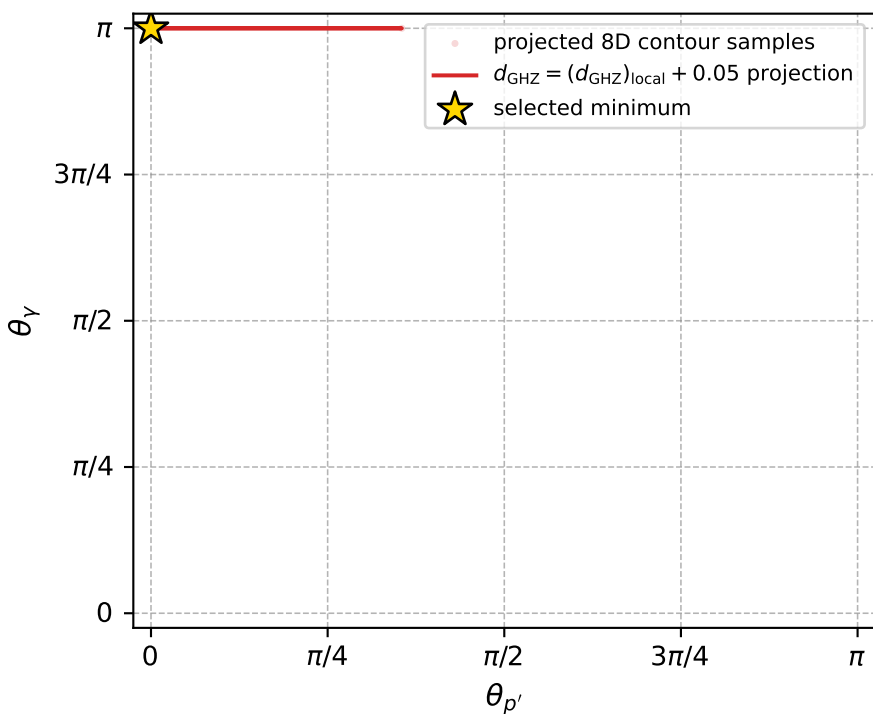


$(h_p, h_Y, h_l) = (+1, +1, -1)$

$(h_p, h_Y, h_l) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 93; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.3185073\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000033$
 above global minimum = $1.0901611\text{e-}08$
 8D contour samples = 117

selected phase-space point:

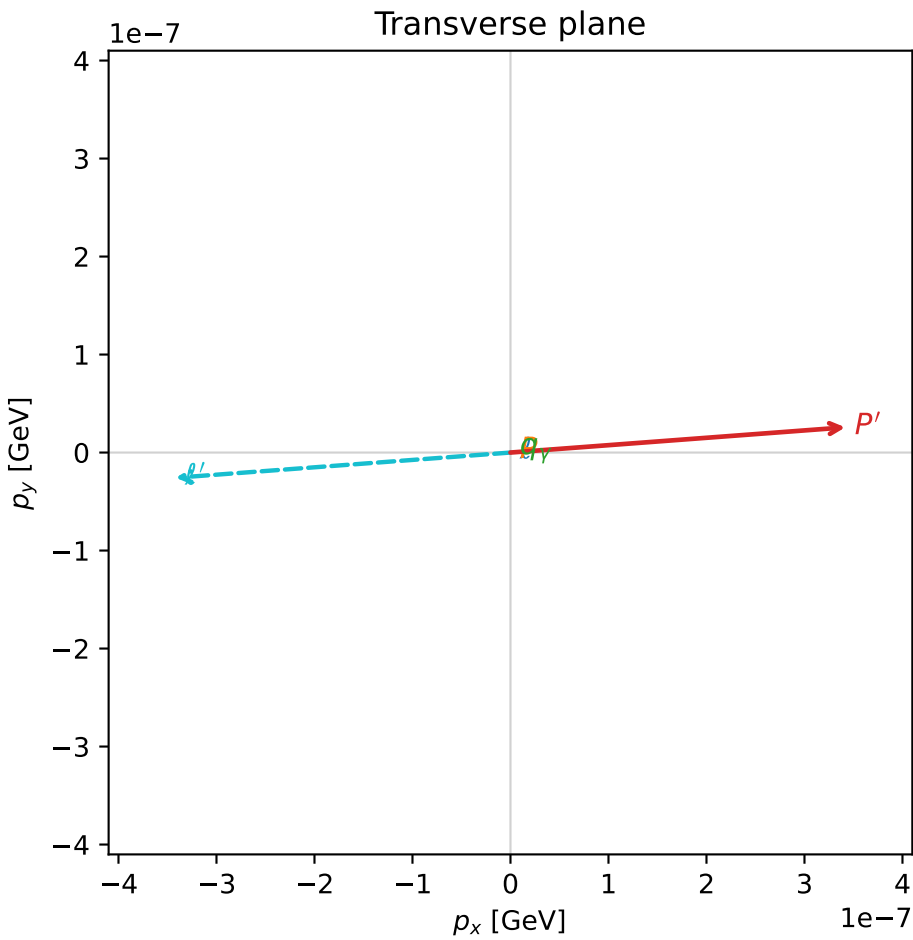
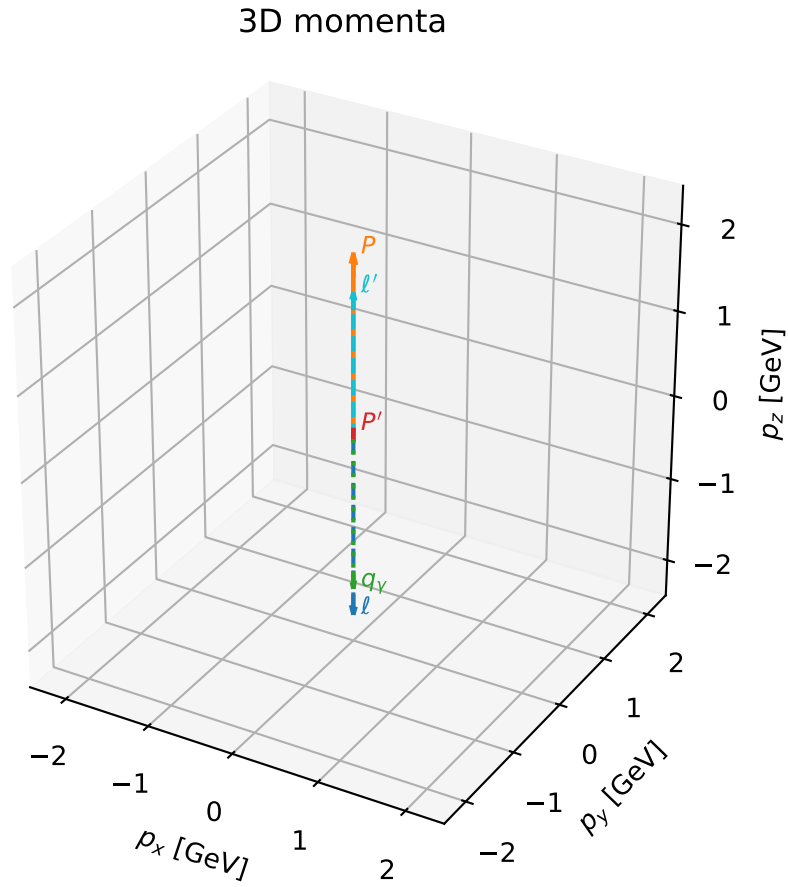
$\text{sqrt_s} = 4.989061$
 $\text{theta_p_out} = 1.434945\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.0591$
 $\text{phi_p_out} = 2.558476$
 $\text{phi_gamma_out} = 4.126371$
 $\text{alpha_e} = 2.670919$
 $\text{alpha_p} = 0.470678$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_97 $d_{\{\text{rm GHZ}\}}=3.45768\text{e-}08$
region: polarization_cluster_03_local_minimum_97
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.363027$ rad
incoming lepton: $-0.711922|+\rangle +0.702259|-\rangle$
proton mixing angle: $\alpha_p=0.77853959$ rad
incoming proton: $0.71194|+\rangle +0.70224|-\rangle$

$s=19.711$, $\sqrt{s}=4.4397$, $|\vec{P}|=2.12076$, $|\vec{P}'|=0.105882$
 $\theta_{P'}=3.23647\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=0.0751304$, $\phi_Y=1.34534$
 $E_Y=1.80081$, $\theta_{\ell'}=2.02129\text{e-}07$, $\phi_{\ell'}=3.21672$
 $Q^2=14.3782$, $x_B=0.79178$, $t=-2.16917$, $W^2=4.66097$, $y=0.964325$

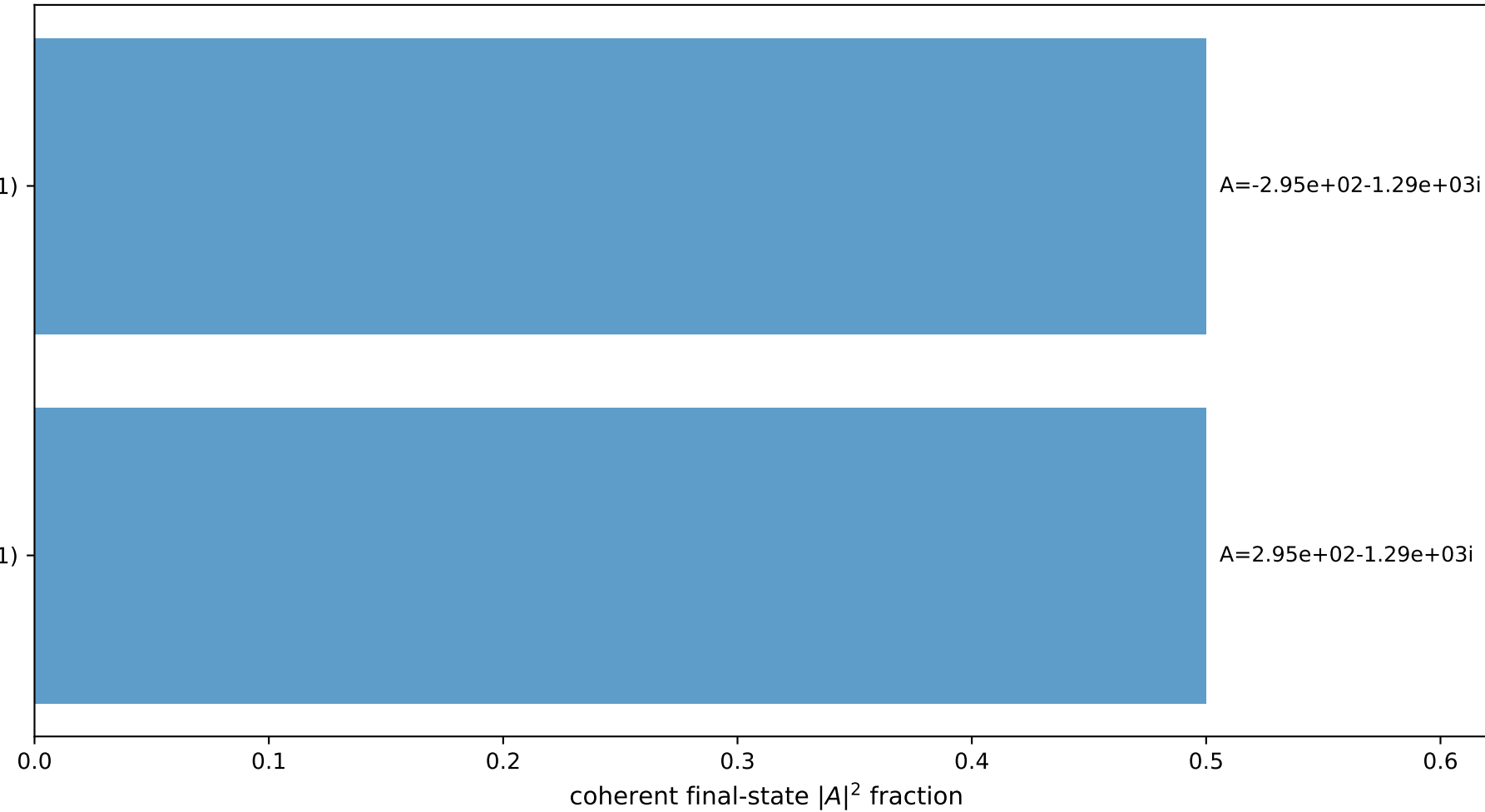
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1208$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1208$
 P $E= 2.3189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1208$
 ℓ' $E= 1.6949$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.6949$
 P' $E= 0.9440$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1059$
 q_Y $E= 1.8008$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8008$



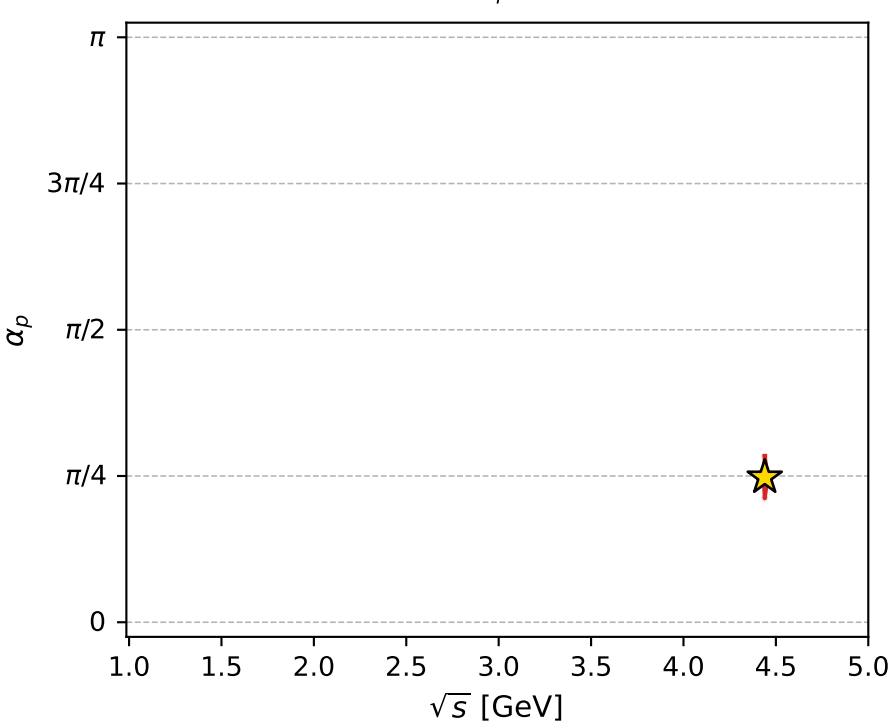
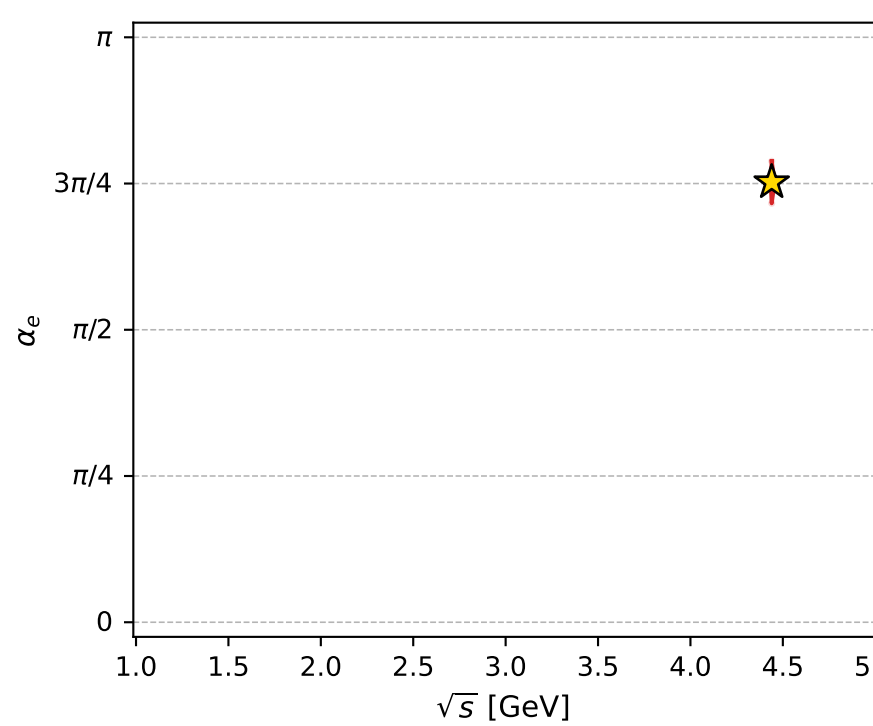
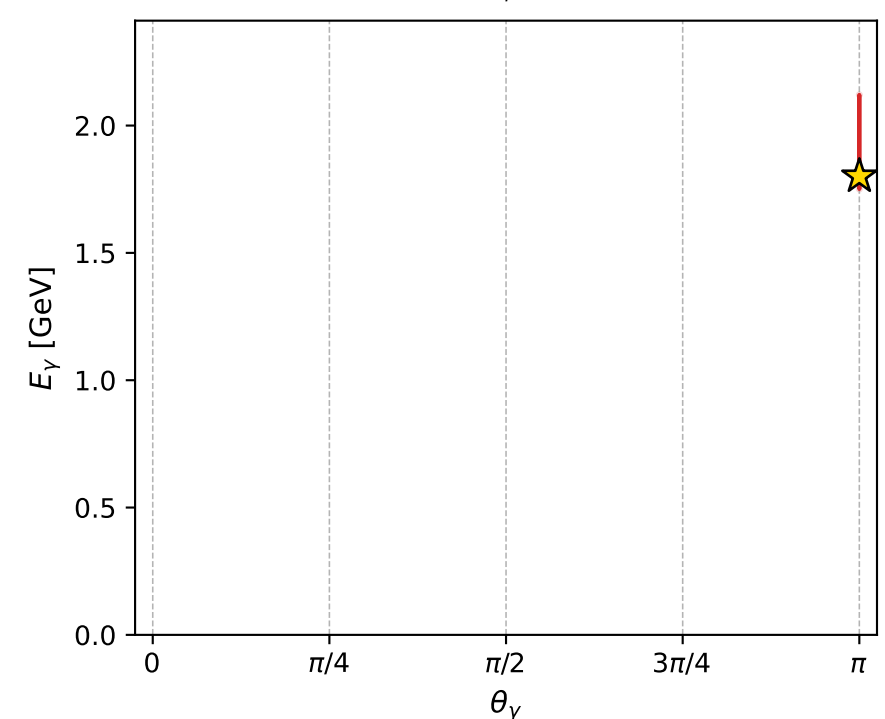
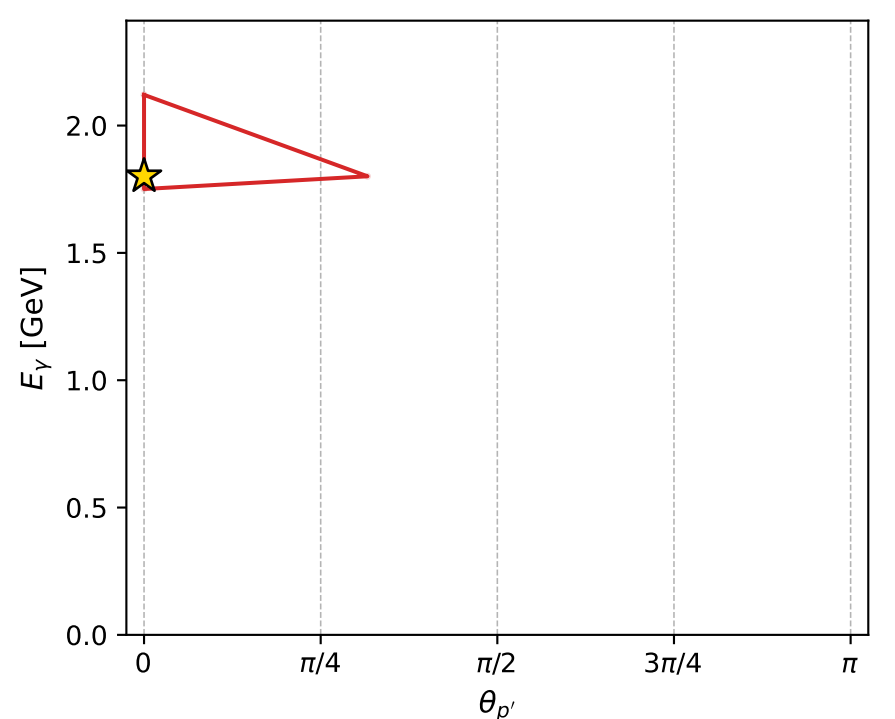
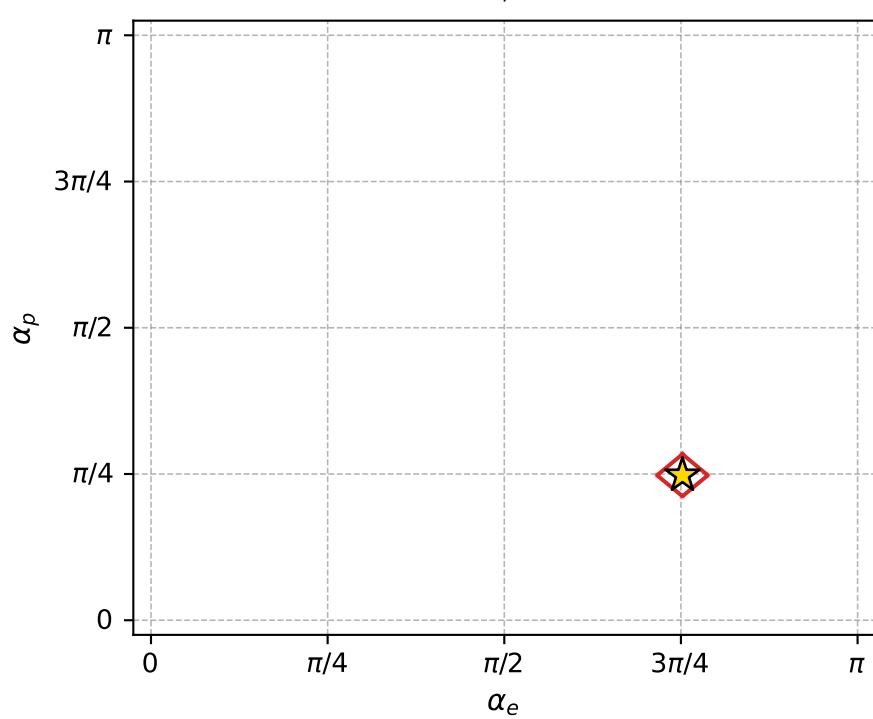
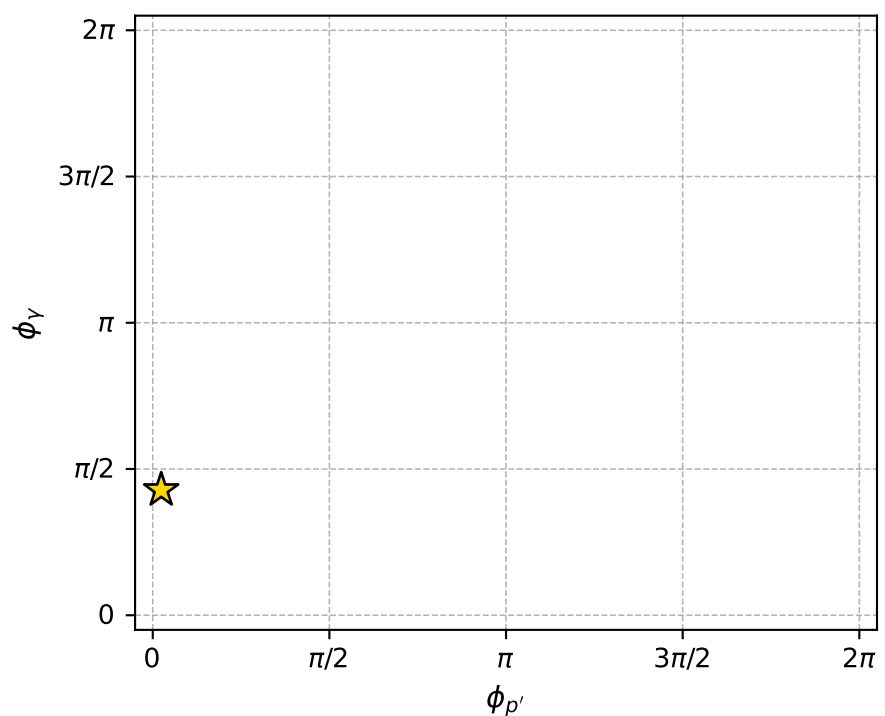
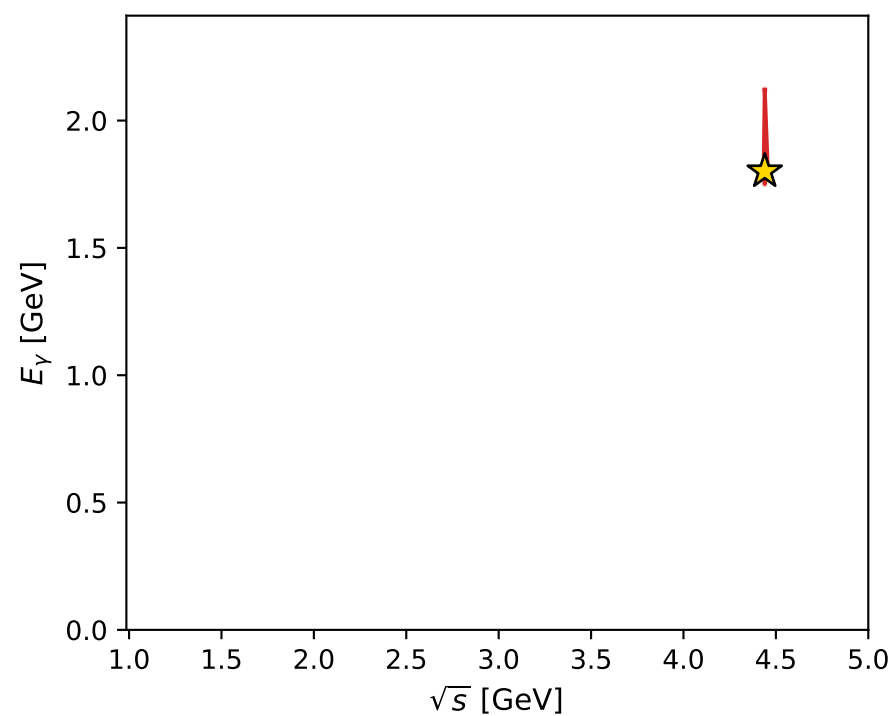
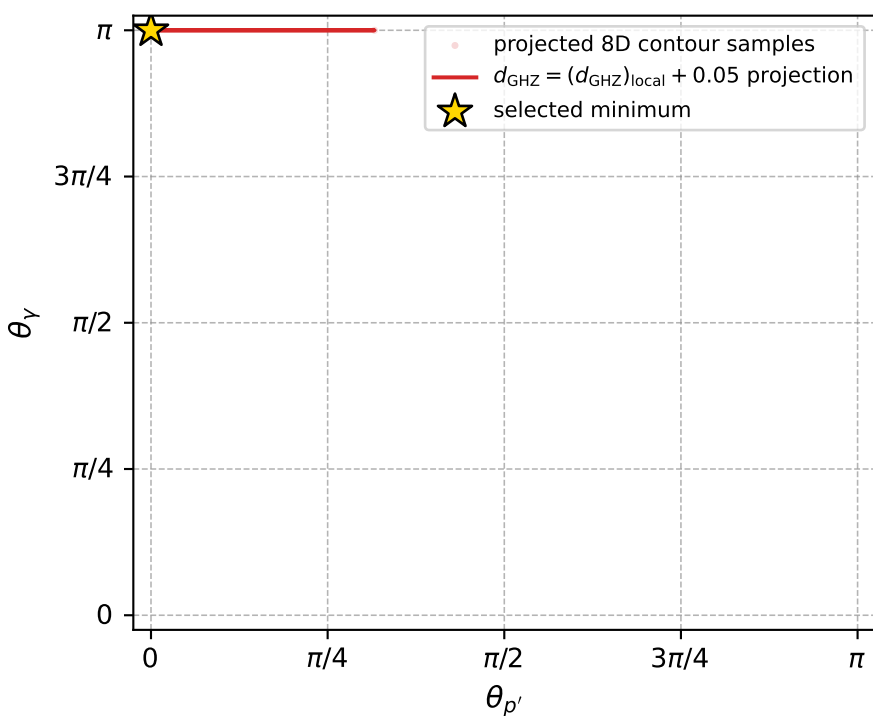
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 97; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.457676\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000035$
 above global minimum = $1.2293298\text{e-}08$
 8D contour samples = 66

selected phase-space point:

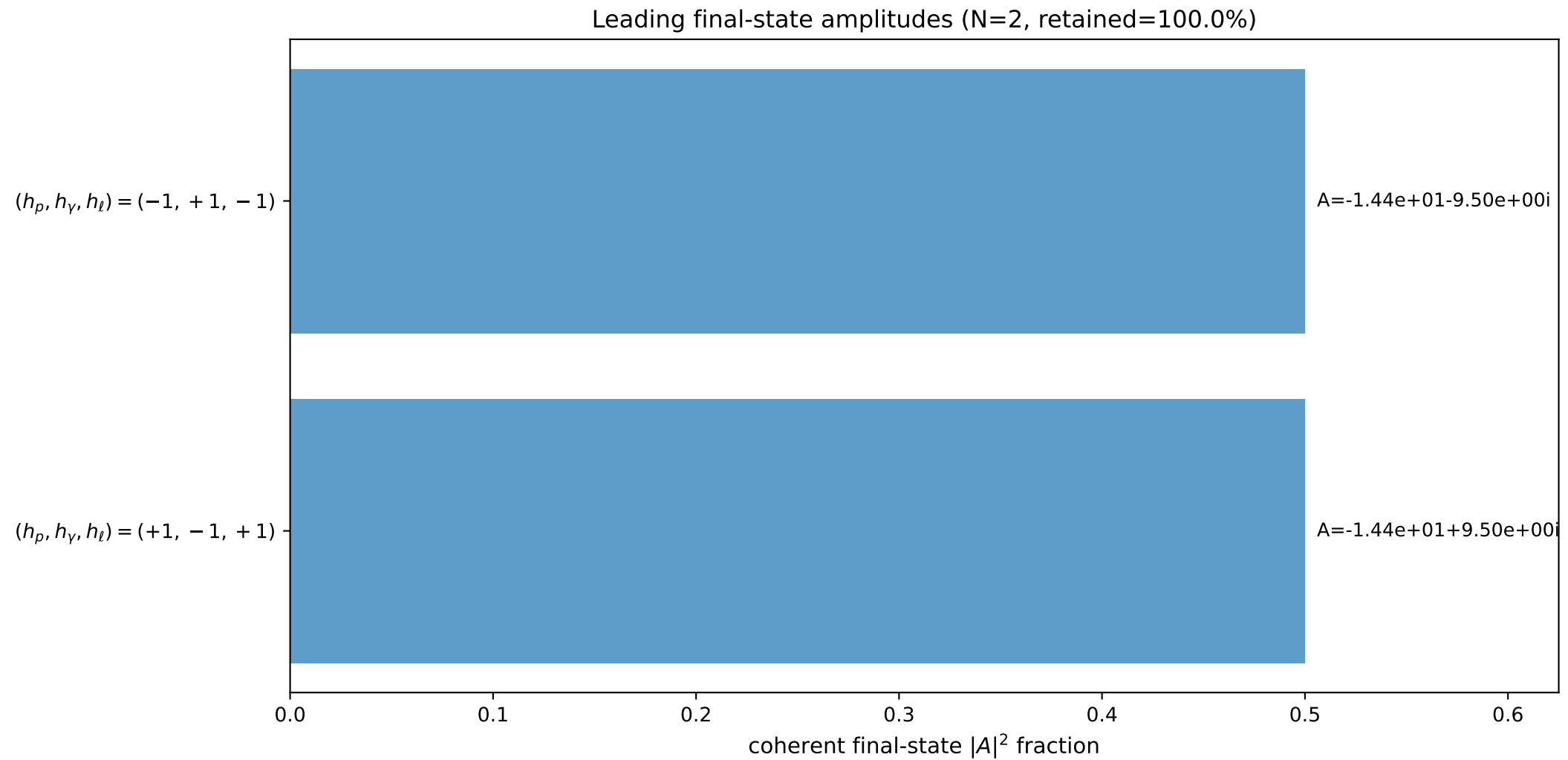
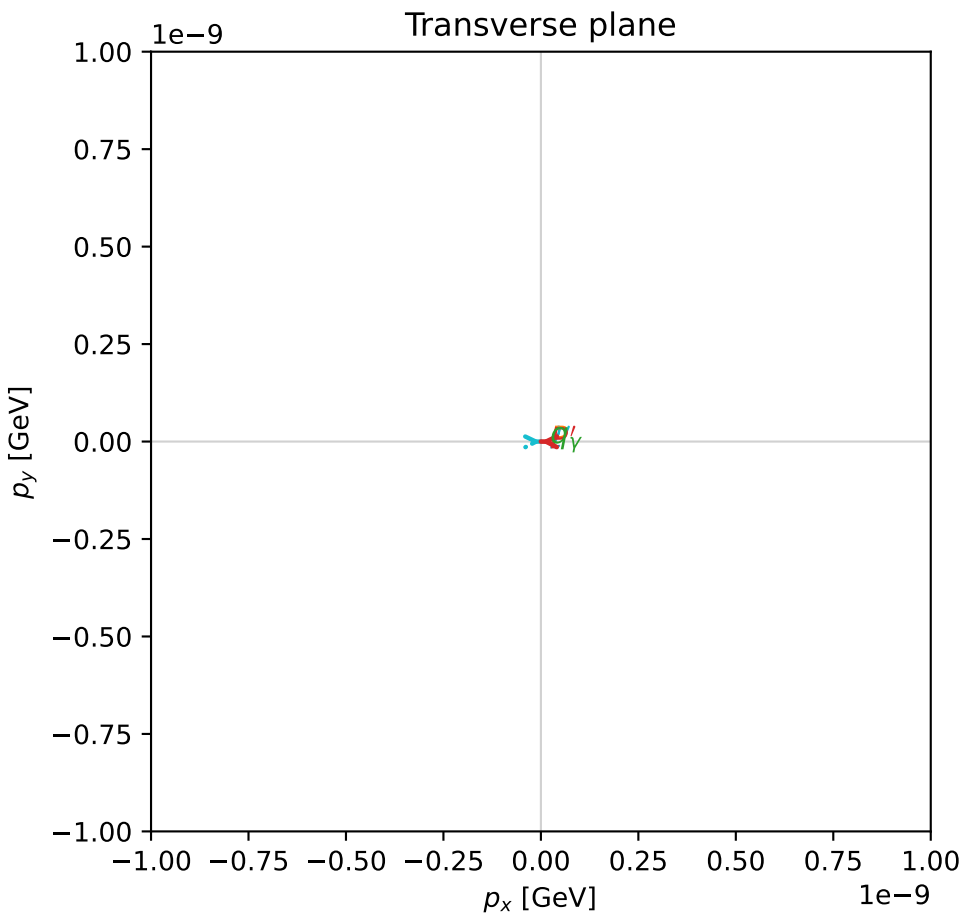
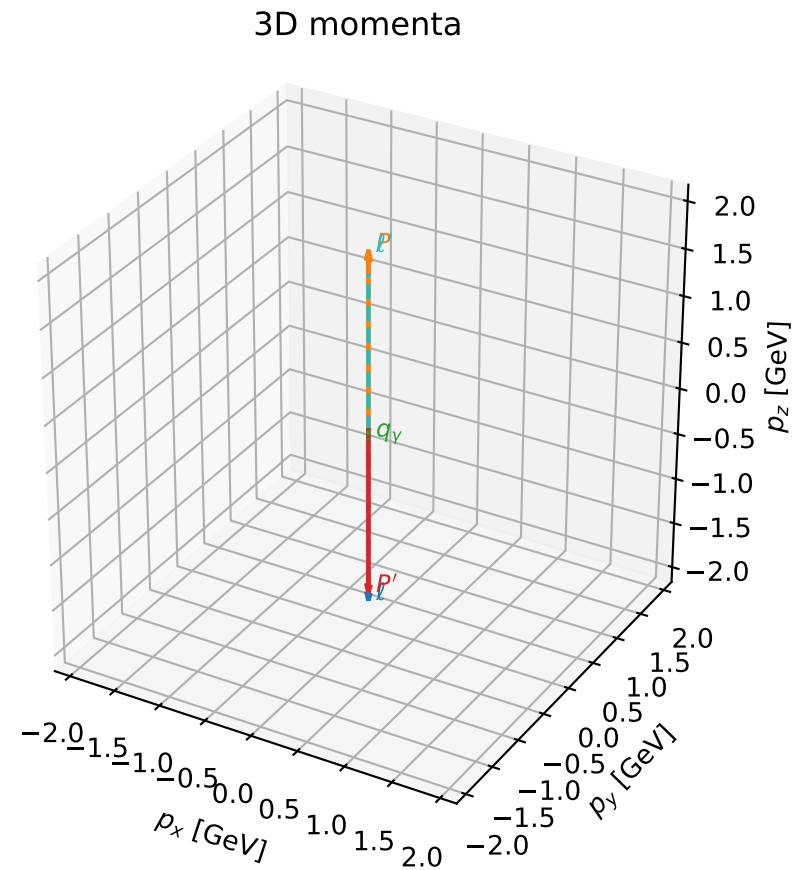
$\text{sqrt_s} = 4.439702$
 $\text{theta_p_out} = 3.236467\text{e-}06$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.800813$
 $\text{phi_p_out} = 0.07513043$
 $\text{phi_gamma_out} = 1.345341$
 $\text{alpha_e} = 2.363027$
 $\text{alpha_p} = 0.7785396$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

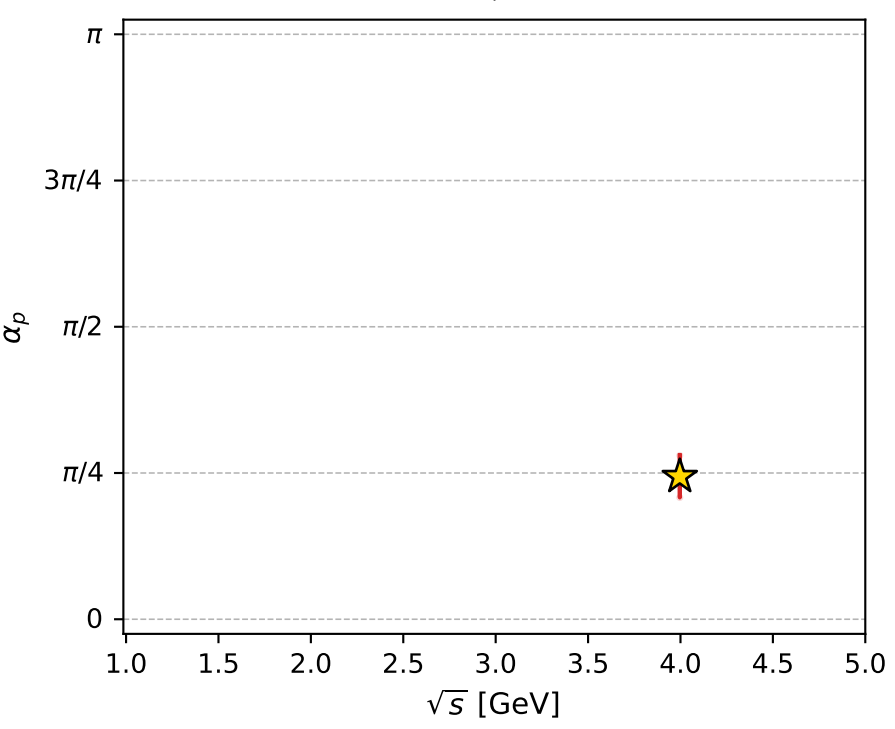
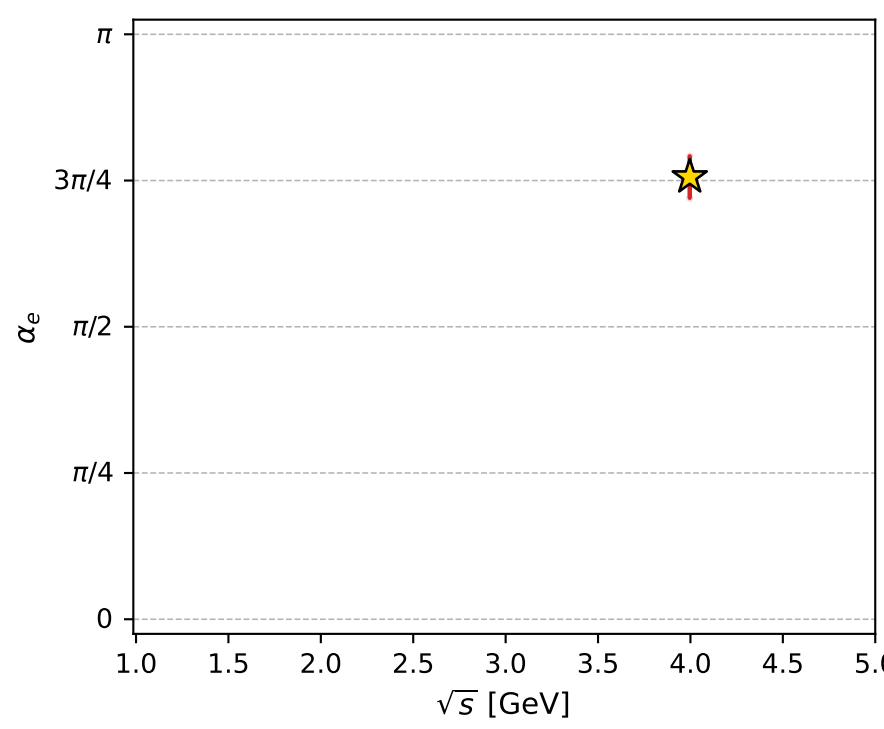
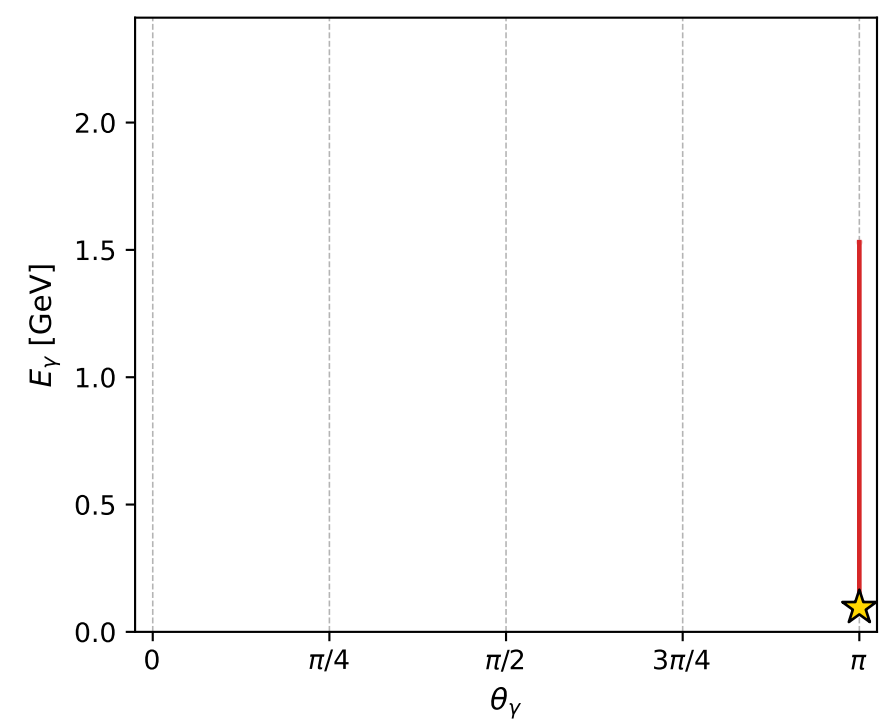
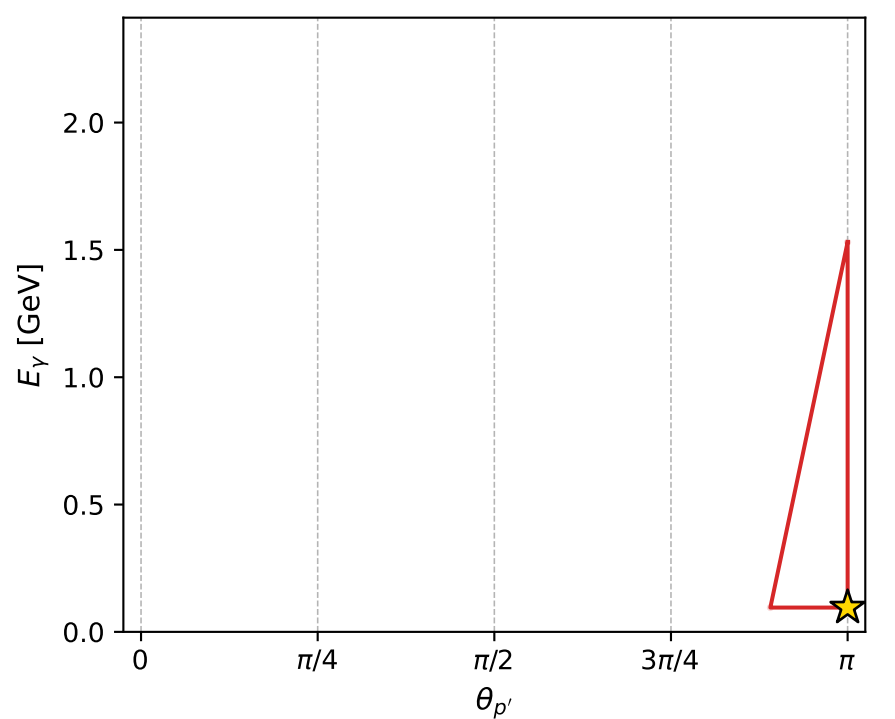
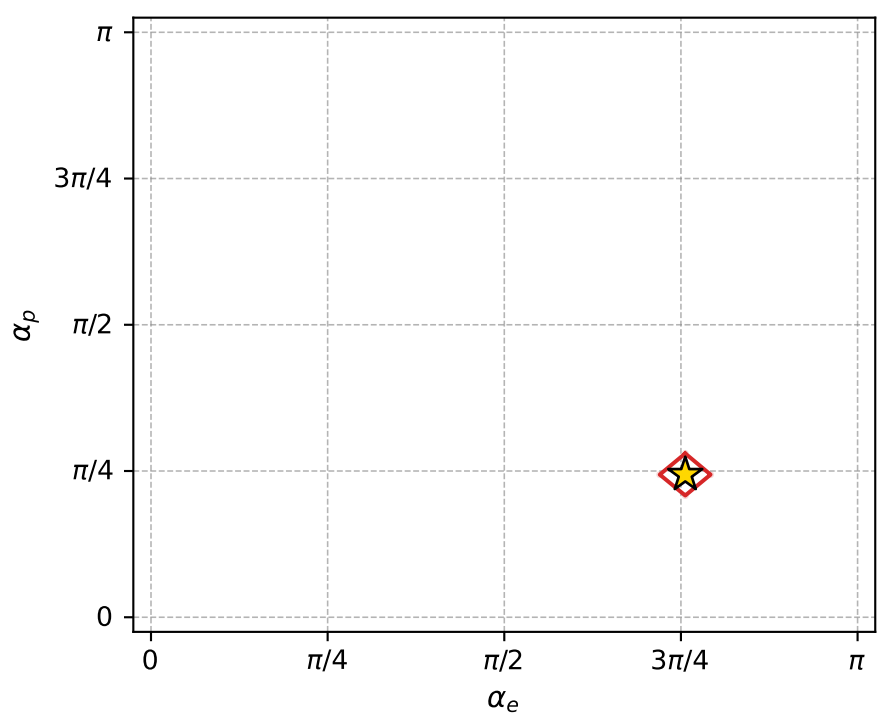
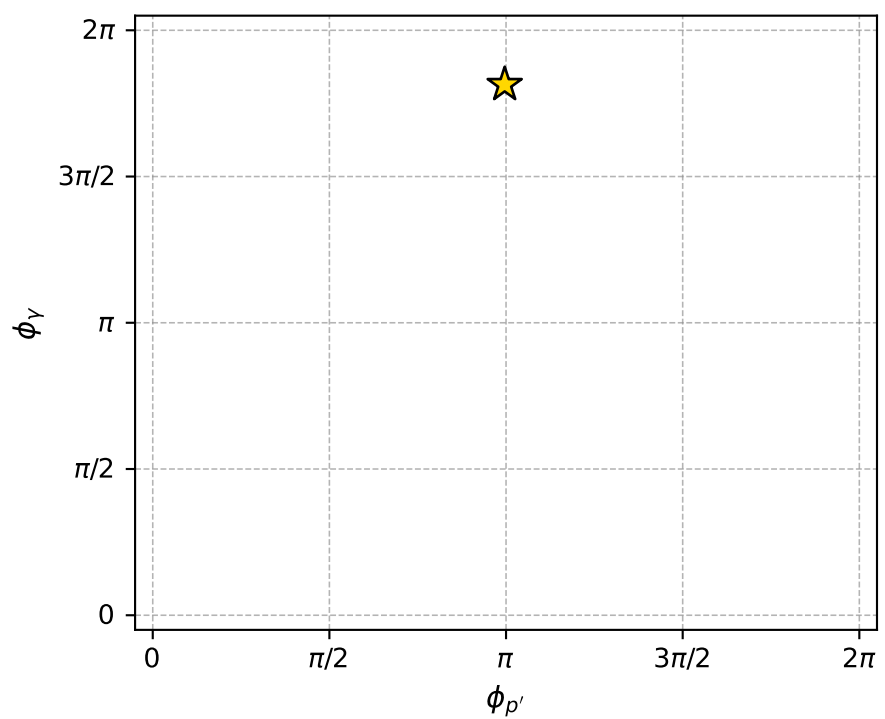
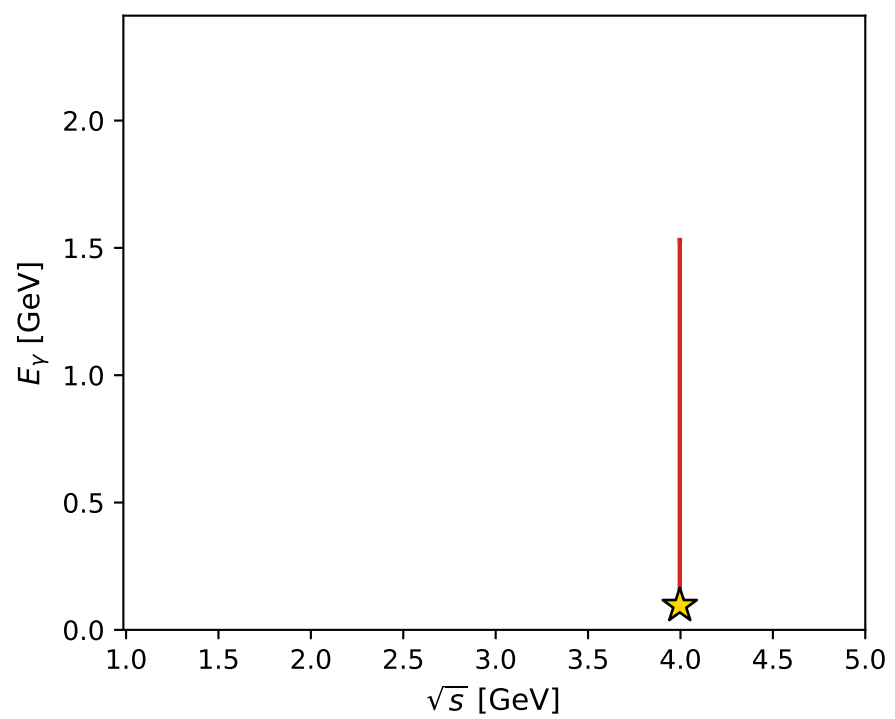
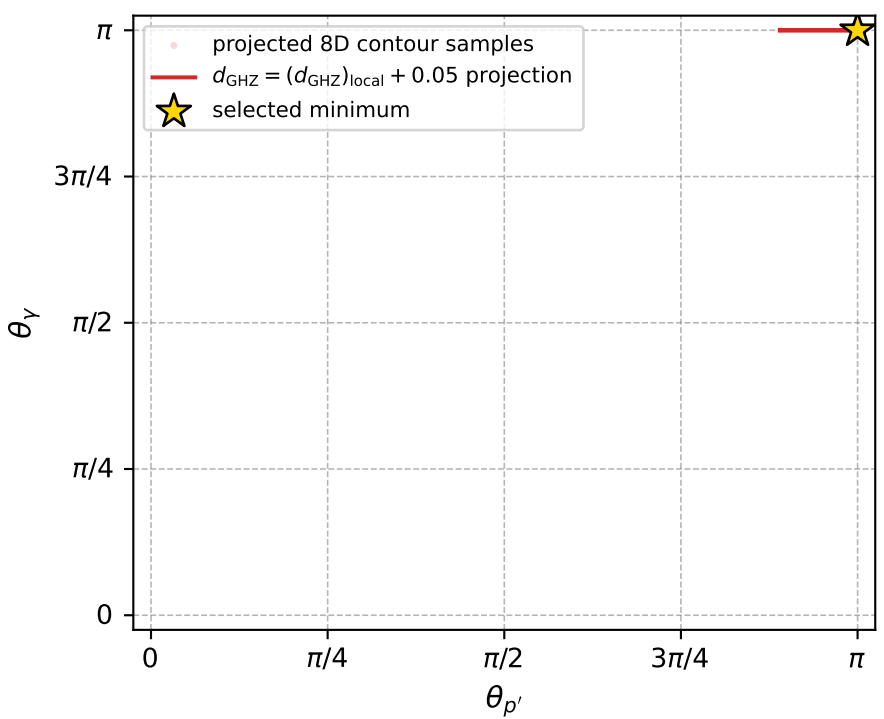
dghz_mixing_angles_polarization_03_local_minimum_102 $d_{\{\text{rm GHZ}\}}=3.6745\text{e-}08$
 region: polarization_cluster_03_local_minimum_102
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3754496$ rad
 incoming lepton: $-0.72059|+\rangle +0.693361|-\rangle$
 proton mixing angle: $\alpha_p=0.7661545$ rad
 incoming proton: $0.720582|+\rangle +0.693369|-\rangle$

$s=15.9699$, $\sqrt{s}=3.99624$, $|\vec{P}|=1.88803$, $|\vec{P}'|=1.78706$
 $\theta_{P'}=3.14159$, $\theta_V=3.14159$, $\phi_{P'}=3.12959$, $\phi_V=5.69987$
 $E_V=0.0954542$, $\theta_{l'}=0$, $\phi_{l'}=0.0182279$
 $Q^2=14.217$, $x_B=0.996905$, $t=-13.4982$, $W^2=0.923985$, $y=0.945067$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8880$
 P $E= 2.1082$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8880$
 l' $E= 1.8825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8825$
 P' $E= 2.0183$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.7871$
 q_V $E= 0.0955$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0955$



electron: polarization cluster P3, local minimum 102; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.6744986\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000037$
 above global minimum = $1.4461524\text{e-}08$
 8D contour samples = 64

selected phase-space point:

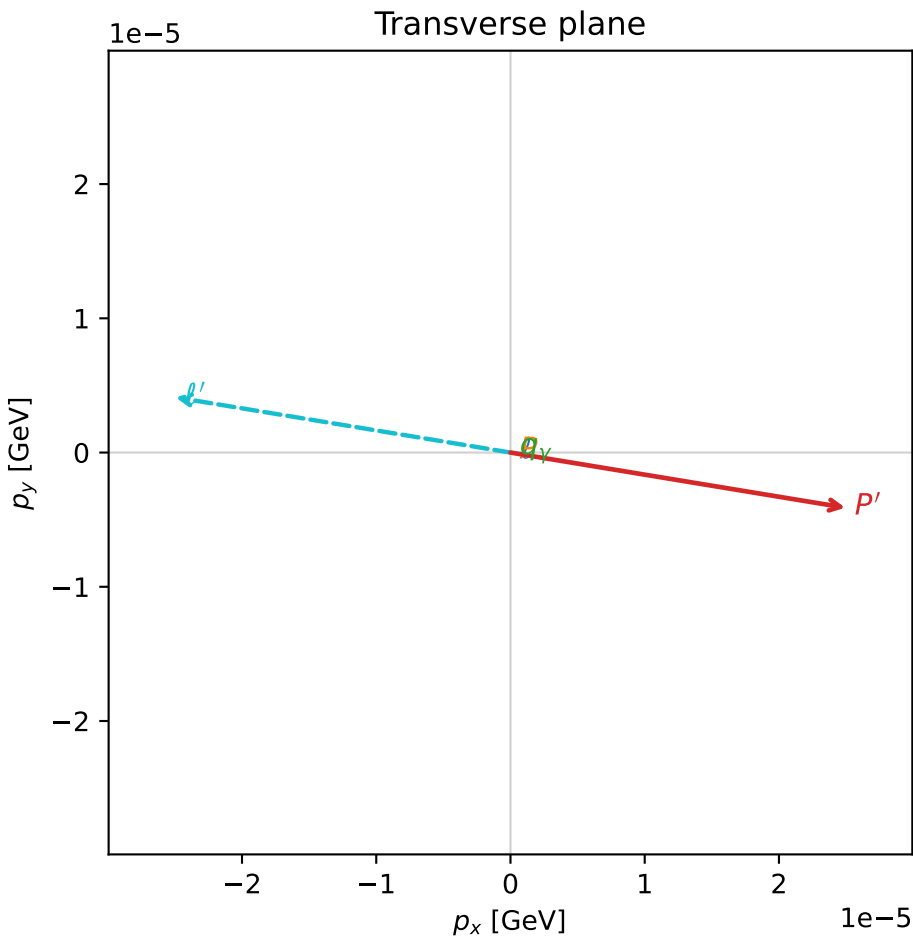
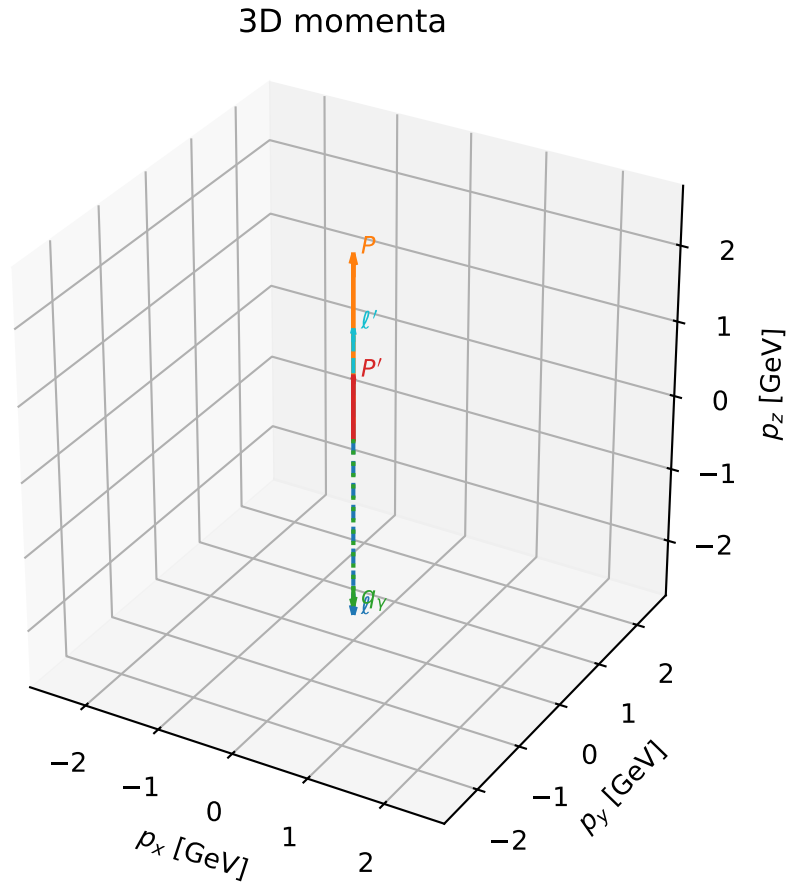
$\text{sqrt}_s = 3.996238$
 $\text{theta}_p_{\text{out}} = 3.141593$
 $\text{theta}_{\gamma_{\text{out}}} = 3.141593$
 $q_{\text{out}} = 0.09545423$
 $\text{phi}_p_{\text{out}} = 3.129588$
 $\text{phi}_{\gamma_{\text{out}}} = 5.699875$
 $\alpha_e = 2.37545$
 $\alpha_p = 0.7661545$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_103 $d_{\{\text{rm GHZ}\}}=3.69193\text{e-}08$
region: polarization_cluster_03_local_minimum_103
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3646323$ rad
incoming lepton: $-0.713048|+\rangle +0.701115|-\rangle$
proton mixing angle: $\alpha_p=0.77696696$ rad
incoming proton: $0.713043|+\rangle +0.70112|-\rangle$

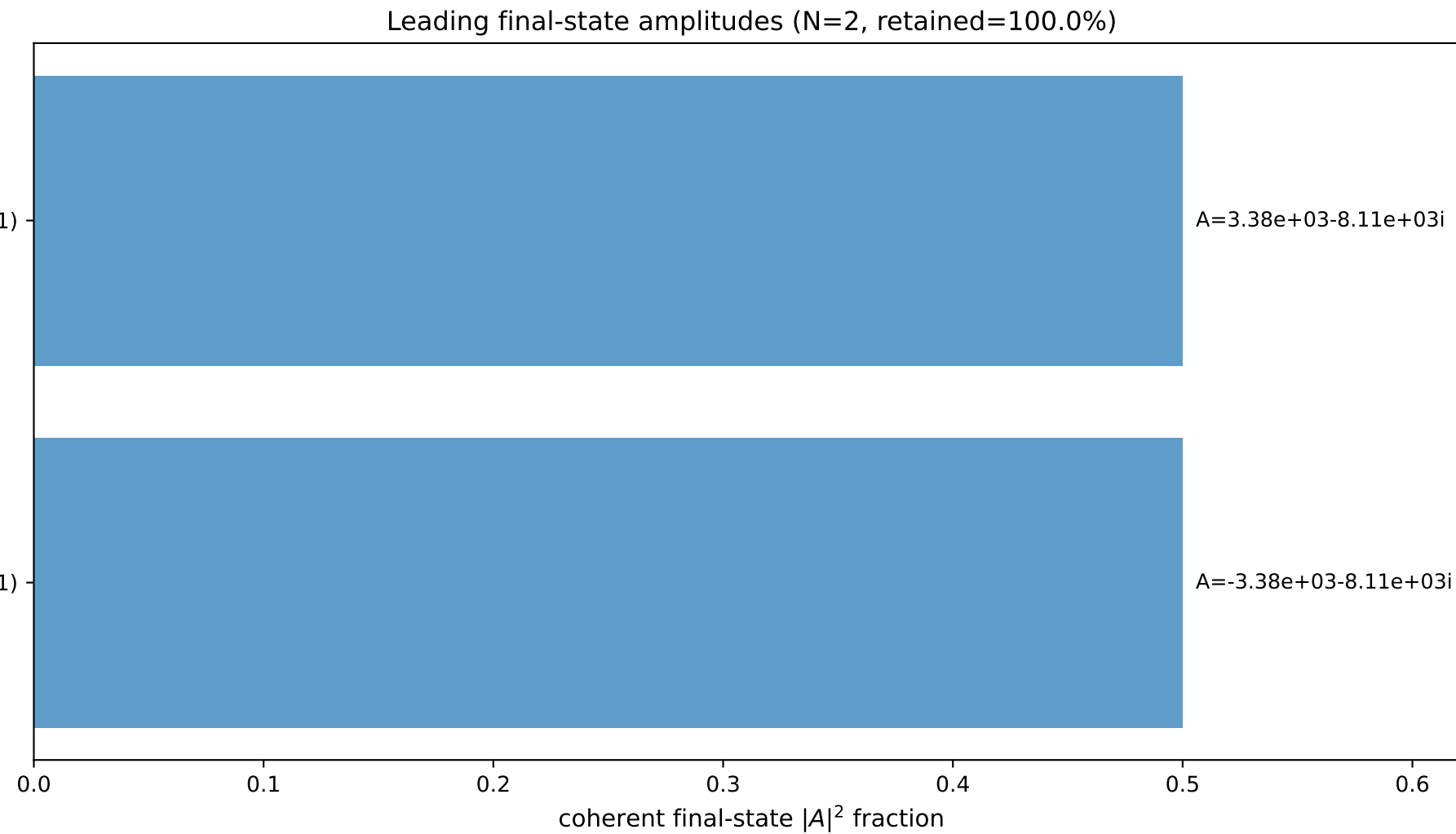
$s=24.9995$, $\sqrt{s}=4.99995$, $|\vec{P}|=2.41199$, $|\vec{P}'|=0.829186$
 $\theta_{P'}=3.04918\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=6.11999$, $\phi_Y=1.17541$
 $E_Y=2.28859$, $\theta_{l'}=1.73244\text{e-}05$, $\phi_{l'}=2.9784$
 $Q^2=14.0803$, $x_B=0.596469$, $t=-0.720361$, $W^2=10.4056$, $y=0.978705$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 1.4594$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4594$
 P' $E= 1.2520$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8292$
 q_Y $E= 2.2886$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2886$

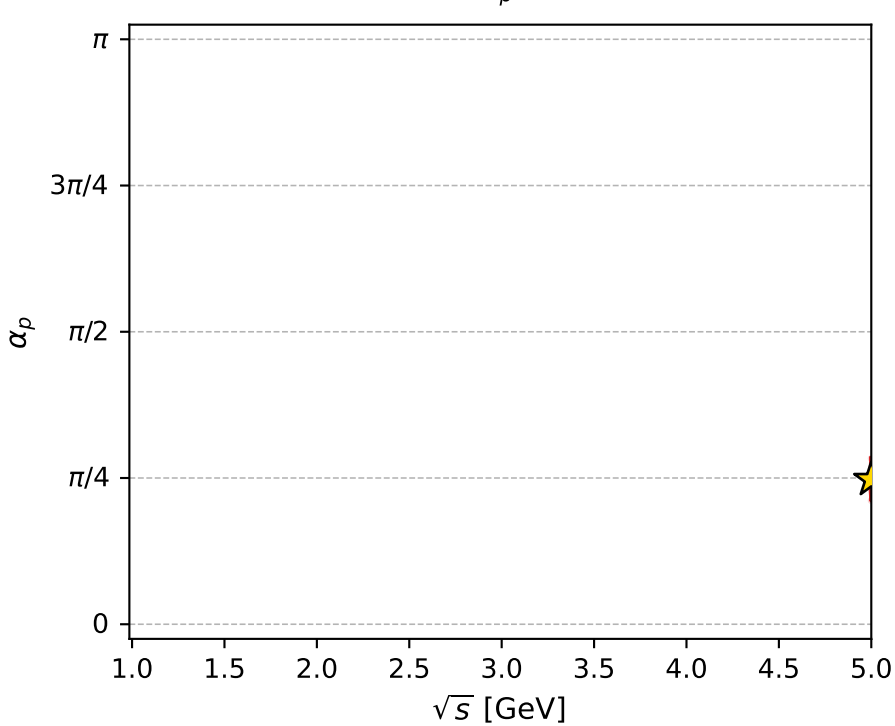
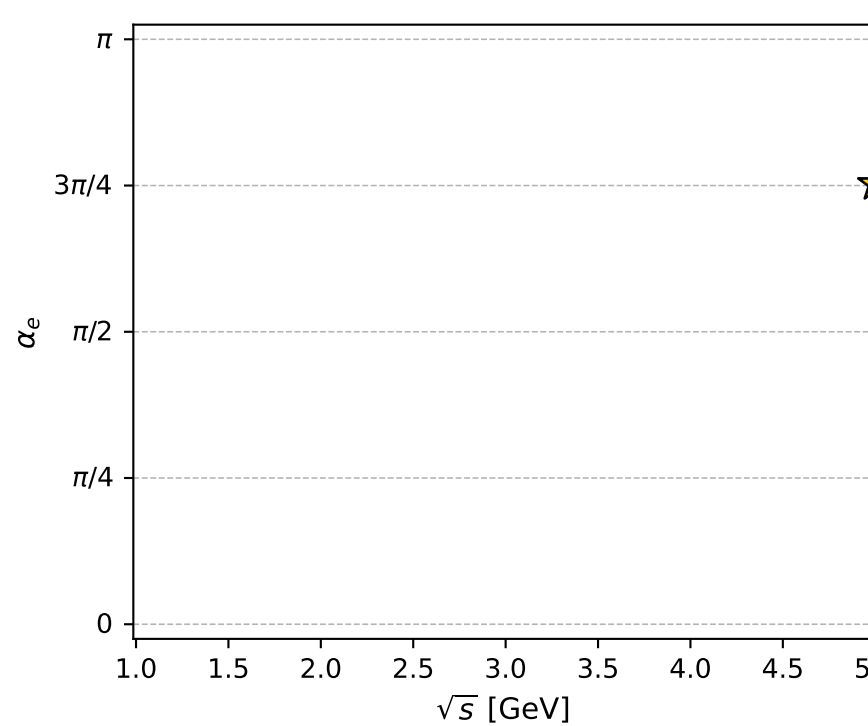
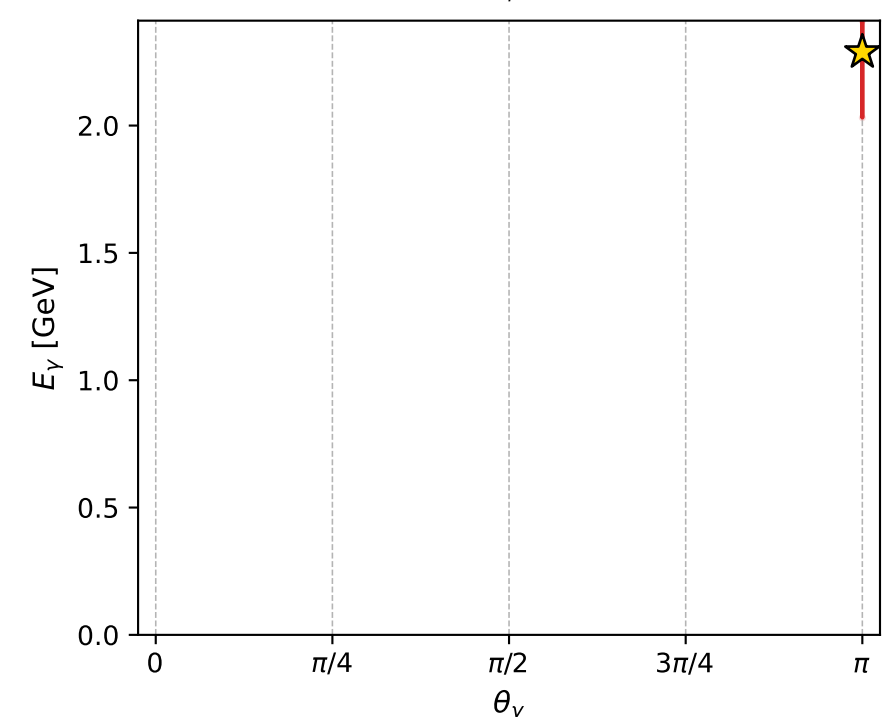
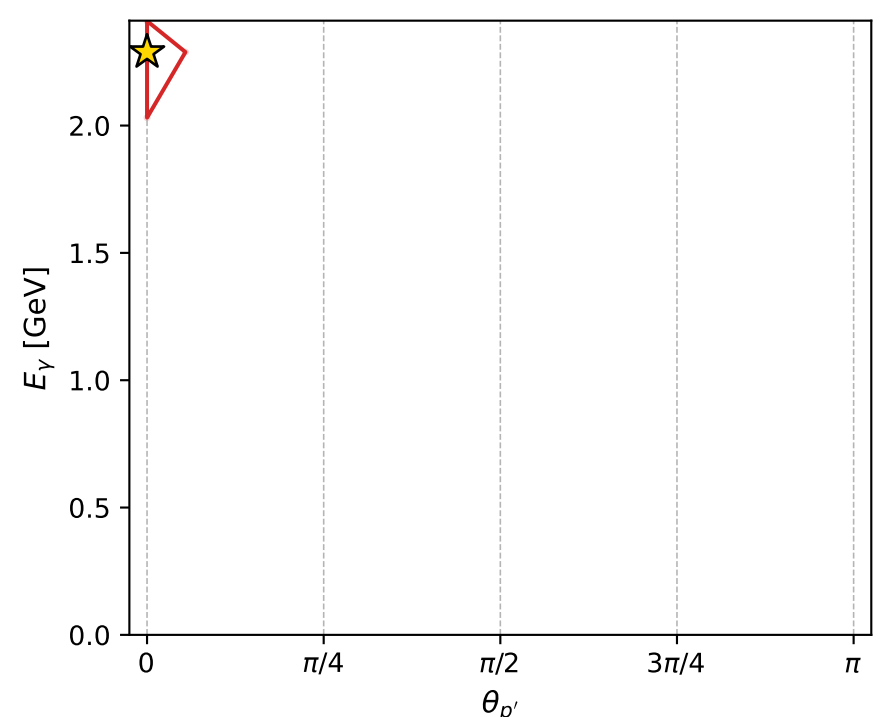
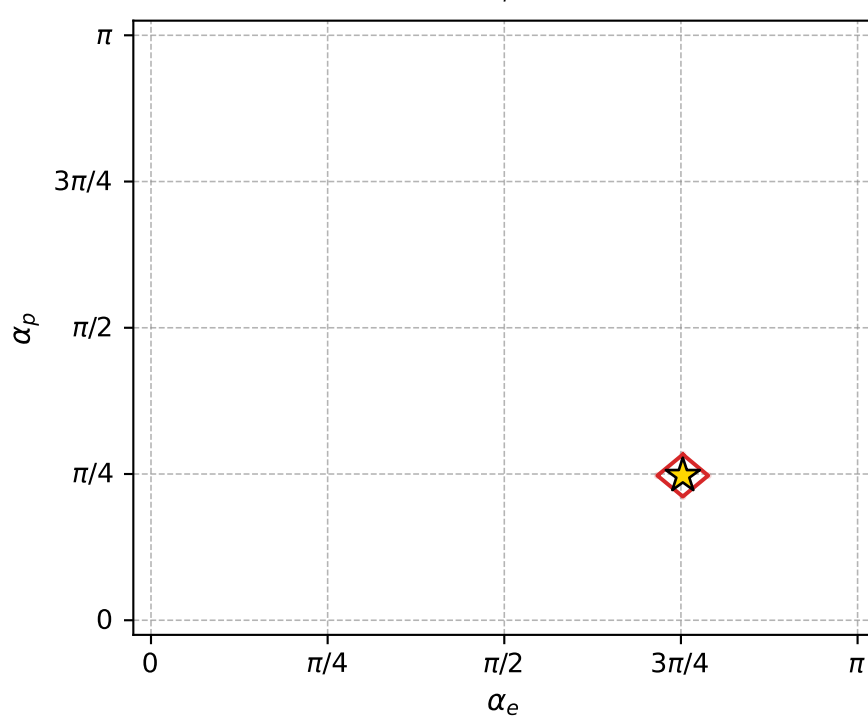
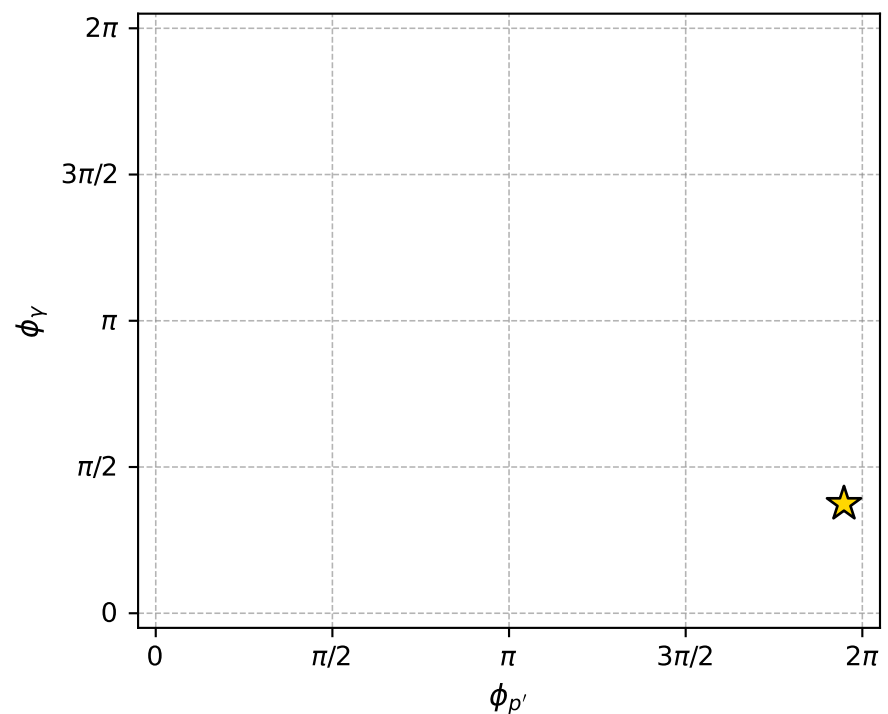
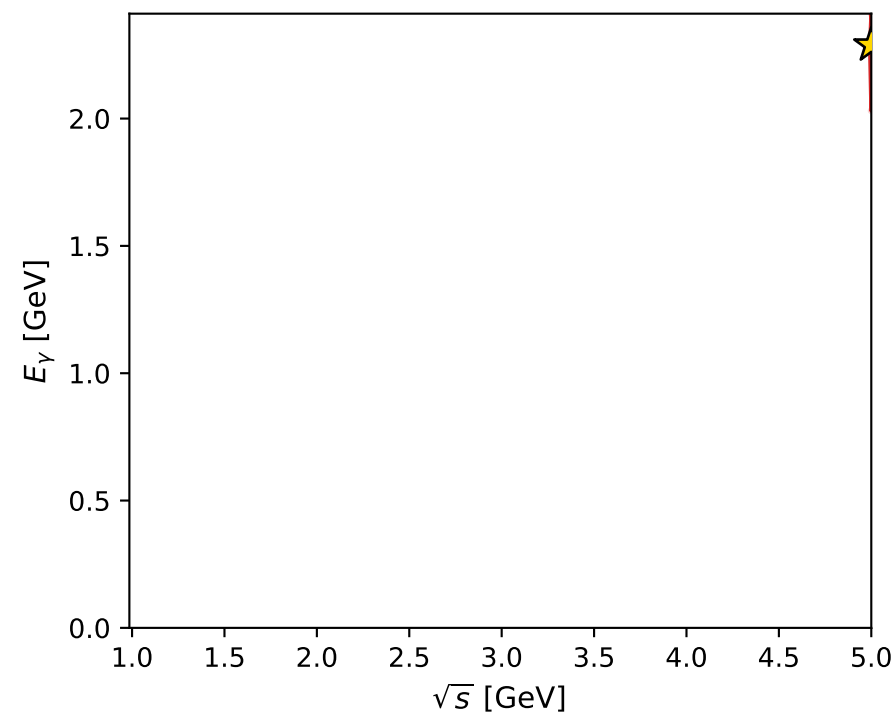
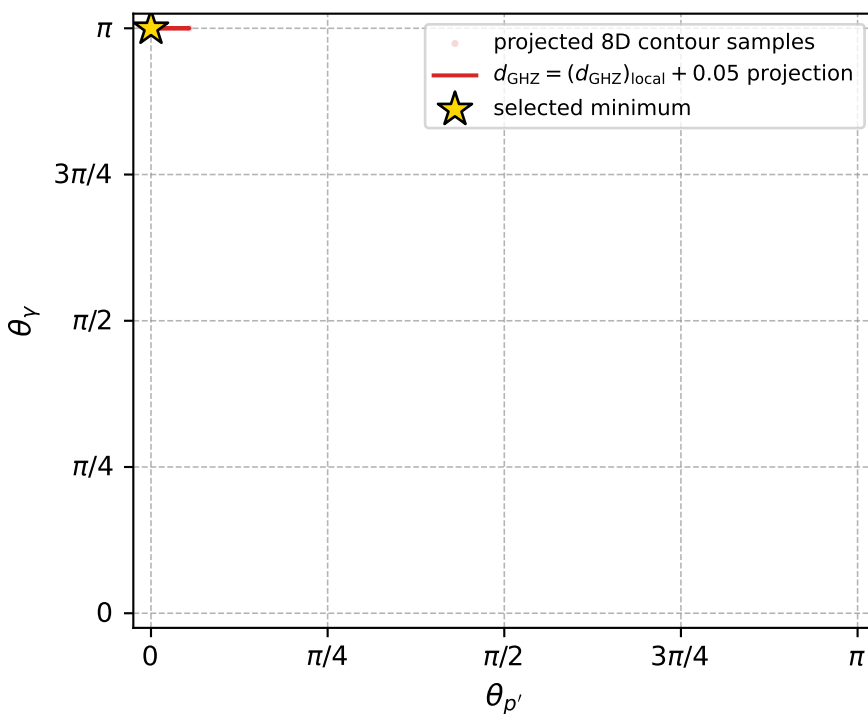


$(h_p, h_Y, h_l) = (+1, +1, -1)$

$(h_p, h_Y, h_l) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 103; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.6919338\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000037$
 above global minimum = $1.4635876\text{e-}08$
 8D contour samples = 33

selected phase-space point:

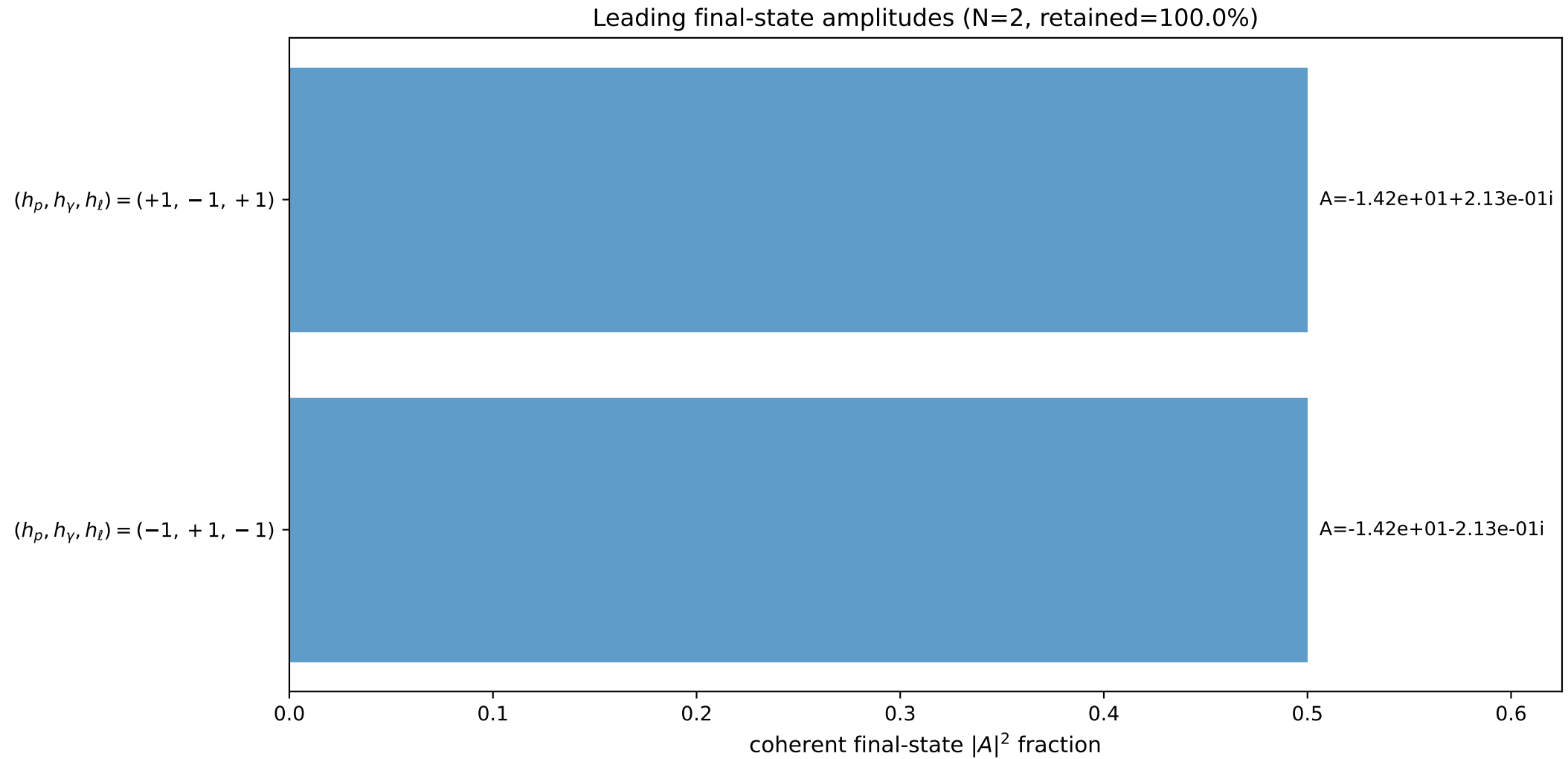
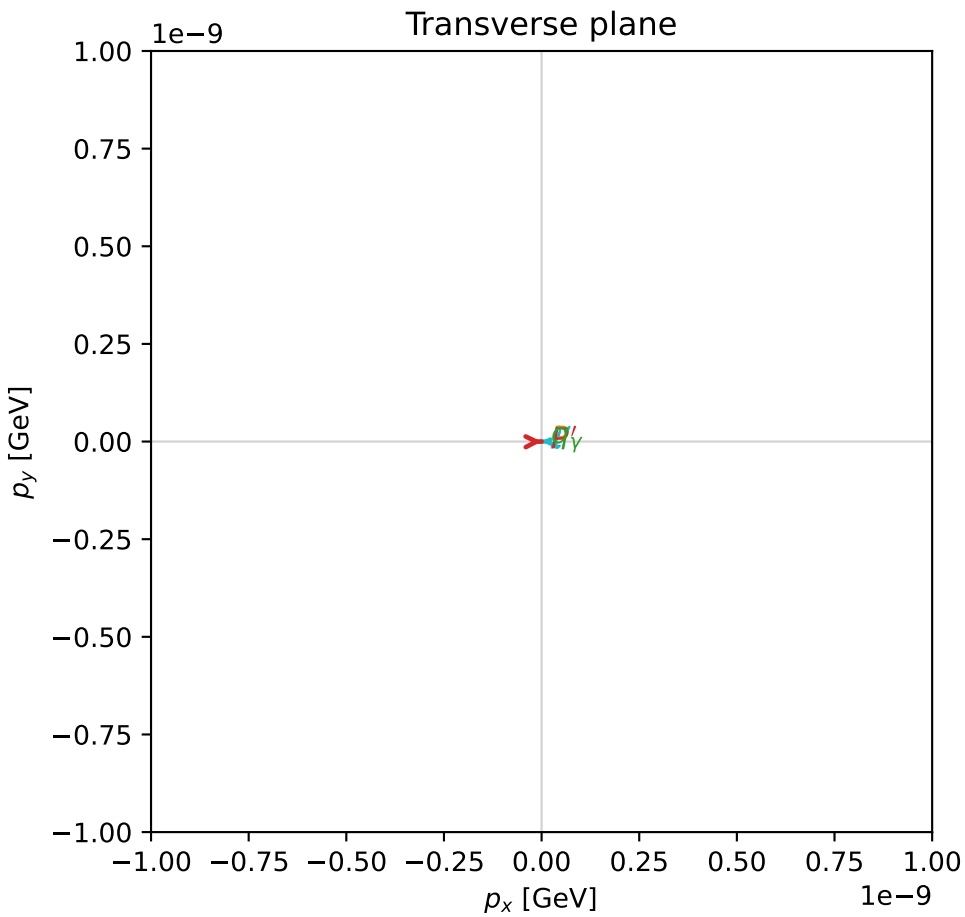
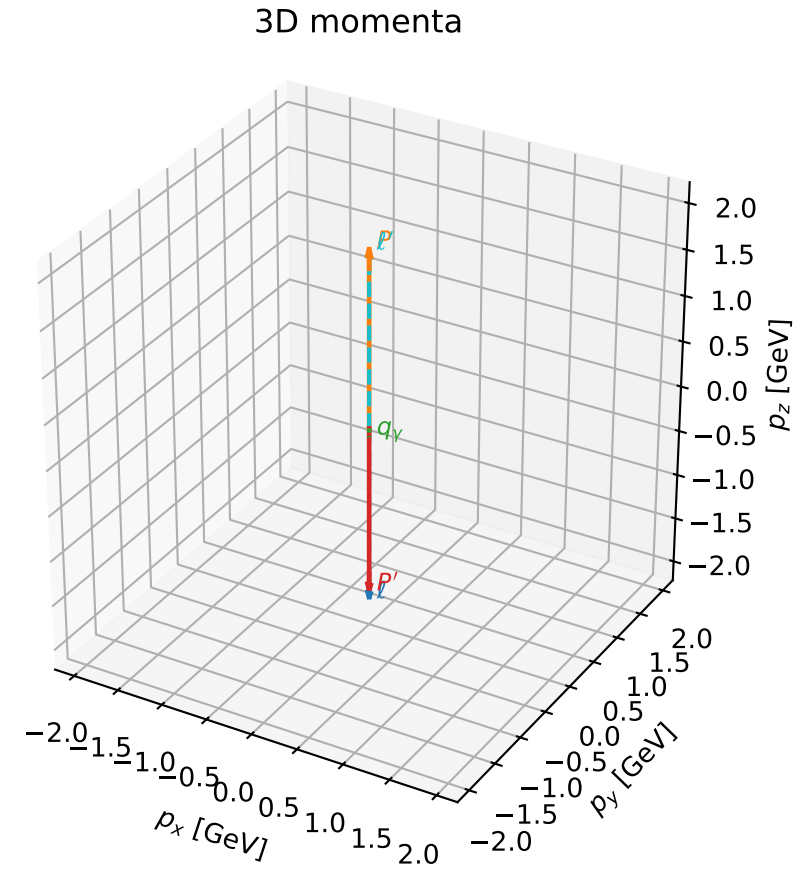
$\text{sqrt_s} = 4.999953$
 $\text{theta_p_out} = 3.04918\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.288592$
 $\text{phi_p_out} = 6.119988$
 $\text{phi_gamma_out} = 1.17541$
 $\text{alpha_e} = 2.364632$
 $\text{alpha_p} = 0.776967$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

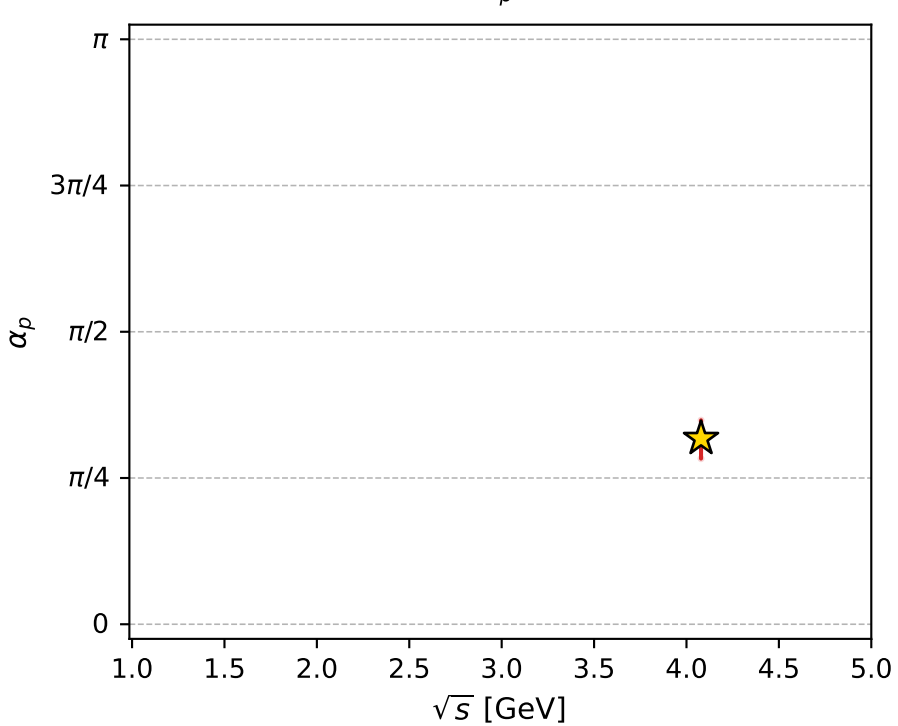
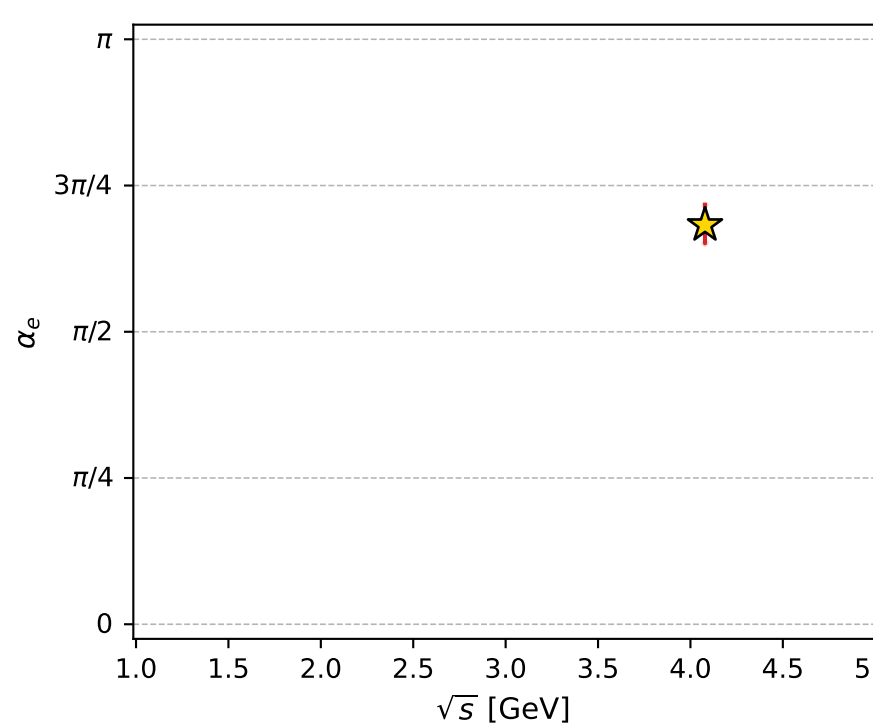
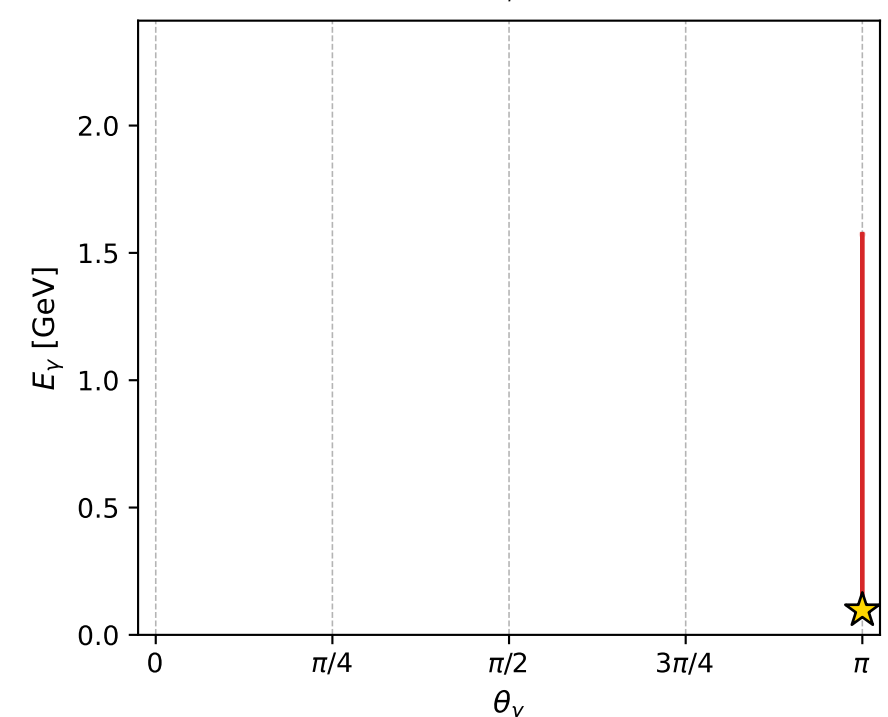
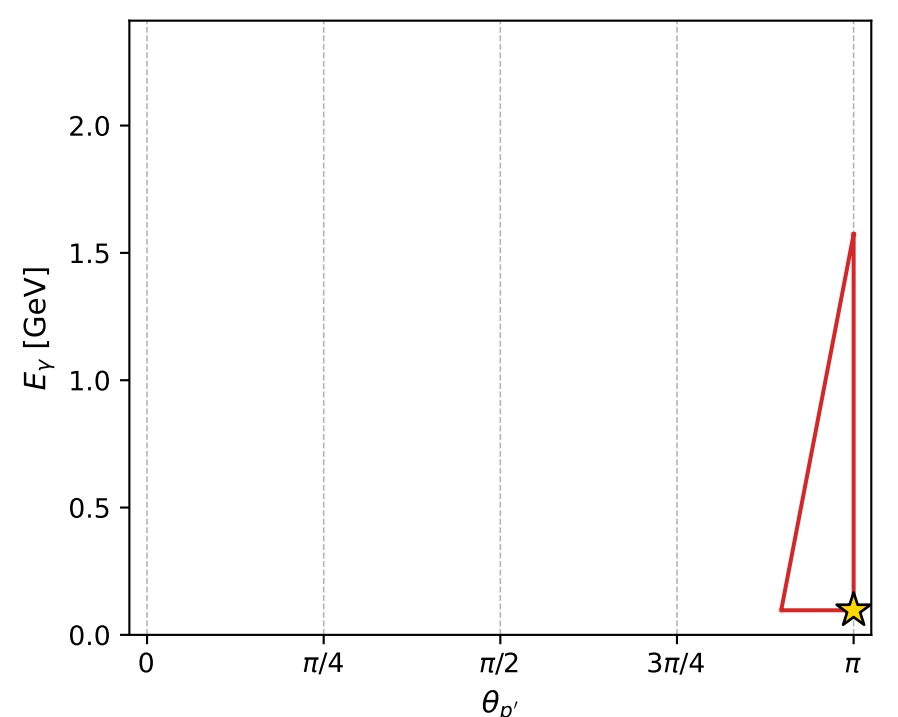
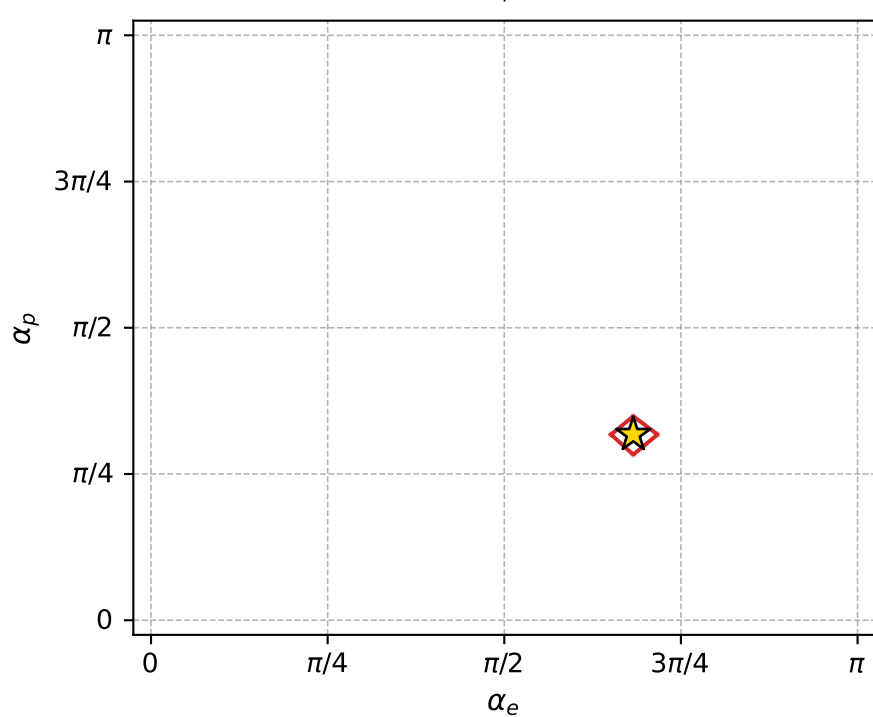
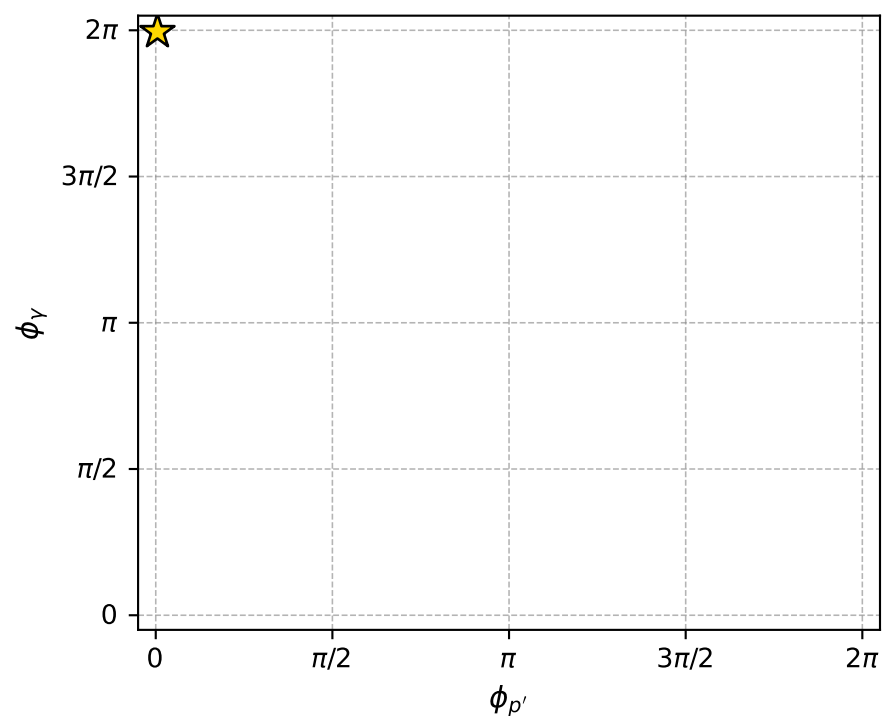
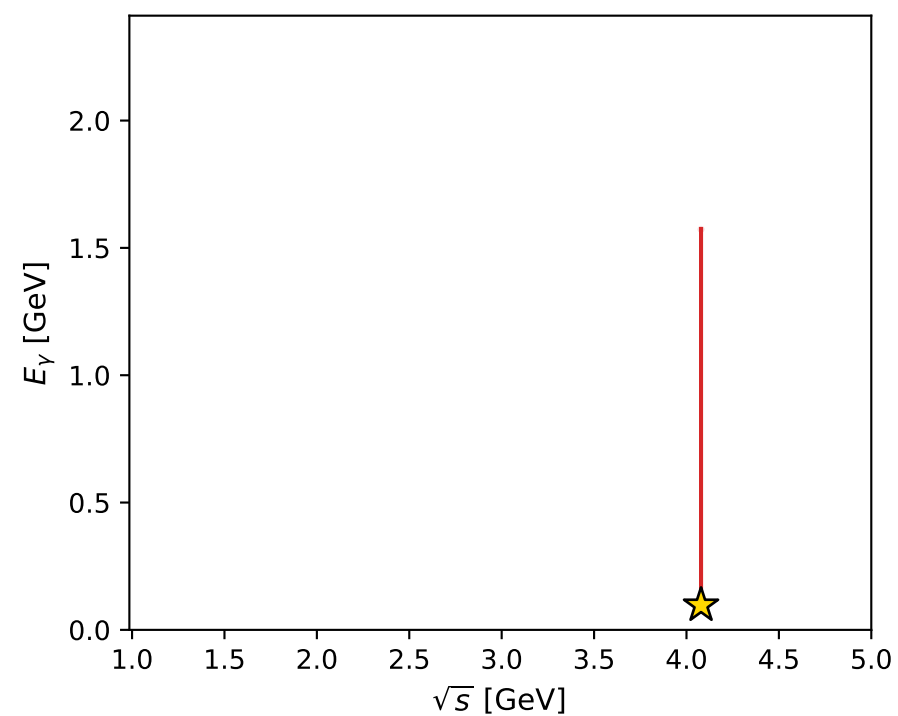
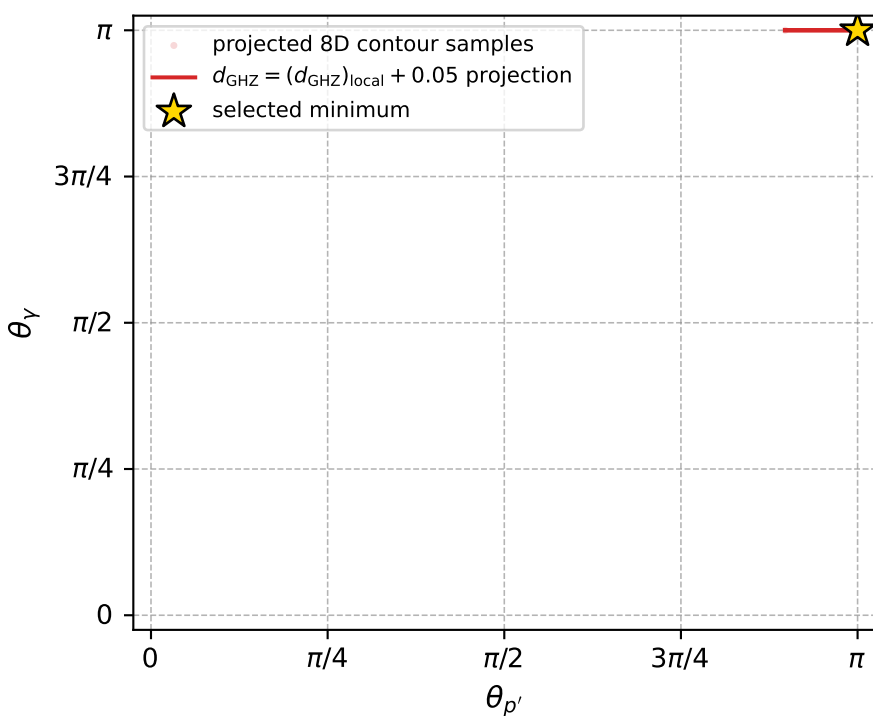
dghz_mixing_angles_polarization_03_local_minimum_106 $d_{\{\text{rm GHZ}\}}=3.85907\text{e-}08$
 region: polarization_cluster_03_local_minimum_106
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.1441764$ rad
 incoming lepton: $-0.542475|+> +0.840072|->$
 proton mixing angle: $\alpha_p=0.99739265$ rad
 incoming proton: $0.542494|+> +0.840059|->$

$s=16.6364$, $\sqrt{s}=4.07877$, $|\vec{P}|=1.93153$, $|\vec{P}'|=1.82957$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=0.0149987$, $\phi_Y=6.26819$
 $E_Y=0.0966003$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.15509$
 $Q^2=14.8818$, $x_B=0.997069$, $t=-14.1375$, $W^2=0.923592$, $y=0.94726$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9315$
 P $E= 2.1472$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9315$
 ℓ' $E= 1.9262$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9262$
 P' $E= 2.0560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8296$
 q_Y $E= 0.0966$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0966$



electron: polarization cluster P3, local minimum 106; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.8590677\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000039$
 above global minimum = $1.6307215\text{e-}08$
 8D contour samples = 60

selected phase-space point:

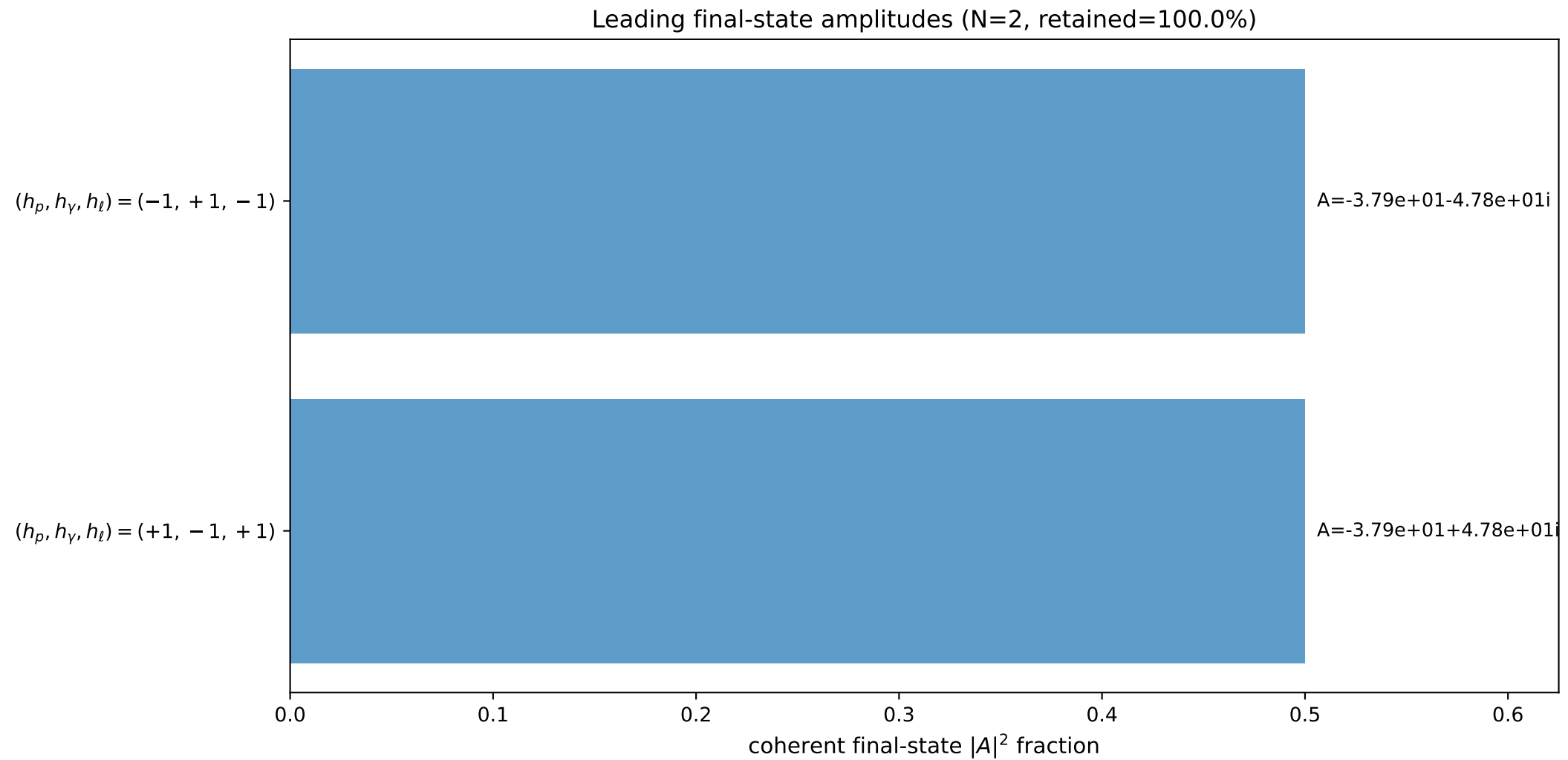
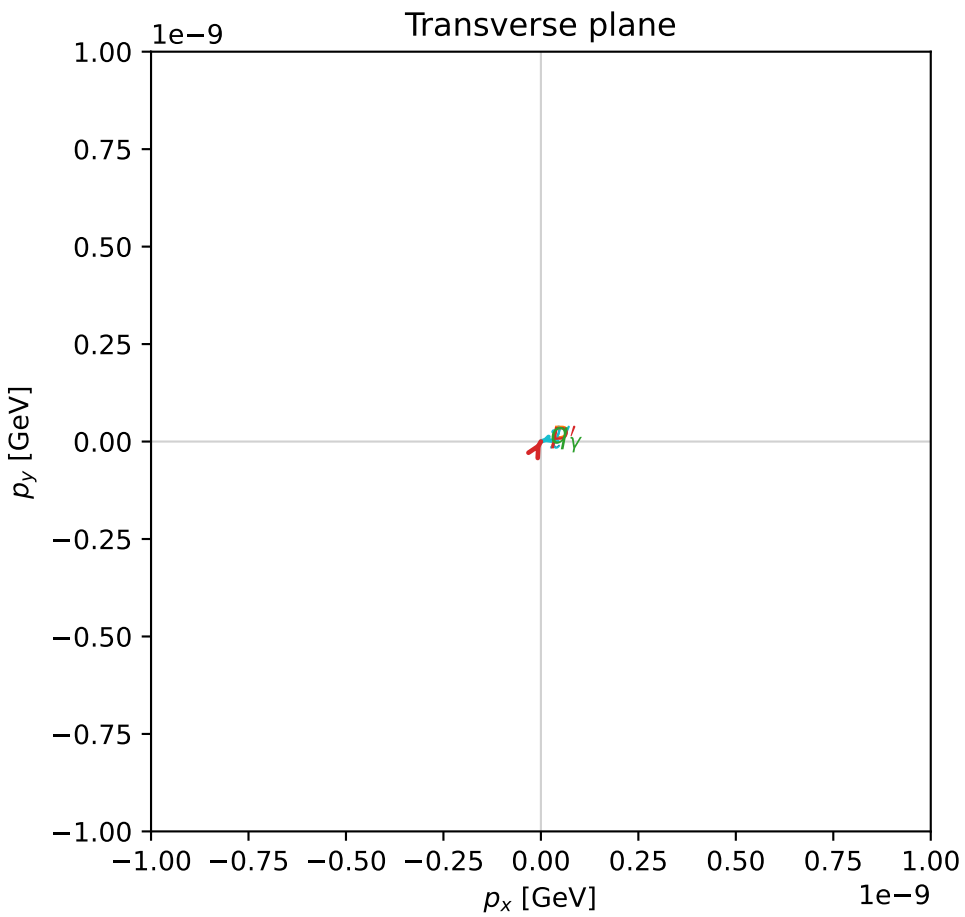
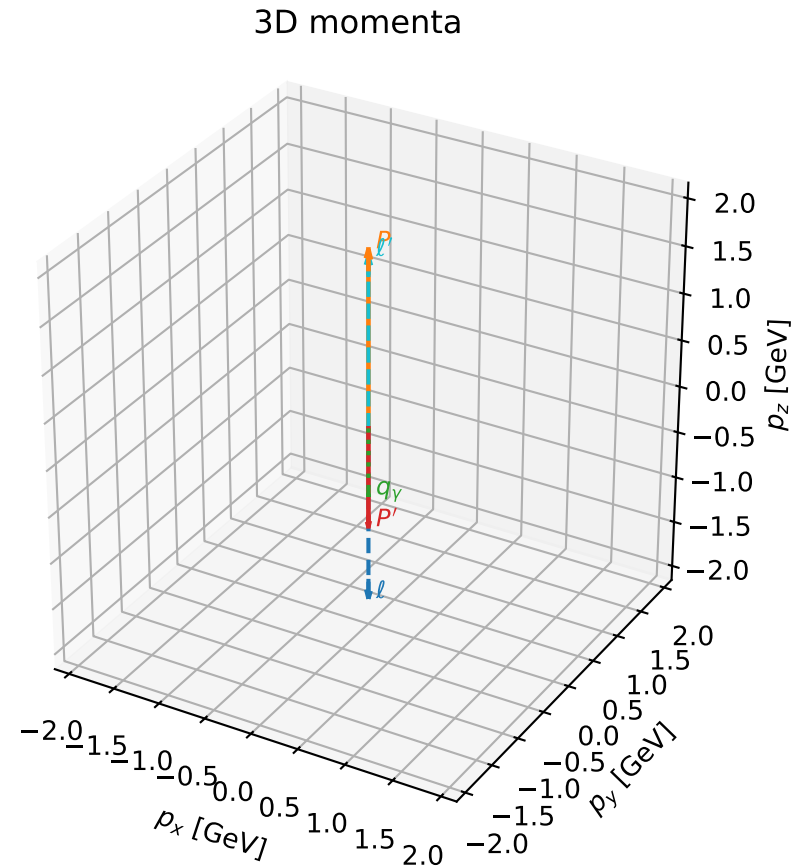
$\text{sqrt_s} = 4.078774$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 0.09660034$
 $\text{phi_p_out} = 0.01499868$
 $\text{phi_gamma_out} = 6.268192$
 $\text{alpha_e} = 2.144176$
 $\text{alpha_p} = 0.9973926$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

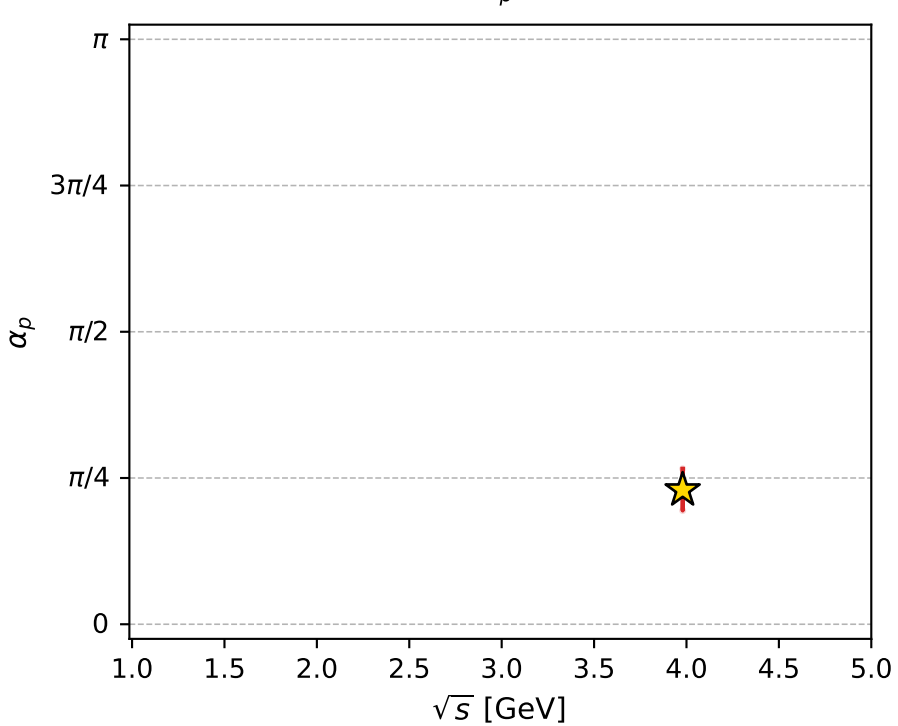
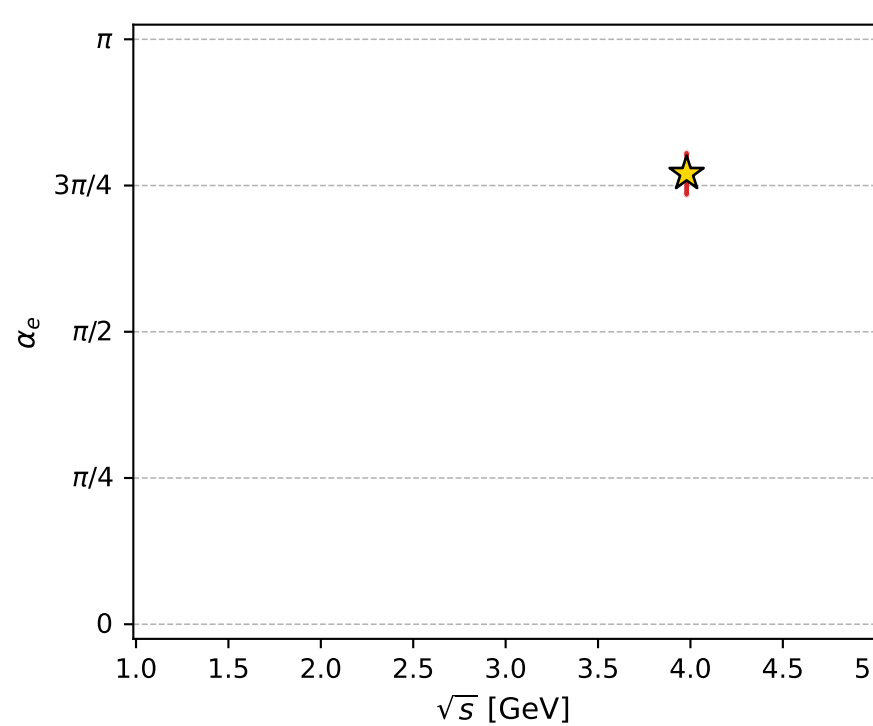
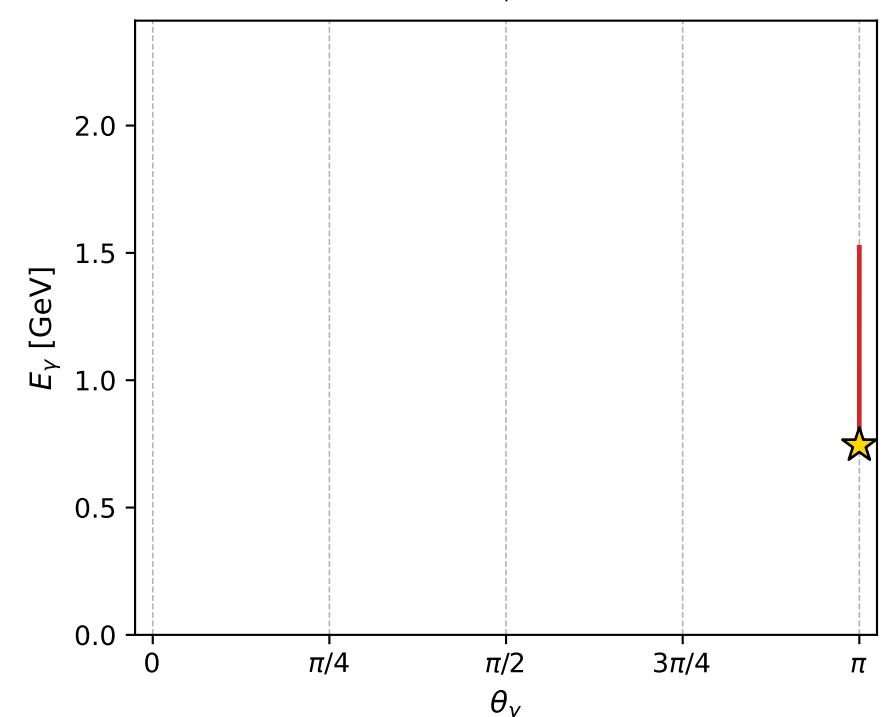
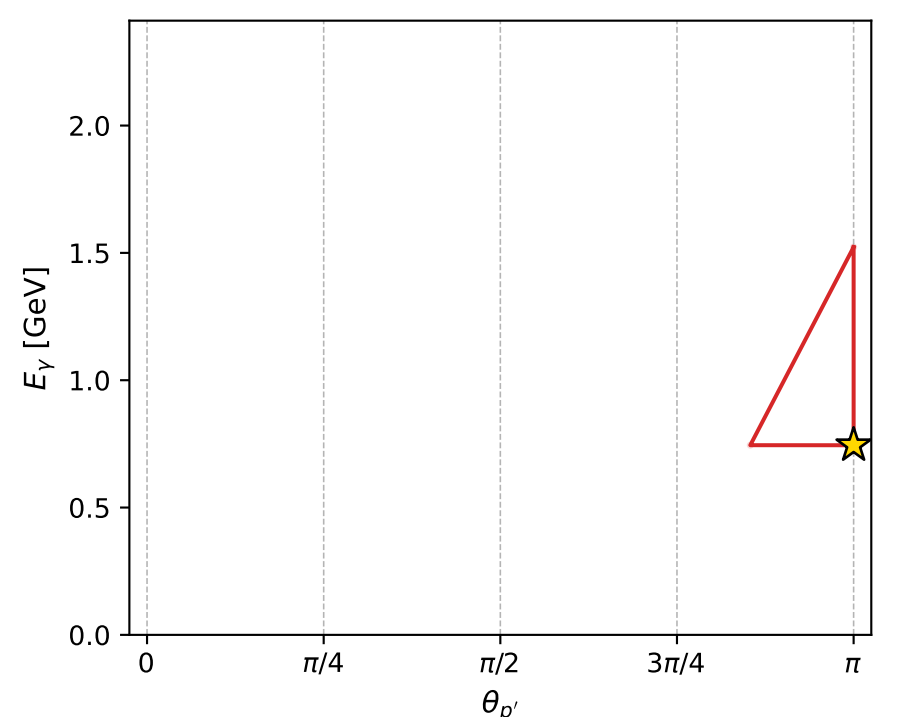
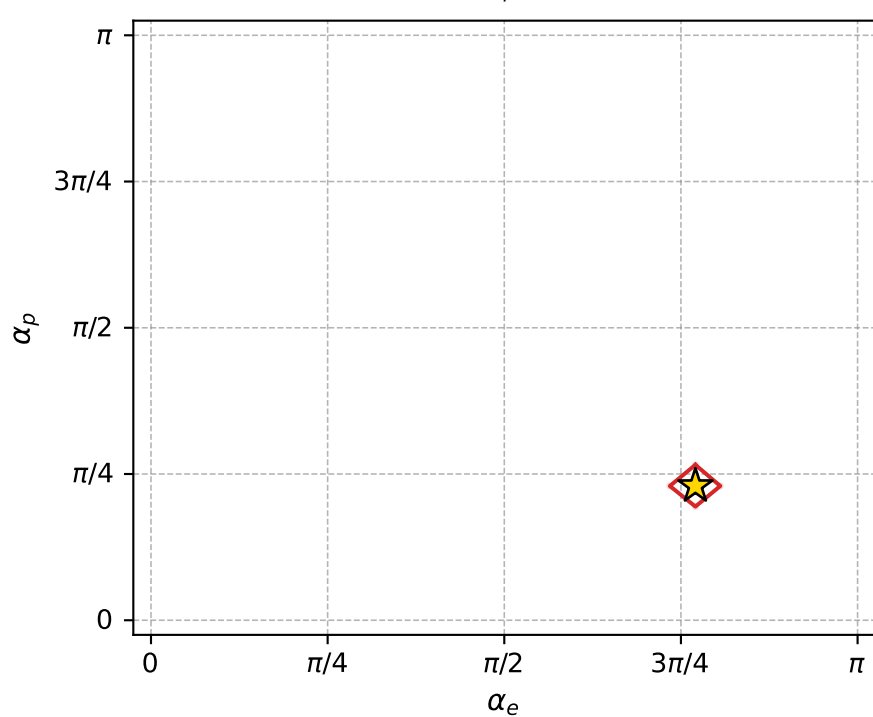
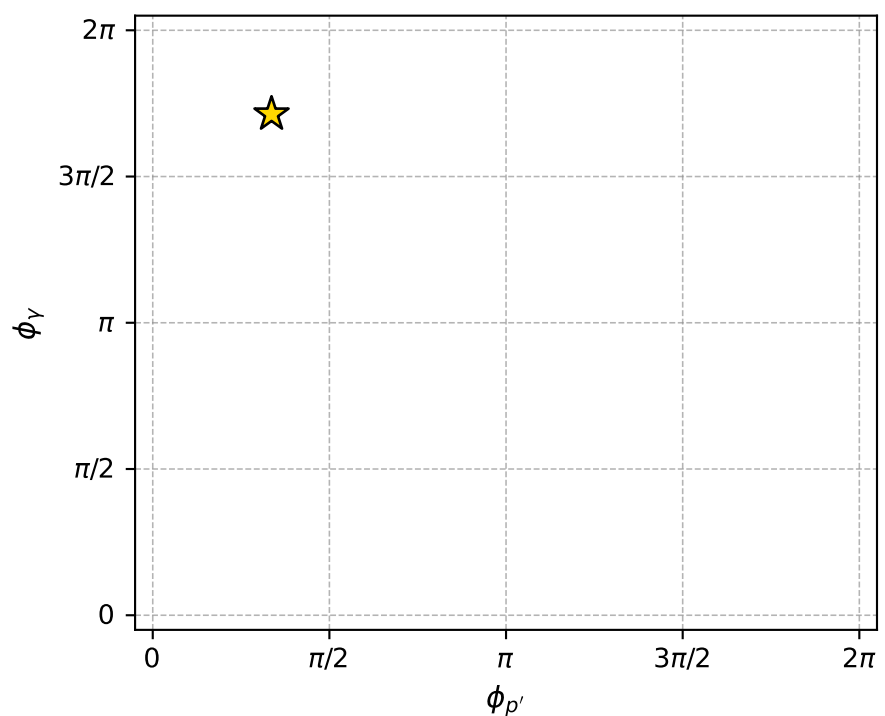
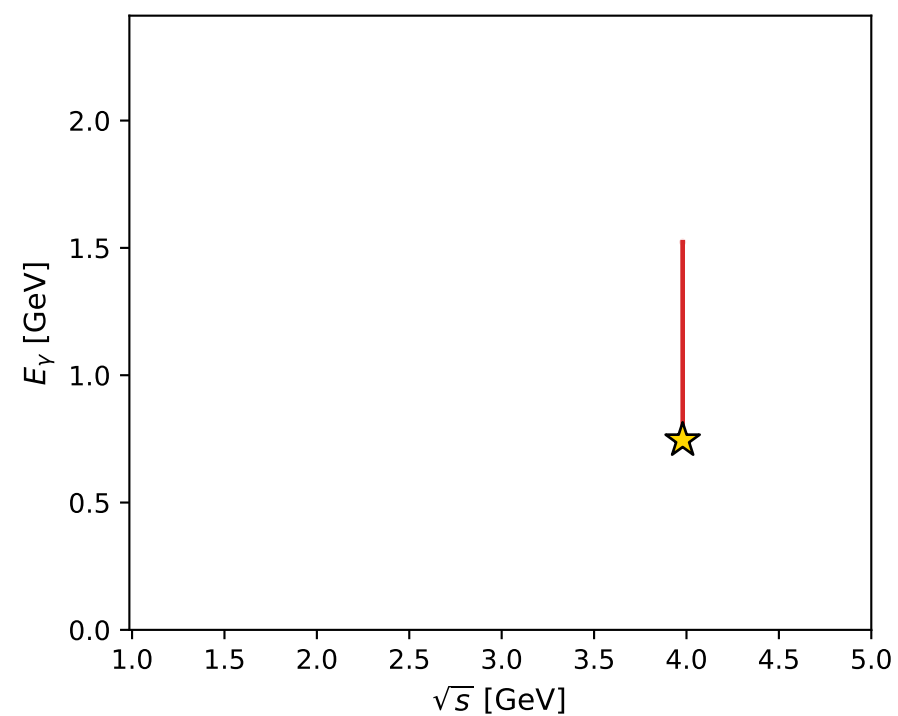
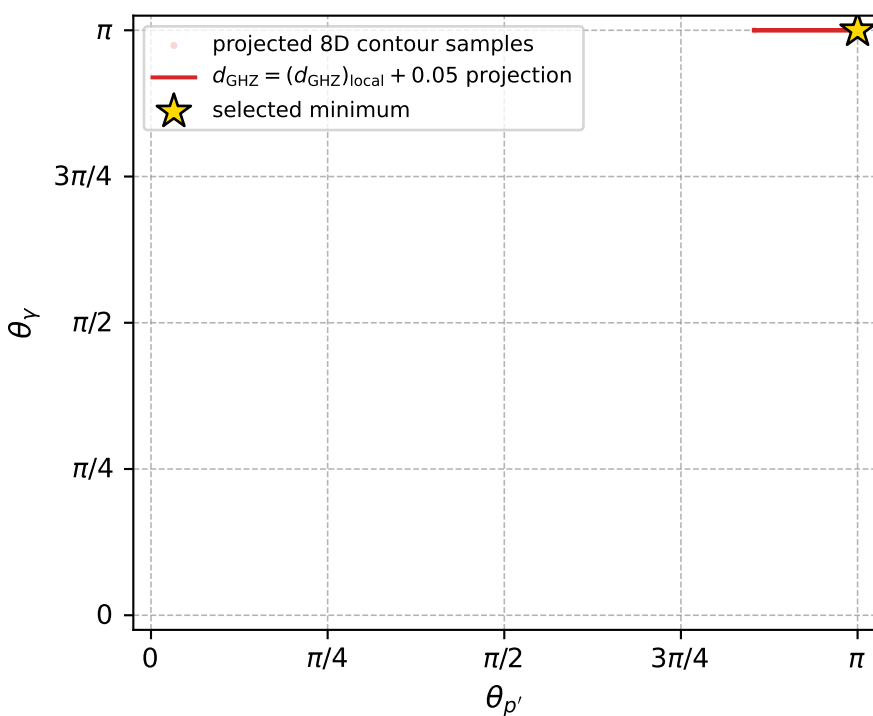
dghz_mixing_angles_polarization_03_local_minimum_107 $d_{\{\text{rm GHZ}\}}=3.86512\text{e-}08$
 region: polarization_cluster_03_local_minimum_107
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.4205789$ rad
 incoming lepton: $-0.751137|+\rangle +0.660146|-\rangle$
 proton mixing angle: $\alpha_p=0.72102631$ rad
 incoming proton: $0.751129|+\rangle +0.660156|-\rangle$

$s=15.8347$, $\sqrt{s}=3.97928$, $|\vec{P}|=1.87909$, $|\vec{P}'|=1.06808$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=1.05624$, $\phi_Y=5.38261$
 $E_Y=0.744851$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.47839$
 $Q^2=13.6266$, $x_B=0.962801$, $t=-8.22517$, $W^2=1.40632$, $y=0.946392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8791$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8791$
 P $E= 2.1002$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8791$
 ℓ' $E= 1.8129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.8129$
 P' $E= 1.4215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0681$
 q_Y $E= 0.7449$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.7449$



electron: polarization cluster P3, local minimum 107; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.8651158\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000039$
 above global minimum = $1.6367696\text{e-}08$
 8D contour samples = 60

selected phase-space point:

$\text{sqrt}_s = 3.979279$
 $\text{theta}_p_{\text{out}} = 3.141593$
 $\text{theta}_{\gamma_{\text{out}}} = 3.141593$
 $q_{\text{out}} = 0.7448514$
 $\text{phi}_p_{\text{out}} = 1.056244$
 $\text{phi}_{\gamma_{\text{out}}} = 5.382606$
 $\alpha_e = 2.420579$
 $\alpha_p = 0.7210263$

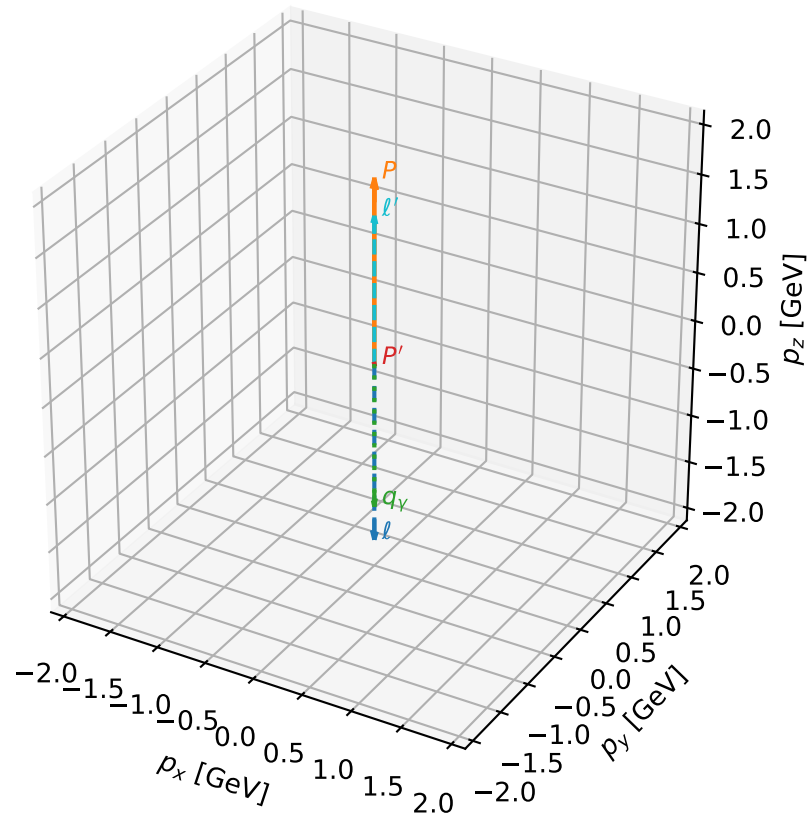
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_118 $d_{\{\text{rm GHZ}\}}=4.43159\text{e-}08$
 region: polarization_cluster_03_local_minimum_118
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3993315$ rad
 incoming lepton: $-0.736942|+\rangle +0.675956|-\rangle$
 proton mixing angle: $\alpha_p=0.74228136$ rad
 incoming proton: $0.736928|+\rangle +0.675971|-\rangle$

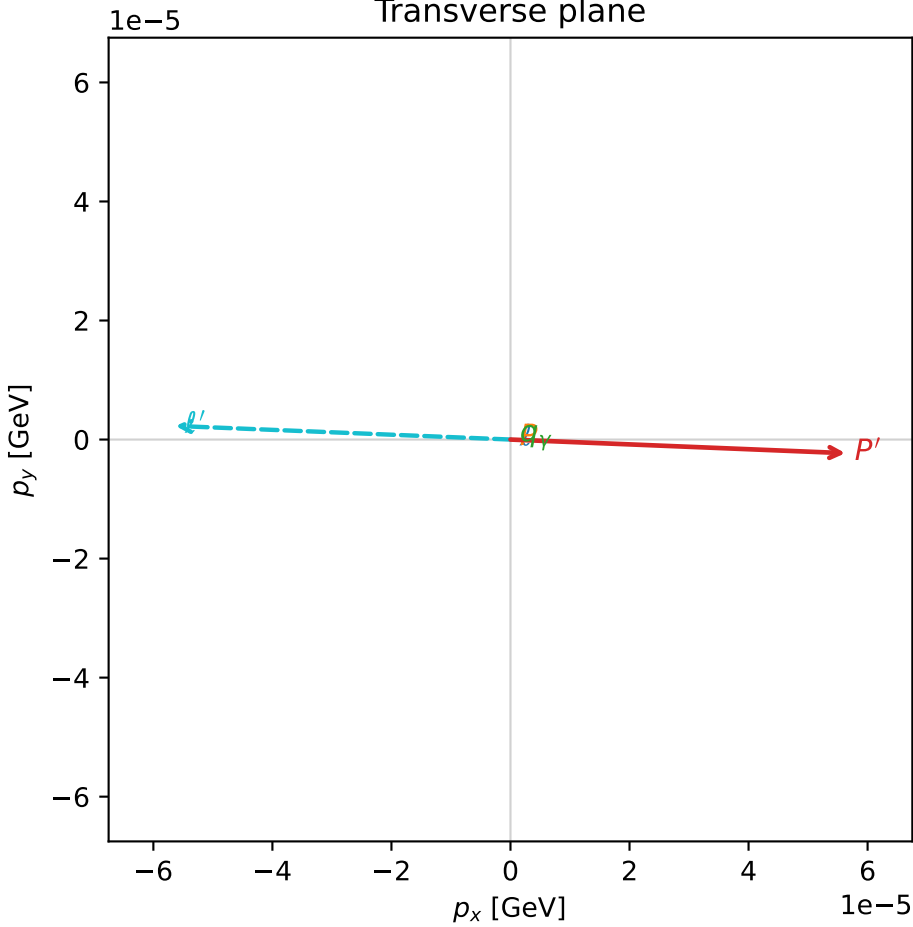
$s=15.5089$, $\sqrt{s}=3.93814$, $|\vec{P}|=1.85736$, $|\vec{P}'|=0.000247174$
 $\theta_{P'}=2.91175$, $\theta_Y=3.14159$, $\phi_{P'}=6.24252$, $\phi_Y=4.56954$
 $E_Y=1.49995$, $\theta_{\ell'}=3.7536\text{e-}05$, $\phi_{\ell'}=3.10092$
 $Q^2=11.1456$, $x_B=0.798465$, $t=-2.14474$, $W^2=3.69303$, $y=0.954178$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8574$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8574$
 P $E= 2.0808$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8574$
 ℓ' $E= 1.5002$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.5002$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0002$
 q_Y $E= 1.4999$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4999$

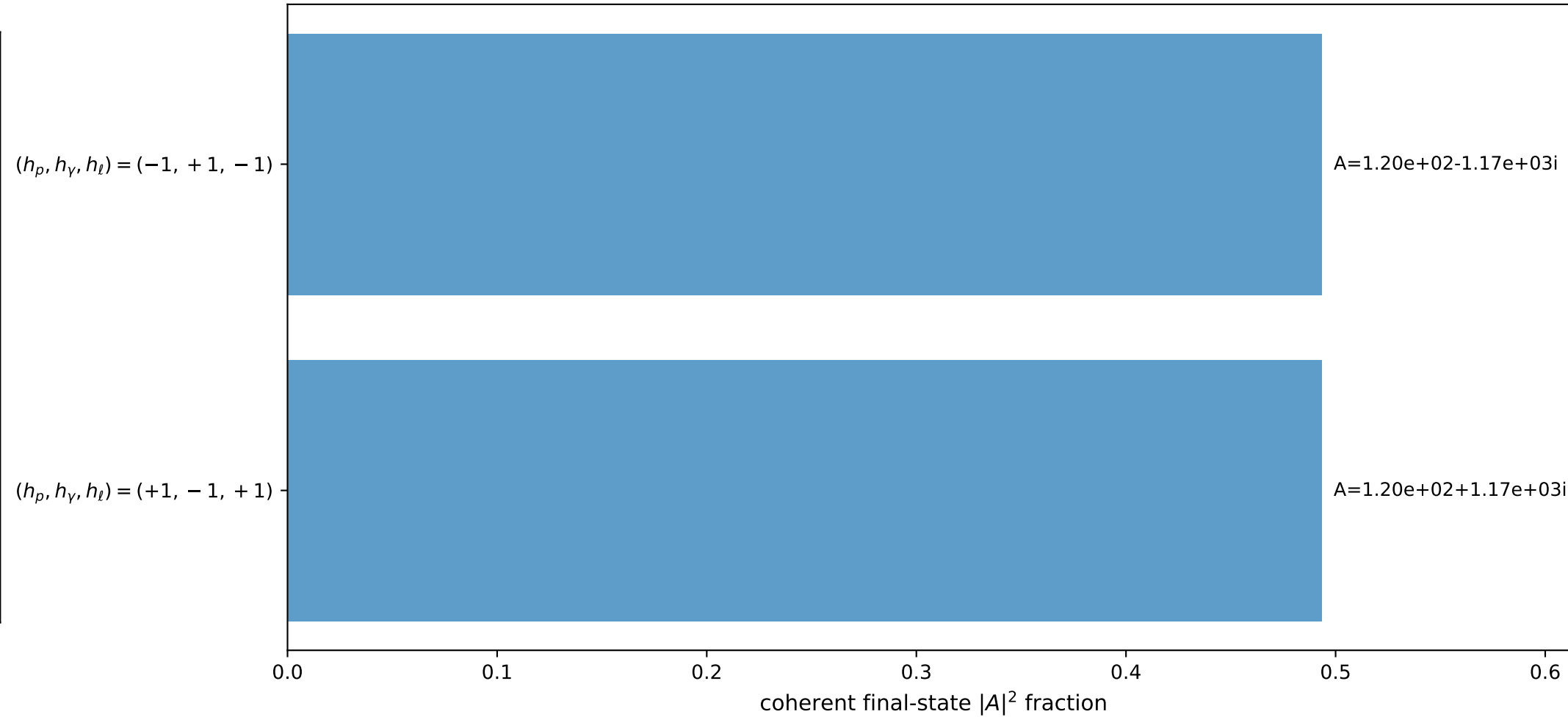
3D momenta



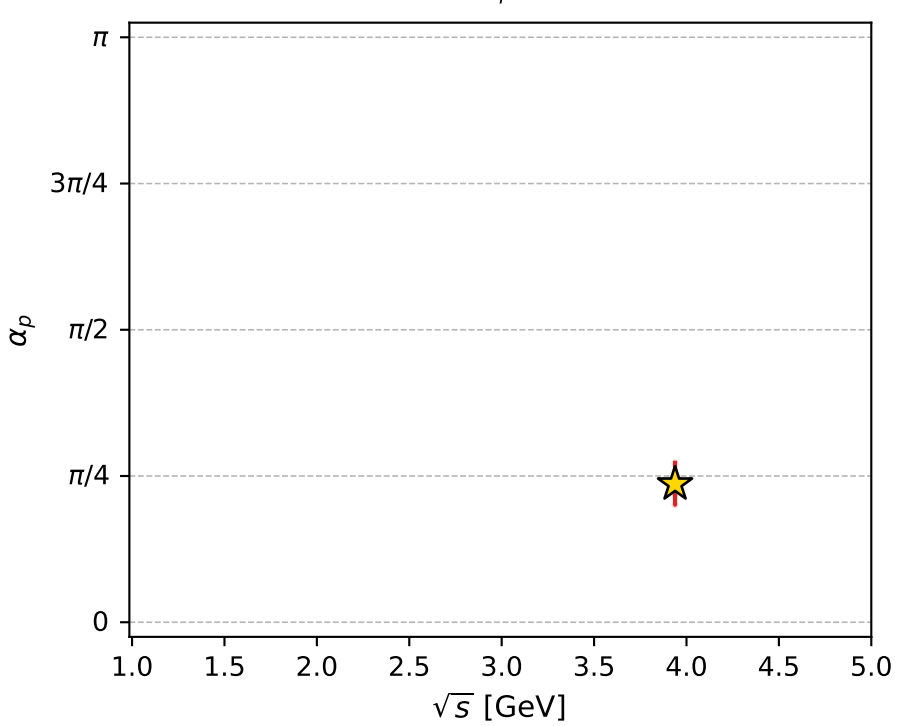
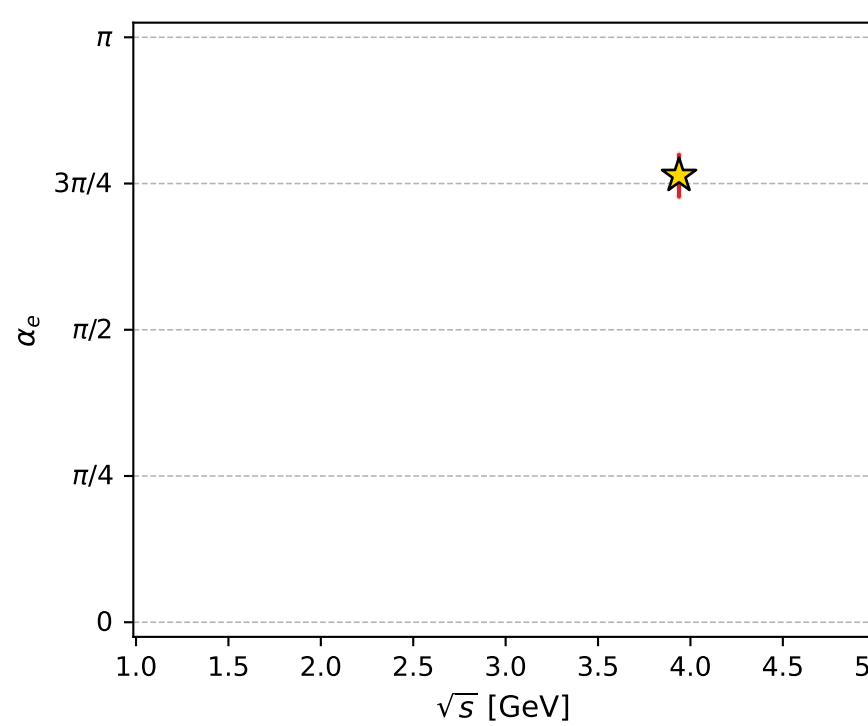
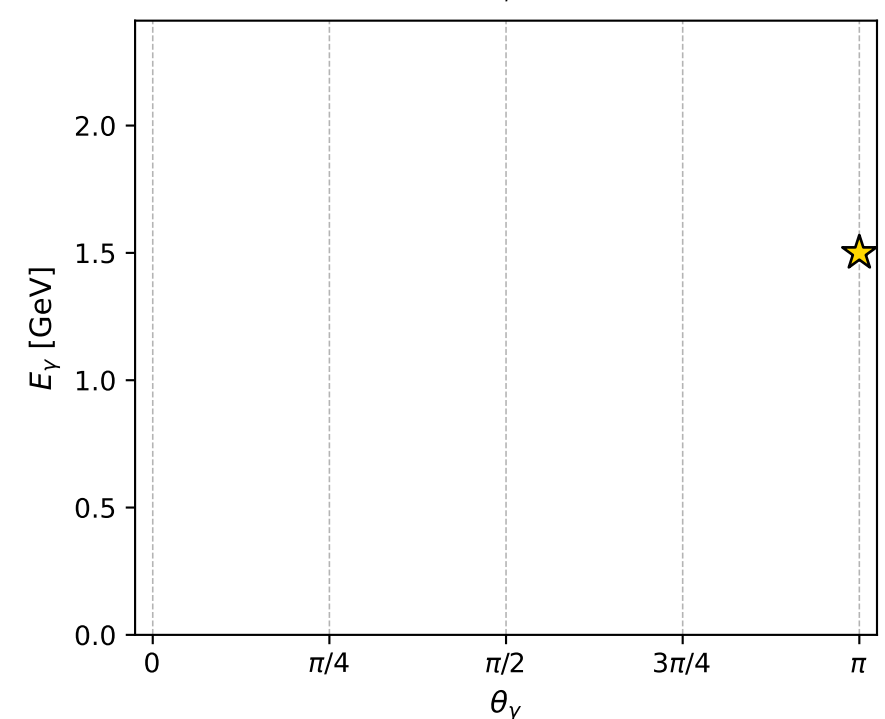
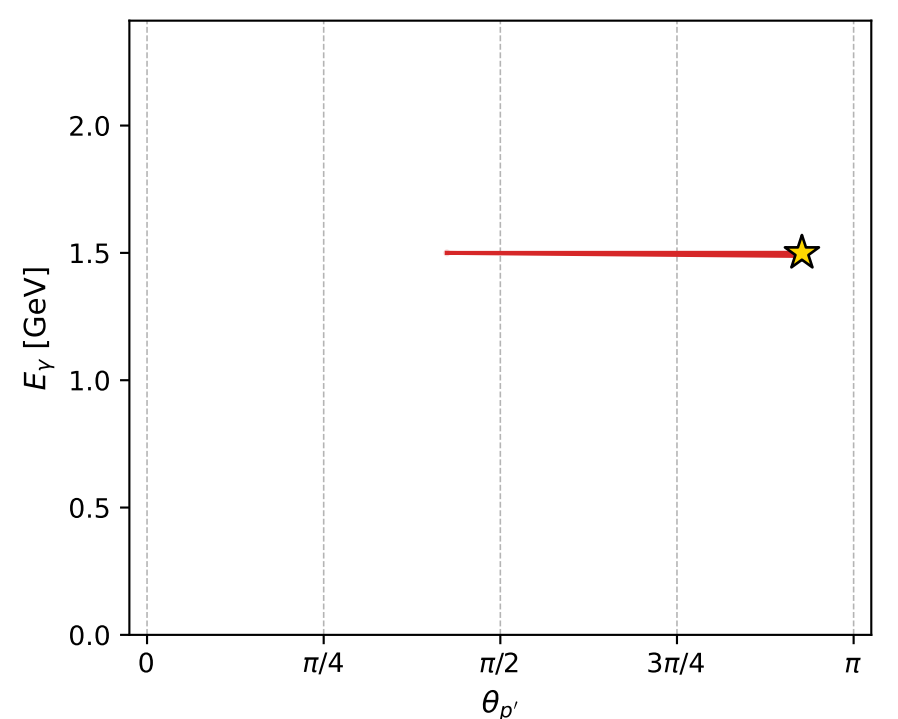
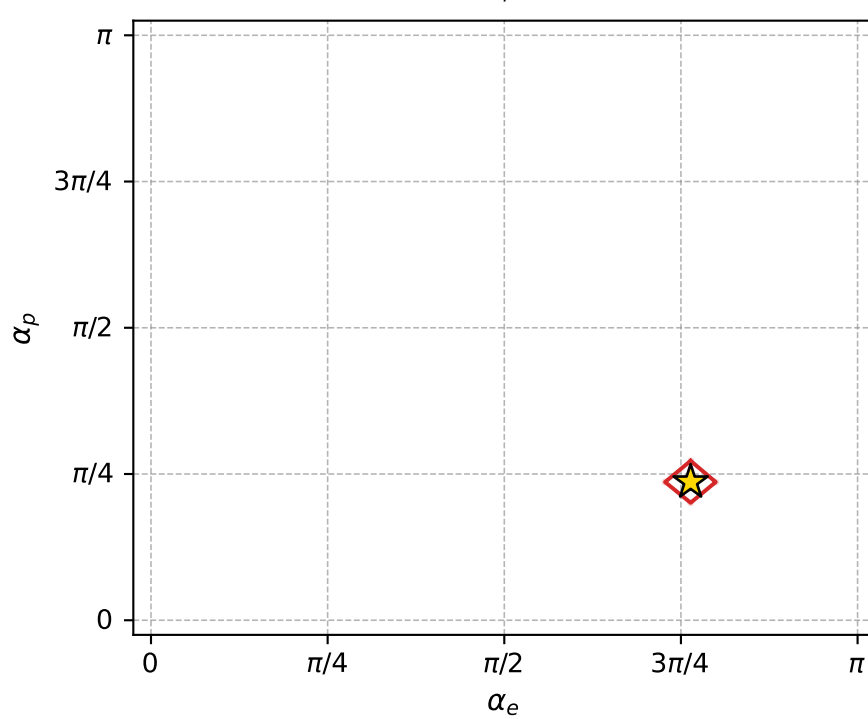
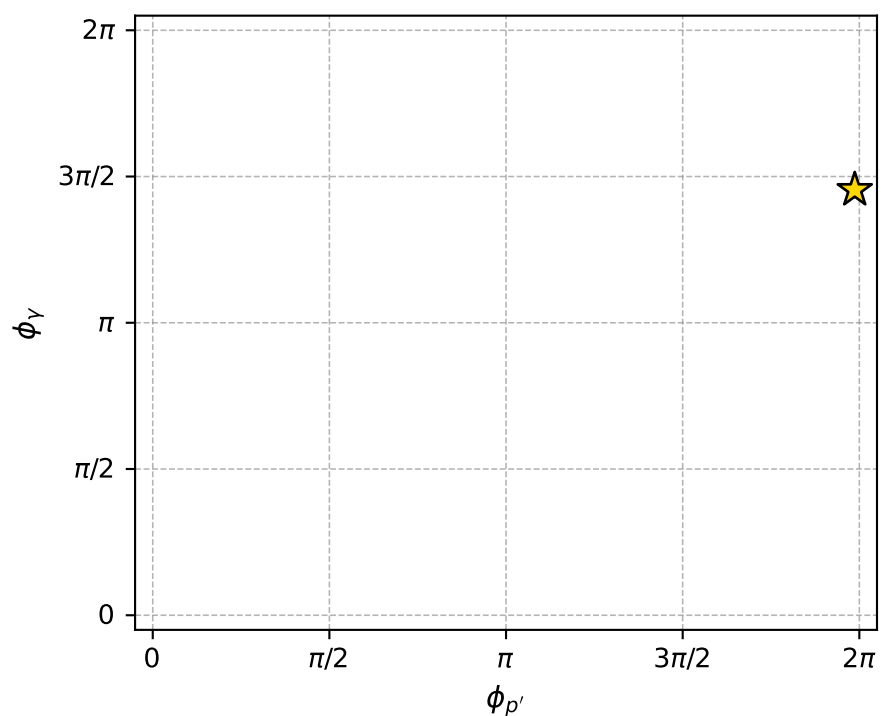
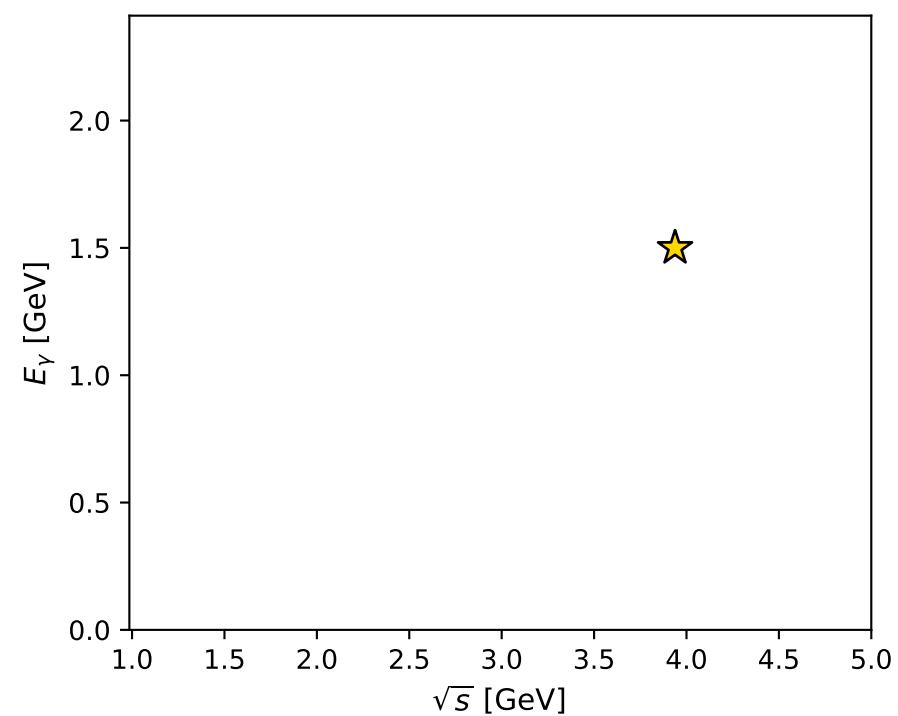
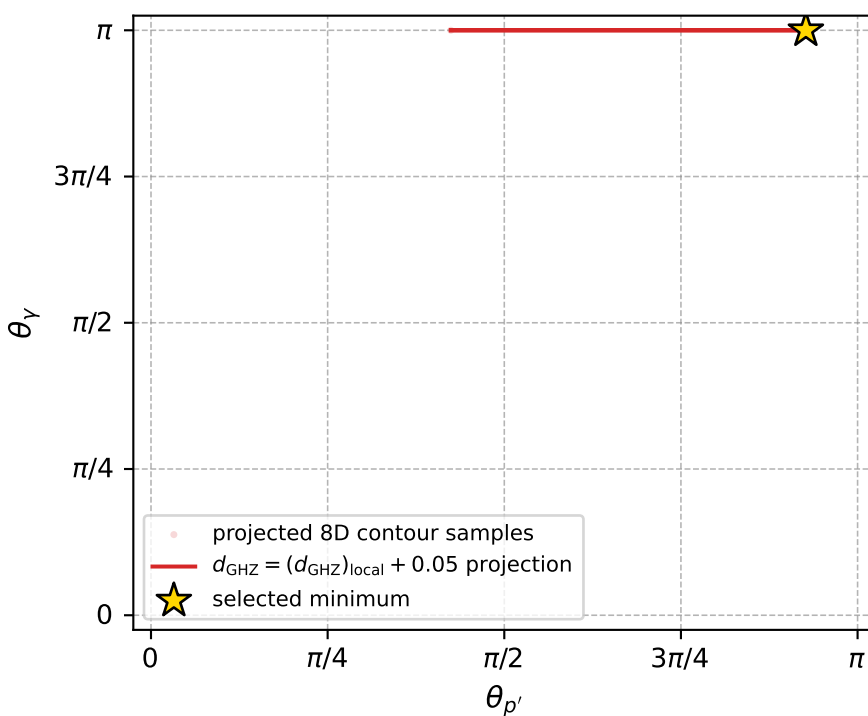
Transverse plane



Leading final-state amplitudes (N=2, retained=98.7%)



electron: polarization cluster P3, local minimum 118; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.4315912\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000044$
 above global minimum = $2.203245\text{e-}08$
 8D contour samples = 122

selected phase-space point:

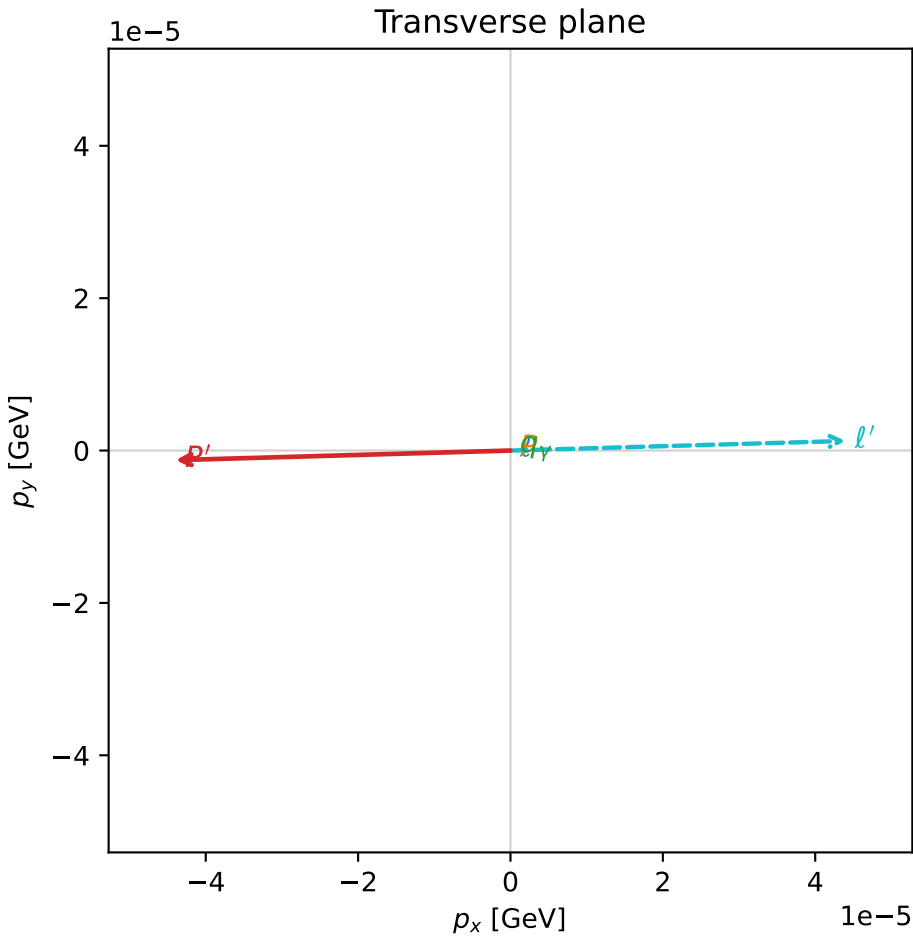
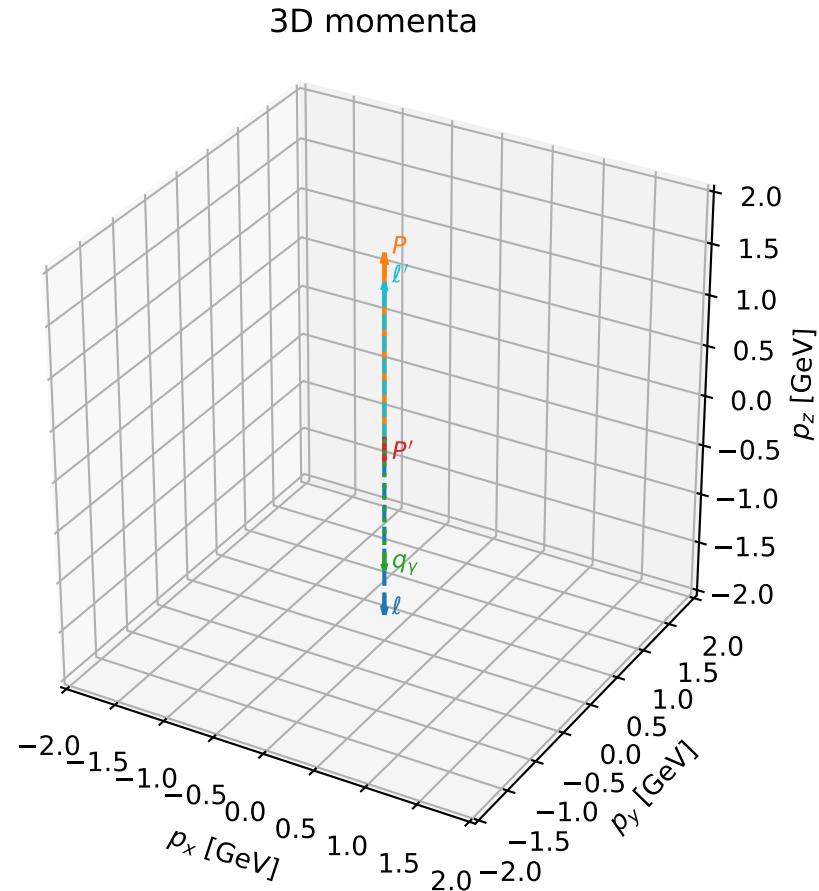
$\text{sqrt_s} = 3.938138$
 $\text{theta_p_out} = 2.911755$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.499949$
 $\text{phi_p_out} = 6.242516$
 $\text{phi_gamma_out} = 4.569536$
 $\text{alpha_e} = 2.399331$
 $\text{alpha_p} = 0.7422814$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

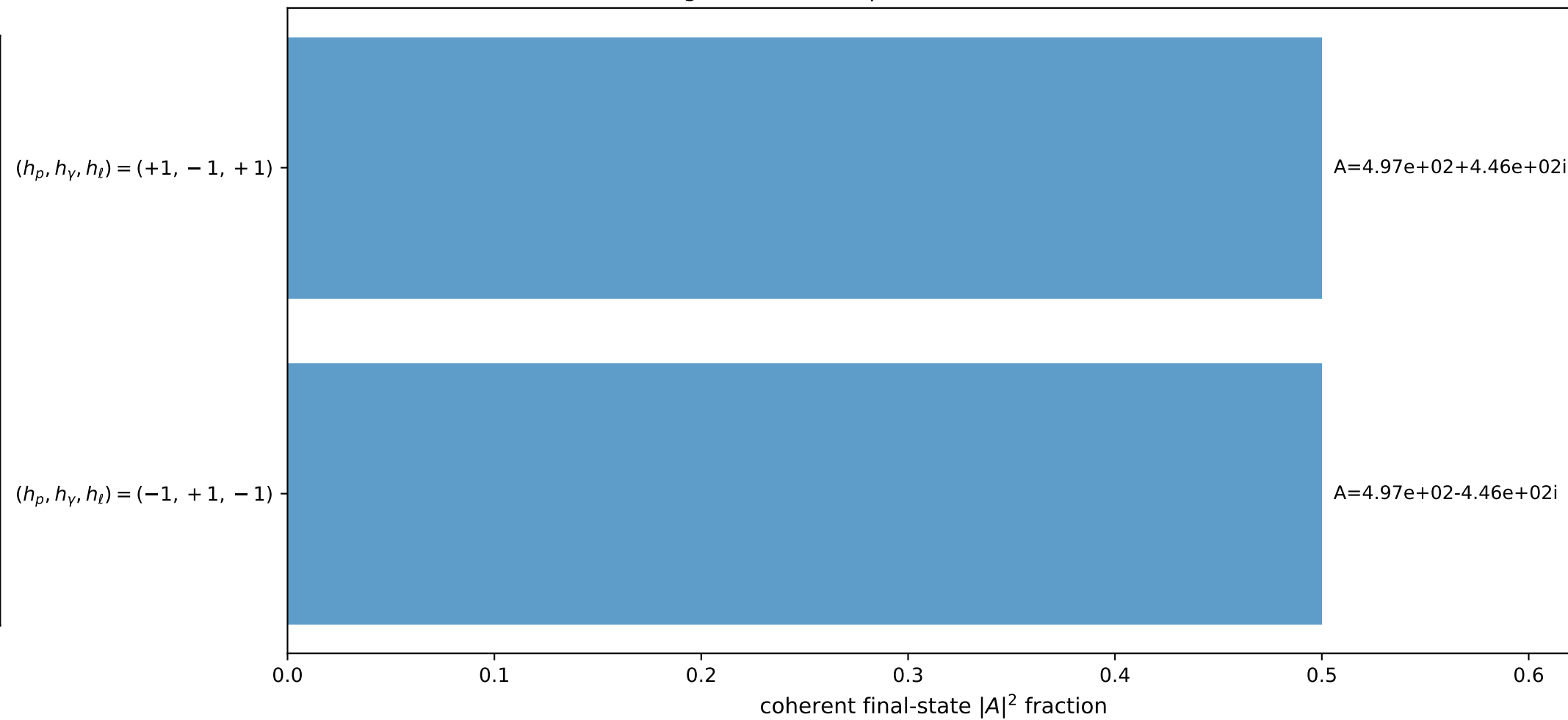
dghz_mixing_angles_polarization_03_local_minimum_124 $d_{\{\text{rm GHZ}\}}=4.80438\text{e-}08$
region: polarization_cluster_03_local_minimum_124
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3690498$ rad
incoming lepton: $-0.716138|+\rangle +0.697959|-\rangle$
proton mixing angle: $\alpha_p=0.77252117$ rad
incoming proton: $0.716153|+\rangle +0.697943|-\rangle$

$s=14.4913$, $\sqrt{s}=3.80674$, $|\vec{P}|=1.78781$, $|\vec{P}'|=0.198316$
 $\theta_{P'}=3.14137$, $\theta_Y=3.14159$, $\phi_{P'}=3.17027$, $\phi_Y=0.760181$
 $E_Y=1.32485$, $\theta_{\ell'}=2.88699\text{e-}05$, $\phi_{\ell'}=0.0286774$
 $Q^2=10.8925$, $x_B=0.843897$, $t=-2.82066$, $W^2=2.89472$, $y=0.948272$

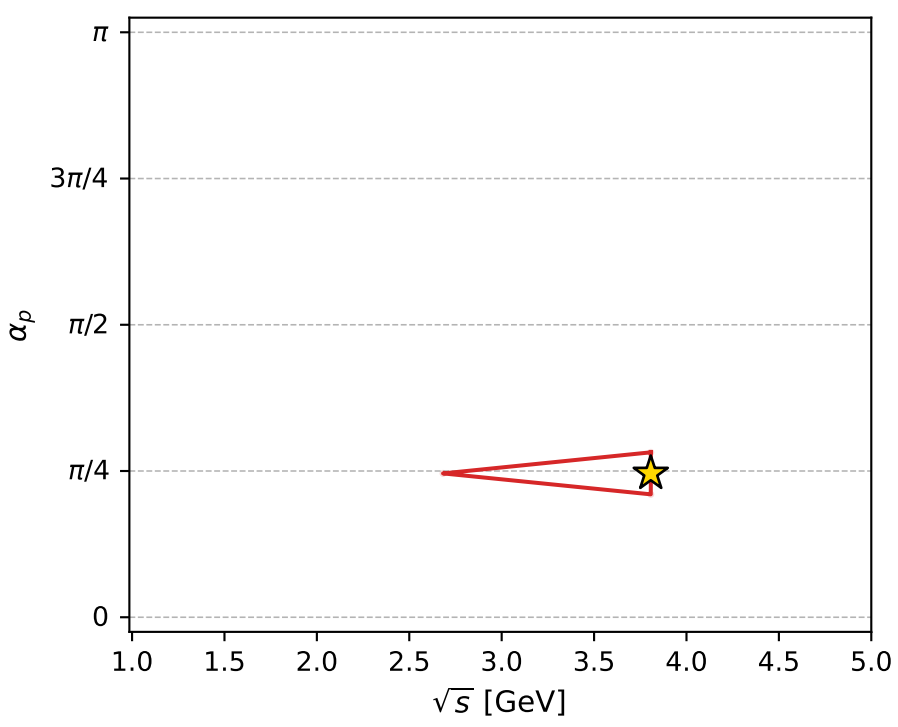
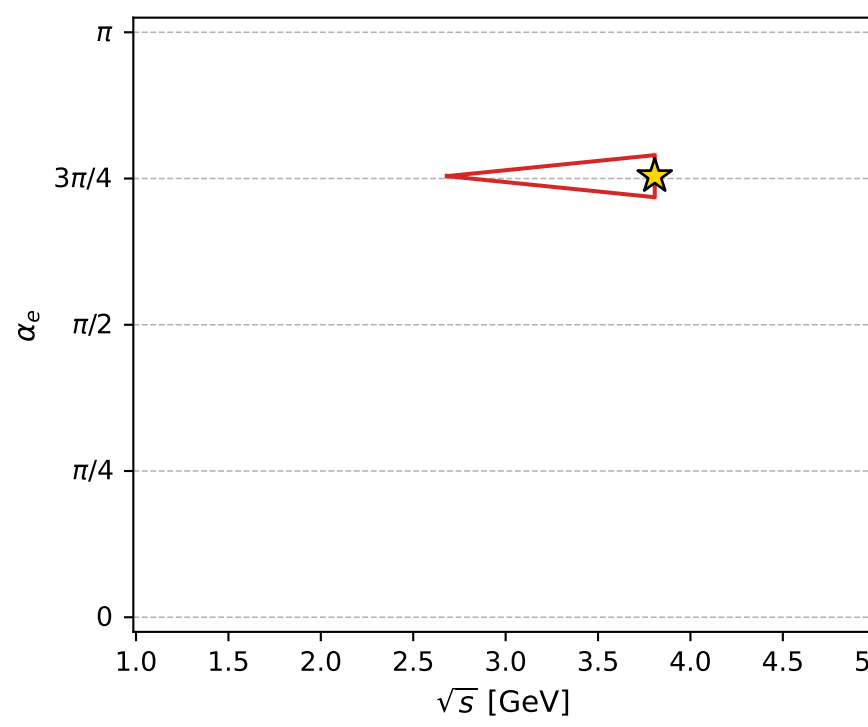
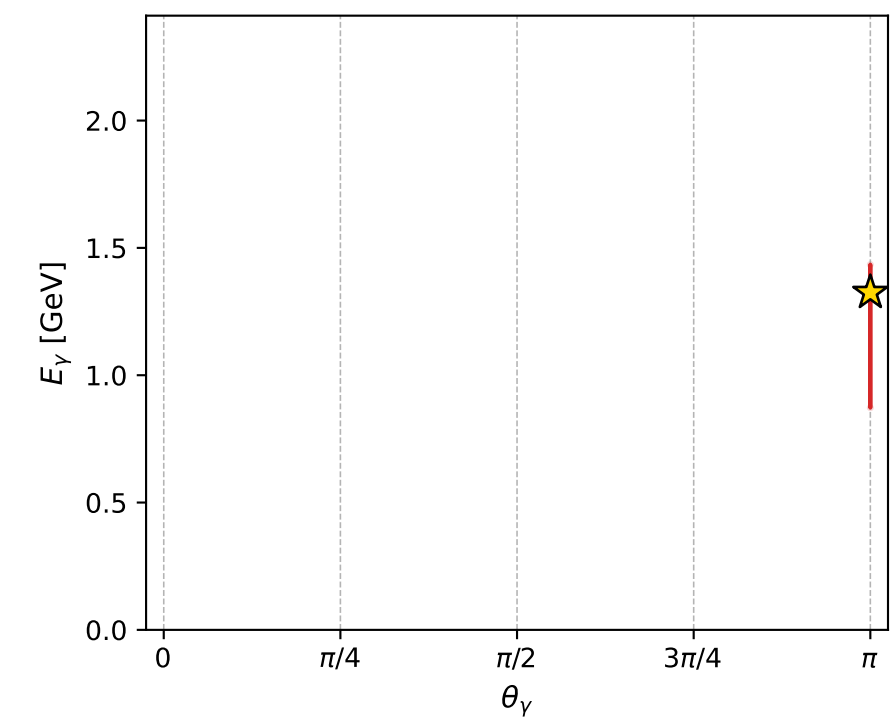
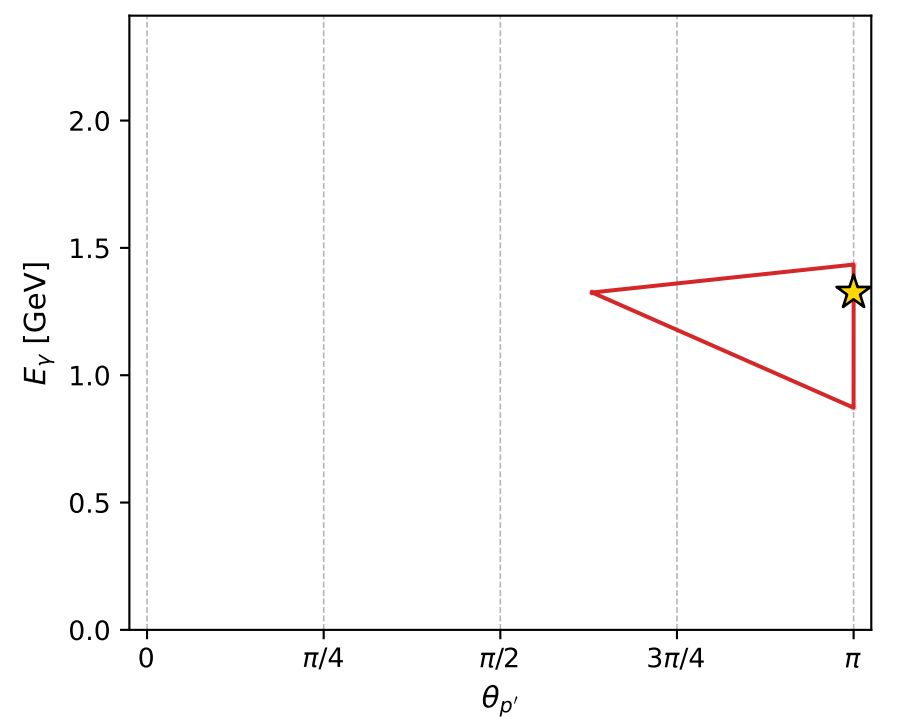
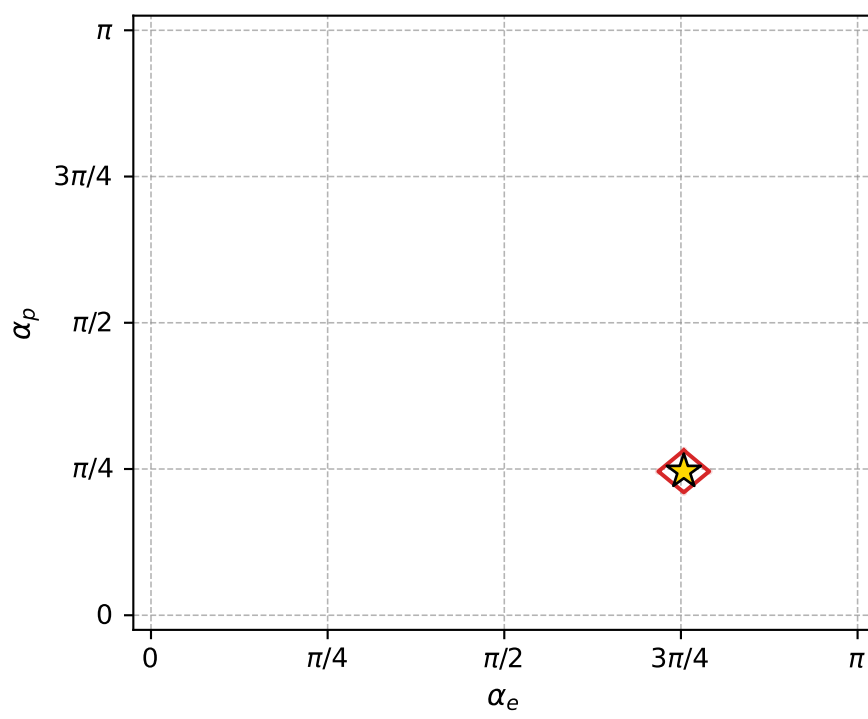
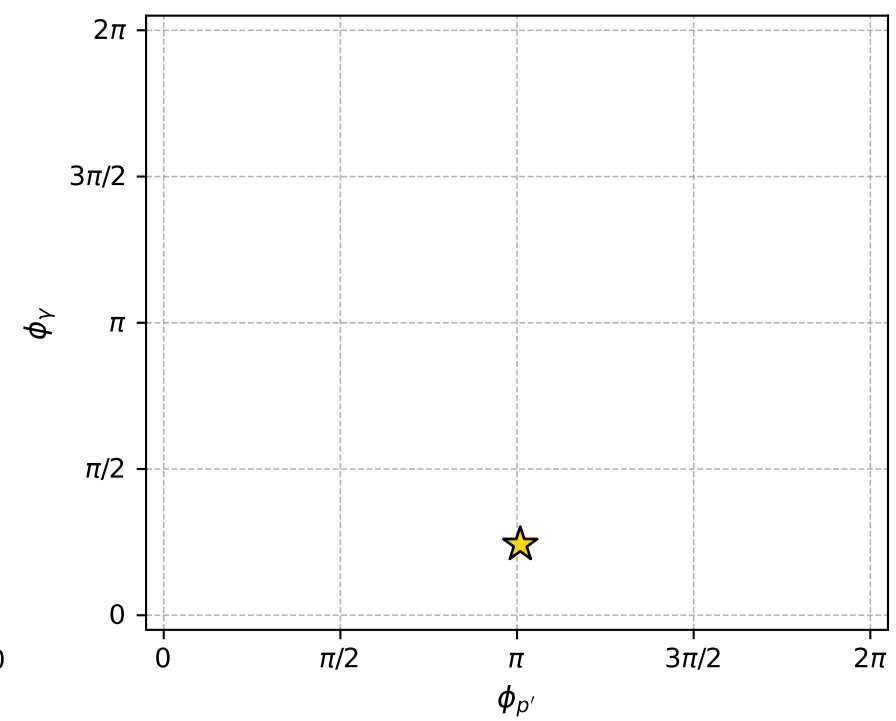
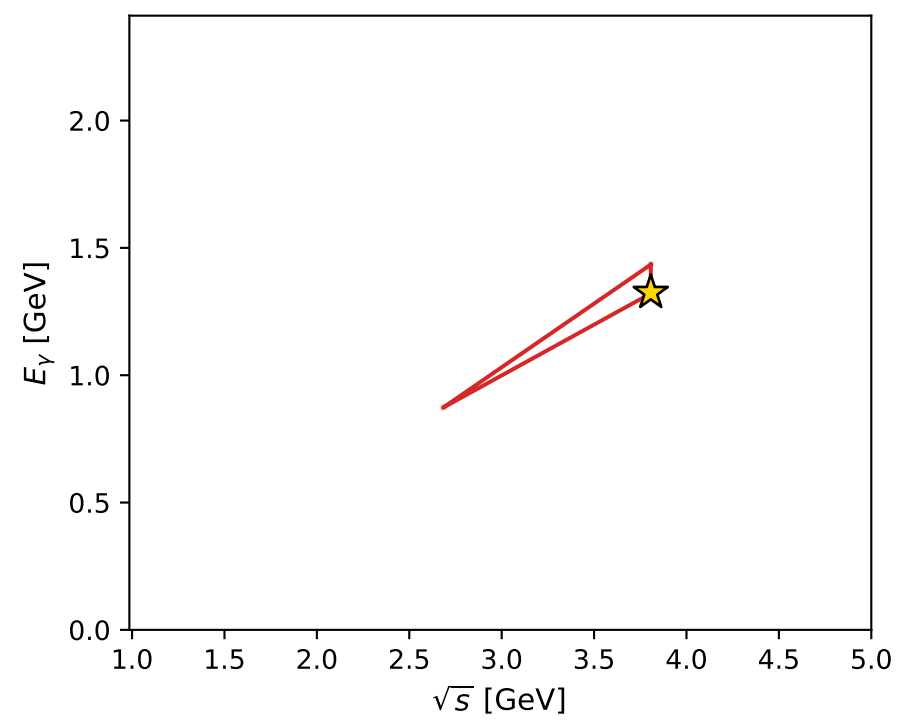
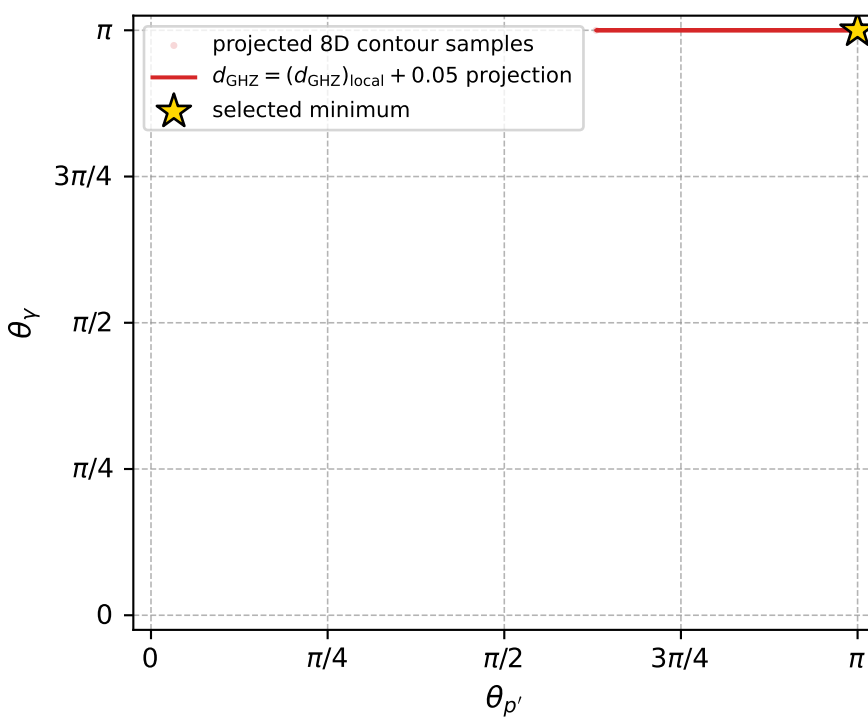
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7878$
 P $E= 2.0189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7878$
 ℓ' $E= 1.5232$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5232$
 P' $E= 0.9587$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1983$
 q_Y $E= 1.3248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3248$



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 124; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.8043829\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000048$
 above global minimum = $2.5760367\text{e-}08$
 8D contour samples = 110

selected phase-space point:

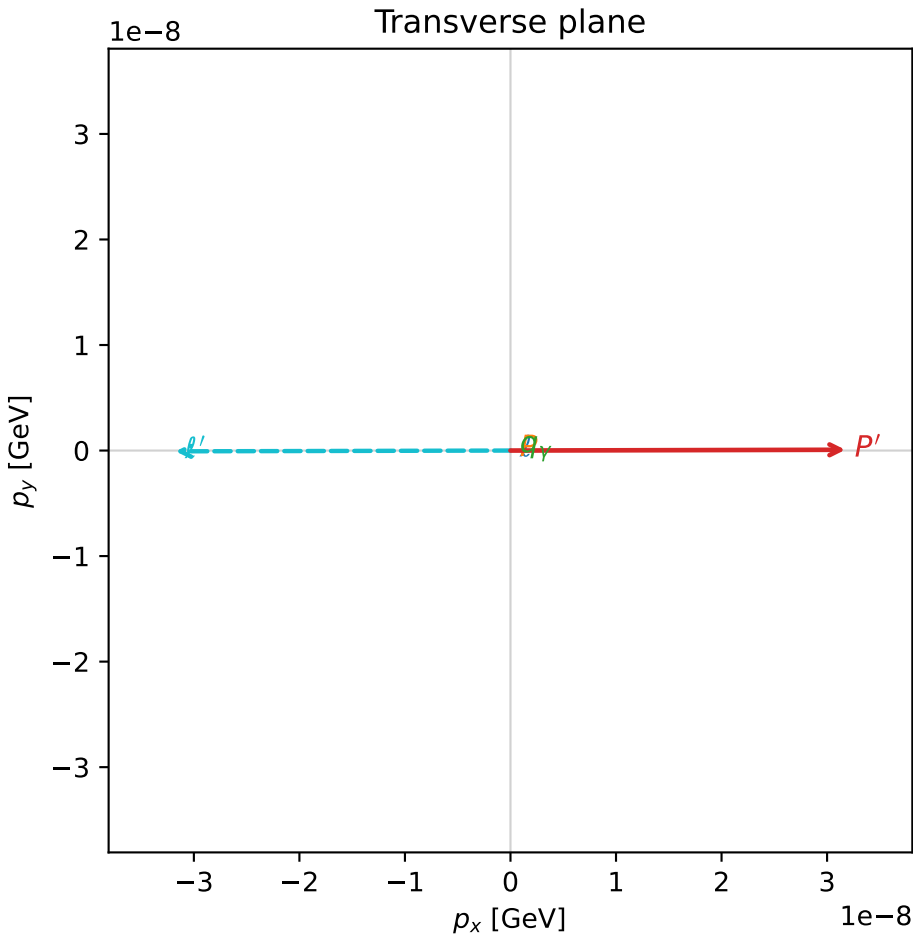
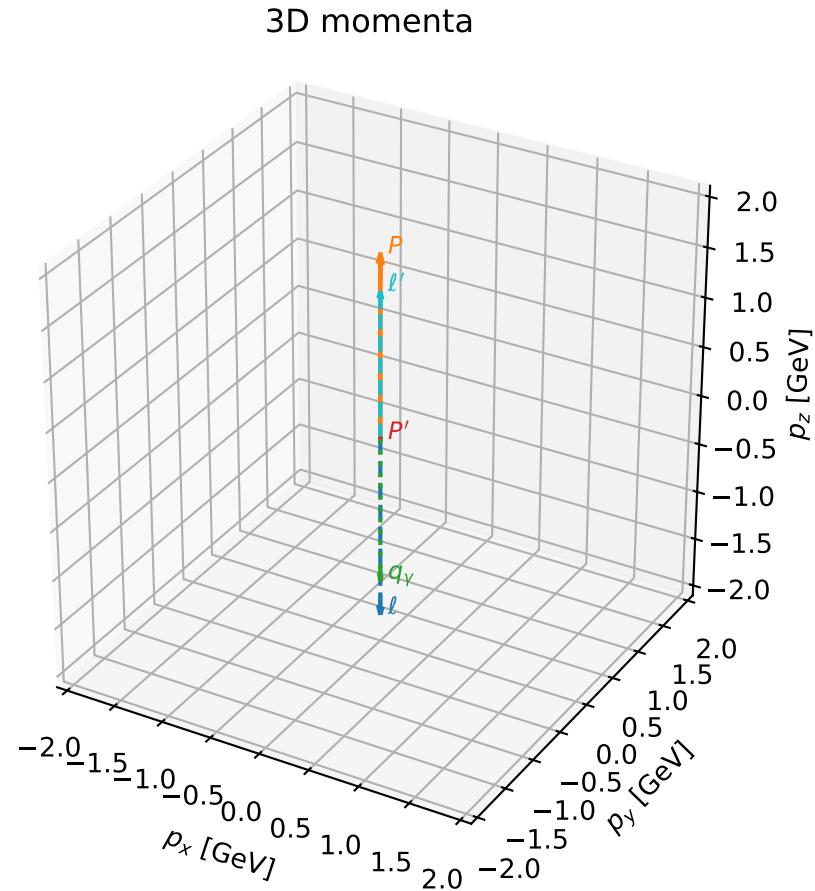
$\text{sqrt_s} = 3.806742$
 $\text{theta_p_out} = 3.141371$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.324845$
 $\text{phi_p_out} = 3.17027$
 $\text{phi_gamma_out} = 0.7601815$
 $\alpha_e = 2.36905$
 $\alpha_p = 0.7725212$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_128 $d_{\{\text{rm GHZ}\}}=5.13615\text{e-}08$
 region: polarization_cluster_03_local_minimum_128
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.5301823$ rad
 incoming lepton: $-0.818839|+\rangle +0.574023|-\rangle$
 proton mixing angle: $\alpha_p=0.61140686$ rad
 incoming proton: $0.818841|+\rangle +0.57402|-\rangle$

$s=15.1318$, $\sqrt{s}=3.88997$, $|\vec{P}|=1.83189$, $|\vec{P}'|=0.000225325$
 $\theta_{P'}=3.14145$, $\theta_Y=3.14159$, $\phi_{P'}=0.00233824$, $\phi_Y=5.84287$
 $E_Y=1.47587$, $\theta_{\ell'}=1.49012\text{e-}08$, $\phi_{\ell'}=3.14393$
 $Q^2=10.8162$, $x_B=0.79623$, $t=-2.10208$, $W^2=3.64791$, $y=0.953148$

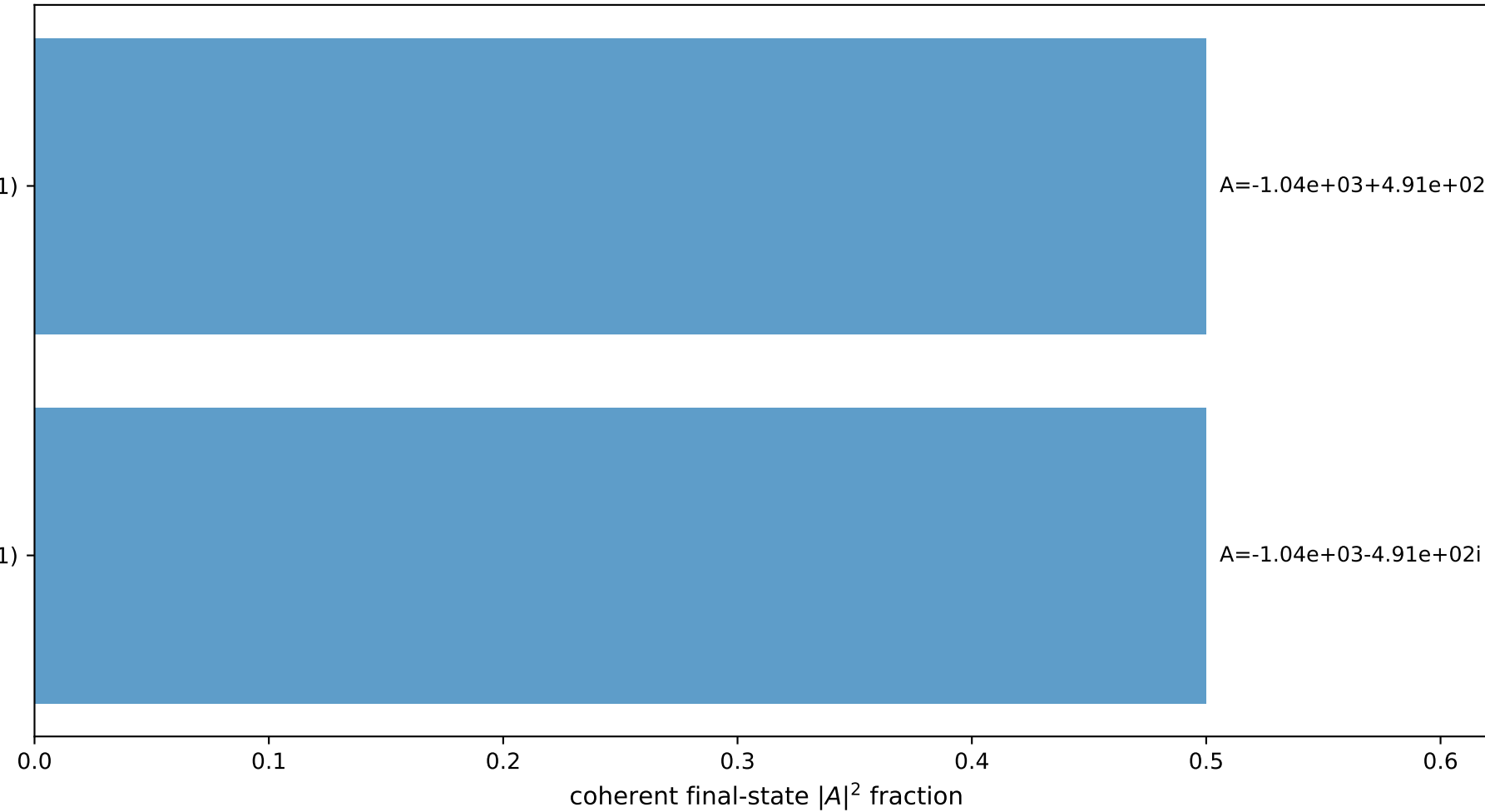
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8319$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8319$
 P $E= 2.0581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8319$
 ℓ' $E= 1.4761$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4761$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_Y $E= 1.4759$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4759$



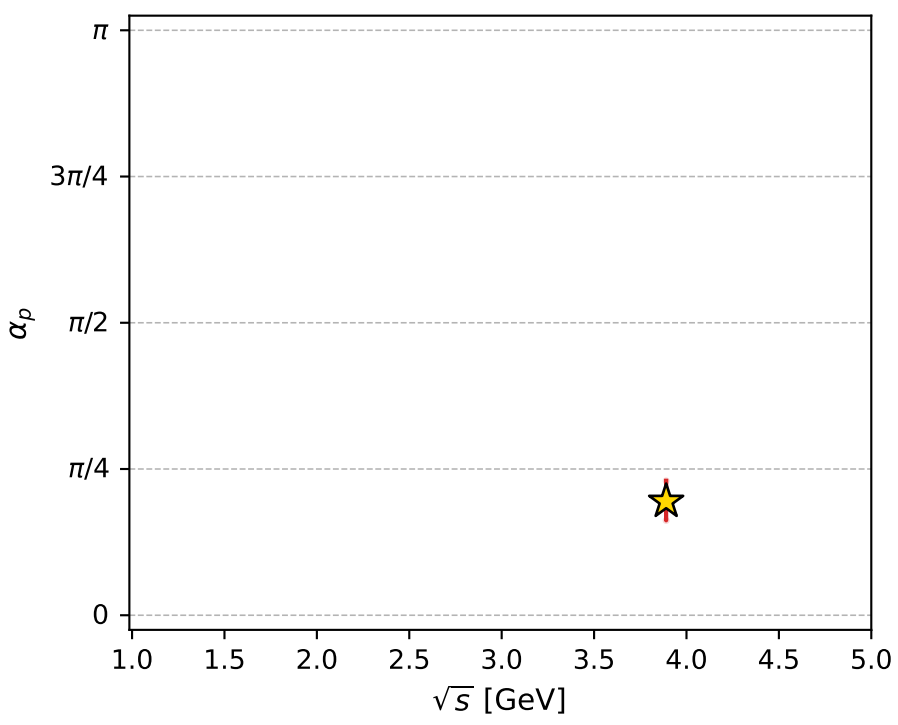
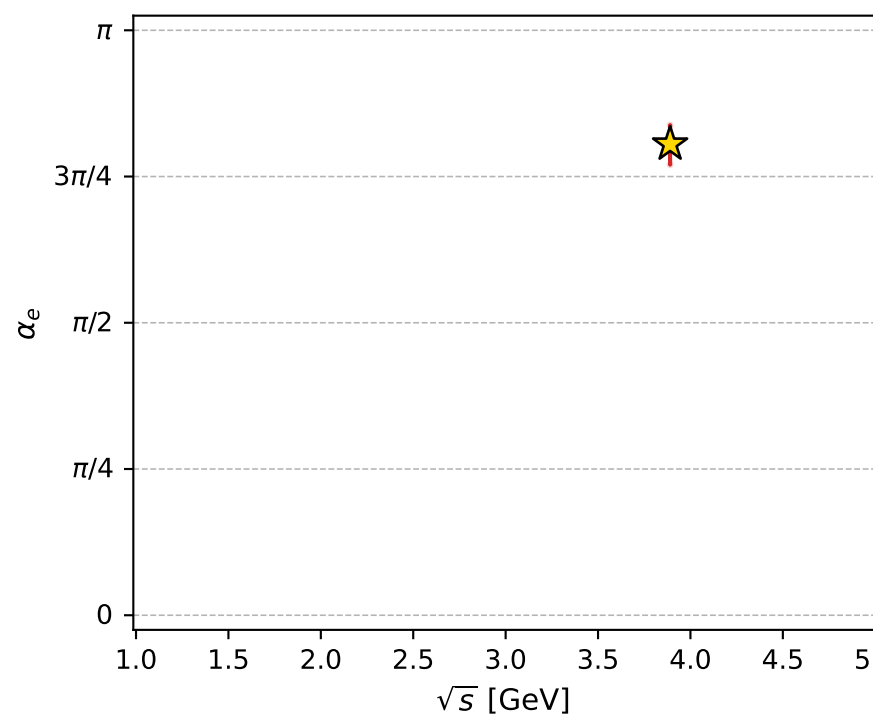
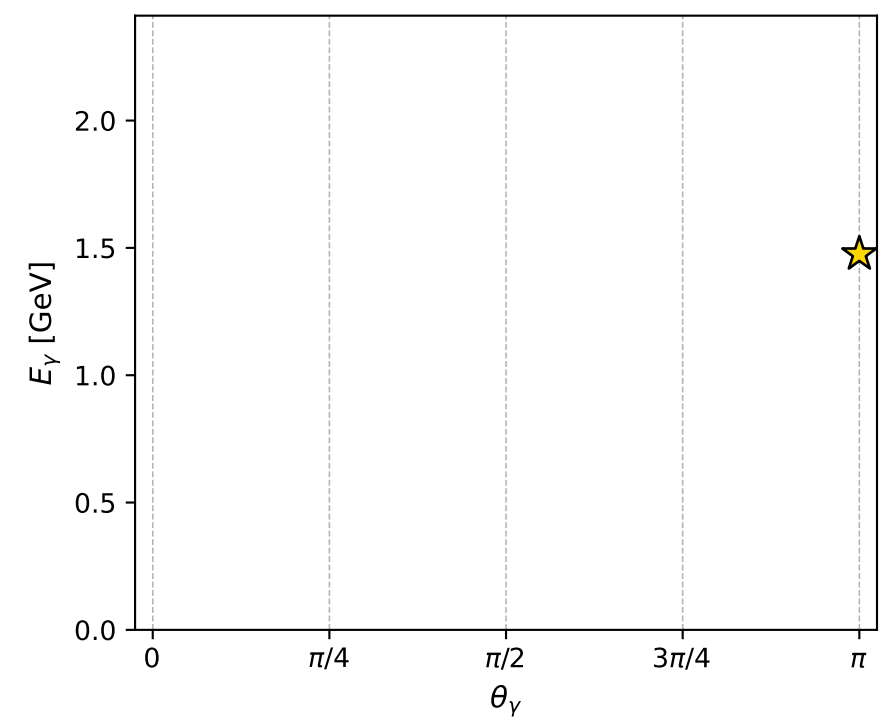
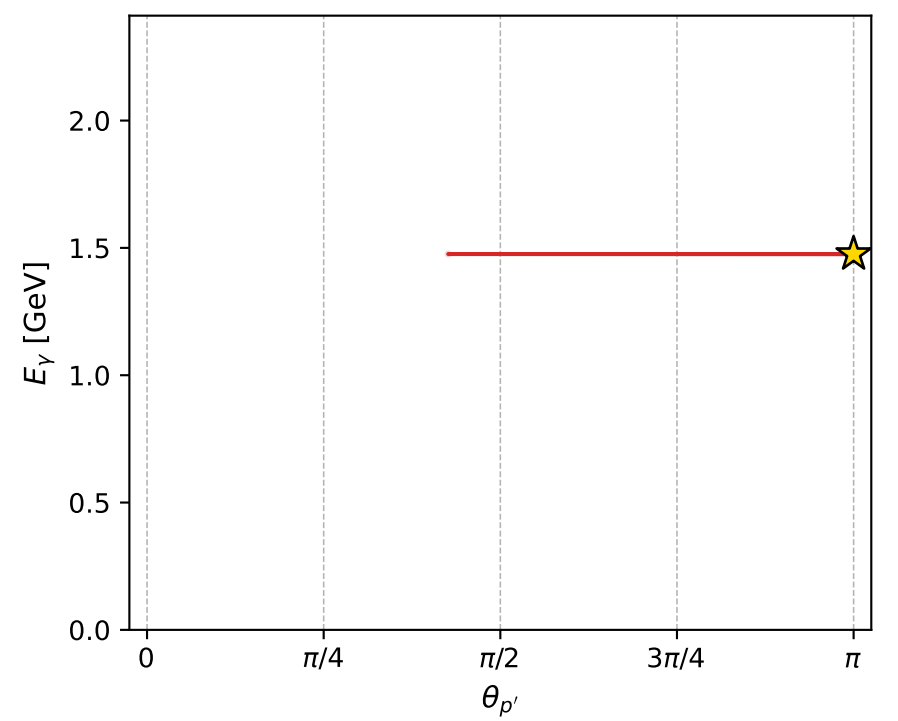
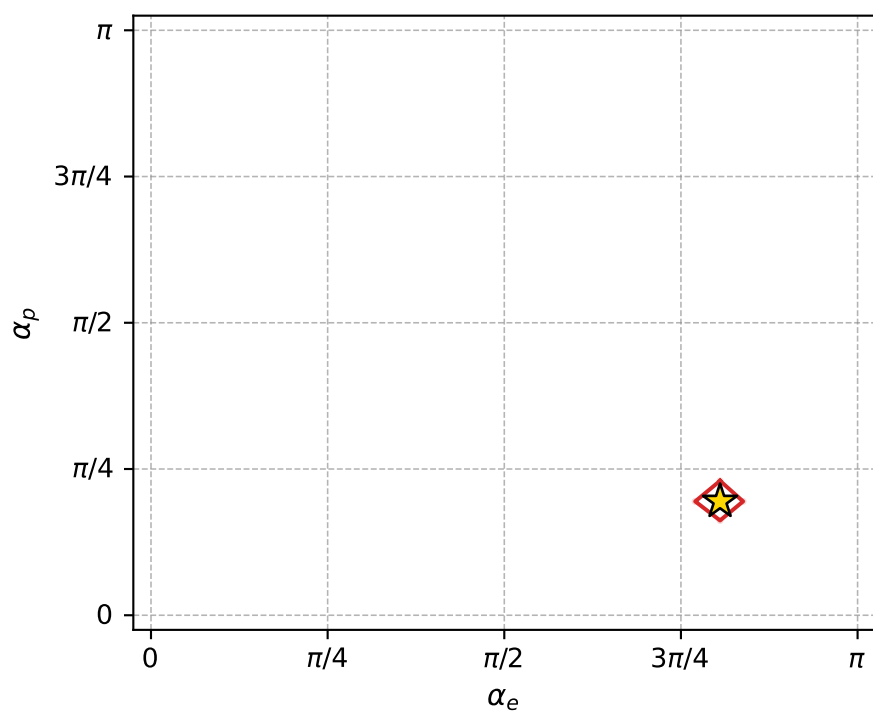
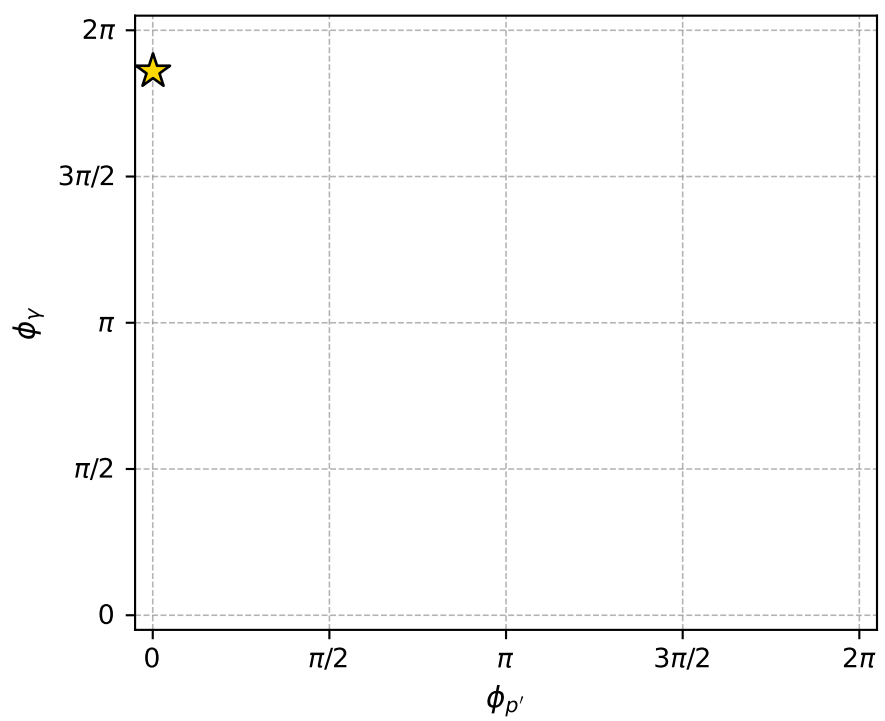
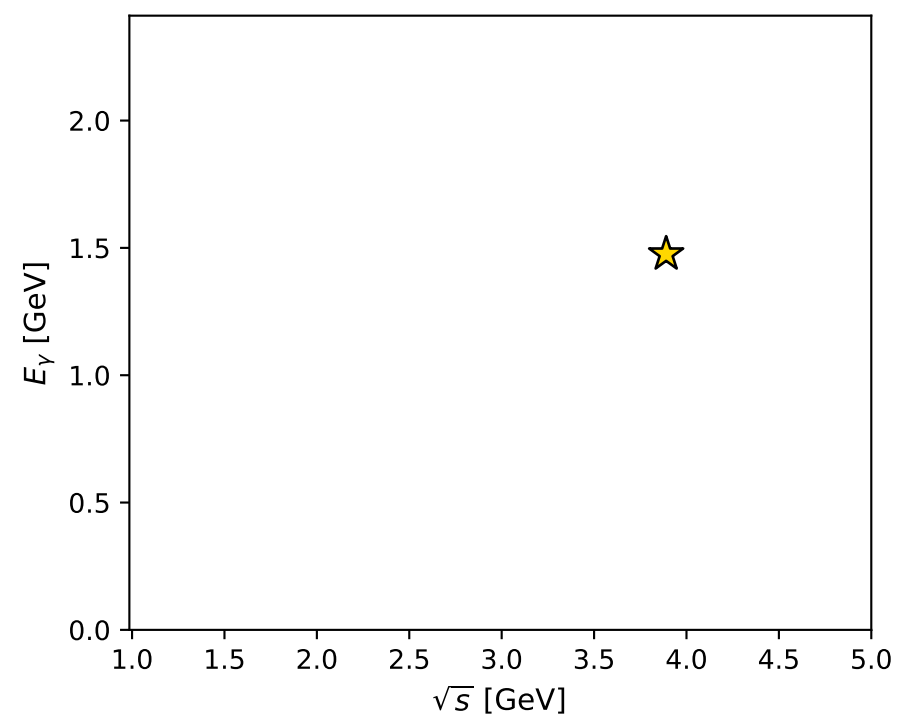
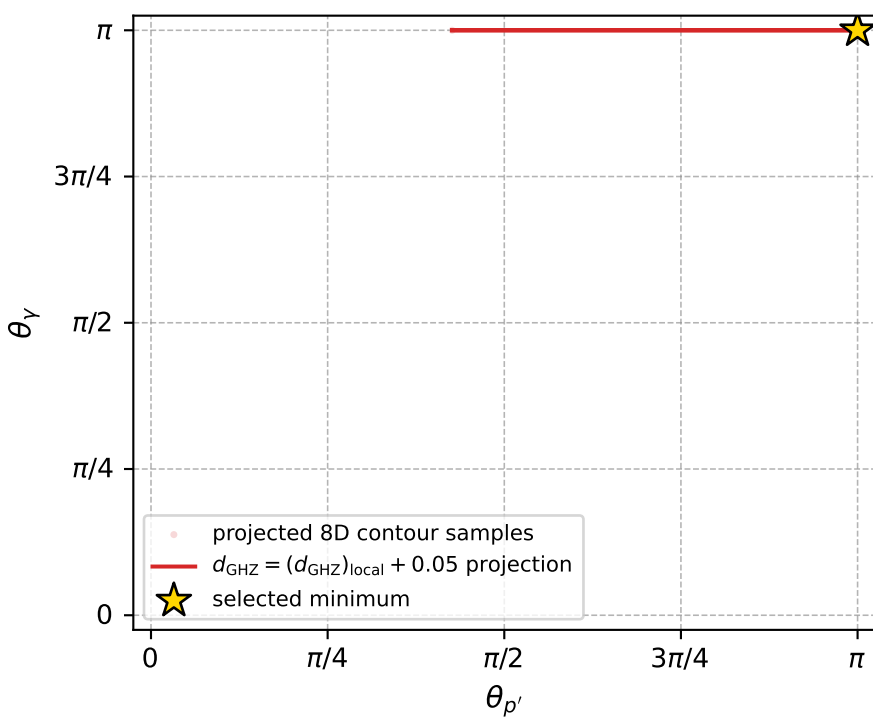
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 128; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.1361475\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000051$
 above global minimum = $2.9078014\text{e-}08$
 8D contour samples = 115

selected phase-space point:

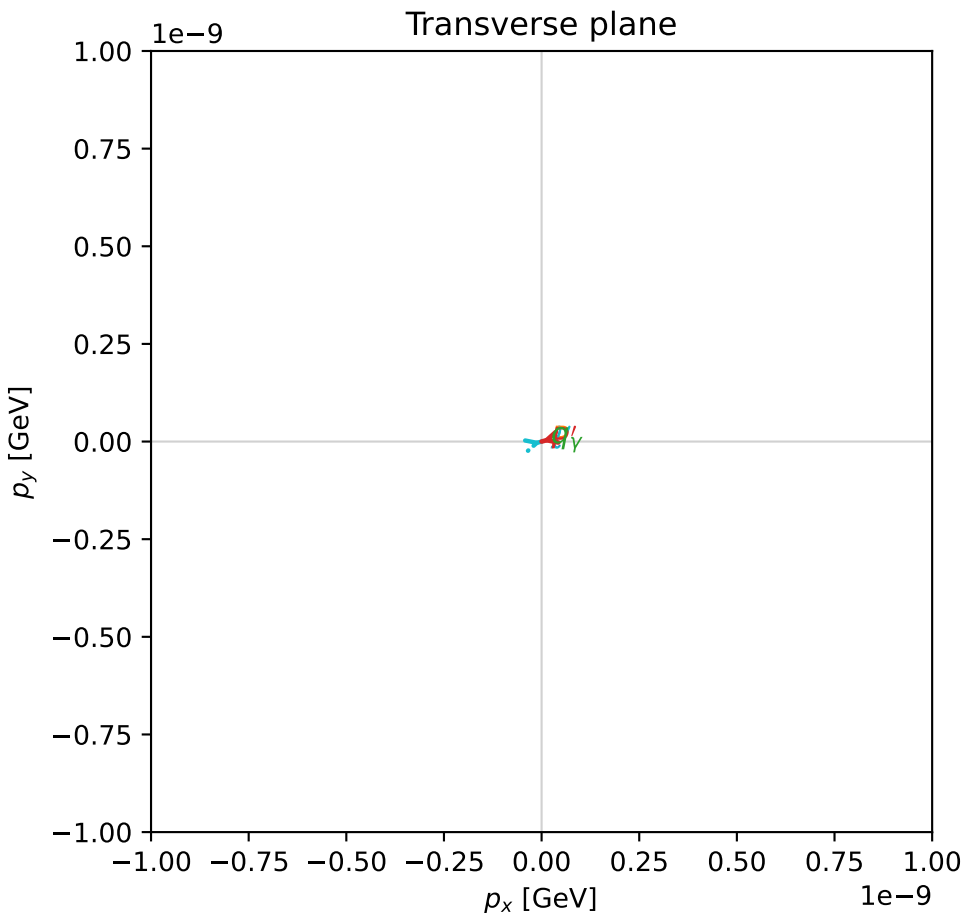
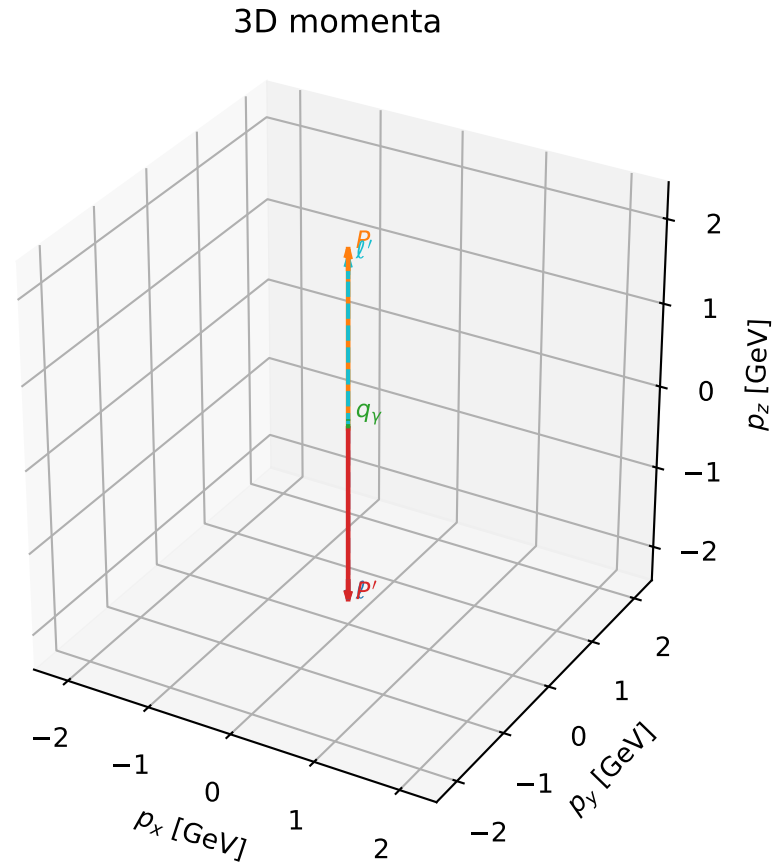
$\text{sqrt_s} = 3.889966$
 $\text{theta_p_out} = 3.141452$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.47587$
 $\text{phi_p_out} = 0.002338237$
 $\text{phi_gamma_out} = 5.842868$
 $\text{alpha_e} = 2.530182$
 $\text{alpha_p} = 0.6114069$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_131 d_{\rm GHZ}=5.31748e-08
region: polarization_cluster_03_local_minimum_131
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.8802171$ rad
incoming lepton: $-0.304507|+\rangle +0.95251|-\rangle$
proton mixing angle: $\alpha_p=0.30941263$ rad
incoming proton: $0.952513|+\rangle +0.304499|-\rangle$

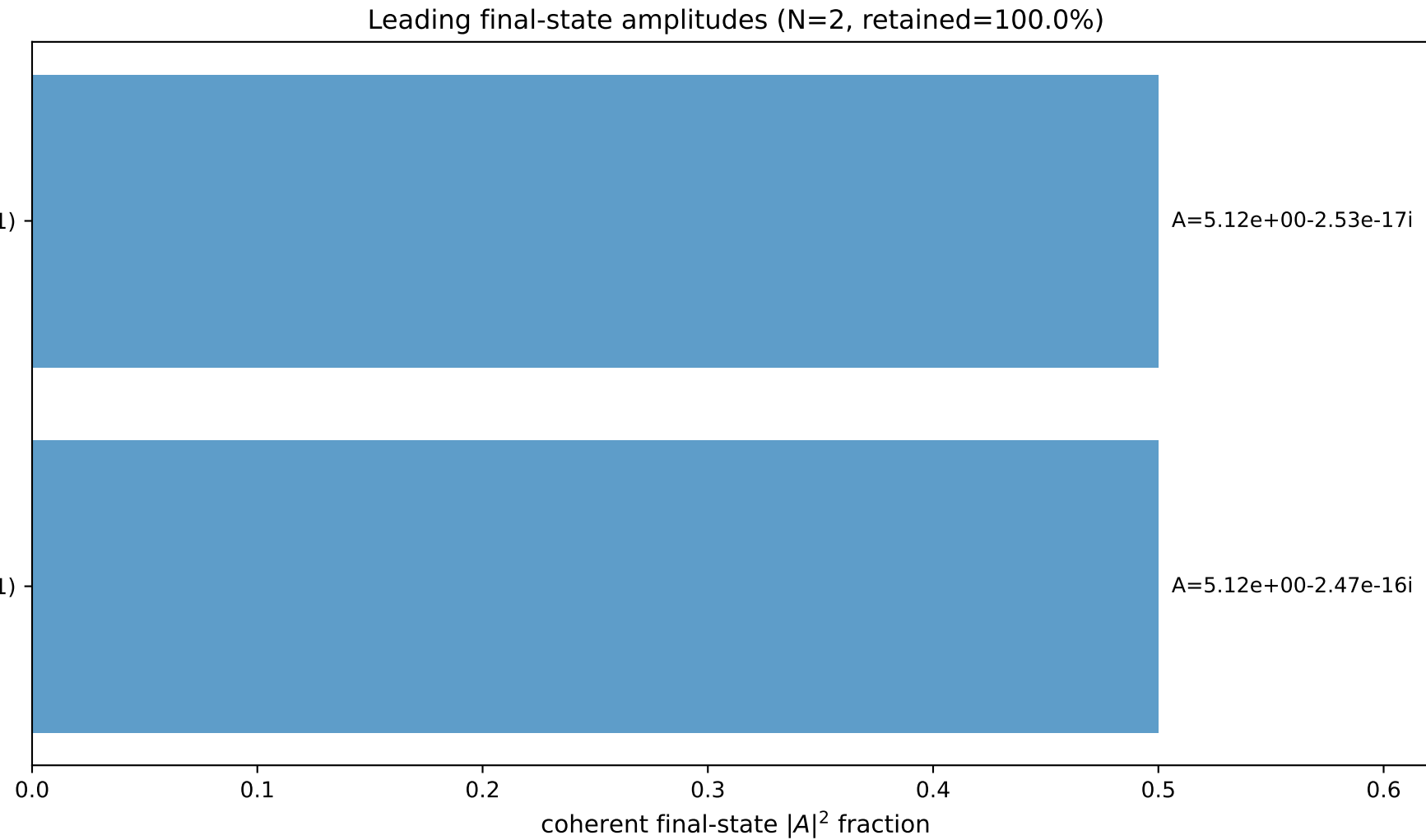
$s=19.5413$, $\sqrt{s}=4.42055$, $|\vec{P}|=2.11076$, $|\vec{P}'|=2.11076$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=3.40678$, $\phi_Y=1.45431$
 $E_Y=0.105533$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.265183$
 $Q^2=16.9302$, $x_B=0.947768$, $t=-17.8212$, $W^2=1.81287$, $y=0.957226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1108$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1108$
 P $E= 2.3098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1108$
 ℓ' $E= 2.0052$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0052$
 P' $E= 2.3098$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1108$
 q_Y $E= 0.1055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1055$

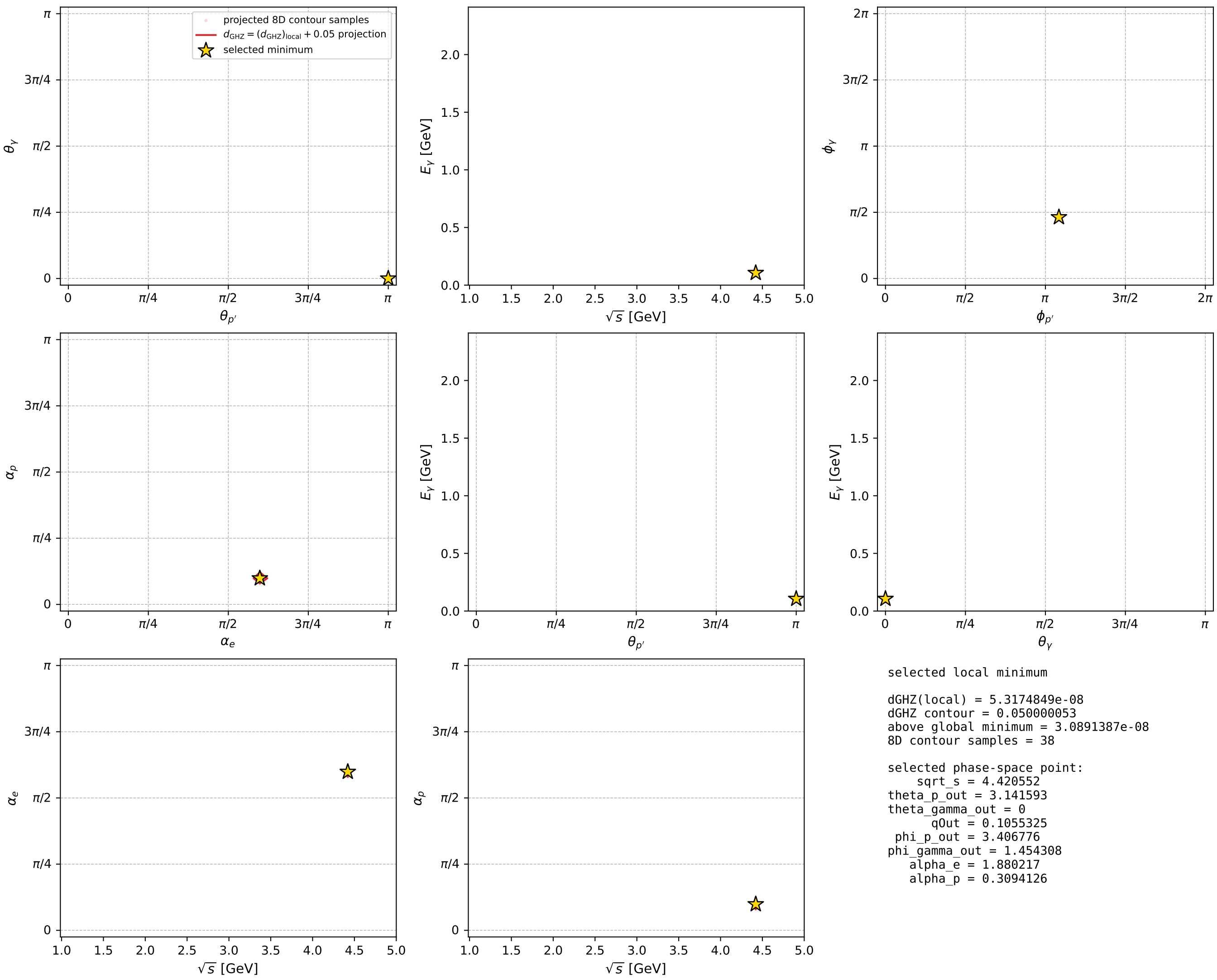


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P3, local minimum 131; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

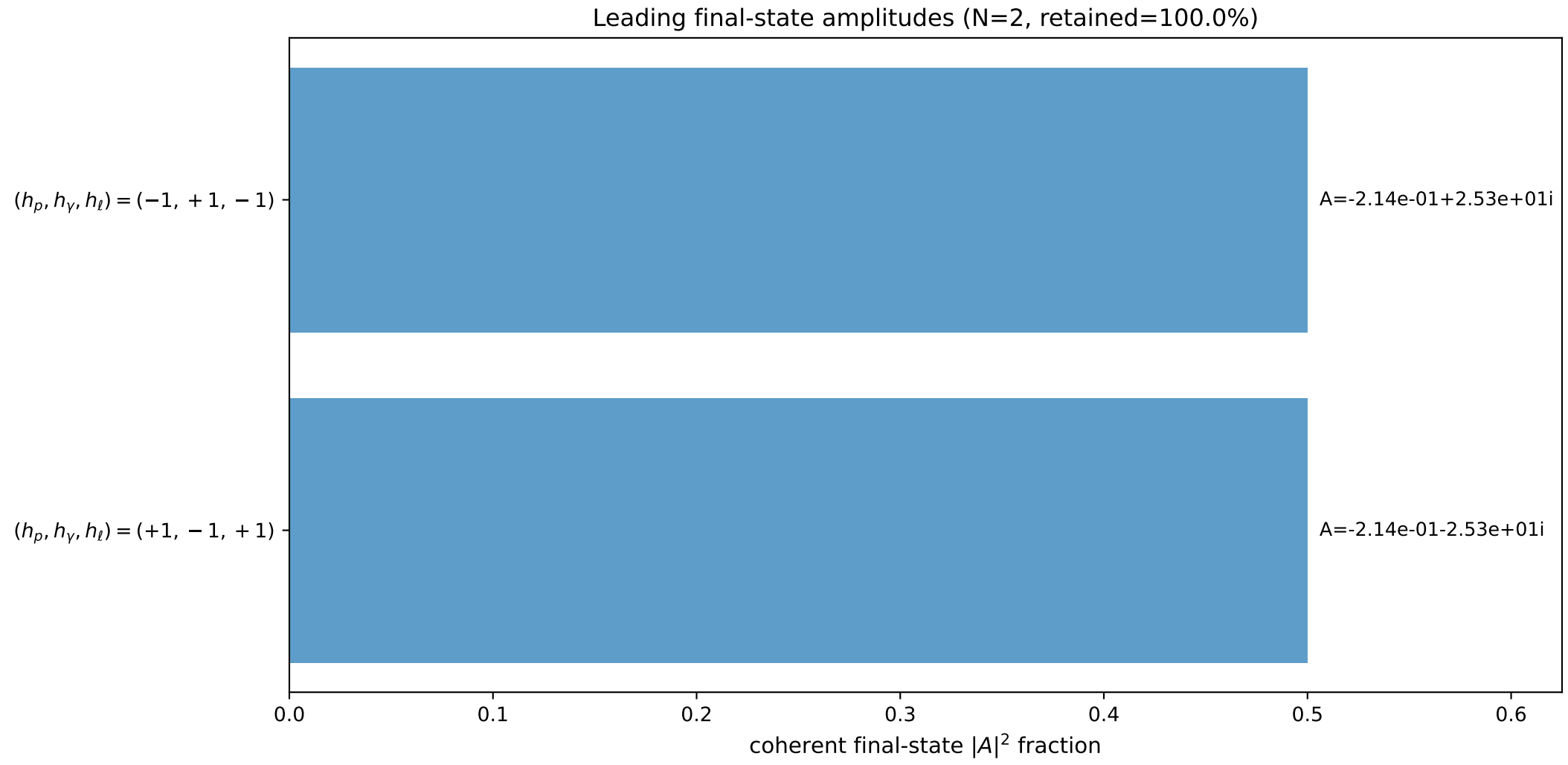
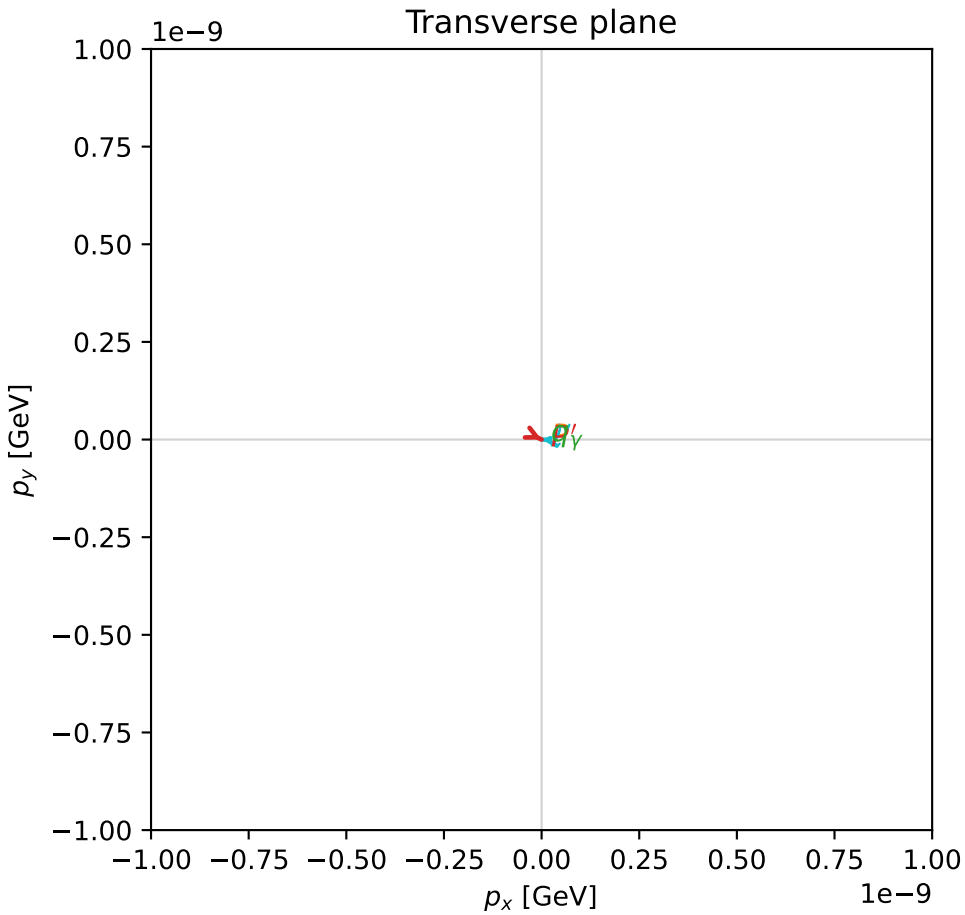
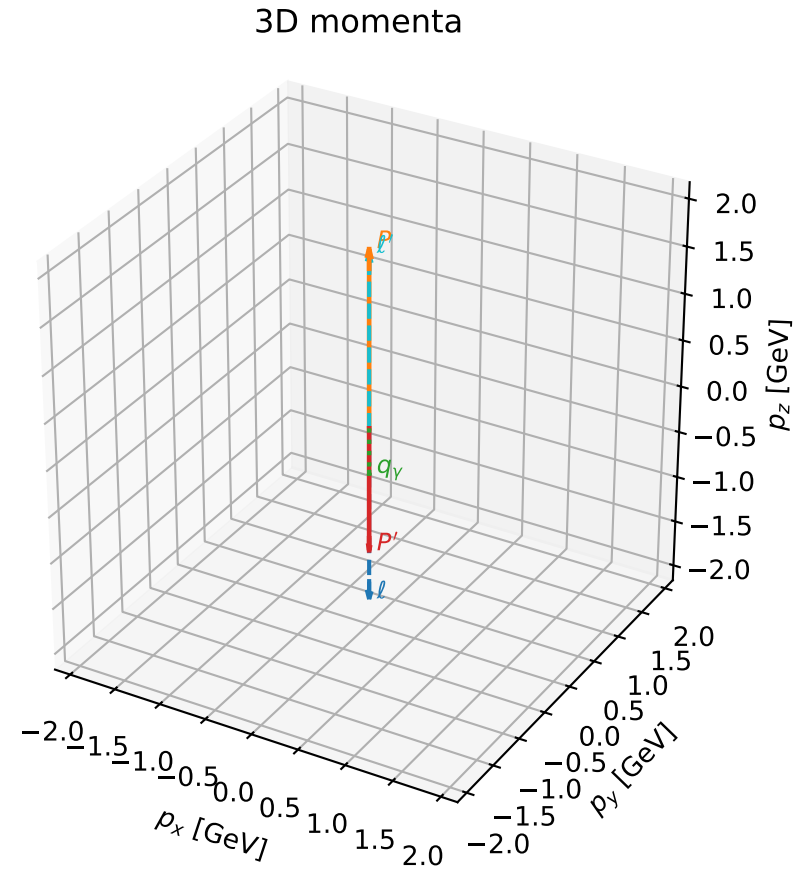


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

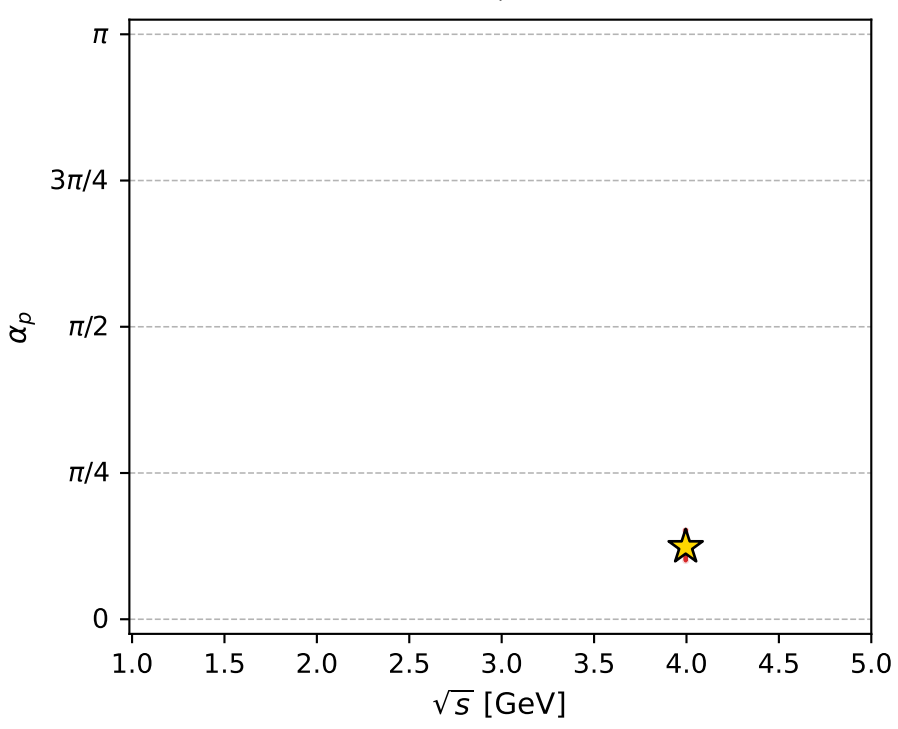
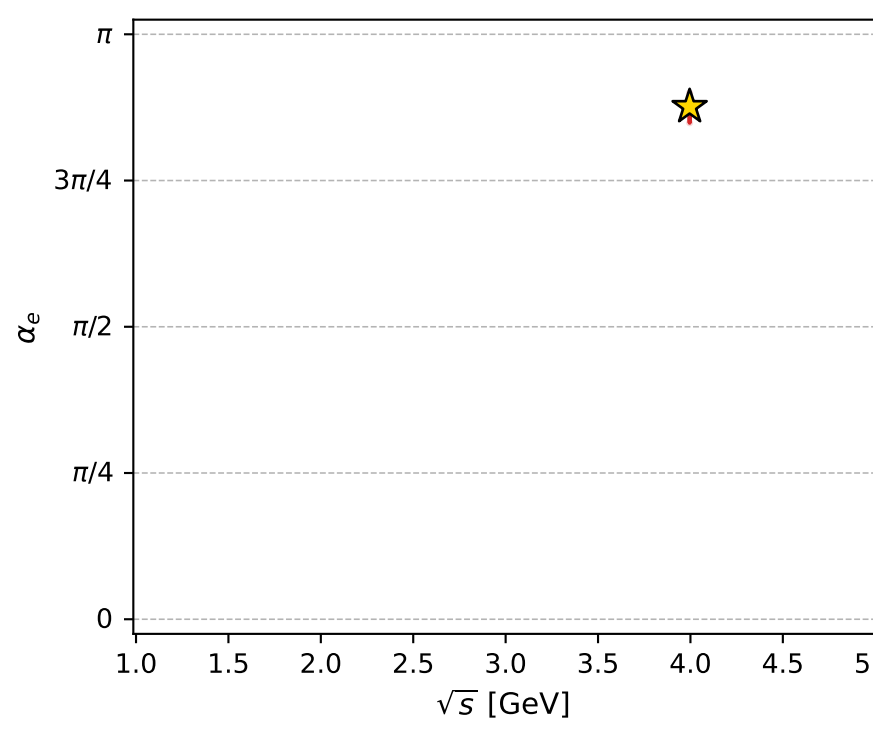
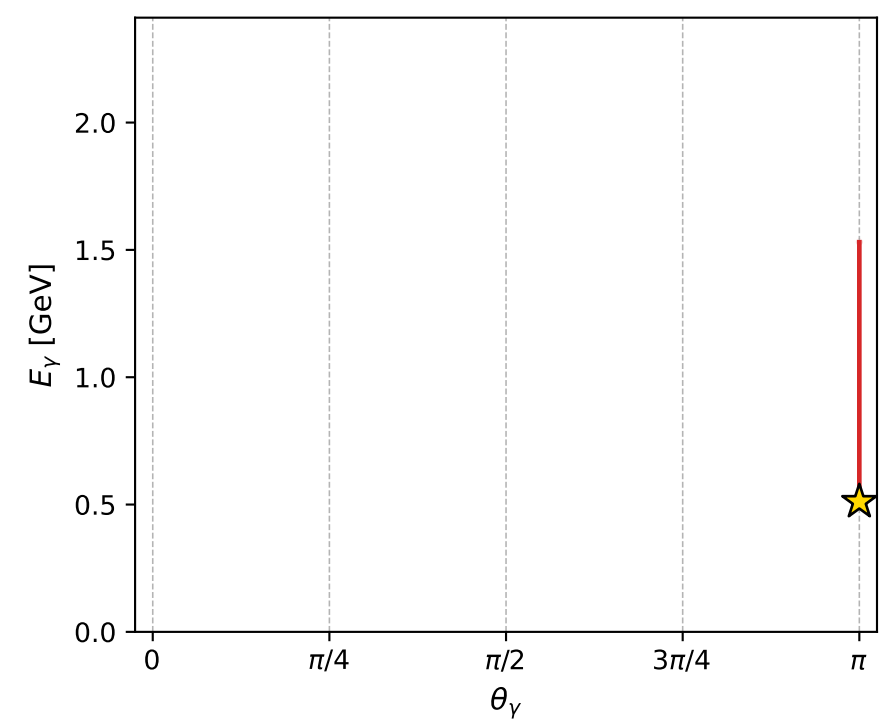
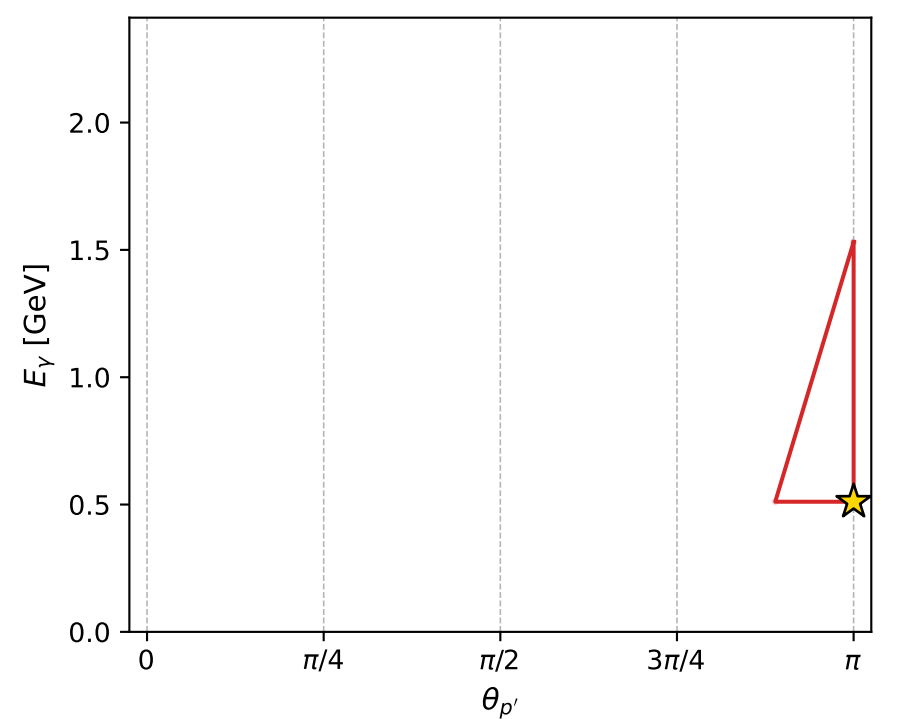
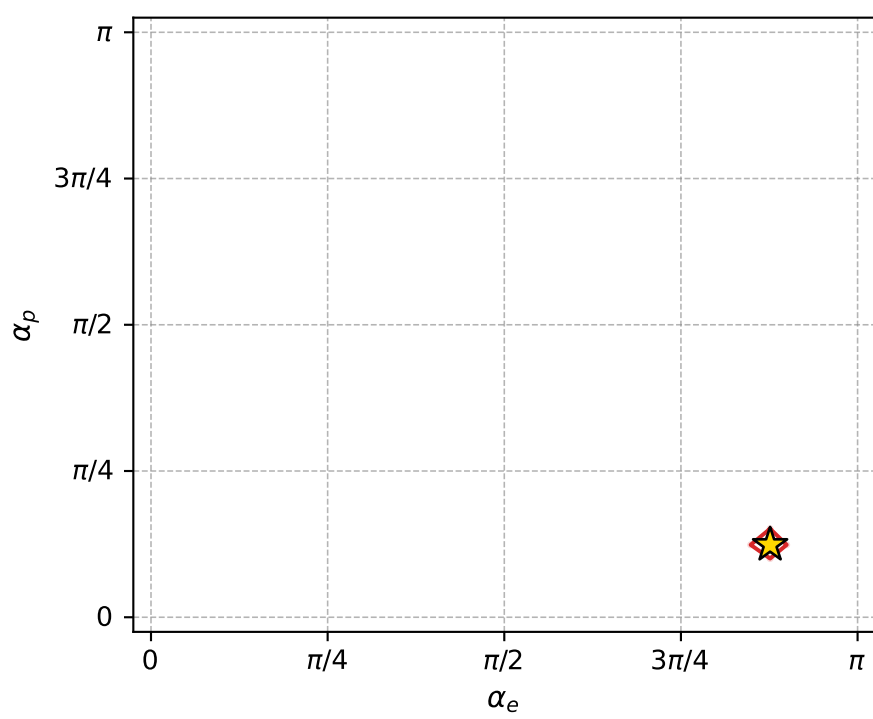
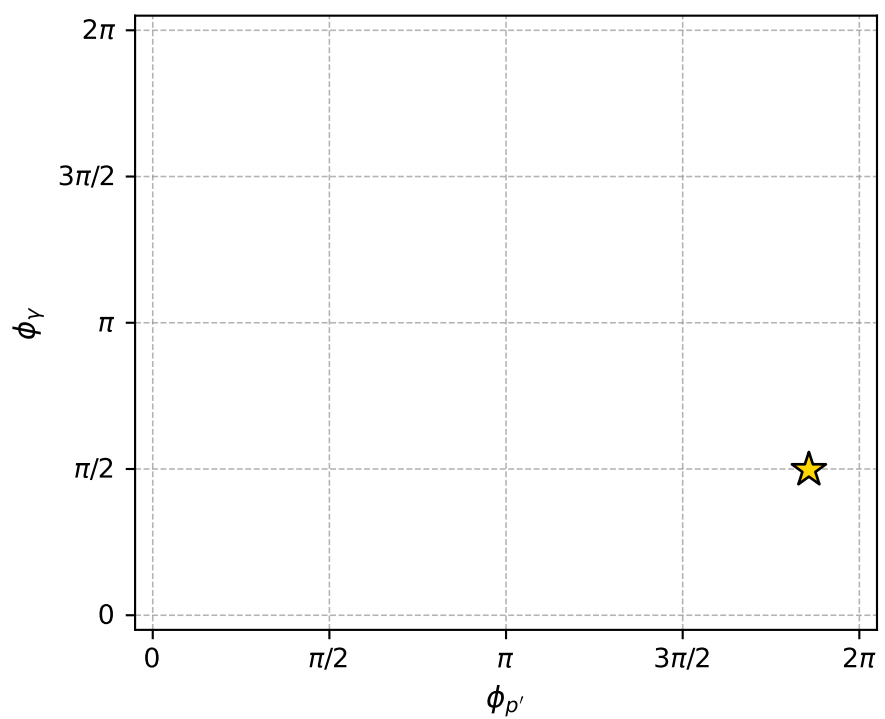
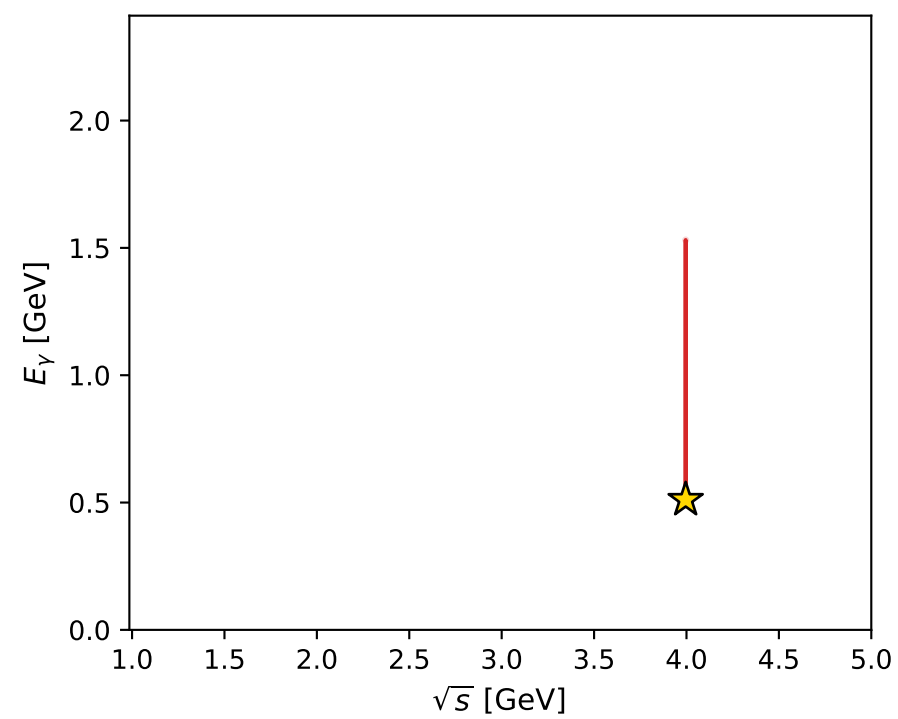
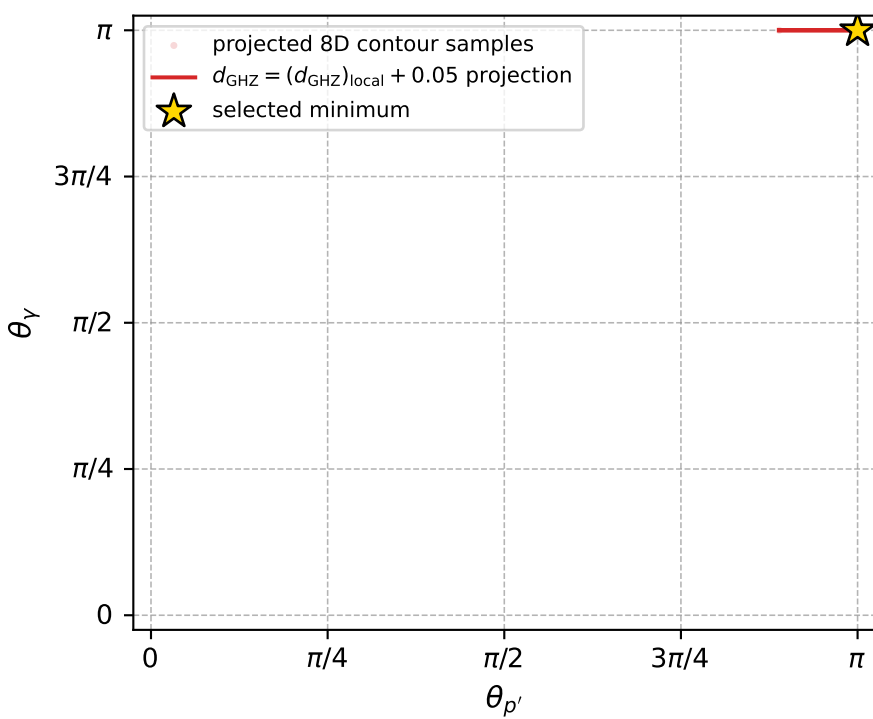
dghz_mixing_angles_polarization_03_local_minimum_132 $d_{\{\text{rm GHZ}\}}=5.32794\text{e-}08$
 region: polarization_cluster_03_local_minimum_132
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.7526505$ rad
 incoming lepton: $-0.925311|+\rangle +0.37921|-\rangle$
 proton mixing angle: $\alpha_p=0.38894583$ rad
 incoming proton: $0.925309|+\rangle +0.379213|-\rangle$

$s=15.965$, $\sqrt{s}=3.99562$, $|\vec{P}|=1.88771$, $|\vec{P}'|=1.33879$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=5.83465$, $\phi_Y=1.56232$
 $E_Y=0.511073$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.08424$
 $Q^2=13.968$, $x_B=0.978806$, $t=-10.1864$, $W^2=1.18229$, $y=0.945994$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8877$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8877$
 P $E= 2.1079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8877$
 ℓ' $E= 1.8499$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8499$
 P' $E= 1.6347$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3388$
 q_Y $E= 0.5111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5111$



electron: polarization cluster P3, local minimum 132; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.3279445\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000053$
 above global minimum = $3.0995983\text{e-}08$
 8D contour samples = 61

selected phase-space point:

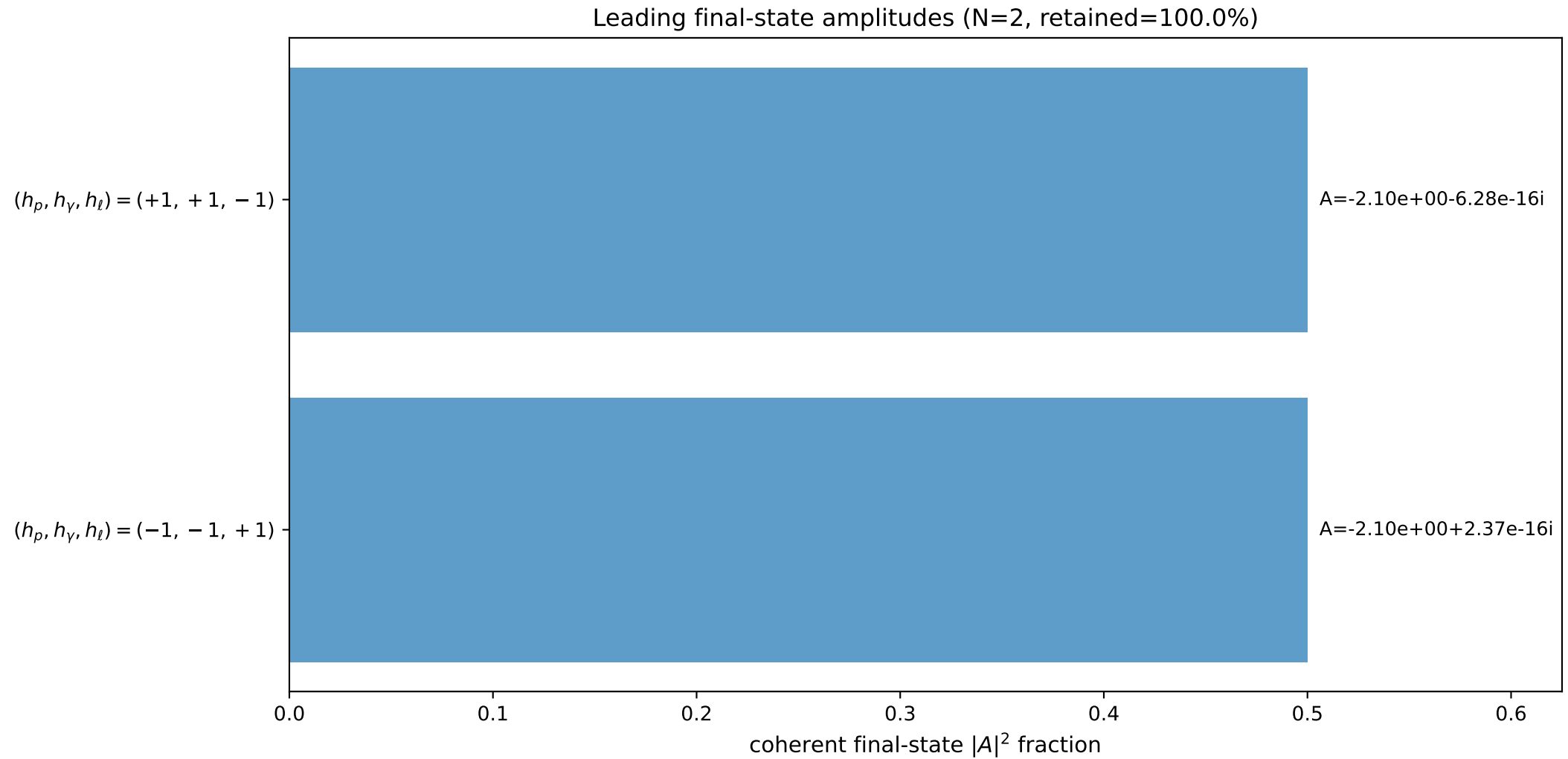
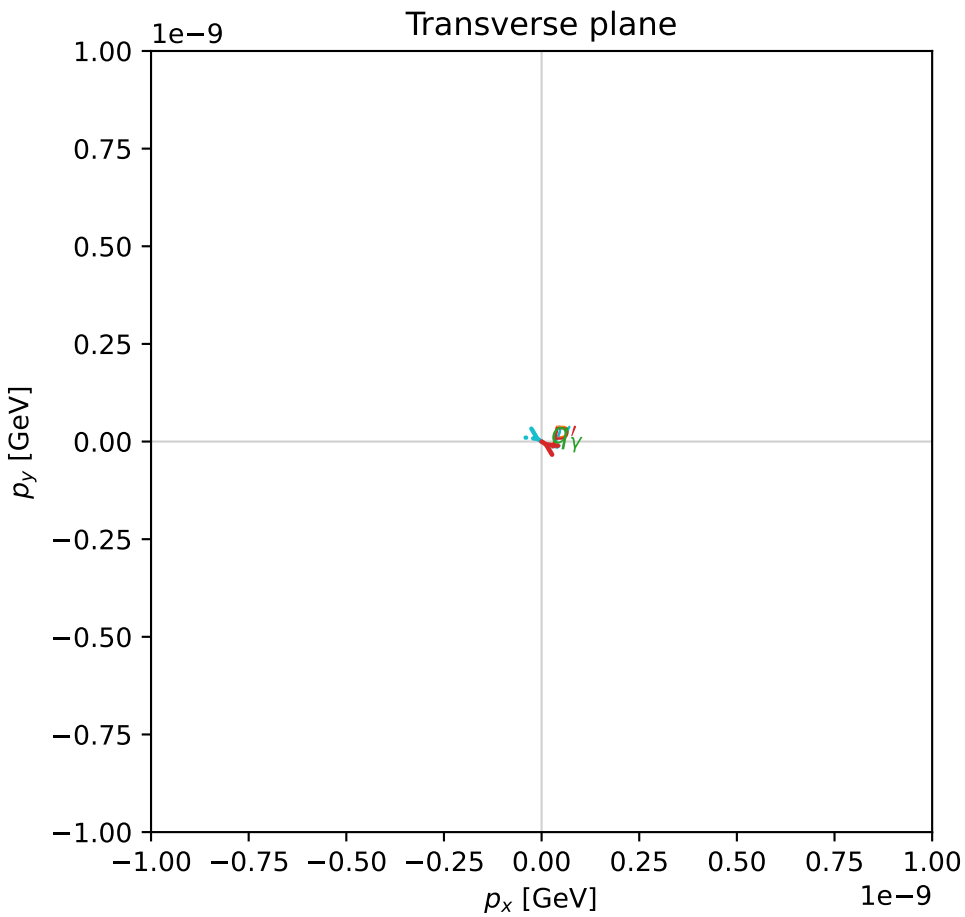
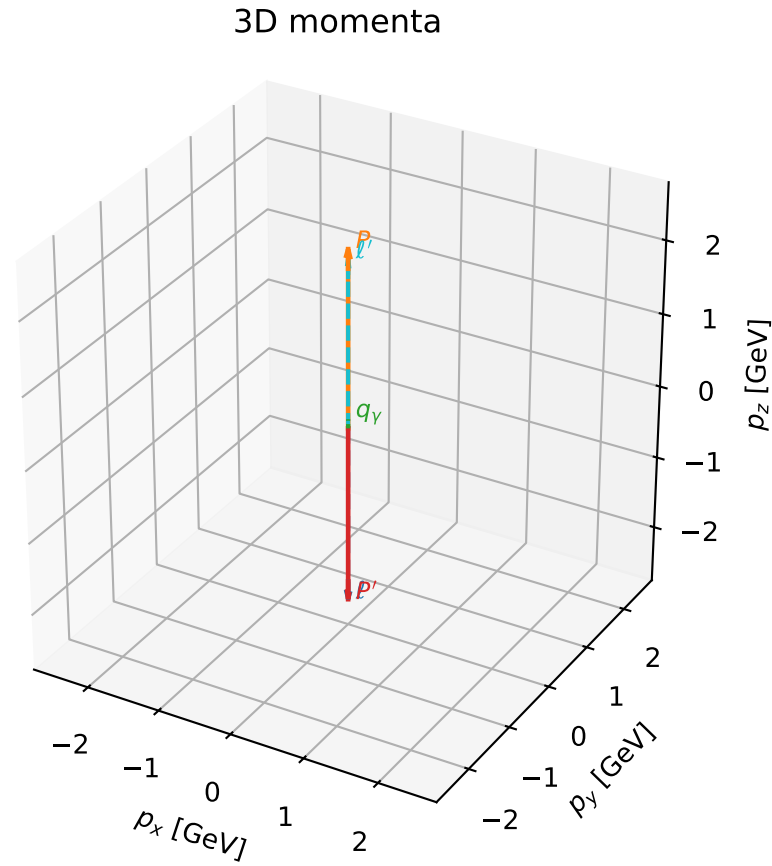
$\text{sqrt_s} = 3.995624$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.511073$
 $\text{phi_p_out} = 5.834648$
 $\text{phi_gamma_out} = 1.562317$
 $\text{alpha_e} = 2.752651$
 $\text{alpha_p} = 0.3889458$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

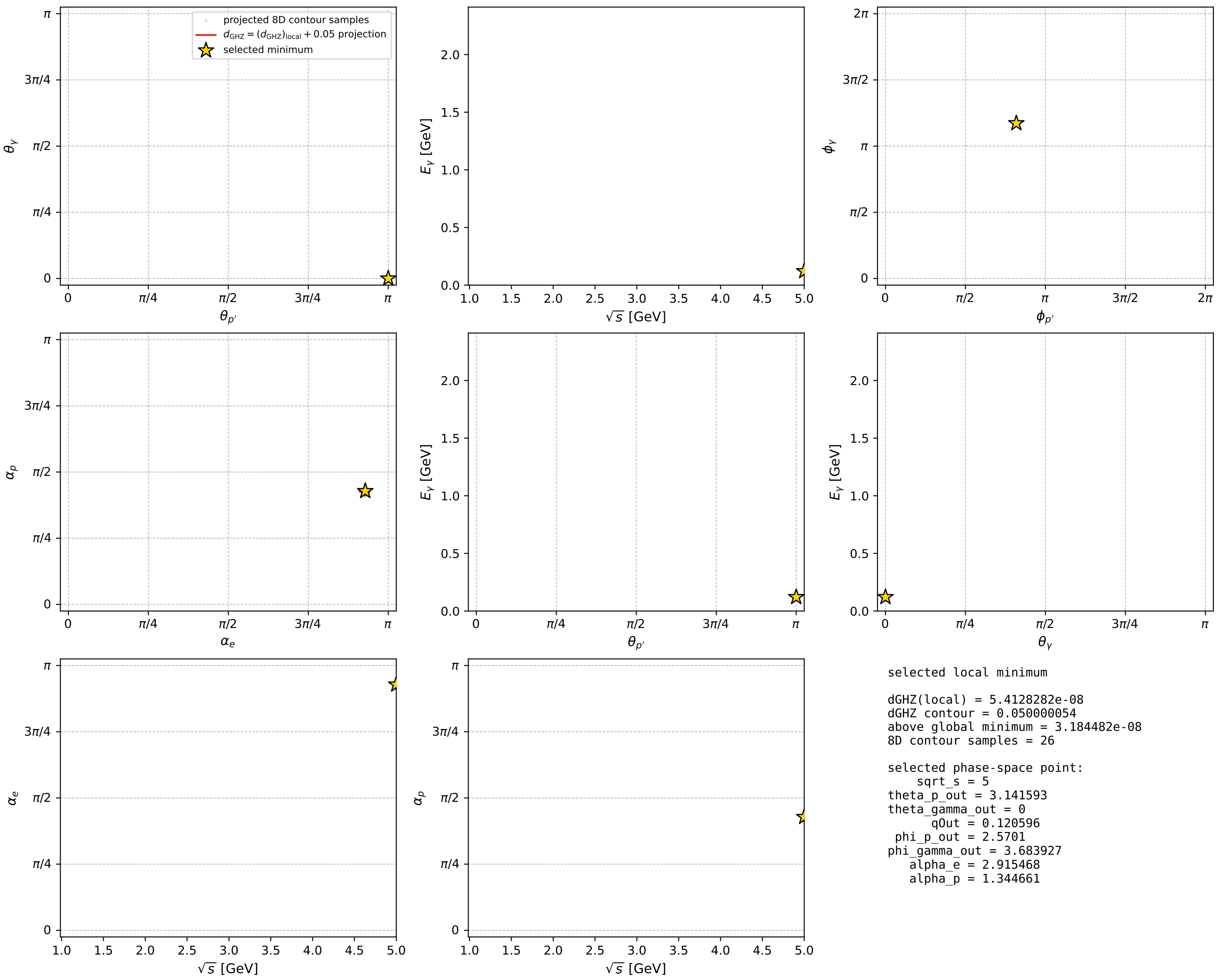
dghz_mixing_angles_polarization_03_local_minimum_134 $d_{\{\text{rm GHZ}\}}=5.41283\text{e-}08$
 region: polarization_cluster_03_local_minimum_134
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.9154679$ rad
 incoming lepton: $-0.974543|+\rangle +0.224203|-\rangle$
 proton mixing angle: $\alpha_p=1.3446609$ rad
 incoming proton: $0.224213|+\rangle +0.97454|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.5701$, $\phi_Y=3.68393$
 $E_Y=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.71169$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$



electron: polarization cluster P3, local minimum 134; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

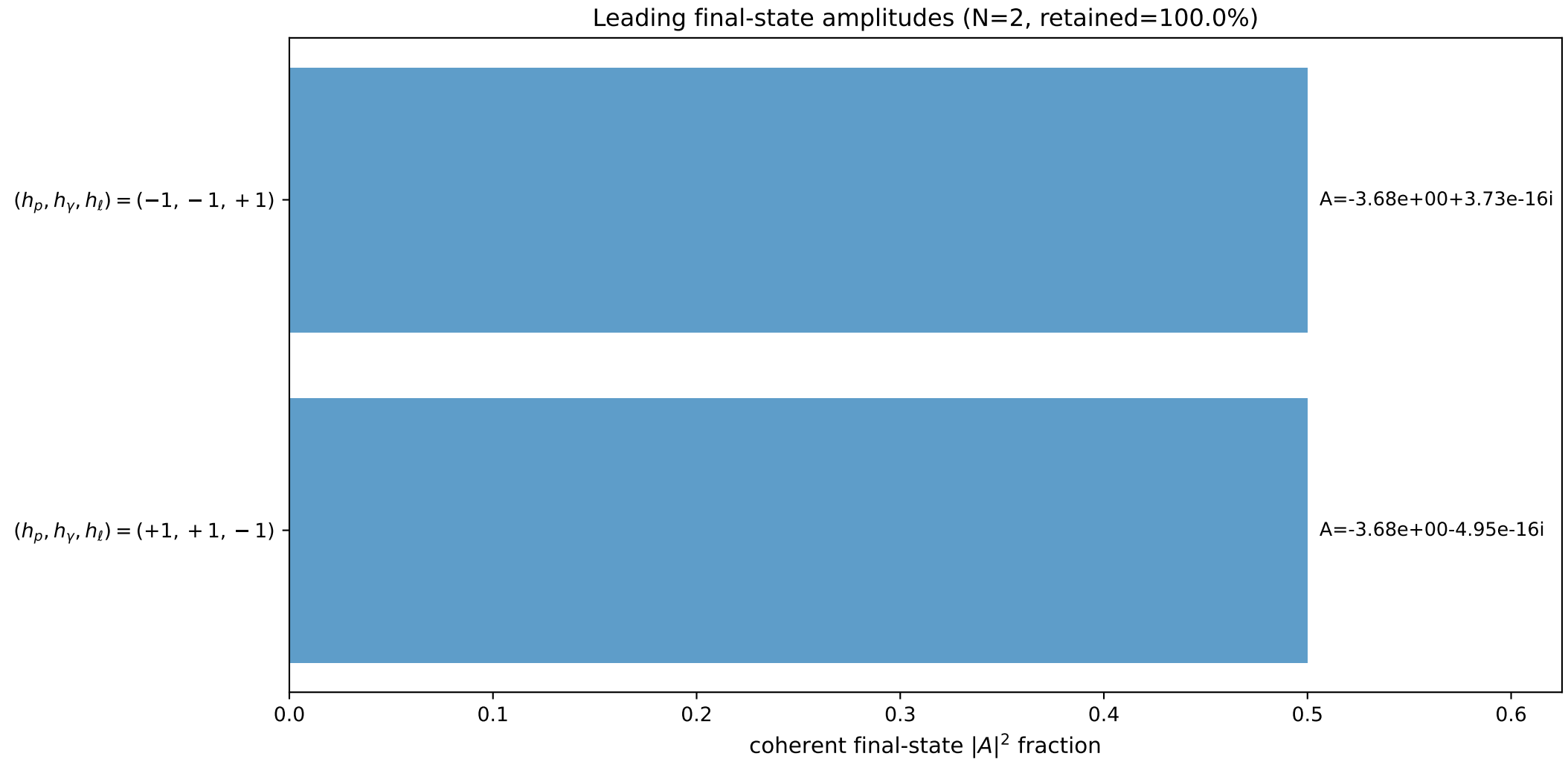
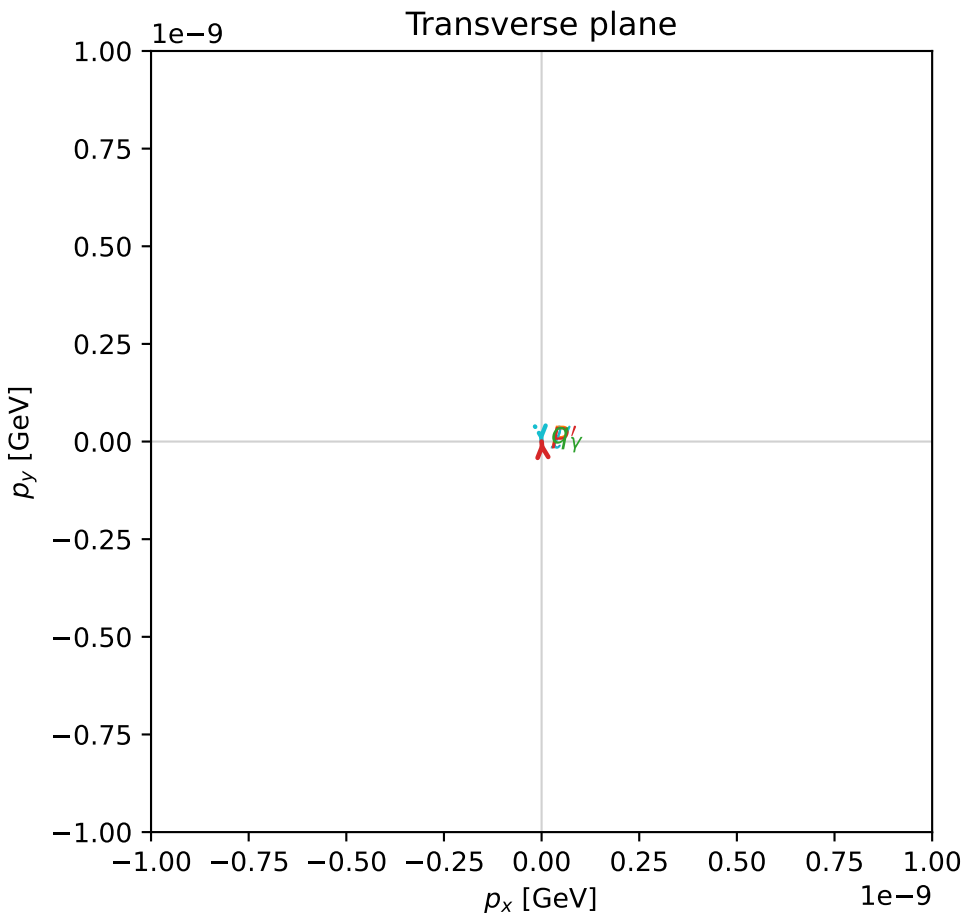
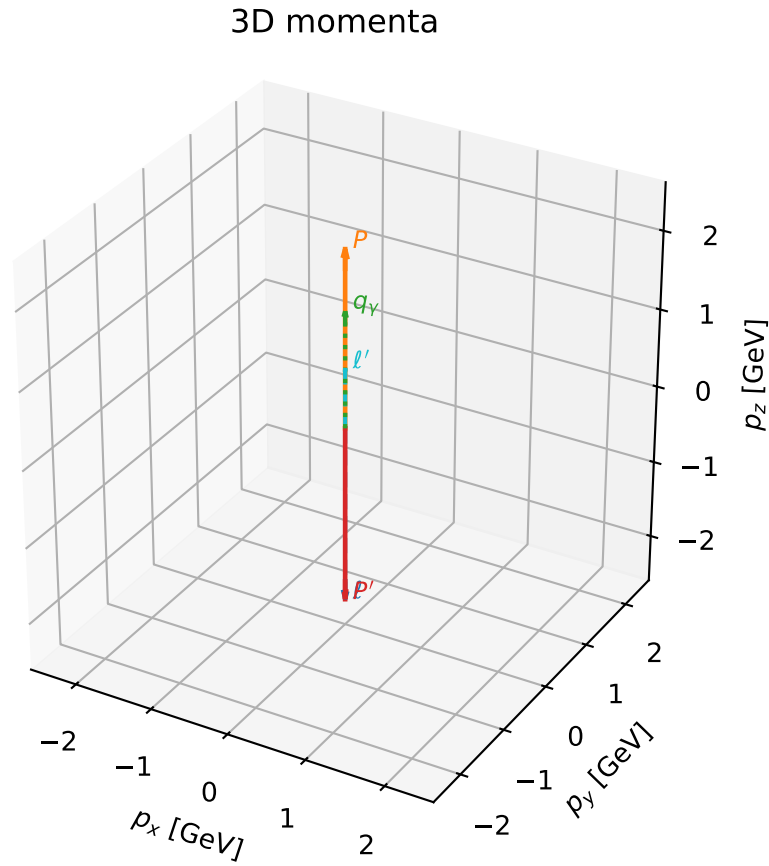


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

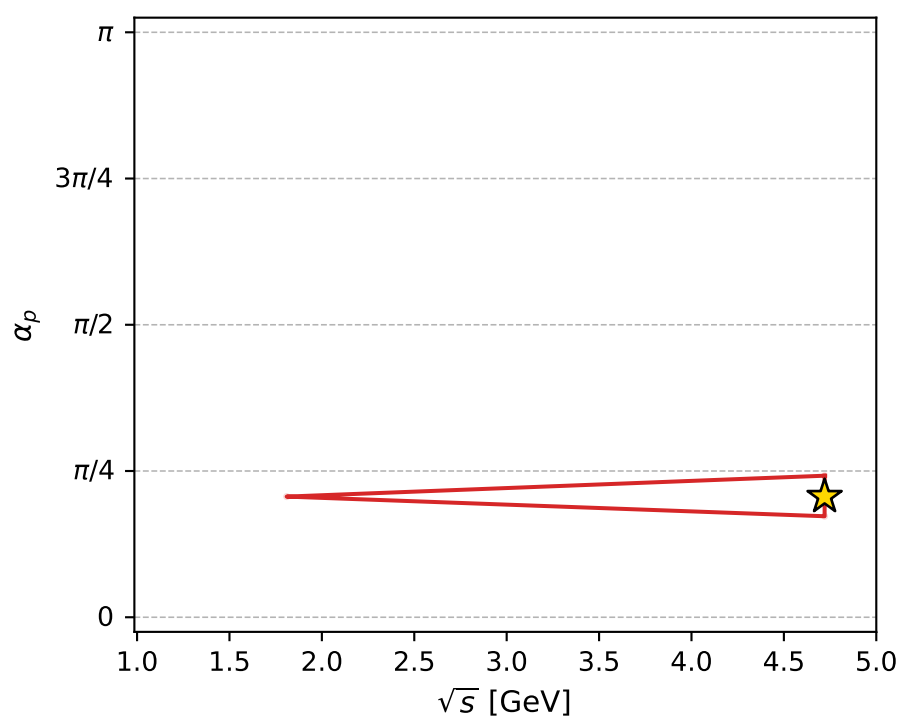
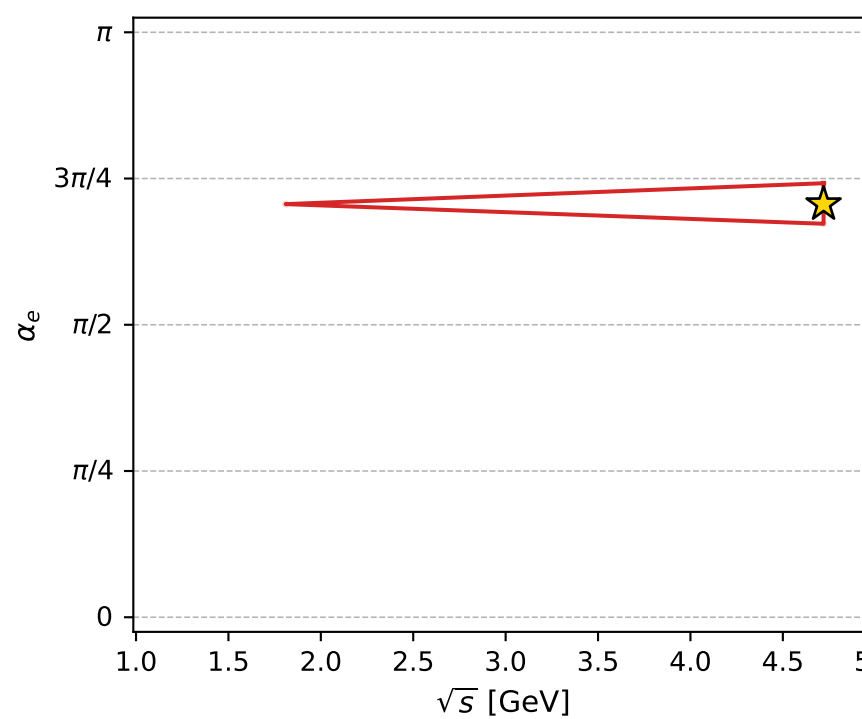
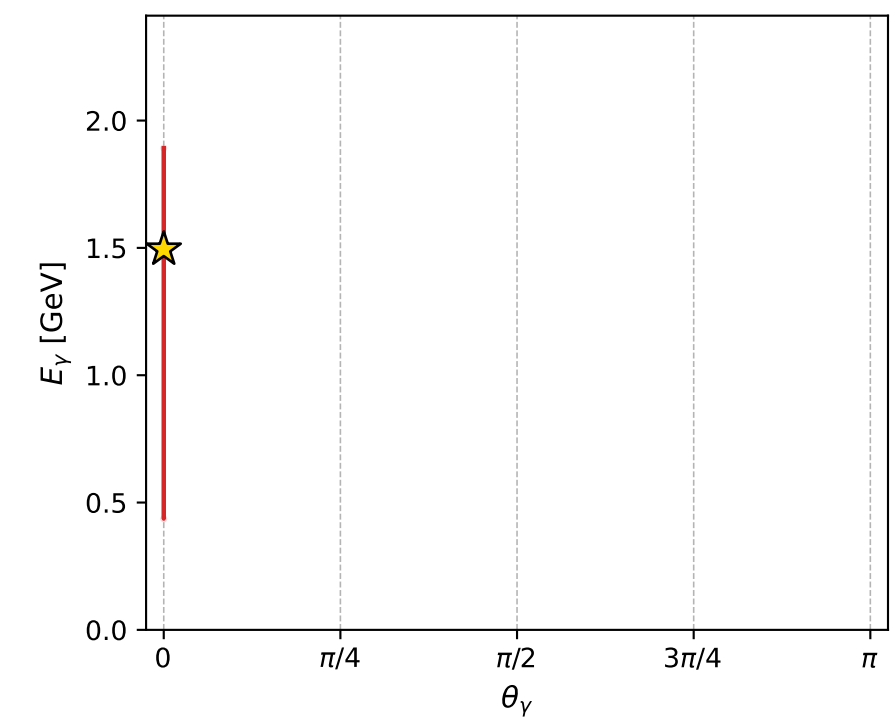
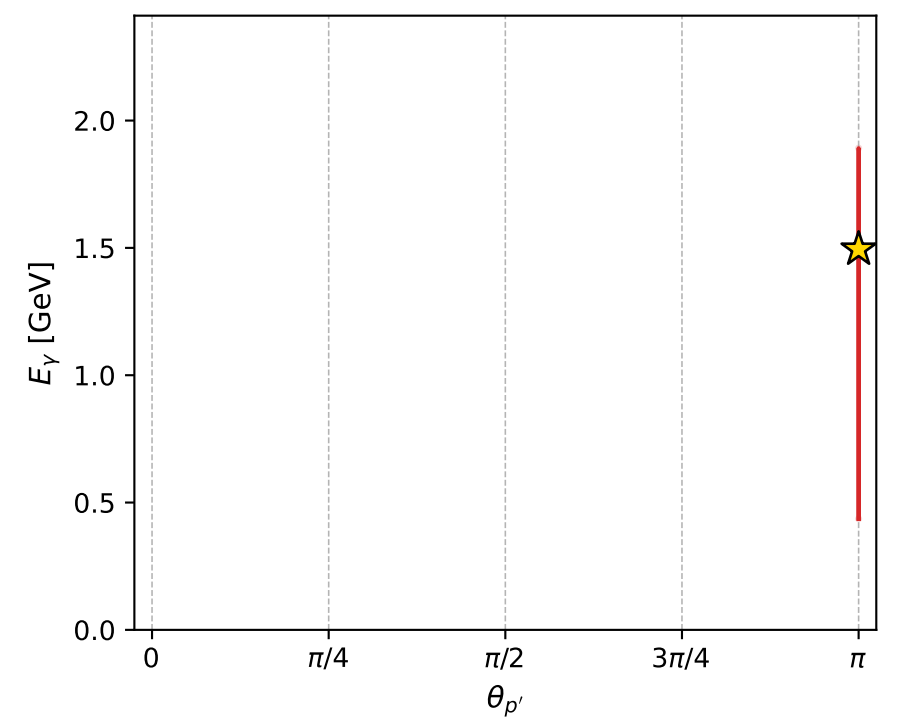
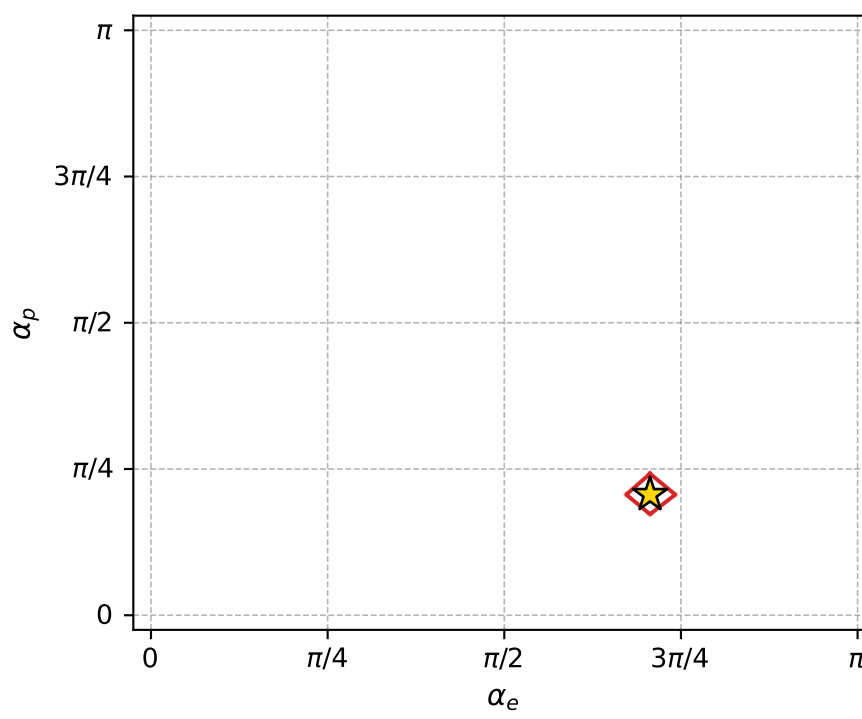
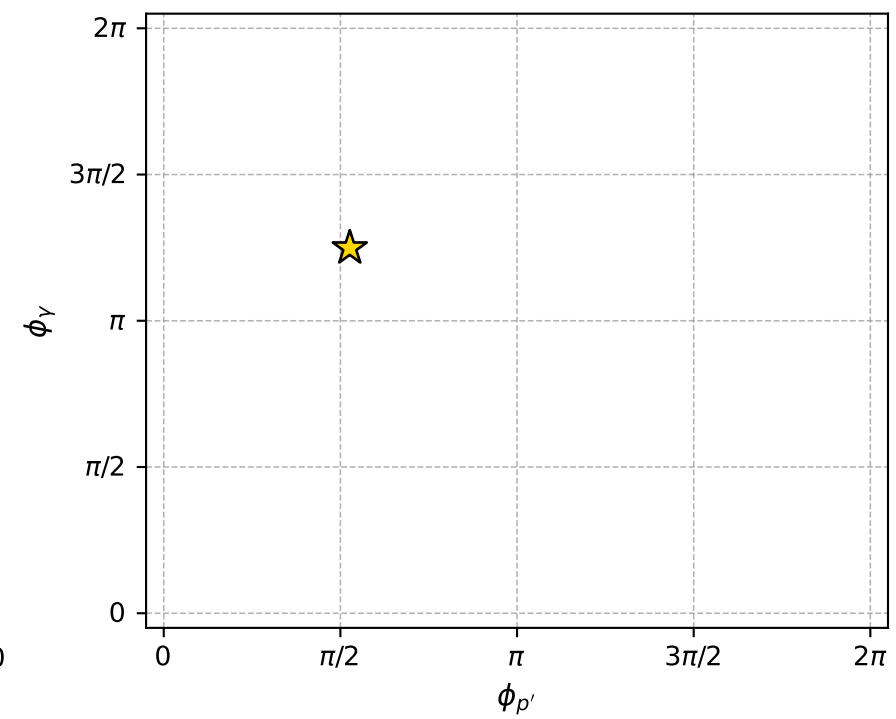
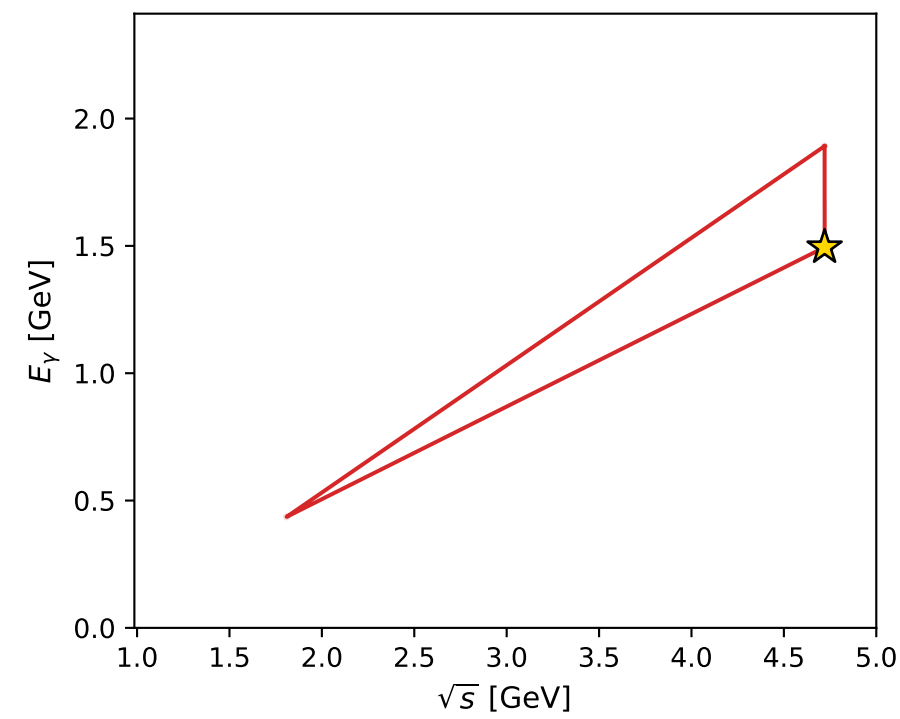
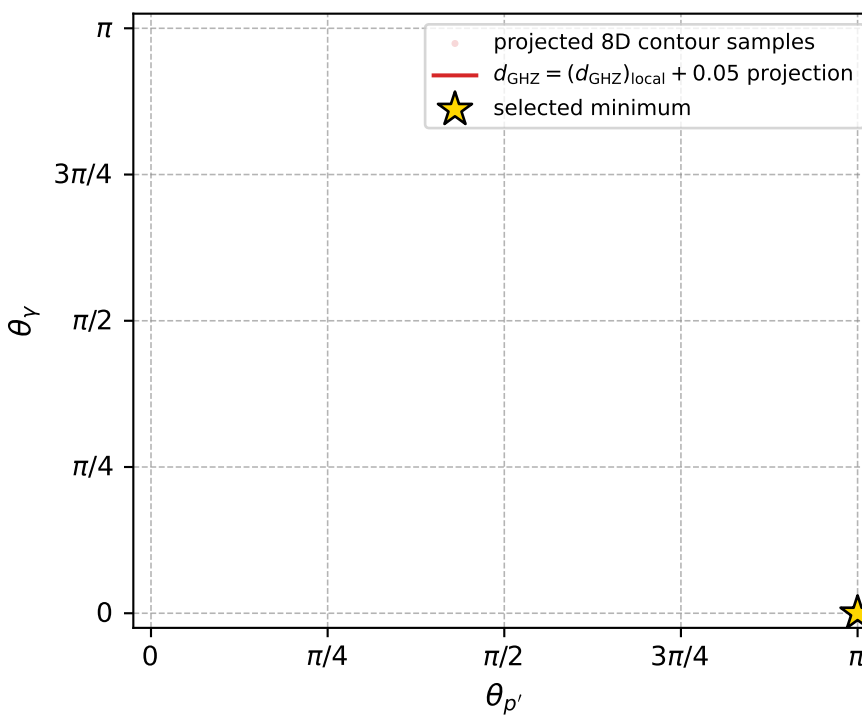
dghz_mixing_angles_polarization_03_local_minimum_143 $d_{\{\text{rm GHZ}\}}=7.74877\text{e-}08$
 region: polarization_cluster_03_local_minimum_143
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2187554$ rad
 incoming lepton: $-0.60356|+> +0.797317|->$
 proton mixing angle: $\alpha_p=0.64796548$ rad
 incoming proton: $0.797313|+> +0.603566|->$

$s=22.2813$, $\sqrt{s}=4.7203$, $|\vec{P}|=2.26695$, $|\vec{P}'|=2.26695$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.65437$, $\phi_Y=3.92312$
 $E_Y=1.49471$, $\theta_{l'}=0$, $\phi_{l'}=4.79596$
 $Q^2=7.00254$, $x_B=0.331661$, $t=-20.5563$, $W^2=14.9908$, $y=0.986548$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2670$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 P $E= 2.4533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2670$
 l' $E= 0.7722$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7722$
 P' $E= 2.4533$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 q_Y $E= 1.4947$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4947$



electron: polarization cluster P3, local minimum 143; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.7487652\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000077$
 above global minimum = $5.5204191\text{e-}08$
 8D contour samples = 78

selected phase-space point:

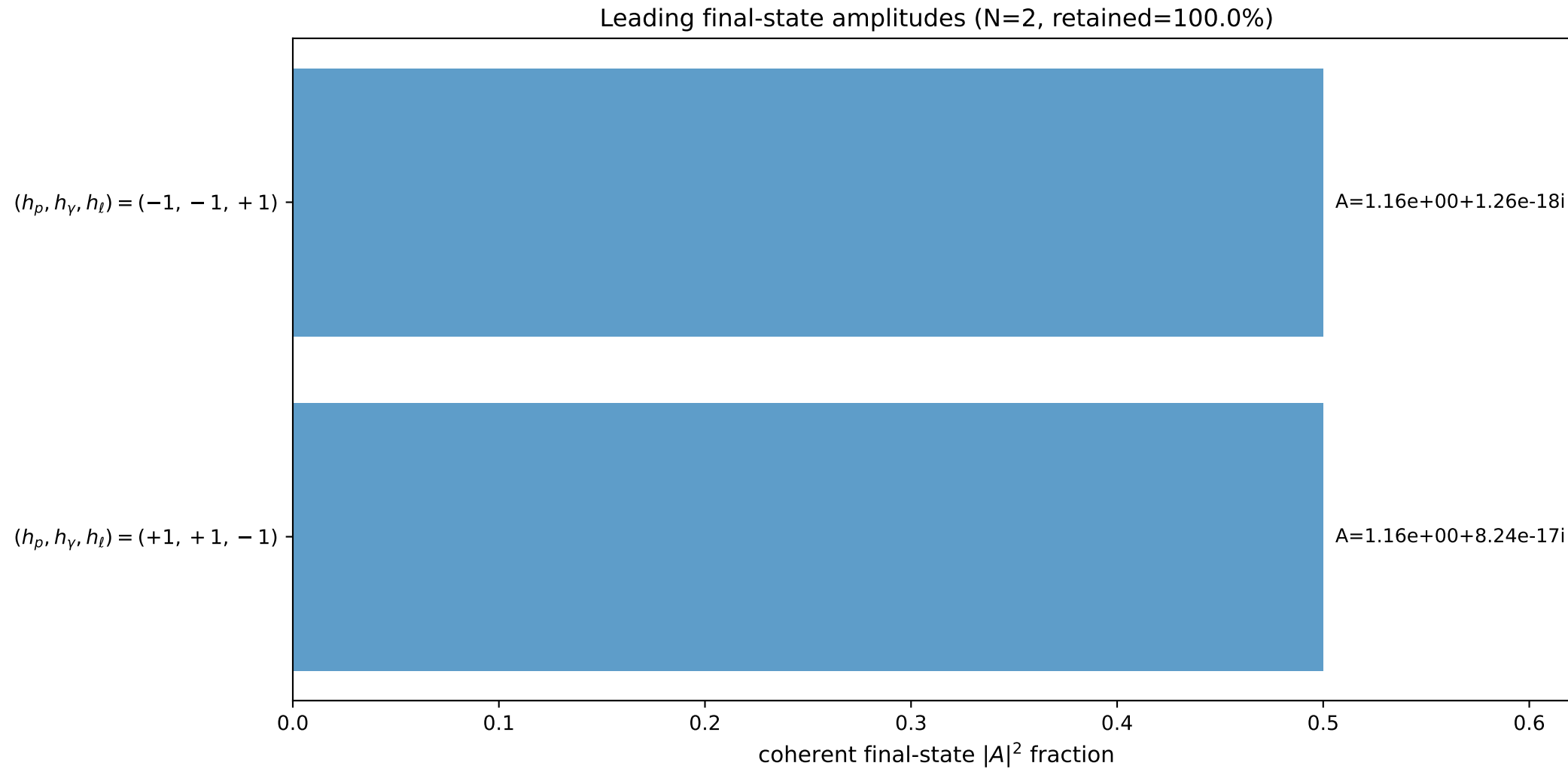
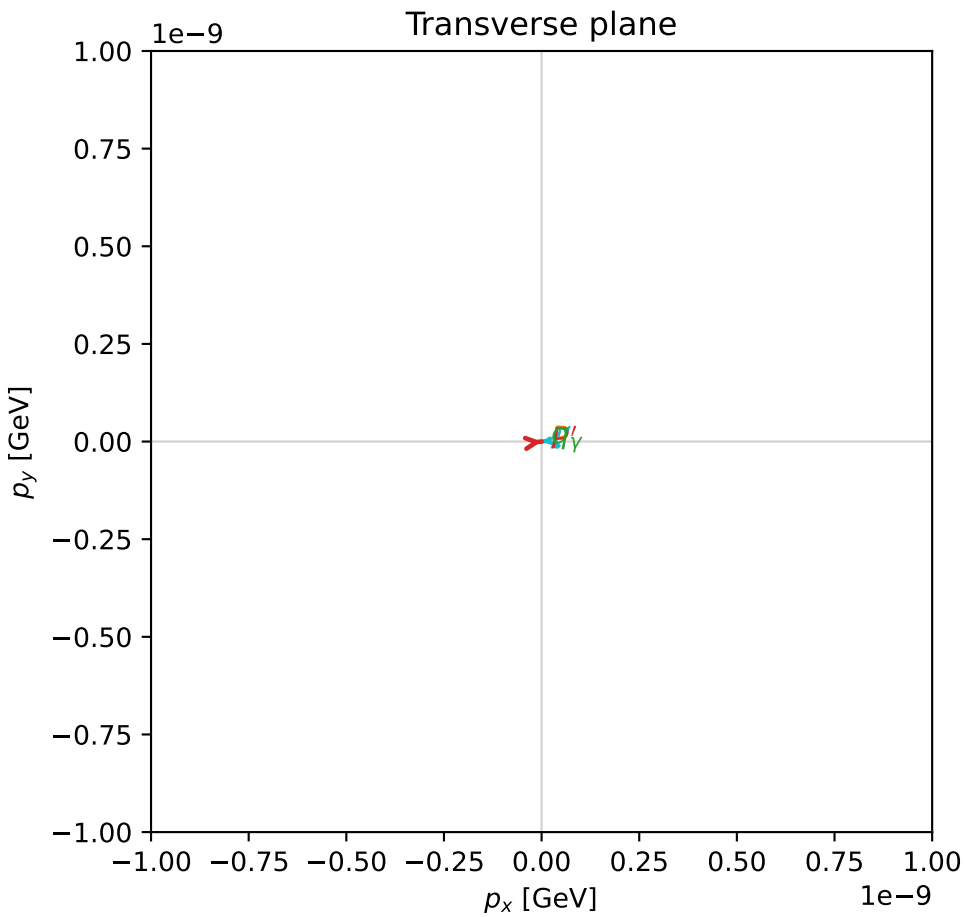
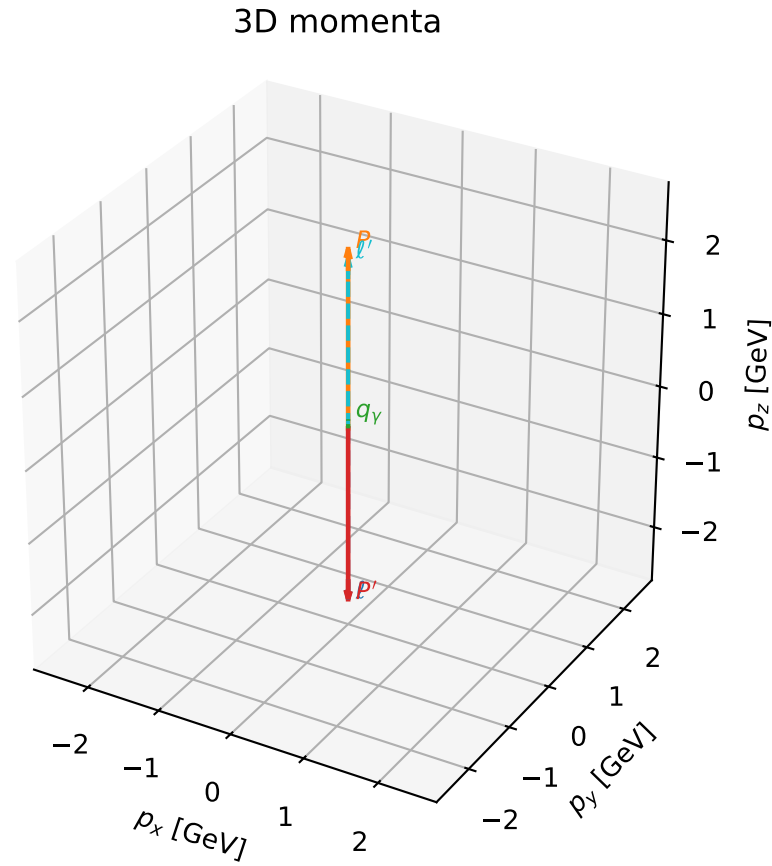
$\text{sqrt_s} = 4.720304$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.494713$
 $\text{phi_p_out} = 1.654368$
 $\text{phi_gamma_out} = 3.923122$
 $\text{alpha_e} = 2.218755$
 $\text{alpha_p} = 0.6479655$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

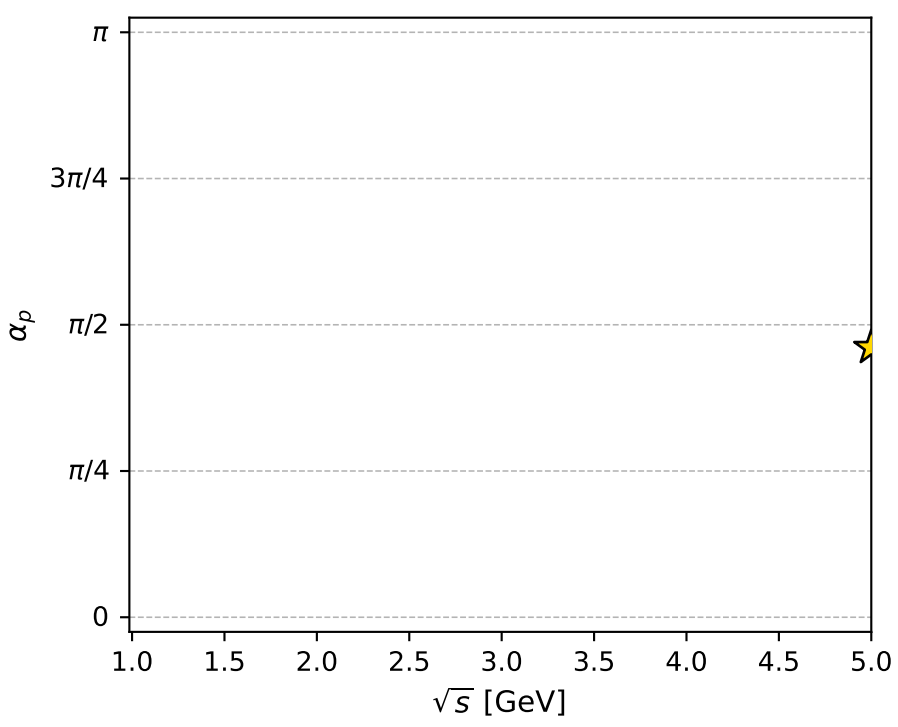
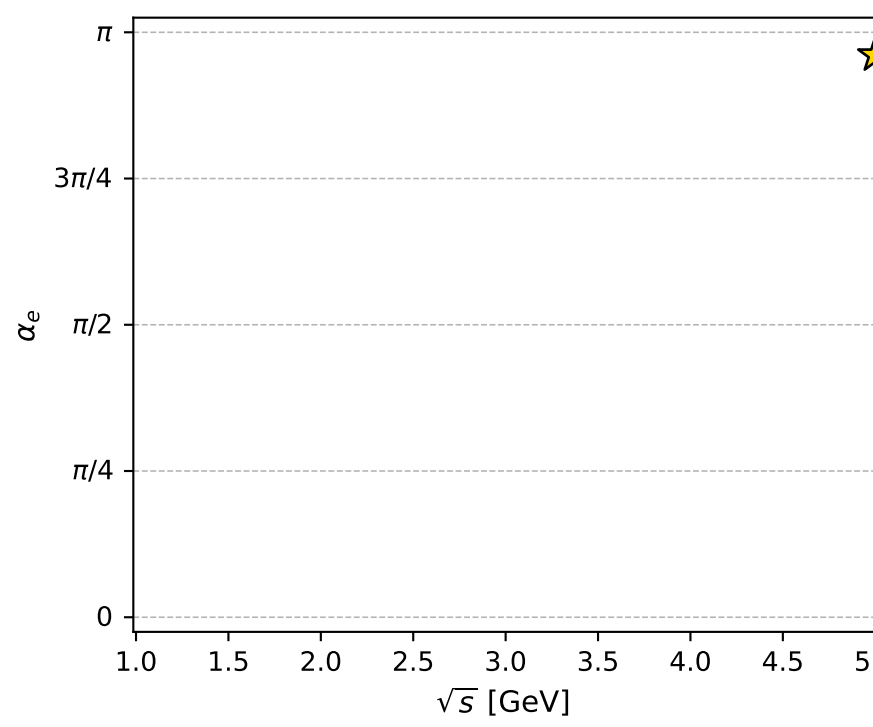
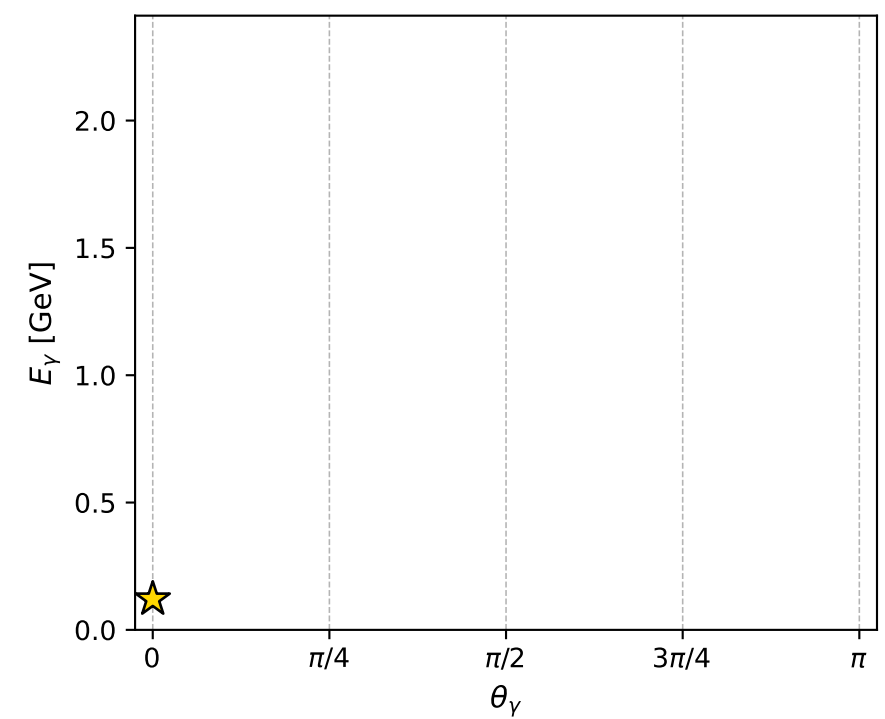
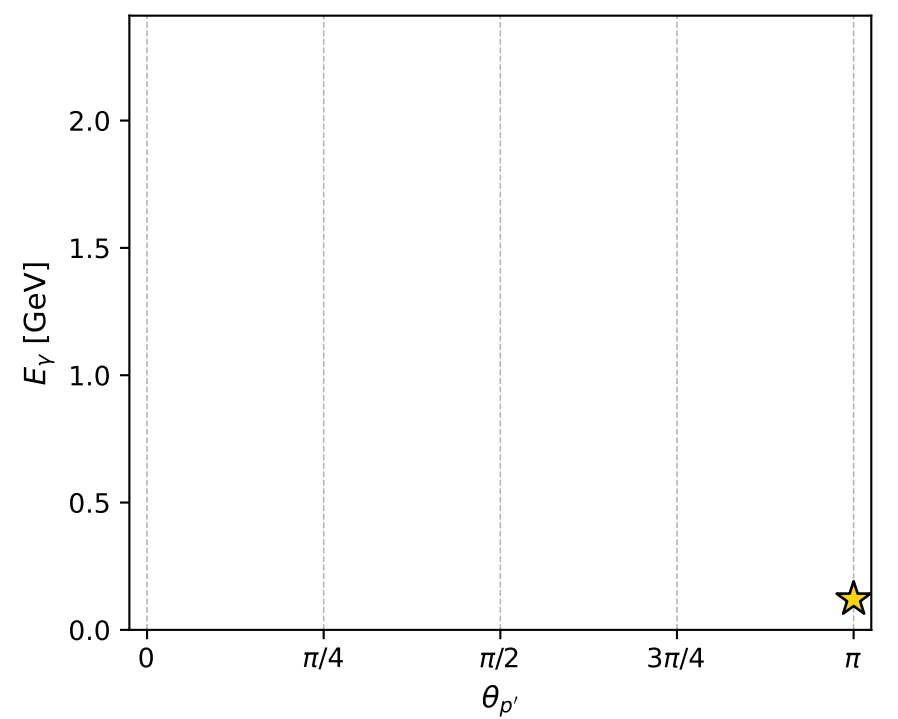
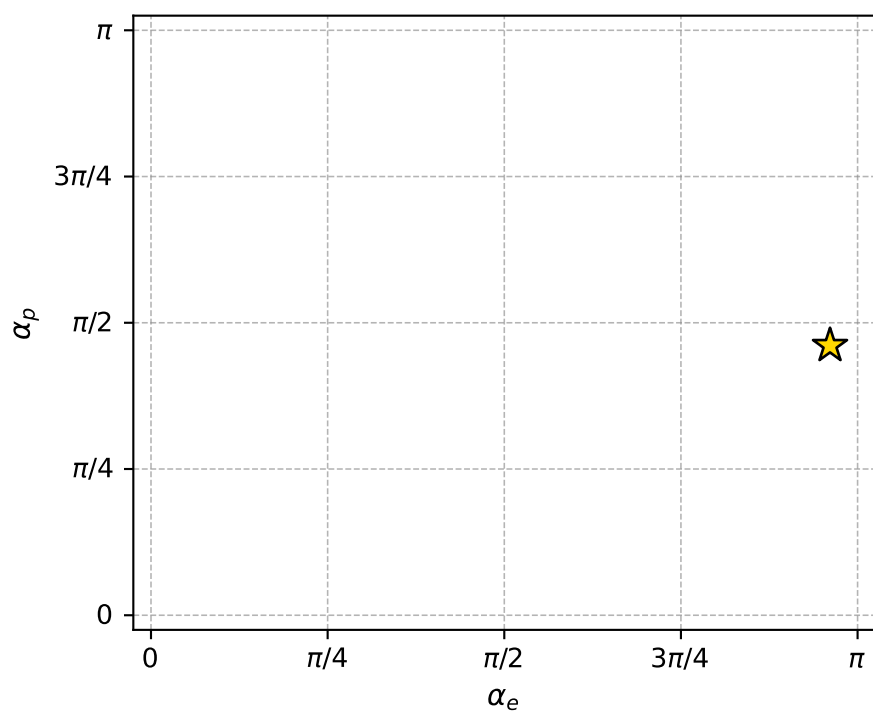
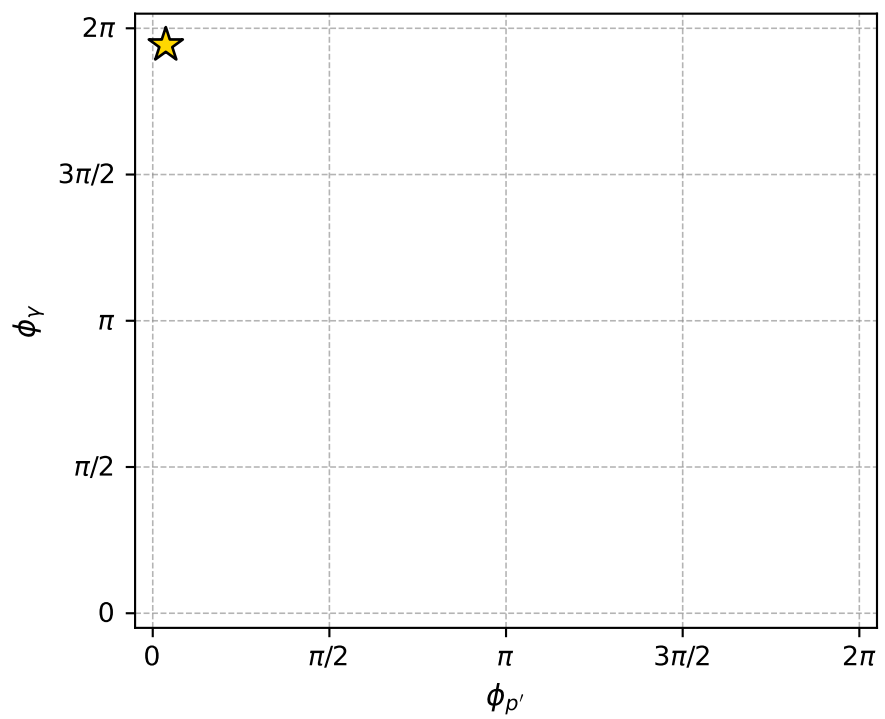
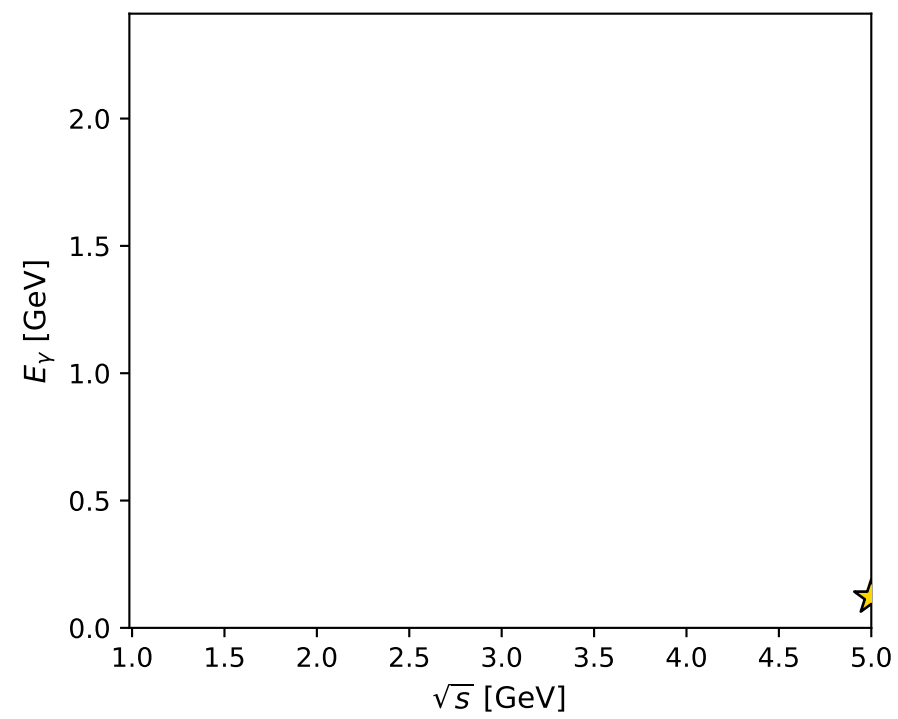
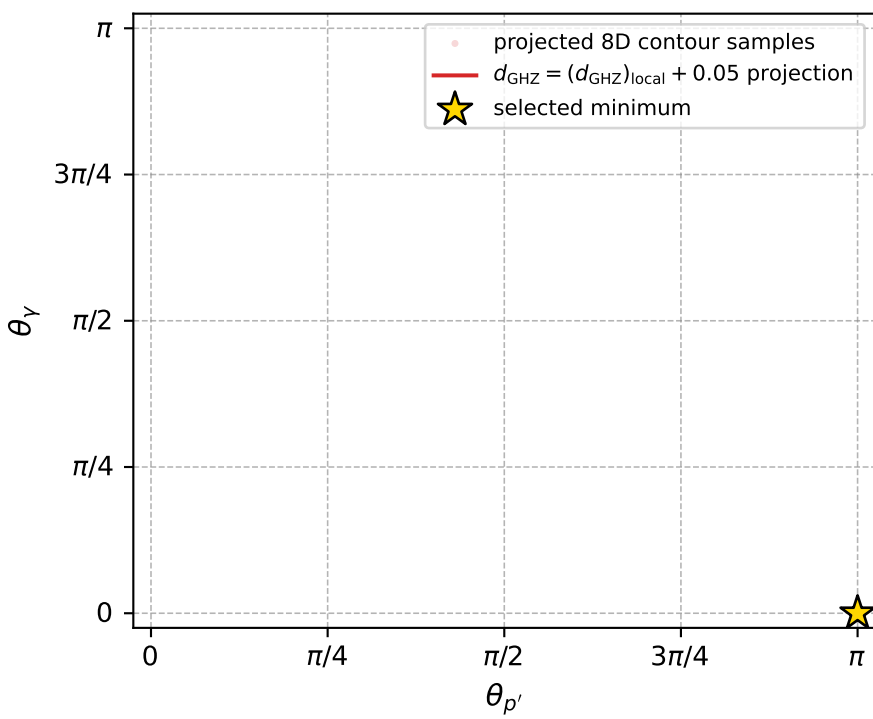
dghz_mixing_angles_polarization_03_local_minimum_154 $d_{\{\text{rm GHZ}\}}=9.73504\text{e-}08$
 region: polarization_cluster_03_local_minimum_154
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=3.0190559$ rad
 incoming lepton: $-0.992502|+\rangle +0.12223|-\rangle$
 proton mixing angle: $\alpha_p=1.4482597$ rad
 incoming proton: $0.12223|+\rangle +0.992502|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.116076$, $\phi_Y=6.10382$
 $E_Y=0.120666$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.25767$
 $Q^2=22.1071$, $x_B=0.948242$, $t=-23.2713$, $W^2=2.08651$, $y=0.966567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2913$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2913$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 0.1207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1207$



electron: polarization cluster P3, local minimum 154; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 9.7350416\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000097$
 above global minimum = $7.5066954\text{e-}08$
 8D contour samples = 31

selected phase-space point:

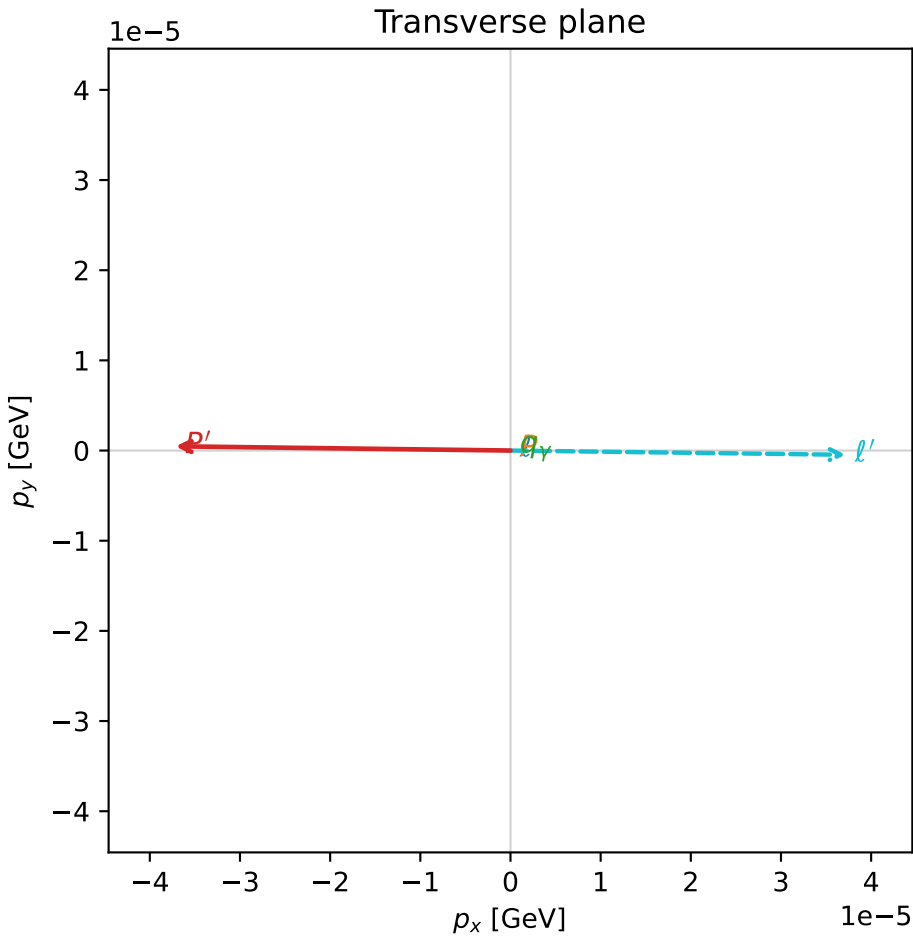
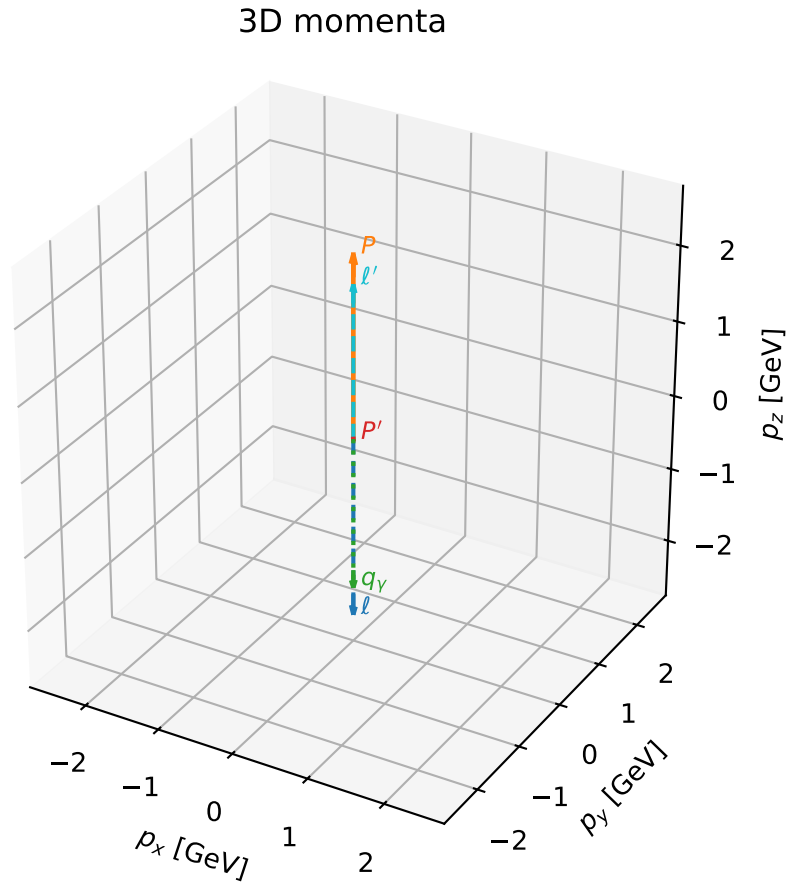
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 0.1206664$
 $\text{phi_p_out} = 0.116076$
 $\text{phi_gamma_out} = 6.103815$
 $\text{alpha_e} = 3.019056$
 $\text{alpha_p} = 1.44826$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_156 $d_{\{\text{rm GHZ}\}}=9.98211\text{e-}08$
 region: polarization_cluster_03_local_minimum_156
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=3.0075642$ rad
 incoming lepton: $-0.991032|+\rangle +0.133628|-\rangle$
 proton mixing angle: $\alpha_p=0.13401798$ rad
 incoming proton: $0.991033|+\rangle +0.133617|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000632829$
 $\theta_{P'}=0.0587284$, $\theta_Y=3.14159$, $\phi_{P'}=3.12895$, $\phi_Y=3.82447$
 $E_Y=2.03132$, $\theta_{\ell'}=1.82912\text{e-}05$, $\phi_{\ell'}=6.27054$
 $Q^2=19.5922$, $x_B=0.837076$, $t=-3.09232$, $W^2=4.69316$, $y=0.97037$

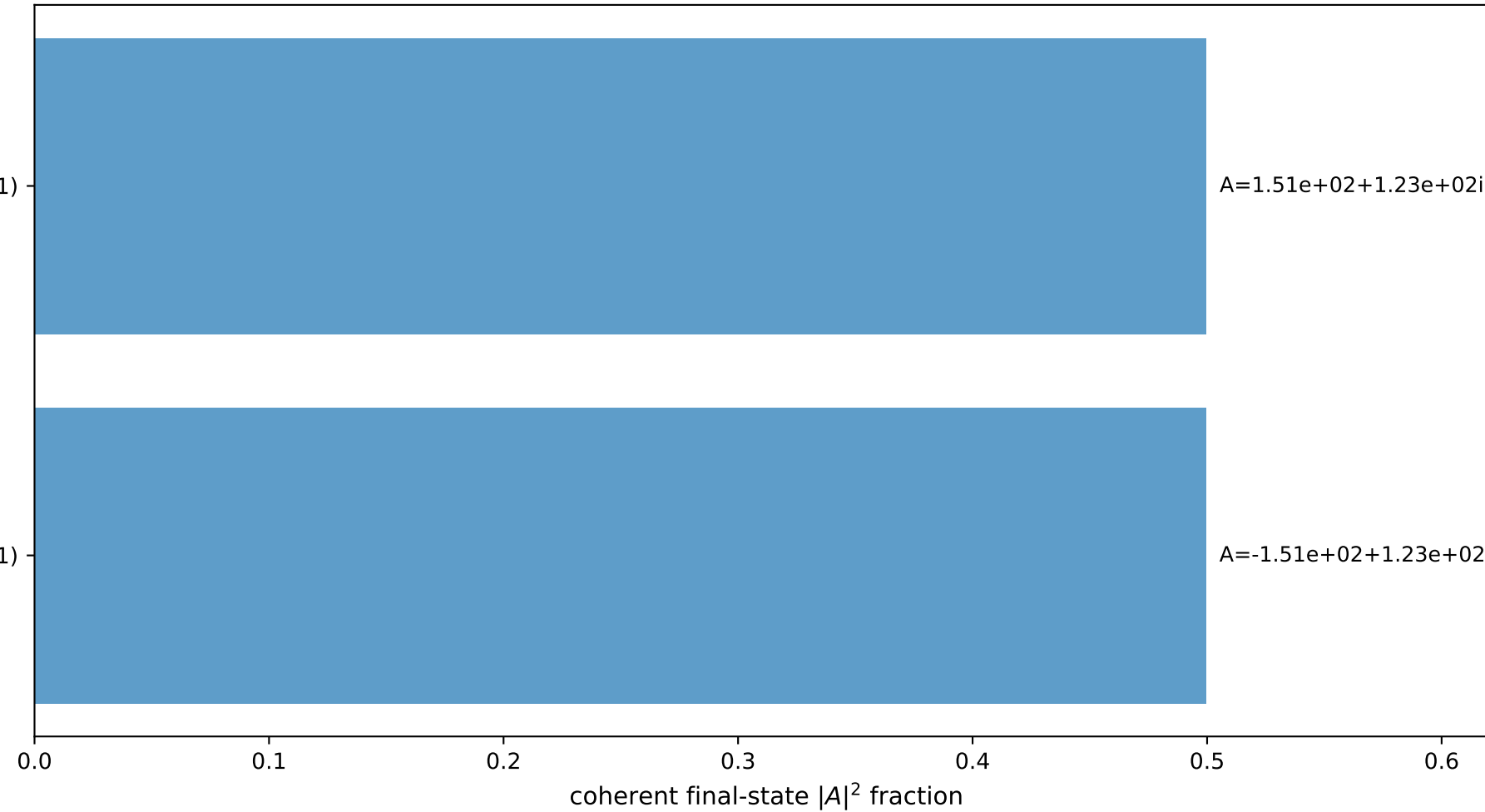
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0307$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 2.0313$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0313$



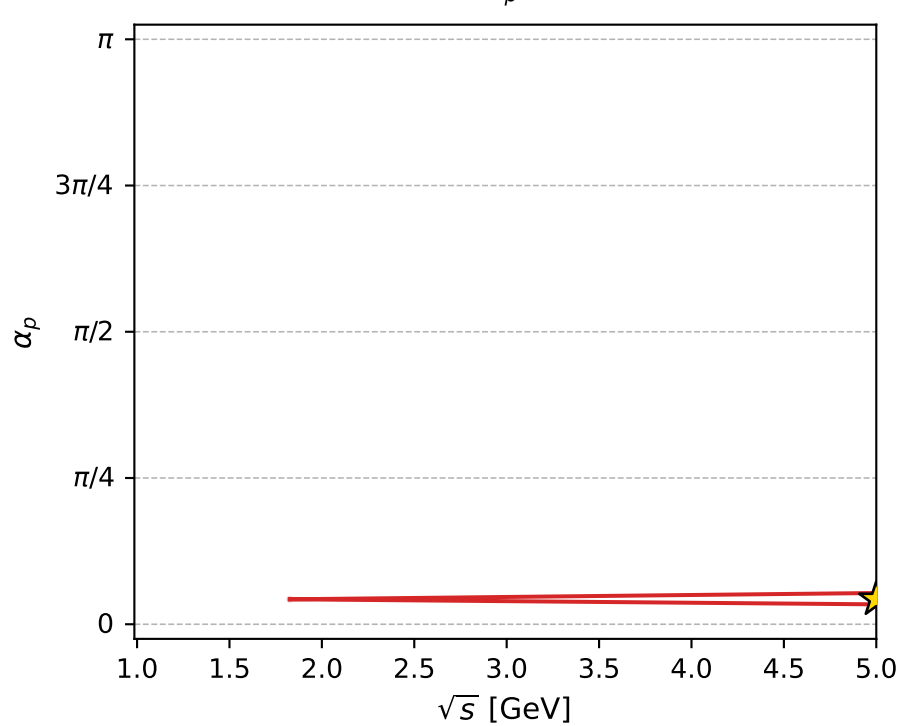
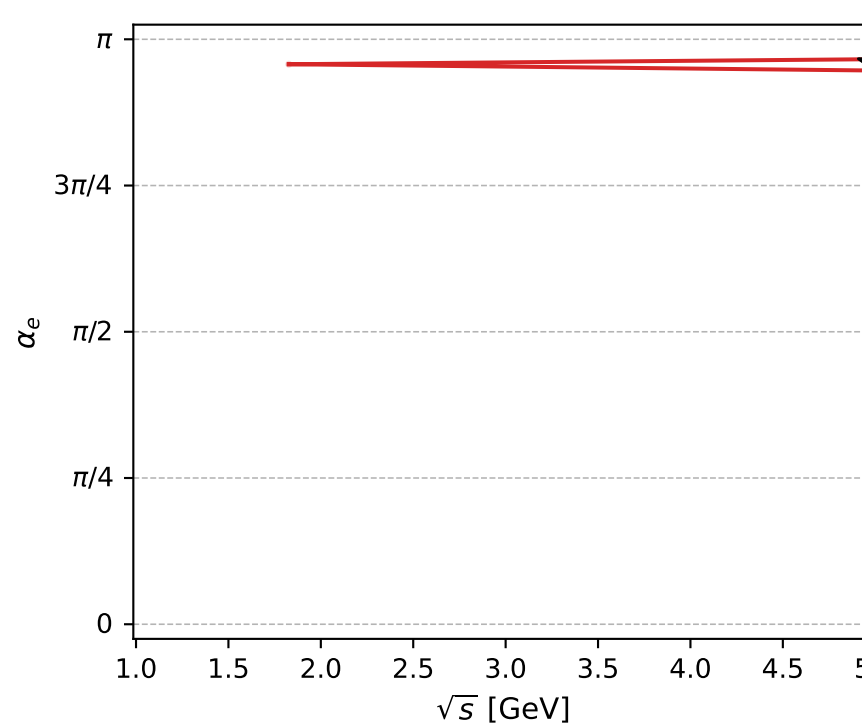
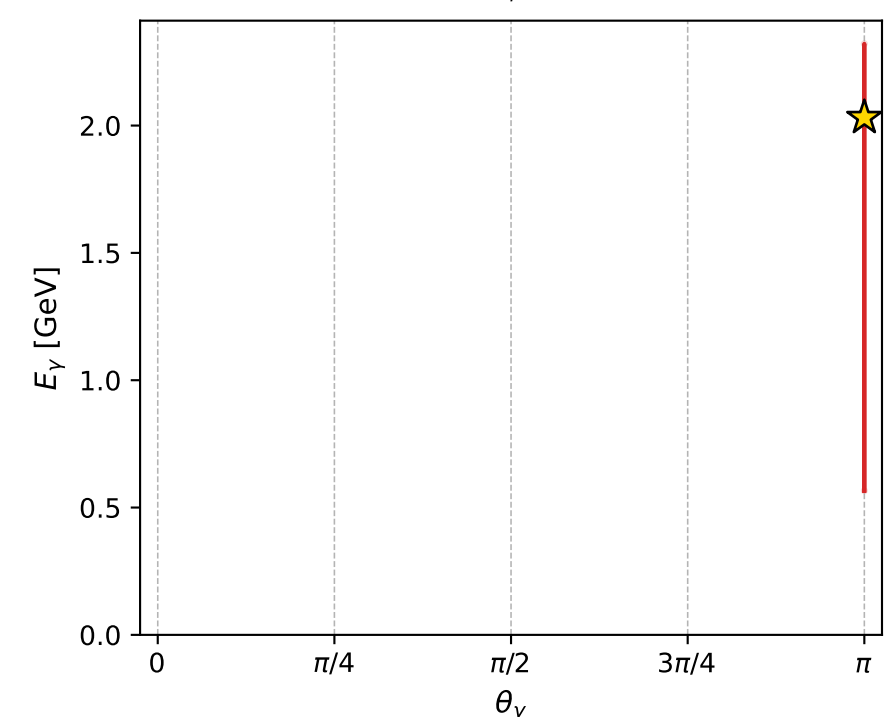
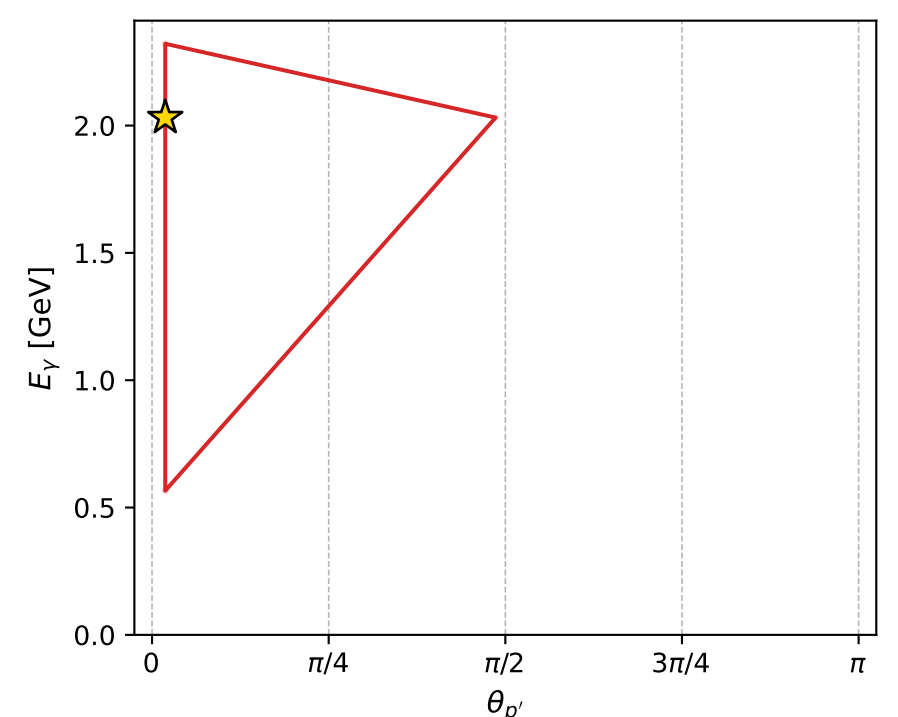
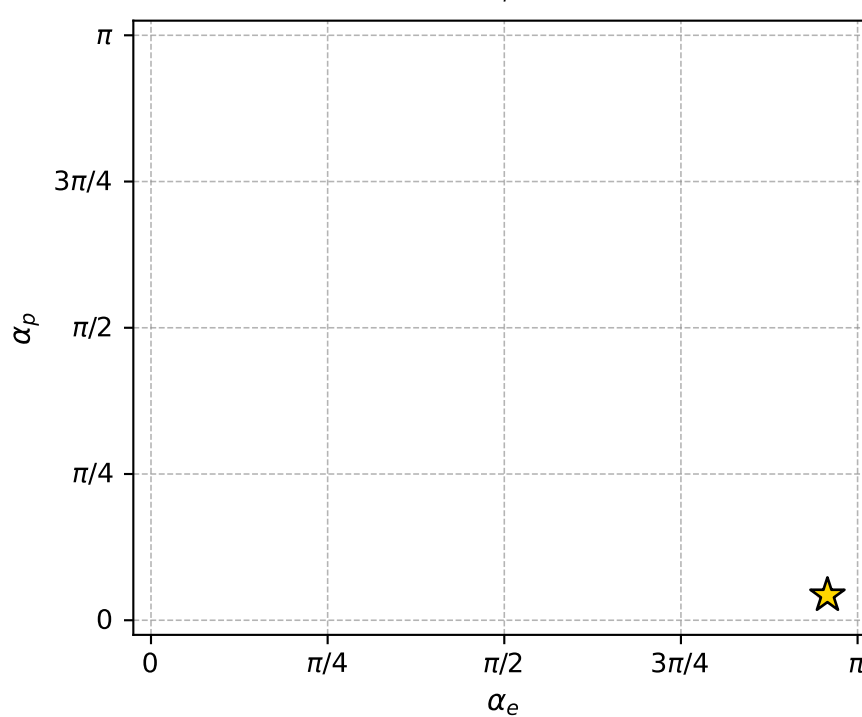
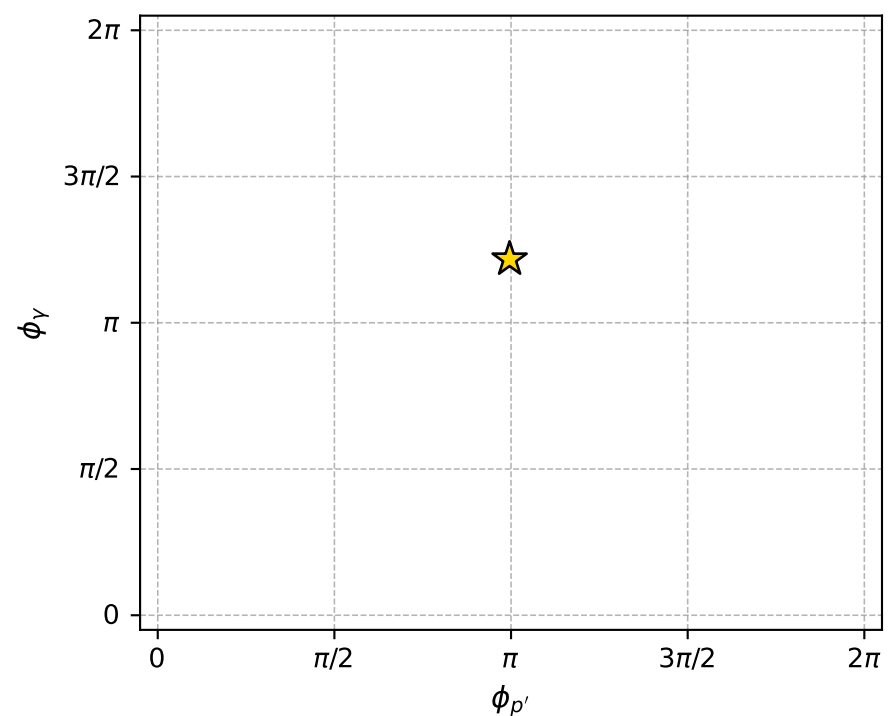
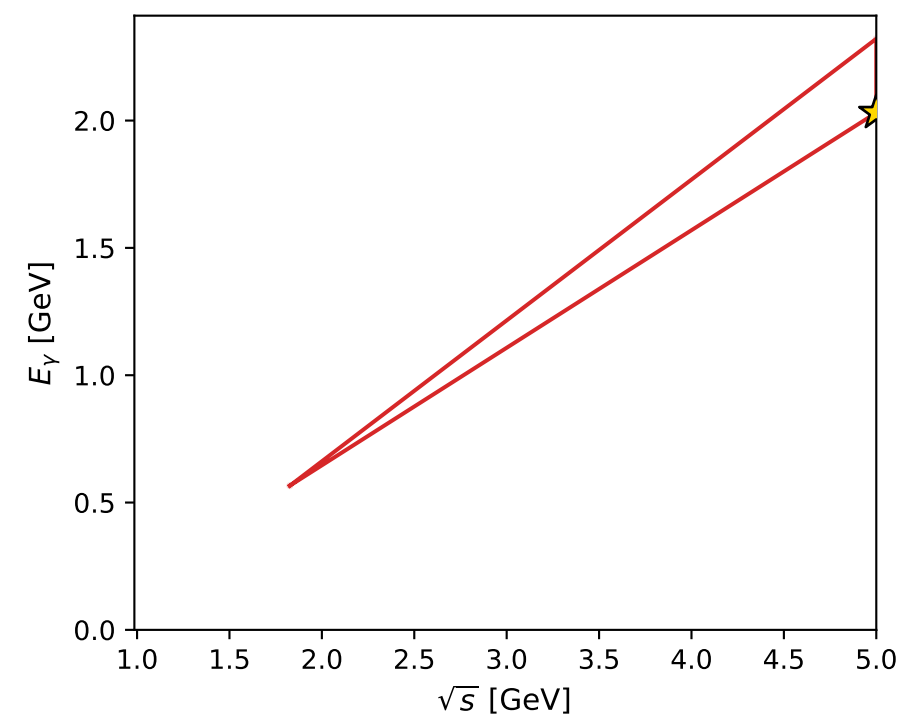
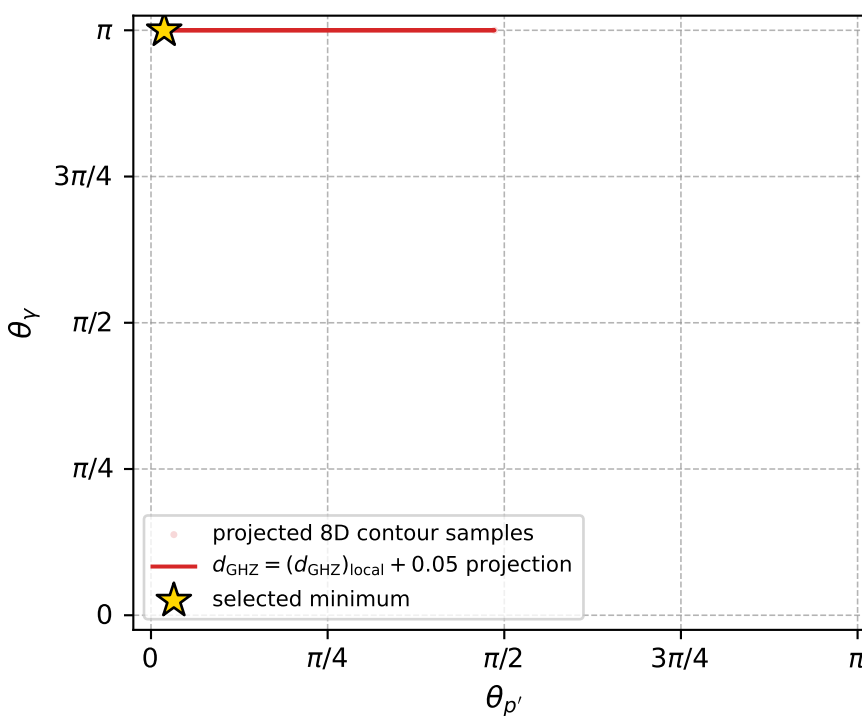
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=99.9%)



electron: polarization cluster P3, local minimum 156; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 9.9821071\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.0500001$
 above global minimum = $7.7537609\text{e-}08$
 8D contour samples = 69

selected phase-space point:

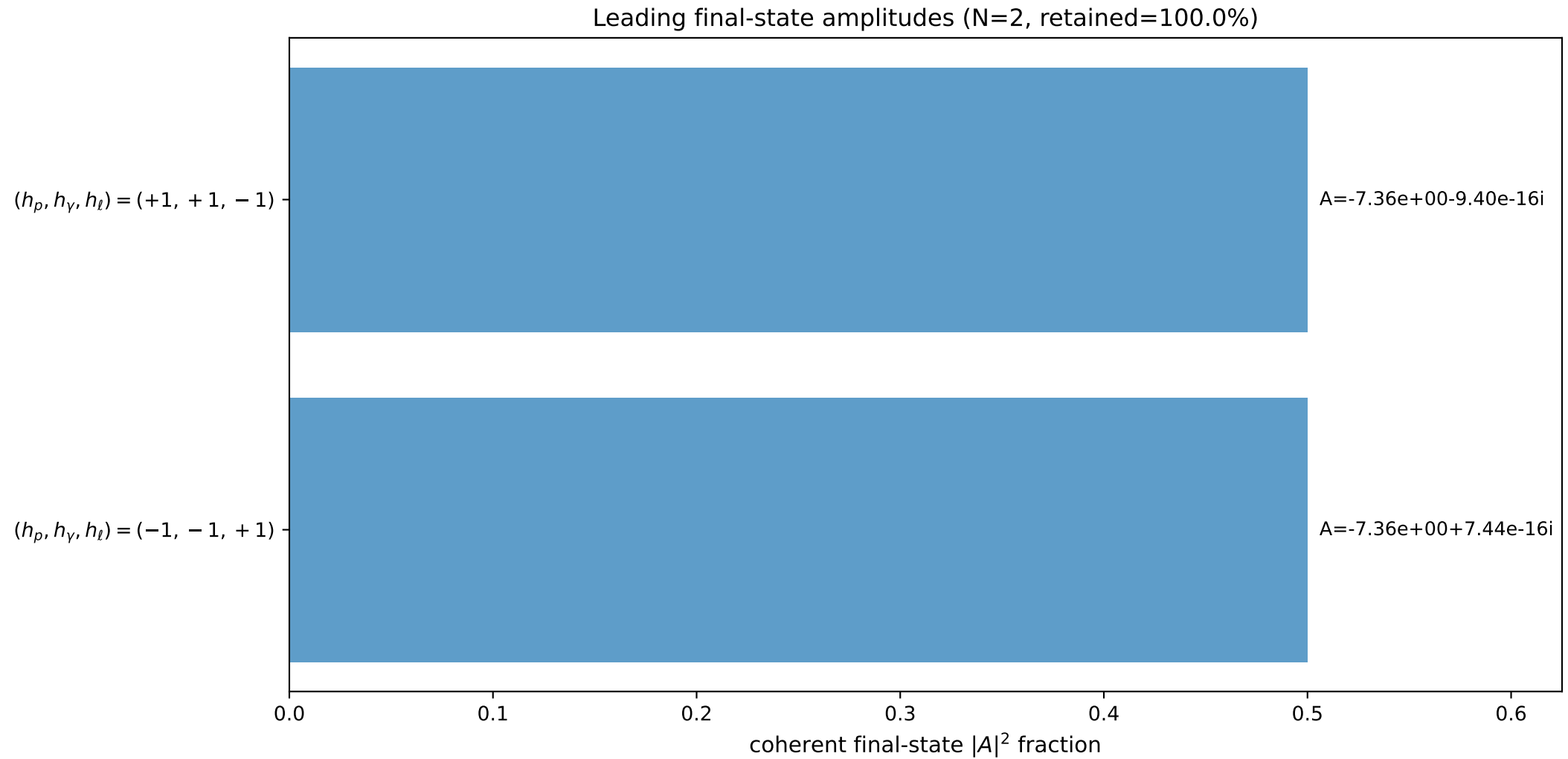
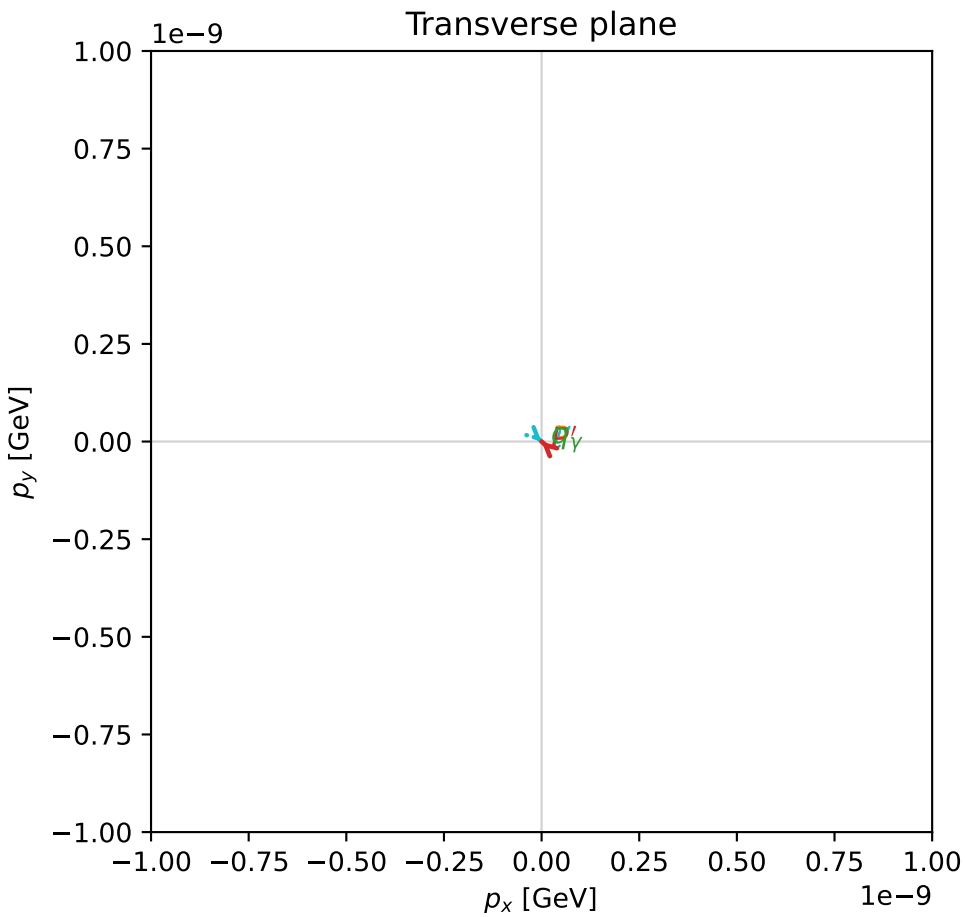
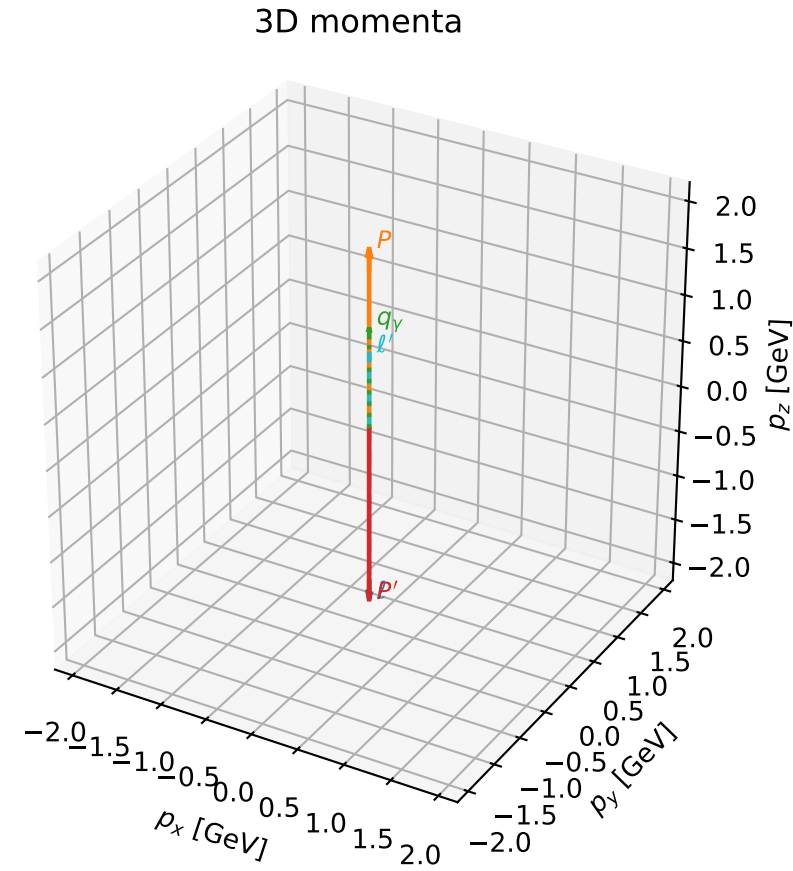
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.05872837$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.031316$
 $\text{phi_p_out} = 3.128946$
 $\text{phi_gamma_out} = 3.824467$
 $\text{alpha_e} = 3.007564$
 $\text{alpha_p} = 0.134018$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

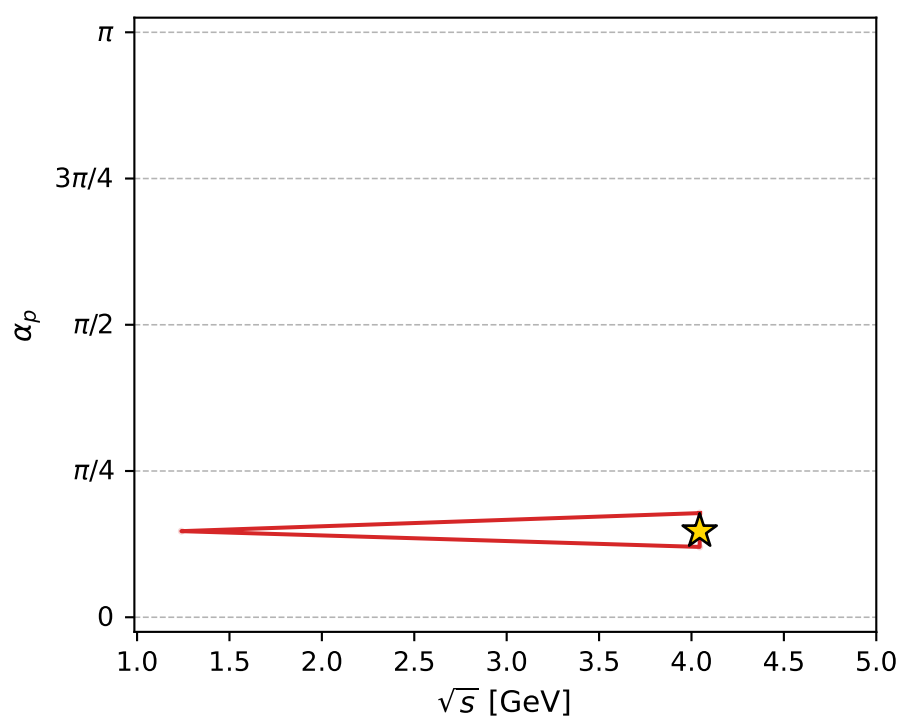
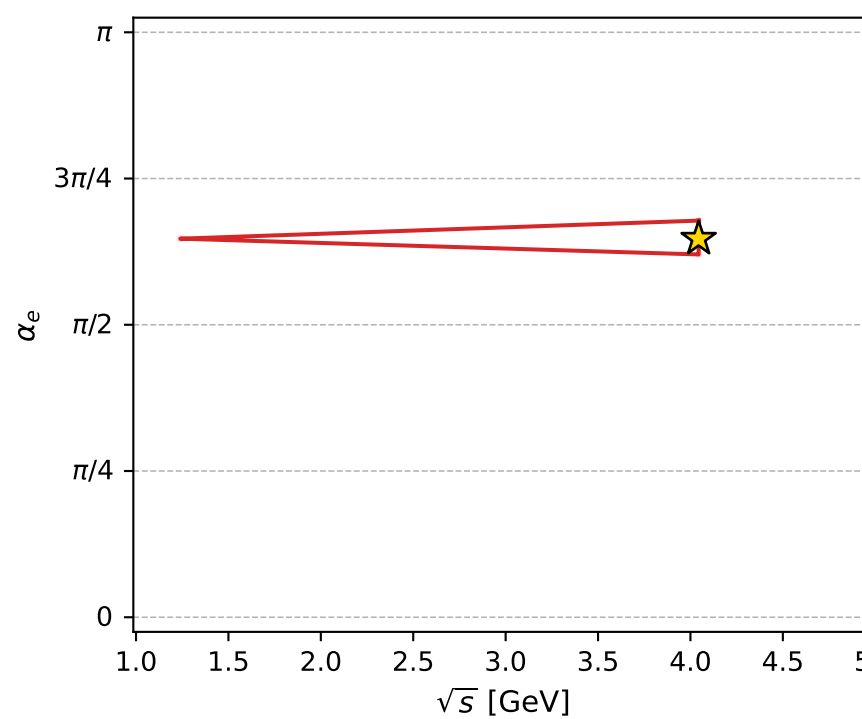
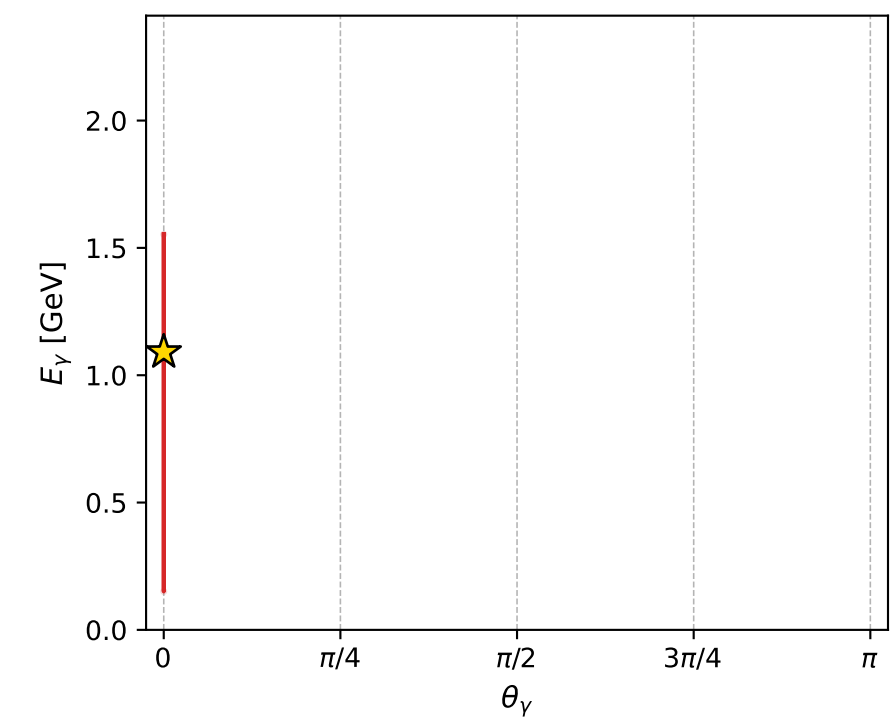
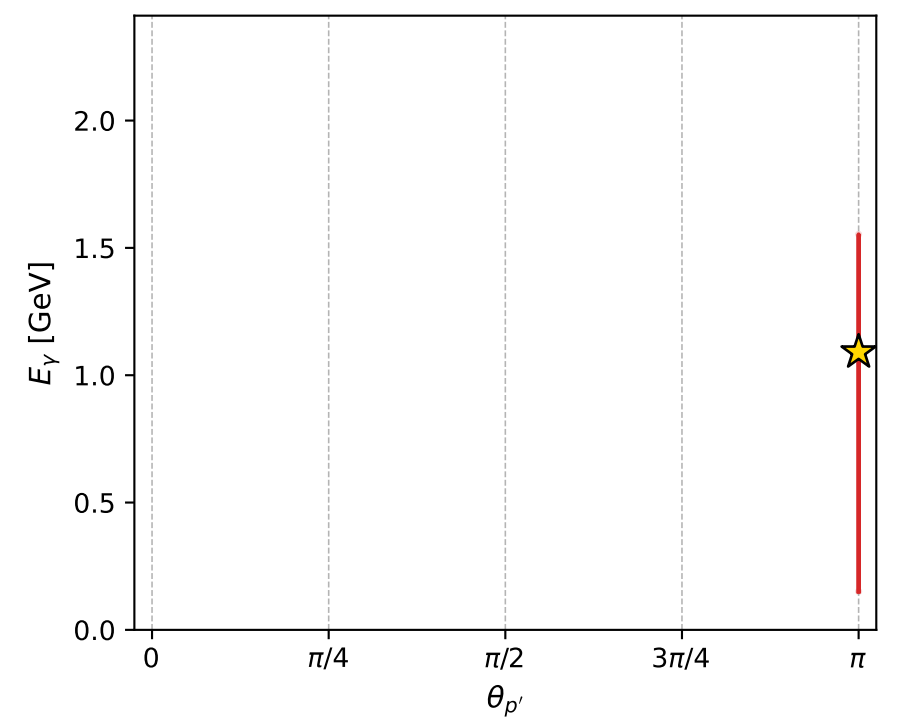
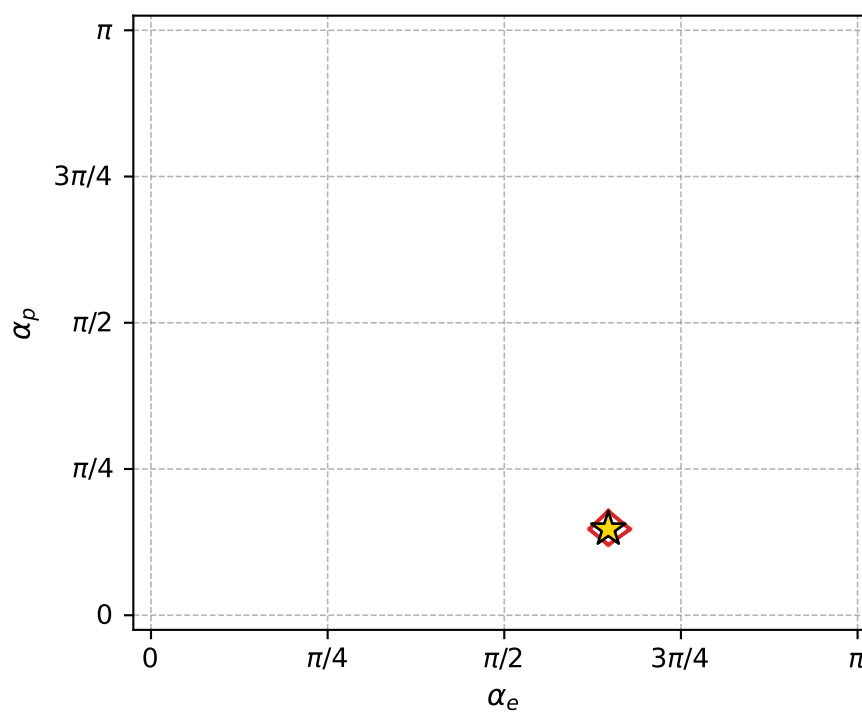
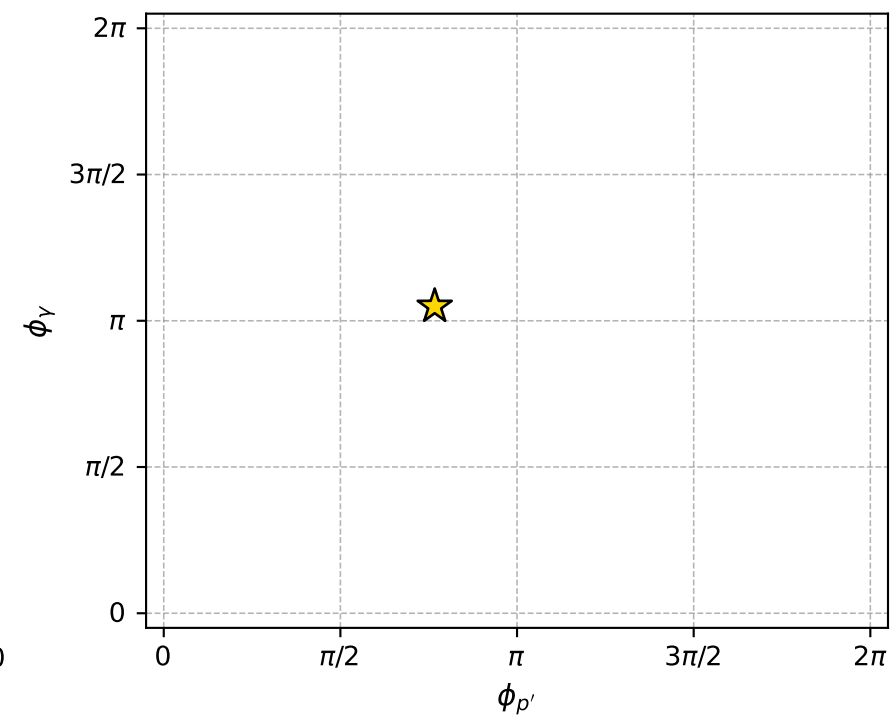
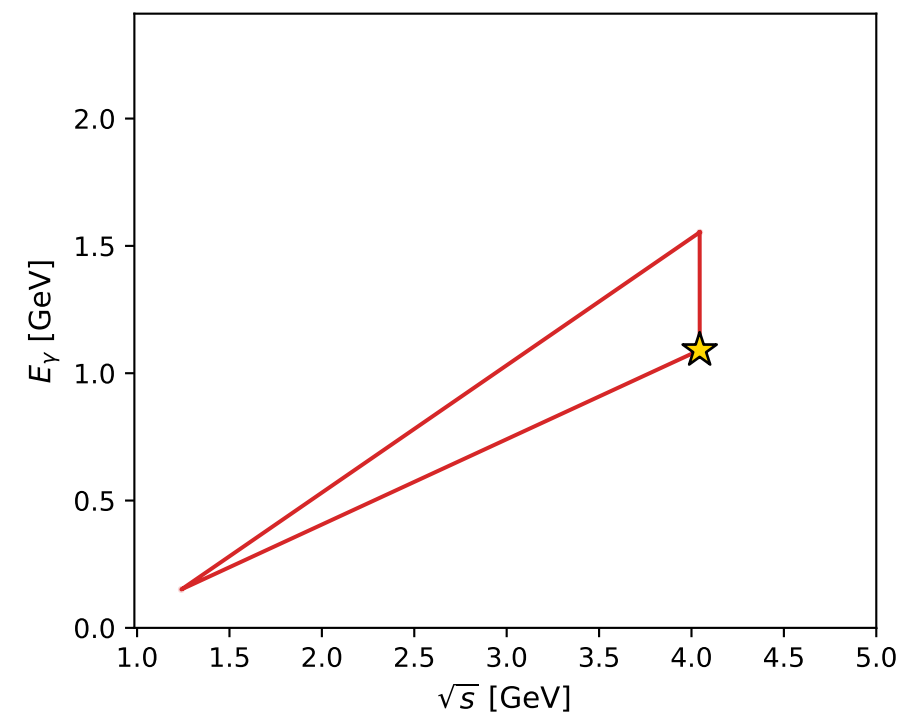
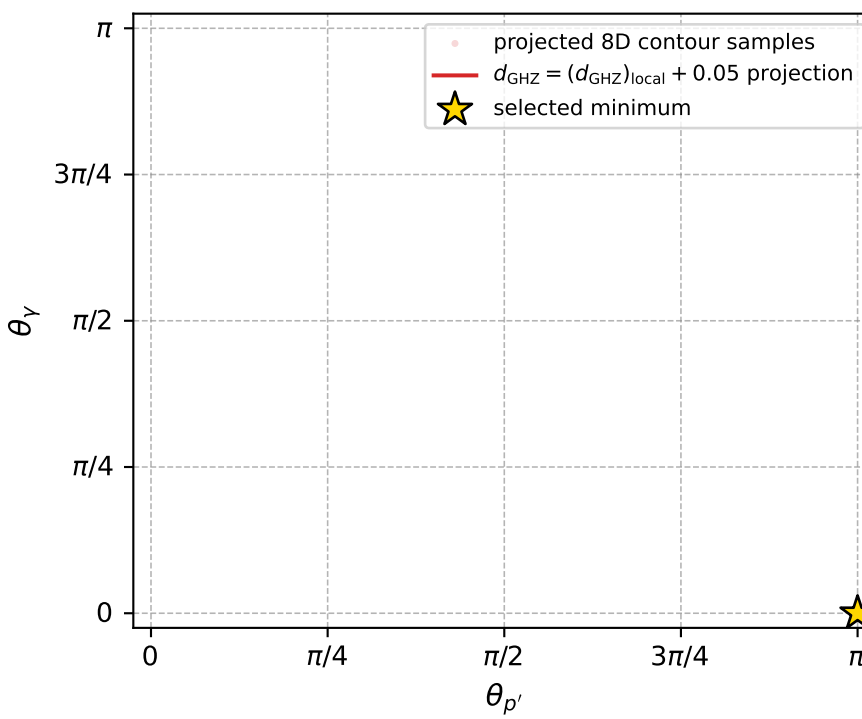
dghz_mixing_angles_polarization_03_local_minimum_158 $d_{\{\text{rm GHZ}\}}=1.03904\text{e-}07$
region: polarization_cluster_03_local_minimum_158
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.0331734$ rad
incoming lepton: $-0.446077|+\rangle +0.894995|-\rangle$
proton mixing angle: $\alpha_p=0.46236846$ rad
incoming proton: $0.894999|+\rangle +0.446069|-\rangle$

$s=16.3566$, $\sqrt{s}=4.04434$, $|\vec{P}|=1.91339$, $|\vec{P}'|=1.91339$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.40943$, $\phi_Y=3.29633$
 $E_Y=1.09092$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.55102$
 $Q^2=6.29483$, $x_B=0.416353$, $t=-14.6443$, $W^2=9.70397$, $y=0.976878$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9134$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 P $E= 2.1309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9134$
 ℓ' $E= 0.8225$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8225$
 P' $E= 2.1309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 q_Y $E= 1.0909$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0909$



electron: polarization cluster P3, local minimum 158; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.0390359\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000104$
 above global minimum = $8.1620129\text{e-}08$
 8D contour samples = 75

selected phase-space point:

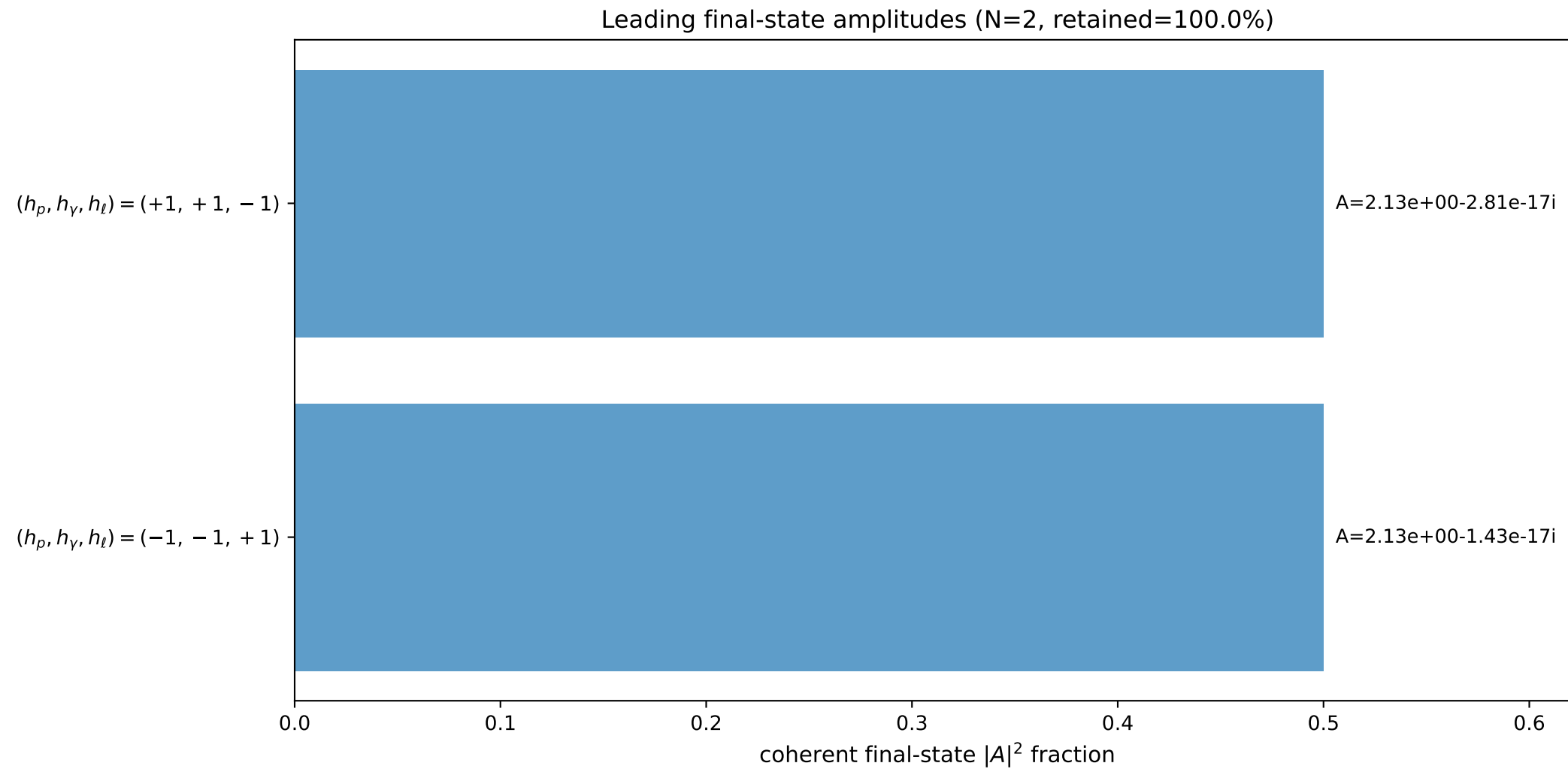
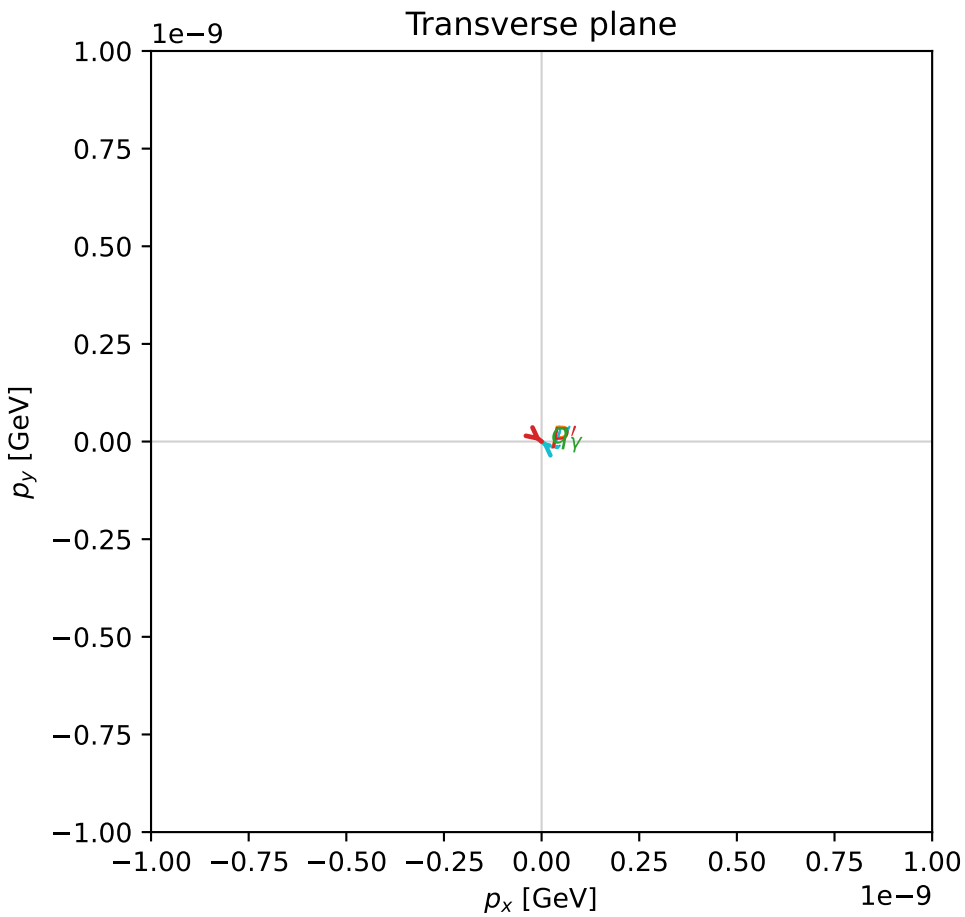
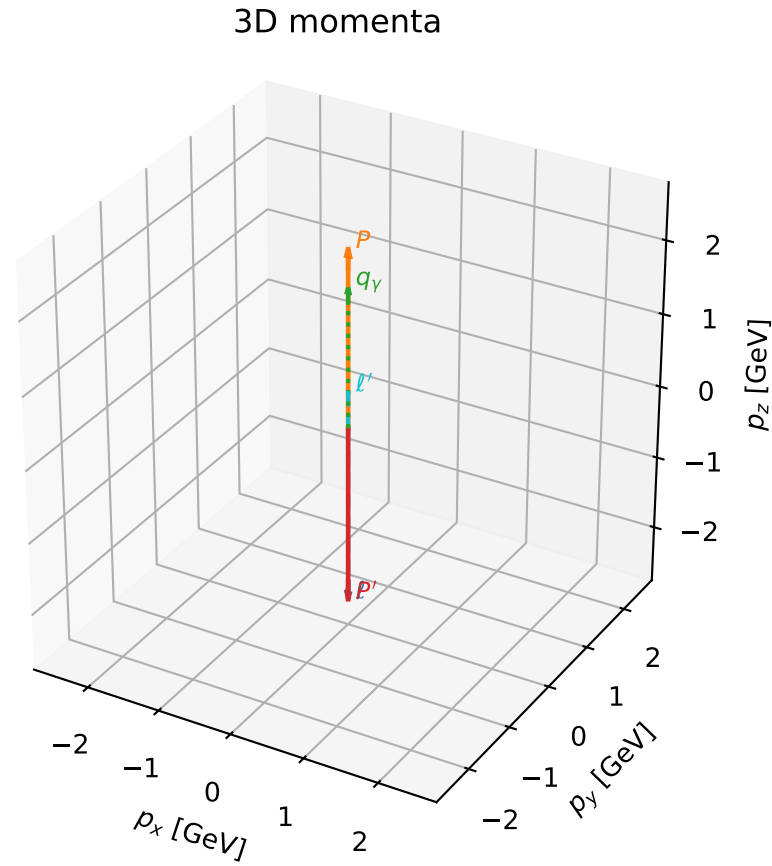
$\text{sqrt_s} = 4.044335$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.090924$
 $\text{phi_p_out} = 2.40943$
 $\text{phi_gamma_out} = 3.296335$
 $\text{alpha_e} = 2.033173$
 $\text{alpha_p} = 0.4623685$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

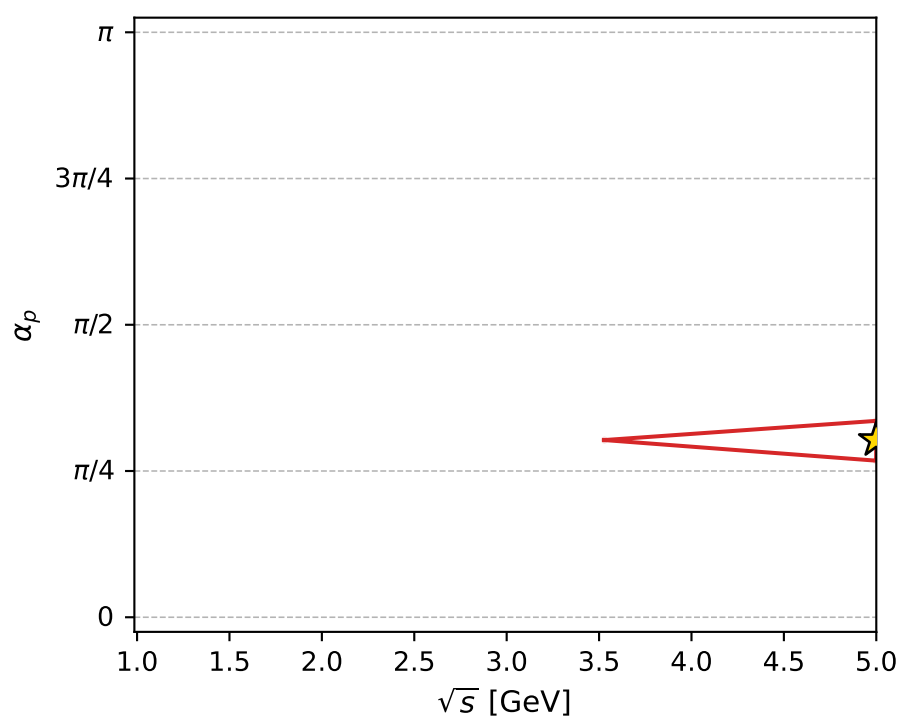
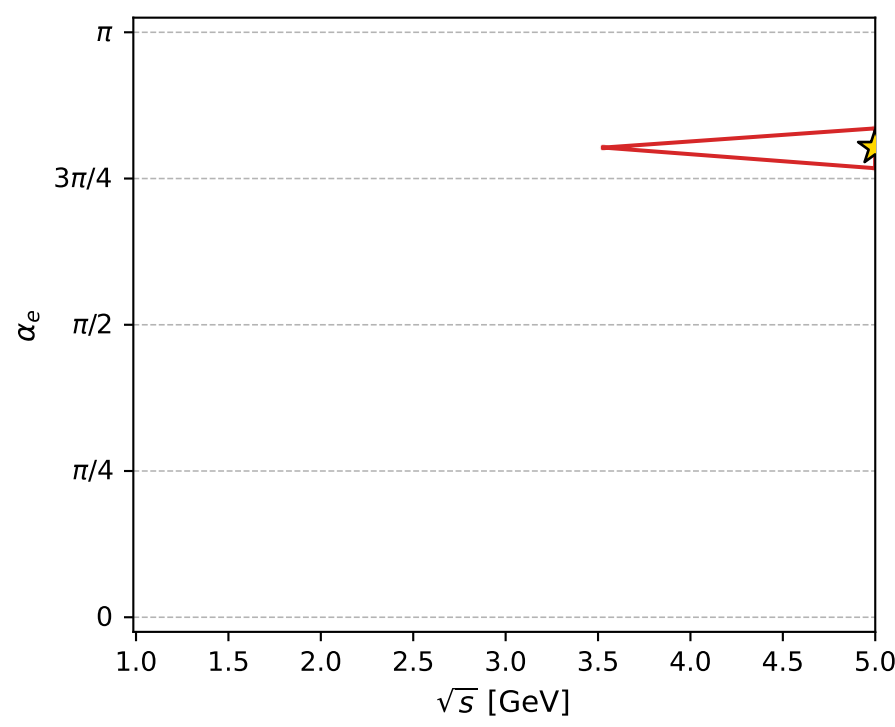
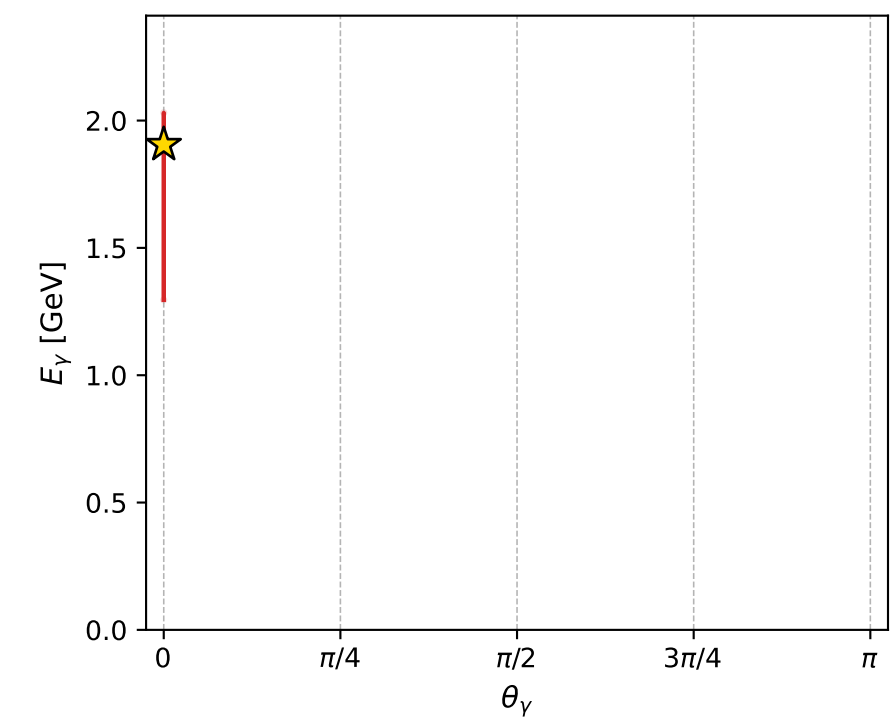
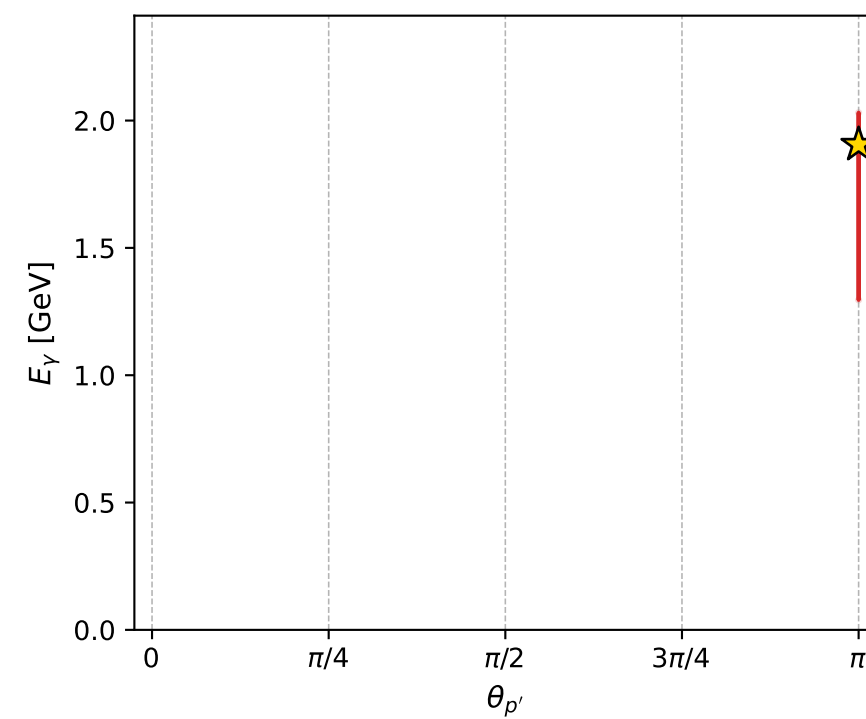
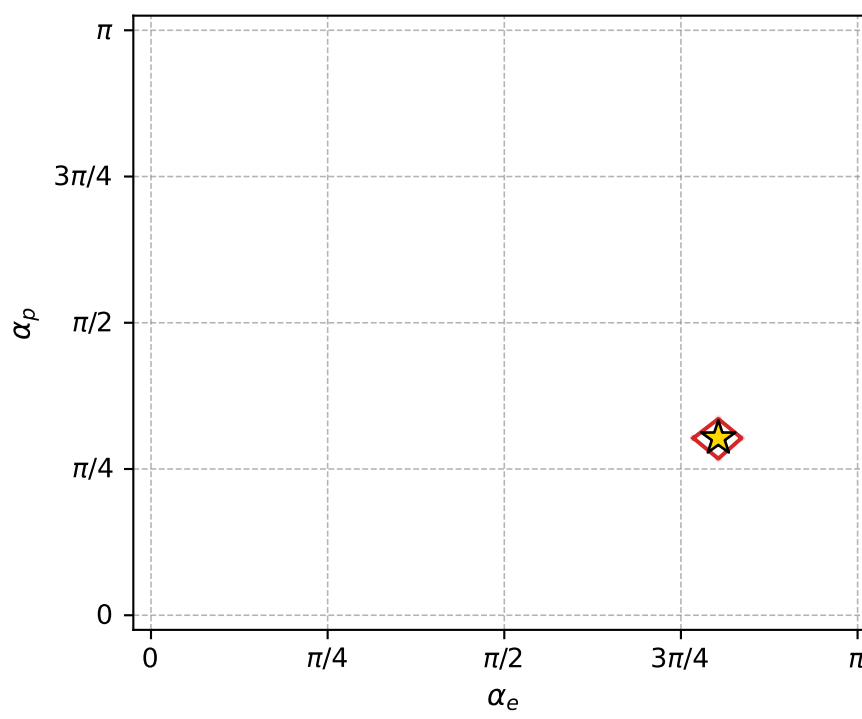
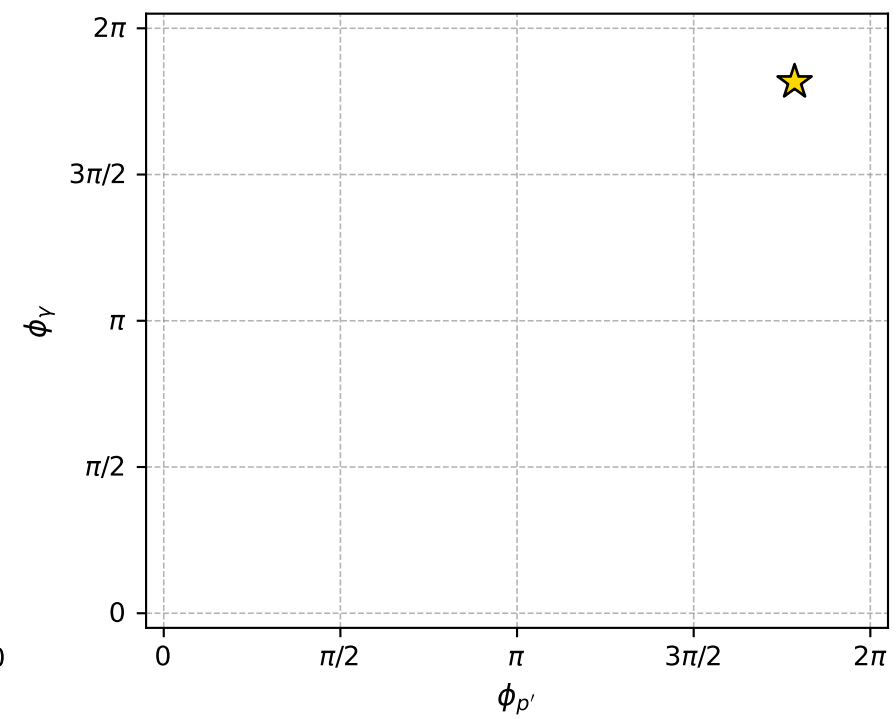
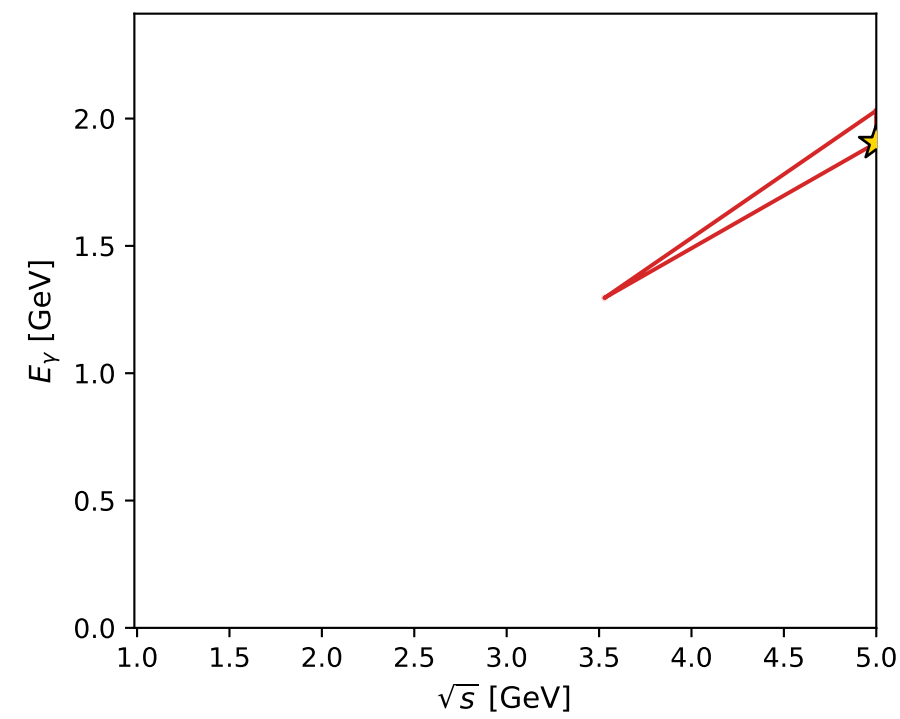
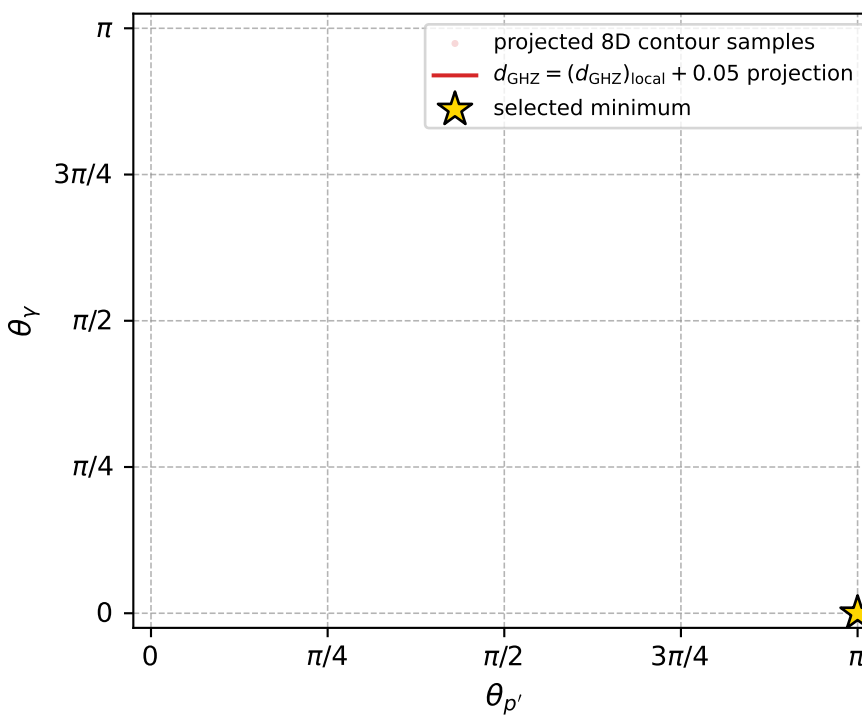
dghz_mixing_angles_polarization_03_local_minimum_161 $d_{\{\text{rm GHZ}\}}=1.13122\text{e-}07$
region: polarization_cluster_03_local_minimum_161
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.5224319$ rad
incoming lepton: $-0.814366|+\rangle +0.580352|-\rangle$
proton mixing angle: $\alpha_p=0.95162257$ rad
incoming proton: $0.580362|+\rangle +0.814358|-\rangle$

$s=24.9928$, $\sqrt{s}=4.99928$, $|\vec{P}|=2.41164$, $|\vec{P}'|=2.41164$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=5.60969$, $\phi_Y=5.70666$
 $E_Y=1.90533$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.4681$
 $Q^2=4.88418$, $x_B=0.204062$, $t=-23.2641$, $W^2=19.9304$, $y=0.992609$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4116$
 P $E= 2.5876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4116$
 ℓ' $E= 0.5063$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.5063$
 P' $E= 2.5876$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4116$
 q_Y $E= 1.9053$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9053$



electron: polarization cluster P3, local minimum 161; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1312189\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000113$
 above global minimum = $9.0838431\text{e-}08$
 8D contour samples = 68

selected phase-space point:

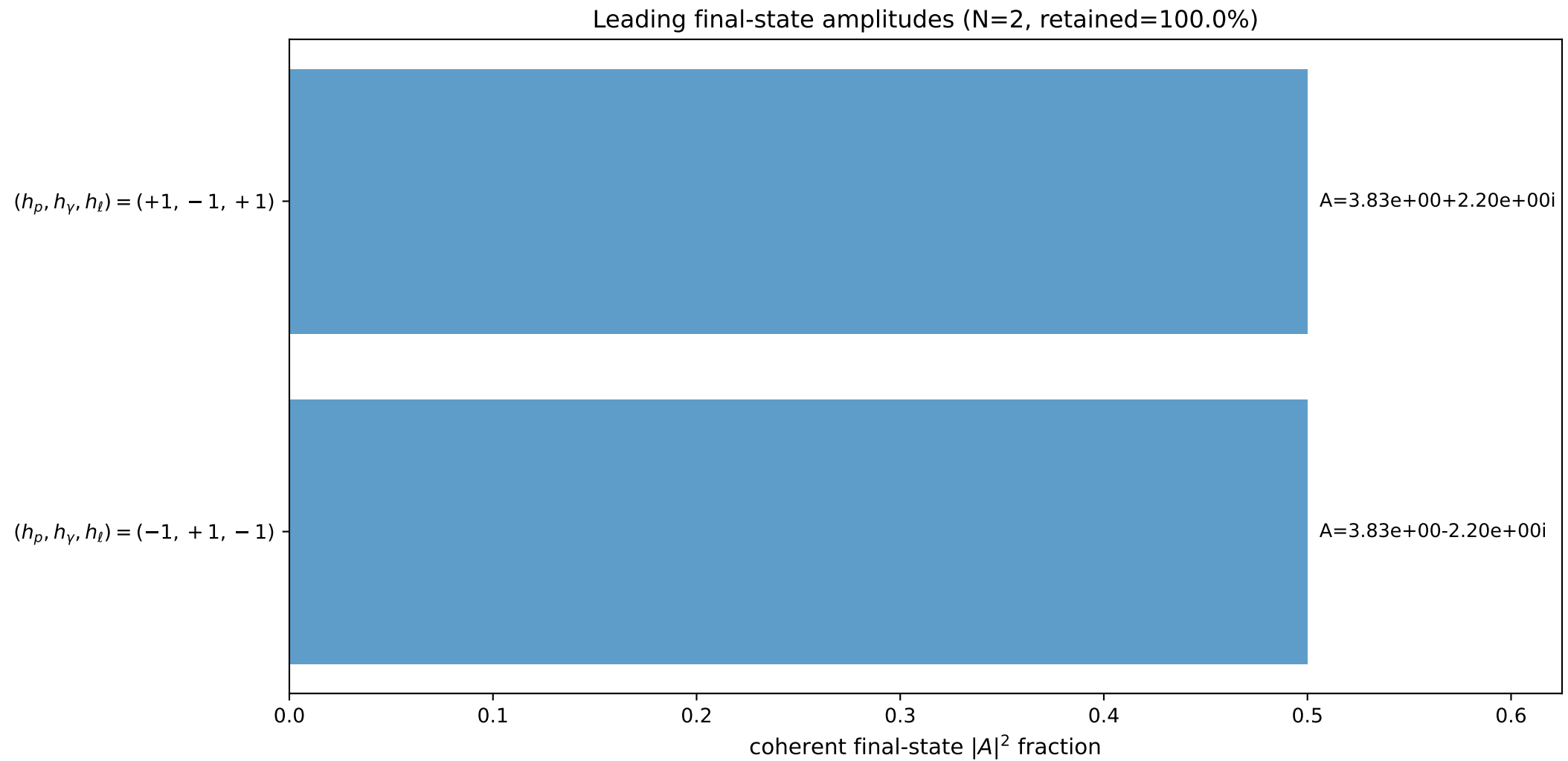
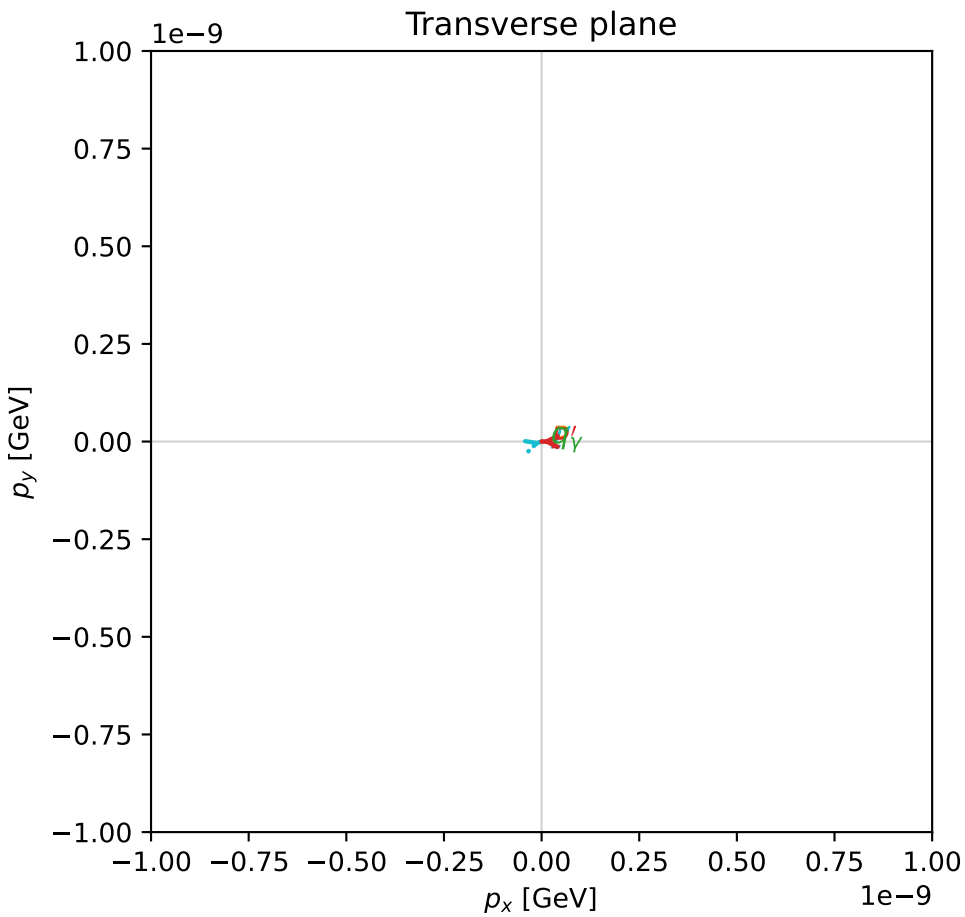
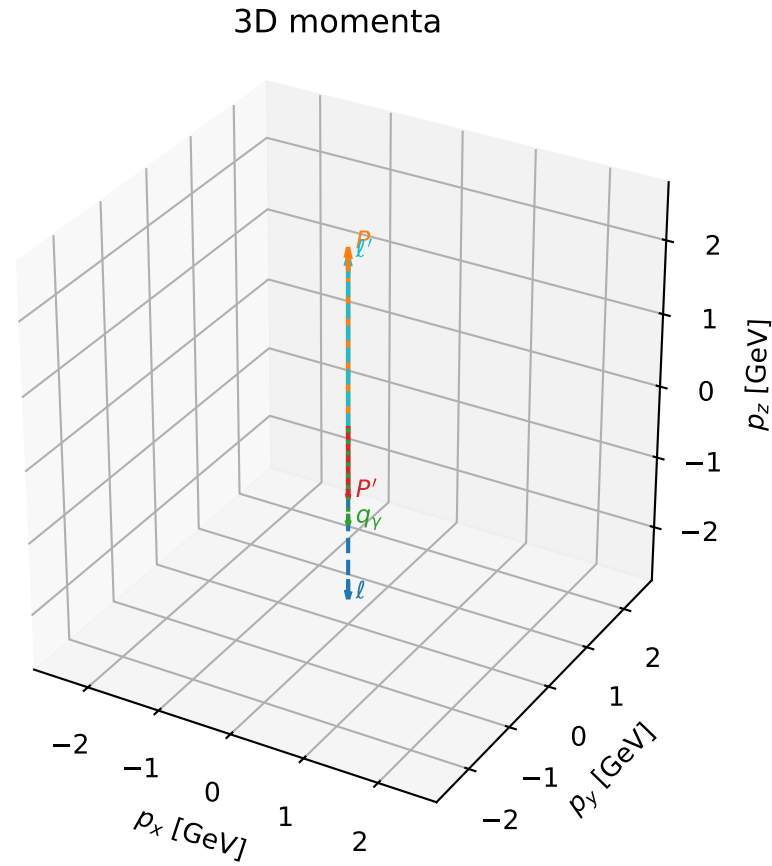
$\text{sqrt_s} = 4.999281$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.905331$
 $\text{phi_p_out} = 5.609693$
 $\text{phi_gamma_out} = 5.706657$
 $\text{alpha_e} = 2.522432$
 $\text{alpha_p} = 0.9516226$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

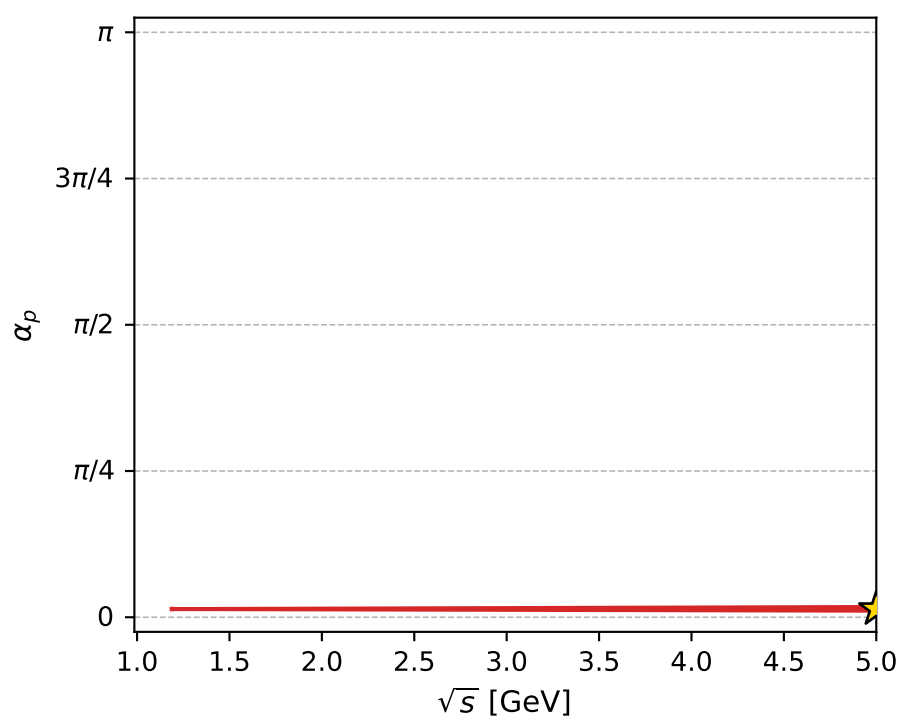
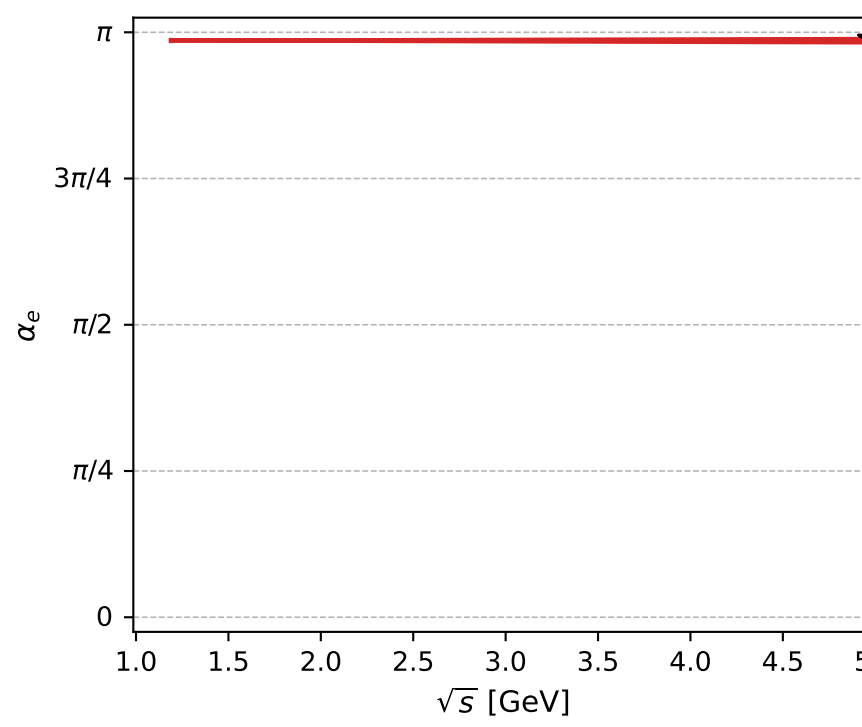
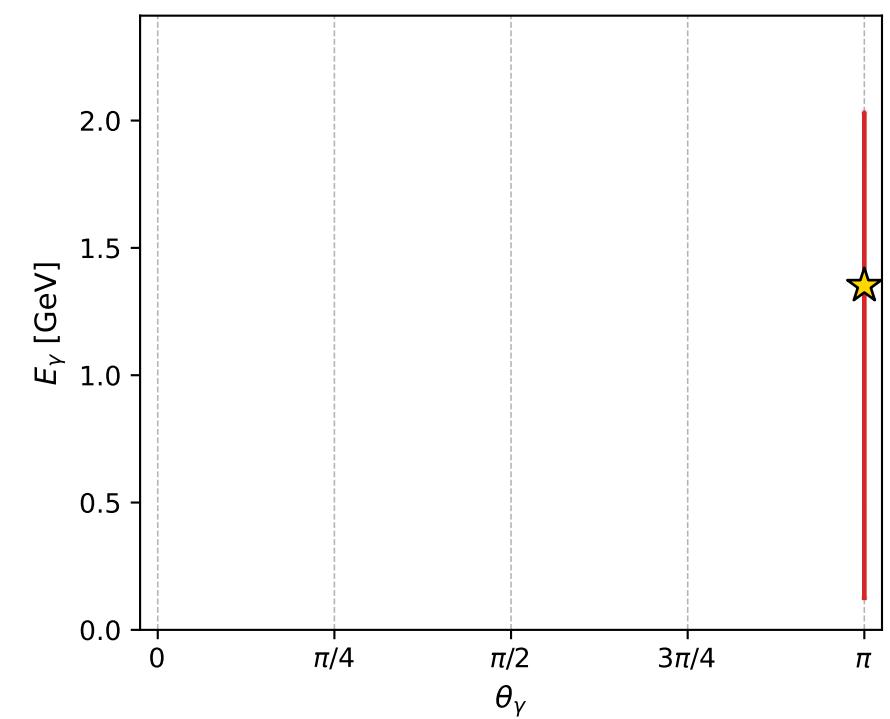
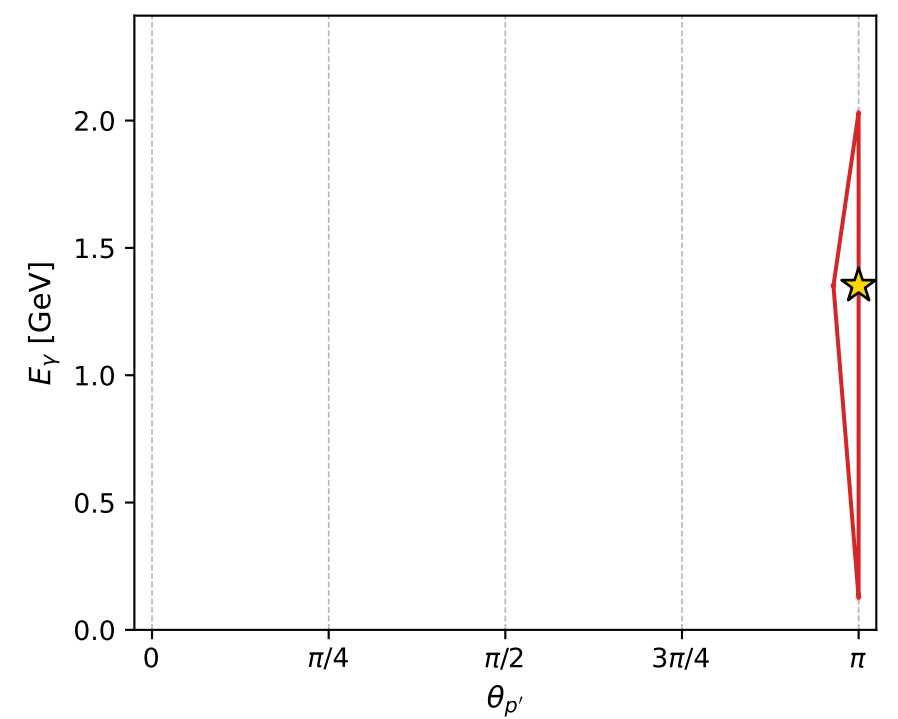
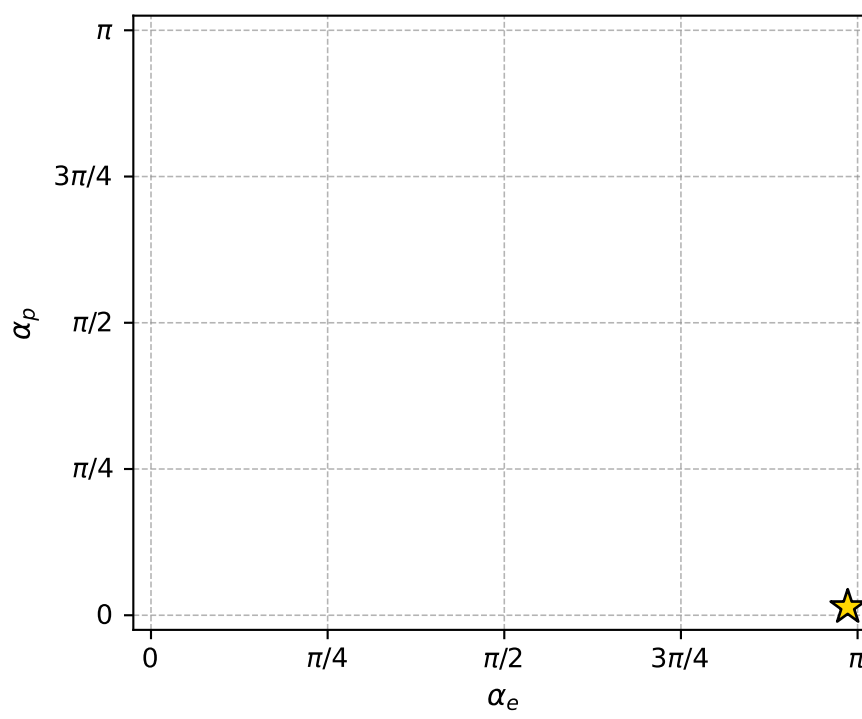
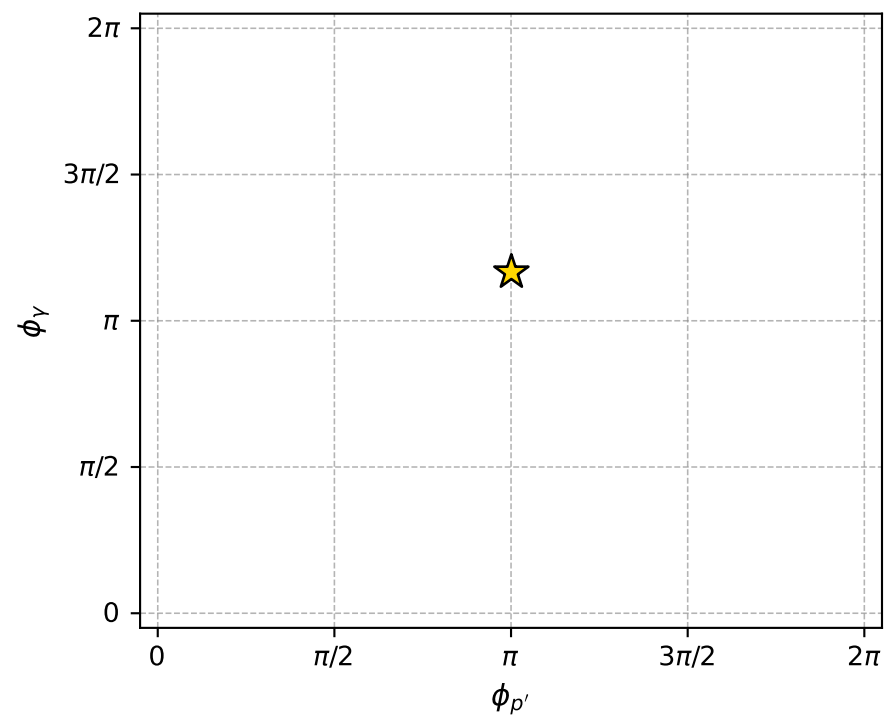
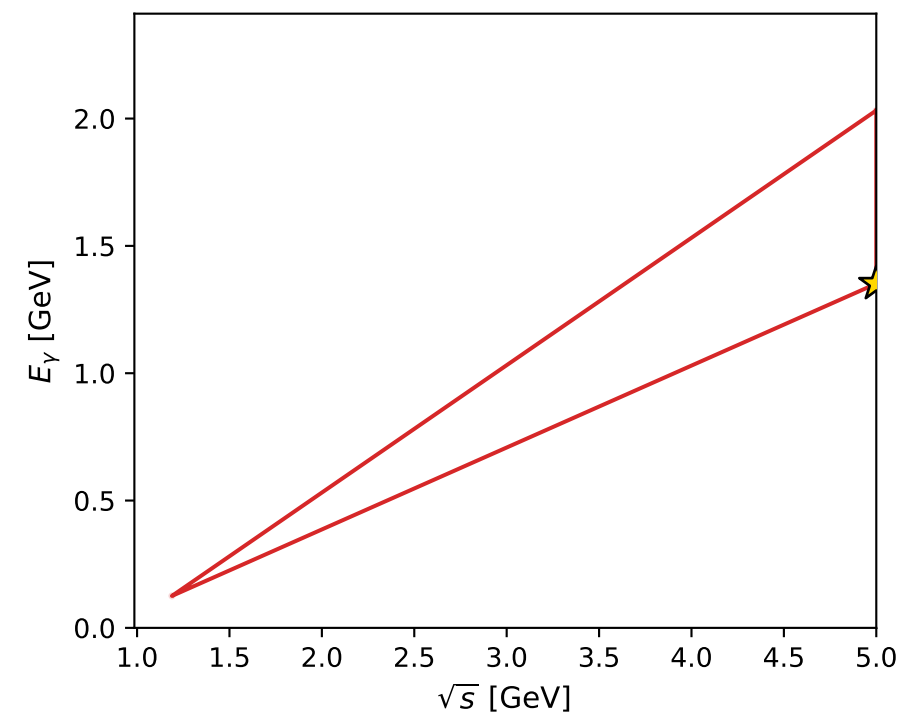
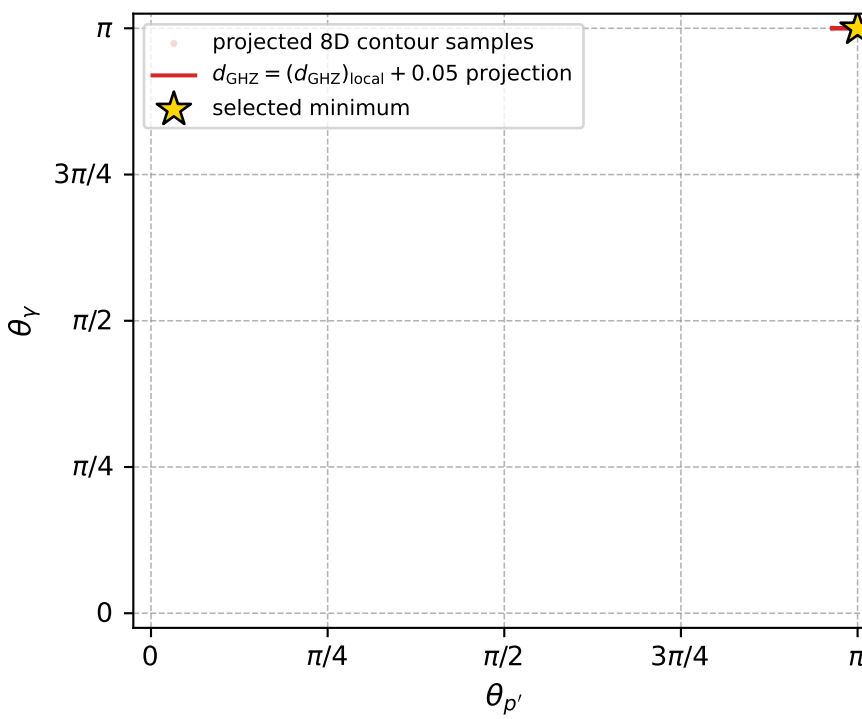
dghz_mixing_angles_polarization_03_local_minimum_174 $d_{\{\text{rm GHZ}\}}=2.67083\text{e-}07$
 region: polarization_cluster_03_local_minimum_174
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=3.0976361$ rad
 incoming lepton: $-0.999034|+\rangle +0.0439424|-\rangle$
 proton mixing angle: $\alpha_p=0.043954198$ rad
 incoming proton: $0.999034|+\rangle +0.04394|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.956774$
 $\theta_{P'}=3.14159$, $\theta_V=3.14159$, $\phi_{P'}=3.14456$, $\phi_V=3.66323$
 $E_V=1.35168$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.307656$
 $Q^2=22.2721$, $x_B=0.955566$, $t=-9.79096$, $W^2=1.9155$, $y=0.966317$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3085$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3085$
 P' $E= 1.3399$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9568$
 q_V $E= 1.3517$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.3517$



electron: polarization cluster P3, local minimum 174; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6708281\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000267$
 above global minimum = $2.4479935\text{e-}07$
 8D contour samples = 36

selected phase-space point:

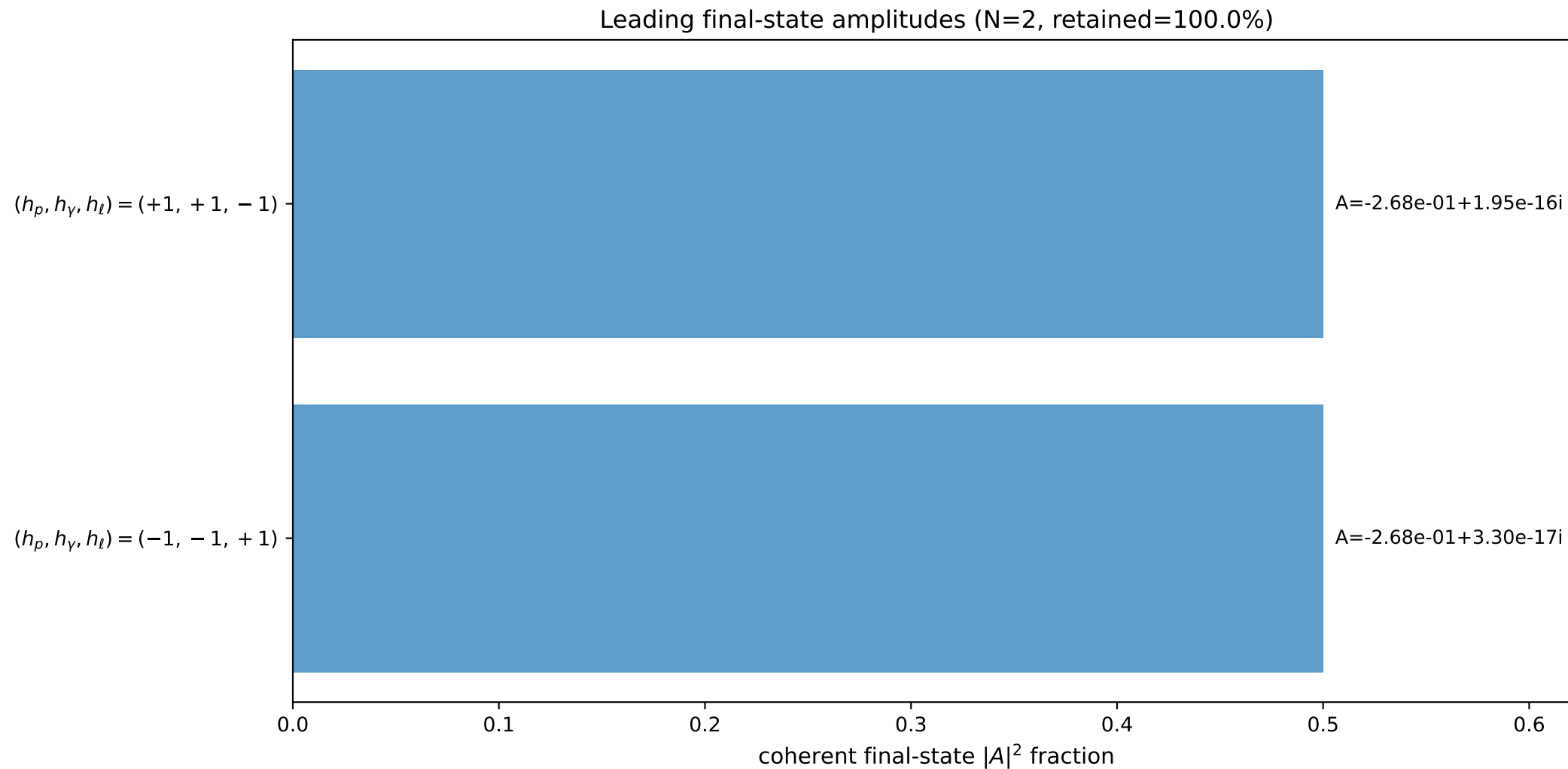
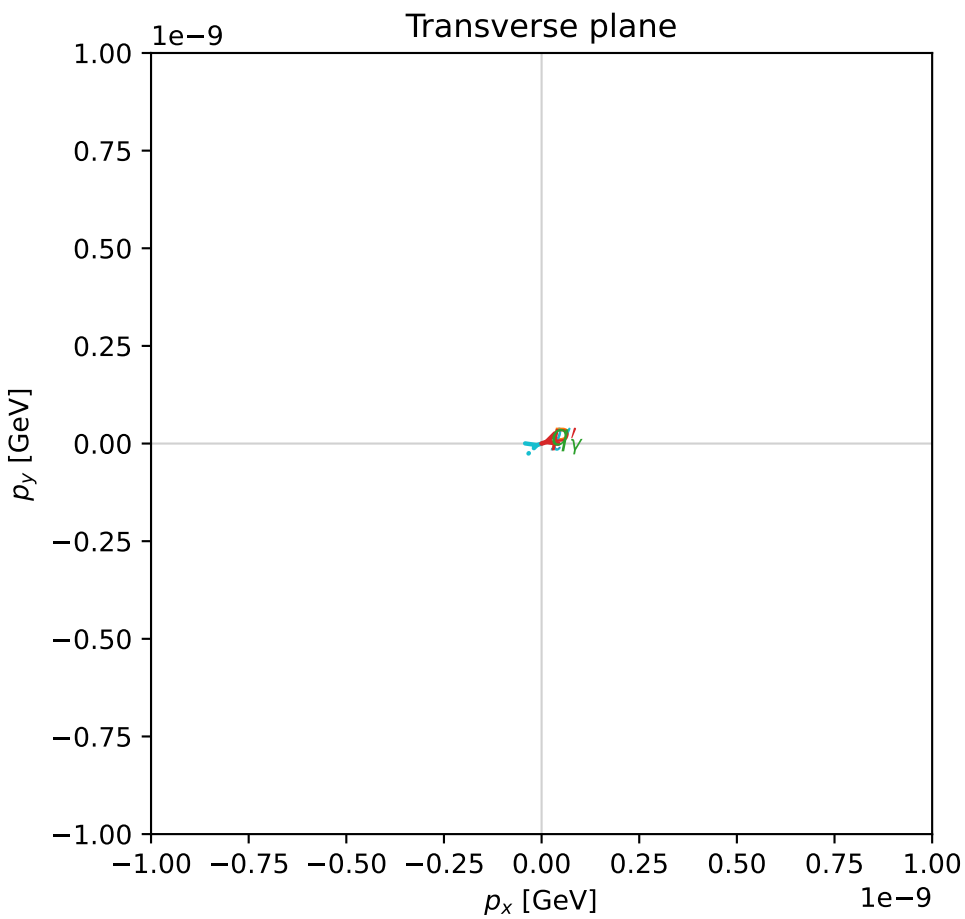
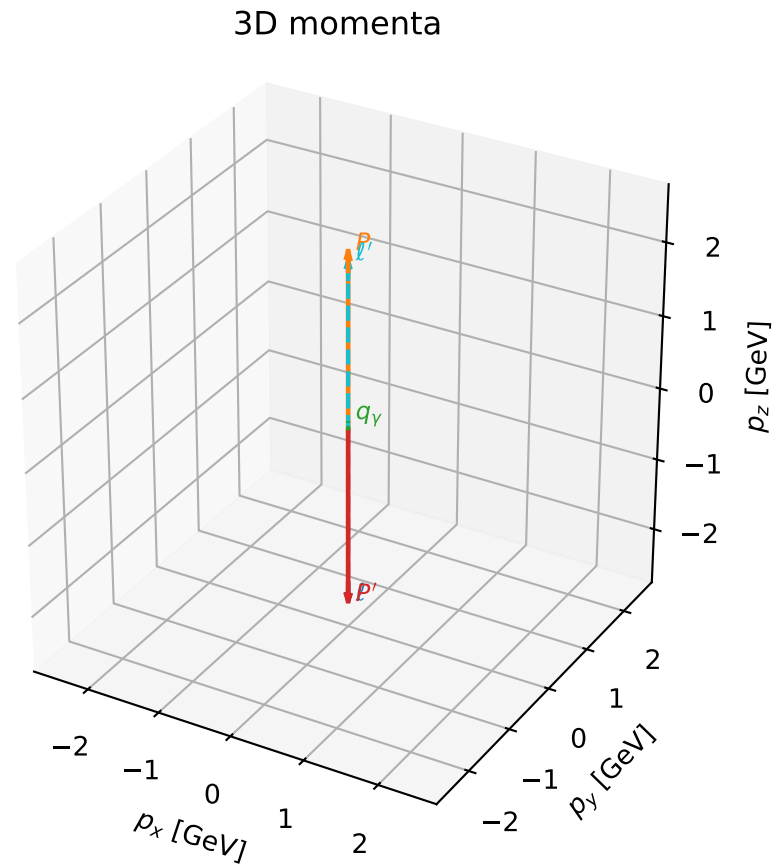
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.351677$
 $\text{phi_p_out} = 3.144557$
 $\text{phi_gamma_out} = 3.663231$
 $\text{alpha_e} = 3.097636$
 $\text{alpha_p} = 0.0439542$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

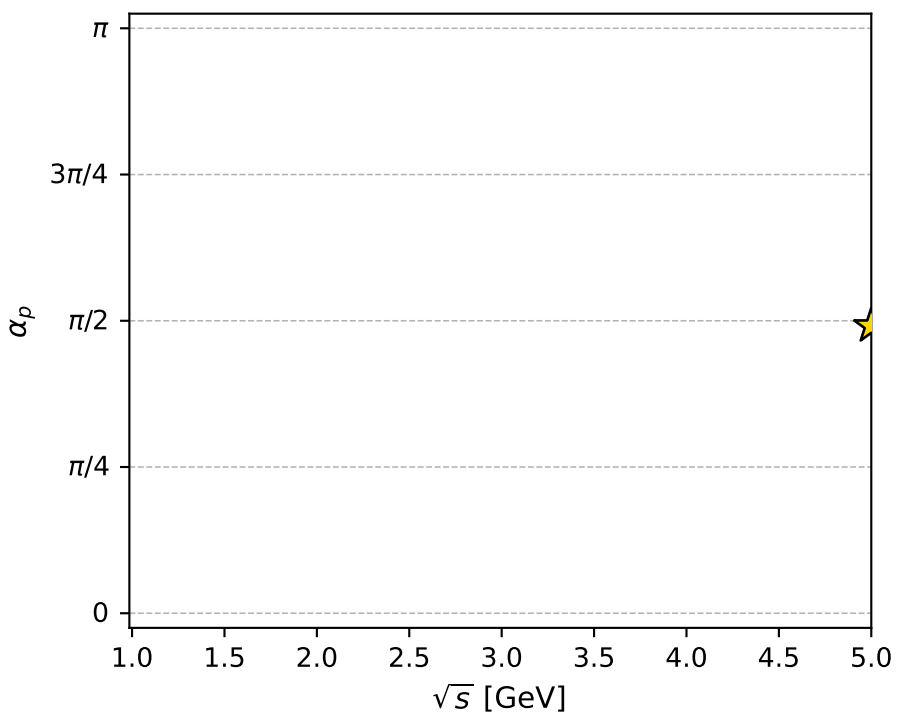
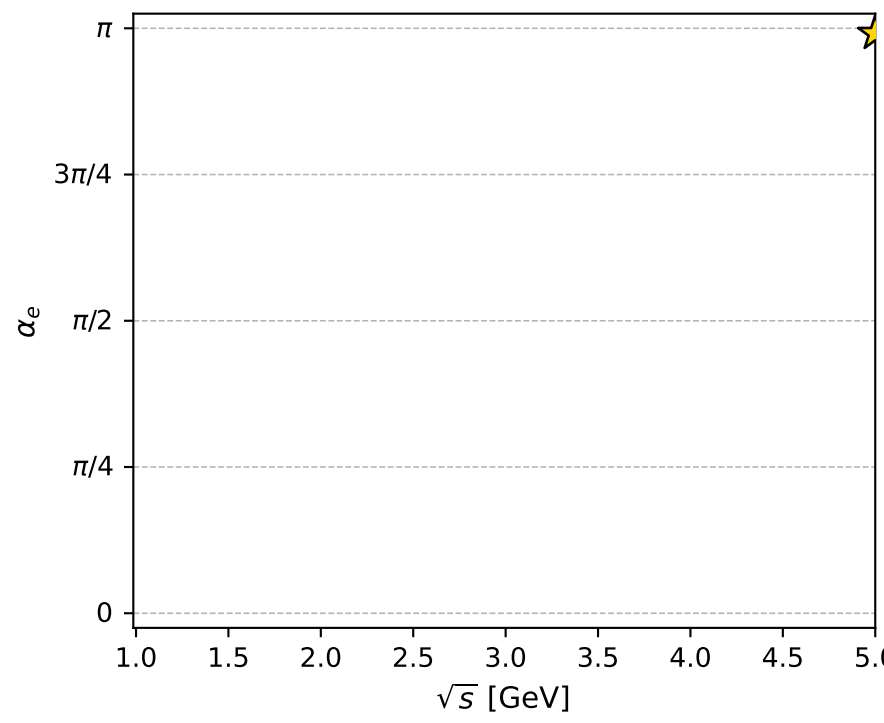
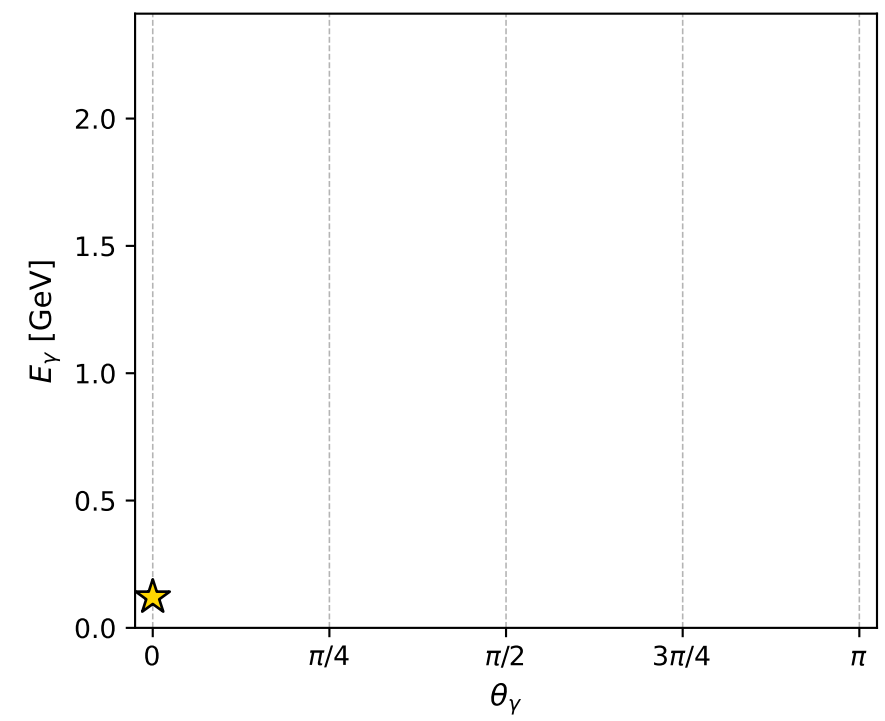
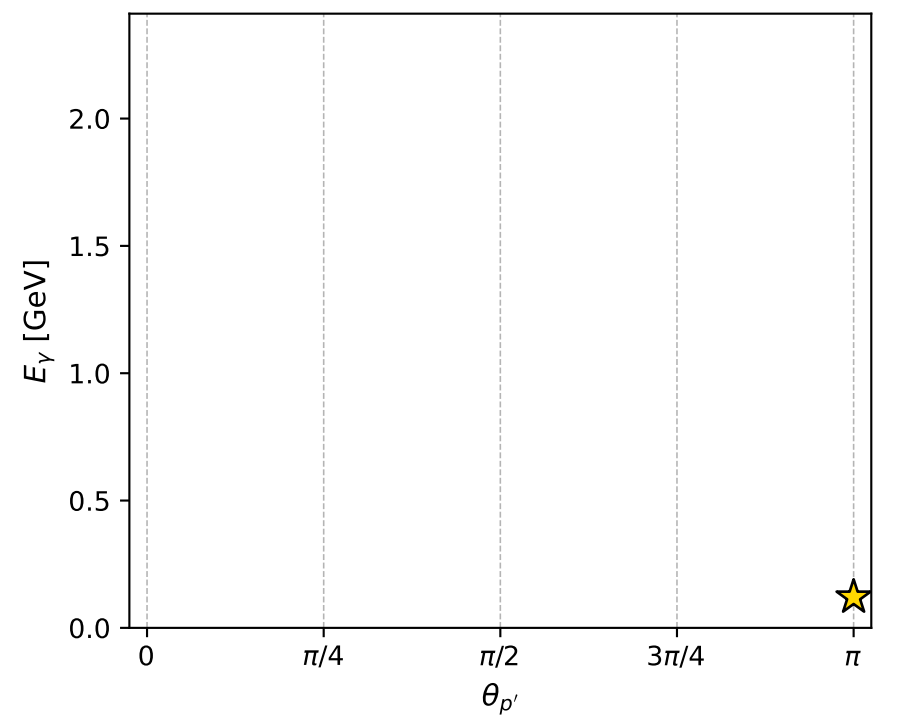
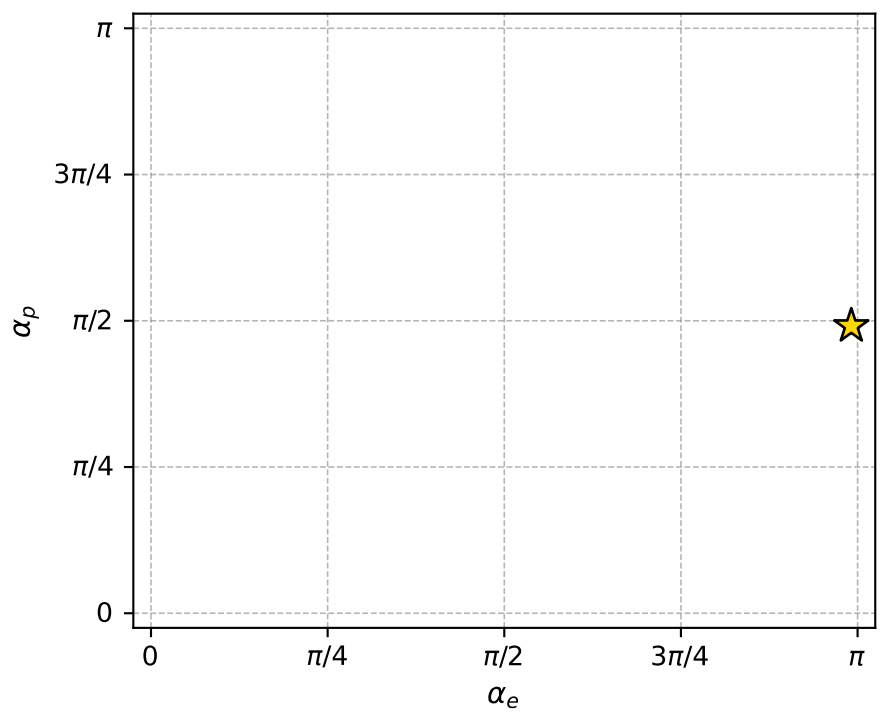
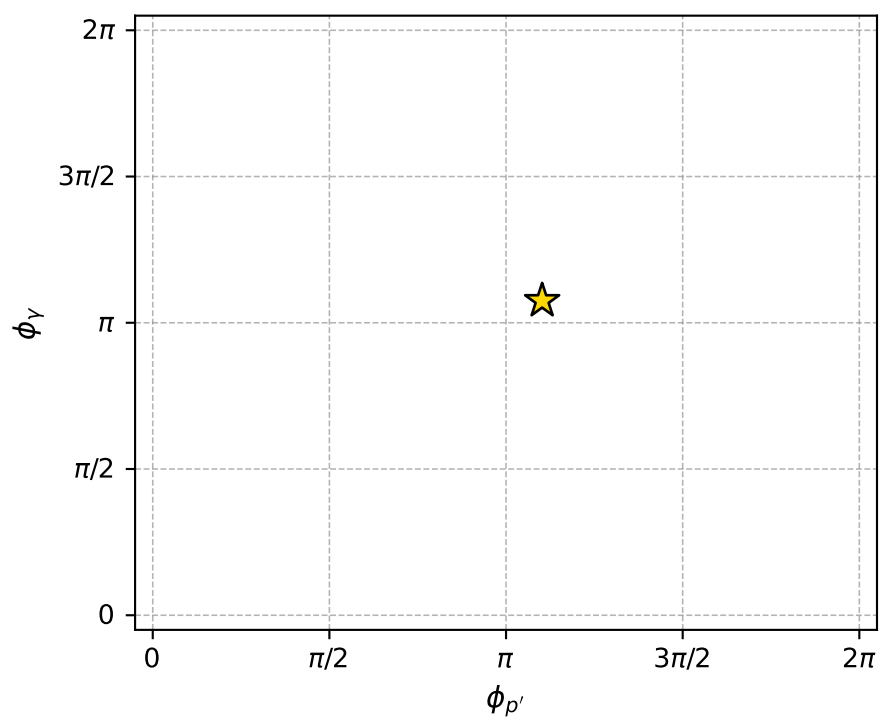
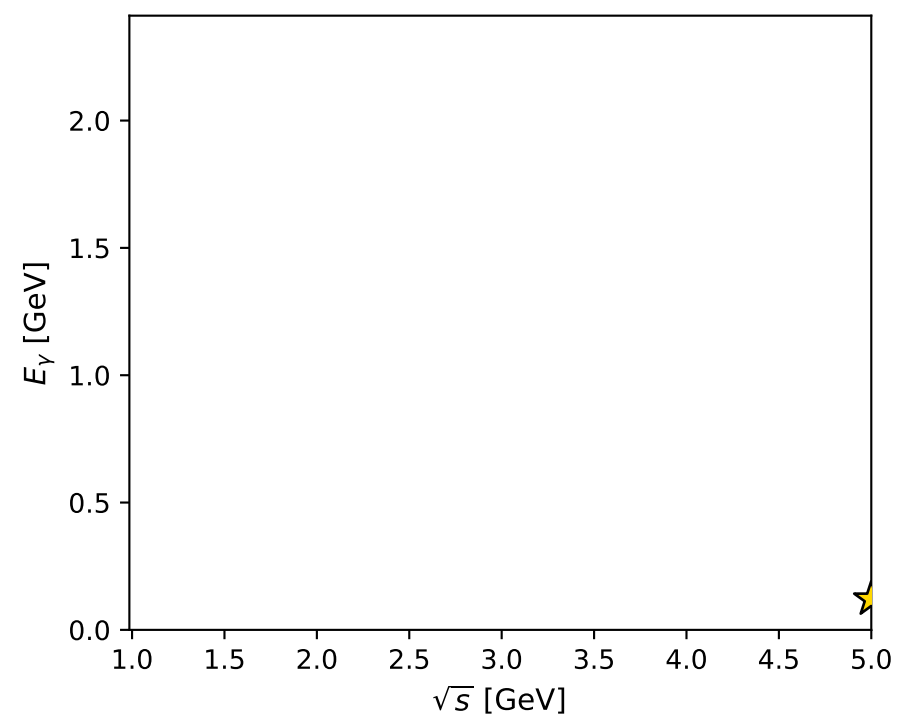
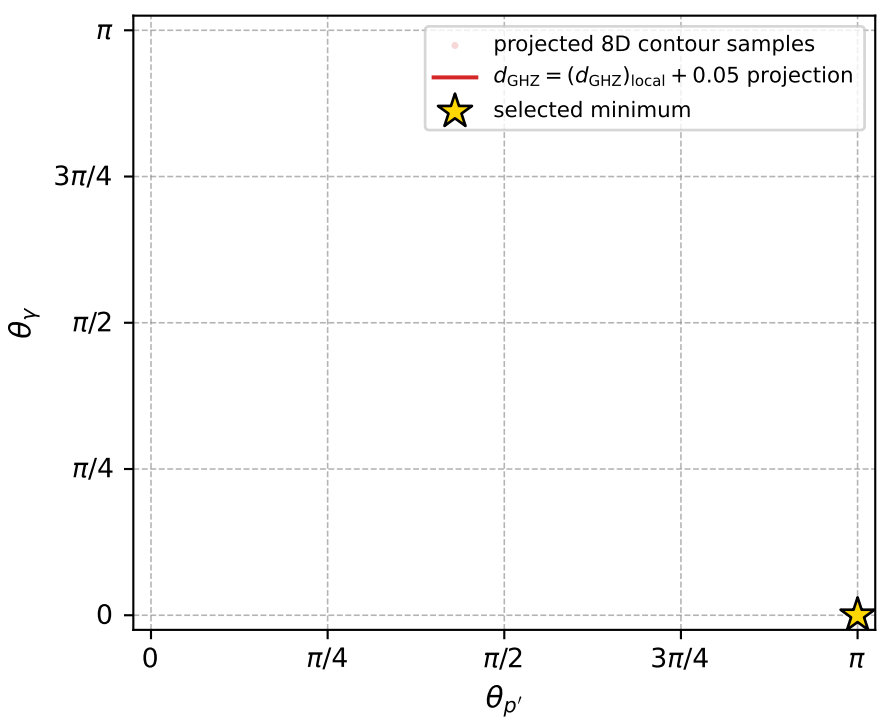
dghz_mixing_angles_polarization_03_local_minimum_178 d_{\rm GHZ}=4.23054e-07
region: polarization_cluster_03_local_minimum_178
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=3.1136594$ rad
incoming lepton: $-0.99961|+\rangle +0.0279296|-\rangle$
proton mixing angle: $\alpha_p=1.5428613$ rad
incoming proton: $0.0279314|+\rangle +0.99961|-\rangle$

$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=2.41181$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=3.46255$, $\phi_Y=3.37801$
 $E_Y=0.120586$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.320954$
 $Q^2=22.104$, $x_B=0.948272$, $t=-23.2674$, $W^2=2.08561$, $y=0.966561$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.2912$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2912$
 P' $E= 2.5878$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4118$
 q_Y $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$



electron: polarization cluster P3, local minimum 178; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.2305389\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000423$
 above global minimum = $4.0077043\text{e-}07$
 8D contour samples = 55

selected phase-space point:

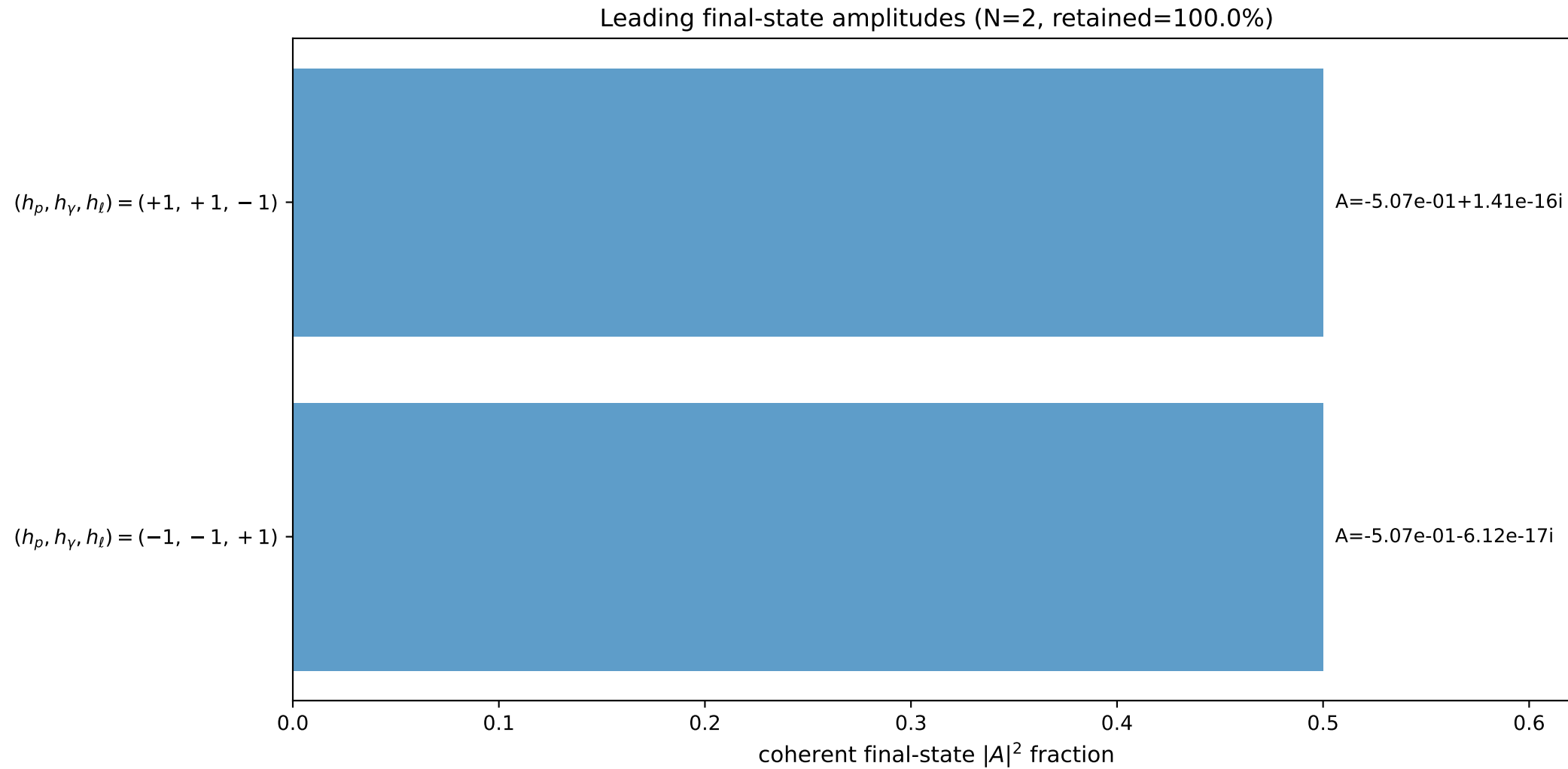
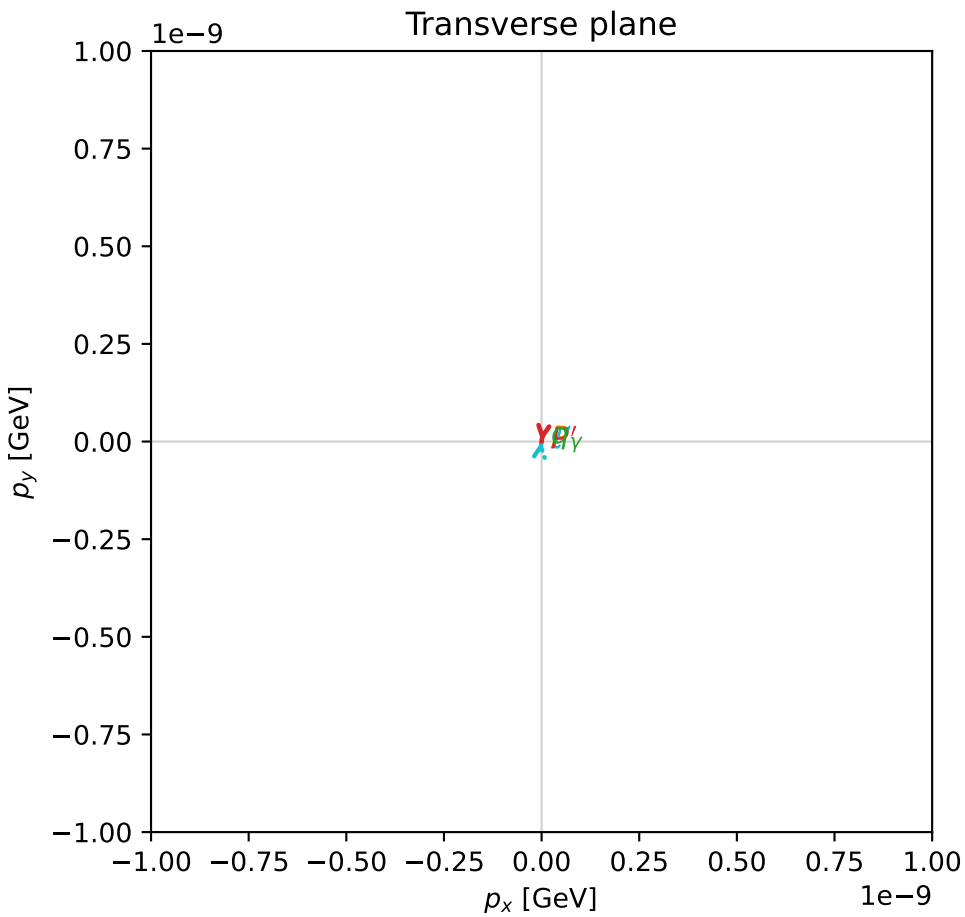
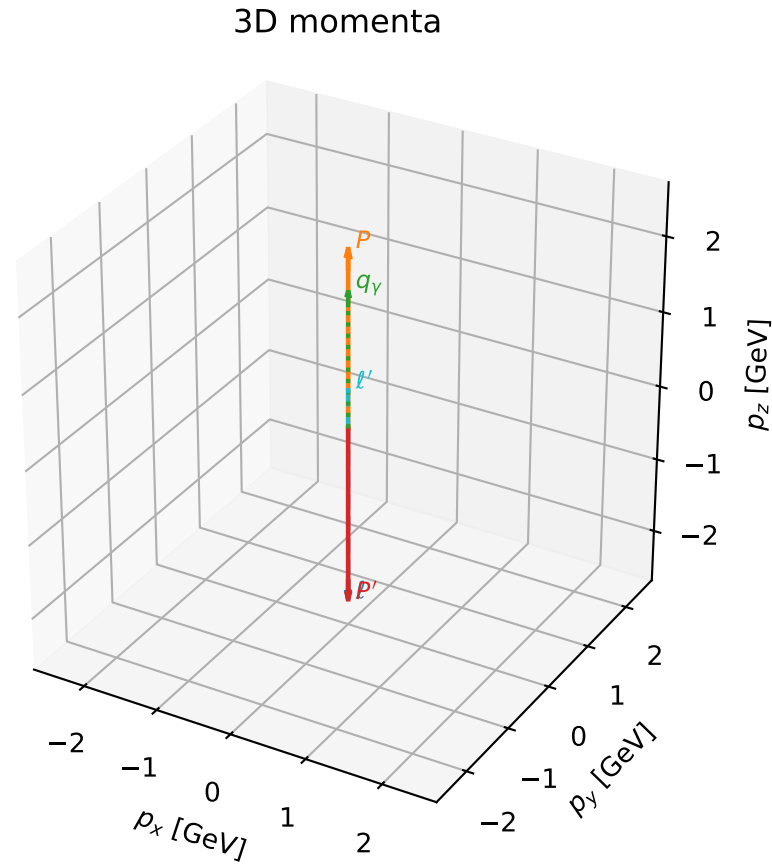
$\text{sqrt_s} = 4.999608$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{qOut} = 0.1205858$
 $\text{phi_p_out} = 3.462547$
 $\text{phi_gamma_out} = 3.378009$
 $\text{alpha_e} = 3.113659$
 $\text{alpha_p} = 1.542861$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

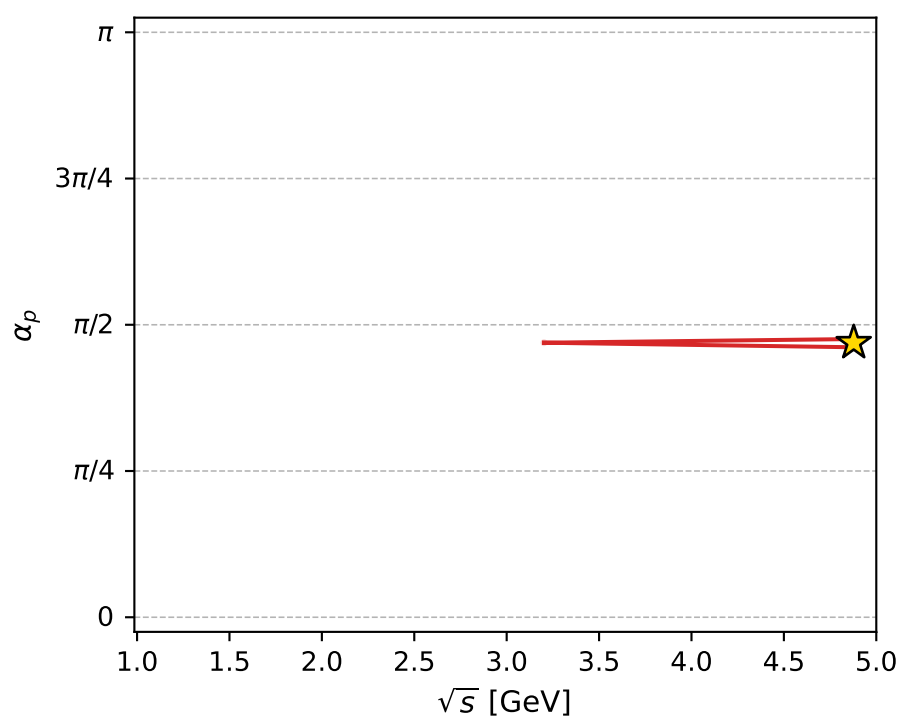
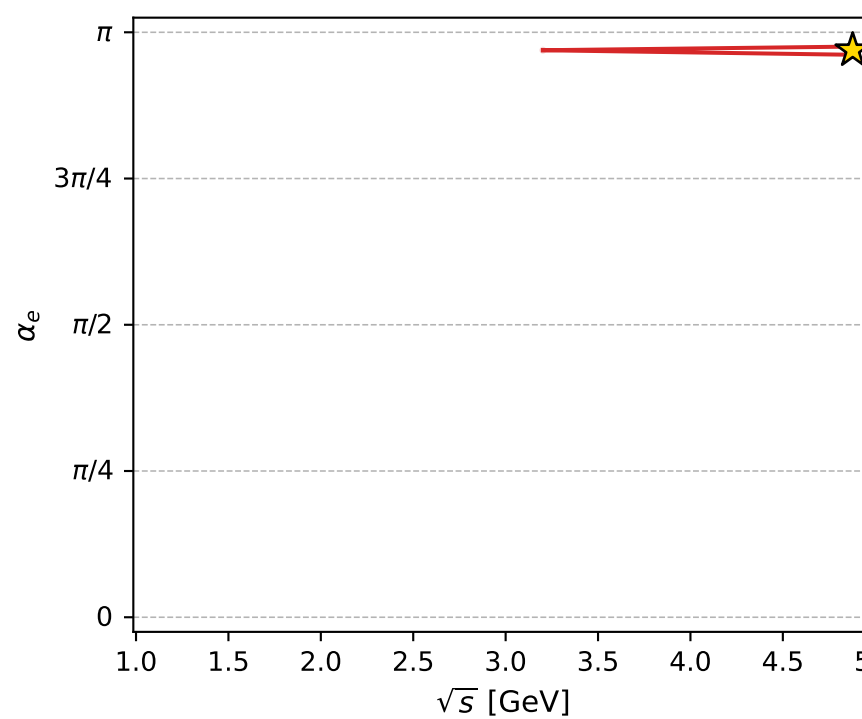
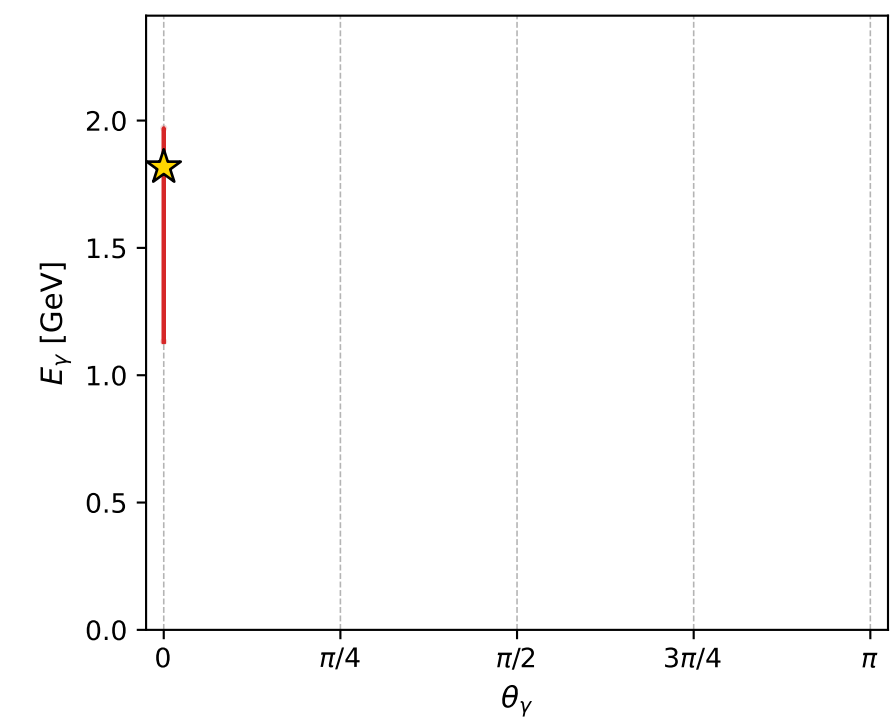
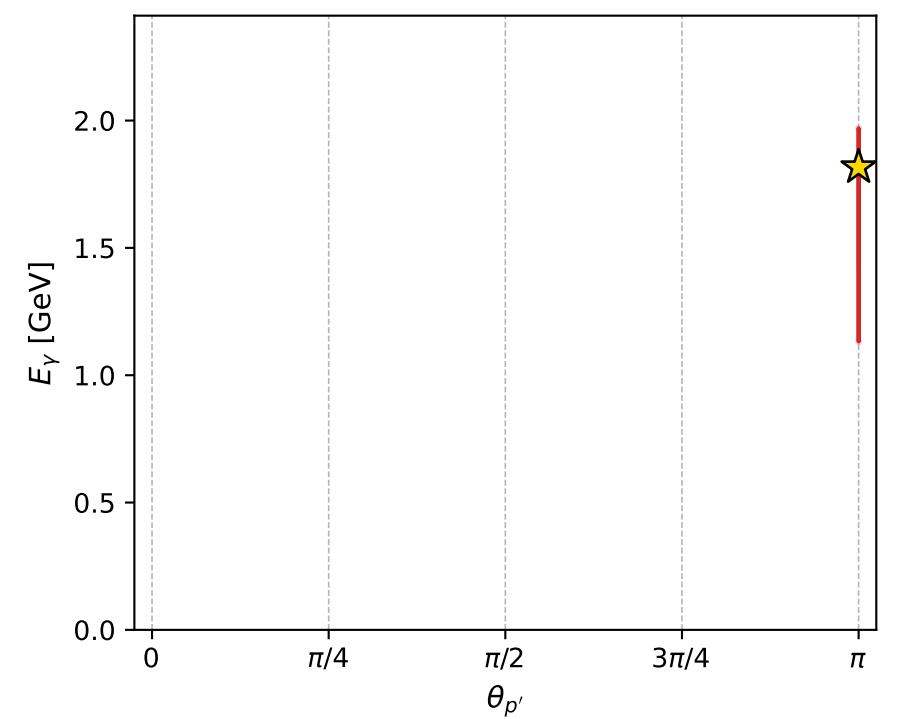
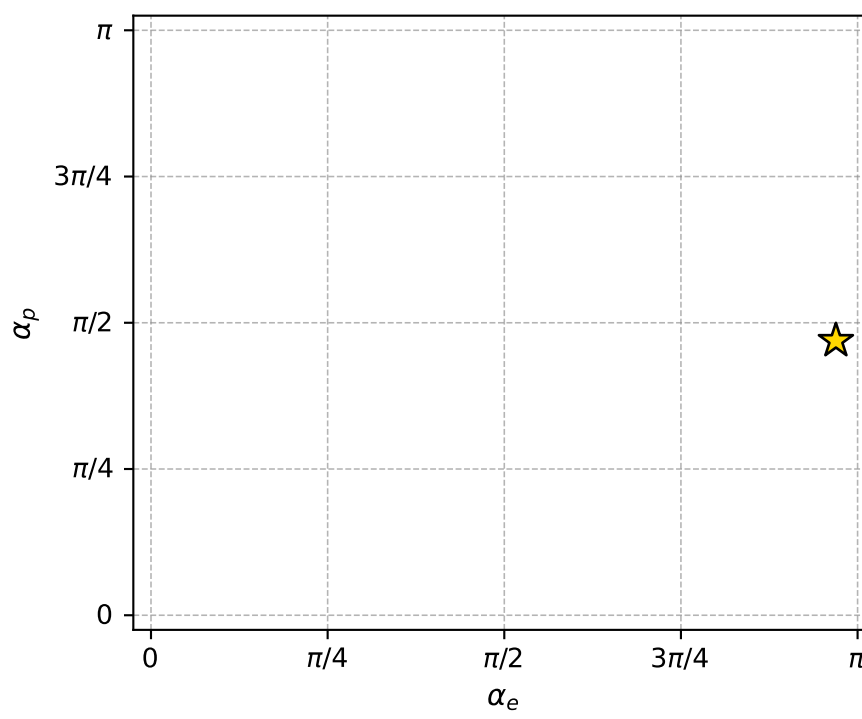
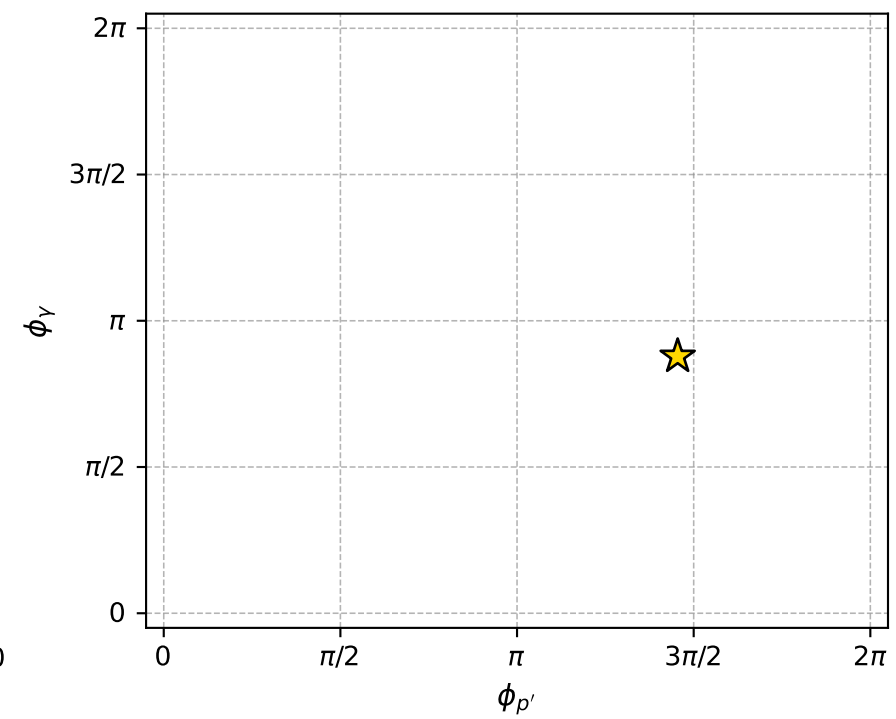
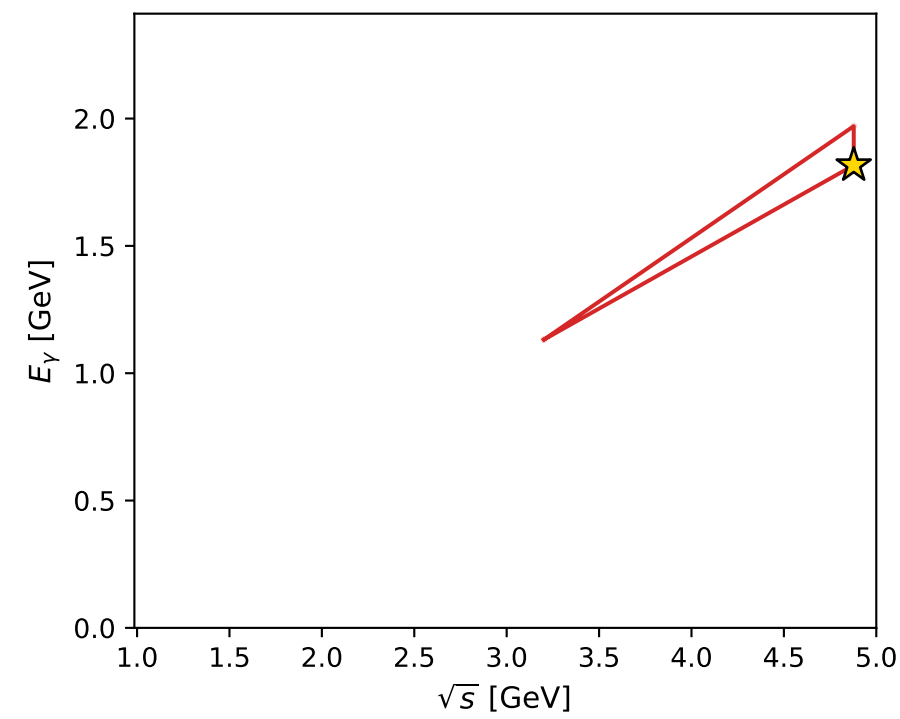
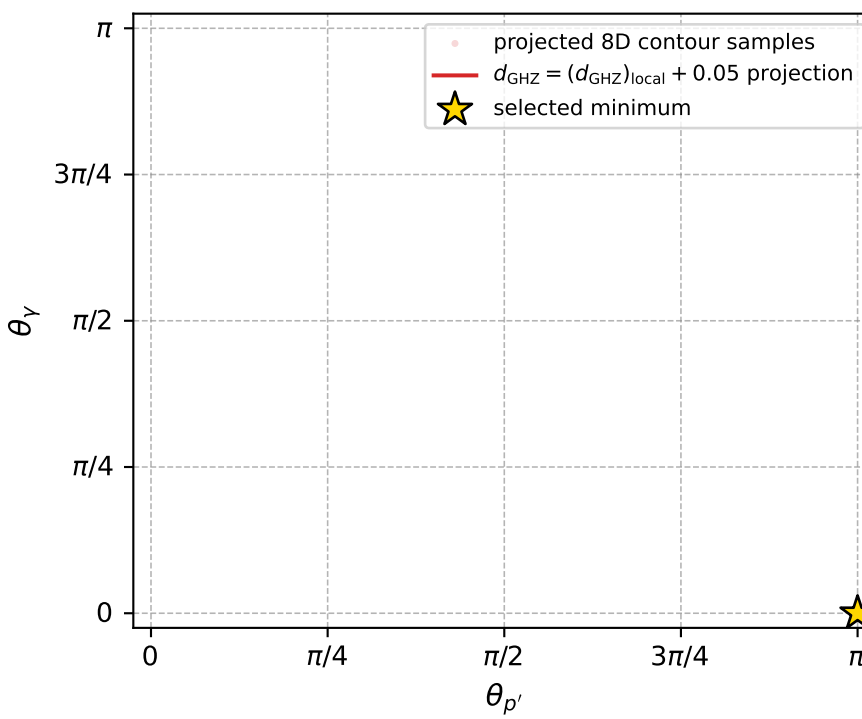
dghz_mixing_angles_polarization_03_local_minimum_181 d_{\rm GHZ}=5.4582e-07
region: polarization_cluster_03_local_minimum_181
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=3.0452158$ rad
incoming lepton: $-0.995359|+\rangle +0.0962277|-\rangle$
proton mixing angle: $\alpha_p=1.4744086$ rad
incoming proton: $0.0962386|+\rangle +0.995358|-\rangle$

$s=23.7939$, $\sqrt{s}=4.8779$, $|\vec{P}|=2.34876$, $|\vec{P}'|=2.34876$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=4.56992$, $\phi_Y=2.75915$
 $E_Y=1.81698$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.42833$
 $Q^2=4.99613$, $x_B=0.219879$, $t=-22.0667$, $W^2=18.6059$, $y=0.991628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3488$
 P $E= 2.5291$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3488$
 ℓ' $E= 0.5318$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5318$
 P' $E= 2.5291$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3488$
 q_Y $E= 1.8170$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8170$



electron: polarization cluster P3, local minimum 181; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.4581997\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000546$
 above global minimum = $5.2353651\text{e-}07$
 8D contour samples = 81

selected phase-space point:

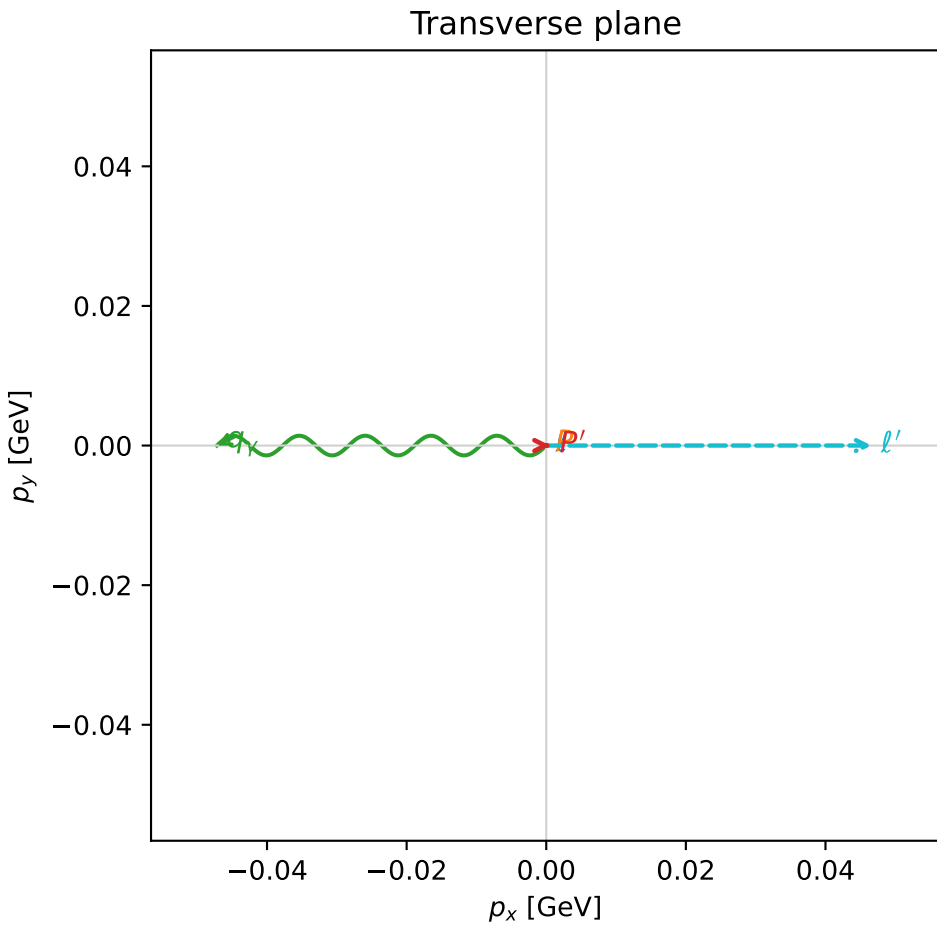
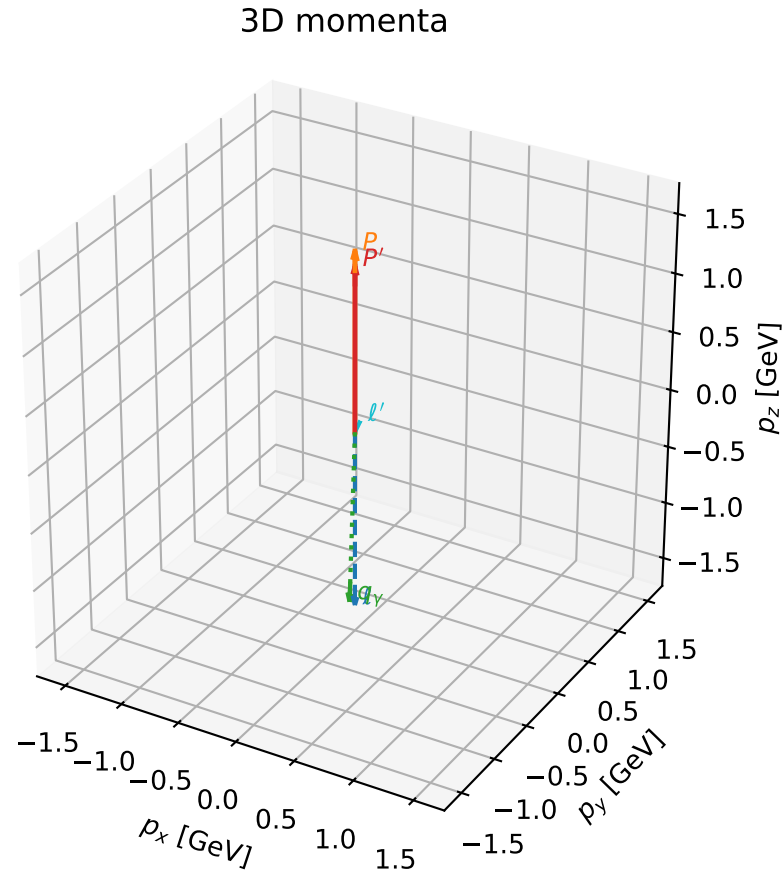
$\text{sqrt_s} = 4.877898$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.816979$
 $\text{phi_p_out} = 4.569922$
 $\text{phi_gamma_out} = 2.759149$
 $\text{alpha_e} = 3.045216$
 $\text{alpha_p} = 1.474409$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_188 d_{\rm GHZ}=1.11719e-05
region: polarization_cluster_03_local_minimum_188
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3764974$ rad
incoming lepton: $-0.721316|+\rangle +0.692606|-\rangle$
proton mixing angle: $\alpha_p=0.84717575$ rad
incoming proton: $0.662102|+\rangle +0.749413|-\rangle$

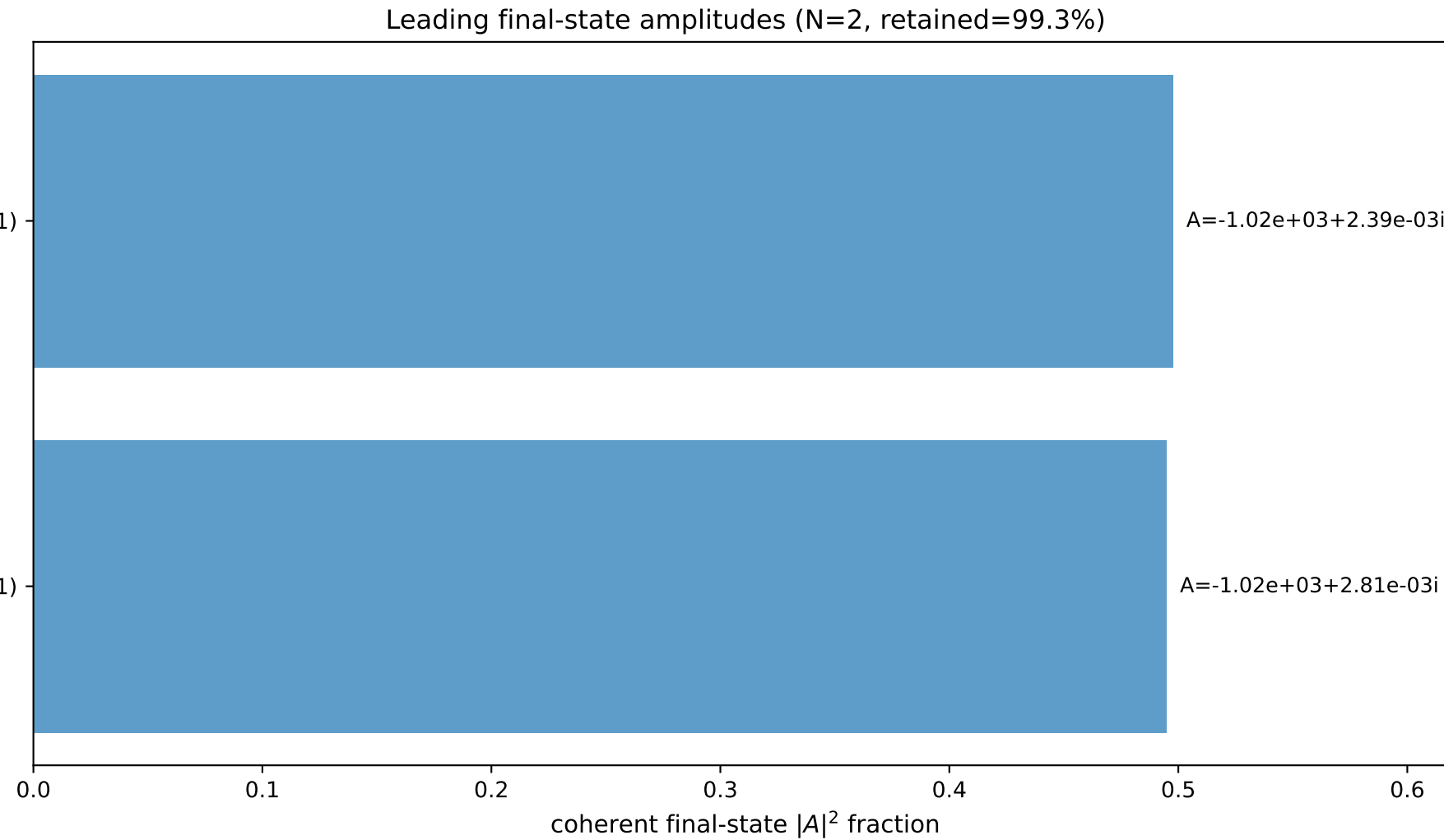
$s=11.0066$, $\sqrt{s}=3.31762$, $|\vec{P}|=1.52621$, $|\vec{P}'|=1.40088$
 $\theta_{P'}=0.000417261$, $\theta_V=3.11038$, $\phi_{P'}=6.35544e-05$, $\phi_V=3.14165$
 $E_V=1.51194$, $\theta_{\ell'}=0.399604$, $\phi_{\ell'}=6.18232e-05$
 $Q^2=0.702332$, $x_B=0.0699923$, $t=-0.00457805$, $W^2=10.2119$, $y=0.990883$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5262$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5262$
 P $E= 1.7914$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5262$
 ℓ' $E= 0.1198$ $p_x= 0.0466$ $p_y= 0.0000$ $p_z= 0.1103$
 P' $E= 1.6859$ $p_x= 0.0006$ $p_y= 0.0000$ $p_z= 1.4009$
 q_V $E= 1.5119$ $p_x= -0.0472$ $p_y= -0.0000$ $p_z= -1.5112$

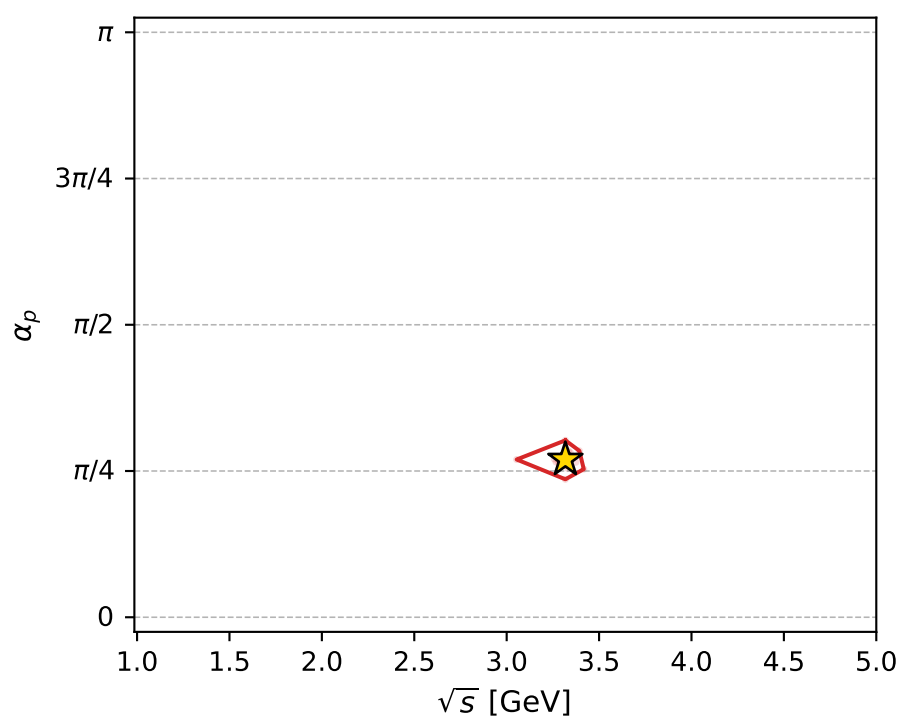
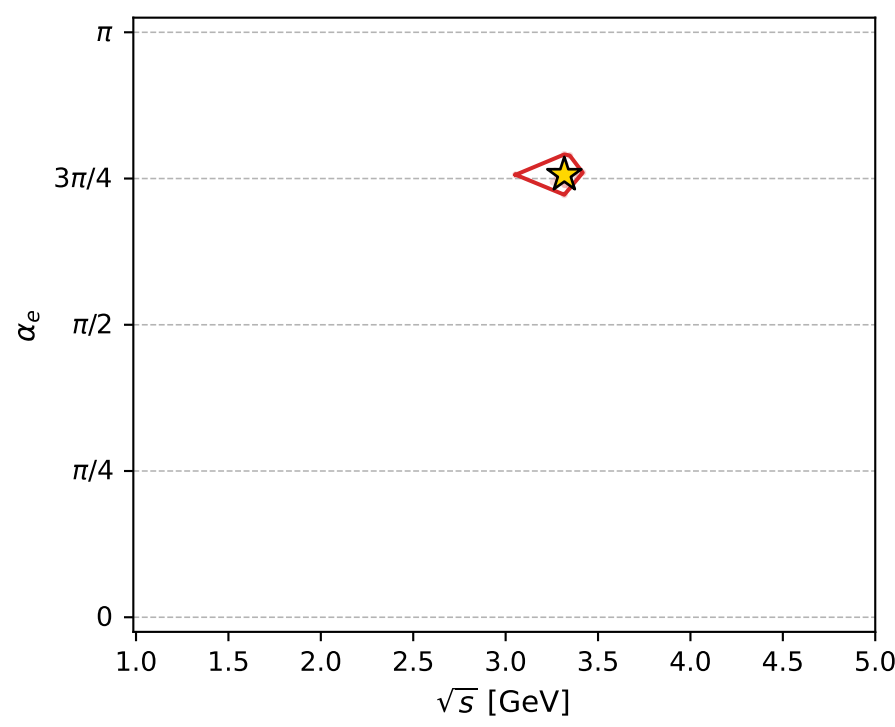
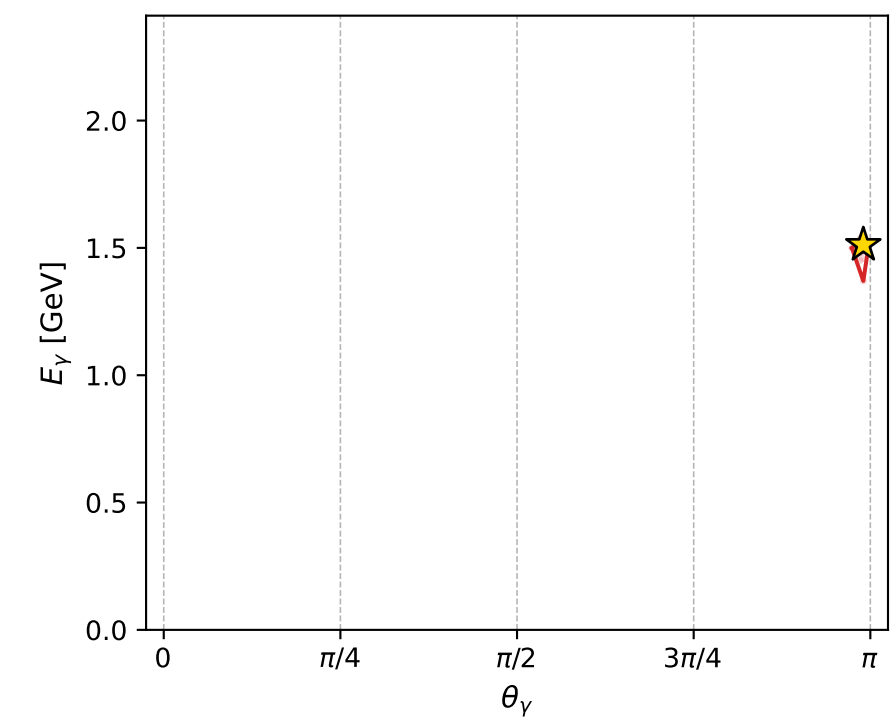
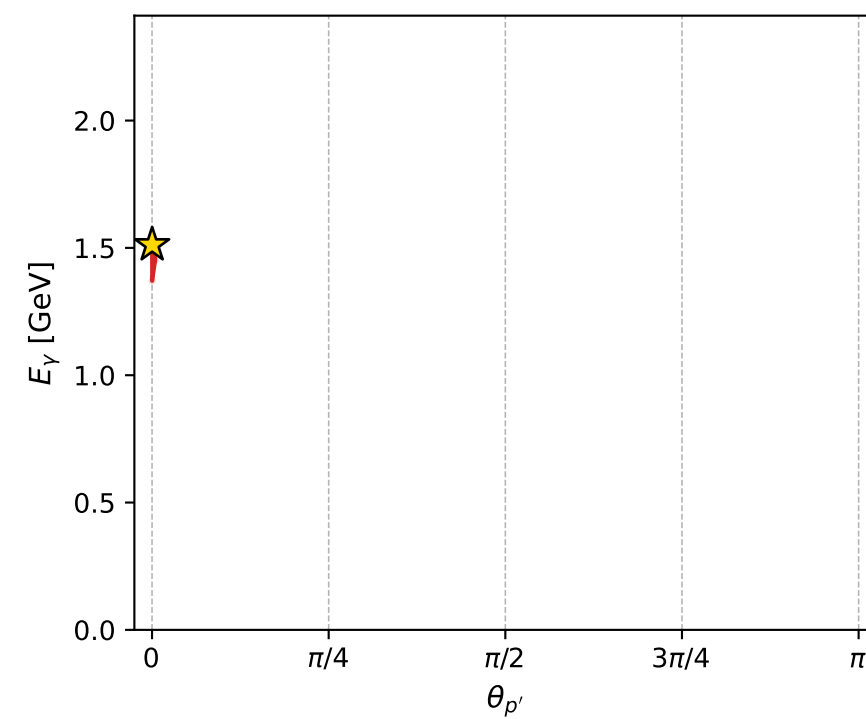
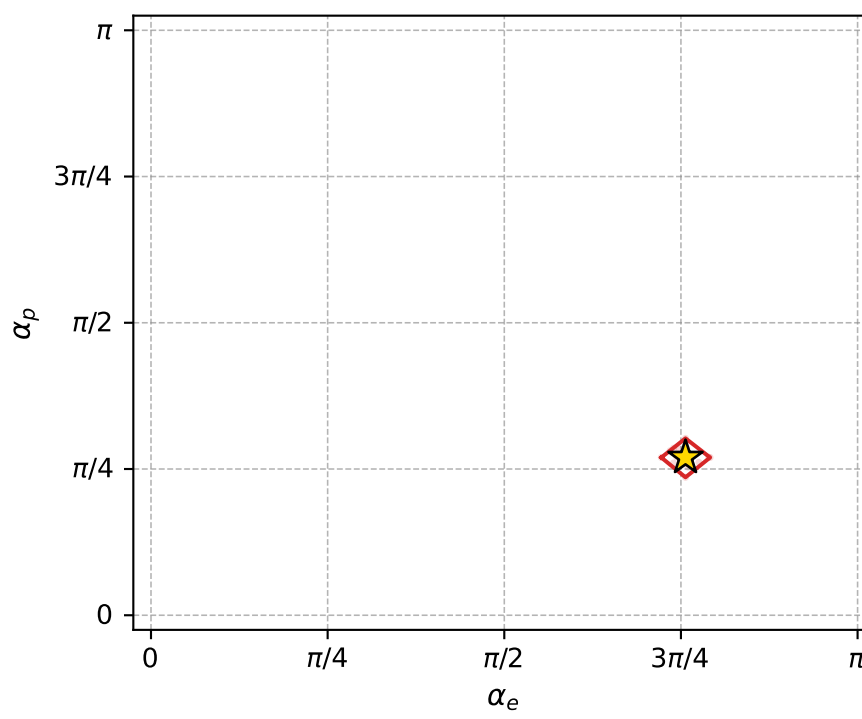
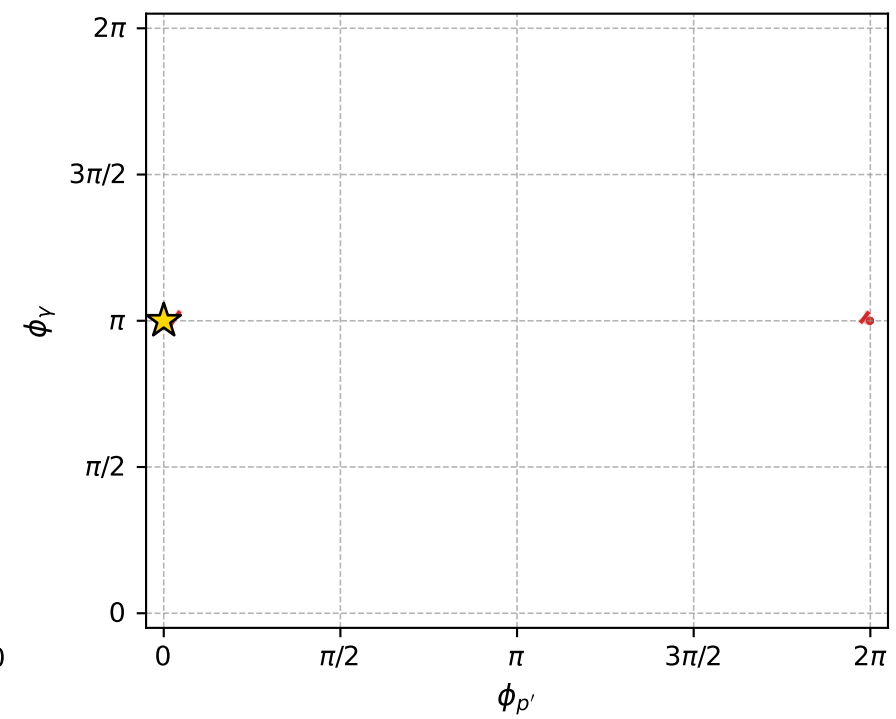
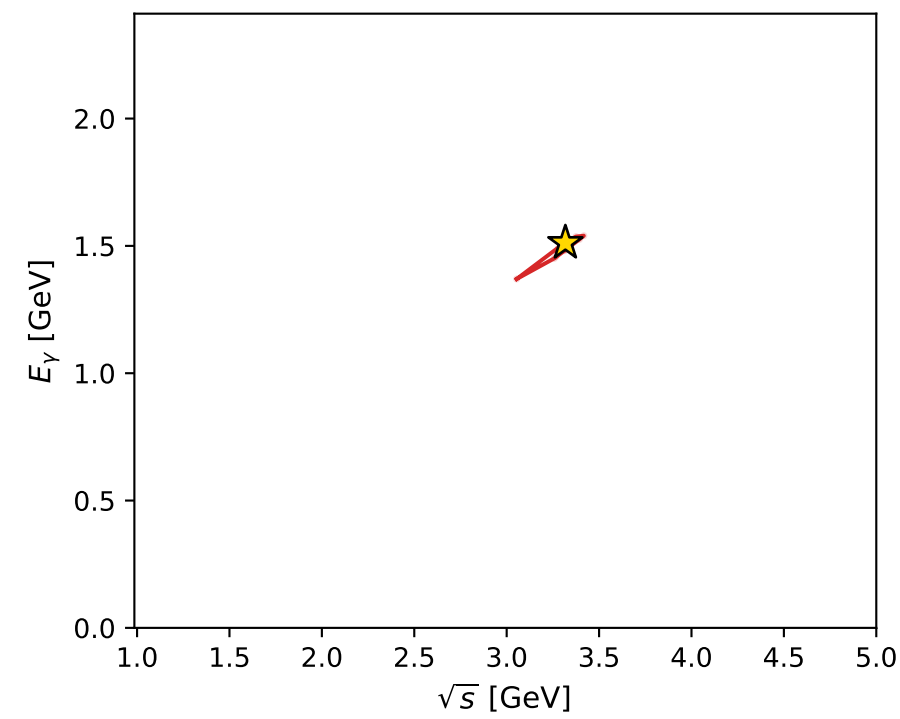
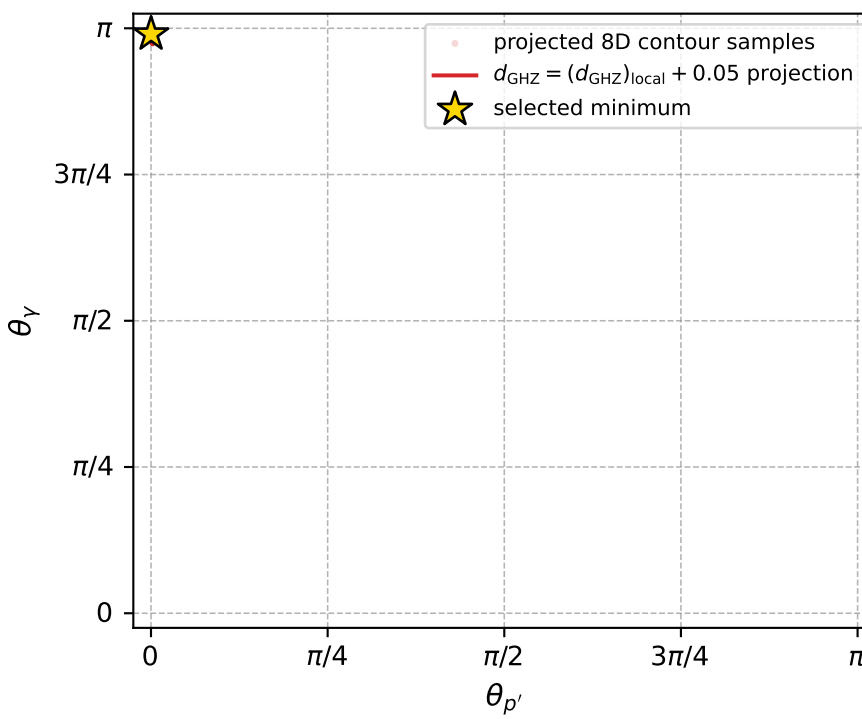


$(h_p, h_V, h_\ell) = (+1, -1, -1)$

$(h_p, h_V, h_\ell) = (-1, +1, +1)$



electron: polarization cluster P3, local minimum 188; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1171927\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050011172$
 above global minimum = $1.1149644\text{e-}05$
 8D contour samples = 140

selected phase-space point:

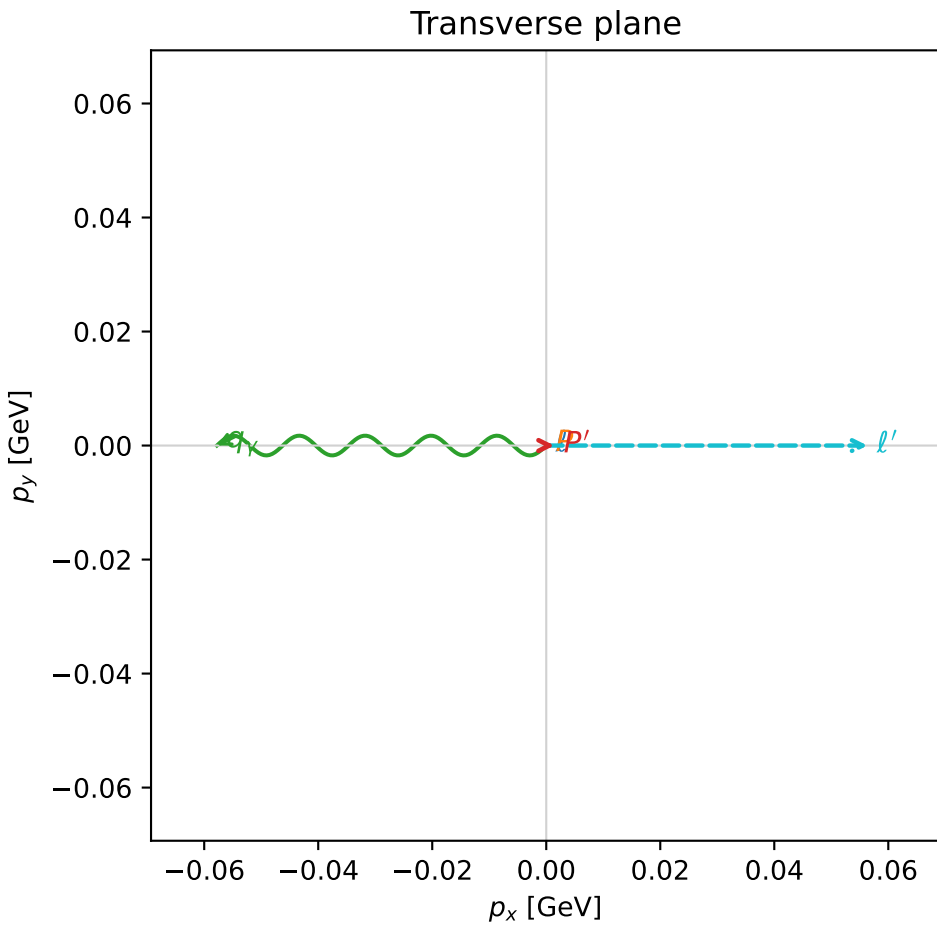
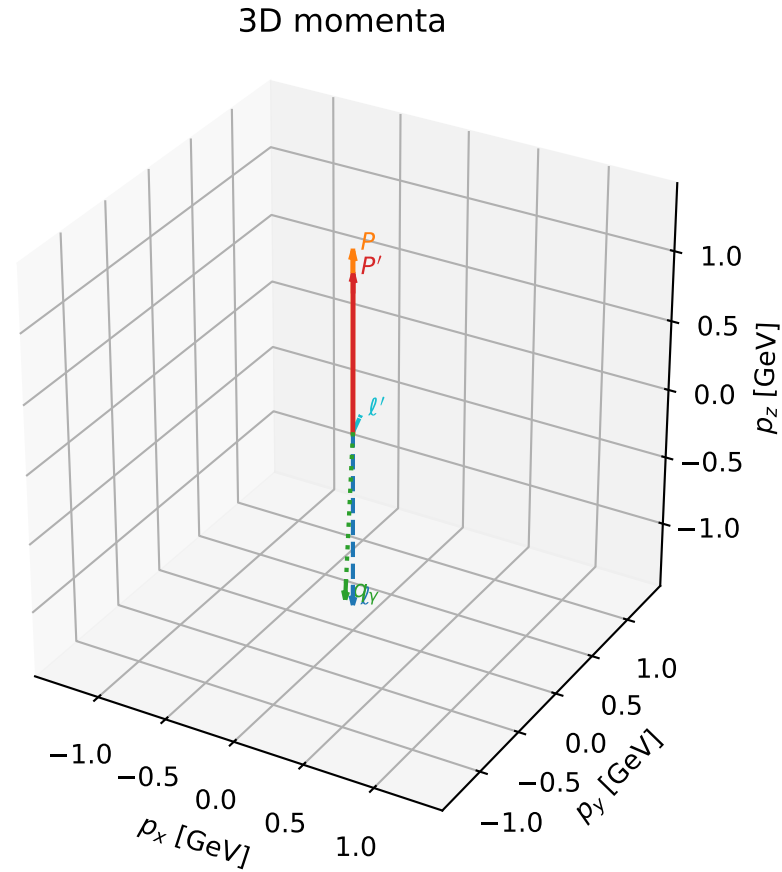
$\text{sqrt_s} = 3.317617$
 $\text{theta_p_out} = 0.0004172612$
 $\text{theta_gamma_out} = 3.110384$
 $\text{q0out} = 1.511941$
 $\text{phi_p_out} = 6.355443\text{e-}05$
 $\text{phi_gamma_out} = 3.141654$
 $\text{alpha_e} = 2.376497$
 $\text{alpha_p} = 0.8471758$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_189 d_{\rm GHZ}=1.25241e-05
region: polarization_cluster_03_local_minimum_189
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.4349155$ rad
incoming lepton: $-0.760524|+\rangle +0.64931|-\rangle$
proton mixing angle: $\alpha_p=0.90852471$ rad
incoming proton: $0.61491|+\rangle +0.788597|-\rangle$

$s=8.26616$, $\sqrt{s}=2.87509$, $|\vec{P}|=1.28454$, $|\vec{P}'|=1.12322$
 $\theta_{P'}=0.00126895$, $\theta_V=3.09582$, $\phi_{P'}=6.28313$, $\phi_V=3.14157$
 $E_V=1.2626$, $\theta_{\ell'}=0.38751$, $\phi_{\ell'}=6.28316$
 $Q^2=0.737755$, $x_B=0.101526$, $t=-0.00984887$, $W^2=7.40874$, $y=0.983798$

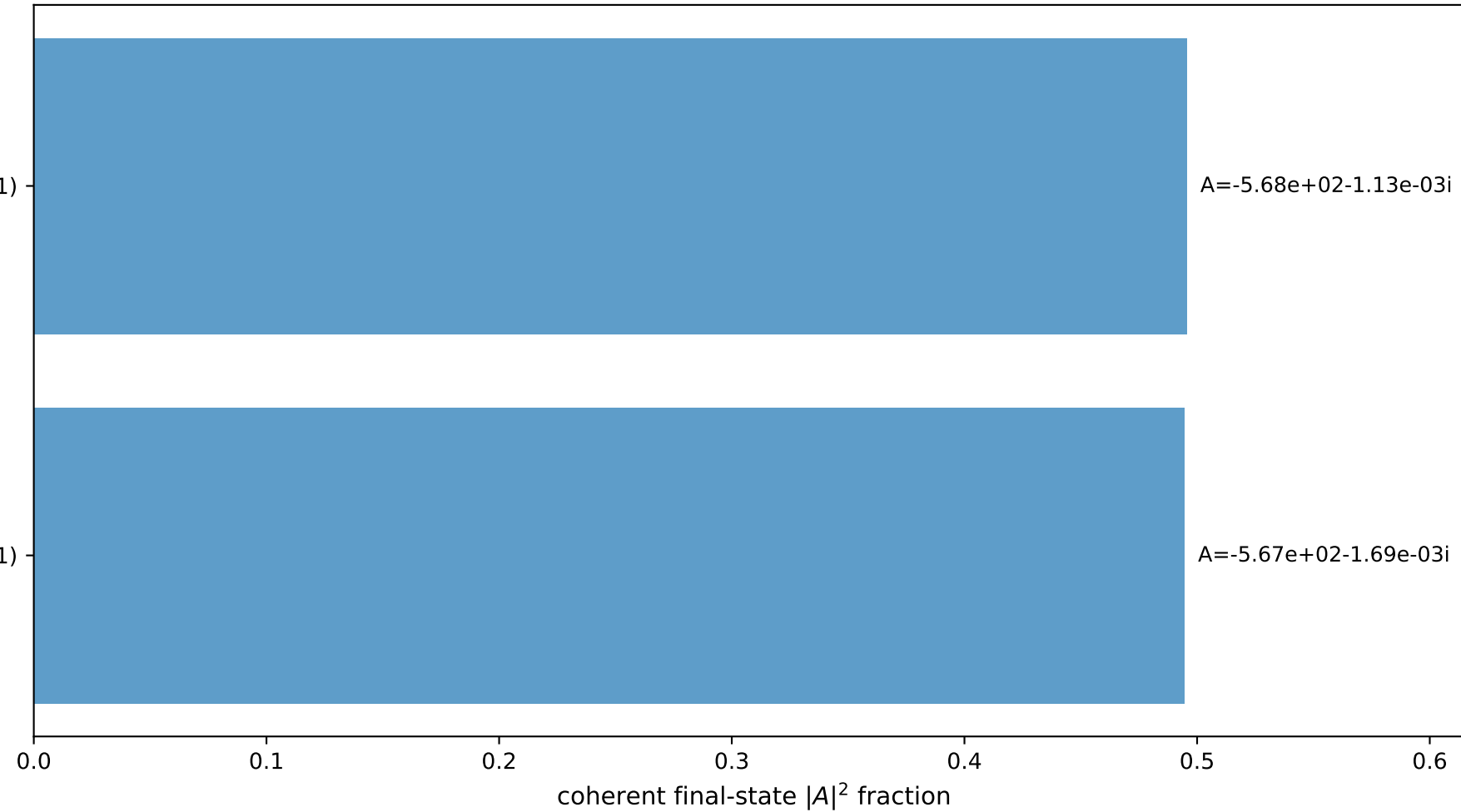
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2845$
 P $E= 1.5906$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2845$
 ℓ' $E= 0.1491$ $p_x= 0.0563$ $p_y= -0.0000$ $p_z= 0.1381$
 P' $E= 1.4634$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1232$
 q_V $E= 1.2626$ $p_x= -0.0578$ $p_y= 0.0000$ $p_z= -1.2613$



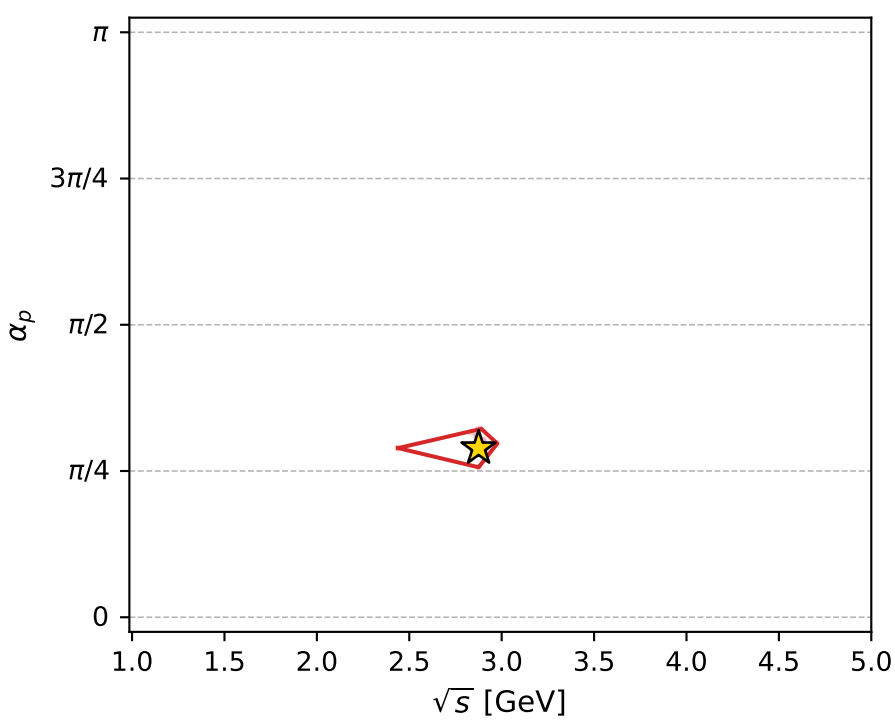
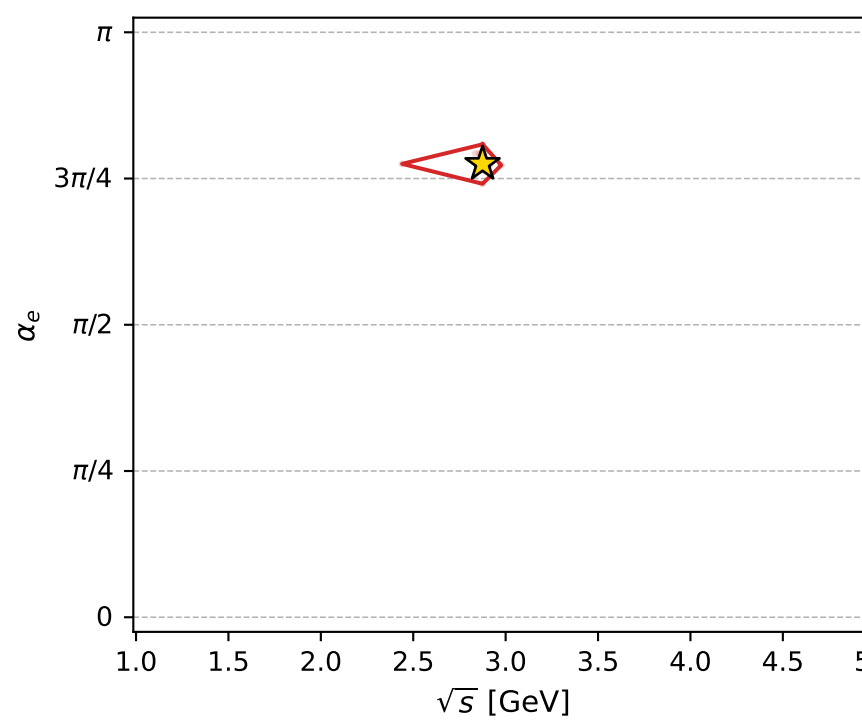
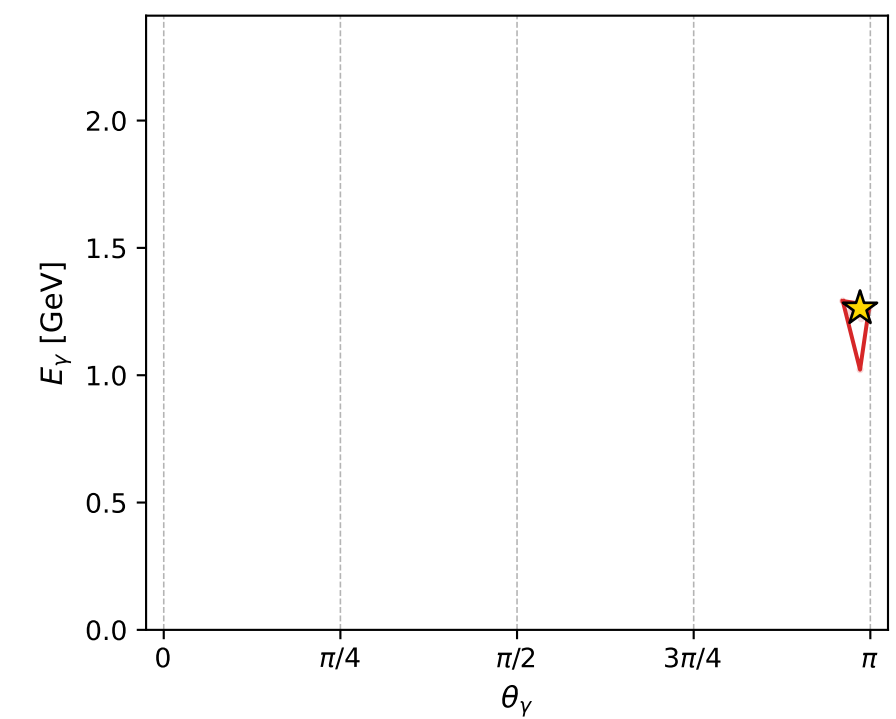
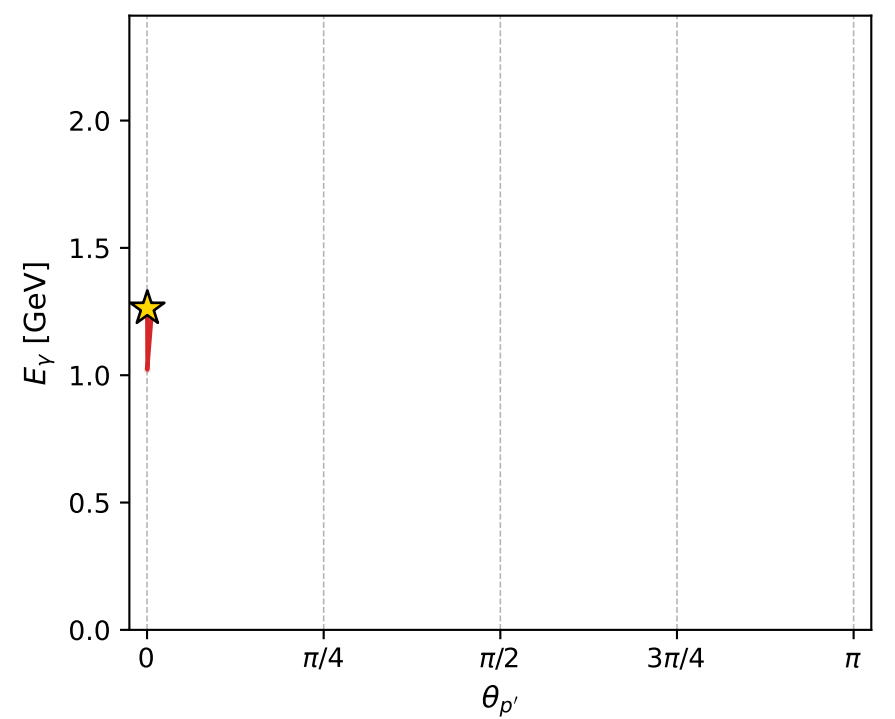
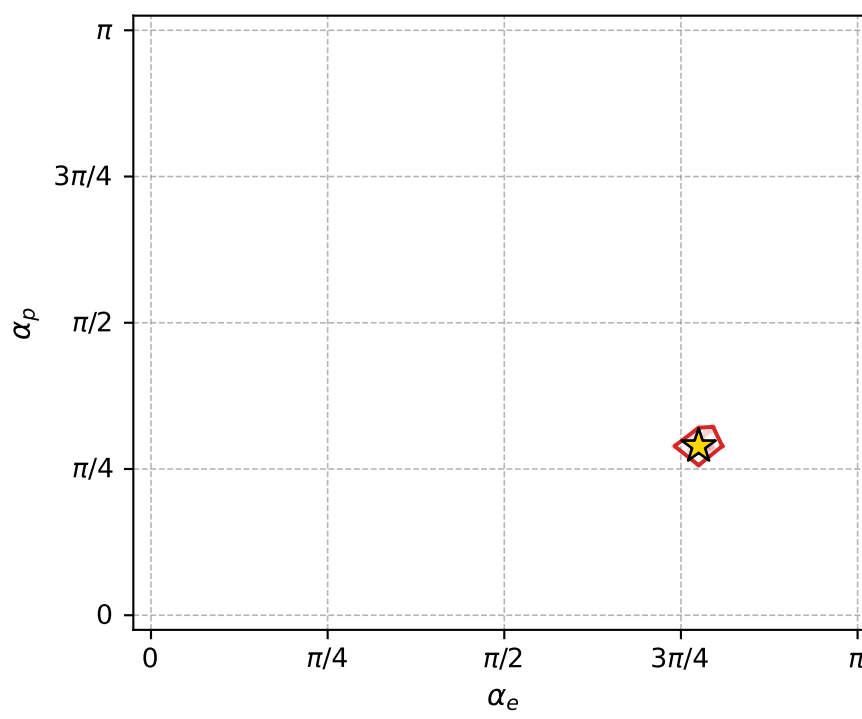
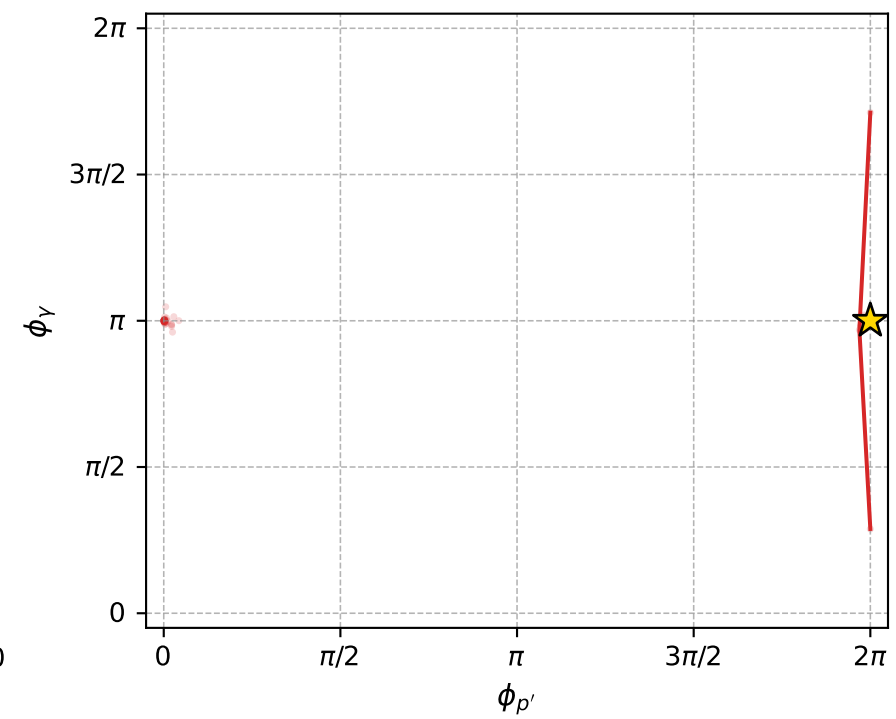
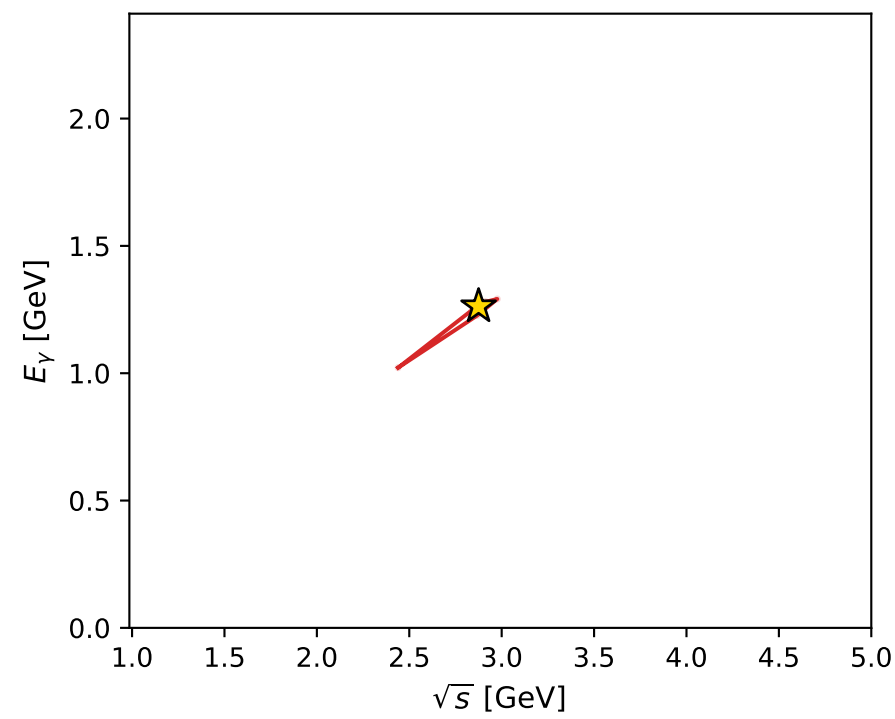
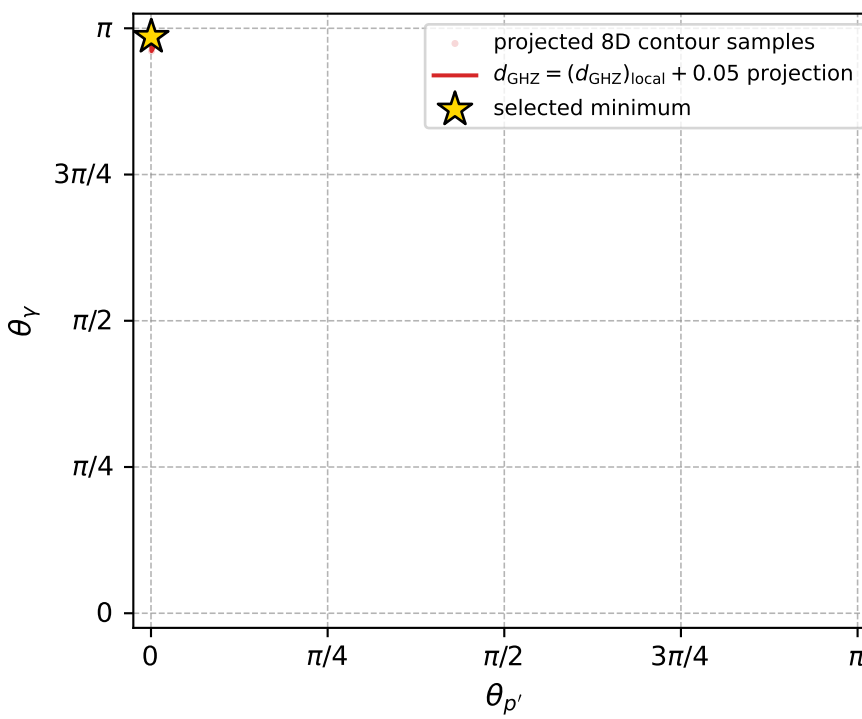
$(h_p, h_V, h_\ell) = (+1, -1, -1)$

$(h_p, h_V, h_\ell) = (-1, +1, +1)$

Leading final-state amplitudes (N=2, retained=99.0%)



electron: polarization cluster P3, local minimum 189; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.2524097\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050012524$
 above global minimum = $1.2501813\text{e-}05$
 8D contour samples = 137

selected phase-space point:

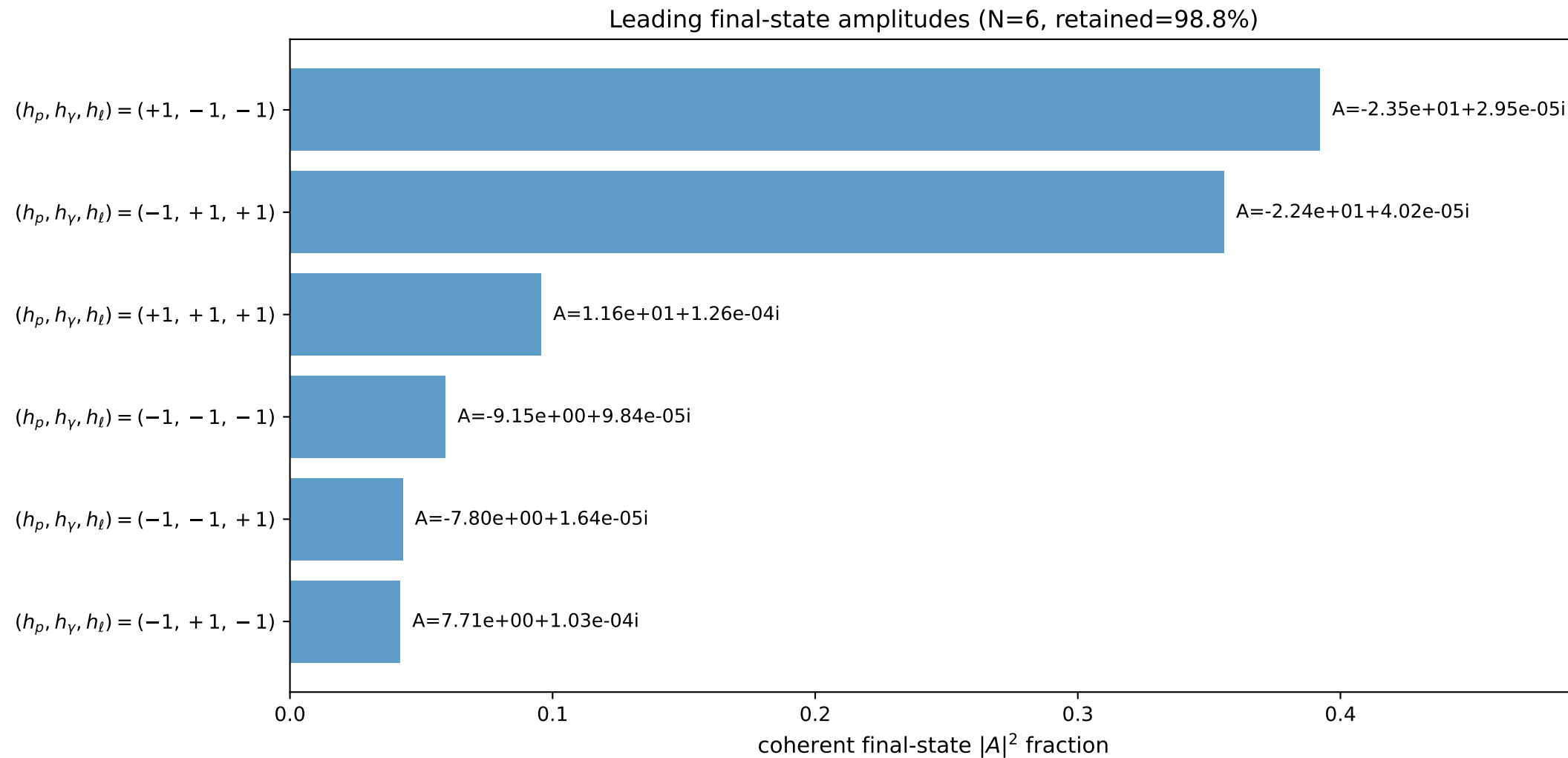
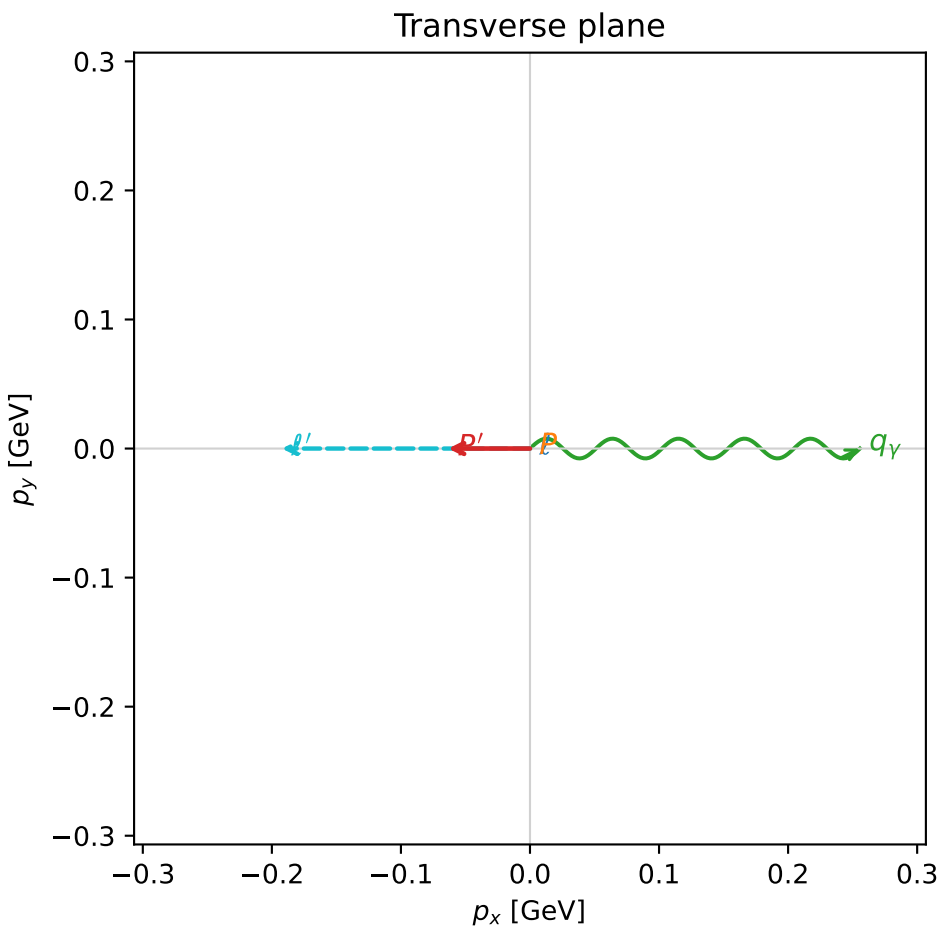
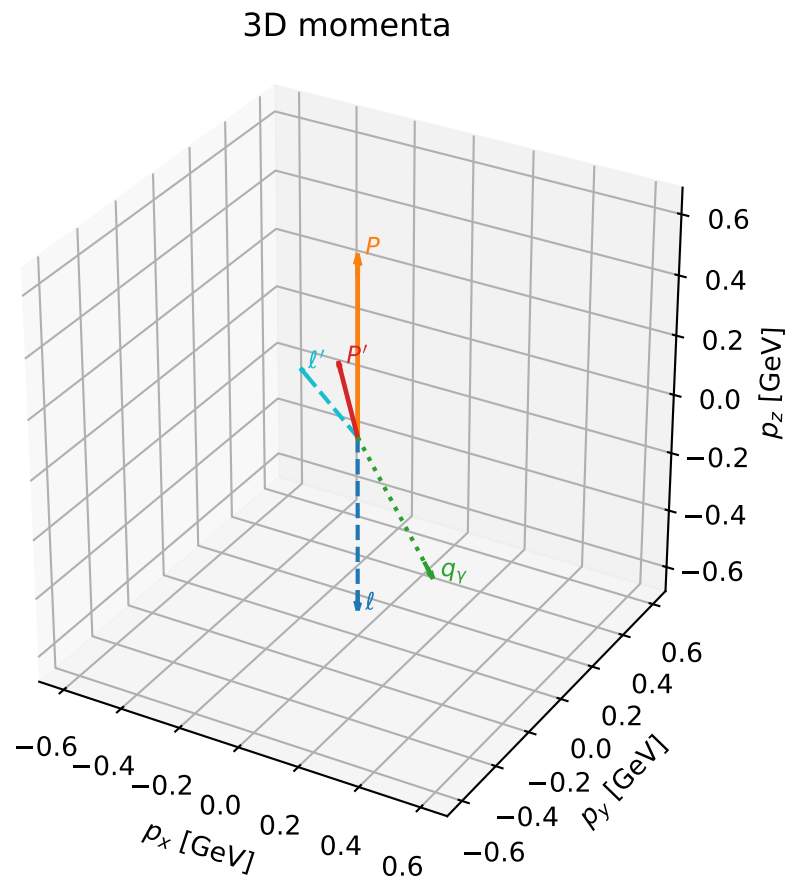
$\text{sqrt_s} = 2.875093$
 $\text{theta_p_out} = 0.001268949$
 $\text{theta_gamma_out} = 3.09582$
 $\text{q0ut} = 1.262602$
 $\text{phi_p_out} = 6.28313$
 $\text{phi_gamma_out} = 3.14157$
 $\text{alpha_e} = 2.434916$
 $\text{alpha_p} = 0.9085247$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

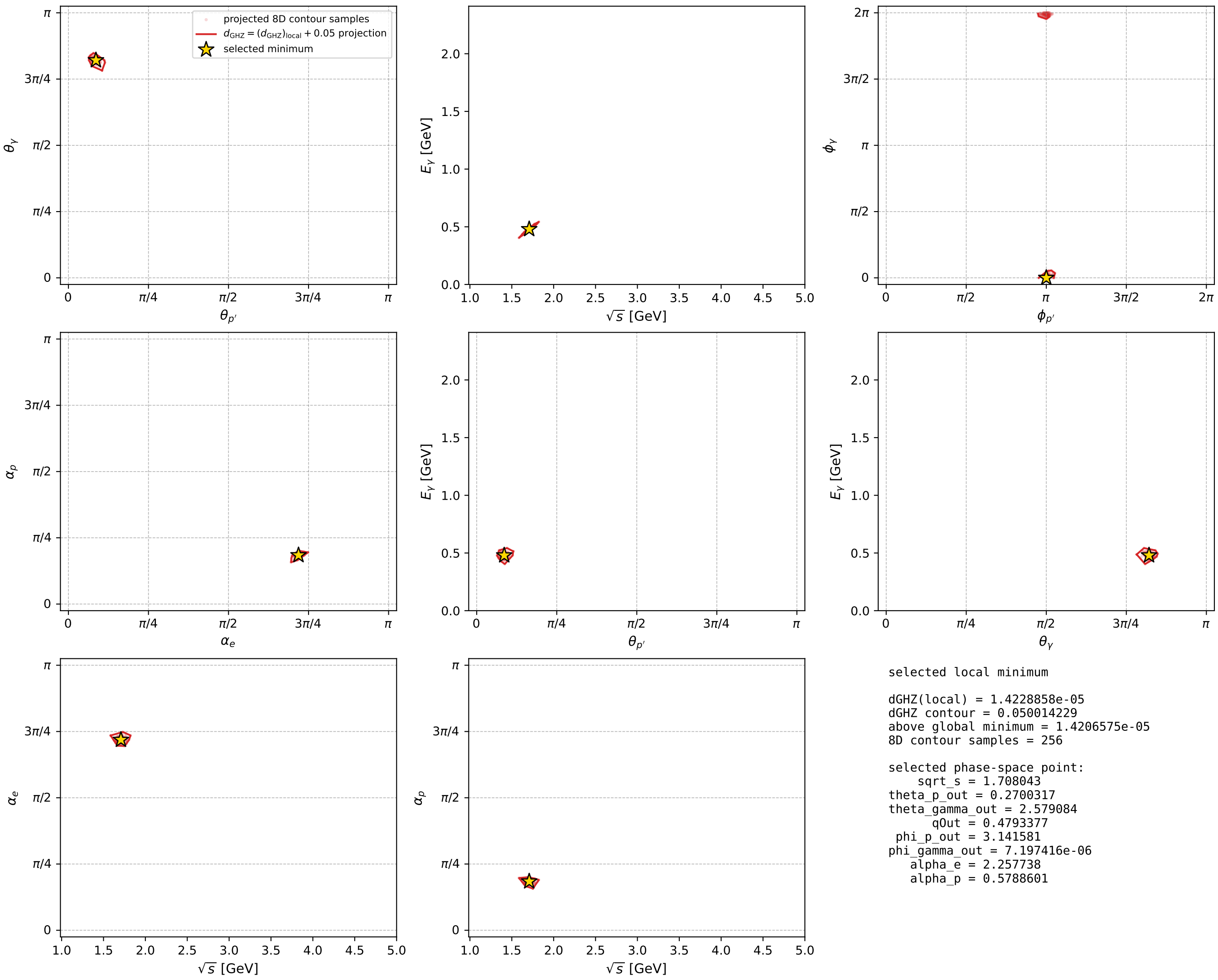
dghz_mixing_angles_polarization_03_local_minimum_190 d_{\rm GHZ}=1.42289e-05
region: polarization_cluster_03_local_minimum_190
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.2577383$ rad
incoming lepton: $-0.634176|+> +0.773189|->$
proton mixing angle: $\alpha_p=0.57886014$ rad
incoming proton: $0.837087|+> +0.54707|->$

$s=2.91741$, $\sqrt{s}=1.70804$, $|\vec{P}|=0.596462$, $|\vec{P}'|=0.237434$
 $\theta_{P'}=0.270032$, $\theta_Y=2.57908$, $\phi_{P'}=3.14158$, $\phi_Y=7.19742e-06$
 $E_Y=0.479338$, $\theta_{\ell'}=0.82778$, $\phi_{\ell'}=3.14161$
 $Q^2=0.52223$, $x_B=0.313128$, $t=-0.11843$, $W^2=2.0254$, $y=0.818517$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5965$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5965$
 P $E= 1.1116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5965$
 ℓ' $E= 0.2611$ $p_x= -0.1923$ $p_y= -0.0000$ $p_z= 0.1767$
 P' $E= 0.9676$ $p_x= -0.0633$ $p_y= 0.0000$ $p_z= 0.2288$
 q_Y $E= 0.4793$ $p_x= 0.2556$ $p_y= 0.0000$ $p_z= -0.4055$



electron: polarization cluster P3, local minimum 190; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

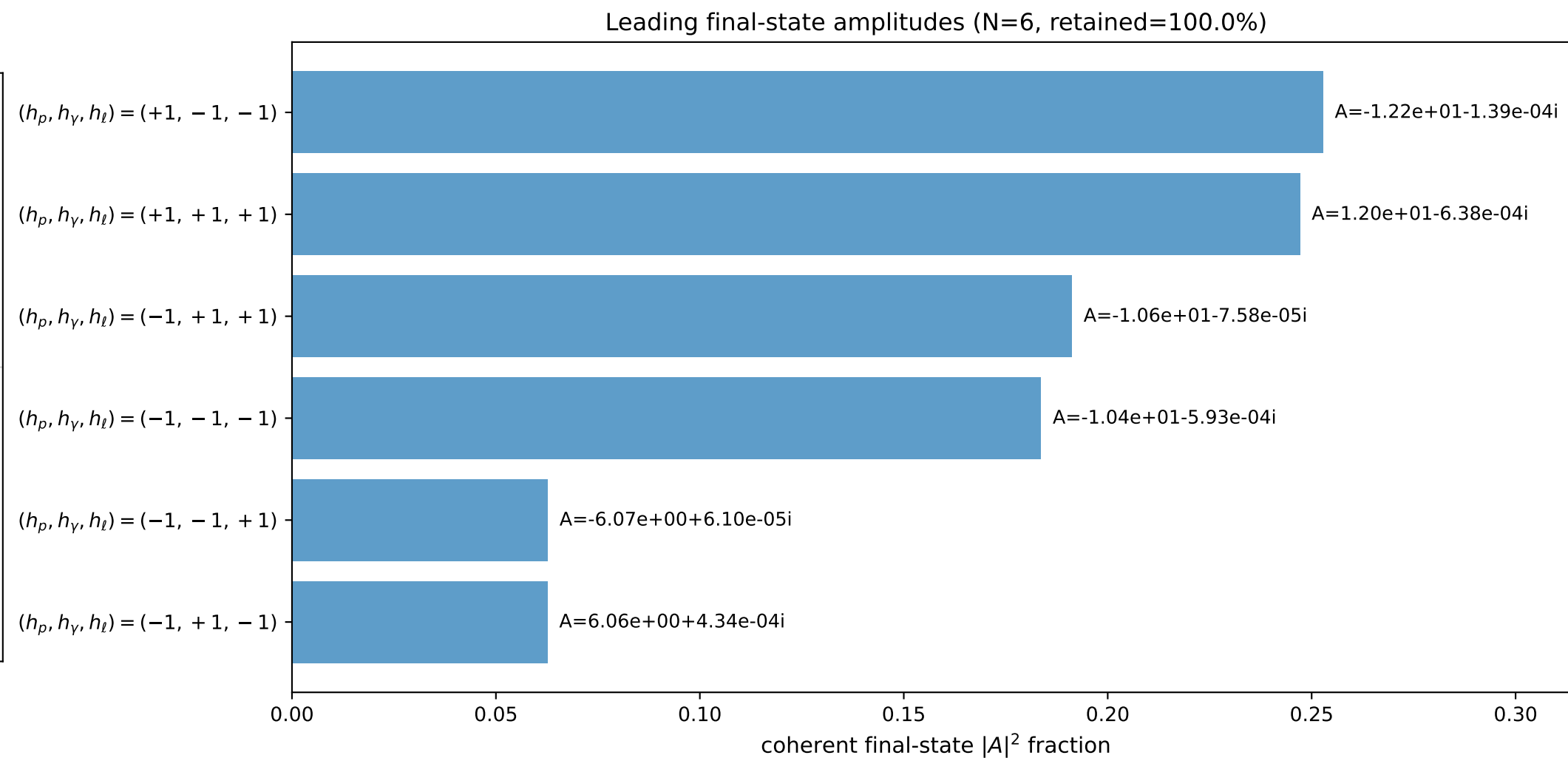
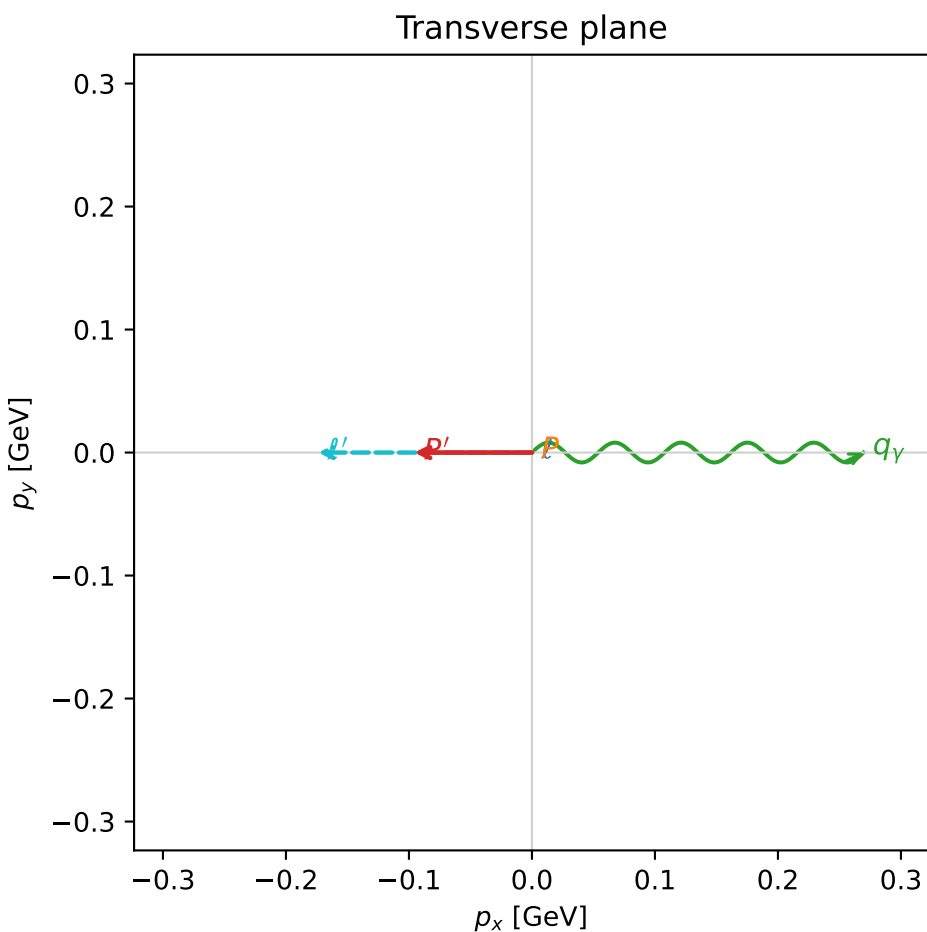
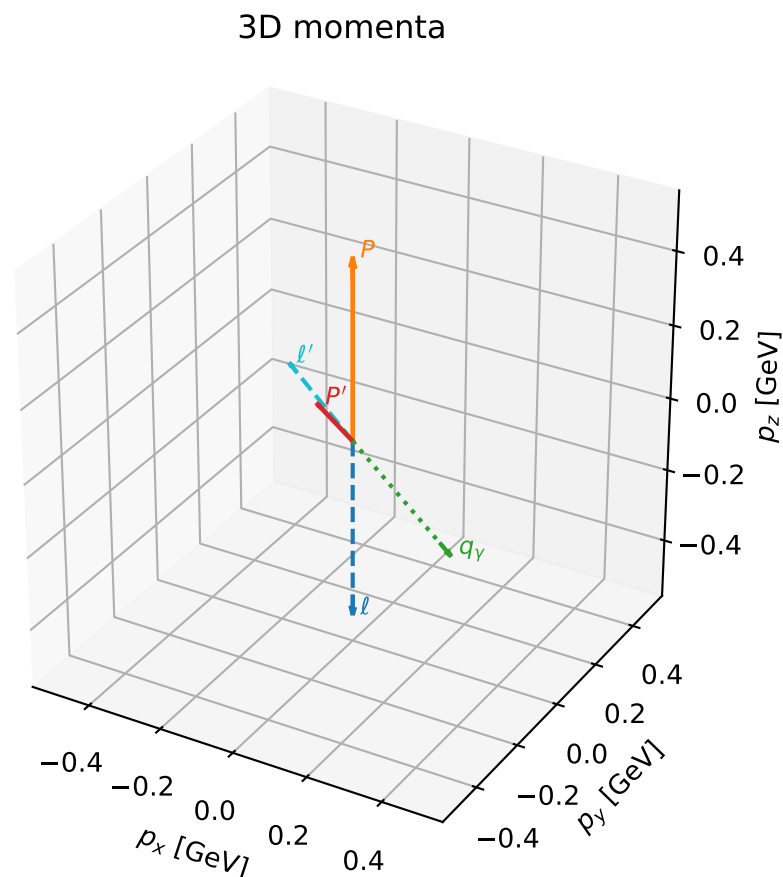


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

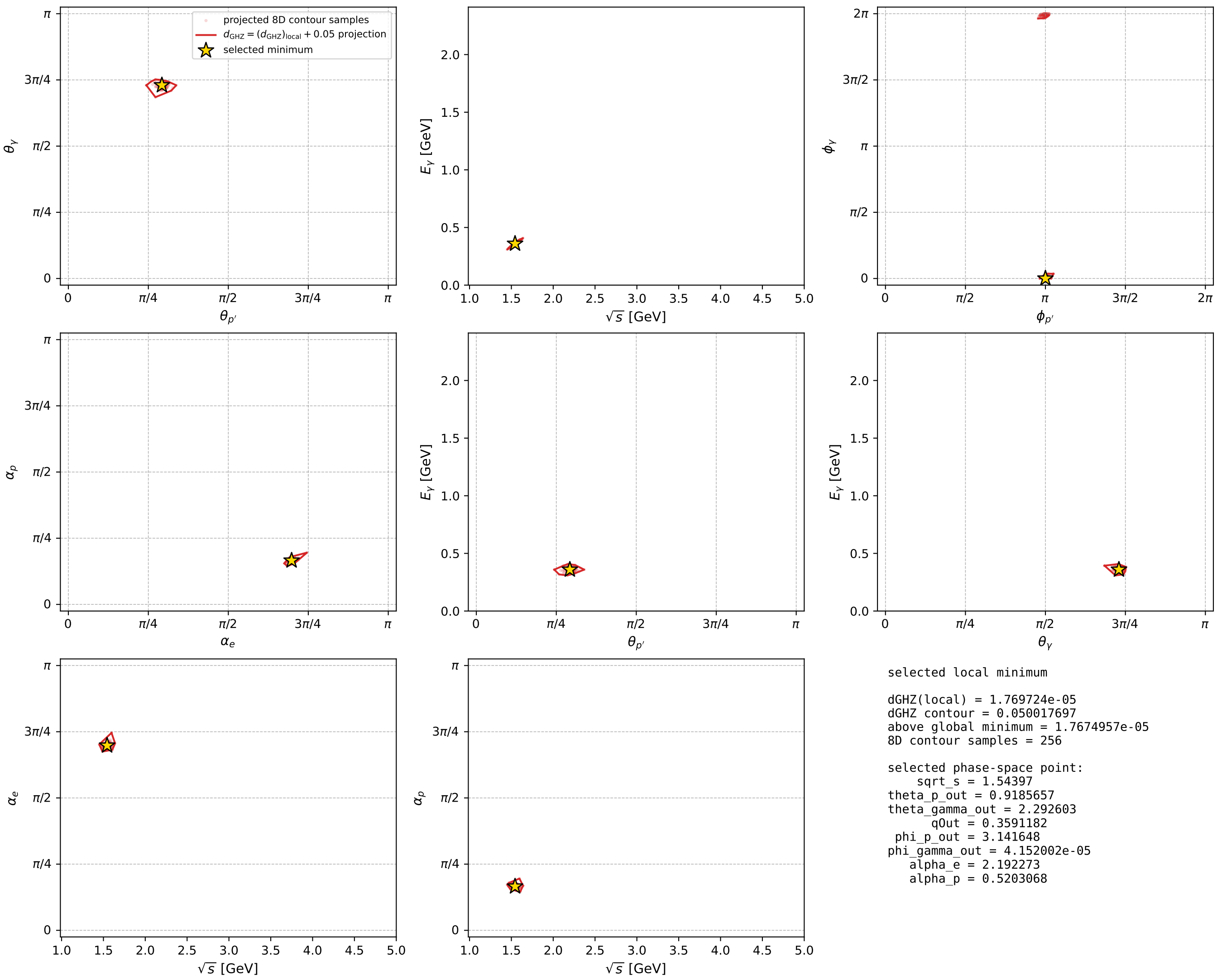
dghz_mixing_angles_polarization_03_local_minimum_192 $d_{\{\text{rm GHZ}\}}=1.76972\text{e-}05$
 region: polarization_cluster_03_local_minimum_192
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.1922727$ rad
 incoming lepton: $-0.582236|+\rangle +0.81302|-\rangle$
 proton mixing angle: $\alpha_p=0.5203068$ rad
 incoming proton: $0.867667|+\rangle +0.497146|-\rangle$

$s=2.38384$, $\sqrt{s}=1.54397$, $|\vec{P}|=0.487056$, $|\vec{P}'|=0.120401$
 $\theta_{P'}=0.918566$, $\theta_Y=2.2926$, $\phi_{P'}=3.14165$, $\phi_Y=4.152\text{e-}05$
 $E_Y=0.359118$, $\theta_{\ell'}=0.813983$, $\phi_{\ell'}=3.14163$
 $Q^2=0.392919$, $x_B=0.339186$, $t=-0.168164$, $W^2=1.64534$, $y=0.770226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4871$
 P $E= 1.0569$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4871$
 ℓ' $E= 0.2392$ $p_x= -0.1739$ $p_y= -0.0000$ $p_z= 0.1642$
 P' $E= 0.9457$ $p_x= -0.0957$ $p_y= -0.0000$ $p_z= 0.0731$
 q_Y $E= 0.3591$ $p_x= 0.2696$ $p_y= 0.0000$ $p_z= -0.2373$



electron: polarization cluster P3, local minimum 192; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

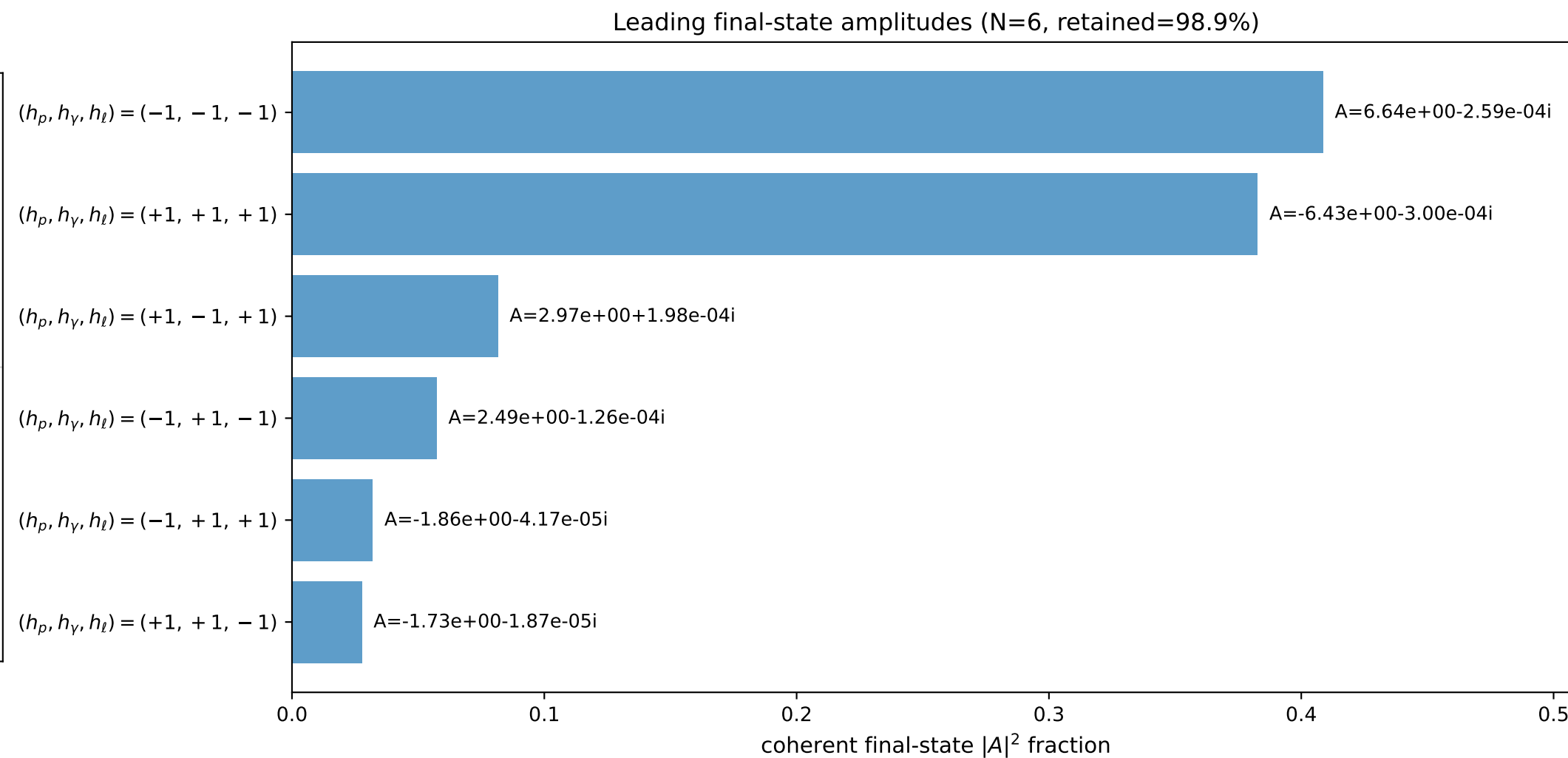
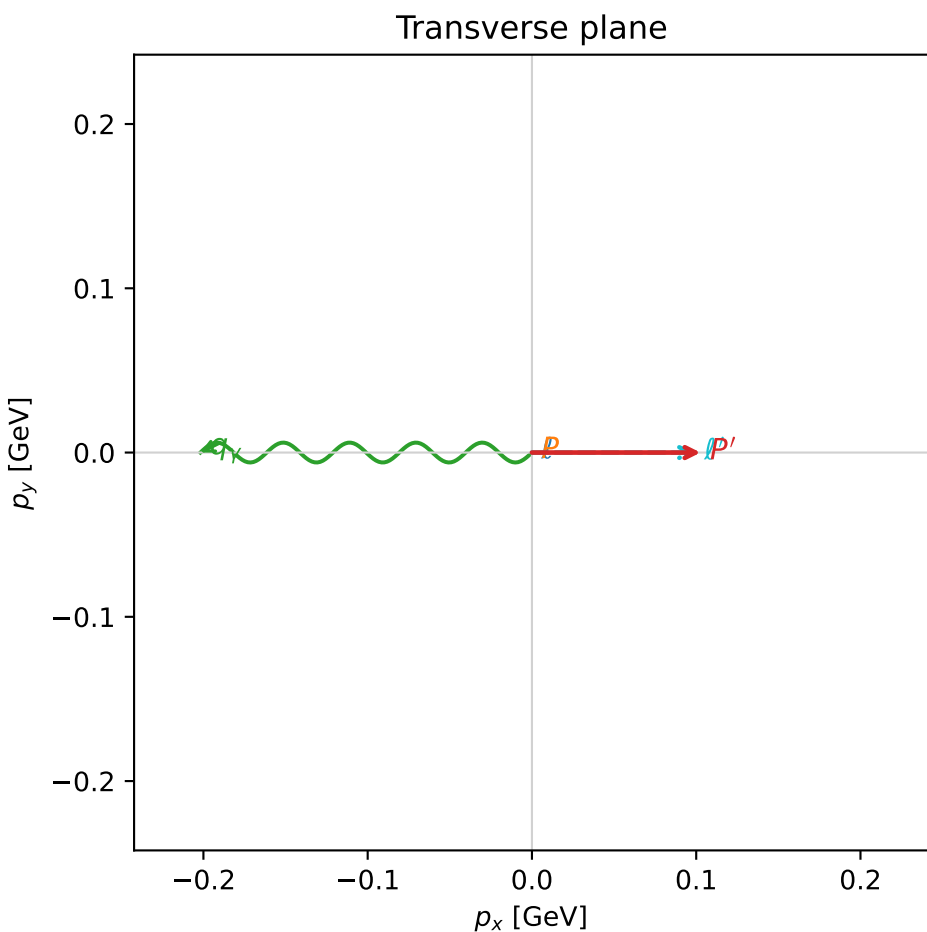
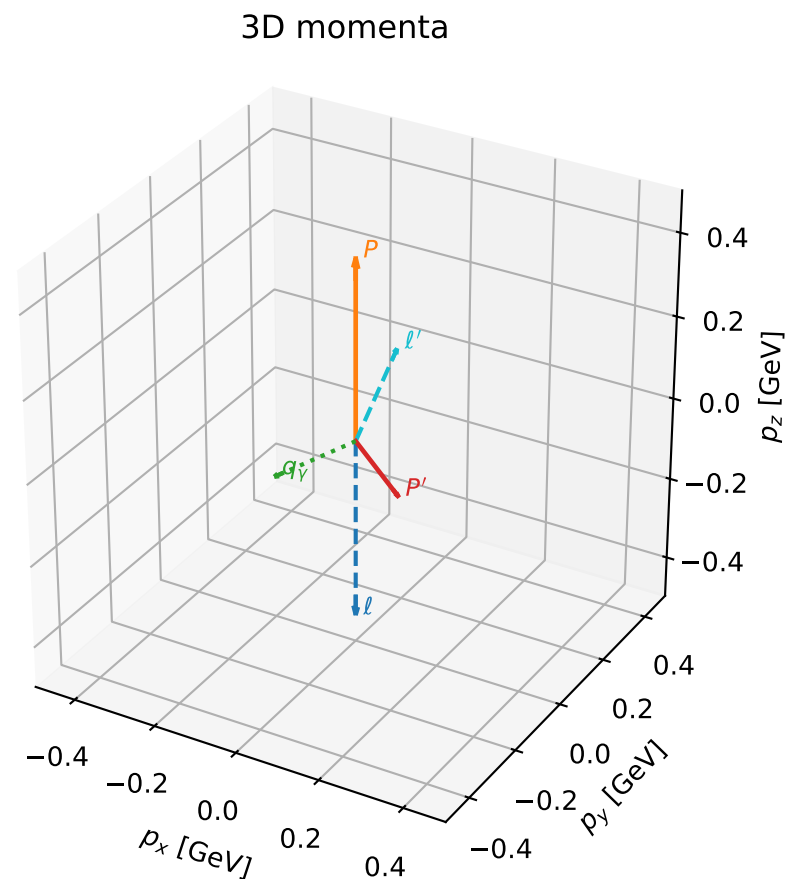


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

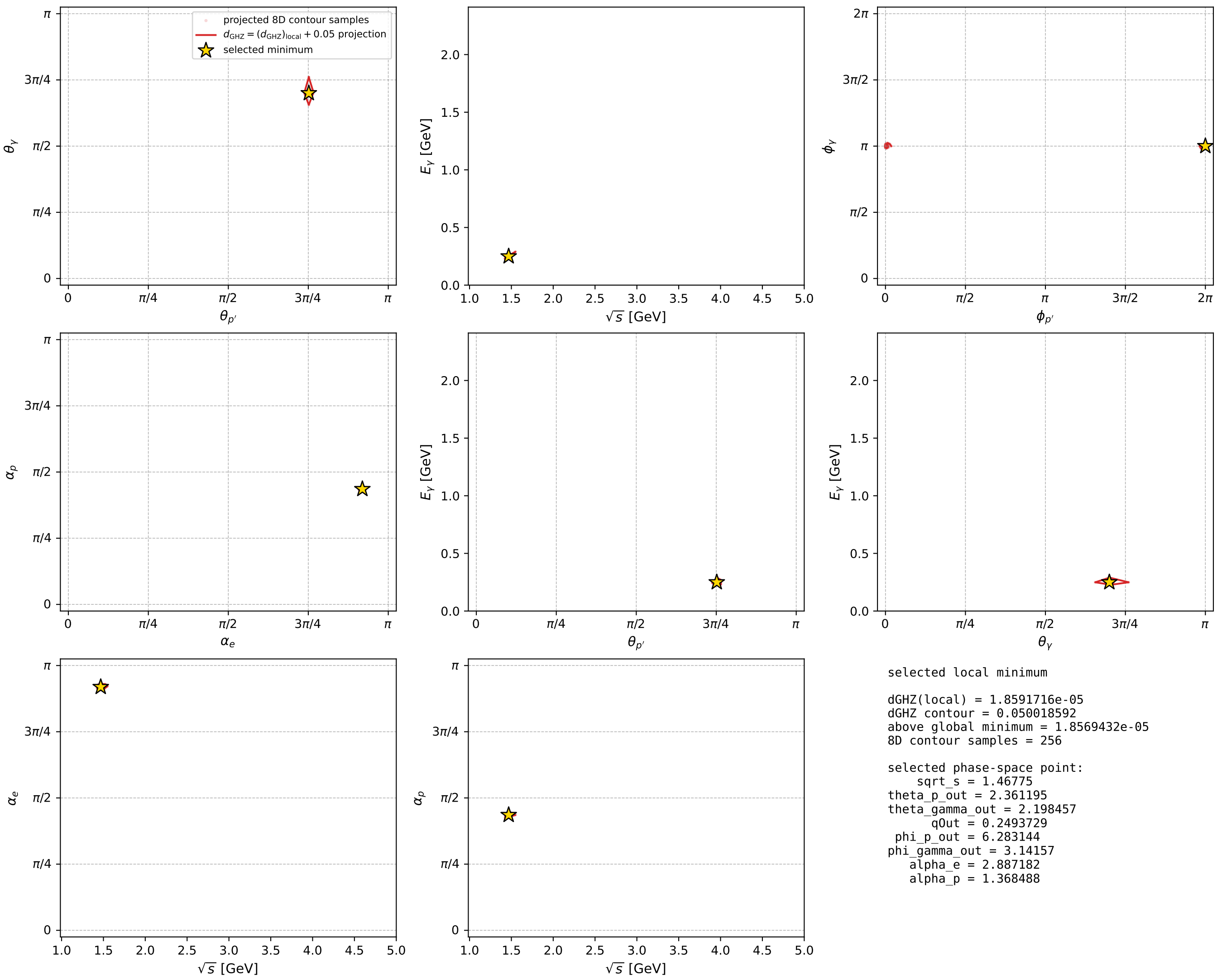
dghz_mixing_angles_polarization_03_local_minimum_194 $d_{\{\text{rm GHZ}\}}=1.85917\text{e-}05$
 region: polarization_cluster_03_local_minimum_194
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.8871823$ rad
 incoming lepton: $-0.967812|+\rangle +0.251675|-\rangle$
 proton mixing angle: $\alpha_p=1.3684879$ rad
 incoming proton: $0.200931|+\rangle +0.979605|-\rangle$

$s=2.15429$, $\sqrt{s}=1.46775$, $|\vec{P}|=0.434149$, $|\vec{P}'|=0.145936$
 $\theta_{P'}=2.36119$, $\theta_Y=2.19846$, $\phi_{P'}=6.28314$, $\phi_Y=3.14157$
 $E_Y=0.249373$, $\theta_{\ell'}=0.377425$, $\phi_{\ell'}=6.28318$
 $Q^2=0.450859$, $x_B=0.482004$, $t=-0.292723$, $W^2=1.36437$, $y=0.733954$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4341$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4341$
 P $E= 1.0336$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4341$
 ℓ' $E= 0.2691$ $p_x= 0.0992$ $p_y= -0.0000$ $p_z= 0.2502$
 P' $E= 0.9493$ $p_x= 0.1027$ $p_y= -0.0000$ $p_z= -0.1037$
 q_Y $E= 0.2494$ $p_x= -0.2018$ $p_y= 0.0000$ $p_z= -0.1464$



electron: polarization cluster P3, local minimum 194; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



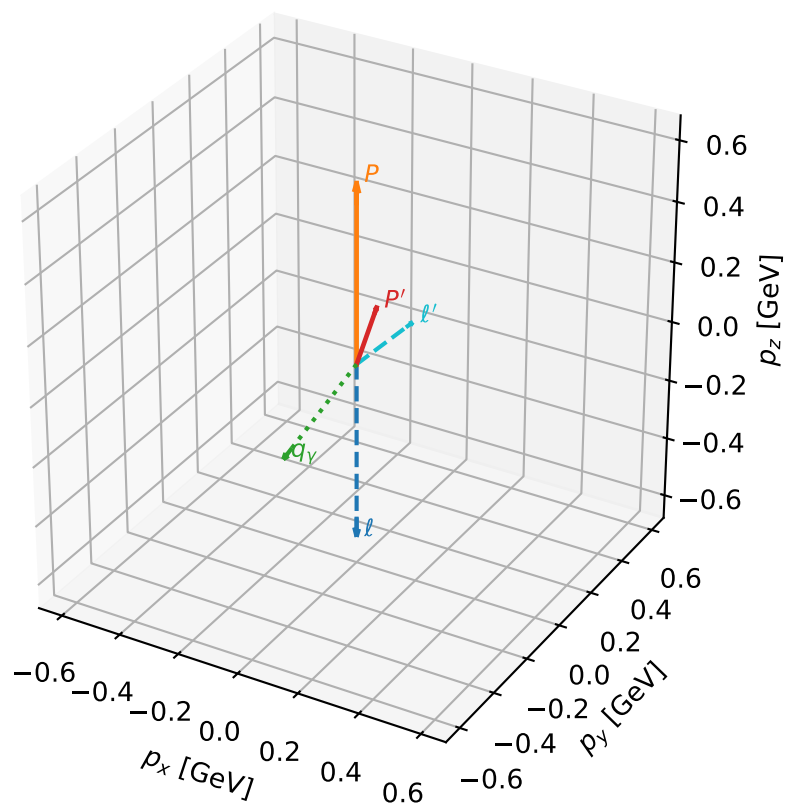
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_197 $d_{\{\text{rm GHZ}\}}=1.98691\text{e-}05$
 region: polarization_cluster_03_local_minimum_197
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.5035531$ rad
 incoming lepton: $-0.803265|+\rangle +0.595622|-\rangle$
 proton mixing angle: $\alpha_p=1.034678$ rad
 incoming proton: $0.510803|+\rangle +0.859698|-\rangle$

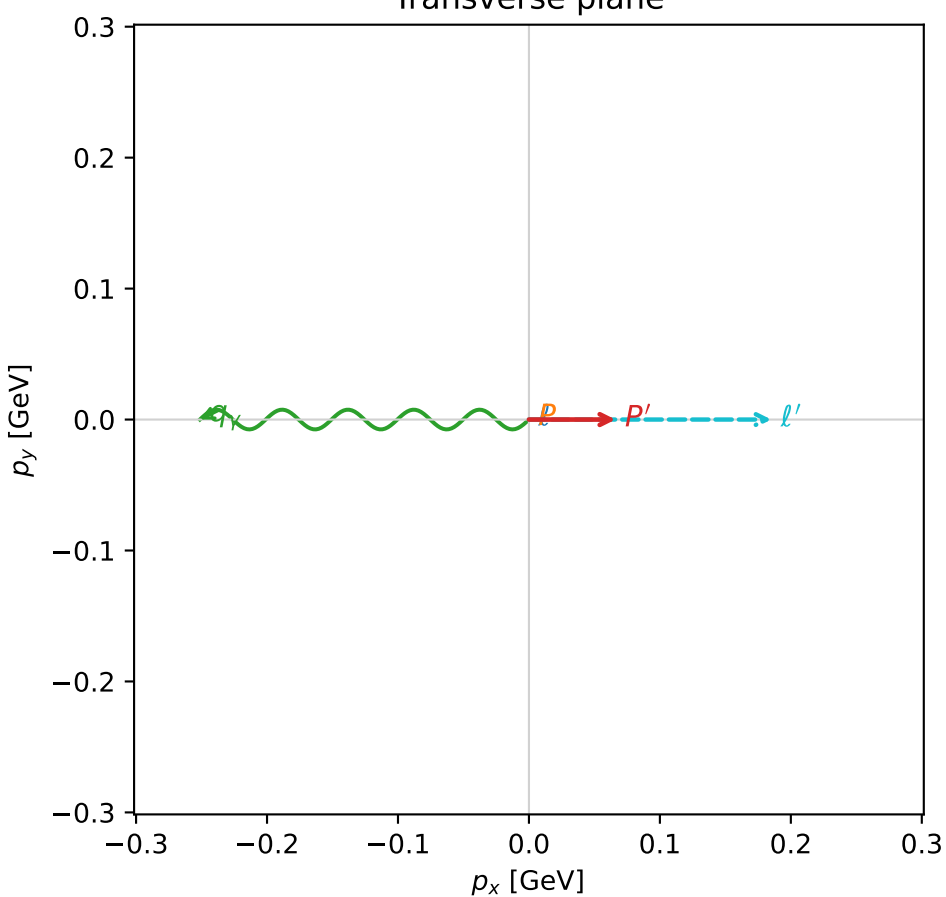
$s=2.89034$, $\sqrt{s}=1.7001$, $|\vec{P}|=0.591287$, $|\vec{P}'|=0.21679$
 $\theta_{p'}=0.310084$, $\theta_V=2.57868$, $\phi_{p'}=6.28315$, $\phi_V=3.14158$
 $E_V=0.470847$, $\theta_{l'}=0.767811$, $\phi_{l'}=6.28317$
 $Q^2=0.541939$, $x_B=0.329208$, $t=-0.131134$, $W^2=1.9841$, $y=0.818801$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5913$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5913$
 P $E= 1.1088$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5913$
 l' $E= 0.2665$ $p_x= 0.1851$ $p_y= -0.0000$ $p_z= 0.1917$
 P' $E= 0.9627$ $p_x= 0.0662$ $p_y= -0.0000$ $p_z= 0.2065$
 q_V $E= 0.4708$ $p_x= -0.2513$ $p_y= 0.0000$ $p_z= -0.3982$

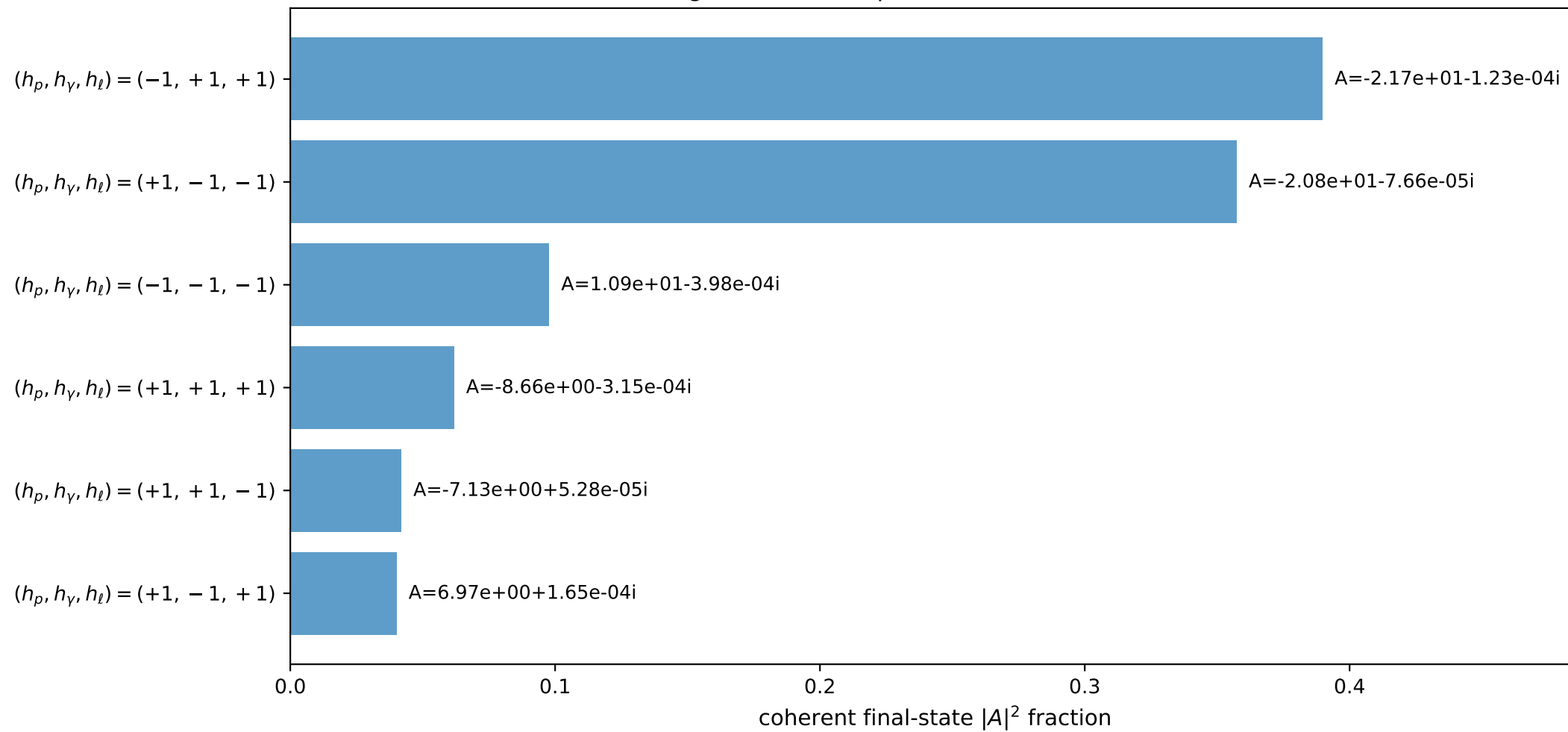
3D momenta



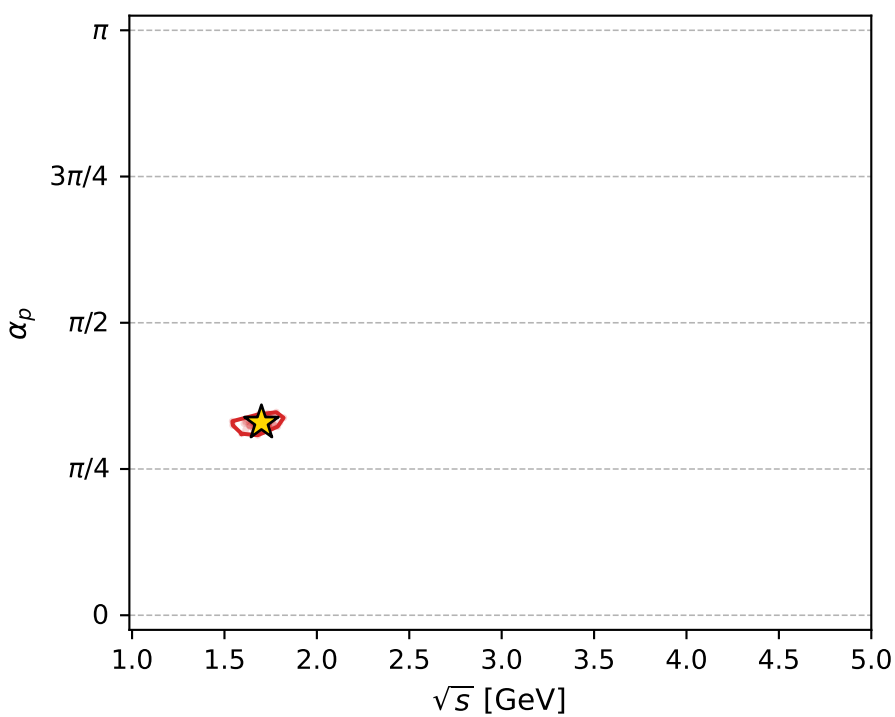
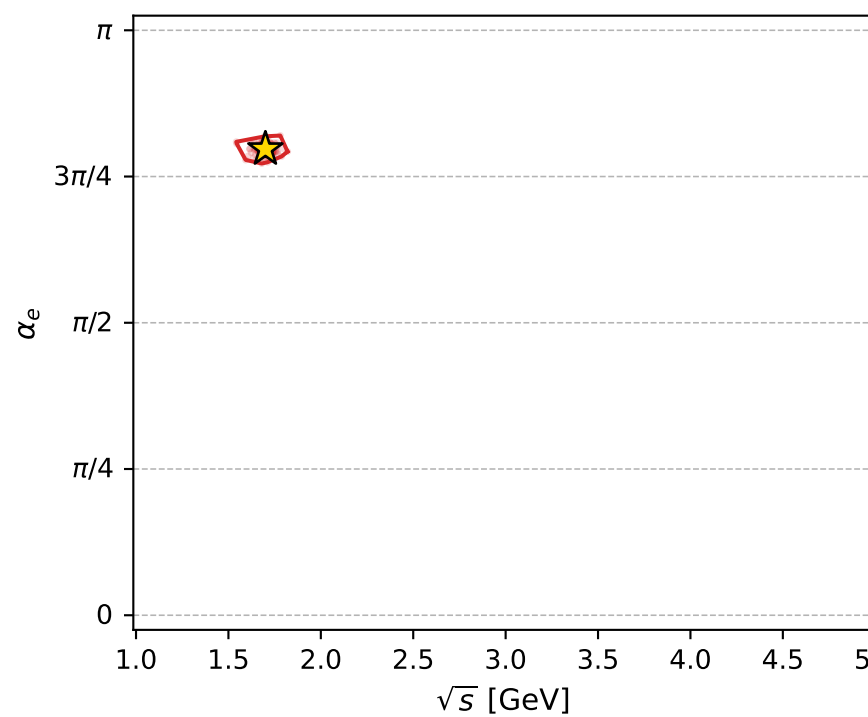
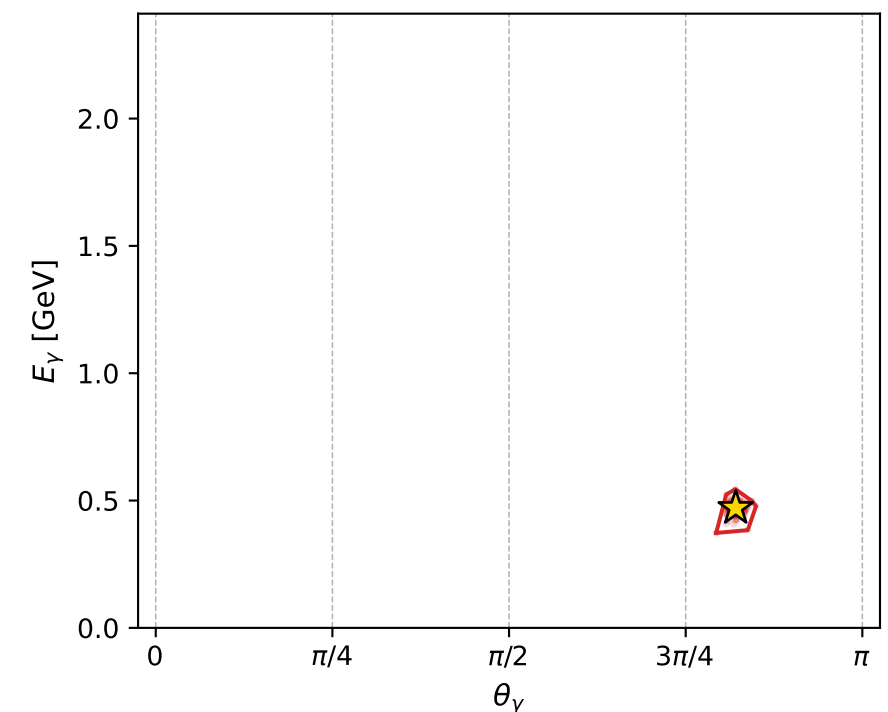
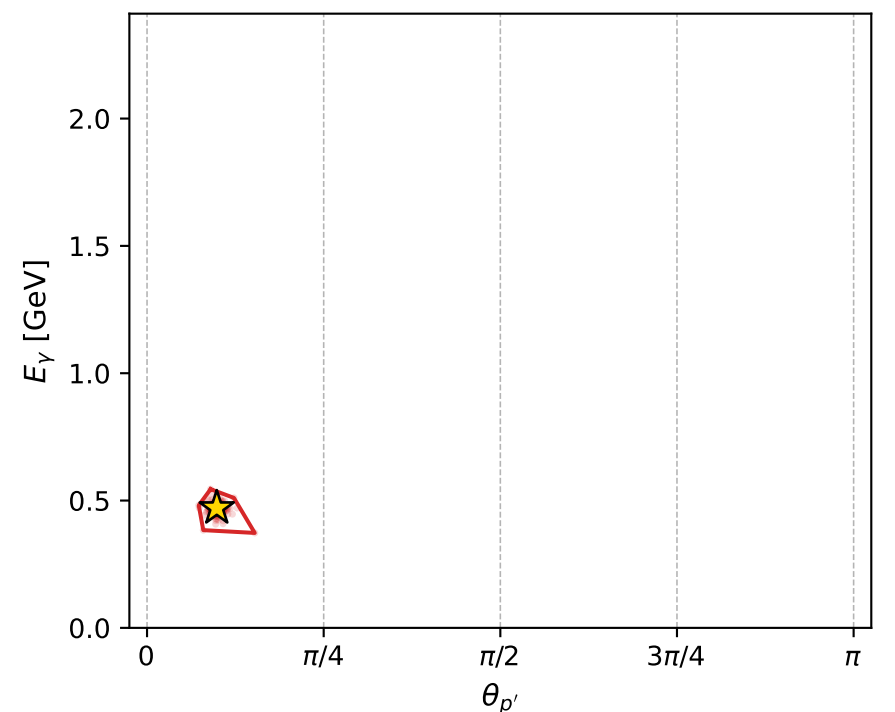
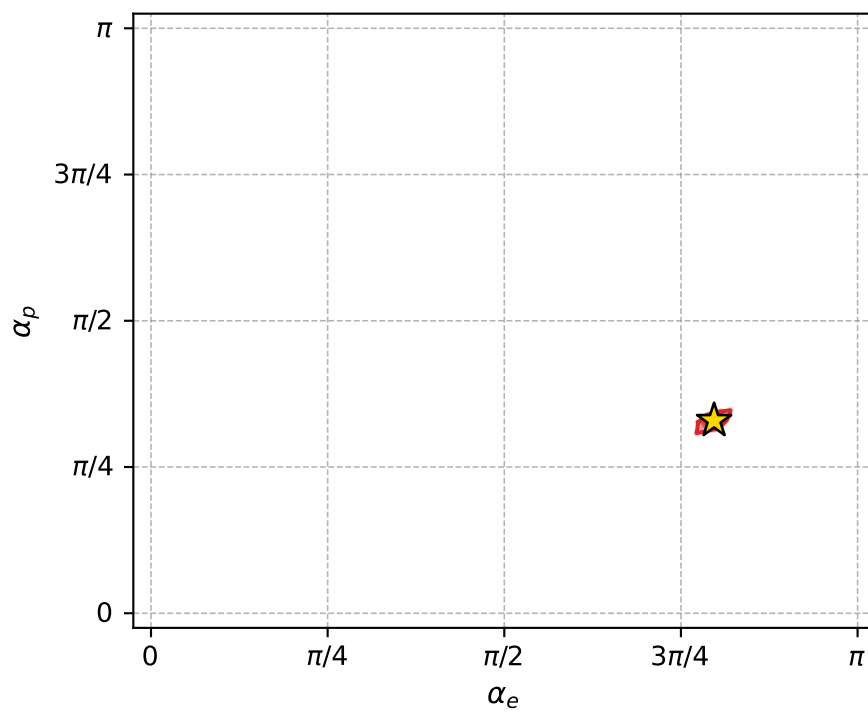
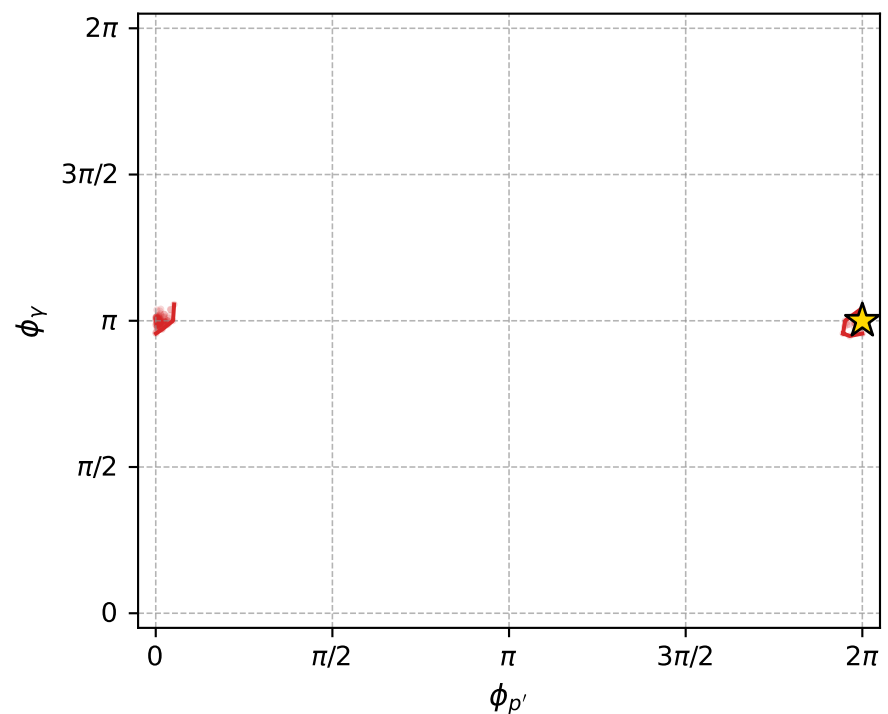
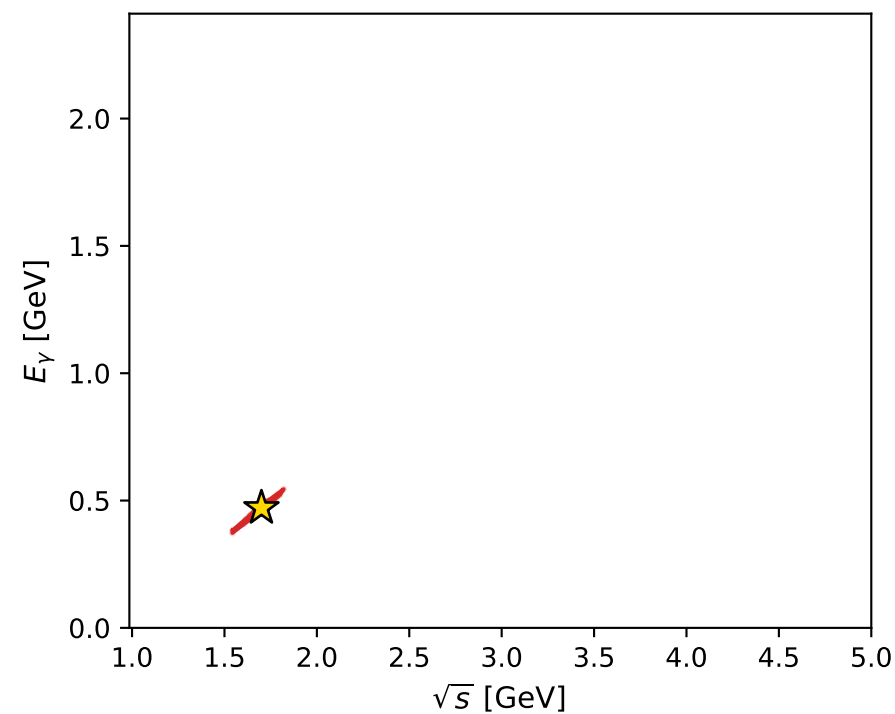
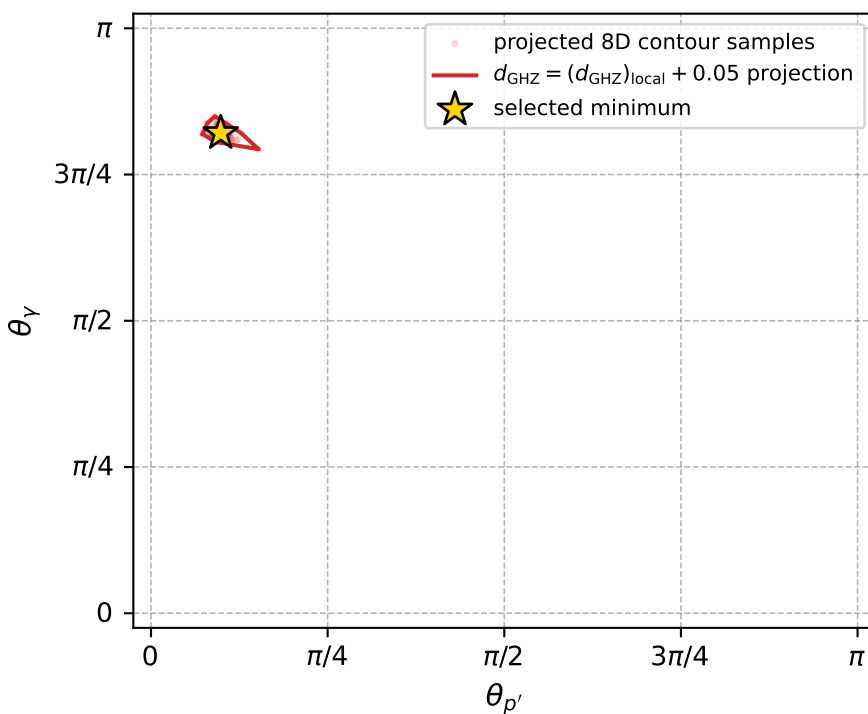
Transverse plane



Leading final-state amplitudes (N=6, retained=98.9%)



electron: polarization cluster P3, local minimum 197; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.9869139\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050019869$
 above global minimum = $1.9846856\text{e-}05$
 8D contour samples = 256

selected phase-space point:

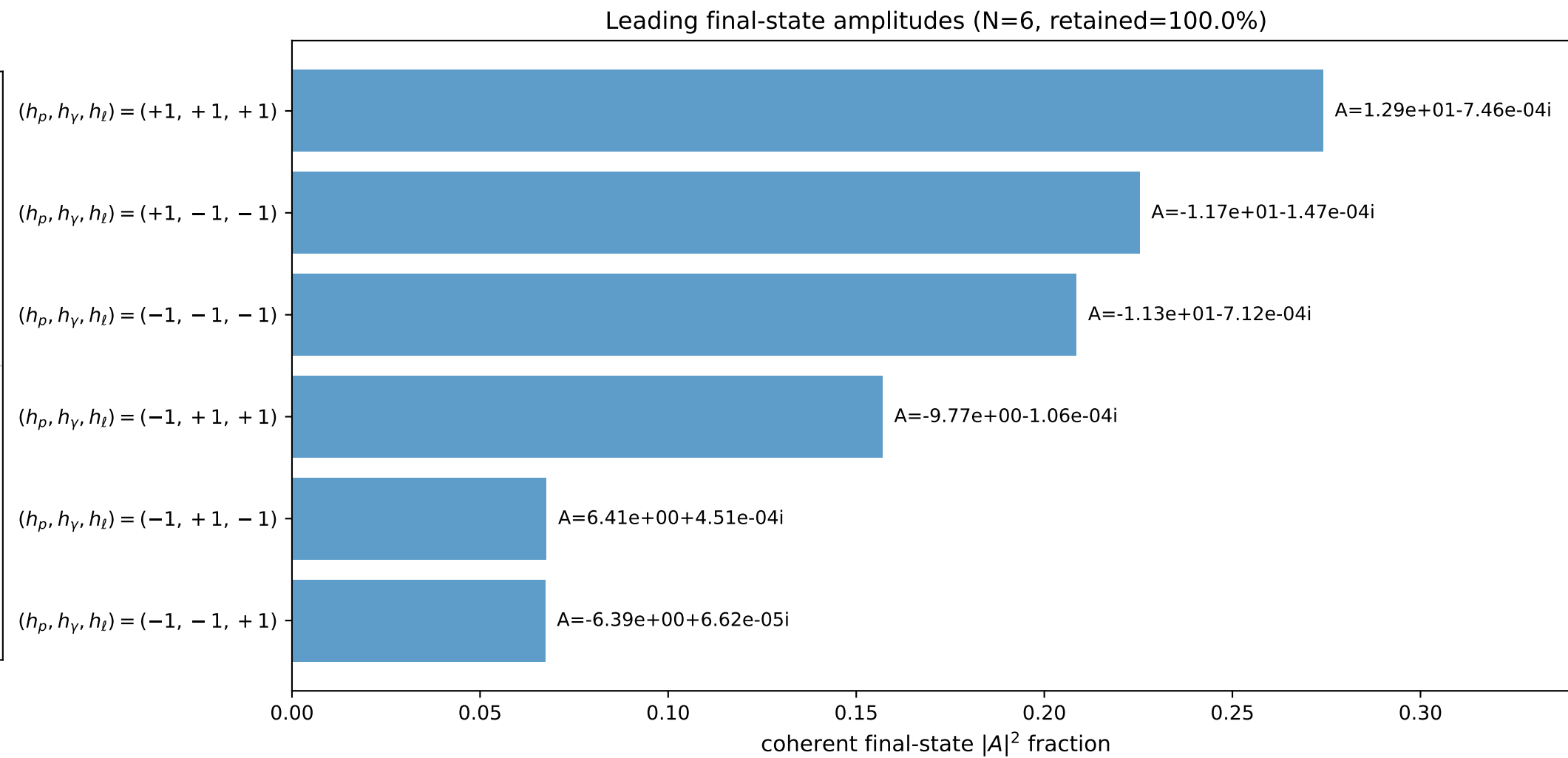
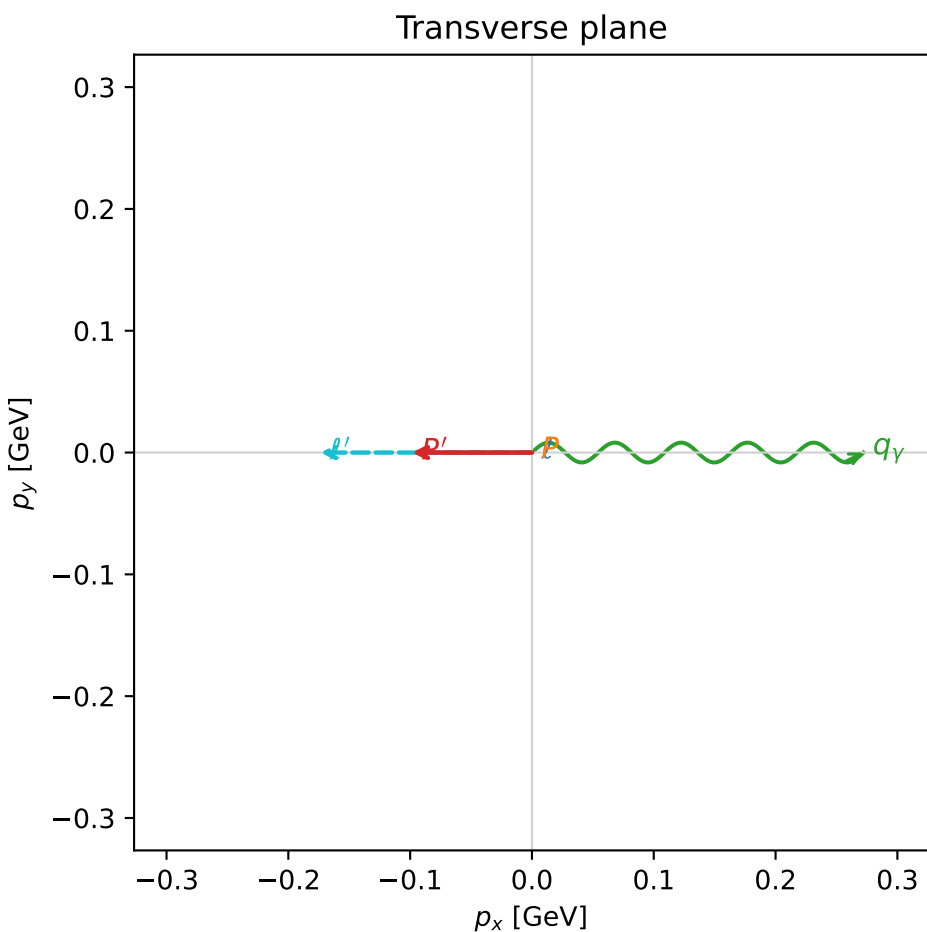
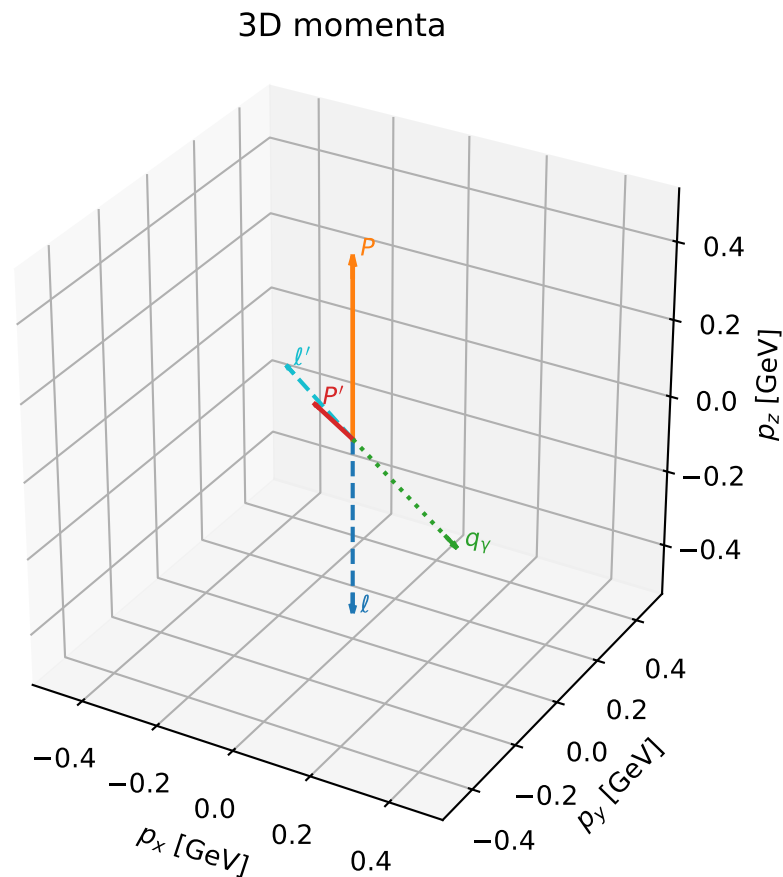
$\text{sqrt_s} = 1.700099$
 $\text{theta_p_out} = 0.3100844$
 $\text{theta_gamma_out} = 2.578679$
 $\text{qOut} = 0.4708474$
 $\text{phi_p_out} = 6.283148$
 $\text{phi_gamma_out} = 3.141575$
 $\text{alpha_e} = 2.503553$
 $\text{alpha_p} = 1.034678$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

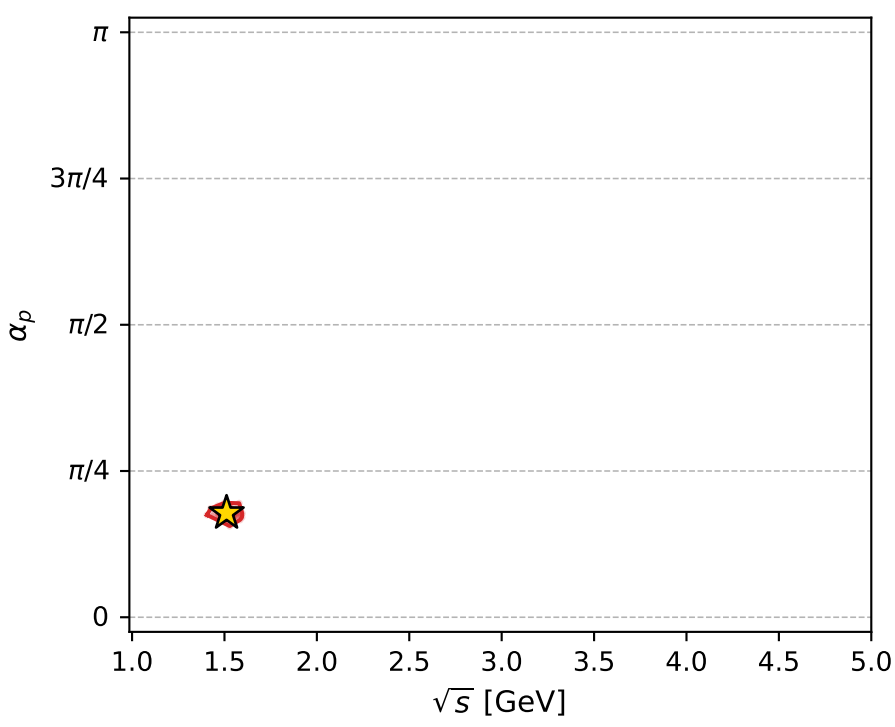
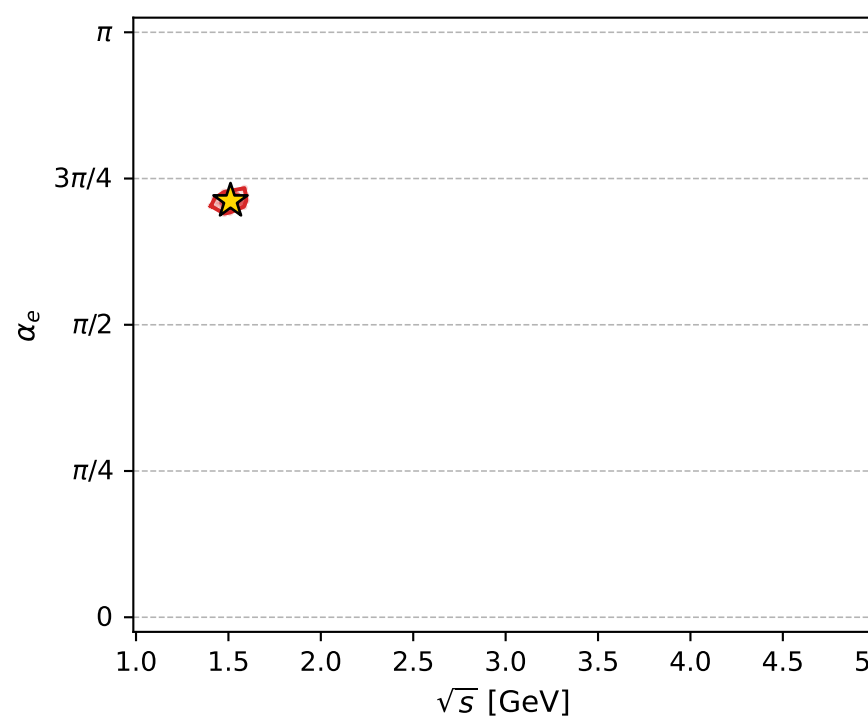
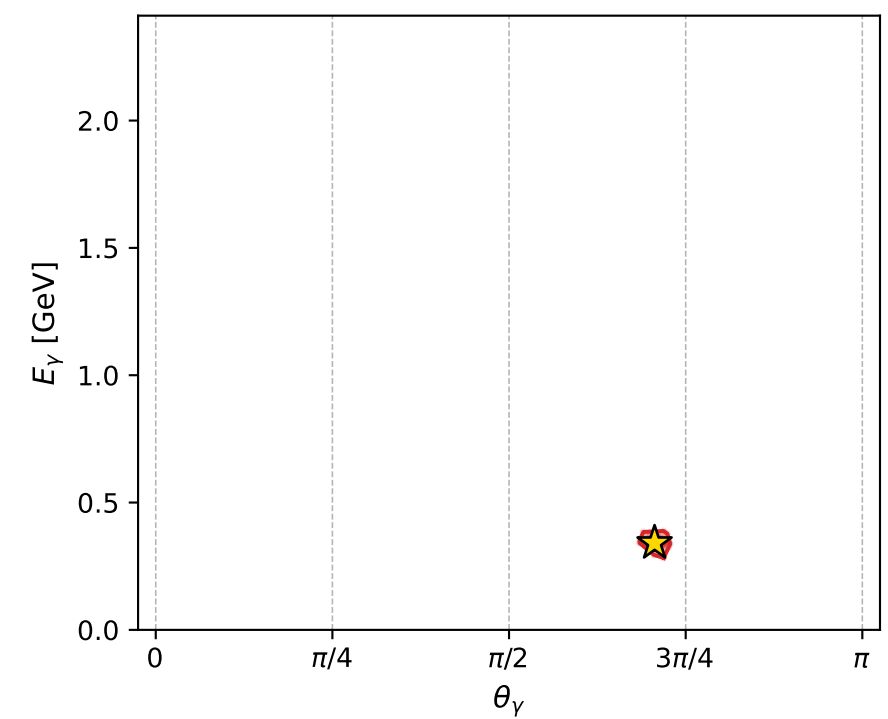
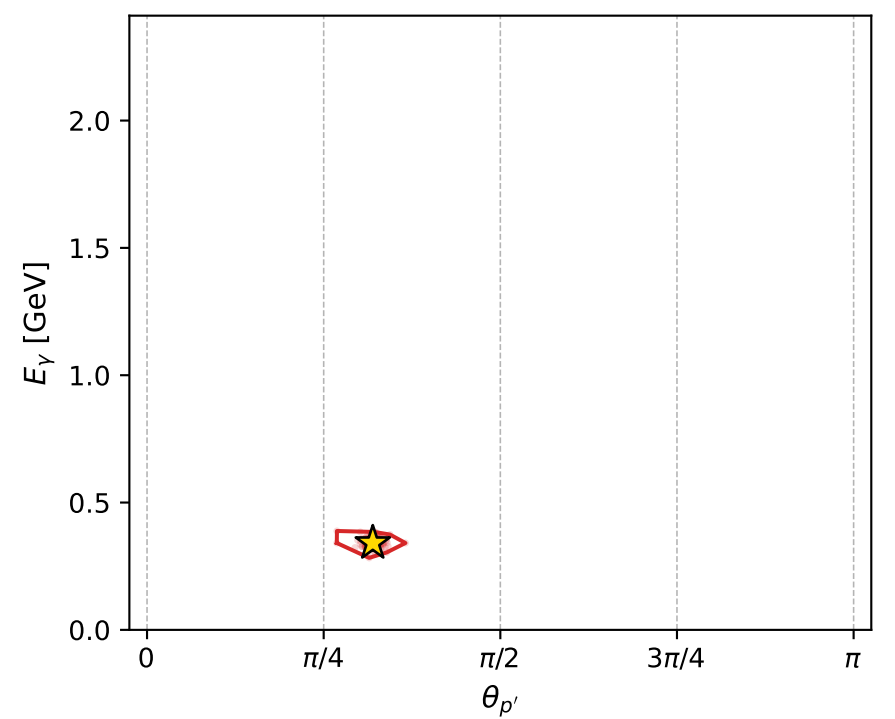
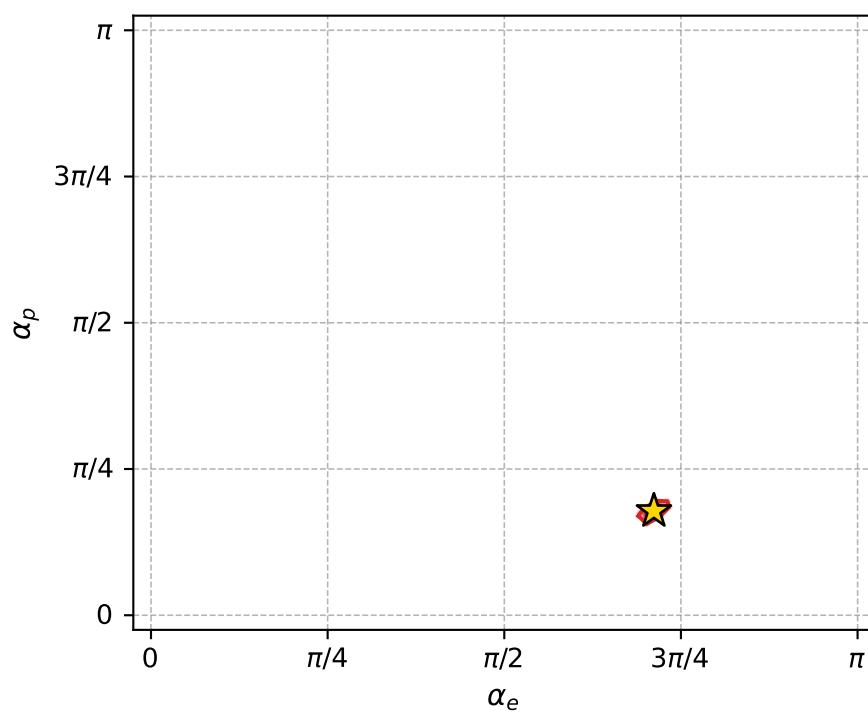
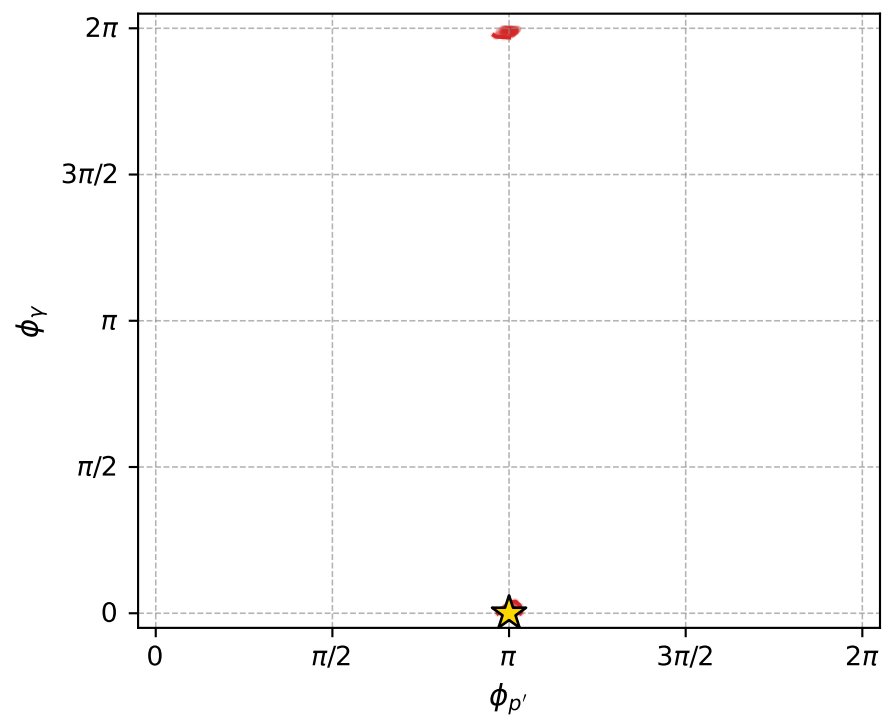
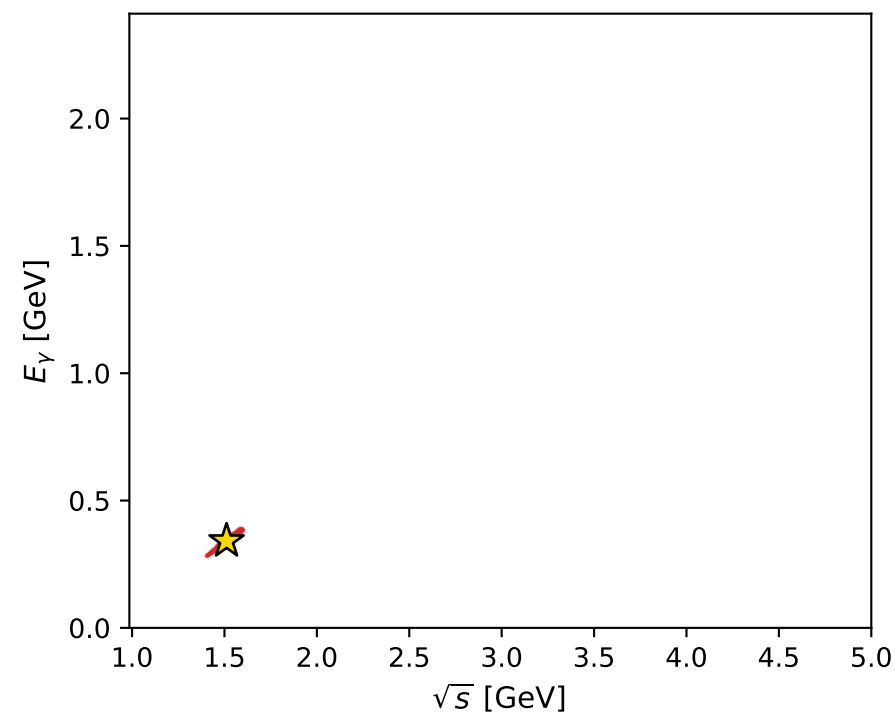
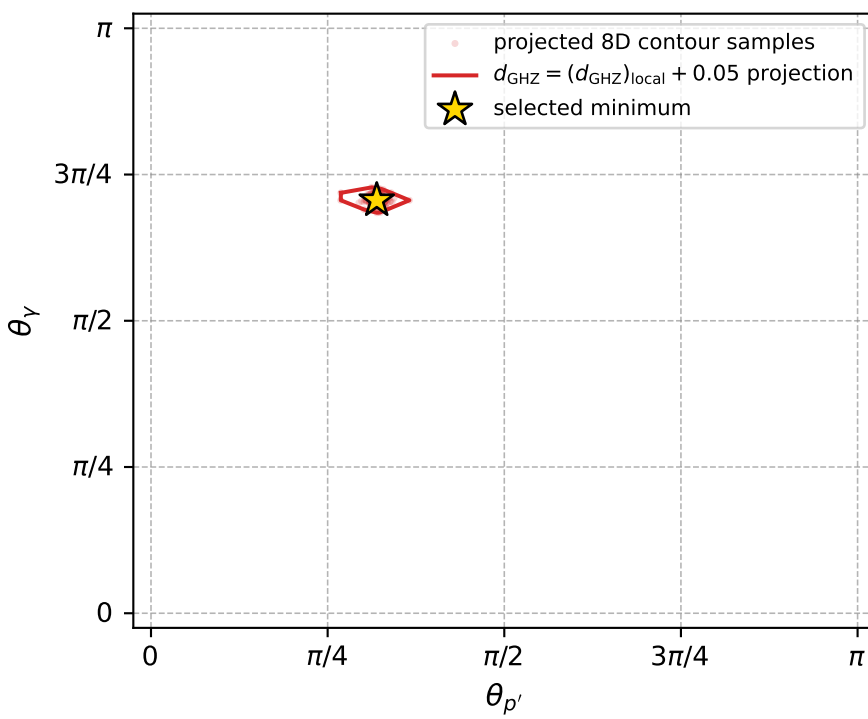
dghz_mixing_angles_polarization_03_local_minimum_201 d_{\rm GHZ}=2.19374e-05
 region: polarization_cluster_03_local_minimum_201
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2360039$ rad
 incoming lepton: $-0.617222|+> +0.786789|->$
 proton mixing angle: $\alpha_p=0.55959252$ rad
 incoming proton: $0.847471|+> +0.530841|->$

$s=2.28407$, $\sqrt{s}=1.51131$, $|\vec{P}|=0.464571$, $|\vec{P}'|=0.116794$
 $\theta_{P'}=1.00372$, $\theta_Y=2.21785$, $\phi_{P'}=3.14165$, $\phi_Y=4.12856e-05$
 $E_Y=0.341149$, $\theta_{\ell'}=0.882251$, $\phi_{\ell'}=3.14162$
 $Q^2=0.341775$, $x_B=0.32057$, $t=-0.160872$, $W^2=1.60421$, $y=0.759242$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4646$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4646$
 P $E= 1.0467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4646$
 ℓ' $E= 0.2249$ $p_x= -0.1737$ $p_y= -0.0000$ $p_z= 0.1429$
 P' $E= 0.9452$ $p_x= -0.0985$ $p_y= -0.0000$ $p_z= 0.0627$
 q_Y $E= 0.3411$ $p_x= 0.2722$ $p_y= 0.0000$ $p_z= -0.2057$



electron: polarization cluster P3, local minimum 201; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.1937395\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050021937$
 above global minimum = $2.1915111\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.511314$
 $\text{theta_p_out} = 1.003716$
 $\text{theta_gamma_out} = 2.217848$
 $\text{q0out} = 0.3411487$
 $\text{phi_p_out} = 3.141654$
 $\text{phi_gamma_out} = 4.128558\text{e-}05$
 $\text{alpha_e} = 2.236004$
 $\text{alpha_p} = 0.5595925$

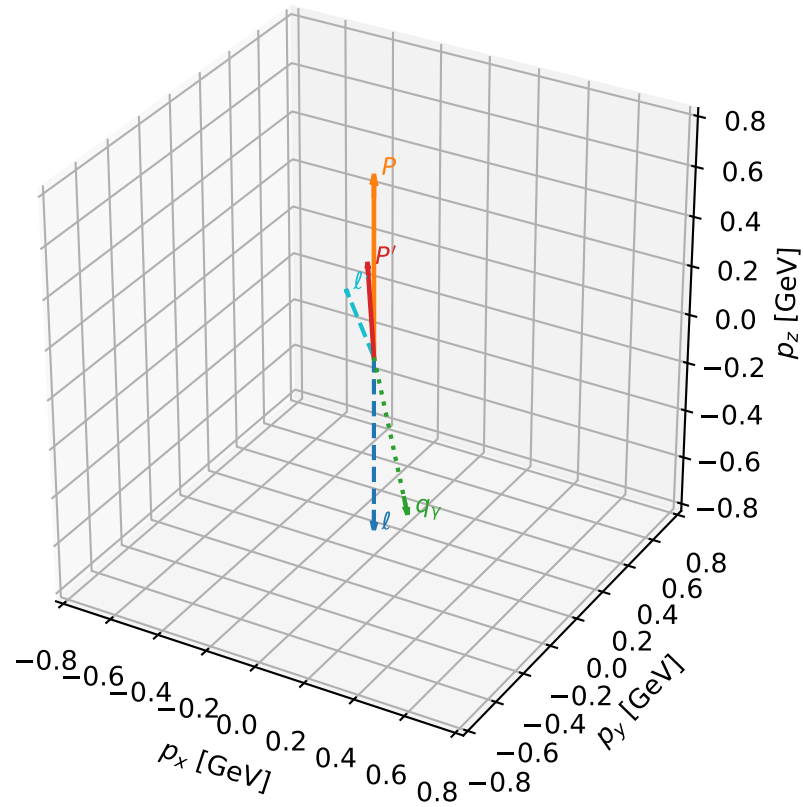
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_204 $d_{\{\text{rm GHZ}\}}=2.47799\text{e-}05$
 region: polarization_cluster_03_local_minimum_204
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.0175873$ rad
 incoming lepton: $-0.432074|+\rangle +0.901838|-\rangle$
 proton mixing angle: $\alpha_p=0.38602922$ rad
 incoming proton: $0.926411|+\rangle +0.376513|-\rangle$

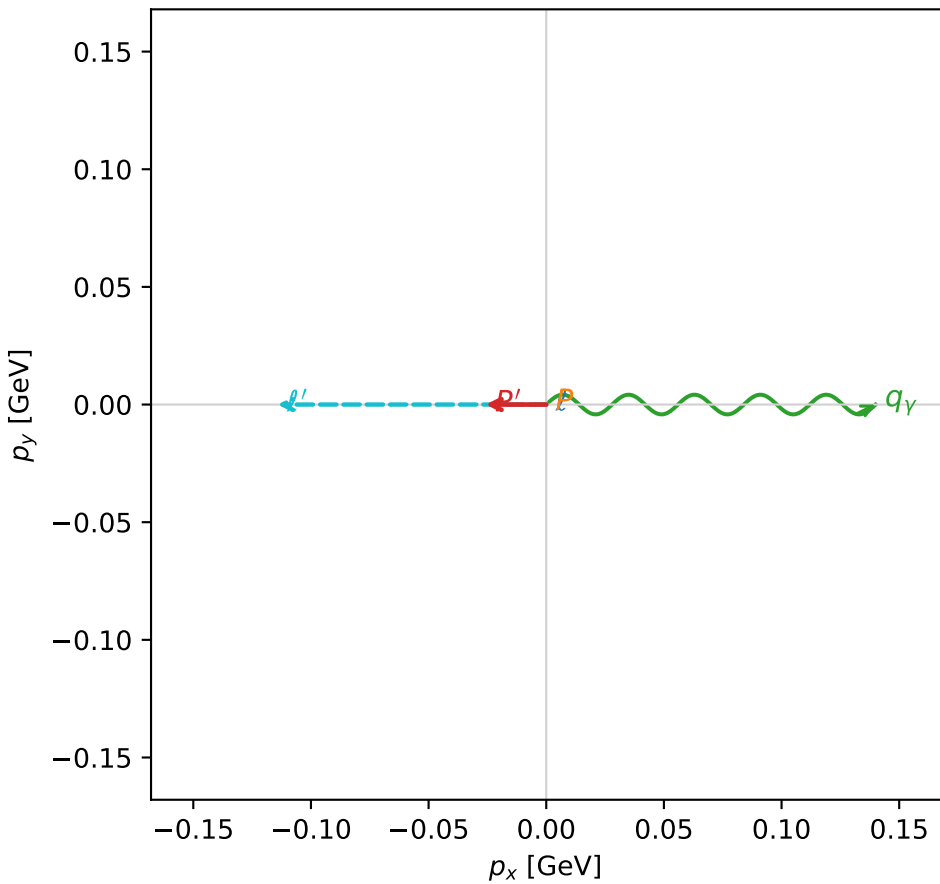
$s=3.63688$, $\sqrt{s}=1.90706$, $|\vec{P}|=0.72285$, $|\vec{P}'|=0.370003$
 $\theta_{P'}=0.0696716$, $\theta_V=2.91723$, $\phi_{P'}=3.14156$, $\phi_V=0$
 $E_V=0.629105$, $\theta_{\ell'}=0.437411$, $\phi_{\ell'}=3.1416$
 $Q^2=0.742871$, $x_B=0.300568$, $t=-0.0948674$, $W^2=2.60853$, $y=0.896454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7228$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7228$
 P $E= 1.1842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7228$
 ℓ' $E= 0.2696$ $p_x= -0.1142$ $p_y= -0.0000$ $p_z= 0.2442$
 P' $E= 1.0083$ $p_x= -0.0258$ $p_y= 0.0000$ $p_z= 0.3691$
 q_V $E= 0.6291$ $p_x= 0.1400$ $p_y= 0.0000$ $p_z= -0.6133$

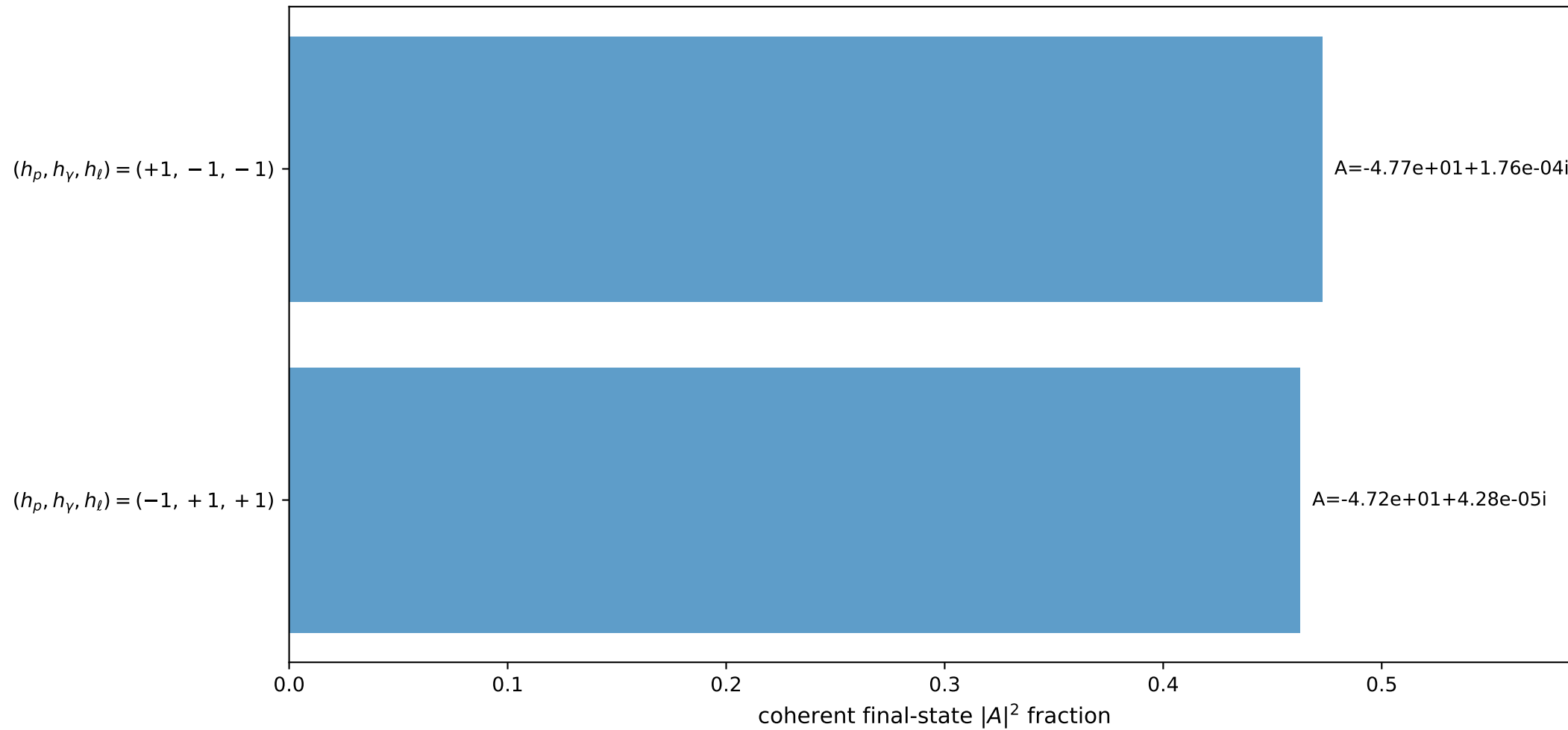
3D momenta



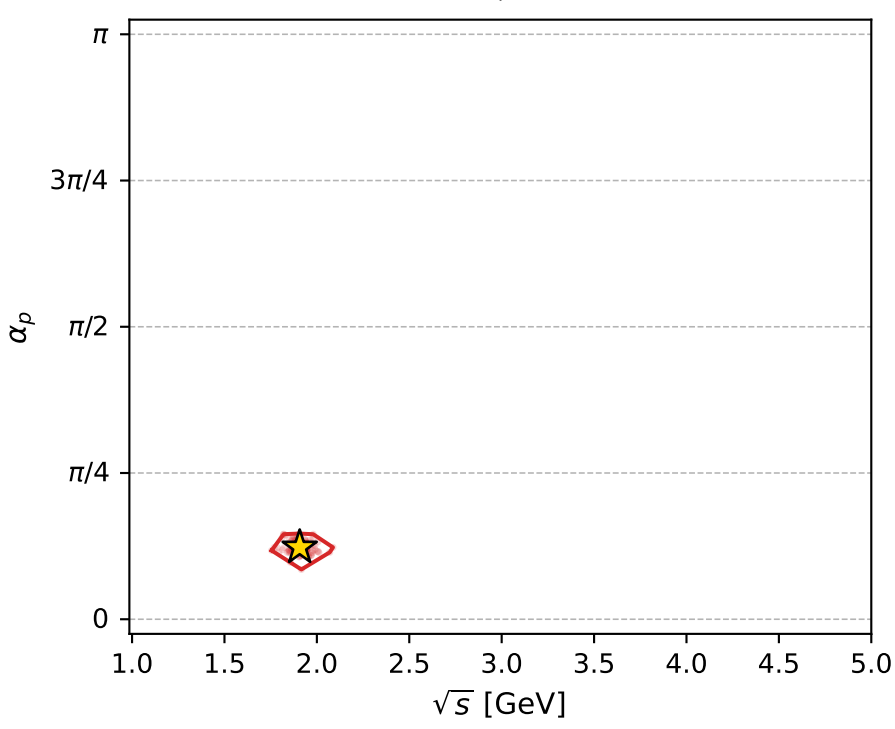
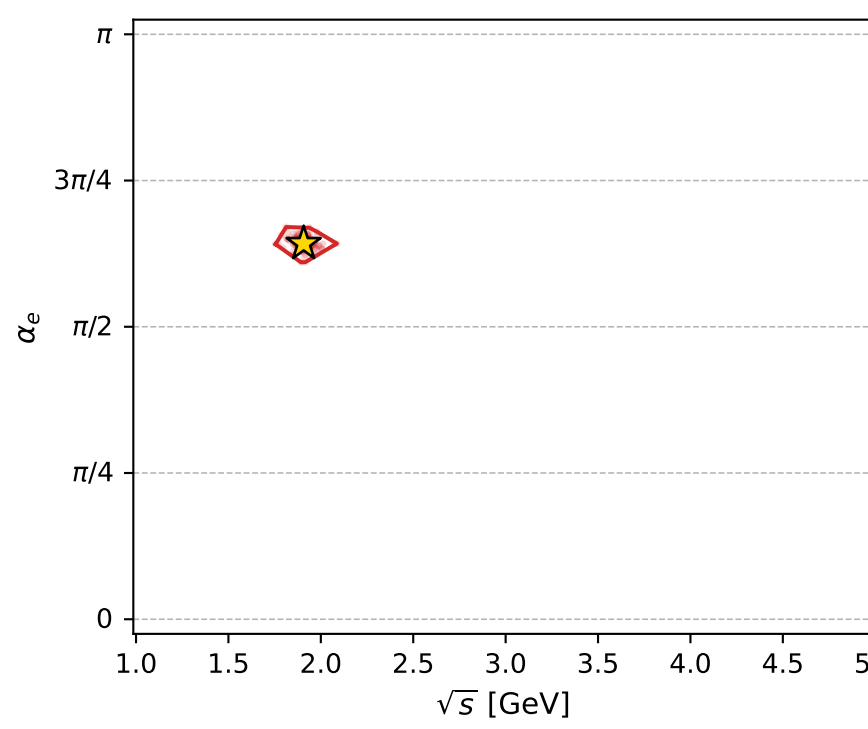
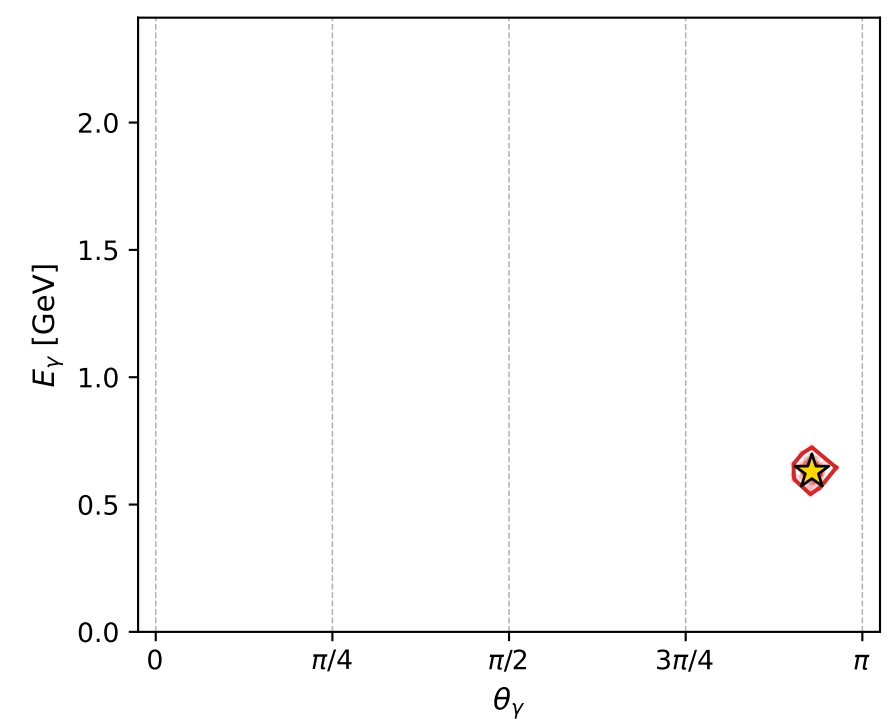
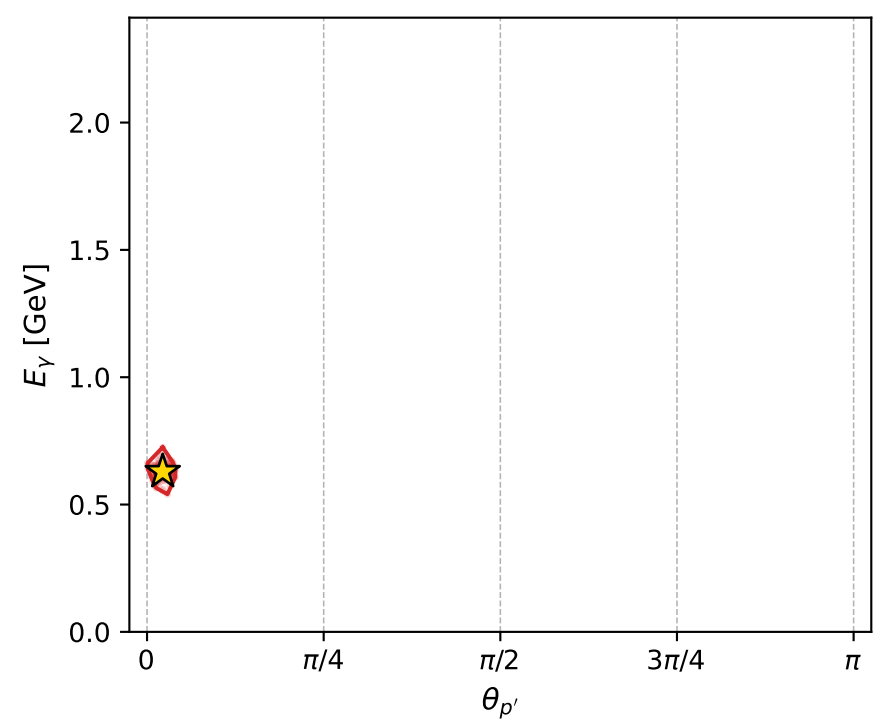
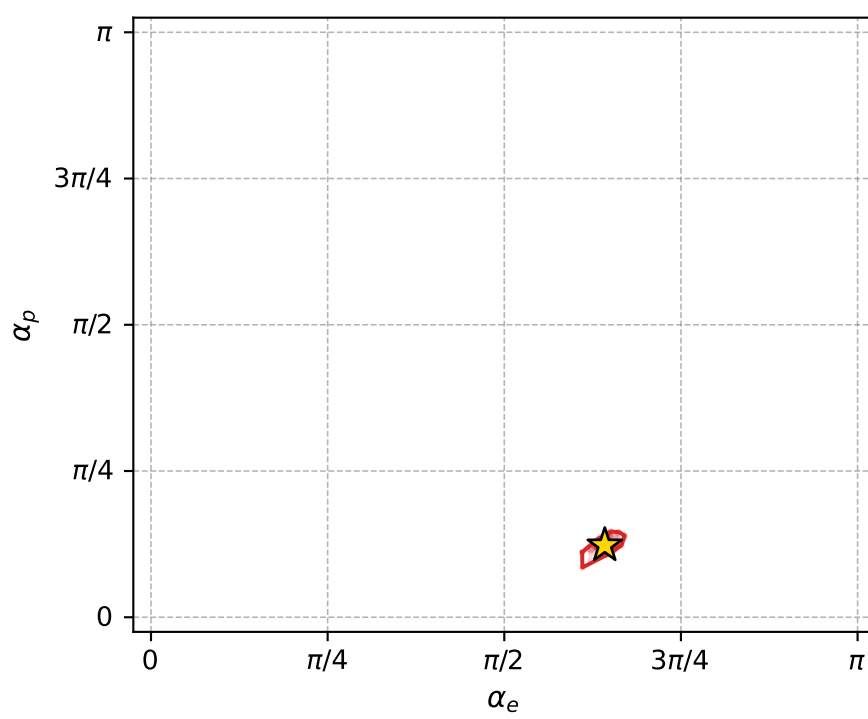
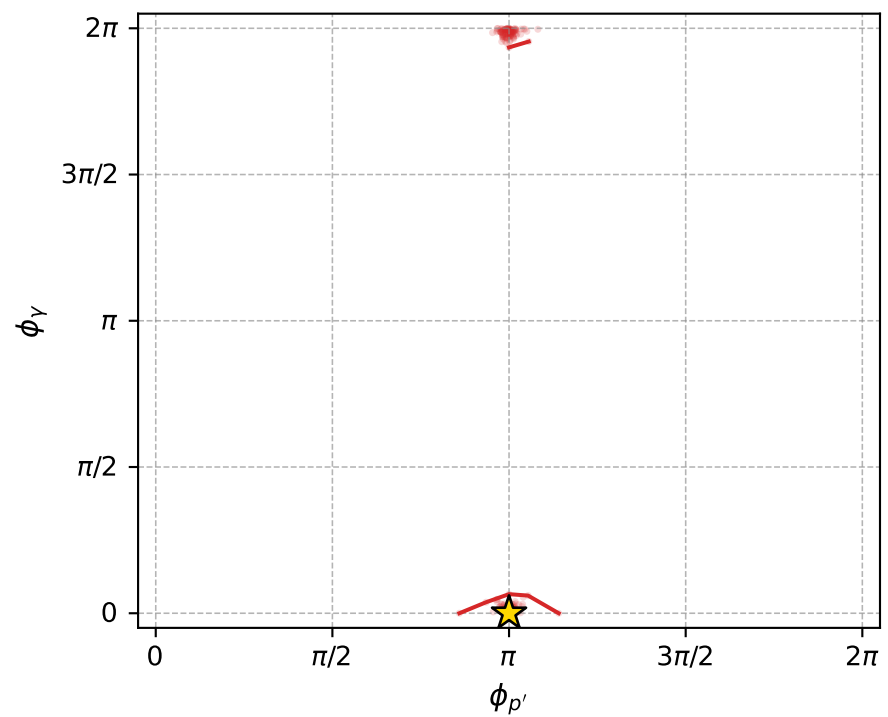
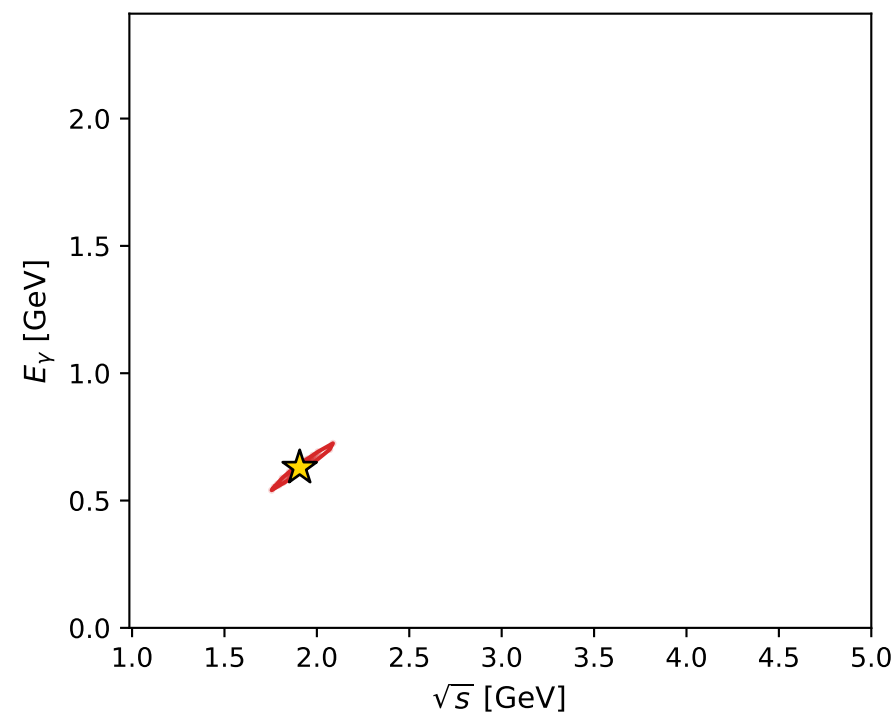
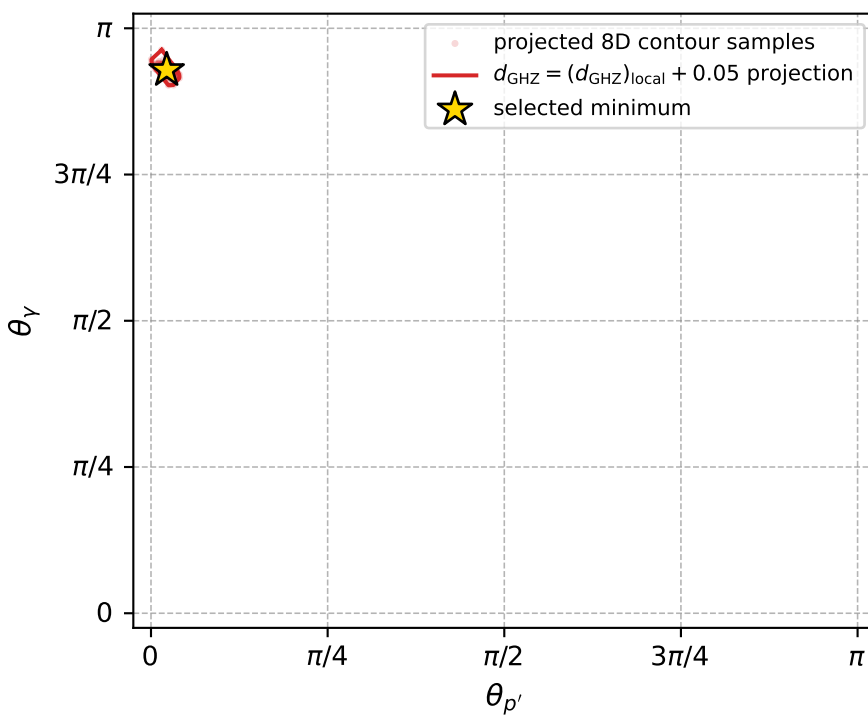
Transverse plane



Leading final-state amplitudes (N=2, retained=93.6%)



electron: polarization cluster P3, local minimum 204; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4779903\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.05002478$
 above global minimum = $2.475762\text{e-}05$
 8D contour samples = 256

selected phase-space point:

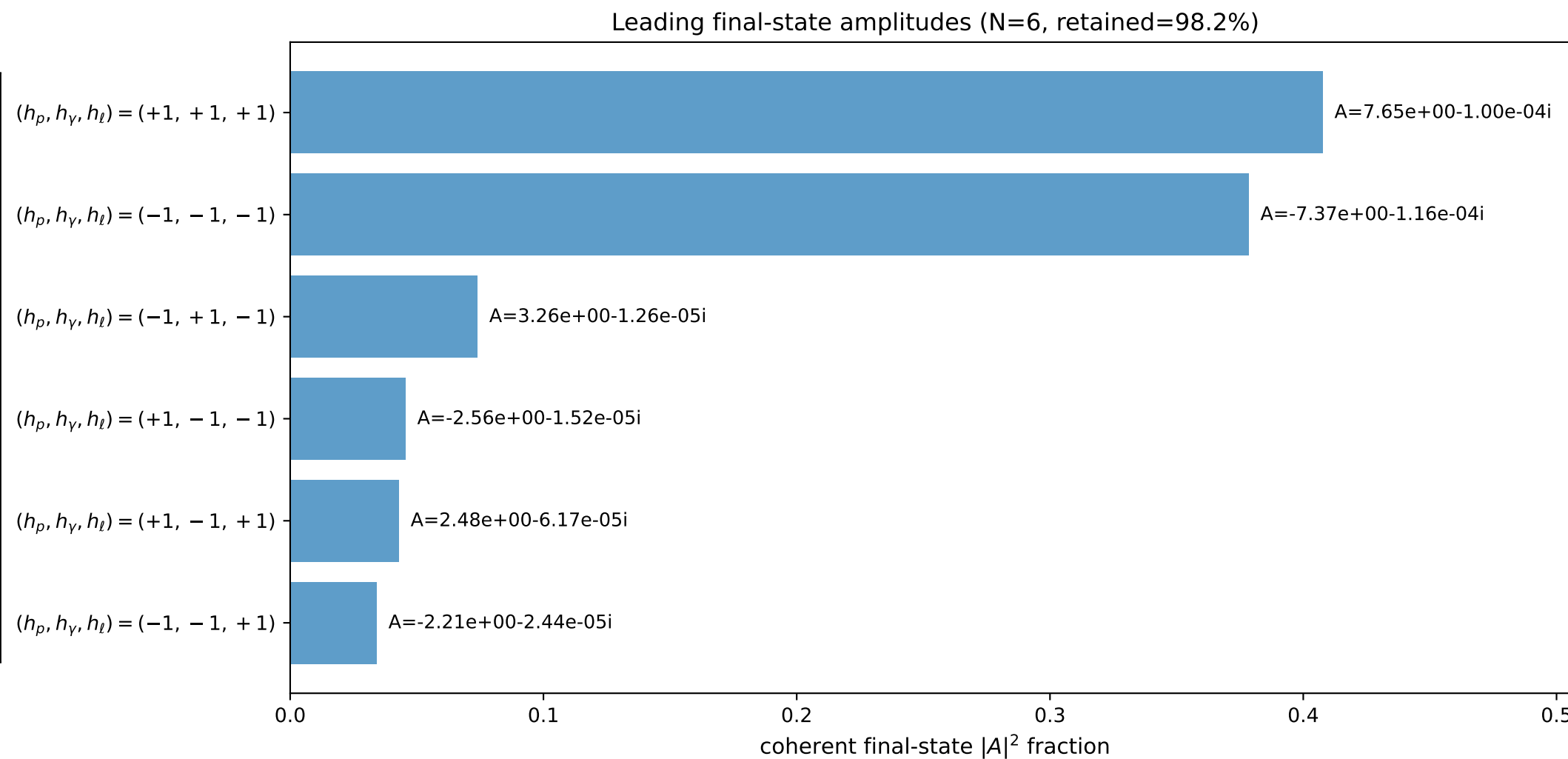
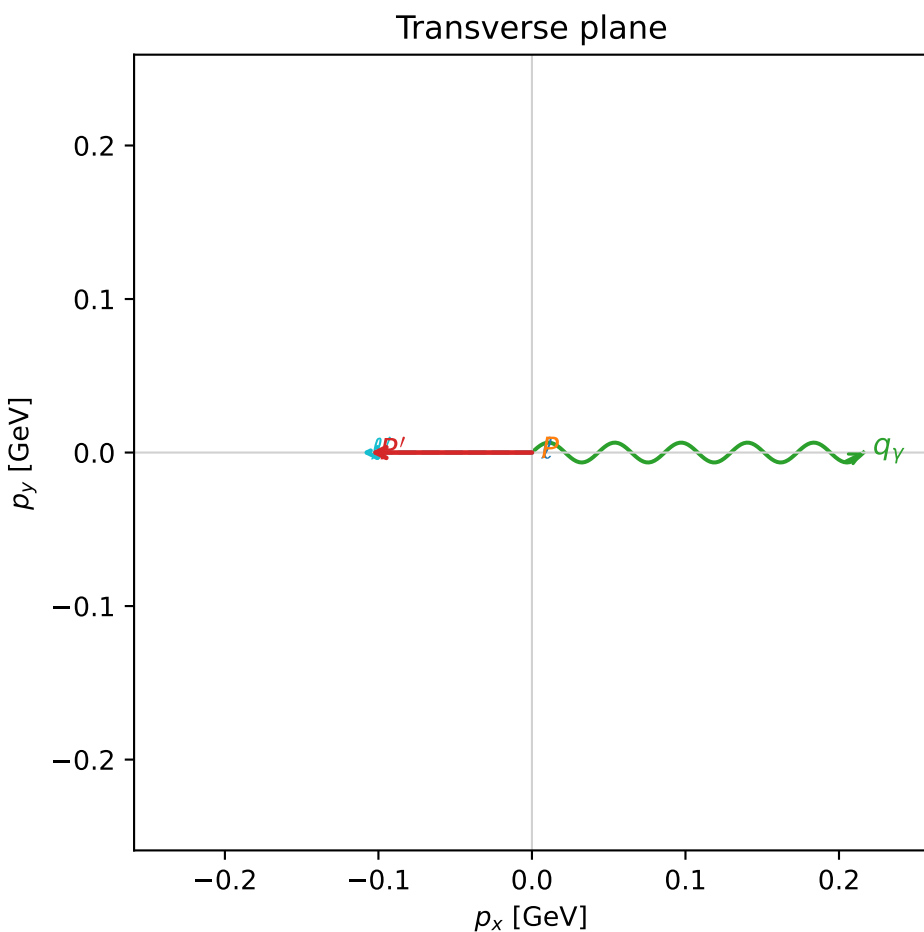
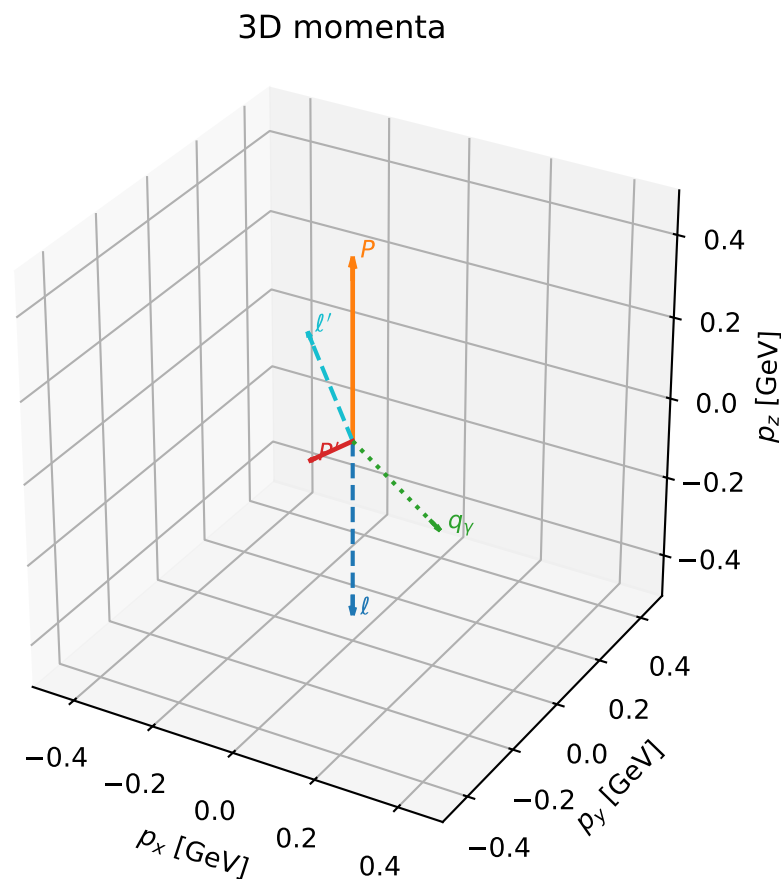
$\text{sqrt_s} = 1.907061$
 $\text{theta_p_out} = 0.06967158$
 $\text{theta_gamma_out} = 2.91723$
 $\text{q0out} = 0.6291055$
 $\text{phi_p_out} = 3.141565$
 $\text{phi_gamma_out} = 0$
 $\text{alpha_e} = 2.017587$
 $\text{alpha_p} = 0.3860292$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

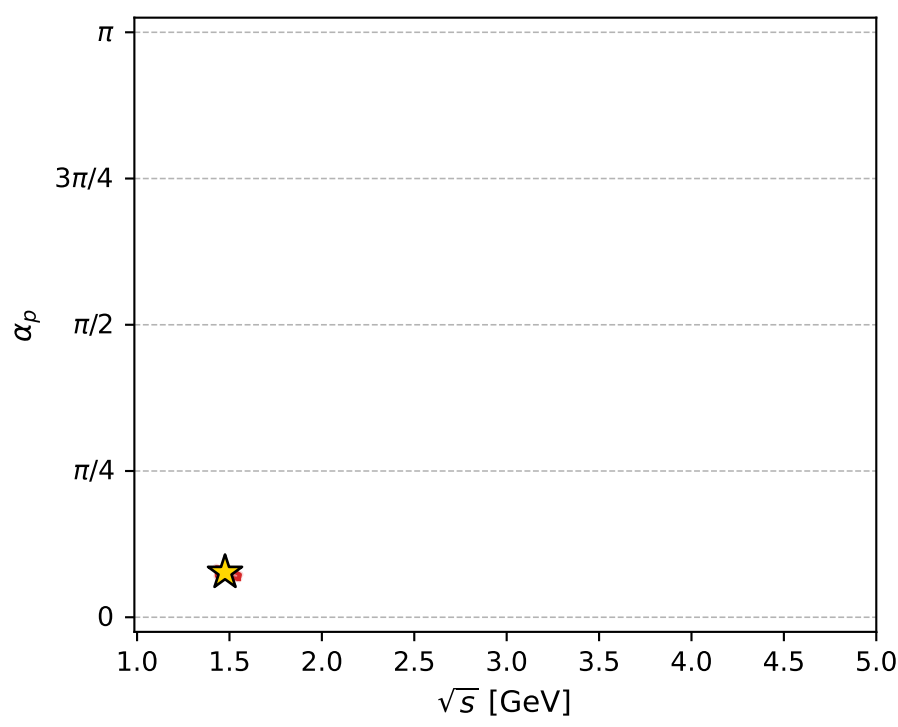
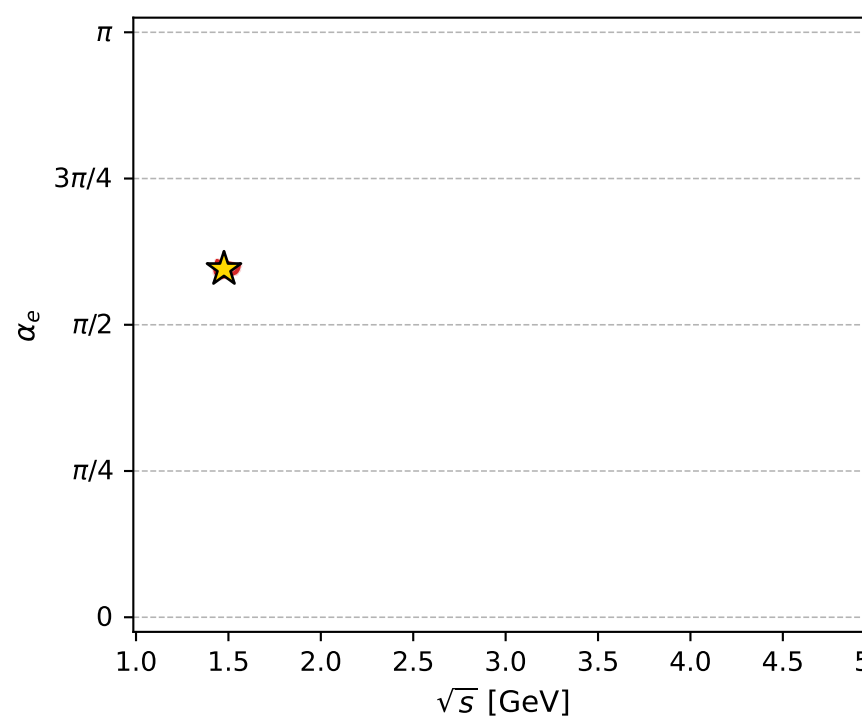
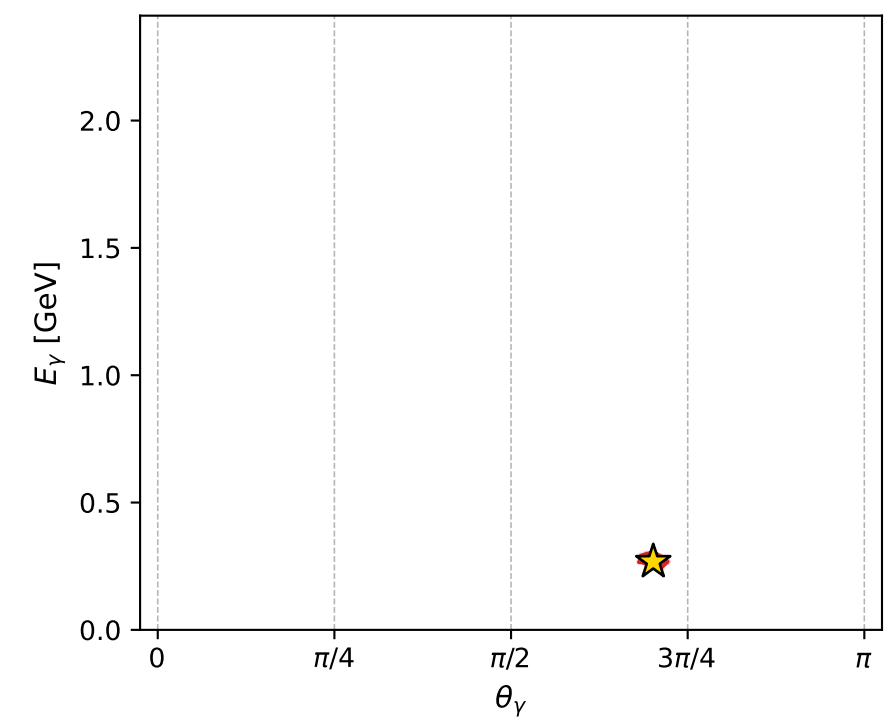
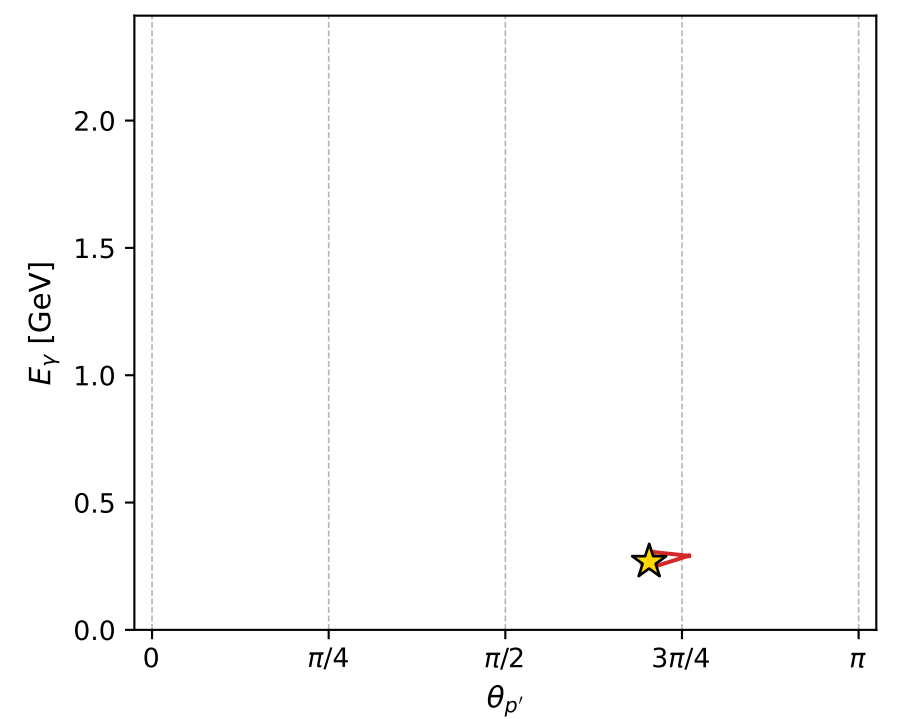
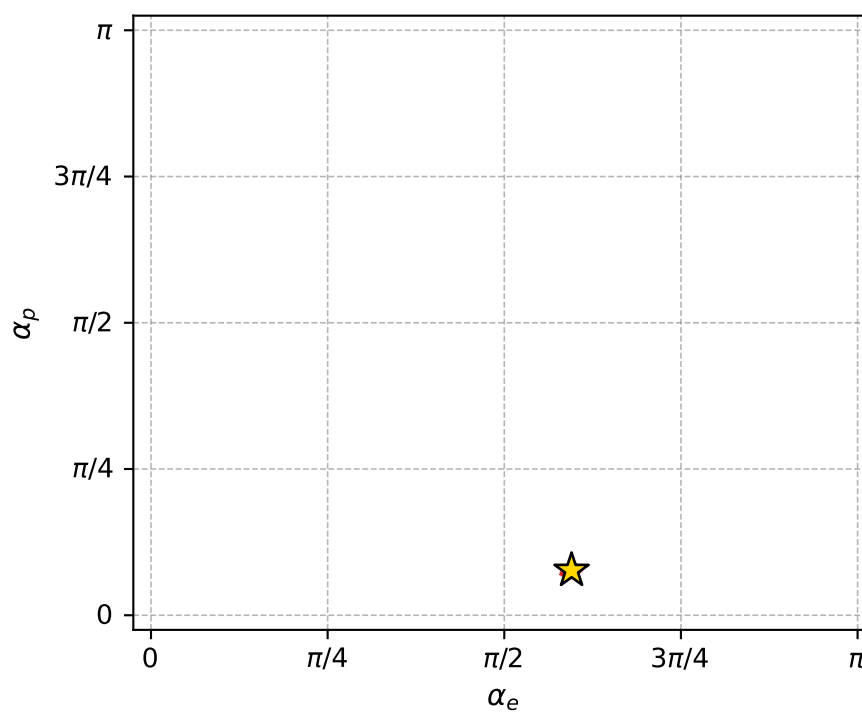
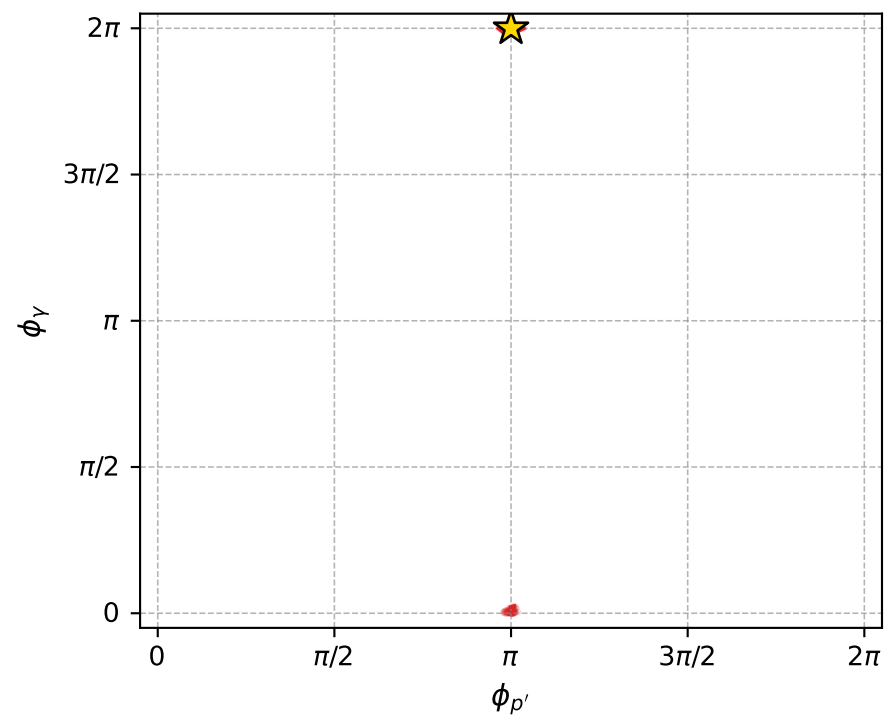
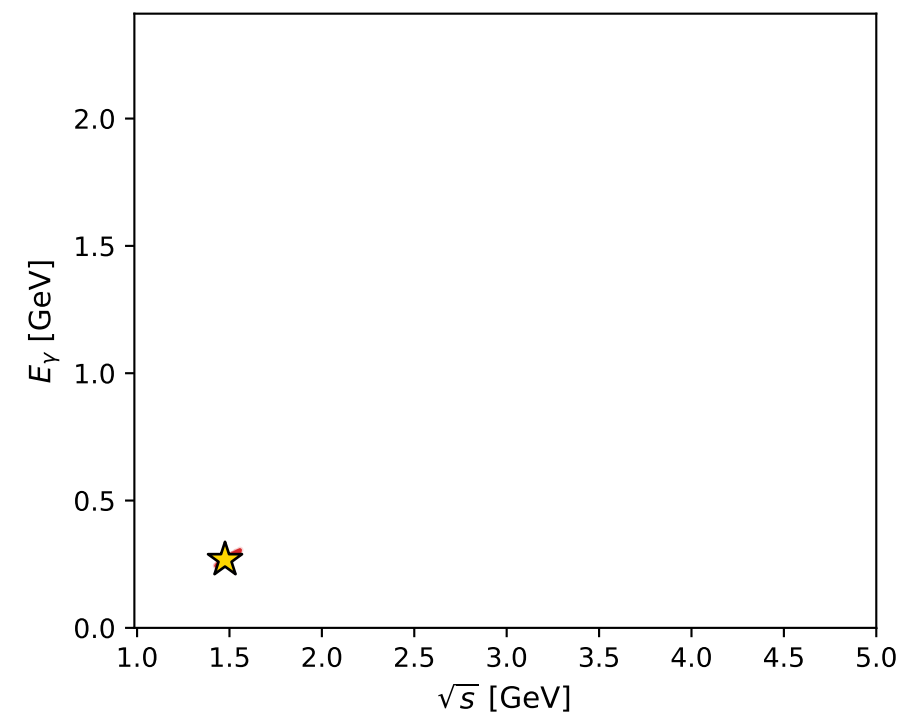
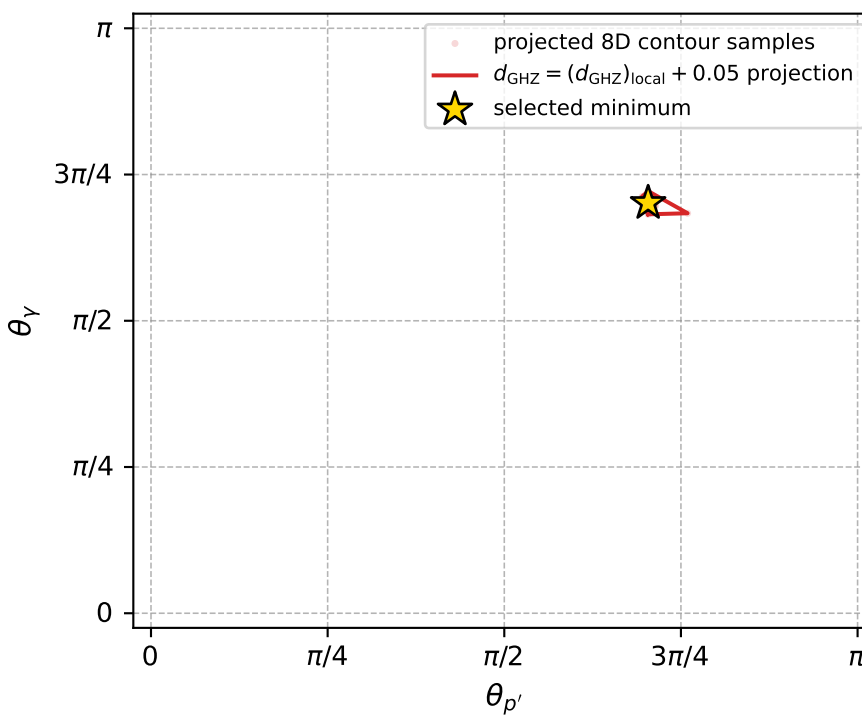
dghz_mixing_angles_polarization_03_local_minimum_208 $d_{\{\text{rm GHZ}\}}=2.65673\text{e-}05$
 region: polarization_cluster_03_local_minimum_208
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.8699449$ rad
 incoming lepton: $-0.294707|+\rangle +0.955588|-\rangle$
 proton mixing angle: $\alpha_p=0.24082942$ rad
 incoming proton: $0.97114|+\rangle +0.238508|-\rangle$

$s=2.17851$, $\sqrt{s}=1.47598$, $|\vec{P}|=0.439934$, $|\vec{P}'|=0.1311$
 $\theta_{P'}=2.21027$, $\theta_Y=2.20313$, $\phi_{P'}=3.14161$, $\phi_Y=6.28318$
 $E_Y=0.267729$, $\theta_{\ell'}=0.438064$, $\phi_{\ell'}=3.14157$
 $Q^2=0.437825$, $x_B=0.453402$, $t=-0.27166$, $W^2=1.40766$, $y=0.743567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4399$
 P $E= 1.0360$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4399$
 ℓ' $E= 0.2611$ $p_x= -0.1108$ $p_y= 0.0000$ $p_z= 0.2365$
 P' $E= 0.9471$ $p_x= -0.1052$ $p_y= -0.0000$ $p_z= -0.0782$
 q_Y $E= 0.2677$ $p_x= 0.2160$ $p_y= -0.0000$ $p_z= -0.1582$



electron: polarization cluster P3, local minimum 208; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6567289\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050026567$
 above global minimum = $2.6545006\text{e-}05$
 8D contour samples = 256

selected phase-space point:

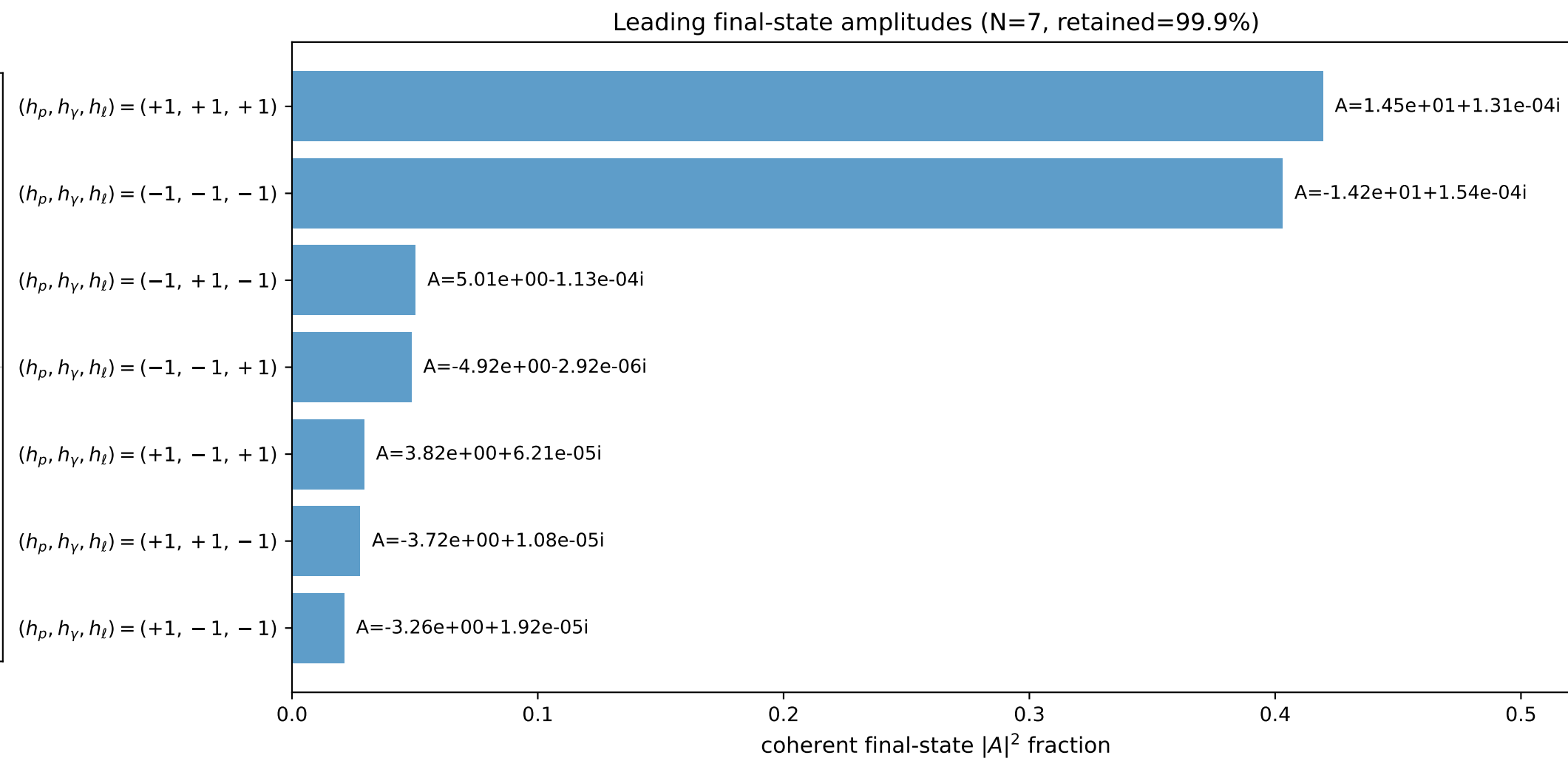
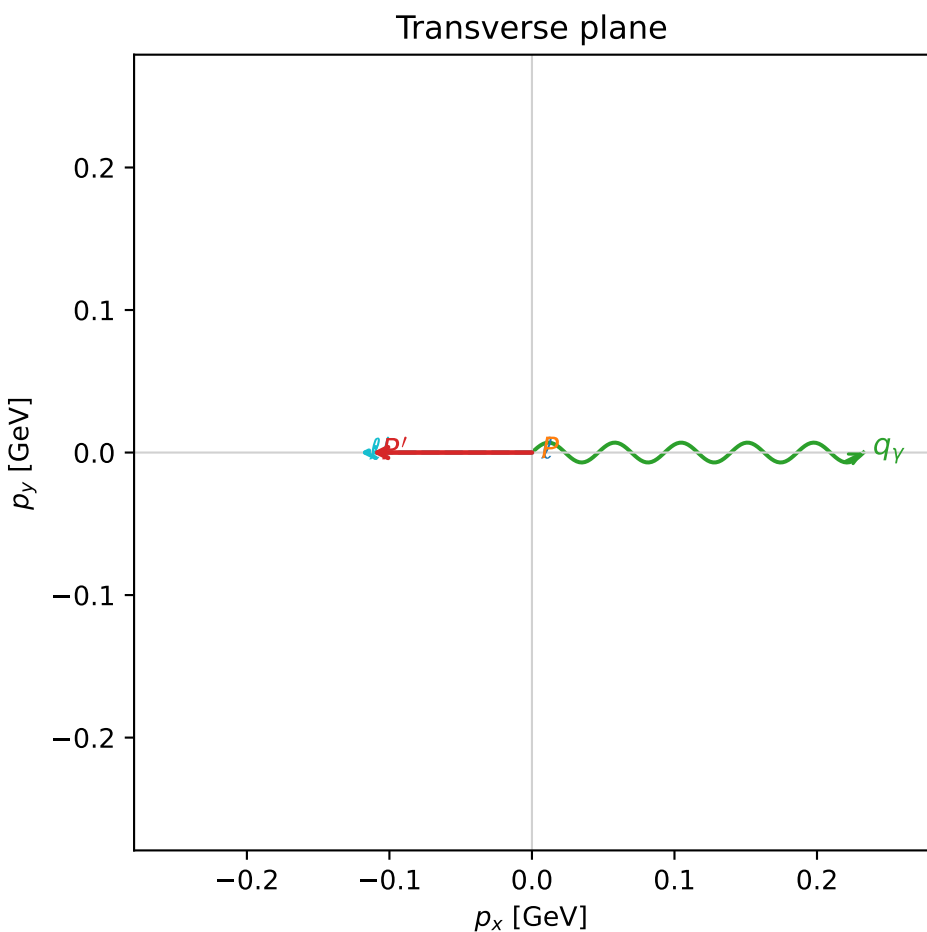
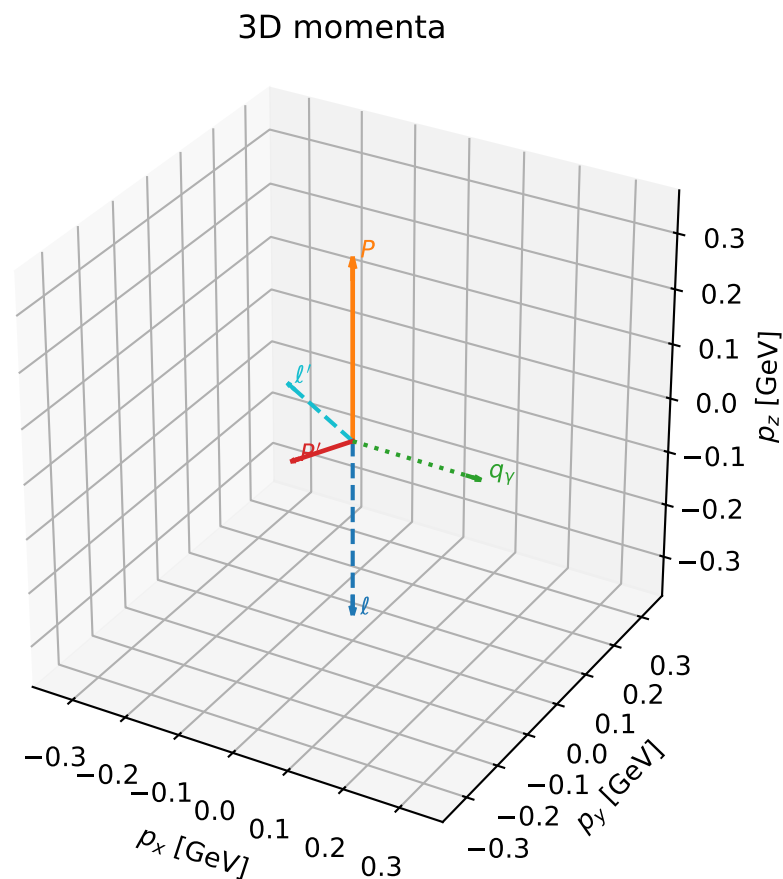
$\text{sqrt_s} = 1.475977$
 $\text{theta_p_out} = 2.210271$
 $\text{theta_gamma_out} = 2.203129$
 $\text{q0ut} = 0.2677292$
 $\text{phi_p_out} = 3.141608$
 $\text{phi_gamma_out} = 6.283181$
 $\text{alpha_e} = 1.869945$
 $\text{alpha_p} = 0.2408294$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

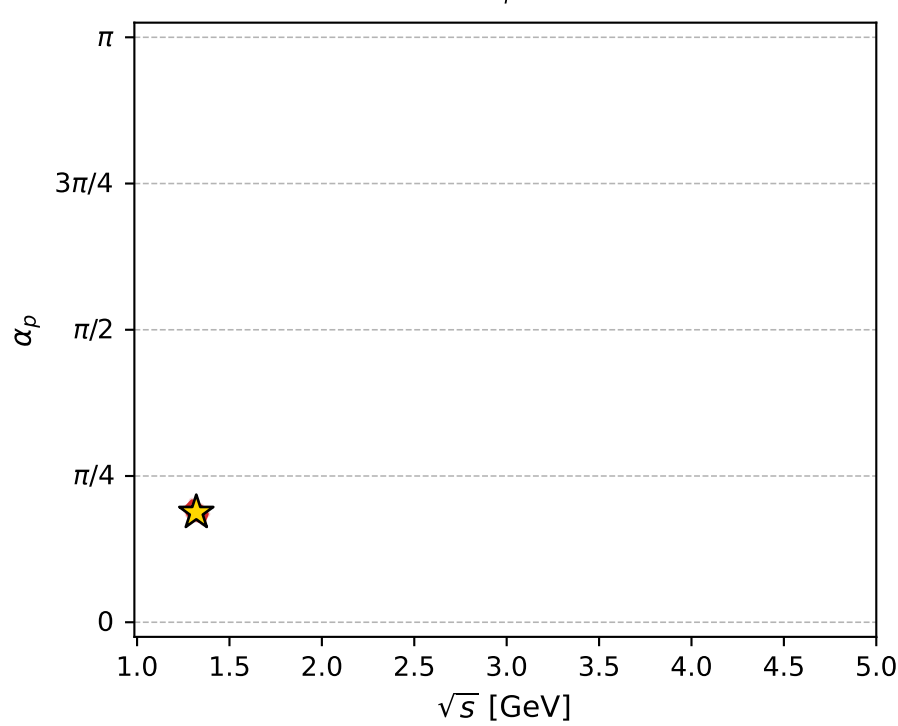
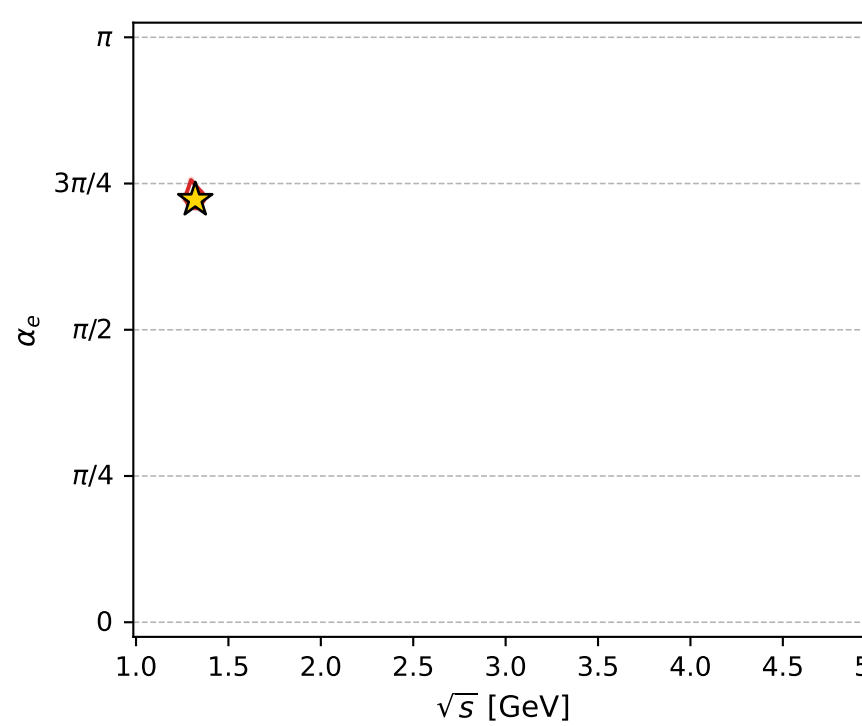
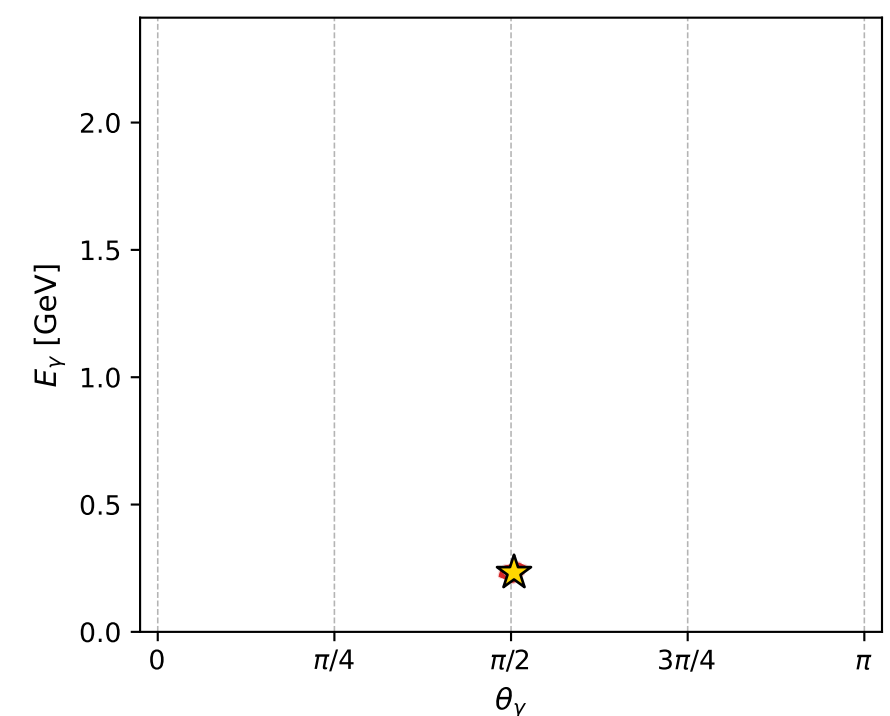
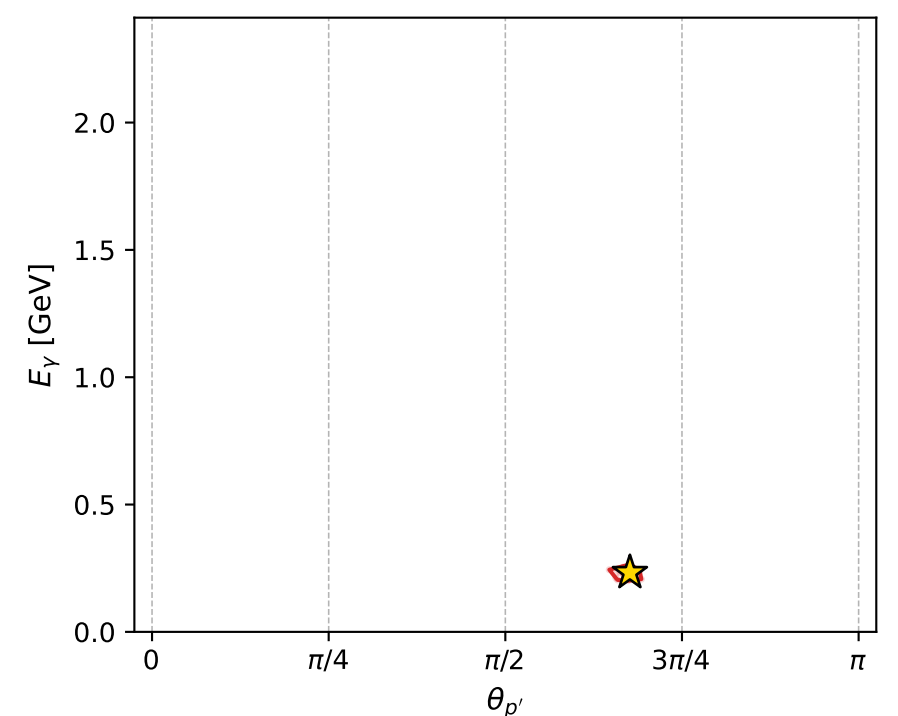
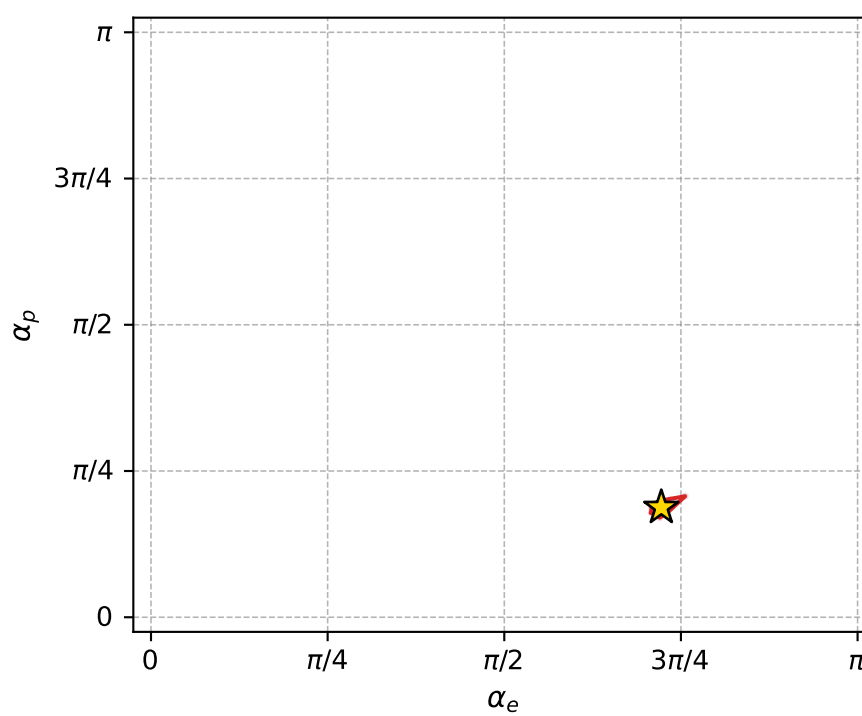
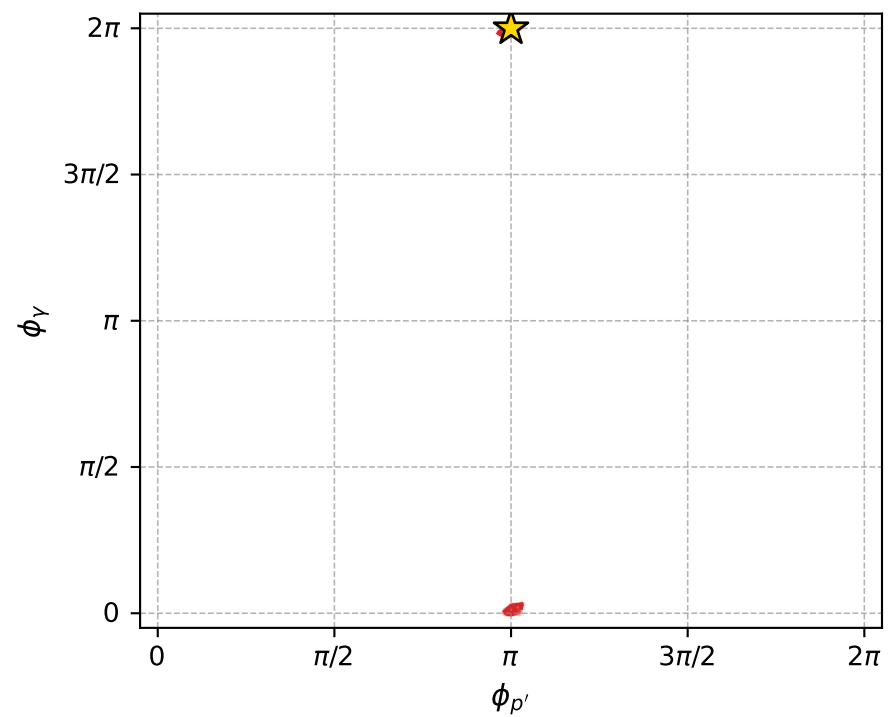
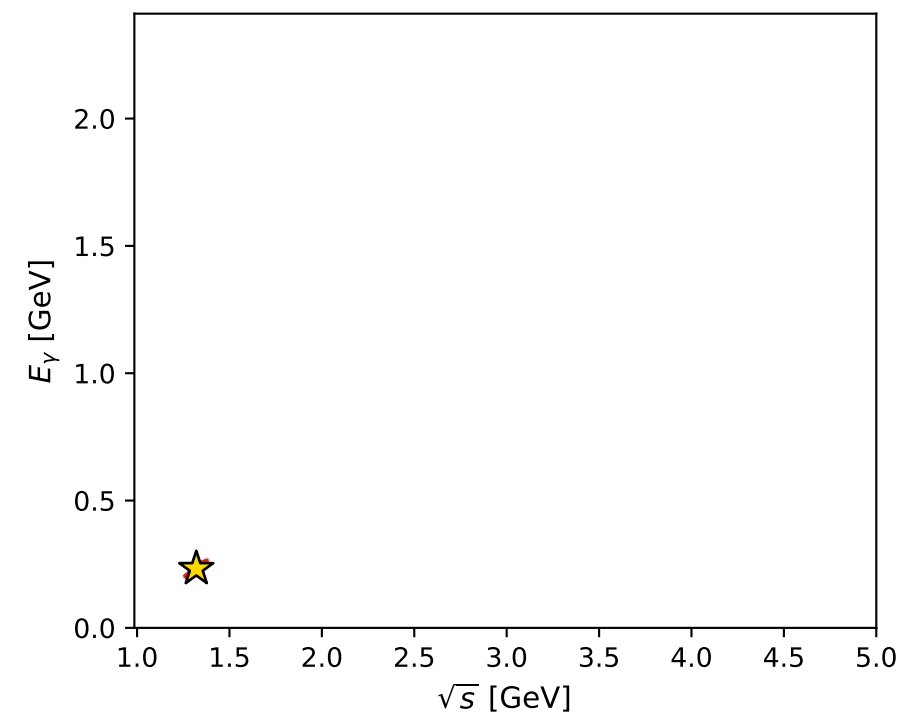
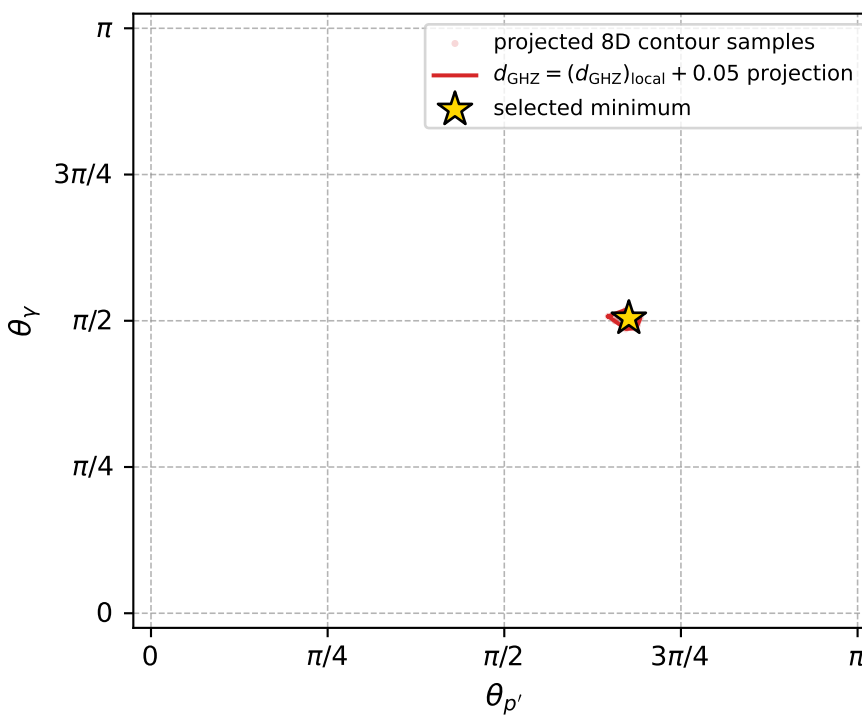
dghz_mixing_angles_polarization_03_local_minimum_209 $d_{\{\text{rm GHZ}\}}=2.80005\text{e-}05$
 region: polarization_cluster_03_local_minimum_209
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2689154$ rad
 incoming lepton: $-0.642778|+\rangle +0.766053|-\rangle$
 proton mixing angle: $\alpha_p=0.58944955$ rad
 incoming proton: $0.831247|+\rangle +0.555904|-\rangle$

$s=1.74362$, $\sqrt{s}=1.32046$, $|\vec{P}|=0.327073$, $|\vec{P}'|=0.131826$
 $\theta_{P'}=2.12469$, $\theta_Y=1.584$, $\phi_{P'}=3.14158$, $\phi_Y=6.28317$
 $E_Y=0.232648$, $\theta_{\ell'}=1.02974$, $\phi_{\ell'}=3.14158$
 $Q^2=0.139337$, $x_B=0.220535$, $t=-0.167582$, $W^2=1.37232$, $y=0.731455$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3271$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3271$
 P $E= 0.9934$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3271$
 ℓ' $E= 0.1406$ $p_x= -0.1205$ $p_y= 0.0000$ $p_z= 0.0724$
 P' $E= 0.9472$ $p_x= -0.1121$ $p_y= 0.0000$ $p_z= -0.0693$
 q_Y $E= 0.2326$ $p_x= 0.2326$ $p_y= -0.0000$ $p_z= -0.0031$



electron: polarization cluster P3, local minimum 209; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8000456\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050028$
 above global minimum = $2.7978173\text{e-}05$
 8D contour samples = 256

selected phase-space point:

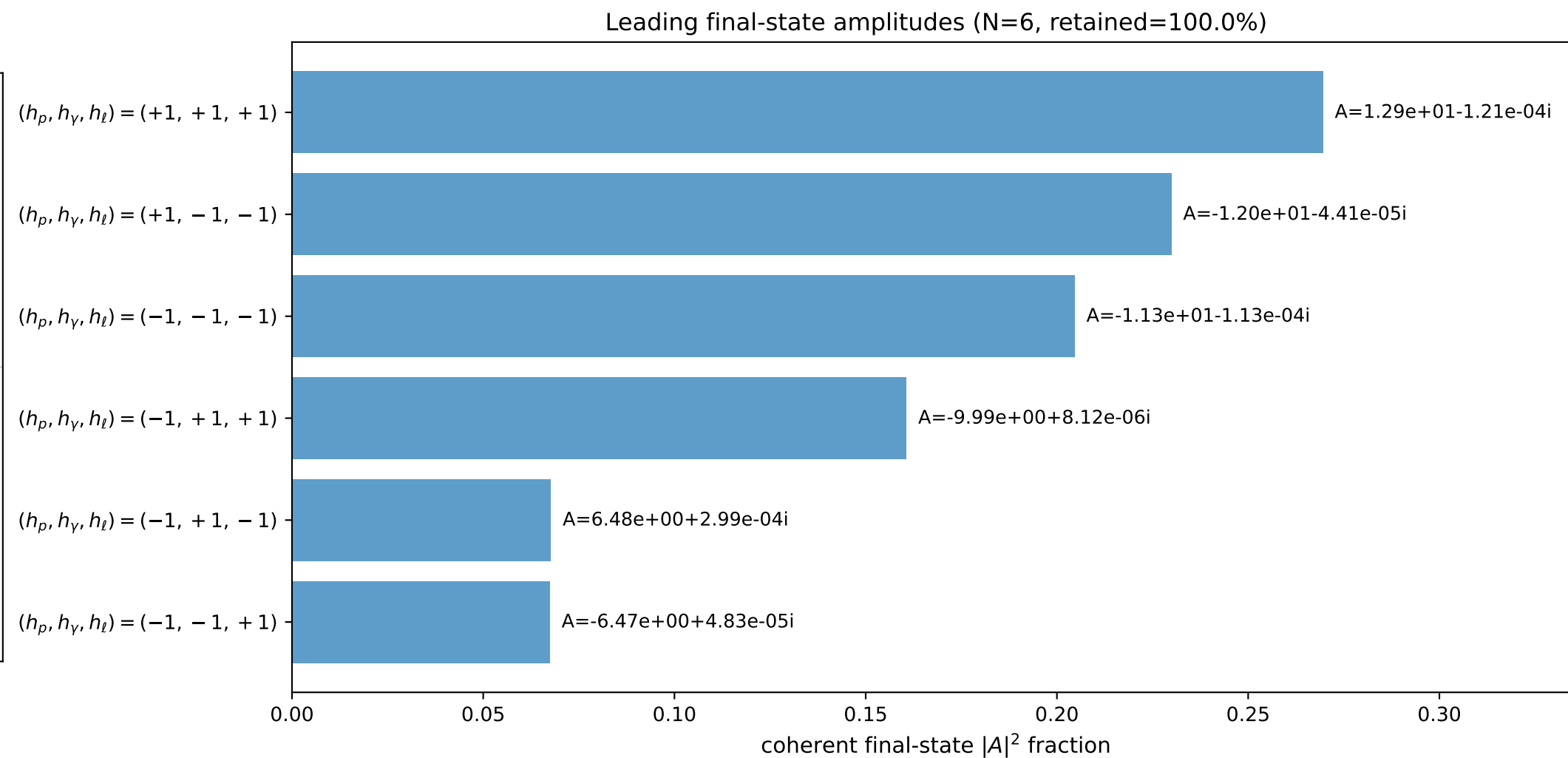
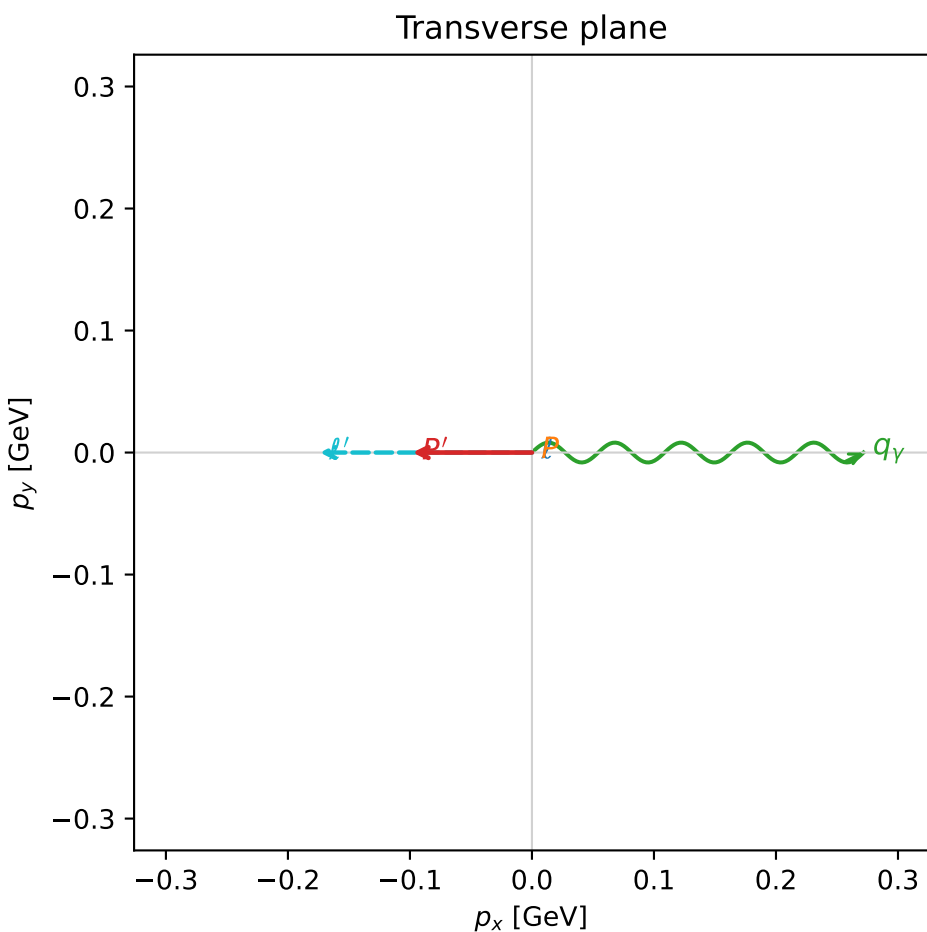
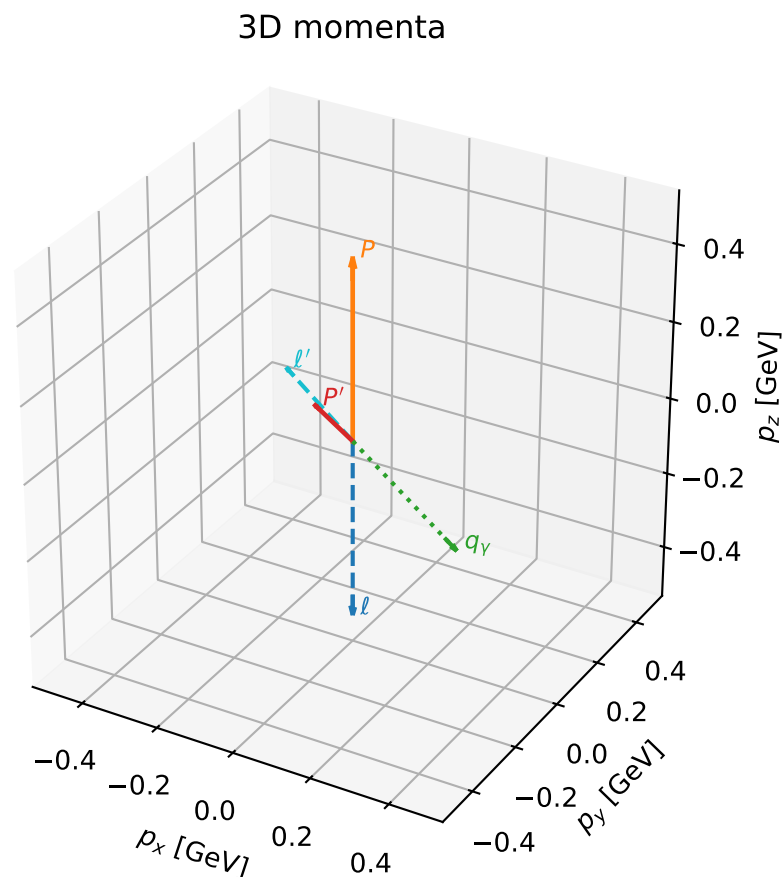
$\text{sqrt_s} = 1.320461$
 $\text{theta_p_out} = 2.124689$
 $\text{theta_gamma_out} = 1.583995$
 $\text{q0out} = 0.2326485$
 $\text{phi_p_out} = 3.141584$
 $\text{phi_gamma_out} = 6.283175$
 $\alpha_e = 2.268915$
 $\alpha_p = 0.5894495$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

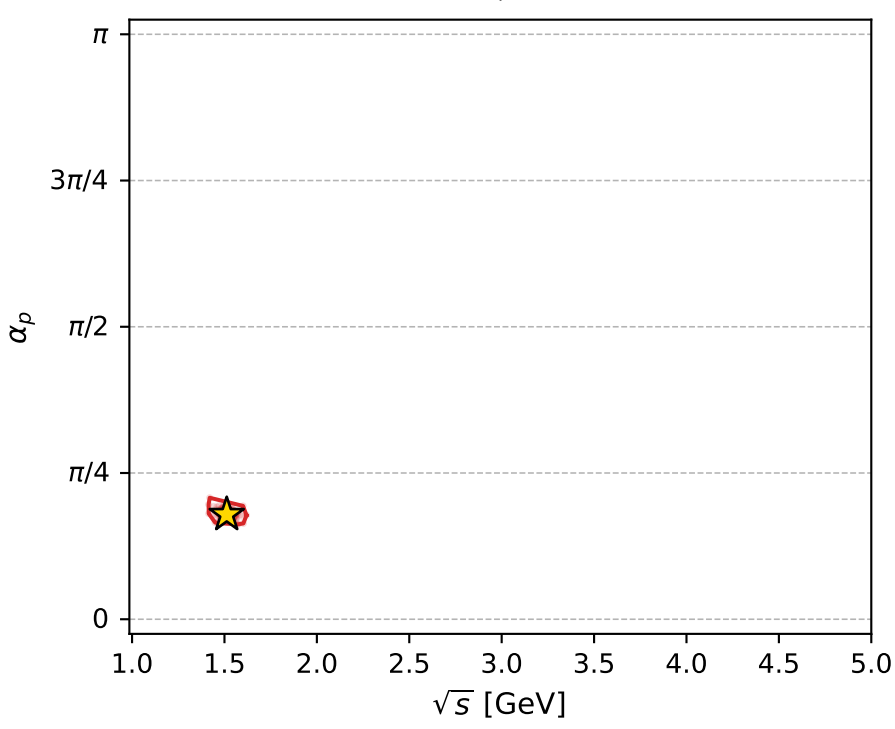
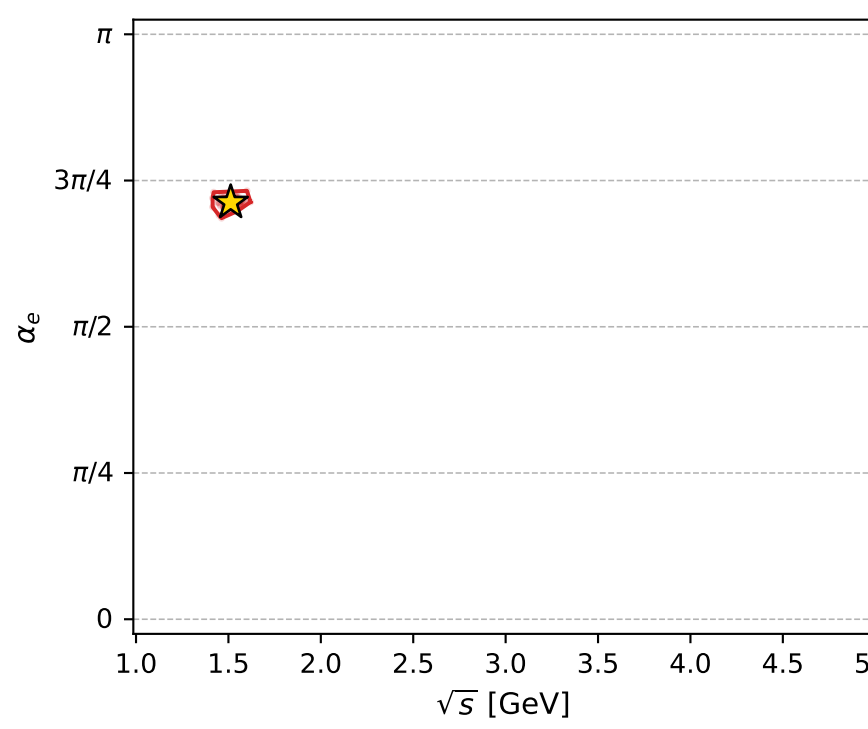
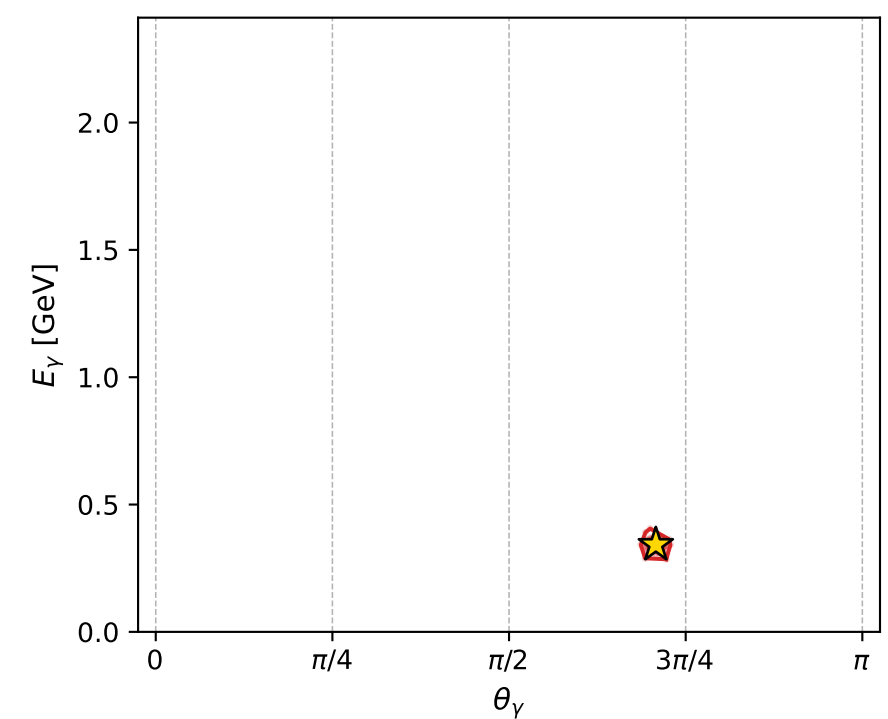
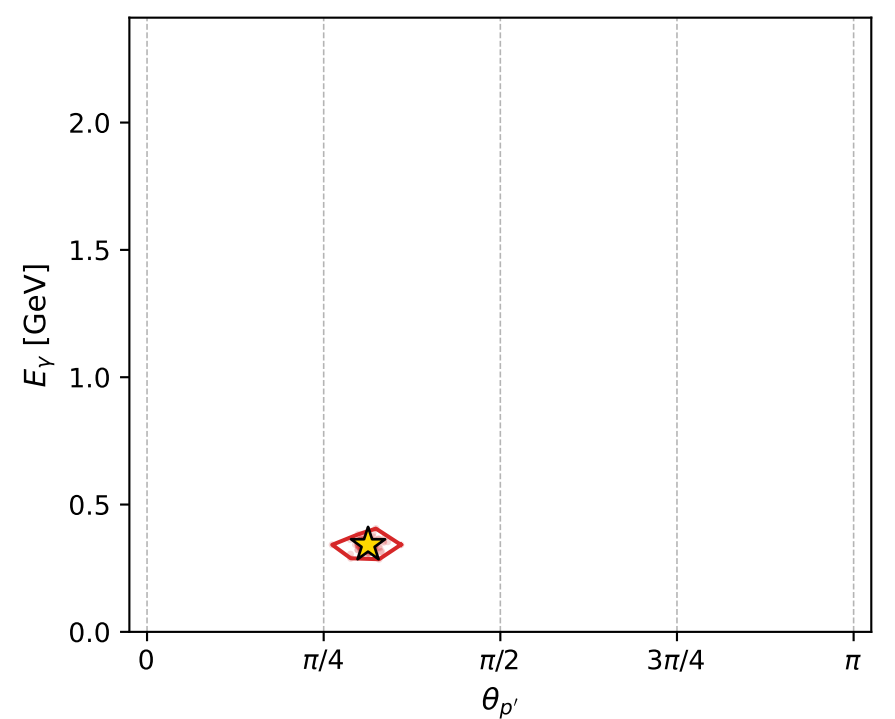
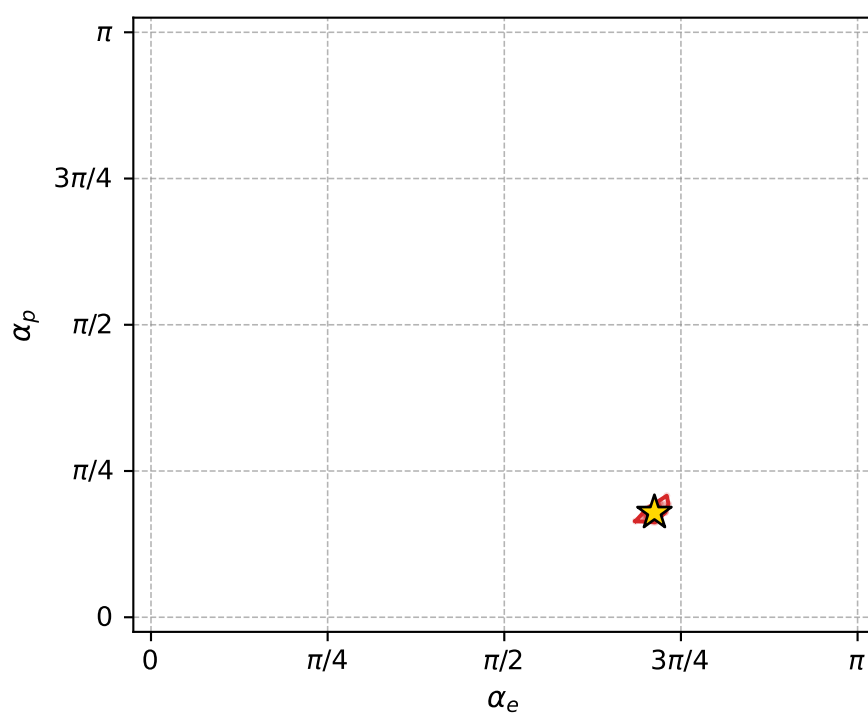
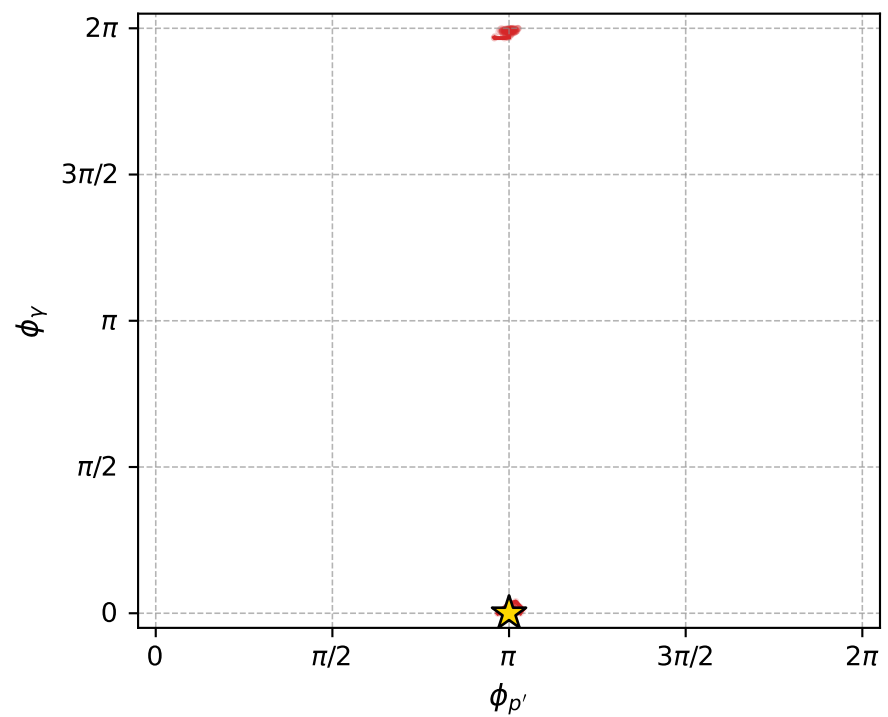
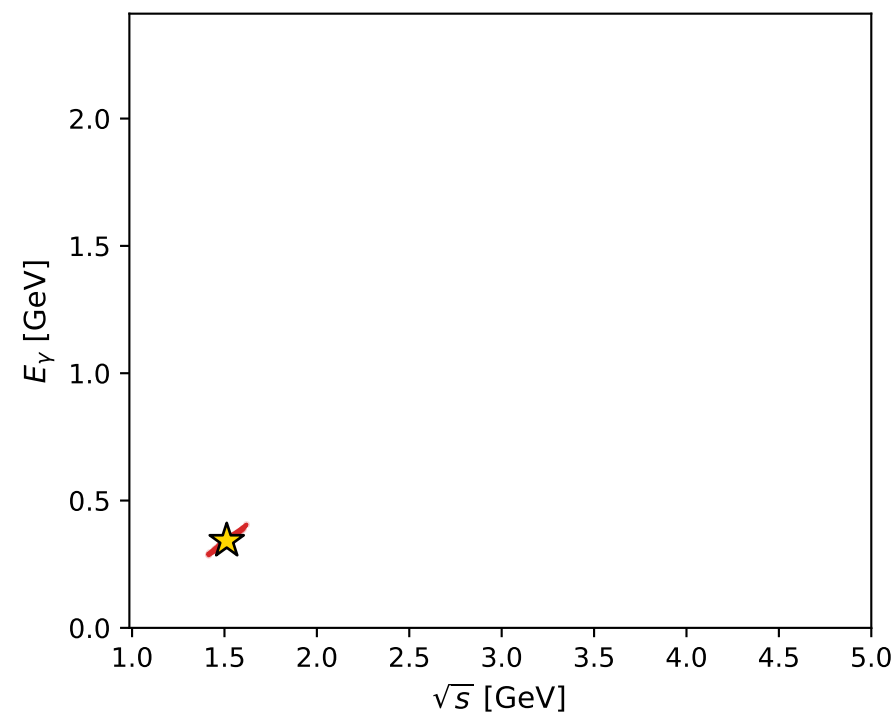
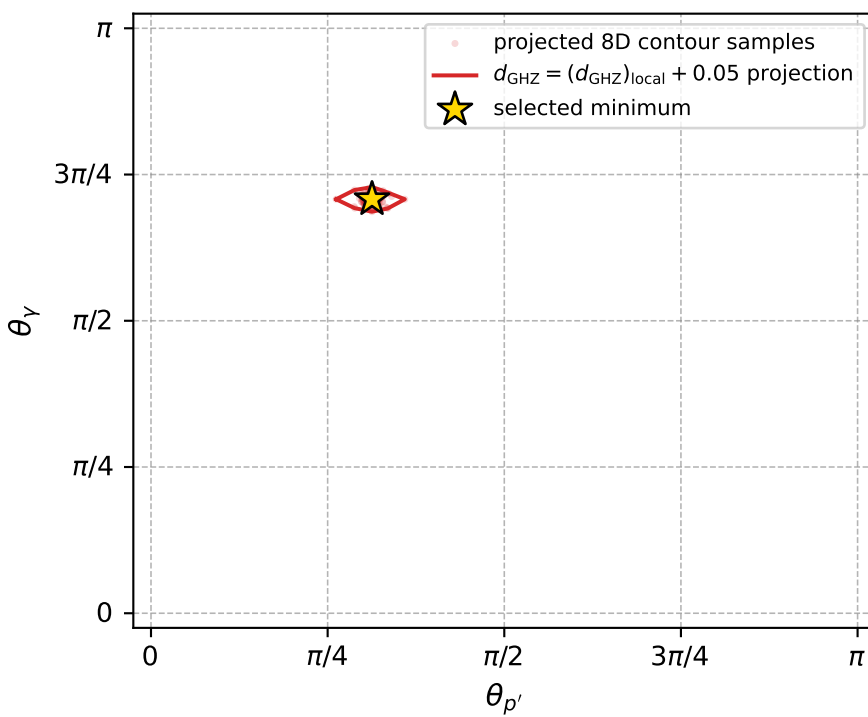
dghz_mixing_angles_polarization_03_local_minimum_211 d_{\rm GHZ}=2.89826e-05
 region: polarization_cluster_03_local_minimum_211
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2383433$ rad
 incoming lepton: $-0.619061|+\rangle +0.785343|-\rangle$
 proton mixing angle: $\alpha_p=0.56208704$ rad
 incoming proton: $0.846145|+\rangle +0.532953|-\rangle$

$s=2.28685$, $\sqrt{s}=1.51223$, $|\vec{P}|=0.465207$, $|\vec{P}'|=0.117589$
 $\theta_{P'}=0.982736$, $\theta_Y=2.22352$, $\phi_{P'}=3.1416$, $\phi_Y=2.52718e-05$
 $E_Y=0.342052$, $\theta_{\ell'}=0.884269$, $\phi_{\ell'}=3.14163$
 $Q^2=0.34179$, $x_B=0.319795$, $t=-0.159212$, $W^2=1.60683$, $y=0.759613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4652$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4652$
 P $E= 1.0470$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4652$
 ℓ' $E= 0.2248$ $p_x= -0.1739$ $p_y= -0.0000$ $p_z= 0.1425$
 P' $E= 0.9453$ $p_x= -0.0978$ $p_y= -0.0000$ $p_z= 0.0652$
 q_Y $E= 0.3421$ $p_x= 0.2717$ $p_y= 0.0000$ $p_z= -0.2077$



electron: polarization cluster P3, local minimum 211; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8982579\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050028983$
 above global minimum = $2.8960296\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.512232$
 $\text{theta_p_out} = 0.9827361$
 $\text{theta_gamma_out} = 2.223521$
 $\text{q0ut} = 0.3420522$
 $\text{phi_p_out} = 3.1416$
 $\text{phi_gamma_out} = 2.527184\text{e-}05$
 $\text{alpha_e} = 2.238343$
 $\text{alpha_p} = 0.562087$

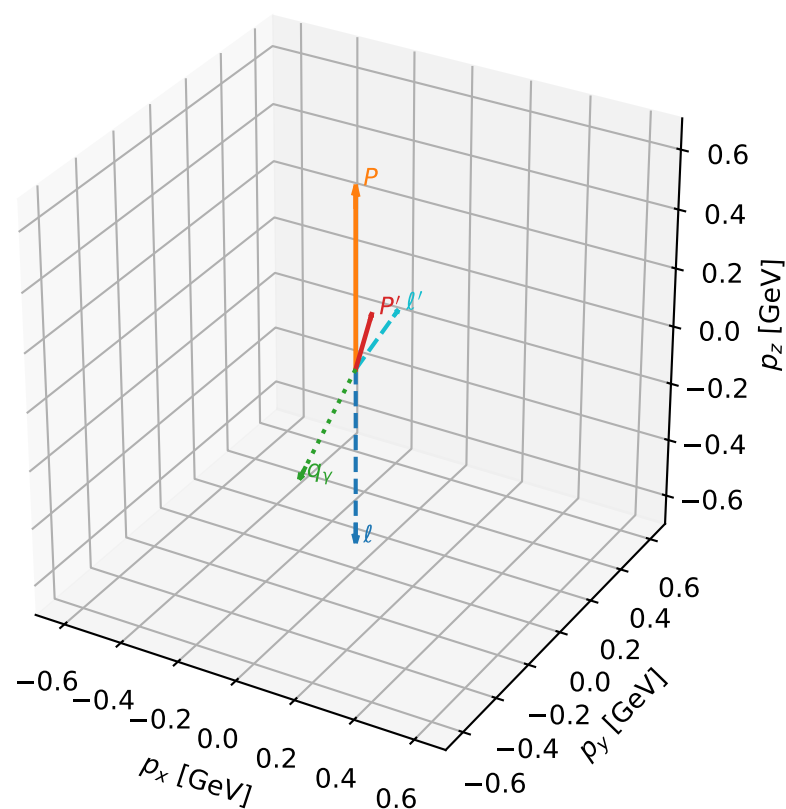
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_212 $d_{\{\text{rm GHZ}\}}=3.17475\text{e-}05$
 region: polarization_cluster_03_local_minimum_212
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.6825441$ rad
 incoming lepton: $-0.896475|+> +0.443095|->$
 proton mixing angle: $\alpha_p=1.1846706$ rad
 incoming proton: $0.376602|+> +0.926375|->$

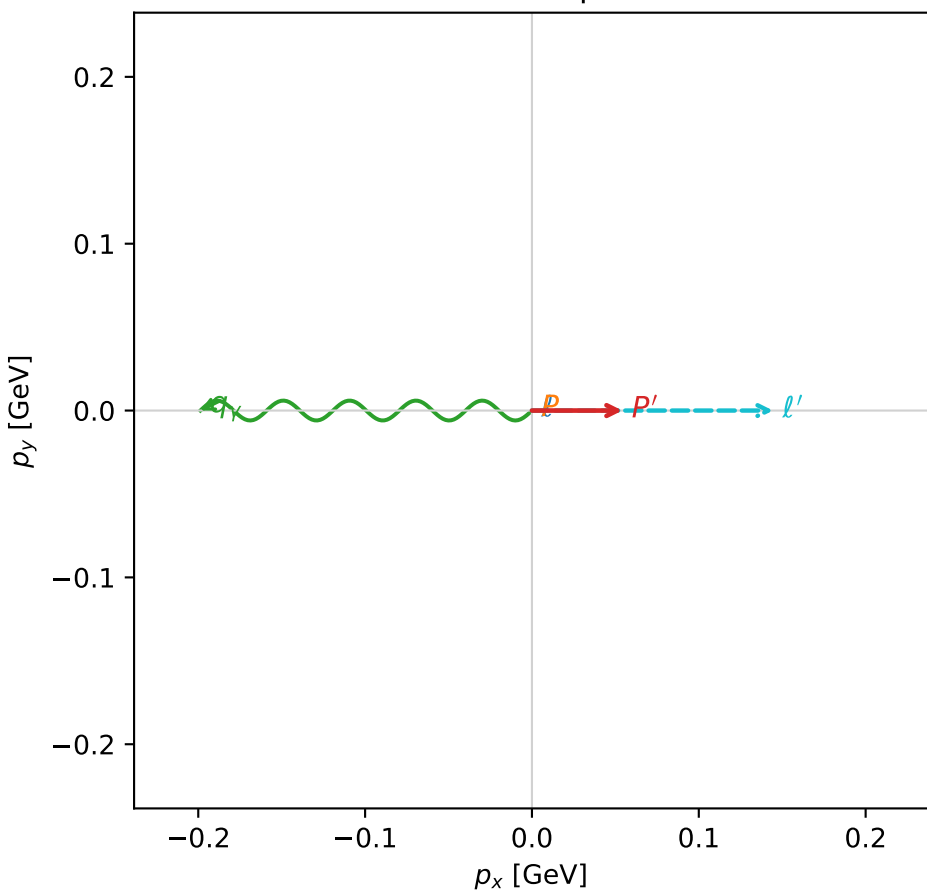
$s=2.99564$, $\sqrt{s}=1.73079$, $|\vec{P}|=0.611222$, $|\vec{P}'|=0.209061$
 $\theta_{p'}=0.262205$, $\theta_Y=2.72142$, $\phi_{p'}=6.2831$, $\phi_Y=3.14155$
 $E_Y=0.487156$, $\theta_{\ell'}=0.536813$, $\phi_{\ell'}=6.28315$
 $Q^2=0.642377$, $x_B=0.360914$, $t=-0.145329$, $W^2=2.01733$, $y=0.841226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6112$
 P $E= 1.1196$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6112$
 ℓ' $E= 0.2826$ $p_x= 0.1445$ $p_y= -0.0000$ $p_z= 0.2429$
 P' $E= 0.9610$ $p_x= 0.0542$ $p_y= -0.0000$ $p_z= 0.2019$
 q_Y $E= 0.4872$ $p_x= -0.1987$ $p_y= 0.0000$ $p_z= -0.4448$

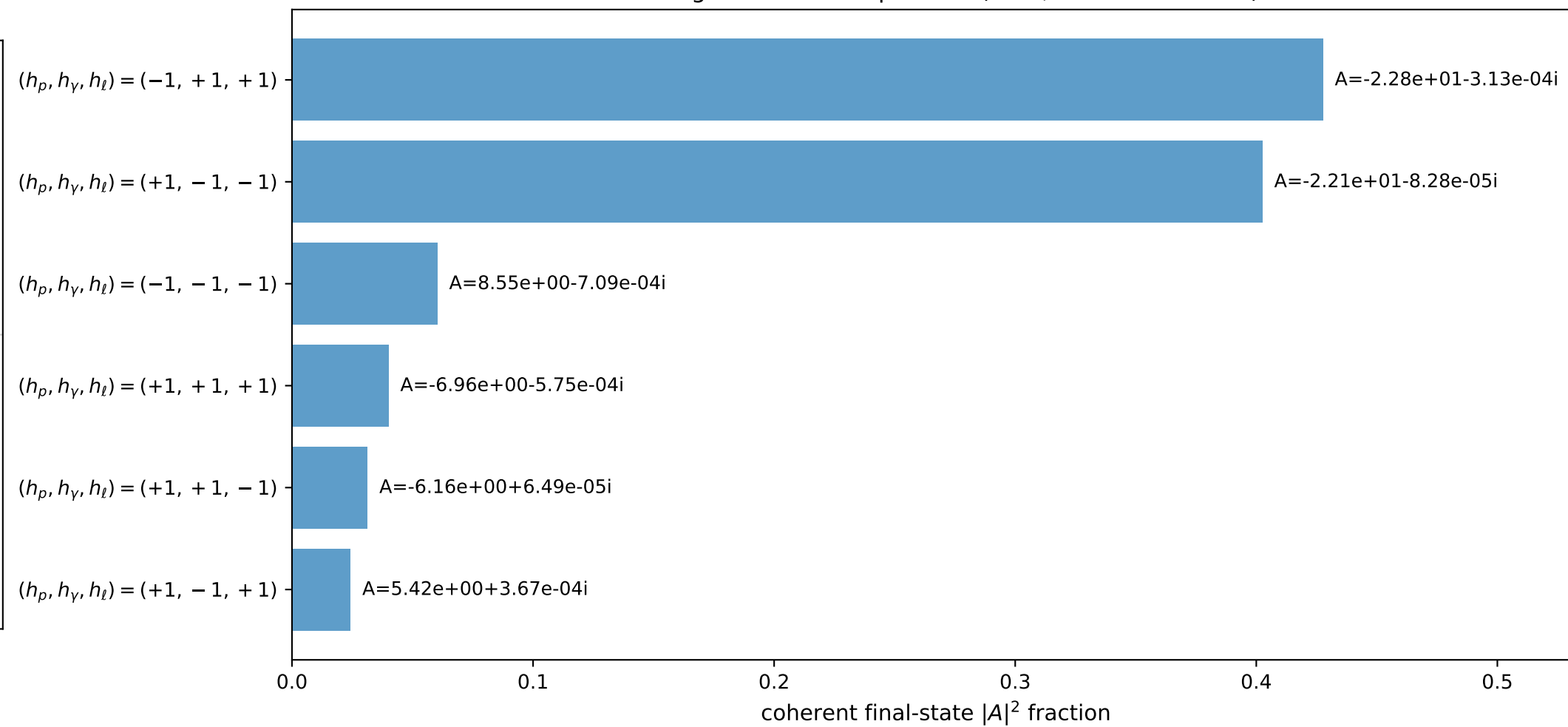
3D momenta



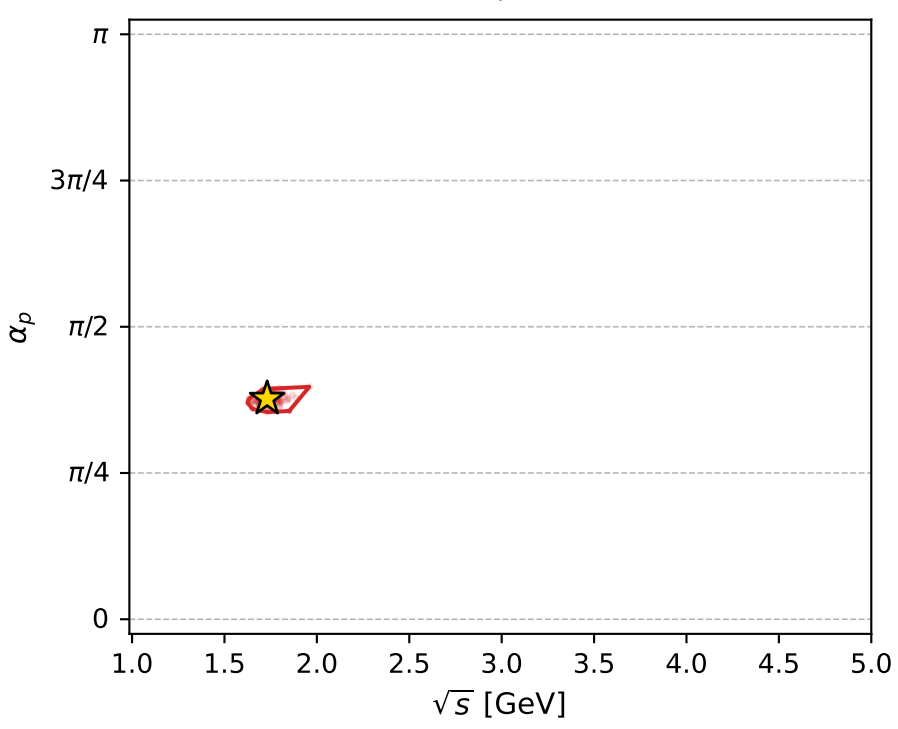
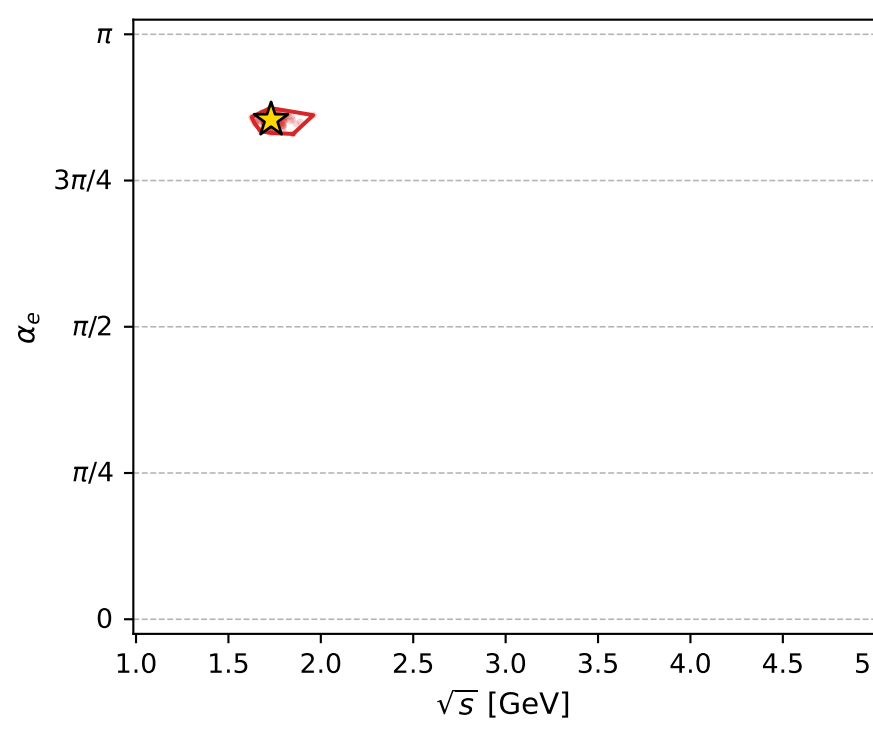
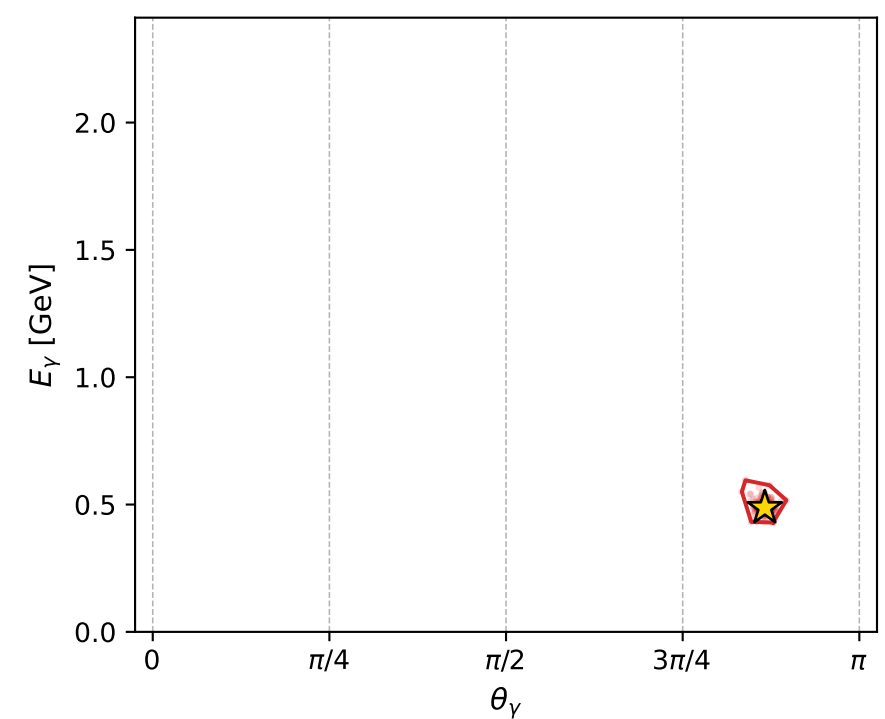
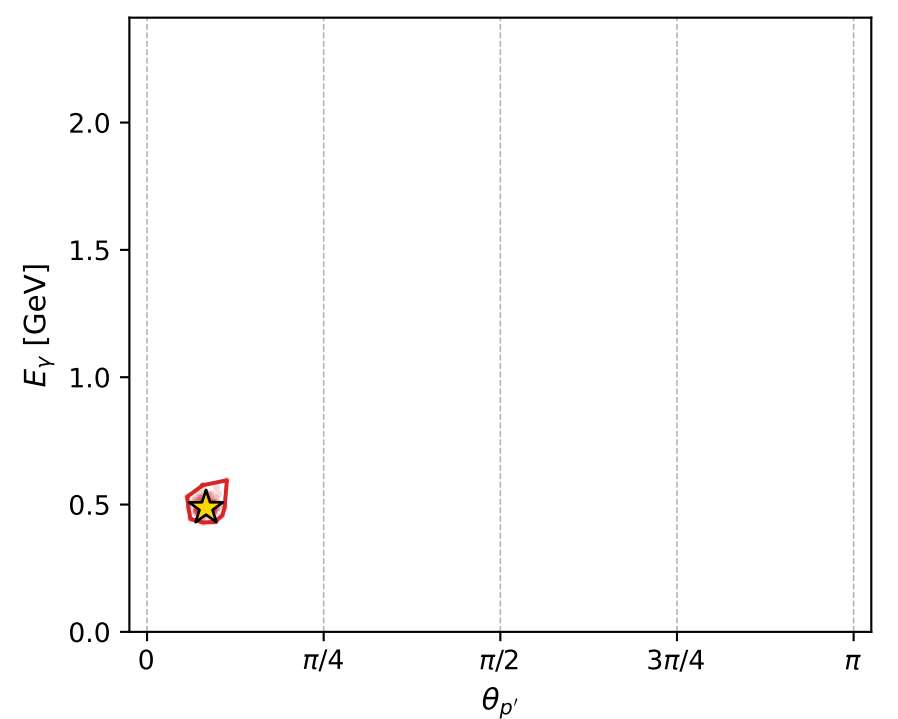
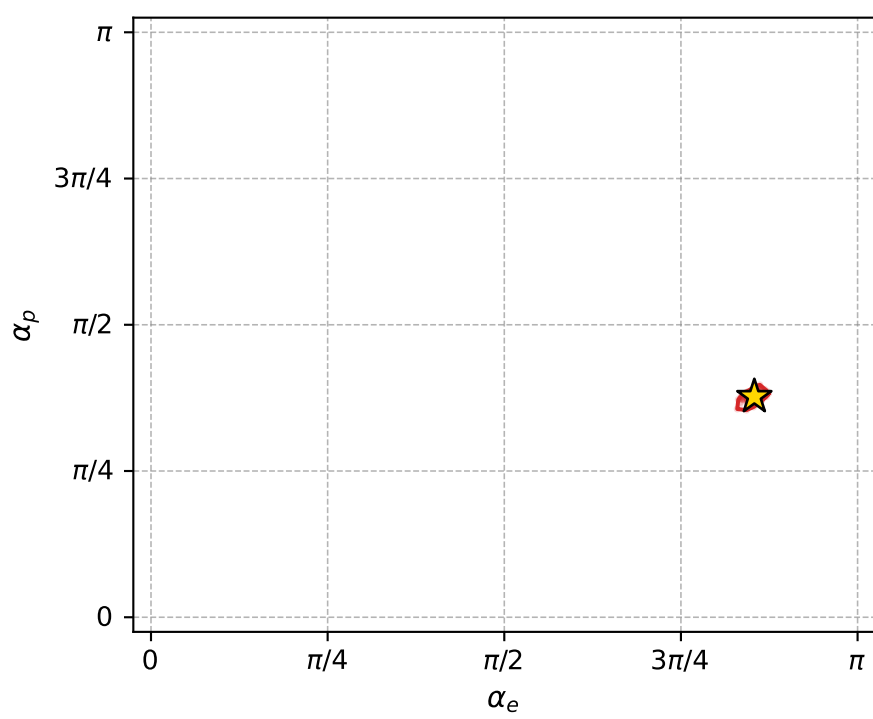
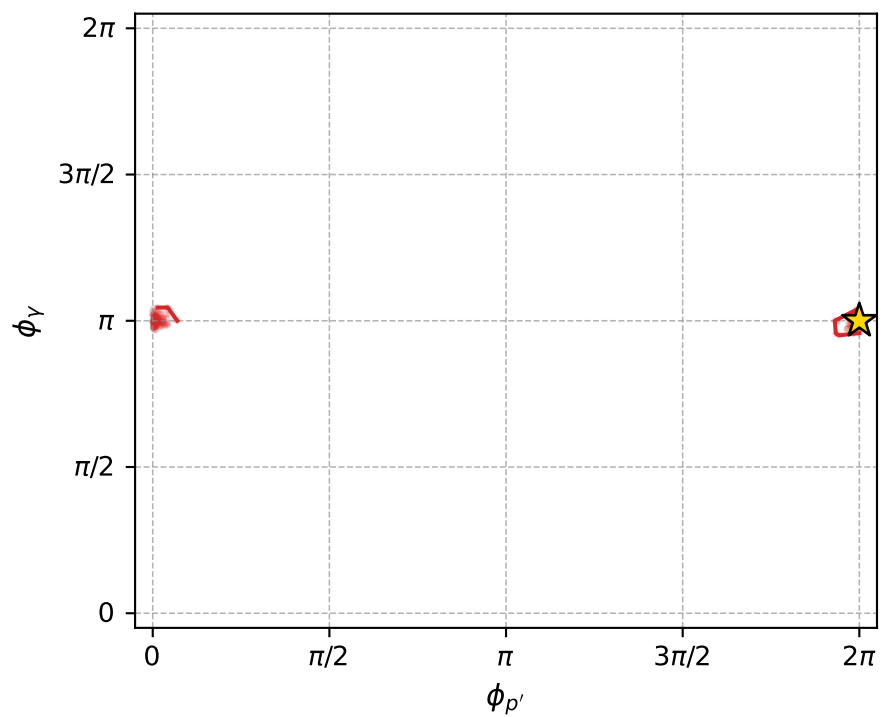
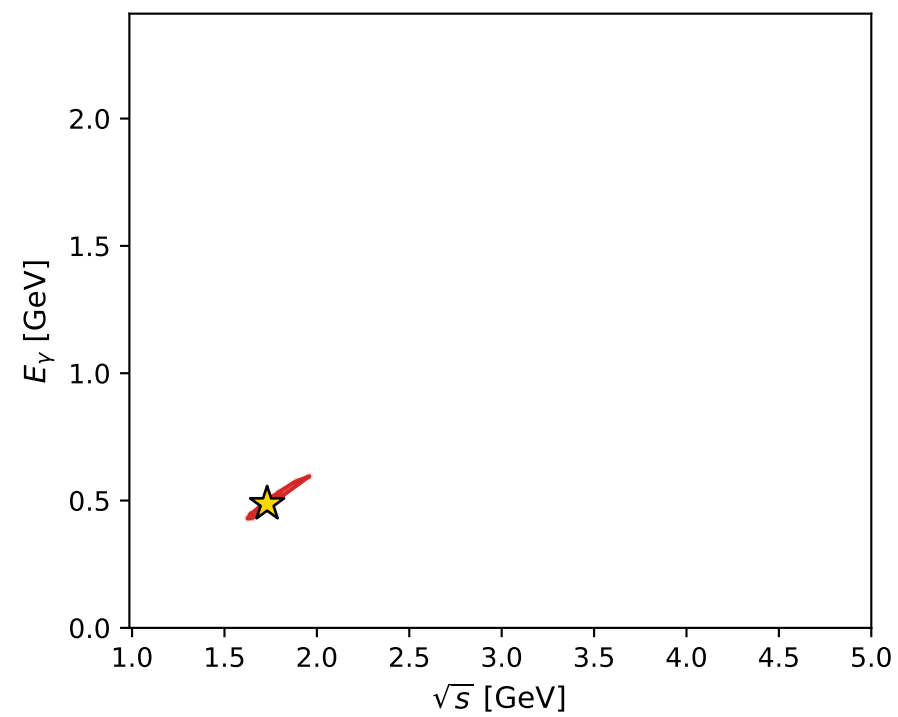
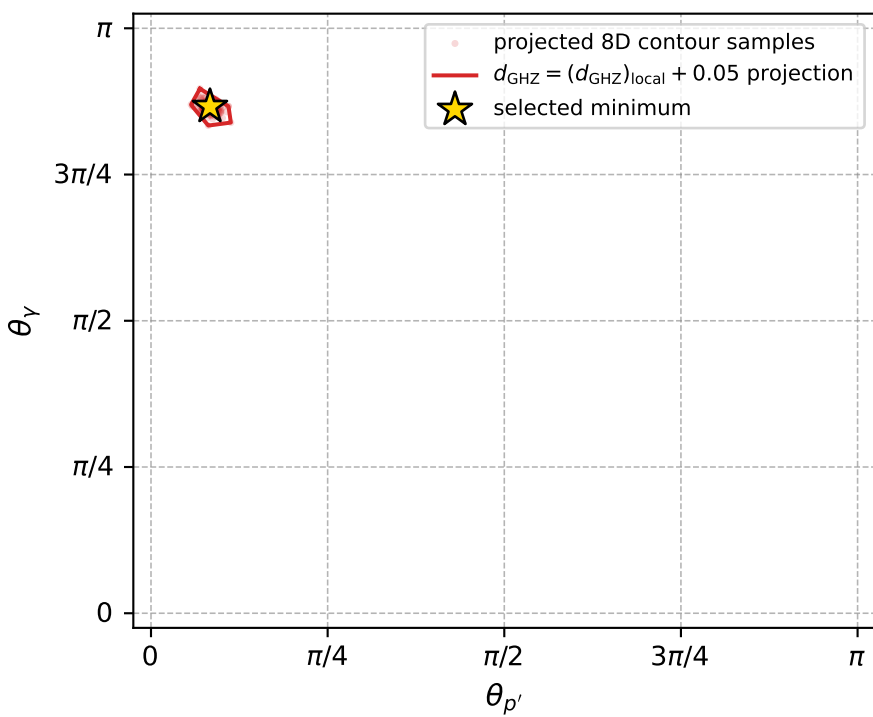
Transverse plane



Leading final-state amplitudes (N=6, retained=98.6%)



electron: polarization cluster P3, local minimum 212; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.1747488\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050031747$
 above global minimum = $3.1725204\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.730791$
 $\text{theta_p_out} = 0.2622053$
 $\text{theta_gamma_out} = 2.721415$
 $\text{q0out} = 0.4871565$
 $\text{phi_p_out} = 6.283102$
 $\text{phi_gamma_out} = 3.141546$
 $\text{alpha_e} = 2.682544$
 $\text{alpha_p} = 1.184671$

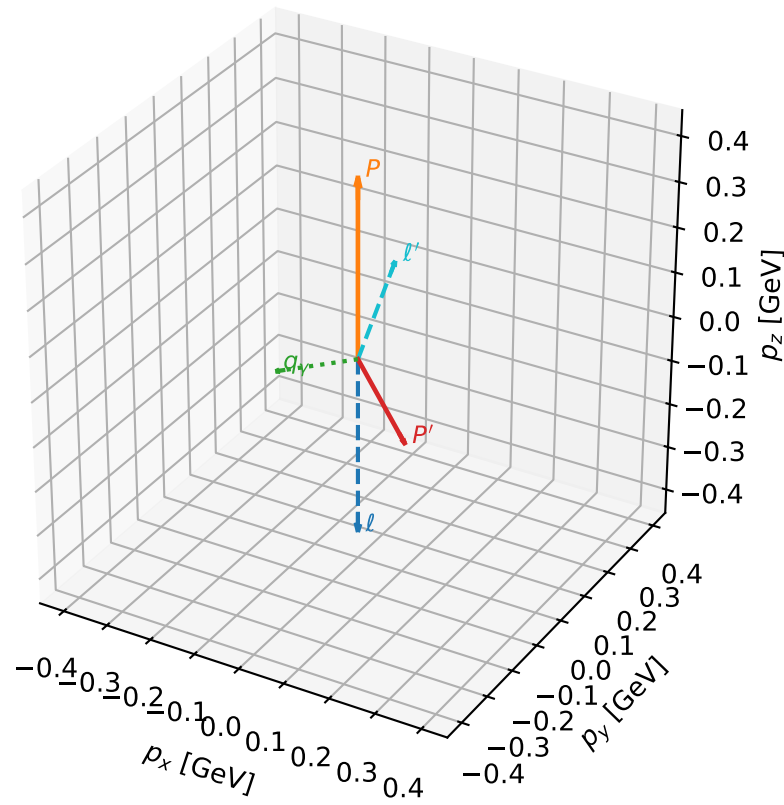
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_214 d_{\rm GHZ}=3.67622e-05
region: polarization_cluster_03_local_minimum_214
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.9233471$ rad
incoming lepton: $-0.976279|+\rangle +0.216517|-\rangle$
proton mixing angle: $\alpha_p=1.3984779$ rad
incoming proton: $0.171467|+\rangle +0.98519|-\rangle$

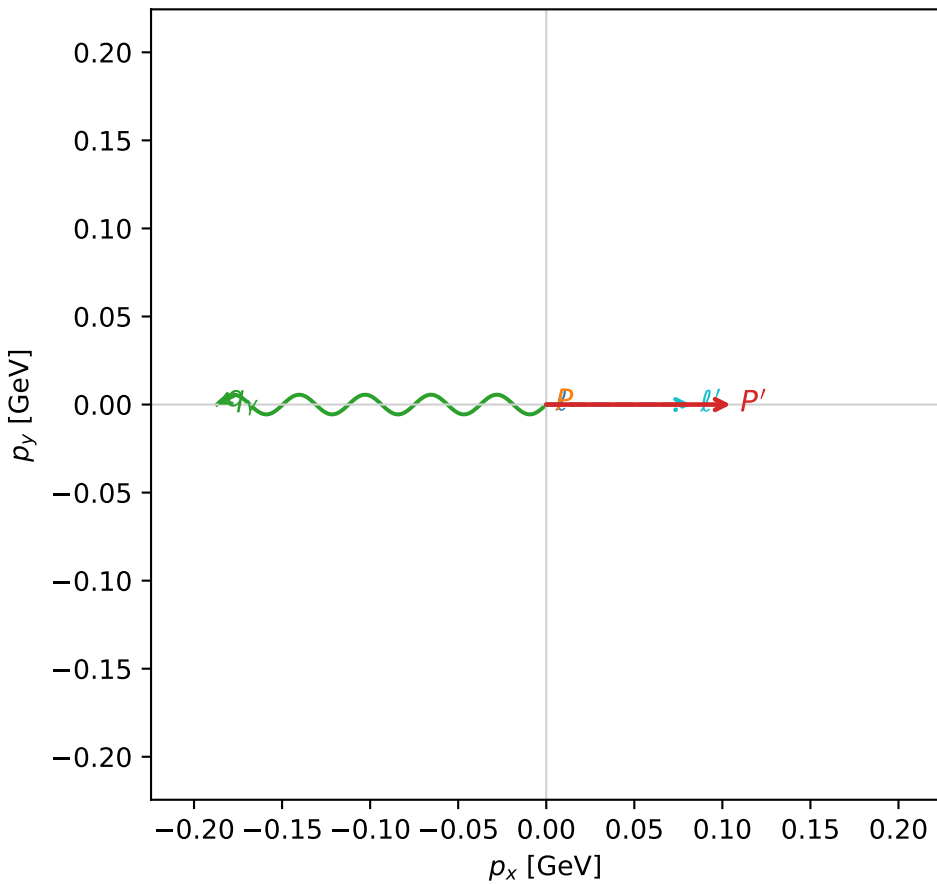
$s=2.00053$, $\sqrt{s}=1.4144$, $|\vec{P}|=0.39617$, $|\vec{P}'|=0.18985$
 $\theta_{P'}=2.55659$, $\theta_V=1.98153$, $\phi_{P'}=2.90377\text{e-}05$, $\phi_V=3.14161$
 $E_V=0.203969$, $\theta_{\ell'}=0.330219$, $\phi_{\ell'}=3.18729\text{e-}06$
 $Q^2=0.390729$, $x_B=0.491753$, $t=-0.314659$, $W^2=1.28368$, $y=0.708996$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3962$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3962$
 P $E= 1.0182$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3962$
 ℓ' $E= 0.2534$ $p_x= 0.0822$ $p_y= 0.0000$ $p_z= 0.2397$
 P' $E= 0.9570$ $p_x= 0.1048$ $p_y= 0.0000$ $p_z= -0.1583$
 q_Y $E= 0.2040$ $p_x= -0.1870$ $p_y= -0.0000$ $p_z= -0.0814$

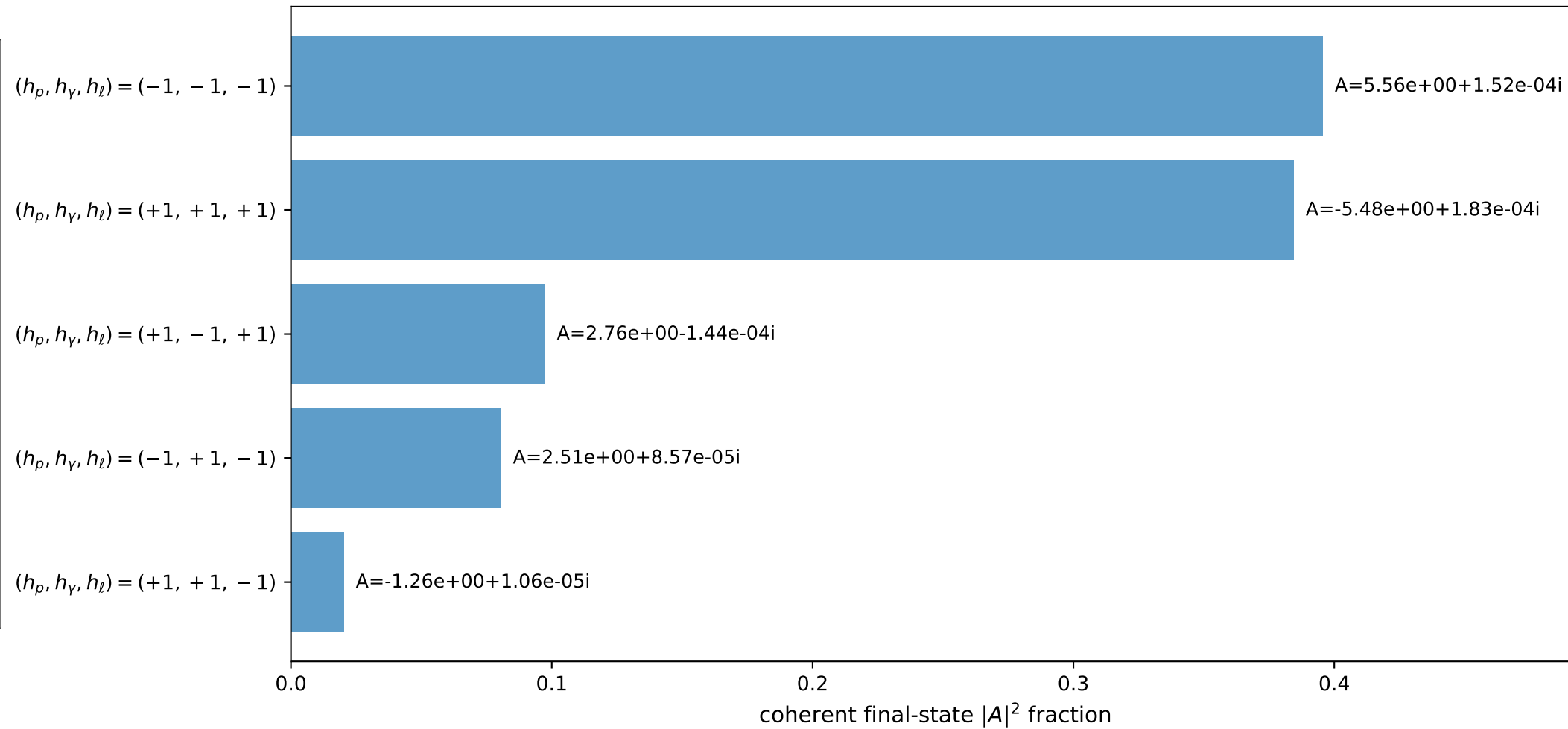
3D momenta



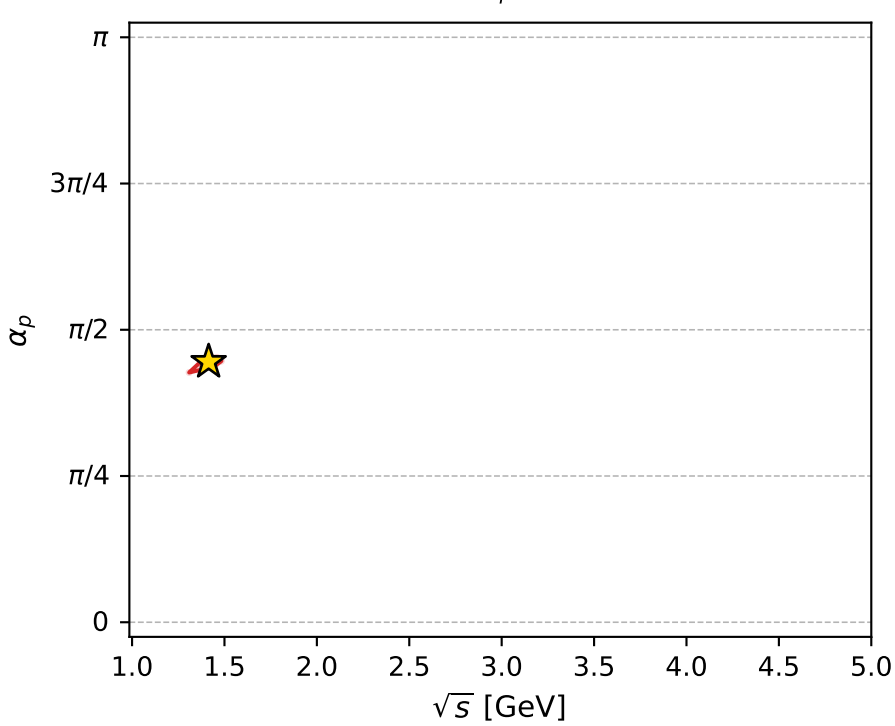
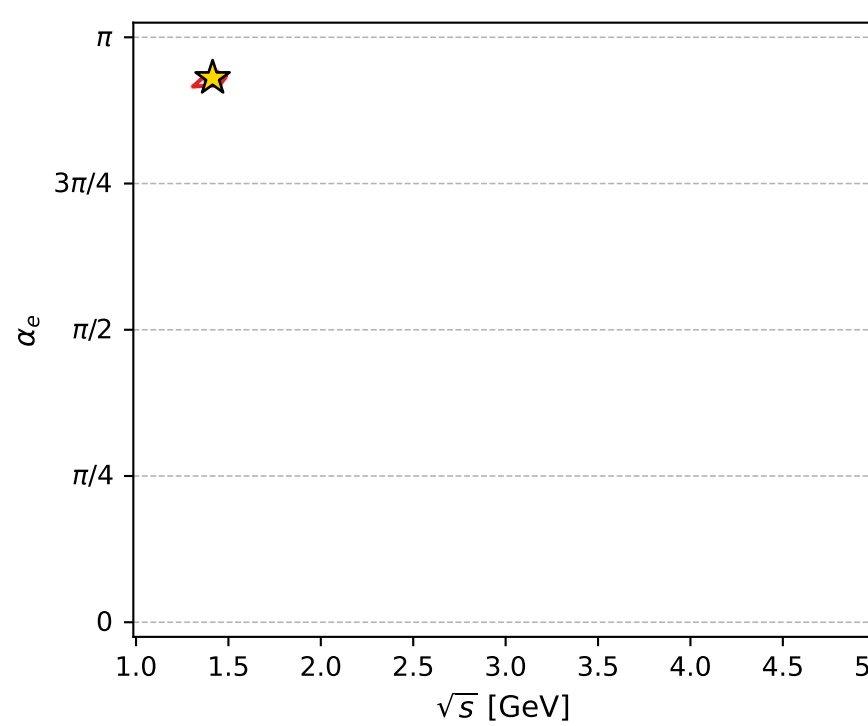
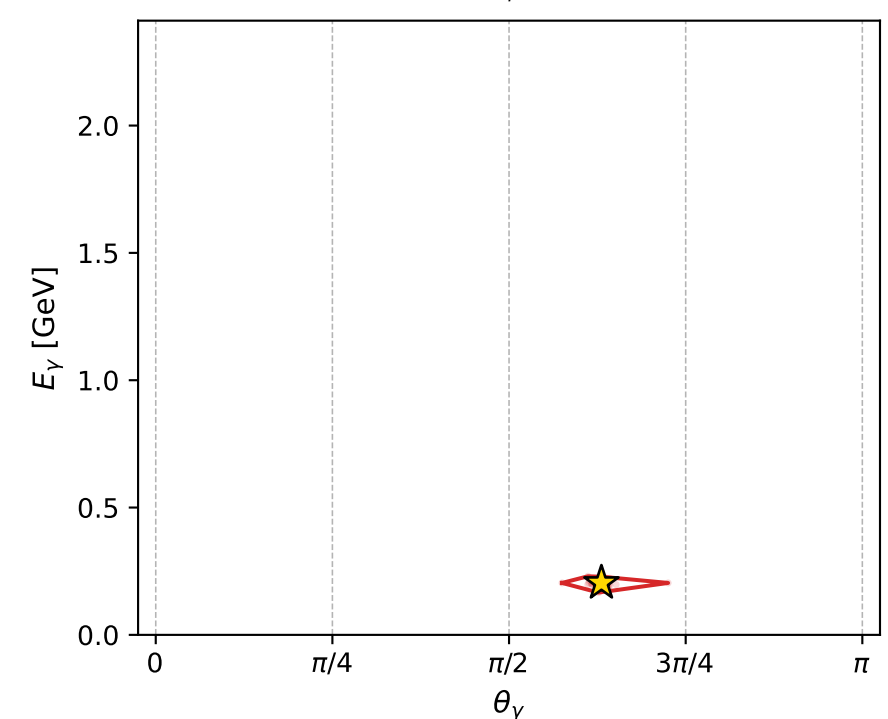
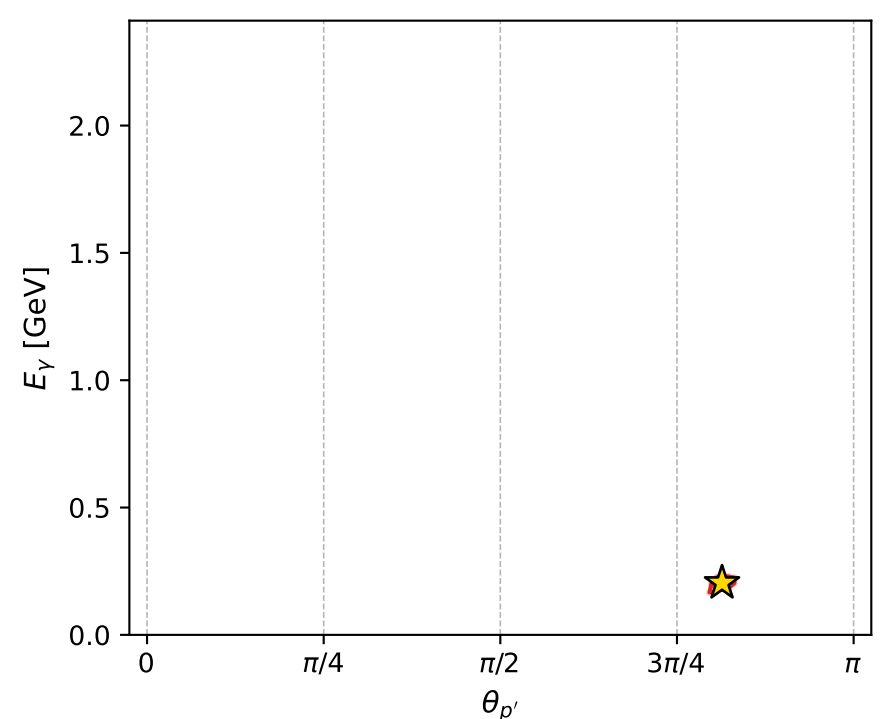
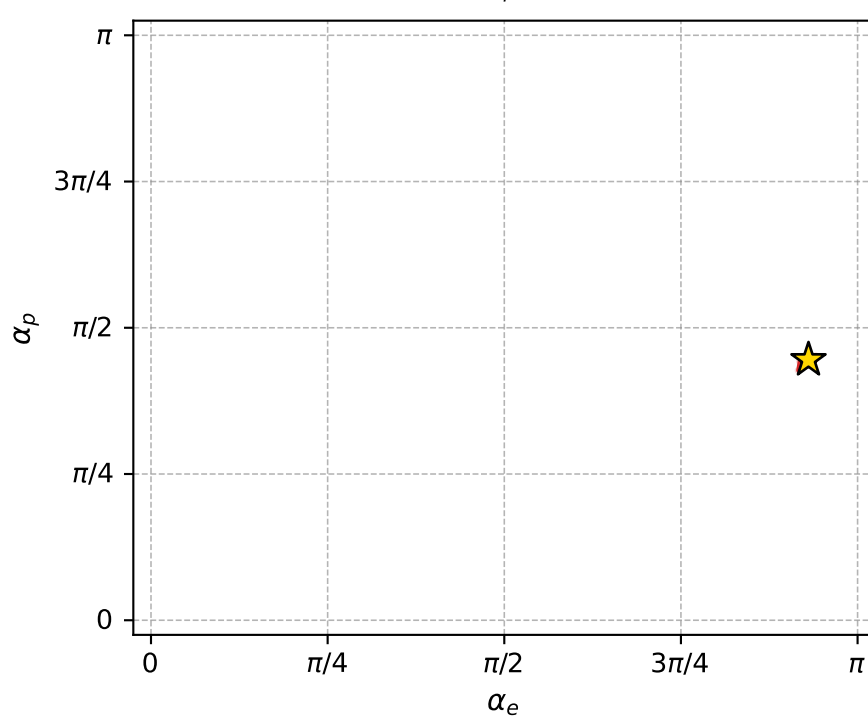
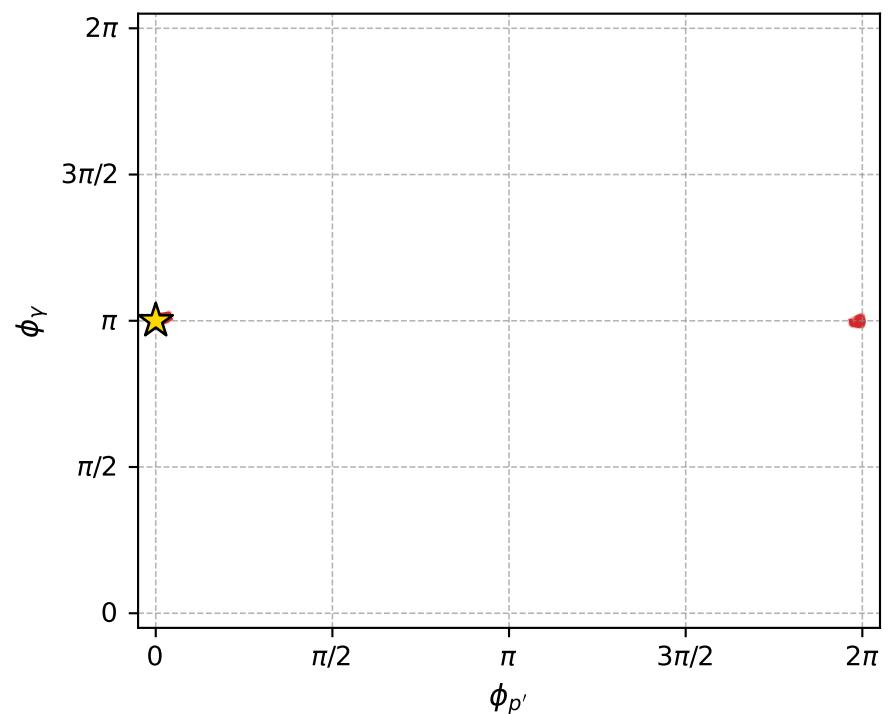
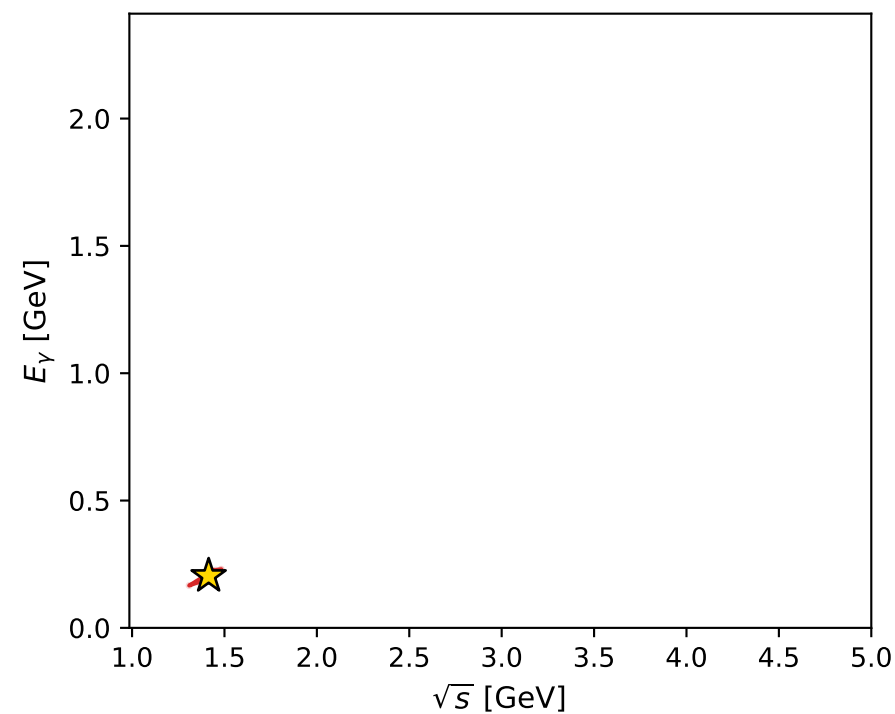
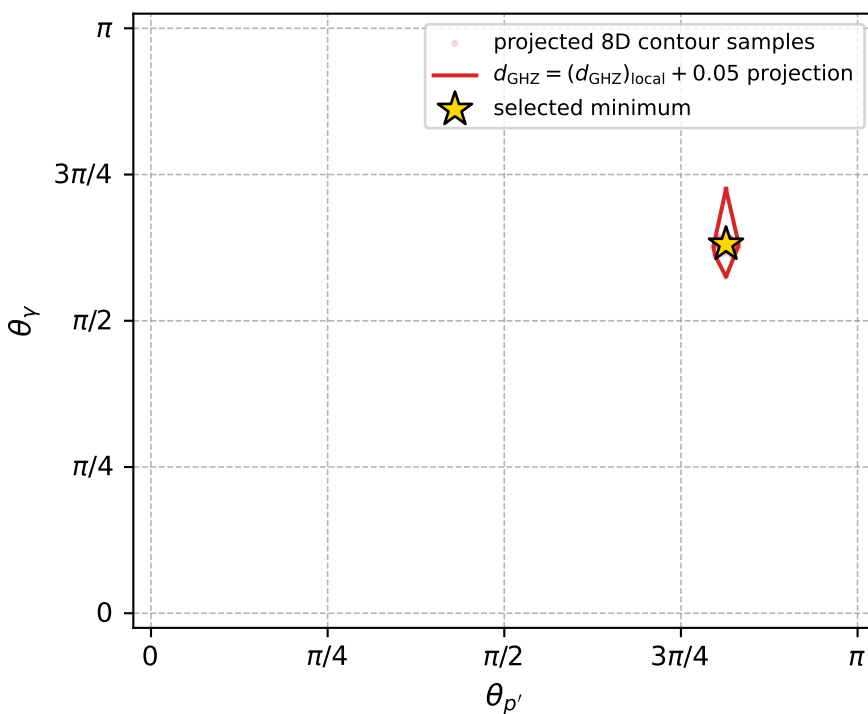
Transverse plane



Leading final-state amplitudes (N=5, retained=97.8%)



electron: polarization cluster P3, local minimum 214; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.6762157\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050036762$
 above global minimum = $3.6739873\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.414402$
 $\text{theta_p_out} = 2.556586$
 $\text{theta_gamma_out} = 1.981526$
 $\text{q0out} = 0.2039694$
 $\text{phi_p_out} = 2.903772\text{e-}05$
 $\text{phi_gamma_out} = 3.14161$
 $\text{alpha_e} = 2.923347$
 $\text{alpha_p} = 1.398478$

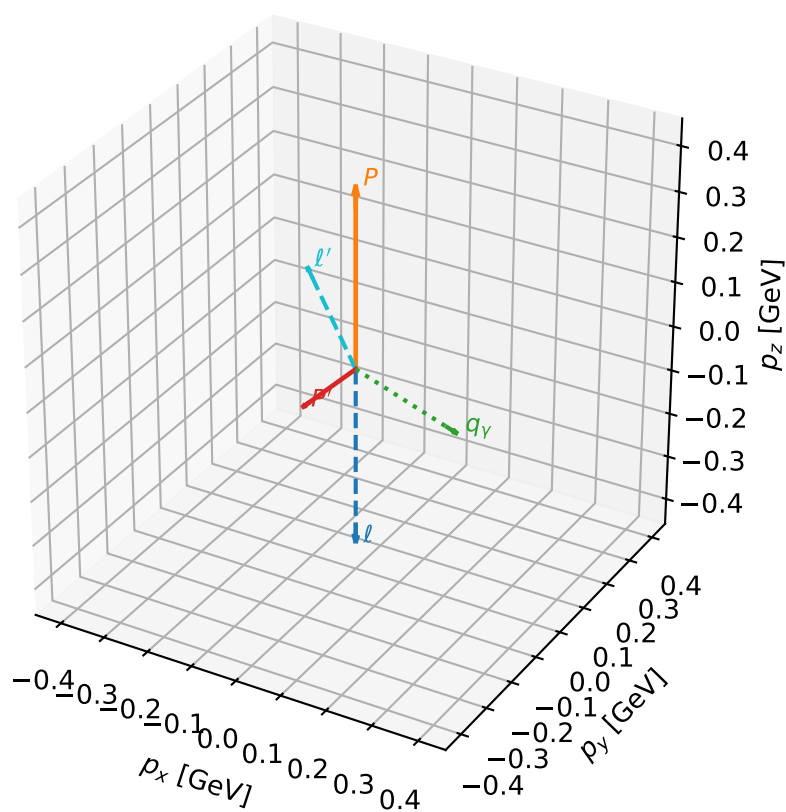
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_222 $d_{\{\text{rm GHZ}\}}=4.55477\text{e-}05$
 region: polarization_cluster_03_local_minimum_222
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.9023971$ rad
 incoming lepton: $-0.325557|+\rangle +0.945522|-\rangle$
 proton mixing angle: $\alpha_p=0.26705424$ rad
 incoming proton: $0.964552|+\rangle +0.263891|-\rangle$

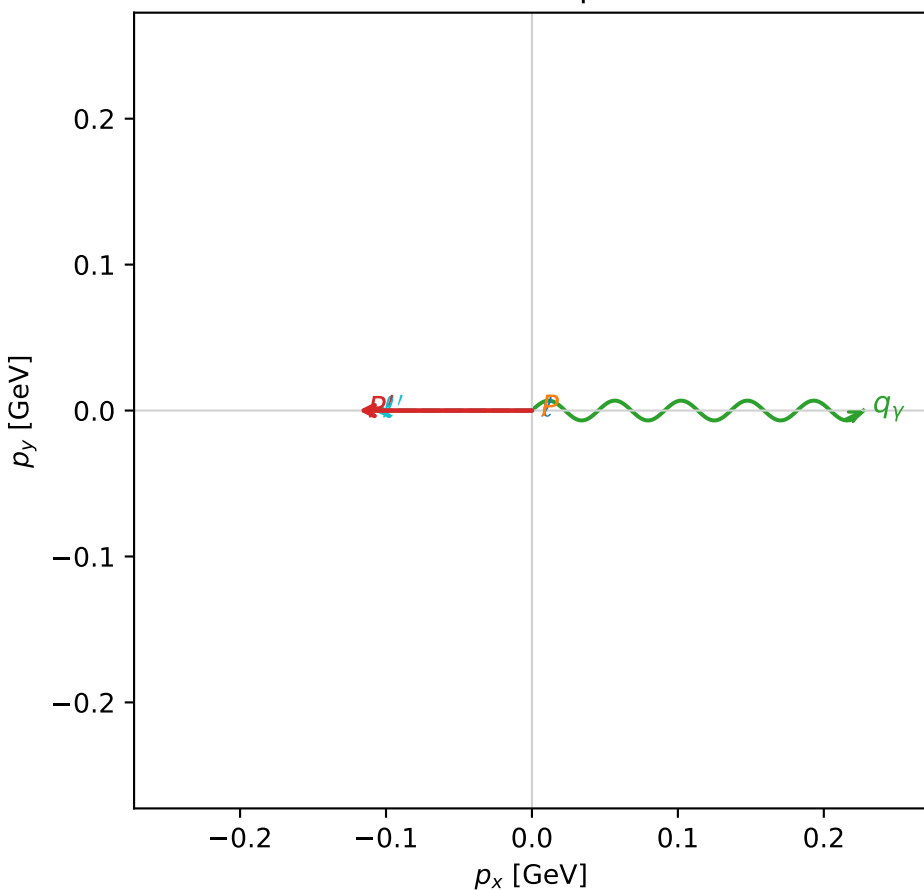
$s=2.01317$, $\sqrt{s}=1.41886$, $|\vec{P}|=0.399378$, $|\vec{P}'|=0.16824$
 $\theta_{P'}=2.3582$, $\theta_V=1.90351$, $\phi_{P'}=3.14165$, $\phi_V=2.99265\text{e-}05$
 $E_V=0.240379$, $\theta_{\ell'}=0.50179$, $\phi_{\ell'}=3.1416$
 $Q^2=0.338055$, $x_B=0.406594$, $t=-0.278597$, $W^2=1.37322$, $y=0.733622$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3994$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3994$
 P $E= 1.0195$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3994$
 ℓ' $E= 0.2255$ $p_x= -0.1085$ $p_y= -0.0000$ $p_z= 0.1977$
 P' $E= 0.9530$ $p_x= -0.1187$ $p_y= -0.0000$ $p_z= -0.1192$
 q_V $E= 0.2404$ $p_x= 0.2272$ $p_y= 0.0000$ $p_z= -0.0785$

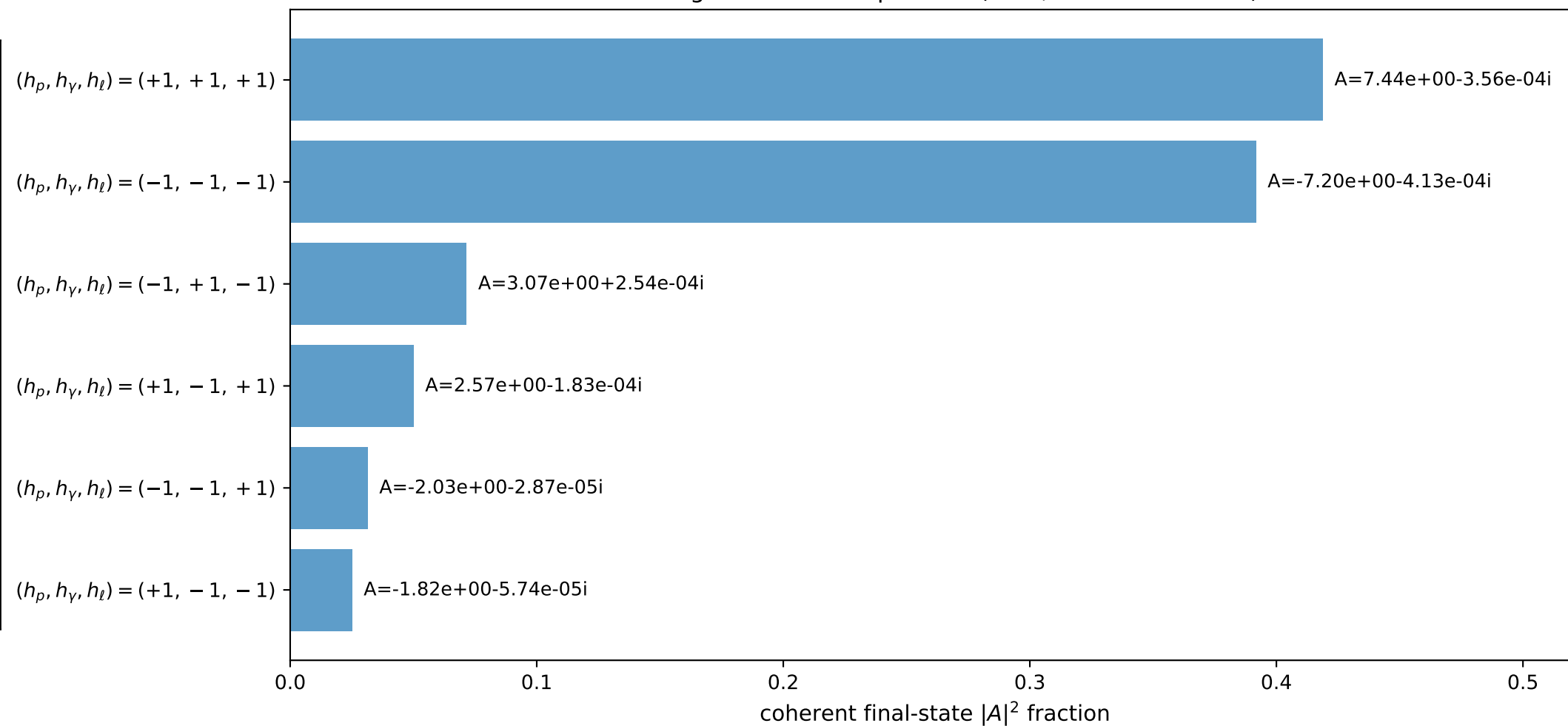
3D momenta



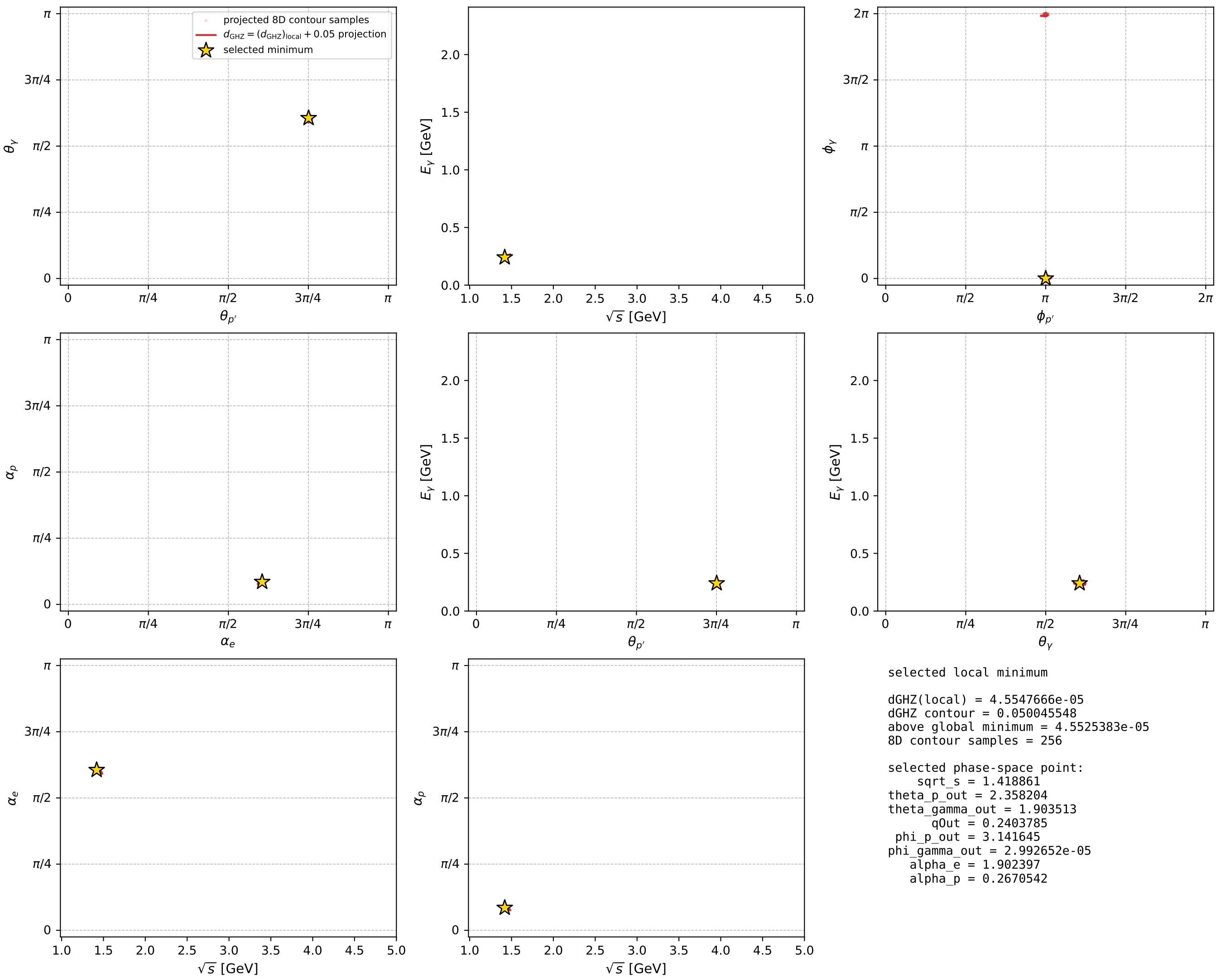
Transverse plane



Leading final-state amplitudes (N=6, retained=98.8%)



electron: polarization cluster P3, local minimum 222; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



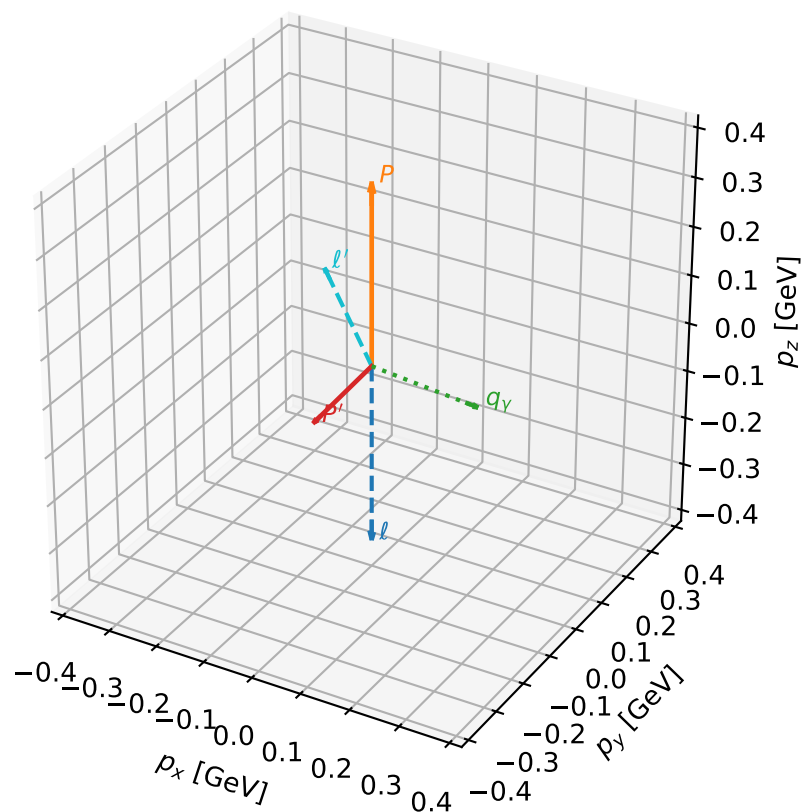
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_229 $d_{\{\text{rm GHZ}\}}=5.49891\text{e-}05$
 region: polarization_cluster_03_local_minimum_229
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.8972105$ rad
 incoming lepton: $-0.320649|+\rangle +0.947198|-\rangle$
 proton mixing angle: $\alpha_p=0.26242738$ rad
 incoming proton: $0.965763|+\rangle +0.259426|-\rangle$

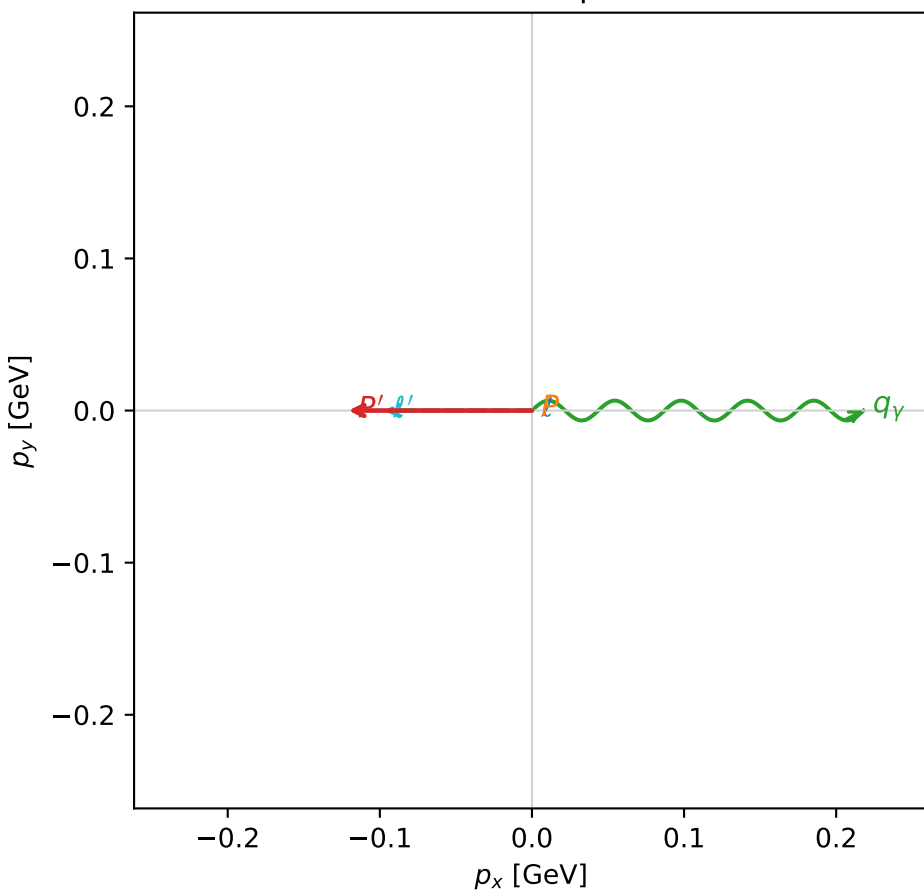
$s=1.89544$, $\sqrt{s}=1.37675$, $|\vec{P}|=0.368839$, $|\vec{P}'|=0.194452$
 $\theta_{P'}=2.47235$, $\theta_Y=1.67099$, $\phi_{P'}=3.14154$, $\phi_Y=6.28315$
 $E_Y=0.219067$, $\theta_{\ell'}=0.508988$, $\phi_{\ell'}=3.14157$
 $Q^2=0.276009$, $x_B=0.372169$, $t=-0.283858$, $W^2=1.34546$, $y=0.730233$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3688$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3688$
 P $E= 1.0079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3688$
 ℓ' $E= 0.1997$ $p_x= -0.0973$ $p_y= 0.0000$ $p_z= 0.1744$
 P' $E= 0.9579$ $p_x= -0.1206$ $p_y= 0.0000$ $p_z= -0.1525$
 q_Y $E= 0.2191$ $p_x= 0.2180$ $p_y= -0.0000$ $p_z= -0.0219$

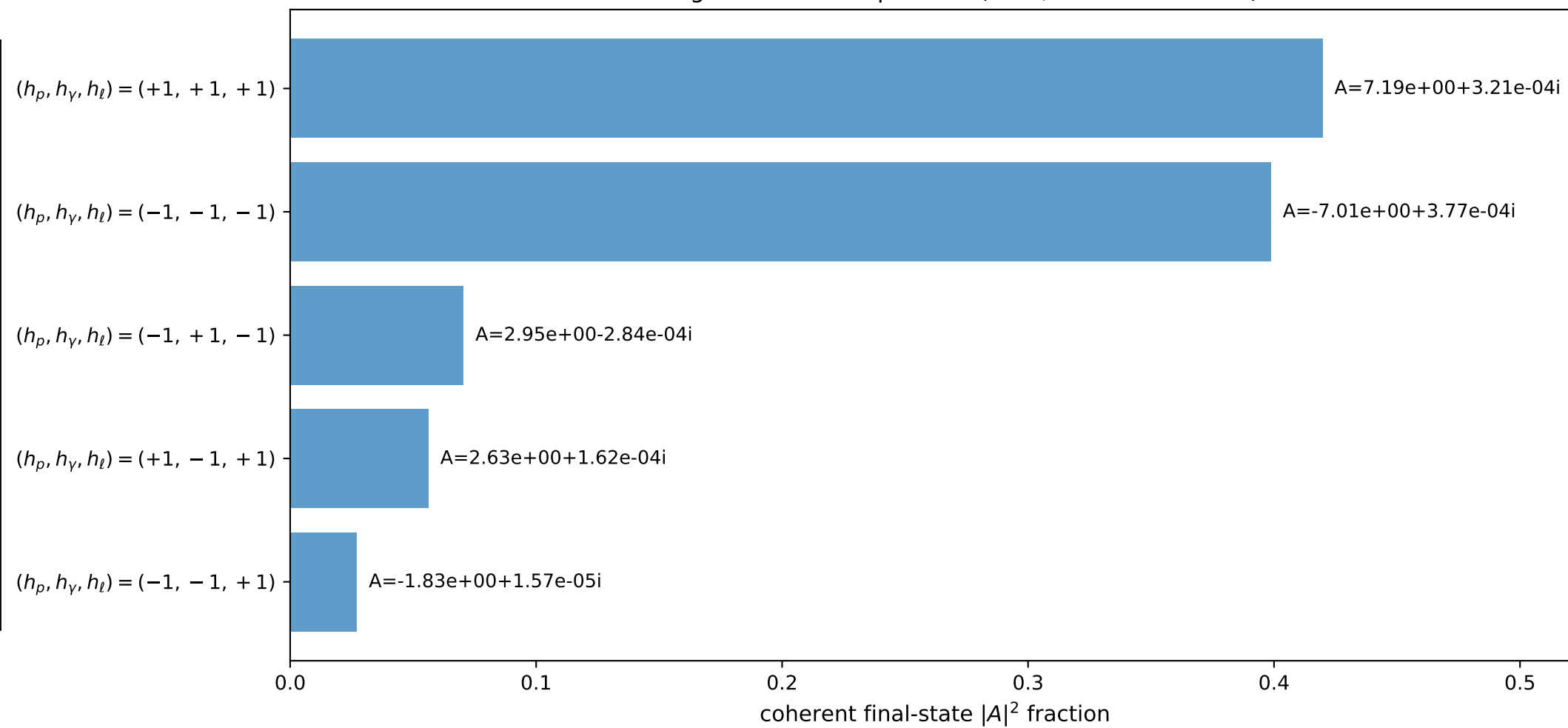
3D momenta



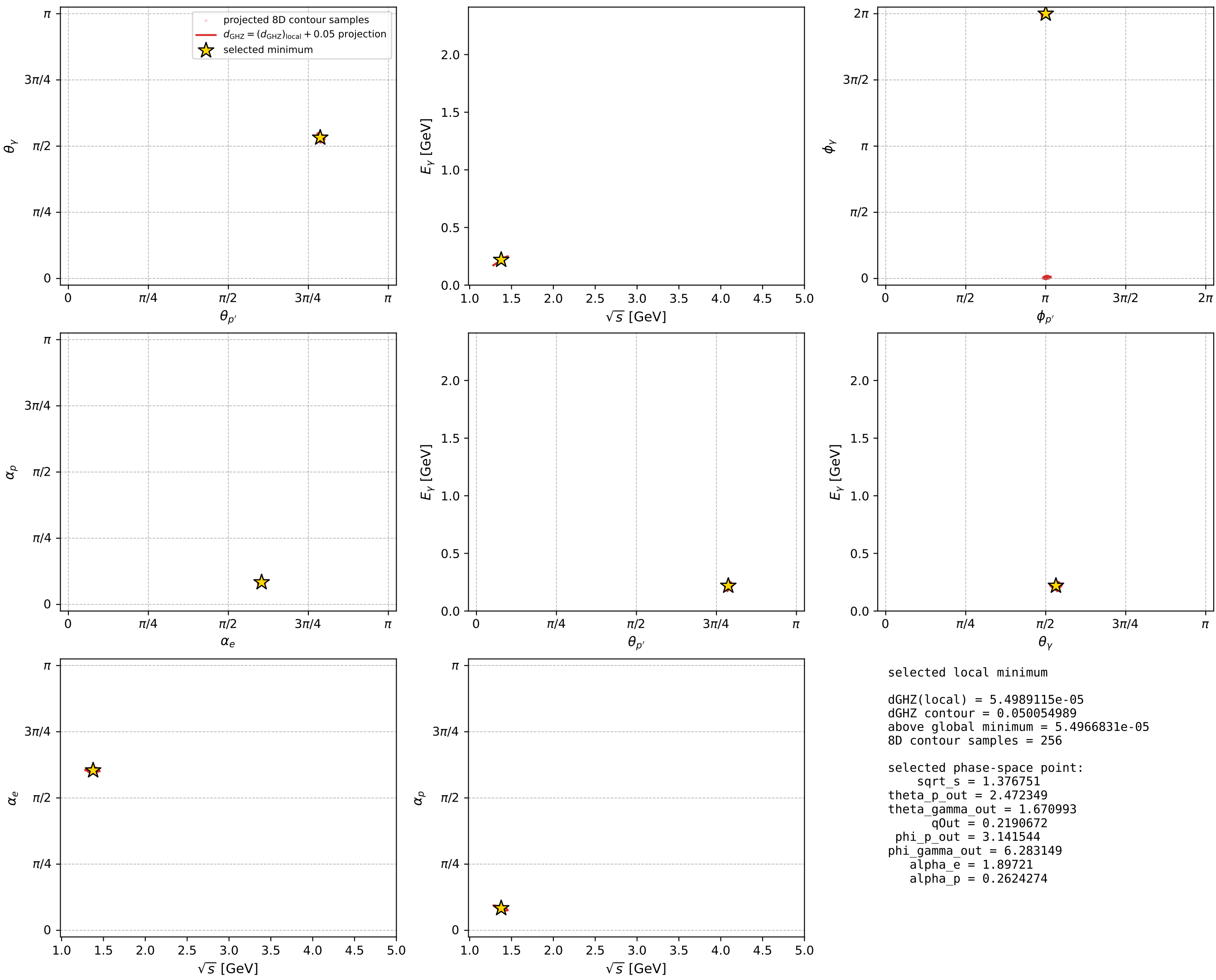
Transverse plane



Leading final-state amplitudes (N=5, retained=97.2%)



electron: polarization cluster P3, local minimum 229; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



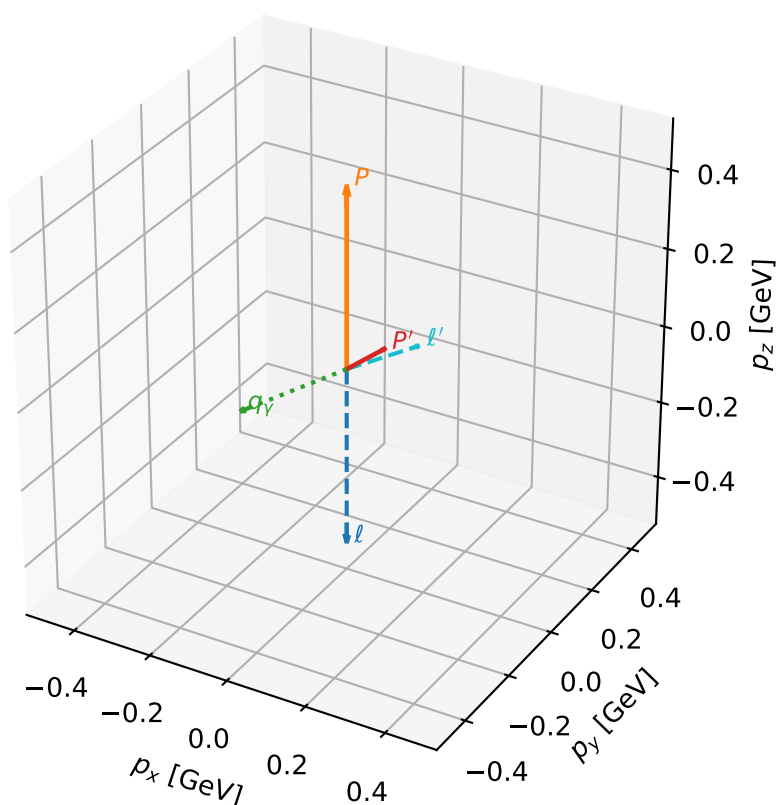
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_237 $d_{\{\text{rm GHZ}\}}=7.19377\text{e-}05$
 region: polarization_cluster_03_local_minimum_237
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3810478$ rad
 incoming lepton: $-0.724461|+\rangle +0.689316|-\rangle$
 proton mixing angle: $\alpha_p=0.93301678$ rad
 incoming proton: $0.595413|+\rangle +0.80342|-\rangle$

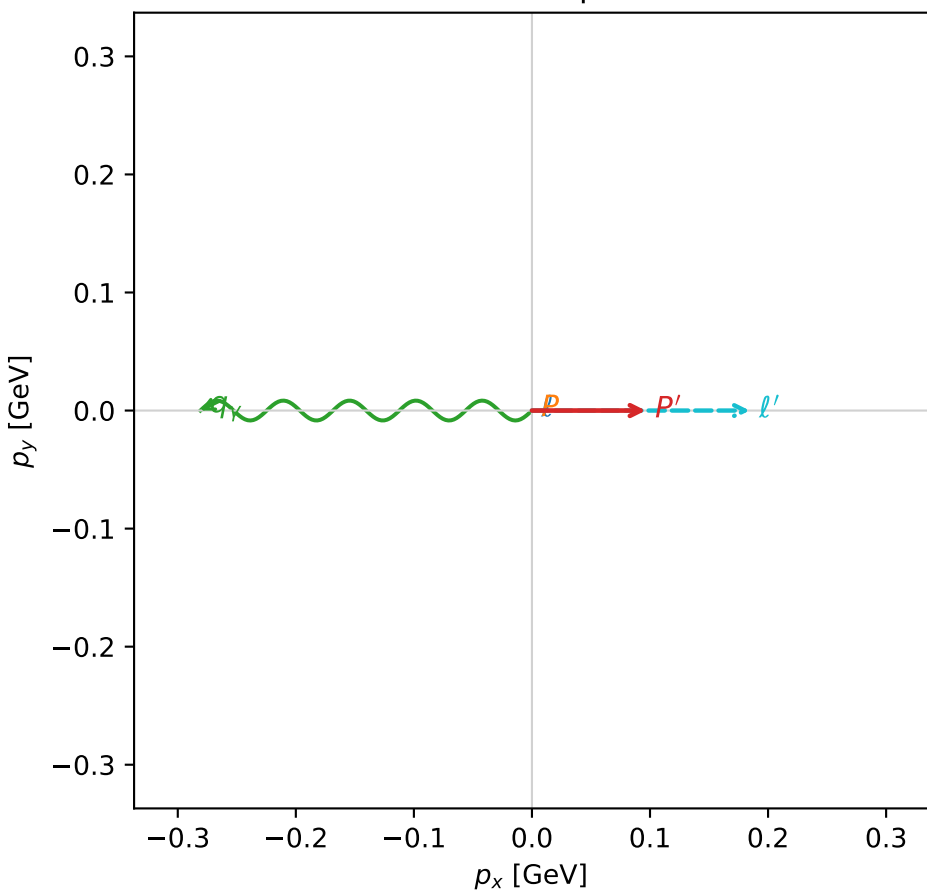
$s=2.25882$, $\sqrt{s}=1.50294$, $|\vec{P}|=0.458761$, $|\vec{P}'|=0.124978$
 $\theta_{P'}=0.880621$, $\theta_Y=2.1714$, $\phi_{P'}=6.28317$, $\phi_Y=3.1416$
 $E_Y=0.340418$, $\theta_{\ell'}=1.02191$, $\phi_{\ell'}=1.22728\text{e-}05$
 $Q^2=0.301907$, $x_B=0.292851$, $t=-0.143492$, $W^2=1.60886$, $y=0.747598$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4588$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4588$
 P $E= 1.0442$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4588$
 ℓ' $E= 0.2162$ $p_x= 0.1845$ $p_y= 0.0000$ $p_z= 0.1128$
 P' $E= 0.9463$ $p_x= 0.0964$ $p_y= -0.0000$ $p_z= 0.0796$
 q_Y $E= 0.3404$ $p_x= -0.2808$ $p_y= -0.0000$ $p_z= -0.1924$

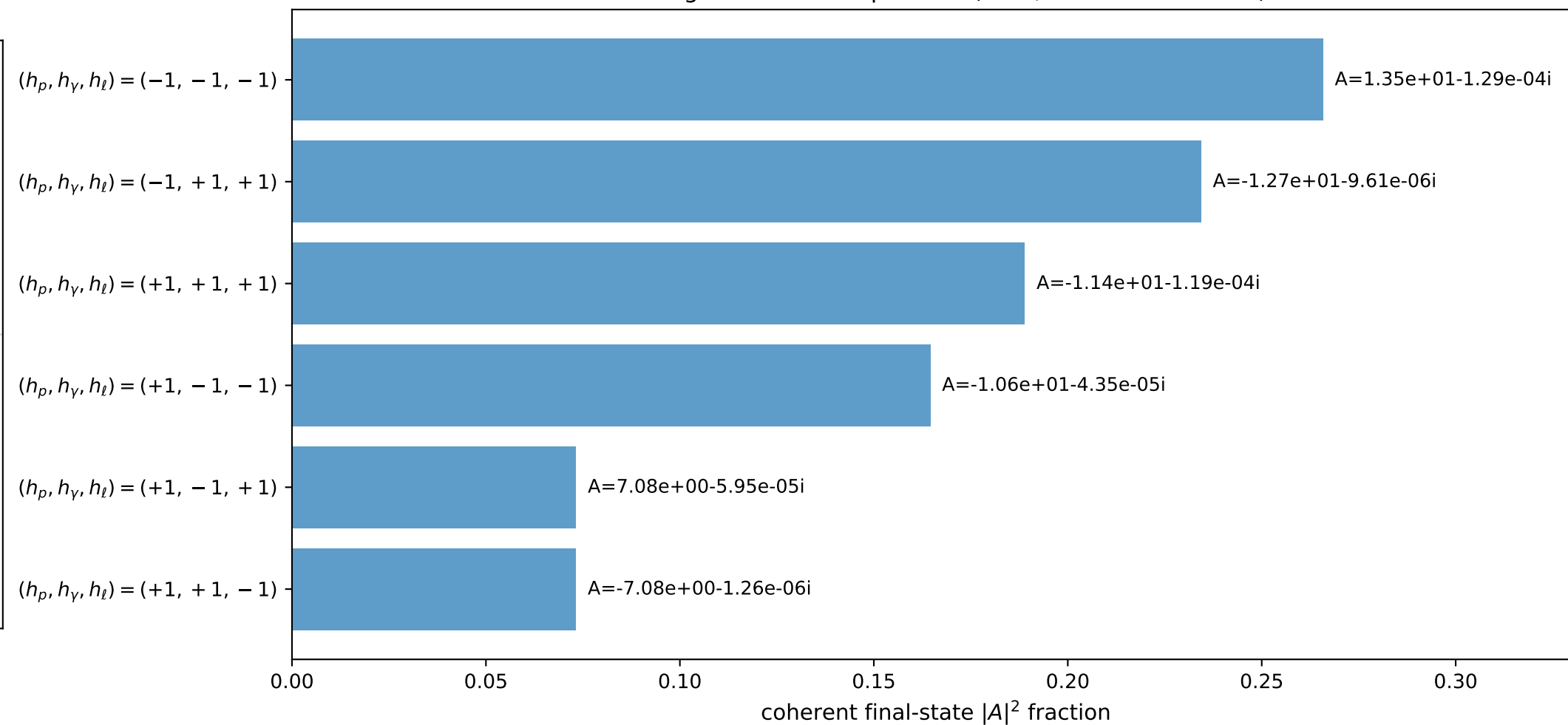
3D momenta



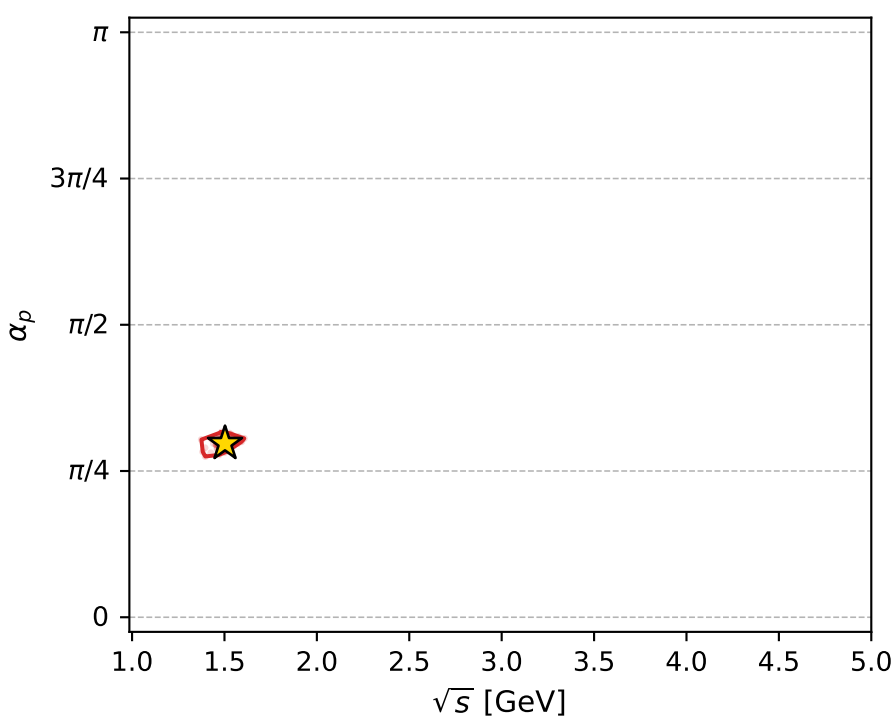
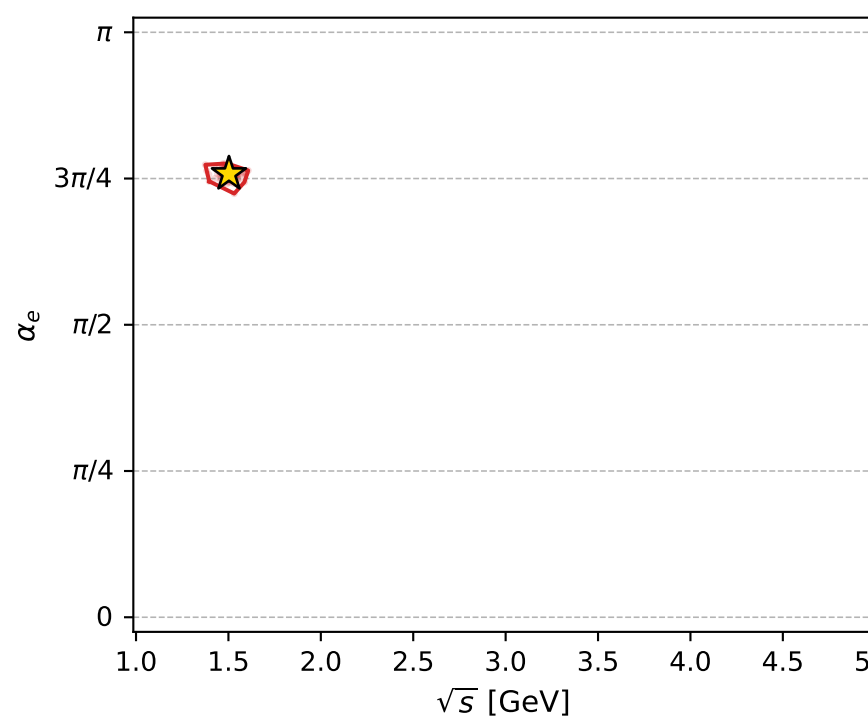
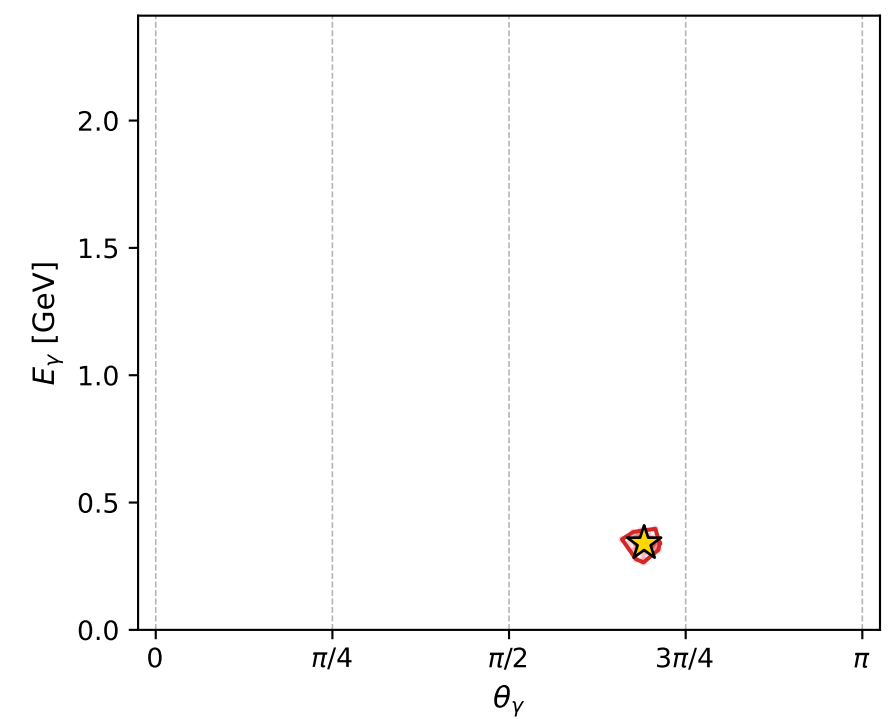
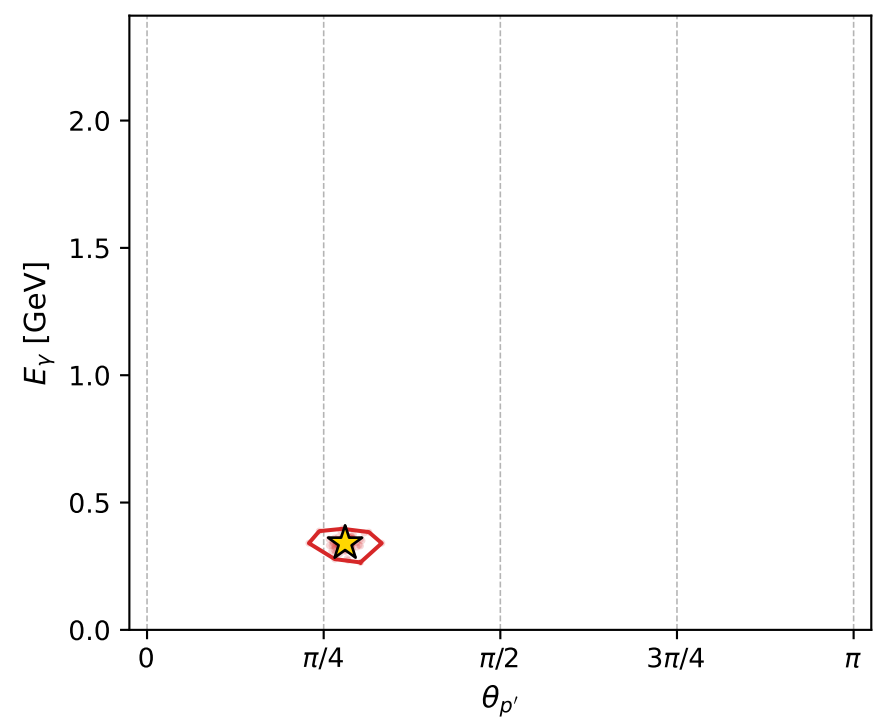
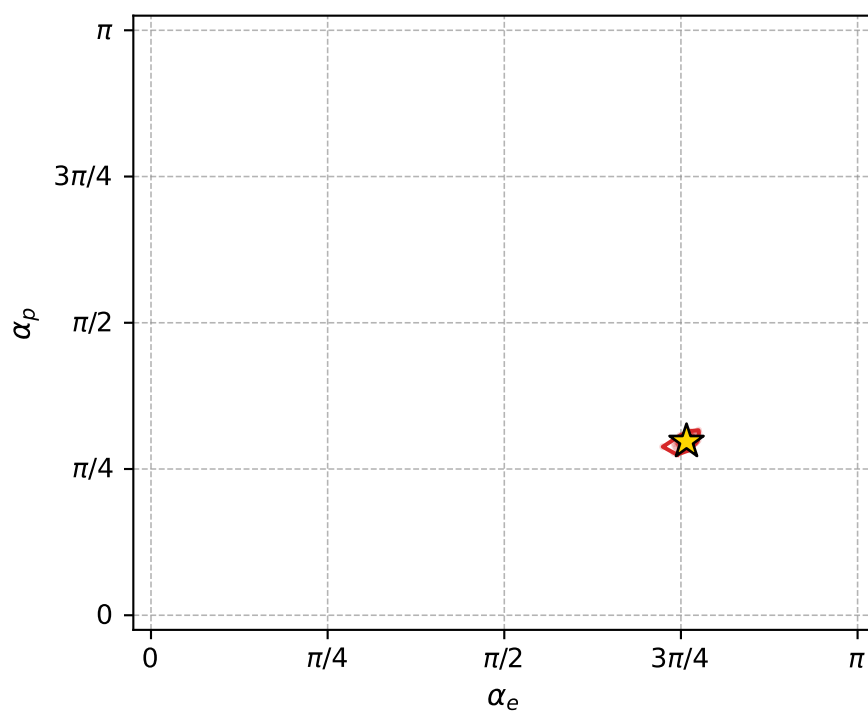
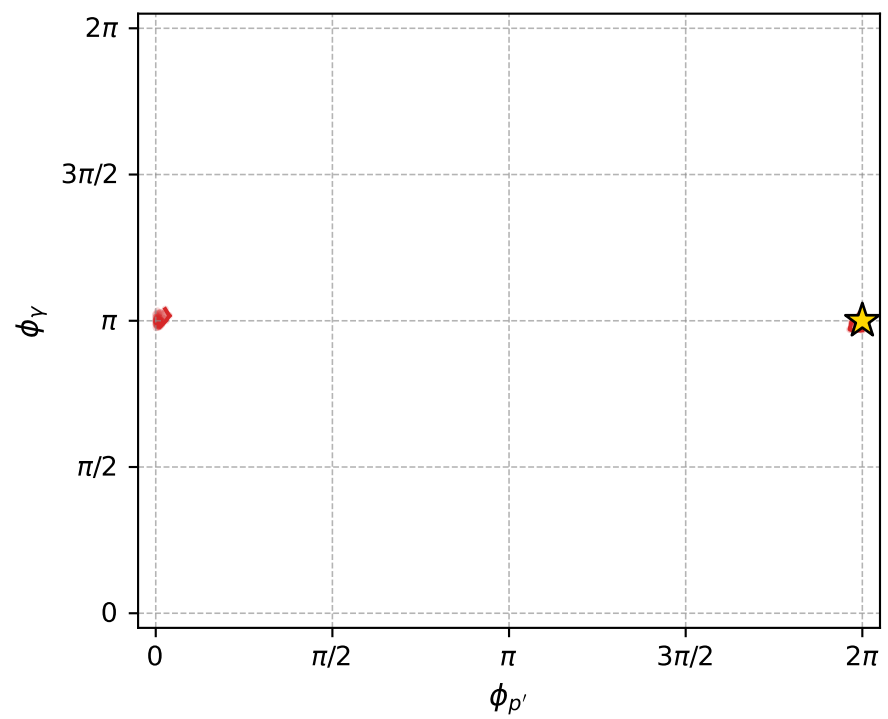
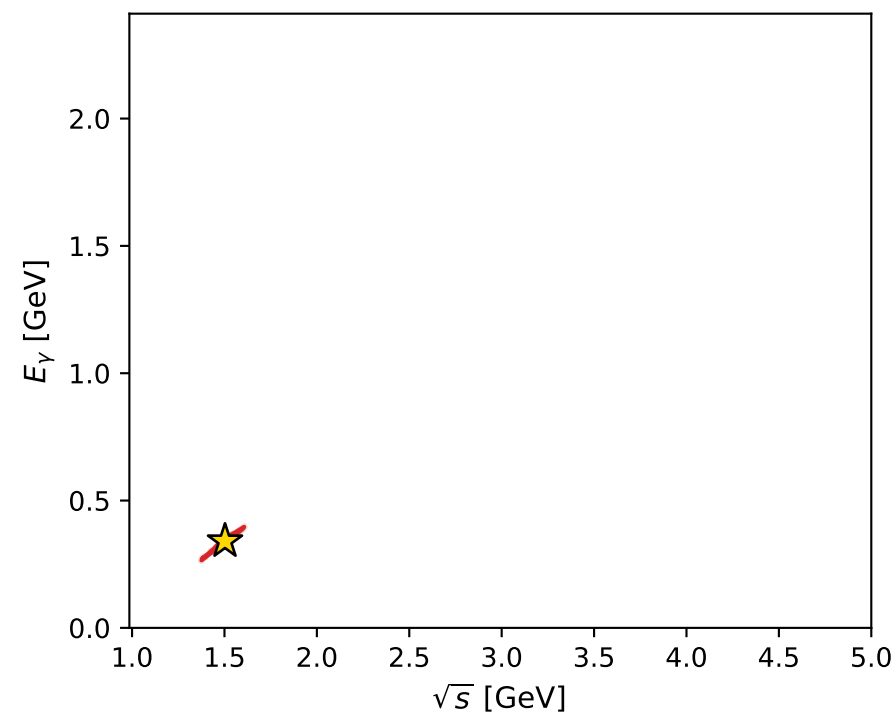
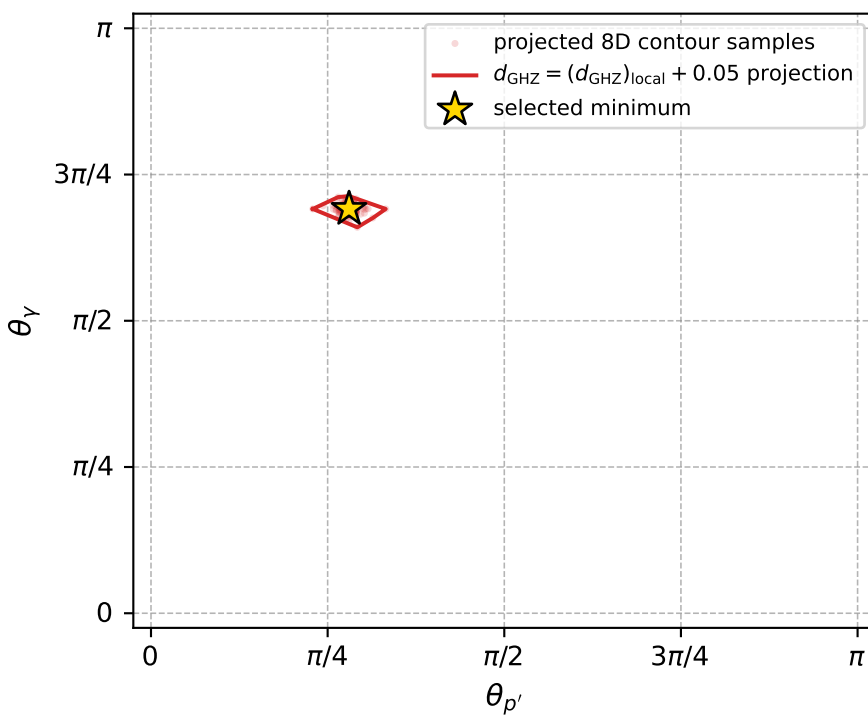
Transverse plane



Leading final-state amplitudes (N=6, retained=100.0%)



electron: polarization cluster P3, local minimum 237; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.1937676\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050071938$
 above global minimum = $7.1915393\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.502939$
 $\text{theta_p_out} = 0.8806209$
 $\text{theta_gamma_out} = 2.171404$
 $\text{qOut} = 0.3404182$
 $\text{phi_p_out} = 6.283174$
 $\text{phi_gamma_out} = 3.141597$
 $\text{alpha_e} = 2.381048$
 $\text{alpha_p} = 0.9330168$

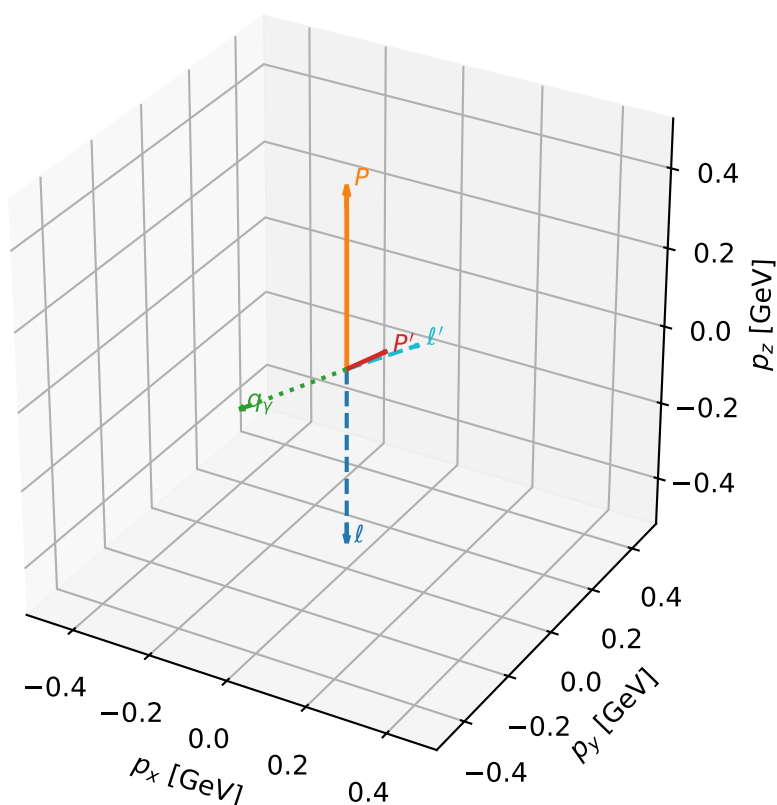
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_239 $d_{\{\text{rm GHZ}\}}=8.09219\text{e-}05$
 region: polarization_cluster_03_local_minimum_239
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3886944$ rad
 incoming lepton: $-0.72971|+\rangle +0.683756|-\rangle$
 proton mixing angle: $\alpha_p=0.93943294$ rad
 incoming proton: $0.590246|+\rangle +0.807224|-\rangle$

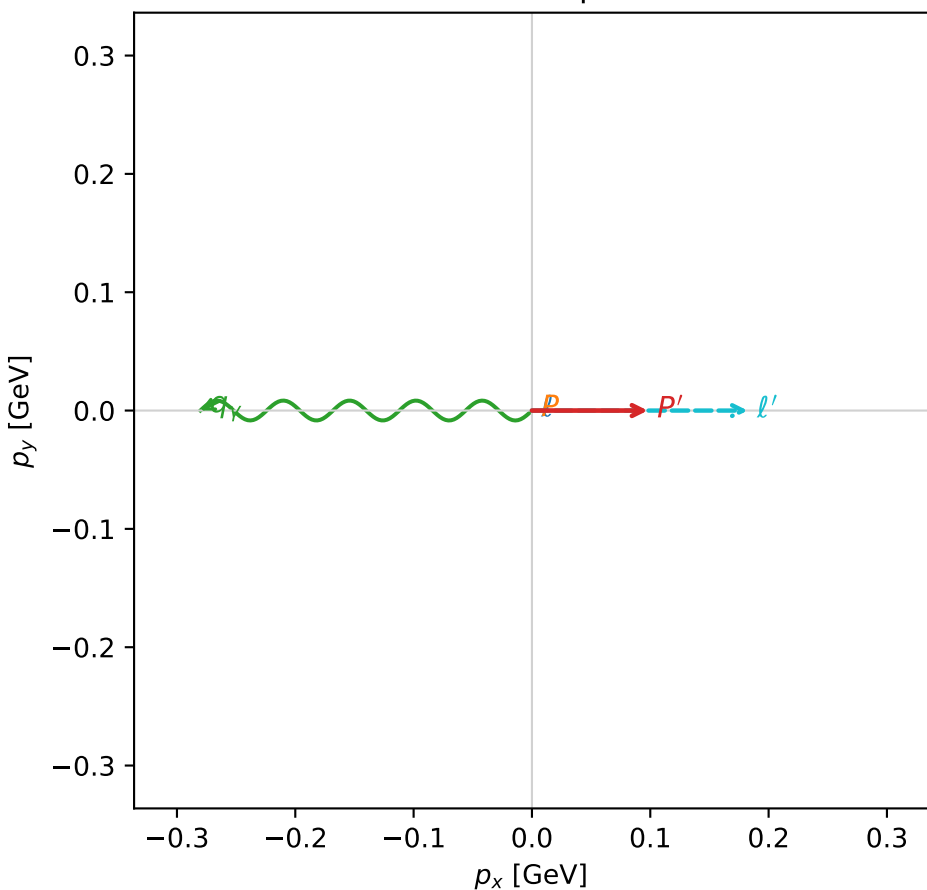
$s=2.24087$, $\sqrt{s}=1.49695$, $|\vec{P}|=0.454598$, $|\vec{P}'|=0.121639$
 $\theta_{P'}=0.935017$, $\theta_Y=2.15655$, $\phi_{P'}=6.28309$, $\phi_Y=3.14147$
 $E_Y=0.336249$, $\theta_{\ell'}=1.01336$, $\phi_{\ell'}=6.28305$
 $Q^2=0.298676$, $x_B=0.293839$, $t=-0.146472$, $W^2=1.59763$, $y=0.746836$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4546$
 P $E= 1.0424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4546$
 ℓ' $E= 0.2148$ $p_x= 0.1823$ $p_y= -0.0000$ $p_z= 0.1137$
 P' $E= 0.9459$ $p_x= 0.0979$ $p_y= -0.0000$ $p_z= 0.0722$
 q_Y $E= 0.3362$ $p_x= -0.2802$ $p_y= 0.0000$ $p_z= -0.1859$

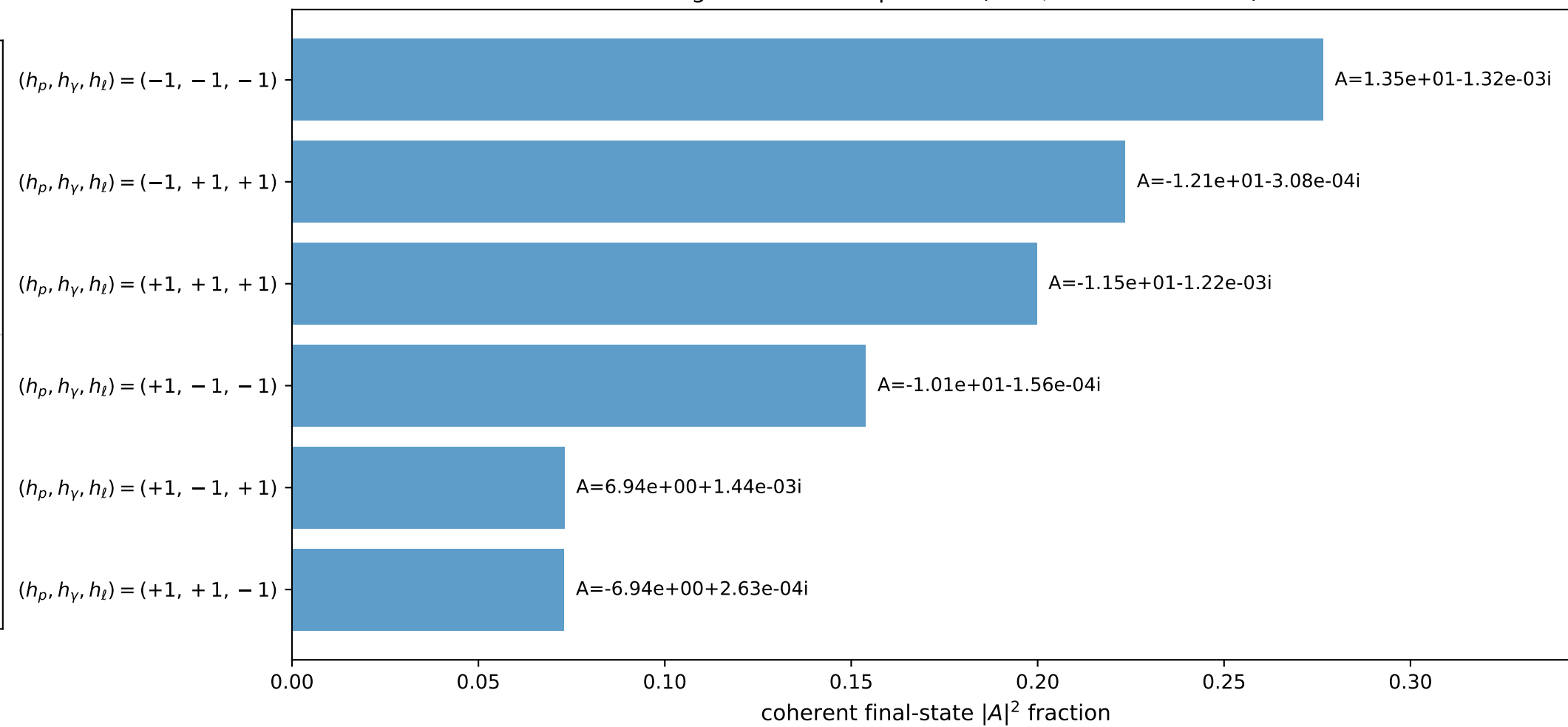
3D momenta



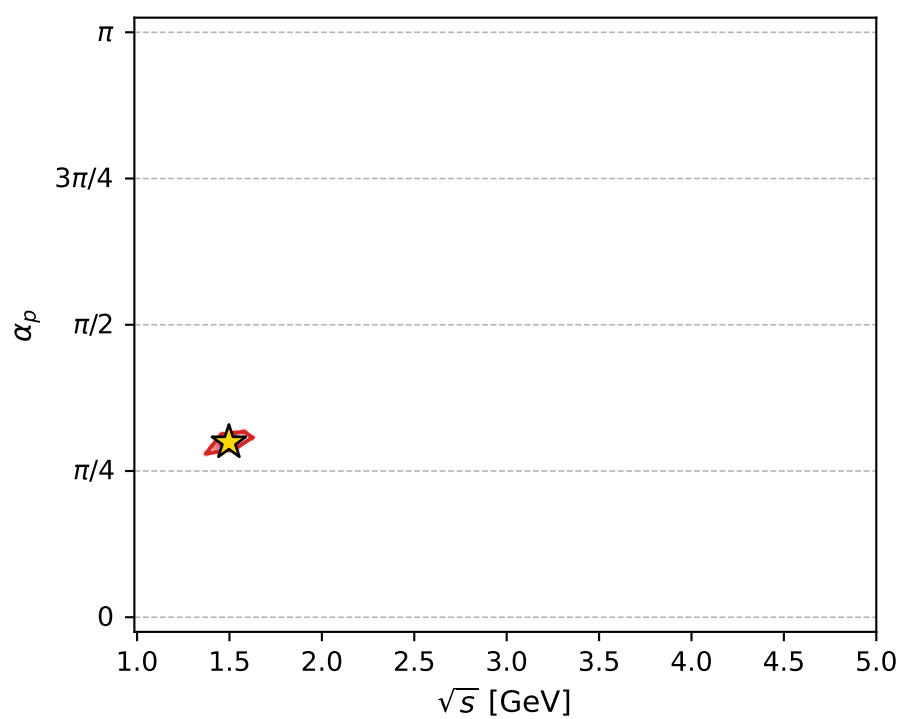
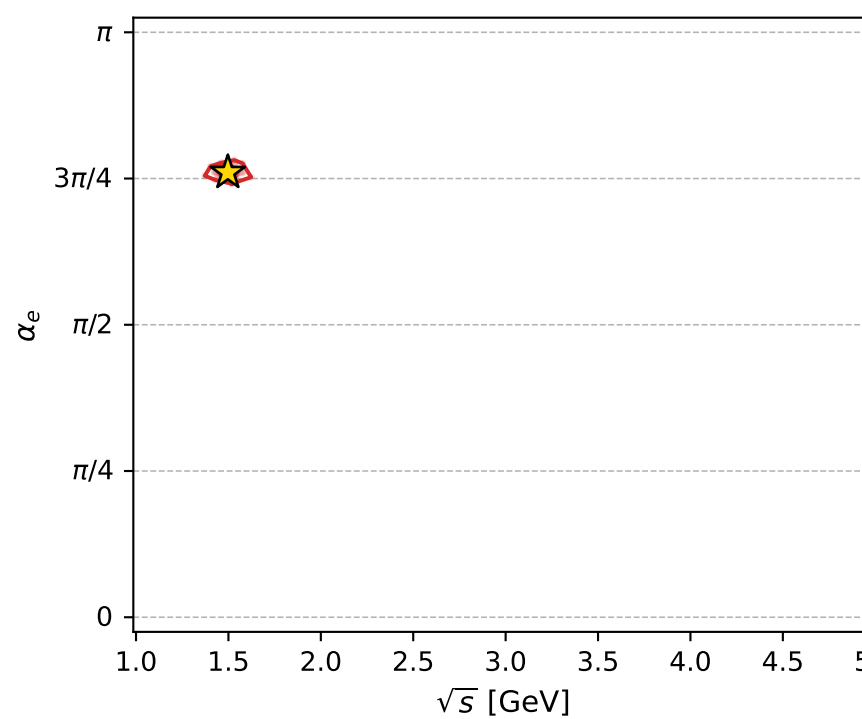
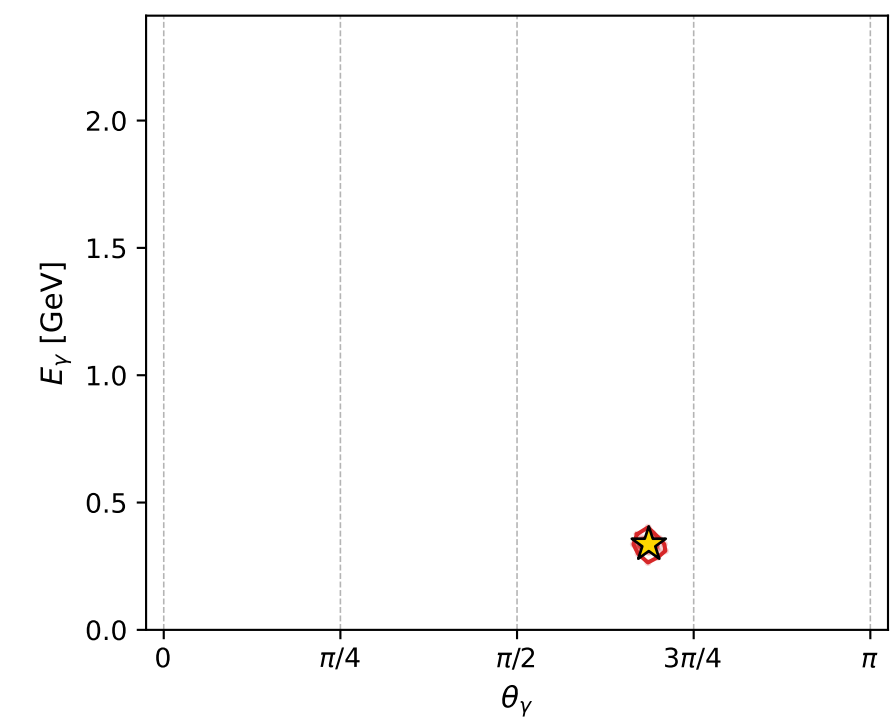
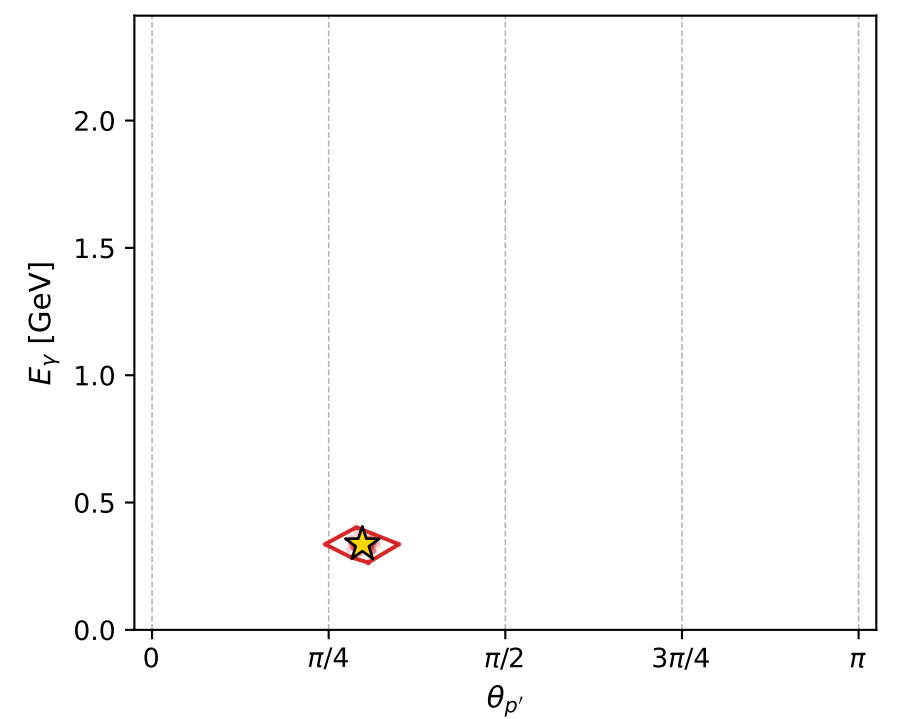
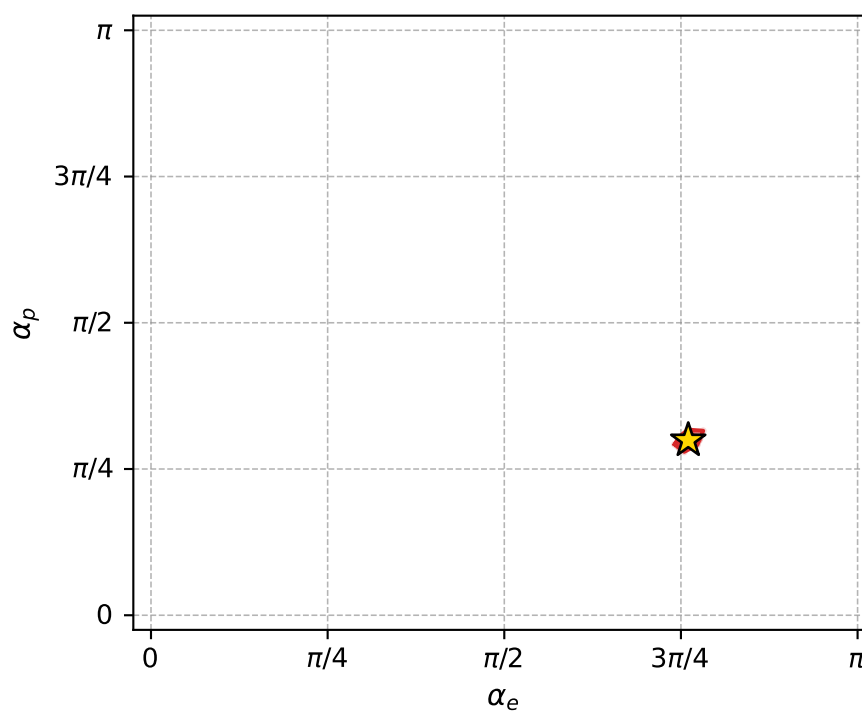
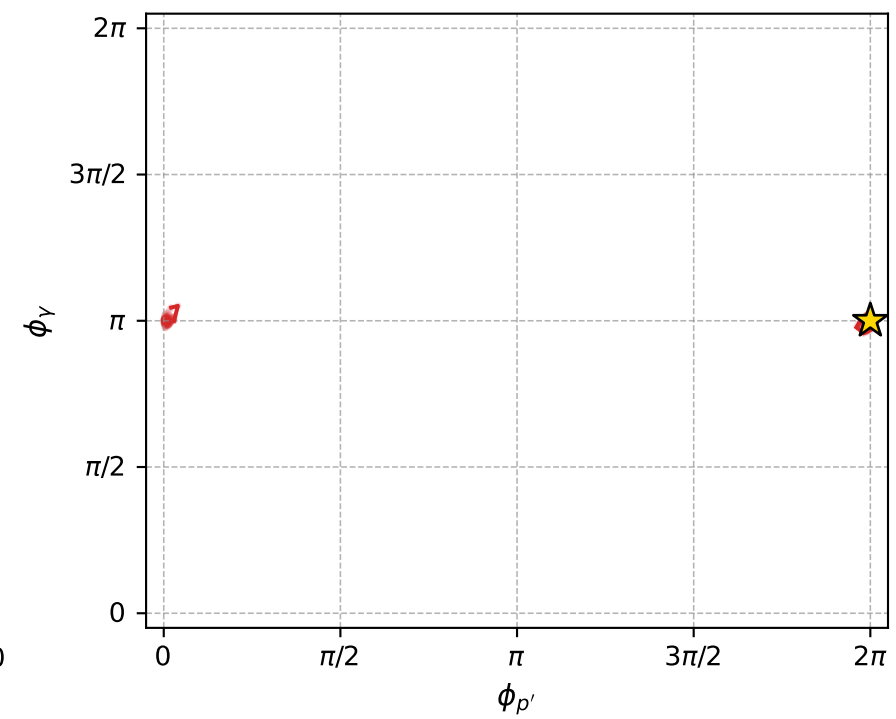
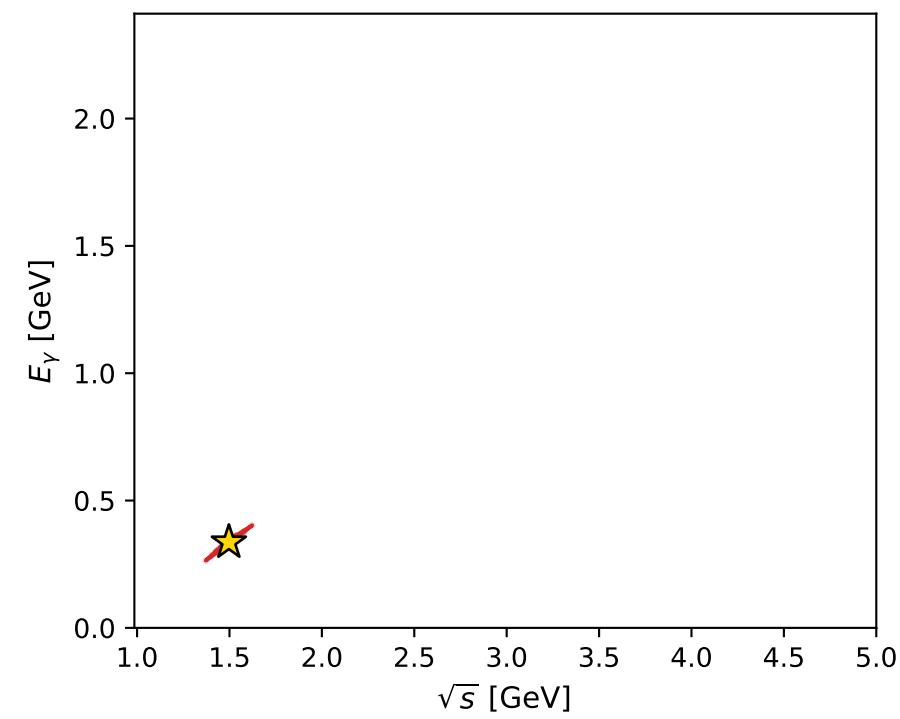
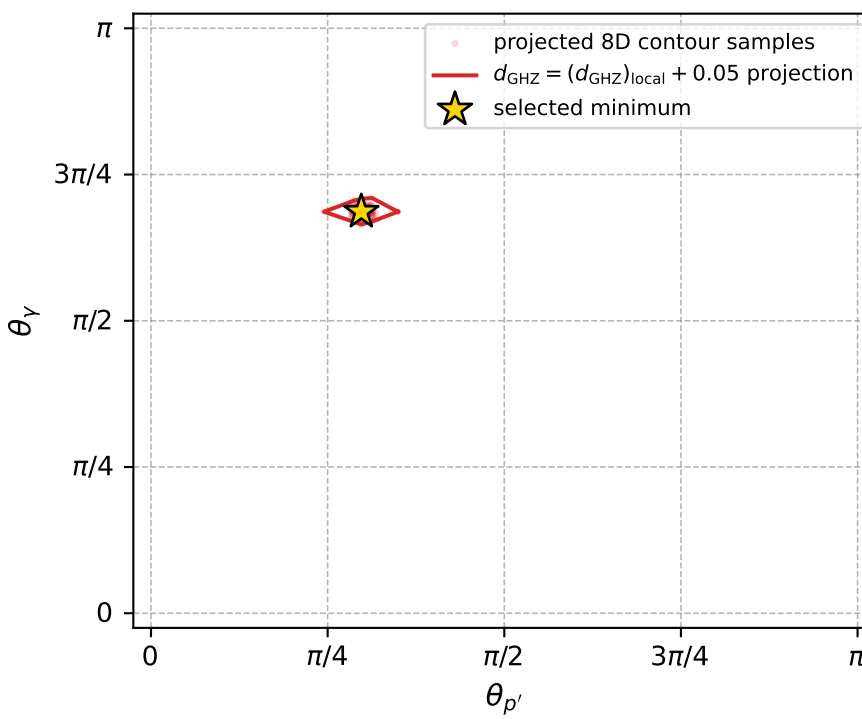
Transverse plane



Leading final-state amplitudes (N=6, retained=99.9%)



electron: polarization cluster P3, local minimum 239; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.0921881\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050080922$
 above global minimum = $8.0899598\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.496953$
 $\text{theta_p_out} = 0.9350174$
 $\text{theta_gamma_out} = 2.156546$
 $\text{q0out} = 0.3362495$
 $\text{phi_p_out} = 6.283087$
 $\text{phi_gamma_out} = 3.14147$
 $\text{alpha_e} = 2.388694$
 $\text{alpha_p} = 0.9394329$

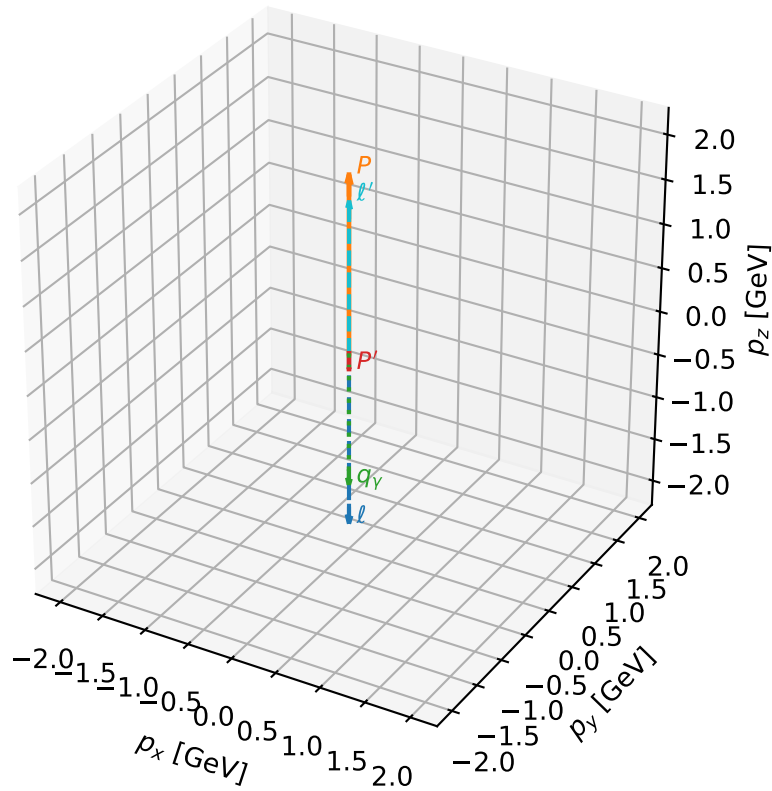
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_240 $d_{\{\text{rm GHZ}\}}=8.30616\text{e-}05$
 region: polarization_cluster_03_local_minimum_240
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.9423254$ rad
 incoming lepton: $-0.363041|+> +0.931773|->$
 proton mixing angle: $\alpha_p=0.37214669$ rad
 incoming proton: $0.931549|+> +0.363616|->$

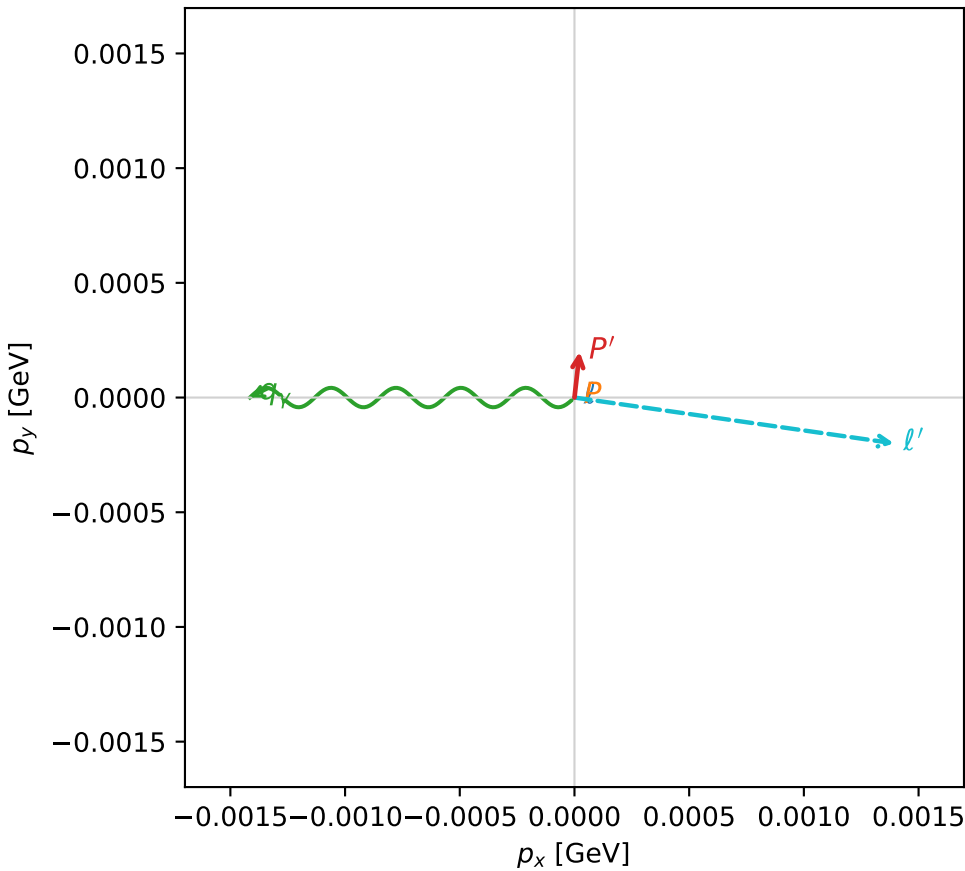
$s=17.6256$, $\sqrt{s}=4.19829$, $|\vec{P}|=1.99436$, $|\vec{P}'|=0.179521$
 $\theta_{p'}=3.14047$, $\theta_Y=3.14067$, $\phi_{p'}=1.45864$, $\phi_Y=3.14159$
 $E_Y=1.53187$, $\theta_{\ell'}=0.000821974$, $\phi_{\ell'}=6.14079$
 $Q^2=13.6525$, $x_B=0.851767$, $t=-3.16598$, $W^2=3.25579$, $y=0.957164$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9944$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9944$
 P $E= 2.2039$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9944$
 ℓ' $E= 1.7114$ $p_x= 0.0014$ $p_y= -0.0002$ $p_z= 1.7114$
 P' $E= 0.9550$ $p_x= 0.0000$ $p_y= 0.0002$ $p_z= -0.1795$
 q_Y $E= 1.5319$ $p_x= -0.0014$ $p_y= 0.0000$ $p_z= -1.5319$

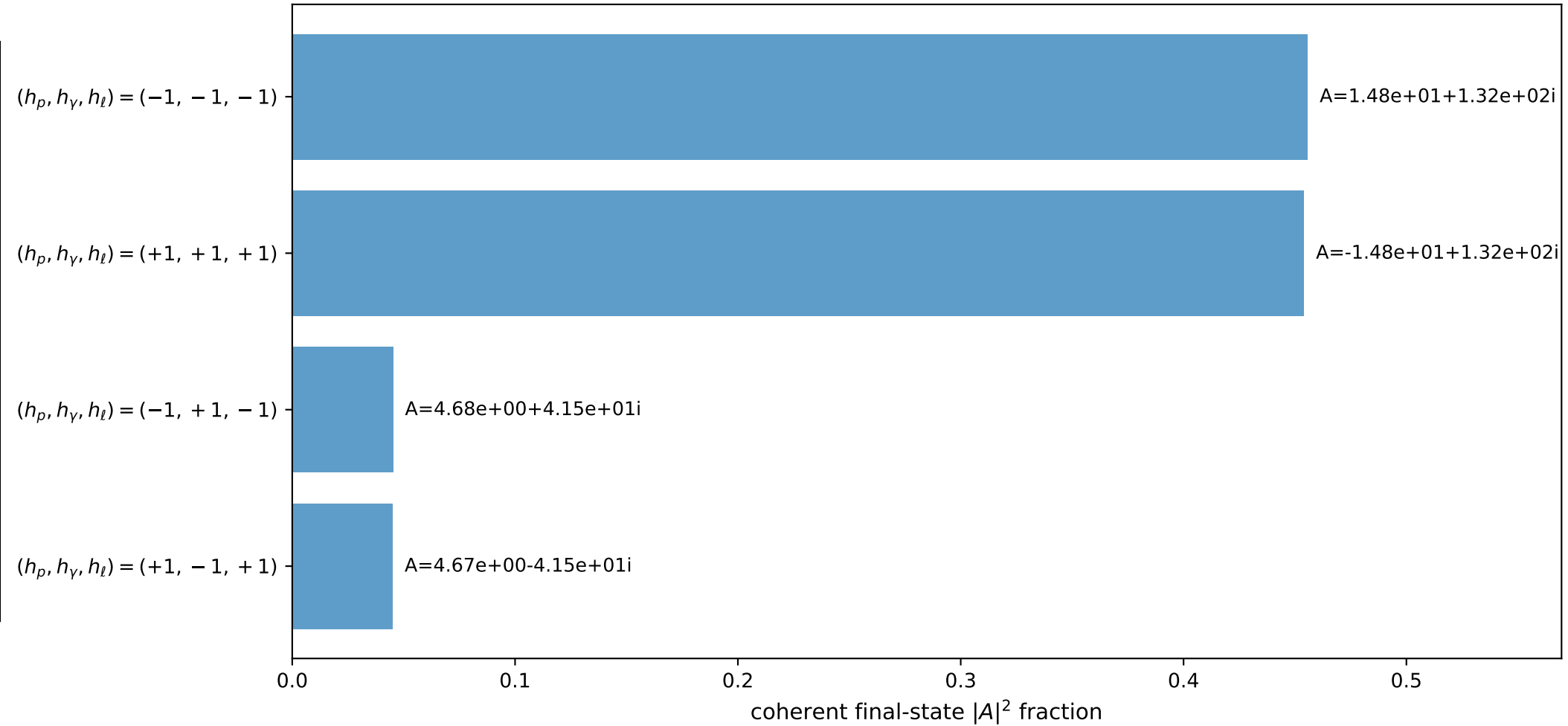
3D momenta



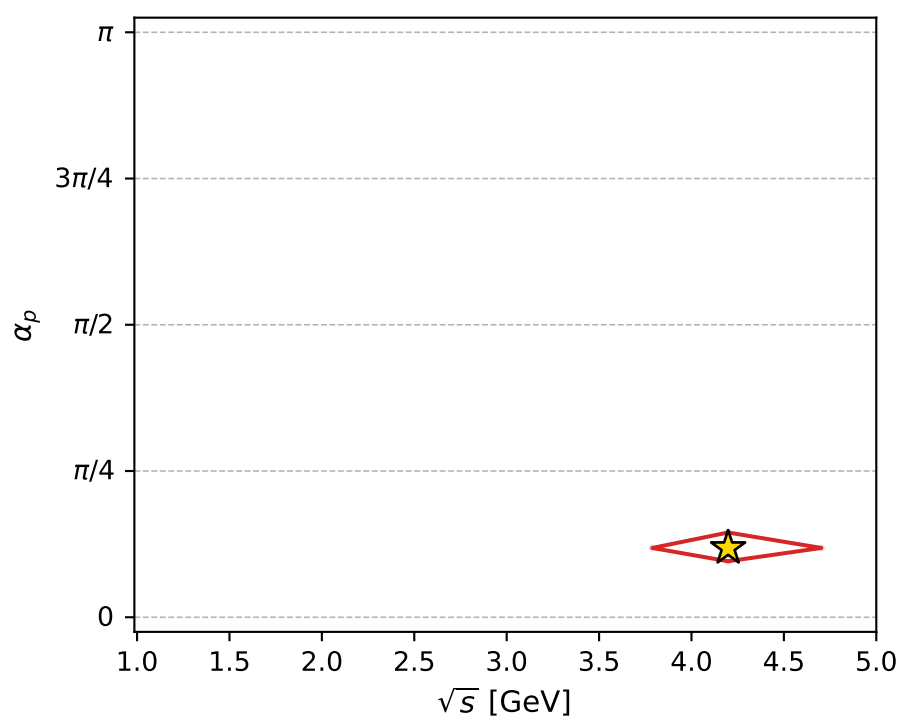
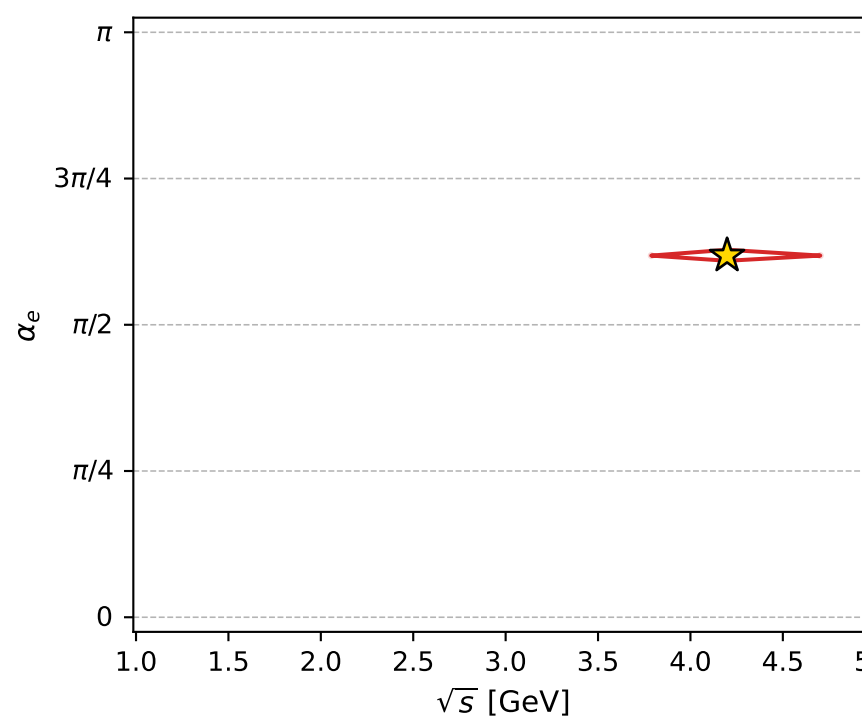
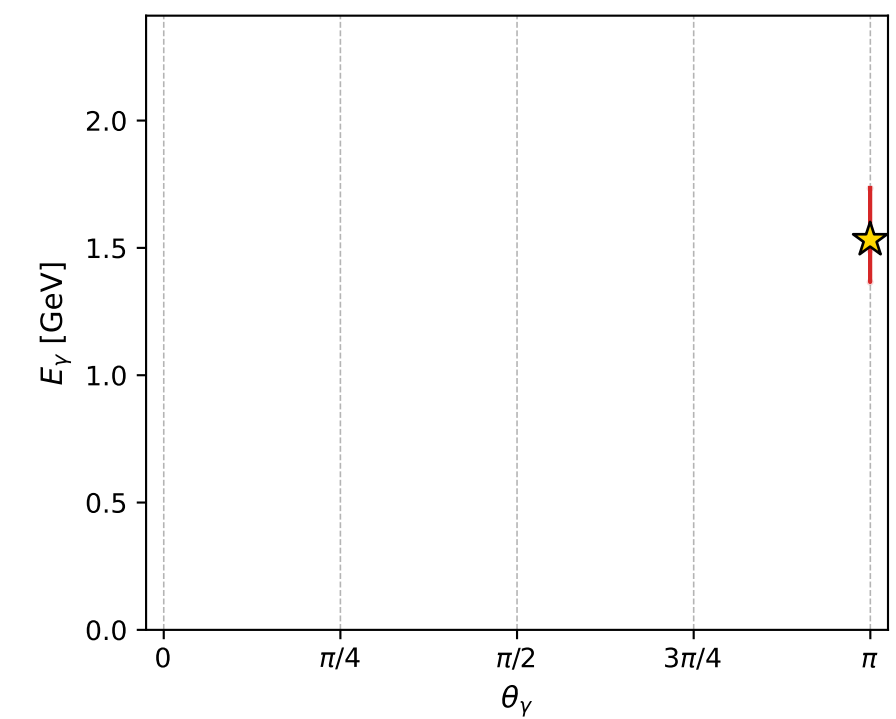
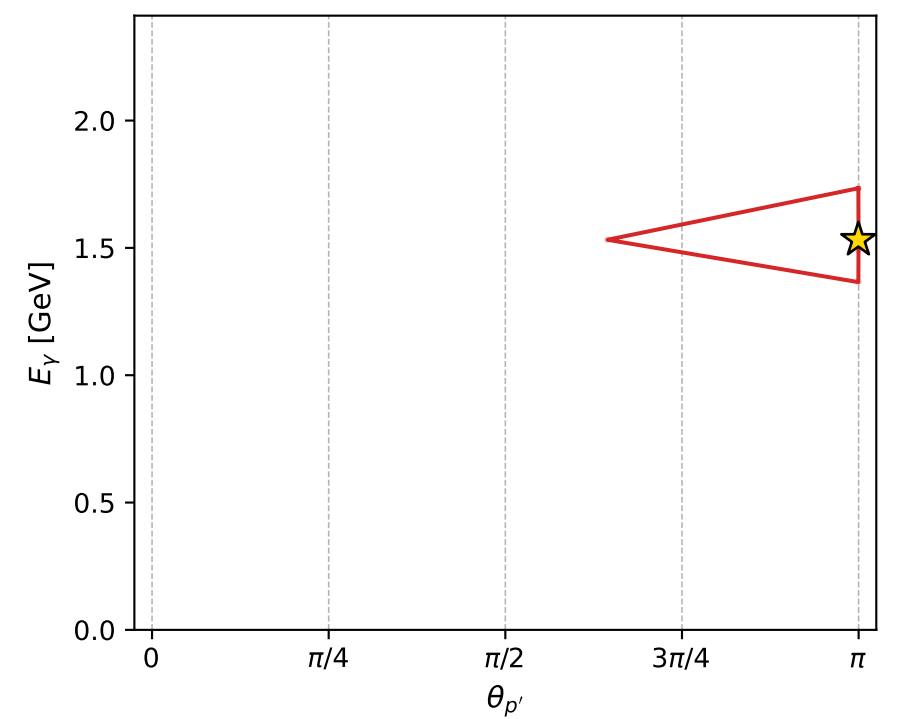
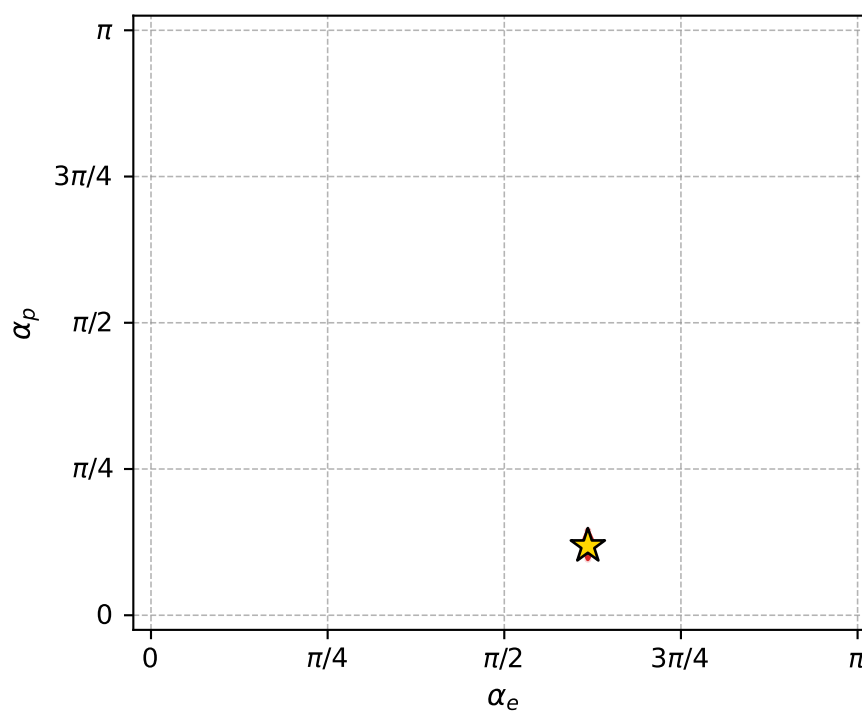
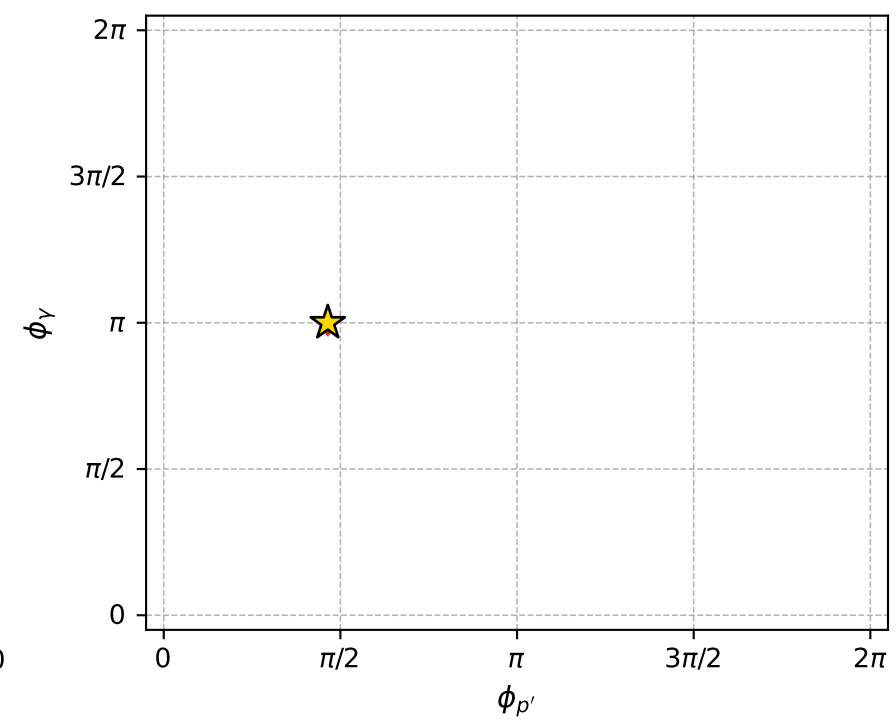
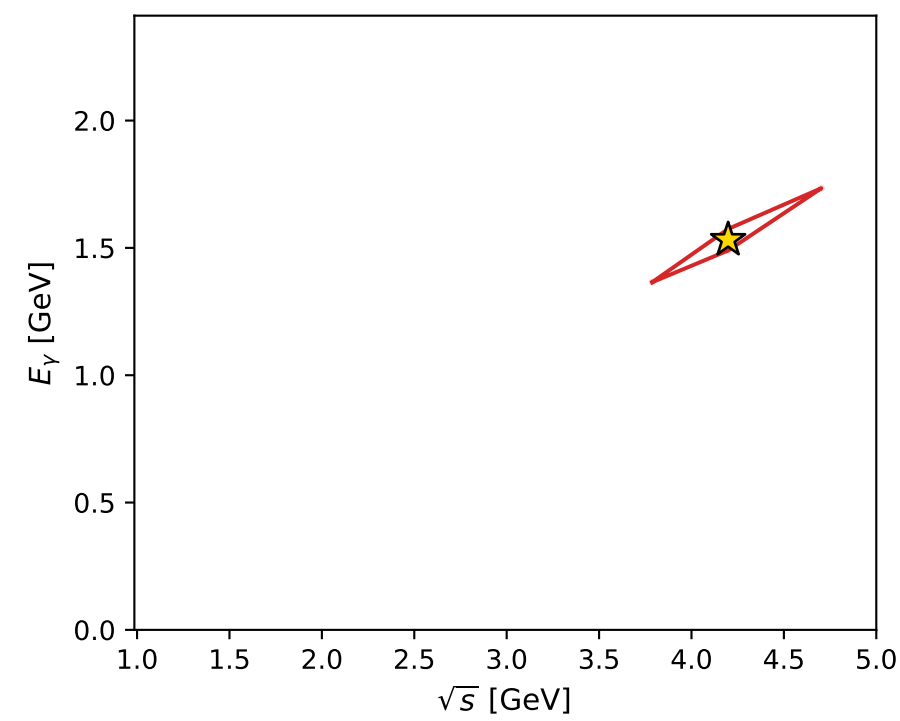
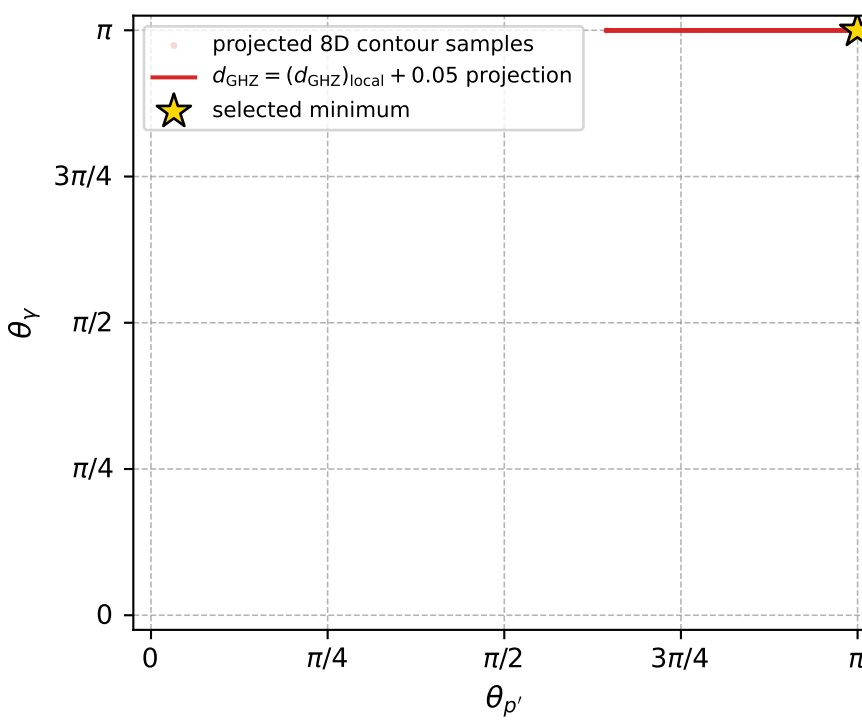
Transverse plane



Leading final-state amplitudes (N=4, retained=100.0%)



electron: polarization cluster P3, local minimum 240; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.3061629\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050083062$
 above global minimum = $8.3039346\text{e-}05$
 8D contour samples = 243

selected phase-space point:

$\text{sqrt_s} = 4.198288$
 $\text{theta_p_out} = 3.140474$
 $\text{theta_gamma_out} = 3.140669$
 $\text{q0ut} = 1.531871$
 $\text{phi_p_out} = 1.458635$
 $\text{phi_gamma_out} = 3.14159$
 $\text{alpha_e} = 1.942325$
 $\text{alpha_p} = 0.3721467$

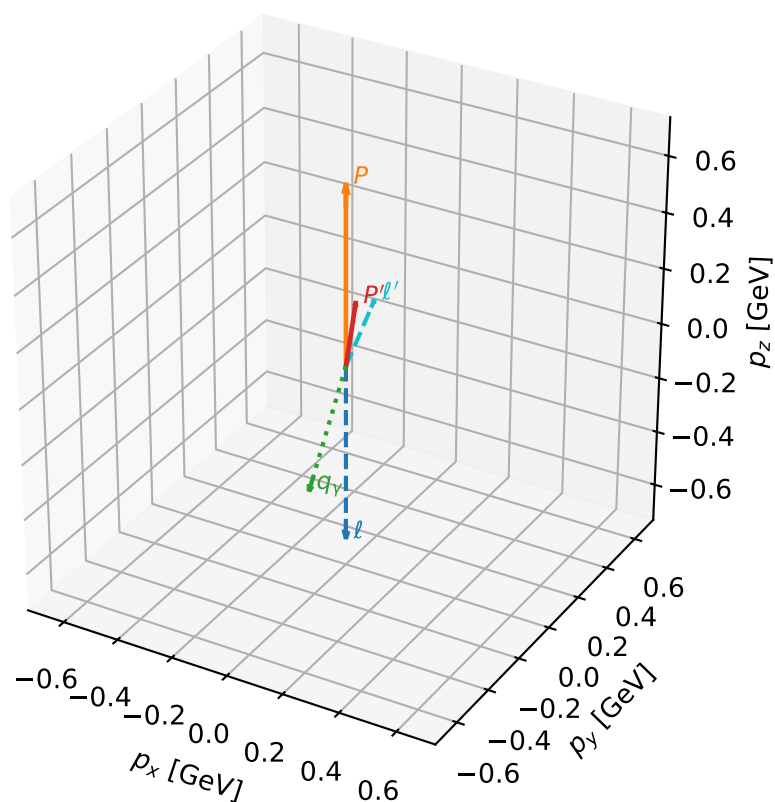
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_246 d_{\rm GHZ}=0.000119626
region: polarization_cluster_03_local_minimum_246
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.8146348$ rad
incoming lepton: $-0.947024|+\rangle +0.321164|-\rangle$
proton mixing angle: $\alpha_p=1.2946563$ rad
incoming proton: $0.272644|+\rangle +0.962115|-\rangle$

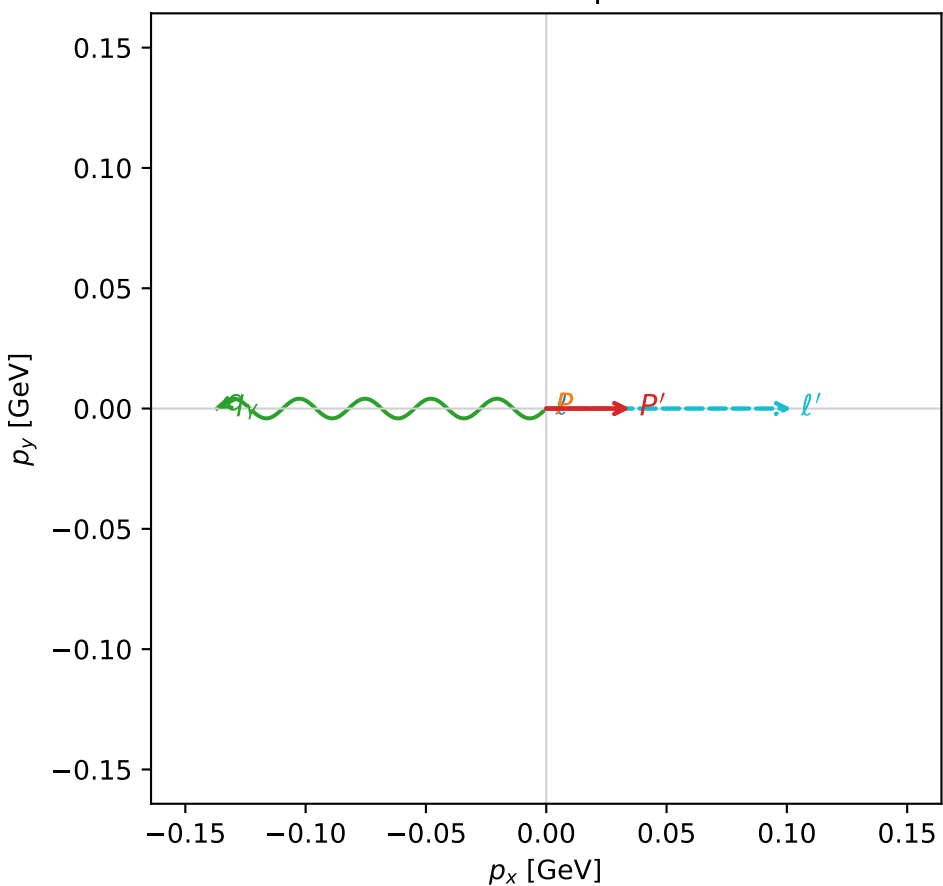
$s=3.14692$, $\sqrt{s}=1.77396$, $|\vec{P}|=0.638989$, $|\vec{P}'|=0.237989$
 $\theta_{p'}=0.147211$, $\theta_V=2.87553$, $\phi_{p'}=3.95953\text{e-}05$, $\phi_V=3.14162$
 $E_V=0.520574$, $\theta_{\ell'}=0.365001$, $\phi_{\ell'}=2.00796\text{e-}05$
 $Q^2=0.706089$, $x_B=0.360311$, $t=-0.136119$, $W^2=2.13342$, $y=0.864401$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6390$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6390$
 P $E= 1.1350$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6390$
 ℓ' $E= 0.2857$ $p_x= 0.1020$ $p_y= 0.0000$ $p_z= 0.2668$
 P' $E= 0.9677$ $p_x= 0.0349$ $p_y= 0.0000$ $p_z= 0.2354$
 q_V $E= 0.5206$ $p_x= -0.1369$ $p_y= -0.0000$ $p_z= -0.5023$

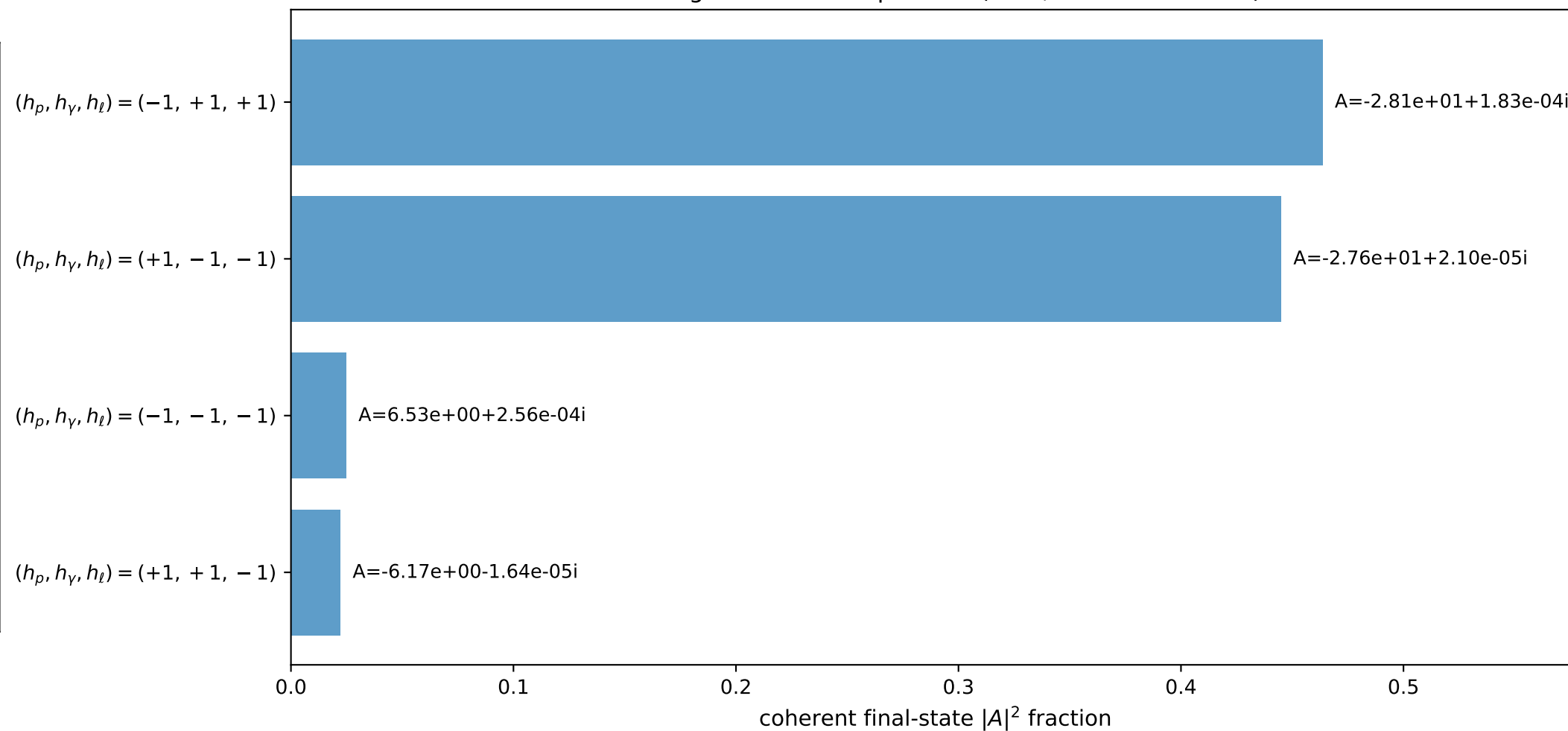
3D momenta



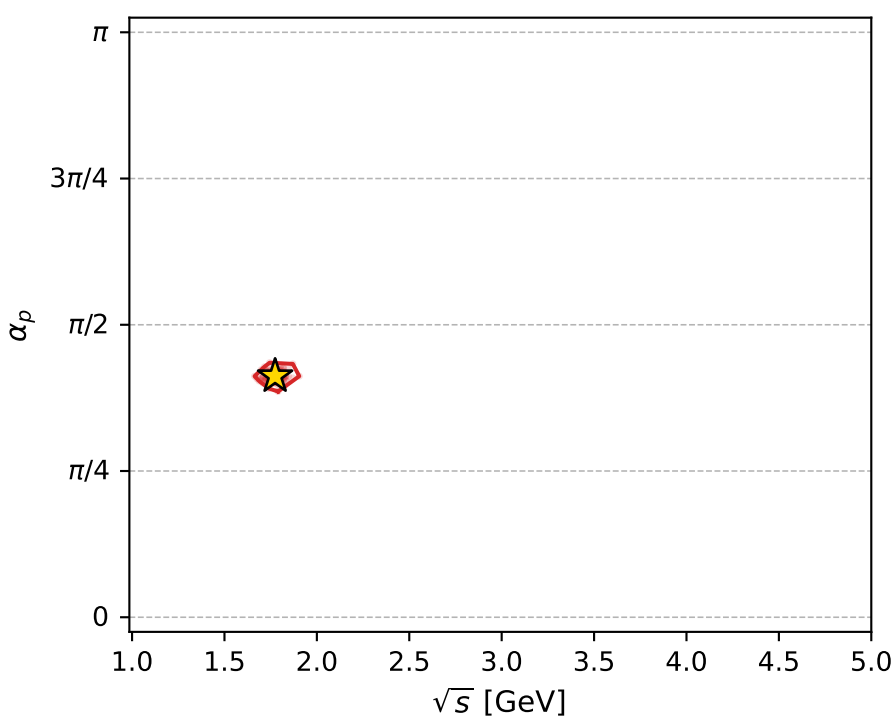
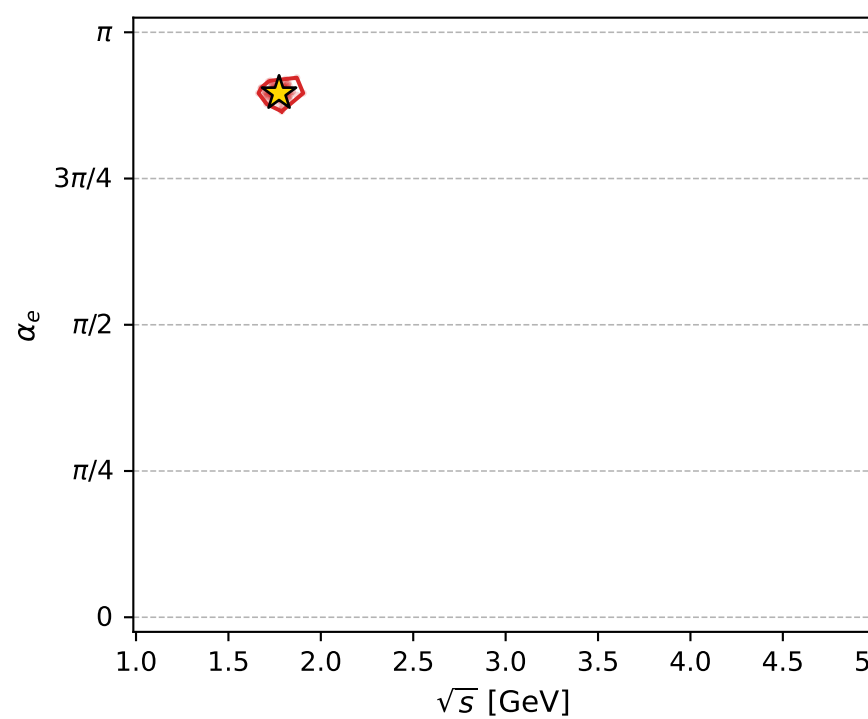
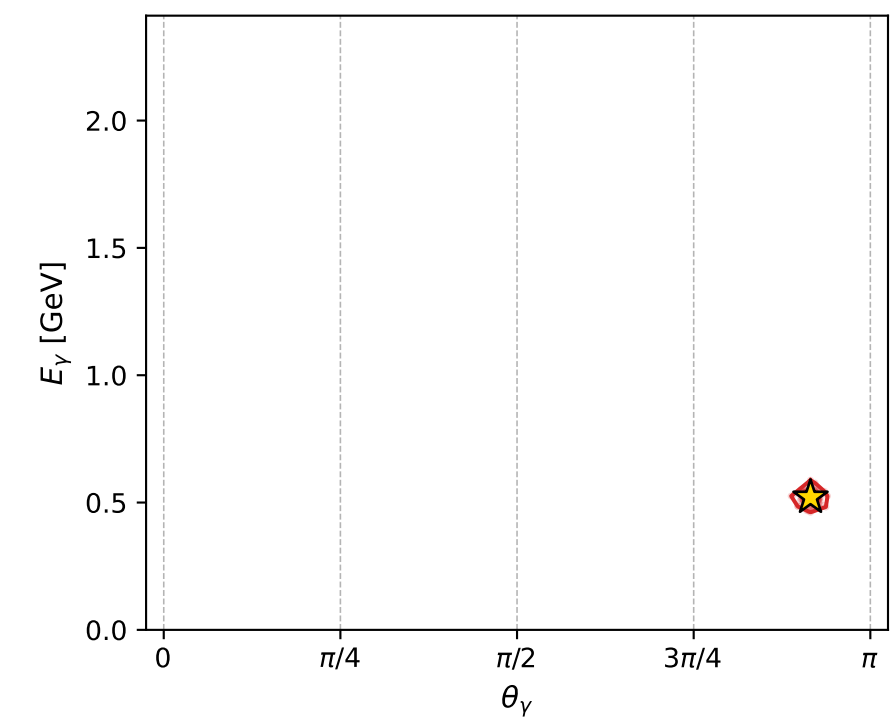
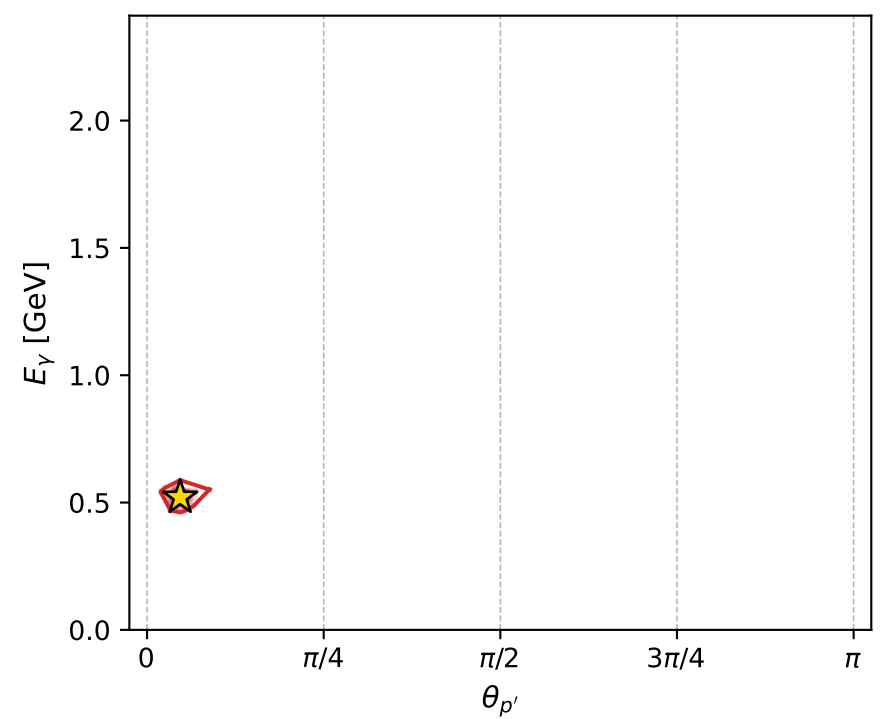
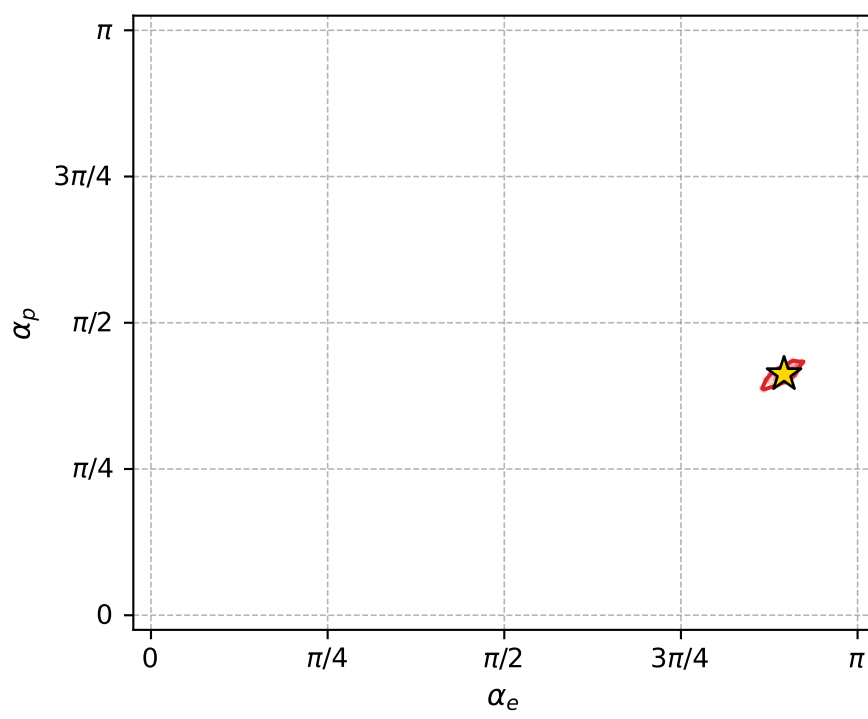
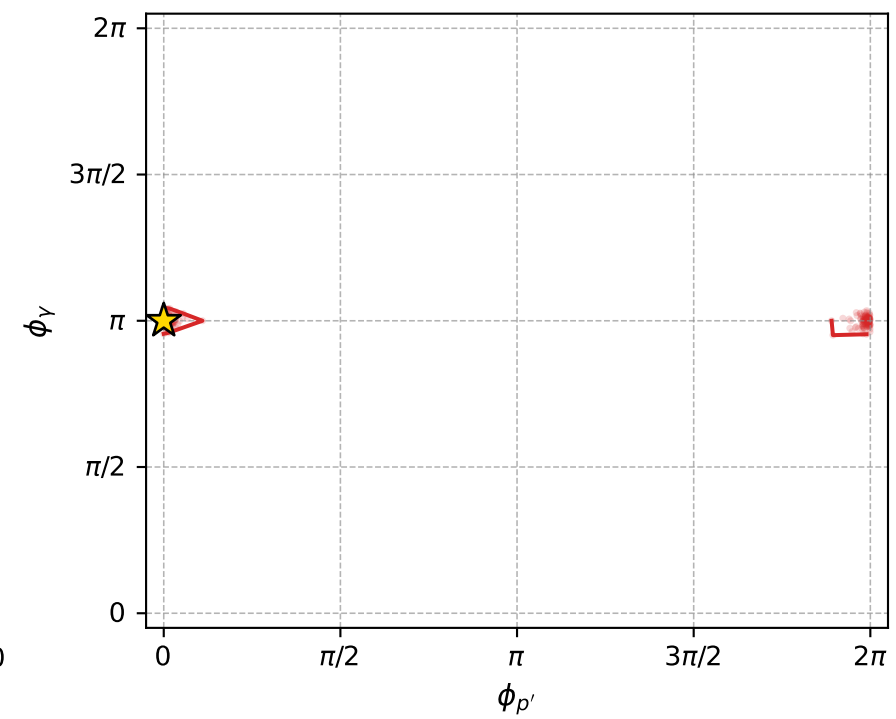
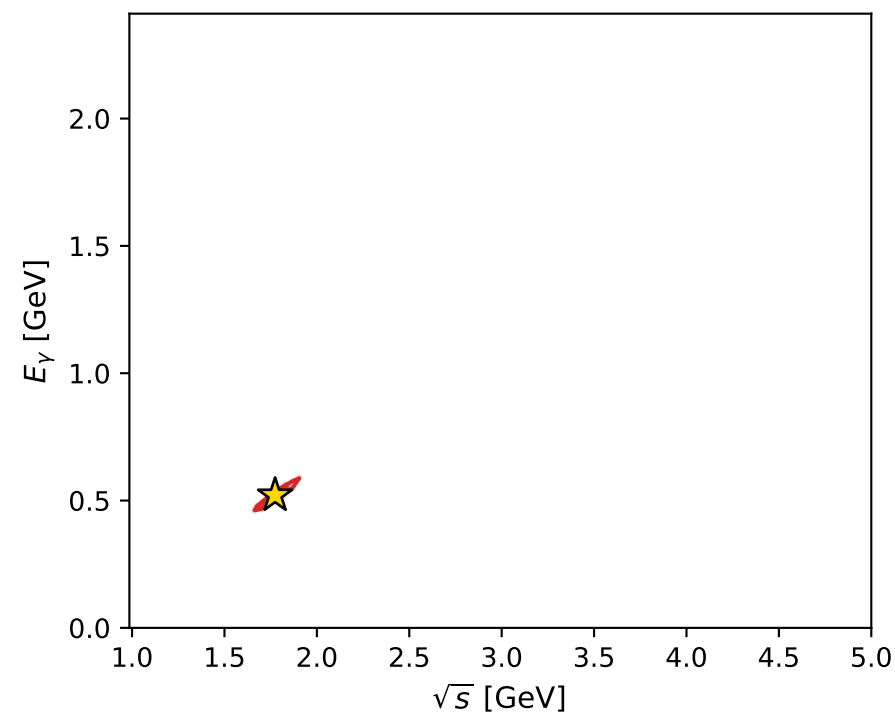
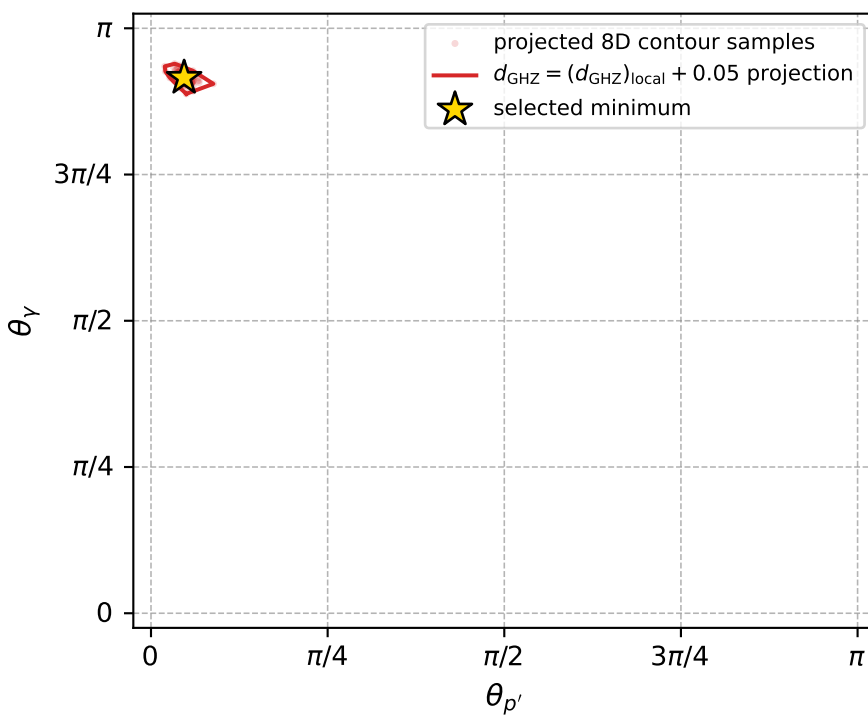
Transverse plane



Leading final-state amplitudes (N=4, retained=95.6%)



electron: polarization cluster P3, local minimum 246; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00011962633$
 $d_{\text{GHZ}} \text{ contour} = 0.050119626$
 above global minimum = 0.00011960405
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.773957$
 $\text{theta_p_out} = 0.1472114$
 $\text{theta_gamma_out} = 2.875533$
 $\text{q0out} = 0.5205745$
 $\text{phi_p_out} = 3.959527\text{e-}05$
 $\text{phi_gamma_out} = 3.141618$
 $\text{alpha_e} = 2.814635$
 $\text{alpha_p} = 1.294656$

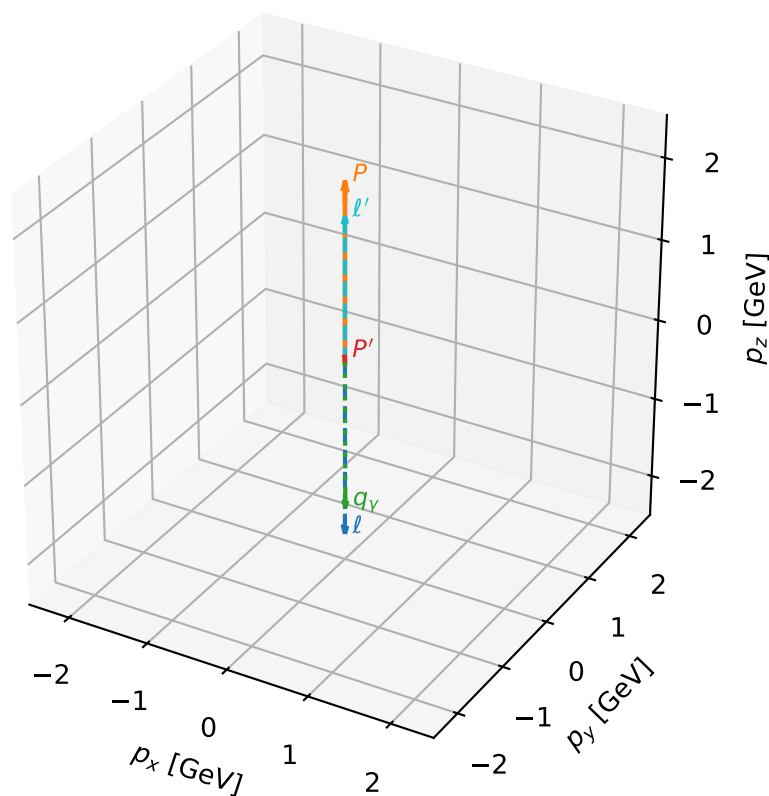
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_248 $d_{\text{GHZ}}=0.000127763$
 region: polarization_cluster_03_local_minimum_248
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.9049644$ rad
 incoming lepton: $-0.972134|+> +0.234426|->$
 proton mixing angle: $\alpha_p=1.3339106$ rad
 incoming proton: $0.234676|+> +0.972074|->$

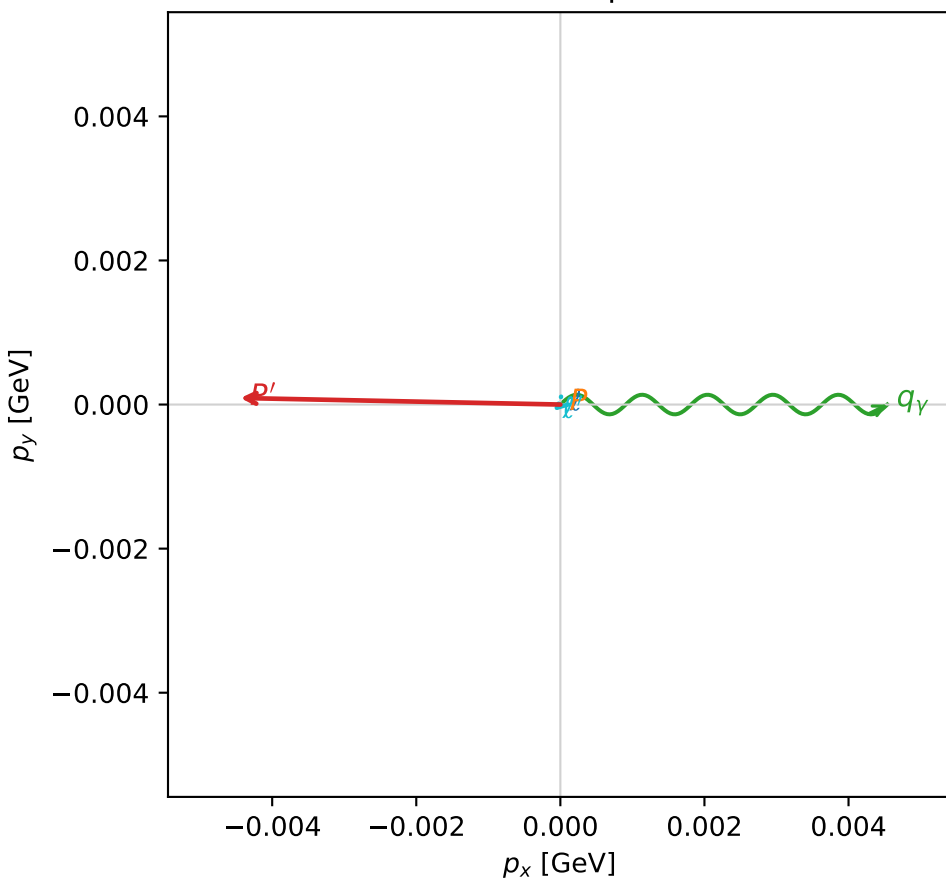
$s=20.8306$, $\sqrt{s}=4.56405$, $|\vec{P}|=2.18564$, $|\vec{P}'|=0.064417$
 $\theta_{p'}=0.0689908$, $\theta_Y=3.13913$, $\phi_{p'}=3.12116$, $\phi_Y=0$
 $E_Y=1.84406$, $\theta_{\ell'}=7.53344\text{e-}05$, $\phi_{\ell'}=3.88476$
 $Q^2=15.5599$, $x_B=0.807696$, $t=-2.43181$, $W^2=4.58449$, $y=0.965605$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1856$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1856$
 P $E= 2.3784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1856$
 ℓ' $E= 1.7798$ $p_x= -0.0001$ $p_y= -0.0001$ $p_z= 1.7798$
 P' $E= 0.9402$ $p_x= -0.0044$ $p_y= 0.0001$ $p_z= 0.0643$
 q_Y $E= 1.8441$ $p_x= 0.0045$ $p_y= 0.0000$ $p_z= -1.8441$

3D momenta



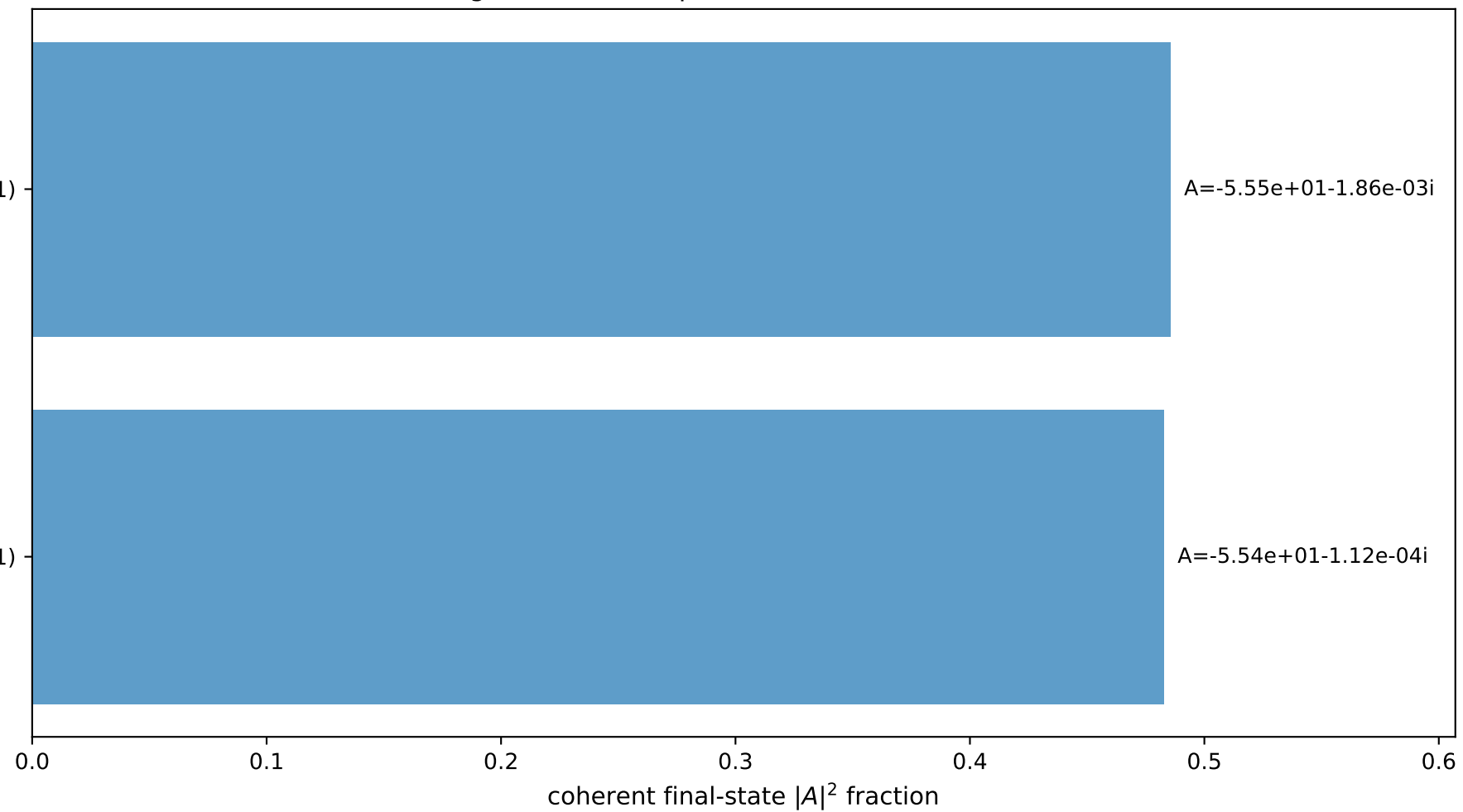
Transverse plane



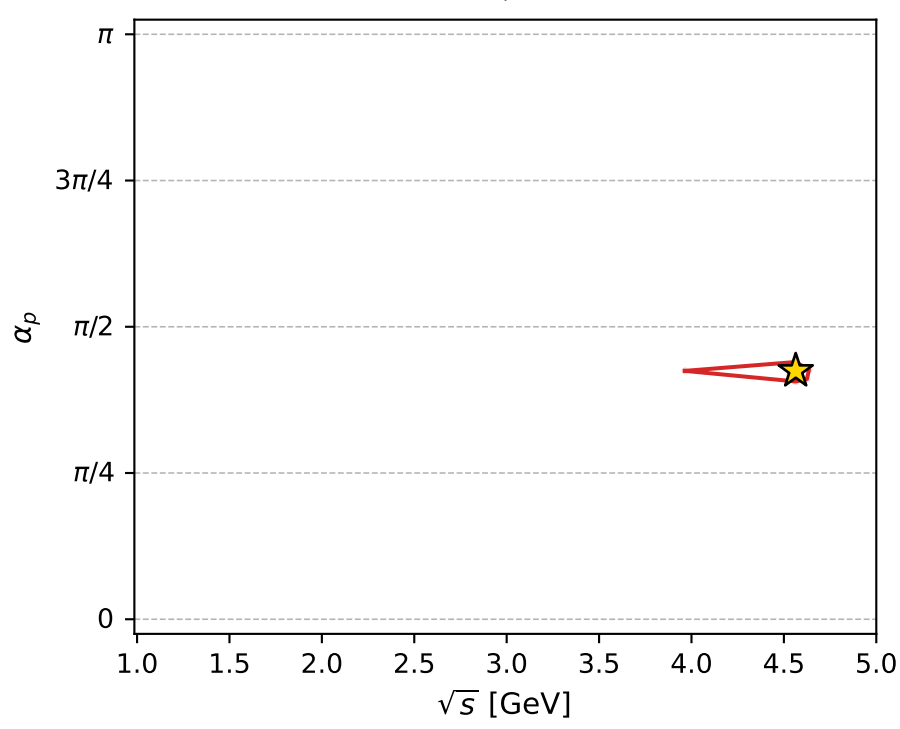
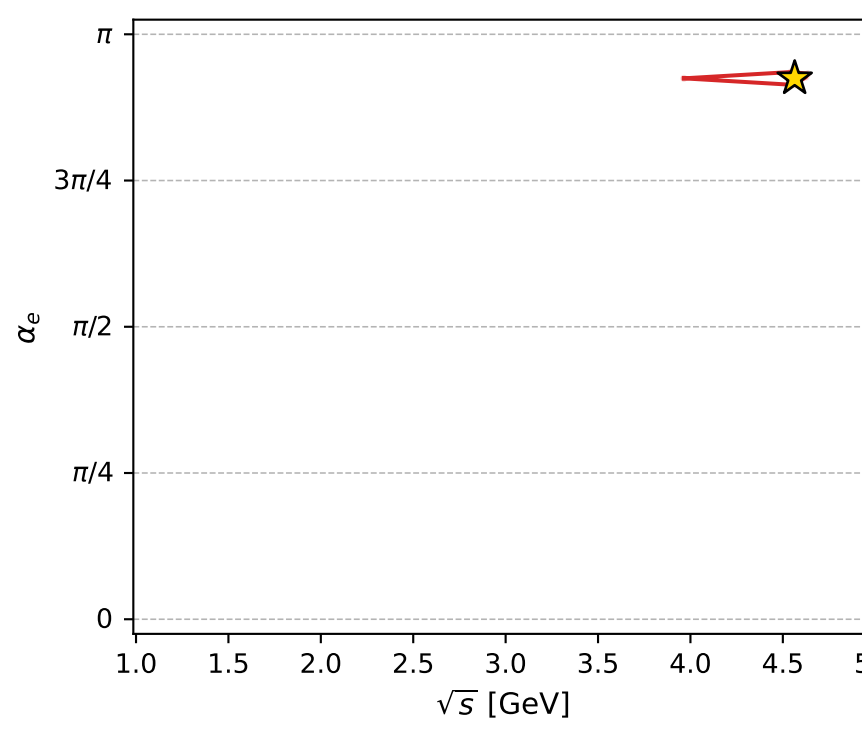
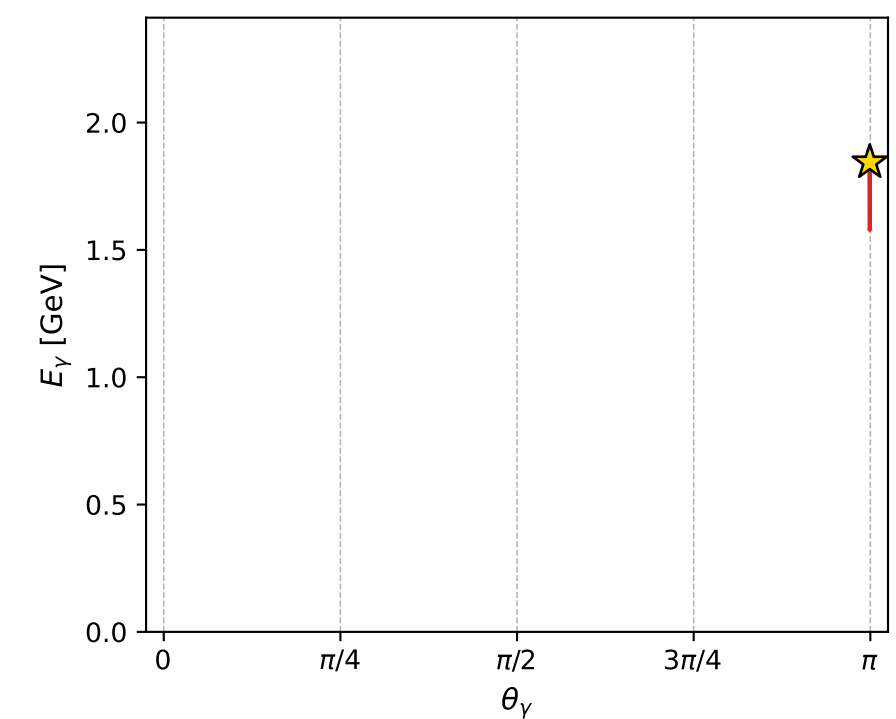
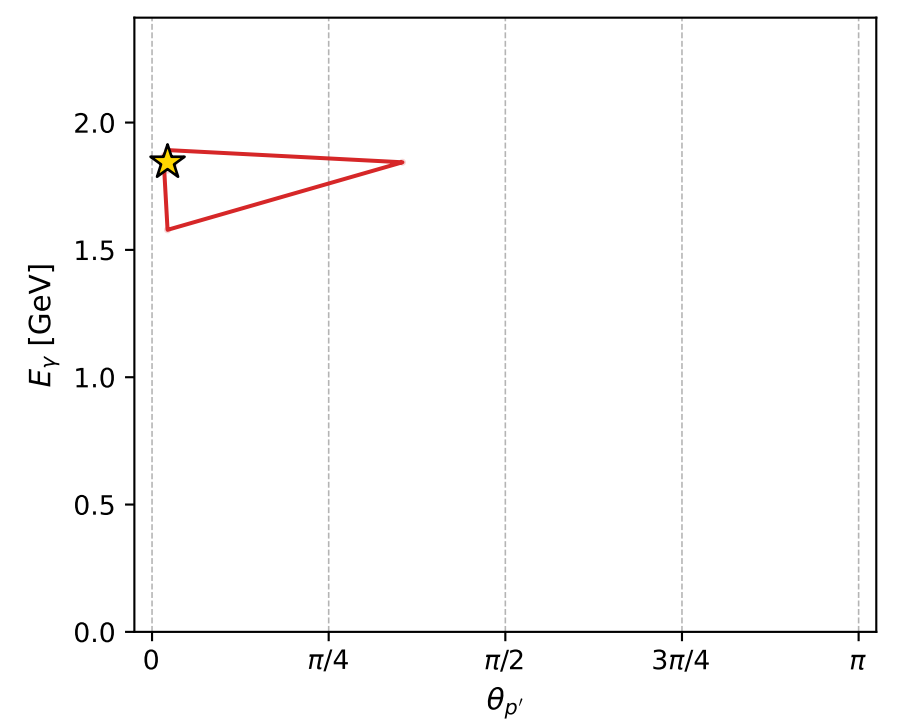
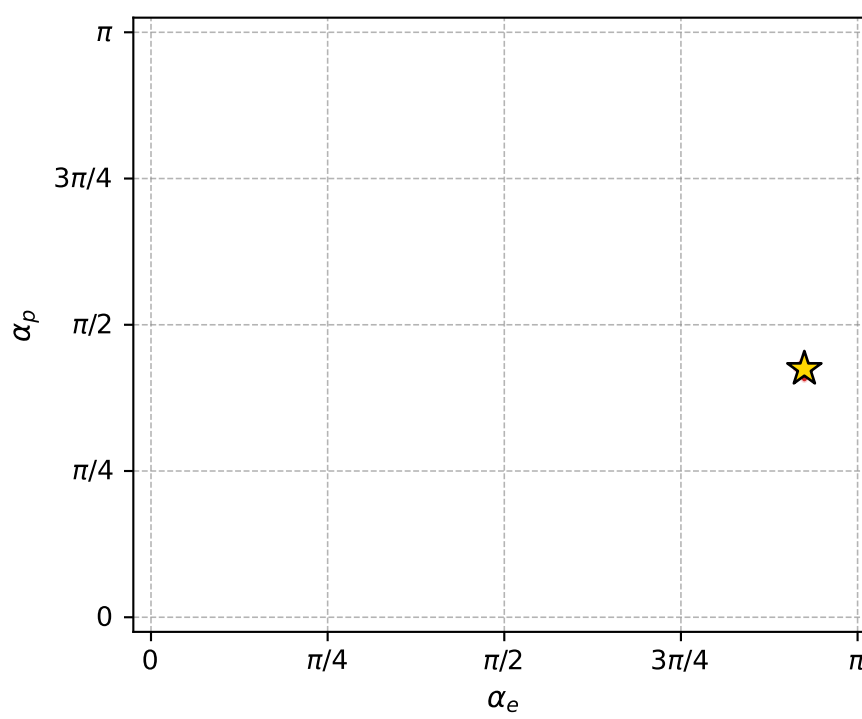
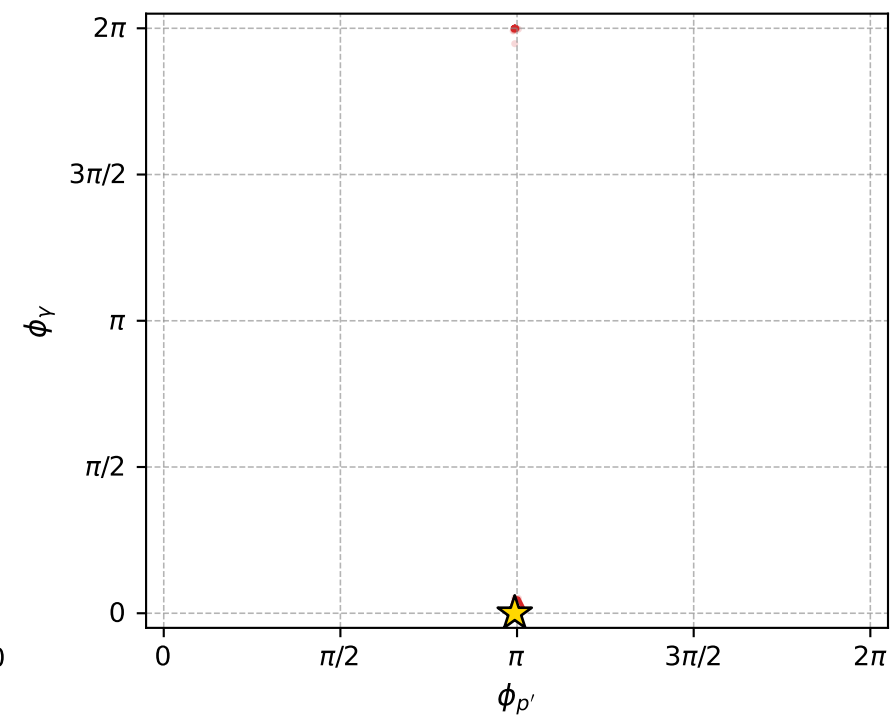
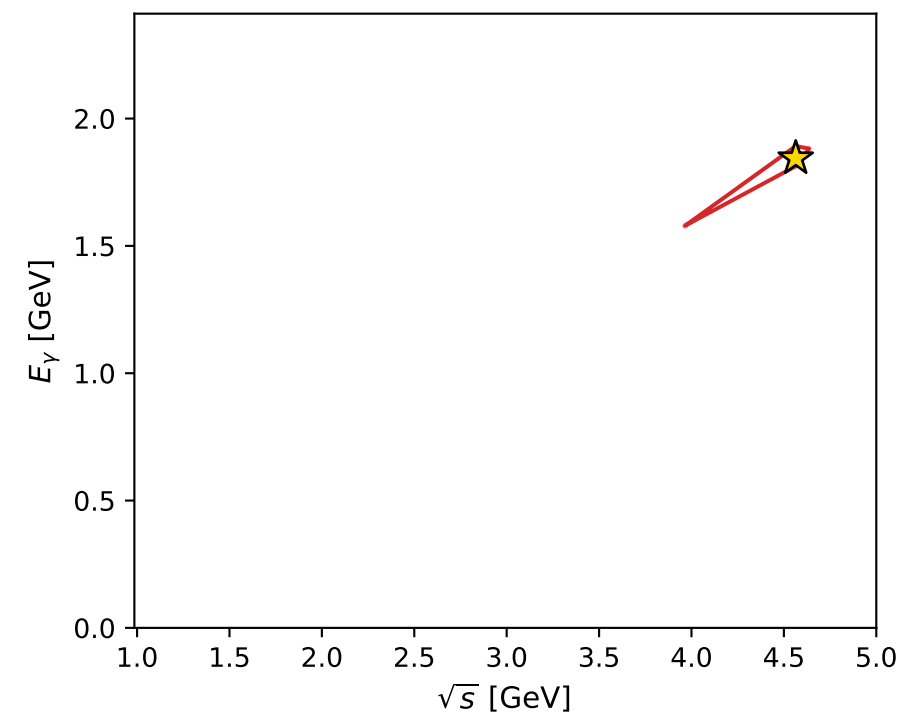
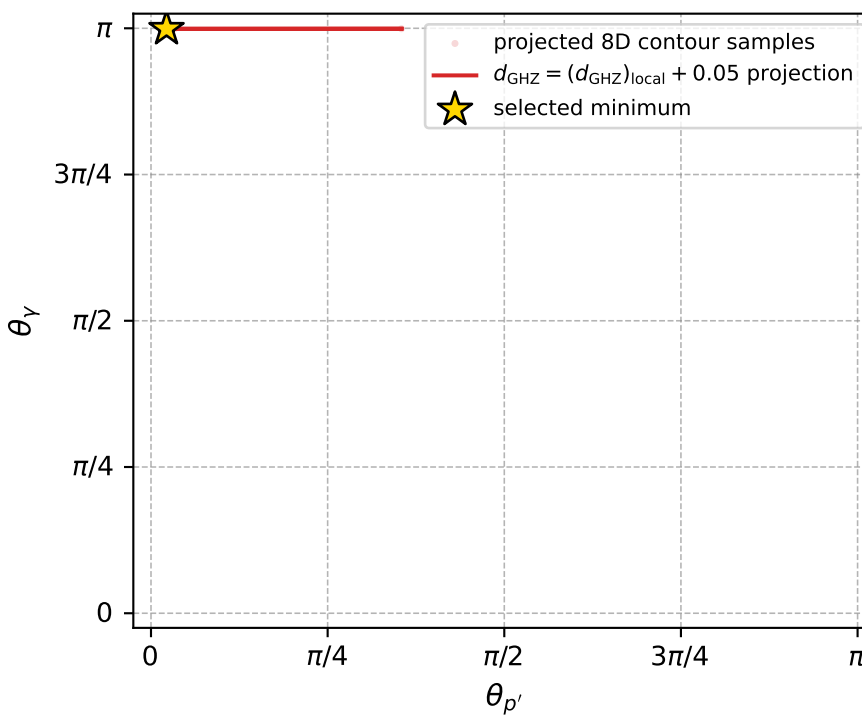
$(h_p, h_Y, h_\ell) = (-1, +1, +1)$

$(h_p, h_Y, h_\ell) = (+1, -1, -1)$

Leading final-state amplitudes (N=2, retained=96.9%)



electron: polarization cluster P3, local minimum 248; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00012776345$
 $d_{\text{GHZ}} \text{ contour} = 0.050127763$
 above global minimum = 0.00012774116
 8D contour samples = 252

selected phase-space point:

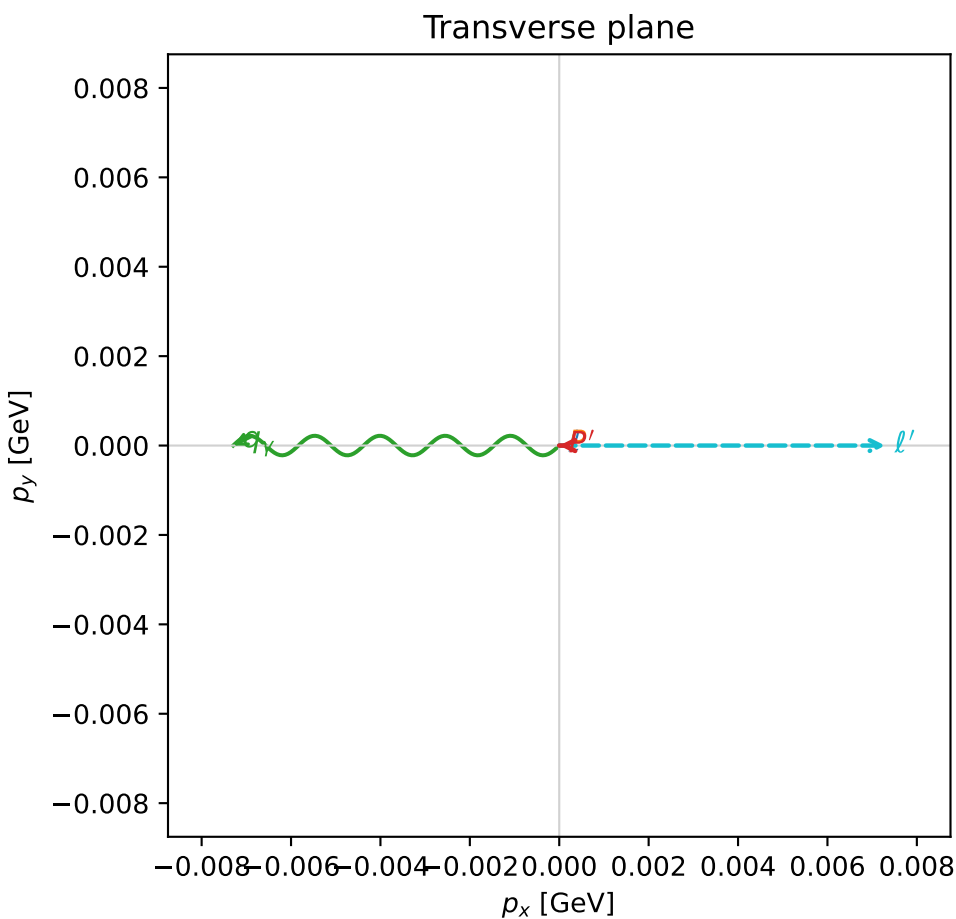
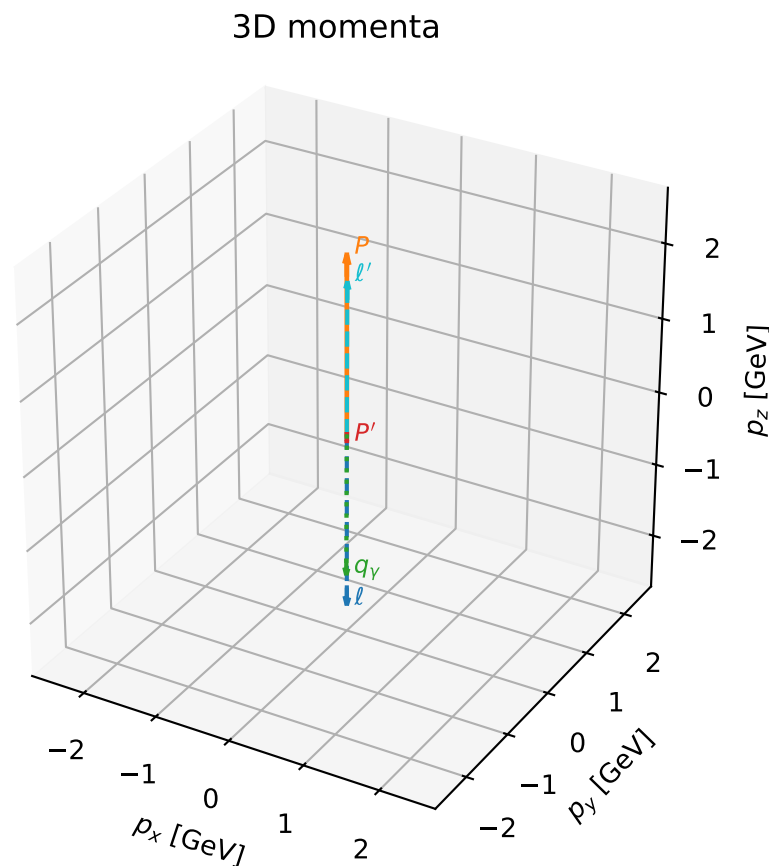
$\text{sqrt_s} = 4.564052$
 $\text{theta_p_out} = 0.06899079$
 $\text{theta_gamma_out} = 3.139132$
 $\text{q0out} = 1.844056$
 $\text{phi_p_out} = 3.121162$
 $\text{phi_gamma_out} = 0$
 $\text{alpha_e} = 2.904964$
 $\text{alpha_p} = 1.333911$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_262 $d_{\text{GHZ}}=0.000177709$
 region: polarization_cluster_03_local_minimum_262
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.6990512$ rad
 incoming lepton: $-0.127904|+> +0.991787|->$
 proton mixing angle: $\alpha_p=0.12870773$ rad
 incoming proton: $0.991729|+> +0.128353|->$

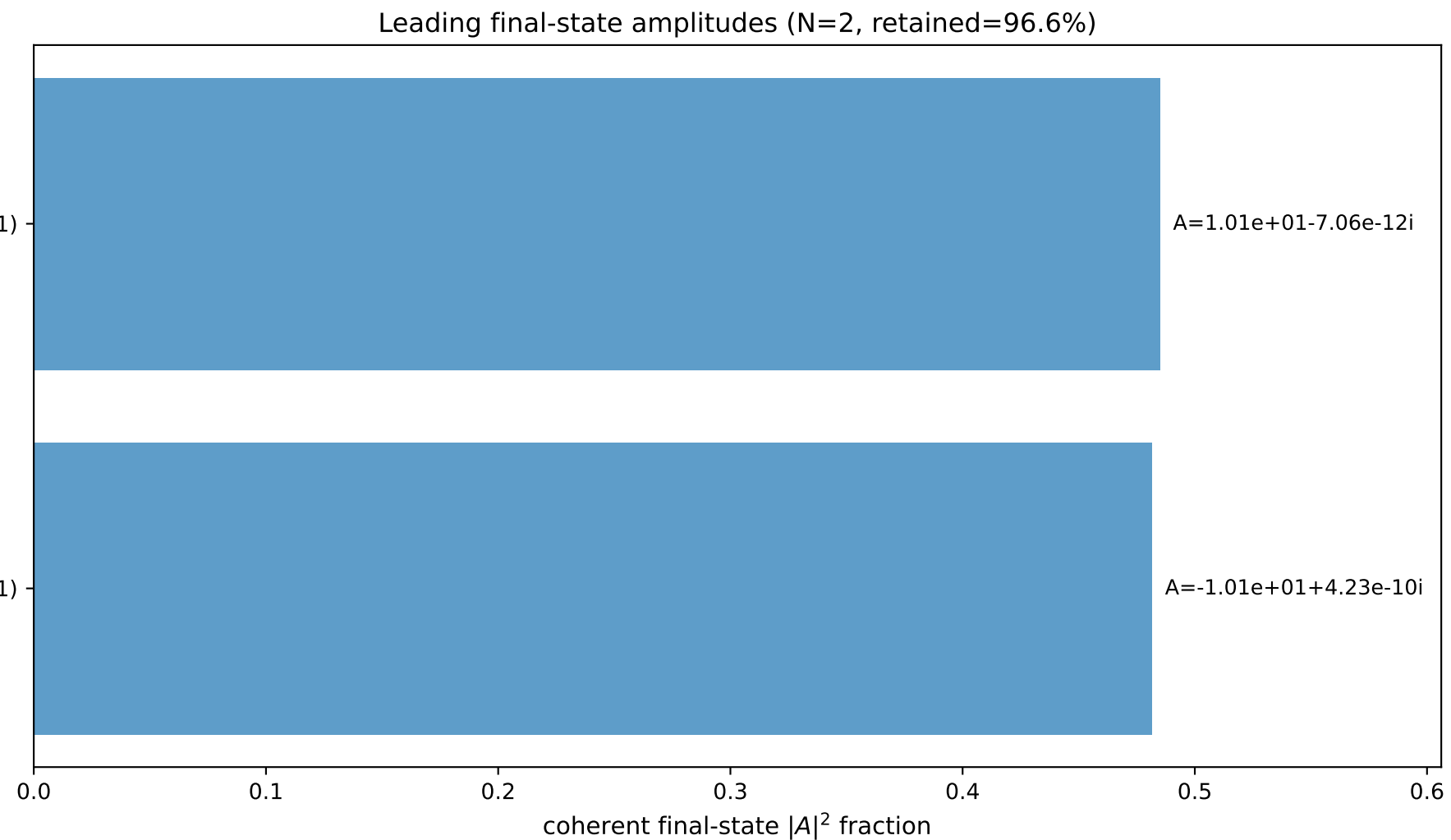
$s=24.2413$, $\sqrt{s}=4.92355$, $|\vec{P}|=2.37242$, $|\vec{P}'|=0.0876034$
 $\theta_{P'}=3.14159$, $\theta_Y=3.13785$, $\phi_{P'}=3.00433$, $\phi_Y=3.14156$
 $E_Y=1.94693$, $\theta_{\ell'}=0.00358498$, $\phi_{\ell'}=6.28316$
 $Q^2=19.307$, $x_B=0.853001$, $t=-3.46271$, $W^2=4.20707$, $y=0.968871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3724$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3724$
 P $E= 2.5511$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3724$
 ℓ' $E= 2.0345$ $p_x= 0.0073$ $p_y= -0.0000$ $p_z= 2.0345$
 P' $E= 0.9421$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0876$
 q_Y $E= 1.9469$ $p_x= -0.0073$ $p_y= 0.0000$ $p_z= -1.9469$

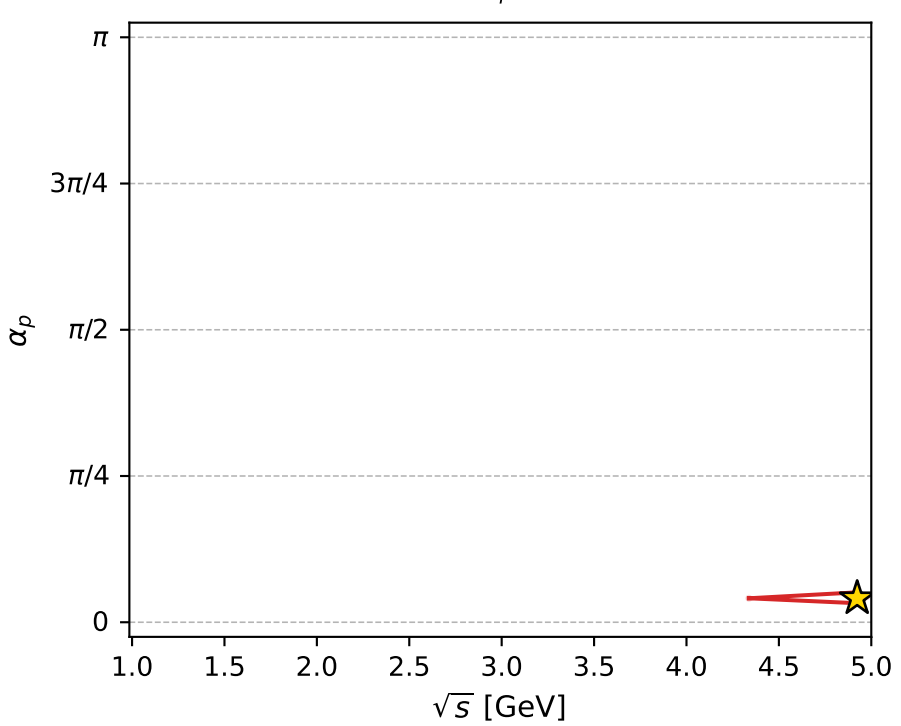
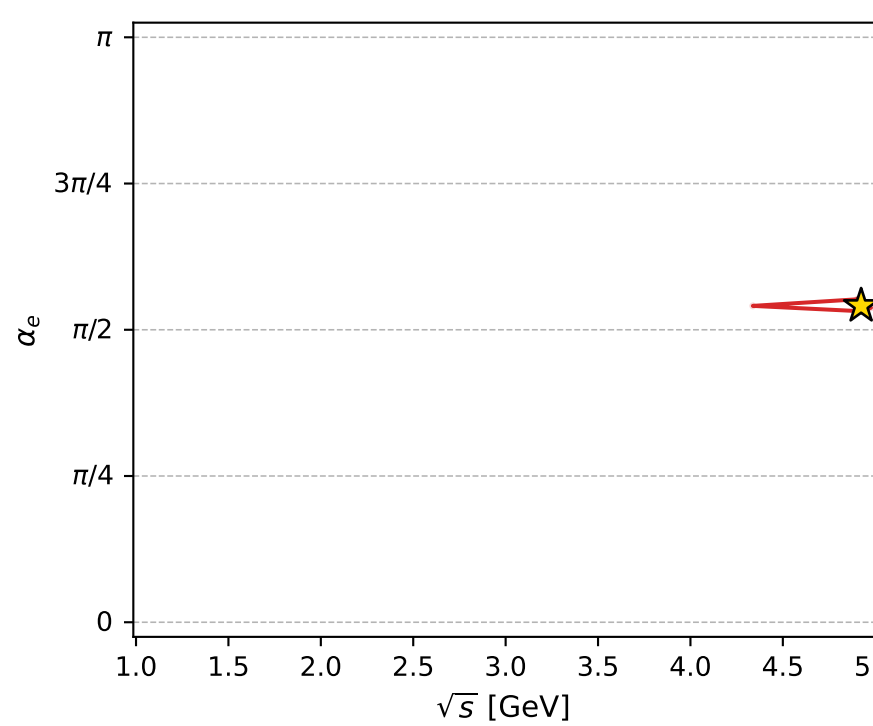
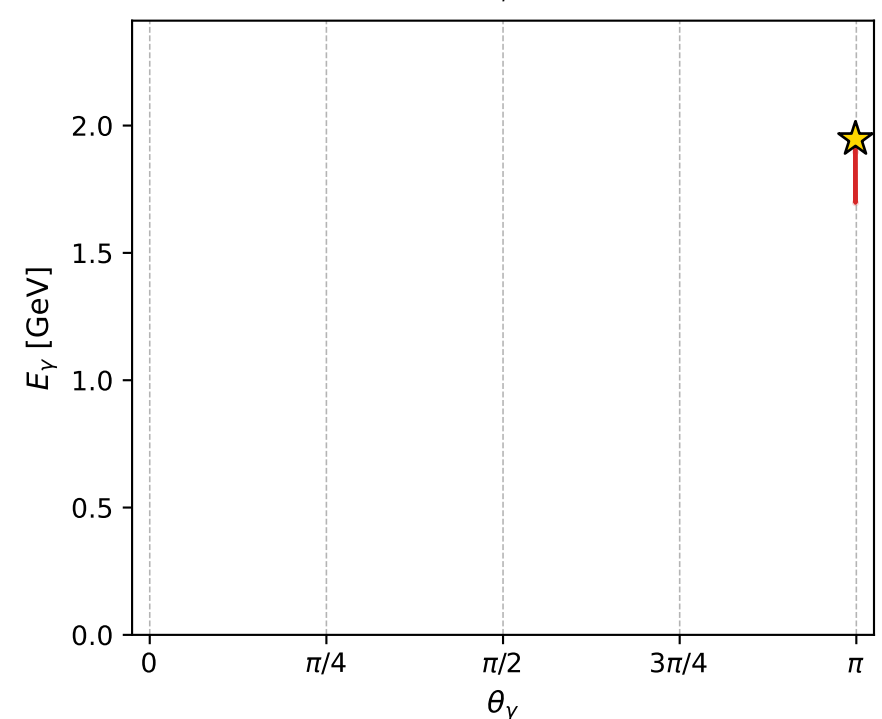
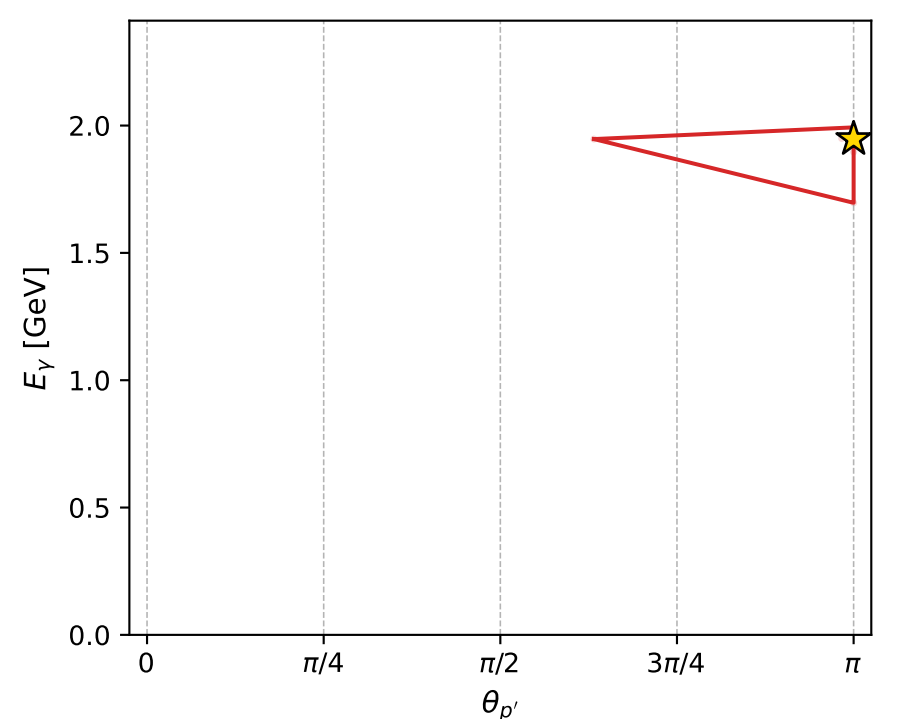
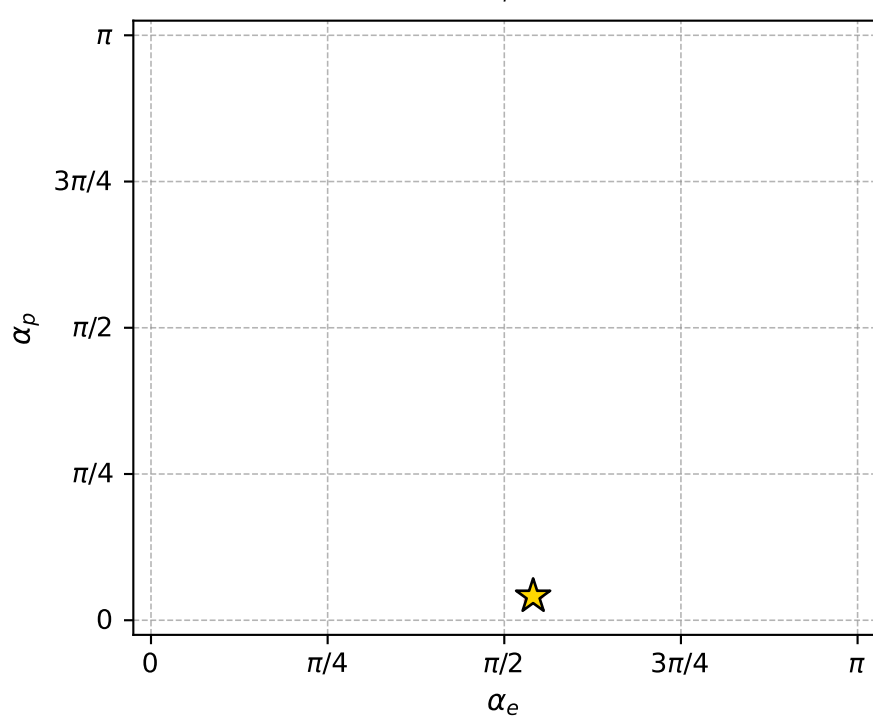
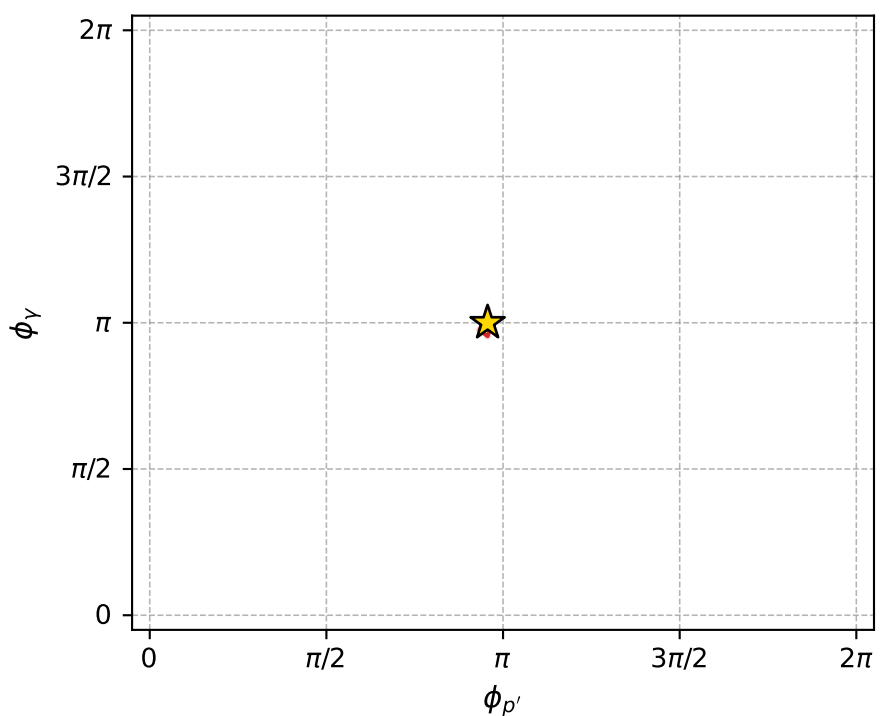
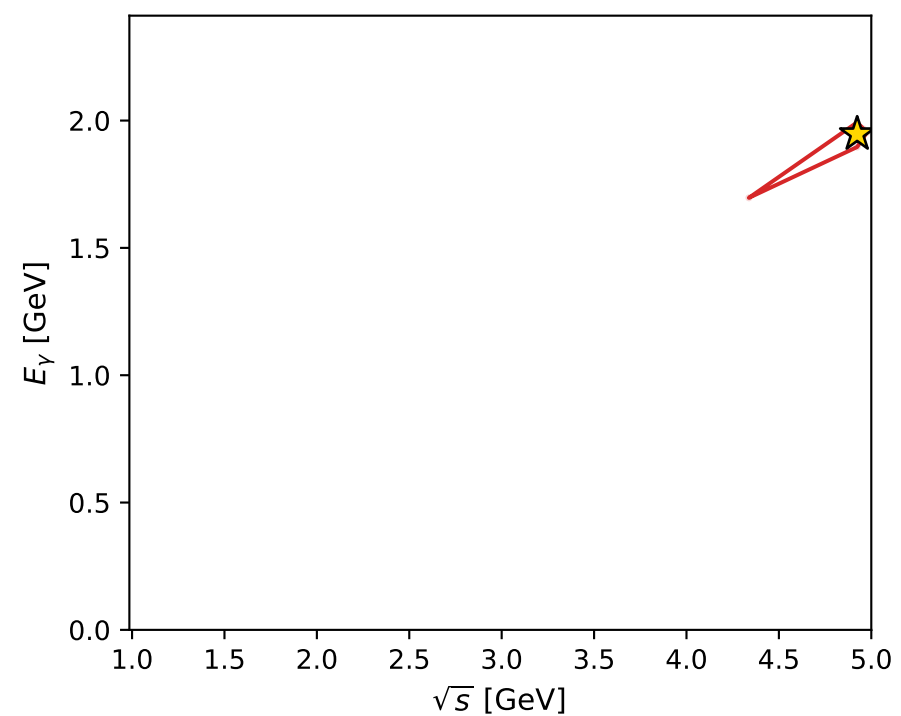
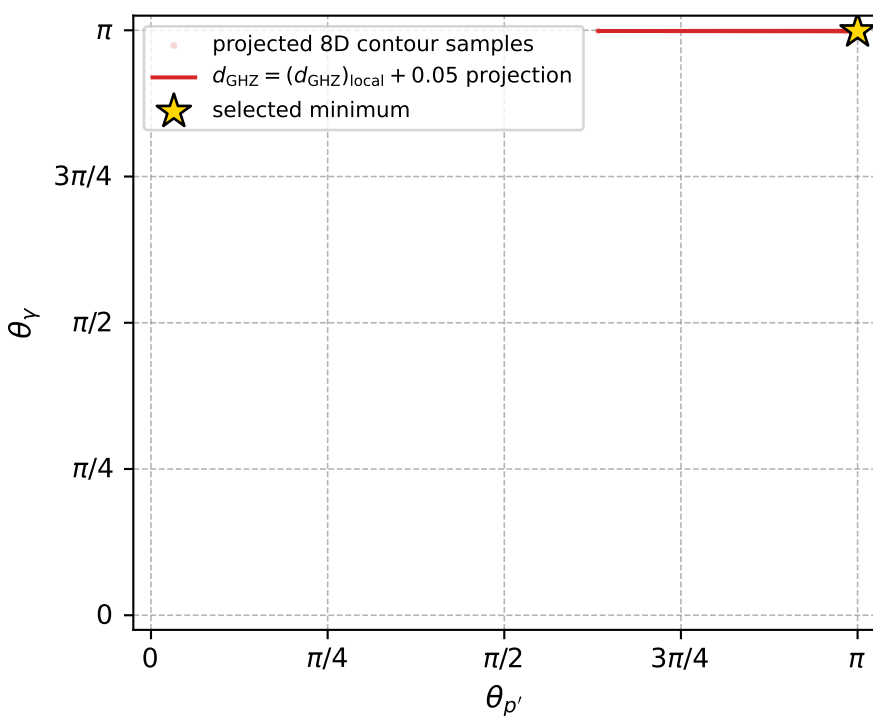


$(h_p, h_Y, h_{\ell}) = (-1, -1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, +1)$



electron: polarization cluster P3, local minimum 262; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00017770893$
 $d_{\text{GHZ}} \text{ contour} = 0.050177709$
 above global minimum = 0.00017768664
 8D contour samples = 128

selected phase-space point:

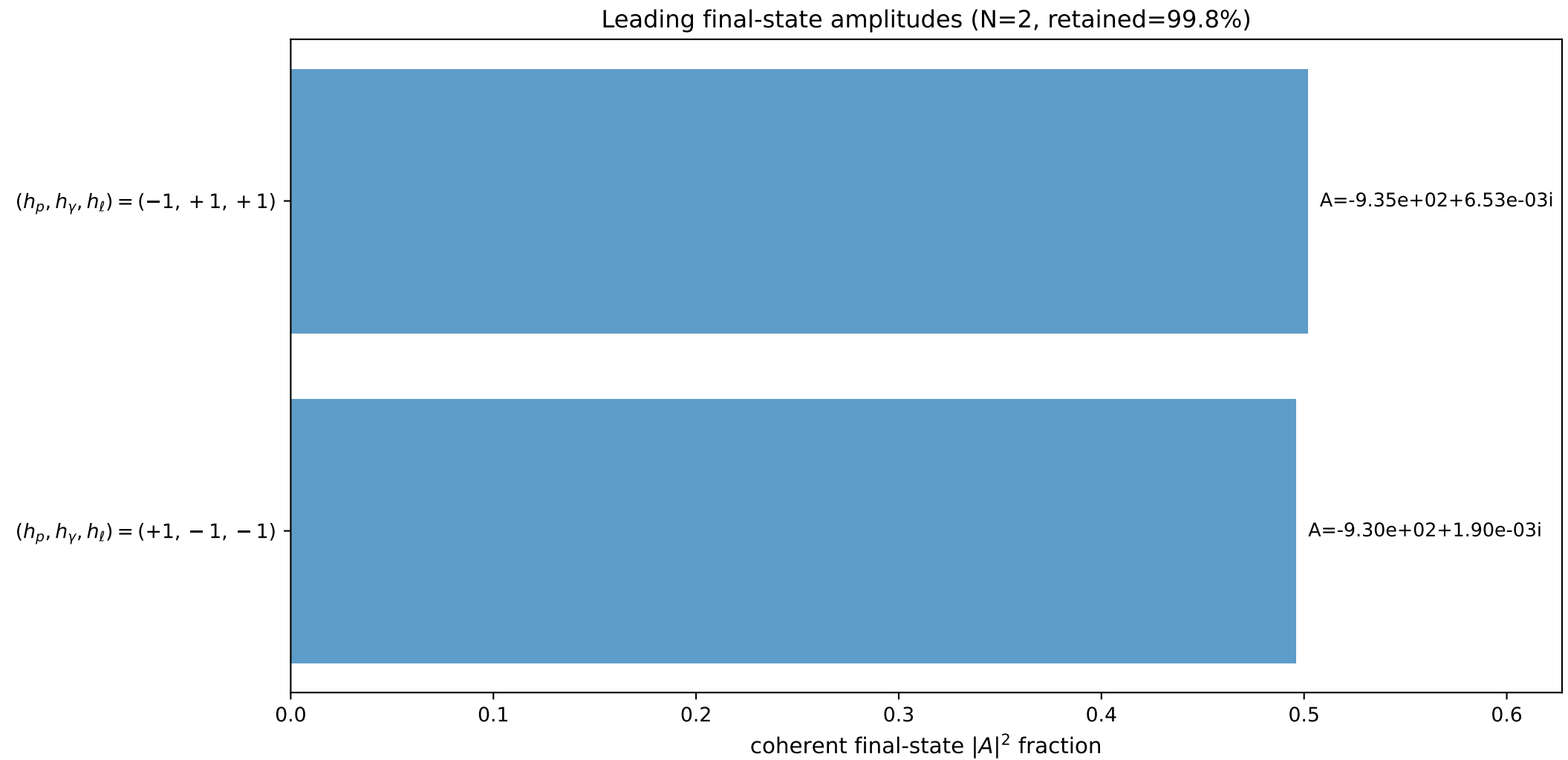
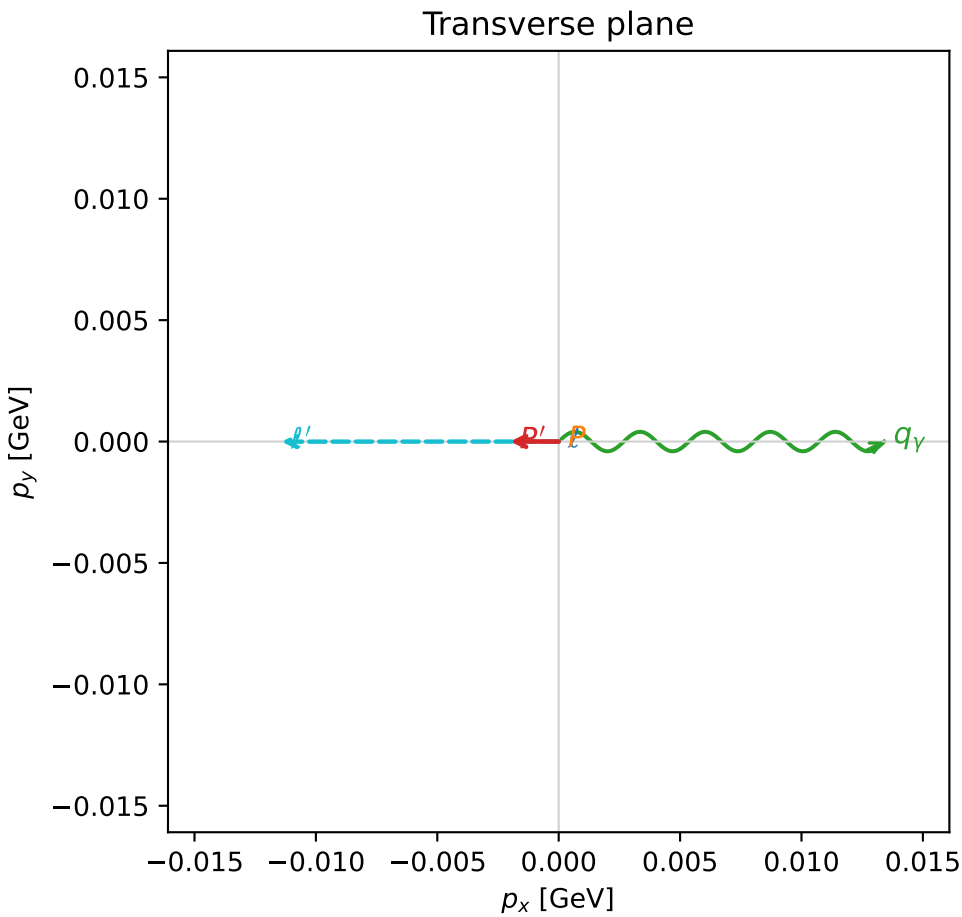
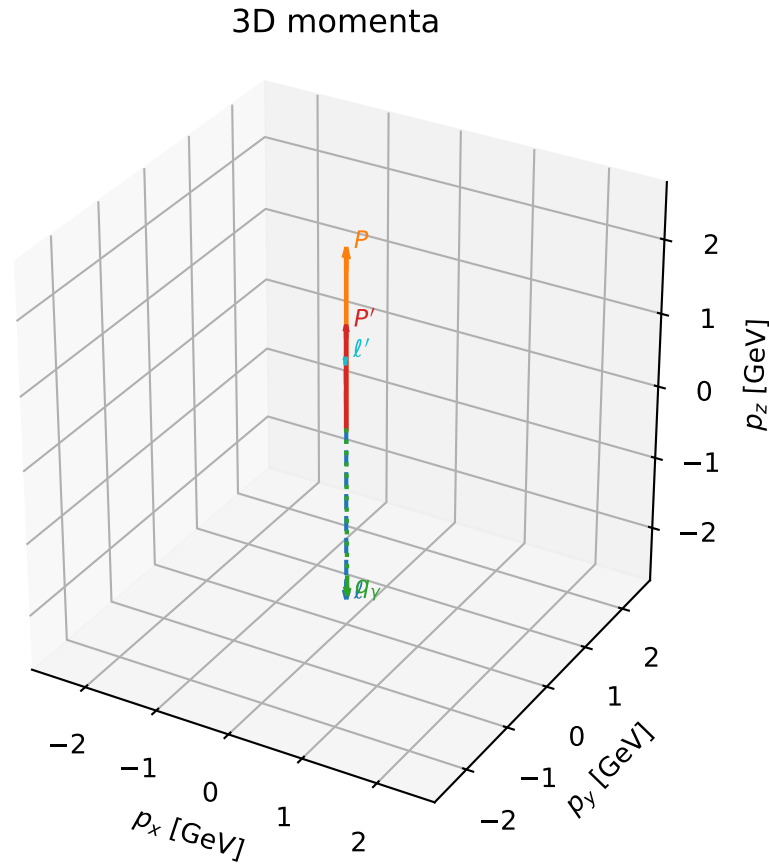
$\text{sqrt_s} = 4.923547$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.137846$
 $\text{q0out} = 1.946931$
 $\text{phi_p_out} = 3.004329$
 $\text{phi_gamma_out} = 3.141565$
 $\text{alpha_e} = 1.699051$
 $\text{alpha_p} = 0.1287077$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

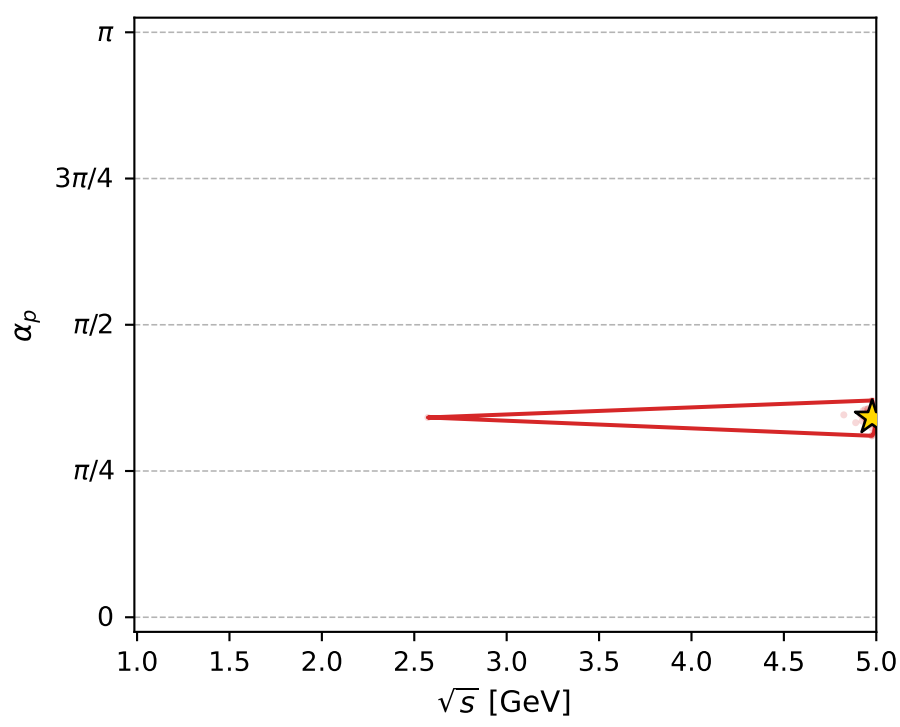
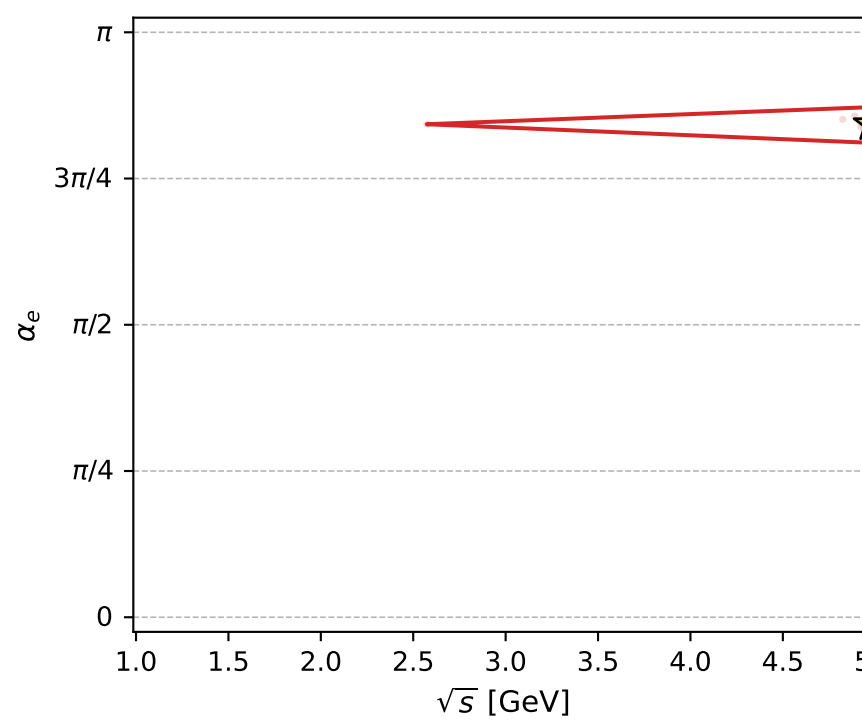
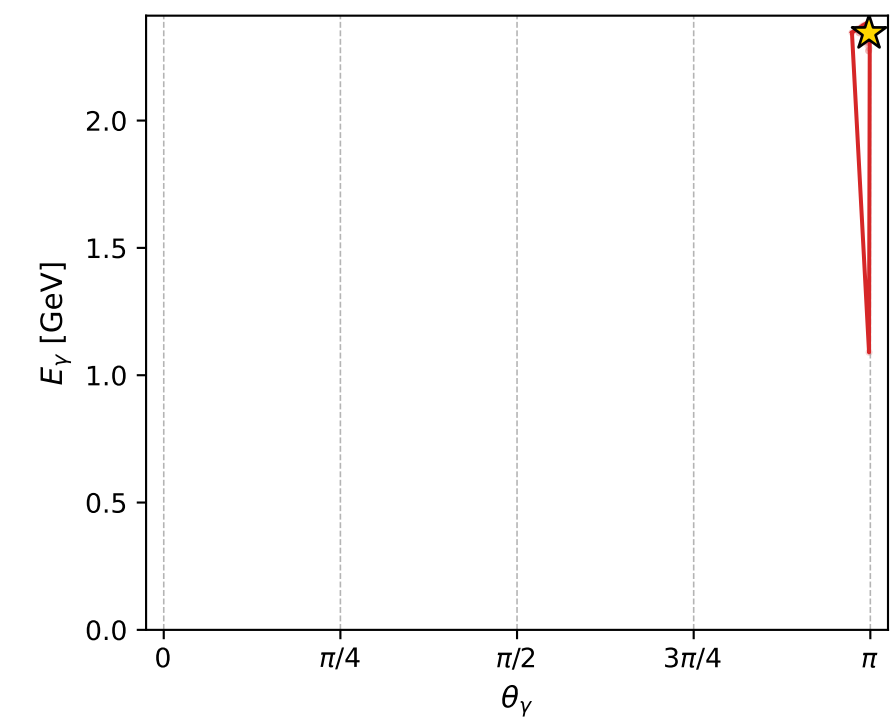
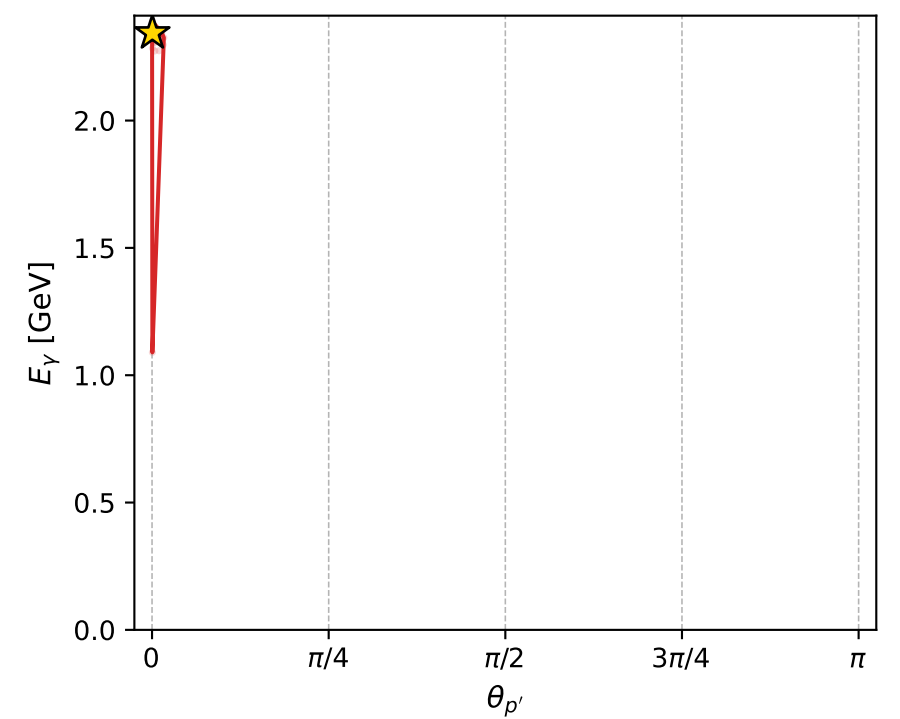
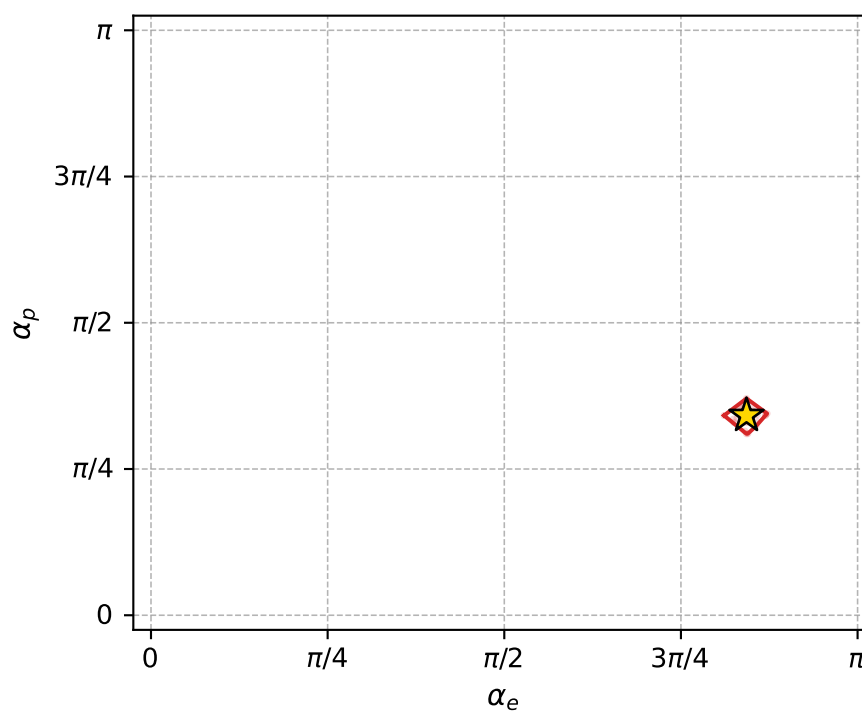
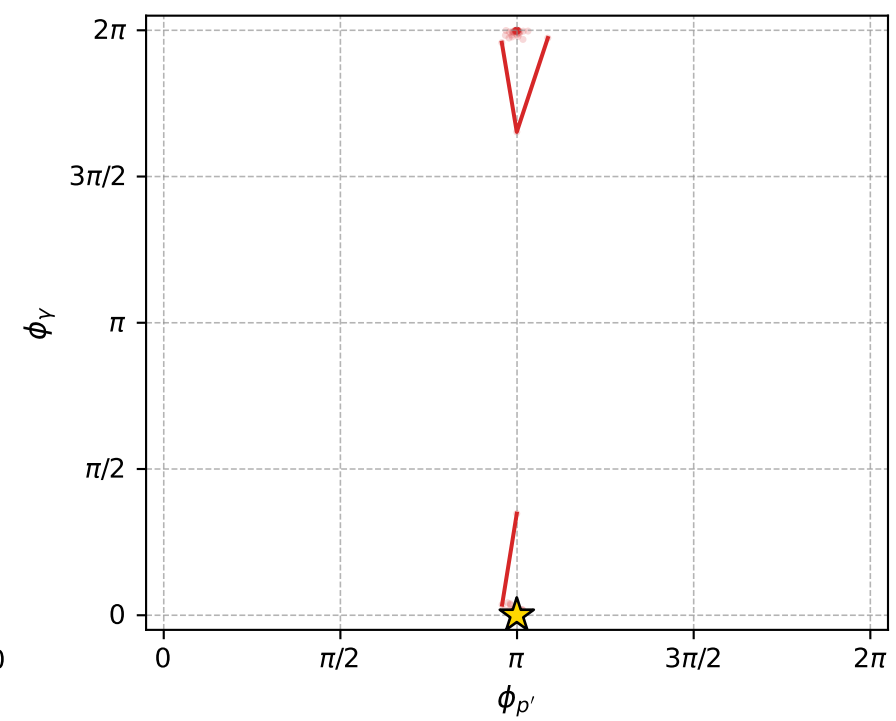
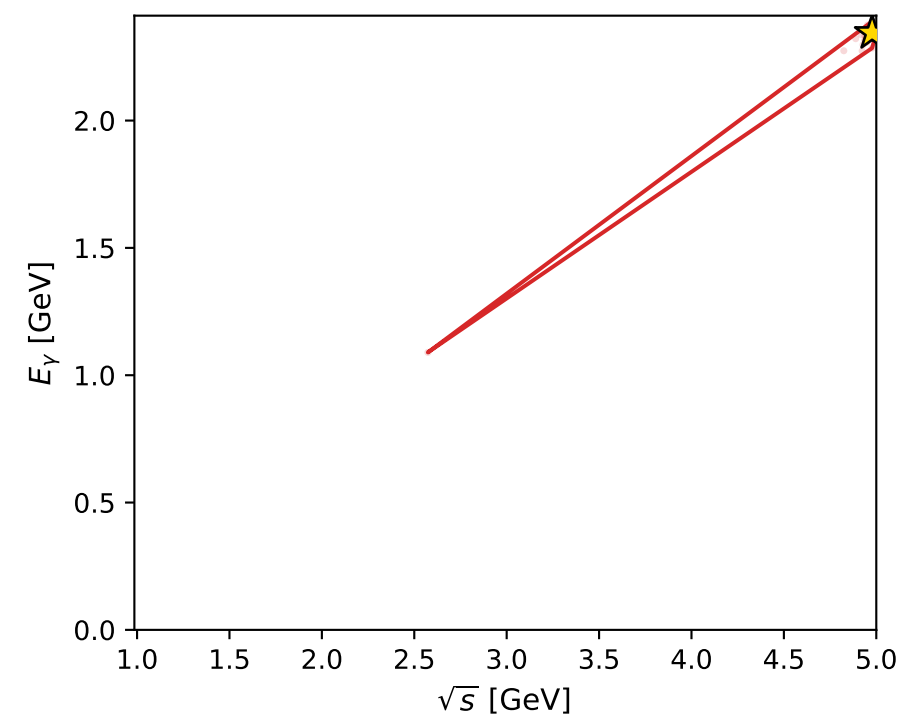
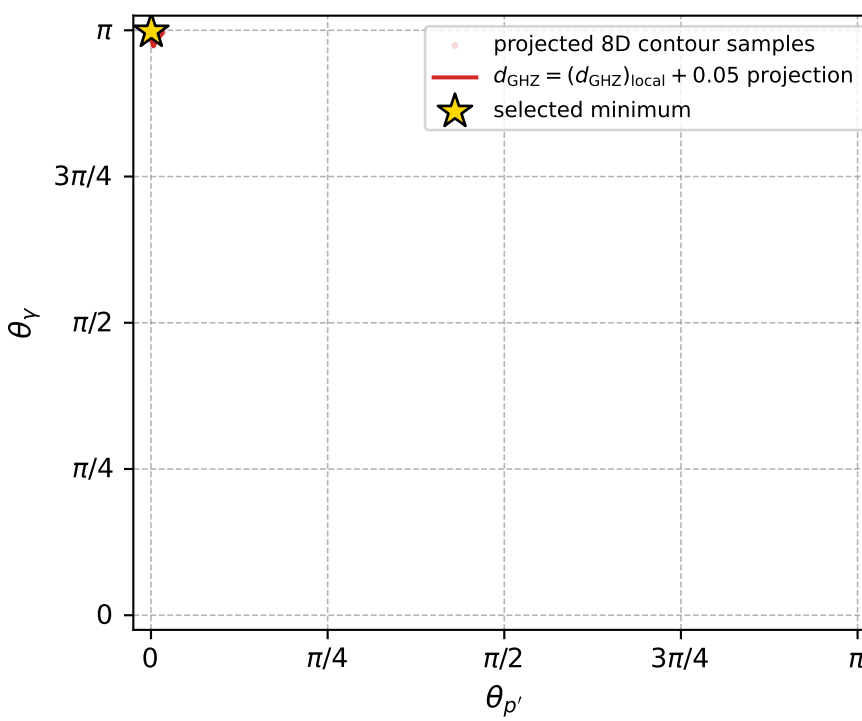
dghz_mixing_angles_polarization_03_local_minimum_267 $d_{\{\text{rm GHZ}\}}=0.000191782$
 region: polarization_cluster_03_local_minimum_267
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.647731$ rad
 incoming lepton: $-0.880509|+\rangle +0.47403|-\rangle$
 proton mixing angle: $\alpha_p=1.0728653$ rad
 incoming proton: $0.477609|+\rangle +0.878573|-\rangle$

$s=24.7744$, $\sqrt{s}=4.97739$, $|\vec{P}|=2.40031$, $|\vec{P}'|=1.37624$
 $\theta_{P'}=0.00145035$, $\theta_Y=3.13587$, $\phi_{P'}=3.13934$, $\phi_Y=0$
 $E_Y=2.34405$, $\theta_{l'}=0.0117929$, $\phi_{l'}=3.14199$
 $Q^2=9.29214$, $x_B=0.394536$, $t=-0.217746$, $W^2=15.1398$, $y=0.985667$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4003$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4003$
 P $E= 2.5771$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4003$
 l' $E= 0.9678$ $p_x= -0.0114$ $p_y= -0.0000$ $p_z= 0.9678$
 P' $E= 1.6655$ $p_x= -0.0020$ $p_y= 0.0000$ $p_z= 1.3762$
 q_Y $E= 2.3441$ $p_x= 0.0134$ $p_y= 0.0000$ $p_z= -2.3440$



electron: polarization cluster P3, local minimum 267; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00019178161$
 $d_{\text{GHZ}} \text{ contour} = 0.050191782$
 above global minimum = 0.00019175932
 8D contour samples = 123

selected phase-space point:

$\text{sqrt}_s = 4.977389$
 $\text{theta}_p_{\text{out}} = 0.001450348$
 $\text{theta}_\gamma_{\text{out}} = 3.135872$
 $q_{\text{out}} = 2.344051$
 $\text{phi}_p_{\text{out}} = 3.139336$
 $\text{phi}_\gamma_{\text{out}} = 0$
 $\alpha_e = 2.647731$
 $\alpha_p = 1.072865$

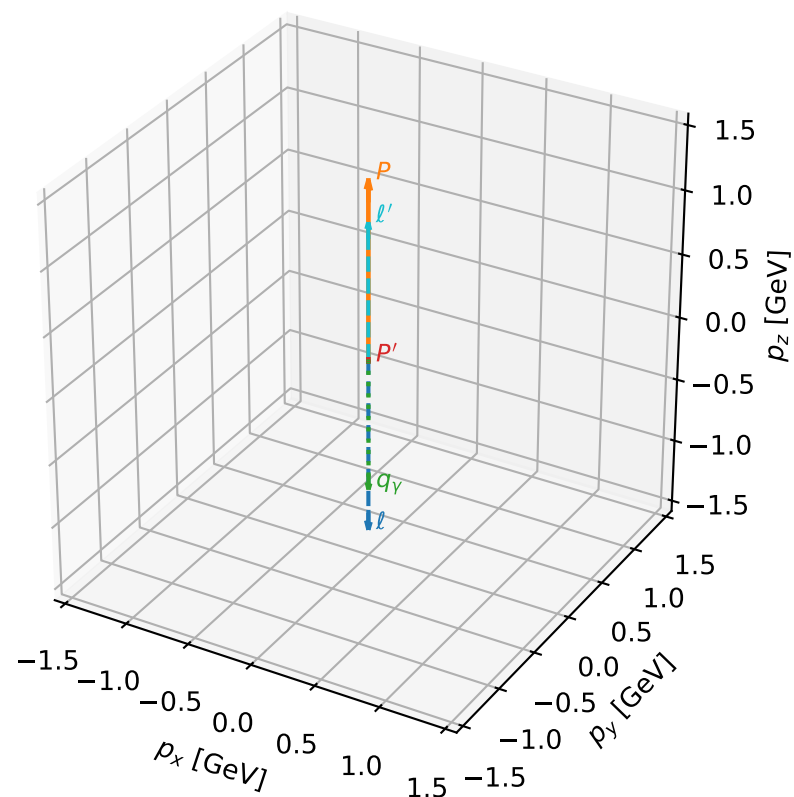
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_278 $d_{\{\text{rm GHZ}\}}=0.000223606$
 region: polarization_cluster_03_local_minimum_278
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.7176681$ rad
 incoming lepton: $-0.911482|+> +0.411341|->$
 proton mixing angle: $\alpha_p=1.1456097$ rad
 incoming proton: $0.412491|+> +0.910962|->$

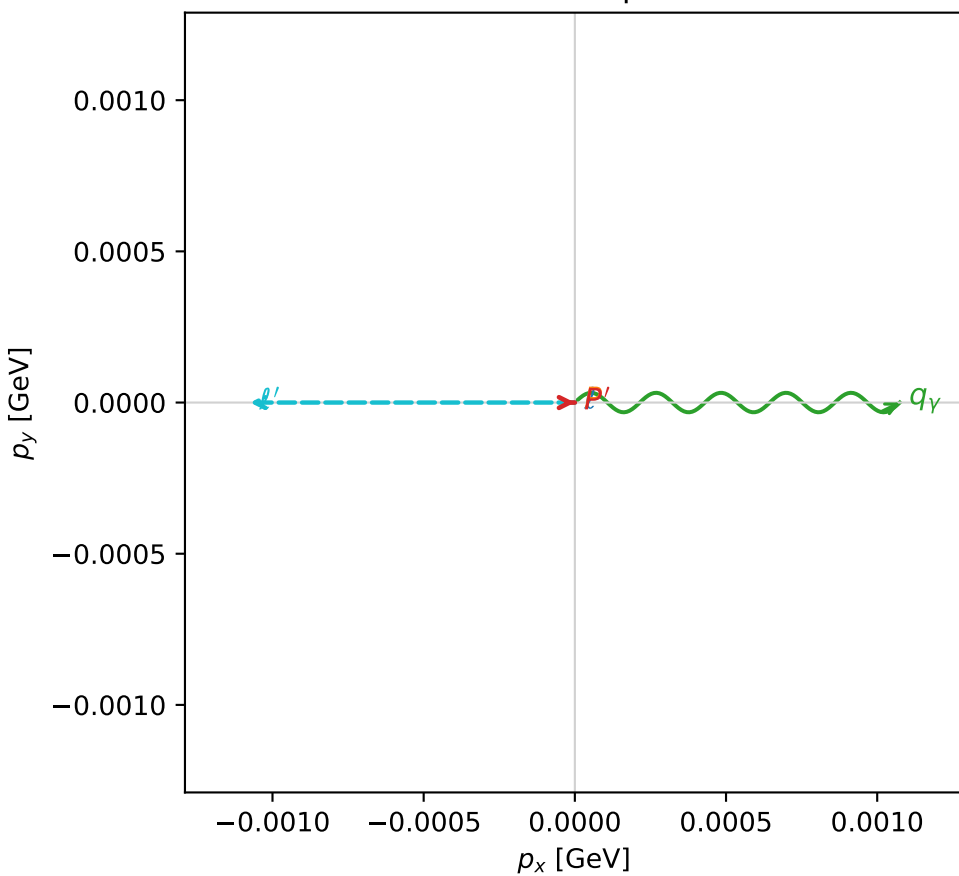
$s=9.28938$, $\sqrt{s}=3.04785$, $|\vec{P}|=1.37959$, $|\vec{P}'|=0.0166545$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14057$, $\phi_{p'}=6.26079$, $\phi_Y=1.63588\text{e-}05$
 $E_Y=1.04652$, $\theta_{l'}=0.00101075$, $\phi_{l'}=3.14161$
 $Q^2=5.86697$, $x_B=0.752591$, $t=-1.41642$, $W^2=2.80857$, $y=0.927008$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.3796$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3796$
 P $E= 1.6683$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3796$
 l' $E= 1.0632$ $p_x= -0.0011$ $p_y= -0.0000$ $p_z= 1.0632$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0167$
 q_Y $E= 1.0465$ $p_x= 0.0011$ $p_y= 0.0000$ $p_z= -1.0465$

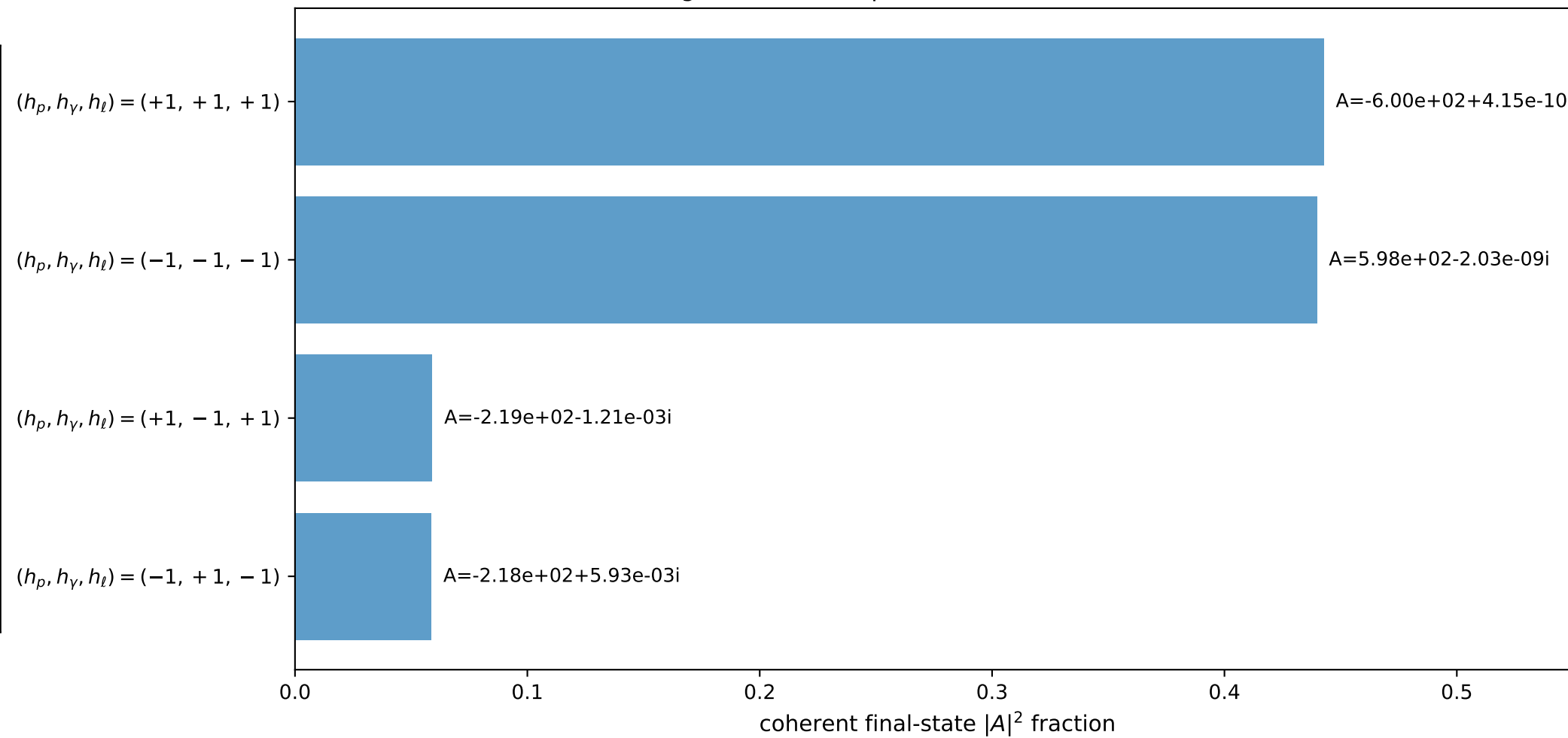
3D momenta



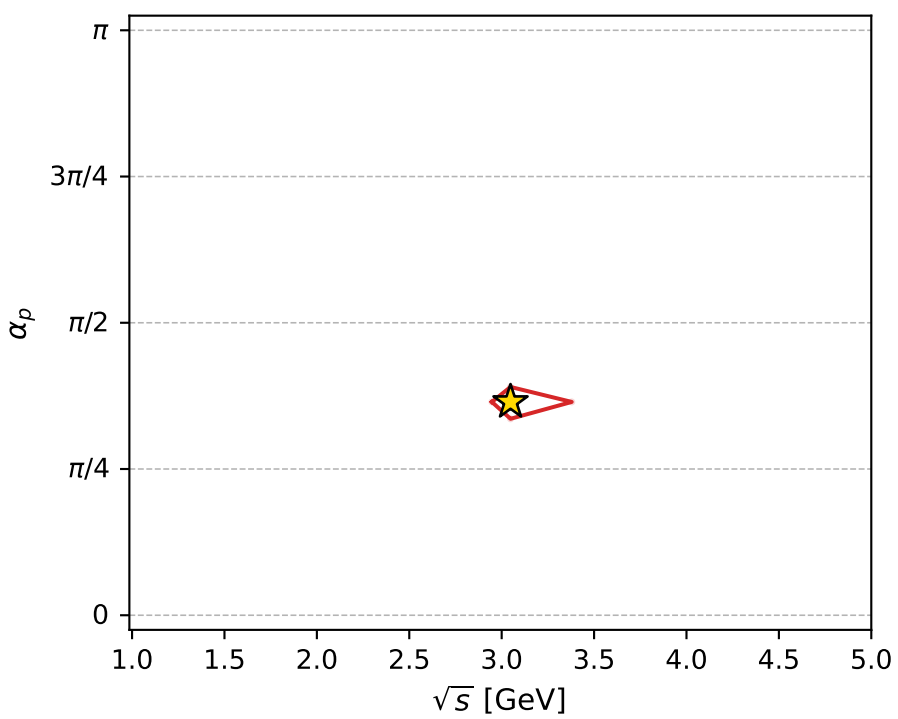
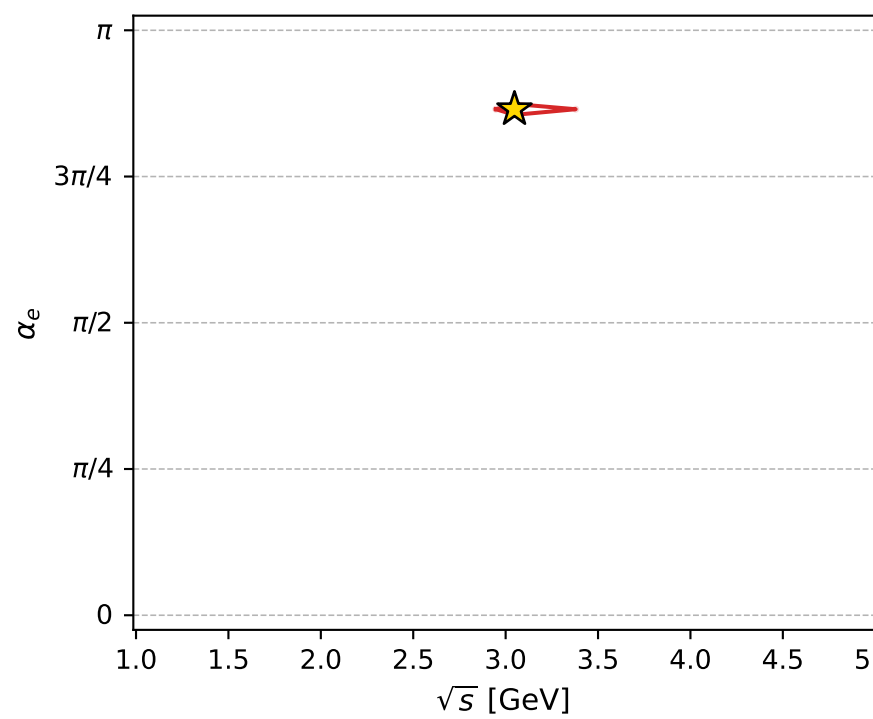
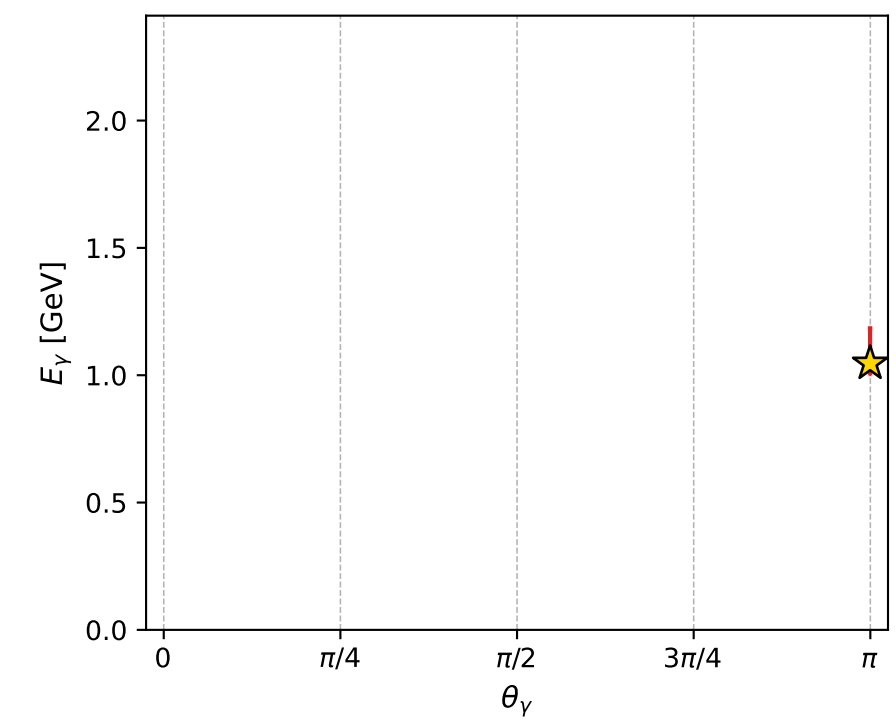
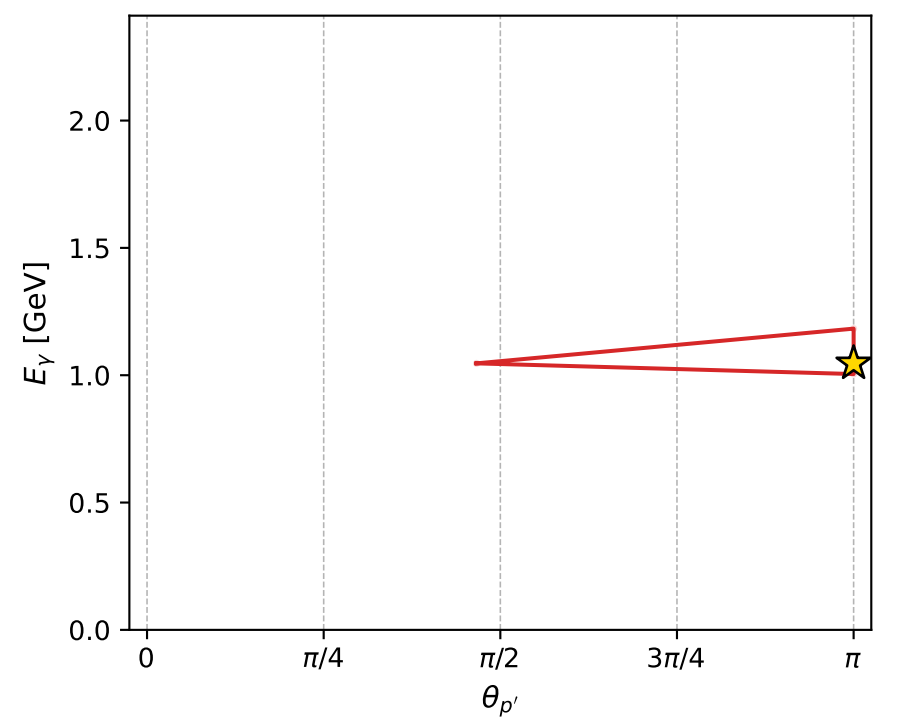
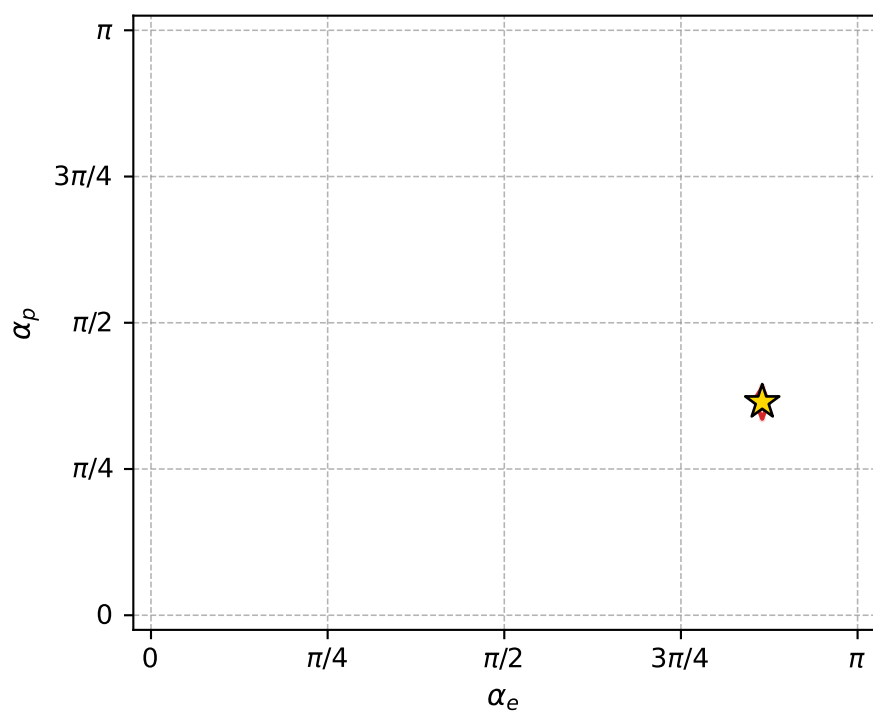
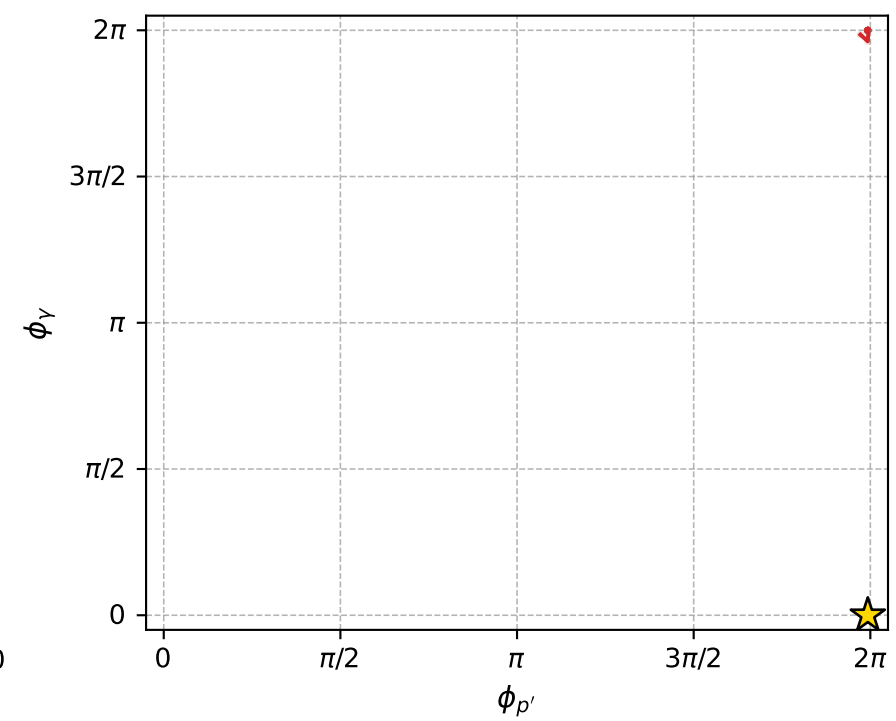
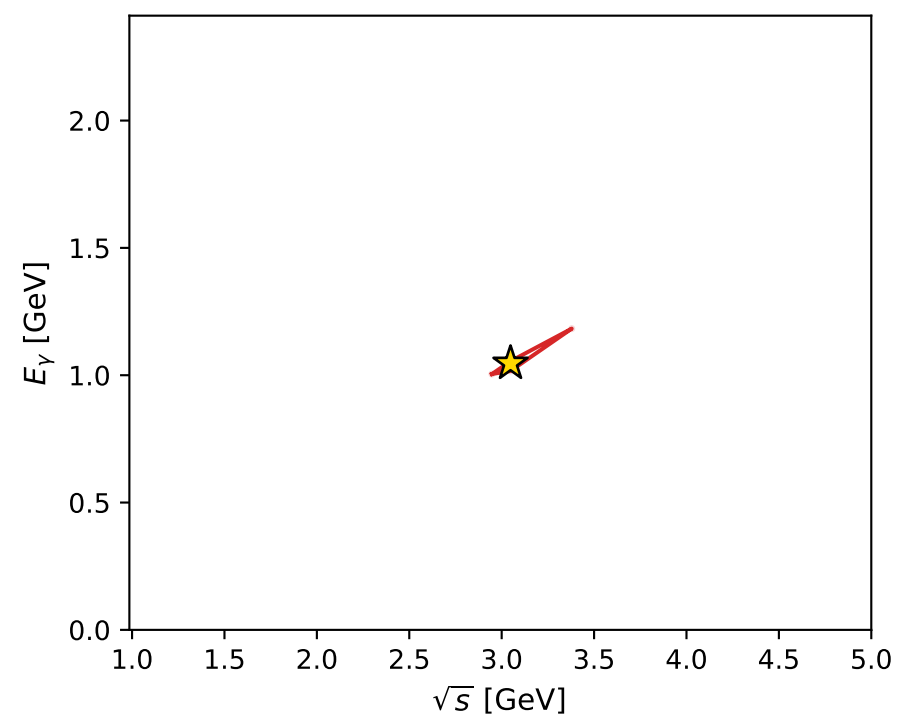
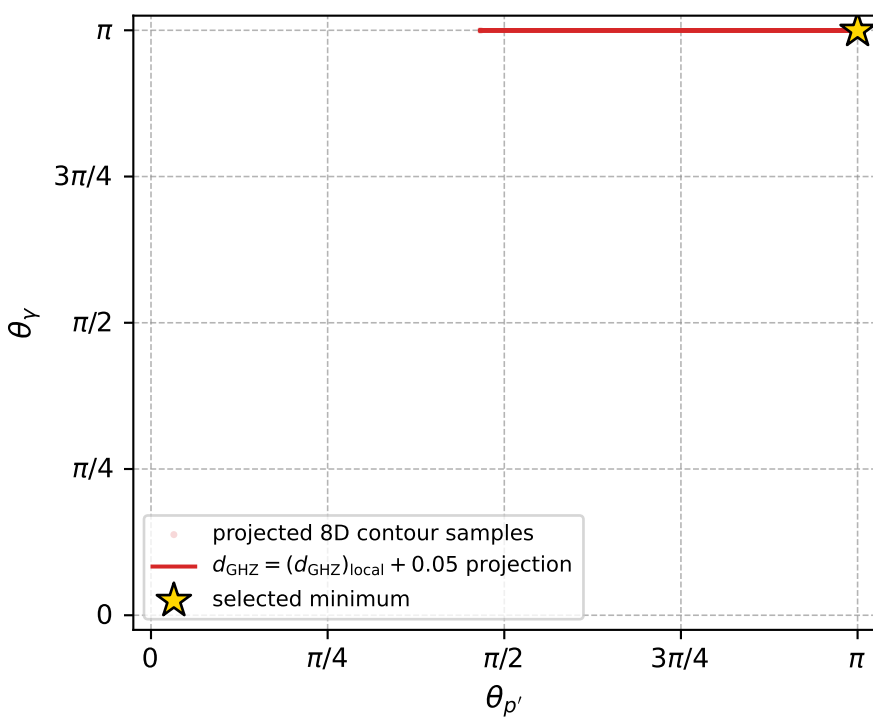
Transverse plane



Leading final-state amplitudes (N=4, retained=100.0%)



electron: polarization cluster P3, local minimum 278; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00022360565$
 $d_{\text{GHZ}} \text{ contour} = 0.050223606$
 above global minimum = 0.00022358337
 8D contour samples = 134

selected phase-space point:
 $\text{sqrt_s} = 3.047848$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.140566$
 $\text{q0ut} = 1.046523$
 $\text{phi_p_out} = 6.260793$
 $\text{phi_gamma_out} = 1.635883\text{e-}05$
 $\text{alpha_e} = 2.717668$
 $\text{alpha_p} = 1.14561$

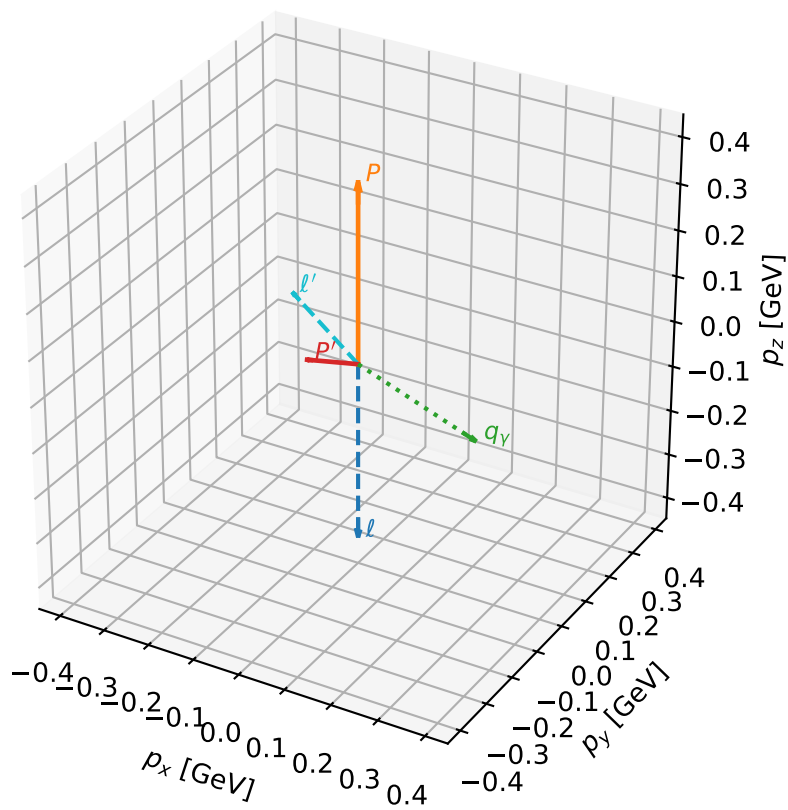
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_279 $d_{\text{GHZ}}=0.000226664$
 region: polarization_cluster_03_local_minimum_279
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.2077528$ rad
 incoming lepton: $-0.594751|+> +0.80391|->$
 proton mixing angle: $\alpha_p=0.53877863$ rad
 incoming proton: $0.858336|+> +0.513088|->$

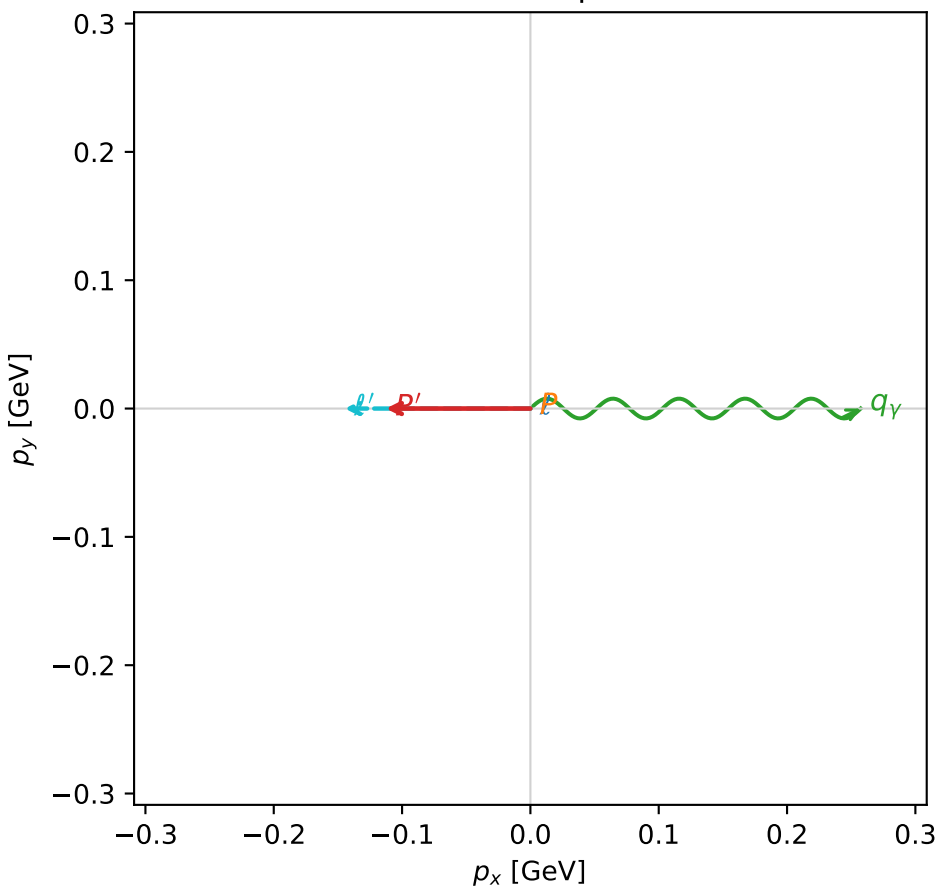
$s=1.97371$, $\sqrt{s}=1.40489$, $|\vec{P}|=0.389308$, $|\vec{P}'|=0.114886$
 $\theta_p=1.76902$, $\theta_Y=1.92101$, $\phi_p=3.14163$, $\phi_Y=4.55554e-05$
 $E_Y=0.274001$, $\theta_{\ell'}=0.892491$, $\phi_{\ell'}=3.14164$
 $Q^2=0.235541$, $x_B=0.291825$, $t=-0.177395$, $W^2=1.45144$, $y=0.737869$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3893$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3893$
 P $E= 1.0156$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3893$
 ℓ' $E= 0.1859$ $p_x= -0.1447$ $p_y= -0.0000$ $p_z= 0.1166$
 P' $E= 0.9450$ $p_x= -0.1126$ $p_y= -0.0000$ $p_z= -0.0226$
 q_Y $E= 0.2740$ $p_x= 0.2574$ $p_y= 0.0000$ $p_z= -0.0940$

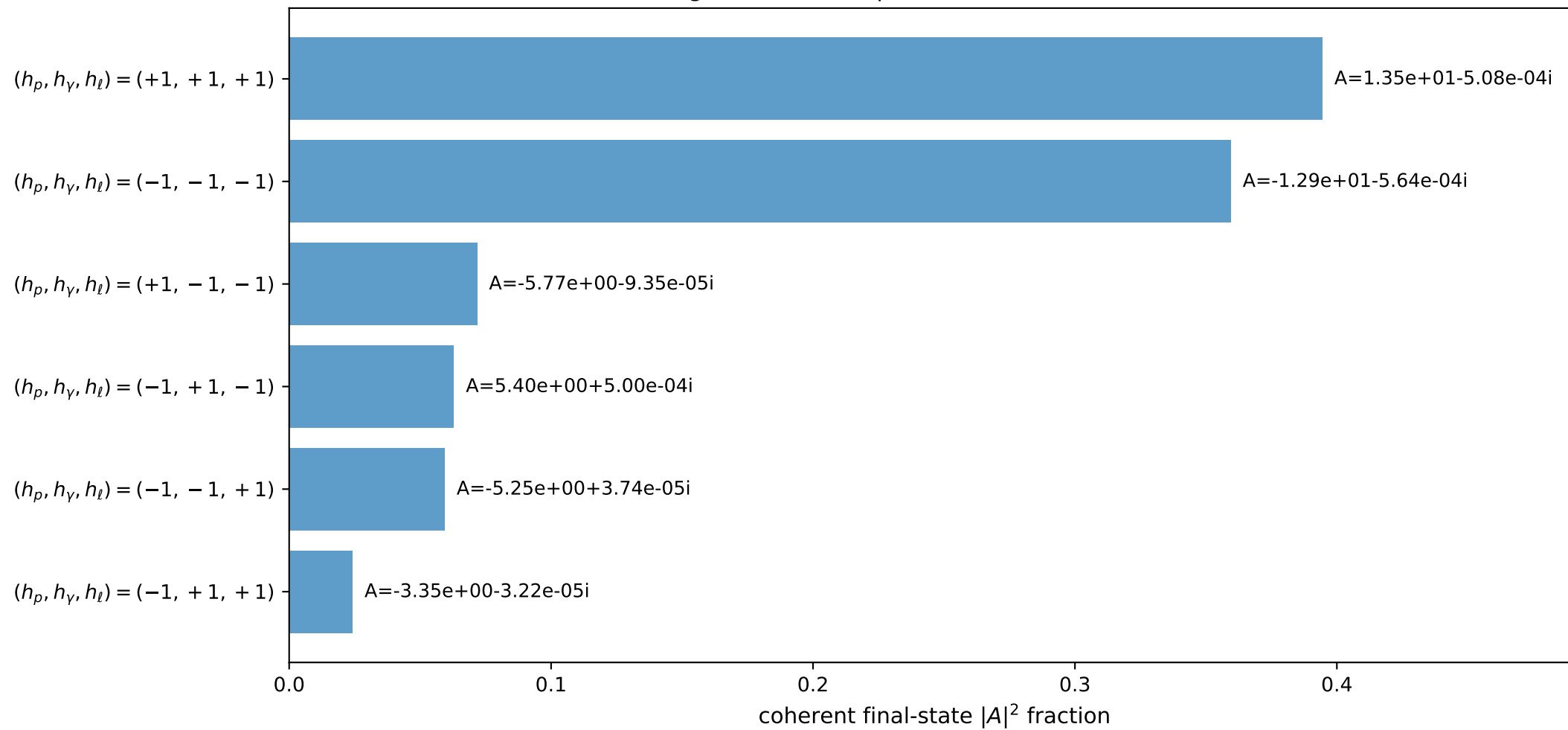
3D momenta



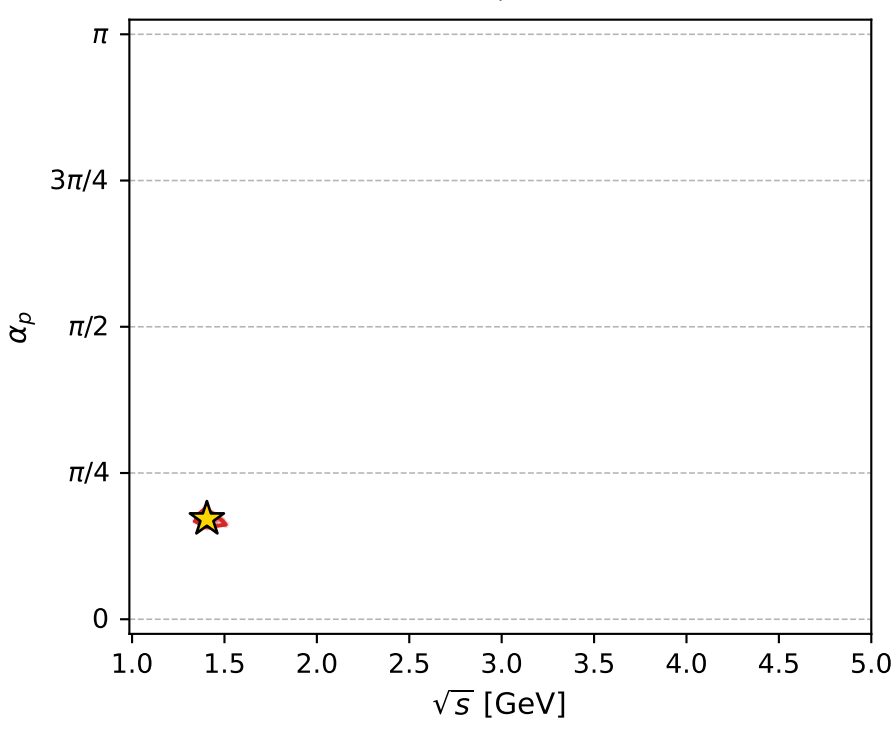
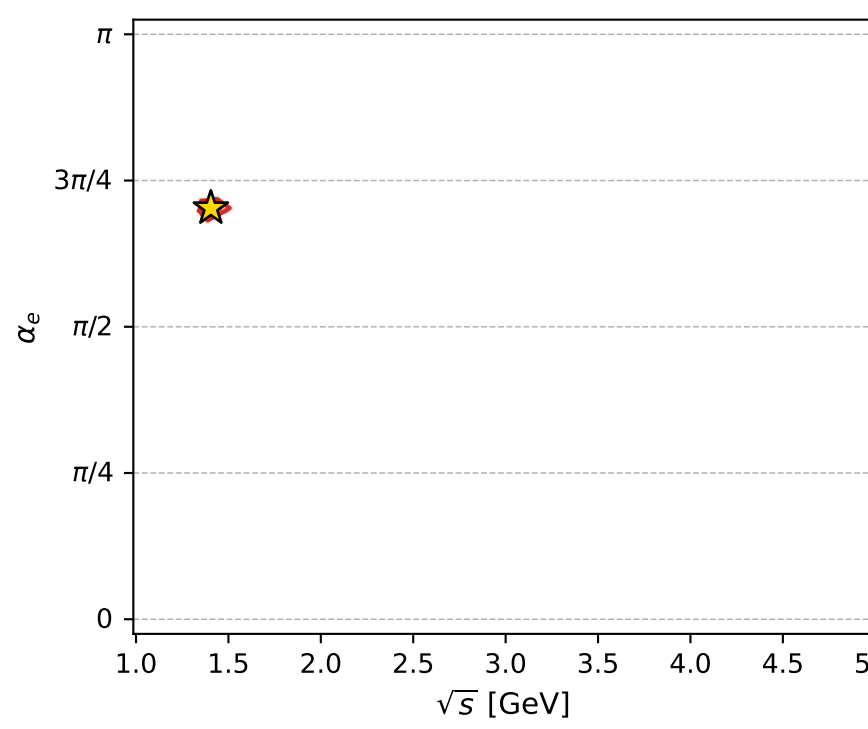
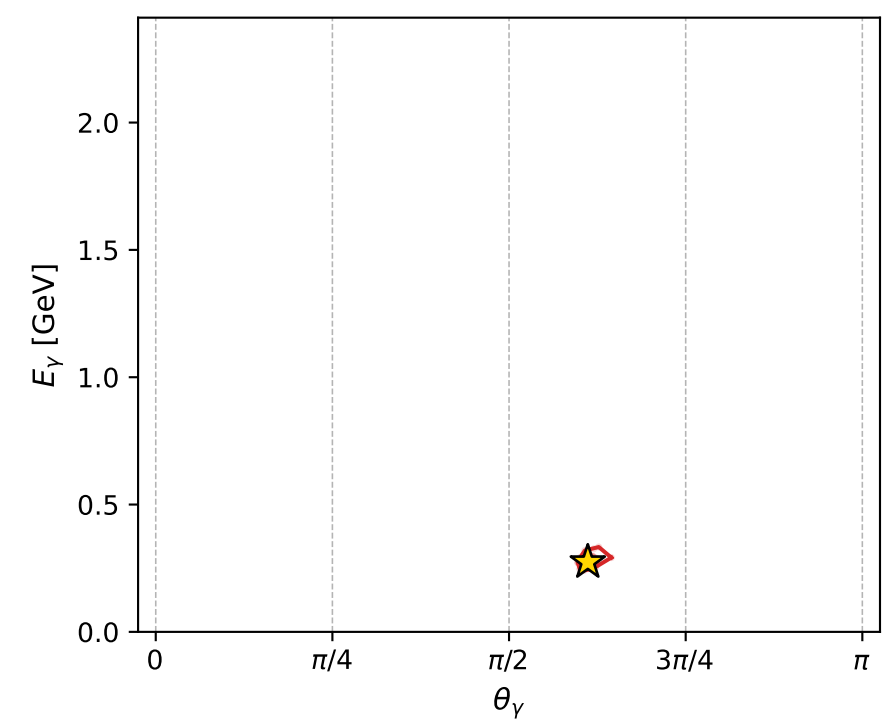
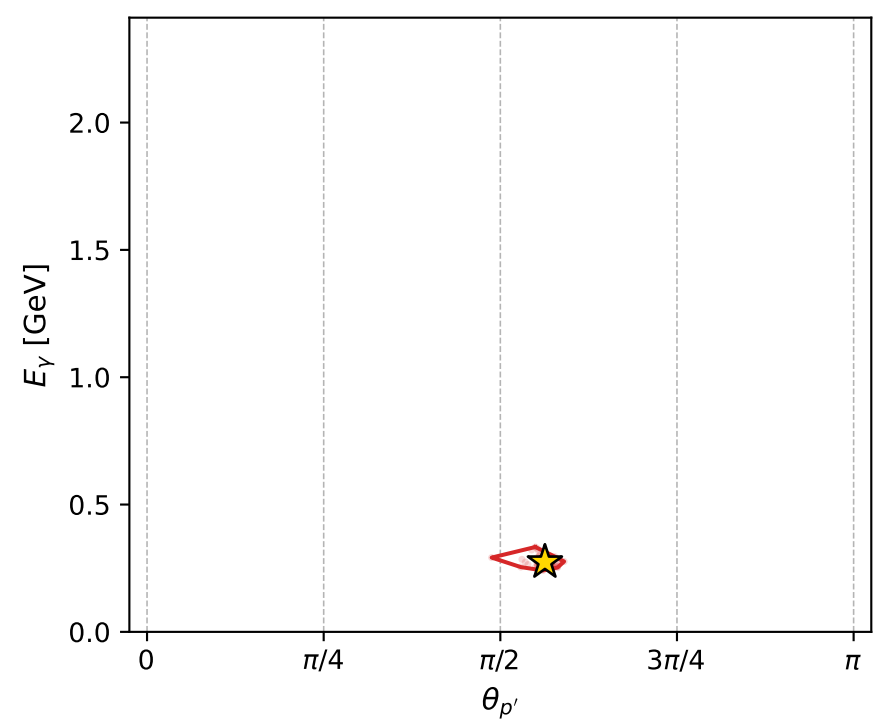
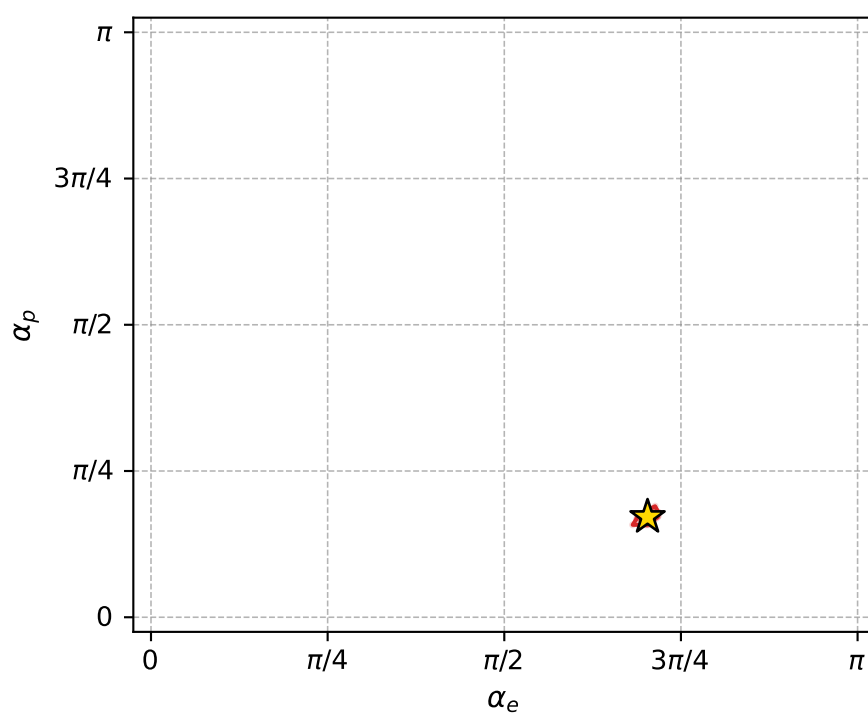
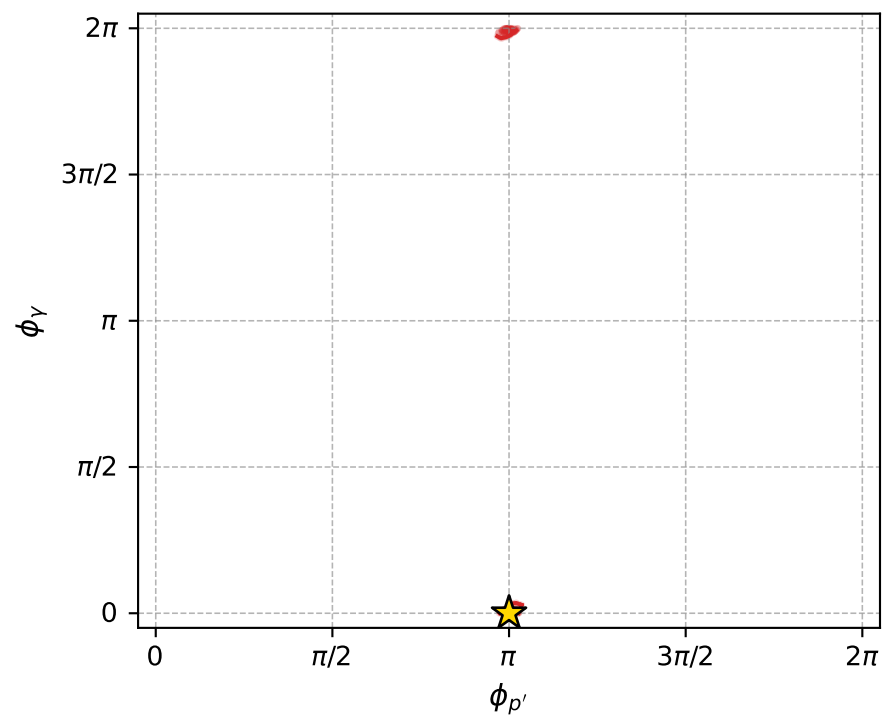
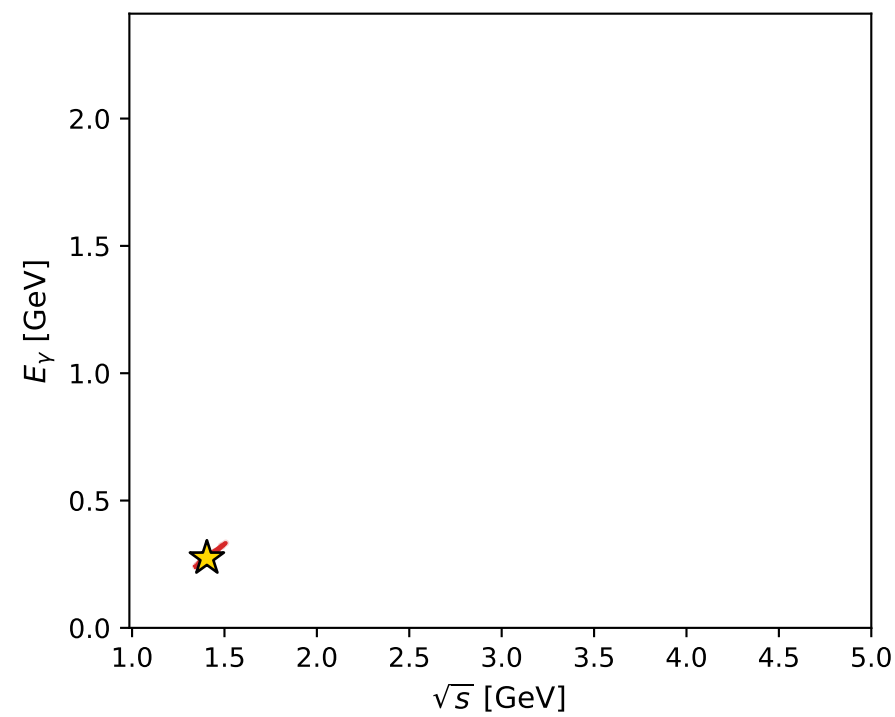
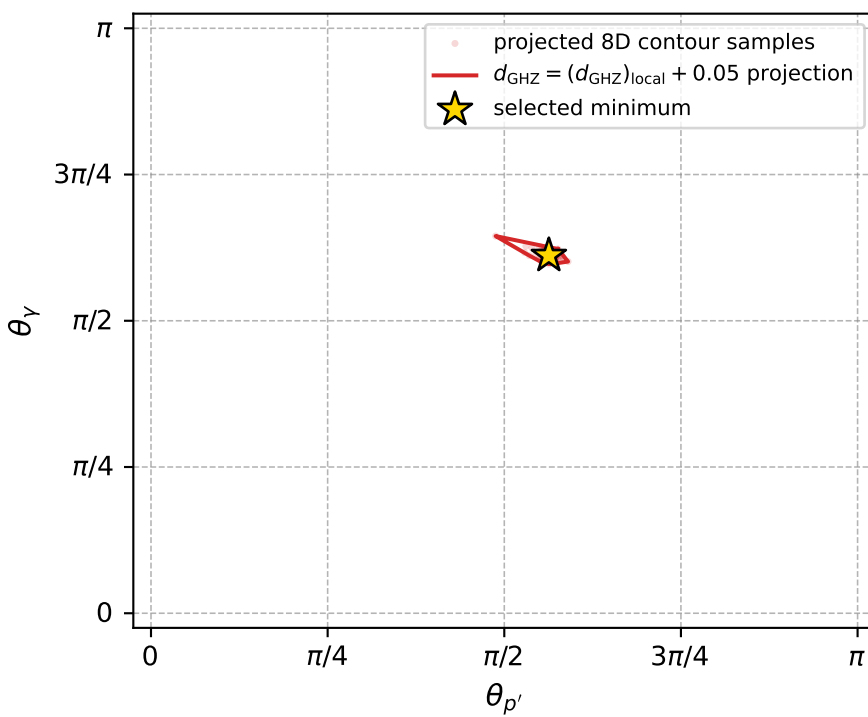
Transverse plane



Leading final-state amplitudes (N=6, retained=97.2%)



electron: polarization cluster P3, local minimum 279; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00022666406$
 $d_{\text{GHZ}} \text{ contour} = 0.050226664$
 above global minimum = 0.00022664178
 8D contour samples = 256

selected phase-space point:

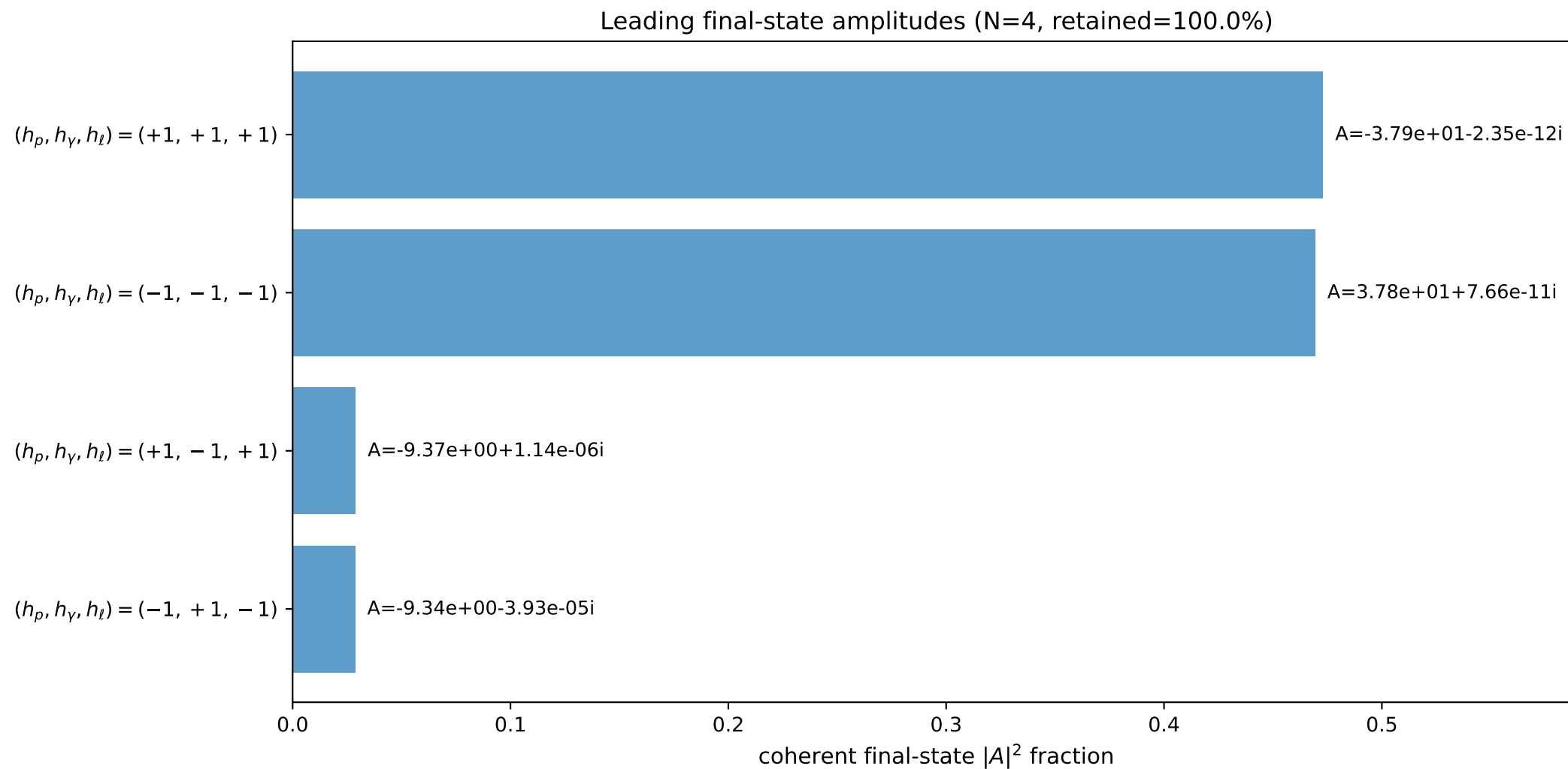
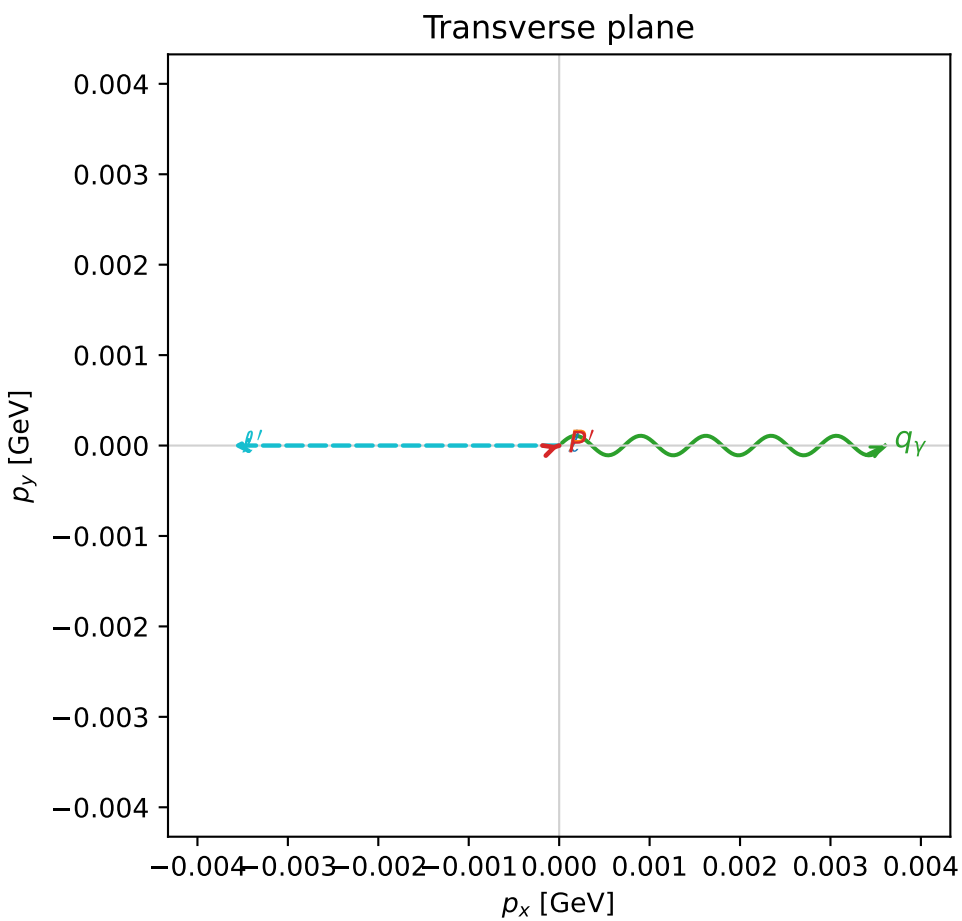
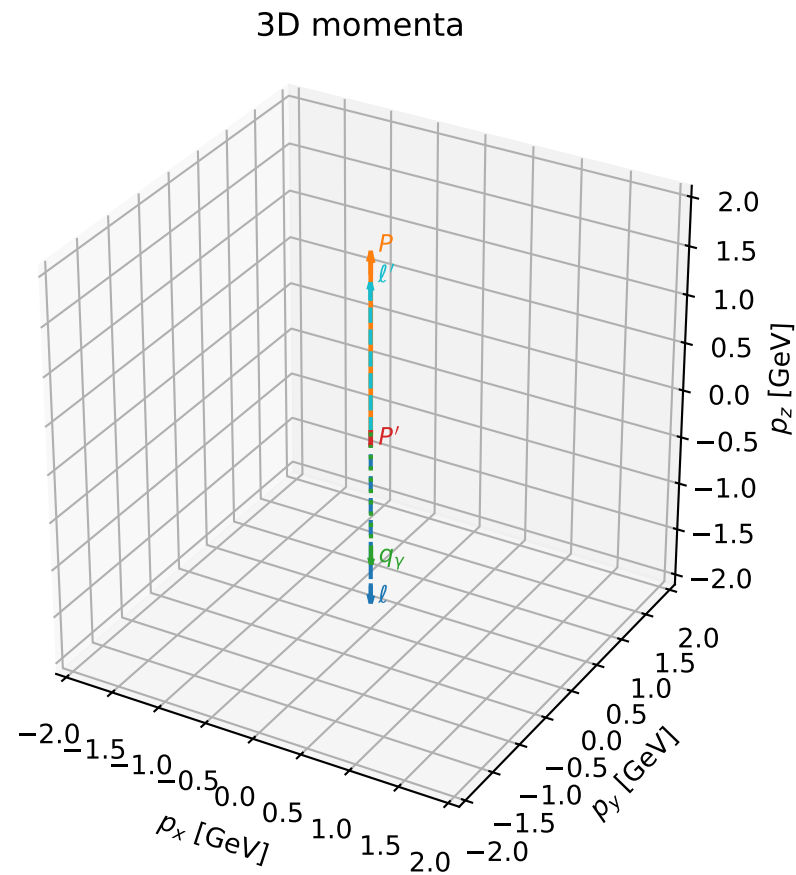
$\text{sqrt_s} = 1.404889$
 $\text{theta_p_out} = 1.76902$
 $\text{theta_gamma_out} = 1.921012$
 $\text{q0ut} = 0.2740008$
 $\text{phi_p_out} = 3.141631$
 $\text{phi_gamma_out} = 4.555535\text{e-}05$
 $\text{alpha_e} = 2.207753$
 $\text{alpha_p} = 0.5387786$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

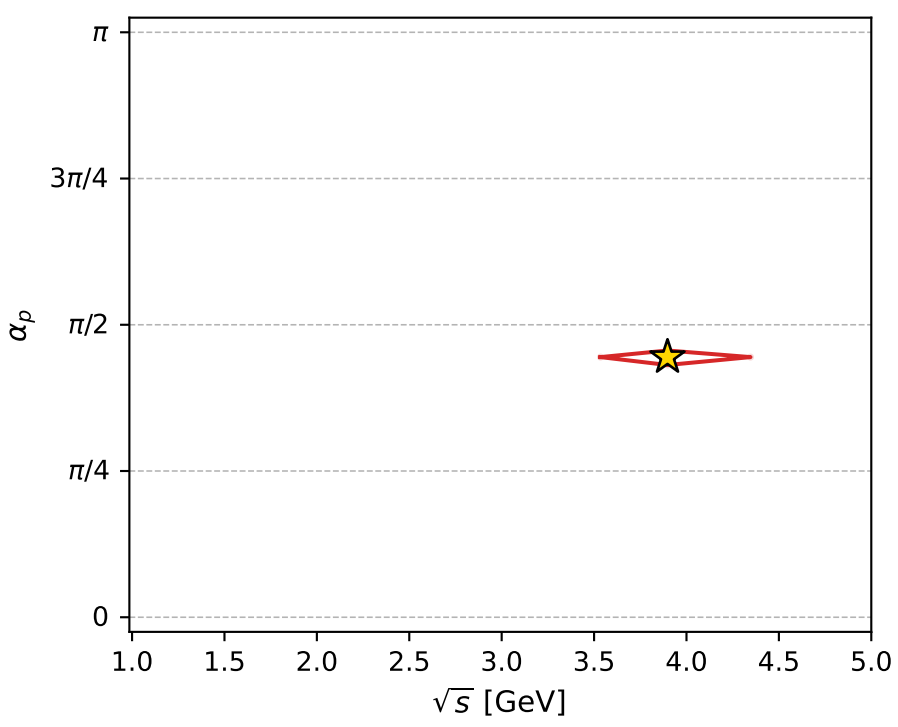
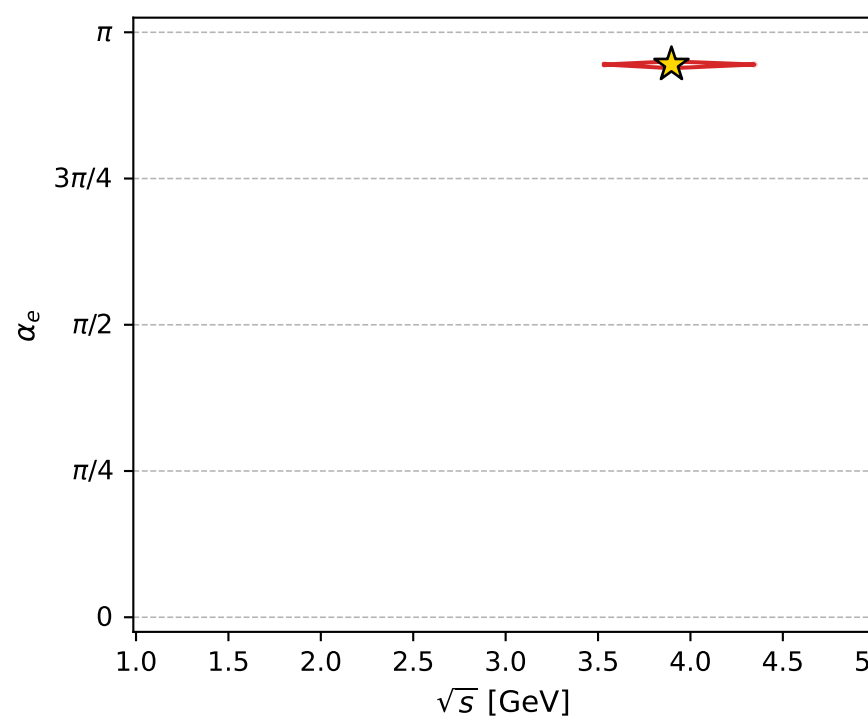
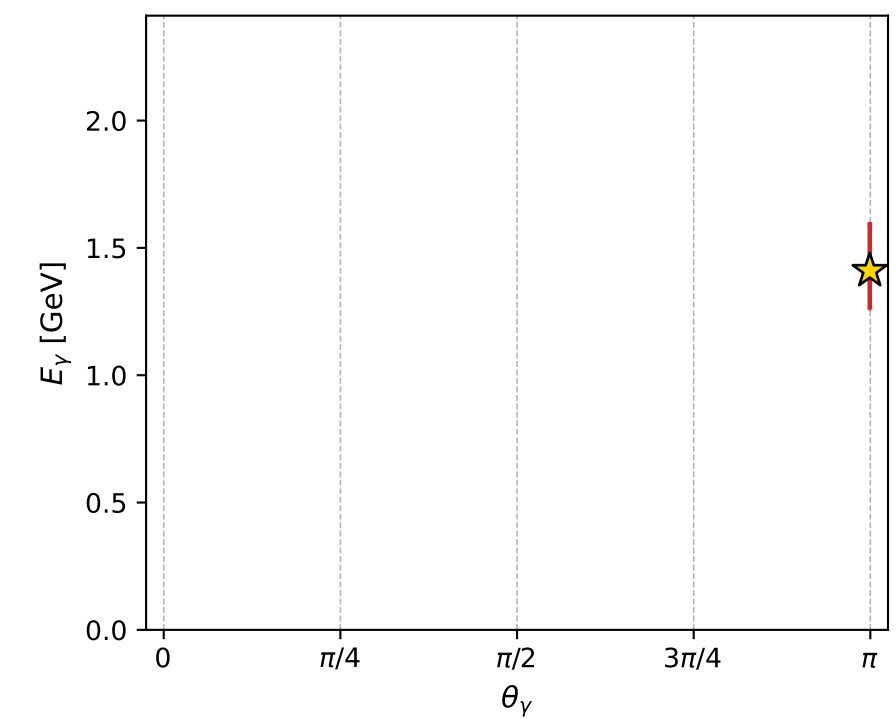
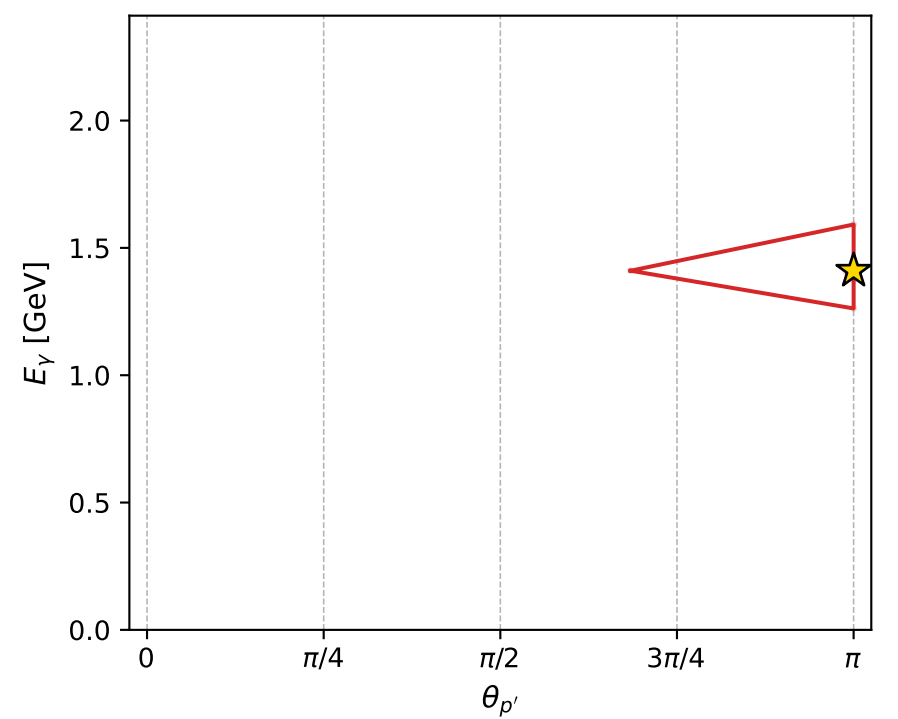
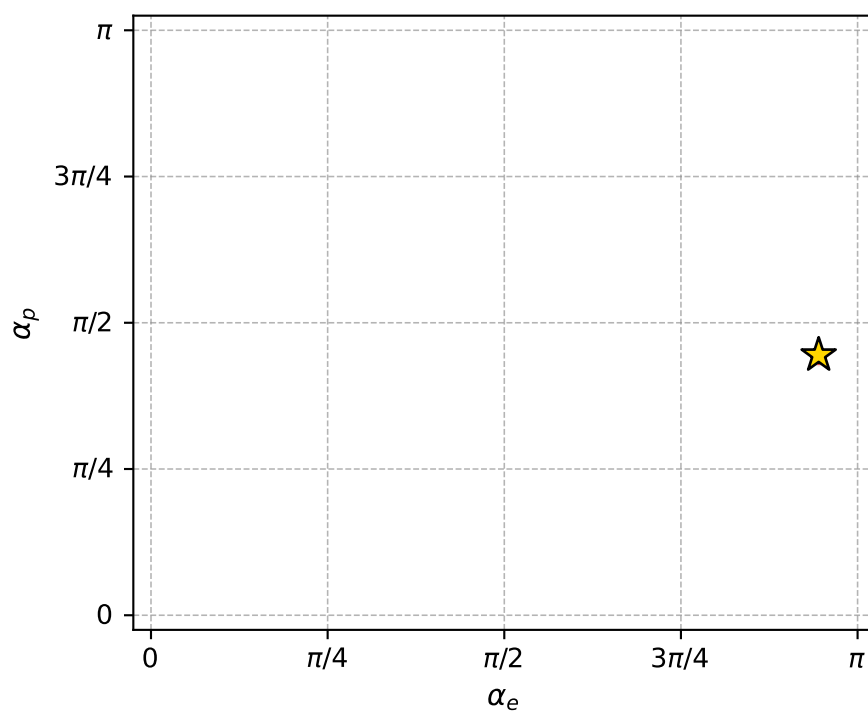
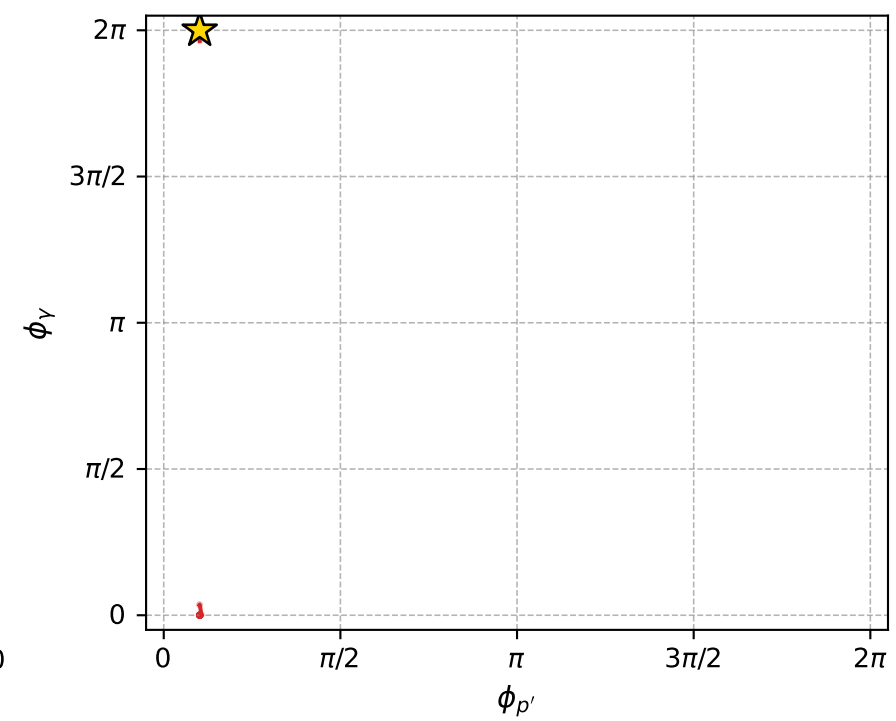
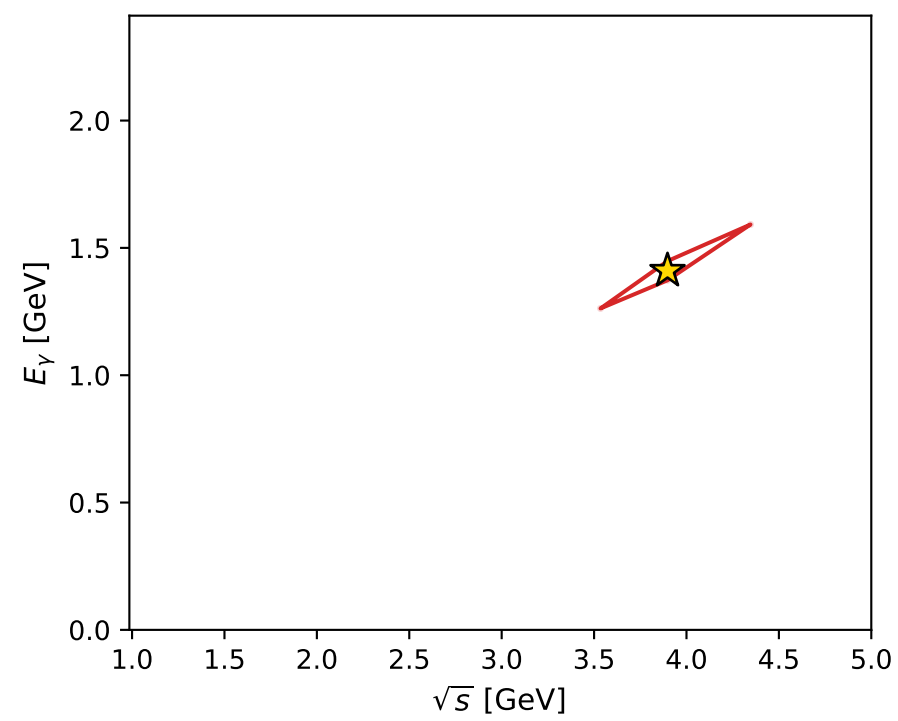
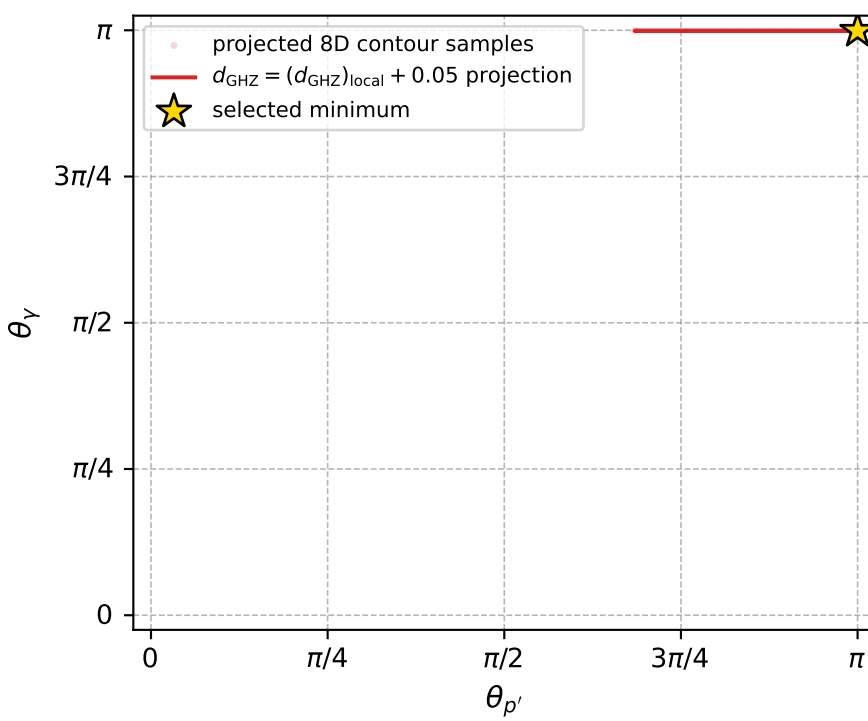
dghz_mixing_angles_polarization_03_local_minimum_281 $d_{\{\text{rm GHZ}\}}=0.000232112$
 region: polarization_cluster_03_local_minimum_281
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.9685984$ rad
 incoming lepton: $-0.985074|+\rangle +0.172133|-\rangle$
 proton mixing angle: $\alpha_p=1.3971439$ rad
 incoming proton: $0.172781|+\rangle +0.98496|-\rangle$

$s=15.1891$, $\sqrt{s}=3.89732$, $|\vec{P}|=1.83578$, $|\vec{P}'|=0.129682$
 $\theta_{p'}=3.14159$, $\theta_V=3.13904$, $\phi_{p'}=0.319777$, $\phi_V=6.28318$
 $E_V=1.41036$, $\theta_{l'}=0.00234065$, $\phi_{l'}=3.14159$
 $Q^2=11.3087$, $x_B=0.830673$, $t=-2.62067$, $W^2=3.18505$, $y=0.951405$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8358$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8358$
 P $E= 2.0615$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8358$
 l' $E= 1.5400$ $p_x= -0.0036$ $p_y= 0.0000$ $p_z= 1.5400$
 P' $E= 0.9469$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1297$
 q_V $E= 1.4104$ $p_x= 0.0036$ $p_y= -0.0000$ $p_z= -1.4104$



electron: polarization cluster P3, local minimum 281; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00023211233$
 $d_{\text{GHZ}} \text{ contour} = 0.050232112$
 above global minimum = 0.00023209004
 8D contour samples = 131

selected phase-space point:

$\text{sqrt_s} = 3.897317$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.139037$
 $\text{q0out} = 1.410357$
 $\text{phi_p_out} = 0.3197771$
 $\text{phi_gamma_out} = 6.283183$
 $\text{alpha_e} = 2.968598$
 $\text{alpha_p} = 1.397144$

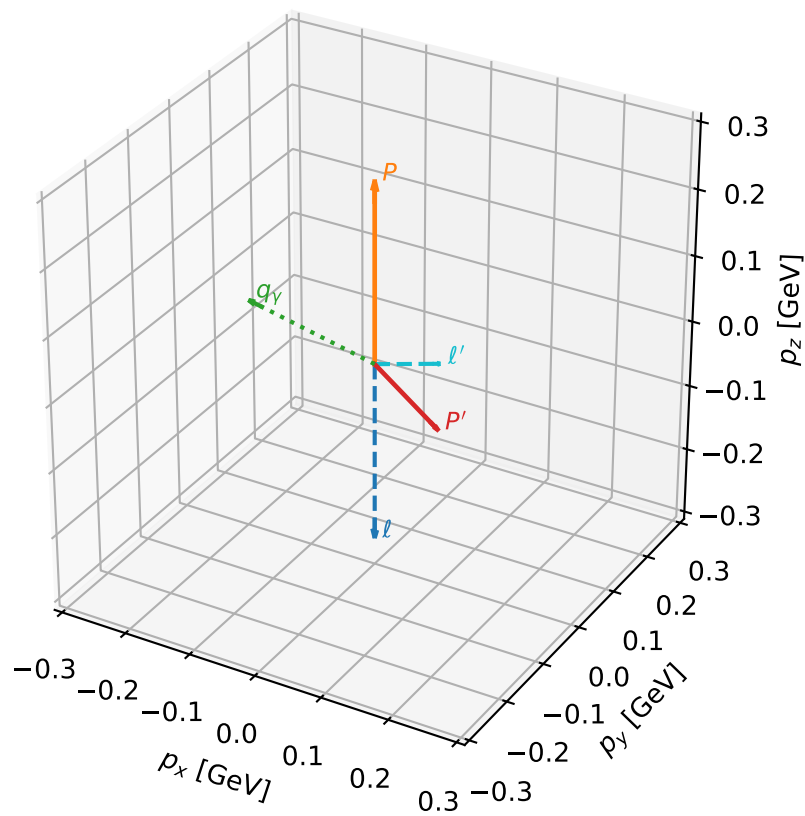
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_284 $d_{\{\text{rm GHZ}\}}=0.000241288$
 region: polarization_cluster_03_local_minimum_284
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.3175277$ rad
 incoming lepton: $-0.679243|+> +0.733913|->$
 proton mixing angle: $\alpha_p=0.87274734$ rad
 incoming proton: $0.642724|+> +0.766098|->$

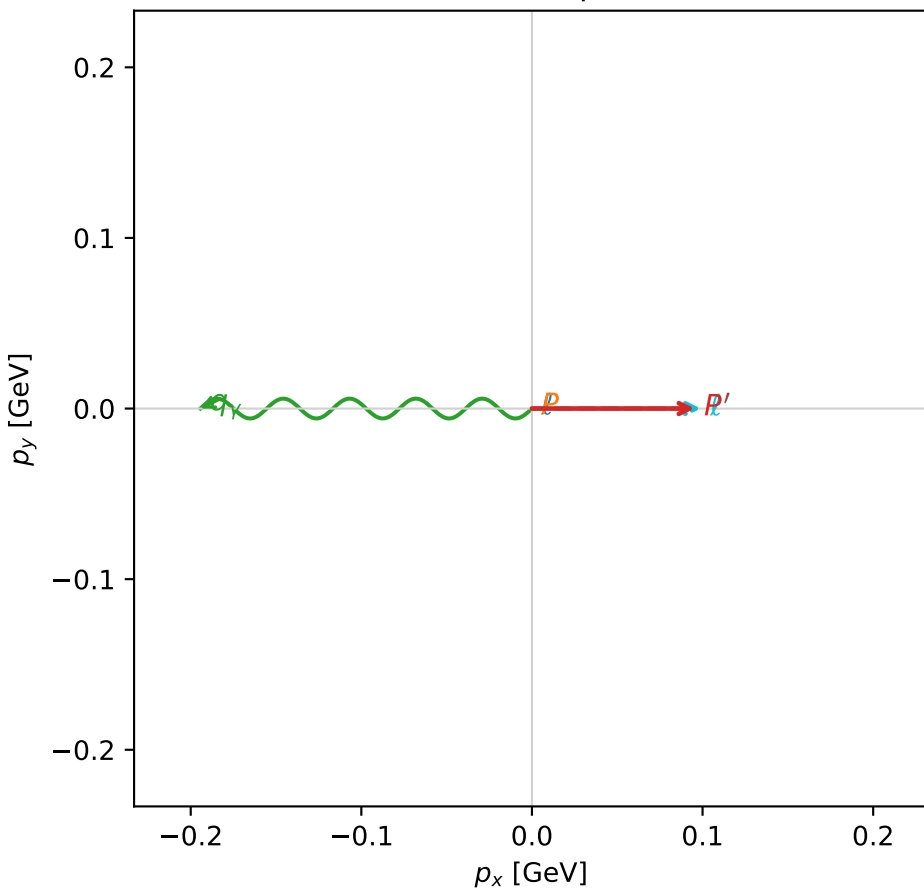
$s=1.55666$, $\sqrt{s}=1.24766$, $|\vec{P}|=0.271235$, $|\vec{P}'|=0.119344$
 $\theta_{P'}=2.21574$, $\theta_Y=1.35477$, $\phi_{P'}=2.60535\text{e-}05$, $\phi_Y=3.1416$
 $E_Y=0.198954$, $\theta_{\ell'}=1.28482$, $\phi_{\ell'}=6.28318$
 $Q^2=0.0717385$, $x_B=0.146056$, $t=-0.125777$, $W^2=1.29928$, $y=0.725704$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2712$
 P $E= 0.9764$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2712$
 ℓ' $E= 0.1031$ $p_x= 0.0990$ $p_y= -0.0000$ $p_z= 0.0291$
 P' $E= 0.9456$ $p_x= 0.0954$ $p_y= 0.0000$ $p_z= -0.0717$
 q_Y $E= 0.1990$ $p_x= -0.1943$ $p_y= -0.0000$ $p_z= 0.0426$

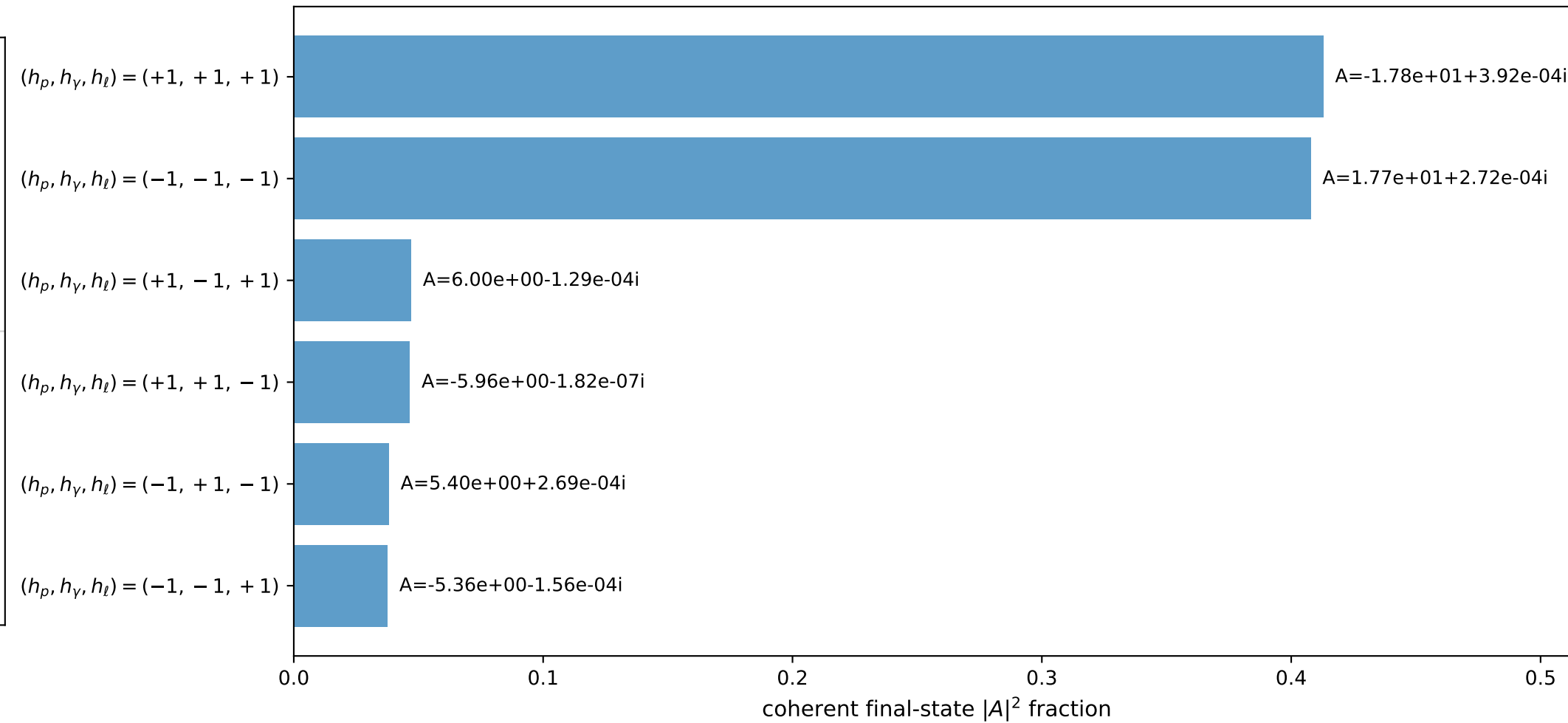
3D momenta



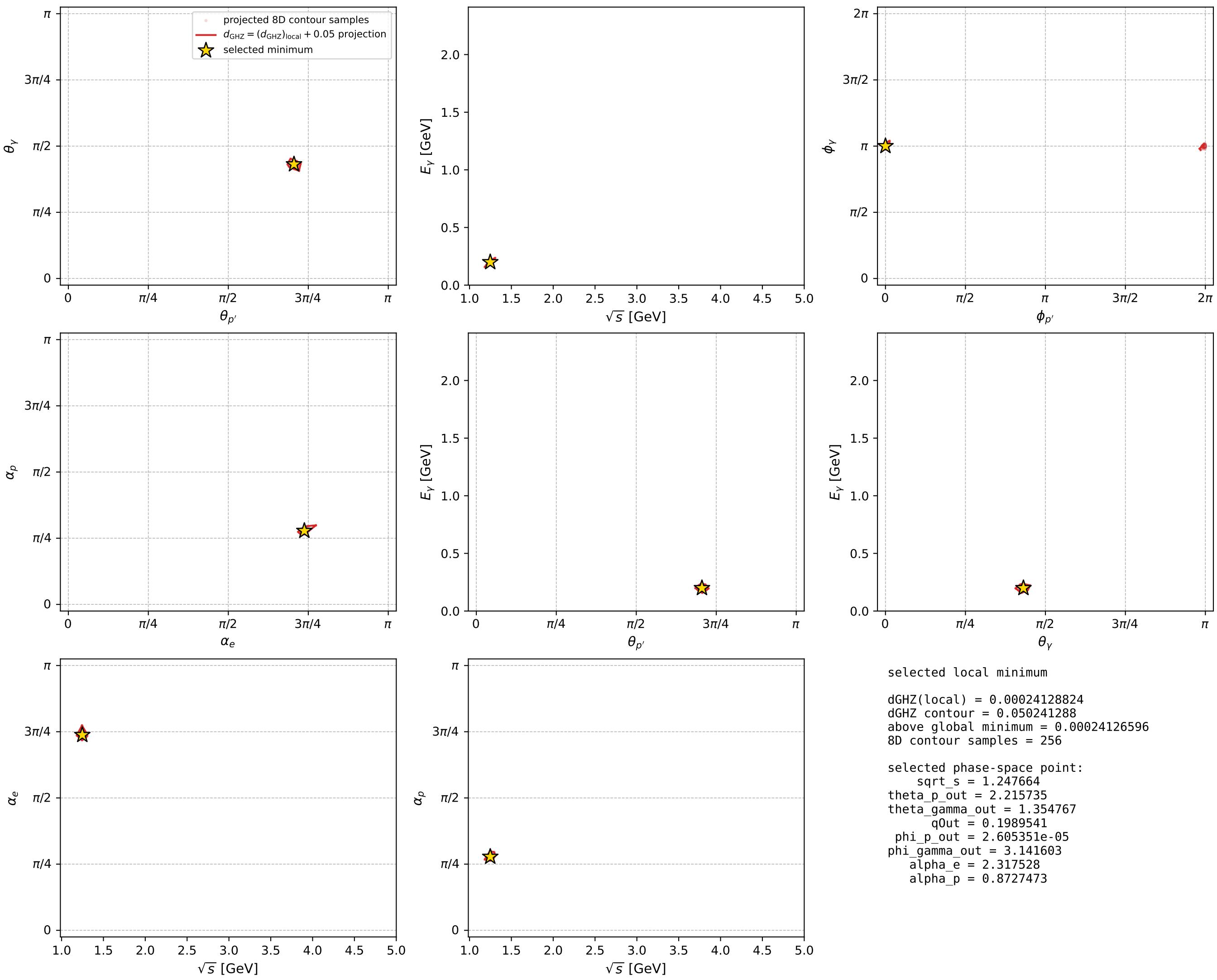
Transverse plane



Leading final-state amplitudes (N=6, retained=99.0%)



electron: polarization cluster P3, local minimum 284; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

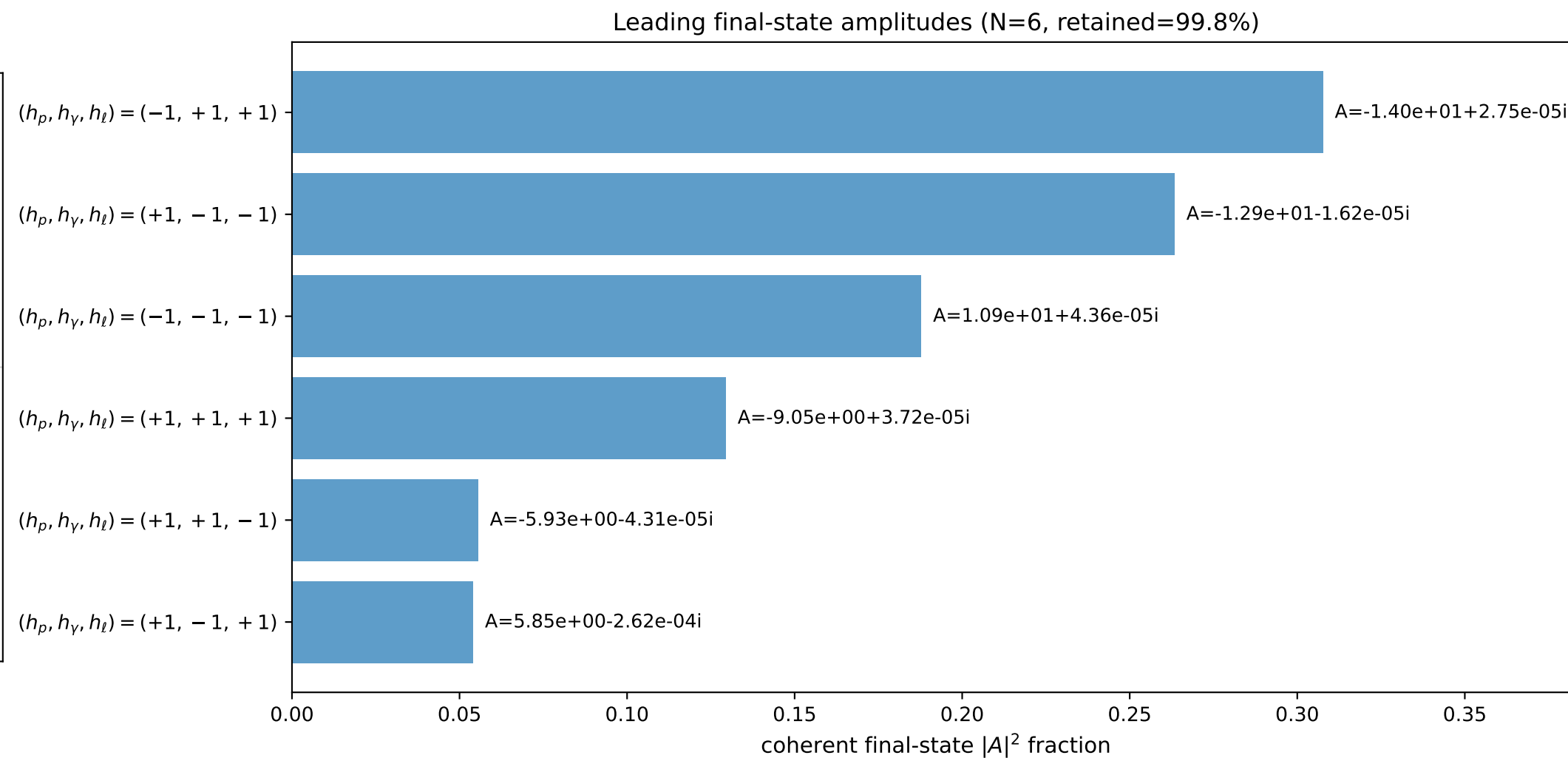
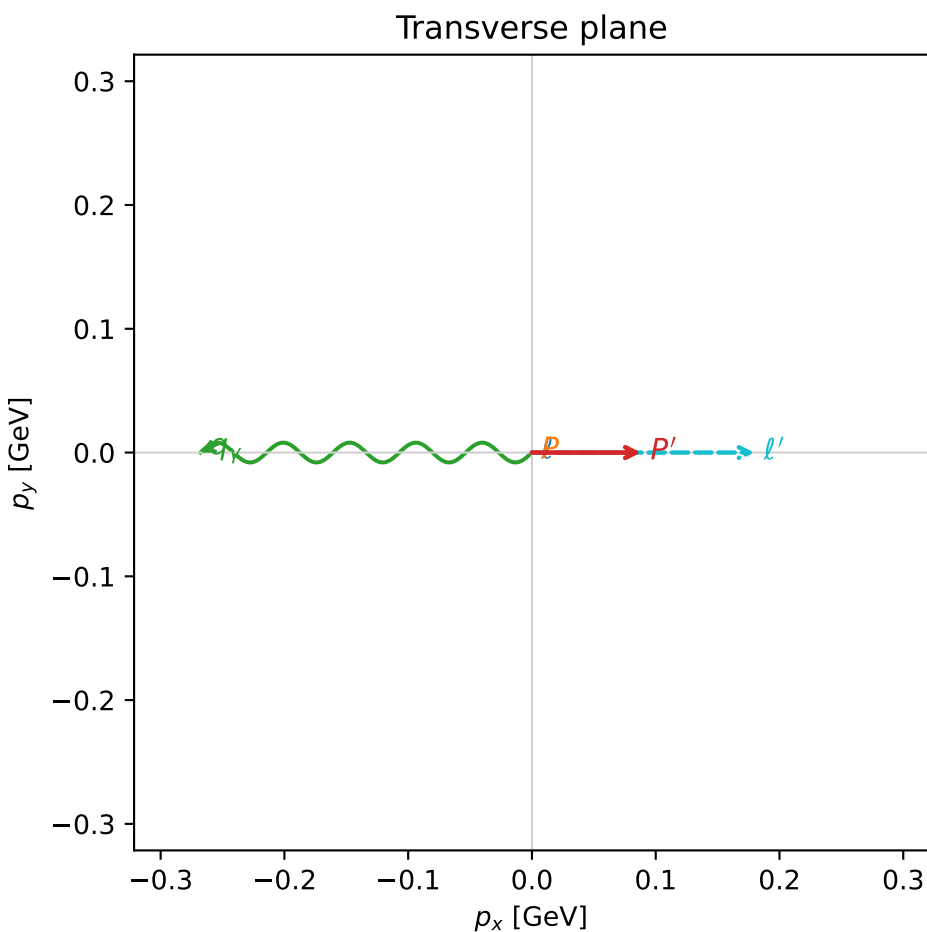
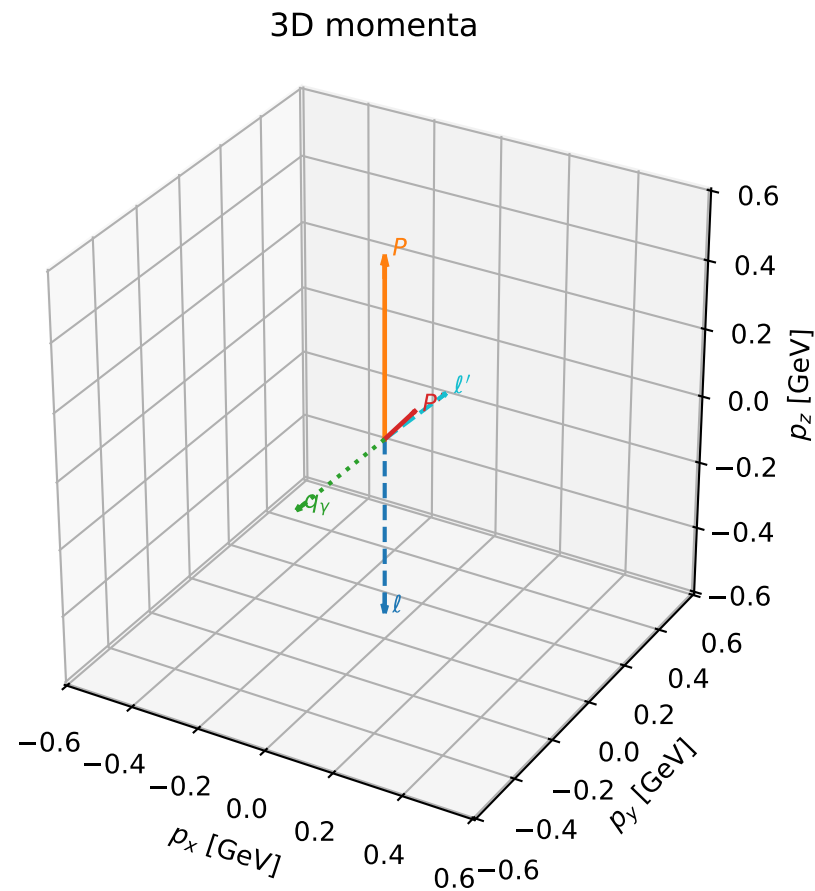


coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

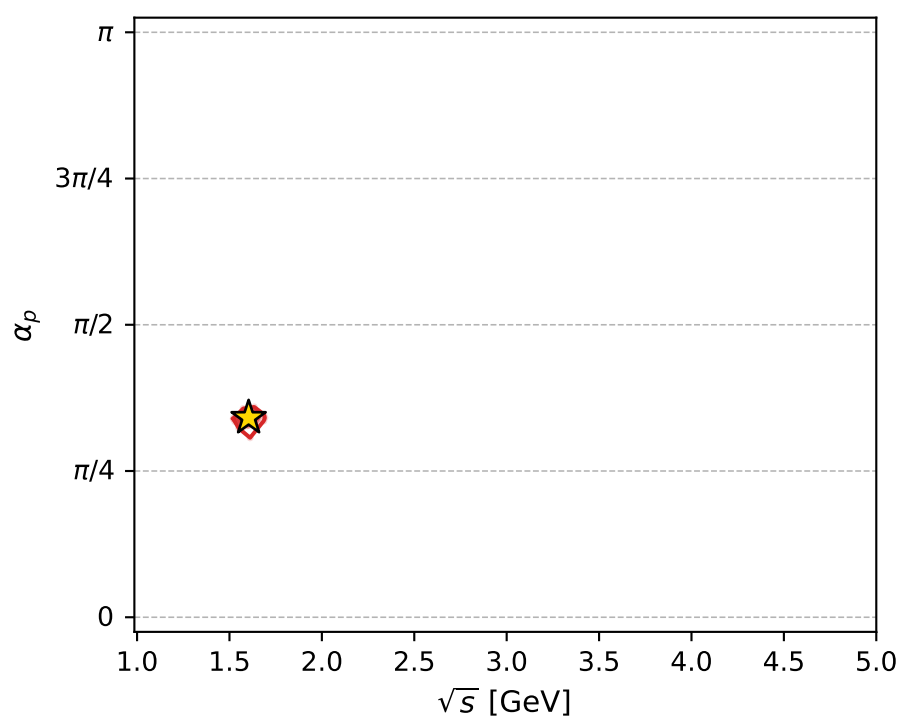
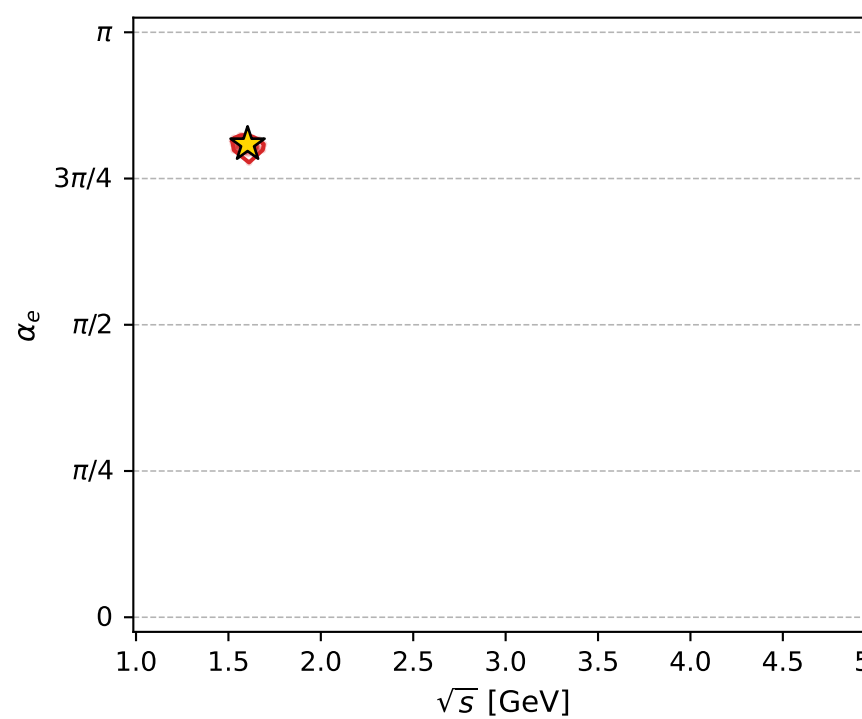
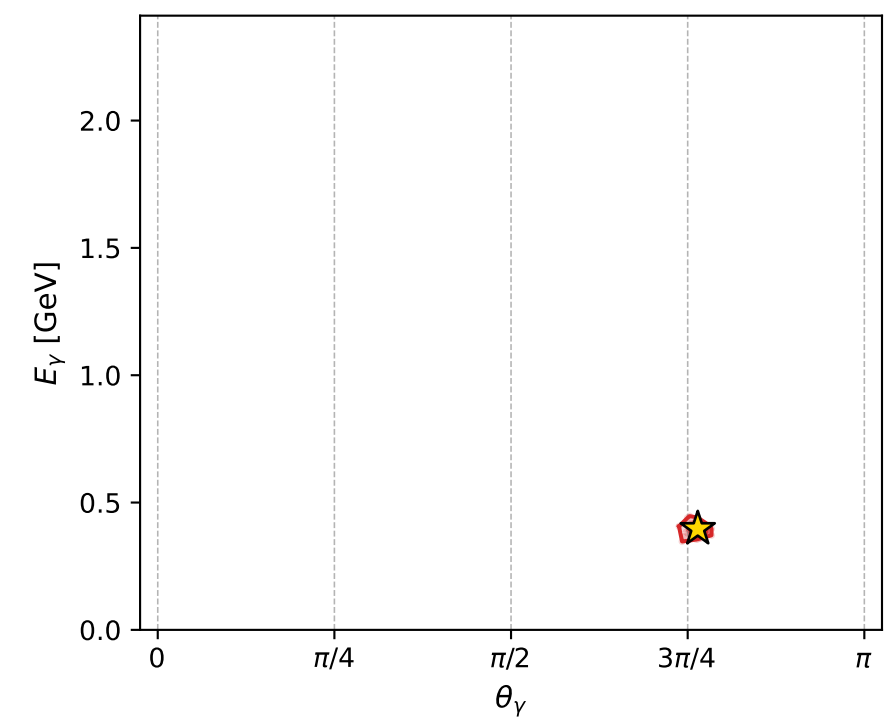
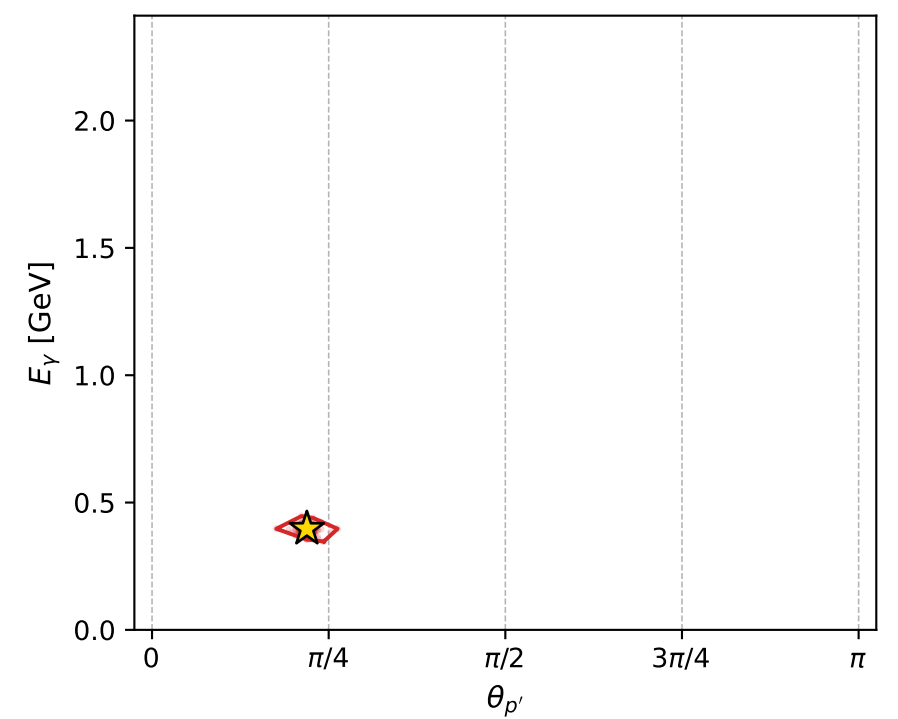
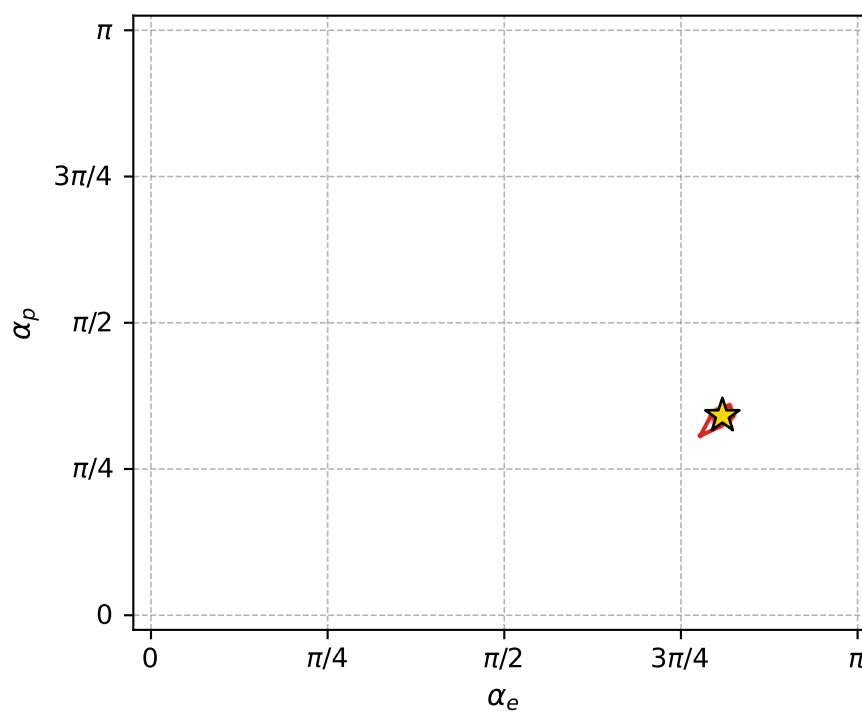
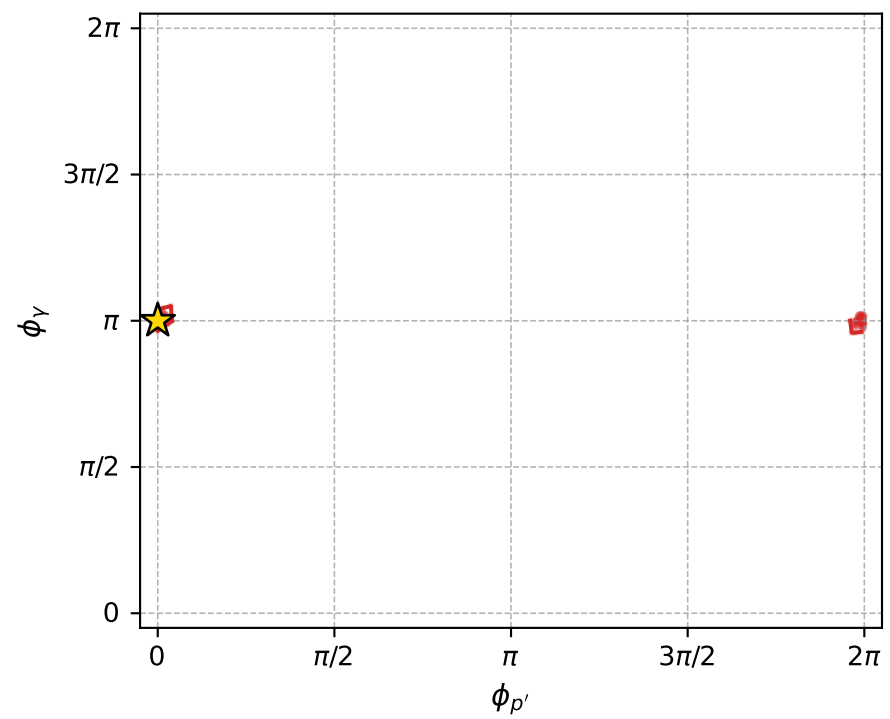
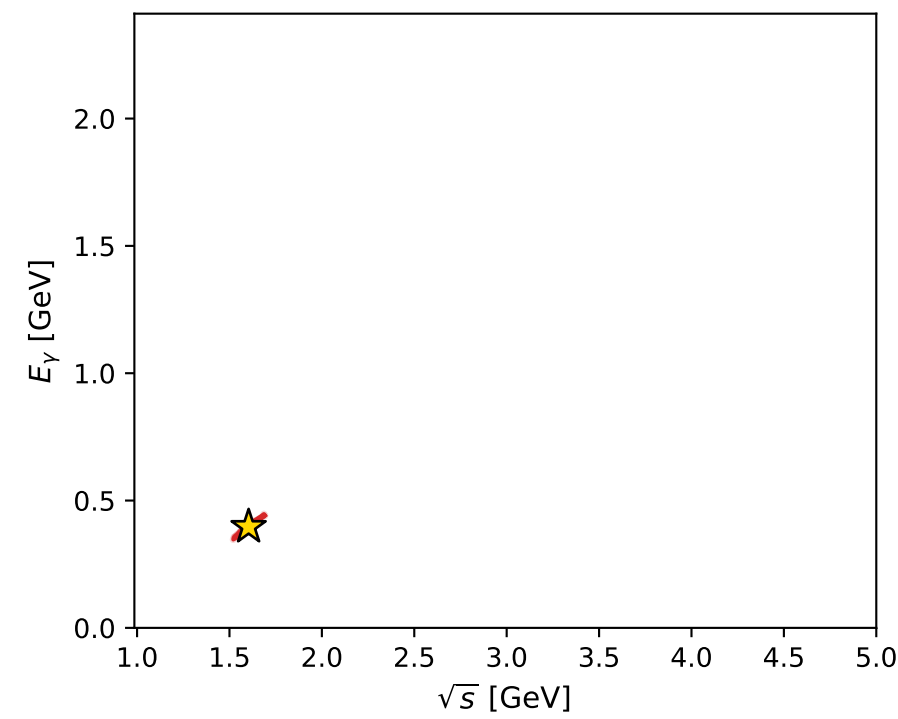
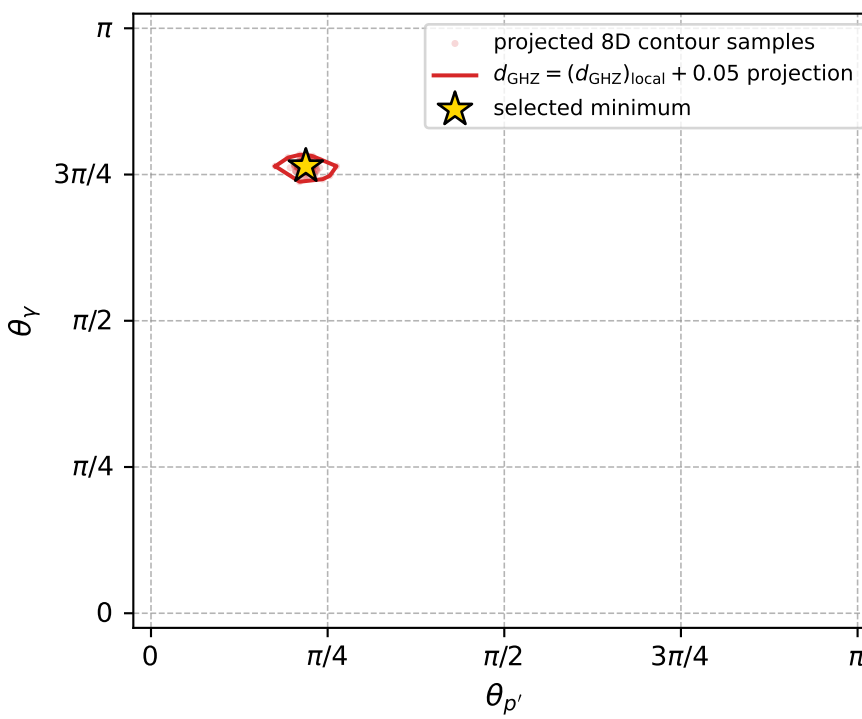
dghz_mixing_angles_polarization_03_local_minimum_287 d_{\rm GHz}=0.000254421
 region: polarization_cluster_03_local_minimum_287
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.5405287$ rad
 incoming lepton: $-0.824734|+\rangle +0.56552|-\rangle$
 proton mixing angle: $\alpha_p=1.0714277$ rad
 incoming proton: $0.478871|+\rangle +0.877885|-\rangle$

$s=2.57179$, $\sqrt{s}=1.60368$, $|\vec{P}|=0.527519$, $|\vec{P}'|=0.138788$
 $\theta_{P'}=0.688459$, $\theta_Y=2.40093$, $\phi_{P'}=2.56357\text{e-}06$, $\phi_Y=3.14162$
 $E_Y=0.396982$, $\theta_{\ell'}=0.768689$, $\phi_{\ell'}=3.51351\text{e-}05$
 $Q^2=0.468742$, $x_B=0.352006$, $t=-0.168093$, $W^2=1.74273$, $y=0.787042$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5275$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5275$
 P $E= 1.0762$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5275$
 ℓ' $E= 0.2585$ $p_x= 0.1797$ $p_y= 0.0000$ $p_z= 0.1858$
 P' $E= 0.9482$ $p_x= 0.0882$ $p_y= 0.0000$ $p_z= 0.1072$
 q_Y $E= 0.3970$ $p_x= -0.2679$ $p_y= -0.0000$ $p_z= -0.2930$



electron: polarization cluster P3, local minimum 287; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00025442131$
 $d_{\text{GHZ}} \text{ contour} = 0.050254421$
 above global minimum = 0.00025439903
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.603679$
 $\text{theta_p_out} = 0.6884594$
 $\text{theta_gamma_out} = 2.400926$
 $\text{q0out} = 0.3969817$
 $\text{phi_p_out} = 2.563571\text{e-}06$
 $\text{phi_gamma_out} = 3.141617$
 $\text{alpha_e} = 2.540529$
 $\text{alpha_p} = 1.071428$

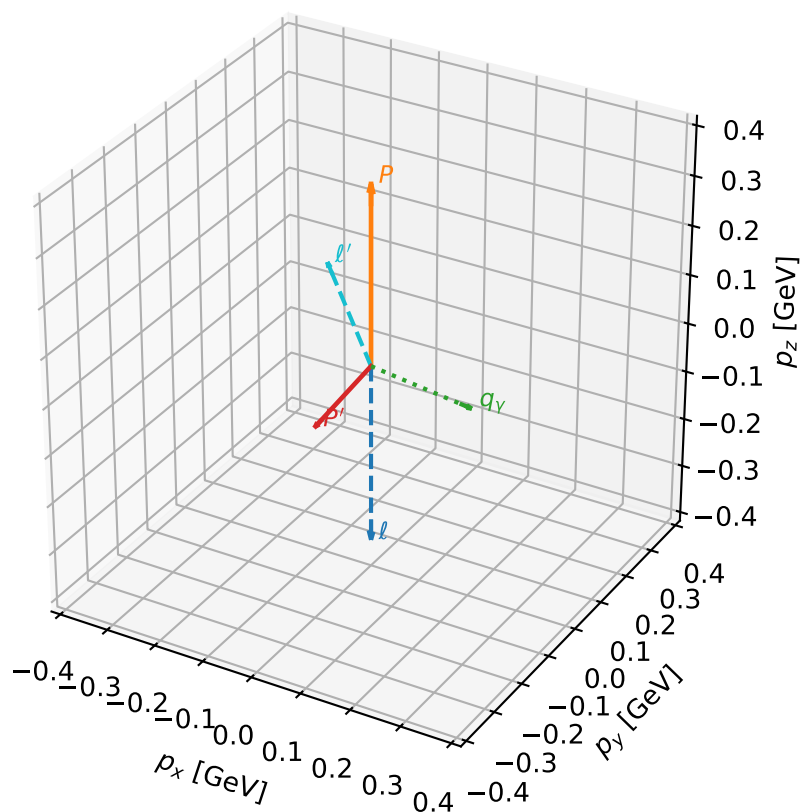
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_290 d_{\rm GHZ}=0.000278129
 region: polarization_cluster_03_local_minimum_290
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.8584687$ rad
 incoming lepton: $-0.283721|+\rangle +0.958907|-\rangle$
 proton mixing angle: $\alpha_p=0.23235983$ rad
 incoming proton: $0.973126|+\rangle +0.230275|-\rangle$

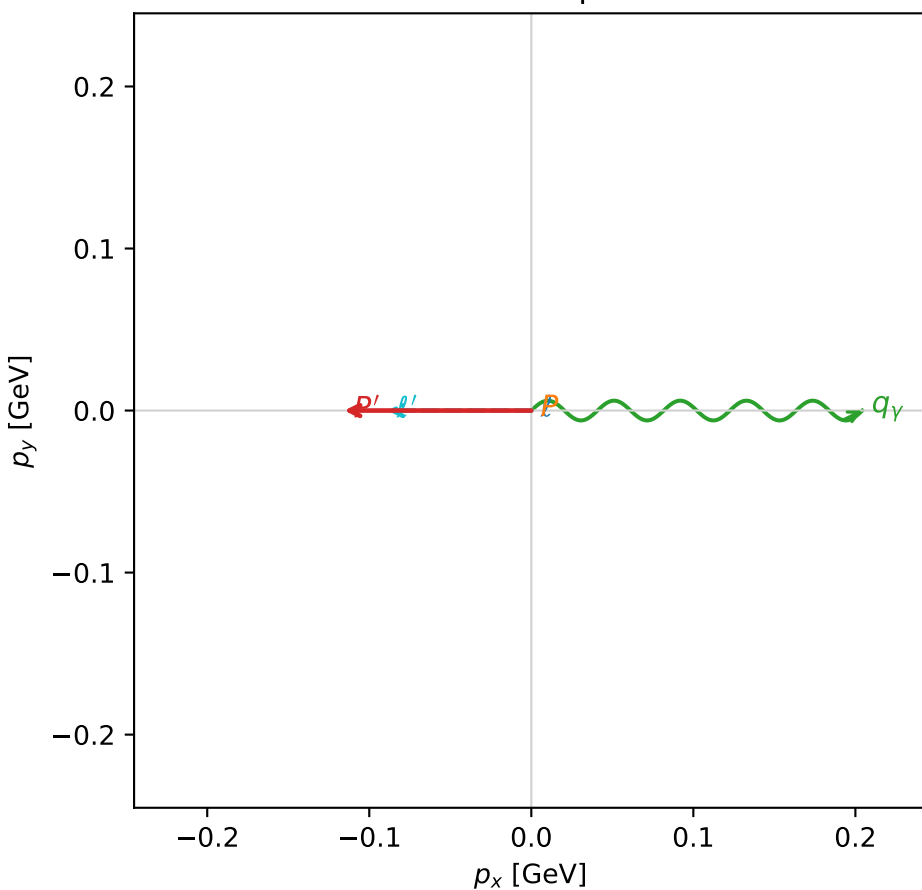
$s=1.87773$, $\sqrt{s}=1.3703$, $|\vec{P}|=0.364111$, $|\vec{P}'|=0.196135$
 $\theta_{P'}=2.5122$, $\theta_Y=1.70329$, $\phi_{P'}=3.14156$, $\phi_Y=6.28315$
 $E_Y=0.206091$, $\theta_{\ell'}=0.446012$, $\phi_{\ell'}=3.14156$
 $Q^2=0.285246$, $x_B=0.396851$, $t=-0.284212$, $W^2=1.31337$, $y=0.720298$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3641$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3641$
 P $E= 1.0062$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3641$
 ℓ' $E= 0.2059$ $p_x= -0.0888$ $p_y= 0.0000$ $p_z= 0.1858$
 P' $E= 0.9583$ $p_x= -0.1155$ $p_y= 0.0000$ $p_z= -0.1586$
 q_Y $E= 0.2061$ $p_x= 0.2043$ $p_y= -0.0000$ $p_z= -0.0272$

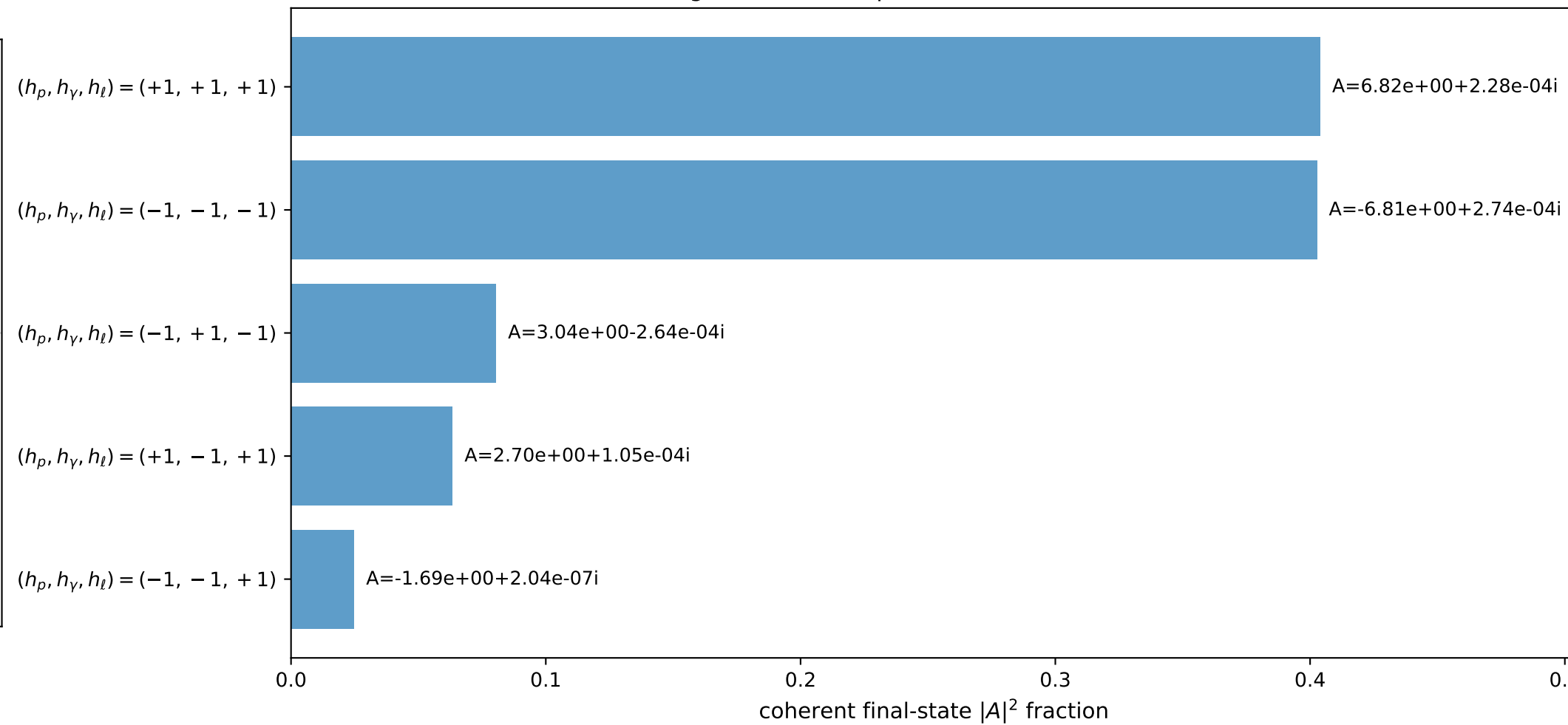
3D momenta



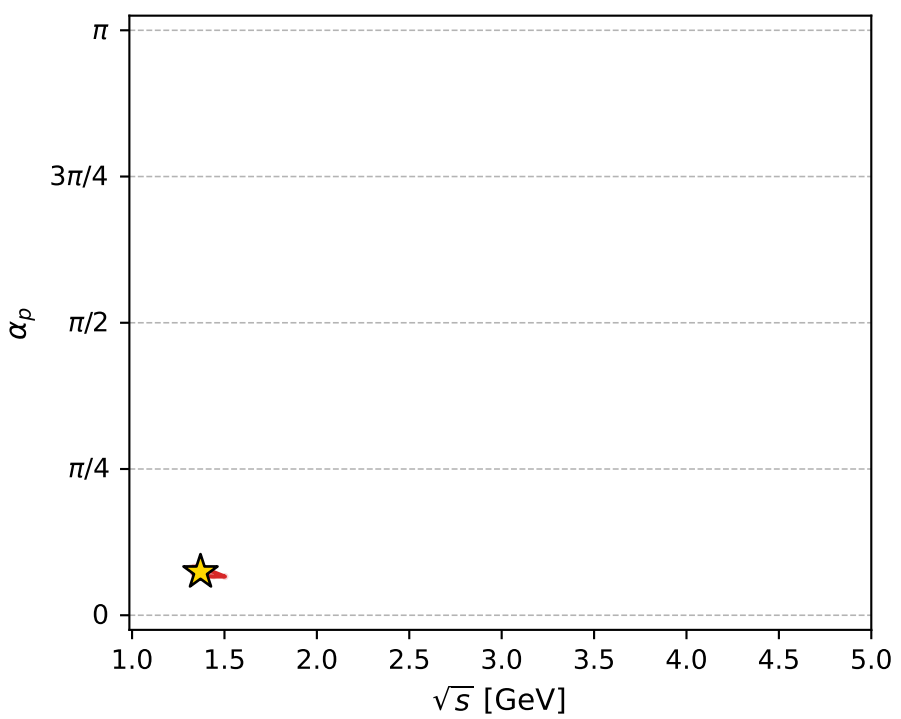
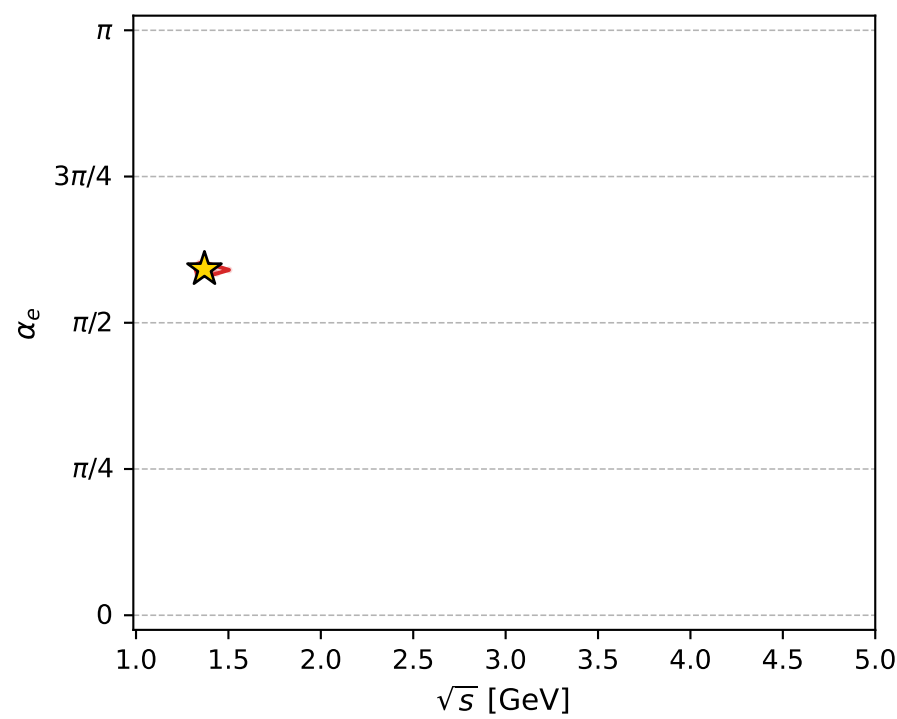
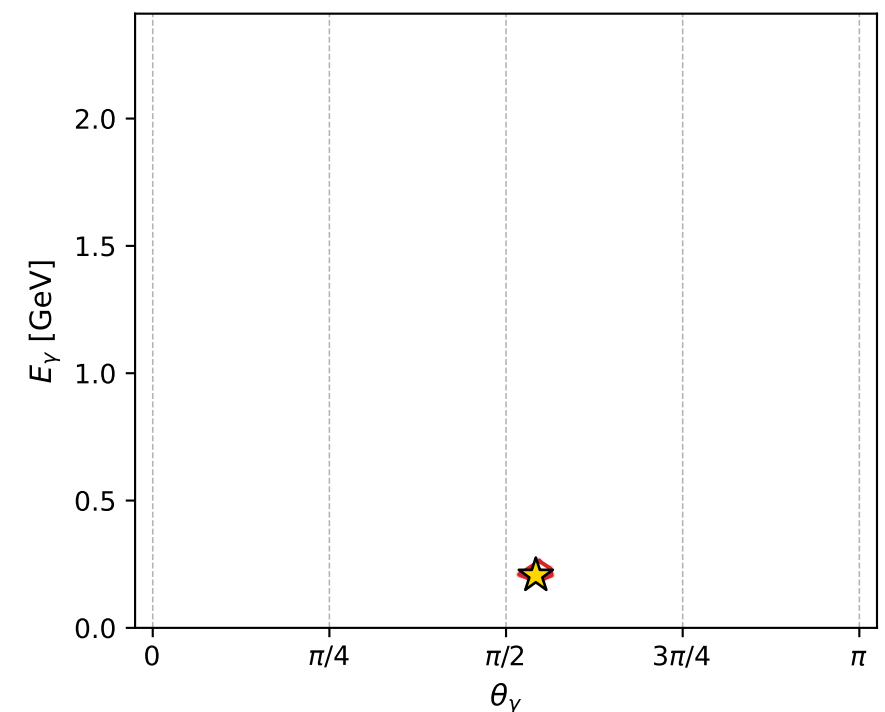
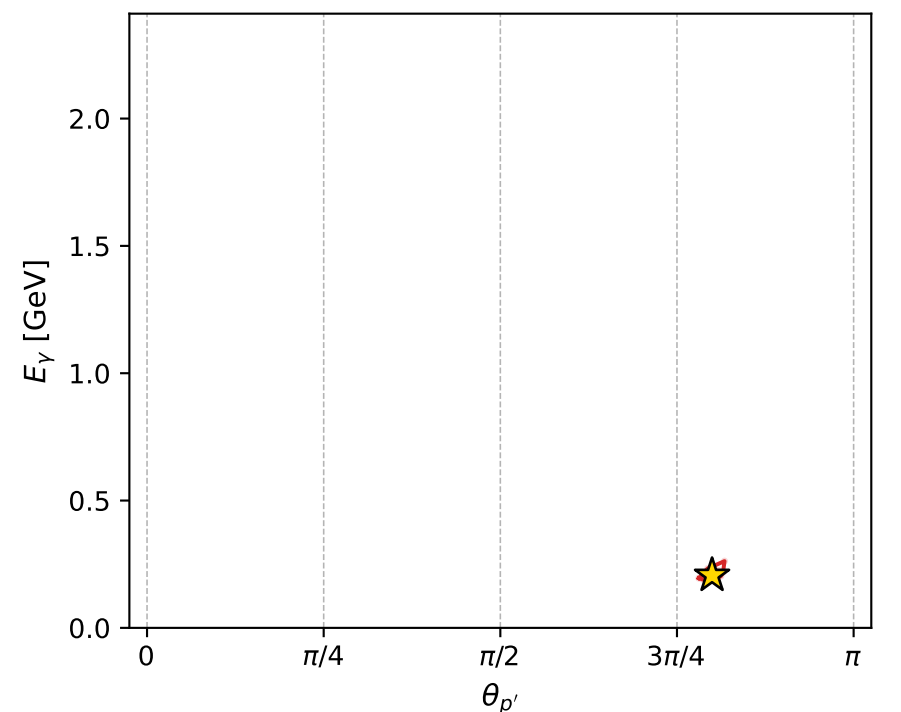
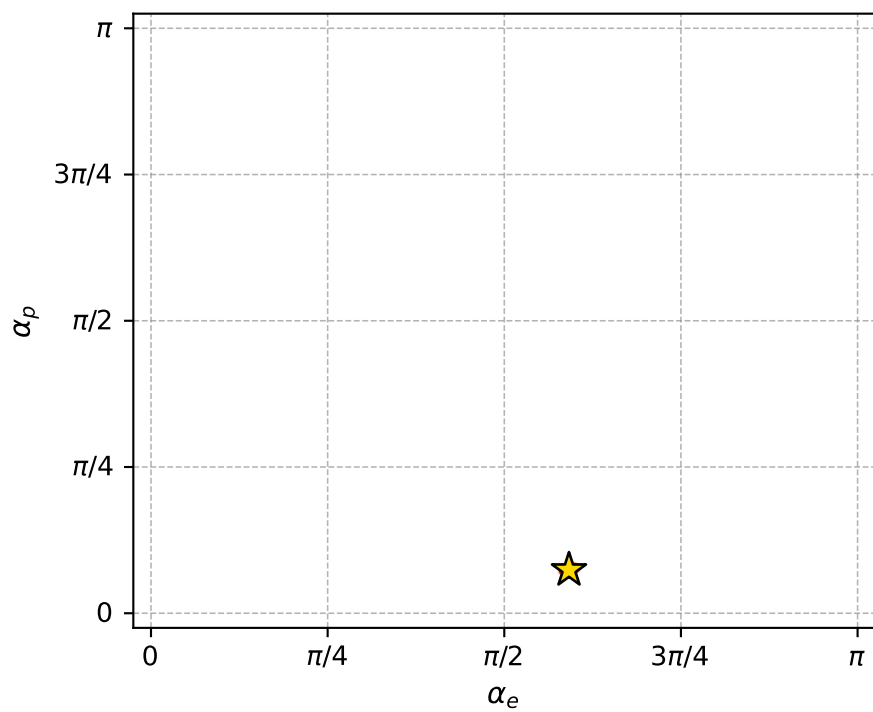
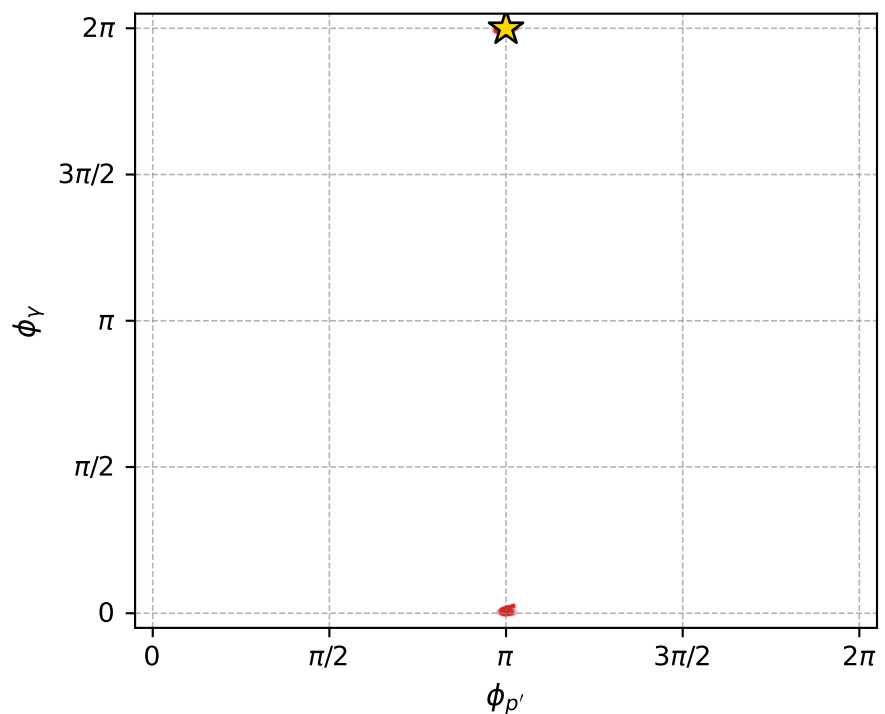
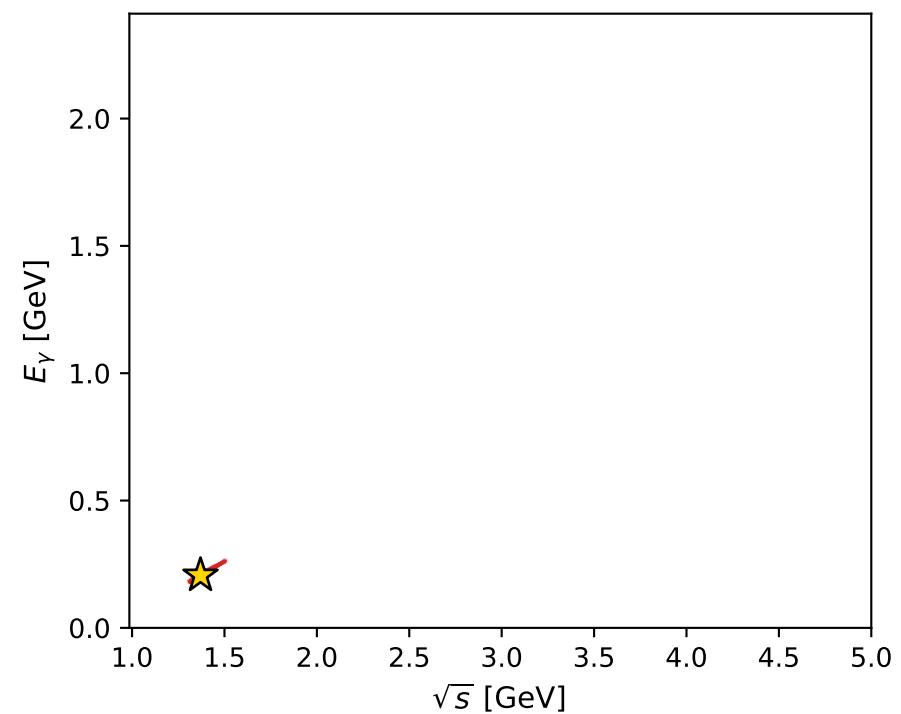
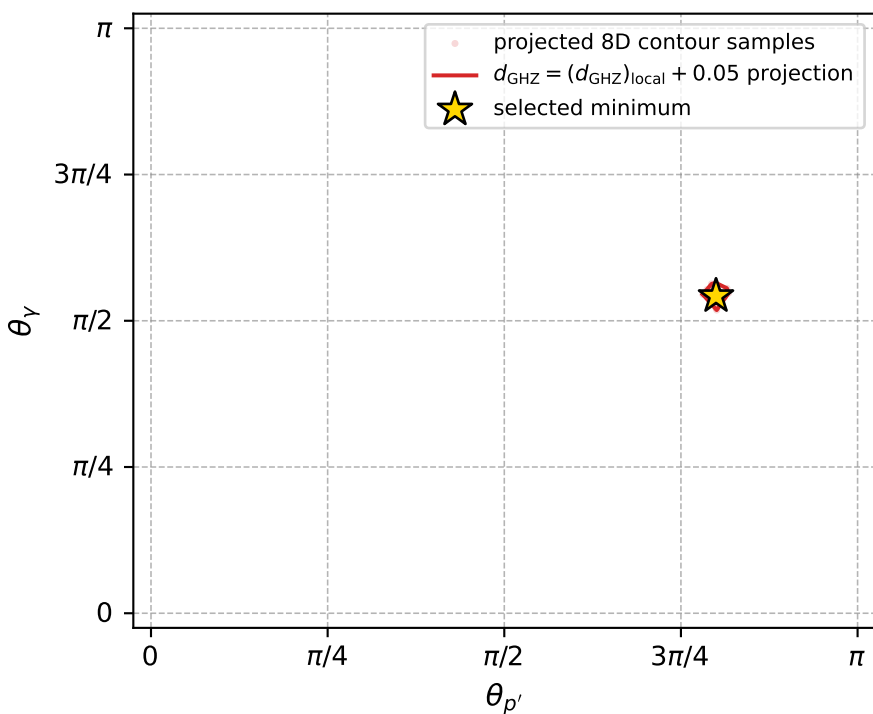
Transverse plane



Leading final-state amplitudes (N=5, retained=97.5%)



electron: polarization cluster P3, local minimum 290; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0002781286$
 $d_{\text{GHZ}} \text{ contour} = 0.050278129$
 above global minimum = 0.00027810631
 8D contour samples = 256

selected phase-space point:

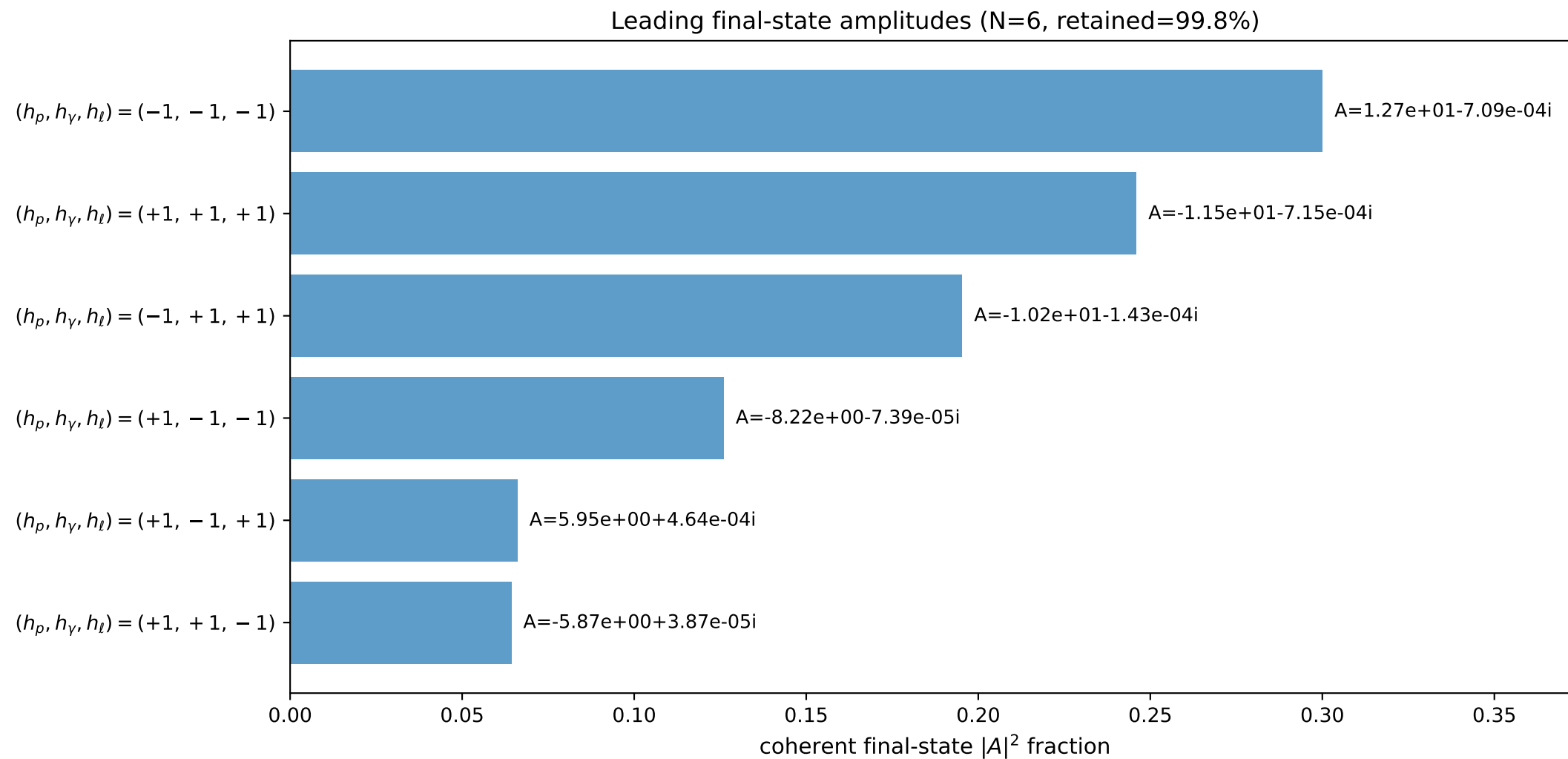
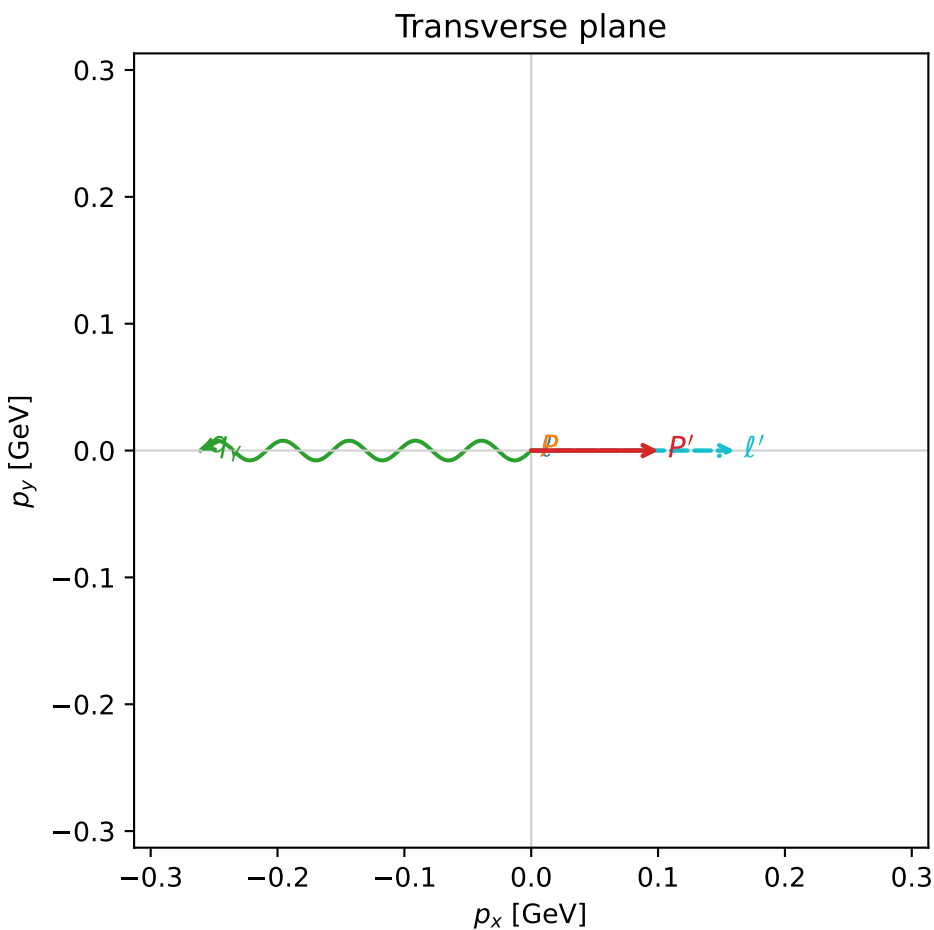
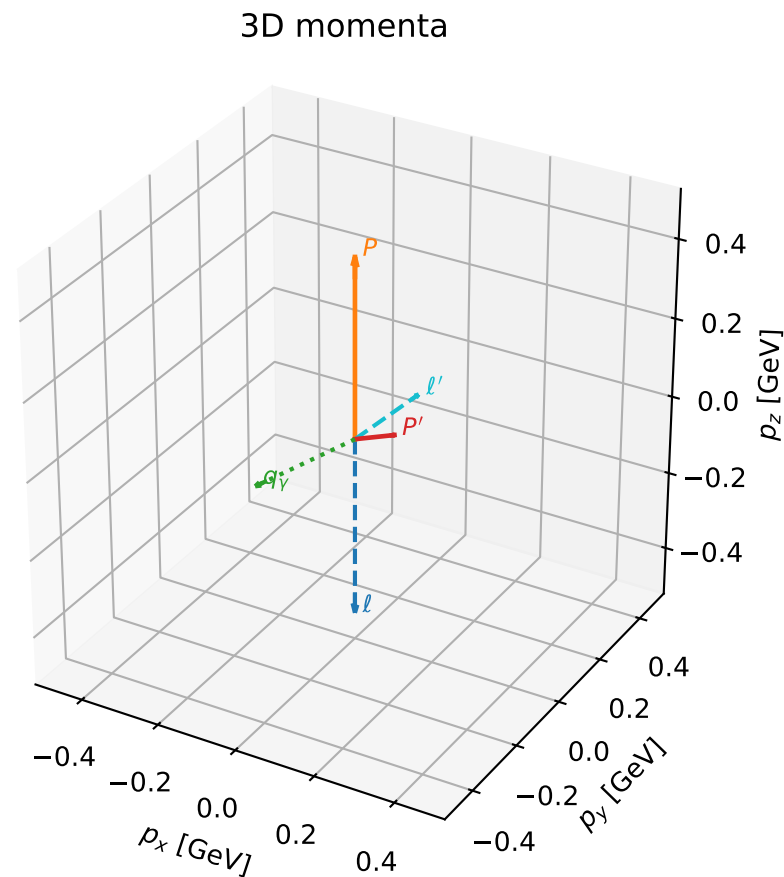
$\text{sqrt_s} = 1.370302$
 $\text{theta_p_out} = 2.5122$
 $\text{theta_gamma_out} = 1.703291$
 $\text{qOut} = 0.2060915$
 $\text{phi_p_out} = 3.141559$
 $\text{phi_gamma_out} = 6.283152$
 $\text{alpha_e} = 1.858469$
 $\text{alpha_p} = 0.2323598$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

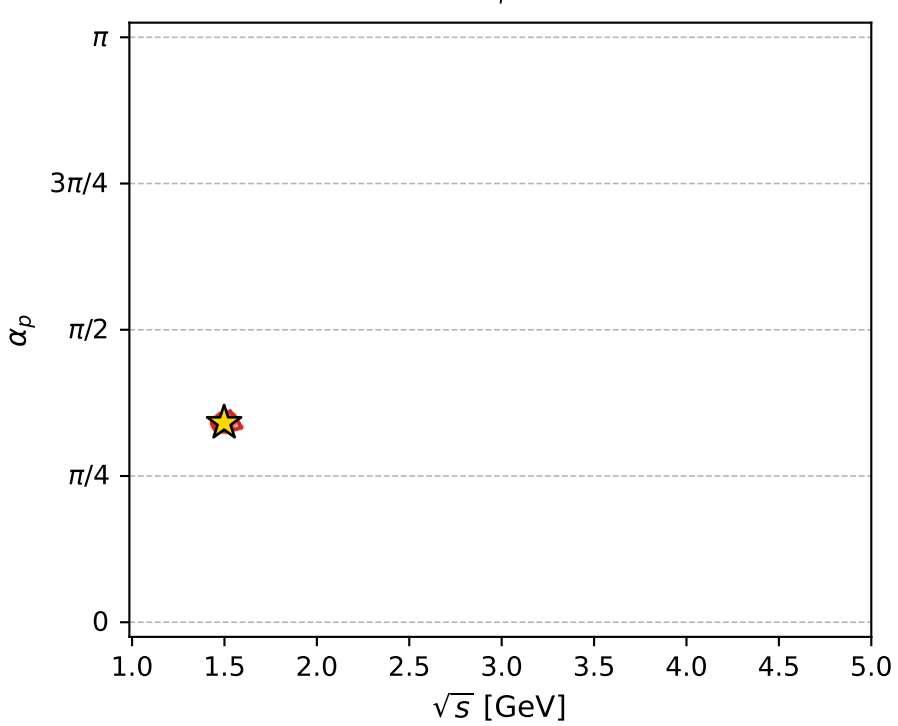
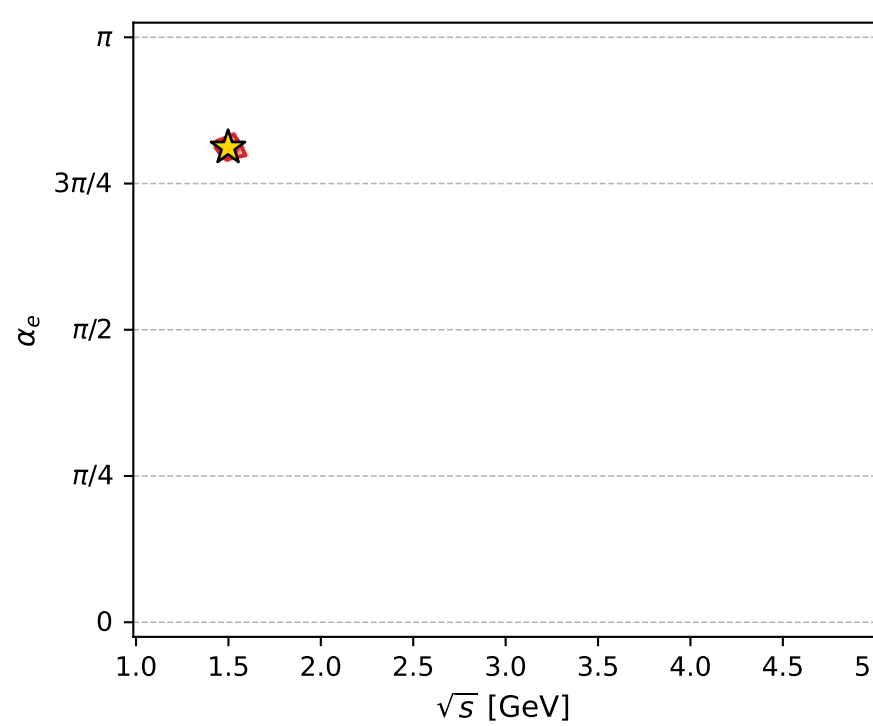
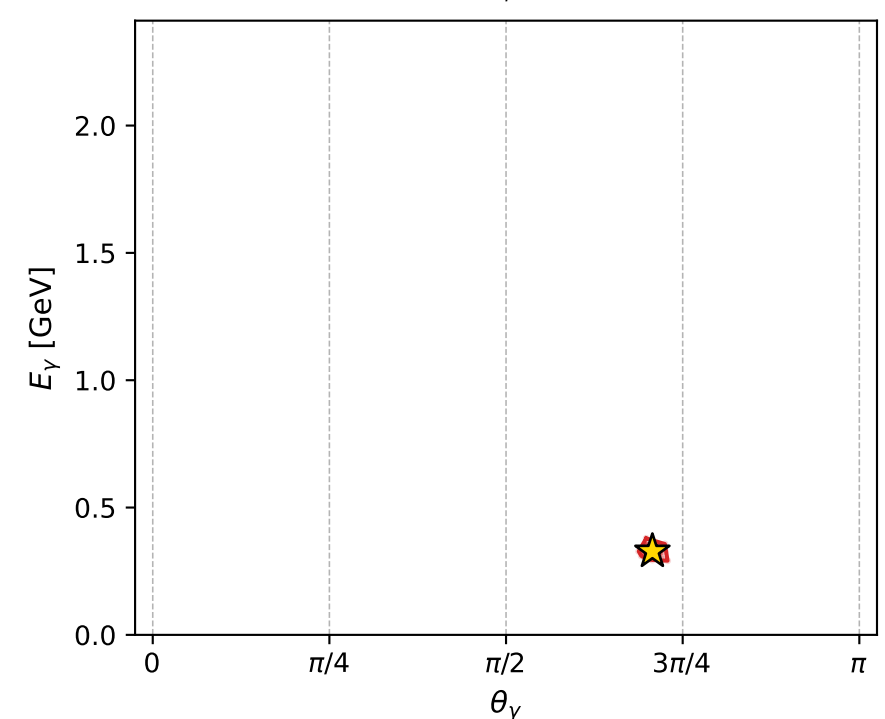
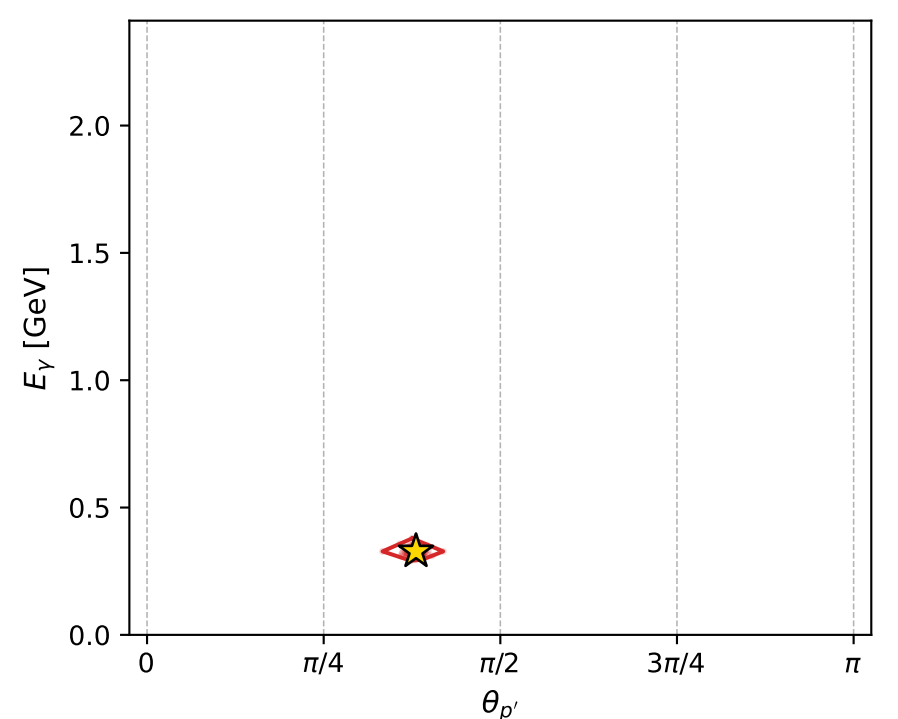
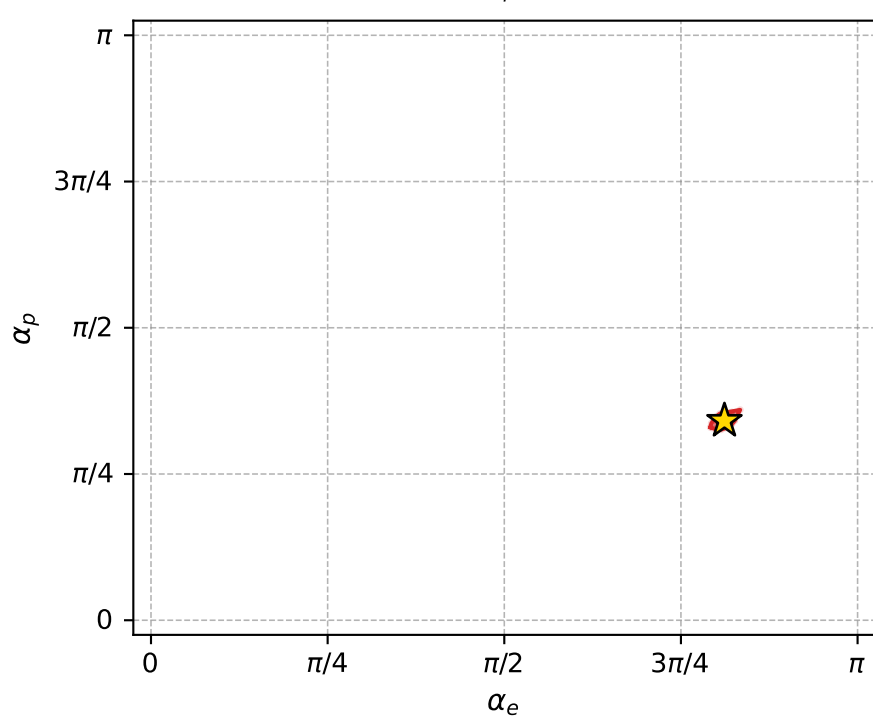
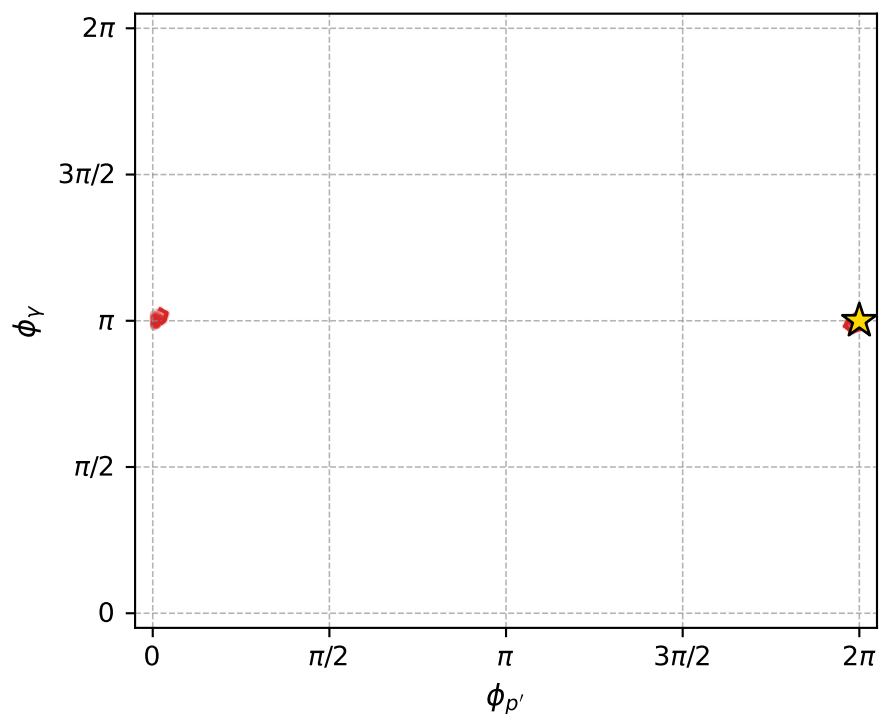
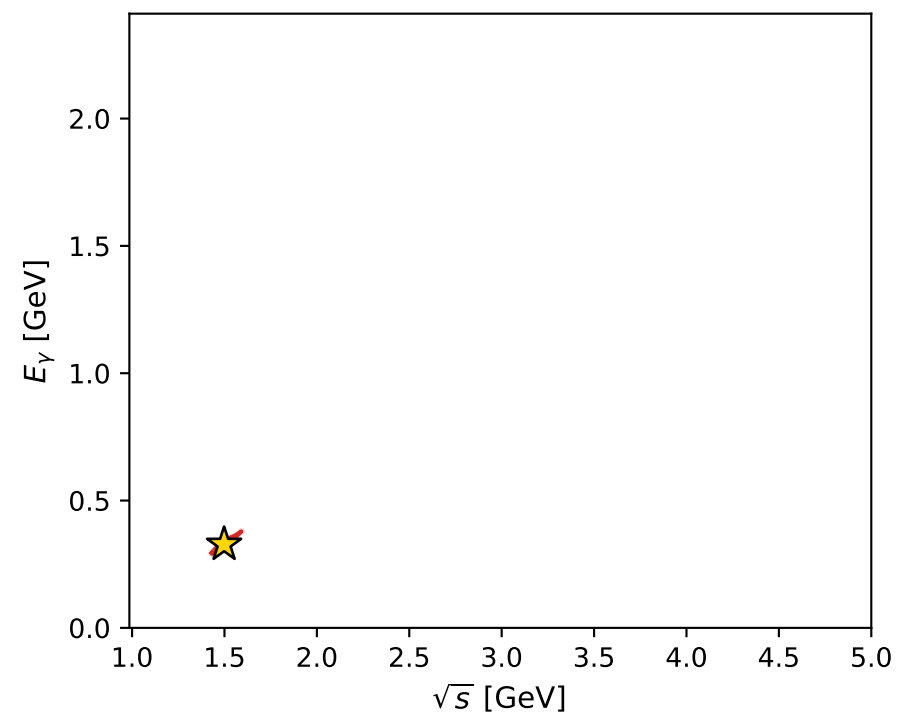
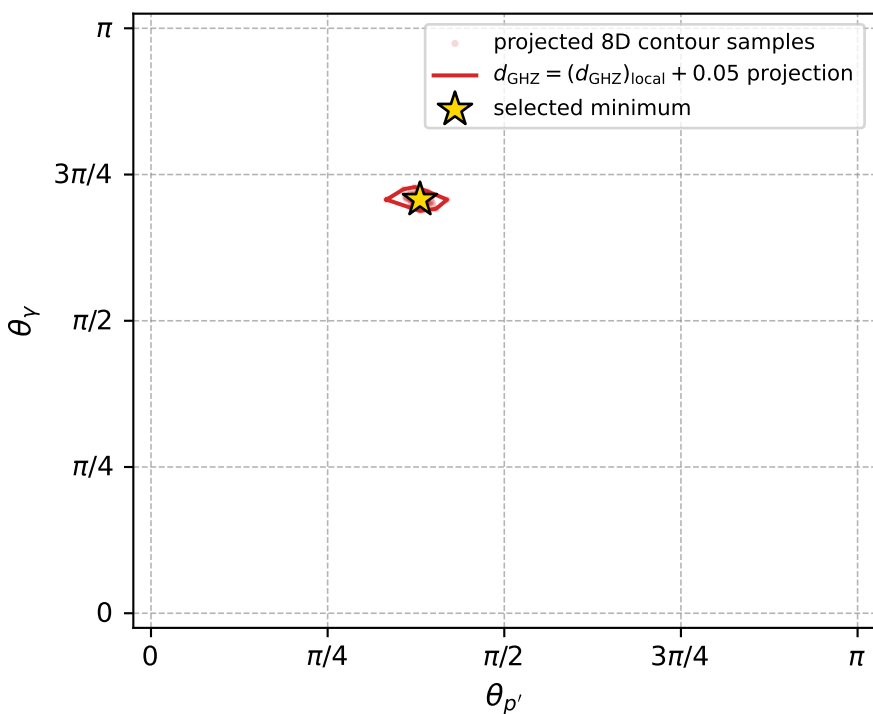
dghz_mixing_angles_polarization_03_local_minimum_299 $d_{\{\text{rm GHZ}\}}=0.000314341$
 region: polarization_cluster_03_local_minimum_299
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.5496543$ rad
 incoming lepton: $-0.829861|+> +0.557971|->$
 proton mixing angle: $\alpha_p=1.0710836$ rad
 incoming proton: $0.479173|+> +0.87772|->$

$s=2.24419$, $\sqrt{s}=1.49806$, $|\vec{P}|=0.455369$, $|\vec{P}'|=0.10794$
 $\theta_{p'}=1.19587$, $\theta_Y=2.2215$, $\phi_{p'}=6.28313$, $\phi_Y=3.14155$
 $E_Y=0.327906$, $\theta_{\ell'}=0.789647$, $\phi_{\ell'}=6.28315$
 $Q^2=0.350694$, $x_B=0.337849$, $t=-0.17331$, $W^2=1.56717$, $y=0.760819$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4554$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4554$
 P $E= 1.0427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4554$
 ℓ' $E= 0.2260$ $p_x= 0.1605$ $p_y= -0.0000$ $p_z= 0.1591$
 P' $E= 0.9442$ $p_x= 0.1004$ $p_y= -0.0000$ $p_z= 0.0395$
 q_Y $E= 0.3279$ $p_x= -0.2609$ $p_y= 0.0000$ $p_z= -0.1986$



electron: polarization cluster P3, local minimum 299; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00031434124$
 $d_{\text{GHZ}} \text{ contour} = 0.050314341$
 above global minimum = 0.00031431896
 8D contour samples = 256

selected phase-space point:

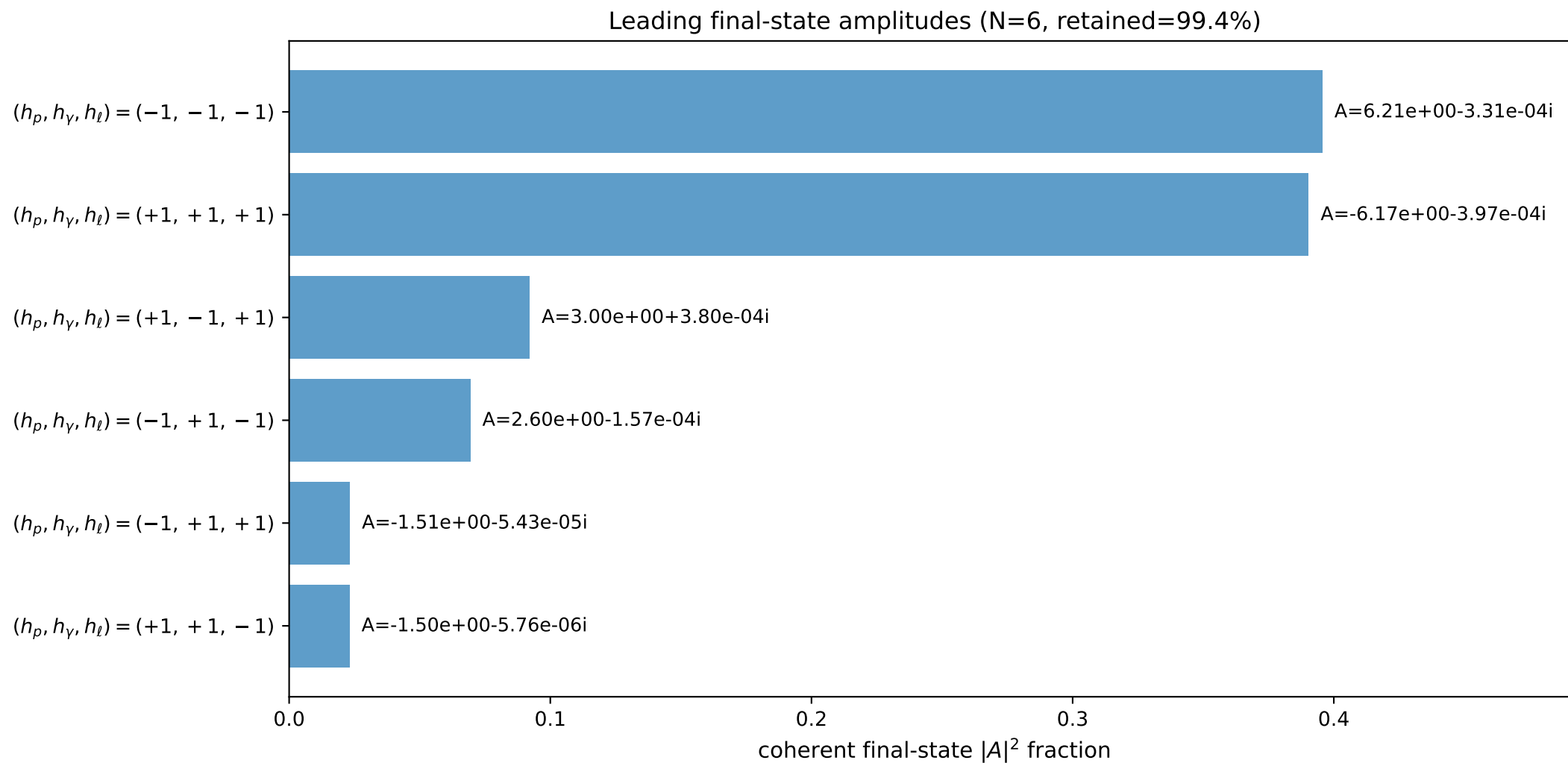
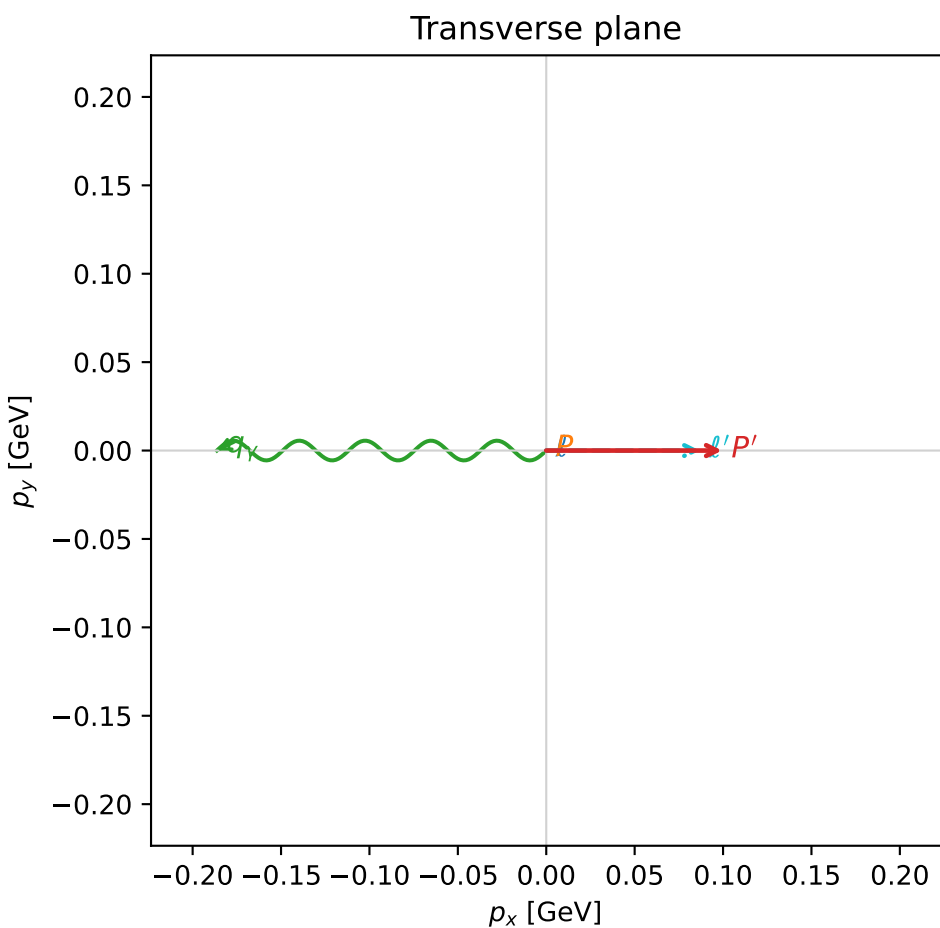
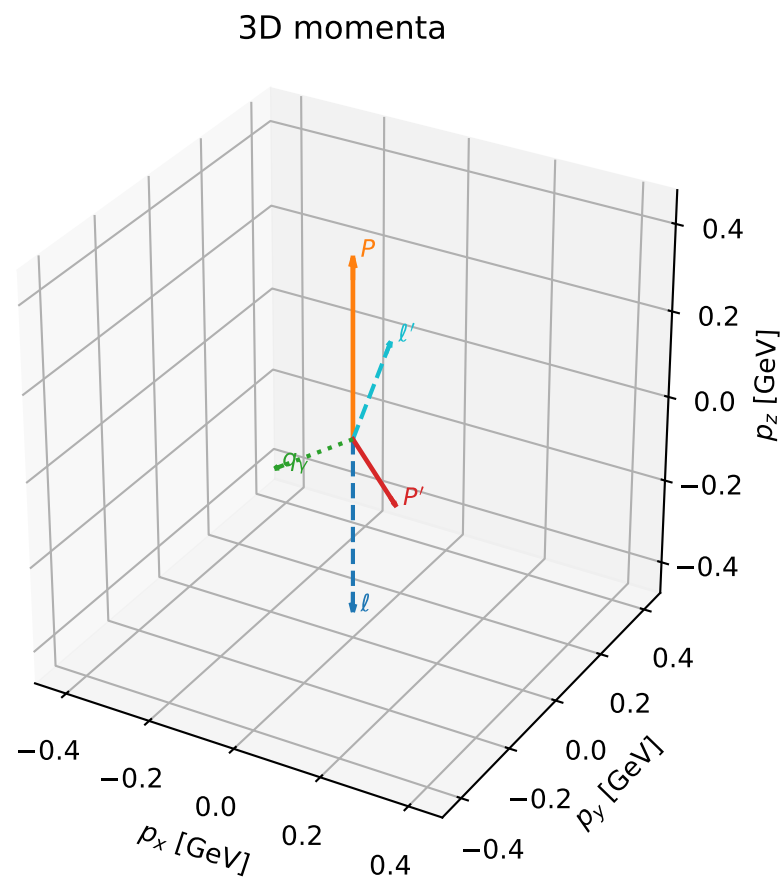
$\text{sqrt_s} = 1.498061$
 $\text{theta_p_out} = 1.19587$
 $\text{theta_gamma_out} = 2.221503$
 $\text{q0out} = 0.3279056$
 $\text{phi_p_out} = 6.283126$
 $\text{phi_gamma_out} = 3.14155$
 $\text{alpha_e} = 2.549654$
 $\text{alpha_p} = 1.071084$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

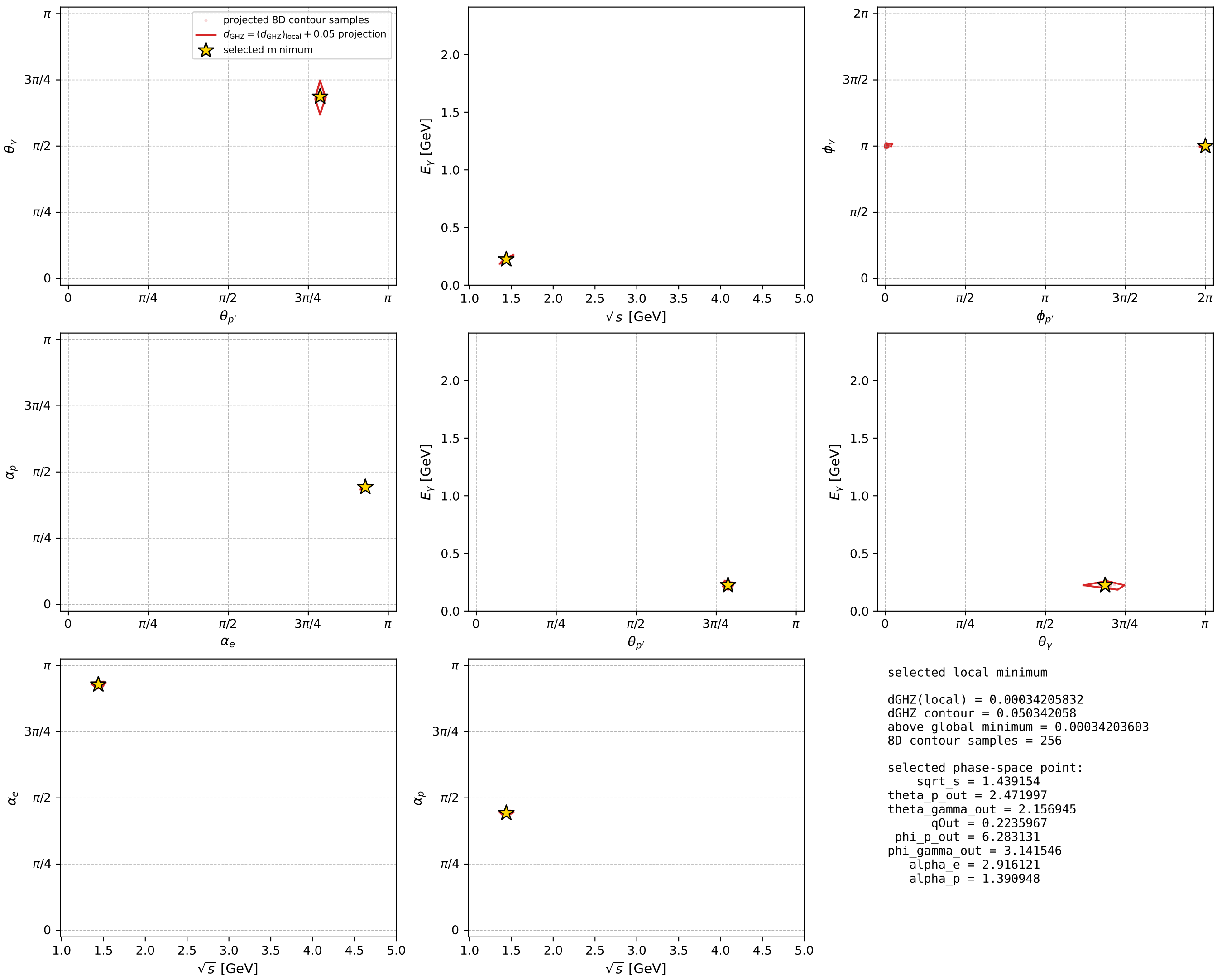
dghz_mixing_angles_polarization_03_local_minimum_306 $d_{\{\text{rm GHZ}\}}=0.000342058$
 region: polarization_cluster_03_local_minimum_306
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.9161208$ rad
 incoming lepton: $-0.974689|+\rangle +0.223566|-\rangle$
 proton mixing angle: $\alpha_p=1.3909478$ rad
 incoming proton: $0.178881|+\rangle +0.983871|-\rangle$

$s=2.07116$, $\sqrt{s}=1.43915$, $|\vec{P}|=0.413896$, $|\vec{P}'|=0.160185$
 $\theta_P=2.472$, $\theta_V=2.15695$, $\phi_{P'}=6.28313$, $\phi_V=3.14155$
 $E_V=0.223597$, $\theta_{\ell'}=0.335256$, $\phi_{\ell'}=6.28315$
 $Q^2=0.424871$, $x_B=0.496123$, $t=-0.295508$, $W^2=1.31135$, $y=0.718851$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4139$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4139$
 P $E= 1.0253$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4139$
 ℓ' $E= 0.2640$ $p_x= 0.0869$ $p_y= -0.0000$ $p_z= 0.2493$
 P' $E= 0.9516$ $p_x= 0.0994$ $p_y= -0.0000$ $p_z= -0.1256$
 q_V $E= 0.2236$ $p_x= -0.1863$ $p_y= 0.0000$ $p_z= -0.1237$



electron: polarization cluster P3, local minimum 306; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



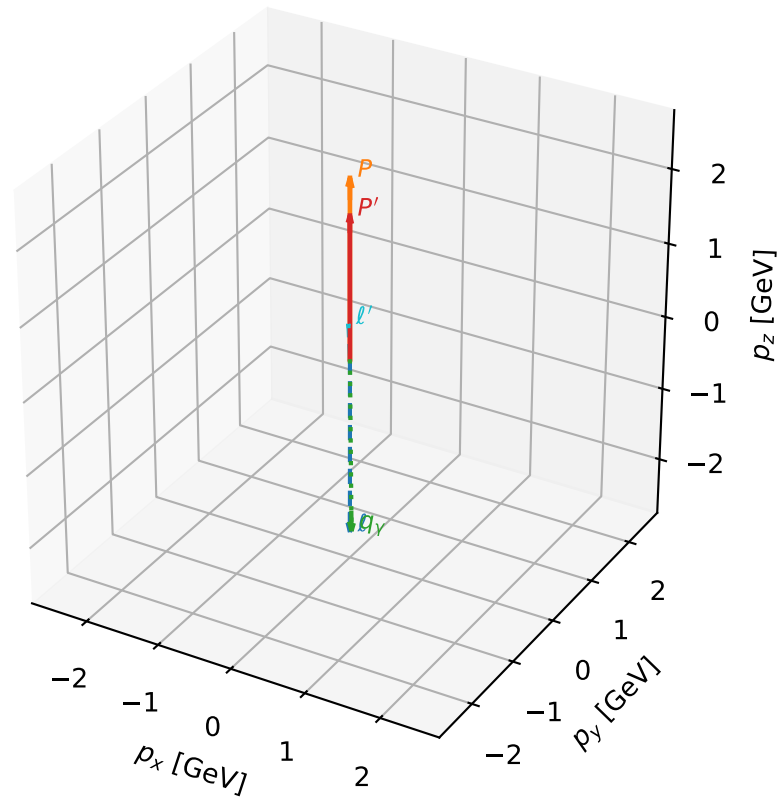
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_316 d_{\rm GHZ}=0.00040247
region: polarization_cluster_03_local_minimum_316
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.555944$ rad
incoming lepton: $-0.833354|+\rangle +0.55274|-\rangle$
proton mixing angle: $\alpha_p=0.97424173$ rad
incoming proton: $0.561796|+\rangle +0.827276|-\rangle$

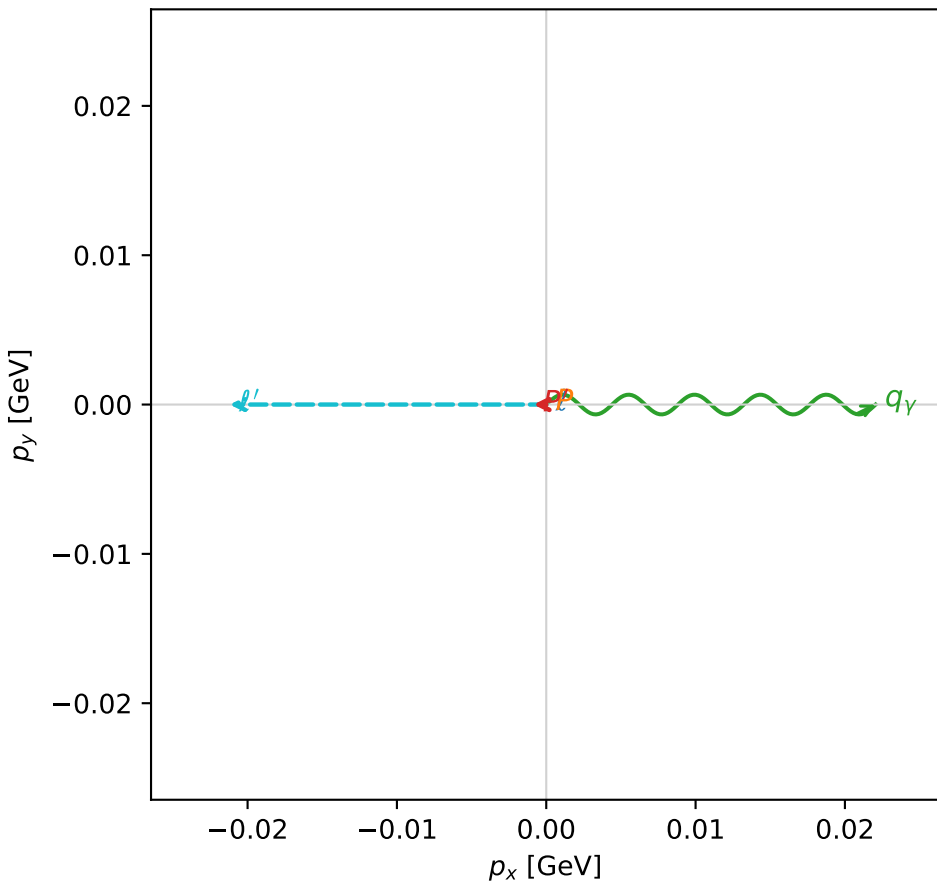
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.91934$
 $\theta_{P'}=0.000445521$, $\theta_V=3.13237$, $\phi_{P'}=3.14125$, $\phi_V=6.28316$
 $E_V=2.39134$, $\theta_{\ell'}=0.0448925$, $\phi_{\ell'}=3.14158$
 $Q^2=4.55525$, $x_B=0.190186$, $t=-0.0386979$, $W^2=20.2762$, $y=0.993012$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.4724$ $p_x= -0.0212$ $p_y= 0.0000$ $p_z= 0.4719$
 P' $E= 2.1363$ $p_x= -0.0009$ $p_y= 0.0000$ $p_z= 1.9193$
 q_V $E= 2.3913$ $p_x= 0.0221$ $p_y= -0.0000$ $p_z= -2.3912$

3D momenta



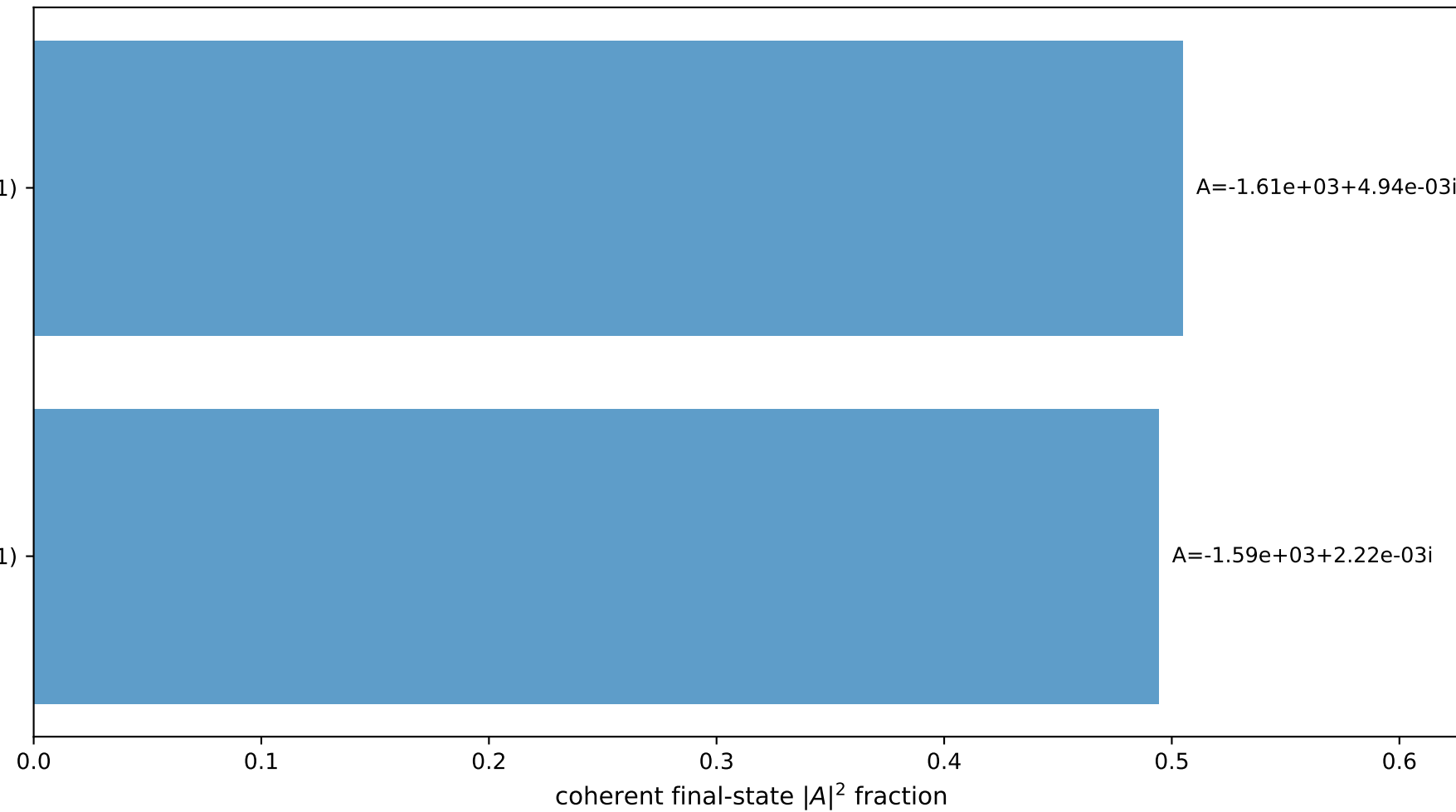
Transverse plane



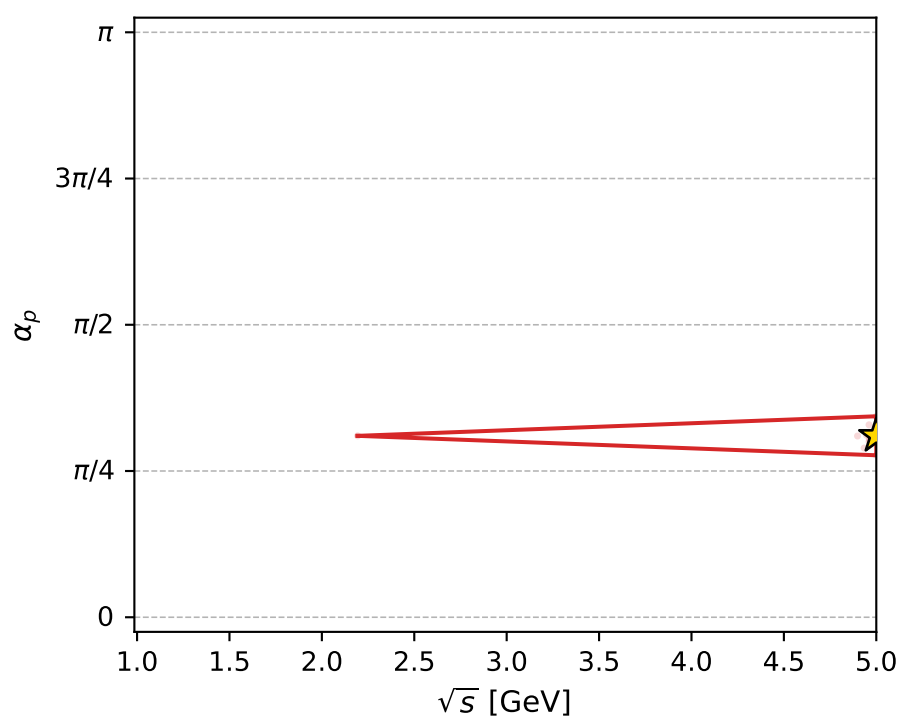
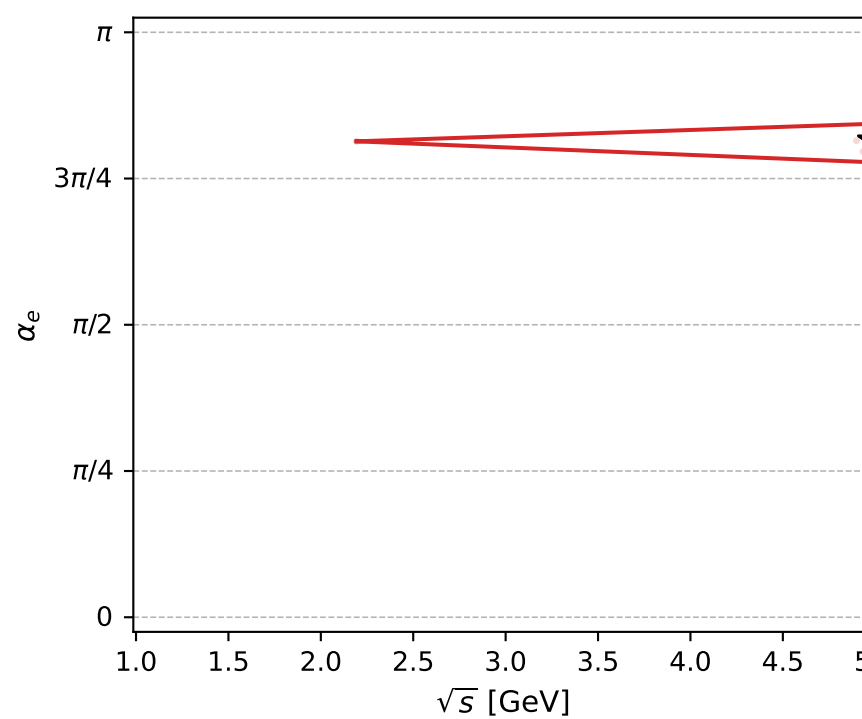
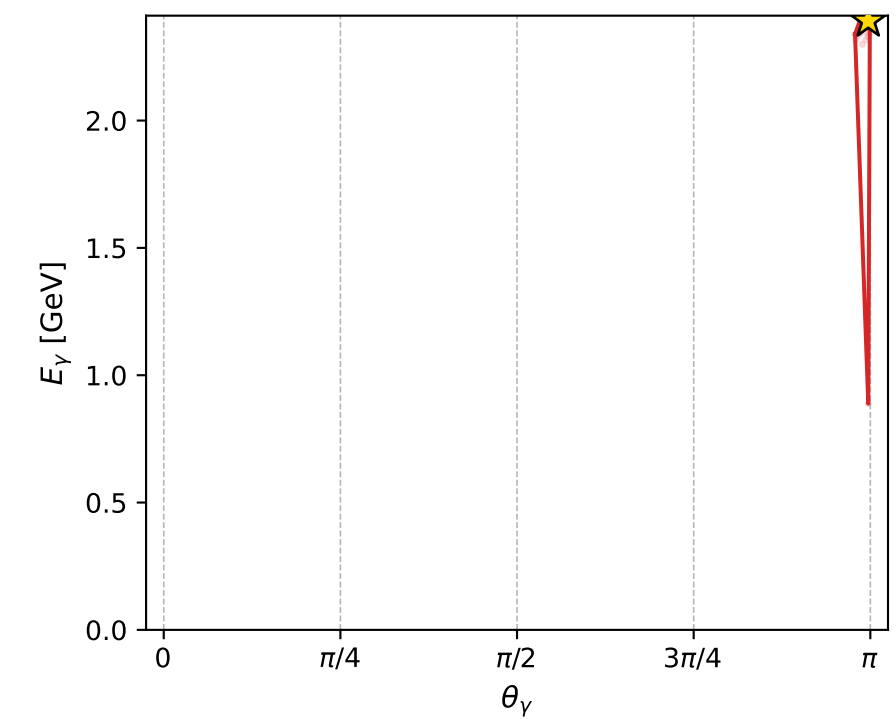
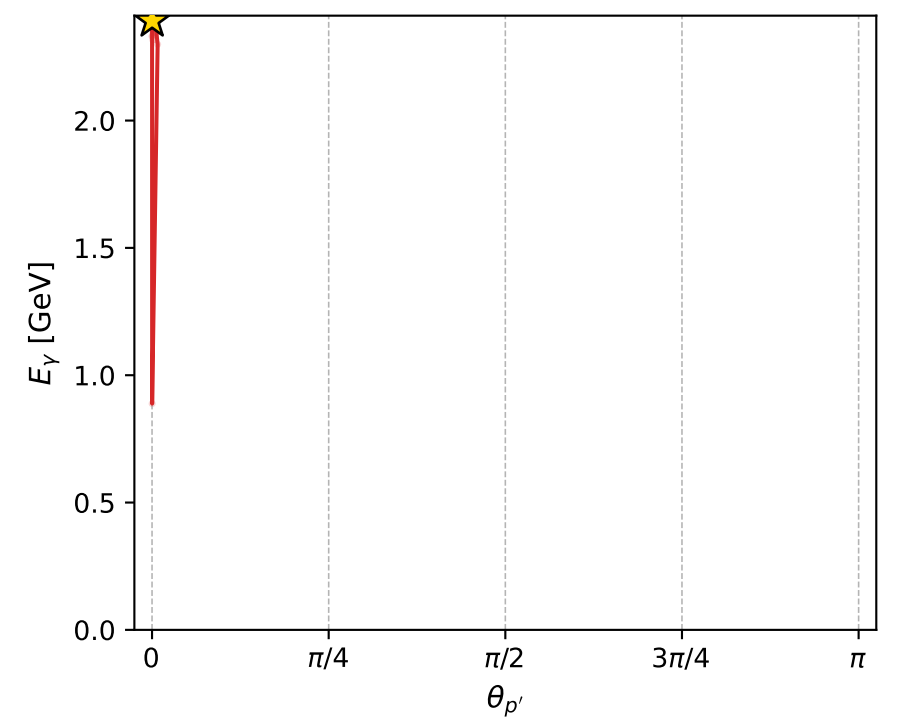
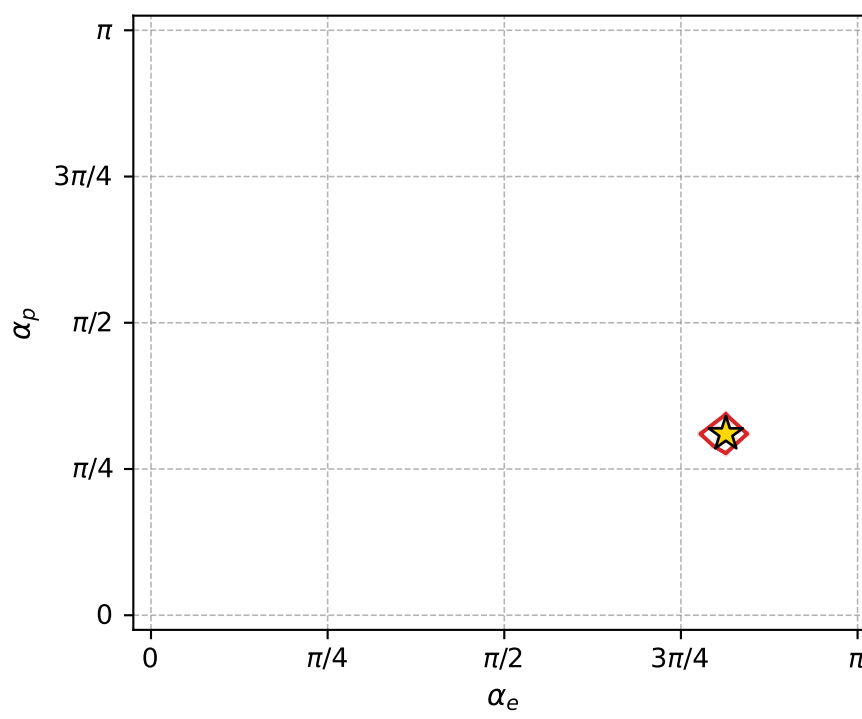
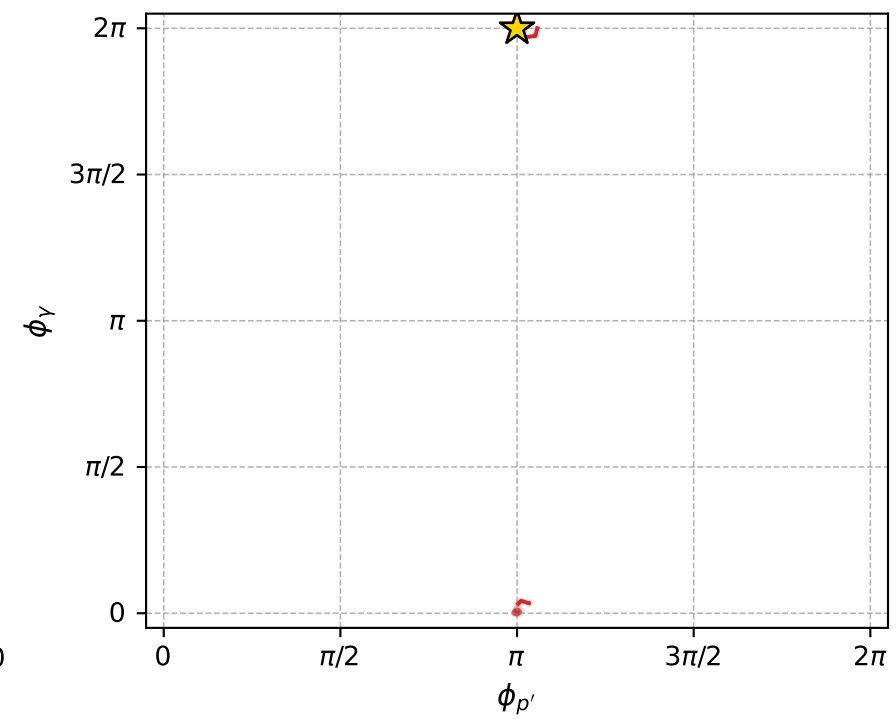
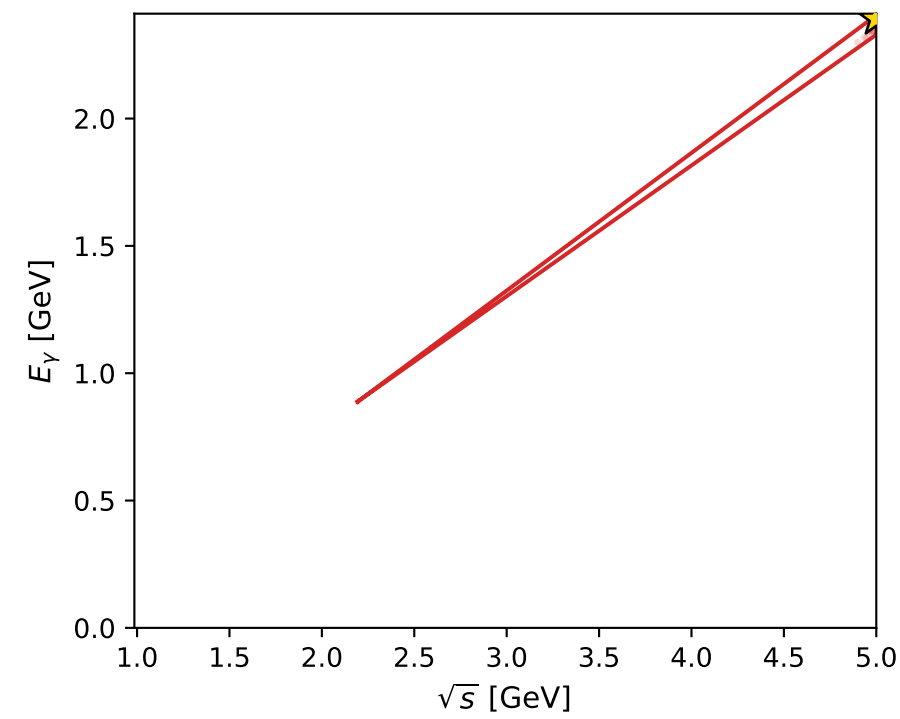
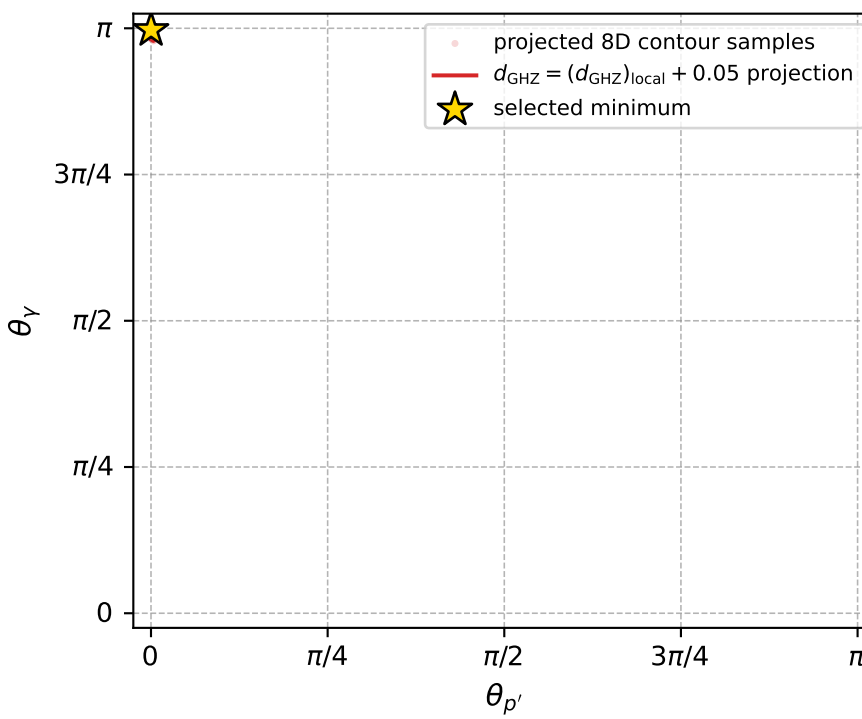
$(h_p, h_V, h_\ell) = (-1, +1, +1)$

$(h_p, h_V, h_\ell) = (+1, -1, -1)$

Leading final-state amplitudes (N=2, retained=99.9%)



electron: polarization cluster P3, local minimum 316; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00040246951$
 $d_{\text{GHZ}} \text{ contour} = 0.05040247$
 above global minimum = 0.00040244722
 8D contour samples = 64

selected phase-space point:

$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0004455212$
 $\text{theta_gamma_out} = 3.13237$
 $\text{q0ut} = 2.39134$
 $\text{phi_p_out} = 3.141248$
 $\text{phi_gamma_out} = 6.283159$
 $\alpha_e = 2.555944$
 $\alpha_p = 0.9742417$

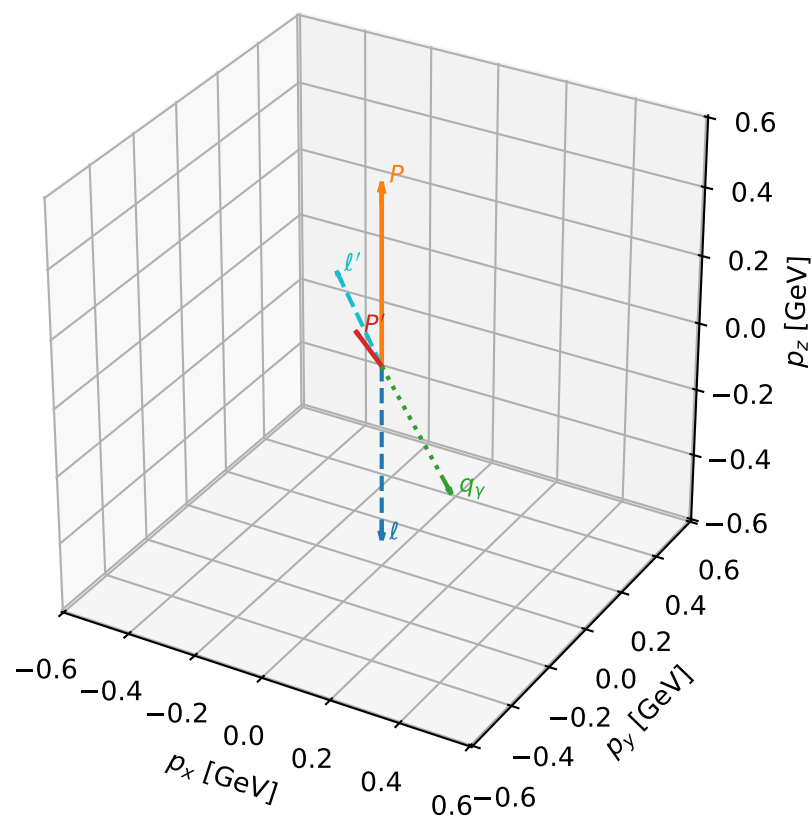
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_320 $d_{\{\text{rm GHZ}\}}=0.000422163$
 region: polarization_cluster_03_local_minimum_320
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.9592544$ rad
 incoming lepton: $-0.378762|+> +0.925494|->$
 proton mixing angle: $\alpha_p=0.32370082$ rad
 incoming proton: $0.948065|+> +0.318077|->$

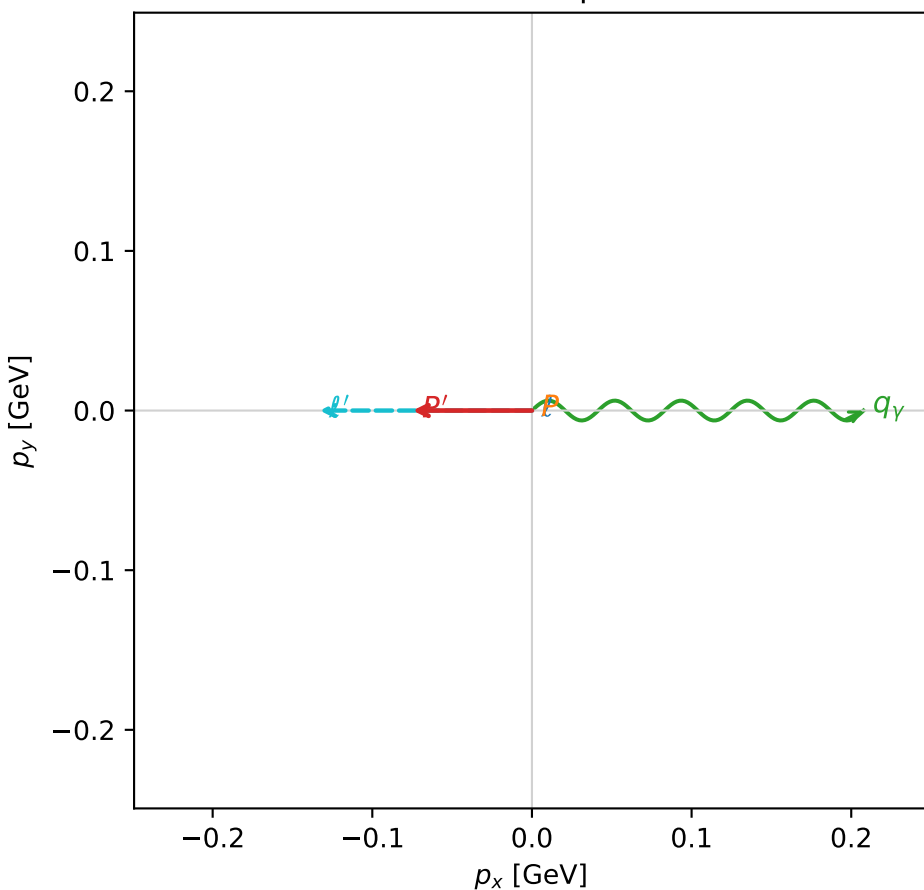
$s=2.56676$, $\sqrt{s}=1.60211$, $|\vec{P}|=0.526466$, $|\vec{P}'|=0.108676$
 $\theta_{P'}=0.757807$, $\theta_Y=2.56677$, $\phi_{P'}=3.14154$, $\phi_Y=6.28316$
 $E_Y=0.381994$, $\theta_{\ell'}=0.50307$, $\phi_{\ell'}=3.14159$
 $Q^2=0.544901$, $x_B=0.404242$, $t=-0.188604$, $W^2=1.6829$, $y=0.799067$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5265$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5265$
 P $E= 1.0756$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5265$
 ℓ' $E= 0.2758$ $p_x= -0.1330$ $p_y= -0.0000$ $p_z= 0.2417$
 P' $E= 0.9443$ $p_x= -0.0747$ $p_y= 0.0000$ $p_z= 0.0789$
 q_Y $E= 0.3820$ $p_x= 0.2077$ $p_y= -0.0000$ $p_z= -0.3206$

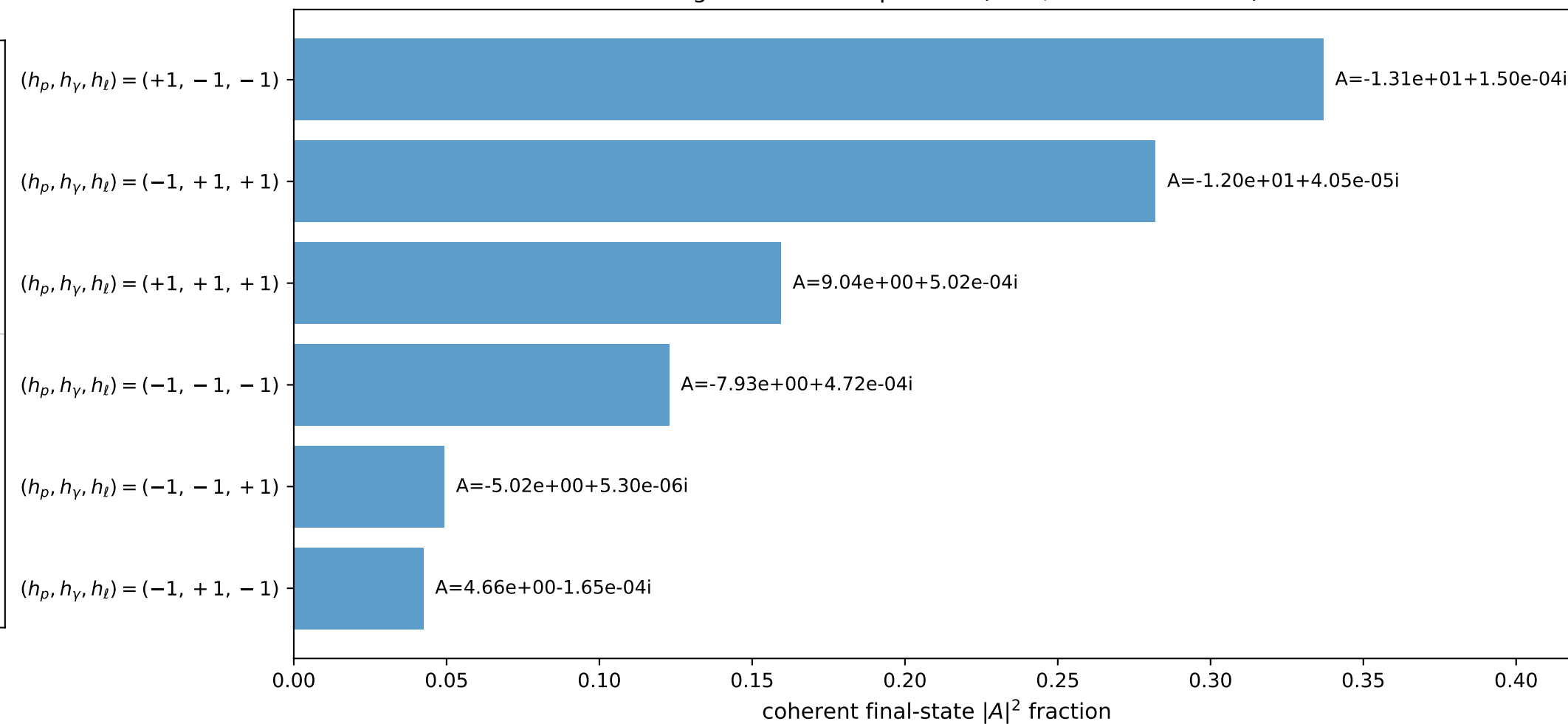
3D momenta



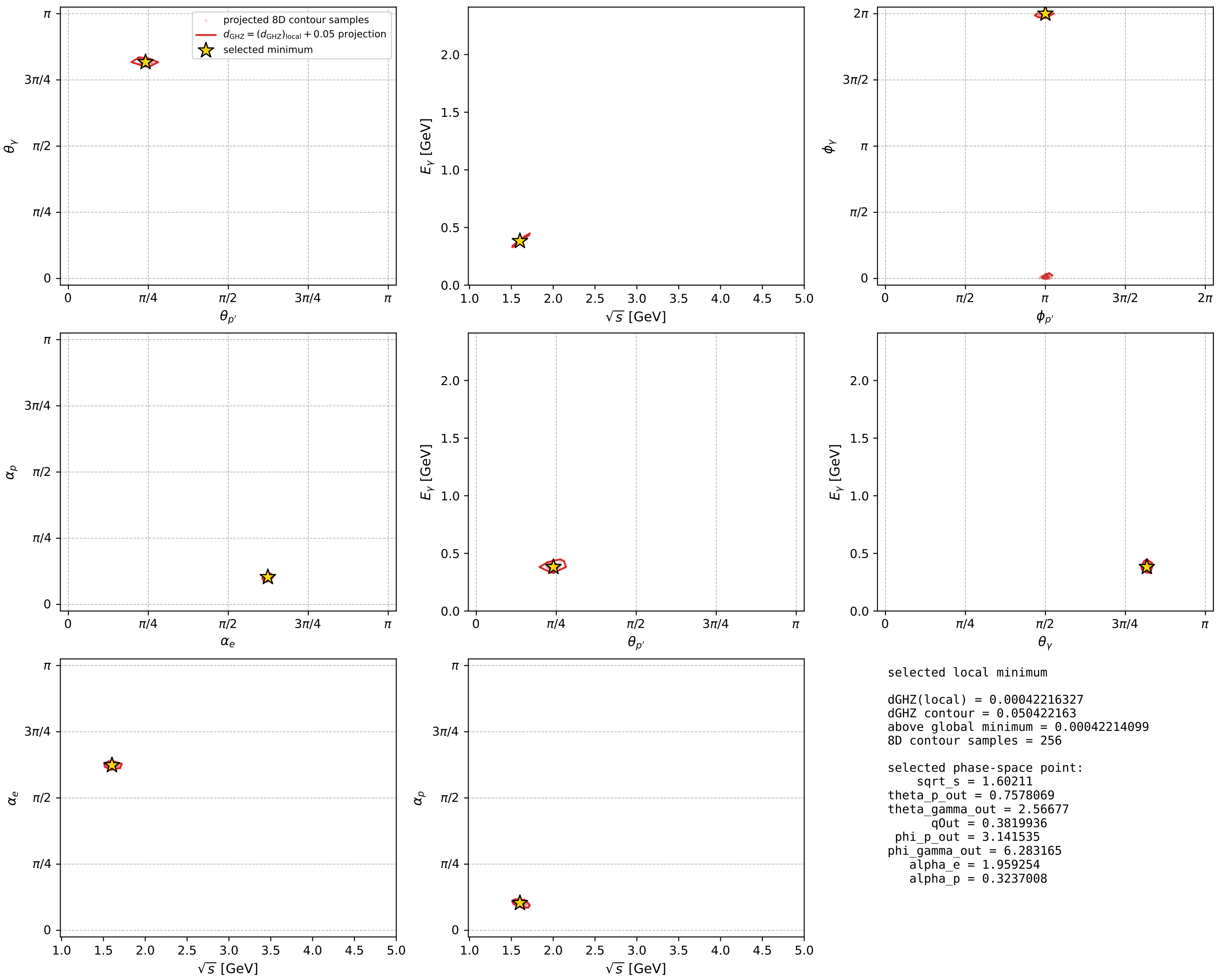
Transverse plane



Leading final-state amplitudes (N=6, retained=99.2%)



electron: polarization cluster P3, local minimum 320; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



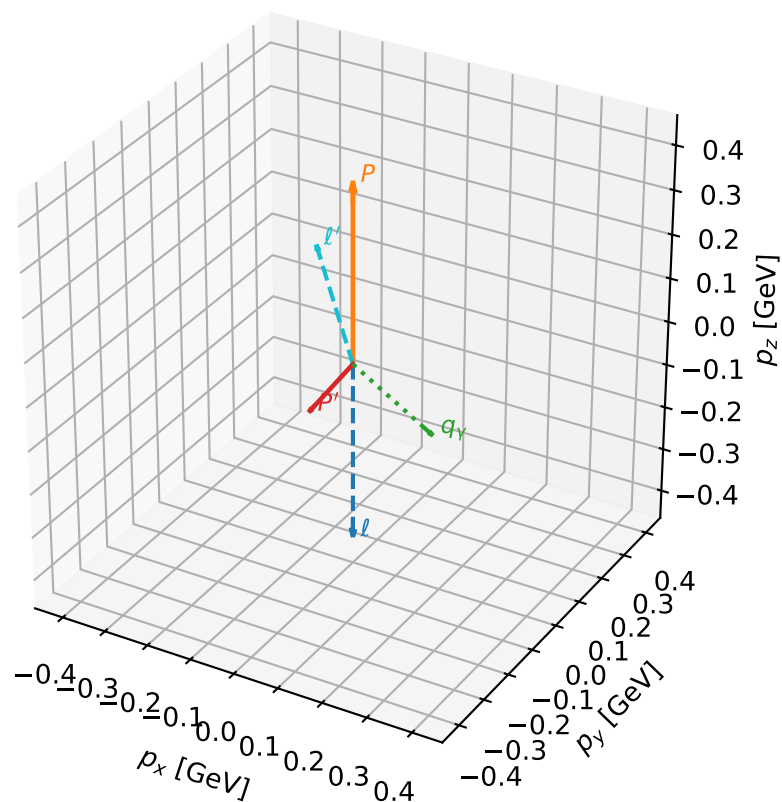
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_328 $d_{\{\text{rm GHZ}\}}=0.000467608$
 region: polarization_cluster_03_local_minimum_328
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.7886912$ rad
 incoming lepton: $-0.216175|+\rangle +0.976355|-\rangle$
 proton mixing angle: $\alpha_p=0.17362968$ rad
 incoming proton: $0.984964|+\rangle +0.172759|-\rangle$

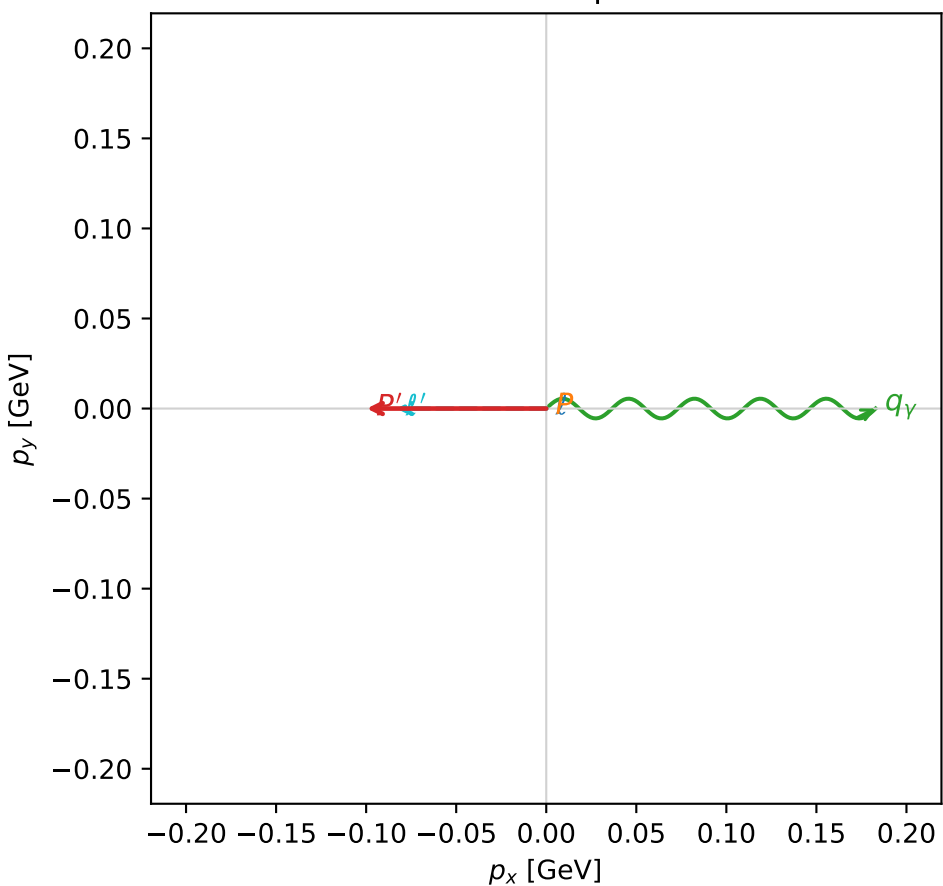
$s=2.03738$, $\sqrt{s}=1.42737$, $|\vec{P}|=0.40548$, $|\vec{P}'|=0.169266$
 $\theta_{P'}=2.51205$, $\theta_Y=2.11333$, $\phi_{P'}=3.14159$, $\phi_Y=6.28318$
 $E_Y=0.213524$, $\theta_{\ell'}=0.32484$, $\phi_{\ell'}=3.14159$
 $Q^2=0.411771$, $x_B=0.499061$, $t=-0.299294$, $W^2=1.29317$, $y=0.712798$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.4055$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.4055$
 P $E=1.0219$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.4055$
 ℓ' $E=0.2607$ $p_x=-0.0832$ $p_y=0.0000$ $p_z=0.2471$
 P' $E=0.9532$ $p_x=-0.0997$ $p_y=0.0000$ $p_z=-0.1368$
 q_Y $E=0.2135$ $p_x=0.1829$ $p_y=-0.0000$ $p_z=-0.1102$

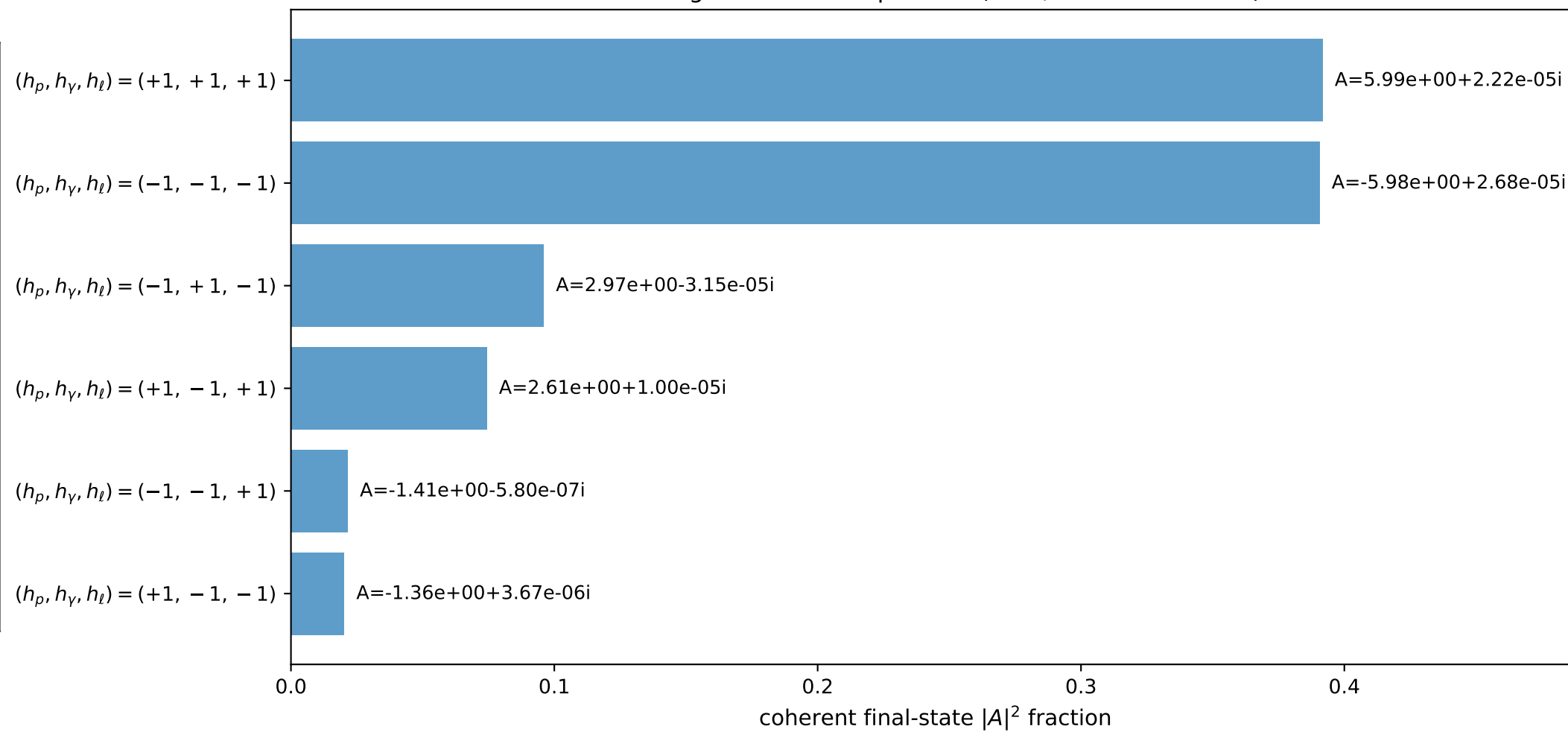
3D momenta



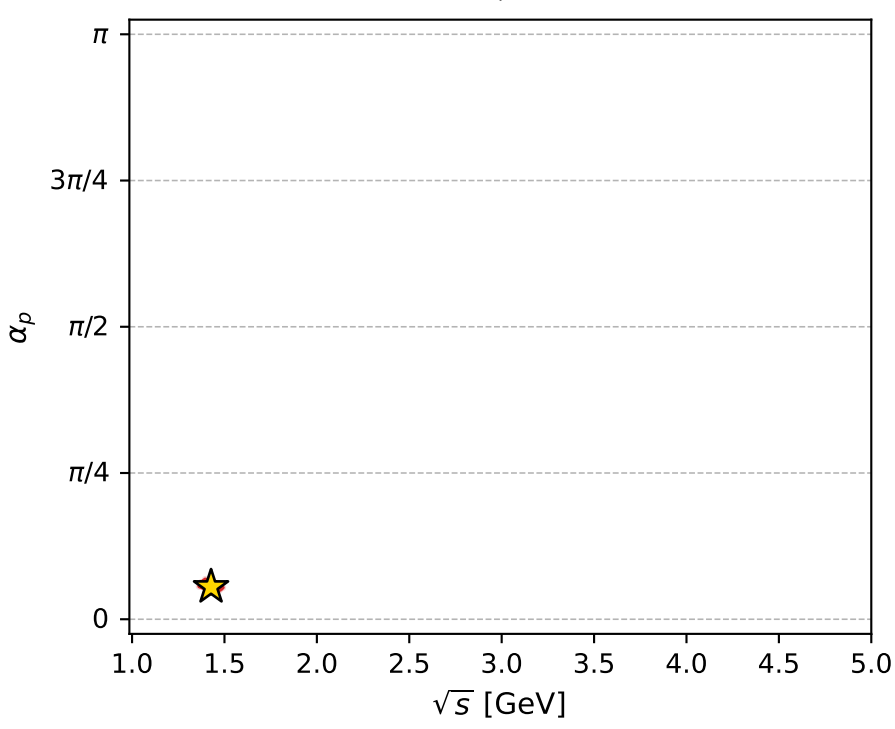
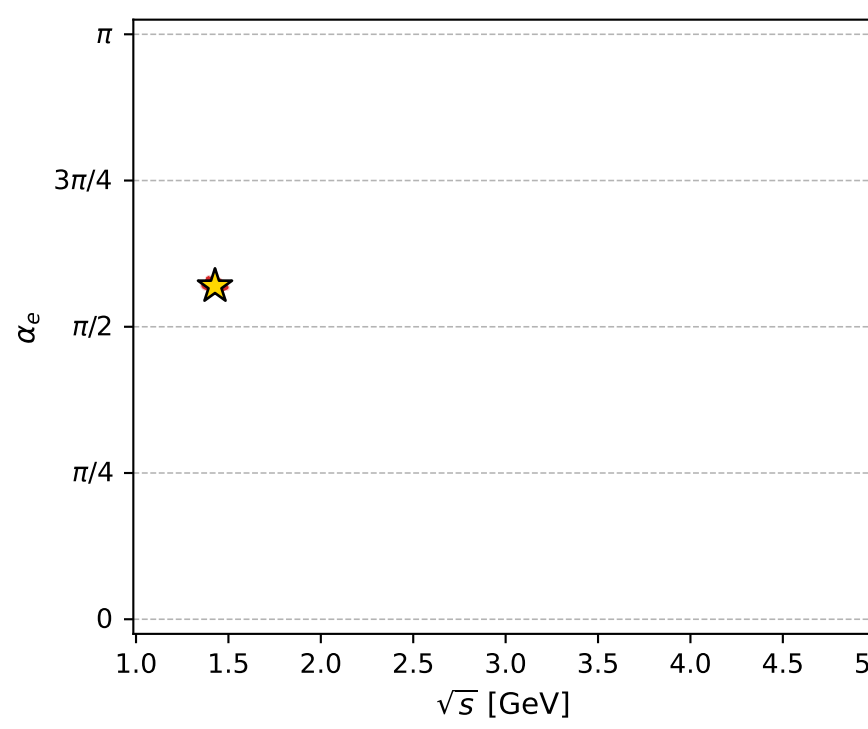
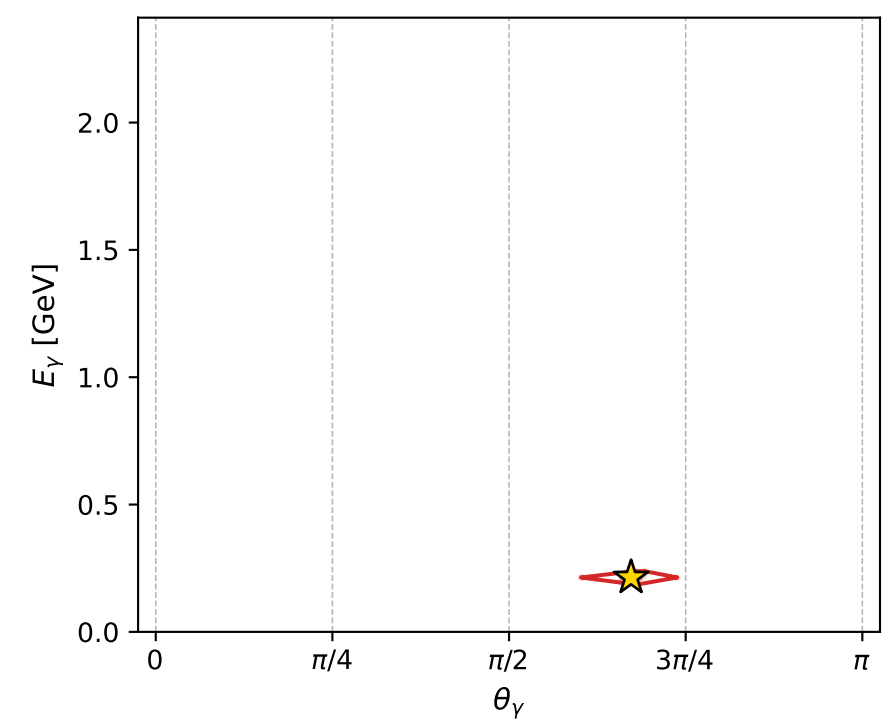
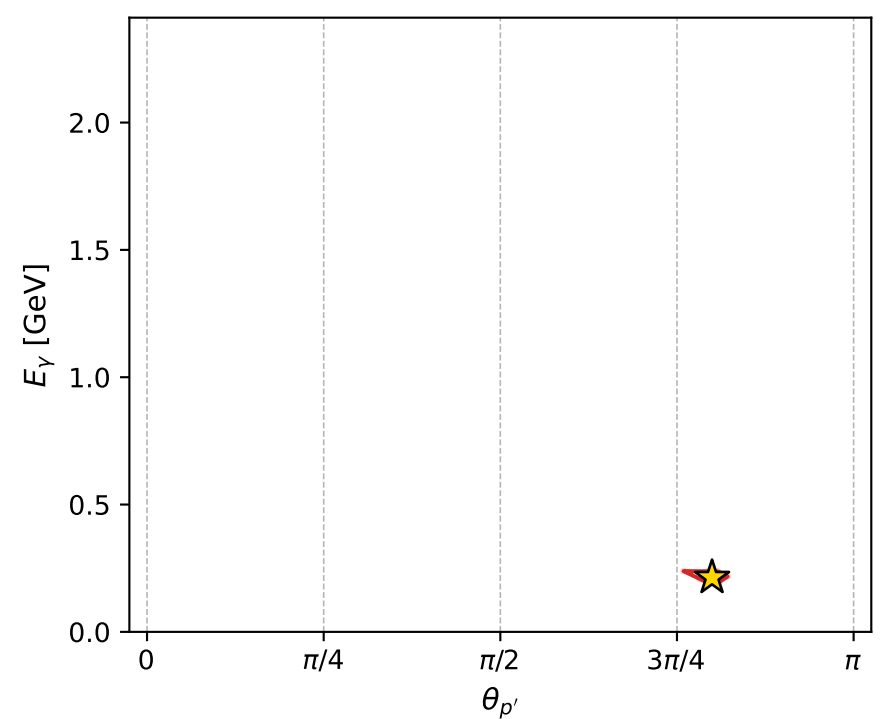
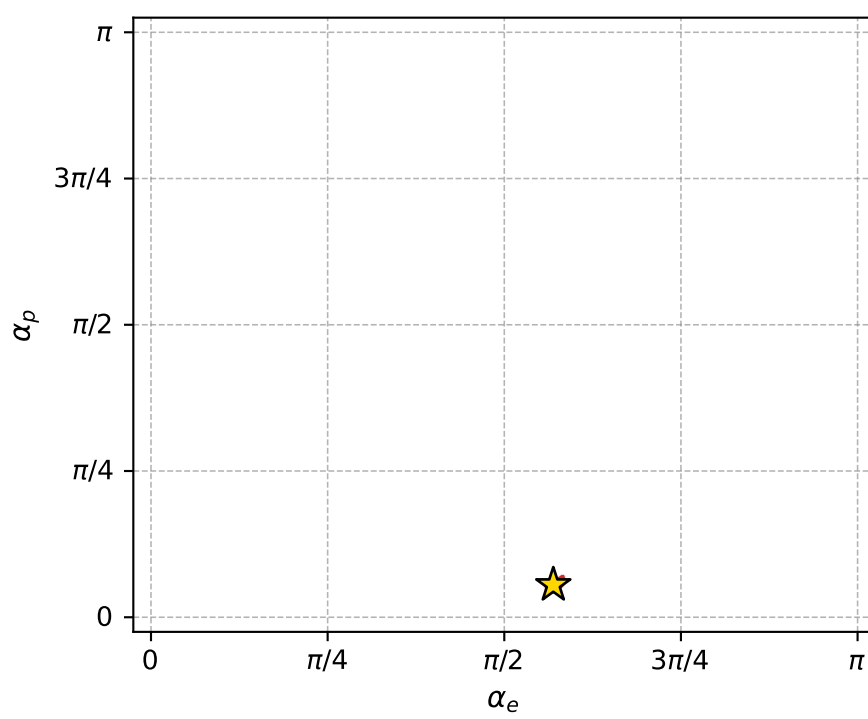
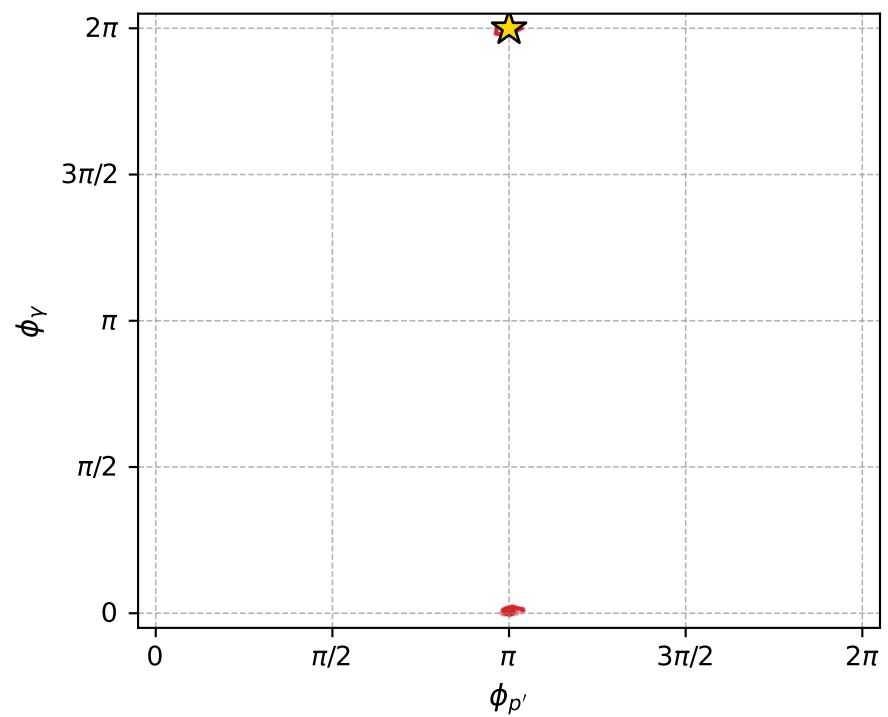
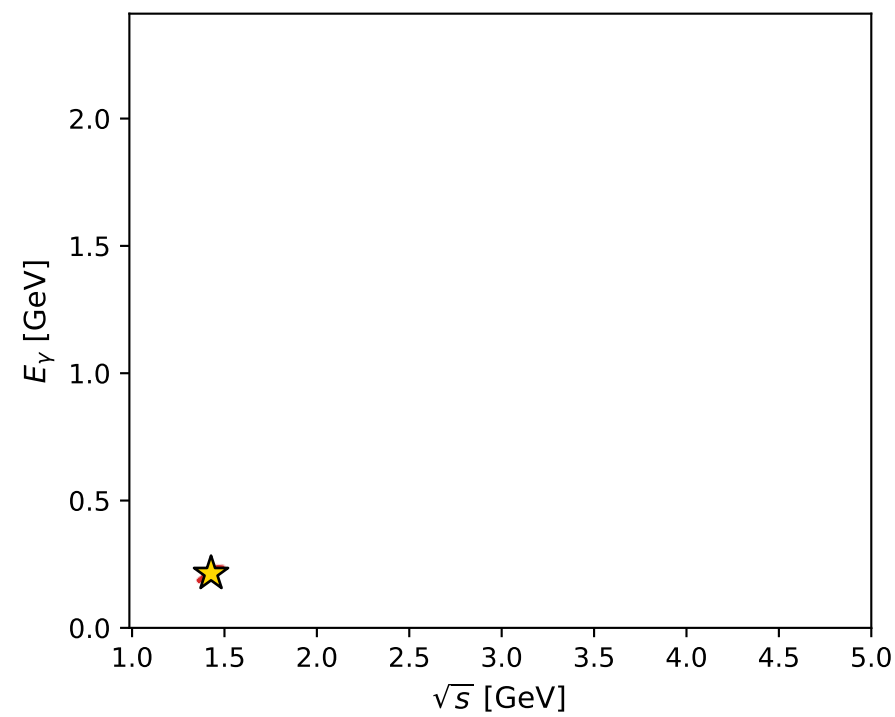
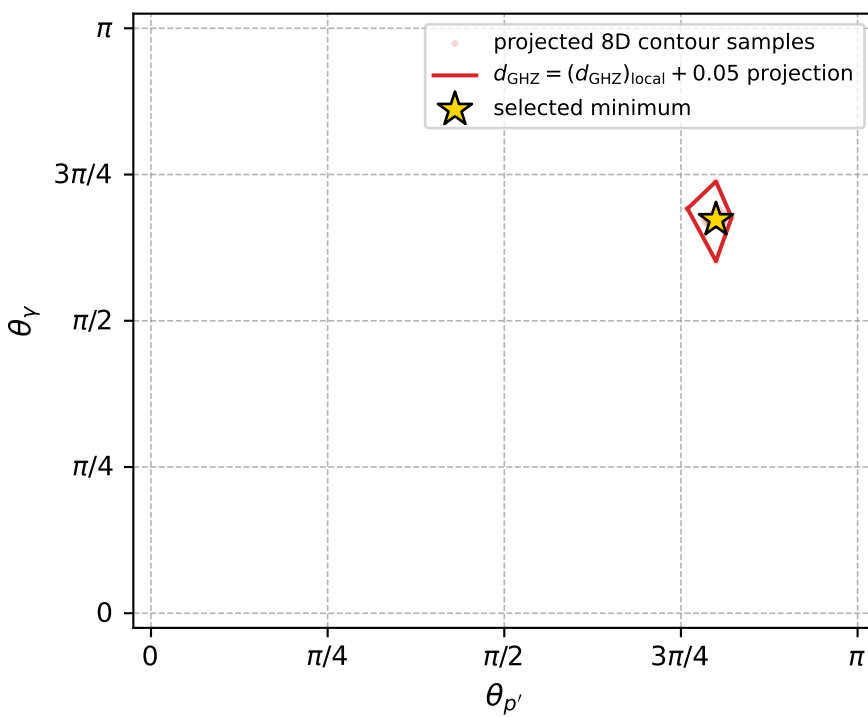
Transverse plane



Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P3, local minimum 328; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00046760782$
 $d_{\text{GHZ}} \text{ contour} = 0.050467608$
 above global minimum = 0.00046758553
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.42737$
 $\text{theta_p_out} = 2.512054$
 $\text{theta_gamma_out} = 2.11333$
 $\text{q0out} = 0.2135236$
 $\text{phi_p_out} = 3.141589$
 $\text{phi_gamma_out} = 6.283181$
 $\text{alpha_e} = 1.788691$
 $\text{alpha_p} = 0.1736297$

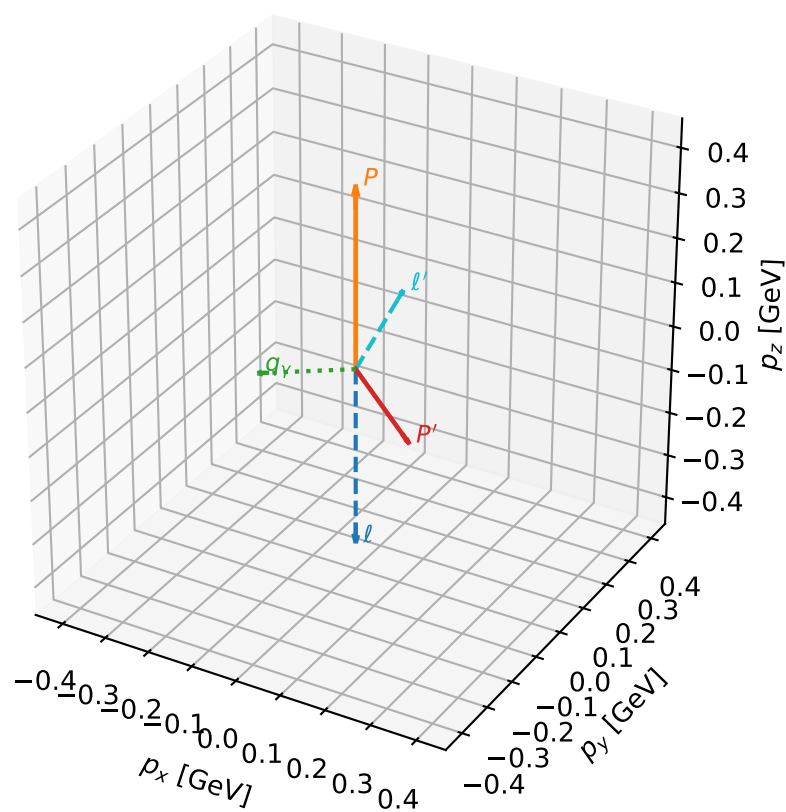
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_329 $d_{\{\text{rm GHZ}\}}=0.000467774$
 region: polarization_cluster_03_local_minimum_329
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.8308229$ rad
 incoming lepton: $-0.952098|+> +0.305792|->$
 proton mixing angle: $\alpha_p=1.3237255$ rad
 incoming proton: $0.244565|+> +0.969633|->$

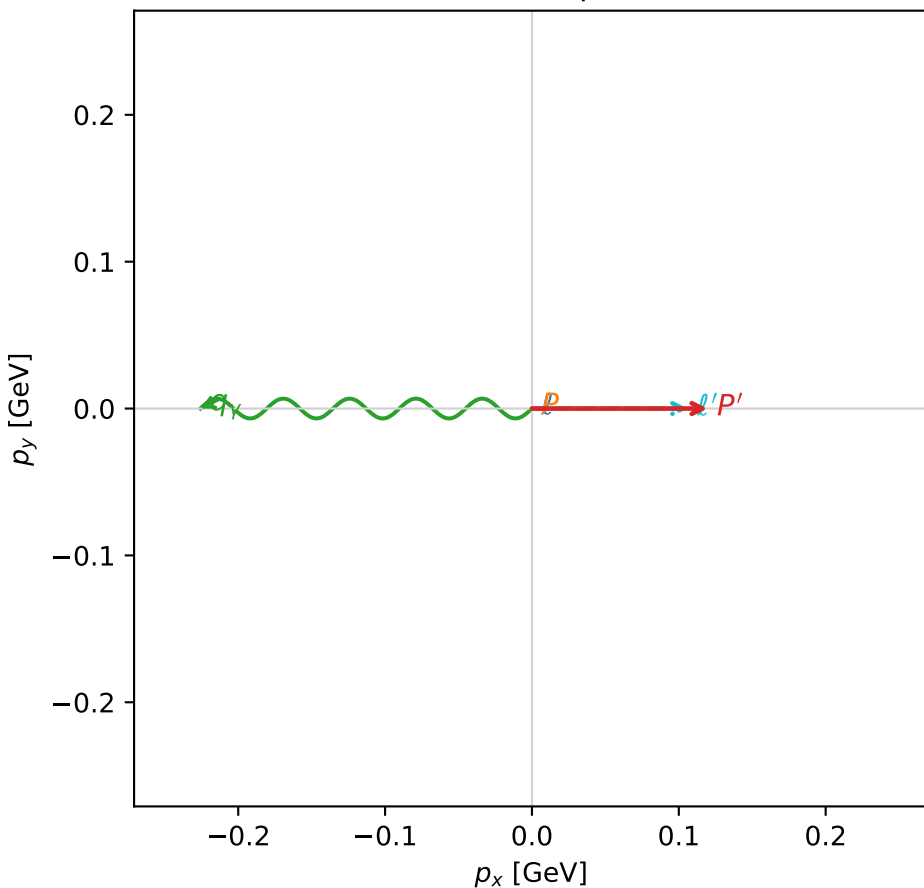
$s=2.02911$, $\sqrt{s}=1.42447$, $|\vec{P}|=0.403402$, $|\vec{P}'|=0.178047$
 $\theta_{p'}=2.40504$, $\theta_Y=1.8892$, $\phi_{p'}=1.13303\text{e-}05$, $\phi_Y=3.1416$
 $E_Y=0.237704$, $\theta_{\ell'}=0.475219$, $\phi_{\ell'}=9.52575\text{e-}06$
 $Q^2=0.353642$, $x_B=0.420047$, $t=-0.296449$, $W^2=1.36811$, $y=0.73256$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4034$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4034$
 P $E= 1.0211$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4034$
 ℓ' $E= 0.2320$ $p_x= 0.1062$ $p_y= 0.0000$ $p_z= 0.2063$
 P' $E= 0.9547$ $p_x= 0.1196$ $p_y= 0.0000$ $p_z= -0.1319$
 q_Y $E= 0.2377$ $p_x= -0.2258$ $p_y= -0.0000$ $p_z= -0.0744$

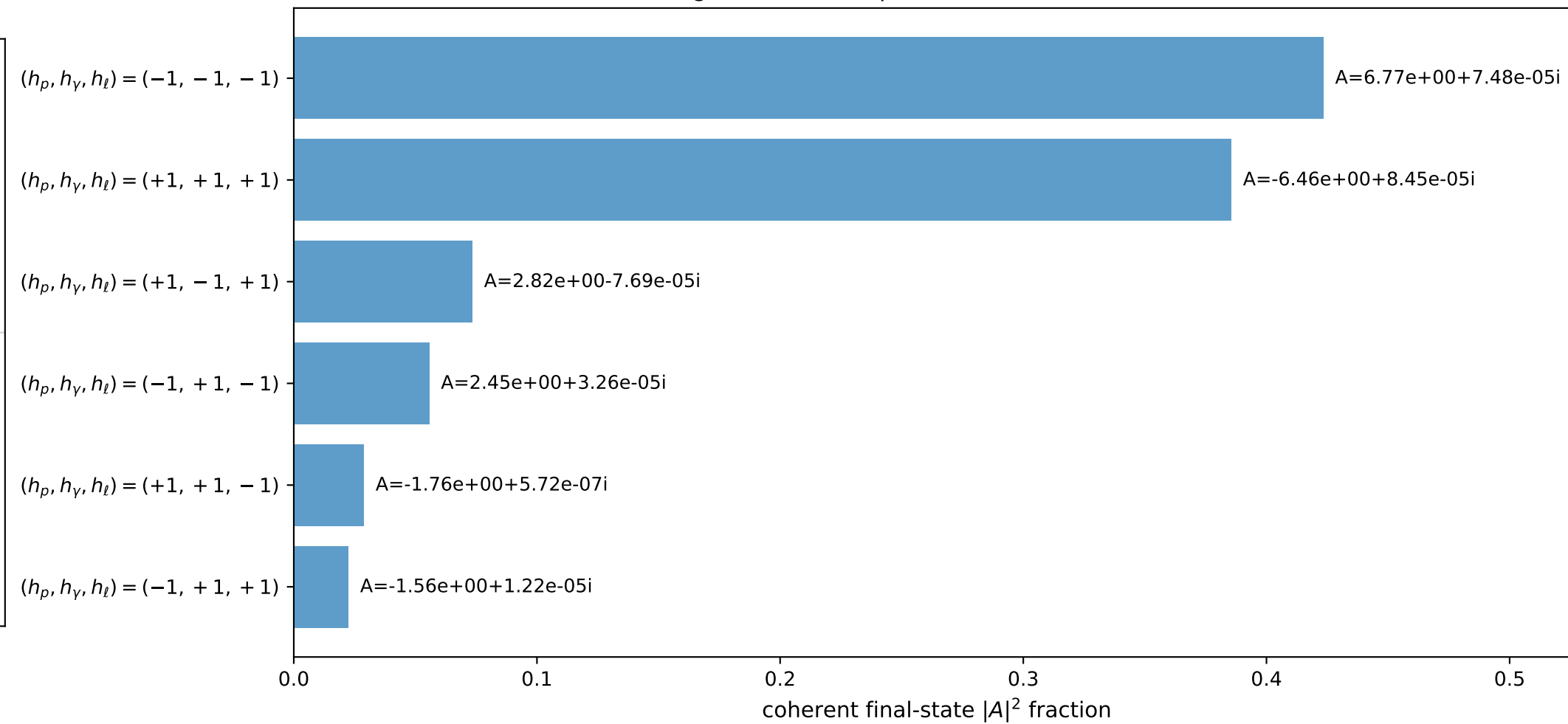
3D momenta



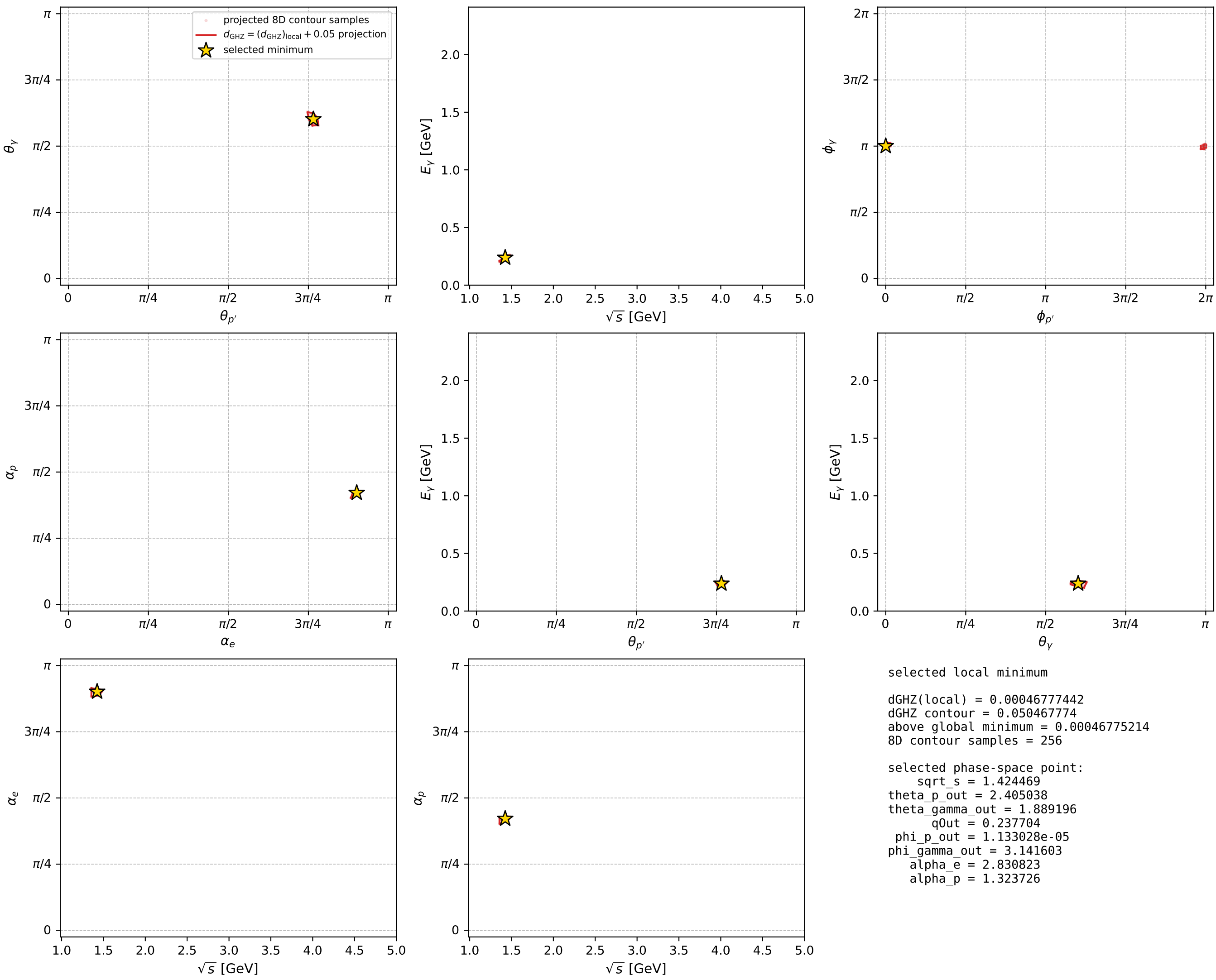
Transverse plane



Leading final-state amplitudes (N=6, retained=98.9%)



electron: polarization cluster P3, local minimum 329; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



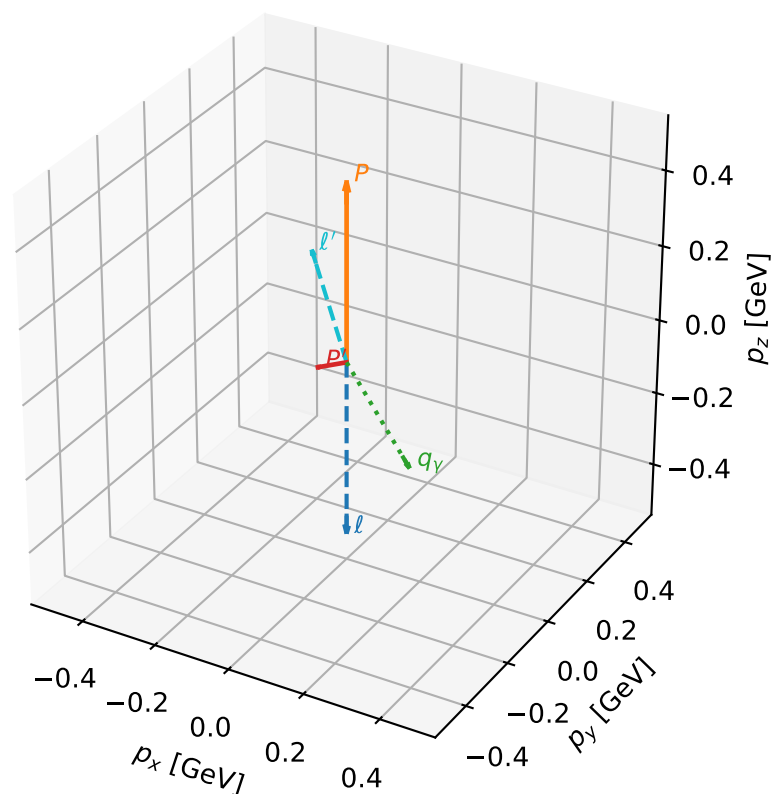
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_345 $d_{\{\text{rm GHZ}\}}=0.000680703$
 region: polarization_cluster_03_local_minimum_345
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.8021119$ rad
 incoming lepton: $-0.229258|+> +0.973366|->$
 proton mixing angle: $\alpha_p=0.18705043$ rad
 incoming proton: $0.982557|+> +0.185962|->$

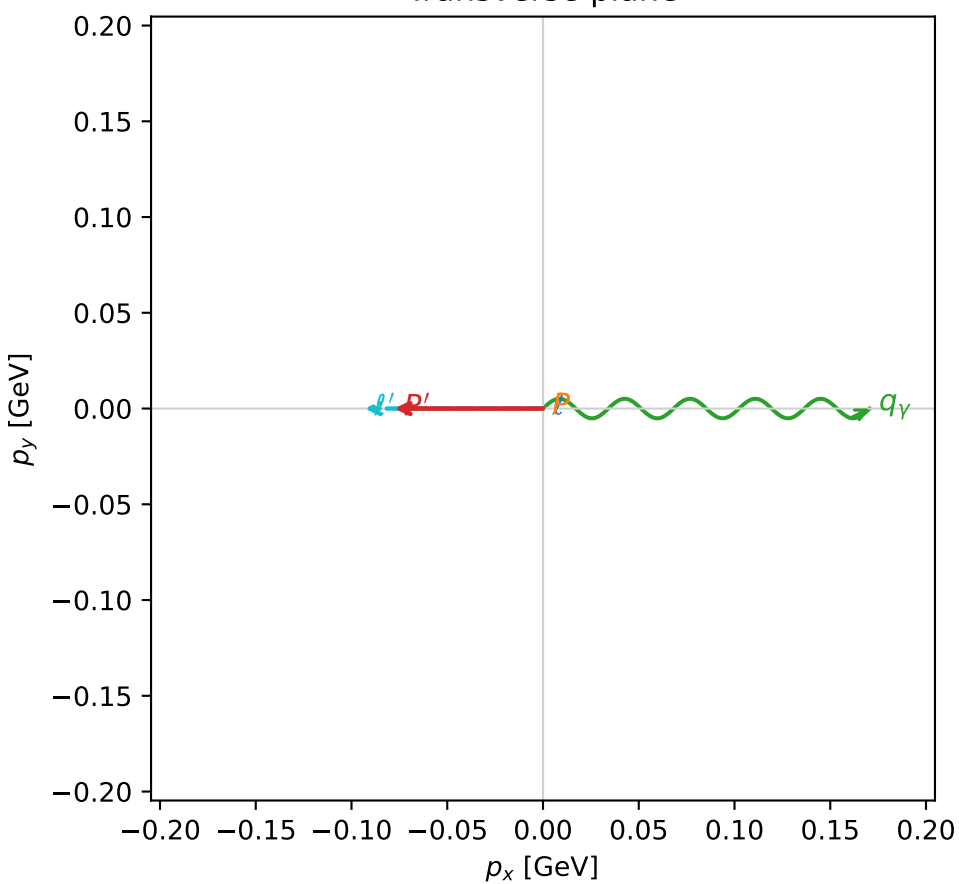
$s=2.31922$, $\sqrt{s}=1.5229$, $|\vec{P}|=0.472577$, $|\vec{P}'|=0.0853878$
 $\theta_{P'}=2.00246$, $\theta_Y=2.51877$, $\phi_{P'}=3.14157$, $\phi_Y=6.28316$
 $E_Y=0.2924$, $\theta_{\ell'}=0.328116$, $\phi_{\ell'}=3.14157$
 $Q^2=0.531026$, $x_B=0.486589$, $t=-0.252626$, $W^2=1.44014$, $y=0.758194$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4726$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4726$
 P $E= 1.0503$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4726$
 ℓ' $E= 0.2886$ $p_x= -0.0930$ $p_y= 0.0000$ $p_z= 0.2732$
 P' $E= 0.9419$ $p_x= -0.0776$ $p_y= 0.0000$ $p_z= -0.0357$
 q_Y $E= 0.2924$ $p_x= 0.1706$ $p_y= -0.0000$ $p_z= -0.2375$

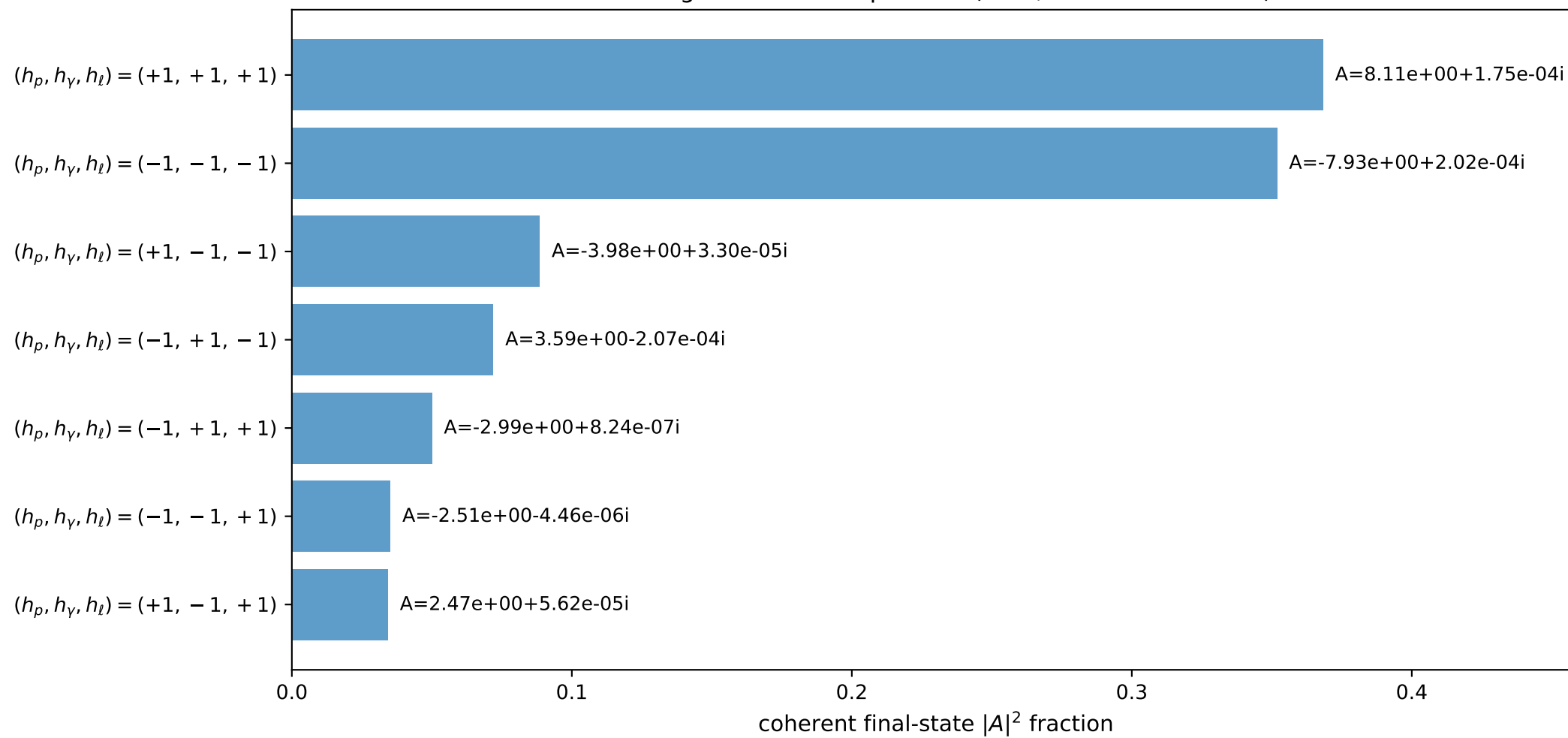
3D momenta



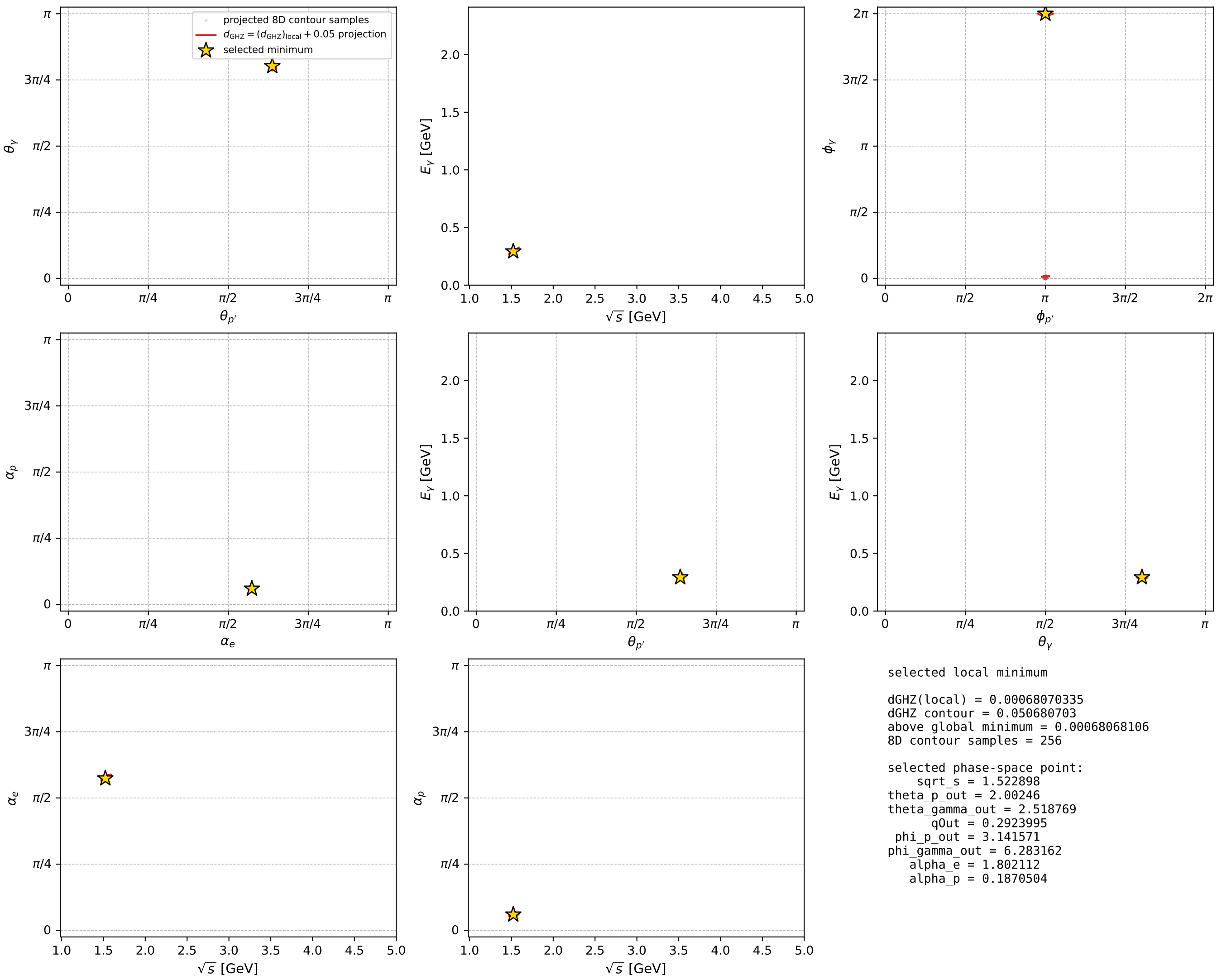
Transverse plane



Leading final-state amplitudes (N=7, retained=100.0%)



electron: polarization cluster P3, local minimum 345; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



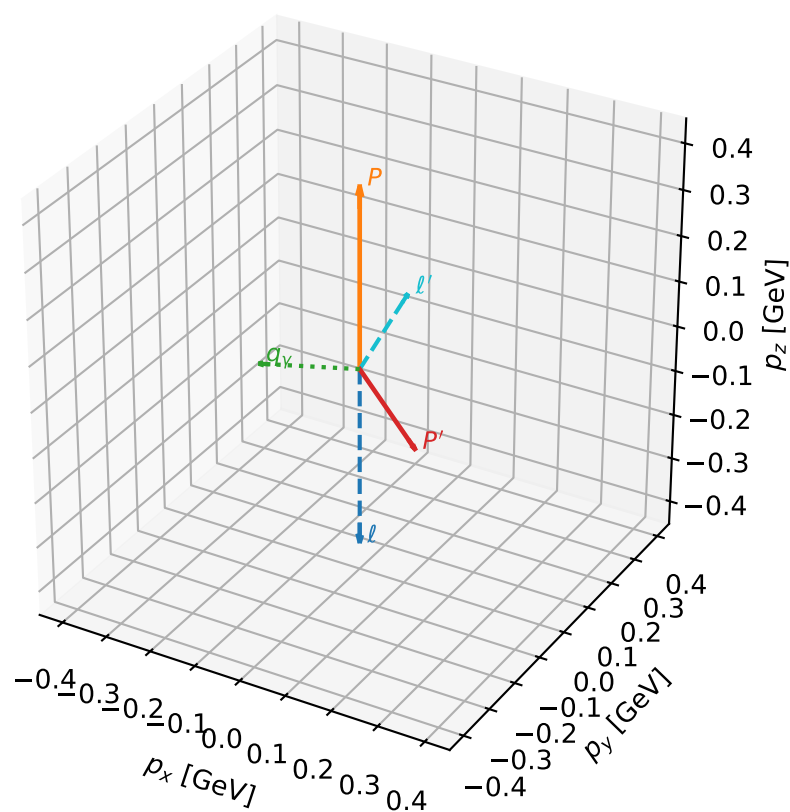
coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_348 $d_{\{\text{rm GHz}\}}=0.000732867$
 region: polarization_cluster_03_local_minimum_348
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.8209935$ rad
 incoming lepton: $-0.949047|+\rangle +0.315135|-\rangle$
 proton mixing angle: $\alpha_p=1.3161743$ rad
 incoming proton: $0.25188|+\rangle +0.967759|-\rangle$

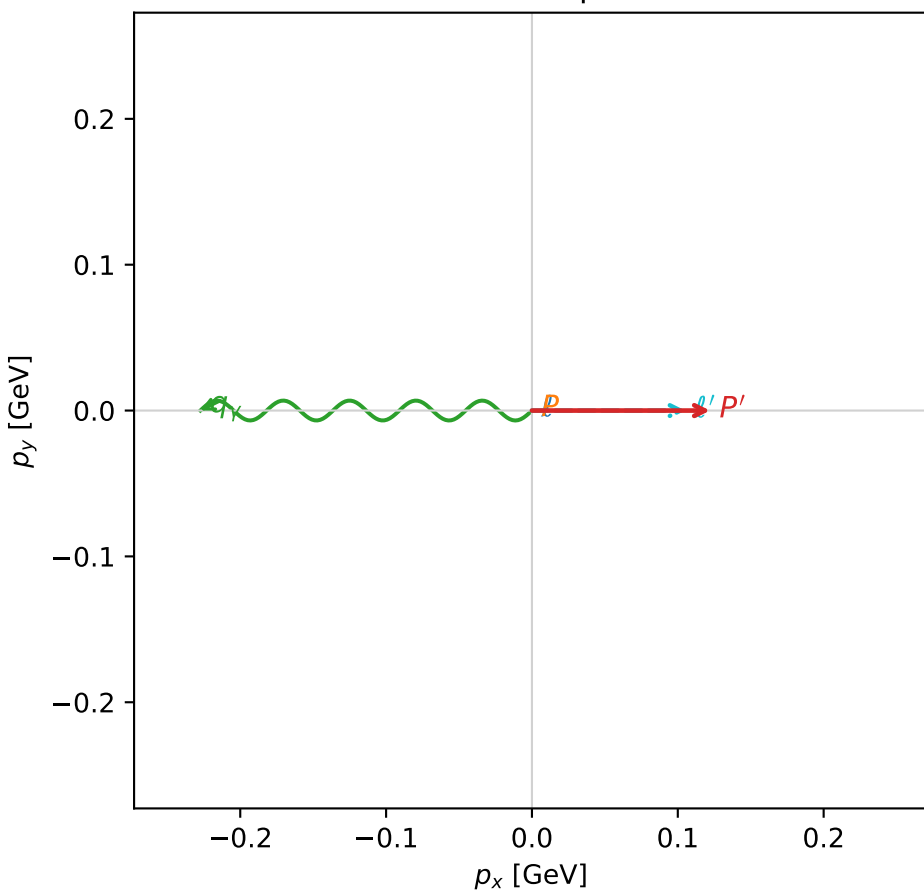
$s=1.99147$, $\sqrt{s}=1.41119$, $|\vec{P}|=0.393859$, $|\vec{P}'|=0.186874$
 $\theta_{P'}=2.42929$, $\theta_Y=1.80087$, $\phi_{P'}=6.28308$, $\phi_Y=3.14151$
 $E_Y=0.233478$, $\theta_{\ell'}=0.495374$, $\phi_{\ell'}=6.28312$
 $Q^2=0.327662$, $x_B=0.402166$, $t=-0.297751$, $W^2=1.36692$, $y=0.73293$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3939$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3939$
 P $E= 1.0173$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3939$
 ℓ' $E= 0.2213$ $p_x= 0.1052$ $p_y= -0.0000$ $p_z= 0.1947$
 P' $E= 0.9564$ $p_x= 0.1221$ $p_y= -0.0000$ $p_z= -0.1414$
 q_Y $E= 0.2335$ $p_x= -0.2273$ $p_y= 0.0000$ $p_z= -0.0532$

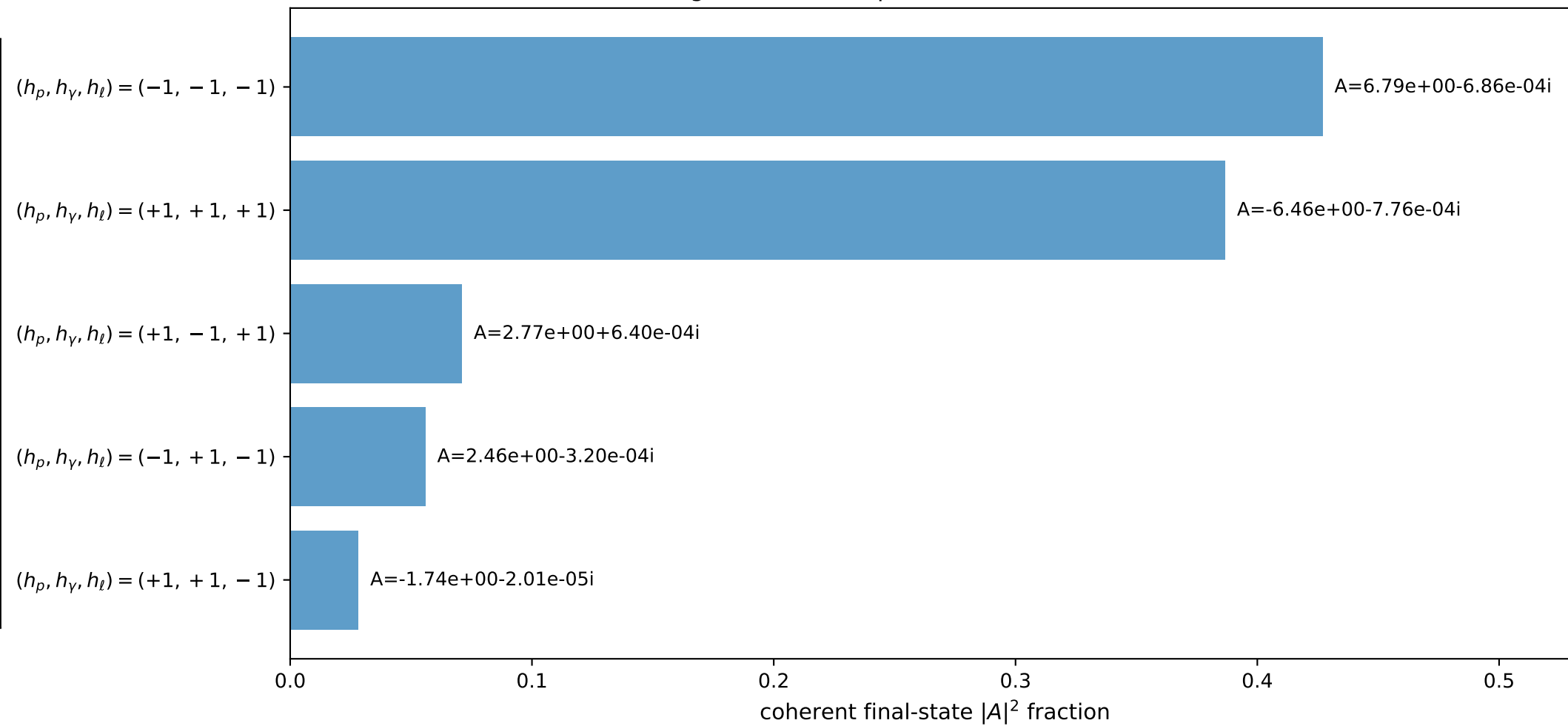
3D momenta



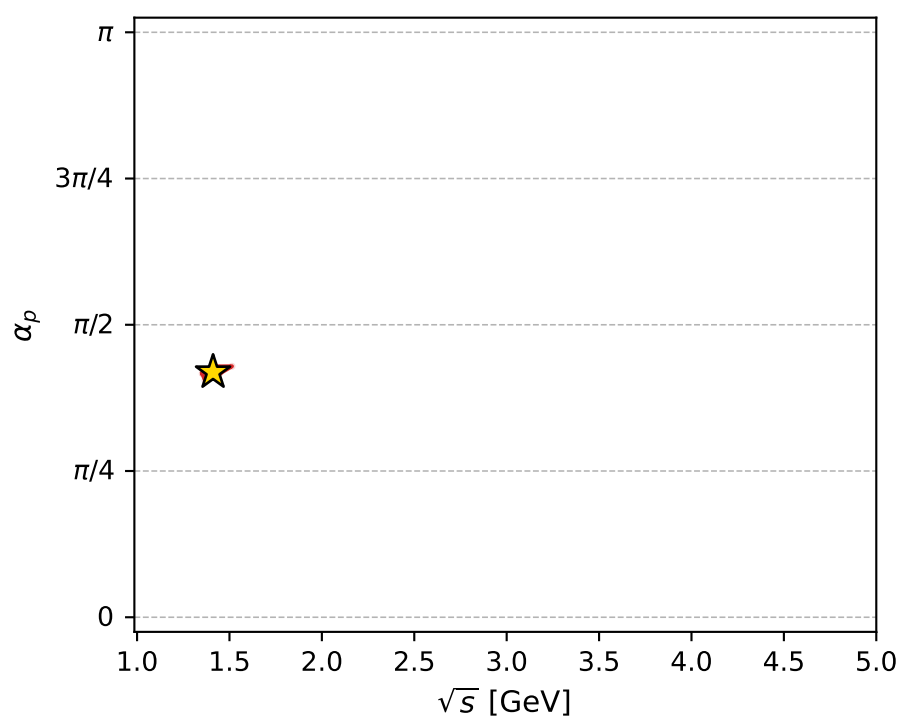
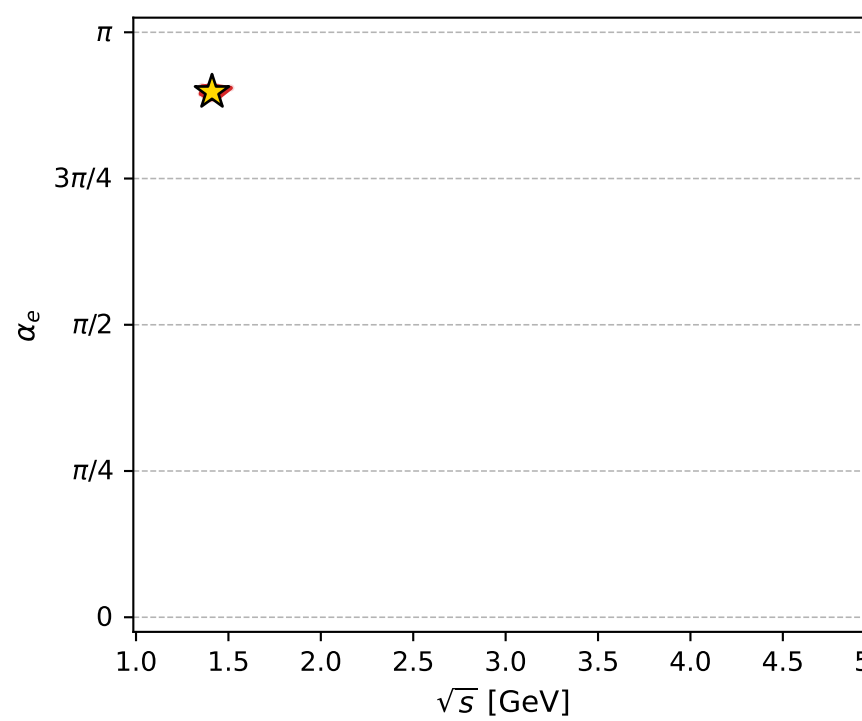
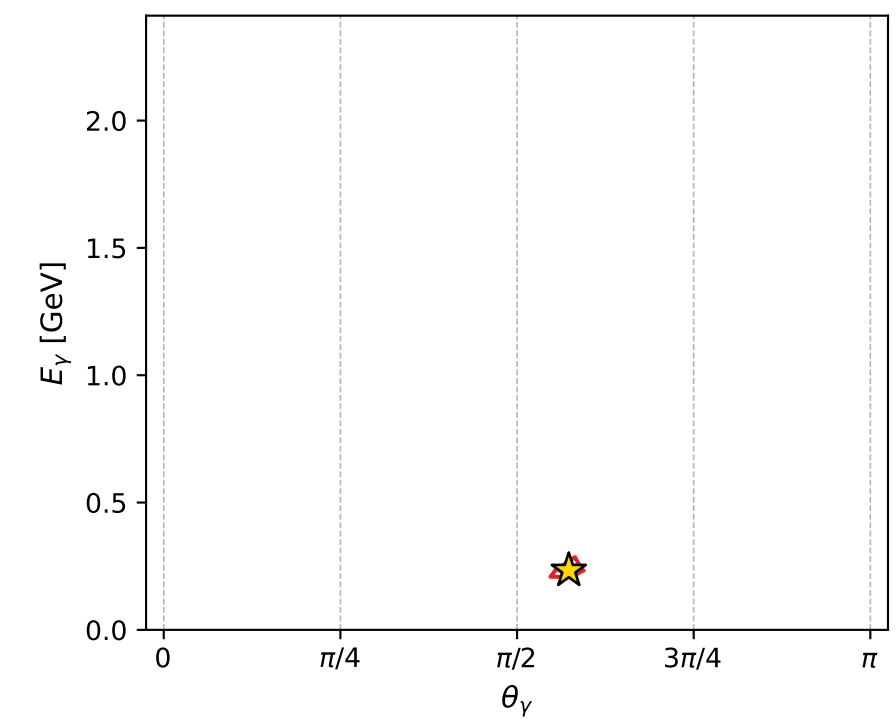
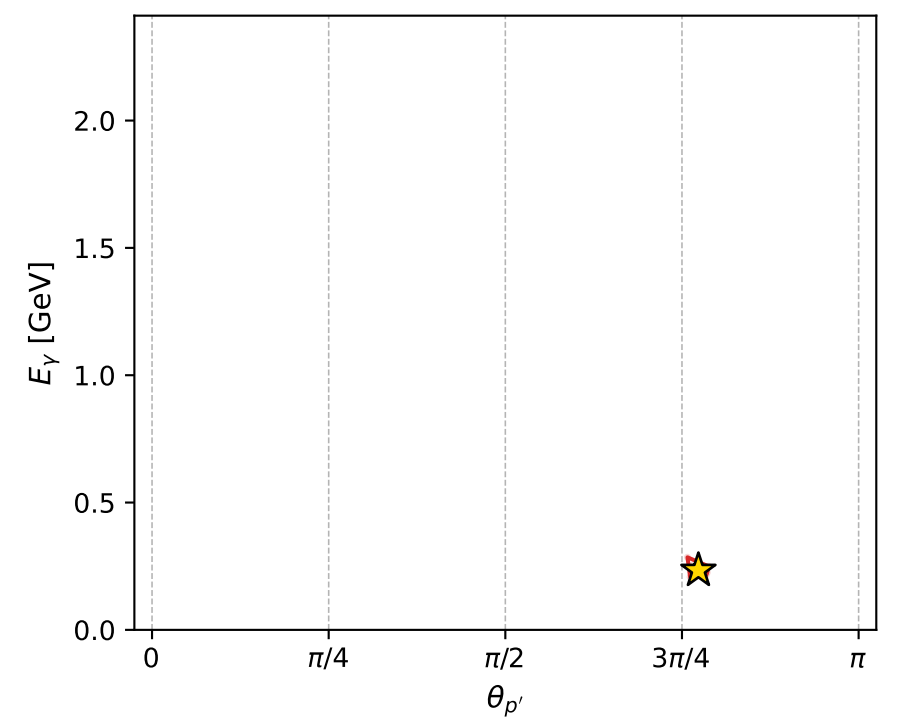
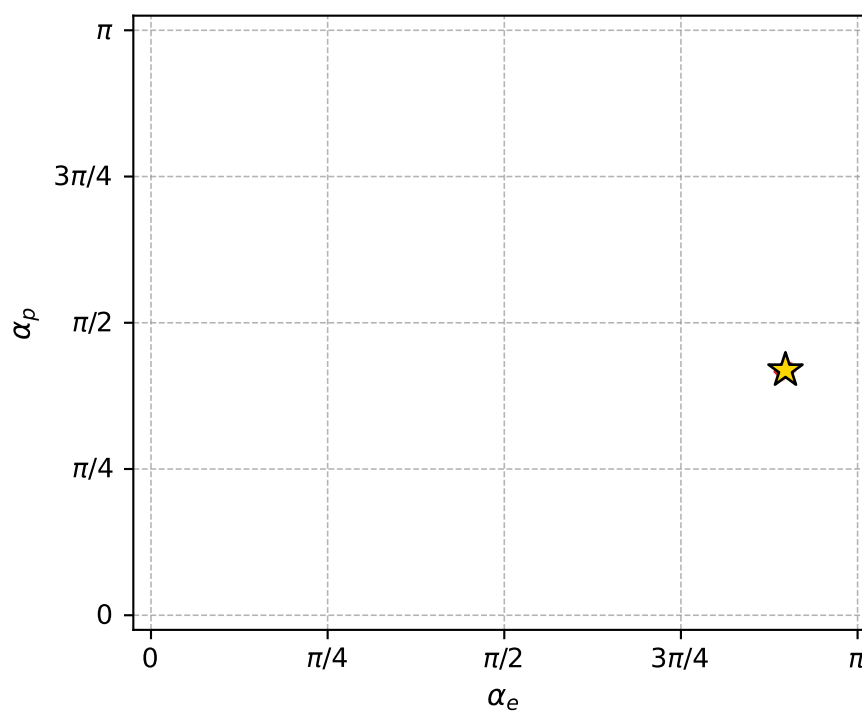
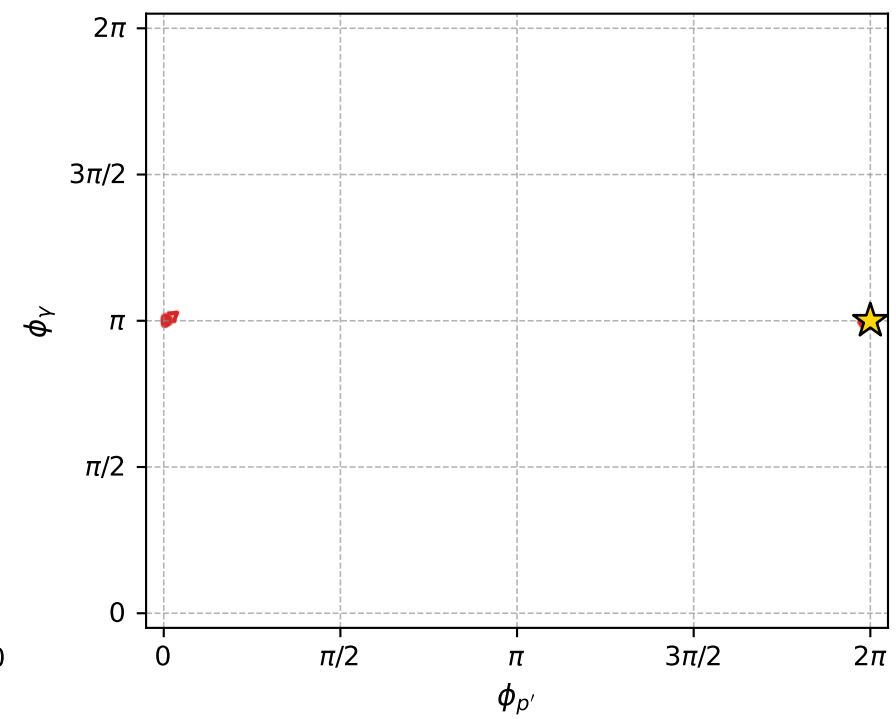
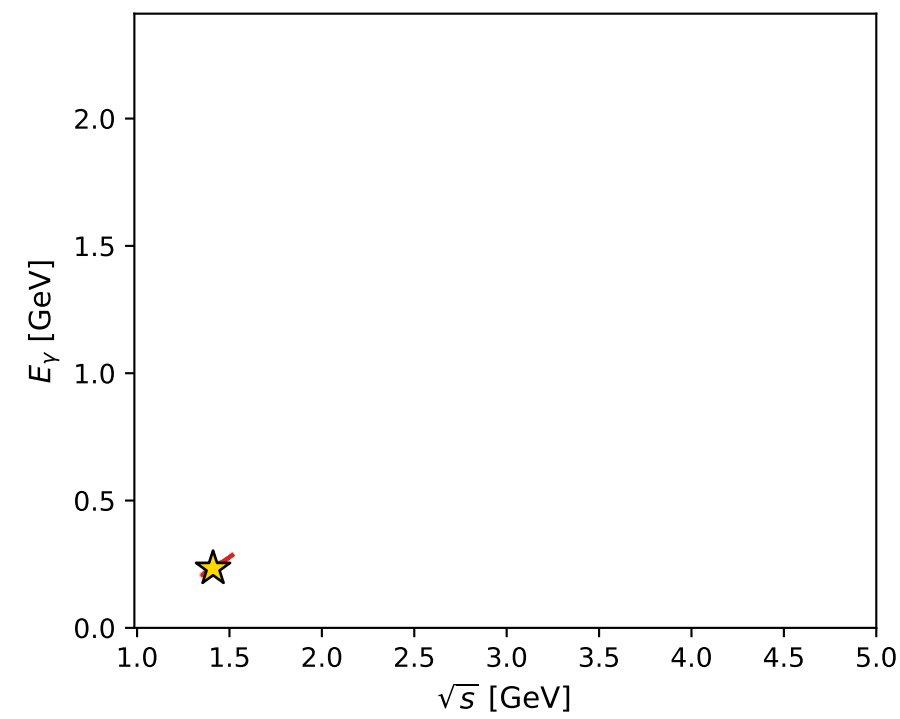
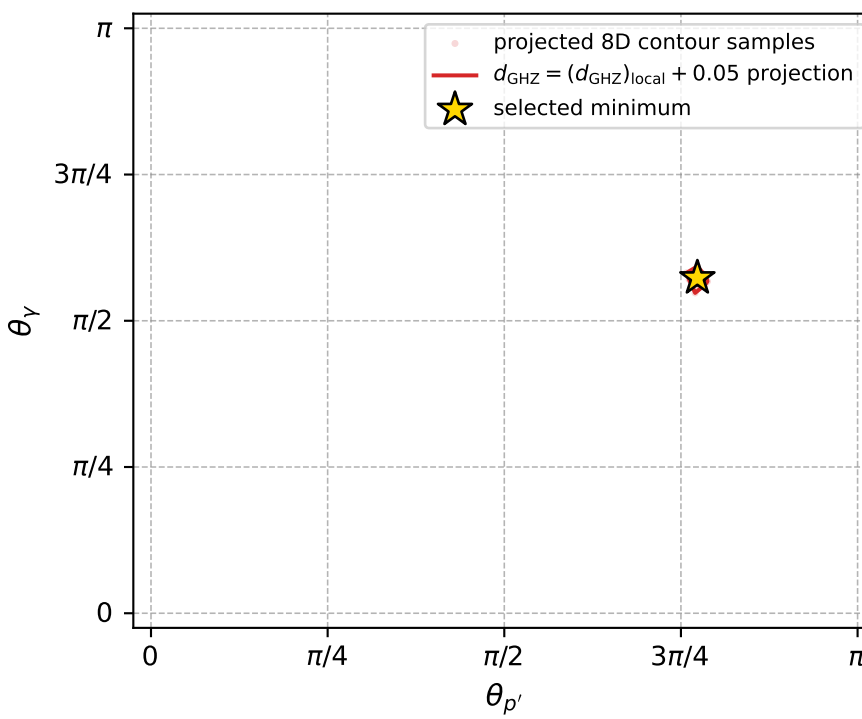
Transverse plane



Leading final-state amplitudes (N=5, retained=96.9%)



electron: polarization cluster P3, local minimum 348; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00073286706$
 $d_{\text{GHZ}} \text{ contour} = 0.050732867$
 above global minimum = 0.00073284477
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.411194$
 $\text{theta_p_out} = 2.429288$
 $\text{theta_gamma_out} = 1.800871$
 $\text{q0out} = 0.2334779$
 $\text{phi_p_out} = 6.283079$
 $\text{phi_gamma_out} = 3.141505$
 $\text{alpha_e} = 2.820993$
 $\text{alpha_p} = 1.316174$

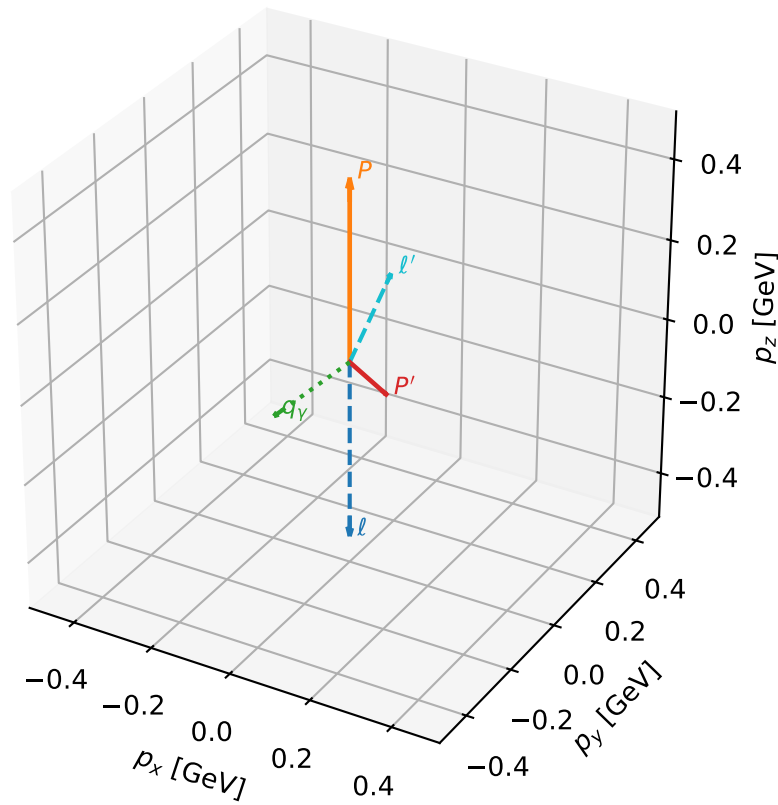
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_352 $d_{\{\text{rm GHZ}\}}=0.000902895$
 region: polarization_cluster_03_local_minimum_352
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.8689525$ rad
 incoming lepton: $-0.963063|+> +0.269275|->$
 proton mixing angle: $\alpha_p=1.3486313$ rad
 incoming proton: $0.220342|+> +0.975423|->$

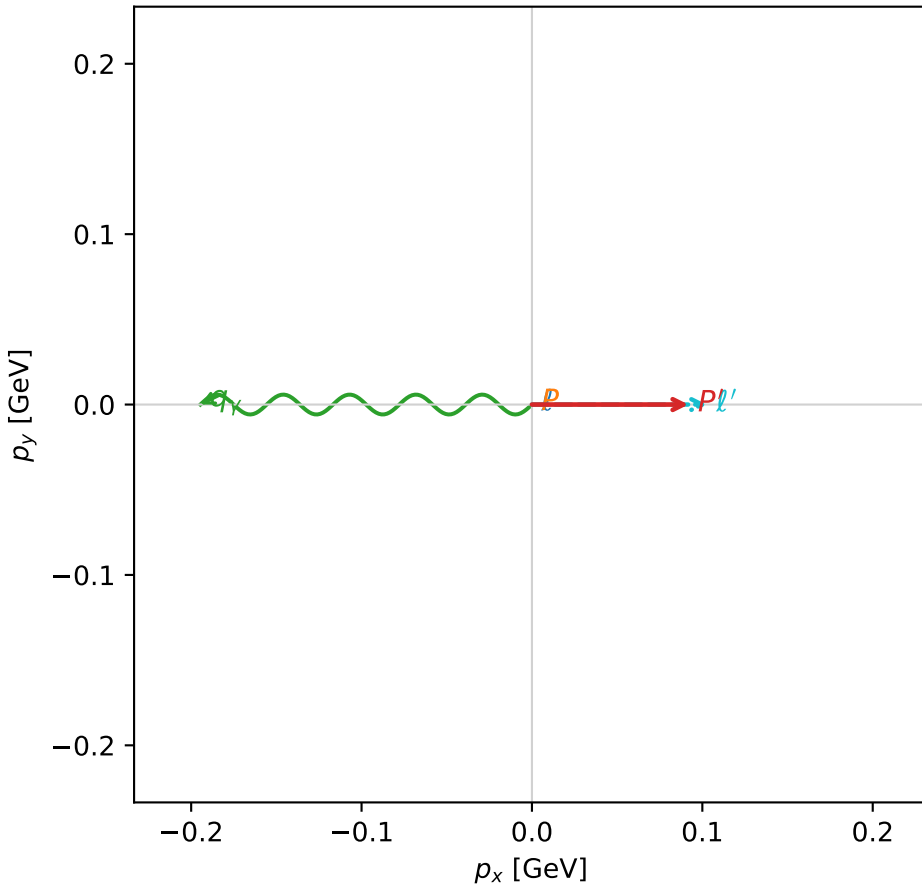
$s=2.21702$, $\sqrt{s}=1.48897$, $|\vec{P}|=0.449029$, $|\vec{P}'|=0.106807$
 $\theta_{P'}=2.10568$, $\theta_Y=2.35674$, $\phi_{P'}=6.28317$, $\phi_Y=3.14157$
 $E_Y=0.275336$, $\theta_{\ell'}=0.390838$, $\phi_{\ell'}=6.28315$
 $Q^2=0.465923$, $x_B=0.465765$, $t=-0.252736$, $W^2=1.41426$, $y=0.748097$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4490$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4490$
 P $E= 1.0399$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4490$
 ℓ' $E= 0.2696$ $p_x= 0.1027$ $p_y= -0.0000$ $p_z= 0.2492$
 P' $E= 0.9441$ $p_x= 0.0919$ $p_y= -0.0000$ $p_z= -0.0544$
 q_Y $E= 0.2753$ $p_x= -0.1946$ $p_y= 0.0000$ $p_z= -0.1948$

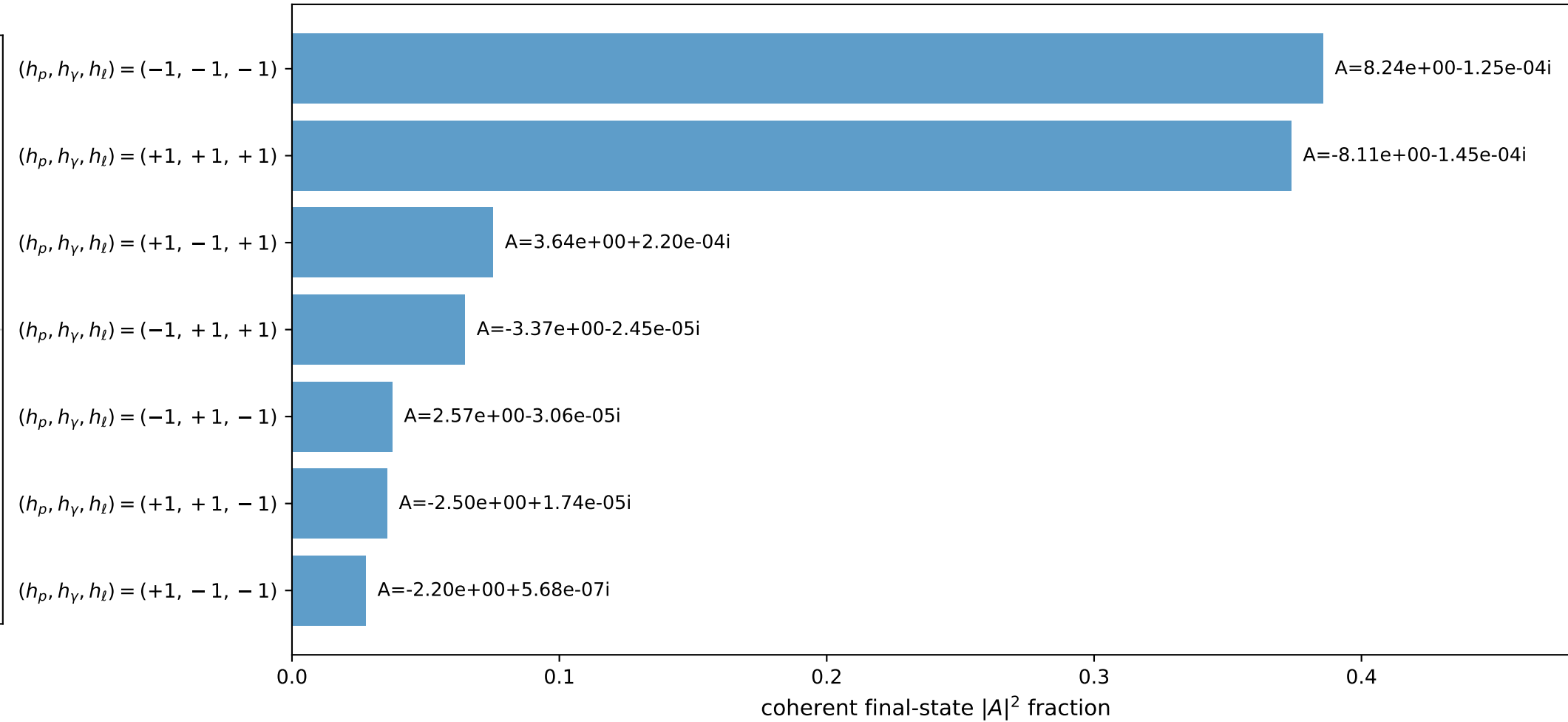
3D momenta



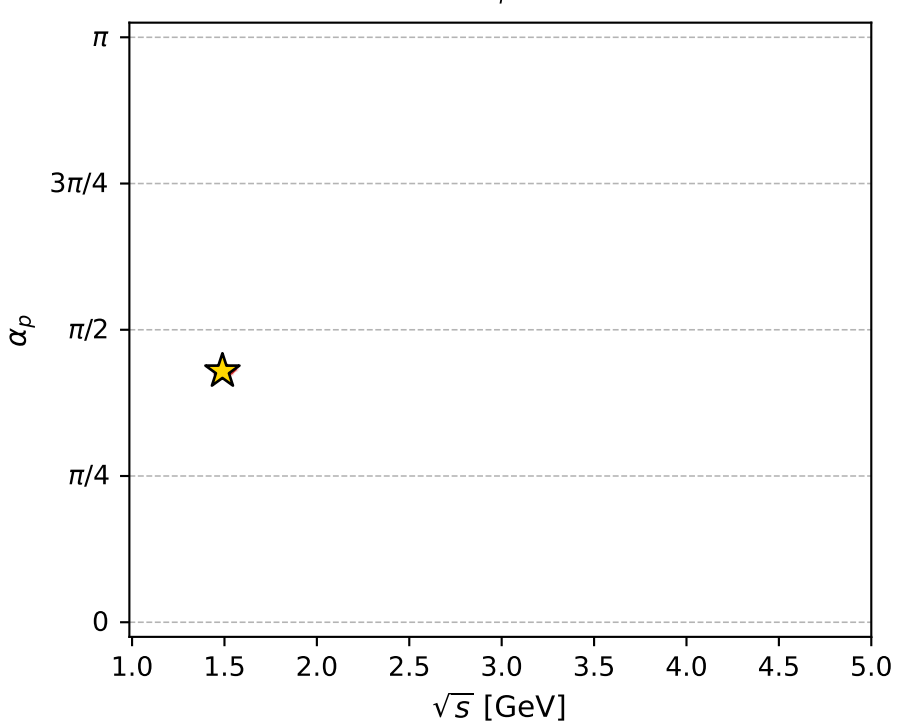
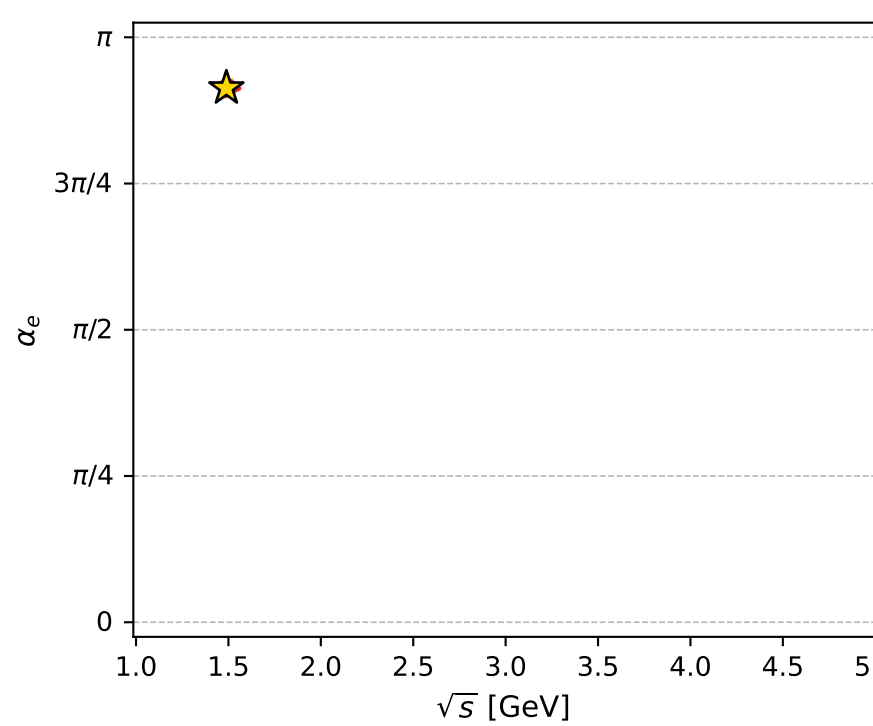
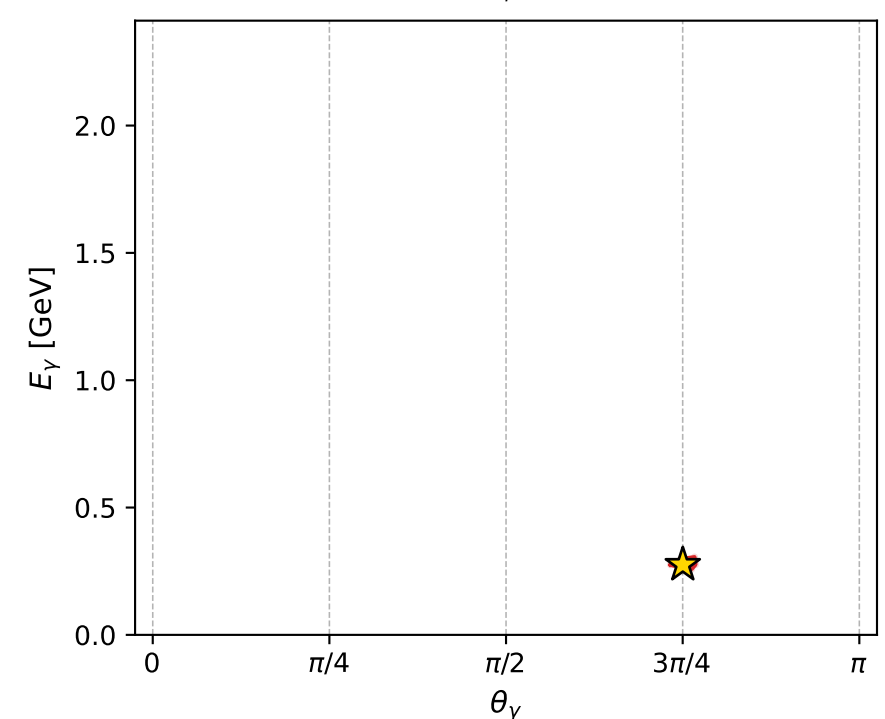
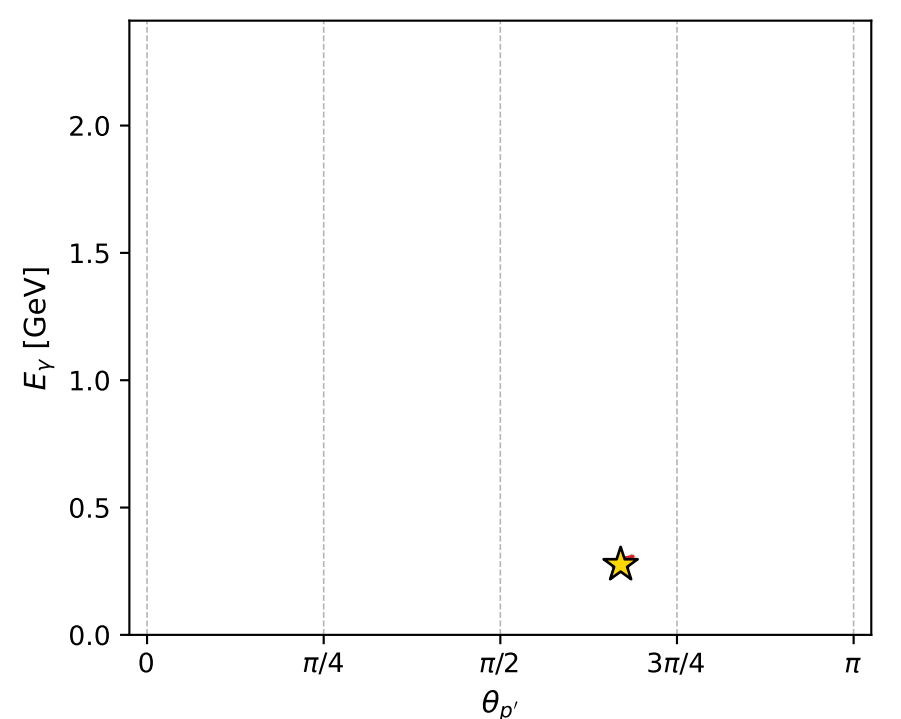
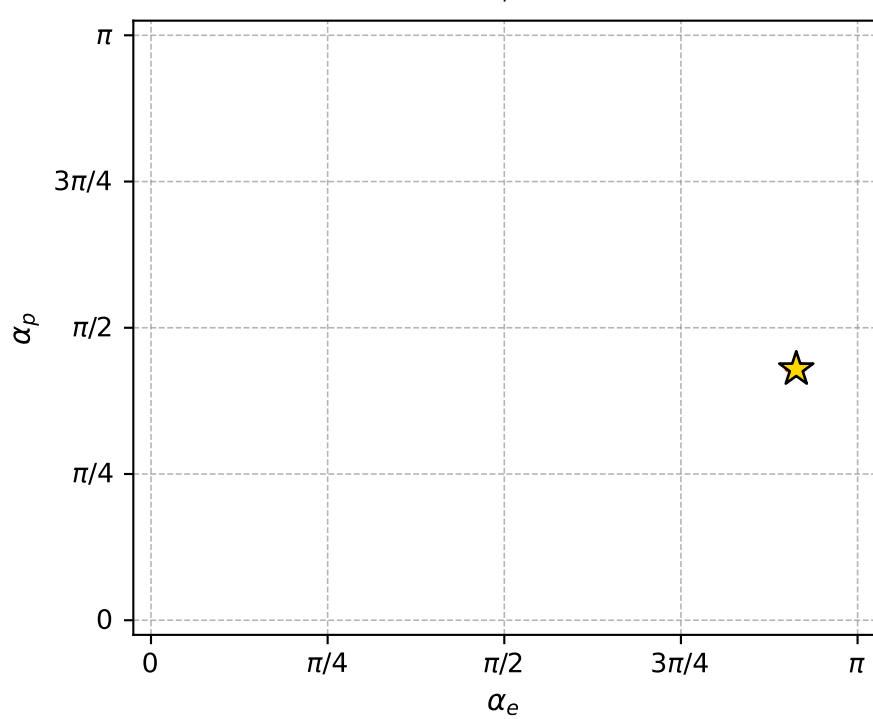
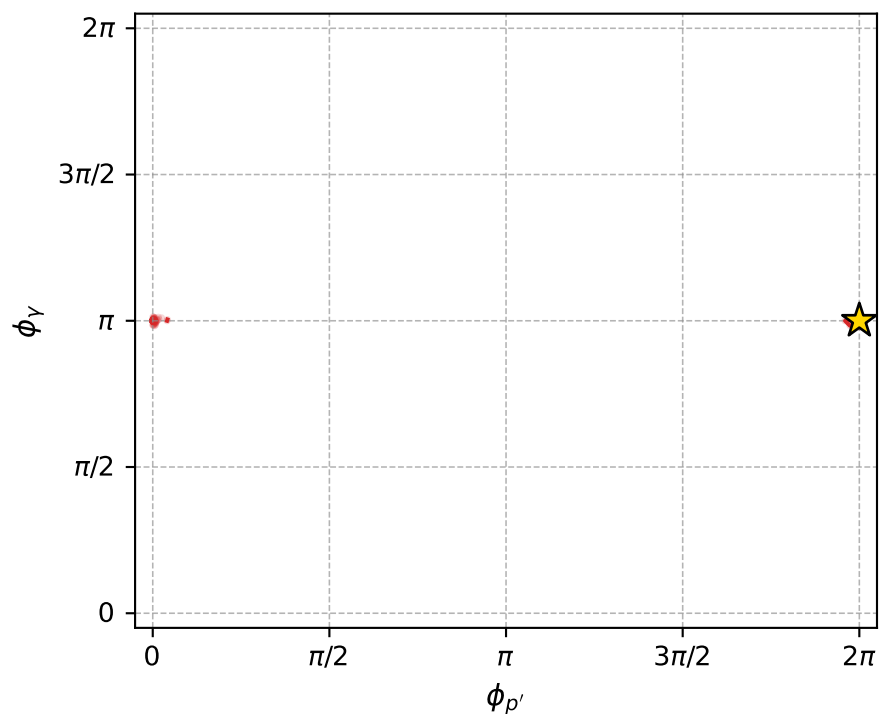
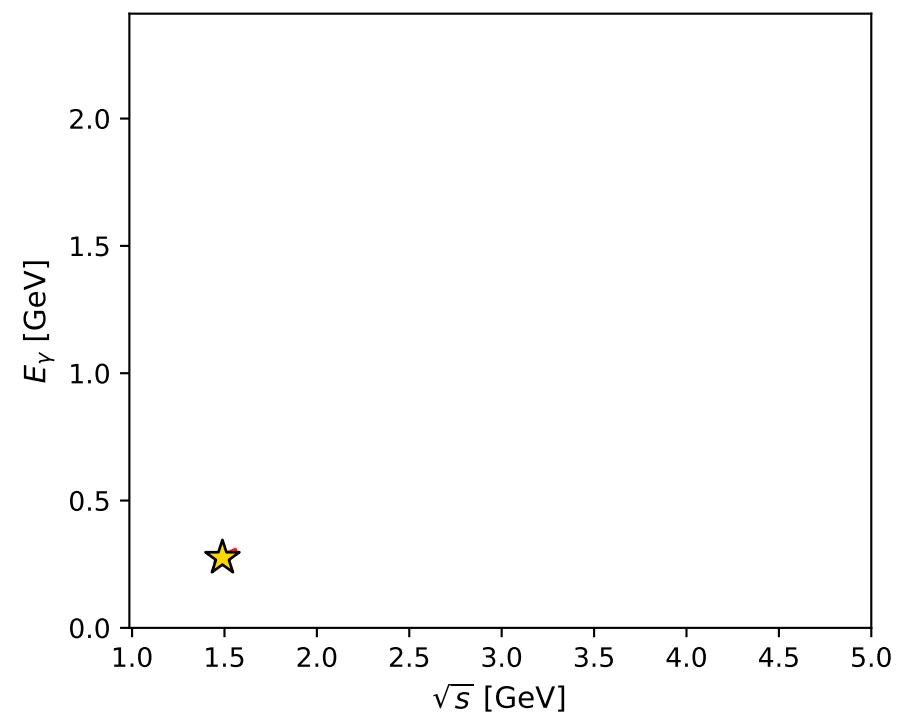
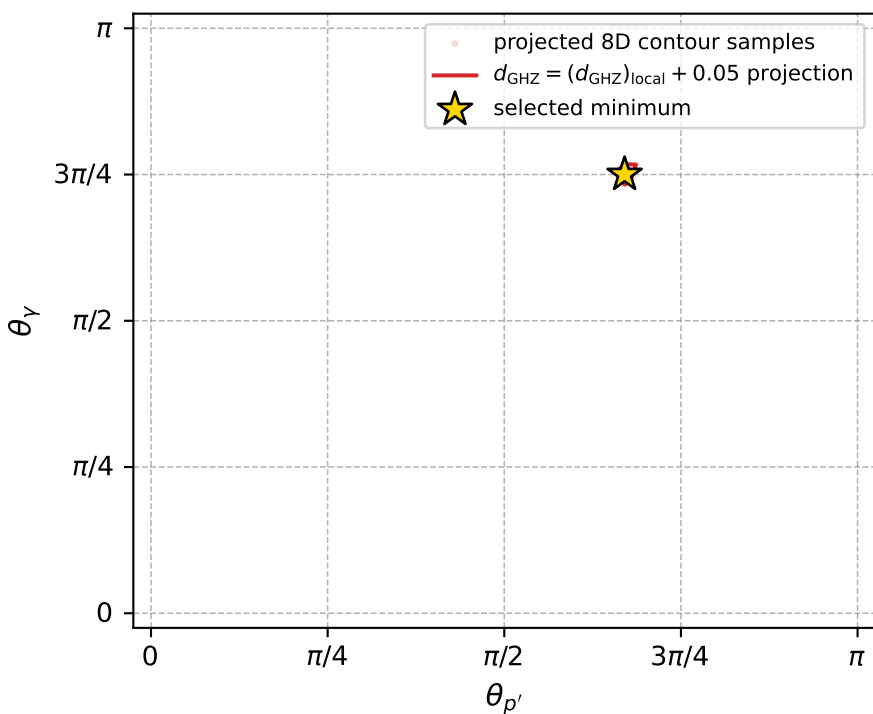
Transverse plane



Leading final-state amplitudes (N=7, retained=99.9%)



electron: polarization cluster P3, local minimum 352; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00090289544$
 $d_{\text{GHZ}} \text{ contour} = 0.050902895$
 above global minimum = 0.00090287316
 8D contour samples = 256

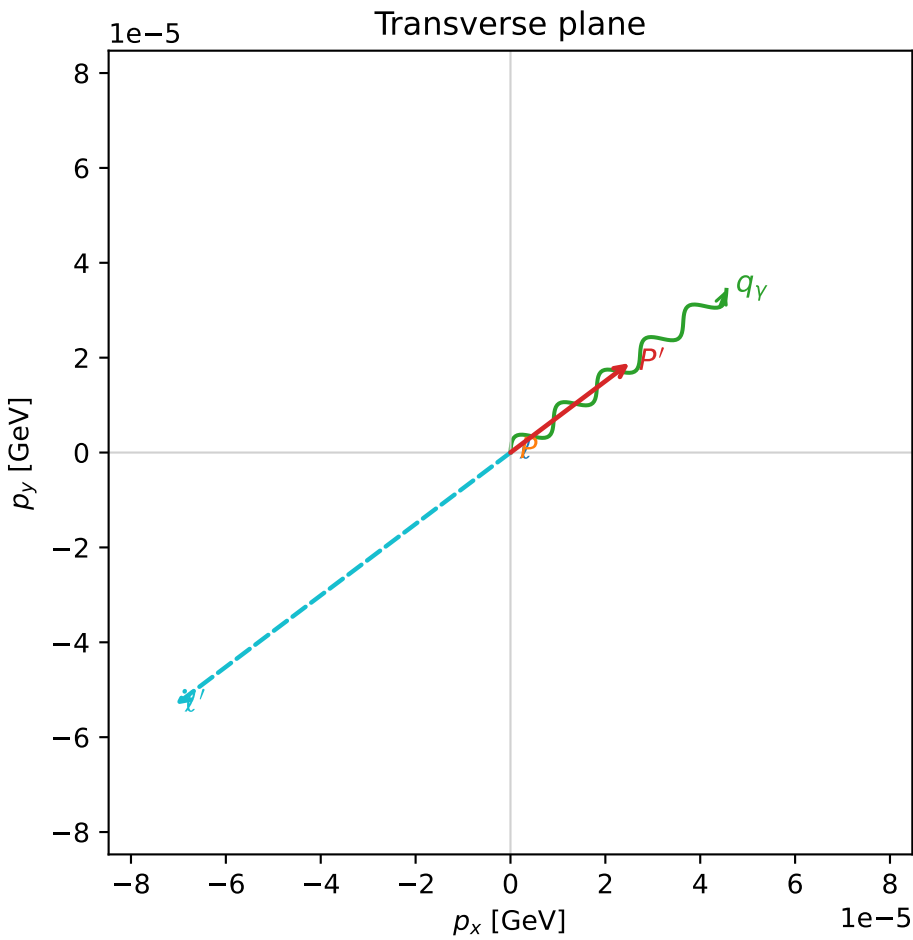
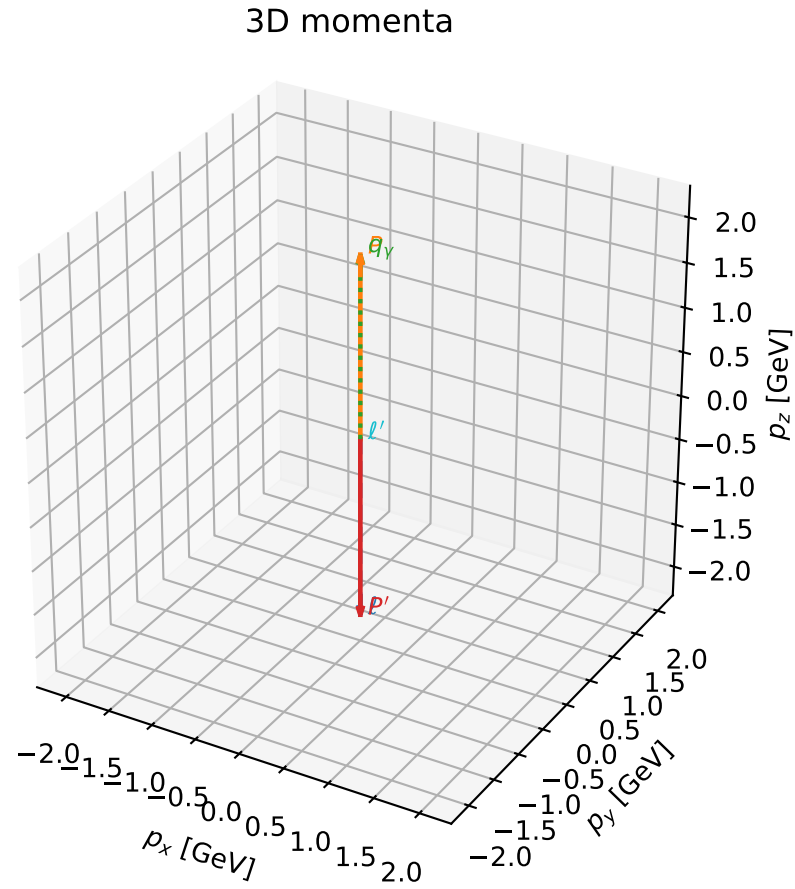
selected phase-space point:
 $\text{sqrt_s} = 1.488967$
 $\text{theta_p_out} = 2.105677$
 $\text{theta_gamma_out} = 2.356741$
 $\text{q0out} = 0.2753357$
 $\text{phi_p_out} = 6.283171$
 $\text{phi_gamma_out} = 3.141567$
 $\text{alpha_e} = 2.868953$
 $\text{alpha_p} = 1.348631$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_03_local_minimum_357 $d_{\{\text{rm GHZ}\}}=0.00125916$
region: polarization_cluster_03_local_minimum_357
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3565227$ rad
incoming lepton: $-0.707339|+\rangle +0.706875|-\rangle$
proton mixing angle: $\alpha_p=0.78632898$ rad
incoming proton: $0.706448|+\rangle +0.707765|-\rangle$

$s=18.3383$, $\sqrt{s}=4.28232$, $|\vec{P}|=2.03843$, $|\vec{P}'|=2.03709$
 $\theta_{P'}=3.14158$, $\theta_Y=2.79619\text{e-}05$, $\phi_{P'}=0.644969$, $\phi_Y=0.64563$
 $E_Y=2.03832$, $\theta_{\ell'}=3.06993$, $\phi_{\ell'}=3.78699$
 $Q^2=0.000426632$, $x_B=2.44525\text{e-}05$, $t=-16.6098$, $W^2=18.3268$, $y=0.999369$

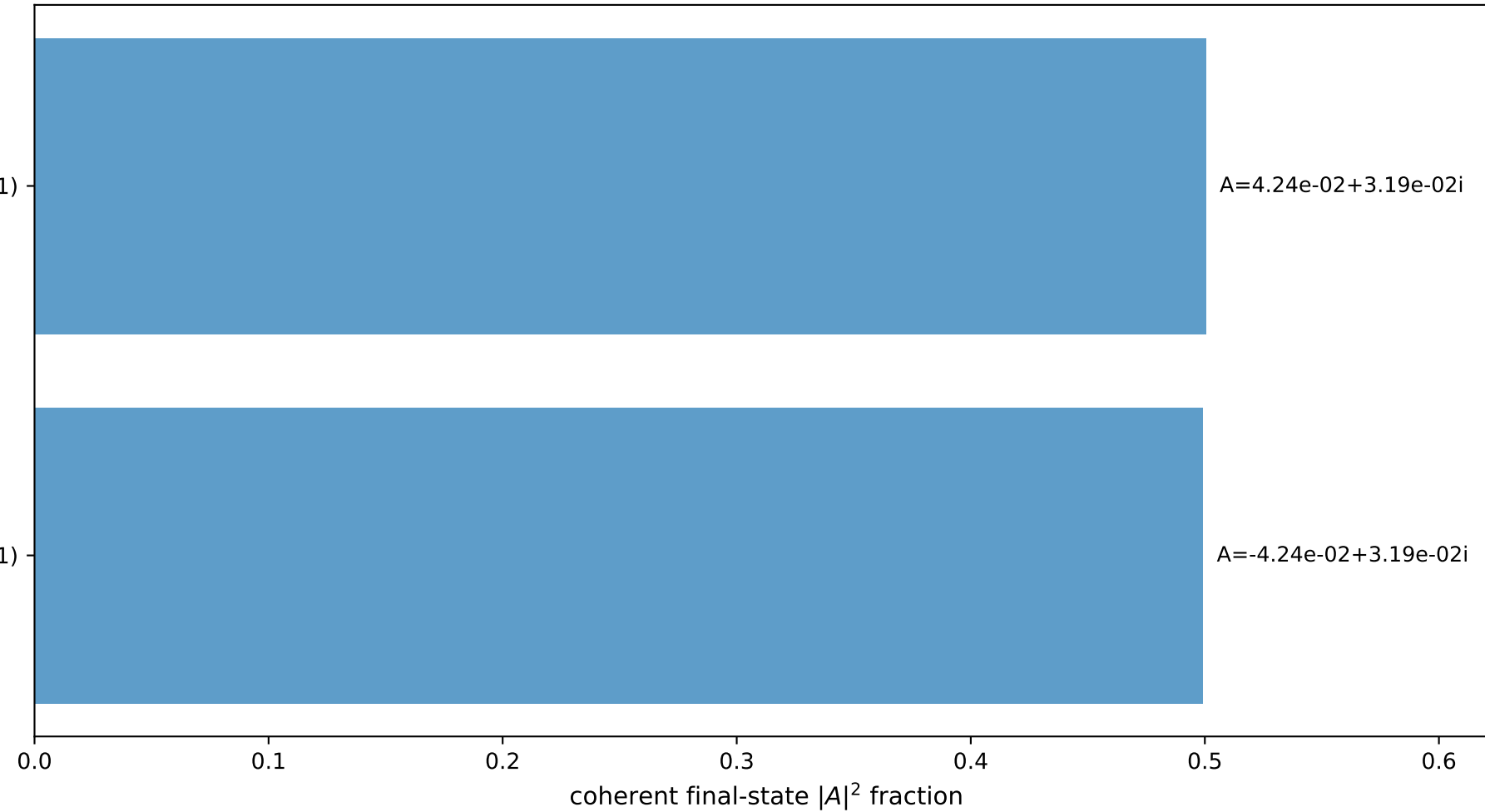
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0384$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0384$
 P $E= 2.2439$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0384$
 ℓ' $E= 0.0013$ $p_x= -0.0001$ $p_y= -0.0001$ $p_z= -0.0012$
 P' $E= 2.2427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0371$
 q_Y $E= 2.0383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0383$



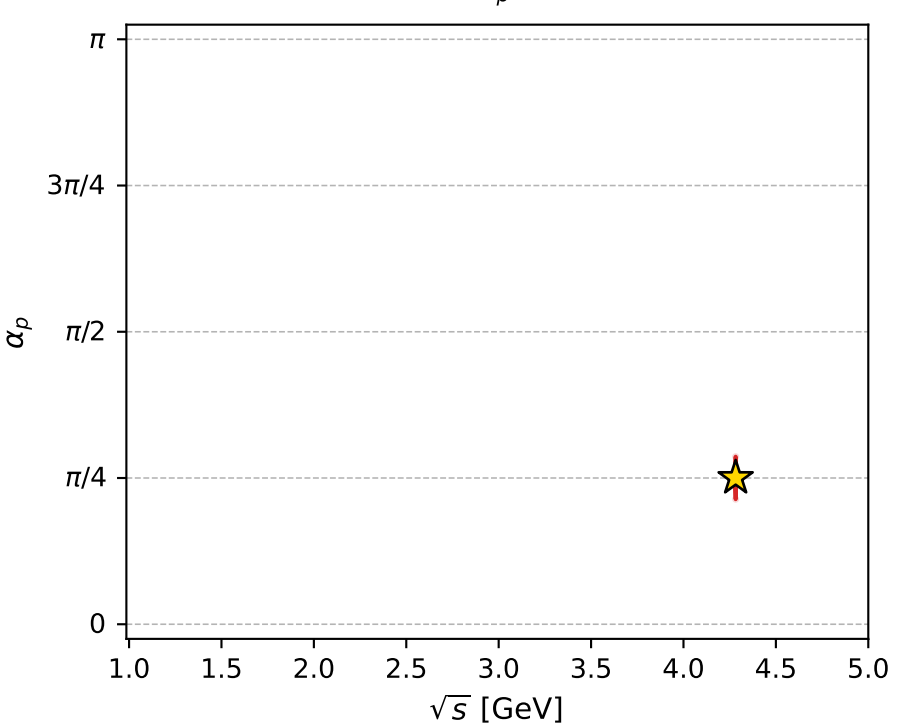
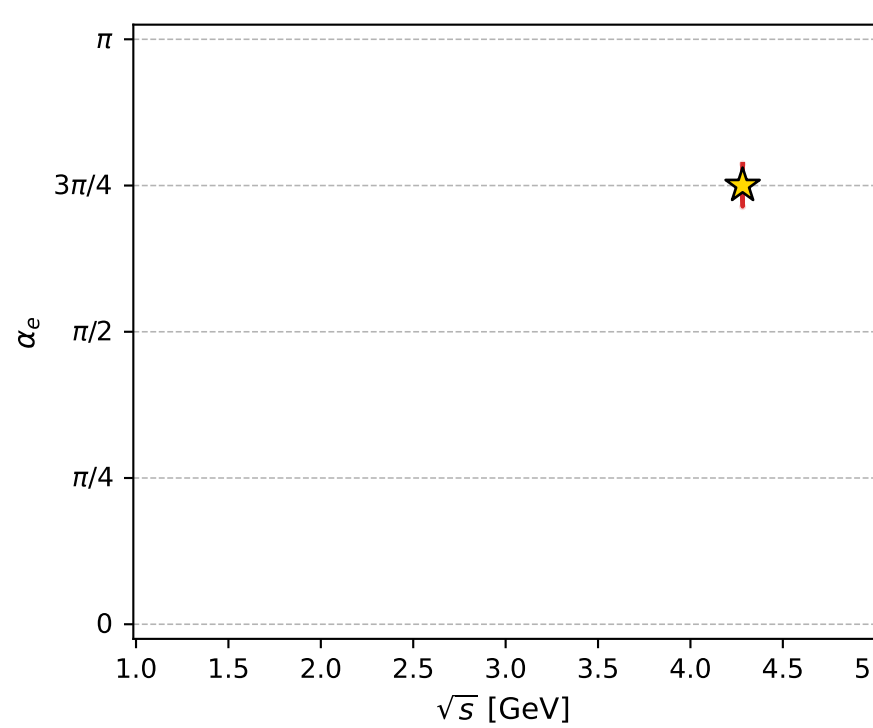
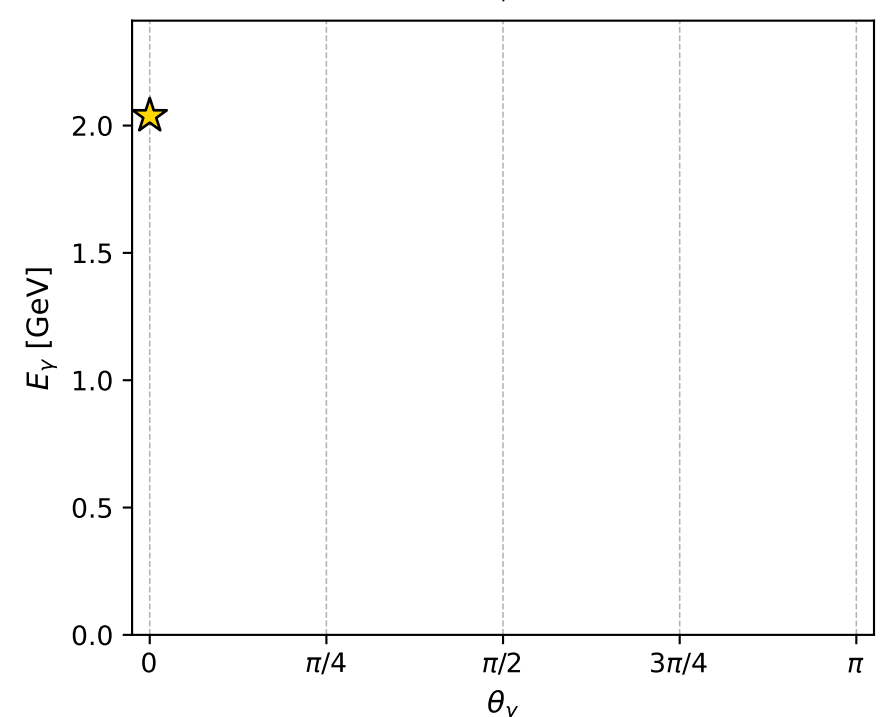
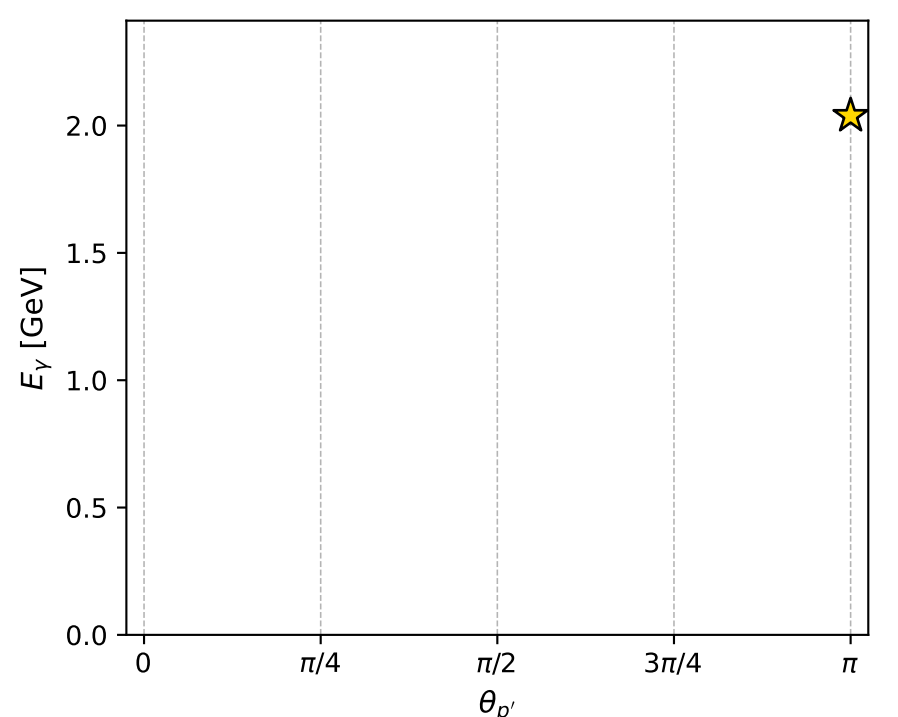
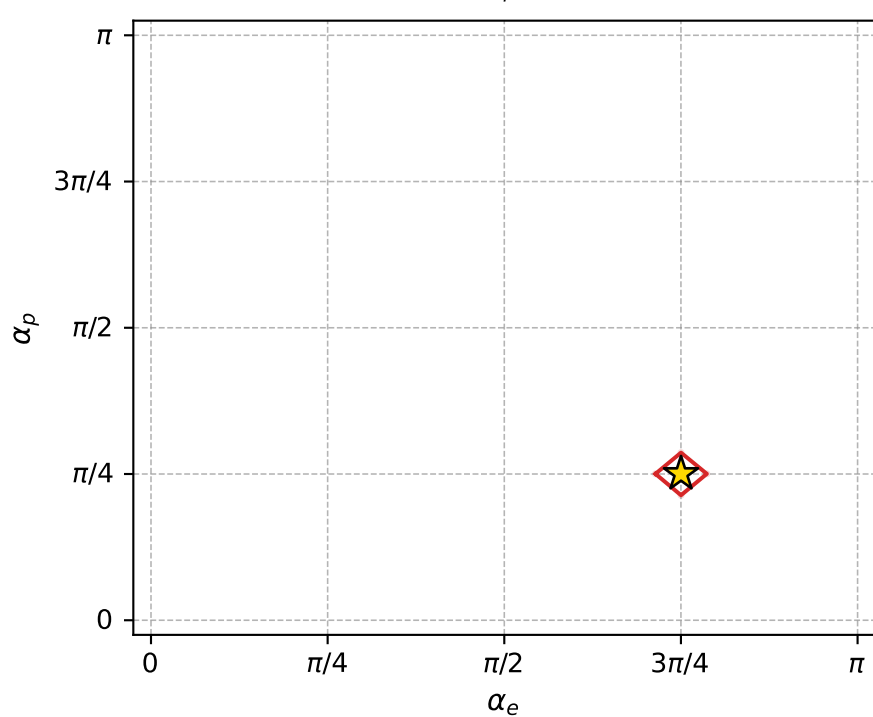
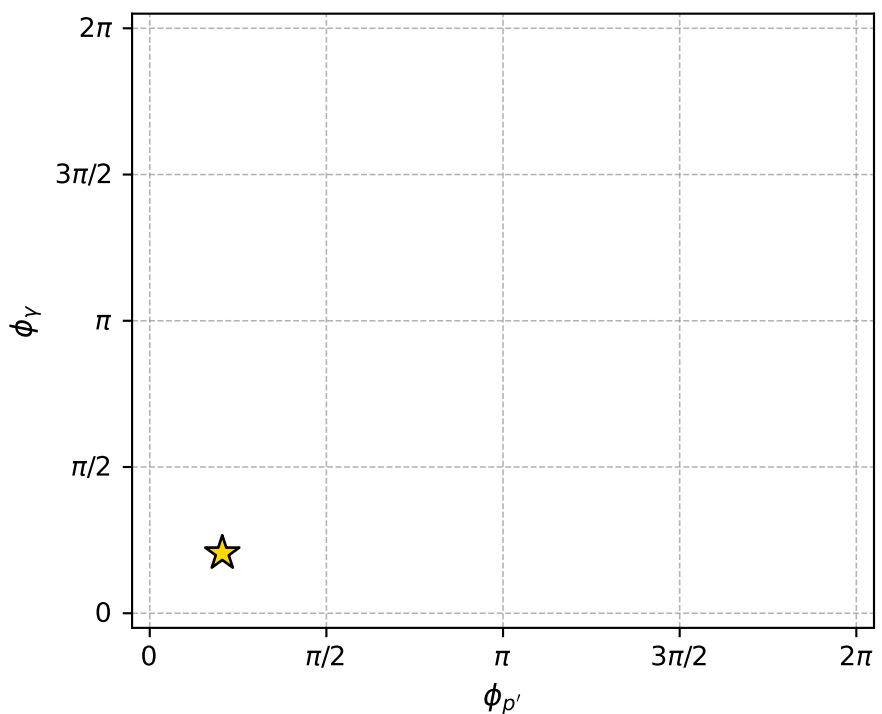
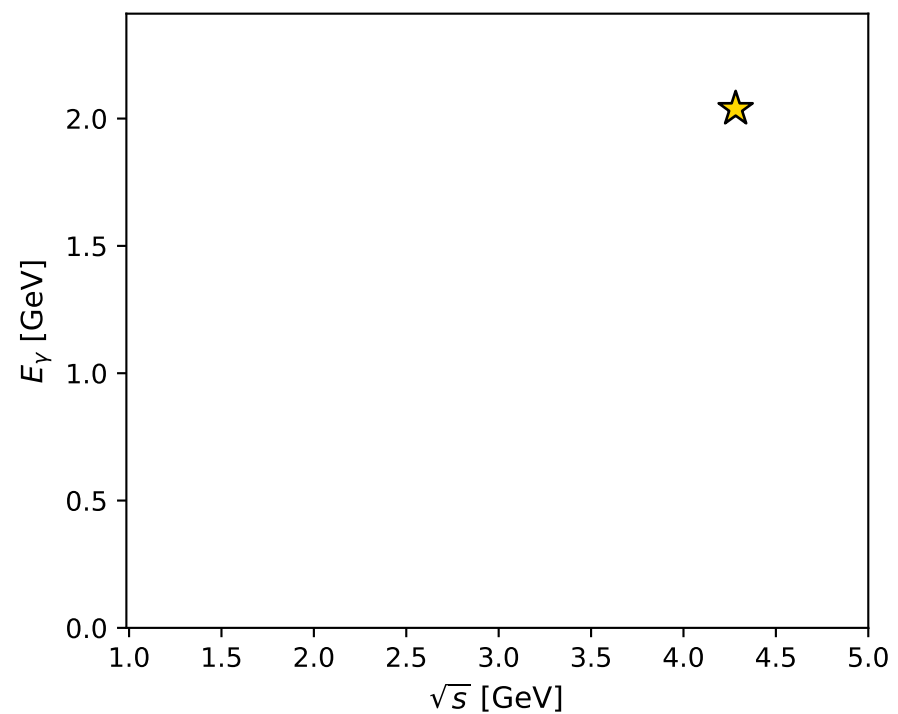
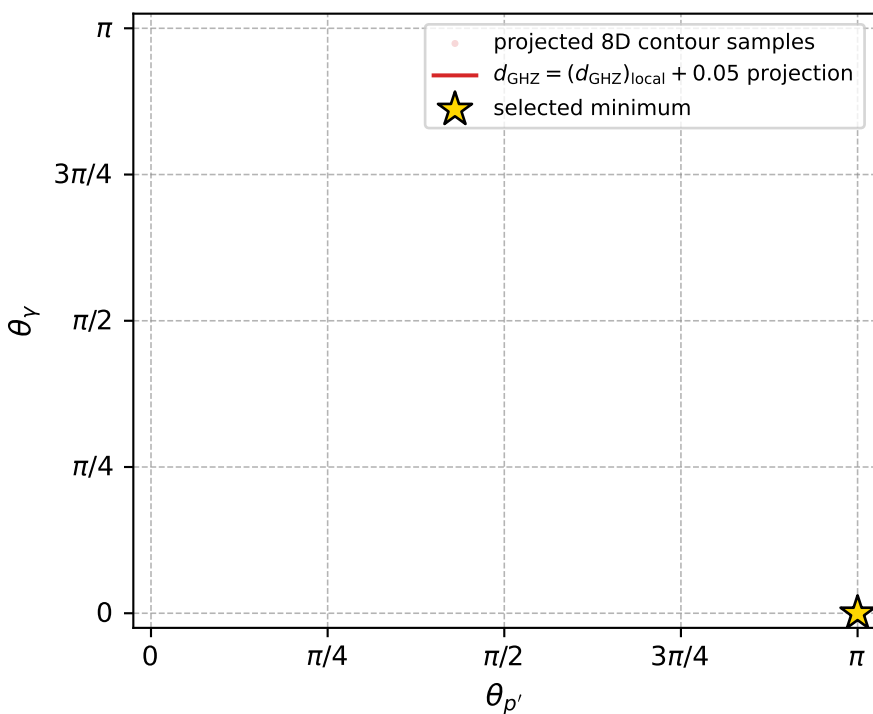
$(h_p, h_Y, h_\ell) = (-1, -1, -1)$

$(h_p, h_Y, h_\ell) = (+1, +1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P3, local minimum 357; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0012591636$
 $d_{\text{GHZ}} \text{ contour} = 0.051259164$
 above global minimum = 0.0012591413
 8D contour samples = 32

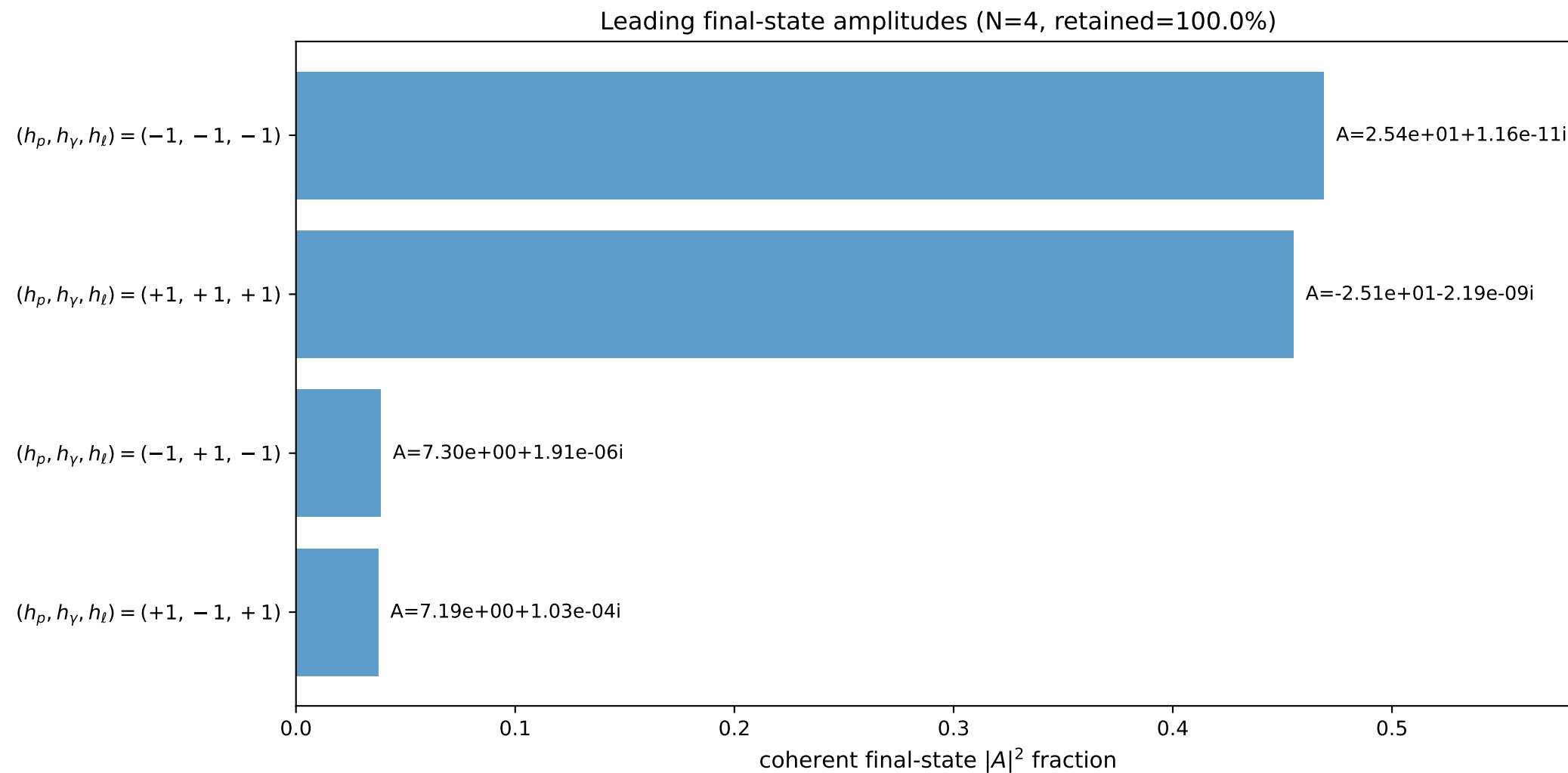
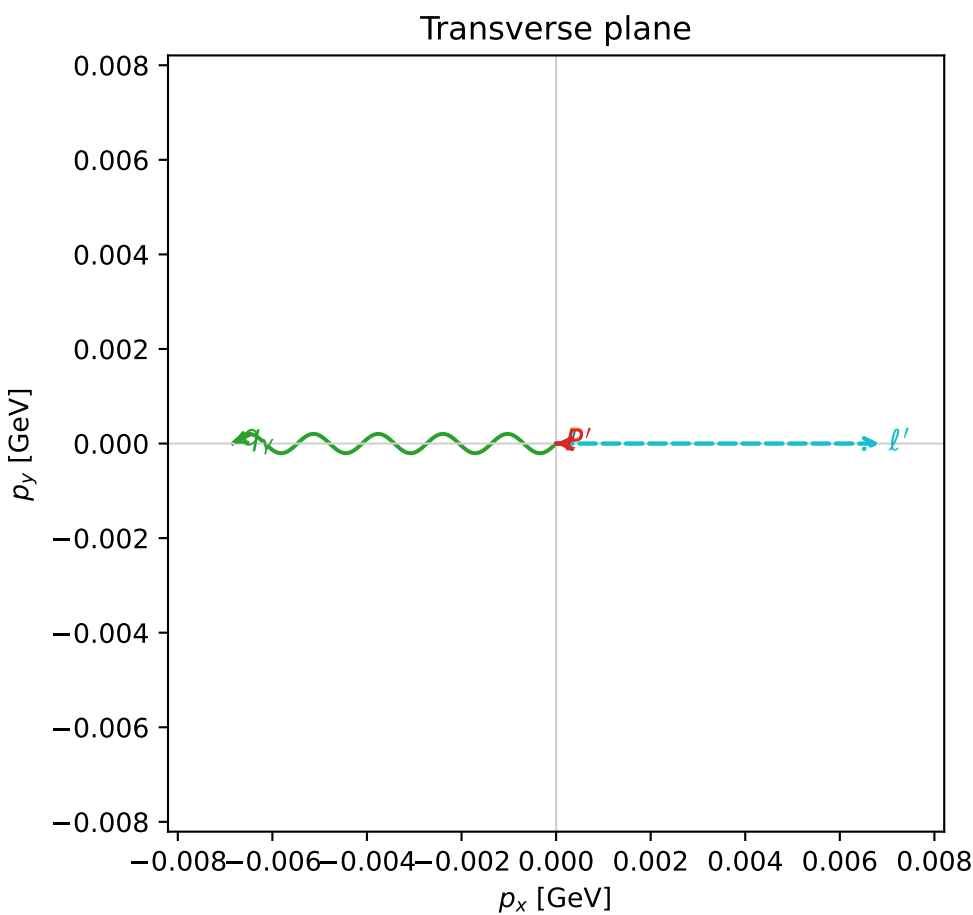
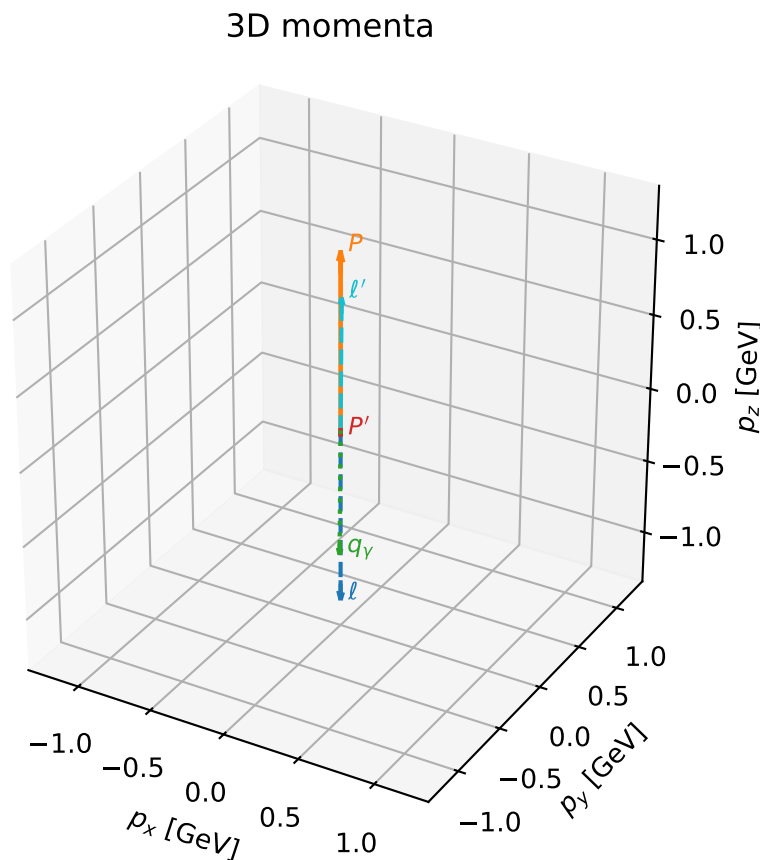
selected phase-space point:
 $\text{sqrt_s} = 4.282322$
 $\text{theta_p_out} = 3.141577$
 $\text{theta_gamma_out} = 2.796193\text{e-}05$
 $\text{q0out} = 2.038317$
 $\text{phi_p_out} = 0.6449693$
 $\text{phi_gamma_out} = 0.6456297$
 $\text{alpha_e} = 2.356523$
 $\text{alpha_p} = 0.786329$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

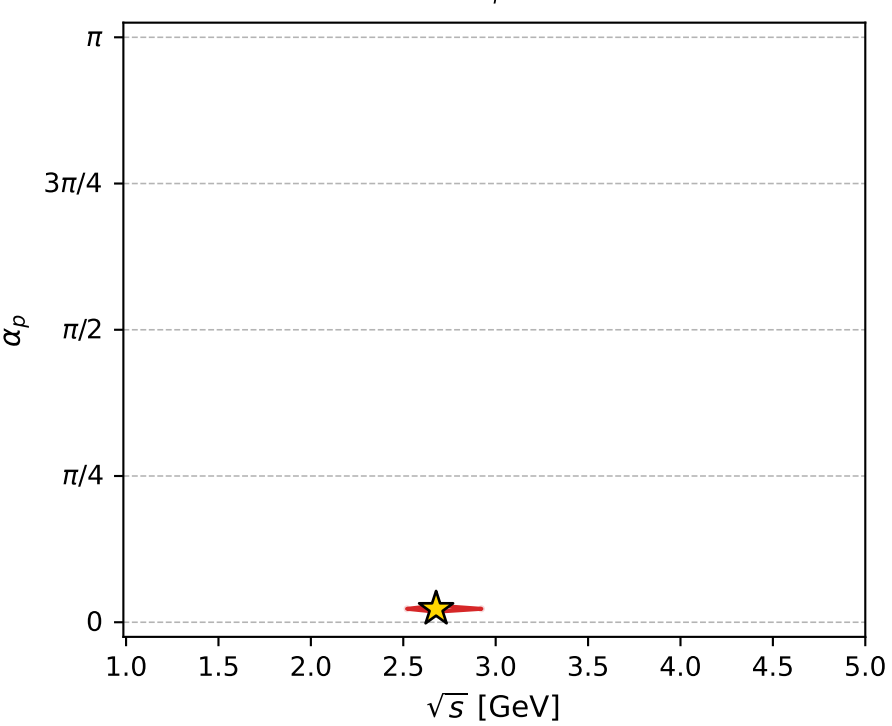
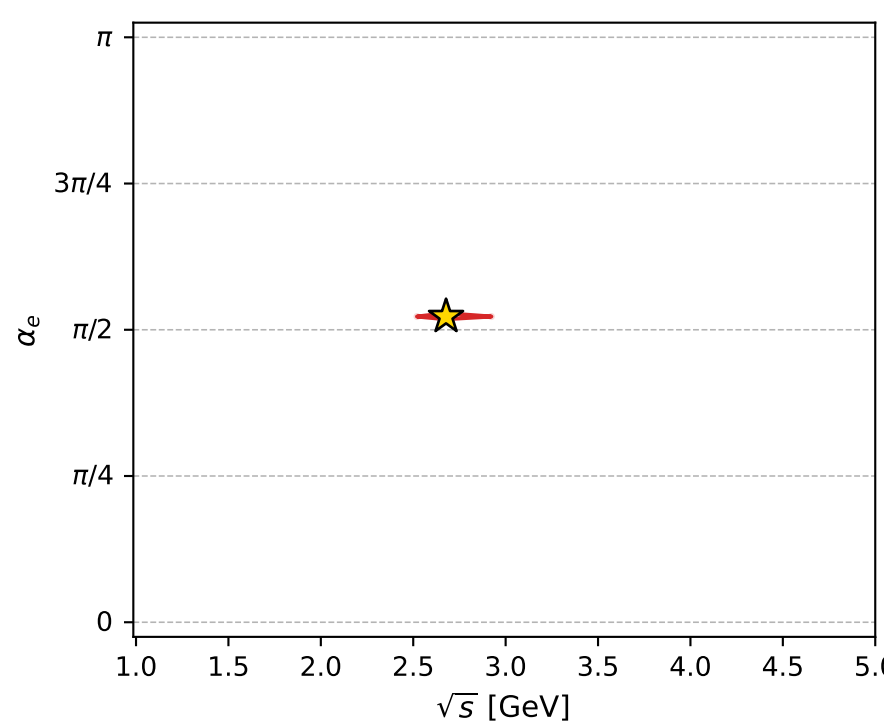
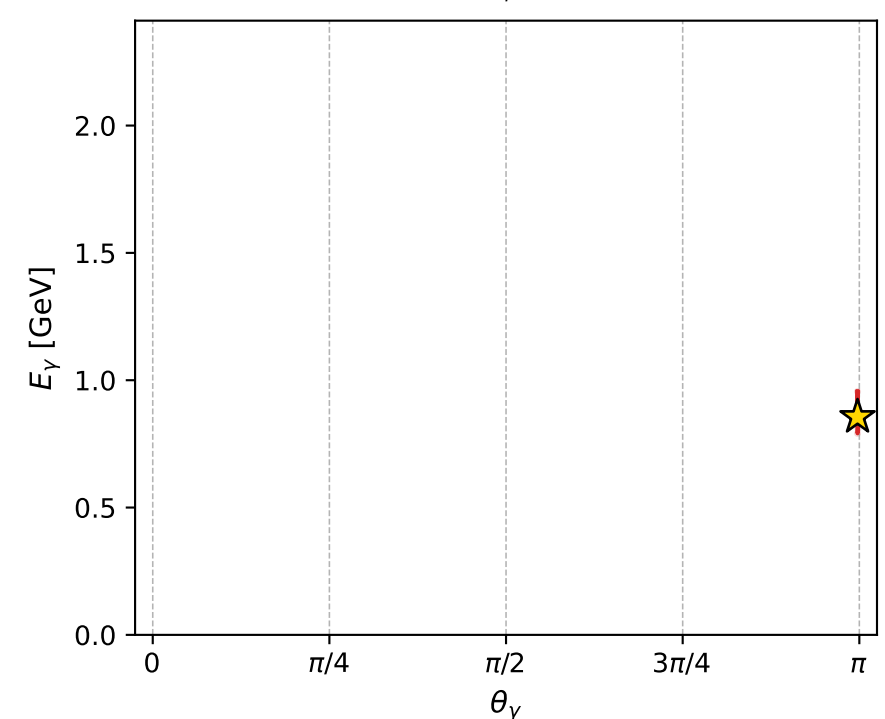
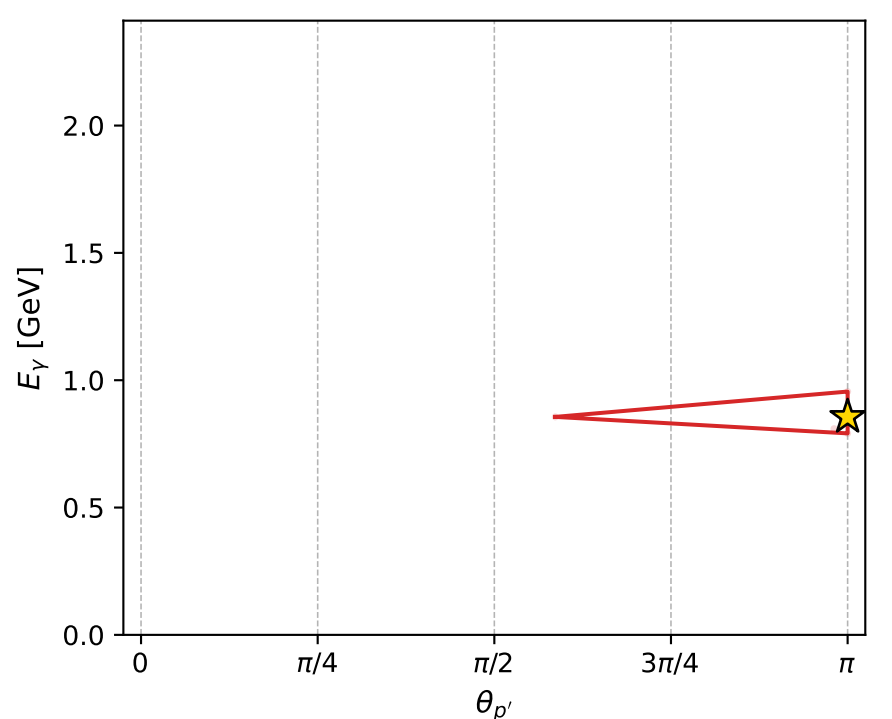
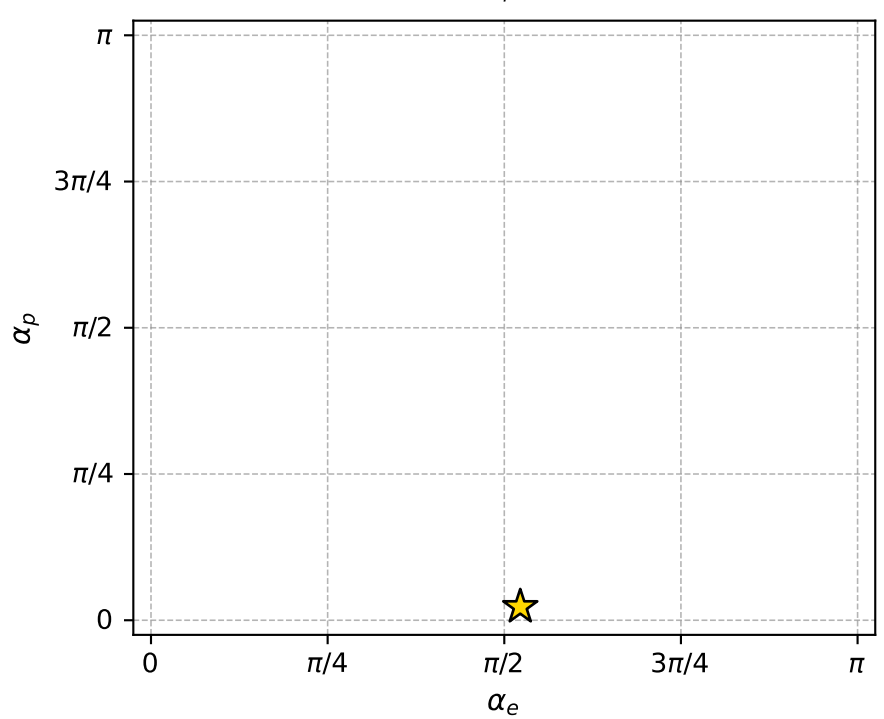
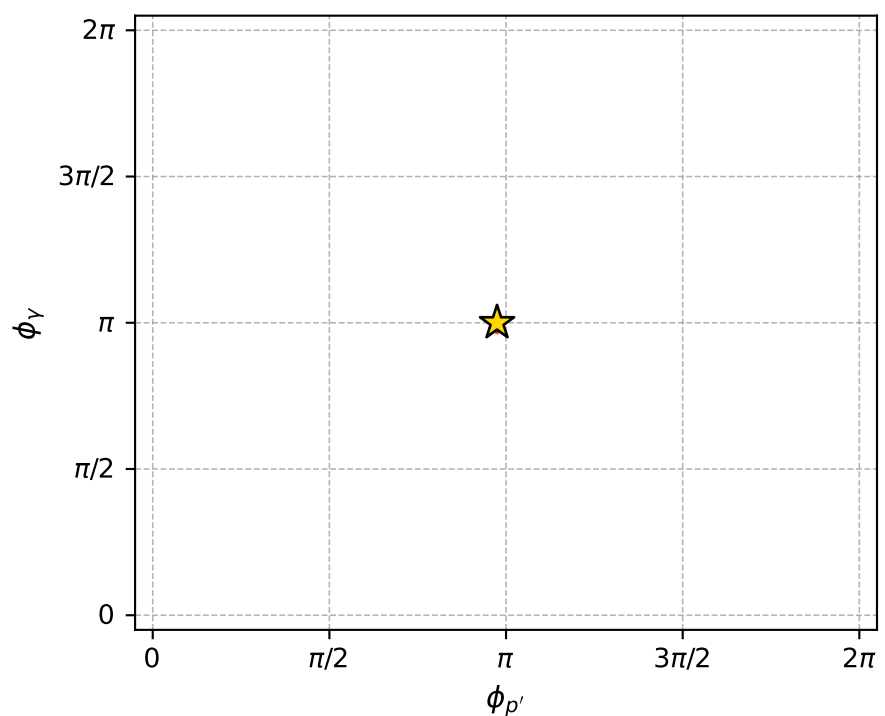
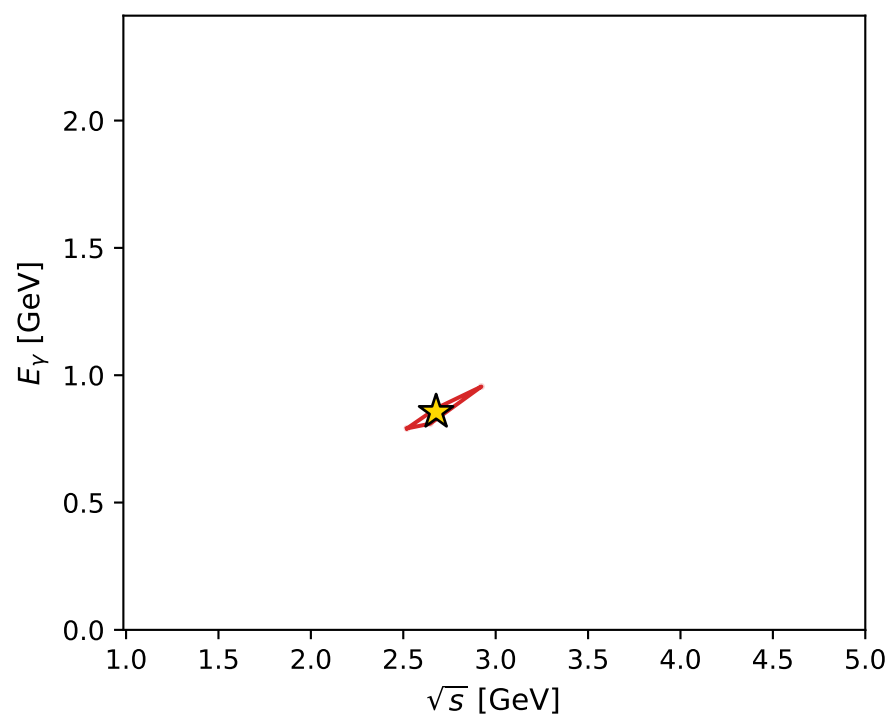
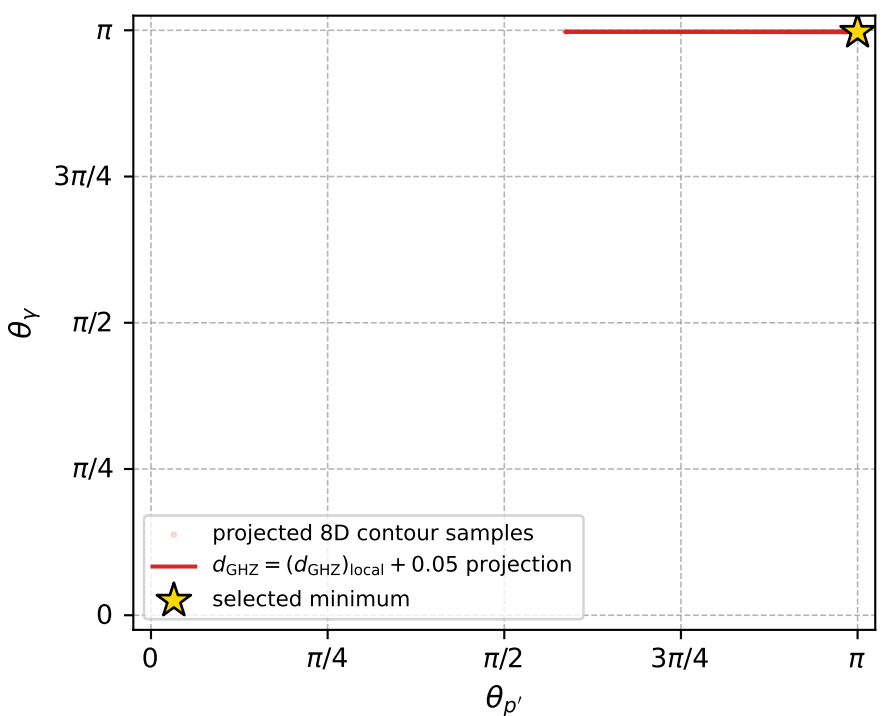
dghz_mixing_angles_polarization_03_local_minimum_362 $d_{\text{GHZ}}=0.00195223$
 region: polarization_cluster_03_local_minimum_362
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.6418069$ rad
 incoming lepton: $-0.070951|+> +0.99748|->$
 proton mixing angle: $\alpha_p=0.072084721$ rad
 incoming proton: $0.997403|+> +0.0720223|->$

$s=7.16942$, $\sqrt{s}=2.67758$, $|\vec{P}|=1.17449$, $|\vec{P}'|=0.0272157$
 $\theta_{P'}=3.14159$, $\theta_Y=3.1336$, $\phi_{P'}=3.06156$, $\phi_Y=3.1416$
 $E_Y=0.855984$, $\theta_{\ell'}=0.00774441$, $\phi_{\ell'}=7.3781e-06$
 $Q^2=4.14917$, $x_B=0.726767$, $t=-1.12522$, $W^2=2.43975$, $y=0.907705$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1745$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1745$
 P $E= 1.5031$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1745$
 ℓ' $E= 0.8832$ $p_x= 0.0068$ $p_y= 0.0000$ $p_z= 0.8832$
 P' $E= 0.9384$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0272$
 q_Y $E= 0.8560$ $p_x= -0.0068$ $p_y= -0.0000$ $p_z= -0.8560$



electron: polarization cluster P3, local minimum 362; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0019522314$
 $d_{\text{GHZ}} \text{ contour} = 0.051952231$
 above global minimum = 0.0019522091
 8D contour samples = 132

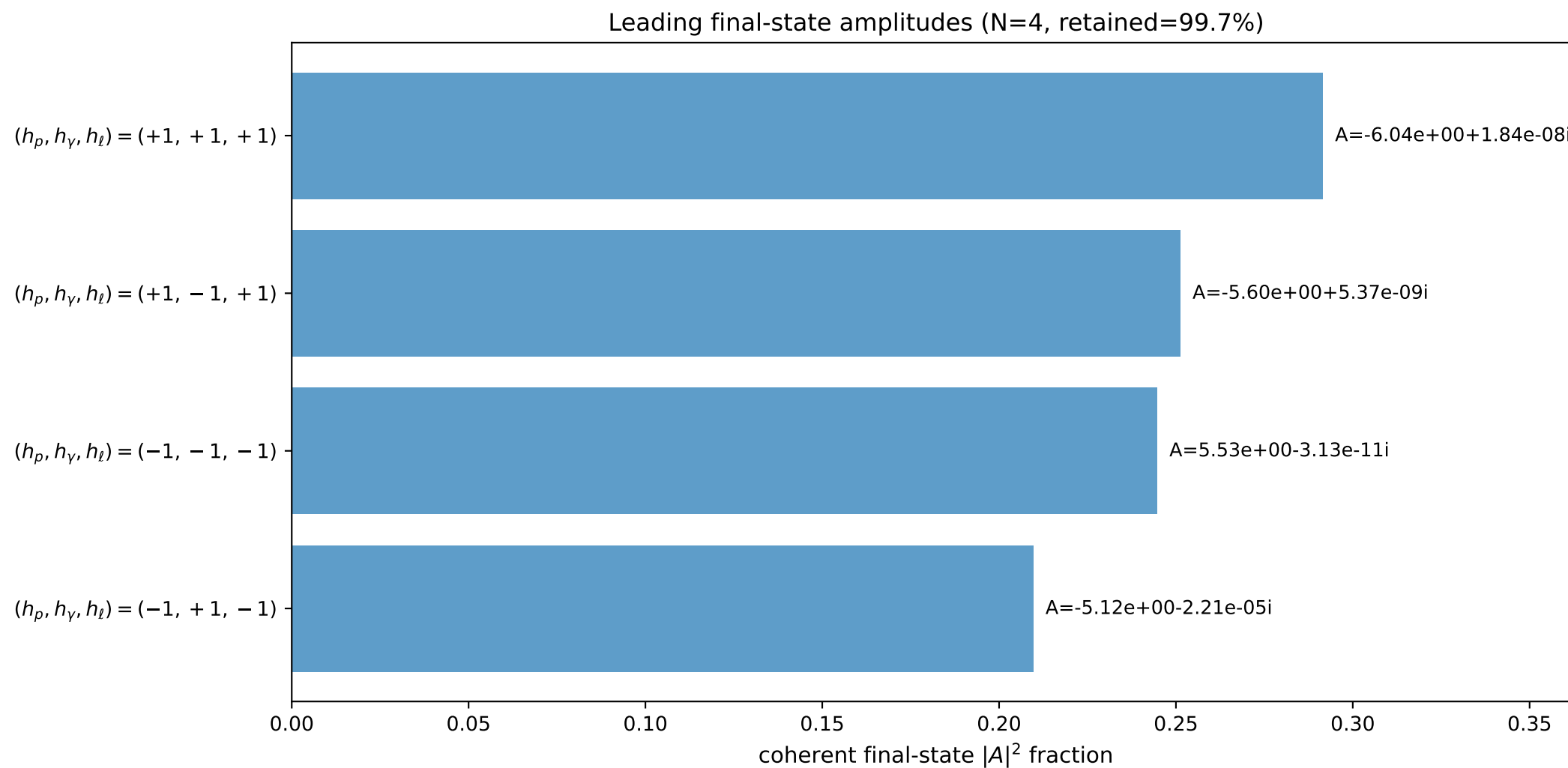
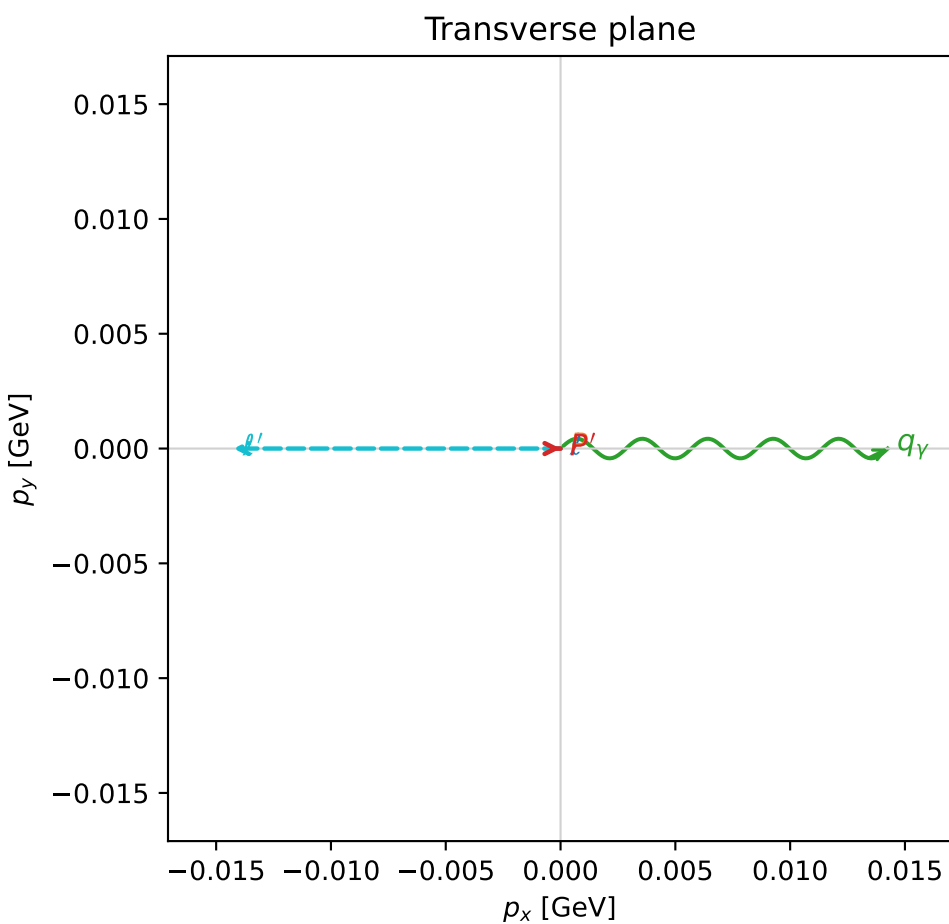
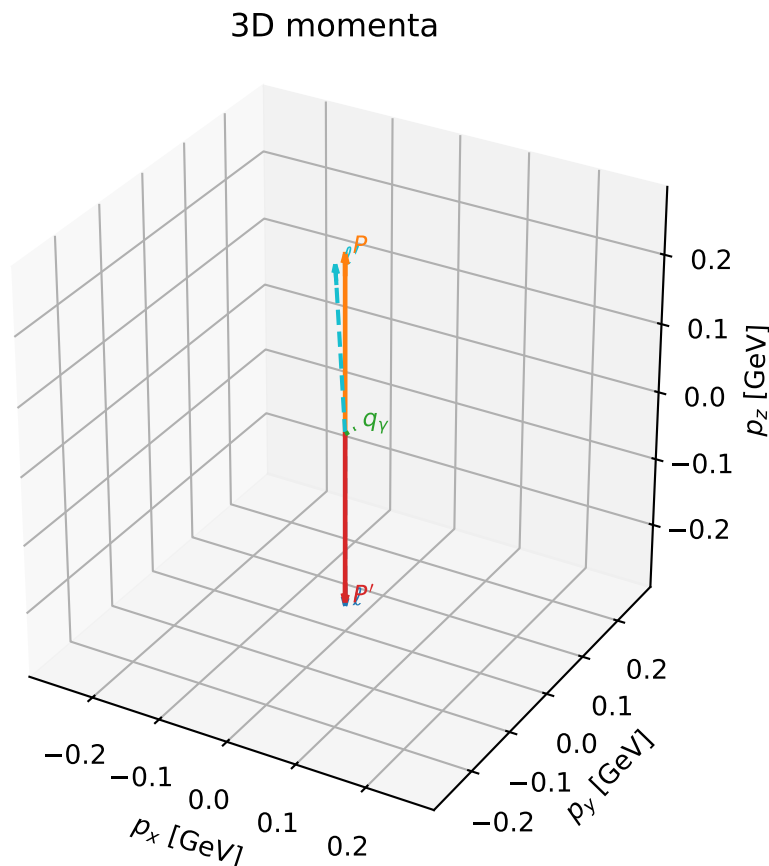
selected phase-space point:
 $\text{sqrt_s} = 2.677577$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.133602$
 $\text{q0out} = 0.8559835$
 $\text{phi_p_out} = 3.06156$
 $\text{phi_gamma_out} = 3.1416$
 $\text{alpha_e} = 1.641807$
 $\text{alpha_p} = 0.07208472$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

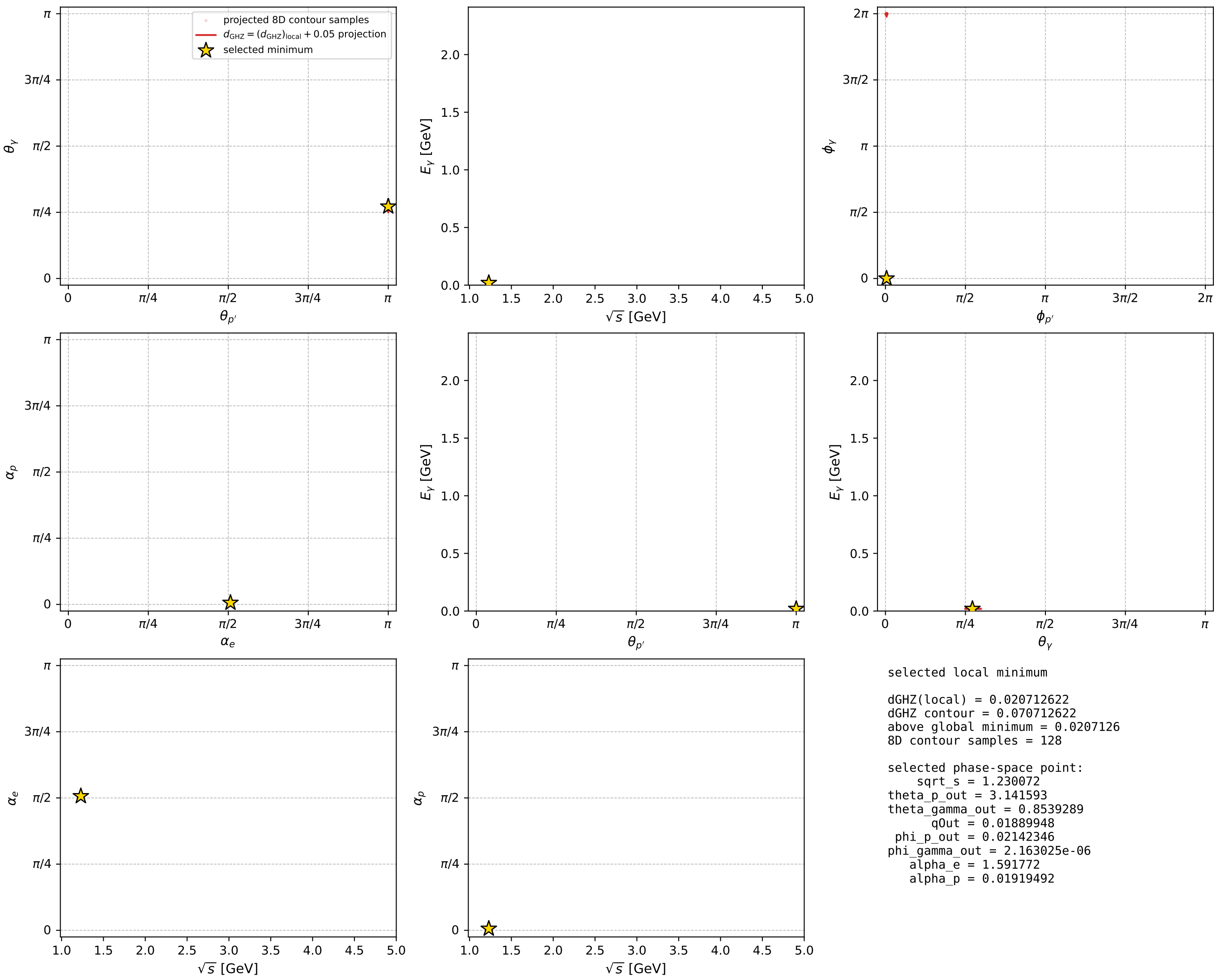
dghz_mixing_angles_polarization_03_local_minimum_368 d_{\rm GHZ}=0.0207126
region: polarization_cluster_03_local_minimum_368
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5917721$ rad
incoming lepton: $-0.0209742|+\rangle +0.99978|-\rangle$
proton mixing angle: $\alpha_p=0.019194918$ rad
incoming proton: $0.999816|+\rangle +0.0191937|-\rangle$

$s=1.51308$, $\sqrt{s}=1.23007$, $|\vec{P}|=0.257396$, $|\vec{P}'|=0.251925$
 $\theta_{p'}=3.14159$, $\theta_Y=0.853929$, $\phi_{p'}=0.0214235$, $\phi_Y=2.16302\text{e-}06$
 $E_Y=0.0188995$, $\theta_{\ell'}=0.0594176$, $\phi_{\ell'}=3.14159$
 $Q^2=0.246811$, $x_B=0.85172$, $t=-0.259406$, $W^2=0.922812$, $y=0.457619$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2574$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2574$
 P $E= 0.9727$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2574$
 ℓ' $E= 0.2399$ $p_x= -0.0142$ $p_y= -0.0000$ $p_z= 0.2395$
 P' $E= 0.9712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2519$
 q_Y $E= 0.0189$ $p_x= 0.0142$ $p_y= 0.0000$ $p_z= 0.0124$



electron: polarization cluster P3, local minimum 368; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

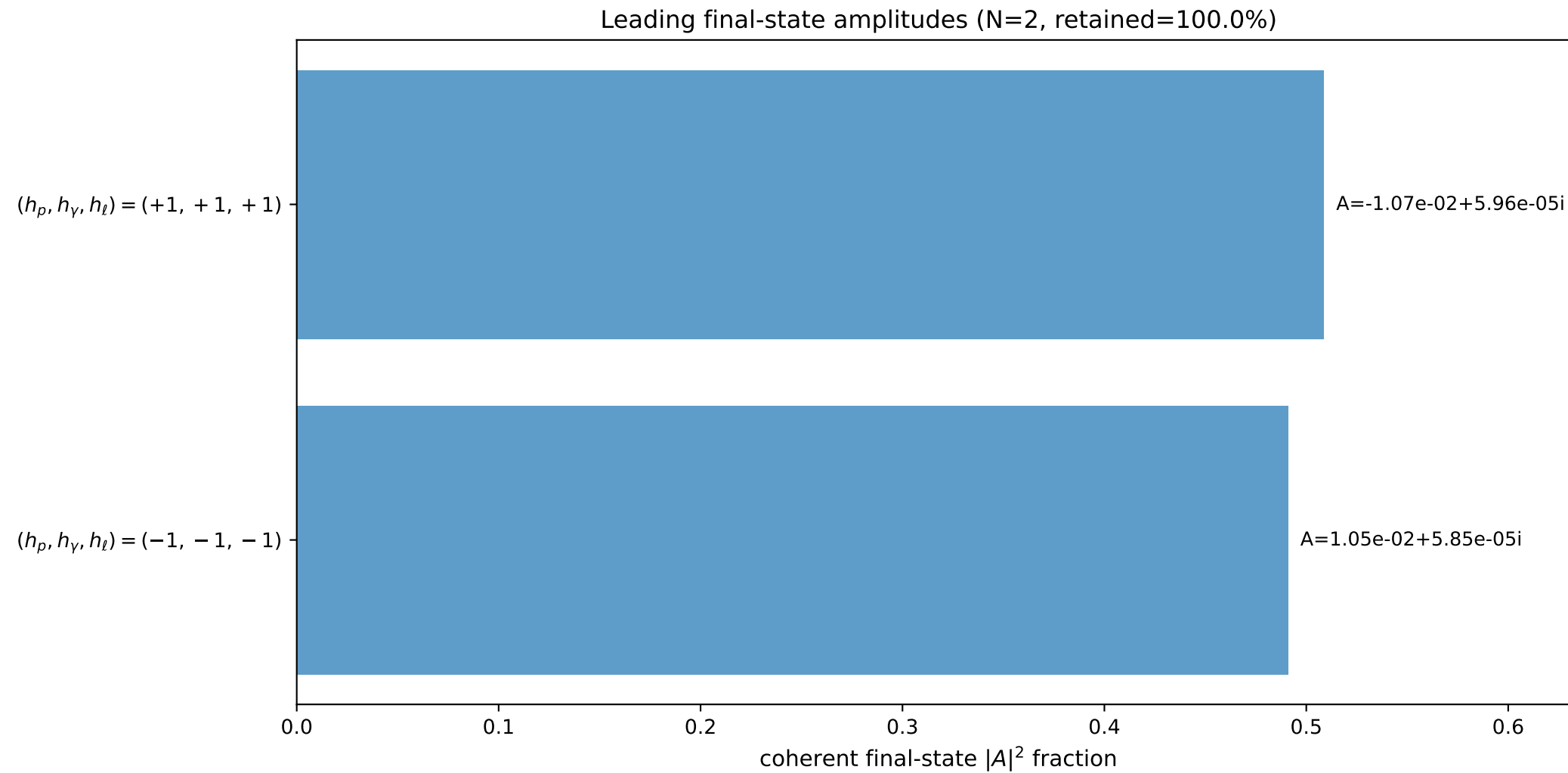
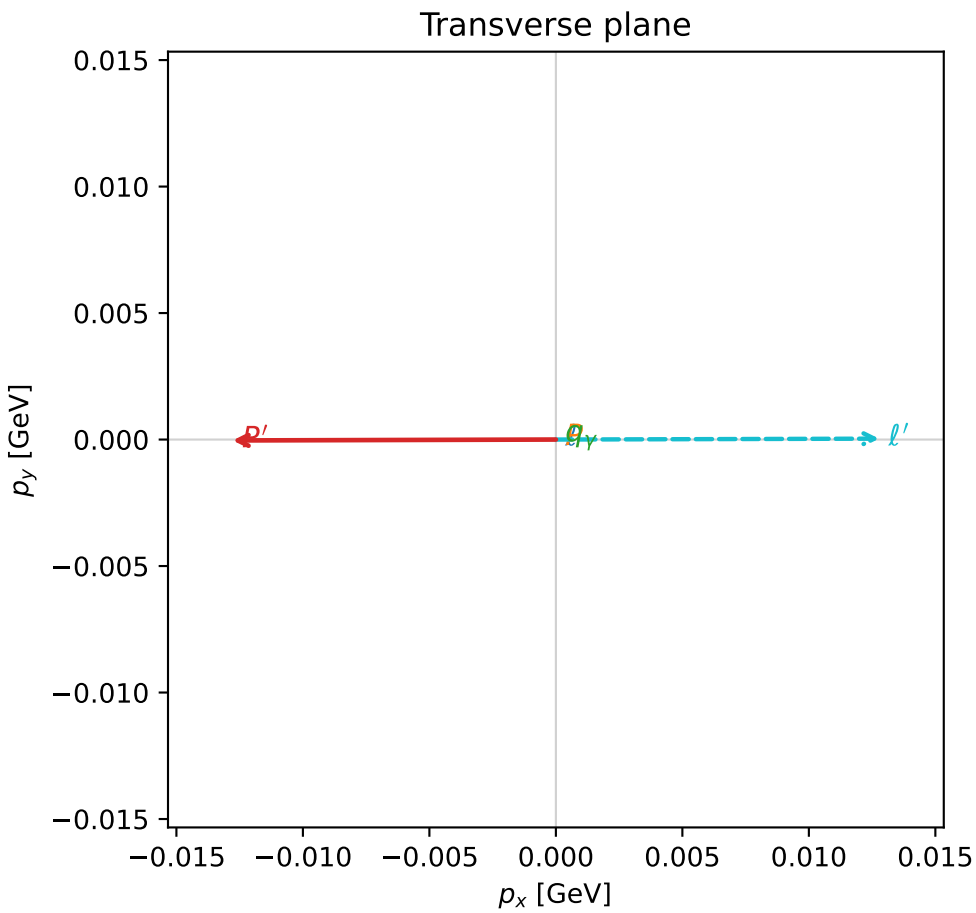
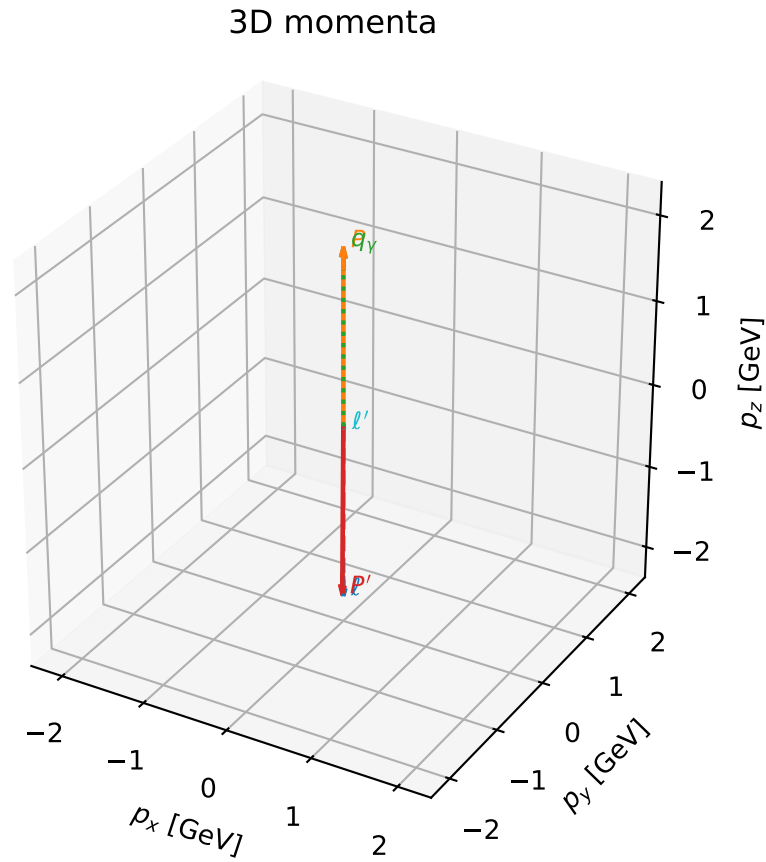


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

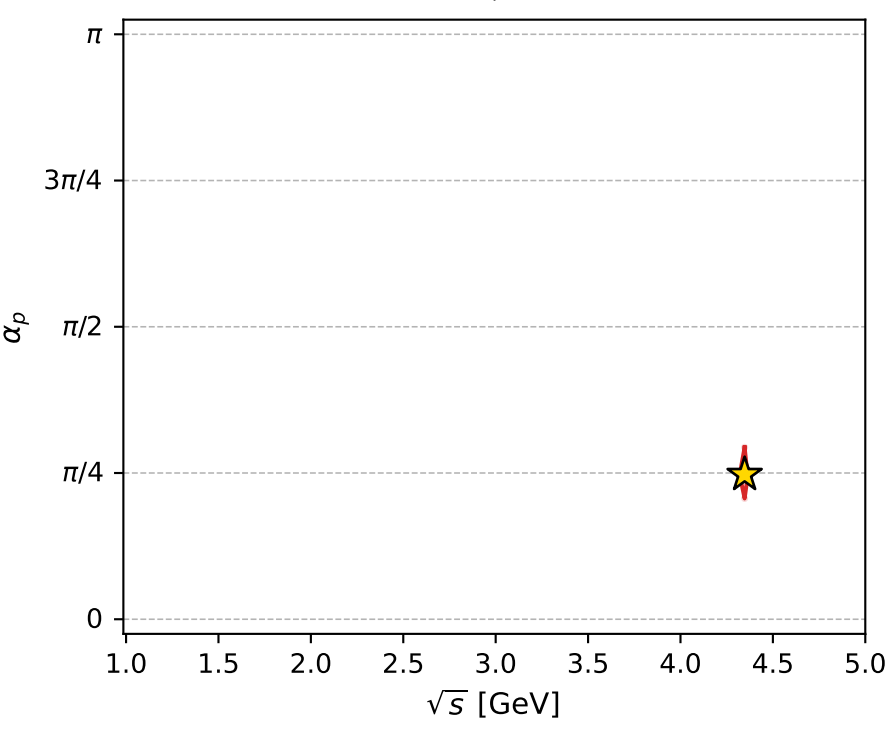
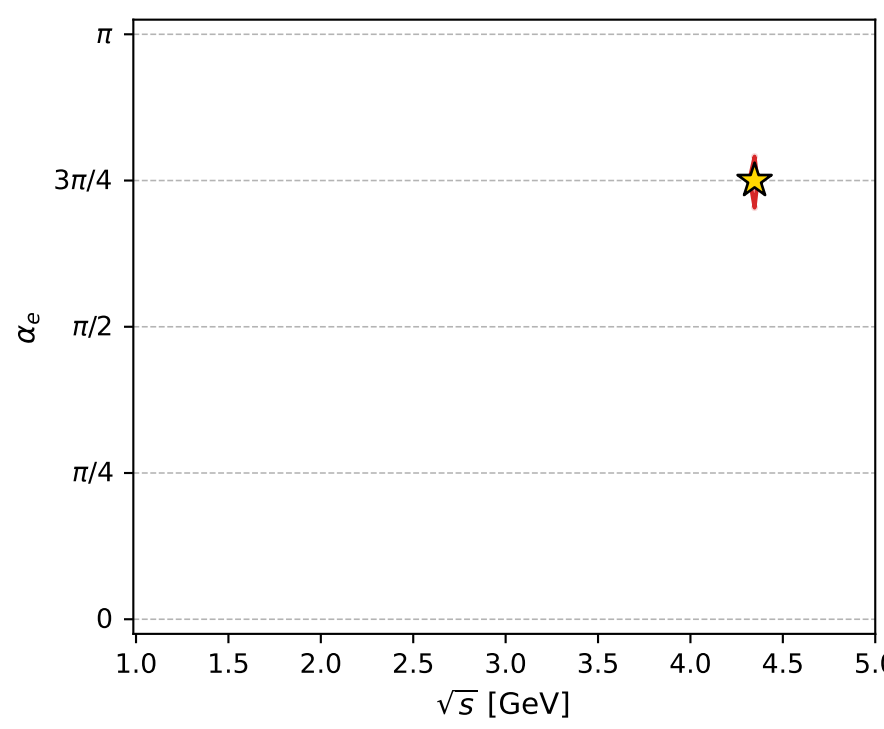
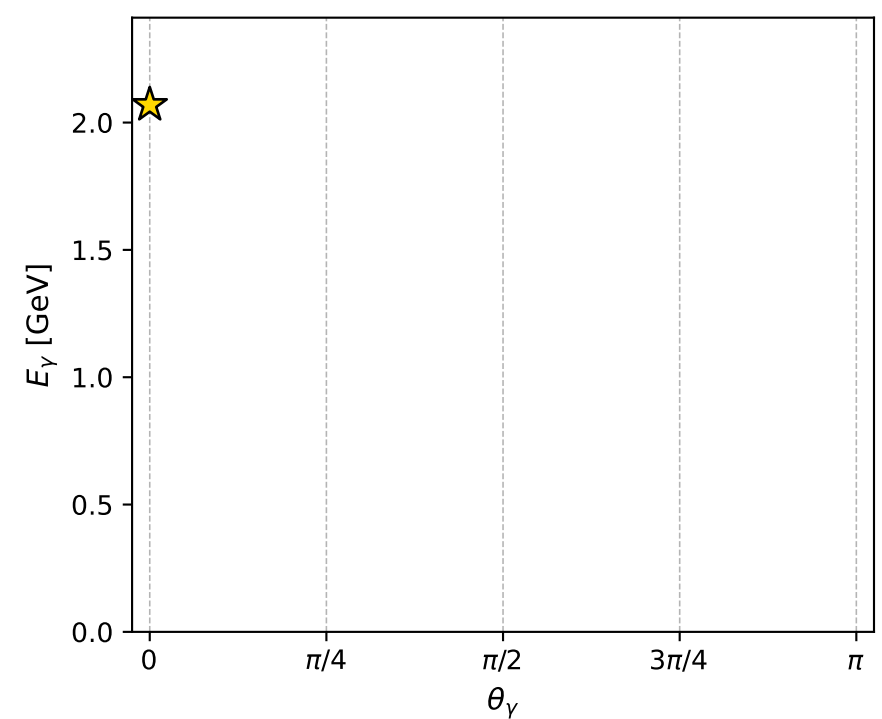
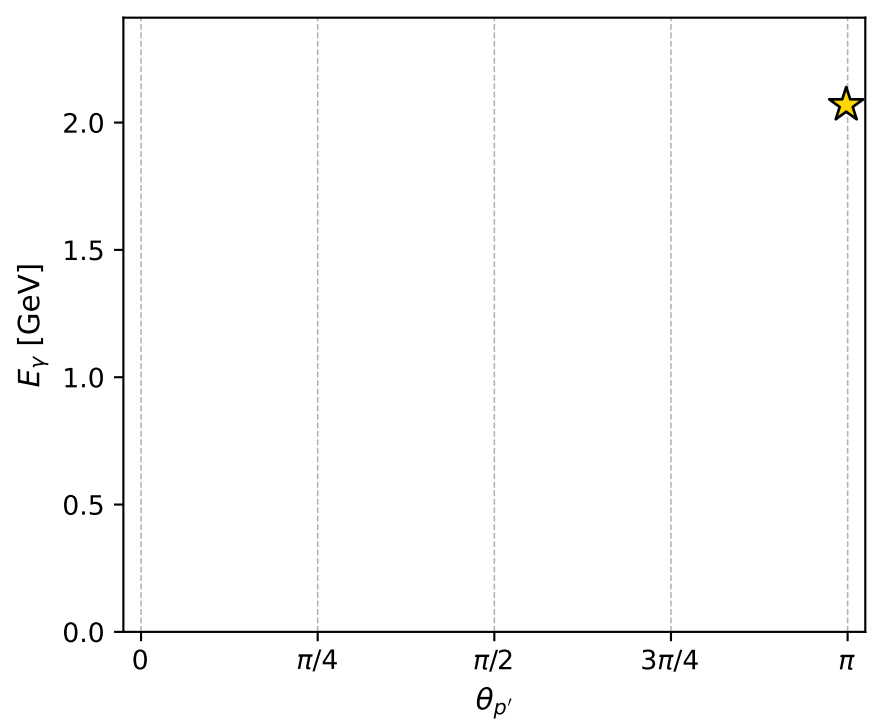
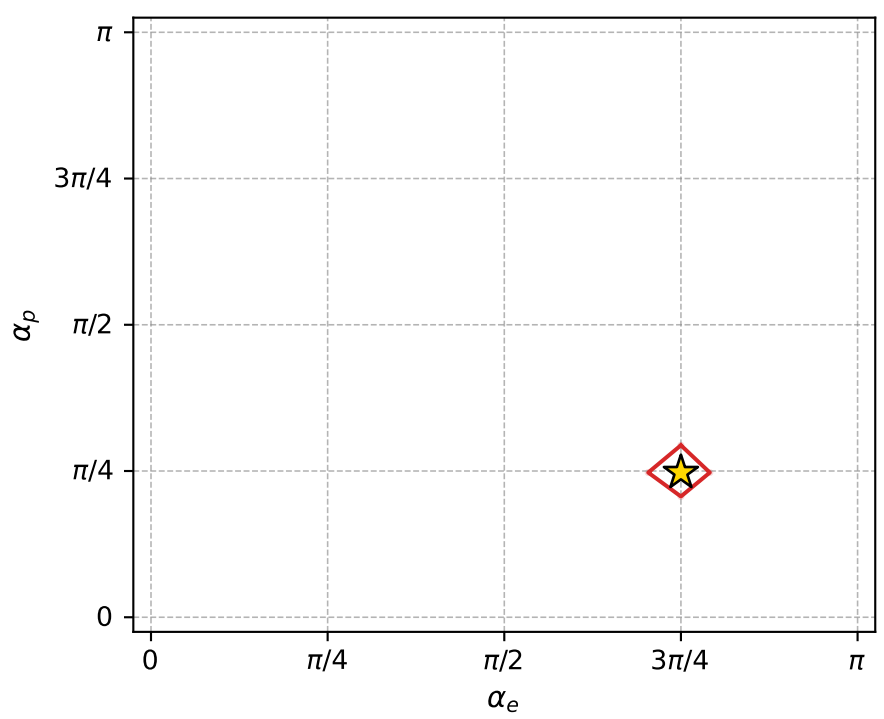
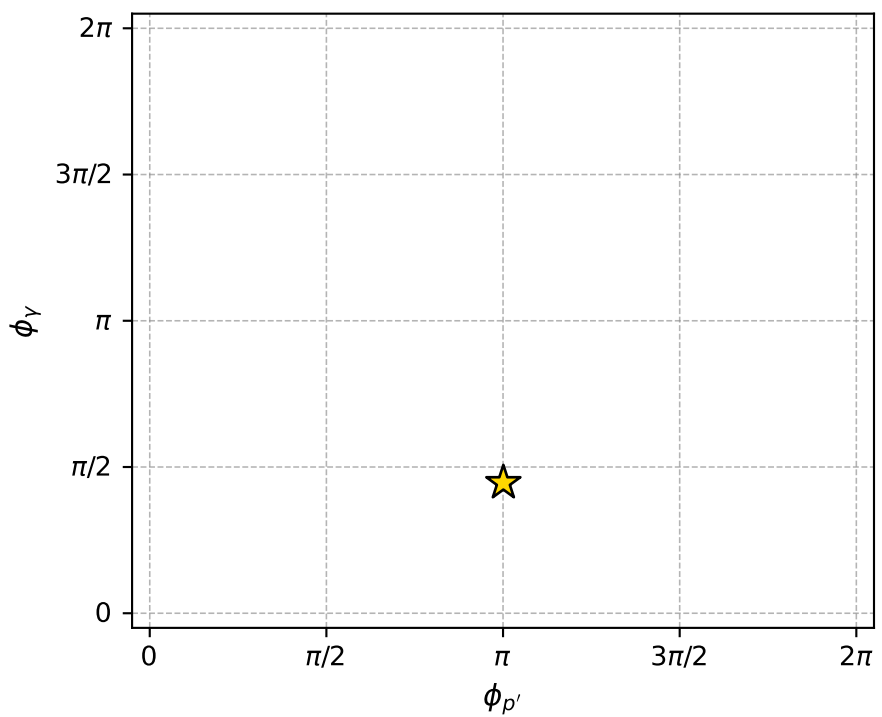
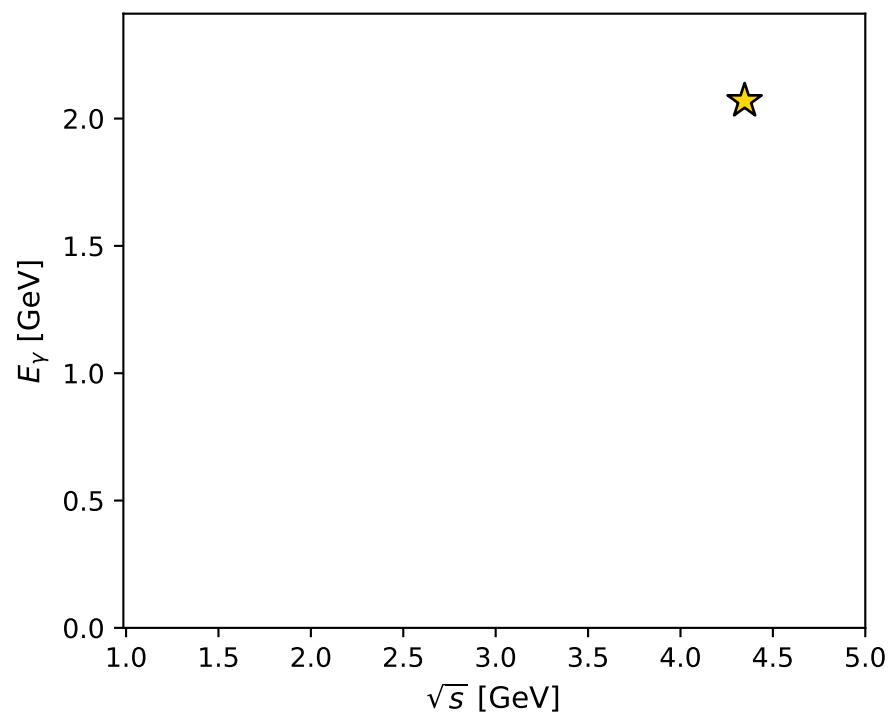
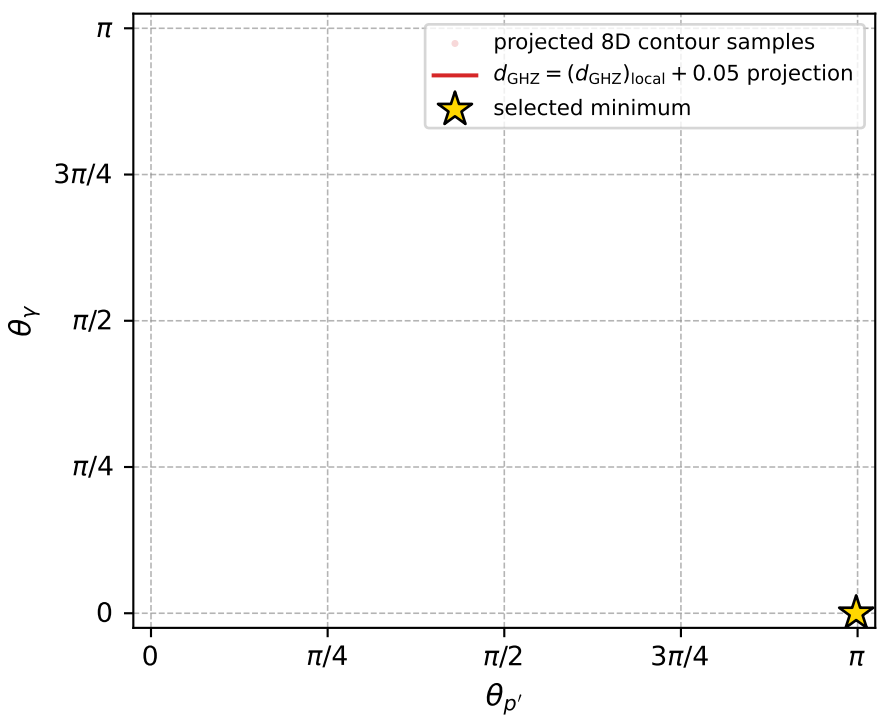
dghz_mixing_angles_polarization_03_local_minimum_372 d_{\rm GHZ}=0.0286865
region: polarization_cluster_03_local_minimum_372
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=2.3560927$ rad
incoming lepton: $-0.707035|+> +0.707179|->$
proton mixing angle: $\alpha_p=0.77718399$ rad
incoming proton: $0.712891|+> +0.701275|->$

$s=18.8971$, $\sqrt{s}=4.34708$, $|\vec{P}|=2.07234$, $|\vec{P}'|=2.04126$
 $\theta_{P'}=3.13533$, $\theta_\gamma=0$, $\phi_{P'}=3.14439$, $\phi_\gamma=1.40154$
 $E_\gamma=2.06954$, $\theta_{\ell'}=2.71784$, $\phi_{\ell'}=0.00279457$
 $Q^2=0.0114083$, $x_B=0.000642414$, $t=-16.9207$, $W^2=18.6269$, $y=0.985637$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0723$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0723$
 P $E= 2.2747$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0723$
 ℓ' $E= 0.0311$ $p_x= 0.0128$ $p_y= 0.0000$ $p_z= -0.0283$
 P' $E= 2.2465$ $p_x= -0.0128$ $p_y= -0.0000$ $p_z= -2.0412$
 q_γ $E= 2.0695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0695$



electron: polarization cluster P3, local minimum 372; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.028686466$
 $d_{\text{GHZ}} \text{ contour} = 0.078686466$
 above global minimum = 0.028686444
 8D contour samples = 117

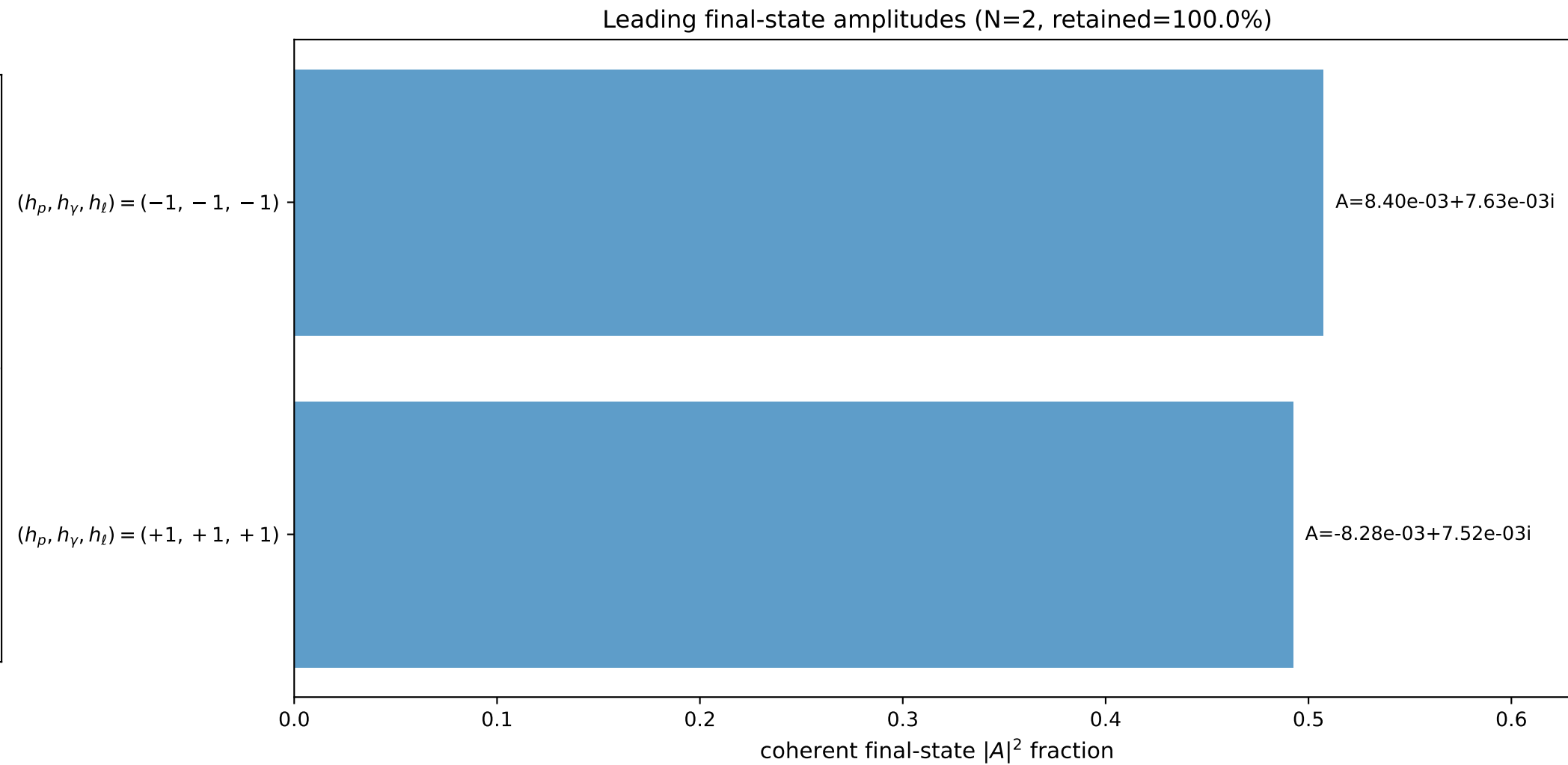
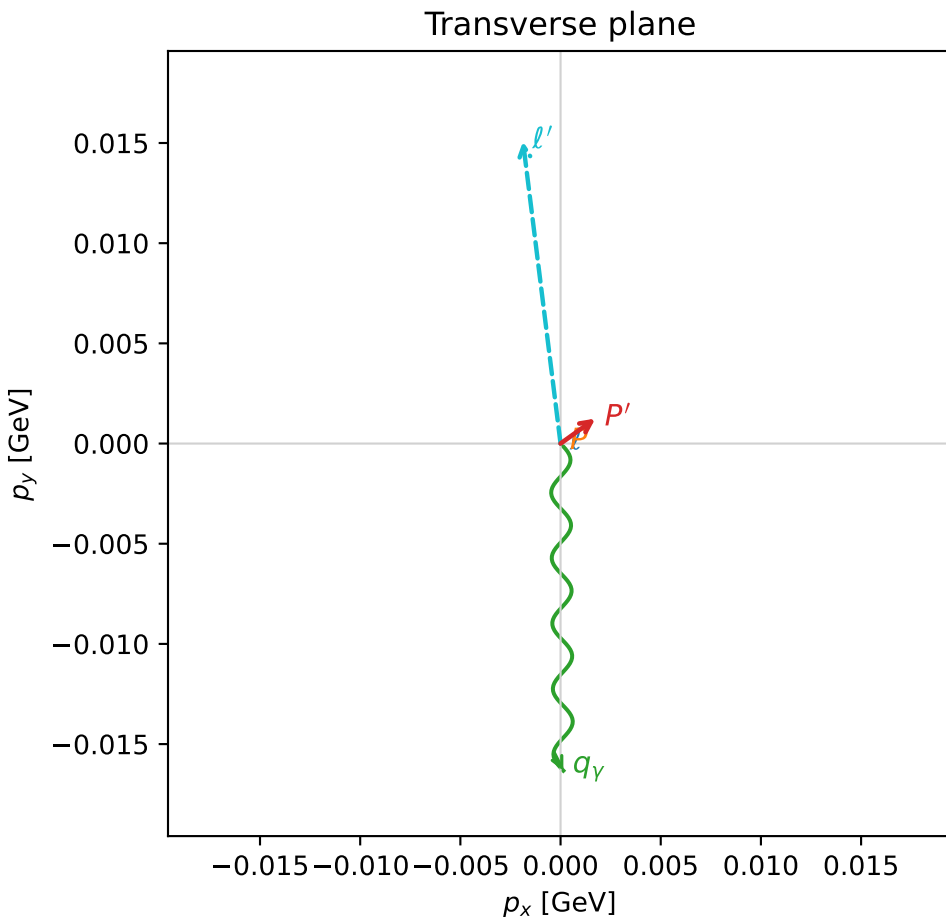
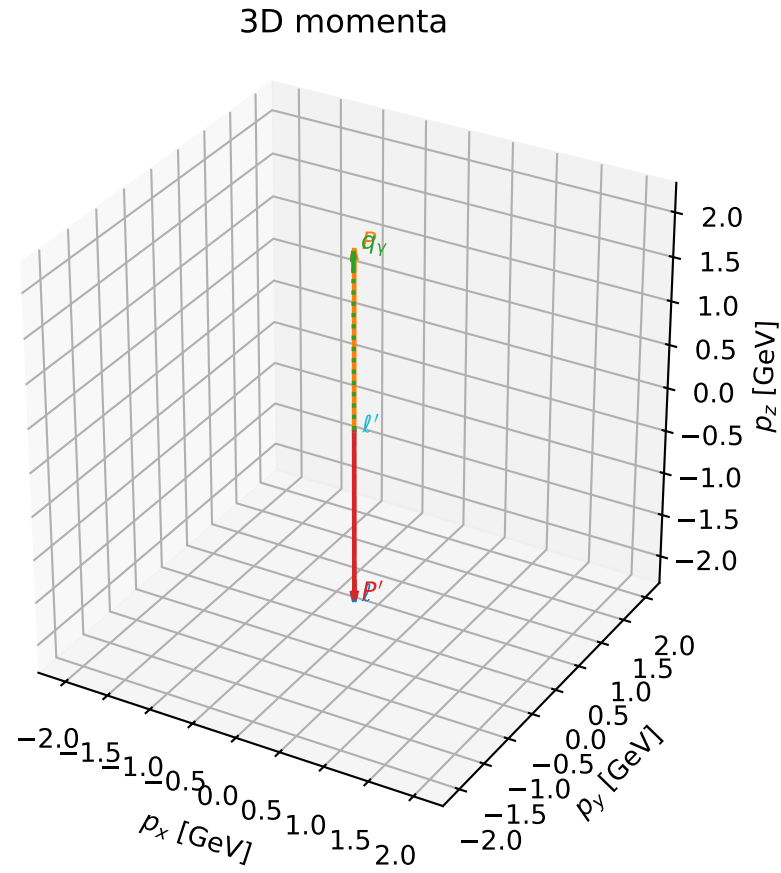
selected phase-space point:
 $\text{sqrt_s} = 4.347082$
 $\text{theta_p_out} = 3.135333$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 2.069544$
 $\text{phi_p_out} = 3.144387$
 $\text{phi_gamma_out} = 1.401538$
 $\text{alpha_e} = 2.356093$
 $\text{alpha_p} = 0.777184$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

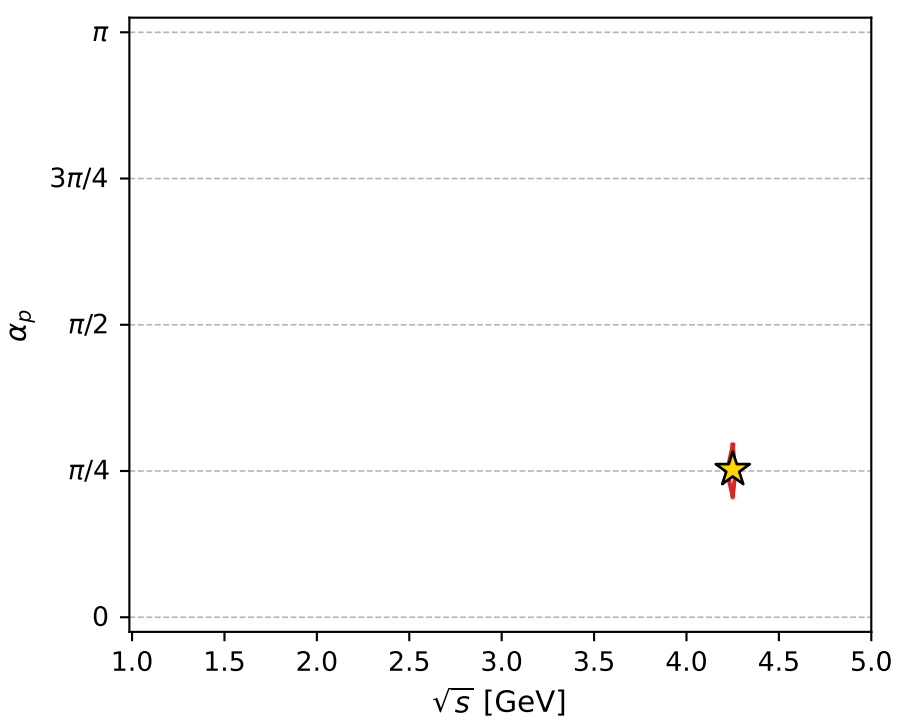
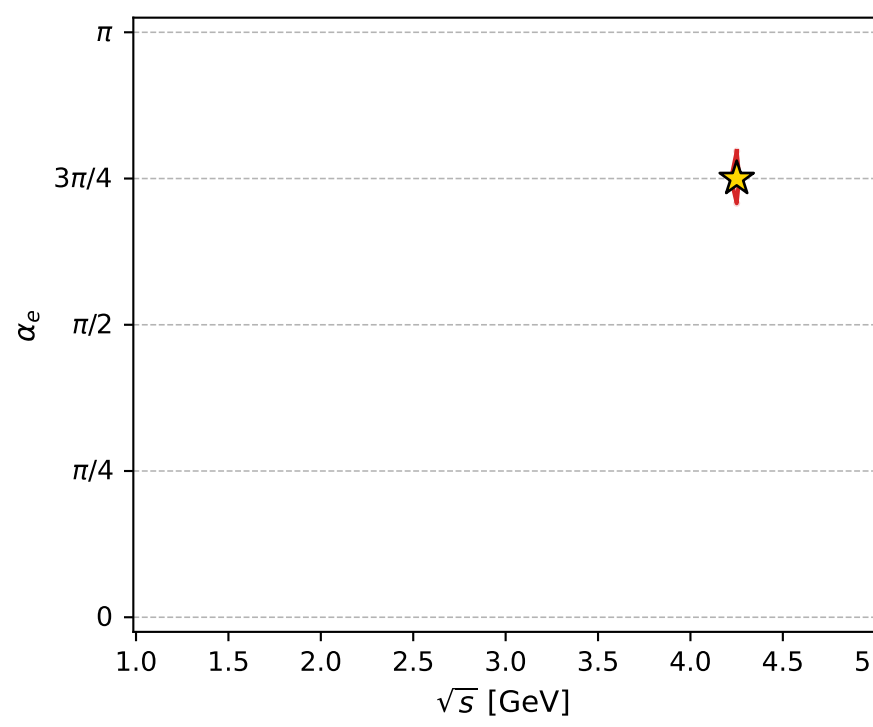
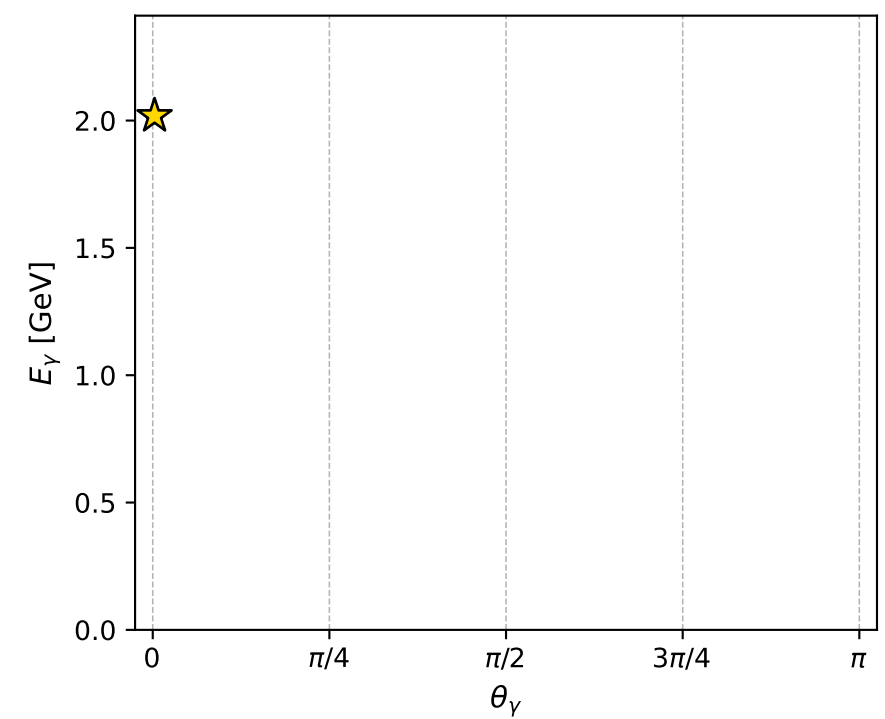
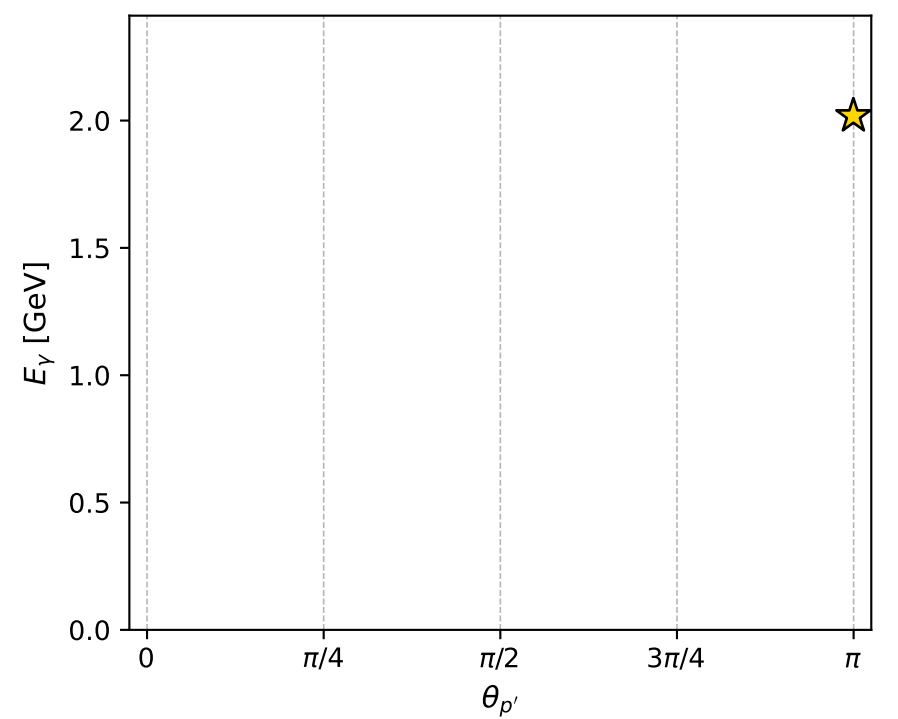
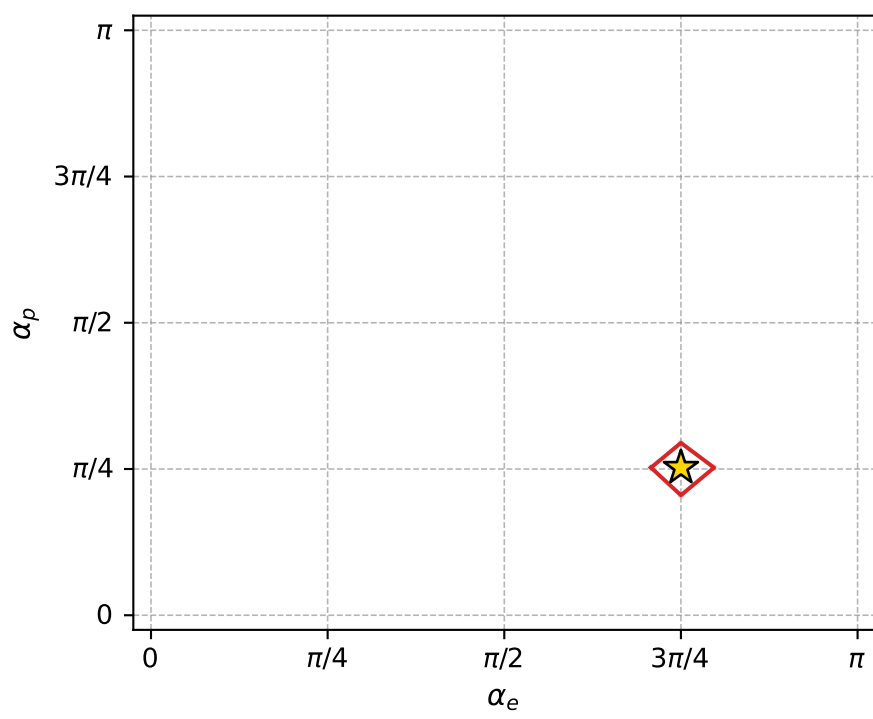
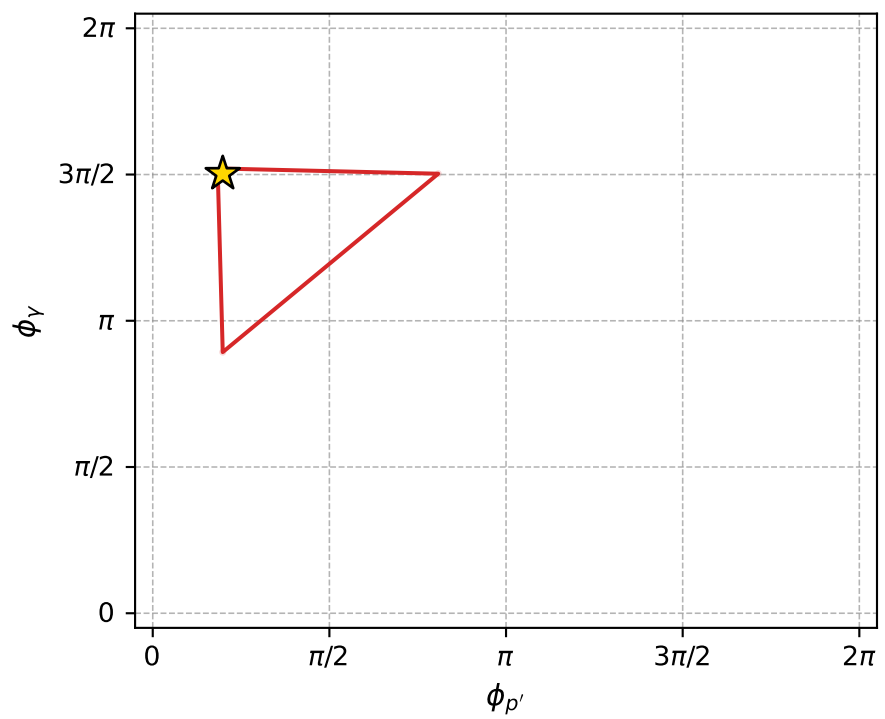
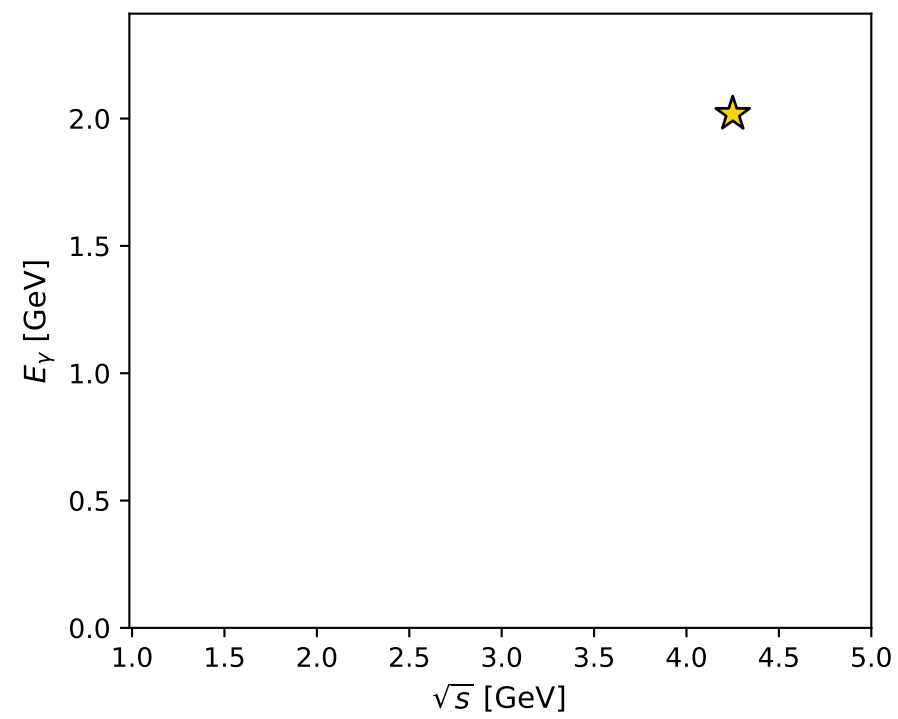
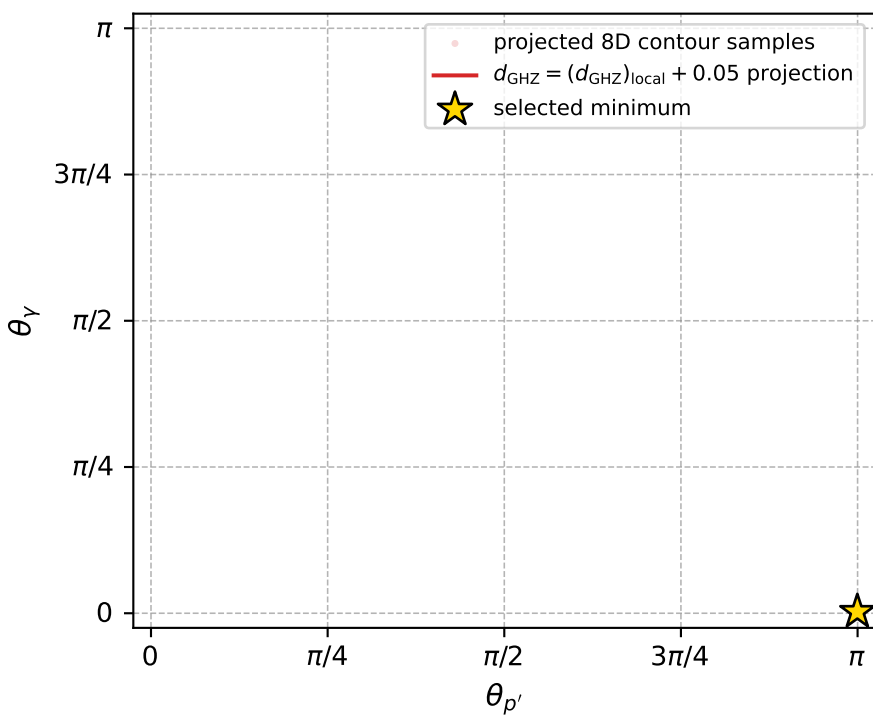
dghz_mixing_angles_polarization_03_local_minimum_374 $d_{\{\text{rm GHZ}\}}=0.0335652$
 region: polarization_cluster_03_local_minimum_374
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=2.356188$ rad
 incoming lepton: $-0.707102|+> +0.707111|->$
 proton mixing angle: $\alpha_p=0.79277071$ rad
 incoming proton: $0.701874|+> +0.712301|->$

$s=18.0658$, $\sqrt{s}=4.25039$, $|\vec{P}|=2.02169$, $|\vec{P}'|=1.98608$
 $\theta_{p'}=3.14052$, $\theta_Y=0.00808957$, $\phi_{p'}=0.622432$, $\phi_Y=4.72104$
 $E_Y=2.01834$, $\theta_{\ell'}=2.70049$, $\phi_{\ell'}=1.69454$
 $Q^2=0.0137918$, $x_B=0.000816221$, $t=-16.0612$, $W^2=17.7632$, $y=0.983193$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0217$
 P $E= 2.2287$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0217$
 ℓ' $E= 0.0356$ $p_x= -0.0019$ $p_y= 0.0151$ $p_z= -0.0322$
 P' $E= 2.1964$ $p_x= 0.0017$ $p_y= 0.0012$ $p_z= -1.9861$
 q_Y $E= 2.0183$ $p_x= 0.0001$ $p_y= -0.0163$ $p_z= 2.0183$



electron: polarization cluster P3, local minimum 374; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.033565205$
 $d_{\text{GHZ}} \text{ contour} = 0.083565205$
 above global minimum = 0.033565183
 8D contour samples = 208

selected phase-space point:
 $\text{sqrt_s} = 4.250389$
 $\text{theta_p_out} = 3.140518$
 $\text{theta_gamma_out} = 0.008089573$
 $\text{q0out} = 2.01834$
 $\text{phi_p_out} = 0.6224318$
 $\text{phi_gamma_out} = 4.72104$
 $\text{alpha_e} = 2.356188$
 $\text{alpha_p} = 0.7927707$